Thermodynamic Computing

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Preface

Digital microelectronics is built on the suppression of thermal noise. In a conventional processor (CPU, GPU, or digital tensor accelerator), information is represented by binary voltage levels ($V_{\text{low}} = 0$, $V_{\text{high}} = V_{\text{DD}}$) separated by an electrostatic energy barrier much larger than the thermal energy scale. At room temperature ($T = 300\,\text{K}$), the thermal energy of a classical degree of freedom is

$$k_{\text{B}}T \;\approx\; 4.142 \times 10^{-21}\,\text{J} \;\approx\; 25.85\,\text{meV},$$

and the corresponding thermal voltage is

$$V_T \;\equiv\; \frac{k_{\text{B}}T}{q_e} \;\approx\; 25.85\,\text{mV},$$

where $q_e \approx 1.602 \times 10^{-19}\,\text{C}$ is the elementary charge. To keep the dynamic bit-error rate below 10^{-18} per gate operation against thermal fluctuations, digital CMOS logic operates at supply voltages $V_{\text{DD}} \sim 0.7\,\text{V} \approx 27\,V_T$. Charging and discharging a gate capacitance $C_g \sim 1\,\text{fF}$ across this voltage swing dissipates

$$E_{\text{switch}} \;=\; \frac{1}{2}C_g V_{\text{DD}}^2 \;\sim\; 2.5 \times 10^{-16}\,\text{J} \;\approx\; 6 \times 10^4\,k_{\text{B}}T,$$

nearly five orders of magnitude above the Landauer limit $k_{\text{B}}T \ln 2 \approx 0.693\,k_{\text{B}}T$ for erasing a single bit. When wire capacitances, clock distribution, instruction decoding, and memory accesses are included, a single 16-bit floating-point multiply-accumulate (MAC) operation on a modern GPU dissipates

$$E_{\text{MAC}} \;\sim\; 0.5\text{–}5\,\text{pJ} \;\approx\; 10^8\text{–}10^9\,k_{\text{B}}T.$$

For deterministic arithmetic, this energy cost buys reliability. For probabilistic workloads, such as sampling from generative models, Bayesian posterior inference, and Monte Carlo simulation, it creates a severe mismatch between algorithm and hardware. A digital processor first spends $\sim 10^9\,k_{\text{B}}T$ per operation to suppress physical noise in its transistors, and then executes millions of deterministic operations in pseudo-random number generators (PRNGs) and neural network layers to simulate random variables and sample from probability distributions $p(\mathbf{x})$.

Thermodynamic Computing. Thermodynamic computing, as developed by Extropic, uses the intrinsic thermal fluctuations of electronic devices directly as the source of randomness and state evolution. By biasing standard CMOS transistors in the subthreshold (weak-inversion) regime ($V_{\text{GS}} < V_{\text{th}}$), where channel current is governed by thermal diffusion over an electrostatic barrier at nanoampere scales, or by operating cryogenic superconducting Josephson junctions near their thermal escape threshold, one builds coupled analog and hybrid circuits whose state trajectories $\mathbf{x}(t)$ obey programmable Langevin, Fokker–Planck, or master equations.

In such hardware, sampling does not require evaluating probability densities in floating-point logic. Instead, the circuit state relaxes under thermal noise toward a stationary distribution $p_\theta(\mathbf{x})$,

or steps through a sequence of conditional distributions. Two published designs illustrate the idea: Non-Equilibrium Analog Thermodynamic Random Networks (NEAT-RNs), which sample Gaussian distributions with subthreshold CMOS circuits [38], and the Denoising Thermodynamic Computer Architecture (DTCA), which runs Denoising Thermodynamic Models (DTMs) on arrays of probabilistic bits [56]. From circuit-level simulations, Jelinčič et al. [56] estimate that the DTCA could reach roughly four orders of magnitude ($\sim 10^4\times$) lower energy per sample than a GPU on small image-generation benchmarks. This book explains where an estimate of that kind comes from and which physical effects limit it.

Scope and Organization. This book develops the physics required to understand thermodynamic computing hardware and algorithms at the level of a first-year graduate course. The material is organized into five parts:

1. **Part I (Chapters 1–3):** Equilibrium statistical mechanics and energy-based models (EBMs): the canonical ensemble and variational free energy, Markov random fields and the Hammersley–Clifford theorem, chromatic block Gibbs sampling, and spin glasses (frustration, the Thouless–Anderson–Palmer equations, and replica symmetry breaking in the Sherrington–Kirkpatrick model). The part ends with the trade-off between expressivity and mixing: sharp, multimodal distributions need high energy barriers, and high barriers make a single monolithic EBM mix exponentially slowly.

2. **Part II (Chapters 4–7):** Continuous-time stochastic processes. We cover Itô and Stratonovich calculus and the Wong–Zakai theorem for circuits with multiplicative noise, derive the fluctuation–dissipation theorem from the Caldeira–Leggett model, and develop the spectral theory of the Fokker–Planck equation (Witten's transformation to a Schrödinger operator and Bakry–Émery log-Sobolev inequalities). The part ends with Kramers' rate for escape over an energy barrier.

3. **Part III (Chapters 8–10):** Stochastic thermodynamics of single trajectories: Sekimoto's first law, Onsager–Machlup path integrals, housekeeping heat, the Crooks and Jarzynski fluctuation theorems, Landauer bounds out of equilibrium, thermodynamic uncertainty relations, and finite-time thermodynamics through the L^2-Wasserstein distance $\mathcal{W}_2(\rho_0, \rho_\tau)$ of optimal transport.

4. **Part IV (Chapters 11–14):** Device and circuit physics. We derive Johnson–Nyquist and shot noise, MOS electrostatics, and the Enz–Krummenacher–Vittoz (EKV) model of subthreshold conduction. We then analyze NEAT-RNs in subthreshold CMOS [38], where detailed balance is broken and the stationary covariance follows from a Lyapunov equation, and finally CMOS probabilistic bits (p-bits) and cryogenic Josephson Quantum Flux Parametrons (QFPs).

5. **Part V (Chapters 15–18):** Score-based diffusion, Anderson's reverse-time theorem, and Denoising Thermodynamic Models (DTMs) [56]. We show that a quadratic forward coupling $\frac{\alpha_t}{2}\left\|\mathbf{x}_{t-1} - c_t\mathbf{x}_t\right\|^2$ makes each conditional denoising step strongly log-concave once α_t exceeds the negative curvature of the learned energy. This removes the Kramers barriers, so each step mixes in a time that does not grow with barrier height. The final chapters treat Total Correlation regularization for sparse two-dimensional hardware layouts, energy budgets for GPUs and subthreshold CMOS, Equilibrium Propagation, and hardware-in-the-loop calibration against Pelgrom threshold-voltage mismatch.

How the Chapters Are Built. The book is self-contained and has no exercises. Instead, each chapter contains at least one long worked computation that solves a concrete system step by step, and ends with a chapter summary that lists what was proved, what was assumed, what is taken up in later chapters, and what remains open. Appendix A collects notation, physical constants, and standing assumptions. Appendices B–D collect standard derivations and proofs that would interrupt the main line: Gaussian identities and Lyapunov equations (Appendix B), stochastic-calculus proofs (Appendix C), and device derivations (Appendix D).

Two model systems run through the entire text. We refer to them throughout by the names below.

- **The RC sampler.** A vector of capacitor node voltages $\mathbf{v}(t) \in \mathbb{R}^N$ coupled by a conductance or transconductance matrix $\mathbf{G}$ and driven by thermal and shot noise:

$$\mathbf{C}\,\mathrm{d}\mathbf{v}_t \;=\; -\mathbf{G}(\mathbf{v}_t - \mathbf{b})\;\mathrm{d}t + \mathbf{B}\;\mathrm{d}\mathbf{W}_t.$$

  Its stationary distribution is Gaussian. We follow it from its equilibrium partition function (Chapter 1), through its Green's function and Kubo susceptibility (Chapter 5), its Hermite spectrum (Chapter 6), its housekeeping heat out of equilibrium (Chapter 8), and its optimal transport schedule (Chapter 10), to its realization in subthreshold CMOS, where asymmetric transconductances ($\mathbf{G} \neq \mathbf{G}^\top$) set the stationary covariance through the Lyapunov equation $\mathbf{A\Sigma} + \mathbf{\Sigma A}^\top + 2\mathbf{D} = \mathbf{0}$ (Chapter 13).

- **The double-well sampler.** The quartic potential $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$ and its binary limit, a spin or probabilistic bit (p-bit) $s \in \{-1, +1\}$, which applies when the barrier $\Delta E = \frac{b^2}{4a}$ is much larger than $k_\mathrm{B}T$. We follow it from Gaussian–Bernoulli marginalization (Chapter 1), chromatic Gibbs sampling (Chapter 2), and spin-glass frustration (Chapter 3), through Kramers escape rates (Chapters 6–7) and Landauer bit erasure (Chapter 9), to CMOS p-bit and Josephson circuits (Chapter 14), reverse-diffusion speciation (Chapter 15), and barrier removal in a Denoising Thermodynamic Model (Chapter 16).

Part I

Equilibrium Statistical Mechanics, Inference, and Disordered Landscapes

Chapter 1

The Canonical Ensemble, Variational Free Energy, and Energy-Based Models

Probabilistic machine learning and classical statistical mechanics rest on the same mathematical object: the canonical ensemble of Boltzmann and Gibbs. An Energy-Based Model (EBM) represents a probability distribution over images, molecular conformations, or latent features by an energy function $E_\theta(x)$ that is exponentiated, weighted by an inverse temperature β, and normalized by a partition function $Z(\theta, \beta)$. This is exactly the equilibrium distribution of a physical system in contact with a heat bath at temperature T. A digital processor does not exploit this correspondence. To draw one sample from an EBM on a graphics processing unit (GPU), transistors switch deterministically across a supply voltage of nearly one volt, with each switching event costing of order 10^4–$10^5\, k_\mathrm{B}T$ precisely so that thermal noise cannot flip a bit. The thermal fluctuations that the sampler needs are then simulated in software with pseudorandom number generators and floating-point arithmetic, at a cost of 10^8 to $10^9\, k_\mathrm{B}T$ per elementary probabilistic update (see the Preface for these estimates).

Thermodynamic computing takes the opposite route. Instead of suppressing thermal fluctuations and then synthesizing pseudorandom noise, one builds electronic or superconducting circuits whose physical equilibrium distribution *is* the target distribution $p_\theta(x)$. Sampling is then carried out by physical thermalization. Published examples include the Non-Equilibrium Analog Thermodynamic Random Networks (NEAT-RNs) of Freitas et al. [38] and the Denoising Thermodynamic Computer Architecture (DTCA) of Jelinčič et al. [56].

This chapter develops the equilibrium statistical mechanics that the rest of the book builds on. We define the canonical ensemble and its free energy (Section 1.1), show that equilibrium minimizes a variational free energy (Section 1.2), and derive the fluctuation–response identities that link derivatives of $\ln Z$ to thermal covariances (Section 1.3). These identities give the maximum-likelihood learning rule for EBMs, whose negative phase requires equilibrium samples and is the step that is expensive on digital hardware (Section 1.4). Finally, we show how integrating out latent variables turns simple quadratic and bilinear couplings into multimodal effective potentials (Section 1.5). The worked computation at the end of the chapter combines the book's two model systems, the RC sampler and the double-well sampler, in a single Gaussian–Bernoulli EBM.

1.1 Microstates, Hamiltonians, and the Canonical Ensemble

Every statistical mechanical theory and every probabilistic generative model begins by specifying two ingredients: the space of microscopic configurations accessible to the system, and an energy functional that assigns a real scalar cost to each configuration. In physical hardware, the choice of configuration space is dictated by the physical degrees of freedom we choose to resolve at the mesoscopic scale.

1.1.1 Configuration Spaces and Reference Measures

Consider a physical circuit or statistical system composed of N coupled degrees of freedom. Depending on the operating regime of the underlying solid-state devices (which we will derive from semiconductor and superconducting device physics in Part IV), the microscopic state x resides in one of three canonical state spaces:

(i) **Discrete Spin Space (Ising / p-bit Configuration Space):** When each elementary unit is a strongly bistable circuit, such as a cross-coupled subthreshold CMOS latch or a Josephson Quantum Flux Parametron biased deep in its bistable regime (both described in Chapter 14), its continuous internal voltage or magnetic flux stays almost always near one of two symmetric potential wells. Coarse-graining each unit to its binary well index yields the discrete hypercube of Ising spin configurations,

$$\Omega = \{-1,+1\}^N := \left\{ \mathbf{s} = (s_1, s_2, \ldots, s_N)^\top : s_i \in \{-1,+1\} \text{ for all } i = 1, \ldots, N \right\}. \tag{1.1}$$

The cardinality of Ω is $|\Omega| = 2^N$. The natural reference measure μ on Ω is the counting measure, so that integrals with respect to μ reduce to finite sums over all 2^N corners of the hypercube:

$$\int_\Omega f(\mathbf{s})\, \mu(\mathrm{d}\mathbf{s}) := \sum_{\mathbf{s}\in\{-1,+1\}^N} f(\mathbf{s}) = \sum_{s_1\in\{-1,+1\}} \sum_{s_2\in\{-1,+1\}} \cdots \sum_{s_N\in\{-1,+1\}} f(s_1, s_2, \ldots, s_N). \tag{1.2}$$

More generally, if each discrete unit possesses $q \geq 2$ metastable states (a Potts spin or q-state clock variable), then $\Omega = \{1, 2, \ldots, q\}^N$ with cardinality q^N.

(ii) **Continuous Euclidean Configuration Space (Analog RC / Voltage Space):** When the elementary units are continuous node voltages $\mathbf{v} = (v_1, v_2, \ldots, v_N)^\top$ on an array of mesoscopic capacitors $C_1, \ldots, C_N$, the state space is the N-dimensional Euclidean space

$$\Omega = \mathbb{R}^N. \tag{1.3}$$

To integrate functions over $\mathbb{R}^N$ while maintaining strict dimensional homogeneity inside transcendental functions such as logarithms and exponentials, we equip $\mathbb{R}^N$ with the dimensionless Lebesgue volume element

$$\mu(\mathrm{d}\mathbf{v}) := \frac{\mathrm{d}^N v}{V_0^N} = \prod_{i=1}^N \frac{\mathrm{d}v_i}{V_0}, \tag{1.4}$$

where $V_0 > 0$ is a fixed reference voltage scale (for example, the thermal voltage $V_T = k_\mathrm{B}T/q_e \approx 25.85\,\mathrm{mV}$ at $T = 300\,\mathrm{K}$, or the elementary single-electron voltage scale q_e/C_node). When working in dimensionless state coordinates $\mathbf{x} = \mathbf{v}/V_0 \in \mathbb{R}^N$, we simply write $\mu(\mathrm{d}\mathbf{x}) = \mathrm{d}^N x$.

(iii) **Hybrid Discrete–Continuous Product Space:** In hybrid samplers and latent-variable Denoising Thermodynamic Models (Chapters 14 and 16), continuous analog nodes $\mathbf{v} \in \mathbb{R}^N$ interact directly with M discrete probabilistic bits (p-bits) $\mathbf{s} \in \{-1, +1\}^M$. The joint microstate is the pair $x = (\mathbf{v}, \mathbf{s})$ in the product space

$$\Omega = \mathbb{R}^N \times \{-1, +1\}^M, \tag{1.5}$$

equipped with the product reference measure $\mu(\mathrm{d}\mathbf{v}, \mathrm{d}\mathbf{s}) = (\mathrm{d}^N v / V_0^N) \otimes \mu_{\text{count}}(\mathrm{d}\mathbf{s})$, so that for any integrable observable $f(\mathbf{v}, \mathbf{s})$,

$$\int_\Omega f(x)\, \mu(\mathrm{d}x) := \sum_{\mathbf{s} \in \{-1,+1\}^M} \int_{\mathbb{R}^N} f(\mathbf{v}, \mathbf{s})\, \frac{\mathrm{d}^N v}{V_0^N}. \tag{1.6}$$

We state definitions and theorems for a general measure space $(\Omega, \mathcal{B}(\Omega), \mu)$, so that each result applies to spin systems, analog circuits, and hybrid architectures without repeating the proof.

1.1.2 Parameterized Energy Functions and the Boltzmann–Gibbs Measure

Suppose our system is governed by a vector of tunable control parameters $\theta \in \Theta \subseteq \mathbb{R}^P$. In a physical circuit, θ represents externally programmable bias voltages, gate transconductances, or capacitive coupling weights set by digital-to-analog converters (DACs); in a machine learning model, θ represents the trainable synaptic weights and biases of a neural or graphical energy function.

Definition 1.1 (Parameterized Hamiltonian, Partition Function, and Canonical Ensemble)**.** Let (Ω, μ) be a configuration space equipped with a σ-finite reference measure μ, and let $\Theta \subseteq \mathbb{R}^P$ be an open parameter domain. A *parameterized energy function* (or *configurational Hamiltonian*) is a measurable function

$$E_\theta : \Omega \to \mathbb{R}, \qquad x \mapsto E_\theta(x), \tag{1.7}$$

carrying physical dimensions of energy (Joules, or electron-volts). Fix an absolute bath temperature $T > 0$, with Boltzmann constant $k_{\mathrm{B}} = 1.380649 \times 10^{-23}\, \mathrm{J} \cdot \mathrm{K}^{-1}$ and inverse thermal energy

$$\beta := \frac{1}{k_{\mathrm{B}} T} > 0. \tag{1.8}$$

We say that E_θ is *thermodynamically admissible* (or *confining*) at inverse temperature β if the unnormalized Boltzmann factor $\exp(-\beta E_\theta(x))$ is μ-integrable over Ω:

$$Z(\theta, \beta) := \int_\Omega \exp(-\beta E_\theta(x))\, \mu(\mathrm{d}x) < \infty. \tag{1.9}$$

The strictly positive normalization constant $Z(\theta, \beta) \in (0, \infty)$ is called the *canonical partition function*. The associated probability density (or probability mass function) with respect to μ,

$$p_\theta(x) = p_{\mathrm{eq}}(x; \theta, \beta) := \frac{1}{Z(\theta, \beta)} \exp(-\beta E_\theta(x)), \qquad x \in \Omega, \tag{1.10}$$

is the *Boltzmann–Gibbs canonical distribution*. For any μ-integrable observable $A : \Omega \to \mathbb{R}$, its canonical ensemble average is written interchangeably as

$$\langle A \rangle_{p_\theta} \equiv \mathbb{E}[A(X)]_{p_\theta} := \int_\Omega A(x)\, p_\theta(x)\, \mu(\mathrm{d}x) = \frac{1}{Z(\theta, \beta)} \int_\Omega A(x) \exp(-\beta E_\theta(x))\, \mu(\mathrm{d}x). \tag{1.11}$$

When Ω is a finite discrete space such as $\{-1,+1\}^N$, any finite-valued function $E_\theta : \{-1,+1\}^N \to \mathbb{R}$ is automatically confining for all $\beta \in (0,\infty)$, because $Z(\theta,\beta)$ is a finite sum of 2^N strictly positive terms. When $\Omega = \mathbb{R}^N$, however, integrability requires $E_\theta(\mathbf{x})$ to grow sufficiently rapidly as $\|\mathbf{x}\| \to \infty$. Specifically, if there exist constants $R > 0$, $c > 0$, and $\alpha > 0$ such that $E_\theta(\mathbf{x}) \geq c\,\|\mathbf{x}\|^\alpha$ for all $\|\mathbf{x}\| \geq R$, then $\exp(-\beta E_\theta(\mathbf{x}))$ decays faster than any polynomial at infinity and $Z(\theta,\beta) < \infty$ for all $\beta > 0$. In physical electronic circuits, this confinement condition is guaranteed by passive capacitive charging energies $\frac{1}{2}\mathbf{v}^\top \mathbf{C}\mathbf{v}$ (Chapters 5 and 13) or by the finite power-supply rails $[-V_{\mathrm{DD}}, +V_{\mathrm{DD}}]$ of the silicon die.

Notation Watch: The Physics–ML–Circuit Rosetta Stone for Energy and Temperature

Statistical physicists, machine learning researchers, and analog circuit designers use slightly different conventions when writing Eq. (1.10). Keeping the translation in mind prevents sign and scaling errors:

1. **Dimensionless vs. Physical Temperature:** In the machine learning literature on Energy-Based Models [1, 50], the inverse temperature β is routinely absorbed into the definition of the parameters θ, effectively setting $\beta = 1$ and treating $E_\theta(x)$ as a dimensionless scalar score. In thermodynamic computing, however, temperature T is a physical control knob (e.g., $T = 300\,\mathrm{K}$ in room-temperature CMOS or $T = 4.2\,\mathrm{K}$ in cryogenic Josephson circuits) whose thermal noise floor $k_\mathrm{B}T$ physically drives state transitions. Throughout this book, we retain $\beta = (k_\mathrm{B}T)^{-1}$ explicitly whenever physical energetics, fluctuation-dissipation relations, or noise amplitudes are at stake, and state clearly when dimensionless normalized units ($\beta = 1$) are used for algebraic convenience.

2. **Sign of the Energy Function:** Both statistical physics and modern EBM theory assign *high* probability to *low* energy states via $p_\theta(x) \propto e^{-\beta E_\theta(x)}$. However, in classical statistics and exponential-family theory, distributions are often written with a plus sign in the exponent, $p(x) = \exp(\boldsymbol{\theta}^\top \boldsymbol{\phi}(x) - A(\boldsymbol{\theta}))$, where $A(\boldsymbol{\theta}) = \ln Z(\boldsymbol{\theta})$ is the log-partition function. Thus, the natural exponential-family parameter $\boldsymbol{\theta}_{\mathrm{stat}}$ corresponds to $-\beta\boldsymbol{\theta}_{\mathrm{phys}}$.

3. **Circuit Thermal Voltage V_T:** In semiconductor device physics (Chapter 12), energies of charge carriers with elementary charge $q_e = 1.602176634 \times 10^{-19}\,\mathrm{C}$ moving across electrostatic potentials V are measured in units of $q_e V$. The thermal energy $k_\mathrm{B}T$ is therefore expressed as a voltage via the *thermal voltage* $V_T := k_\mathrm{B}T/q_e$, so that Boltzmann factors take the form $\exp(-V/V_T) = \exp(-q_e V/k_\mathrm{B}T)$.

A fuller dictionary is given in Section A.3.

An immediate structural property of the canonical distribution is that absolute energy scales are unobservable within equilibrium sampling; only energy *differences* between microstates determine relative probabilities.

Proposition 1.2 (Gauge Invariance Under Global Energy Shifts)**.** *Let $E_\theta : \Omega \to \mathbb{R}$ be a thermodynamically admissible energy function at inverse temperature $\beta > 0$, and let $c(\theta) \in \mathbb{R}$ be an arbitrary configuration-independent scalar that may depend on the parameters θ. Define the shifted energy function*

$$\tilde{E}_\theta(x) := E_\theta(x) + c(\theta), \qquad \forall\, x \in \Omega. \tag{1.12}$$

Then the partition function transforms multiplicatively as $\tilde{Z}(\theta,\beta) = e^{-\beta c(\theta)} Z(\theta,\beta)$, and the canonical

distribution is strictly invariant:

$$\tilde{p}_\theta(x) = p_\theta(x) \qquad \forall\, x \in \Omega. \tag{1.13}$$

Conversely, if two continuous energy functions $E_1, E_2 : \Omega \to \mathbb{R}$ induce the exact same positive Boltzmann distribution $p_1(x) = p_2(x)$ for all $x \in \Omega$, then $E_1(x) - E_2(x)$ is constant on Ω.

Proof. Substituting $\tilde{E}_\theta(x) = E_\theta(x) + c(\theta)$ into the definition of the partition function (1.9) and factoring the x-independent constant $\exp(-\beta c(\theta))$ outside the μ-integral yields

$$\begin{aligned}\tilde{Z}(\theta,\beta) &= \int_\Omega \exp\big(-\beta \tilde{E}_\theta(x)\big)\, \mu(\mathrm{d}x) \\ &= \int_\Omega \exp\big(-\beta E_\theta(x) - \beta c(\theta)\big)\, \mu(\mathrm{d}x) \\ &= \exp\big(-\beta c(\theta)\big) \int_\Omega \exp\big(-\beta E_\theta(x)\big)\, \mu(\mathrm{d}x) = \exp\big(-\beta c(\theta)\big) Z(\theta,\beta).\end{aligned} \tag{1.14}$$

Substituting Eq. (1.14) into the Boltzmann-Gibbs density (1.10), the factors of $\exp(-\beta c(\theta))$ in the numerator and denominator cancel identically:

$$\tilde{p}_\theta(x) = \frac{\exp\big(-\beta E_\theta(x) - \beta c(\theta)\big)}{\exp\big(-\beta c(\theta)\big) Z(\theta,\beta)} = \frac{\exp\big(-\beta E_\theta(x)\big)}{Z(\theta,\beta)} = p_\theta(x). \tag{1.15}$$

For the converse, suppose $p_1(x) = p_2(x)$ for all $x \in \Omega$. Taking the natural logarithm of both sides of $Z_1^{-1} e^{-\beta E_1(x)} = Z_2^{-1} e^{-\beta E_2(x)}$ gives

$$-\beta E_1(x) - \ln Z_1(\beta) = -\beta E_2(x) - \ln Z_2(\beta) \implies E_1(x) - E_2(x) = -\frac{1}{\beta} \ln \frac{Z_1(\beta)}{Z_2(\beta)}, \tag{1.16}$$

where the right-hand side is independent of $x \in \Omega$. □

In circuit hardware, Proposition 1.2 corresponds to the physical fact that shifting the common-mode ground reference potential of an isolated energy function does not alter the relative probabilities of microstates. This gauge freedom reappears in Chapter 18, where it identifies the parameter directions that hardware calibration cannot and need not correct.

1.1.3 Helmholtz Free Energy, Internal Energy, and Canonical Entropy

While the partition function $Z(\theta,\beta)$ is dimensionless and multiplies over independent subsystems, thermodynamic potentials carry units of energy and add over independent subsystems. Taking $k_\mathrm{B}T$ times a logarithm converts one into the other.

Definition 1.3 (Helmholtz Free Energy, Internal Energy, and Gibbs–Shannon Entropy)**.** Let $p_\theta(x) = Z(\theta,\beta)^{-1} e^{-\beta E_\theta(x)}$ be a canonical ensemble on (Ω, μ). We define three thermodynamic functionals:

(i) The *Helmholtz free energy*:

$$F(\theta,\beta) := -k_\mathrm{B}T \ln Z(\theta,\beta) = -\frac{1}{\beta} \ln \int_\Omega \exp\big(-\beta E_\theta(x)\big)\, \mu(\mathrm{d}x). \tag{1.17}$$

(ii) The *internal (mean) energy*:

$$U(\theta,\beta) := \mathbb{E}[E_\theta(X)]_{p_\theta} = \int_\Omega E_\theta(x)\, p_\theta(x)\, \mu(\mathrm{d}x). \tag{1.18}$$

(iii) The *Gibbs–Shannon entropy* (for any probability density q with respect to μ):

$$S(q) := -k_{\mathrm{B}} \int_{\Omega} q(x) \ln q(x)\, \mu(\mathrm{d}x) = -k_{\mathrm{B}} \mathbb{E}[\ln q(X)]_q\,, \tag{1.19}$$

with the canonical equilibrium entropy denoted by $S(\theta, \beta) := S(p_\theta)$.

Using the definition of $F(\theta, \beta)$, we can rewrite the canonical Boltzmann-Gibbs density (1.10) in the compact, self-normalized exponential form

$$p_\theta(x) = \exp\Big(\beta\big[F(\theta, \beta) - E_\theta(x)\big]\Big). \tag{1.20}$$

Equation (1.20) gives $F(\theta, \beta)$ a direct meaning: a microstate x has probability density greater than unity (relative to μ) if and only if its energy $E_\theta(x)$ lies below the Helmholtz free energy $F(\theta, \beta)$, and is exponentially suppressed whenever $E_\theta(x) - F(\theta, \beta) \gg k_{\mathrm{B}}T$.

The following identities link Z, F, U, S, and the thermal energy fluctuations of the canonical ensemble.

Theorem 1.4 (Canonical Thermodynamic Identities and Energy Fluctuations)**.** *Assume that $E_\theta : \Omega \to \mathbb{R}$ is thermodynamically admissible on an open interval of inverse temperatures $\beta \in (\beta_{\min}, \beta_{\max}) \subseteq (0, \infty)$ such that $\int_\Omega |E_\theta(x)|^2\, e^{-\beta E_\theta(x)}\, \mu(\mathrm{d}x) < \infty$. Then $\ln Z(\theta, \beta)$ and $F(\theta, \beta)$ are twice continuously differentiable with respect to β (and T), and satisfy the following exact identities:*

(i) ***Fundamental Free-Energy Decomposition:***

$$F(\theta, \beta) = U(\theta, \beta) - TS(\theta, \beta). \tag{1.21}$$

(ii) ***First β-Derivative and Internal Energy:***

$$\frac{\partial \ln Z(\theta, \beta)}{\partial \beta} = -\frac{\partial\big(\beta F(\theta, \beta)\big)}{\partial \beta} = -U(\theta, \beta). \tag{1.22}$$

(iii) ***Temperature Derivative of Free Energy and Canonical Entropy:***

$$\frac{\partial F(\theta, T)}{\partial T} = -S(\theta, T), \qquad S(\theta, \beta) = k_{\mathrm{B}}\big(\ln Z(\theta, \beta) + \beta U(\theta, \beta)\big). \tag{1.23}$$

(iv) ***Second β-Derivative, Energy Variance, and Heat Capacity:***

$$\frac{\partial^2 \ln Z(\theta, \beta)}{\partial \beta^2} = -\frac{\partial U(\theta, \beta)}{\partial \beta} = \mathrm{Var}_{p_\theta}(E_\theta(X)) = \mathbb{E}\Big[\big(E_\theta(X) - U(\theta, \beta)\big)^2\Big]_{p_\theta} \geq 0, \tag{1.24}$$

and the heat capacity at fixed parameters θ, $C_\theta(\theta, T) := \frac{\partial U(\theta, T)}{\partial T} = T\frac{\partial S(\theta, T)}{\partial T}$, is given by

$$C_\theta(\theta, T) = k_{\mathrm{B}}\beta^2\, \mathrm{Var}_{p_\theta}(E_\theta(X)) = \frac{\mathrm{Var}_{p_\theta}(E_\theta(X))}{k_{\mathrm{B}}T^2} \geq 0. \tag{1.25}$$

Proof. We prove each identity step by step.

Proof of (i): Taking the natural logarithm of the canonical density $p_\theta(x) = Z(\theta,\beta)^{-1}\exp(-\beta E_\theta(x))$ gives

$$\ln p_\theta(x) = -\beta E_\theta(x) - \ln Z(\theta,\beta). \tag{1.26}$$

Substituting Eq. (1.26) into the definition of the canonical Gibbs-Shannon entropy (1.19) and using the normalization $\int_\Omega p_\theta(x)\,\mu(\mathrm{d}x) = 1$, we obtain

$$\begin{aligned} S(\theta,\beta) &= -k_\mathrm{B}\int_\Omega p_\theta(x)\ln p_\theta(x)\,\mu(\mathrm{d}x) \\ &= -k_\mathrm{B}\int_\Omega p_\theta(x)\big(-\beta E_\theta(x) - \ln Z(\theta,\beta)\big)\,\mu(\mathrm{d}x) \\ &= k_\mathrm{B}\beta\int_\Omega E_\theta(x)\,p_\theta(x)\,\mu(\mathrm{d}x) + k_\mathrm{B}\ln Z(\theta,\beta)\int_\Omega p_\theta(x)\,\mu(\mathrm{d}x) \\ &= \frac{1}{T}U(\theta,\beta) + k_\mathrm{B}\ln Z(\theta,\beta), \end{aligned} \tag{1.27}$$

where we used $k_\mathrm{B}\beta = k_\mathrm{B}/(k_\mathrm{B}T) = 1/T$. Multiplying both sides of Eq. (1.27) by T and recalling from Eq. (1.17) that $F(\theta,\beta) = -k_\mathrm{B}T\ln Z(\theta,\beta)$, we find

$$TS(\theta,\beta) = U(\theta,\beta) + k_\mathrm{B}T\ln Z(\theta,\beta) = U(\theta,\beta) - F(\theta,\beta), \tag{1.28}$$

which rearranges immediately to $F(\theta,\beta) = U(\theta,\beta) - TS(\theta,\beta)$, establishing (i) and the second part of Eq. (1.23).

Proof of (ii): By the moment integrability hypothesis $\int_\Omega |E_\theta(x)|\,e^{-\beta E_\theta(x)}\,\mu(\mathrm{d}x) < \infty$, the dominated convergence theorem permits differentiation with respect to β inside the μ-integral defining $Z(\theta,\beta)$:

$$\frac{\partial Z(\theta,\beta)}{\partial\beta} = \int_\Omega \frac{\partial}{\partial\beta}\exp(-\beta E_\theta(x))\,\mu(\mathrm{d}x) = -\int_\Omega E_\theta(x)\exp(-\beta E_\theta(x))\,\mu(\mathrm{d}x). \tag{1.29}$$

Applying the chain rule to $\ln Z(\theta,\beta)$ and substituting Eq. (1.29), we obtain

$$\begin{aligned} \frac{\partial \ln Z(\theta,\beta)}{\partial\beta} &= \frac{1}{Z(\theta,\beta)}\frac{\partial Z(\theta,\beta)}{\partial\beta} \\ &= -\int_\Omega E_\theta(x)\frac{\exp(-\beta E_\theta(x))}{Z(\theta,\beta)}\,\mu(\mathrm{d}x) \\ &= -\int_\Omega E_\theta(x)\,p_\theta(x)\,\mu(\mathrm{d}x) = -U(\theta,\beta). \end{aligned} \tag{1.30}$$

Since $\beta F(\theta,\beta) = -\ln Z(\theta,\beta)$ by Eq. (1.17), this proves Eq. (1.22).

Proof of (iii): Expressing the Helmholtz free energy as a function of T via $\beta(T) = (k_\mathrm{B}T)^{-1}$ with $\frac{\mathrm{d}\beta}{\mathrm{d}T} = -(k_\mathrm{B}T^2)^{-1} = -k_\mathrm{B}\beta^2$, we differentiate $F(\theta,T) = -k_\mathrm{B}T\ln Z(\theta,\beta(T))$ using the product and chain rules:

$$\begin{aligned} \frac{\partial F(\theta,T)}{\partial T} &= -k_\mathrm{B}\ln Z(\theta,\beta) - k_\mathrm{B}T\frac{\partial \ln Z(\theta,\beta)}{\partial\beta}\frac{\mathrm{d}\beta}{\mathrm{d}T} \\ &= -k_\mathrm{B}\ln Z(\theta,\beta) - (k_\mathrm{B}T)\big(-U(\theta,\beta)\big)\left(-\frac{1}{k_\mathrm{B}T^2}\right) \\ &= -k_\mathrm{B}\ln Z(\theta,\beta) - \frac{U(\theta,\beta)}{T} = -S(\theta,T), \end{aligned} \tag{1.31}$$

where the final equality uses Eq. (1.27).

Proof of (iv): Differentiating $\frac{\partial \ln Z}{\partial \beta} = -U(\theta,\beta)$ a second time with respect to β using the quotient rule on $Z(\theta,\beta)^{-1}\int_\Omega E_\theta(x)e^{-\beta E_\theta(x)}\,\mu(\mathrm{d}x)$ gives

$$\begin{aligned}\frac{\partial^2 \ln Z(\theta,\beta)}{\partial\beta^2} &= -\frac{\partial}{\partial\beta}\left[\frac{1}{Z(\theta,\beta)}\int_\Omega E_\theta(x)\exp(-\beta E_\theta(x))\,\mu(\mathrm{d}x)\right]\\ &= -\frac{1}{Z(\theta,\beta)}\int_\Omega E_\theta(x)(-E_\theta(x))\exp(-\beta E_\theta(x))\,\mu(\mathrm{d}x)\\ &\quad + \frac{1}{Z(\theta,\beta)^2}\frac{\partial Z(\theta,\beta)}{\partial\beta}\int_\Omega E_\theta(x)\exp(-\beta E_\theta(x))\,\mu(\mathrm{d}x)\\ &= \int_\Omega E_\theta(x)^2\,p_\theta(x)\,\mu(\mathrm{d}x) - \left(\int_\Omega E_\theta(x)\,p_\theta(x)\,\mu(\mathrm{d}x)\right)^2\\ &= \mathbb{E}\Big[E_\theta(X)^2\Big]_{p_\theta} - \left(\mathbb{E}[E_\theta(X)]_{p_\theta}\right)^2 = \mathrm{Var}_{p_\theta}\left(E_\theta(X)\right) \geq 0. \end{aligned} \tag{1.32}$$

Finally, applying the chain rule $\frac{\partial}{\partial T} = \frac{\mathrm{d}\beta}{\mathrm{d}T}\frac{\partial}{\partial\beta} = -(k_\mathrm{B}T^2)^{-1}\frac{\partial}{\partial\beta}$ to $U(\theta,T)$ yields the heat capacity

$$C_\theta(\theta,T) = \frac{\partial U(\theta,T)}{\partial T} = -\frac{1}{k_\mathrm{B}T^2}\frac{\partial U(\theta,\beta)}{\partial\beta} = \frac{1}{k_\mathrm{B}T^2}\frac{\partial^2 \ln Z(\theta,\beta)}{\partial\beta^2} = k_\mathrm{B}\beta^2\,\mathrm{Var}_{p_\theta}\left(E_\theta(X)\right), \tag{1.33}$$

completing the proof. □

Equation (1.21), $F = U - TS$, expresses the competition between energy and entropy that runs through the whole book. At low temperature ($T \to 0$, $\beta \to \infty$), the entropic term $-TS$ vanishes and the canonical distribution concentrates on the global minima (ground states) of $E_\theta(x)$. At high temperature ($T \to \infty$, $\beta \to 0$), the entropic term dominates and, on a finite state space, spreads probability uniformly over Ω. At any finite temperature, the equilibrium distribution balances low energy against high entropy.

We now evaluate these quantities for the two model systems that recur throughout the book: the RC sampler, a network of capacitor node voltages with a quadratic energy, and the double-well sampler, whose binary limit is a single probabilistic bit.

Example 1.5 (The RC Sampler: Equilibrium Statistics of a Coupled Gaussian Network)**.** Consider a network of N continuous voltage nodes $\mathbf{v} = (v_1,\ldots,v_N)^\top \in \mathbb{R}^N$ coupled through a symmetric, positive-definite stiffness (or effective inverse-covariance) matrix $\mathbf{K} = \mathbf{K}^\top \succ 0$ around a programmable reference bias vector $\mathbf{b} \in \mathbb{R}^N$, with quadratic energy

$$E_\theta(\mathbf{v}) = \frac{1}{2}(\mathbf{v}-\mathbf{b})^\top\mathbf{K}(\mathbf{v}-\mathbf{b}). \tag{1.34}$$

Which circuit matrix plays the role of $\mathbf{K}$ depends on the noise. For a passive RC network driven only by its own thermal noise, $\mathbf{K}$ is the capacitance matrix $\mathbf{C}$ and the conductances set only the relaxation rates; with engineered noise, $\mathbf{K}$ can instead be set by the conductance matrix $\mathbf{G}$ (Section 5.6). This chapter only needs $\mathbf{K}$ as a stiffness matrix. We compute the partition function $Z(\theta,\beta)$, Helmholtz free energy $F(\theta,\beta)$, internal energy $U(\theta,\beta)$, and entropy $S(\theta,\beta)$ step by step.

To evaluate the partition function with respect to the dimensionless volume element $\mu(\mathrm{d}\mathbf{v}) = \mathrm{d}^N v/V_0^N$, we shift coordinates to the displacement vector $\mathbf{u} := \mathbf{v} - \mathbf{b}$, which has unit Jacobian $\det(\frac{\partial\mathbf{v}}{\partial\mathbf{u}}) = 1$:

$$Z(\theta,\beta) = \int_{\mathbb{R}^N}\exp\left(-\frac{\beta}{2}(\mathbf{v}-\mathbf{b})^\top\mathbf{K}(\mathbf{v}-\mathbf{b})\right)\frac{\mathrm{d}^N v}{V_0^N} = \int_{\mathbb{R}^N}\exp\left(-\frac{\beta}{2}\mathbf{u}^\top\mathbf{K}\mathbf{u}\right)\frac{\mathrm{d}^N u}{V_0^N}. \tag{1.35}$$

Because $\mathbf{K}$ is real, symmetric, and positive definite, the spectral theorem guarantees an orthogonal eigendecomposition $\mathbf{K} = \mathbf{O}\mathbf{\Lambda}\mathbf{O}^\top$, where $\mathbf{O}^\top\mathbf{O} = \mathbf{I}_N$ and $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_N)$ with strictly positive eigenvalues $\lambda_i > 0$. Changing variables to normal modes $\mathbf{y} := \mathbf{O}^\top\mathbf{u}$ (with $|\det \mathbf{O}| = 1$, so $\mathrm{d}^N u = \mathrm{d}^N y$) factorizes the N-dimensional integral into a product of N one-dimensional Gaussian integrals:

$$\begin{aligned} Z(\theta, \beta) &= \int_{\mathbb{R}^N} \exp\left(-\frac{\beta}{2}\sum_{i=1}^{N} \lambda_i y_i^2\right) \frac{\mathrm{d}^N y}{V_0^N} = \prod_{i=1}^{N}\left[\frac{1}{V_0}\int_{-\infty}^{\infty} \exp\left(-\frac{\beta\lambda_i}{2} y_i^2\right) \mathrm{d}y_i\right] \\ &= \prod_{i=1}^{N}\left[\frac{1}{V_0}\sqrt{\frac{2\pi}{\beta\lambda_i}}\right] = \left(\frac{2\pi}{\beta V_0^2}\right)^{N/2}\left(\prod_{i=1}^{N}\lambda_i\right)^{-1/2} = \det\left(\frac{\beta V_0^2}{2\pi}\mathbf{K}\right)^{-1/2}, \end{aligned} \tag{1.36}$$

where we used $\det \mathbf{K} = \prod_{i=1}^{N}\lambda_i$. Consequently, the canonical equilibrium distribution (1.10) is the multivariate Gaussian $\mathcal{N}(\mathbf{b}, \mathbf{\Sigma}_{\mathrm{eq}})$ with equilibrium covariance matrix

$$\mathbf{\Sigma}_{\mathrm{eq}} = \mathrm{Cov}_{p_\theta}(\mathbf{V}) = \frac{1}{\beta}\mathbf{K}^{-1} = k_{\mathrm{B}}T\,\mathbf{K}^{-1}. \tag{1.37}$$

From Eq. (1.36), the Helmholtz free energy is

$$F(\theta, \beta) = -\frac{1}{\beta}\ln Z(\theta, \beta) = \frac{1}{2\beta}\ln\det\left(\frac{\beta V_0^2}{2\pi}\mathbf{K}\right) = \frac{k_{\mathrm{B}}T}{2}\ln\det\left(\frac{V_0^2}{2\pi k_{\mathrm{B}}T}\mathbf{K}\right). \tag{1.38}$$

The free energy does not depend on the mean vector $\mathbf{b}$, because translating a harmonic well in unbounded $\mathbb{R}^N$ does not change the volume of accessible states. Applying Theorem 1.4(ii) to $\ln Z(\theta, \beta) = -\frac{N}{2}\ln\beta + \frac{N}{2}\ln(2\pi/V_0^2) - \frac{1}{2}\ln\det\mathbf{K}$, we obtain the internal energy:

$$U(\theta, \beta) = -\frac{\partial \ln Z(\theta, \beta)}{\partial\beta} = \frac{N}{2\beta} = \frac{N}{2}k_{\mathrm{B}}T. \tag{1.39}$$

This is the classical *Equipartition Theorem*: each quadratic degree of freedom contributes exactly $\frac{1}{2}k_{\mathrm{B}}T$ to the mean thermal energy, regardless of how strong or weak the stiffness eigenvalues λ_i of $\mathbf{K}$ are. Differentiating a second time via Theorem 1.4(iv) yields the energy variance and heat capacity:

$$\mathrm{Var}_{p_\theta}(E_\theta(\mathbf{V})) = -\frac{\partial U}{\partial\beta} = \frac{N}{2\beta^2} = \frac{N}{2}(k_{\mathrm{B}}T)^2, \qquad C_\theta = \frac{N}{2}k_{\mathrm{B}}. \tag{1.40}$$

Finally, by Theorem 1.4(iii), the canonical entropy of the Gaussian network is

$$S(\theta, \beta) = \frac{U - F}{T} = \frac{k_{\mathrm{B}}}{2}\ln\det\left(\frac{2\pi e k_{\mathrm{B}}T}{V_0^2}\mathbf{K}^{-1}\right) = \frac{k_{\mathrm{B}}}{2}\ln\det\left(\frac{2\pi e}{V_0^2}\mathbf{\Sigma}_{\mathrm{eq}}\right). \tag{1.41}$$

A circuit whose stationary Gaussian covariance is programmable is the goal of thermodynamic linear-algebra proposals [3] and of the NEAT-RN circuits of Freitas et al. [38]. In the equilibrium case treated here, programming $\mathbf{K}$ in hardware gives thermal voltage fluctuations with covariance $k_{\mathrm{B}}T\mathbf{K}^{-1}$, so the circuit samples $\mathcal{N}(\mathbf{b}, k_{\mathrm{B}}T\mathbf{K}^{-1})$ and, in effect, inverts $\mathbf{K}$. (In a NEAT-RN, which is out of equilibrium, the covariance follows instead from a Lyapunov equation; see Chapter 13.) The time needed to do so is set by the slowest relaxation mode of the network, which Chapters 5 and 6 compute.

Example 1.6 (The Double-Well Sampler in Its Binary Limit: A Single p-Bit in a Local Field)**.** The double-well sampler is a particle in the quartic potential $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$. When the barrier between the wells is much larger than $k_{\mathrm{B}}T$, only the sign of x matters. The system then

reduces to a single probabilistic bit (p-bit) $s \in \{-1, +1\}$ in a programmable local field. This field is proportional to the tilt of the double well, and to keep the notation light we again call it $h \in \mathbb{R}$. The energy function is

$$E_h(s) = -hs, \qquad s \in \{-1, +1\}\,. \tag{1.42}$$

Summing over the two microstates $s = +1$ and $s = -1$ gives the partition function and Helmholtz free energy:

$$Z(h, \beta) = e^{-\beta E_h(+1)} + e^{-\beta E_h(-1)} = e^{\beta h} + e^{-\beta h} = 2\cosh(\beta h), \tag{1.43}$$

$$F(h, \beta) = -k_{\mathrm{B}} T \ln Z(h, \beta) = -k_{\mathrm{B}} T \ln\bigl(2\cosh(\beta h)\bigr). \tag{1.44}$$

The canonical probabilities of the $+1$ and -1 states are given by the logistic sigmoid function $\sigma(u) := (1 + e^{-u})^{-1}$:

$$p_h(s = \pm 1) = \frac{e^{\pm \beta h}}{e^{\beta h} + e^{-\beta h}} = \frac{1}{1 + e^{\mp 2\beta h}} = \sigma(\pm 2\beta h) = \frac{1}{2}\bigl(1 \pm \tanh(\beta h)\bigr). \tag{1.45}$$

The equilibrium magnetization (mean spin value) and internal energy follow immediately:

$$m(h, \beta) := \mathbb{E}[S]_{p_h} = (+1)p_h(+1) + (-1)p_h(-1) = \frac{e^{\beta h} - e^{-\beta h}}{e^{\beta h} + e^{-\beta h}} = \tanh(\beta h), \tag{1.46}$$

$$U(h, \beta) = \mathbb{E}[-hS]_{p_h} = -h \tanh(\beta h). \tag{1.47}$$

Using the identity $1 - \tanh^2(u) = \operatorname{sech}^2(u)$ and $S^2 = (\pm 1)^2 = 1$, the spin variance is

$$\mathrm{Var}_{p_h}(S) = \mathbb{E}\Bigl[S^2\Bigr]_{p_h} - \bigl(\mathbb{E}[S]_{p_h}\bigr)^2 = 1 - \tanh^2(\beta h) = \operatorname{sech}^2(\beta h), \tag{1.48}$$

and the energy variance is $\mathrm{Var}_{p_h}(E_h(S)) = h^2 \operatorname{sech}^2(\beta h)$. The Helmholtz free energy (1.44) has two simple limits:

- **High-temperature / zero-field limit ($|\beta h| \ll 1$):** Expanding $\ln(2\cosh u) = \ln 2 + \frac{1}{2}u^2 - \frac{1}{12}u^4 + \mathcal{O}(u^6)$ yields
$$F(h, \beta) = -k_{\mathrm{B}} T \ln 2 - \frac{h^2}{2k_{\mathrm{B}} T} + \mathcal{O}(\beta^3 h^4). \tag{1.49}$$
At $h = 0$, the internal energy is $U = 0$ and the free energy is purely entropic: $F(0, \beta) = -TS(0, \beta) = -k_{\mathrm{B}} T \ln 2$. Its magnitude $k_{\mathrm{B}} T \ln 2$ is the Landauer cost of erasing one bit (Section 9.3).

- **Low-temperature / strong-field limit ($|\beta h| \gg 1$):** Writing $2\cosh(\beta h) = e^{\beta|h|}(1 + e^{-2\beta|h|})$, we obtain
$$F(h, \beta) = -\,|h| - k_{\mathrm{B}} T e^{-2\beta|h|} + \mathcal{O}\bigl(k_{\mathrm{B}} T e^{-4\beta|h|}\bigr), \tag{1.50}$$
showing that the p-bit freezes into the ground state $s^* = \mathrm{sgn}(h)$ with ground-state energy $-\,|h|$ and exponentially suppressed thermal excitations.

1.2 The Gibbs–Bogoliubov–Feynman Variational Principle

In Section 1.1, the canonical distribution $p_\theta(x) = Z(\theta, \beta)^{-1} e^{-\beta E_\theta(x)}$ was introduced axiomatically as the equilibrium law of a system in contact with a heat bath at temperature T. A physical circuit, however, is often prepared far from equilibrium, for instance just after a digital controller clamps

some input nodes or resets a capacitor array. Its instantaneous probability distribution $q(x)$ is then far from $p_\theta(x)$. Why does the physical system relax toward $p_\theta(x)$, and how can we quantify its thermodynamic distance from equilibrium at any intermediate instant?

To answer both questions, we extend the Helmholtz free energy from a function of the parameters (θ, β) to a *functional* $\mathcal{F}[q]$ defined over the entire space $\mathcal{P}(\Omega)$ of probability distributions on Ω.

1.2.1 Non-Equilibrium Variational Free Energy and Kullback–Leibler Divergence

Definition 1.7 (Variational Free Energy Functional and Relative Entropy)**.** Let (Ω, μ) be a configuration space, let $E_\theta : \Omega \to \mathbb{R}$ be an energy function, and let $q \in \mathcal{P}(\Omega)$ be any probability density (or mass function) with respect to μ such that $\mathbb{E}[|E_\theta(X)|]_q < \infty$ and $S(q) = -k_{\mathrm{B}} \int_\Omega q(x) \ln q(x)\, \mu(\mathrm{d}x)$ is well-defined.

(i) The *non-equilibrium variational free energy* (or *Gibbs functional*) of the distribution q is defined as

$$\mathcal{F}[q] := \mathbb{E}[E_\theta(X)]_q - TS(q) = \int_\Omega q(x) E_\theta(x)\, \mu(\mathrm{d}x) + k_{\mathrm{B}}T \int_\Omega q(x) \ln q(x)\, \mu(\mathrm{d}x). \tag{1.51}$$

(ii) Given a second probability density $p \in \mathcal{P}(\Omega)$, the *Kullback–Leibler (KL) divergence* (or *relative entropy*) of q with respect to p is defined as

$$D_{\mathrm{KL}}(q \,\|\, p) := \begin{cases} \displaystyle\int_{\mathrm{supp}(q)} q(x) \ln \frac{q(x)}{p(x)}\, \mu(\mathrm{d}x), & \text{if } \mathrm{supp}(q) \subseteq \mathrm{supp}(p), \\ +\infty, & \text{otherwise,} \end{cases} \tag{1.52}$$

with the standard continuity convention $0 \ln 0 = 0$ and $0 \ln(0/0) = 0$.

Before linking $\mathcal{F}[q]$ and $D_{\mathrm{KL}}(q \,\|\, p_\theta)$, we need the basic positivity property of the Kullback–Leibler divergence.

Lemma 1.8 (Gibbs' Inequality)**.** *For any two probability densities $q, p \in \mathcal{P}(\Omega)$ with respect to μ,*

$$D_{\mathrm{KL}}(q \,\|\, p) \geq 0, \tag{1.53}$$

with equality $D_{\mathrm{KL}}(q \,\|\, p) = 0$ if and only if $q(x) = p(x)$ for μ-almost every $x \in \Omega$.

Proof. If $\mathrm{supp}(q)$ is not contained in $\mathrm{supp}(p)$ (up to a μ-null set), then $D_{\mathrm{KL}}(q \,\|\, p) = +\infty > 0$ by definition. Assume therefore that $p(x) > 0$ for μ-almost every $x \in \mathrm{supp}(q) := \{x \in \Omega : q(x) > 0\}$. Consider the real-valued function $f(u) := u - 1 - \ln u$ on $(0, \infty)$. Its first and second derivatives are

$$f'(u) = 1 - \frac{1}{u}, \qquad f''(u) = \frac{1}{u^2} > 0 \quad \forall\, u \in (0, \infty). \tag{1.54}$$

Since $f''(u) > 0$ everywhere on $(0, \infty)$ and $f'(1) = 0$, f is strictly convex on $(0, \infty)$ and achieves its unique global minimum at $u = 1$, where $f(1) = 0$. Therefore, for all $u > 0$,

$$-\ln u \geq 1 - u, \qquad \text{with equality if and only if } u = 1. \tag{1.55}$$

For each $x \in \mathrm{supp}(q)$, set $u(x) = p(x)/q(x) > 0$. Multiplying Eq. (1.55) by $q(x) > 0$ and integrating over $\mathrm{supp}(q)$ with respect to μ yields

$$D_{\mathrm{KL}}(q \,\|\, p) = \int_{\mathrm{supp}(q)} q(x) \left(-\ln \frac{p(x)}{q(x)} \right) \mu(\mathrm{d}x)$$

$$\geq \int_{\mathrm{supp}(q)} q(x)\left(1-\frac{p(x)}{q(x)}\right)\mu(\mathrm{d}x)$$
$$= \int_{\mathrm{supp}(q)} q(x)\,\mu(\mathrm{d}x) - \int_{\mathrm{supp}(q)} p(x)\,\mu(\mathrm{d}x) = 1 - \int_{\mathrm{supp}(q)} p(x)\,\mu(\mathrm{d}x). \tag{1.56}$$

Since $p(x) \geq 0$ on Ω and $\int_\Omega p(x)\,\mu(\mathrm{d}x) = 1$, we have $\int_{\mathrm{supp}(q)} p(x)\,\mu(\mathrm{d}x) \leq \int_\Omega p(x)\,\mu(\mathrm{d}x) = 1$. Substituting this into Eq. (1.56) gives $D_{\mathrm{KL}}(q\,\|\,p) \geq 1 - 1 = 0$.

To characterize equality, observe that $D_{\mathrm{KL}}(q\,\|\,p) = 0$ holds if and only if both inequalities in the chain above are equalities: first, $u(x) = p(x)/q(x) = 1$ for μ-almost every $x \in \mathrm{supp}(q)$, which means $q(x) = p(x)$ μ-a.e. on $\mathrm{supp}(q)$; and second, $\int_{\mathrm{supp}(q)} p(x)\,\mu(\mathrm{d}x) = 1$, which implies $p(x) = 0 = q(x)$ μ-a.e. on $\Omega \setminus \mathrm{supp}(q)$. Thus $q(x) = p(x)$ μ-almost everywhere on Ω. □

The variational principle for the canonical ensemble follows directly.

Theorem 1.9 (Variational Principle for the Canonical Ensemble)**.** *Let $E_\theta : \Omega \to \mathbb{R}$ be a thermodynamically admissible energy function at inverse temperature $\beta = (k_{\mathrm{B}}T)^{-1} > 0$, with canonical partition function $Z(\theta,\beta)$, Helmholtz free energy $F(\theta,\beta) = -k_{\mathrm{B}}T \ln Z(\theta,\beta)$, and Boltzmann-Gibbs distribution $p_\theta(x) = Z(\theta,\beta)^{-1}e^{-\beta E_\theta(x)}$. Then for every probability density $q \in \mathcal{P}(\Omega)$ with finite mean energy and entropy, the variational free energy $\mathcal{F}[q]$ decomposes exactly as*

$$\mathcal{F}[q] = F(\theta,\beta) + k_{\mathrm{B}}T\, D_{\mathrm{KL}}(q\,\|\,p_\theta)\,. \tag{1.57}$$

Consequently, for all $q \in \mathcal{P}(\Omega)$,

$$\mathcal{F}[q] \geq F(\theta,\beta), \tag{1.58}$$

and the canonical Boltzmann-Gibbs distribution p_θ is the unique global minimizer of $\mathcal{F}[q]$ over $\mathcal{P}(\Omega)$:

$$F(\theta,\beta) = \min_{q\in\mathcal{P}(\Omega)} \mathcal{F}[q], \qquad p_\theta = \arg\min_{q\in\mathcal{P}(\Omega)} \mathcal{F}[q]. \tag{1.59}$$

Proof. Starting from the definition of the variational free energy functional (1.51), we express the energy $E_\theta(x)$ in terms of the canonical density $p_\theta(x)$ and the Helmholtz free energy $F(\theta,\beta)$. Taking the logarithm of $p_\theta(x) = \exp(\beta[F(\theta,\beta) - E_\theta(x)])$ from Eq. (1.20) and solving for $E_\theta(x)$ yields the point-wise identity

$$E_\theta(x) = F(\theta,\beta) - k_{\mathrm{B}}T \ln p_\theta(x) \qquad \forall\, x \in \Omega. \tag{1.60}$$

Substituting Eq. (1.60) into the first integral of $\mathcal{F}[q]$ in Eq. (1.51) gives

$$\begin{aligned}
\mathcal{F}[q] &= \int_\Omega q(x)E_\theta(x)\,\mu(\mathrm{d}x) + k_{\mathrm{B}}T\int_\Omega q(x)\ln q(x)\,\mu(\mathrm{d}x)\\
&= \int_\Omega q(x)\left[F(\theta,\beta) - k_{\mathrm{B}}T\ln p_\theta(x)\right]\mu(\mathrm{d}x) + k_{\mathrm{B}}T\int_\Omega q(x)\ln q(x)\,\mu(\mathrm{d}x)\\
&= F(\theta,\beta)\underbrace{\int_\Omega q(x)\,\mu(\mathrm{d}x)}_{=1} + k_{\mathrm{B}}T\int_\Omega q(x)\left[\ln q(x) - \ln p_\theta(x)\right]\mu(\mathrm{d}x)\\
&= F(\theta,\beta) + k_{\mathrm{B}}T\int_\Omega q(x)\ln\frac{q(x)}{p_\theta(x)}\,\mu(\mathrm{d}x)\\
&= F(\theta,\beta) + k_{\mathrm{B}}T\,D_{\mathrm{KL}}(q\,\|\,p_\theta)\,.
\end{aligned} \tag{1.61}$$

This proves the exact identity (1.57). Since $k_{\mathrm{B}}T > 0$ and $D_{\mathrm{KL}}(q\,\|\,p_\theta) \geq 0$ by Gibbs' inequality (Lemma 1.8), it follows immediately that $\mathcal{F}[q] \geq F(\theta,\beta)$ for all $q \in \mathcal{P}(\Omega)$, with equality $\mathcal{F}[q] = F(\theta,\beta)$ if and only if $D_{\mathrm{KL}}(q\,\|\,p_\theta) = 0$, which by Lemma 1.8 holds if and only if $q = p_\theta$ μ-almost everywhere. □

Physics Insight: Physical Thermalization as Variational Inference

Theorem 1.9 connects statistical physics, Bayesian variational inference, and thermodynamic hardware:

- **In Variational Bayesian Inference:** Suppose we wish to approximate an intractable target posterior distribution $p_\theta(z) = Z^{-1}e^{-E_\theta(z)}$ (where $\beta = 1$ and Z is the marginal likelihood or "model evidence") using a tractable family of distributions $q_\phi(z) \in \mathcal{Q}$. Multiplying Eq. (1.57) by -1 for $\beta = 1$ yields

$$\ln Z(\theta) = -F(\theta) = \underbrace{-\mathbb{E}[E_\theta(Z)]_{q_\phi} + \frac{1}{k_\mathrm{B}} S(q_\phi)}_{:=\mathrm{ELBO}(q_\phi)=-\mathcal{F}[q_\phi]} + D_{\mathrm{KL}}(q_\phi \,\|\, p_\theta)\,. \tag{1.62}$$

  In machine learning, $-\mathcal{F}[q_\phi]$ is called the *Evidence Lower Bound (ELBO)* because $D_{\mathrm{KL}}(q_\phi \,\|\, p_\theta) \geq 0$ implies $\mathrm{ELBO}(q_\phi) \leq \ln Z(\theta)$. Maximizing the ELBO over q_ϕ is identical to minimizing the variational free energy $\mathcal{F}[q_\phi]$.

- **In Physical Thermodynamic Hardware:** When an analog electronic circuit with energy landscape $E_\theta(\mathbf{v})$ is coupled to a thermal bath at temperature T, the time evolution of its instantaneous probability density $\rho_t(\mathbf{v})$ is governed by the Fokker-Planck equation (Chapter 6). We will prove, through the Fokker–Planck H-theorem (Section 6.3) and the Jordan–Kinderlehrer–Otto (JKO) scheme of optimal transport (Section 10.4), that the circuit dynamics is the steepest-descent gradient flow of $\mathcal{F}[\rho_t]$ in the Wasserstein geometry on probability distributions, with

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t] = k_\mathrm{B}T\frac{\mathrm{d}}{\mathrm{d}t}D_{\mathrm{KL}}(\rho_t \,\|\, p_\theta) = -T\dot{\Sigma}_{\mathrm{tot}}(t) \leq 0, \tag{1.63}$$

  where $\dot{\Sigma}_{\mathrm{tot}} \geq 0$ is the total entropy production rate (Chapter 8). In Section 9.3 we will also show that the excess $\mathcal{F}[\rho_t] - F(\theta,\beta) = k_\mathrm{B}T D_{\mathrm{KL}}(\rho_t \,\|\, p_\theta)$ is the maximum work that can be extracted from the non-equilibrium distribution ρ_t as it relaxes isothermally to equilibrium; in free relaxation this amount is dissipated as heat instead. A circuit relaxing toward thermal equilibrium is therefore carrying out variational inference, with the Second Law guaranteeing that the variational objective decreases.

1.2.2 The Gibbs–Bogoliubov–Feynman Inequality and Mean-Field Theory

Whenever the exact partition function $Z(\theta,\beta)$ of an interacting system cannot be evaluated in closed form, Theorem 1.9 still yields non-perturbative bounds on $F(\theta,\beta)$, obtained by comparing with the canonical distribution p_0 of a solvable reference Hamiltonian $E_0(x)$.

Corollary 1.10 (Gibbs–Bogoliubov–Feynman Variational Bound)**.** *Let $E_\theta(x) = E_0(x) + \Delta E(x)$ be a decomposition of the Hamiltonian into a solvable reference Hamiltonian $E_0(x)$ with partition function $Z_0(\beta)$, free energy $F_0(\beta) = -k_\mathrm{B}T\ln Z_0(\beta)$, and canonical distribution $p_0(x) = Z_0(\beta)^{-1}e^{-\beta E_0(x)}$, plus a perturbation $\Delta E(x)$. Then the exact Helmholtz free energy $F(\theta,\beta)$ is bounded from above and below by*

$$F_0(\beta) + \mathbb{E}[\Delta E(X)]_{p_\theta} \leq F(\theta,\beta) \leq F_0(\beta) + \mathbb{E}[\Delta E(X)]_{p_0}\,. \tag{1.64}$$

Proof. Evaluating the variational free energy $\mathcal{F}[q]$ of the full system $E_\theta = E_0 + \Delta E$ at the reference trial distribution $q = p_0$, and noting that $\mathcal{F}_0[p_0] = \mathbb{E}[E_0]_{p_0} - TS(p_0) = F_0(\beta)$ by Theorem 1.4(i), we

have

$$\begin{aligned}\mathcal{F}[p_0] &= \mathbb{E}[E_0(X) + \Delta E(X)]_{p_0} - TS(p_0) \\ &= \left(\mathbb{E}[E_0(X)]_{p_0} - TS(p_0)\right) + \mathbb{E}[\Delta E(X)]_{p_0} \\ &= F_0(\beta) + \mathbb{E}[\Delta E(X)]_{p_0}\,. \end{aligned} \tag{1.65}$$

By the variational inequality $F(\theta,\beta) \le \mathcal{F}[p_0]$ of Theorem 1.9, we immediately obtain the upper bound $F(\theta,\beta) \le F_0(\beta) + \mathbb{E}[\Delta E(X)]_{p_0}$.

To obtain the lower bound in Eq. (1.64), we simply reverse the roles of the two Hamiltonians: write $E_0(x) = E_\theta(x) + (-\Delta E(x))$ and apply the upper bound with E_θ as the reference Hamiltonian and $-\Delta E$ as the perturbation:

$$F_0(\beta) \le F(\theta,\beta) + \mathbb{E}[-\Delta E(X)]_{p_\theta} = F(\theta,\beta) - \mathbb{E}[\Delta E(X)]_{p_\theta}\,. \tag{1.66}$$

Rearranging this inequality gives $F_0(\beta) + \mathbb{E}[\Delta E(X)]_{p_\theta} \le F(\theta,\beta)$, completing the proof. □

As an application of Theorem 1.9, we derive the Weiss mean-field equations for an interacting network of N Ising spins. They serve as the baseline for the Thouless–Anderson–Palmer (TAP) correction in Chapter 3.

Example 1.11 (Variational Mean-Field Theory for an Interacting Ising Network)**.** Consider a network of N coupled Ising spins $\mathbf{s} \in \{-1,+1\}^N$ with pairwise symmetric couplings $J_{ij} = J_{ji} \in \mathbb{R}$ ($J_{ii} = 0$) and local external fields $h_i \in \mathbb{R}$:

$$E_\theta(\mathbf{s}) = -\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} J_{ij}s_i s_j - \sum_{i=1}^{N} h_i s_i. \tag{1.67}$$

Because of the interaction terms $J_{ij}s_is_j$, computing $Z(\theta,\beta)$ exactly requires summing over 2^N configurations. In the naive mean-field approximation, we restrict the variational minimization of $\mathcal{F}[q]$ in Theorem 1.9 to the submanifold of fully factorized (product) distributions over the N spins:

$$q_{\mathbf{m}}(\mathbf{s}) = \prod_{i=1}^{N} q_i(s_i), \qquad q_i(s_i) := \frac{1+m_i s_i}{2}, \qquad s_i \in \{-1,+1\}\,, \tag{1.68}$$

parameterized by the variational magnetization vector $\mathbf{m} = (m_1,\dots,m_N)^\top \in [-1,+1]^N$. Each q_i is normalized and has mean m_i:

$$\sum_{s_i\in\{-1,+1\}} q_i(s_i) = \frac{1+m_i}{2} + \frac{1-m_i}{2} = 1, \tag{1.69}$$

$$\mathbb{E}[S_i]_{q_{\mathbf{m}}} = (+1)\frac{1+m_i}{2} + (-1)\frac{1-m_i}{2} = m_i. \tag{1.70}$$

Moreover, because $q_{\mathbf{m}}$ is a product measure, distinct spins $i \ne j$ are statistically independent under $q_{\mathbf{m}}$, so $\mathbb{E}[S_iS_j]_{q_{\mathbf{m}}} = \mathbb{E}[S_i]_{q_{\mathbf{m}}}\,\mathbb{E}[S_j]_{q_{\mathbf{m}}} = m_im_j$ for all $i \ne j$. Consequently, the expected energy under $q_{\mathbf{m}}$ is

$$\mathbb{E}[E_\theta(\mathbf{S})]_{q_{\mathbf{m}}} = -\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} J_{ij}m_im_j - \sum_{i=1}^{N} h_i m_i. \tag{1.71}$$

Similarly, the logarithm of the product distribution is additive, $\ln q_{\mathbf{m}}(\mathbf{s}) = \sum_{i=1}^{N} \ln q_i(s_i)$, so the Gibbs-Shannon entropy decomposes into a sum of N single-spin binary entropies:

$$\begin{aligned} S(q_{\mathbf{m}}) &= -k_{\mathrm{B}} \sum_{\mathbf{s}\in\{-1,+1\}^N} \left(\prod_{k=1}^{N} q_k(s_k) \right) \sum_{i=1}^{N} \ln q_i(s_i) \\ &= -k_{\mathrm{B}} \sum_{i=1}^{N} \sum_{s_i\in\{-1,+1\}} q_i(s_i) \ln q_i(s_i) \\ &= -k_{\mathrm{B}} \sum_{i=1}^{N} \left[\frac{1+m_i}{2} \ln\left(\frac{1+m_i}{2}\right) + \frac{1-m_i}{2} \ln\left(\frac{1-m_i}{2}\right) \right]. \end{aligned} \tag{1.72}$$

Combining Eqs. (1.71) and (1.72), the variational free energy $\mathcal{F}_{\mathrm{MF}}(\mathbf{m}) := \mathcal{F}[q_{\mathbf{m}}] = \mathbb{E}[E_\theta]_{q_{\mathbf{m}}} - TS(q_{\mathbf{m}})$ becomes an explicit function of $\mathbf{m} \in (-1,1)^N$:

$$\begin{aligned} \mathcal{F}_{\mathrm{MF}}(\mathbf{m}) = &-\frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} J_{ij} m_i m_j - \sum_{i=1}^{N} h_i m_i \\ &+ k_{\mathrm{B}} T \sum_{i=1}^{N} \left[\frac{1+m_i}{2} \ln\left(\frac{1+m_i}{2}\right) + \frac{1-m_i}{2} \ln\left(\frac{1-m_i}{2}\right) \right]. \end{aligned} \tag{1.73}$$

By Theorem 1.9, $F(\theta,\beta) \le \mathcal{F}_{\mathrm{MF}}(\mathbf{m})$ for every $\mathbf{m} \in [-1,+1]^N$. To find the tightest variational bound (and the best factorized approximation to p_θ in KL divergence), we differentiate $\mathcal{F}_{\mathrm{MF}}(\mathbf{m})$ with respect to each m_i. Using the symmetry $J_{ij} = J_{ji}$ and the derivative

$$\begin{aligned} \frac{\partial}{\partial m_i} \left[\frac{1+m_i}{2} \ln\left(\frac{1+m_i}{2}\right) + \frac{1-m_i}{2} \ln\left(\frac{1-m_i}{2}\right) \right] &= \frac{1}{2} \ln\left(\frac{1+m_i}{2}\right) + \frac{1}{2} - \frac{1}{2} \ln\left(\frac{1-m_i}{2}\right) - \frac{1}{2} \\ &= \frac{1}{2} \ln\left(\frac{1+m_i}{1-m_i}\right) = \operatorname{artanh}(m_i), \end{aligned} \tag{1.74}$$

the stationarity condition $\frac{\partial \mathcal{F}_{\mathrm{MF}}}{\partial m_i} = 0$ yields

$$-\sum_{j=1}^{N} J_{ij} m_j - h_i + k_{\mathrm{B}} T \operatorname{artanh}(m_i) = 0 \implies m_i = \tanh\left(\beta \Big(h_i + \sum_{j=1}^{N} J_{ij} m_j \Big) \right). \tag{1.75}$$

Comparing Eq. (1.75) with the exact single-p-bit magnetization $m = \tanh(\beta h)$ derived in Example 1.6, we see the physical meaning of variational mean-field theory: each spin s_i responds as if it were an isolated p-bit immersed in an effective local field $h_i^{\mathrm{eff}} = h_i + \sum_{j=1}^{N} J_{ij} m_j$ generated by the average magnetizations of its neighbors.

1.3 Exponential Families, Cumulant Generation, and Static Fluctuation–Response

In both physical circuits and machine learning models, the Hamiltonian $E_\theta(x)$ is rarely an unstructured black-box function of the control parameters θ. Instead, $E_\theta(x)$ is typically a linear combination of elementary interaction terms (local node biases, pairwise conductances, or higher-order clique potentials), each weighted by a tunable parameter θ_a. Such models belong to the class of *exponential families*, for which the log-partition function $\ln Z(\theta,\beta)$ acts as a universal cumulant generating function.

1.3.1 Linearly Parameterized Hamiltonians and Natural Exponential Families

Definition 1.12 (Linearly Parameterized Energy-Based Model / Exponential Family)**.** Let $\boldsymbol{\phi} = (\phi_1, \phi_2, \ldots, \phi_P)^\top : \Omega \to \mathbb{R}^P$ be a vector of measurable observables (called *sufficient statistics* in statistics or *conjugate generalized displacements/potentials* in statistical mechanics), and let $E_{\text{base}} : \Omega \to \mathbb{R}$ be a fixed parameter-independent base energy. A *linearly parameterized Energy-Based Model* has Hamiltonian

$$E_\theta(x) = E_{\text{base}}(x) + \sum_{a=1}^{P} \theta_a \phi_a(x) = E_{\text{base}}(x) + \boldsymbol{\theta}^\top \boldsymbol{\phi}(x), \qquad \boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^P, \tag{1.76}$$

where the natural parameter domain Θ at inverse temperature $\beta > 0$ is the set of parameters for which the partition function converges:

$$\Theta(\beta) := \left\{ \boldsymbol{\theta} \in \mathbb{R}^P : Z(\boldsymbol{\theta}, \beta) = \int_\Omega \exp\Big(-\beta\big[E_{\text{base}}(x) + \boldsymbol{\theta}^\top \boldsymbol{\phi}(x)\big]\Big)\, \mu(\mathrm{d}x) < \infty \right\}. \tag{1.77}$$

The family of canonical distributions $\{p_\theta : \boldsymbol{\theta} \in \Theta(\beta)\}$ is called an *exponential family* in canonical form. We define the *log-partition function* (or *cumulant generating function*) as

$$\psi(\boldsymbol{\theta}, \beta) := \ln Z(\boldsymbol{\theta}, \beta) = -\beta F(\boldsymbol{\theta}, \beta). \tag{1.78}$$

Both model systems are linearly parameterized exponential families:

- For the RC sampler (Example 1.5), expanding $E_\theta(\mathbf{v}) = \frac{1}{2}\mathbf{v}^\top \mathbf{K}\mathbf{v} - (\mathbf{K}\mathbf{b})^\top \mathbf{v} + \frac{1}{2}\mathbf{b}^\top \mathbf{K}\mathbf{b}$ shows that the sufficient statistics are the linear node voltages $\phi_i(\mathbf{v}) = v_i$ (conjugate to external bias currents $-(\mathbf{K}\mathbf{b})_i$) and the quadratic products $\phi_{ij}(\mathbf{v}) = \frac{1}{2} v_i v_j$ (conjugate to the stiffness/conductance matrix entries K_{ij}).
- For a network of coupled p-bits, the many-unit version of the double-well sampler in its binary limit (Example 1.11), the sufficient statistics are the single-spin magnetizations $\phi_i(\mathbf{s}) = -s_i$ (conjugate to local fields h_i) and the two-spin correlations $\phi_{ij}(\mathbf{s}) = -s_i s_j$ (conjugate to pairwise synaptic couplings J_{ij}).

1.3.2 The Cumulant Hierarchy and Static Fluctuation–Response Theorem

When we vary a control parameter θ_a in a physical circuit, for example a gate bias voltage, how does the ensemble average $\mathbb{E}[\phi_b(X)]_{p_\theta}$ of another observable respond? The derivatives of $\ln Z(\theta, \beta)$ and $F(\theta, \beta)$ with respect to θ answer this: first derivatives are equilibrium averages, and second derivatives are equilibrium covariances. We first state the result for a general smooth parameterization $E_\theta(x)$ (which may be nonlinear in θ, as in deep EBMs) and then specialize to the linear exponential family (1.76).

Theorem 1.13 (Static Fluctuation–Response Theorem and Cumulant Hierarchy)**.** *Let $\theta \mapsto E_\theta(x)$ be three times continuously differentiable on an open domain $\Theta \subseteq \mathbb{R}^P$, and assume that E_θ and its partial derivatives with respect to θ satisfy local dominated integrability against $\exp(-\beta E_\theta(x))$ on Θ.*

(i) ***General Parameterization (Generalized Forces and Hessian):*** *For any $a, b \in \{1, \ldots, P\}$, the first derivatives of $\ln Z(\theta, \beta)$ and $F(\theta, \beta)$ are the canonical expectations of the generalized force $\frac{\partial E_\theta}{\partial \theta_a}$:*

$$\frac{\partial \ln Z(\theta, \beta)}{\partial \theta_a} = -\beta \mathbb{E}\left[\frac{\partial E_\theta(X)}{\partial \theta_a}\right]_{p_\theta}, \qquad \frac{\partial F(\theta, \beta)}{\partial \theta_a} = \mathbb{E}\left[\frac{\partial E_\theta(X)}{\partial \theta_a}\right]_{p_\theta}. \tag{1.79}$$

The second derivatives (the Hessian of $\ln Z$ *and* F*) satisfy*

$$\frac{\partial^2 \ln Z(\theta,\beta)}{\partial\theta_a\partial\theta_b} = -\beta\mathbb{E}\left[\frac{\partial^2 E_\theta(X)}{\partial\theta_a\partial\theta_b}\right]_{p_\theta} + \beta^2 \operatorname{Cov}_{p_\theta}\left(\frac{\partial E_\theta(X)}{\partial\theta_a},\, \frac{\partial E_\theta(X)}{\partial\theta_b}\right), \tag{1.80}$$

$$\frac{\partial^2 F(\theta,\beta)}{\partial\theta_a\partial\theta_b} = \mathbb{E}\left[\frac{\partial^2 E_\theta(X)}{\partial\theta_a\partial\theta_b}\right]_{p_\theta} - \beta \operatorname{Cov}_{p_\theta}\left(\frac{\partial E_\theta(X)}{\partial\theta_a},\, \frac{\partial E_\theta(X)}{\partial\theta_b}\right). \tag{1.81}$$

Moreover, for any parameter-independent observable $A:\Omega\to\mathbb{R}$*, its static linear response (susceptibility) to a perturbation in* θ_b *is given by the thermal covariance*

$$\frac{\partial}{\partial\theta_b}\mathbb{E}[A(X)]_{p_\theta} = -\beta \operatorname{Cov}_{p_\theta}\left(A(X),\, \frac{\partial E_\theta(X)}{\partial\theta_b}\right). \tag{1.82}$$

(ii) ***Linear Exponential Family*** *(*$E_\theta(x) = E_{\text{base}}(x) + \sum_{a=1}^{P}\theta_a\phi_a(x)$*): When* E_θ *is linear in* $\boldsymbol{\theta}$*, all second and higher derivatives of* E_θ *with respect to* $\boldsymbol{\theta}$ *vanish (*$\frac{\partial^2 E_\theta}{\partial\theta_a\partial\theta_b} = 0$*), and the derivatives of* $\psi(\boldsymbol{\theta},\beta) = \ln Z(\boldsymbol{\theta},\beta)$ *generate the exact central cumulants of* $\boldsymbol{\phi}(X)$*:*

$$\frac{\partial \ln Z(\boldsymbol{\theta},\beta)}{\partial\theta_a} = -\beta\mathbb{E}[\phi_a(X)]_{p_\theta}, \tag{1.83}$$

$$\frac{\partial^2 \ln Z(\boldsymbol{\theta},\beta)}{\partial\theta_a\partial\theta_b} = -\beta\frac{\partial\mathbb{E}[\phi_a(X)]_{p_\theta}}{\partial\theta_b} = \beta^2 \operatorname{Cov}_{p_\theta}(\phi_a(X),\phi_b(X)), \tag{1.84}$$

$$\frac{\partial^3 \ln Z(\boldsymbol{\theta},\beta)}{\partial\theta_a\partial\theta_b\partial\theta_c} = -\beta^3\mathbb{E}\big[(\phi_a(X)-\langle\phi_a\rangle)(\phi_b(X)-\langle\phi_b\rangle)(\phi_c(X)-\langle\phi_c\rangle)\big]_{p_\theta}. \tag{1.85}$$

Proof. We carry out each differentiation under the integral sign explicitly.

Step 1: First derivative. Differentiating $Z(\theta,\beta) = \int_\Omega \exp(-\beta E_\theta(x))\,\mu(\mathrm{d}x)$ with respect to θ_a yields

$$\frac{\partial Z(\theta,\beta)}{\partial\theta_a} = \int_\Omega \frac{\partial}{\partial\theta_a}\exp(-\beta E_\theta(x))\,\mu(\mathrm{d}x) = -\beta\int_\Omega \frac{\partial E_\theta(x)}{\partial\theta_a}\exp(-\beta E_\theta(x))\,\mu(\mathrm{d}x). \tag{1.86}$$

Dividing by $Z(\theta,\beta)$ gives the first identity in Eq. (1.79):

$$\frac{\partial \ln Z(\theta,\beta)}{\partial\theta_a} = \frac{1}{Z(\theta,\beta)}\frac{\partial Z(\theta,\beta)}{\partial\theta_a} = -\beta\int_\Omega \frac{\partial E_\theta(x)}{\partial\theta_a}p_\theta(x)\,\mu(\mathrm{d}x) = -\beta\mathbb{E}\left[\frac{\partial E_\theta(X)}{\partial\theta_a}\right]_{p_\theta}. \tag{1.87}$$

Multiplying Eq. (1.87) by $-k_{\mathrm{B}}T = -1/\beta$ gives $\frac{\partial F}{\partial\theta_a} = \mathbb{E}\left[\frac{\partial E_\theta}{\partial\theta_a}\right]_{p_\theta}$.

Step 2: Derivative of the canonical density $p_\theta(x)$ **and observable response (1.82).** To differentiate ensemble averages systematically, we first compute $\frac{\partial p_\theta(x)}{\partial\theta_b}$ for fixed $x\in\Omega$. Writing $p_\theta(x) = \exp(-\beta E_\theta(x) - \ln Z(\theta,\beta))$ and applying the chain rule along with Eq. (1.87), we obtain the score identity:

$$\begin{aligned}\frac{\partial p_\theta(x)}{\partial\theta_b} &= p_\theta(x)\left(-\beta\frac{\partial E_\theta(x)}{\partial\theta_b} - \frac{\partial \ln Z(\theta,\beta)}{\partial\theta_b}\right)\\ &= -\beta\, p_\theta(x)\left(\frac{\partial E_\theta(x)}{\partial\theta_b} - \mathbb{E}\left[\frac{\partial E_\theta(X)}{\partial\theta_b}\right]_{p_\theta}\right),\end{aligned} \tag{1.88}$$

or equivalently, in terms of the logarithmic derivative (the parameter score function),

$$\frac{\partial \ln p_\theta(x)}{\partial \theta_b} = -\beta \left(\frac{\partial E_\theta(x)}{\partial \theta_b} - \mathbb{E}\Big[\frac{\partial E_\theta(X)}{\partial \theta_b} \Big]_{p_\theta} \right). \tag{1.89}$$

Now let $A(x)$ be any parameter-independent observable. Differentiating $\mathbb{E}[A(X)]_{p_\theta} = \int_\Omega A(x) p_\theta(x)\, \mu(\mathrm{d}x)$ with respect to θ_b and substituting Eq. (1.88) gives

$$\begin{aligned}
\frac{\partial}{\partial \theta_b} \mathbb{E}[A(X)]_{p_\theta} &= \int_\Omega A(x) \frac{\partial p_\theta(x)}{\partial \theta_b}\, \mu(\mathrm{d}x) \\
&= -\beta \int_\Omega A(x) \left(\frac{\partial E_\theta(x)}{\partial \theta_b} - \mathbb{E}\Big[\frac{\partial E_\theta(X)}{\partial \theta_b} \Big]_{p_\theta} \right) p_\theta(x)\, \mu(\mathrm{d}x) \\
&= -\beta \left(\mathbb{E}\Big[A(X) \frac{\partial E_\theta(X)}{\partial \theta_b} \Big]_{p_\theta} - \mathbb{E}[A(X)]_{p_\theta}\, \mathbb{E}\Big[\frac{\partial E_\theta(X)}{\partial \theta_b} \Big]_{p_\theta} \right) \\
&= -\beta\, \mathrm{Cov}_{p_\theta}\left(A(X),\, \frac{\partial E_\theta(X)}{\partial \theta_b} \right),
\end{aligned} \tag{1.90}$$

which proves Eq. (1.82).

Step 3: Second derivative. To compute $\frac{\partial^2 \ln Z}{\partial \theta_a \partial \theta_b}$, we differentiate Eq. (1.87) with respect to θ_b. Because $\frac{\partial E_\theta(x)}{\partial \theta_a}$ may itself depend on θ_b, the product rule inside the integral yields two terms, one from differentiating $\frac{\partial E_\theta(x)}{\partial \theta_a}$ and one from differentiating $p_\theta(x)$ via Eq. (1.88):

$$\begin{aligned}
\frac{\partial^2 \ln Z(\theta,\beta)}{\partial \theta_a \partial \theta_b} &= -\beta \frac{\partial}{\partial \theta_b} \int_\Omega \frac{\partial E_\theta(x)}{\partial \theta_a} p_\theta(x)\, \mu(\mathrm{d}x) \\
&= -\beta \int_\Omega \left[\frac{\partial^2 E_\theta(x)}{\partial \theta_a \partial \theta_b} p_\theta(x) + \frac{\partial E_\theta(x)}{\partial \theta_a} \frac{\partial p_\theta(x)}{\partial \theta_b} \right] \mu(\mathrm{d}x) \\
&= -\beta \mathbb{E}\left[\frac{\partial^2 E_\theta(X)}{\partial \theta_a \partial \theta_b} \right]_{p_\theta} + \beta^2 \int_\Omega \frac{\partial E_\theta(x)}{\partial \theta_a} \left(\frac{\partial E_\theta(x)}{\partial \theta_b} - \mathbb{E}\Big[\frac{\partial E_\theta(X)}{\partial \theta_b} \Big]_{p_\theta} \right) p_\theta(x)\, \mu(\mathrm{d}x) \\
&= -\beta \mathbb{E}\left[\frac{\partial^2 E_\theta(X)}{\partial \theta_a \partial \theta_b} \right]_{p_\theta} + \beta^2\, \mathrm{Cov}_{p_\theta}\left(\frac{\partial E_\theta(X)}{\partial \theta_a},\, \frac{\partial E_\theta(X)}{\partial \theta_b} \right).
\end{aligned} \tag{1.91}$$

This proves Eq. (1.80), and multiplying by $-1/\beta$ gives Eq. (1.81).

Step 4: Specialization to linear exponential families and third cumulant. When $E_\theta(x) = E_{\mathrm{base}}(x) + \sum_{a=1}^{P} \theta_a \phi_a(x)$, we have $\frac{\partial E_\theta(x)}{\partial \theta_a} = \phi_a(x)$ and $\frac{\partial^2 E_\theta(x)}{\partial \theta_a \partial \theta_b} = 0$. Equations (1.83) and (1.84) follow immediately. To obtain the third derivative (1.85), write $\delta\phi_a(x) := \phi_a(x) - \mathbb{E}[\phi_a(X)]_{p_\theta}$, so that $\frac{\partial \mathbb{E}[\phi_a]_{p_\theta}}{\partial \theta_c} = -\beta \mathbb{E}[\delta\phi_a\, \delta\phi_c]_{p_\theta}$ by Eq. (1.84). Differentiating Eq. (1.84) with respect to θ_c and using $\frac{\partial(\delta\phi_a)}{\partial \theta_c} = -\frac{\partial \langle \phi_a \rangle}{\partial \theta_c} = \beta \mathbb{E}[\delta\phi_a\, \delta\phi_c]_{p_\theta}$ along with Eq. (1.88) gives

$$\begin{aligned}
\frac{\partial^3 \ln Z}{\partial \theta_a \partial \theta_b \partial \theta_c} &= \beta^2 \frac{\partial}{\partial \theta_c} \int_\Omega \delta\phi_a(x)\, \delta\phi_b(x)\, p_\theta(x)\, \mu(\mathrm{d}x) \\
&= \beta^2 \int_\Omega \left[\frac{\partial(\delta\phi_a)}{\partial \theta_c} \delta\phi_b(x) + \delta\phi_a(x) \frac{\partial(\delta\phi_b)}{\partial \theta_c} \right] p_\theta(x)\, \mu(\mathrm{d}x) + \beta^2 \int_\Omega \delta\phi_a(x)\, \delta\phi_b(x) \frac{\partial p_\theta(x)}{\partial \theta_c}\, \mu(\mathrm{d}x)
\end{aligned}$$

$$= 0 + 0 - \beta^3 \int_\Omega \delta\phi_a(x)\,\delta\phi_b(x)\,\delta\phi_c(x)\,p_\theta(x)\,\mu(\mathrm{d}x)$$
$$= -\beta^3 \mathbb{E}\big[(\phi_a(X) - \langle\phi_a\rangle)(\phi_b(X) - \langle\phi_b\rangle)(\phi_c(X) - \langle\phi_c\rangle)\big]_{p_\theta}, \tag{1.92}$$

where the first two integrals vanished because $\mathbb{E}[\delta\phi_b]_{p_\theta} = \mathbb{E}[\delta\phi_a]_{p_\theta} = 0$ and $\frac{\partial(\delta\phi_a)}{\partial\theta_c}$ is independent of x. This completes the proof. □

1.3.3 Convexity of the Log-Partition Function and Legendre Duality

In matrix notation, Eqs. (1.83) and (1.84) state that for a linearly parameterized EBM $E_\theta(x) = E_{\text{base}}(x) + \boldsymbol{\theta}^\top \boldsymbol{\phi}(x)$, the gradient and Hessian of the log-partition function $\psi(\boldsymbol{\theta}) = \ln Z(\boldsymbol{\theta}, \beta)$ are

$$\nabla_{\boldsymbol{\theta}} \ln Z(\boldsymbol{\theta}, \beta) = -\beta \mathbb{E}[\boldsymbol{\phi}(X)]_{p_\theta}, \qquad \nabla^2_{\boldsymbol{\theta}} \ln Z(\boldsymbol{\theta}, \beta) = \beta^2 \operatorname{Cov}_{p_\theta}(\boldsymbol{\phi}(X)). \tag{1.93}$$

Because a covariance matrix is always positive semidefinite, Eq. (1.93) directly determines the convexity of the log-partition function.

Corollary 1.14 (Convexity of the Log-Partition Function and Concavity of Free Energy)**.** *For any linearly parameterized Energy-Based Model* (1.76)*, the natural parameter domain $\Theta(\beta) \subseteq \mathbb{R}^P$ is a convex set. On the interior of $\Theta(\beta)$, the log-partition function $\boldsymbol{\theta} \mapsto \ln Z(\boldsymbol{\theta}, \beta)$ is convex and the Helmholtz free energy $\boldsymbol{\theta} \mapsto F(\boldsymbol{\theta}, \beta)$ is concave. Furthermore, $\ln Z(\boldsymbol{\theta}, \beta)$ is* strictly *convex if and only if the family is* minimal*, that is, if no non-zero linear combination $\mathbf{u}^\top \boldsymbol{\phi}(x)$ is μ-almost everywhere constant on Ω.*

Proof. Let $\boldsymbol{\theta}_0, \boldsymbol{\theta}_1 \in \Theta(\beta)$ and $\lambda \in (0,1)$, and set $\boldsymbol{\theta}_\lambda = (1-\lambda)\boldsymbol{\theta}_0 + \lambda\boldsymbol{\theta}_1$. By Hölder's inequality with conjugate exponents $p = (1-\lambda)^{-1}$ and $q = \lambda^{-1}$ (satisfying $1/p + 1/q = 1$),

$$Z(\boldsymbol{\theta}_\lambda, \beta) = \int_\Omega \Big[\exp\big(-\beta E_{\boldsymbol{\theta}_0}(x)\big)\Big]^{1-\lambda} \Big[\exp\big(-\beta E_{\boldsymbol{\theta}_1}(x)\big)\Big]^{\lambda} \mu(\mathrm{d}x)$$
$$\le \big[Z(\boldsymbol{\theta}_0, \beta)\big]^{1-\lambda} \big[Z(\boldsymbol{\theta}_1, \beta)\big]^{\lambda} < \infty, \tag{1.94}$$

which shows both that $\boldsymbol{\theta}_\lambda \in \Theta(\beta)$ (so $\Theta(\beta)$ is convex) and, upon taking logarithms, that $\ln Z(\boldsymbol{\theta}_\lambda, \beta) \le (1-\lambda)\ln Z(\boldsymbol{\theta}_0, \beta) + \lambda \ln Z(\boldsymbol{\theta}_1, \beta)$.

On the interior of $\Theta(\beta)$, for any vector $\mathbf{u} \in \mathbb{R}^P$, evaluating the quadratic form of the Hessian (1.93) gives

$$\mathbf{u}^\top \big[\nabla^2_{\boldsymbol{\theta}} \ln Z(\boldsymbol{\theta}, \beta)\big]\mathbf{u} = \beta^2 \sum_{a=1}^{P}\sum_{b=1}^{P} u_a \operatorname{Cov}_{p_\theta}(\phi_a(X), \phi_b(X)) u_b$$
$$= \beta^2 \operatorname{Var}_{p_\theta}(\mathbf{u}^\top \boldsymbol{\phi}(X)) \ge 0. \tag{1.95}$$

Thus $\nabla^2_{\boldsymbol{\theta}} \ln Z(\boldsymbol{\theta}, \beta) \succeq 0$, proving convexity of $\ln Z$ and concavity of $F = -k_{\mathrm{B}} T \ln Z$. Finally, for any $\mathbf{u} \neq \mathbf{0}$, $\operatorname{Var}_{p_\theta}(\mathbf{u}^\top \boldsymbol{\phi}(X)) = 0$ if and only if $\mathbf{u}^\top \boldsymbol{\phi}(X) = c$ holds p_θ-almost everywhere (and hence μ-almost everywhere, since $p_\theta(x) > 0$ everywhere on Ω). Thus $\nabla^2_{\boldsymbol{\theta}} \ln Z(\boldsymbol{\theta}, \beta) \succ 0$ if and only if the family is minimal. □

Strict convexity of $\psi(\boldsymbol{\theta}) = \ln Z(\boldsymbol{\theta}, \beta)$ in a minimal exponential family implies that the gradient map $\boldsymbol{\theta} \mapsto \nabla_{\boldsymbol{\theta}} \psi(\boldsymbol{\theta}) = -\beta \mathbb{E}[\boldsymbol{\phi}(X)]_{p_\theta}$ is globally injective. This establishes a one-to-one Legendre-Fenchel duality between the *natural parameters* $\boldsymbol{\theta}$ (fields and couplings) and the *expectation parameters* $\boldsymbol{\eta} := \mathbb{E}[\boldsymbol{\phi}(X)]_{p_\theta}$ (magnetizations and correlations).

Proposition 1.15 (Legendre–Fenchel Duality Between Free Energy and Canonical Entropy)**.** *Consider a minimal linear exponential family with* $E_{\mathrm{base}}(x) = 0$ *and* $E_\theta(x) = \boldsymbol{\theta}^\top \boldsymbol{\phi}(x)$*. Define the expectation parameter domain* $\mathcal{M} := \left\{ \boldsymbol{\eta}(\boldsymbol{\theta}) = \mathbb{E}[\boldsymbol{\phi}(X)]_{p_\theta} : \boldsymbol{\theta} \in \operatorname{int} \Theta(\beta) \right\}$*. Then:*

(i) For each $\boldsymbol{\eta} \in \mathcal{M}$*, there exists a unique natural parameter* $\boldsymbol{\theta}(\boldsymbol{\eta}) \in \operatorname{int} \Theta(\beta)$ *such that* $\mathbb{E}[\boldsymbol{\phi}(X)]_{p_{\boldsymbol{\theta}(\boldsymbol{\eta})}} = \boldsymbol{\eta}$*.*

(ii) The Legendre-Fenchel conjugate of the Helmholtz free energy $F(\boldsymbol{\theta}, \beta)$ *with respect to* $\boldsymbol{\theta}$ *is the negative canonical entropy* $-TS(\boldsymbol{\eta})$*:*

$$\sup_{\boldsymbol{\theta} \in \operatorname{int} \Theta(\beta)} \left\{ F(\boldsymbol{\theta}, \beta) - \boldsymbol{\theta}^\top \boldsymbol{\eta} \right\} = -TS(p_{\boldsymbol{\theta}(\boldsymbol{\eta})}), \tag{1.96}$$

and the inverse Jacobian of the moment map is the inverse susceptibility matrix:

$$\frac{\partial \boldsymbol{\eta}}{\partial \boldsymbol{\theta}} = -\beta \operatorname{Cov}_{p_\theta}(\boldsymbol{\phi}(X)), \qquad \frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{\eta}} = -k_{\mathrm{B}} T \left[\operatorname{Cov}_{p_{\boldsymbol{\theta}(\boldsymbol{\eta})}}(\boldsymbol{\phi}(X)) \right]^{-1} = T \nabla^2_{\boldsymbol{\eta}} S(p_{\boldsymbol{\theta}(\boldsymbol{\eta})}). \tag{1.97}$$

Proof. Define the objective function $g_{\boldsymbol{\eta}}(\boldsymbol{\theta}) := F(\boldsymbol{\theta}, \beta) - \boldsymbol{\theta}^\top \boldsymbol{\eta}$ on $\operatorname{int} \Theta(\beta)$. By Corollary 1.14, because the family is minimal, $\nabla^2_{\boldsymbol{\theta}} F(\boldsymbol{\theta}, \beta) = -\beta \operatorname{Cov}_{p_\theta}(\boldsymbol{\phi}(X)) \prec 0$ is strictly negative definite everywhere on $\operatorname{int} \Theta(\beta)$. Thus $g_{\boldsymbol{\eta}}(\boldsymbol{\theta})$ is strictly concave in $\boldsymbol{\theta}$, and any stationary point is its unique global maximizer. Differentiating $g_{\boldsymbol{\eta}}(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$ via Eq. (1.79) gives

$$\nabla_{\boldsymbol{\theta}} g_{\boldsymbol{\eta}}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} F(\boldsymbol{\theta}, \beta) - \boldsymbol{\eta} = \mathbb{E}[\boldsymbol{\phi}(X)]_{p_\theta} - \boldsymbol{\eta} = \mathbf{0}, \tag{1.98}$$

which has the unique solution $\boldsymbol{\theta} = \boldsymbol{\theta}(\boldsymbol{\eta})$ for any $\boldsymbol{\eta} \in \mathcal{M}$, proving (i). Evaluating $g_{\boldsymbol{\eta}}(\boldsymbol{\theta})$ at $\boldsymbol{\theta} = \boldsymbol{\theta}(\boldsymbol{\eta})$ and noting that the internal energy is $U(\boldsymbol{\theta}(\boldsymbol{\eta}), \beta) = \mathbb{E}\left[\boldsymbol{\theta}(\boldsymbol{\eta})^\top \boldsymbol{\phi}(X) \right]_{p_{\boldsymbol{\theta}(\boldsymbol{\eta})}} = \boldsymbol{\theta}(\boldsymbol{\eta})^\top \boldsymbol{\eta}$, Theorem 1.4(i) gives

$$g_{\boldsymbol{\eta}}(\boldsymbol{\theta}(\boldsymbol{\eta})) = F(\boldsymbol{\theta}(\boldsymbol{\eta}), \beta) - U(\boldsymbol{\theta}(\boldsymbol{\eta}), \beta) = -TS(p_{\boldsymbol{\theta}(\boldsymbol{\eta})}), \tag{1.99}$$

which establishes Eq. (1.96). Finally, differentiating $-TS(p_{\boldsymbol{\theta}(\boldsymbol{\eta})}) = F(\boldsymbol{\theta}(\boldsymbol{\eta}), \beta) - \boldsymbol{\theta}(\boldsymbol{\eta})^\top \boldsymbol{\eta}$ with respect to $\boldsymbol{\eta}$ using the chain rule and $\nabla_{\boldsymbol{\theta}} F(\boldsymbol{\theta}(\boldsymbol{\eta}), \beta) = \boldsymbol{\eta}$ yields

$$-T \nabla_{\boldsymbol{\eta}} S(p_{\boldsymbol{\theta}(\boldsymbol{\eta})}) = \left(\frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{\eta}} \right)^\top \left[\nabla_{\boldsymbol{\theta}} F(\boldsymbol{\theta}(\boldsymbol{\eta}), \beta) - \boldsymbol{\eta} \right] - \boldsymbol{\theta}(\boldsymbol{\eta}) = -\boldsymbol{\theta}(\boldsymbol{\eta}), \tag{1.100}$$

so $\boldsymbol{\theta}(\boldsymbol{\eta}) = T \nabla_{\boldsymbol{\eta}} S(p_{\boldsymbol{\theta}(\boldsymbol{\eta})})$. Differentiating once more with respect to $\boldsymbol{\eta}$ and applying the inverse function theorem to $\frac{\partial \boldsymbol{\eta}}{\partial \boldsymbol{\theta}} = -\beta \operatorname{Cov}_{p_\theta}(\boldsymbol{\phi}(X))$ yields Eq. (1.97). □

Heuristic Derivation: Why Susceptibilities Diverge Only When Fluctuations Correlate Globally

Consider the static susceptibility matrix $\chi_{ab} := -\frac{\partial \langle \phi_a \rangle}{\partial \theta_b} = \beta \operatorname{Cov}_{p_\theta}(\phi_a, \phi_b)$ from Eq. (1.84). Take $\phi_a(\mathbf{s}) = -M(\mathbf{s})$, where $M(\mathbf{s}) = \sum_{i=1}^N s_i$ is the total magnetization of an N-spin hardware array, so that θ_a is a uniform external field h. Then the magnetic susceptibility per spin is

$$\chi := \frac{1}{N} \frac{\partial \langle M \rangle}{\partial h} = \frac{\beta}{N} \operatorname{Var}_{p_\theta}(M) = \frac{\beta}{N} \sum_{i=1}^N \sum_{j=1}^N \operatorname{Cov}_{p_\theta}(S_i, S_j). \tag{1.101}$$

Each individual spin has bounded variance $\operatorname{Var}(S_i) \leq 1$. Suppose the spins sit on a d-

dimensional lattice and their correlations decay beyond a finite correlation length ξ, so that $|\mathrm{Cov}_{p_\theta}(S_i, S_j)| \le Ce^{-\operatorname{dist}(i,j)/\xi}$. Then for each spin i the sum over j in Eq. (1.101) is bounded by a constant of order ξ^d, independent of N. Consequently, χ stays finite as $N \to \infty$.
A phase transition, signalled by a diverging second derivative of F/N, therefore requires $\xi \to \infty$ as $N \to \infty$: $\mathrm{Cov}_{p_\theta}(S_i, S_j)$ must remain non-negligible across the whole chip, so that of order N^2 terms in the double sum (1.101) contribute. This link between parameter sensitivity, correlation length, and mixing time reappears in Chapter 3 (spin-glass phase transitions) and Chapter 17 (Total Correlation regularization on sparse 2D silicon grids).

1.4 Maximum-Likelihood Learning of Energy-Based Models

Having established the forward map from an energy function $E_\theta(x)$ to its canonical equilibrium distribution $p_\theta(x)$, we now turn to the inverse problem, which is the central problem of both generative machine learning and hardware calibration:

> *Given an empirical target distribution $p_{\mathrm{data}}(x)$ (represented by a dataset of observed configurations $\mathcal{D} = \{x^{(1)}, x^{(2)}, \dots, x^{(M)}\}$ drawn i.i.d. from p_{data}), how should we adjust the physical parameters $\theta \in \Theta$ of the Hamiltonian $E_\theta(x)$ so that the canonical equilibrium distribution $p_\theta(x)$ of our thermodynamic computer matches $p_{\mathrm{data}}(x)$?*

1.4.1 KL Divergence Minimization and the Contrastive Gradient

The natural information-theoretic measure of discrepancy between the target data distribution p_{data} and the model equilibrium distribution p_θ is the forward Kullback-Leibler divergence $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta)$. Expanding it with Definition 1.7(ii) gives

$$\begin{aligned} D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta) &= \int_\Omega p_{\mathrm{data}}(x) \ln \frac{p_{\mathrm{data}}(x)}{p_\theta(x)} \, \mu(\mathrm{d}x) \\ &= \underbrace{\int_\Omega p_{\mathrm{data}}(x) \ln p_{\mathrm{data}}(x) \, \mu(\mathrm{d}x)}_{=-S(p_{\mathrm{data}})/k_{\mathrm{B}} \text{ (independent of } \theta)} - \underbrace{\int_\Omega p_{\mathrm{data}}(x) \ln p_\theta(x) \, \mu(\mathrm{d}x)}_{:=\mathcal{L}(\theta) \text{ (expected log-likelihood)}} . \end{aligned} \tag{1.102}$$

Because the data entropy $S(p_{\mathrm{data}})$ does not depend on the model parameters θ, minimizing $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta)$ with respect to θ is mathematically identical to maximizing the expected log-likelihood functional

$$\mathcal{L}(\theta) := \mathbb{E}[\ln p_\theta(X)]_{p_{\mathrm{data}}} = -\beta \mathbb{E}[E_\theta(X)]_{p_{\mathrm{data}}} - \ln Z(\theta, \beta) = \beta(F(\theta, \beta) - \mathbb{E}[E_\theta(X)]_{p_{\mathrm{data}}}). \tag{1.103}$$

The last form of Eq. (1.103) has a direct reading: maximizing $\mathcal{L}(\theta)$ lowers the average energy of the data $\mathbb{E}[E_\theta]_{p_{\mathrm{data}}}$ relative to the equilibrium free energy $F(\theta, \beta)$. The same statement follows from Theorem 1.9, which gives $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta) = \beta(\mathcal{F}[p_{\mathrm{data}}] - F(\theta, \beta))$ with $\mathcal{F}[p_{\mathrm{data}}] = \mathbb{E}[E_\theta]_{p_{\mathrm{data}}} - TS(p_{\mathrm{data}})$; the entropy term does not depend on θ.

We now derive the gradient and Hessian of $\mathcal{L}(\theta)$ with respect to θ.

Theorem 1.16 (Contrastive Gradient and Fisher Hessian of EBMs)**.** *Let $\theta \mapsto E_\theta(x)$ satisfy the differentiability and integrability hypotheses of Theorem 1.13, and let $p_{\mathrm{data}} \in \mathcal{P}(\Omega)$ be a target distribution such that $\mathbb{E}[\|\nabla_\theta E_\theta(X)\|]_{p_{\mathrm{data}}} < \infty$.*

(i) ***Contrastive Log-Likelihood Gradient:*** *The gradient of the expected log-likelihood* $\mathcal{L}(\theta) = \mathbb{E}[\ln p_\theta(X)]_{p_{\text{data}}} = -D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta) + \text{const}$ *with respect to* θ *is given by*

$$\nabla_\theta \mathcal{L}(\theta) = -\nabla_\theta D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta) = -\beta\Big(\underbrace{\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_{\text{data}}}}_{\textit{Positive (Clamped) Phase}} - \underbrace{\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_\theta}}_{\textit{Negative (Free / Thermodynamic) Phase}} \Big). \tag{1.104}$$

(ii) ***Moment-Matching Condition for Linear Exponential Families:*** *If* $E_\theta(x) = E_{\text{base}}(x) + \sum_{a=1}^{P} \theta_a \phi_a(x)$*, then* $\mathcal{L}(\boldsymbol{\theta})$ *is globally concave on* $\Theta(\beta)$*, with gradient and Hessian*

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_a} = -\beta\Big(\mathbb{E}[\phi_a(X)]_{p_{\text{data}}} - \mathbb{E}[\phi_a(X)]_{p_\theta}\Big), \tag{1.105}$$

$$\frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_a \partial \theta_b} = -\frac{\partial^2 \ln Z(\boldsymbol{\theta}, \beta)}{\partial \theta_a \partial \theta_b} = -\beta^2 \operatorname{Cov}_{p_\theta}(\phi_a(X), \phi_b(X)). \tag{1.106}$$

Consequently, any stationary point $\boldsymbol{\theta}^* \in \Theta(\beta)$ *of* $\mathcal{L}(\boldsymbol{\theta})$ *is a global maximizer of the log-likelihood and satisfies the exact moment-matching condition*

$$\mathbb{E}[\phi_a(X)]_{p_{\boldsymbol{\theta}^*}} = \mathbb{E}[\phi_a(X)]_{p_{\text{data}}} \qquad \forall\, a = 1, \ldots, P. \tag{1.107}$$

Proof. We differentiate $\mathcal{L}(\theta)$ in Eq. (1.103) component by component with respect to θ_a. Since $p_{\text{data}}(x)$ does not depend on θ, differentiating the first term under the p_{data}-integral yields

$$\frac{\partial}{\partial \theta_a}\left(-\beta \int_\Omega E_\theta(x)\, p_{\text{data}}(x)\, \mu(\mathrm{d}x)\right) = -\beta \int_\Omega \frac{\partial E_\theta(x)}{\partial \theta_a}\, p_{\text{data}}(x)\, \mu(\mathrm{d}x) = -\beta \mathbb{E}\left[\frac{\partial E_\theta(X)}{\partial \theta_a}\right]_{p_{\text{data}}}. \tag{1.108}$$

For the second term $-\ln Z(\theta, \beta)$ in Eq. (1.103), we invoke the static Fluctuation-Response identity (1.79) proven in Theorem 1.13(i):

$$-\frac{\partial \ln Z(\theta, \beta)}{\partial \theta_a} = +\beta \mathbb{E}\left[\frac{\partial E_\theta(X)}{\partial \theta_a}\right]_{p_\theta}. \tag{1.109}$$

Adding Eqs. (1.108) and (1.109) establishes the contrastive gradient formula (1.104).

When $E_\theta(x) = E_{\text{base}}(x) + \sum_{a=1}^{P} \theta_a \phi_a(x)$, we have $\frac{\partial E_\theta(x)}{\partial \theta_a} = \phi_a(x)$, which immediately reduces Eq. (1.104) to Eq. (1.105). Differentiating Eq. (1.105) with respect to θ_b, the data expectation $\mathbb{E}[\phi_a(X)]_{p_{\text{data}}}$ is constant in $\boldsymbol{\theta}$ and vanishes, leaving only the derivative of $-\ln Z(\boldsymbol{\theta}, \beta)$, which by Theorem 1.13(ii) equals $-\beta^2 \operatorname{Cov}_{p_\theta}(\phi_a(X), \phi_b(X))$. Since $\operatorname{Cov}_{p_\theta}(\boldsymbol{\phi}(X)) \succeq 0$, the Hessian $\nabla^2_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$ is negative semidefinite everywhere on $\Theta(\beta)$, proving global concavity of $\mathcal{L}(\boldsymbol{\theta})$ and the optimality of the moment-matching condition (1.107). □

1.4.2 Physical Meaning of the Positive and Negative Phases

Consider how gradient ascent on the log-likelihood, $\Delta\theta = \eta \nabla_\theta \mathcal{L}(\theta)$ with learning rate $\eta > 0$, reshapes the energy landscape $E_\theta(x)$:

$$\Delta\theta = -\eta\beta\Big(\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_{\text{data}}} - \mathbb{E}[\nabla_\theta E_\theta(X)]_{p_\theta}\Big). \tag{1.110}$$

To see the effect of this parameter change on the energy $E_\theta(x)$ of a configuration $x \in \Omega$, take the first-order Taylor expansion $\Delta E_\theta(x) \approx (\nabla_\theta E_\theta(x))^\top \Delta\theta$:

1. **The Positive (Clamped) Phase ($-\eta\beta\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_{\text{data}}}$):** During this phase, the system is clamped to configurations $x_{\text{data}} \sim p_{\text{data}}$ observed from the training dataset. The term $-\eta\beta\nabla_\theta E_\theta(x_{\text{data}})$ pushes θ in the direction that *decreases* $E_\theta(x_{\text{data}})$, carving deep energy valleys at the locations of real data samples.

2. **The Negative (Free / Thermodynamic) Phase ($+\eta\beta\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_\theta}$):** If we only executed the positive phase, we could trivially minimize $\mathbb{E}[E_\theta(X)]_{p_{\text{data}}}$ by sending $E_\theta(x) \to -\infty$ everywhere. What prevents this collapse is the partition function $Z(\theta, \beta)$, whose derivative supplies the negative phase. During the negative phase, the system is disconnected from external data clamping and allowed to thermalize freely under its own Hamiltonian E_θ, generating equilibrium samples $x_{\text{model}} \sim p_\theta$. The update $+\eta\beta\nabla_\theta E_\theta(x_{\text{model}})$ *raises* the energy of configurations wherever the model currently assigns high probability. This fills in spurious energy wells (the model's "fantasies") until the valleys of p_θ coincide with the support of p_{data}.

In standard digital machine learning, computing the positive phase expectation $\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_{\text{data}}}$ over a mini-batch of data is straightforward because the data points $x^{(m)} \sim p_{\text{data}}$ are already given. Computing the negative phase expectation $\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_\theta}$, however, requires drawing independent equilibrium samples from the model's current Boltzmann distribution $p_\theta(x)$. Because $Z(\theta, \beta)$ is intractable, digital implementations must resort to Markov Chain Monte Carlo (MCMC) methods (Chapter 2) or Langevin dynamics (Chapter 5).

To avoid running a long Markov chain from scratch at every gradient step, Hinton [50] introduced *k-step Contrastive Divergence (CD-k)*, which initializes a short Markov chain at a training data point $x^{(0)} \sim p_{\text{data}}$, applies k steps of a transition kernel T_θ (typically $k = 1$ to 10), and replaces the equilibrium expectation $\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_\theta}$ with the k-step distribution p_k:

$$\nabla_\theta \mathcal{L}_{\text{CD-}k}(\theta) := -\beta\Big(\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_{\text{data}}} - \mathbb{E}[\nabla_\theta E_\theta(X)]_{p_k}\Big). \tag{1.111}$$

Alternatively, *Persistent Contrastive Divergence (PCD)* maintains a persistent population of Markov chains across parameter updates, relying on the assumption that a small learning rate η changes p_θ slowly enough that the chains remain near equilibrium. Both methods fail when $E_\theta(x)$ develops several modes separated by free-energy barriers $\Delta F \gg k_{\text{B}}T$: as Chapters 3 and 7 show, the mixing time then grows as $\tau_{\text{mix}} \propto \exp(\beta\Delta F)$. Short CD-$k$ chains started near the data never reach distant spurious wells, which therefore go unpenalized, while PCD chains become trapped in metastable valleys.

Hardware Realization: Why the Negative Phase Is Expensive on Digital Hardware, and How an Analog Sampler Executes It

Why is the negative phase $\mathbb{E}[\nabla_\theta E_\theta(X)]_{p_\theta}$ so expensive on a digital processor such as a GPU, and what changes when the sampling is done by a physical circuit?
Consider a single update of a node i coupled to d neighbors in a graphical model. On a synchronous digital processor [52, 56]:

1. **Memory movement:** Fetching d 16-bit weights θ_{ij} and the neighbor states from on-chip SRAM or High-Bandwidth Memory (HBM) costs $E_{\text{mem}} \sim 5$–$50\,\text{pJ}$ per word ($1\,\text{pJ} \approx 2.4 \times 10^8\, k_{\text{B}}T$ at $T = 300\,\text{K}$), spent mostly on charging long interconnect wires.

2. **Arithmetic:** Computing the local field $\sum_j \theta_{ij} x_j$ and evaluating $\tanh(\cdot)$ or $\exp(\cdot)$ switches many strong-inversion CMOS gates ($V_{\text{DD}} \approx 0.7$–$0.9\,\text{V}$), at ~ 0.5–$2\,\text{pJ}$ per multiply-accumulate (MAC) operation.

3. **Pseudorandom numbers:** Generating one uniform or Gaussian pseudorandom number (for example with a Philox generator followed by a Box–Muller transform) takes tens to hundreds of integer and floating-point operations, roughly 1–10 pJ ($\sim 10^9\, k_{\mathrm{B}}T$), to imitate a single thermal kick.

In an analog sampler built from subthreshold CMOS, of the kind described by Extropic [30, 56], the same update looks different:

- **No weight fetching:** The parameters θ_{ij} are held locally as bias voltages or currents of transconductance cells next to the state nodes. By Kirchhoff's current law, the local field current $I_i = \sum_j G_{ij} v_j$ is summed on the node itself, with no transfer over a memory bus.
- **Boltzmann statistics from device physics:** A MOSFET biased in weak inversion ($V_{\mathrm{GS}} < V_{\mathrm{th}}$, Chapter 12) conducts a drain current set by thermally activated diffusion over a source–channel barrier, $I_{\mathrm{D}} = I_0 \exp((V_{\mathrm{GS}} - V_{\mathrm{th}})/(nV_T))$. The exponential is not computed; it is the Boltzmann distribution of the electrons themselves.
- **Noise at no arithmetic cost:** The fluctuations needed for sampling come from the thermal and shot noise of the transistors (Chapters 11 and 13). As a representative estimate made in this book (Chapter 17 develops such budgets in detail), a node capacitance $C_{\mathrm{node}} \sim 0.5$–$2$ fF equilibrates in $\tau_{\mathrm{step}} \sim 5$–$20$ ns at a cost of ~ 10–50 aJ $\approx 2.5 \times 10^3$–$10^4\, k_{\mathrm{B}}T$ per update.

On these representative numbers a single analog update is five to six orders of magnitude cheaper than its digital counterpart. System-level comparisons, which include control, readout, and the number of updates needed per sample, give smaller ratios: for the DTCA, Jelinčič et al. [56] estimate roughly 10^4 lower energy per sample than a GPU on small image-generation benchmarks.

1.5 Latent Variables and Hybrid Effective Potentials

In a silicon circuit, individual devices implement simple local energy terms. Networks of capacitors and conductances implement positive-definite quadratic forms, as in the RC sampler, whose Boltzmann distribution is a unimodal Gaussian. Transconductance cells and current-summing nodes implement bilinear couplings $v_i W_{ij} s_j$ between continuous voltages $\mathbf{v} \in \mathbb{R}^N$ and p-bit states $\mathbf{s} \in \{-1, +1\}^M$, or $J_{ij} s_i s_j$ between p-bits. If a chip offers only quadratic self-energies and bilinear couplings, how can it represent a non-Gaussian, multimodal distribution over a continuous visible space $\mathbb{R}^N$?

The answer is *marginalization over latent (hidden) variables*. When we partition the microscopic degrees of freedom of a physical system into a visible subset $v \in \Omega_V$ (the variables whose marginal statistics we observe or clamp to data) and a hidden or auxiliary subset $h \in \Omega_H$ (internal degrees of freedom that fluctuate in thermal equilibrium with v), integrating out the hidden variables h replaces the bare joint Hamiltonian $E_\theta(v, h)$ with a temperature-dependent *effective free-energy potential* $F_{\mathrm{eff}}(v; \theta, \beta)$. Even when $E_\theta(v, h)$ is strictly convex in v for every fixed h, the thermal fluctuations of h induce an effective attractive interaction that can bend $F_{\mathrm{eff}}(v; \theta, \beta)$ into a non-convex landscape with multiple local minima.

1.5.1 Joint Ensembles, Marginal Distributions, and the Effective Potential

Let the total state space be a product space $\Omega = \Omega_V \times \Omega_H$ equipped with the product reference measure $\mu(\mathrm{d}v, \mathrm{d}h) = \mu_V(\mathrm{d}v) \otimes \mu_H(\mathrm{d}h)$.

Definition 1.17 (Latent-Variable EBM and Effective Free-Energy Potential)**.** Let $E_\theta : \Omega_V \times \Omega_H \to \mathbb{R}$ be a joint Hamiltonian that is thermodynamically admissible at inverse temperature $\beta = (k_{\mathrm{B}}T)^{-1} > 0$, with joint partition function

$$Z(\theta, \beta) = \int_{\Omega_V} \int_{\Omega_H} \exp(-\beta E_\theta(v, h))\, \mu_H(\mathrm{d}h)\, \mu_V(\mathrm{d}v) < \infty, \tag{1.112}$$

and joint canonical distribution

$$p_\theta(v, h) = \frac{1}{Z(\theta, \beta)} \exp(-\beta E_\theta(v, h)). \tag{1.113}$$

For each fixed visible configuration $v \in \Omega_V$, we define:

(i) The *clamped (conditional) partition function*:

$$Z_H(v; \theta, \beta) := \int_{\Omega_H} \exp(-\beta E_\theta(v, h))\, \mu_H(\mathrm{d}h). \tag{1.114}$$

(ii) The *effective free-energy potential* (or *potential of mean force*) over the visible space Ω_V:

$$F_{\mathrm{eff}}(v; \theta, \beta) := -k_{\mathrm{B}}T \ln Z_H(v; \theta, \beta) = -\frac{1}{\beta} \ln \int_{\Omega_H} \exp(-\beta E_\theta(v, h))\, \mu_H(\mathrm{d}h). \tag{1.115}$$

From Definition 1.17, the marginal distribution $p_\theta(v)$ of the visible variables and the conditional posterior distribution $p_\theta(h \mid v)$ of the hidden variables given v take standard canonical forms:

$$p_\theta(v) = \int_{\Omega_H} p_\theta(v, h)\, \mu_H(\mathrm{d}h) = \frac{Z_H(v; \theta, \beta)}{Z(\theta, \beta)} = \frac{1}{Z(\theta, \beta)} \exp(-\beta F_{\mathrm{eff}}(v; \theta, \beta)), \tag{1.116}$$

$$p_\theta(h \mid v) = \frac{p_\theta(v, h)}{p_\theta(v)} = \frac{\exp(-\beta E_\theta(v, h))}{Z_H(v; \theta, \beta)} = \exp\Big(\beta\big[F_{\mathrm{eff}}(v; \theta, \beta) - E_\theta(v, h)\big]\Big), \tag{1.117}$$

where the full partition function is the visible integral of the effective Boltzmann factor:

$$Z(\theta, \beta) = \int_{\Omega_V} \exp(-\beta F_{\mathrm{eff}}(v; \theta, \beta))\, \mu_V(\mathrm{d}v). \tag{1.118}$$

Equation (1.116) shows that the marginal distribution $p_\theta(v)$ is itself an exact canonical Boltzmann-Gibbs distribution on Ω_V, governed not by a bare mechanical energy, but by the clamped Helmholtz free energy $F_{\mathrm{eff}}(v; \theta, \beta)$ of the hidden subsystem H held at fixed v. Applying Theorem 1.4(i) to the conditional ensemble $p_\theta(h \mid v)$ at fixed v, we can decompose $F_{\mathrm{eff}}(v; \theta, \beta)$ into a conditional internal energy and a conditional entropy:

$$F_{\mathrm{eff}}(v; \theta, \beta) = \mathbb{E}[E_\theta(v, H)]_{p_\theta(h|v)} - TS(p_\theta(\cdot \mid v)). \tag{1.119}$$

Visible configurations v that leave the hidden variables free to fluctuate, with high conditional entropy $S(p_\theta(\cdot \mid v))$, therefore have lower effective free energy $F_{\mathrm{eff}}(v; \theta, \beta)$. The resulting drive toward such configurations is called an *entropic force*.

1.5.2 First and Second Derivatives of the Effective Potential: How Fluctuations Carve Non-Convex Wells

Because $F_{\text{eff}}(v;\theta,\beta) = -k_{\text{B}}T \ln Z_H(v;\theta,\beta)$ is the Helmholtz free energy of the conditional ensemble $p_\theta(h \mid v)$, we can apply the static Fluctuation-Response Theorem (Theorem 1.13) directly to $F_{\text{eff}}(v;\theta,\beta)$, treating either the model parameters θ or the continuous visible state $\mathbf{v} \in \mathbb{R}^N$ itself as parameters of the conditional partition function $Z_H(\mathbf{v};\theta,\beta)$.

Theorem 1.18 (Derivatives of the Effective Potential and Curvature Reduction by Latent Fluctuations)**.** *Let $\Omega_V = \mathbb{R}^N$ and let $E_\theta(\mathbf{v},h)$ be twice continuously differentiable with respect to $(\mathbf{v},\theta)$, satisfying dominated integrability over Ω_H.*

(i) ***Effective Mean Force (Visible Gradient):*** *For any visible state $\mathbf{v} \in \mathbb{R}^N$, the gradient of the effective free-energy potential with respect to $\mathbf{v}$ equals the conditional expectation of the bare force:*

$$\nabla_{\mathbf{v}} F_{\text{eff}}(\mathbf{v};\theta,\beta) = \mathbb{E}[\nabla_{\mathbf{v}} E_\theta(\mathbf{v},H)]_{p_\theta(h|\mathbf{v})}\,. \tag{1.120}$$

(ii) ***Effective Stiffness Matrix (Visible Hessian):*** *For any $\mathbf{v} \in \mathbb{R}^N$, the Hessian matrix of $F_{\text{eff}}(\mathbf{v};\theta,\beta)$ with respect to $\mathbf{v}$ is given by*

$$\nabla^2_{\mathbf{v}} F_{\text{eff}}(\mathbf{v};\theta,\beta) = \mathbb{E}\Big[\nabla^2_{\mathbf{v}} E_\theta(\mathbf{v},H)\Big]_{p_\theta(h|\mathbf{v})} - \beta\,\mathrm{Cov}_{p_\theta(h|\mathbf{v})}\Big(\nabla_{\mathbf{v}} E_\theta(\mathbf{v},H),\, \nabla_{\mathbf{v}} E_\theta(\mathbf{v},H)^\top\Big). \tag{1.121}$$

(iii) ***Latent-Variable Contrastive Learning Gradient:*** *For any target visible data distribution $p_{\text{data}}(\mathbf{v})$, the gradient of the marginal log-likelihood $\mathcal{L}(\theta) = \mathbb{E}[\ln p_\theta(\mathbf{V})]_{p_{\text{data}}}$ with respect to θ is*

$$\begin{aligned}\nabla_\theta \mathcal{L}(\theta) &= -\beta\Big(\mathbb{E}[\nabla_\theta F_{\text{eff}}(\mathbf{V};\theta,\beta)]_{p_{\text{data}}(\mathbf{v})} - \mathbb{E}[\nabla_\theta F_{\text{eff}}(\mathbf{V};\theta,\beta)]_{p_\theta(\mathbf{v})}\Big) \\ &= -\beta\Big(\underbrace{\mathbb{E}\Big[\mathbb{E}[\nabla_\theta E_\theta(\mathbf{V},H)]_{p_\theta(h|\mathbf{V})}\Big]_{p_{\text{data}}(\mathbf{v})}}_{\textit{Clamped Positive Phase}} - \underbrace{\mathbb{E}[\nabla_\theta E_\theta(\mathbf{V},H)]_{p_\theta(\mathbf{v},h)}}_{\textit{Unclamped Joint Negative Phase}}\Big).\end{aligned} \tag{1.122}$$

Proof. Viewing $\mathbf{v} \in \mathbb{R}^N$ as a parameter vector in the conditional partition function $Z_H(\mathbf{v};\theta,\beta) = \int_{\Omega_H} e^{-\beta E_\theta(\mathbf{v},h)}\,\mu_H(\mathrm{d}h)$ with free energy $F_{\text{eff}}(\mathbf{v};\theta,\beta) = -k_{\text{B}}T \ln Z_H(\mathbf{v};\theta,\beta)$, we apply Eqs. (1.79) and (1.81) of Theorem 1.13(i) with θ_a, θ_b replaced by v_i, v_j and p_θ replaced by the conditional distribution $p_\theta(h \mid \mathbf{v})$. This yields Eqs. (1.120) and (1.121) directly.

To prove (iii), we apply Theorem 1.16(i) to the visible marginal distribution

$$p_\theta(\mathbf{v}) = Z(\theta,\beta)^{-1} e^{-\beta F_{\text{eff}}(\mathbf{v};\theta,\beta)},$$

obtaining the first line of Eq. (1.122). Next, applying Eq. (1.79) to

$$F_{\text{eff}}(\mathbf{v};\theta,\beta) = -k_{\text{B}}T \ln Z_H(\mathbf{v};\theta,\beta)$$

with respect to θ gives

$$\nabla_\theta F_{\text{eff}}(\mathbf{v};\theta,\beta) = \mathbb{E}[\nabla_\theta E_\theta(\mathbf{v},H)]_{p_\theta(h|\mathbf{v})}\,. \tag{1.123}$$

Taking the expectation of Eq. (1.123) over $\mathbf{V} \sim p_{\text{data}}(\mathbf{v})$ gives the positive-phase term in Eq. (1.122). Taking the expectation of Eq. (1.123) over $\mathbf{V} \sim p_\theta(\mathbf{v})$ and using the law of total expectation $p_\theta(\mathbf{v})p_\theta(h \mid \mathbf{v}) = p_\theta(\mathbf{v},h)$ gives

$$\mathbb{E}\Big[\mathbb{E}[\nabla_\theta E_\theta(\mathbf{V},H)]_{p_\theta(h|\mathbf{V})}\Big]_{p_\theta(\mathbf{v})} = \int_{\Omega_V} p_\theta(\mathbf{v}) \left(\int_{\Omega_H} \nabla_\theta E_\theta(\mathbf{v},h)\, p_\theta(h \mid \mathbf{v})\, \mu_H(\mathrm{d}h)\right) \mu_V(\mathrm{d}\mathbf{v}) \tag{1.124}$$

$$= \mathbb{E}[\nabla_\theta E_\theta(\mathbf{V},H)]_{p_\theta(\mathbf{v},h)}\,, \tag{1.125}$$

which yields the unclamped joint negative phase in Eq. (1.122) and completes the proof. □

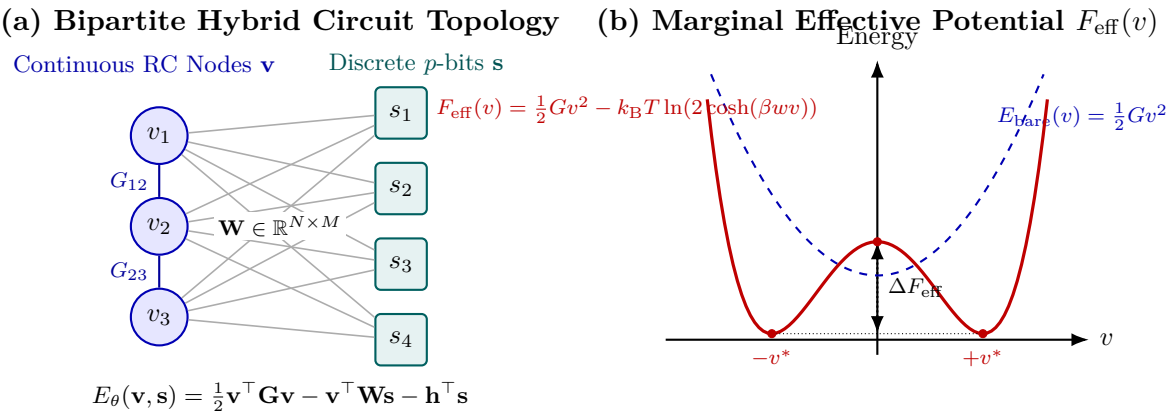


Figure 1.1: Marginalizing over p-bits turns a convex quadratic energy into a multimodal effective potential. (a) A hybrid bipartite circuit coupling N continuous capacitor voltage nodes $\mathbf{v} \in \mathbb{R}^N$ (with stiffness matrix $\mathbf{G} \succ 0$) to M conditionally independent discrete p-bits $\mathbf{s} \in \{-1,+1\}^M$ via bilinear transconductance weights $\mathbf{W}$. (b) Integrating out a single p-bit subtracts the free-energy contribution $k_B T \ln(2\cosh(\beta w v))$ from the bare convex parabola $\frac{1}{2}Gv^2$. Whenever $\beta w^2 > G$, the fluctuation-induced negative curvature $-\beta w^2 \operatorname{sech}^2(\beta w v)$ at $v = 0$ overcomes the bare stiffness G and turns the origin into a barrier of height ΔF_{eff} separating two symmetric wells at $\pm v^*$.

The second-derivative identity (1.121) has two parts:

$$\nabla^2_{\mathbf{v}} F_{\text{eff}}(\mathbf{v}) = \underbrace{\mathbb{E}\Big[\nabla^2_{\mathbf{v}} E_\theta(\mathbf{v},H)\Big]_{p_\theta(h|\mathbf{v})}}_{\text{Bare mechanical stiffness}} - \underbrace{\beta\,\mathrm{Cov}_{p_\theta(h|\mathbf{v})}\Big(\nabla_{\mathbf{v}} E_\theta(\mathbf{v},H),\, \nabla_{\mathbf{v}} E_\theta(\mathbf{v},H)^\top\Big)}_{\text{Fluctuation-induced softening } \succeq 0}. \tag{1.126}$$

Because the covariance matrix of the coupling force $\nabla_{\mathbf{v}} E_\theta(\mathbf{v},H)$ under $p_\theta(h \mid \mathbf{v})$ is positive semidefinite, subtracting $\beta\,\mathrm{Cov}_{p_\theta(h|\mathbf{v})}(\nabla_{\mathbf{v}} E_\theta)$ can never raise the curvature of F_{eff} in any direction above the bare curvature $\mathbb{E}[\nabla^2_{\mathbf{v}} E_\theta]$. If the coupling between $\mathbf{v}$ and H is strong enough that this softening exceeds the bare stiffness in some region of $\mathbb{R}^N$, the Hessian of F_{eff} acquires negative eigenvalues there. The bottom of the bare potential then turns into a local maximum or saddle point, and the single minimum splits into several wells.

This mechanism, illustrated in Fig. 1.1, is a basic design principle of hybrid thermodynamic hardware [31, 56]. The following worked computation carries it out for a circuit that couples the RC sampler to an array of p-bits, and shows that integrating out the p-bits produces the double-well sampler.

Worked computation 1.19 (The Gaussian–Bernoulli EBM: Conditionals, Effective Potentials, Induced Couplings, and Learning Gradients)**.** Consider a hybrid thermodynamic circuit consisting of N continuous analog voltage nodes $\mathbf{v} = (v_1,\dots,v_N)^\top \in \mathbb{R}^N$, as in the RC sampler, coupled to M probabilistic bits $\mathbf{s} = (s_1,\dots,s_M)^\top \in \{-1,+1\}^M$, the binary limit of the double-well sampler. The joint state space is $\Omega = \mathbb{R}^N \times \{-1,+1\}^M$ with reference measure $\mu(\mathrm{d}\mathbf{v},\mathrm{d}\mathbf{s}) = (\mathrm{d}^N v / V_0^N) \otimes \mu_{\text{count}}(\mathrm{d}\mathbf{s})$ (for algebraic simplicity we set the reference voltage scale to $V_0 = 1$). The system is governed by the joint Gaussian-Bernoulli Hamiltonian

$$E_\theta(\mathbf{v},\mathbf{s}) = \frac{1}{2}(\mathbf{v}-\mathbf{b})^\top\mathbf{G}(\mathbf{v}-\mathbf{b}) - \mathbf{v}^\top\mathbf{W}\mathbf{s} - \mathbf{h}^\top\mathbf{s}, \tag{1.127}$$

where the parameter tuple $\theta = (\mathbf{G},\mathbf{b},\mathbf{W},\mathbf{h})$ consists of:

- a symmetric, positive-definite stiffness matrix $\mathbf{G} = \mathbf{G}^\top \succ 0$ in $\mathbb{R}^{N\times N}$ (it plays the role of $\mathbf{K}$ in Example 1.5; see Section 5.6 for when a conductance matrix can set it),
- a continuous voltage bias vector $\mathbf{b} \in \mathbb{R}^N$,
- a bilinear coupling (transconductance) matrix $\mathbf{W} \in \mathbb{R}^{N\times M}$ with columns $\mathbf{w}_j \equiv \mathbf{W}_{:,j} \in \mathbb{R}^N$ $(j = 1, \dots, M)$, and
- a discrete p-bit bias vector $\mathbf{h} = (h_1, \dots, h_M)^\top \in \mathbb{R}^M$.

There are no direct couplings $J_{jk}s_j s_k$ between the p-bits in Eq. (1.127); the p-bits interact only through the continuous nodes $\mathbf{v}$, forming the bipartite graph of Fig. 1.1(a). We derive (1) both conditional distributions, (2) the effective free energy $F_{\text{eff}}(\mathbf{v})$ and its double-well bifurcation, (3) the effective Ising Hamiltonian $F_{\text{eff}}(\mathbf{s})$ obtained by integrating out $\mathbf{v}$, and (4) the contrastive learning rules.

Part 1: Exact Conditional Distributions $p_\theta(\mathbf{s} \mid \mathbf{v})$ and $p_\theta(\mathbf{v} \mid \mathbf{s})$.

First, clamp the continuous voltages $\mathbf{v} \in \mathbb{R}^N$ and find the conditional distribution of the p-bits $\mathbf{s}$. Writing the bilinear coupling as a sum over the M columns $\mathbf{w}_j$ of $\mathbf{W}$, $\mathbf{v}^\top\mathbf{W}\mathbf{s} = \sum_{j=1}^M (\mathbf{w}_j^\top\mathbf{v})s_j$, the joint energy (1.127) decomposes as

$$E_\theta(\mathbf{v}, \mathbf{s}) = \frac{1}{2}(\mathbf{v} - \mathbf{b})^\top\mathbf{G}(\mathbf{v} - \mathbf{b}) - \sum_{j=1}^M (\mathbf{w}_j^\top\mathbf{v} + h_j)s_j. \tag{1.128}$$

The clamped partition function over $\mathbf{s} \in \{-1, +1\}^M$ at fixed $\mathbf{v}$ factorizes into a product of M independent two-state sums:

$$\begin{aligned} Z_S(\mathbf{v}; \theta, \beta) &= \sum_{\mathbf{s}\in\{-1,+1\}^M} \exp\big(-\beta E_\theta(\mathbf{v}, \mathbf{s})\big) \\ &= \exp\left(-\frac{\beta}{2}(\mathbf{v} - \mathbf{b})^\top\mathbf{G}(\mathbf{v} - \mathbf{b})\right) \sum_{s_1=\pm1} \cdots \sum_{s_M=\pm1} \prod_{j=1}^M \exp\Big(\beta(\mathbf{w}_j^\top\mathbf{v} + h_j)s_j\Big) \\ &= \exp\left(-\frac{\beta}{2}(\mathbf{v} - \mathbf{b})^\top\mathbf{G}(\mathbf{v} - \mathbf{b})\right) \prod_{j=1}^M \left[\sum_{s_j\in\{-1,+1\}} \exp\Big(\beta(\mathbf{w}_j^\top\mathbf{v} + h_j)s_j\Big)\right] \\ &= \exp\left(-\frac{\beta}{2}(\mathbf{v} - \mathbf{b})^\top\mathbf{G}(\mathbf{v} - \mathbf{b})\right) \prod_{j=1}^M \Big[2\cosh\Big(\beta(\mathbf{w}_j^\top\mathbf{v} + h_j)\Big)\Big]. \end{aligned} \tag{1.129}$$

Dividing $\exp(-\beta E_\theta(\mathbf{v}, \mathbf{s}))$ by $Z_S(\mathbf{v}; \theta, \beta)$, the quadratic $\mathbf{v}$-factor cancels identically and the conditional distribution of $\mathbf{s}$ given $\mathbf{v}$ factorizes completely across the M p-bits:

$$p_\theta(\mathbf{s} \mid \mathbf{v}) = \prod_{j=1}^M p_\theta(s_j \mid \mathbf{v}), \qquad p_\theta(s_j \mid \mathbf{v}) = \frac{\exp\big(\beta(\mathbf{w}_j^\top\mathbf{v} + h_j)s_j\big)}{2\cosh\big(\beta(\mathbf{w}_j^\top\mathbf{v} + h_j)\big)} = \sigma\Big(2\beta s_j(\mathbf{w}_j^\top\mathbf{v} + h_j)\Big). \tag{1.130}$$

Thus, conditioned on the continuous analog nodes $\mathbf{v}$, all M p-bits are statistically independent and can be sampled simultaneously in parallel. From Example 1.6, the conditional mean and conditional covariance of the p-bits are

$$\mathbb{E}[\mathbf{S} \mid \mathbf{v}]_{p_\theta} = \tanh\big(\beta(\mathbf{W}^\top\mathbf{v} + \mathbf{h})\big), \tag{1.131}$$

$$\mathrm{Cov}_{p_\theta}(\mathbf{S} \mid \mathbf{v}) = \mathrm{diag}\Big(\mathrm{sech}^2\big(\beta(\mathbf{w}_1^\top\mathbf{v} + h_1)\big), \dots, \mathrm{sech}^2\big(\beta(\mathbf{w}_M^\top\mathbf{v} + h_M)\big)\Big), \tag{1.132}$$

where $\tanh(\cdot)$ is applied component-wise.

Second, clamp the p-bits $\mathbf{s} \in \{-1,+1\}^M$ and find the conditional distribution $p_\theta(\mathbf{v} \mid \mathbf{s})$ of the continuous voltages. We complete the square in $\mathbf{v}$ inside Eq. (1.127). Let $\mathbf{u} := \mathbf{v} - \mathbf{b}$, so that $\mathbf{v} = \mathbf{u} + \mathbf{b}$. Substituting this into $E_\theta(\mathbf{v}, \mathbf{s})$ gives

$$E_\theta(\mathbf{v}, \mathbf{s}) = \frac{1}{2}\mathbf{u}^\top \mathbf{G}\mathbf{u} - \mathbf{u}^\top(\mathbf{W}\mathbf{s}) - \mathbf{b}^\top \mathbf{W}\mathbf{s} - \mathbf{h}^\top \mathbf{s}. \tag{1.133}$$

Since $\mathbf{G} = \mathbf{G}^\top \succ 0$ is invertible, we can write the linear term as $\mathbf{u}^\top(\mathbf{W}\mathbf{s}) = \mathbf{u}^\top \mathbf{G}(\mathbf{G}^{-1}\mathbf{W}\mathbf{s})$ and complete the quadratic form:

$$\begin{aligned}\frac{1}{2}\mathbf{u}^\top \mathbf{G}\mathbf{u} - \mathbf{u}^\top \mathbf{G}(\mathbf{G}^{-1}\mathbf{W}\mathbf{s}) &= \frac{1}{2}(\mathbf{u} - \mathbf{G}^{-1}\mathbf{W}\mathbf{s})^\top \mathbf{G}(\mathbf{u} - \mathbf{G}^{-1}\mathbf{W}\mathbf{s}) - \frac{1}{2}(\mathbf{G}^{-1}\mathbf{W}\mathbf{s})^\top \mathbf{G}(\mathbf{G}^{-1}\mathbf{W}\mathbf{s}) \\ &= \frac{1}{2}(\mathbf{v} - \boldsymbol{\mu}(\mathbf{s}))^\top \mathbf{G}(\mathbf{v} - \boldsymbol{\mu}(\mathbf{s})) - \frac{1}{2}\mathbf{s}^\top \mathbf{W}^\top \mathbf{G}^{-1}\mathbf{W}\mathbf{s}, \end{aligned} \tag{1.134}$$

where we defined the $\mathbf{s}$-dependent conditional mean voltage vector

$$\boldsymbol{\mu}(\mathbf{s}) := \mathbf{b} + \mathbf{G}^{-1}\mathbf{W}\mathbf{s}. \tag{1.135}$$

Substituting Eq. (1.134) back into Eq. (1.133) yields the exact decomposition of the joint Hamiltonian:

$$E_\theta(\mathbf{v}, \mathbf{s}) = \frac{1}{2}(\mathbf{v} - \boldsymbol{\mu}(\mathbf{s}))^\top \mathbf{G}(\mathbf{v} - \boldsymbol{\mu}(\mathbf{s})) - \frac{1}{2}\mathbf{s}^\top \mathbf{W}^\top \mathbf{G}^{-1}\mathbf{W}\mathbf{s} - (\mathbf{h} + \mathbf{W}^\top \mathbf{b})^\top \mathbf{s}. \tag{1.136}$$

At fixed $\mathbf{s}$, the last two terms in Eq. (1.136) are independent of $\mathbf{v}$ and cancel upon normalization. Integrating the Gaussian factor over $\mathbf{v} \in \mathbb{R}^N$ as in Example 1.5 shows that the conditional distribution of $\mathbf{v}$ given $\mathbf{s}$ is a multivariate Gaussian whose covariance $k_\mathrm{B}T\mathbf{G}^{-1}$ is independent of $\mathbf{s}$ and whose mean $\boldsymbol{\mu}(\mathbf{s})$ is shifted linearly by the p-bit state $\mathbf{s}$:

$$p_\theta(\mathbf{v} \mid \mathbf{s}) = \mathcal{N}(\boldsymbol{\mu}(\mathbf{s}),\, k_\mathrm{B}T\mathbf{G}^{-1}) = \det\left(\frac{\beta}{2\pi}\mathbf{G}\right)^{1/2} \exp\left(-\frac{\beta}{2}(\mathbf{v} - \mathbf{b} - \mathbf{G}^{-1}\mathbf{W}\mathbf{s})^\top \mathbf{G}(\mathbf{v} - \mathbf{b} - \mathbf{G}^{-1}\mathbf{W}\mathbf{s})\right). \tag{1.137}$$

Physically, if $\mathbf{G}$ is realized as a conductance matrix, each p-bit $s_j \in \{-1,+1\}$ injects a current $w_{ij}s_j$ into capacitor node i, shifting the equilibrium node voltages by $\mathbf{G}^{-1}\mathbf{W}\mathbf{s}$.

Part 2: Continuous Marginal $p_\theta(\mathbf{v})$ and Effective Free Energy $F_\mathrm{eff}(\mathbf{v})$.

Taking $-k_\mathrm{B}T \ln Z_S(\mathbf{v}; \theta, \beta)$ from Eq. (1.129) gives the effective free-energy potential (1.115) over $\mathbb{R}^N$:

$$F_\mathrm{eff}(\mathbf{v}; \theta, \beta) = \frac{1}{2}(\mathbf{v} - \mathbf{b})^\top \mathbf{G}(\mathbf{v} - \mathbf{b}) - \frac{1}{\beta}\sum_{j=1}^{M} \ln\Big(2\cosh\Big(\beta(\mathbf{w}_j^\top \mathbf{v} + h_j)\Big)\Big). \tag{1.138}$$

The marginal distribution $p_\theta(\mathbf{v}) \propto \exp(-\beta F_\mathrm{eff}(\mathbf{v}))$ can also be written as a Gaussian mixture over all 2^M corners $\mathbf{s} \in \{-1,+1\}^M$ of the p-bit hypercube:

$$p_\theta(\mathbf{v}) = \sum_{\mathbf{s} \in \{-1,+1\}^M} p_\theta(\mathbf{s})\, \mathcal{N}(\mathbf{v};\, \mathbf{b} + \mathbf{G}^{-1}\mathbf{W}\mathbf{s},\, k_\mathrm{B}T\mathbf{G}^{-1}). \tag{1.139}$$

Coupling N Gaussian nodes to M p-bits therefore lets the circuit represent a mixture of up to 2^M Gaussian components in $\mathbb{R}^N$ with only $N + M$ units and NM bilinear couplings.

We compute the gradient and Hessian of $F_\mathrm{eff}(\mathbf{v})$ directly and check them against Theorem 1.18. Differentiating Eq. (1.138) with respect to $\mathbf{v}$ gives the effective mean force:

$$\nabla_\mathbf{v} F_\mathrm{eff}(\mathbf{v}) = \mathbf{G}(\mathbf{v} - \mathbf{b}) - \sum_{j=1}^{M} \tanh\Big(\beta(\mathbf{w}_j^\top \mathbf{v} + h_j)\Big)\mathbf{w}_j = \mathbf{G}(\mathbf{v} - \mathbf{b}) - \mathbf{W}\mathbb{E}[\mathbf{S} \mid \mathbf{v}]_{p_\theta}, \tag{1.140}$$

which matches $\mathbb{E}[\nabla_{\mathbf{v}} E_\theta(\mathbf{v},\mathbf{S})]_{p_\theta(\mathbf{s}|\mathbf{v})}$ from Eq. (1.120). Differentiating Eq. (1.140) a second time with respect to $\mathbf{v}$ using $\frac{\mathrm{d}}{\mathrm{d}u}\tanh(u) = \operatorname{sech}^2(u)$ gives the effective Hessian matrix:

$$\nabla^2_{\mathbf{v}} F_{\text{eff}}(\mathbf{v}) = \mathbf{G} - \beta \sum_{j=1}^{M} \operatorname{sech}^2\Big(\beta(\mathbf{w}_j^\top \mathbf{v} + h_j)\Big)\mathbf{w}_j\mathbf{w}_j^\top = \mathbf{G} - \beta \mathbf{W}\,\mathrm{Cov}_{p_\theta}(\mathbf{S} \mid \mathbf{v})\mathbf{W}^\top, \tag{1.141}$$

which reproduces the general curvature-softening formula (1.121) since $\nabla_{\mathbf{v}} E_\theta(\mathbf{v},\mathbf{s}) = \mathbf{G}(\mathbf{v}-\mathbf{b}) - \mathbf{W}\mathbf{s}$ has conditional covariance $\mathbf{W}\,\mathrm{Cov}(\mathbf{S} \mid \mathbf{v})\mathbf{W}^\top$.

To see how Eq. (1.138) produces the quartic potential of the double-well sampler, consider the one-dimensional symmetric case ($N = 1$, $M = 1$, $b = 0$, $h = 0$) with scalar conductance $G > 0$ and scalar coupling weight $w \in \mathbb{R}$:

$$F_{\text{eff}}(v) = \frac{1}{2}Gv^2 - \frac{1}{\beta}\ln\big(2\cosh(\beta w v)\big). \tag{1.142}$$

Expand $\ln\cosh(u)$ in a Maclaurin series around $u = 0$. Since $\frac{\mathrm{d}}{\mathrm{d}u}\ln\cosh(u) = \tanh(u) = u - \frac{1}{3}u^3 + \frac{2}{15}u^5 - \mathcal{O}(u^7)$, integrating term by term from 0 to u yields

$$\ln\cosh(u) = \int_0^u \tanh(t)\,\mathrm{d}t = \frac{1}{2}u^2 - \frac{1}{12}u^4 + \frac{1}{45}u^6 - \mathcal{O}(u^8). \tag{1.143}$$

Substituting $u = \beta w v$ into Eq. (1.143) and inserting the expansion into Eq. (1.142), we obtain the fourth-order (Landau) expansion of the effective free energy:

$$\begin{aligned} F_{\text{eff}}(v) &= -k_{\mathrm{B}}T\ln 2 + \frac{1}{2}Gv^2 - \frac{1}{\beta}\left[\frac{1}{2}(\beta w v)^2 - \frac{1}{12}(\beta w v)^4 + \mathcal{O}\big((\beta w v)^6\big)\right] \\ &= -k_{\mathrm{B}}T\ln 2 - \frac{\beta w^2 - G}{2}v^2 + \frac{\beta^3 w^4}{12}v^4 - \mathcal{O}(\beta^5 w^6 v^6). \end{aligned} \tag{1.144}$$

Comparing Eq. (1.144) with the double-well potential $E(v) = \frac{\tilde{a}}{4}v^4 - \frac{\tilde{b}}{2}v^2 + \text{const}$ (Eq. (A.3) with $h = 0$), we identify the effective coefficients

$$\tilde{b} = \beta w^2 - G, \qquad \tilde{a} = \frac{1}{3}\beta^3 w^4 > 0. \tag{1.145}$$

From Eqs. (1.141) and (1.144), the curvature at the origin $v = 0$ is $F''_{\text{eff}}(0) = G - \beta w^2 = -\tilde{b}$. The effective potential therefore undergoes a pitchfork bifurcation at the critical coupling $\beta w^2 = G$:

- **Subcritical (Unimodal) Regime ($\beta w^2 \le G$):** Because $\operatorname{sech}^2(u) \le 1$ for all $u \in \mathbb{R}$, we have $F''_{\text{eff}}(v) = G - \beta w^2 \operatorname{sech}^2(\beta w v) \ge G - \beta w^2 \ge 0$ for all $v \in \mathbb{R}$. Thus $F_{\text{eff}}(v)$ is globally convex on $\mathbb{R}$ with a single minimum at $v^* = 0$.
- **Supercritical (Bistable Double-Well) Regime ($\beta w^2 > G$):** At the origin, $F''_{\text{eff}}(0) = G - \beta w^2 < 0$, so $v = 0$ becomes a local maximum (an energy barrier top). The two symmetric global minima $\pm v^*$ satisfy the stationarity condition $F'_{\text{eff}}(v^*) = 0$:

  $$G v^* = w\tanh(\beta w v^*). \tag{1.146}$$

  Near the bifurcation threshold ($0 < \tilde{b}/G \ll 1$), solving the quartic approximation $-\tilde{b}v^* + \tilde{a}(v^*)^3 = 0$ gives the approximate minima and barrier height $\Delta F_{\text{eff}} := F_{\text{eff}}(0) - F_{\text{eff}}(\pm v^*)$:

  $$v^* \approx \pm\sqrt{\frac{\tilde{b}}{\tilde{a}}} = \pm\sqrt{\frac{3(\beta w^2 - G)}{\beta^3 w^4}}, \qquad \Delta F_{\text{eff}} \approx \frac{\tilde{b}^2}{4\tilde{a}} = \frac{3(\beta w^2 - G)^2}{4\beta^3 w^4}. \tag{1.147}$$

In the strong-coupling / low-temperature limit ($\beta w^2 \gg G$), $\tanh(\beta w v^*) \to \pm 1$, so the two wells settle at $v^* \to \pm w/G$ with asymptotic barrier height

$$\Delta F_{\text{eff}} = F_{\text{eff}}(0) - F_{\text{eff}}(w/G) \to \frac{w^2}{2G} - k_{\text{B}}T \ln 2. \tag{1.148}$$

Part 3: Discrete Marginal $p_\theta(\mathbf{s})$ and Induced Ising Couplings.

Now integrate out the continuous voltages $\mathbf{v} \in \mathbb{R}^N$ instead, to obtain the marginal distribution $p_\theta(\mathbf{s})$ over the discrete p-bits $\mathbf{s} \in \{-1, +1\}^M$? Using the completed-square decomposition (1.136) of $E_\theta(\mathbf{v}, \mathbf{s})$, we integrate $\exp(-\beta E_\theta(\mathbf{v}, \mathbf{s}))$ over $\mathbf{v} \in \mathbb{R}^N$ at fixed $\mathbf{s}$:

$$\begin{aligned} Z_V(\mathbf{s}; \theta, \beta) &= \int_{\mathbb{R}^N} \exp(-\beta E_\theta(\mathbf{v}, \mathbf{s}))\, \mathrm{d}^N v \\ &= \exp\left(\frac{\beta}{2}\mathbf{s}^\top \mathbf{W}^\top \mathbf{G}^{-1}\mathbf{W}\mathbf{s} + \beta(\mathbf{h} + \mathbf{W}^\top \mathbf{b})^\top \mathbf{s}\right) \int_{\mathbb{R}^N} \exp\left(-\frac{\beta}{2}(\mathbf{v} - \boldsymbol{\mu}(\mathbf{s}))^\top \mathbf{G}(\mathbf{v} - \boldsymbol{\mu}(\mathbf{s}))\right) \mathrm{d}^N v \\ &= \det\left(\frac{\beta}{2\pi}\mathbf{G}\right)^{-1/2} \exp\left(\frac{\beta}{2}\mathbf{s}^\top \mathbf{W}^\top \mathbf{G}^{-1}\mathbf{W}\mathbf{s} + \beta(\mathbf{h} + \mathbf{W}^\top \mathbf{b})^\top \mathbf{s}\right). \end{aligned} \tag{1.149}$$

Taking $-k_{\text{B}}T \ln Z_V(\mathbf{s}; \theta, \beta)$ gives the effective Hamiltonian of the p-bits on $\{-1, +1\}^M$:

$$F_{\text{eff}}(\mathbf{s}; \theta, \beta) = -\frac{1}{2}\mathbf{s}^\top \mathbf{J}_{\text{eff}}\mathbf{s} - (\mathbf{h}_{\text{eff}})^\top \mathbf{s} + \frac{k_{\text{B}}T}{2} \ln\det\left(\frac{\beta}{2\pi}\mathbf{G}\right), \tag{1.150}$$

where the induced effective Ising coupling matrix $\mathbf{J}_{\text{eff}} \in \mathbb{R}^{M \times M}$ and effective local field $\mathbf{h}_{\text{eff}} \in \mathbb{R}^M$ are

$$\mathbf{J}_{\text{eff}} := \mathbf{W}^\top \mathbf{G}^{-1}\mathbf{W}, \qquad \mathbf{h}_{\text{eff}} := \mathbf{h} + \mathbf{W}^\top \mathbf{b}. \tag{1.151}$$

Although the p-bits have no direct pairwise connections in (1.127), integrating out the continuous network $\mathbf{v}$ produces an all-to-all, positive-semidefinite Hebbian interaction $\mathbf{J}_{\text{eff}} = \mathbf{W}^\top \mathbf{G}^{-1}\mathbf{W}$ (since $(J_{\text{eff}})_{jk} = \mathbf{w}_j^\top \mathbf{G}^{-1}\mathbf{w}_k$ and $s_j^2 = +1$ makes the diagonal entries $\frac{1}{2}(J_{\text{eff}})_{jj}s_j^2 = \frac{1}{2}\mathbf{w}_j^\top \mathbf{G}^{-1}\mathbf{w}_j$ constant energy offsets). In particular, when $\mathbf{G} = G\mathbf{I}_N$, $(J_{\text{eff}})_{jk} = G^{-1}\sum_{i=1}^N W_{ij}W_{ik}$ is the Hopfield outer-product (associative-memory) matrix. For random $\mathbf{W}$ these couplings have random signs and frustrated loops, the spin-glass setting of Chapter 3.

Part 4: Contrastive Learning Gradients for the Hybrid Parameters.

Finally, suppose the continuous nodes $\mathbf{v}$ are clamped to an empirical data distribution $p_{\text{data}}(\mathbf{v})$ while the p-bits $\mathbf{s}$ act as latent variables. Differentiating $E_\theta(\mathbf{v}, \mathbf{s})$ in Eq. (1.127) with respect to each parameter in $\theta = (\mathbf{G}, \mathbf{b}, \mathbf{W}, \mathbf{h})$ gives the generalized forces:

$$\nabla_{\mathbf{G}} E_\theta(\mathbf{v}, \mathbf{s}) = \frac{1}{2}(\mathbf{v} - \mathbf{b})(\mathbf{v} - \mathbf{b})^\top, \qquad \nabla_{\mathbf{b}} E_\theta(\mathbf{v}, \mathbf{s}) = -\mathbf{G}(\mathbf{v} - \mathbf{b}), \tag{1.152}$$

$$\nabla_{\mathbf{W}} E_\theta(\mathbf{v}, \mathbf{s}) = -\mathbf{v}\mathbf{s}^\top, \qquad \nabla_{\mathbf{h}} E_\theta(\mathbf{v}, \mathbf{s}) = -\mathbf{s}. \tag{1.153}$$

Substituting Eqs. (1.152) and (1.153) into the latent-variable contrastive gradient theorem (1.122) of Theorem 1.18(iii) and using $\mathbb{E}[\mathbf{S} \mid \mathbf{v}]_{p_\theta} = \tanh(\beta(\mathbf{W}^\top \mathbf{v} + \mathbf{h}))$ from Eq. (1.131), we obtain the learning gradients:

$$\nabla_{\mathbf{G}} \mathcal{L}(\theta) = -\frac{\beta}{2}\Big(\mathbb{E}\Big[(\mathbf{V} - \mathbf{b})(\mathbf{V} - \mathbf{b})^\top\Big]_{p_{\text{data}}(\mathbf{v})} - \mathbb{E}\Big[(\mathbf{V} - \mathbf{b})(\mathbf{V} - \mathbf{b})^\top\Big]_{p_\theta(\mathbf{v},\mathbf{s})}\Big), \tag{1.154}$$

$$\nabla_{\mathbf{b}} \mathcal{L}(\theta) = \beta\mathbf{G}\Big(\mathbb{E}[\mathbf{V}]_{p_{\text{data}}(\mathbf{v})} - \mathbb{E}[\mathbf{V}]_{p_\theta(\mathbf{v},\mathbf{s})}\Big), \tag{1.155}$$

$$\nabla_{\mathbf{W}}\mathcal{L}(\theta) = \beta\Big(\mathbb{E}\Big[\mathbf{V}\tanh\big(\beta(\mathbf{W}^\top\mathbf{V}+\mathbf{h})\big)^\top\Big]_{p_{\text{data}}(\mathbf{v})} - \mathbb{E}\Big[\mathbf{V}\mathbf{S}^\top\Big]_{p_\theta(\mathbf{v},\mathbf{s})}\Big), \tag{1.156}$$

$$\nabla_{\mathbf{h}}\mathcal{L}(\theta) = \beta\Big(\mathbb{E}\Big[\tanh\big(\beta(\mathbf{W}^\top\mathbf{V}+\mathbf{h})\big)\Big]_{p_{\text{data}}(\mathbf{v})} - \mathbb{E}[\mathbf{S}]_{p_\theta(\mathbf{v},\mathbf{s})}\Big). \tag{1.157}$$

The positive-phase expectations in Eqs. (1.156) and (1.157) require *no* MCMC sampling over $\mathbf{s}$: because the p-bits are conditionally independent given $\mathbf{v}$, their clamped posterior expectation is evaluated in one step as $\tanh(\beta(\mathbf{W}^\top\mathbf{v}+\mathbf{h}))$. On hardware, both the clamped positive phase and the unclamped negative phase $\mathbb{E}\Big[\mathbf{V}\mathbf{S}^\top\Big]_{p_\theta(\mathbf{v},\mathbf{s})}$ could be estimated as time-averaged voltage–spin cross-correlations measured on the chip.

Summary

Proved in this chapter:

- Invariance of the Boltzmann–Gibbs distribution under configuration-independent energy shifts (Proposition 1.2).
- The identities linking $\ln Z$ and $F = -k_{\mathrm{B}}T\ln Z$ to the internal energy, entropy, energy variance, and heat capacity (Theorem 1.4), evaluated for the RC sampler (Example 1.5) and for a single p-bit (Example 1.6).
- Gibbs' inequality, the variational decomposition $\mathcal{F}[q] = F + k_{\mathrm{B}}TD_{\mathrm{KL}}(q\,\|\,p_\theta)$, the Gibbs–Bogoliubov–Feynman bounds, and the Weiss mean-field equations (Lemma 1.8, Theorem 1.9, Corollary 1.10, and Example 1.11).
- The static fluctuation–response theorem, $\nabla^2_{\boldsymbol{\theta}}\ln Z = \beta^2\,\mathrm{Cov}_{p_\theta}(\boldsymbol{\phi})$, and convexity of $\ln Z$ in the natural parameters (Theorem 1.13 and Corollary 1.14).
- The contrastive gradient of the log-likelihood (Theorem 1.16), the curvature-softening identity for effective potentials (Theorem 1.18), and the closed-form analysis of the Gaussian–Bernoulli EBM (Worked computation 1.19).

Assumed:

- A heat bath at temperature T, a finite partition function, and enough integrability of E_θ and its parameter derivatives to differentiate under the integral sign.

Taken up later:

- How coupling to a bath drives an arbitrary initial density to p_θ: Langevin dynamics (Chapter 5), the Fokker–Planck H-theorem (Chapter 6), and proofs in Chapter C.
- Multivariate Gaussian identities, Schur complements, and Wick's theorem (Chapter B).
- Device physics of subthreshold CMOS and Josephson p-bits (Chapters 12 to 14 and Chapter D).

Open:

- Generative modeling with non-equilibrium steady states in active circuits where $\mathbf{G} \neq \mathbf{G}^\top$ breaks detailed balance. The stationary distribution is then not the Boltzmann distribution of an energy built from $\mathbf{G}$; for linear circuits its covariance follows from a Lyapunov equation (Chapters 13 and 18; [37, 38]).

Bibliographic notes

The mathematical formulation of the canonical ensemble and the variational characterization of equilibrium free energy date back to J. Willard Gibbs; modern statistical-physics expositions of the Gibbs-Bogoliubov-Feynman variational principle, cumulant expansions, and linear response theory can be found in Mézard and Montanari [76], Sekimoto [98], and Kubo [62]. The connection between Ising spin networks, Boltzmann machines, and maximum-likelihood learning via positive (clamped) and negative (free) phases was introduced by Ackley et al. [1], while Contrastive Divergence (CD-k) was formulated by Hinton [50]. The energy cost of digital CMOS computing, in particular the dominance of memory movement over arithmetic, was quantified by Horowitz [52]. Thermodynamic computing and sampling with physical noise are developed in Conte et al. [21], Extropic Corporation [30], Coles et al. [20], Aifer et al. [3], Extropic Corporation [31], Jelinčič et al. [56], and Freitas et al. [38], with probabilistic bit (p-bit) hardware pioneered by Camsari et al. [18] and Borders et al. [14].

Chapter 2

Markov Random Fields, Factor Graphs, and Chromatic Gibbs Sampling

In Chapter 1 we saw that a confining energy $E_\theta(x)$ on a configuration space Ω defines a Boltzmann–Gibbs distribution $p_\theta(x) = Z(\theta,\beta)^{-1}\exp(-\beta E_\theta(x))$, and that training an Energy-Based Model requires equilibrium samples from p_θ in the negative phase. So far the state $x = (x_1,\dots,x_N)^\top$ has been treated as a single point in Ω. If every variable interacted directly with every other, updating one variable x_i would require the current values of all $N-1$ others. On a chip this is impractical: all-to-all wiring among $N \sim 10^5$–10^6 stochastic units needs routing area that grows as N^2, its parasitic capacitance slows the analog circuits, and long wires pick up crosstalk.

Physical circuits are therefore *sparse* and *local*. Each capacitor node or probabilistic bit (p-bit) is wired to a small, bounded set of neighbors. The theory of *Markov random fields* (MRFs) and *factor graphs* translates this wiring into statistics. A missing wire between nodes i and j means that X_i and X_j are conditionally independent given the rest of the network, and each node is shielded from the rest of the chip by its immediate neighborhood, its *Markov blanket*. A local update therefore needs only local information.

Locality does not by itself make parallel updates correct. If two coupled neighbors are updated in the same clock cycle, each using the other's old value, the combined update converges to the wrong distribution. The cure is to color the graph so that no two neighbors share a color, and to update one color class at a time. Nodes of the same color are conditionally independent given the other colors, so a whole class can be sampled in one clock phase without error.

The chapter develops these ideas in order: the Markov properties and the Hammersley–Clifford theorem (Section 2.1), factor graphs and their local conditionals (Section 2.2), Markov chains and Gibbs kernels (Section 2.3), the failure of synchronous updates (Section 2.4), and chromatic block Gibbs sampling (Section 2.5). A worked computation on small Ising cells (Worked computation 2.28) compares chromatic and synchronous sampling exactly.

2.1 Undirected Graphical Models and the Hammersley–Clifford Theorem

Consider a system of N random variables $\mathbf{X} = (X_1, X_2, \dots, X_N)^\top$ taking values in a product configuration space

$$\Omega = \prod_{i=1}^{N} \Omega_i, \qquad \mu(\mathrm{d}\mathbf{x}) = \bigotimes_{i=1}^{N} \mu_i(\mathrm{d}x_i), \tag{2.1}$$

where each single-site state space (Ω_i, μ_i) may be discrete (such as $\Omega_i = \{-1, +1\}$ with counting measure) or continuous (such as $\Omega_i = \mathbb{R}$ with Lebesgue measure). For any subset of indices $A \subseteq \mathcal{V} := \{1, 2, \dots, N\}$, we denote the subvector of variables indexed by A as $\mathbf{x}_A := (x_i)_{i \in A} \in \Omega_A := \prod_{i \in A} \Omega_i$, and we write $\mathbf{x}_{-i} \equiv \mathbf{x}_{\mathcal{V} \setminus \{i\}}$ for the configuration of all variables except node i.

2.1.1 Graph Topology, Cliques, and Markov Properties

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite, simple, undirected graph with vertex set $\mathcal{V} = \{1, \dots, N\}$ and edge set $\mathcal{E} \subseteq \binom{\mathcal{V}}{2}$ (containing unordered pairs $\{i, j\}$ with $i \neq j$, which we write simply as $(i, j) \in \mathcal{E}$).

Definition 2.1 (Neighborhoods, Boundaries, Separation, and Cliques)**.** Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph.

(i) For each vertex $i \in \mathcal{V}$, its *neighborhood* (or *boundary*) is the set of vertices directly connected to i by an edge:

$$\partial i \equiv \mathcal{N}(i) := \{ j \in \mathcal{V} : (i, j) \in \mathcal{E} \}. \tag{2.2}$$

For any subset $A \subseteq \mathcal{V}$, its boundary is $\partial A := \{ j \in \mathcal{V} \setminus A : \exists\, i \in A \text{ with } (i, j) \in \mathcal{E} \}$, and its closure is $\bar{A} := A \cup \partial A$.

(ii) Let $A, B, S \subseteq \mathcal{V}$ be pairwise disjoint subsets of vertices. We say that S *separates* A and B in $\mathcal{G}$ if every path in $\mathcal{G}$ from any vertex $i \in A$ to any vertex $j \in B$ contains at least one vertex in S.

(iii) A subset of vertices $C \subseteq \mathcal{V}$ is called a *clique* if every pair of distinct vertices $i, j \in C$ is connected by an edge $(i, j) \in \mathcal{E}$ (by convention, the empty set $\emptyset$ and all singletons $\{i\}$ are cliques). A clique C is a *maximal clique* if no proper superset $C' \supsetneq C$ is also a clique. We denote the set of all cliques of $\mathcal{G}$ by $\mathcal{C}(\mathcal{G})$ and the set of maximal cliques by $\mathcal{C}_{\max}(\mathcal{G})$.

Given three random vectors $\mathbf{X}_A, \mathbf{X}_B, \mathbf{X}_S$ with joint density $p(\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_S)$, $\mathbf{X}_A$ is *conditionally independent* of $\mathbf{X}_B$ given $\mathbf{X}_S$, written $\mathbf{X}_A \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_S$, if and only if for all $\mathbf{x}_S$ with $p(\mathbf{x}_S) > 0$, the conditional density factorizes as

$$p(\mathbf{x}_A, \mathbf{x}_B \mid \mathbf{x}_S) = p(\mathbf{x}_A \mid \mathbf{x}_S)\, p(\mathbf{x}_B \mid \mathbf{x}_S), \tag{2.3}$$

or equivalently, there exist functions $g(\mathbf{x}_A, \mathbf{x}_S)$ and $h(\mathbf{x}_B, \mathbf{x}_S)$ such that

$$p(\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_S) = g(\mathbf{x}_A, \mathbf{x}_S)\, h(\mathbf{x}_B, \mathbf{x}_S).$$

We can now state the three standard ways of encoding conditional independence on an undirected graph $\mathcal{G}$.

Definition 2.2 (Pairwise, Local, and Global Markov Properties)**.** Let $p(\mathbf{x})$ be a joint probability density on $\Omega = \prod_{i=1}^{N} \Omega_i$ with respect to μ, and let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph on $\mathcal{V} = \{1, \dots, N\}$.

(i) **Pairwise Markov Property (P):** For every pair of non-adjacent vertices $i, j \in \mathcal{V}$ with $i \neq j$ and $(i, j) \notin \mathcal{E}$,

$$X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_{\mathcal{V}\setminus\{i,j\}}. \tag{2.4}$$

(ii) **Local Markov Property (L):** For every vertex $i \in \mathcal{V}$, conditioned on its immediate graph neighborhood ∂i, X_i is conditionally independent of all remaining variables in the graph:

$$X_i \perp\!\!\!\perp \mathbf{X}_{\mathcal{V}\setminus(\{i\}\cup\partial i)} \mid \mathbf{X}_{\partial i}, \tag{2.5}$$

or equivalently, the single-site conditional density satisfies $p(x_i \mid \mathbf{x}_{-i}) = p(x_i \mid \mathbf{x}_{\partial i})$. The neighborhood ∂i is called the *Markov blanket* of node i.

(iii) **Global Markov Property (G):** For any three pairwise disjoint subsets $A, B, S \subseteq \mathcal{V}$ such that S separates A and B in $\mathcal{G}$,

$$\mathbf{X}_A \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_S. \tag{2.6}$$

A probability distribution satisfying the Global Markov Property on $\mathcal{G}$ is called a *Markov Random Field (MRF)* with respect to $\mathcal{G}$.

How are the three Markov properties related? In general, (G) is stronger than (L), which is stronger than (P). However, whenever $p(\mathbf{x}) > 0$ everywhere on Ω, the three properties are equivalent. Positivity holds for every Boltzmann–Gibbs distribution $p_\theta(\mathbf{x}) = Z^{-1}e^{-\beta E_\theta(\mathbf{x})}$ at finite temperature $T > 0$ with $|E_\theta(\mathbf{x})| < \infty$.

Proposition 2.3 (Equivalence of Markov Properties Under Positivity)**.** *For any probability density p on Ω, the implications* (G) $\implies$ (L) $\implies$ (P) *hold unconditionally. If in addition $p(\mathbf{x}) > 0$ for all $\mathbf{x} \in \Omega$, then* (P) $\implies$ (G)*, so that the Pairwise, Local, and Global Markov properties are all equivalent:*

$$\text{(G)} \iff \text{(L)} \iff \text{(P)}. \tag{2.7}$$

Proof. **Proof of** (G) $\implies$ (L)**:** Fix any vertex $i \in \mathcal{V}$, and set $A = \{i\}$, $S = \partial i$, and $B = \mathcal{V}\setminus(\{i\}\cup\partial i)$. Every path in $\mathcal{G}$ starting at $i \in A$ and ending at a vertex in B must leave i via some edge $(i, k) \in \mathcal{E}$. By definition of ∂i, the first vertex k along this path lies in $S = \partial i$. Hence $S = \partial i$ separates A and B in $\mathcal{G}$, so (G) directly implies $X_i \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_{\partial i}$, which is (L).

Proof of (L) $\implies$ (P)**:** Fix two distinct vertices $i \neq j$ with $(i, j) \notin \mathcal{E}$. Since $(i, j) \notin \mathcal{E}$, vertex j does not belong to ∂i, so $\partial i \subseteq \mathcal{V} \setminus \{i, j\}$. Partition the remaining vertices outside $\{i\} \cup \partial i$ into $\{j\}$ and $R := \mathcal{V} \setminus (\{i, j\} \cup \partial i)$. By (L), $X_i \perp\!\!\!\perp (X_j, \mathbf{X}_R) \mid \mathbf{X}_{\partial i}$, which means the conditional density factorizes as

$$p(x_i, x_j, \mathbf{x}_R \mid \mathbf{x}_{\partial i}) = p(x_i \mid \mathbf{x}_{\partial i})\, p(x_j, \mathbf{x}_R \mid \mathbf{x}_{\partial i}). \tag{2.8}$$

Conditioning on the full complement $\mathbf{x}_{\mathcal{V}\setminus\{i,j\}} = (\mathbf{x}_{\partial i}, \mathbf{x}_R)$, we compute $p(x_i \mid x_j, \mathbf{x}_{\mathcal{V}\setminus\{i,j\}})$ using Eq. (2.8):

$$p(x_i \mid x_j, \mathbf{x}_R, \mathbf{x}_{\partial i}) = \frac{p(x_i, x_j, \mathbf{x}_R \mid \mathbf{x}_{\partial i})}{p(x_j, \mathbf{x}_R \mid \mathbf{x}_{\partial i})} = \frac{p(x_i \mid \mathbf{x}_{\partial i})\, p(x_j, \mathbf{x}_R \mid \mathbf{x}_{\partial i})}{p(x_j, \mathbf{x}_R \mid \mathbf{x}_{\partial i})} = p(x_i \mid \mathbf{x}_{\partial i}), \tag{2.9}$$

which is independent of x_j. Thus $X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_{\mathcal{V}\setminus\{i,j\}}$, establishing (P).

Proof of (P) $\implies$ (G) **under** $p(\mathbf{x}) > 0$**:** Assume $p(\mathbf{x}) > 0$ everywhere on Ω. We first establish the *Intersection Lemma* for strictly positive densities: if U, V, W, Z are disjoint subsets of variables

such that $U \perp\!\!\!\perp V \mid (W, Z)$ and $U \perp\!\!\!\perp W \mid (V, Z)$, then $U \perp\!\!\!\perp (V, W) \mid Z$. Indeed, the two conditional independences mean

$$p(\mathbf{x}_U \mid \mathbf{x}_V, \mathbf{x}_W, \mathbf{x}_Z) = f(\mathbf{x}_U, \mathbf{x}_W, \mathbf{x}_Z) \quad \text{and} \quad p(\mathbf{x}_U \mid \mathbf{x}_V, \mathbf{x}_W, \mathbf{x}_Z) = g(\mathbf{x}_U, \mathbf{x}_V, \mathbf{x}_Z), \tag{2.10}$$

where division by marginal densities is valid everywhere because $p > 0$. Since $f(\mathbf{x}_U, \mathbf{x}_W, \mathbf{x}_Z) = g(\mathbf{x}_U, \mathbf{x}_V, \mathbf{x}_Z)$ holds for all $(\mathbf{x}_V, \mathbf{x}_W)$, fixing a reference value $\mathbf{x}_W = \mathbf{x}_W^*$ shows that $g(\mathbf{x}_U, \mathbf{x}_V, \mathbf{x}_Z) = f(\mathbf{x}_U, \mathbf{x}_W^*, \mathbf{x}_Z) := h(\mathbf{x}_U, \mathbf{x}_Z)$ depends neither on $\mathbf{x}_V$ nor on $\mathbf{x}_W$. Thus $p(\mathbf{x}_U \mid \mathbf{x}_V, \mathbf{x}_W, \mathbf{x}_Z) = h(\mathbf{x}_U, \mathbf{x}_Z) = p(\mathbf{x}_U \mid \mathbf{x}_Z)$, proving $U \perp\!\!\!\perp (V, W) \mid Z$.

Now let $A, B, S \subseteq \mathcal{V}$ be disjoint nonempty sets such that S separates A and B in $\mathcal{G}$. We proceed by reverse induction on $|S| \leq N - 2$.

- *Base case ($|S| = N - 2$):* Then $|A| = |B| = 1$, say $A = \{i\}$ and $B = \{j\}$, and $S = \mathcal{V} \setminus \{i, j\}$. Since S separates i and j, there is no direct edge between i and j, so $(i, j) \notin \mathcal{E}$. The Pairwise Markov Property (P) immediately gives $X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_S$.

- *Inductive step:* Suppose (G) holds for all separating sets of cardinality strictly greater than $|S|$, and suppose $|A \cup B \cup S| \leq N$. First suppose $A \cup B \cup S = \mathcal{V}$. If $|B| > 1$, pick any $j \in B$ and set $B' = B \setminus \{j\}$. Both $S \cup \{j\}$ and $S \cup B'$ have size strictly greater than $|S|$ and separate A from B' and A from $\{j\}$, respectively. By the induction hypothesis, $\mathbf{X}_A \perp\!\!\!\perp \mathbf{X}_{B'} \mid (\mathbf{X}_S, X_j)$ and $\mathbf{X}_A \perp\!\!\!\perp X_j \mid (\mathbf{X}_S, \mathbf{X}_{B'})$. Applying the Intersection Lemma yields $\mathbf{X}_A \perp\!\!\!\perp (\mathbf{X}_{B'}, X_j) \mid \mathbf{X}_S$, i.e., $\mathbf{X}_A \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_S$. The same argument reduces $|A| > 1$ to $|A| = 1$. Finally, if $A \cup B \cup S \subsetneq \mathcal{V}$, pick any vertex $k \in \mathcal{V} \setminus (A \cup B \cup S)$. The set $S \cup \{k\}$ still separates A from B and is larger than S, so the induction hypothesis gives $\mathbf{X}_A \perp\!\!\!\perp \mathbf{X}_B \mid (\mathbf{X}_S, X_k)$. Next, because removing S disconnects A from B, the vertex k cannot be joined to both A and B in $\mathcal{G} \setminus S$. Suppose it is not joined to B (the other case is symmetric). Then $A \cup S$ separates B from $\{k\}$, and since $|A \cup S| > |S|$ the induction hypothesis gives $\mathbf{X}_B \perp\!\!\!\perp X_k \mid (\mathbf{X}_A, \mathbf{X}_S)$. The Intersection Lemma, applied with $U = B$, $V = A$, $W = \{k\}$, $Z = S$, combines the two statements into $\mathbf{X}_B \perp\!\!\!\perp (\mathbf{X}_A, X_k) \mid \mathbf{X}_S$, and marginalizing over X_k gives $\mathbf{X}_A \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_S$.

This completes the induction and proves (P) $\implies$ (G). □

Why is the positivity hypothesis $p(\mathbf{x}) > 0$ essential in Proposition 2.3? Suppose $N = 3$ with $\mathcal{E} = \emptyset$ (three isolated vertices with no edges) and consider the zero-temperature deterministic distribution where $X_1 = X_2 = X_3 = \pm 1$ with probability $1/2$ each. For any pair (i, j), conditioning on the third variable X_k fixes $X_i = X_j = X_k$ deterministically, so $X_i \perp\!\!\!\perp X_j \mid X_k$ holds trivially and (P) is satisfied. Yet X_1 and X_2 are maximally correlated unconditionally (X_1 is not independent of (X_2, X_3) given $\partial 1 = \emptyset$), violating (L) and (G). At any finite physical temperature $T > 0$, thermal fluctuations prevent deterministic locking ($p(\mathbf{x}) > 0$ everywhere), eliminating this pathology.

2.1.2 Gibbs Random Fields and the Hammersley–Clifford Theorem

While Definition 2.2 characterizes a graphical model through conditional independence statements, statistical physics specifies a system through its Hamiltonian $E_\theta(\mathbf{x})$. How does the graph structure $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ constrain the algebraic form of $E_\theta(\mathbf{x})$?

Definition 2.4 (Gibbs Random Field and Clique Potentials)**.** A probability density $p(\mathbf{x})$ on Ω is called a *Gibbs distribution* (or *Gibbs Random Field*) with respect to the undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$

at inverse temperature $\beta > 0$ if its energy function decomposes as a sum of *clique potentials* $E_C : \Omega_C \to \mathbb{R}$ supported solely on the cliques $C \in \mathcal{C}(\mathcal{G})$:

$$E_\theta(\mathbf{x}) = \sum_{C \in \mathcal{C}(\mathcal{G})} E_C(\mathbf{x}_C), \qquad p(\mathbf{x}) = \frac{1}{Z(\beta)} \exp\left(-\beta \sum_{C \in \mathcal{C}(\mathcal{G})} E_C(\mathbf{x}_C) \right), \tag{2.11}$$

where each $E_C(\mathbf{x}_C)$ depends only on the variables $\mathbf{x}_C = (x_i)_{i \in C}$ belonging to the clique C. (By grouping non-maximal cliques into maximal cliques containing them, the sum in Eq. (2.11) can equivalently be restricted to maximal cliques $C \in \mathcal{C}_{\max}(\mathcal{G})$.)

The central theorem of undirected graphical models, due to Hammersley and Clifford [45], says that the two descriptions agree. We prove it by Möbius inversion over the subsets of $\mathcal{V}$, which constructs the clique potentials explicitly.

Theorem 2.5 (Hammersley–Clifford Theorem)**.** *Let $p(\mathbf{x}) > 0$ be a strictly positive probability distribution on $\Omega = \prod_{i=1}^{N} \Omega_i$, and let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph on $\mathcal{V} = \{1, \dots, N\}$. Then $p(\mathbf{x})$ is a Markov Random Field with respect to $\mathcal{G}$ (satisfying the Pairwise, Local, and Global Markov properties) if and only if $p(\mathbf{x})$ is a Gibbs Random Field whose Hamiltonian decomposes over the cliques of $\mathcal{G}$ as in Eq. (2.11).*

Specifically, fix an arbitrary reference configuration $\mathbf{x}^* = (x_1^*, \dots, x_N^*) \in \Omega$, and for any configuration $\mathbf{x} \in \Omega$ and subset $B \subseteq \mathcal{V}$, denote by $(\mathbf{x}_B, \mathbf{x}^*_{\mathcal{V} \setminus B}) \in \Omega$ the hybrid configuration whose i-th component is x_i for $i \in B$ and x_i^* for $i \notin B$. Define for every subset $A \subseteq \mathcal{V}$ the canonical Möbius potential*:*

$$U_A(\mathbf{x}_A) := -k_{\mathrm{B}} T \sum_{B \subseteq A} (-1)^{|A \setminus B|} \ln p(\mathbf{x}_B, \mathbf{x}^*_{\mathcal{V} \setminus B}). \tag{2.12}$$

Then:

(i) For every positive distribution $p(\mathbf{x}) > 0$, $\sum_{A \subseteq \mathcal{V}} U_A(\mathbf{x}_A) = -k_{\mathrm{B}} T \ln p(\mathbf{x})$.

(ii) $U_A(\mathbf{x}_A) = 0$ whenever $x_i = x_i^*$ for at least one $i \in A$ (for $A \neq \emptyset$).

(iii) If $p(\mathbf{x})$ satisfies the Pairwise Markov Property on $\mathcal{G}$, then $U_A(\mathbf{x}_A) = 0$ identically for every subset $A \subseteq \mathcal{V}$ that is not *a clique of $\mathcal{G}$, yielding the exact clique decomposition*

$$p(\mathbf{x}) = p(\mathbf{x}^*) \exp\left(-\beta \sum_{C \in \mathcal{C}(\mathcal{G}) \setminus \{\emptyset\}} U_C(\mathbf{x}_C) \right). \tag{2.13}$$

Proof. **Step 1: Proof that every Gibbs Random Field is a Markov Random Field.** Suppose $p(\mathbf{x})$ has the clique decomposition (2.11). By Proposition 2.3, since $p(\mathbf{x}) > 0$, it suffices to verify the Pairwise Markov Property (P) for any non-adjacent pair $i \neq j$ with $(i, j) \notin \mathcal{E}$. Because i and j are not connected by an edge, no clique $C \in \mathcal{C}(\mathcal{G})$ can contain both i and j simultaneously ($\{i, j\} \not\subseteq C$). Therefore, we can partition the set of cliques $\mathcal{C}(\mathcal{G})$ into two disjoint collections: $\mathcal{C}_i := \{C \in \mathcal{C}(\mathcal{G}) : i \in C\}$ (none of which contain j) and $\mathcal{C}_{\setminus i} := \{C \in \mathcal{C}(\mathcal{G}) : i \notin C\}$ (none of which contain i). This splits the total energy into a sum of two functions, one independent of x_j and one independent of x_i:

$$E_\theta(\mathbf{x}) = \underbrace{\sum_{C \in \mathcal{C}_i} E_C(x_i, \mathbf{x}_{C \setminus \{i\}})}_{\text{independent of } x_j} + \underbrace{\sum_{C \in \mathcal{C}_{\setminus i}} E_C(\mathbf{x}_C)}_{\text{independent of } x_i}. \tag{2.14}$$

Exponentiating $-\beta E_\theta(\mathbf{x})$ shows that $p(x_i, x_j, \mathbf{x}_{\mathcal{V}\setminus\{i,j\}}) = g(x_i, \mathbf{x}_{\mathcal{V}\setminus\{i,j\}})h(x_j, \mathbf{x}_{\mathcal{V}\setminus\{i,j\}})$, which implies $X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_{\mathcal{V}\setminus\{i,j\}}$. Thus p is a Markov Random Field on $\mathcal{G}$.

Step 2: Proof of the Möbius inversion identity (i) and normalization property (ii). We now prove that summing the canonical potentials $U_A(\mathbf{x}_A)$ defined in Eq. (2.12) over all subsets $A \subseteq \mathcal{V}$ reconstructs $-k_\mathrm{B}T \ln p(\mathbf{x})$. Interchanging the order of summation over $A \subseteq \mathcal{V}$ and $B \subseteq A$ gives

$$\begin{aligned}\sum_{A\subseteq\mathcal{V}} U_A(\mathbf{x}_A) &= -k_\mathrm{B}T \sum_{A\subseteq\mathcal{V}} \sum_{B\subseteq A} (-1)^{|A\setminus B|} \ln p(\mathbf{x}_B, \mathbf{x}^*_{\mathcal{V}\setminus B}) \\ &= -k_\mathrm{B}T \sum_{B\subseteq\mathcal{V}} \ln p(\mathbf{x}_B, \mathbf{x}^*_{\mathcal{V}\setminus B}) \left[\sum_{A:B\subseteq A\subseteq\mathcal{V}} (-1)^{|A\setminus B|}\right]. \end{aligned} \tag{2.15}$$

For each fixed subset $B \subseteq \mathcal{V}$, any superset A satisfying $B \subseteq A \subseteq \mathcal{V}$ is uniquely written as $A = B \sqcup D$, where $D = A \setminus B$ ranges over all subsets of the complement $\mathcal{V} \setminus B$. If $B \subsetneq \mathcal{V}$, then $m := |\mathcal{V} \setminus B| \geq 1$, and the inner bracket in Eq. (2.15) evaluates to zero by the binomial theorem:

$$\sum_{A:B\subseteq A\subseteq\mathcal{V}} (-1)^{|A\setminus B|} = \sum_{D\subseteq\mathcal{V}\setminus B} (-1)^{|D|} = \sum_{k=0}^{m} \binom{m}{k} (-1)^k = (1-1)^m = 0 \qquad (\text{for } B \subsetneq \mathcal{V}). \tag{2.16}$$

If $B = \mathcal{V}$, then $D = \emptyset$ is the only term in the sum, yielding $(-1)^0 = 1$. Thus the inner bracket in Eq. (2.15) equals the Kronecker delta $\delta_{B,\mathcal{V}}$, and the entire outer sum collapses to the single term $B = \mathcal{V}$:

$$\sum_{A\subseteq\mathcal{V}} U_A(\mathbf{x}_A) = -k_\mathrm{B}T \ln p(\mathbf{x}_\mathcal{V}, \mathbf{x}^*_\emptyset) = -k_\mathrm{B}T \ln p(\mathbf{x}), \tag{2.17}$$

which proves (i). Note also that for $A = \emptyset$, $U_\emptyset = -k_\mathrm{B}T \ln p(\mathbf{x}^*)$, which accounts for the prefactor $p(\mathbf{x}^*)$ in Eq. (2.13).

To prove (ii), suppose $A \neq \emptyset$ and there is some index $i \in A$ such that $x_i = x_i^*$. Partition the subsets $B \subseteq A$ into pairs $(K, K \cup \{i\})$ where K ranges over all subsets of $A \setminus \{i\}$:

$$U_A(\mathbf{x}_A) = -k_\mathrm{B}T \sum_{K\subseteq A\setminus\{i\}} (-1)^{|A\setminus(K\cup\{i\})|} \Big[\ln p(\mathbf{x}_{K\cup\{i\}}, \mathbf{x}^*_{\mathcal{V}\setminus(K\cup\{i\})}) - \ln p(\mathbf{x}_K, \mathbf{x}^*_{\mathcal{V}\setminus K})\Big], \tag{2.18}$$

where the relative minus sign arises because $|A \setminus K| = |A \setminus (K \cup \{i\})| + 1$. When $x_i = x_i^*$, the two arguments $(\mathbf{x}_{K\cup\{i\}}, \mathbf{x}^*_{\mathcal{V}\setminus(K\cup\{i\})})$ and $(\mathbf{x}_K, \mathbf{x}^*_{\mathcal{V}\setminus K})$ are identical component by component, so every bracket in Eq. (2.18) vanishes identically and $U_A(\mathbf{x}_A) = 0$.

Step 3: Proof of (iii) that $U_A(\mathbf{x}_A) \equiv 0$ whenever A is not a clique. Suppose $p(\mathbf{x})$ satisfies the Pairwise Markov Property on $\mathcal{G}$, and let $A \subseteq \mathcal{V}$ be a subset that is *not* a clique of $\mathcal{G}$. By definition of a clique, there exist two distinct vertices $i, j \in A$ such that $(i, j) \notin \mathcal{E}$. Every subset $B \subseteq A$ can be uniquely written in one of four forms: K, $K \cup \{i\}$, $K \cup \{j\}$, or $K \cup \{i, j\}$, where $K := B \setminus \{i, j\}$ ranges over all subsets of $A \setminus \{i, j\}$. Grouping the sum in Eq. (2.12) into these quadruplets and noting that $|A \setminus (K \cup \{i\})| = |A \setminus (K \cup \{j\})| = |A \setminus K| - 1$ while $|A \setminus (K \cup \{i, j\})| = |A \setminus K| - 2$, we obtain

$$\begin{aligned}U_A(\mathbf{x}_A) = -k_\mathrm{B}T \sum_{K\subseteq A\setminus\{i,j\}} (-1)^{|A\setminus K|} \Big[&\ln p(\mathbf{x}_{K\cup\{i,j\}}, \mathbf{x}^*_{\mathcal{V}\setminus(K\cup\{i,j\})}) - \ln p(\mathbf{x}_{K\cup\{i\}}, \mathbf{x}^*_{\mathcal{V}\setminus(K\cup\{i\})}) \\ &- \ln p(\mathbf{x}_{K\cup\{j\}}, \mathbf{x}^*_{\mathcal{V}\setminus(K\cup\{j\})}) + \ln p(\mathbf{x}_K, \mathbf{x}^*_{\mathcal{V}\setminus K})\Big] \end{aligned}$$

$$= -k_{\mathrm{B}}T \sum_{K \subseteq A \setminus \{i,j\}} (-1)^{|A \setminus K|} \ln \left[\frac{p(x_i, x_j, \tilde{\mathbf{x}}_{-ij})\, p(x_i^*, x_j^*, \tilde{\mathbf{x}}_{-ij})}{p(x_i, x_j^*, \tilde{\mathbf{x}}_{-ij})\, p(x_i^*, x_j, \tilde{\mathbf{x}}_{-ij})} \right], \tag{2.19}$$

where for each fixed $K \subseteq A \setminus \{i,j\}$, $\tilde{\mathbf{x}}_{-ij} := (\mathbf{x}_K, \mathbf{x}^*_{\mathcal{V} \setminus (K \cup \{i,j\})})$ is the common configuration on $\mathcal{V} \setminus \{i,j\}$.

Now apply the Pairwise Markov Property $X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_{\mathcal{V} \setminus \{i,j\}}$ to the numerator and denominator inside the logarithm of Eq. (2.19). Dividing both top and bottom by $p(\tilde{\mathbf{x}}_{-ij})^2 > 0$ and factoring the joint conditional distribution as $p(x_i, x_j \mid \tilde{\mathbf{x}}_{-ij}) = p(x_i \mid \tilde{\mathbf{x}}_{-ij})\, p(x_j \mid \tilde{\mathbf{x}}_{-ij})$, we find

$$\begin{aligned} \frac{p(x_i, x_j, \tilde{\mathbf{x}}_{-ij})\, p(x_i^*, x_j^*, \tilde{\mathbf{x}}_{-ij})}{p(x_i, x_j^*, \tilde{\mathbf{x}}_{-ij})\, p(x_i^*, x_j, \tilde{\mathbf{x}}_{-ij})} &= \frac{p(x_i, x_j \mid \tilde{\mathbf{x}}_{-ij})\, p(x_i^*, x_j^* \mid \tilde{\mathbf{x}}_{-ij})}{p(x_i, x_j^* \mid \tilde{\mathbf{x}}_{-ij})\, p(x_i^*, x_j \mid \tilde{\mathbf{x}}_{-ij})} \\ &= \frac{p(x_i \mid \tilde{\mathbf{x}}_{-ij})\, p(x_j \mid \tilde{\mathbf{x}}_{-ij})\, p(x_i^* \mid \tilde{\mathbf{x}}_{-ij})\, p(x_j^* \mid \tilde{\mathbf{x}}_{-ij})}{p(x_i \mid \tilde{\mathbf{x}}_{-ij})\, p(x_j^* \mid \tilde{\mathbf{x}}_{-ij})\, p(x_i^* \mid \tilde{\mathbf{x}}_{-ij})\, p(x_j \mid \tilde{\mathbf{x}}_{-ij})} = 1. \end{aligned} \tag{2.20}$$

Because $\ln 1 = 0$, every term in the sum (2.19) vanishes identically. Thus $U_A(\mathbf{x}_A) = 0$ for every non-clique $A \notin \mathcal{C}(\mathcal{G})$, and only cliques $C \in \mathcal{C}(\mathcal{G})$ contribute to $\sum_{A \subseteq \mathcal{V}} U_A(\mathbf{x}_A) = -k_{\mathrm{B}}T \ln p(\mathbf{x})$, completing the proof of the Hammersley-Clifford Theorem. □

Physics Insight: The Hammersley–Clifford Theorem as a Principle of Silicon Circuit Layout

Theorem 2.5 has two direct consequences for circuit design.

1. *Conditional independence means local couplings.* The theorem guarantees that if a target probabilistic model possesses a sparse conditional independence graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, its Hamiltonian $E_\theta(\mathbf{x})$ never requires long-range interaction terms between non-adjacent vertices $(i,j) \notin \mathcal{E}$. Every statistical dependency is generated by local clique potentials $U_C(\mathbf{x}_C)$ wired solely among mutually adjacent nodes on the chip.

2. *Local conditional forces.* For any vertex $i \in \mathcal{V}$, splitting the sum over cliques into those containing i and those not containing i gives

$$E_\theta(\mathbf{x}) = \sum_{C \in \mathcal{C}(\mathcal{G}) : i \in C} E_C(x_i, \mathbf{x}_{C \setminus \{i\}}) + \sum_{C \in \mathcal{C}(\mathcal{G}) : i \notin C} E_C(\mathbf{x}_C). \tag{2.21}$$

   When computing the single-node conditional distribution $p_\theta(x_i \mid \mathbf{x}_{-i})$ (for a discrete p-bit update) or the local drift force $-\frac{\partial E_\theta(\mathbf{x})}{\partial x_i}$ (for a continuous Langevin capacitor node), the second sum in Eq. (2.21) cancels out or differentiates to zero. Since every clique C containing i is a subset of $\{i\} \cup \partial i$, the physical update of node i depends only on the voltages and spin states of its neighbors ∂i.

2.2 Hypergraphical Models, Factor Graphs, and Higher-Order Interactions

While an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ captures pairwise adjacency, it is structurally ambiguous whenever cliques of size three or larger are present. Consider three spins $s_1, s_2, s_3 \in \{-1, +1\}$ connected in a triangle $\mathcal{K}_3$. The triangle may stand for three pairwise couplings $-J_{12}s_1s_2 - J_{23}s_2s_3 - J_{31}s_3s_1$, or for an irreducible three-body interaction $-K_{123}s_1s_2s_3$ (such as a three-spin

parity constraint). Both energies have the same undirected graph $\mathcal{K}_3$, yet the circuits that compute their local fields are different.

To represent interactions among k variables (k-*local* interactions) explicitly, we replace undirected graphs by *hypergraphs* and *bipartite factor graphs*. This is also the representation used by Extropic's open-source sampling library `THRML` [31].

2.2.1 Bipartite Factor Graphs and Exact Local Conditionals

Definition 2.6 (Hypergraphical Model and Bipartite Factor Graph)**.** Let $\mathcal{V} = \{1, 2, \dots, N\}$ index the system variables $\mathbf{x} = (x_1, \dots, x_N)^\top \in \Omega = \prod_{i=1}^N \Omega_i$, and let $\mathcal{F} = \{a_1, a_2, \dots, a_M\}$ be a finite index set of *factors* (or *interaction hyperedges*). Each factor $a \in \mathcal{F}$ is associated with a non-empty subset of variables $\partial a \subseteq \mathcal{V}$ called its *scope* (or *hyperedge*), of arity (locality) $k_a := |\partial a|$, and a local factor energy function $E_a : \Omega_{\partial a} \to \mathbb{R}$.

(i) The *bipartite factor graph* is the undirected bipartite graph $\mathcal{G}_{\mathrm{FG}} = (\mathcal{V}, \mathcal{F}, \mathcal{E}_{\mathrm{FG}})$ whose two vertex parts are the variable nodes $i \in \mathcal{V}$ (drawn as circles) and the factor nodes $a \in \mathcal{F}$ (drawn as filled squares), with an edge $(i, a) \in \mathcal{E}_{\mathrm{FG}}$ if and only if $i \in \partial a$.

(ii) For each variable node $i \in \mathcal{V}$, we denote the set of factors incident to i by

$$\partial i := \{a \in \mathcal{F} : i \in \partial a\}, \tag{2.22}$$

and the induced *Markov blanket* of variable i in the hypergraph by

$$\mathcal{N}_{\mathcal{F}}(i) := \left(\bigcup_{a \in \partial i} \partial a \right) \setminus \{i\}. \tag{2.23}$$

(iii) The total Hamiltonian and canonical Boltzmann-Gibbs distribution of the factor graph are

$$E_\theta(\mathbf{x}) = \sum_{a \in \mathcal{F}} E_a(\mathbf{x}_{\partial a}), \qquad p_\theta(\mathbf{x}) = \frac{1}{Z(\theta, \beta)} \exp\left(-\beta \sum_{a \in \mathcal{F}} E_a(\mathbf{x}_{\partial a}) \right) = \frac{1}{Z(\theta, \beta)} \prod_{a \in \mathcal{F}} \psi_a(\mathbf{x}_{\partial a}), \tag{2.24}$$

where $\psi_a(\mathbf{x}_{\partial a}) := \exp(-\beta E_a(\mathbf{x}_{\partial a})) > 0$ is the positive *factor potential*.

We first derive the single-variable conditional distribution $p_\theta(x_i \mid \mathbf{x}_{-i})$ on an arbitrary factor graph.

Proposition 2.7 (Exact Single-Site Conditional on a Factor Graph)**.** *For any bipartite factor graph $\mathcal{G}_{\mathrm{FG}} = (\mathcal{V}, \mathcal{F}, \mathcal{E}_{\mathrm{FG}})$ with canonical distribution (2.24), define the* local clamping energy *of variable $i \in \mathcal{V}$ as the sum of factor energies over the incident factors ∂i:*

$$E_i(x_i \mid \mathbf{x}_{\mathcal{N}_{\mathcal{F}}(i)}) := \sum_{a \in \partial i} E_a(x_i, \mathbf{x}_{\partial a \setminus \{i\}}). \tag{2.25}$$

Then the conditional distribution of X_i given all remaining variables $\mathbf{X}_{-i} = \mathbf{x}_{-i}$ depends only on the Markov blanket $\mathbf{x}_{\mathcal{N}_{\mathcal{F}}(i)}$ and is given by

$$p_\theta(x_i \mid \mathbf{x}_{-i}) = p_\theta(x_i \mid \mathbf{x}_{\mathcal{N}_{\mathcal{F}}(i)}) = \frac{\exp\Big(-\beta E_i(x_i \mid \mathbf{x}_{\mathcal{N}_{\mathcal{F}}(i)})\Big)}{\int_{\Omega_i} \exp\Big(-\beta E_i(x_i' \mid \mathbf{x}_{\mathcal{N}_{\mathcal{F}}(i)})\Big)\, \mu_i(\mathrm{d}x_i')}. \tag{2.26}$$

Proof. Partition the factor set $\mathcal{F}$ into the incident factors $\partial i = \{a \in \mathcal{F} : i \in \partial a\}$ and the non-incident factors $\mathcal{F} \setminus \partial i = \{a \in \mathcal{F} : i \notin \partial a\}$. Then the total Hamiltonian splits as

$$E_\theta(x_i, \mathbf{x}_{-i}) = \sum_{a \in \partial i} E_a(x_i, \mathbf{x}_{\partial a \setminus \{i\}}) + \sum_{b \in \mathcal{F} \setminus \partial i} E_b(\mathbf{x}_{\partial b}) = E_i(x_i \mid \mathbf{x}_{\mathcal{N}_{\mathcal{F}}(i)}) + E_{\setminus i}(\mathbf{x}_{-i}), \tag{2.27}$$

where $E_{\setminus i}(\mathbf{x}_{-i})$ is completely independent of x_i. Substituting Eq. (2.27) into Bayes' rule $p_\theta(x_i \mid \mathbf{x}_{-i}) = p_\theta(x_i, \mathbf{x}_{-i}) / \int_{\Omega_i} p_\theta(x_i', \mathbf{x}_{-i})\, \mu_i(\mathrm{d}x_i')$, both the global partition function $Z(\theta, \beta)^{-1}$ and the factor $\exp(-\beta E_{\setminus i}(\mathbf{x}_{-i}))$ factor out of the integral in the denominator and cancel against the numerator, leaving Eq. (2.26). □

2.2.2 Two Standard Factor-Graph Families: k-Local Ising and Potts Models

We now specialize Proposition 2.7 to two families that `THRML` [31] supports directly. Gaussian (continuous) nodes are treated in Example 2.27.

Example 2.8 (Higher-Order k-Local Ising / p-Spin Factor Graphs)**.** Let $\Omega = \{-1, +1\}^N$ be an N-spin system whose factor set $\mathcal{F}$ consists of interaction hyperedges $a \subseteq \mathcal{V}$ of arbitrary cardinality $k_a = |a| \geq 1$ (e.g., unary biases $|a| = 1$, pairwise edges $|a| = 2$, three-body plaquettes $|a| = 3$, etc.), with coupling coefficients $J_a \in \mathbb{R}$ and monomial factor energies

$$E_a(\mathbf{s}_a) = -J_a \prod_{j \in a} s_j, \qquad E_\theta(\mathbf{s}) = -\sum_{a \in \mathcal{F}} J_a \prod_{j \in a} s_j. \tag{2.28}$$

For any spin $i \in \mathcal{V}$ and any incident hyperedge $a \in \partial i$, we can factor $s_i \in \{-1, +1\}$ out of the product $\prod_{j \in a} s_j = s_i \prod_{j \in a \setminus \{i\}} s_j$. The local clamping energy (2.25) of spin i therefore takes the linear-in-s_i form

$$E_i(s_i \mid \mathbf{s}_{-i}) = -s_i\, H_i^{\mathrm{eff}}(\mathbf{s}_{-i}), \tag{2.29}$$

where the *effective local field* acting on spin i is the sum over all incident hyperedges $a \in \partial i$:

$$H_i^{\mathrm{eff}}(\mathbf{s}_{-i}) := -\frac{1}{2}\Big(E_\theta(s_i = +1, \mathbf{s}_{-i}) - E_\theta(s_i = -1, \mathbf{s}_{-i})\Big) = \sum_{a \in \partial i} J_a \prod_{j \in a \setminus \{i\}} s_j. \tag{2.30}$$

In particular, when $\mathcal{F}$ contains unary fields ($J_{\{i\}} = h_i$), pairwise couplings ($J_{\{i,j\}} = J_{ij}$), and three-body couplings ($J_{\{i,j,k\}} = K_{ijk}$), the effective local field is

$$H_i^{\mathrm{eff}}(\mathbf{s}_{-i}) = h_i + \sum_{j \in \partial_2 i} J_{ij} s_j + \sum_{\{j,k\} \in \partial_3 i} K_{ijk} s_j s_k, \tag{2.31}$$

where $\partial_2 i$ is the set of spins that share a pairwise factor with i, and $\partial_3 i$ the set of pairs $\{j, k\}$ that share a three-body factor with i. Substituting Eq. (2.29) into Proposition 2.7, the single-spin conditional distribution has the same form as that of an isolated p-bit in the field $H_i^{\mathrm{eff}}(\mathbf{s}_{-i})$ (Example 1.6), with $\sigma(u) = (1 + e^{-u})^{-1}$ the logistic function:

$$p_\theta(s_i \mid \mathbf{s}_{-i}) = \frac{\exp\big(\beta s_i H_i^{\mathrm{eff}}(\mathbf{s}_{-i})\big)}{2\cosh\big(\beta H_i^{\mathrm{eff}}(\mathbf{s}_{-i})\big)} = \sigma\Big(2\beta s_i H_i^{\mathrm{eff}}(\mathbf{s}_{-i})\Big), \qquad \mathbb{E}[S_i \mid \mathbf{s}_{-i}]_{p_\theta} = \tanh\big(\beta H_i^{\mathrm{eff}}(\mathbf{s}_{-i})\big). \tag{2.32}$$

Example 2.9 (q-State Potts and Categorical Factor Models)**.** Suppose each variable $\sigma_i \in \{1, 2, \ldots, q\}$ is a discrete categorical (Potts) variable with $q \geq 2$ states, used to represent tokenized discrete variables or multi-level logic in `THRML` [31, 56]. Using the one-hot basis representation $\mathbf{e}(\sigma_i) \in \{0, 1\}^q$

with $[\mathbf{e}(\sigma_i)]_\alpha = \delta_{\sigma_i,\alpha}$ for $\alpha \in \{1,\dots,q\}$, a pairwise Potts Hamiltonian with local bias vectors $\mathbf{h}_i \in \mathbb{R}^q$ and coupling matrices $\mathbf{J}_{ij} = \mathbf{J}_{ji}^\top \in \mathbb{R}^{q\times q}$ (here ∂i denotes the variables coupled to i, as in Definition 2.1) is

$$E_\theta(\boldsymbol{\sigma}) = -\frac{1}{2}\sum_{i=1}^{N}\sum_{j\in\partial i}\mathbf{e}(\sigma_i)^\top \mathbf{J}_{ij}\mathbf{e}(\sigma_j) - \sum_{i=1}^{N}\mathbf{h}_i^\top \mathbf{e}(\sigma_i) = -\frac{1}{2}\sum_{i=1}^{N}\sum_{j\in\partial i}(J_{ij})_{\sigma_i,\sigma_j} - \sum_{i=1}^{N} h_{i,\sigma_i}. \tag{2.33}$$

By Proposition 2.7, the local clamping energy of node i in state $\sigma_i = \alpha \in \{1,\dots,q\}$ is $E_i(\alpha \mid \boldsymbol{\sigma}_{-i}) = -\big(h_{i,\alpha} + \sum_{j\in\partial i}(J_{ij})_{\alpha,\sigma_j}\big)$, and the single-site conditional is the Boltzmann *softmax* distribution over the q categories:

$$p_\theta(\sigma_i = \alpha \mid \boldsymbol{\sigma}_{-i}) = \frac{\exp\Big(\beta\big[h_{i,\alpha} + \sum_{j\in\partial i}(J_{ij})_{\alpha,\sigma_j}\big]\Big)}{\sum_{\gamma=1}^{q}\exp\Big(\beta\big[h_{i,\gamma} + \sum_{j\in\partial i}(J_{ij})_{\gamma,\sigma_j}\big]\Big)}. \tag{2.34}$$

2.3 Discrete-Time Markov Chains, Detailed Balance, and Gibbs Kernels

Given a factor graph with canonical distribution $p_\theta(x)$, how do we construct a discrete-time stochastic dynamical process $X^{(0)} \to X^{(1)} \to X^{(2)} \to \cdots$ whose distribution $\rho^{(t)}(x)$ converges to $p_\theta(x)$ as $t \to \infty$? This is the subject of Markov chain Monte Carlo (MCMC), which also describes clocked updates in digital and analog sampling hardware.

2.3.1 Markov Transition Operators, Global Balance, and Detailed Balance

Let Ω be a finite configuration space of cardinality $|\Omega|$ (for continuous state spaces $\Omega = \mathbb{R}^N$, sums $\sum_{x\in\Omega}$ are replaced by integrals $\int_\Omega \mu(\mathrm{d}x)$).

Notation Watch: Column-Stochastic Operator Convention vs. Row-Stochastic Convention

In probability textbooks, state distributions $\boldsymbol{\pi}^{(t)}$ are sometimes written as row vectors evolving from the left via $\boldsymbol{\pi}^{(t+1)} = \boldsymbol{\pi}^{(t)}\mathbf{K}$, where $\sum_{x'} K_{x,x'} = 1$. In statistical physics, quantum mechanics, and partial differential equations, however, probability densities $\rho_t(x)$ are usually treated as vectors (or functions) on which linear evolution operators act *from the left*: $\rho_{t+\Delta t} = \mathbf{P}\rho_t$ in discrete time, matching the Fokker-Planck equation $\partial_t\rho_t = \mathcal{L}_{\mathrm{FP}}\rho_t$ in continuous time (Chapter 6).

This book uses the left-acting convention throughout: we write $P(x' \mid x) \equiv P(x \to x')$ for the conditional probability of transitioning *from* initial state x *to* target state x', and arrange these into the matrix $P_{x',x} := P(x' \mid x)$. Because the probabilities of leaving state x must sum to unity, $\mathbf{P}$ is a *column-stochastic* matrix:

$$P(x' \mid x) \geq 0 \quad \forall\, x, x' \in \Omega, \qquad \sum_{x'\in\Omega} P(x' \mid x) = 1 \quad \forall\, x \in \Omega, \tag{2.35}$$

and a probability distribution $\boldsymbol{\rho}^{(t)} = (\rho^{(t)}(x))_{x\in\Omega}$ evolves forward in time by left matrix multiplication:

$$\rho^{(t+1)}(x') = (\mathbf{P}\rho^{(t)})(x') = \sum_{x\in\Omega} P(x' \mid x)\,\rho^{(t)}(x). \tag{2.36}$$

Definition 2.10 (Global Balance, Detailed Balance, Irreducibility, and Aperiodicity)**.** Let $\mathbf{P} = (P(x' \mid x))_{x',x\in\Omega}$ be a Markov transition kernel on Ω, and let $p_{\mathrm{eq}} \in \mathcal{P}(\Omega)$ be a strictly positive target distribution ($p_{\mathrm{eq}}(x) > 0$ for all $x \in \Omega$).

(i) **Stationarity (Global Balance):** We say that p_{eq} is a *stationary* (or *invariant*) distribution of $\mathbf{P}$ if $\mathbf{P}p_{\mathrm{eq}} = p_{\mathrm{eq}}$, that is,

$$\sum_{x\in\Omega} P(x' \mid x)\, p_{\mathrm{eq}}(x) = p_{\mathrm{eq}}(x') \qquad \forall\, x' \in \Omega. \tag{2.37}$$

Using $\sum_{x\in\Omega} P(x \mid x') = 1$ to rewrite the right-hand side as $p_{\mathrm{eq}}(x') = \sum_{x\in\Omega} P(x \mid x')p_{\mathrm{eq}}(x')$, global balance (2.37) is equivalent to the zero-net-flux (Kirchhoff current) condition at every state x':

$$\sum_{x\in\Omega\setminus\{x'\}} \Big[\underbrace{P(x' \mid x)\, p_{\mathrm{eq}}(x)}_{\text{Flux } x\to x'} - \underbrace{P(x \mid x')\, p_{\mathrm{eq}}(x')}_{\text{Flux } x'\to x}\Big] = 0 \qquad \forall\, x' \in \Omega. \tag{2.38}$$

(ii) **Detailed Balance (Microscopic Reversibility):** We say that $\mathbf{P}$ satisfies *detailed balance* with respect to p_{eq} if every individual pairwise probability flux in Eq. (2.38) vanishes term by term:

$$P(x' \mid x)\, p_{\mathrm{eq}}(x) = P(x \mid x')\, p_{\mathrm{eq}}(x') \qquad \forall\, x, x' \in \Omega. \tag{2.39}$$

(iii) **Irreducibility and Aperiodicity (Ergodicity):** The chain $\mathbf{P}$ is *irreducible* if for every pair $x, x' \in \Omega$, there exists an integer $t \geq 1$ such that the t-step transition probability $P^{(t)}(x' \mid x) = (\mathbf{P}^t)_{x',x} > 0$. It is *aperiodic* if $\gcd\left\{t \geq 1 : P^{(t)}(x \mid x) > 0\right\} = 1$ for all $x \in \Omega$. A finite-state chain that is irreducible and aperiodic is called *ergodic*; equivalently, there is a finite integer $t_0 \geq 1$ such that $P^{(t_0)}(x' \mid x) > 0$ for all $x, x' \in \Omega$.

Detailed balance is a stronger condition than global balance: summing Eq. (2.39) over $x \in \Omega$ immediately gives $\sum_x P(x' \mid x)p_{\mathrm{eq}}(x) = \sum_x P(x \mid x')p_{\mathrm{eq}}(x') = p_{\mathrm{eq}}(x')$, which is global balance (2.37). Physically, global balance without detailed balance corresponds to a *non-equilibrium steady state (NESS)* in which probability circulates around closed loops $x \to x' \to x'' \to x$ with non-zero steady-state currents $\mathbf{J}_{\mathrm{ss}} \neq \mathbf{0}$ (as we will study in Chapters 6, 8 and 13), whereas detailed balance (2.39) corresponds to true thermal equilibrium with zero probability currents ($\mathbf{J}_{\mathrm{eq}} = \mathbf{0}$).

Detailed balance also makes the transition operator $\mathbf{P}$ self-adjoint in a suitable inner product, which gives exact convergence rates.

Theorem 2.11 (Spectral Decomposition and Exponential Convergence of Reversible Markov Chains)**.** *Let $\mathbf{P}$ be a Markov transition matrix on a finite state space Ω satisfying detailed balance (2.39) with respect to $p_{\mathrm{eq}} > 0$.*

*(i) **Hermitian Symmetrization:** Define the diagonal multiplication matrix $\mathbf{D}_{\mathrm{eq}} := \mathrm{diag}(p_{\mathrm{eq}}(x))_{x\in\Omega}$ and the similarity-transformed matrix*

$$\mathbf{S} := \mathbf{D}_{\mathrm{eq}}^{-1/2}\mathbf{P}\mathbf{D}_{\mathrm{eq}}^{1/2}, \qquad S(x',x) = p_{\mathrm{eq}}(x')^{-1/2}P(x' \mid x)\, p_{\mathrm{eq}}(x)^{1/2}. \tag{2.40}$$

Then $\mathbf{S}$ is a real symmetric matrix ($S(x',x) = S(x,x')$). Consequently, $\mathbf{P}$ is diagonalizable with all real eigenvalues $1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{|\Omega|} \geq -1$, and possesses right eigenvectors $\boldsymbol{\psi}_k$ and left eigenvectors $\boldsymbol{\phi}_k^\top$ ($\mathbf{P}\boldsymbol{\psi}_k = \lambda_k\boldsymbol{\psi}_k$, $\boldsymbol{\phi}_k^\top\mathbf{P} = \lambda_k\boldsymbol{\phi}_k^\top$) related by $\psi_k(x) = p_{\mathrm{eq}}(x)\phi_k(x)$ and satisfying the p_{eq}-orthonormality and completeness relations

$$\langle \phi_j,\, \phi_k \rangle_{p_{\mathrm{eq}}} := \sum_{x\in\Omega} p_{\mathrm{eq}}(x)\phi_j(x)\phi_k(x) = \delta_{jk}, \qquad P^{(t)}(x' \mid x) = p_{\mathrm{eq}}(x') \sum_{k=1}^{|\Omega|} \lambda_k^t\, \phi_k(x')\phi_k(x), \tag{2.41}$$

where the leading eigenpair for $\lambda_1 = 1$ is $\phi_1(x) = 1$ and $\psi_1(x) = p_{\mathrm{eq}}(x)$.

(ii) ***Spectral Gap and χ^2 / Total Variation Mixing Bound:*** *If $\mathbf{P}$ is irreducible and aperiodic, then the second largest eigenvalue modulus is strictly less than unity:*

$$\lambda_* := \max_{k\geq 2} |\lambda_k| = \max(\lambda_2, \left|\lambda_{|\Omega|}\right|) < 1, \qquad \gamma_{\text{spec}} := 1 - \lambda_* > 0, \tag{2.42}$$

where γ_{spec} is the absolute spectral gap. *For any initial distribution $\rho^{(0)} \in \mathcal{P}(\Omega)$ and $\rho^{(t)} = \mathbf{P}^t\rho^{(0)}$, the total variation distance $\left\|\rho^{(t)} - p_{\text{eq}}\right\|_{\text{TV}} := \frac{1}{2}\sum_{x\in\Omega}\left|\rho^{(t)}(x) - p_{\text{eq}}(x)\right|$ and the Pearson χ^2-divergence $\chi^2(\rho^{(t)} \,\|\, p_{\text{eq}}) := \sum_{x\in\Omega} p_{\text{eq}}(x)\left(\frac{\rho^{(t)}(x)}{p_{\text{eq}}(x)} - 1\right)^2$ decay exponentially at rate γ_{spec}:*

$$2\left\|\rho^{(t)} - p_{\text{eq}}\right\|_{\text{TV}} \leq \sqrt{\chi^2(\rho^{(t)} \,\|\, p_{\text{eq}})} \leq (1-\gamma_{\text{spec}})^t\sqrt{\chi^2(\rho^{(0)} \,\|\, p_{\text{eq}})} \leq \frac{e^{-\gamma_{\text{spec}}t}}{\sqrt{\min_{x\in\Omega} p_{\text{eq}}(x)}}. \tag{2.43}$$

Thus the ε-mixing time $\tau_{\text{mix}}(\varepsilon) := \min\left\{t \geq 0 : \sup_{\rho^{(0)}}\left\|\mathbf{P}^t\rho^{(0)} - p_{\text{eq}}\right\|_{\text{TV}} \leq \varepsilon\right\}$ satisfies

$$\tau_{\text{mix}}(\varepsilon) \leq \frac{1}{\gamma_{\text{spec}}} \ln\left(\frac{1}{2\varepsilon\sqrt{\min_{x\in\Omega} p_{\text{eq}}(x)}}\right). \tag{2.44}$$

Proof. **Proof of (i):** Using detailed balance $P(x' \mid x)p_{\text{eq}}(x) = P(x \mid x')p_{\text{eq}}(x')$, we multiply both sides by $p_{\text{eq}}(x')^{-1/2}p_{\text{eq}}(x)^{-1/2}$ to obtain

$$S(x', x) = p_{\text{eq}}(x')^{-1/2}P(x' \mid x)\, p_{\text{eq}}(x)^{1/2} = p_{\text{eq}}(x)^{-1/2}P(x \mid x')\, p_{\text{eq}}(x')^{1/2} = S(x, x'), \tag{2.45}$$

proving that $\mathbf{S} = \mathbf{S}^\top$ is real symmetric. By the finite-dimensional spectral theorem, $\mathbf{S}$ admits an orthonormal eigenbasis $\{\mathbf{u}_k\}_{k=1}^{|\Omega|}$ in $\mathbb{R}^{|\Omega|}$ with real eigenvalues $\lambda_k \in \mathbb{R}$: $\mathbf{S}\mathbf{u}_k = \lambda_k\mathbf{u}_k$ and $\mathbf{u}_j^\top\mathbf{u}_k = \delta_{jk}$. Defining $\phi_k(x) := p_{\text{eq}}(x)^{-1/2}u_k(x)$ and $\psi_k(x) := p_{\text{eq}}(x)^{1/2}u_k(x) = p_{\text{eq}}(x)\phi_k(x)$, we have $\mathbf{P}\boldsymbol{\psi}_k = \mathbf{D}_{\text{eq}}^{1/2}\mathbf{S}\mathbf{u}_k = \lambda_k\boldsymbol{\psi}_k$, $\boldsymbol{\phi}_k^\top\mathbf{P} = \mathbf{u}_k^\top\mathbf{S}\mathbf{D}_{\text{eq}}^{-1/2} = \lambda_k\boldsymbol{\phi}_k^\top$, and $\langle\phi_j,\, \phi_k\rangle_{p_{\text{eq}}} = \sum_x p_{\text{eq}}(x)p_{\text{eq}}(x)^{-1}u_j(x)u_k(x) = \mathbf{u}_j^\top\mathbf{u}_k = \delta_{jk}$. Moreover, for any eigenvalue λ_k, choosing x^* to maximize $|\phi_k(x)|$ gives

$$|\lambda_k|\,|\phi_k(x^*)| = \left|\sum_{x'\in\Omega}\phi_k(x')P(x' \mid x^*)\right| \leq \sum_{x'\in\Omega}|\phi_k(x^*)|\,P(x' \mid x^*) = |\phi_k(x^*)|\,, \tag{2.46}$$

so $|\lambda_k| \leq 1$, i.e., $\lambda_k \in [-1, 1]$. From $\mathbf{S}^t = \sum_{k=1}^{|\Omega|}\lambda_k^t\mathbf{u}_k\mathbf{u}_k^\top$, multiplying on the left by $\mathbf{D}_{\text{eq}}^{1/2}$ and on the right by $\mathbf{D}_{\text{eq}}^{-1/2}$ yields the t-step spectral propagator (2.41).

Proof of (ii): When $\mathbf{P}$ is irreducible and aperiodic, the Perron-Frobenius theorem applied to the strictly positive matrix $\mathbf{P}^{t_0}$ implies that $\lambda_1 = 1$ is a simple eigenvalue and every other eigenvalue satisfies $|\lambda_k| < 1$ for $k \geq 2$, so $\gamma_{\text{spec}} = 1 - \lambda_* > 0$.

Now let $\rho^{(0)} \in \mathcal{P}(\Omega)$ be any initial distribution, and define the relative density $h^{(t)}(x) := \rho^{(t)}(x)/p_{\text{eq}}(x)$. Expanding $h^{(0)}(x)$ in the p_{eq}-orthonormal basis $\{\phi_k\}_{k=1}^{|\Omega|}$ gives $h^{(0)}(x) = \sum_{k=1}^{|\Omega|} c_k\phi_k(x)$, where $c_k = \left\langle\phi_k,\, h^{(0)}\right\rangle_{p_{\text{eq}}} = \sum_{x\in\Omega}\phi_k(x)\rho^{(0)}(x)$. In particular, since $\phi_1(x) = 1$, the $k = 1$ coefficient is $c_1 = \sum_{x\in\Omega}\rho^{(0)}(x) = 1$. Applying the t-step spectral propagator (2.41) to $\rho^{(0)}(x) = p_{\text{eq}}(x)h^{(0)}(x)$ and dividing by $p_{\text{eq}}(x')$ separates the equilibrium mode $k = 1$ ($\lambda_1 = 1$) from the transient modes $k \geq 2$:

$$h^{(t)}(x') - 1 = \frac{\rho^{(t)}(x')}{p_{\text{eq}}(x')} - 1 = \sum_{k=2}^{|\Omega|} c_k\lambda_k^t\,\phi_k(x'). \tag{2.47}$$

Using the p_{eq}-orthonormality $\langle \phi_j,\, \phi_k \rangle_{p_{\mathrm{eq}}} = \delta_{jk}$ (Parseval's identity in $\ell^2(\Omega, p_{\mathrm{eq}})$), the χ^2-divergence at step t is

$$\begin{aligned}\chi^2(\rho^{(t)} \,\|\, p_{\mathrm{eq}}) &= \sum_{x\in\Omega} p_{\mathrm{eq}}(x)\big(h^{(t)}(x)-1\big)^2 = \sum_{k=2}^{|\Omega|} c_k^2 \lambda_k^{2t} \\ &\le \lambda_*^{2t} \sum_{k=2}^{|\Omega|} c_k^2 = (1-\gamma_{\mathrm{spec}})^{2t}\, \chi^2(\rho^{(0)} \,\|\, p_{\mathrm{eq}}). \end{aligned} \tag{2.48}$$

By the Cauchy-Schwarz inequality,

$$\begin{aligned} 2\left\|\rho^{(t)} - p_{\mathrm{eq}}\right\|_{\mathrm{TV}} &= \sum_{x\in\Omega} p_{\mathrm{eq}}(x)^{1/2} \cdot p_{\mathrm{eq}}(x)^{1/2} \left|h^{(t)}(x)-1\right| \\ &\le \left(\sum_{x\in\Omega} p_{\mathrm{eq}}(x)\right)^{1/2} \left(\sum_{x\in\Omega} p_{\mathrm{eq}}(x)\big(h^{(t)}(x)-1\big)^2\right)^{1/2} = \sqrt{\chi^2(\rho^{(t)} \,\|\, p_{\mathrm{eq}})}. \end{aligned} \tag{2.49}$$

Finally, for any initial state $\rho^{(0)}(x) = \delta_{x,x_0}$ concentrated at a single configuration $x_0 \in \Omega$, we have $\chi^2(\delta_{x_0} \,\|\, p_{\mathrm{eq}}) = p_{\mathrm{eq}}(x_0)^{-1} - 1 \le (\min_x p_{\mathrm{eq}}(x))^{-1}$. Combining this with Eqs. (2.48) and (2.49) and $1-\gamma_{\mathrm{spec}} \le e^{-\gamma_{\mathrm{spec}}}$ yields Eqs. (2.43) and (2.44). □

2.3.2 Metropolis–Hastings vs. Single-Site Glauber (Gibbs) Kernels

How do we construct concrete transition kernels $P(x' \mid x)$ that satisfy detailed balance (2.39) with respect to $p_\theta(x) = Z(\theta,\beta)^{-1} e^{-\beta E_\theta(x)}$ without ever evaluating the intractable partition function $Z(\theta,\beta)$? The point is that in the detailed-balance ratio

$$\frac{P(x' \mid x)}{P(x \mid x')} = \frac{p_\theta(x')}{p_\theta(x)} = \exp\Big(-\beta\big[E_\theta(x') - E_\theta(x)\big]\Big) = e^{-\beta \Delta E_\theta(x \to x')}, \tag{2.50}$$

the partition function $Z(\theta,\beta)$ cancels identically. Two standard families of local kernels exploit Eq. (2.50):

1. **The Metropolis–Hastings Kernel:** Let $q(x' \mid x)$ be an arbitrary proposal distribution (for example, choosing a single spin $i \in \{1,\dots,N\}$ uniformly at random and proposing to flip $s_i \to -s_i$, which is symmetric: $q(x' \mid x) = q(x \mid x') = 1/N$). The Metropolis-Hastings acceptance probability for moving from x to $x' \neq x$ is

$$A_{\mathrm{MH}}(x \to x') := \min\left(1,\, \frac{p_\theta(x')\, q(x \mid x')}{p_\theta(x)\, q(x' \mid x)}\right) \overset{q \text{ sym}}{=} \min\Big(1,\, e^{-\beta[E_\theta(x') - E_\theta(x)]}\Big), \tag{2.51}$$

   with transition probability $P_{\mathrm{MH}}(x' \mid x) = q(x' \mid x) A_{\mathrm{MH}}(x \to x')$ for $x' \neq x$ and rejection self-loop probability $P_{\mathrm{MH}}(x \mid x) = 1 - \sum_{x' \neq x} P_{\mathrm{MH}}(x' \mid x)$. To verify detailed balance for any $x \neq x'$, simply use the identity $u \min(1, v/u) = \min(u,v) = v \min(1, u/v)$ for $u = p_\theta(x) q(x' \mid x)$ and $v = p_\theta(x') q(x \mid x')$:

$$P_{\mathrm{MH}}(x' \mid x)\, p_\theta(x) = \min\big(p_\theta(x) q(x' \mid x),\, p_\theta(x') q(x \mid x')\big) = P_{\mathrm{MH}}(x \mid x')\, p_\theta(x'). \tag{2.52}$$

2. **The Single-Site Gibbs (Heat-Bath / Glauber) Kernel:** Rather than proposing a state change and then running a separate accept/reject comparator, the single-site Gibbs update

for variable $i \in \mathcal{V}$ simply draws the new state $X_i' \sim p_\theta(x_i' \mid \mathbf{x}_{-i})$ directly from its exact local conditional distribution (2.26) while holding all other variables $\mathbf{x}_{-i}$ fixed:

$$P_i(\mathbf{x}' \mid \mathbf{x}) := \delta_{\mathbf{x}'_{-i},\mathbf{x}_{-i}}\, p_\theta(x_i' \mid \mathbf{x}_{-i}). \tag{2.53}$$

For a binary spin $s_i \in \{-1,+1\}$ in a factor graph with effective local field $H_i^{\text{eff}}(\mathbf{s}_{-i})$ (Example 2.8), Eq. (2.53) is the *Glauber (heat-bath) dynamics*, which sets $s_i' = +1$ with logistic probability $\sigma(2\beta H_i^{\text{eff}}(\mathbf{s}_{-i})) = (1+e^{-2\beta H_i^{\text{eff}}})^{-1}$ and $s_i' = -1$ with probability $\sigma(-2\beta H_i^{\text{eff}}(\mathbf{s}_{-i}))$.

Two properties of the single-site (and block) Gibbs kernel make it the natural update rule for sampling hardware.

Proposition 2.12 (Detailed Balance and Projection Idempotence of the Gibbs Kernel)**.** *For any subset of variables $A \subseteq \mathcal{V}$ (with complement $-A \equiv \mathcal{V} \setminus A$), define the* block Gibbs kernel P_A *that resamples $\mathbf{X}_A' \sim p_\theta(\mathbf{x}_A' \mid \mathbf{x}_{-A})$ while holding $\mathbf{x}_{-A}$ fixed:*

$$P_A(\mathbf{x}' \mid \mathbf{x}) := \delta_{\mathbf{x}'_{-A},\mathbf{x}_{-A}}\, p_\theta(\mathbf{x}_A' \mid \mathbf{x}_{-A}). \tag{2.54}$$

Then:

(i) P_A satisfies detailed balance with respect to $p_\theta(\mathbf{x})$: $P_A(\mathbf{x}' \mid \mathbf{x})p_\theta(\mathbf{x}) = P_A(\mathbf{x} \mid \mathbf{x}')p_\theta(\mathbf{x}')$.

(ii) P_A is an idempotent projection operator ($\mathbf{P}_A^2 = \mathbf{P}_A$), so its only eigenvalues are 1 and 0.

Proof. For (i), multiply $P_A(\mathbf{x}' \mid \mathbf{x})$ by $p_\theta(\mathbf{x}) = p_\theta(\mathbf{x}_A \mid \mathbf{x}_{-A})p_\theta(\mathbf{x}_{-A})$ and use the Kronecker delta $\delta_{\mathbf{x}'_{-A},\mathbf{x}_{-A}}$ to interchange $\mathbf{x}_{-A}$ and $\mathbf{x}'_{-A}$:

$$\begin{aligned} P_A(\mathbf{x}' \mid \mathbf{x})\, p_\theta(\mathbf{x}) &= \delta_{\mathbf{x}'_{-A},\mathbf{x}_{-A}}\, p_\theta(\mathbf{x}_A' \mid \mathbf{x}_{-A})\, p_\theta(\mathbf{x}_A \mid \mathbf{x}_{-A})\, p_\theta(\mathbf{x}_{-A}) \\ &= \delta_{\mathbf{x}_{-A},\mathbf{x}'_{-A}}\, p_\theta(\mathbf{x}_A \mid \mathbf{x}'_{-A})\, p_\theta(\mathbf{x}_A' \mid \mathbf{x}'_{-A})\, p_\theta(\mathbf{x}'_{-A}) = P_A(\mathbf{x} \mid \mathbf{x}')\, p_\theta(\mathbf{x}'), \end{aligned} \tag{2.55}$$

which is symmetric under $\mathbf{x} \leftrightarrow \mathbf{x}'$, proving detailed balance.

For (ii), we compute the two-step matrix product $(\mathbf{P}_A^2)(\mathbf{x}'' \mid \mathbf{x})$ by summing over the intermediate state $\mathbf{x}' = (\mathbf{x}_A', \mathbf{x}'_{-A})$ and using normalization $\sum_{\mathbf{x}_A'} p_\theta(\mathbf{x}_A' \mid \mathbf{x}_{-A}) = 1$:

$$\begin{aligned} \sum_{\mathbf{x}' \in \Omega} P_A(\mathbf{x}'' \mid \mathbf{x}')\, P_A(\mathbf{x}' \mid \mathbf{x}) &= \sum_{\mathbf{x}_A'} \sum_{\mathbf{x}'_{-A}} \delta_{\mathbf{x}''_{-A},\mathbf{x}'_{-A}}\, p_\theta(\mathbf{x}_A'' \mid \mathbf{x}'_{-A})\, \delta_{\mathbf{x}'_{-A},\mathbf{x}_{-A}}\, p_\theta(\mathbf{x}_A' \mid \mathbf{x}_{-A}) \\ &= \delta_{\mathbf{x}''_{-A},\mathbf{x}_{-A}}\, p_\theta(\mathbf{x}_A'' \mid \mathbf{x}_{-A}) \underbrace{\sum_{\mathbf{x}_A'} p_\theta(\mathbf{x}_A' \mid \mathbf{x}_{-A})}_{=1} = P_A(\mathbf{x}'' \mid \mathbf{x}), \end{aligned} \tag{2.56}$$

proving $\mathbf{P}_A^2 = \mathbf{P}_A$. □

A single block kernel P_A is therefore never ergodic on its own: it leaves $\mathbf{x}_{-A}$ frozen, so one application already does everything P_A can do, and repeating it changes nothing. Convergence to p_θ requires composing kernels for blocks that together cover $\mathcal{V}$. To analyze such compositions it is convenient to track the *density ratio* $h^{(t)}(\mathbf{x}) := \rho^{(t)}(\mathbf{x})/p_\theta(\mathbf{x})$ rather than $\rho^{(t)}$ itself, and to work in the Hilbert space $L^2(p_\theta)$ of real functions on Ω with inner product $\langle f, g\rangle_{p_\theta} := \sum_{\mathbf{x}} p_\theta(\mathbf{x}) f(\mathbf{x}) g(\mathbf{x})$ (the same space used in Theorem 2.11). In this language a block Gibbs update has a transparent meaning: it replaces the density ratio by its conditional expectation.

Lemma 2.13 (Block Gibbs Updates Act as Conditional Expectations)**.** *Let $\rho \in \mathcal{P}(\Omega)$, $h = \rho/p_\theta$, and $\rho' = \mathbf{P}_A \rho$. Then $h' := \rho'/p_\theta$ is*

$$h'(\mathbf{x}) = (\Pi_A h)(\mathbf{x}) := \mathbb{E}[h(\mathbf{X}) \mid \mathbf{X}_{-A} = \mathbf{x}_{-A}]_{p_\theta} = \sum_{\mathbf{y}_A \in \Omega_A} p_\theta(\mathbf{y}_A \mid \mathbf{x}_{-A})\, h(\mathbf{y}_A, \mathbf{x}_{-A}), \tag{2.57}$$

and Π_A is an orthogonal projection in $L^2(p_\theta)$ onto the subspace of functions that do not depend on $\mathbf{x}_A$. It fixes constants and maps the mean-zero subspace $L^2_0(p_\theta) := \left\{ f : \mathbb{E}[f]_{p_\theta} = 0 \right\}$ into itself.

Proof. By Eqs. (2.36) and (2.54), the Kronecker delta collapses the sum over $\mathbf{x}_{-A}$:

$$\rho'(\mathbf{x}') = \sum_{\mathbf{x}_A} p_\theta(\mathbf{x}'_A \mid \mathbf{x}'_{-A})\, \rho(\mathbf{x}_A, \mathbf{x}'_{-A}) = p_\theta(\mathbf{x}'_A \mid \mathbf{x}'_{-A}) \sum_{\mathbf{x}_A} p_\theta(\mathbf{x}_A, \mathbf{x}'_{-A})\, h(\mathbf{x}_A, \mathbf{x}'_{-A}). \tag{2.58}$$

Writing $p_\theta(\mathbf{x}_A, \mathbf{x}'_{-A}) = p_\theta(\mathbf{x}_A \mid \mathbf{x}'_{-A}) p_\theta(\mathbf{x}'_{-A})$ and $p_\theta(\mathbf{x}'_A \mid \mathbf{x}'_{-A}) p_\theta(\mathbf{x}'_{-A}) = p_\theta(\mathbf{x}')$, we get $\rho'(\mathbf{x}') = p_\theta(\mathbf{x}') \sum_{\mathbf{x}_A} p_\theta(\mathbf{x}_A \mid \mathbf{x}'_{-A}) h(\mathbf{x}_A, \mathbf{x}'_{-A})$, which is Eq. (2.57). The right-hand side depends only on $\mathbf{x}'_{-A}$, and if h already depends only on $\mathbf{x}_{-A}$ then $\Pi_A h = h$, so $\Pi_A^2 = \Pi_A$. Self-adjointness follows from the tower property of conditional expectation:

$$\langle f,\, \Pi_A g \rangle_{p_\theta} = \mathbb{E}[f\, \mathbb{E}[g \mid \mathbf{X}_{-A}]]_{p_\theta} = \mathbb{E}[\mathbb{E}[f \mid \mathbf{X}_{-A}]\, \mathbb{E}[g \mid \mathbf{X}_{-A}]]_{p_\theta} = \langle \Pi_A f,\, g \rangle_{p_\theta}\,. \tag{2.59}$$

A self-adjoint idempotent is an orthogonal projection. Finally $\Pi_A 1 = 1$ and $\mathbb{E}[\Pi_A f]_{p_\theta} = \mathbb{E}[f]_{p_\theta}$, so $L^2_0(p_\theta)$ is invariant. □

Since $\chi^2(\rho \,\|\, p_\theta) = \|h - 1\|^2_{p_\theta}$ and an orthogonal projection never increases a norm, Lemma 2.13 already shows that every block Gibbs update is non-expansive in χ^2. The remaining question is when a composition of such projections is strictly contracting.

2.3.3 Composing Gibbs Kernels: Scan Orders, Reversibility, and Ergodicity

There are two standard ways to combine the single-site kernels $P_1, \dots, P_N$ of Eq. (2.53), and they differ in a structural property that is easy to overlook.

Proposition 2.14 (Random Scan, Systematic Scan, and Ergodicity)**.** *Let $A_1, \dots, A_m$ be blocks covering $\mathcal{V}$, with block Gibbs kernels $P_{A_1}, \dots, P_{A_m}$ for a strictly positive p_θ.*

(i) The random-scan *kernel $P_{\mathrm{RS}} := \sum_{k=1}^m w_k P_{A_k}$ with weights $w_k > 0$, $\sum_k w_k = 1$, satisfies detailed balance with respect to p_θ.*

(ii) The systematic-scan *(sweep) kernel $P_{\mathrm{SS}} := P_{A_m} \cdots P_{A_2} P_{A_1}$ leaves p_θ stationary. Its time reversal with respect to p_θ, defined by $\hat{P}(\mathbf{x} \mid \mathbf{x}') := P(\mathbf{x}' \mid \mathbf{x}) p_\theta(\mathbf{x}) / p_\theta(\mathbf{x}')$, is the reversed sweep $\hat{P}_{\mathrm{SS}} = P_{A_1} P_{A_2} \cdots P_{A_m}$. Hence P_{SS} satisfies detailed balance if and only if the forward and reversed sweeps coincide; the palindromic sweep $P_{A_1} \cdots P_{A_m} \cdots P_{A_1}$ always does.*

(iii) The systematic-scan kernel is strictly positive, $P_{\mathrm{SS}}(\mathbf{x}' \mid \mathbf{x}) > 0$ for all $\mathbf{x}, \mathbf{x}'$, whenever the blocks partition $\mathcal{V}$. In particular it is irreducible and aperiodic.

Proof. (i) Detailed balance is linear in the kernel: summing $w_k P_{A_k}(\mathbf{x}' \mid \mathbf{x}) p_\theta(\mathbf{x}) = w_k P_{A_k}(\mathbf{x} \mid \mathbf{x}') p_\theta(\mathbf{x}')$ over k gives the claim.

(ii) Stationarity holds because each factor preserves p_θ: $P_{A_1} p_\theta = p_\theta$, then $P_{A_2} p_\theta = p_\theta$, and so on. For the time reversal, take $m = 2$ (the general case follows by induction). Using detailed balance of P_{A_1} and then of P_{A_2} inside the sum over the intermediate state $\mathbf{x}'$,

$$(P_{A_2} P_{A_1})(\mathbf{x}'' \mid \mathbf{x})\, p_\theta(\mathbf{x}) = \sum_{\mathbf{x}'} P_{A_2}(\mathbf{x}'' \mid \mathbf{x}')\, P_{A_1}(\mathbf{x}' \mid \mathbf{x})\, p_\theta(\mathbf{x}) = \sum_{\mathbf{x}'} P_{A_2}(\mathbf{x}'' \mid \mathbf{x}')\, p_\theta(\mathbf{x}')\, P_{A_1}(\mathbf{x} \mid \mathbf{x}')$$

$$= \sum_{\mathbf{x}'} p_\theta(\mathbf{x}'')\, P_{A_2}(\mathbf{x}' \mid \mathbf{x}'')\, P_{A_1}(\mathbf{x} \mid \mathbf{x}') = (P_{A_1} P_{A_2})(\mathbf{x} \mid \mathbf{x}'')\, p_\theta(\mathbf{x}''). \tag{2.60}$$

Thus $\hat{P}_{\mathrm{SS}} = P_{A_1} P_{A_2}$, and detailed balance $\hat{P}_{\mathrm{SS}} = P_{\mathrm{SS}}$ holds exactly when the two orders agree. For a palindrome the reversed product is the same product, so detailed balance holds.

(iii) When the blocks partition $\mathcal{V}$, a sweep resamples every variable exactly once. Starting from $\mathbf{x}$, the probability of arriving at $\mathbf{x}'$ is the product of the block conditionals evaluated along the unique path that writes $\mathbf{x}'_{A_1}$, then $\mathbf{x}'_{A_2}$, and so on:

$$P_{\mathrm{SS}}(\mathbf{x}' \mid \mathbf{x}) = \prod_{k=1}^{m} p_\theta(\mathbf{x}'_{A_k} \mid \mathbf{x}'_{A_1}, \ldots, \mathbf{x}'_{A_{k-1}}, \mathbf{x}_{A_{k+1}}, \ldots, \mathbf{x}_{A_m}) > 0, \tag{2.61}$$

since every conditional of a strictly positive distribution is strictly positive. A strictly positive matrix is irreducible ($t = 1$ in Definition 2.10) and aperiodic ($P_{\mathrm{SS}}(\mathbf{x} \mid \mathbf{x}) > 0$). □

For two coupled spins, $A_1 = \{1\}$ and $A_2 = \{2\}$, the forward sweep $(P_2 P_1)(\mathbf{s}'' \mid \mathbf{s}) = p_\theta(s_1'' \mid s_2) p_\theta(s_2'' \mid s_1'')$ depends on the initial state only through s_2, while the reversed sweep $(P_1 P_2)(\mathbf{s}'' \mid \mathbf{s}) = p_\theta(s_2'' \mid s_1) p_\theta(s_1'' \mid s_2'')$ depends only on s_1. These differ whenever the coupling is non-zero, so the ordinary sweep of a coupled system is stationary but not reversible. This matters for the spectral theory: Theorem 2.11 does not apply directly to P_{SS}, whose eigenvalues may be complex. We return to this point in Section 2.5.

The proof of Theorem 2.11(ii) invoked the Perron–Frobenius theorem. The part we need has a short self-contained proof.

Lemma 2.15 (Perron–Frobenius for Strictly Positive Stochastic Matrices)**.** *Let $\mathbf{Q}$ be a column-stochastic matrix on a finite Ω with $Q(x' \mid x) > 0$ for all x, x'. If $\boldsymbol{\phi} \in \mathbb{C}^{|\Omega|}$ is a left eigenvector, $\sum_{x'} \phi(x') Q(x' \mid x) = \lambda \phi(x)$ for all x, with $|\lambda| = 1$, then $\lambda = 1$ and $\boldsymbol{\phi}$ is constant. Consequently, if $\mathbf{P}$ is stochastic and $\mathbf{P}^{t_0}$ is strictly positive for some $t_0 \geq 1$, the only eigenvalue of $\mathbf{P}$ on the unit circle is $\lambda = 1$, and its left eigenspace consists of constants.*

Proof. Choose x^* with $|\phi(x^*)| = \max_x |\phi(x)| > 0$. Then

$$|\phi(x^*)| = |\lambda \phi(x^*)| = \Big| \sum_{x'} \phi(x') Q(x' \mid x^*) \Big| \leq \sum_{x'} |\phi(x')|\, Q(x' \mid x^*) \leq |\phi(x^*)| \sum_{x'} Q(x' \mid x^*) = |\phi(x^*)|\,. \tag{2.62}$$

Both inequalities must be equalities. Since every weight $Q(x' \mid x^*)$ is strictly positive, equality in the second one forces $|\phi(x')| = |\phi(x^*)|$ for every x'. Equality in the triangle inequality (the first one) forces all the complex numbers $\phi(x') Q(x' \mid x^*)$ to share a common phase, so $\phi(x') = \phi(x^*)$ for all x': $\boldsymbol{\phi}$ is constant. Inserting a constant into the eigenvalue equation and using $\sum_{x'} Q(x' \mid x) = 1$ gives $\lambda = 1$.

For the consequence, let $\boldsymbol{\phi}^\top \mathbf{P} = \lambda \boldsymbol{\phi}^\top$ with $|\lambda| = 1$. Then $\boldsymbol{\phi}^\top \mathbf{P}^{t_0} = \lambda^{t_0} \boldsymbol{\phi}^\top$ with $|\lambda^{t_0}| = 1$, and the first part applied to $\mathbf{Q} = \mathbf{P}^{t_0}$ shows that $\boldsymbol{\phi}$ is constant. A constant left vector satisfies $\boldsymbol{\phi}^\top \mathbf{P} = \boldsymbol{\phi}^\top$, so $\lambda = 1$. □

For a reversible $\mathbf{P}$, which is diagonalizable by Theorem 2.11(i), the lemma shows that $\lambda_1 = 1$ has a one-dimensional eigenspace and that $|\lambda_k| < 1$ for $k \geq 2$, which is exactly what part (ii) of that theorem used.

2.3.4 Variational Form of the Spectral Gap

For reversible chains the gap $1-\lambda_2$ has a variational characterization that parallels the Poincaré inequality of continuous diffusions (Section 6.5, Eq. (6.82)). It is the tool we use to read off gaps in examples.

Proposition 2.16 (Dirichlet Form and the Discrete Poincaré Inequality)**.** *Let* $\mathbf{P}$ *satisfy detailed balance with respect to* $p_{\mathrm{eq}} > 0$*, and define the backward operator* $(\mathcal{K}f)(x) := \sum_{x'} P(x' \mid x) f(x')$ *and the* Dirichlet form

$$\mathcal{D}(f,f) := \frac{1}{2} \sum_{x,x' \in \Omega} p_{\mathrm{eq}}(x)\, P(x' \mid x)\, \big(f(x') - f(x)\big)^2. \tag{2.63}$$

Then $\mathcal{D}(f,f) = \langle f, (\mathbf{I} - \mathcal{K}) f \rangle_{p_{\mathrm{eq}}}$*, and the relaxation gap is*

$$1 - \lambda_2 = \min_{f \text{ non-constant}} \frac{\mathcal{D}(f,f)}{\mathrm{Var}_{p_{\mathrm{eq}}}(f)}, \tag{2.64}$$

so that $\mathrm{Var}_{p_{\mathrm{eq}}}(f) \le (1-\lambda_2)^{-1} \mathcal{D}(f,f)$ *for every observable* f*.*

Proof. Expand the square in Eq. (2.63):

$$2\mathcal{D}(f,f) = \sum_{x,x'} p_{\mathrm{eq}}(x) P(x' \mid x) f(x')^2 + \sum_{x,x'} p_{\mathrm{eq}}(x) P(x' \mid x) f(x)^2 - 2 \sum_{x,x'} p_{\mathrm{eq}}(x) P(x' \mid x) f(x) f(x'). \tag{2.65}$$

In the first sum, detailed balance replaces $p_{\mathrm{eq}}(x)P(x' \mid x)$ by $p_{\mathrm{eq}}(x')P(x \mid x')$, and $\sum_x P(x \mid x') = 1$ leaves $\sum_{x'} p_{\mathrm{eq}}(x') f(x')^2 = \|f\|^2_{p_{\mathrm{eq}}}$. The second sum equals $\|f\|^2_{p_{\mathrm{eq}}}$ by normalization. The third is $2\sum_x p_{\mathrm{eq}}(x) f(x) (\mathcal{K}f)(x) = 2\langle f, \mathcal{K}f \rangle_{p_{\mathrm{eq}}}$. Hence $\mathcal{D}(f,f) = \|f\|^2_{p_{\mathrm{eq}}} - \langle f, \mathcal{K}f \rangle_{p_{\mathrm{eq}}}$. The left eigenvectors ϕ_k of Theorem 2.11 are exactly the eigenfunctions of $\mathcal{K}$ ($\mathcal{K}\phi_k = \lambda_k \phi_k$) and form a p_{eq}-orthonormal basis with $\phi_1 = 1$. Both $\mathcal{D}$ and $\mathrm{Var}_{p_{\mathrm{eq}}}$ are unchanged by adding a constant to f, so take $f = \sum_{k \ge 2} a_k \phi_k$ with zero mean. Then $\mathcal{D}(f,f) = \sum_{k \ge 2} (1-\lambda_k) a_k^2 \ge (1-\lambda_2) \sum_{k \ge 2} a_k^2 = (1-\lambda_2)\, \mathrm{Var}_{p_{\mathrm{eq}}}(f)$, with equality at $f = \phi_2$. □

Equation (2.63) has a direct physical reading: $\mathcal{D}(f,f)$ is the equilibrium rate at which the observable f changes per step, and Eq. (2.64) says that a small gap means some observable has a large equilibrium variance yet changes rarely. In the Ising examples below this observable is the magnetization, and the small gap is the discrete counterpart of barrier crossing in the double-well sampler, the quartic double well whose binary limit is the p-bit of Example 1.6 (Kramers escape, Chapter 7).

Example 2.17 (Glauber versus Metropolis Flip Probabilities for an Ising Spin)**.** Let spin i feel the effective field H_i^{eff} of Eq. (2.30). Flipping $s_i \to -s_i$ changes the energy by $\Delta E_i = 2 s_i H_i^{\mathrm{eff}}$. The heat-bath kernel draws s_i' from Eq. (2.32) regardless of the current value, so the probability that the spin actually flips is $p_\theta(-s_i \mid \mathbf{s}_{-i}) = (1 + e^{\beta \Delta E_i})^{-1}$. Metropolis with a deterministic proposal $s_i \to -s_i$ flips with probability $\min(1, e^{-\beta \Delta E_i})$. Writing $u = \beta \Delta E_i$,

$$w_{\mathrm{G}}(u) = \frac{1}{1+e^{u}}, \qquad w_{\mathrm{M}}(u) = \min\big(1, e^{-u}\big), \qquad \frac{w_{\mathrm{G}}(u)}{w_{\mathrm{M}}(u)} = \frac{1}{1+e^{-|u|}} \in \big[\tfrac{1}{2}, 1\big). \tag{2.66}$$

Both satisfy $w(u)/w(-u) = e^{-u}$, which is detailed balance. Metropolis flips more often, by at most a factor of two (at $u = 0$, where Glauber flips with probability $1/2$ and Metropolis with probability 1). The practical advantage of the heat-bath rule for hardware is different: it never needs to know the old value s_i. A p-bit that outputs $+1$ with probability $\sigma(2\beta H_i^{\mathrm{eff}})$, independent of its previous output, implements Eq. (2.53) with no comparator and no memory of its own state (Example 1.6, Section 14.1).

2.4 The Synchronous Update Pathology

The single-site heat-bath kernel is exact, local, and needs no partition function. The obvious way to run it on a chip with N p-bits is to latch the current configuration $\mathbf{s}$, let every node compute its local field from the latched values, and let all N nodes draw new states in the same clock cycle. Each node then samples from its exact conditional distribution. The joint update is nevertheless wrong. This section shows exactly how it is wrong: it is a reversible Markov chain, but reversible with respect to a different distribution, and that distribution can get even the sign of the nearest-neighbor correlations wrong.

Throughout this section we use the pairwise Ising energy of Eq. (1.67),

$$E_\theta(\mathbf{s}) = -\frac{1}{2}\sum_{i,j=1}^{N} J_{ij}s_i s_j - \sum_{i=1}^{N} h_i s_i, \qquad J_{ij} = J_{ji}, \quad J_{ii} = 0, \tag{2.67}$$

with local field $H_i(\mathbf{s}) := h_i + \sum_j J_{ij}s_j$, which is the effective field Eq. (2.31) without three-body terms. It depends only on $\mathbf{s}_{\partial i}$ and not on s_i.

Definition 2.18 (Synchronous (Little) Dynamics)**.** The *synchronous* or *fully parallel* Gibbs kernel updates every variable at once from its single-site conditional evaluated at the old configuration:

$$P_{\text{sync}}(\mathbf{x}' \mid \mathbf{x}) := \prod_{i=1}^{N} p_\theta(x_i' \mid \mathbf{x}_{-i}). \tag{2.68}$$

For the Ising energy (2.67) this reads

$$P_{\text{sync}}(\mathbf{s}' \mid \mathbf{s}) = \prod_{i=1}^{N} \frac{\exp(\beta s_i' H_i(\mathbf{s}))}{2\cosh(\beta H_i(\mathbf{s}))}. \tag{2.69}$$

Compare Eq. (2.68) with the block Gibbs kernel Eq. (2.54) for $A = \mathcal{V}$, which would draw the whole configuration from p_θ in one step. The synchronous kernel replaces the joint distribution $p_\theta(\mathbf{x}')$ by a product of conditionals, each conditioned on the *old* neighbors. The smallest example already shows everything that goes wrong.

Example 2.19 (Two Coupled Spins: Synchronous versus Sequential Updates)**.** Take $N = 2$, $E_\theta(s_1, s_2) = -Js_1s_2$, $h = 0$, and set $t := \tanh(\beta J)$. The target distribution and the single-site conditionals are

$$p_\theta(s_1, s_2) = \frac{e^{\beta J s_1 s_2}}{4\cosh(\beta J)} = \frac{1 + t\, s_1 s_2}{4}, \qquad q(a \mid b) := p_\theta(s_i = a \mid s_j = b) = \frac{e^{\beta J ab}}{2\cosh(\beta J)} = \frac{1 + t\, ab}{2}, \tag{2.70}$$

where we used $e^{\beta J ab} = \cosh(\beta J)(1 + t\, ab)$ for $ab = \pm 1$. Order the states as $(++, +-, -+, --)$, with columns labeling the initial state and rows the final state.

Synchronous kernel. By Eq. (2.68), $P_{\text{sync}}(\mathbf{s}' \mid \mathbf{s}) = q(s_1' \mid s_2)\, q(s_2' \mid s_1)$. From $\mathbf{s} = (+, -)$, for instance, s_1' copies the old $s_2 = -1$ with probability $(1 + t)/2$ and s_2' copies the old $s_1 = +1$ with probability $(1 + t)/2$, so the transition $(+, -) \to (-, +)$ has probability $(1 + t)^2/4$. Filling in all sixteen entries,

$$\mathbf{P}_{\text{sync}} = \frac{1}{4}\begin{pmatrix} (1+t)^2 & 1-t^2 & 1-t^2 & (1-t)^2 \\ 1-t^2 & (1-t)^2 & (1+t)^2 & 1-t^2 \\ 1-t^2 & (1+t)^2 & (1-t)^2 & 1-t^2 \\ (1-t)^2 & 1-t^2 & 1-t^2 & (1+t)^2 \end{pmatrix}. \tag{2.71}$$

Each column sums to $\frac{1}{4}[(1+t)^2+2(1-t^2)+(1-t)^2]=1$. The matrix is symmetric, so its rows also sum to one, and the uniform distribution $\pi_{\text{sync}}(\mathbf{s})=1/4$ is stationary. Because all entries are positive (for $|t|<1$), Lemma 2.15 says this stationary law is unique. The eigenfunctions are the four functions 1, s_1+s_2, s_1-s_2, s_1s_2, which span all functions of two spins; applying $\mathcal{K}f(\mathbf{s})=\sum_{\mathbf{s}'}P_{\text{sync}}(\mathbf{s}'\mid\mathbf{s})f(\mathbf{s}')$ and using $\sum_a a\,q(a\mid b)=tb$ gives

$$\mathcal{K}(s_1+s_2)=t\,(s_2+s_1),\qquad \mathcal{K}(s_1-s_2)=t\,(s_2-s_1)=-t\,(s_1-s_2),\qquad \mathcal{K}(s_1s_2)=t^2s_2s_1, \tag{2.72}$$

so $\operatorname{spec}(\mathbf{P}_{\text{sync}})=\{1,t,-t,t^2\}$.

Three things have gone wrong. First, the chain samples the uniform distribution, so $\langle s_1s_2\rangle_{\text{sync}}=0$ instead of the correct $\langle s_1s_2\rangle_{p_\theta}=t$, for either sign of J. The error in distribution is

$$D_{\text{KL}}(\pi_{\text{sync}}\,\|\,p_\theta)=\sum_{\mathbf{s}}\frac{1}{4}\ln\frac{1/4}{p_\theta(\mathbf{s})}=-\ln 4-\frac{1}{2}\ln\frac{e^{\beta J}}{4\cosh\beta J}-\frac{1}{2}\ln\frac{e^{-\beta J}}{4\cosh\beta J}=\ln\cosh(\beta J), \tag{2.73}$$

which grows like $\beta\,|J|-\ln 2$ at strong coupling. Second, detailed balance with respect to the target fails: $P_{\text{sync}}(+-\mid++)\,p_\theta(++)=\frac{(1-t^2)(1+t)}{16}$, whereas $P_{\text{sync}}(++\mid+-)\,p_\theta(+-)=\frac{(1-t^2)(1-t)}{16}$. Even global balance fails, since $(\mathbf{P}_{\text{sync}}p_\theta)(++)=\frac{(1+t)(1-t+t^2)}{4}\neq p_\theta(++)$ for $0<|t|<1$. Third, the eigenvalue $-t$ belongs to the anti-phase mode s_1-s_2. For strong coupling ($t\to 1$) the configuration $(+,-)$ goes to $(-,+)$ with probability $(1+t)^2/4\to 1$ and back again: each spin copies the other's *old* value, and the pair oscillates with period two instead of aligning.

Sequential (two-phase) kernel. Now update spin 1 first and spin 2 second, $\mathbf{P}_{\text{chr}}=\mathbf{P}_2\mathbf{P}_1$. Spin 2 now sees the *new* value of spin 1: $(P_2P_1)(\mathbf{s}''\mid\mathbf{s})=q(s_1''\mid s_2)\,q(s_2''\mid s_1'')$. In the same basis,

$$\mathbf{P}_{\text{chr}}=\frac{1}{4}\begin{pmatrix}(1+t)^2 & 1-t^2 & (1+t)^2 & 1-t^2\\ 1-t^2 & (1-t)^2 & 1-t^2 & (1-t)^2\\ (1-t)^2 & 1-t^2 & (1-t)^2 & 1-t^2\\ 1-t^2 & (1+t)^2 & 1-t^2 & (1+t)^2\end{pmatrix}. \tag{2.74}$$

Columns one and three coincide (and so do two and four) because the old s_1 is overwritten before it is read. The matrix has rank two, so two eigenvalues vanish, and its trace $\frac{1}{4}[2(1+t)^2+2(1-t)^2]=1+t^2$ fixes the remaining pair to $\{1,t^2\}$. The stationary vector is exactly $p_\theta=\frac{1}{4}(1+t,1-t,1-t,1+t)^\top$: for the first row, $\frac{1}{16}[(1+t)^2((1+t)+(1-t))+(1-t^2)((1-t)+(1+t))]=\frac{(1+t)}{4}$, and the other rows are checked the same way. The two-phase sampler is exact, has no negative eigenvalue, and its spectral gap $1-t^2=\operatorname{sech}^2(\beta J)$ is the same for ferromagnetic and antiferromagnetic coupling.

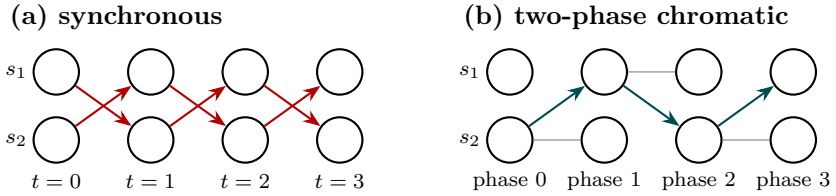


Figure 2.1: Information flow for two coupled spins. (a) Synchronous updates: each new value is drawn from the conditional given the neighbor's *old* value, so the arrows cross, and two independent chains are interleaved (Corollary 2.21(ii)). (b) Two-phase chromatic updates: one spin is updated per phase from the neighbor's *current* value (gray lines: held values), forming a single chain that samples p_θ exactly.

The general structure behind Example 2.19 was found by Little in the context of neural networks and worked out by Peretto [67, 87].

Theorem 2.20 (Little–Peretto Stationary Distribution of Synchronous Dynamics)**.** *For the pairwise Ising energy* (2.67) *with symmetric couplings, the synchronous kernel* (2.69) *satisfies detailed balance with respect to*

$$\pi_{\text{sync}}(\mathbf{s}) = \frac{1}{Z_{\text{sync}}}\, e^{\beta \mathbf{h}^\top \mathbf{s}} \prod_{i=1}^{N} 2\cosh(\beta H_i(\mathbf{s})) = \frac{1}{Z_{\text{sync}}} \exp(-\beta E_{\text{sync}}(\mathbf{s})), \tag{2.75}$$

$$E_{\text{sync}}(\mathbf{s}) := -\mathbf{h}^\top \mathbf{s} - k_{\text{B}} T \sum_{i=1}^{N} \ln 2\cosh(\beta H_i(\mathbf{s})), \tag{2.76}$$

with $Z_{\text{sync}} = \sum_{\mathbf{s},\mathbf{s}'} \exp(\beta[\mathbf{h}^\top \mathbf{s} + \mathbf{h}^\top \mathbf{s}' + \mathbf{s}'^\top \mathbf{J}\mathbf{s}])$*. It is the unique stationary distribution of* P_{sync} *at any* $T > 0$*, and* P_{sync} *satisfies detailed balance with respect to* p_θ *if and only if* $p_\theta = \pi_{\text{sync}}$*.*

Proof. Multiply Eq. (2.69) by Eq. (2.75). The factors $2\cosh(\beta H_i(\mathbf{s}))$ cancel one by one, leaving

$$P_{\text{sync}}(\mathbf{s}' \mid \mathbf{s})\, \pi_{\text{sync}}(\mathbf{s}) = \frac{1}{Z_{\text{sync}}} \exp\Big(\beta \mathbf{h}^\top \mathbf{s} + \beta \sum_i s_i' H_i(\mathbf{s})\Big) = \frac{1}{Z_{\text{sync}}} \exp\Big(\beta\big[\mathbf{h}^\top \mathbf{s} + \mathbf{h}^\top \mathbf{s}' + \mathbf{s}'^\top \mathbf{J}\mathbf{s}\big]\Big), \tag{2.77}$$

where we used $\sum_i s_i' H_i(\mathbf{s}) = \sum_i s_i' h_i + \sum_{i,j} s_i' J_{ij} s_j$. Because $\mathbf{J} = \mathbf{J}^\top$, the scalar $\mathbf{s}'^\top \mathbf{J}\mathbf{s} = \mathbf{s}^\top \mathbf{J}\mathbf{s}'$ is symmetric under $\mathbf{s} \leftrightarrow \mathbf{s}'$, and so is the whole right-hand side. This is detailed balance with respect to π_{sync}. Summing Eq. (2.77) over $\mathbf{s}$ and $\mathbf{s}'$ gives the stated Z_{sync}. Every entry of Eq. (2.69) is positive, so by Lemma 2.15 the stationary distribution is unique. Finally, if P_{sync} were in detailed balance with p_θ, then p_θ would be stationary, and uniqueness forces $p_\theta = \pi_{\text{sync}}$. □

The energy E_{sync} explains the period-two behavior at low temperature. As $T \to 0$ the single-site term behaves as $k_{\text{B}} T \ln 2\cosh(\beta H) \to |H|$, and therefore

$$E_{\text{sync}}(\mathbf{s}) \xrightarrow{T \to 0} -\mathbf{h}^\top \mathbf{s} - \sum_{i=1}^{N} |H_i(\mathbf{s})|\,. \tag{2.78}$$

The true energy at $\mathbf{h} = 0$ is $-\frac{1}{2}\sum_i s_i H_i(\mathbf{s})$, which rewards each spin for being *aligned* with its field. The synchronous energy rewards each spin only for having a *strong* field, whatever its own sign. For a ferromagnet on a bipartite lattice, the checkerboard state has fields of maximal magnitude (every spin is anti-aligned with them), so it has the same E_{sync} as the two ordered states. Under synchronous dynamics a checkerboard flips to its complement in one step and back in the next, which is the period-two orbit of Example 2.19 on a lattice.

We call π_{sync} a *pseudo-likelihood steady state* because of the structure of Eq. (2.75): it is assembled from the single-site normalizers $2\cosh(\beta H_i)$, which are exactly the denominators of the conditionals whose product forms Besag's pseudo-likelihood $\prod_i p_\theta(s_i \mid \mathbf{s}_{-i})$ [11]. A synchronous sampler effectively trusts the local conditionals one at a time and never enforces their joint consistency.

Physics Insight: Synchronous Dynamics Samples a Doubled Bipartite System

Equation (2.77) has a useful reading. Define a two-layer system with spins $\mathbf{s}$ (layer one) and $\mathbf{s}'$ (layer two), with no couplings inside a layer and the coupling matrix $\mathbf{J}$ between layers:

$$\tilde{p}(\mathbf{s}, \mathbf{s}') = \frac{1}{Z_{\text{sync}}} \exp(\beta[\mathbf{h}^\top \mathbf{s} + \mathbf{h}^\top \mathbf{s}' + \mathbf{s}'^\top \mathbf{J}\mathbf{s}]). \tag{2.79}$$

This is the Ising model on the *bipartite double cover* of $\mathcal{G}$: every node i is split into i and i', and every edge (i, j) becomes the two edges (i, j') and (j, i'). Its layer-two conditional is

$\tilde{p}(\mathbf{s}' \mid \mathbf{s}) = \prod_i e^{\beta s_i' H_i(\mathbf{s})}/2\cosh(\beta H_i(\mathbf{s})) = P_{\text{sync}}(\mathbf{s}' \mid \mathbf{s})$, and its layer-one marginal is π_{sync}. So the synchronous chain is an exact sampler, but of the wrong model: it treats the state at time $t+1$ and the state at time t as two coupled replicas, and the physical correlations between neighbors at equal times are replaced by correlations between a node and its neighbors one clock tick earlier. The doubled system is bipartite even when $\mathcal{G}$ is not, which is why the synchronous chain shows no frustration even when the target is frustrated (Worked computation 2.28).

Three exact consequences of Theorem 2.20 make the size of the error visible.

Corollary 2.21 (Signatures of the Synchronous Pathology)**.** *Under the hypotheses of Theorem 2.20:*

(i) ***Blindness to the sign of*** $\mathbf{J}$***.*** *At* $\mathbf{h} = 0$*,* π_{sync} *is unchanged by* $\mathbf{J} \to -\mathbf{J}$*. A synchronous sampler returns identical statistics for a ferromagnet and the corresponding antiferromagnet.*

(ii) ***Bipartite graphs.*** *If* $\mathcal{G}$ *is bipartite with parts* A *and* B *(all couplings between* A *and* B*), then for any* $\mathbf{h}$*,*

$$\pi_{\text{sync}}(\mathbf{s}_A, \mathbf{s}_B) = p_\theta(\mathbf{s}_A)\, p_\theta(\mathbf{s}_B), \tag{2.80}$$

the product of the exact sublattice marginals. All correlations within a sublattice are correct, and all correlations between sublattices are destroyed. If some $J_{ij} \neq 0$*, then* $\pi_{\text{sync}} \neq p_\theta$*.*

(iii) ***General graphs.*** *At* $\mathbf{h} = 0$ *and* $\mathbf{J} \neq 0$*, the synchronous sampler is wrong for at least one of the two models* $\pm\mathbf{J}$*.*

Proof. (i) At $\mathbf{h} = 0$, $H_i(\mathbf{s}) = \sum_j J_{ij} s_j$ changes sign under $\mathbf{J} \to -\mathbf{J}$, and cosh is even.

(ii) For $i \in A$, $H_i(\mathbf{s}) = h_i + \sum_{j \in B} J_{ij} s_j$ depends only on $\mathbf{s}_B$; for $j \in B$, H_j depends only on $\mathbf{s}_A$. So Eq. (2.75) splits as

$$\pi_{\text{sync}}(\mathbf{s}) \propto \Big[e^{\beta \mathbf{h}_A^\top \mathbf{s}_A} \prod_{j \in B} 2\cosh\big(\beta H_j(\mathbf{s}_A)\big)\Big]\Big[e^{\beta \mathbf{h}_B^\top \mathbf{s}_B} \prod_{i \in A} 2\cosh\big(\beta H_i(\mathbf{s}_B)\big)\Big]. \tag{2.81}$$

On the other hand, the exact marginal of $\mathbf{s}_B$ is obtained by summing the spins of A one at a time, since they do not interact with each other:

$$p_\theta(\mathbf{s}_B) = \frac{1}{Z}\sum_{\mathbf{s}_A} e^{\beta[\mathbf{h}_A^\top \mathbf{s}_A + \mathbf{h}_B^\top \mathbf{s}_B + \mathbf{s}_A^\top \mathbf{J}_{AB} \mathbf{s}_B]} = \frac{1}{Z} e^{\beta \mathbf{h}_B^\top \mathbf{s}_B} \prod_{i \in A} \sum_{s_i = \pm 1} e^{\beta s_i H_i(\mathbf{s}_B)} = \frac{1}{Z} e^{\beta \mathbf{h}_B^\top \mathbf{s}_B} \prod_{i \in A} 2\cosh\big(\beta H_i(\mathbf{s}_B)\big), \tag{2.82}$$

and likewise for $p_\theta(\mathbf{s}_A)$. The two brackets are therefore proportional to the exact marginals, and normalization gives Eq. (2.80) with $Z_{\text{sync}} = Z^2$. To see that p_θ is not a product, take $i \in A$, $j \in B$ with $J_{ij} \neq 0$. Under p_θ, $p_\theta(s_i = +1 \mid \mathbf{s}_{-i}) = \sigma(2\beta H_i(\mathbf{s}))$ changes when s_j flips, because H_i shifts by $2J_{ij}$ and σ is strictly increasing. Under the product law the conditional of s_i given everything else does not depend on $\mathbf{s}_B$ at all.

(iii) Suppose the sampler were exact for both $\mathbf{J}$ and $-\mathbf{J}$. By (i), $p_{\mathbf{J}} = \pi_{\text{sync}} = p_{-\mathbf{J}}$, so $e^{\beta \mathbf{s}^\top \mathbf{J} \mathbf{s}/2}/Z(\mathbf{J}) = e^{-\beta \mathbf{s}^\top \mathbf{J} \mathbf{s}/2}/Z(-\mathbf{J})$ for every $\mathbf{s}$, which means $\mathbf{s}^\top \mathbf{J} \mathbf{s} = 2\sum_{i<j} J_{ij} s_i s_j$ is a constant c. Multiplying by $s_k s_l$ $(k \neq l)$ and averaging over all 2^N configurations, the left side gives $2J_{kl}$ (the Walsh functions $s_i s_j$ are orthonormal under the uniform average) and the right side gives $c \cdot 0$. Hence $\mathbf{J} = 0$. □

Part (ii) has a simple dynamical explanation. On a bipartite graph, $\mathbf{s}_A(t+1)$ depends only on $\mathbf{s}_B(t)$, and $\mathbf{s}_B(t+1)$ only on $\mathbf{s}_A(t)$. The synchronous chain therefore consists of two chains that never interact: $\mathbf{s}_A(0) \to \mathbf{s}_B(1) \to \mathbf{s}_A(2) \to \cdots$ and $\mathbf{s}_B(0) \to \mathbf{s}_A(1) \to \mathbf{s}_B(2) \to \cdots$. Each of them is an exact two-phase sampler of p_θ, but at any fixed time the A half of the configuration comes from one chain and the B half from the other. They are independent, hence Eq. (2.80). The cure is equally simple: keep only one of the two interleaved chains, that is, update A and B in alternate phases. This is the two-color case of the next section.

For higher-order factors the detailed-balance computation Eq. (2.77) breaks down, because the exponent $\beta \sum_i s_i' H_i^{\text{eff}}(\mathbf{s})$ is no longer symmetric under $\mathbf{s} \leftrightarrow \mathbf{s}'$: with a three-body term $K s_i s_j s_k$, the synchronous exponent contains $K(s_i' s_j s_k + s_j' s_i s_k + s_k' s_i s_j)$, which is not invariant under exchange of the old and new configurations. The synchronous chain then still has a unique stationary law (its kernel is positive), but that law has no closed form, need not satisfy detailed balance with respect to anything, and is again not p_θ in general.

Heuristic Derivation: Stale Fields and the Size of the Error

The pathology is a timing error. A heat-bath update of spin i is exact only if the field H_i it uses is the current field. In the synchronous scheme, spin i uses the field produced by its neighbors' old values, while the neighbors are themselves being replaced in the same step. The error is therefore controlled by how much the neighbors change during one step. At high temperature ($\beta\,|J_{ij}| \ll 1$) the fields are weak and the error in correlations is small: from Eq. (2.73), $D_{\mathrm{KL}}(\pi_{\text{sync}} \,\|\, p_\theta) = \ln\cosh(\beta J) \approx \frac{1}{2}(\beta J)^2$ per edge. At strong coupling the neighbors react maximally to each other's stale values and the error becomes of order $\beta\,|J|$ per edge. Physical asynchronous p-bit networks, in which nodes flip at random uncorrelated times, behave like random-scan Gibbs samplers only if the local field seen by a node is refreshed faster than its neighbors flip. If two coupled nodes routinely sample within the settling time of each other's field, a fraction of updates is effectively synchronous and the bias above reappears in proportion.

Hardware Realization: Clocked Arrays, Asynchronous p-Bits, and Field Settling

Early p-bit networks [14, 18] avoid the synchronous pathology by running asynchronously: each p-bit is a free-running stochastic element, and the analog synapse that forms H_i is designed to respond much faster than the typical interval between flips of the node's neighbors. This approximates random-scan Gibbs sampling without a global clock, at the cost of a timing condition that must hold across the whole array and that degrades as the flip rate is pushed up. A clocked sampling array, such as the denoising architecture described by Jelincic et al. [56] (Section 17.1), replaces this statistical timing condition by a deterministic schedule. The array is divided into color classes, the field of every node in the active class is computed from neighbors that are held fixed during the phase, and the class samples at once. The field must settle within the phase: if the local field is formed by charging a node capacitance through the synapse network with time constant τ_{RC}, the phase must last several τ_{RC} so that the residual error $e^{-t_{\text{phase}}/\tau_{RC}}$ in H_i is negligible compared with $k_{\mathrm{B}}T$-scale field differences. The remaining phase time is set by how long the p-bit needs to produce a fresh, decorrelated sample.

2.5 k-Colorable Factor Graphs and Chromatic Block Gibbs Sampling

Section 2.4 showed that updating coupled neighbors in the same step is the source of the error. The remedy is to update in the same step only nodes that are *not* coupled. The Markov properties of Section 2.1 guarantee that such nodes are conditionally independent given everything else, so drawing them simultaneously and independently is the same as drawing them jointly from their exact block conditional. Partitioning the whole graph into such sets is a graph-coloring problem.

2.5.1 Interaction Graphs, Proper Colorings, and Chromatic Number

For a factor graph the relevant notion of adjacency is *appears in a common factor*. Two variables that share a three-body factor are coupled even though no pairwise edge is drawn between them.

Definition 2.22 (Interaction Graph, Proper Coloring, and Chromatic Number)**.** Let $(\mathcal{V}, \mathcal{F})$ be a factor graph (Definition 2.6).

(i) Its *interaction graph* $\mathcal{G}_{\mathcal{F}} = (\mathcal{V}, \mathcal{E}_{\mathcal{F}})$ has an edge (i, j), $i \neq j$, whenever some factor $a \in \mathcal{F}$ contains both: $\{i, j\} \subseteq \partial a$. Equivalently, $j \in \mathcal{N}_{\mathcal{F}}(i)$ of Eq. (2.23).

(ii) A *proper k-coloring* is a map $c : \mathcal{V} \to \{1, \dots, k\}$ such that $c(i) \neq c(j)$ for every edge $(i, j) \in \mathcal{E}_{\mathcal{F}}$. Its *color classes* $\mathcal{V}_c := c^{-1}(c)$ partition $\mathcal{V}$, and each is an *independent set*: no two of its vertices are adjacent.

(iii) The *chromatic number* $\chi(\mathcal{G}_{\mathcal{F}})$ is the smallest k for which a proper k-coloring exists.

The scope ∂a of every factor is a clique of $\mathcal{G}_{\mathcal{F}}$, so $\chi(\mathcal{G}_{\mathcal{F}}) \geq \max_a |\partial a|$: a k-body interaction always needs at least k colors. Standard cases:

- *Bipartite graphs*, $\chi = 2$: the square and simple-cubic lattices (color by the parity of $i_x + i_y (+ i_z)$), any tree, and the visible–hidden graph of a restricted Boltzmann machine (a pairwise model with couplings only between a visible and a hidden layer).
- The *triangular lattice*, $\chi = 3$: every elementary triangle needs three colors. Drawing it as the square lattice plus the diagonal bonds $(i_x, i_y) \to (i_x + 1, i_y + 1)$, the coloring $c = (i_x + i_y) \bmod 3$ works, since the three bond vectors change $i_x + i_y$ by 1, 1, and 2, none of which is divisible by 3.
- The *King's graph* (square lattice plus both diagonals), $\chi = 4$: every 2×2 block is a complete graph $\mathcal{K}_4$. The coloring is constructed in Worked computation 2.28.
- The complete graph $\mathcal{K}_N$ of a fully connected Ising model (a fully connected Boltzmann machine), $\chi = N$: no two nodes can be updated together, and chromatic Gibbs offers no parallelism at all.

The last case is the graph-theoretic form of the wiring argument at the start of the chapter: dense connectivity removes parallelism. Bounded degree, by contrast, always leaves a bounded number of colors.

Lemma 2.23 (Greedy Coloring Bound)**.** *If every vertex of $\mathcal{G}_{\mathcal{F}}$ has at most Δ neighbors, then $\chi(\mathcal{G}_{\mathcal{F}}) \leq \Delta + 1$.*

Proof. List the vertices in any order and color them one by one, giving each vertex the smallest color in $\{1, \dots, \Delta + 1\}$ not already used by one of its previously colored neighbors. A vertex has at most Δ neighbors, so at most Δ colors are excluded and one of the $\Delta + 1$ colors is always free. Every edge joins a vertex to an earlier neighbor whose color it avoided, so the coloring is proper. □

Brooks' theorem sharpens this to $\chi \le \Delta$ unless the graph is complete or an odd cycle. For lattices the degree bound is loose: the King's graph has $\Delta = 8$ but $\chi = 4$. The number of colors, and hence the number of clock phases, is a property of the lattice geometry rather than of the degree alone.

2.5.2 Conditional Independence Within a Color Class

Proposition 2.24 (Factorization of the Conditional on an Independent Set)**.** *Let p_θ be the factor-graph distribution (2.24), and let $A \subseteq \mathcal{V}$ be an independent set of $\mathcal{G}_\mathcal{F}$. Then*

$$p_\theta(\mathbf{x}_A \mid \mathbf{x}_{-A}) = \prod_{i \in A} p_\theta(x_i \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)}), \tag{2.83}$$

where each factor on the right is the single-site conditional (2.26), and every $\mathcal{N}_\mathcal{F}(i)$ with $i \in A$ lies in $\mathcal{V} \setminus A$. Consequently the block Gibbs kernel P_A equals the product of the single-site kernels P_i, $i \in A$, taken in any order, and the P_i with $i \in A$ commute with each other.

Proof. Because A is independent, no factor contains two vertices of A. Split $\mathcal{F}$ into the families $\partial i = \{a : i \in \partial a\}$ for $i \in A$, which are pairwise disjoint, and the remaining factors $\mathcal{F}_{\text{rest}}$, which contain no vertex of A. Then

$$E_\theta(\mathbf{x}) = \sum_{i \in A} E_i(x_i \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)}) + E_{\text{rest}}(\mathbf{x}_{-A}), \tag{2.84}$$

with E_i the local clamping energy (2.25). Each $\mathcal{N}_\mathcal{F}(i)$ excludes the other vertices of A (they are not neighbors of i) and excludes i itself, so it lies in $\mathcal{V} \setminus A$. By Bayes' rule,

$$\begin{aligned} p_\theta(\mathbf{x}_A \mid \mathbf{x}_{-A}) &= \frac{e^{-\beta E_{\text{rest}}(\mathbf{x}_{-A})} \prod_{i \in A} e^{-\beta E_i(x_i \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)})}}{e^{-\beta E_{\text{rest}}(\mathbf{x}_{-A})} \int \prod_{i \in A} e^{-\beta E_i(y_i \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)})} \prod_{i \in A} \mu_i(\mathrm{d}y_i)} \\ &= \prod_{i \in A} \frac{e^{-\beta E_i(x_i \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)})}}{\int_{\Omega_i} e^{-\beta E_i(y_i \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)})} \mu_i(\mathrm{d}y_i)}, \end{aligned} \tag{2.85}$$

where the integral over Ω_A of a product of single-variable functions factorizes into a product of single-variable integrals. Each factor is Eq. (2.26). For the kernels: applying P_i for $i \in A$ in some order resamples each x_i from $p_\theta(x_i \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)})$, and since $\mathcal{N}_\mathcal{F}(i)$ contains no other vertex of A, none of these conditionals is affected by the earlier draws. The composite kernel is therefore $\delta_{\mathbf{x}'_{-A}, \mathbf{x}_{-A}} \prod_{i \in A} p_\theta(x'_i \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)}) = P_A(\mathbf{x}' \mid \mathbf{x})$ for every order. □

Proposition 2.24 is the exact statement behind parallel hardware updates. Within an independent set, *all at once* and *one after another* are the same Markov kernel, because no node in the set reads the state of any other. If the set contains two nodes that share a factor, this fails, and the parallel update becomes a partial version of the synchronous kernel of Section 2.4.

Definition 2.25 (Chromatic Block Gibbs Sweep)**.** Given a proper k-coloring with classes $\mathcal{V}_1, \dots, \mathcal{V}_k$, the *chromatic block Gibbs sweep* is

$$\mathbf{P}_{\text{chr}} := \mathbf{P}_{\mathcal{V}_k} \cdots \mathbf{P}_{\mathcal{V}_2} \mathbf{P}_{\mathcal{V}_1}. \tag{2.86}$$

In phase c, every node $i \in \mathcal{V}_c$ draws $x_i' \sim p_\theta(x_i' \mid \mathbf{x}_{\mathcal{N}_\mathcal{F}(i)})$ simultaneously and independently, using the current values of its neighbors, all of which belong to other classes.

By Proposition 2.24, $\mathbf{P}_{\text{chr}}$ is literally the sequential single-site systematic scan $\mathbf{P}_{\text{SS}}$ of Proposition 2.14, taken in an order that lists the color classes one after another. Parallel execution changes the wall-clock time of a sweep, not its statistics.

Theorem 2.26 (Exactness and Convergence of Chromatic Block Gibbs Sampling)**.** *Let $p_\theta > 0$ on a finite Ω, and let $\mathcal{V}_1, \dots, \mathcal{V}_k$ be the classes of a proper coloring of $\mathcal{G}_\mathcal{F}$. Write $\Pi_c := \Pi_{\mathcal{V}_c}$ for the conditional-expectation projections of Lemma 2.13.*

(i) $\mathbf{P}_{\text{chr}}$ *is strictly positive and p_θ is its unique stationary distribution. Its time reversal is the reversed sweep $\mathbf{P}_{\mathcal{V}_1} \cdots \mathbf{P}_{\mathcal{V}_k}$.*

(ii) The density ratio obeys $h^{(t+1)} = \Pi_k \cdots \Pi_1 h^{(t)}$, and with the sweep contraction

$$\kappa := \|\Pi_k \cdots \Pi_1\|_{L_0^2(p_\theta) \to L_0^2(p_\theta)} < 1, \tag{2.87}$$

we have $\chi^2(\rho^{(t)} \,\|\, p_\theta) \le \kappa^{2t}\, \chi^2(\rho^{(0)} \,\|\, p_\theta)$ for every initial distribution.

(iii) $\mathbf{P}_{\text{chr}}$ *and the operator $\Pi_k \cdots \Pi_1$ have the same spectrum. Its non-trivial eigenvalues satisfy $|\lambda| \le \lambda_* \le \kappa < 1$, where λ_* is the largest modulus among them; for $k \ge 3$ they may be complex.*

(iv) For two colors, $A = \mathcal{V}_1$ and $B = \mathcal{V}_2$, let

$$\varrho := \sup \left\{ \mathbb{E}[f(\mathbf{X}_A) g(\mathbf{X}_B)]_{p_\theta} : \mathbb{E}[f]_{p_\theta} = \mathbb{E}[g]_{p_\theta} = 0,\ \mathbb{E}\Big[f^2\Big]_{p_\theta} = \mathbb{E}\Big[g^2\Big]_{p_\theta} = 1 \right\} \tag{2.88}$$

be the maximal correlation *between the two color classes. Then $\kappa = \varrho$, every non-zero eigenvalue of $\mathbf{P}_{\text{chr}}$ is real and lies in $[0,1]$, the second largest is $\lambda_2 = \varrho^2$, and $\chi^2(\rho^{(t+1)} \,\|\, p_\theta) \le \varrho^4\, \chi^2(\rho^{(t)} \,\|\, p_\theta)$ for all $t \ge 1$.*

Proof. (i) This is Proposition 2.14(ii)–(iii) with blocks $\mathcal{V}_c$, together with Lemma 2.15 for uniqueness.

(ii) Apply Lemma 2.13 once per phase: phase one maps h to $\Pi_1 h$, phase two maps that to $\Pi_2 \Pi_1 h$, and so on. Since $\chi^2(\rho \,\|\, p_\theta) = \|h - 1\|_{p_\theta}^2$ and every Π_c fixes the constant 1, $h^{(t+1)} - 1 = \Pi_k \cdots \Pi_1 (h^{(t)} - 1)$ with $h^{(t)} - 1 \in L_0^2(p_\theta)$, which gives the bound with κ. To show $\kappa < 1$, note that $L_0^2(p_\theta)$ is finite-dimensional, so the norm is attained at some f with $\|f\|_{p_\theta} = 1$. For an orthogonal projection, Pythagoras gives $\|g\|^2 = \|\Pi g\|^2 + \|g - \Pi g\|^2$, so $\|\Pi g\| \le \|g\|$ with equality only if $\Pi g = g$. If $\kappa = 1$, the chain

$$1 = \|\Pi_k \cdots \Pi_1 f\| \le \|\Pi_{k-1} \cdots \Pi_1 f\| \le \cdots \le \|\Pi_1 f\| \le \|f\| = 1 \tag{2.89}$$

consists of equalities, which forces $\Pi_1 f = f$, then $\Pi_2 f = f$, and so on. Then f does not depend on $\mathbf{x}_{\mathcal{V}_c}$ for any c, so it is constant, and a constant in $L_0^2(p_\theta)$ is zero. This contradicts $\|f\| = 1$.

(iii) Writing $\rho = \mathbf{D}_{\text{eq}} h$ with $\mathbf{D}_{\text{eq}} = \operatorname{diag}(p_\theta)$, part (ii) says $\mathbf{P}_{\text{chr}} \mathbf{D}_{\text{eq}} h = \mathbf{D}_{\text{eq}} \Pi_k \cdots \Pi_1 h$ for all h, so $\mathbf{P}_{\text{chr}} = \mathbf{D}_{\text{eq}} (\Pi_k \cdots \Pi_1) \mathbf{D}_{\text{eq}}^{-1}$ and the spectra agree. The eigenvalue 1 belongs to the constants; all other eigenvectors can be taken in the invariant subspace $L_0^2(p_\theta)$, where the spectral radius is bounded by the operator norm κ. A product of three or more non-commuting projections is not self-adjoint and need not have real eigenvalues.

(iv) Now $\Pi_1 = \Pi_A$ projects onto the functions of $\mathbf{x}_B$ alone, $\mathcal{H}_B$, and $\Pi_2 = \Pi_B$ onto $\mathcal{H}_A$. For mean-zero $g \in \mathcal{H}_A$, the vector $\Pi_A g$ lies in $\mathcal{H}_B$, so

$$\|\Pi_A g\|_{p_\theta} = \sup_{u \in \mathcal{H}_B,\, \|u\| = 1} \langle u,\, \Pi_A g \rangle_{p_\theta} = \sup_{u \in \mathcal{H}_B,\, \|u\| = 1} \langle u,\, g \rangle_{p_\theta} \le \varrho\, \|g\|_{p_\theta}\,, \tag{2.90}$$

where $\langle u,\, \Pi_A g\rangle = \langle \Pi_A u,\, g\rangle = \langle u,\, g\rangle$ for $u \in \mathcal{H}_B$, and u may be taken mean-zero because g is. Equality holds for the maximizer $g = f^*$ of Eq. (2.88); the same argument with A and B exchanged bounds Π_B on mean-zero elements of $\mathcal{H}_B$. For any $f \in L^2_0$, $\|\Pi_B\Pi_A f\| \le \varrho\,\|\Pi_A f\| \le \varrho\,\|f\|$, with equality when f is the maximizing mean-zero function of $\mathbf{x}_B$ in Eq. (2.88), which Π_A leaves unchanged and Π_B contracts by exactly ϱ; so $\kappa = \varrho$. Every eigenvector of $\Pi_B\Pi_A$ with non-zero eigenvalue lies in its range $\mathcal{H}_A$, on which $\Pi_B\Pi_A$ acts as $T := \Pi_B\Pi_A\Pi_B$. This operator is self-adjoint and positive semi-definite on $\mathcal{H}_A$, since $\langle g,\, Tg\rangle = \|\Pi_A g\|^2 \ge 0$, and by Eq. (2.90) its largest eigenvalue on mean-zero functions is $\sup_g \|\Pi_A g\|^2 / \|g\|^2 = \varrho^2$. After the first sweep, $h^{(t)} - 1 \in \mathcal{H}_A \cap L^2_0$, so $\left\|h^{(t+1)} - 1\right\| = \left\|T(h^{(t)} - 1)\right\| \le \varrho^2 \left\|h^{(t)} - 1\right\|$; squaring gives the last claim. □

Part (iv) gives the two-color sampler a clean physical meaning. The rate of a sweep is set by the largest equal-time correlation that any observable of one sublattice can have with any observable of the other. When the sublattices are almost independent (high temperature), $\varrho \approx 0$ and one sweep nearly equilibrates the system. When they are locked together (an ordered phase), $\varrho \to 1$ and the gap $1 - \varrho^2$ closes. The non-negativity of the spectrum also means the two-color sampler has no oscillating modes, in contrast with the eigenvalue $-t$ of the synchronous chain in Example 2.19.

Example 2.27 (The RC Sampler: Chromatic and Synchronous Updates of a Gaussian Network)**.** For the RC sampler, the coupled linear RC circuit of Example 1.5, written with its conductance matrix $\mathbf{G}$ and energy $E_\theta(\mathbf{v}) = \frac{1}{2}(\mathbf{v} - \mathbf{b})^\top \mathbf{G}(\mathbf{v} - \mathbf{b})$, the precision matrix is $\mathbf{\Lambda} = \beta\mathbf{G}$. Split it into its diagonal and off-diagonal parts, $\mathbf{\Lambda} = \mathbf{D} + \mathbf{R}$. Completing the square in v_i gives the single-site conditional

$$v_i \mid \mathbf{v}_{-i} \sim \mathcal{N}\Big(b_i - \frac{1}{G_{ii}} \sum_{j \ne i} G_{ij}(v_j - b_j),\ \frac{k_{\mathrm{B}}T}{G_{ii}}\Big), \tag{2.91}$$

which depends only on the nodes wired to i ($G_{ij} \ne 0$). A chromatic sweep over the conductance graph is exact by Theorem 2.26. The synchronous update, in centered coordinates $\mathbf{u} = \mathbf{v} - \mathbf{b}$, is the noisy Jacobi iteration $\mathbf{u}' = -\mathbf{D}^{-1}\mathbf{R}\mathbf{u} + \mathbf{D}^{-1/2}\boldsymbol{\xi}$ with $\boldsymbol{\xi} \sim \mathcal{N}(0, \mathbf{I})$. It converges only if the spectral radius of $\mathbf{D}^{-1}\mathbf{R}$ is below one, which holds for instance when $\mathbf{G}$ is strictly diagonally dominant; $\mathbf{\Lambda} \succ 0$ alone does not guarantee it, and otherwise the synchronous iteration diverges. Assume convergence. Repeating the doubled-system argument of Section 2.4, the pair $(\mathbf{u}, \mathbf{u}')$ has joint precision $\left(\begin{smallmatrix} \mathbf{D} & \mathbf{R} \\ \mathbf{R} & \mathbf{D} \end{smallmatrix}\right)$, whose marginal covariance for $\mathbf{u}$ is the inverse Schur complement

$$\mathbf{\Sigma}_{\text{sync}} = \left(\mathbf{D} - \mathbf{R}\mathbf{D}^{-1}\mathbf{R}\right)^{-1} \ne \mathbf{\Lambda}^{-1} = (\mathbf{D} + \mathbf{R})^{-1} \quad (\mathbf{R} \ne 0). \tag{2.92}$$

One checks directly that $\mathbf{\Sigma}_{\text{sync}}$ solves the stationarity condition $\mathbf{\Sigma} = \mathbf{M}\mathbf{\Sigma}\mathbf{M}^\top + \mathbf{D}^{-1}$ with $\mathbf{M} = -\mathbf{D}^{-1}\mathbf{R}$, while inserting $\mathbf{\Sigma} = \mathbf{\Lambda}^{-1}$ leaves the residual $\mathbf{D}^{-1}\mathbf{R}\mathbf{D}^{-1}$. For two nodes with $\mathbf{D} = \mathbf{I}$ and $\mathbf{R} = r\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $|r| < 1$: the exact covariance is $(1 - r^2)^{-1}\left(\begin{smallmatrix} 1 & -r \\ -r & 1 \end{smallmatrix}\right)$, whereas $\mathbf{\Sigma}_{\text{sync}} = (1 - r^2)^{-1}\mathbf{I}$. The variances are right and the cross-covariance is lost, exactly as in Corollary 2.21(ii). The pathology is not a property of discrete spins; it affects any clocked sampler of a coupled network.

2.5.3 Parallel Cost of a Sweep

Let τ_{phase} be the time a hardware node needs to produce a fresh sample from its conditional once its neighbors are fixed. A sequential sweep updates one node at a time and takes $N\tau_{\text{phase}}$. A chromatic sweep updates all of $\mathcal{V}_c$ in one phase and takes

$$T_{\text{sweep}} = k\,\tau_{\text{phase}}, \qquad \frac{T^{\text{seq}}_{\text{sweep}}}{T^{\text{chr}}_{\text{sweep}}} = \frac{N}{k}, \tag{2.93}$$

independent of N. Because the two sweeps are the same Markov kernel, the number of sweeps needed to reach accuracy ε is the same, roughly $\ln(1/\varepsilon)/(1-\lambda_*)$ for the sweep spectral radius λ_*, and the whole factor N/k is gained in wall-clock time. This is the precise sense in which local, colorable hardware completes a sweep in a time independent of the system size. It does not make hard distributions easy: $1-\lambda_*$ is a property of p_θ and the sweep, and it can be exponentially small in β (Worked computation 2.28, Chapters 3 and 7).

Notation Watch: Steps, Sweeps, and Phases

Three units of discrete time appear in the literature and are easily confused. In the Monte Carlo literature a *step* is often one single-site update, and N steps make one *Monte Carlo sweep*; Glauber's original rates are per site per unit time. In hardware, the natural unit is a clock *phase*, in which one color class is updated, and k phases make one sweep. In this book, t in $\mathbf{P}_{\text{chr}}^{t}$ always counts sweeps, mixing times τ_{mix} and the entries t_{mix} of Table 2.1 are in sweeps, and wall-clock time is obtained by multiplying by $T_{\text{sweep}} = k\tau_{\text{phase}}$. When comparing with continuous-time Langevin relaxation (Chapter 6), a spectral gap γ per sweep corresponds to a relaxation rate $-\ln(1-\gamma)/T_{\text{sweep}} \approx \gamma/T_{\text{sweep}}$ for small γ.

Hardware Realization: Chromatic Schedules in `THRML` and on a Sampling Die

In `THRML` [31] a model is a factor graph whose nodes may be spins, categorical (Potts) variables, or continuous variables, and a sampling program is an ordered list of *blocks*. A block is a set of nodes that are updated together; for the update to be exact it must be an independent set of the interaction graph, so in practice the blocks are the color classes of Definition 2.22. To update a block, each node gathers the current states of its neighbors through the factors it belongs to, forms the parameters of its conditional (an effective field for a spin, Eq. (2.32); a vector of logits for a Potts variable, Eq. (2.34); a mean and variance for a Gaussian node, Eq. (2.91)), and draws a sample. All nodes of the block do this at once. Nodes clamped to data during the positive phase of learning (Section 1.4.2) are simply left out of the list of free blocks.

On a sampling die the same schedule becomes a clock sequence. As a representative estimate, using the die size of the design study in Section 17.1, take a locally connected King's-graph lattice with $k = 4$ colors and $N \approx 2.5 \times 10^5$ p-bits. Such a die updates about 6×10^4 nodes per phase and completes a sweep in four phases. A sequential scan would need 2.5×10^5 phases for the same sweep, so the gain per sweep is $N/k \approx 6 \times 10^4$. If the phase time is of order nanoseconds (an order of magnitude for a p-bit that must let its input field settle and then produce one decorrelated sample, Section 2.4), one sweep takes of order 10 ns regardless of die size. The engineering numbers that matter are therefore the phase time and the spectral gap of the model being sampled, not the node count.

Worked computation 2.28 (Chromatic versus Synchronous Gibbs Sampling on Bipartite and King's-Graph Ising Cells)**.** We compare the exact chromatic sampler and the synchronous sampler on three small Ising cells that are the building blocks of the lattices used in sampling hardware: the four-spin square plaquette (bipartite, two colors), the 2×2 King's cell (the complete graph $\mathcal{K}_4$, four colors), and the 3×3 King's cell (nine spins, four colors). Each node is a p-bit, the binary limit of the double-well sampler (Example 1.6). Throughout, $\mathbf{h} = 0$, every edge carries the same coupling J, and

$$K := \beta J, \qquad t := \tanh K, \qquad \tau := \tanh 2K = \frac{2t}{1+t^2}. \tag{2.94}$$

$K > 0$ is a ferromagnet and $K < 0$ an antiferromagnet. The small cells allow everything to be computed exactly: the plaquette by hand, the King's cells by exact diagonalization of the 16×16 and 512×512 transition matrices.

Part 1: The cells and their colorings.

Plaquette. Label the corners $1, 2, 3, 4$ around the square, with edges $(1,2), (2,3), (3,4), (4,1)$. It is bipartite with $A = \{1, 3\}$ and $B = \{2, 4\}$, the two sublattices of the square lattice.

King's lattice. Sites are $(i_x, i_y) \in \mathbb{Z}^2$, and two sites are neighbors when they differ by $(d_x, d_y) \in \{-1, 0, 1\}^2 \setminus \{(0,0)\}$: four axial and four diagonal neighbors. Define

$$c(i_x, i_y) := (i_x \bmod 2) + 2\,(i_y \bmod 2) + 1 \in \{1, 2, 3, 4\}\,. \tag{2.95}$$

This is a proper coloring: for a neighbor, at least one of d_x, d_y equals ± 1 and changes the parity of that coordinate, which changes the corresponding binary digit of $c - 1$, so the colors differ. Four colors are also necessary, because the sites $(0,0), (1,0), (0,1), (1,1)$ are pairwise neighbors and form a $\mathcal{K}_4$. Hence $\chi = 4$, and each color class is a square sublattice of spacing 2 containing a quarter of the sites. In the 2×2 cell every class is a single spin, so the chromatic sweep is an ordinary sequential scan of the four spins. In the 3×3 cell (Fig. 2.2) the classes have sizes 4, 2, 2, 1, and the cell has 12 axial and 8 diagonal edges.

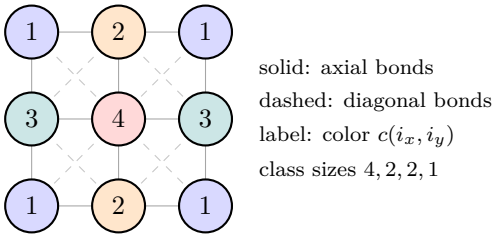


Figure 2.2: The 3×3 King's cell with the four-coloring (2.95). No bond joins two sites with the same label, so each color class can be updated in one parallel phase.

Gauge symmetry and frustration. On a bipartite graph at $\mathbf{h} = 0$, the relabeling $g(\mathbf{s}_A, \mathbf{s}_B) := (\mathbf{s}_A, -\mathbf{s}_B)$ flips the sign of every bond, so $E_{-J}(g(\mathbf{s})) = E_J(\mathbf{s})$ and $p_{-J}(g(\mathbf{s})) = p_J(\mathbf{s})$. The block conditionals transform the same way, so the chromatic kernels obey $P_{\text{chr}}^{(-J)}(g(\mathbf{s}') \mid g(\mathbf{s})) = P_{\text{chr}}^{(J)}(\mathbf{s}' \mid \mathbf{s})$: the two transition matrices are conjugate by a permutation and have identical spectra. A ferromagnet and an antiferromagnet on a bipartite cell are the same sampling problem. On the King's cells this fails. Flipping one spin changes the sign of the two bonds a closed loop uses at that spin, so the product of the couplings around any loop is invariant under such relabelings. Around a triangle this product is J^3, which has different signs for $\pm J$. The antiferromagnet on a triangle is *frustrated*: no configuration satisfies all three bonds. Frustration is the subject of Section 3.1; here it will show up as a qualitative difference between the two signs.

Part 2: Exact equilibrium of the plaquette.

With the transfer matrix $\mathbf{T} = \left(\begin{smallmatrix} e^{K} & e^{-K} \\ e^{-K} & e^{K} \end{smallmatrix}\right)$, whose eigenvalues are $\lambda_{\pm} = 2\cosh K, 2\sinh K$ with eigenvectors $(1, \pm 1)/\sqrt{2}$, the closed ring of four spins has

$$Z = \operatorname{Tr} \mathbf{T}^4 = (2\cosh K)^4 + (2\sinh K)^4 = 16\cosh^4 K\,(1 + t^4). \tag{2.96}$$

All four bonds are equivalent, so the nearest-neighbor correlation is a quarter of $\partial_K \ln Z$:

$$c_{12} := \langle s_1 s_2 \rangle = \frac{1}{4}\frac{\partial \ln Z}{\partial K} = \frac{16\cosh^3 K \sinh K + 16\sinh^3 K \cosh K}{16\cosh^4 K + 16\sinh^4 K} = \frac{t + t^3}{1 + t^4}. \tag{2.97}$$

For the diagonal pair, insert σ_z at sites 1 and 3: $\langle s_1 s_3\rangle = \mathrm{Tr}(\sigma_z \mathbf{T}^2 \sigma_z \mathbf{T}^2)/Z$. In the eigenbasis of $\mathbf{T}$, σ_z exchanges the two eigenvectors, so the trace equals $2\lambda_+^2\lambda_-^2$ and

$$c_{13} := \langle s_1 s_3\rangle = \frac{2\lambda_+^2\lambda_-^2}{\lambda_+^4+\lambda_-^4} = \frac{2t^2}{1+t^4}. \tag{2.98}$$

Under $K \to -K$, c_{12} changes sign and c_{13} does not, as the gauge symmetry requires.

Part 3: Exact spectrum of the two-phase chromatic sampler on the plaquette.

Let $m_A = s_1 + s_3$ and $m_B = s_2 + s_4$. In phase A, spins 1 and 3 are drawn independently in the same field Jm_B; in phase B, spins 2 and 4 in the field Jm_A. Since $m \in \{-2, 0, 2\}$, two identities hold for every value:

$$\tanh(Km) = \frac{m}{2}\,\tau, \qquad \tanh^2(Km_B) = \tau^2\,\frac{1+s_2s_4}{2}, \tag{2.99}$$

the second because $m_B = \pm 2$ exactly when $s_2 s_4 = +1$. By Theorem 2.26(iv), the non-zero eigenvalues of the 16×16 matrix $\mathbf{P}_{\mathrm{chr}} = \mathbf{P}_B\mathbf{P}_A$ are those of $Tg = \mathbb{E}[\mathbb{E}[g(\mathbf{S}'_A) \mid \mathbf{S}_B] \mid \mathbf{S}_A]$ acting on the four-dimensional space of functions of (s_1, s_3), spanned by 1, m_A, $d_A := s_1 - s_3$, and $s_1 s_3$. Using $\mathbb{E}[s_1 \mid \mathbf{s}_B] = \tanh(Km_B)$ and conditional independence of s_1, s_3:

$$\mathbb{E}[m_A \mid \mathbf{s}_B] = 2\tanh(Km_B) = \tau\, m_B, \qquad Tm_A = \tau\,\mathbb{E}[m_B \mid \mathbf{s}_A] = \tau^2 m_A, \tag{2.100}$$

$$\mathbb{E}[d_A \mid \mathbf{s}_B] = \tanh(Km_B) - \tanh(Km_B) = 0, \qquad Td_A = 0, \tag{2.101}$$

$$\mathbb{E}[s_1 s_3 \mid \mathbf{s}_B] = \tanh^2(Km_B) = \frac{\tau^2}{2}(1+s_2s_4), \qquad T(s_1s_3) = \frac{\tau^2}{2} + \frac{\tau^4}{4} + \frac{\tau^4}{4}\,s_1s_3. \tag{2.102}$$

The last line, restricted to mean-zero functions, has eigenvalue $\tau^4/4$. The non-zero spectrum of the sweep is therefore

$$\mathrm{spec}(\mathbf{P}_{\mathrm{chr}}) \setminus \{0\} = \Big\{1,\ \tanh^2 2K,\ \tfrac{1}{4}\tanh^4 2K\Big\}, \qquad \gamma_{\mathrm{chr}} = 1 - \tanh^2 2K = \mathrm{sech}^2\, 2K. \tag{2.103}$$

Exact diagonalization of the 16×16 matrix at $K = 0.5$ returns $\{1, 0.580026, 0.084107\}$ and twelve zeros, matching $\tanh^2 1 = 0.580026$ and $\frac{1}{4}\tanh^4 1 = 0.084107$. As a check on Eq. (2.88), the correlation coefficient of m_A and m_B is

$$\frac{\mathrm{Cov}(m_A, m_B)}{\mathrm{Var}(m_A)} = \frac{4c_{12}}{2+2c_{13}} = \frac{2t(1+t^2)/(1+t^4)}{(1+t^2)^2/(1+t^4)} = \frac{2t}{1+t^2} = \tau, \tag{2.104}$$

so the sublattice magnetizations achieve the maximal correlation $\varrho = \tau$ and $\lambda_2 = \varrho^2$, as the theorem states. At strong coupling $\gamma_{\mathrm{chr}} = 4/(e^{2K}+e^{-2K})^2 \approx 4e^{-4|K|}$. This is an Arrhenius law with barrier $4\,|J|$, the energy cost of flipping one spin of an ordered plaquette (two broken bonds), which is the first step of any path between the two ground states. The result is the same for both signs of K, as the gauge symmetry requires.

How sharp is the bound of Theorem 2.26? Start from the all-up configuration, for which $p_\theta(\uparrow\uparrow\uparrow\uparrow) = e^{4K}/Z$ and $\chi^2(\delta_\uparrow \,\|\, p_\theta) = Ze^{-4K} - 1$. Part (ii) with $\kappa = \varrho = \tau$ gives χ^2 after one sweep at most $\tau^2\chi^2(\delta_\uparrow \,\|\, p_\theta)$, and part (iv) gives a further factor τ^4 per sweep after that. With the Cauchy–Schwarz step Eq. (2.49),

$$\left\|\rho^{(t)} - p_\theta\right\|_{\mathrm{TV}} \le \frac{1}{2}\,\tau^{2t-1}\sqrt{Ze^{-4K}-1}. \tag{2.105}$$

At $K = 1$, $Ze^{-4K} = 2.2205$ and $\tau = \tanh 2 = 0.96403$, so the bound drops below 0.01 after $t = 56$ sweeps; exact iteration of the 16×16 matrix gives 54. At $K = 0.5$ the bound gives 9 against the

exact 8, and at $K = 0.25$ both give 4. Using only the operator norm κ for every sweep, without the self-adjoint structure of part (iv), would give $\left\|\rho^{(t)} - p_\theta\right\|_{\mathrm{TV}} \le \frac{1}{2}\tau^t\sqrt{Ze^{-4K} - 1}$ and roughly twice as many sweeps (110 at $K = 1$). The two-color structure is what makes the estimate tight.

Part 4: The synchronous sampler on the plaquette.

By Corollary 2.21(ii), $\pi_{\mathrm{sync}}(\mathbf{s}) = p_\theta(s_1, s_3)\, p_\theta(s_2, s_4)$, with $p_\theta(s_1, s_3) = \frac{1}{4}(1 + c_{13}s_1s_3)$. The diagonal correlations are exact, and the nearest-neighbor correlation is $\langle s_1 s_2\rangle_{\mathrm{sync}} = 0$ for every K. To compute the error in distribution, note that $\ln p_\theta(\mathbf{s}) = -\ln Z + K\sum_{\mathrm{bonds}} s_i s_j$, and each bond joins A to B, so its average under the product law vanishes. Hence $\mathbb{E}[\ln p_\theta]_{\pi_{\mathrm{sync}}} = -\ln Z$ and

$$D_{\mathrm{KL}}(\pi_{\mathrm{sync}} \,\|\, p_\theta) = \ln Z - S_A - S_B, \qquad S_A = S_B = \ln 4 - \tfrac{1}{2}\left[(1+c_{13})\ln(1+c_{13}) + (1-c_{13})\ln(1-c_{13})\right], \tag{2.106}$$

where S_A is the Shannon entropy of $p_\theta(s_1, s_3)$ in nats. With $\ln Z = 4\ln 2 + 4\ln\cosh K + \ln(1 + t^4)$ from Eq. (2.96),

$$D_{\mathrm{KL}}(\pi_{\mathrm{sync}} \,\|\, p_\theta) = 4\ln\cosh K + \ln(1 + t^4) + (1 + c_{13})\ln(1 + c_{13}) + (1 - c_{13})\ln(1 - c_{13}). \tag{2.107}$$

For weak coupling, $t \approx K$ and $c_{13} \approx 2K^2$, so the terms are $2K^2$, $\mathcal{O}(K^4)$, and (the last two together) $\approx c_{13}^2 = \mathcal{O}(K^4)$, giving $D_{\mathrm{KL}}(\pi_{\mathrm{sync}} \,\|\, p_\theta) \approx 2K^2$, four times the two-spin value $\frac{1}{2}K^2$ of Eq. (2.73), one per bond. For strong coupling, $t, c_{13} \to 1$ and the sum tends to $(4\,|K| - 4\ln 2) + \ln 2 + 2\ln 2 + 0 = 4\,|K| - \ln 2$, growing without bound. The synchronous chain also has an oscillating mode. Since $\mathbb{E}[m_A(t+1) \mid \mathbf{s}(t)] = \tau\, m_B(t)$ and $\mathbb{E}[m_B(t+1) \mid \mathbf{s}(t)] = \tau\, m_A(t)$, the functions $m_A \pm m_B$ are eigenfunctions of its backward operator with eigenvalues $\pm\tau$. The eigenvalue $-\tau \to -1$ is the checkerboard flip-flop described after Theorem 2.20.

Part 5: The 2×2 King's cell ($\mathcal{K}_4$).

With $M = \sum_{i=1}^4 s_i$, the identity $M^2 = 4 + 2\sum_{i<j} s_i s_j$ gives $E_\theta = -J\sum_{i<j} s_i s_j = -\frac{J}{2}(M^2 - 4)$. The levels are $M = \pm 4$ (two states, energy $-6J$), $M = \pm 2$ (eight states, energy 0), and $M = 0$ (six states, energy $+2J$). Therefore $Z = 2e^{6K} + 8 + 6e^{-2K}$, and since all six pairs are equivalent, $\langle s_i s_j\rangle = (\langle M^2\rangle - 4)/12$ with $\langle M^2\rangle = (32e^{6K} + 32)/Z$. Simplifying,

$$c_{\mathrm{ex}}(K) = \frac{32e^{6K} + 32 - 4Z}{12\,Z} = \frac{e^{6K} - e^{-2K}}{e^{6K} + 4 + 3e^{-2K}}. \tag{2.108}$$

For the antiferromagnet, $c_{\mathrm{ex}} \to -\frac{1}{3}$ as $K \to -\infty$. The ground states are the six configurations with $M = 0$, in each of which four pairs are anti-aligned and two are aligned, so $\langle s_i s_j\rangle = (2 - 4)/6 = -\frac{1}{3}$: frustration prevents the bonds from all being satisfied.

For the synchronous sampler, the field on spin i is $J(M - s_i)$. In the class $M = \pm 4$ all four fields have magnitude $3\,|J|$; for $M = \pm 2$, the three majority spins feel $|J|$ and the minority spin feels $3\,|J|$; for $M = 0$ every spin feels $|J|$. By Eq. (2.75) the class weights are $w_4 = (2\cosh 3K)^4$, $w_2 = (2\cosh K)^3(2\cosh 3K)$, $w_0 = (2\cosh K)^4$ per configuration. The average of $s_i s_j$ within the classes is 1, 0, $-\frac{1}{3}$ respectively, so

$$c_{\mathrm{sync}}(K) = \frac{2w_4 - 6w_0/3}{2w_4 + 8w_2 + 6w_0} = \frac{r^4 - 1}{r^4 + 4r + 3}, \qquad r := \frac{\cosh 3K}{\cosh K} = 2\cosh 2K - 1, \tag{2.109}$$

using $\cosh 3K = 4\cosh^3 K - 3\cosh K$. This is even in K (Corollary 2.21(i)) and tends to $+1$ at strong coupling of either sign. For the ferromagnet the synchronous answer is too small at intermediate coupling and becomes nearly right at strong coupling, where both distributions concentrate on the two ordered states. For the antiferromagnet it has the wrong sign: by Eq. (2.78) the synchronous

sampler favors the states with the largest field magnitudes, which are the ordered states $M = \pm 4$, and for $K < 0$ these are the *highest*-energy states of p_θ.

The chromatic sweep of $\mathcal{K}_4$ is a product of four non-commuting projections, and its spectrum is complex. At $K = -0.25$, exact diagonalization gives the leading non-trivial eigenvalues $0.0845 \pm 0.1427\,i$ (modulus 0.1658), followed by the real eigenvalue 0.0953. The asymptotic rate is set by the modulus, as in Theorem 2.26(iii).

Part 6: Exact numerics for all three cells.

Table 2.1 collects the results. For the chromatic sweep we list λ_*, the largest modulus among the non-trivial eigenvalues of $\mathbf{P}_{\text{chr}}$, the gap $1 - \lambda_*$, and t_{mix}, the number of sweeps after which the total variation distance from p_θ first drops below 0.01 when started from the all-up configuration. For the synchronous sampler we list the stationary nearest-neighbor correlation (averaged over all bonds of the cell) and $D_{\text{KL}}(\pi_{\text{sync}} \,\|\, p_\theta)$. The plaquette rows agree with Eqs. (2.97), (2.103) and (2.107), and the $\mathcal{K}_4$ correlations with Eqs. (2.108) and (2.109).

		Chromatic sweep (exact)			Nearest-neighbor $\langle s_i s_j \rangle$ and error		
Cell	K	λ_*	$1-\lambda_*$	t_{mix}	exact	synchronous	$D_{\text{KL}}(\pi_{\text{sync}}\|p_\theta)$
Plaquette	± 0.25	0.2136	0.786	4	± 0.259	0.000	0.142
($\chi = 2$)	± 0.50	0.5800	0.420	8	± 0.536	0.000	0.697
	± 1.00	0.9293	0.0707	54	± 0.900	0.000	2.925
2×2 King	$+0.25$	0.4283	0.572	6	$+0.376$	$+0.141$	0.092
($\mathcal{K}_4$, $\chi = 4$)	-0.25	0.1658	0.834	3	-0.155	$+0.141$	0.399
	$+0.50$	0.8464	0.154	25	$+0.783$	$+0.592$	0.081
	-0.50	0.3314	0.669	5	-0.219	$+0.592$	2.910
	$+1.00$	0.9955	4.5×10^{-3}	869	$+0.989$	$+0.984$	0.001
	-1.00	0.6169	0.383	10	-0.282	$+0.984$	9.059
3×3 King	$+0.25$	0.7677	0.232	16	$+0.477$	$+0.289$	0.221
($\chi = 4$)	-0.25	0.2650	0.735	4	-0.157	$+0.289$	2.580
	$+0.50$	0.9937	6.3×10^{-3}	617	$+0.912$	$+0.908$	0.023
	-0.50	0.4984	0.502	7	-0.231	$+0.908$	14.91
	$+1.00$	$1 - 1.4 \times 10^{-6}$	1.4×10^{-6}	2.9×10^{6}	$+0.997$	$+0.997$	6×10^{-5}
	-1.00	0.8628	0.137	28	-0.313	$+0.997$	29.62

Table 2.1: Exact comparison of chromatic block Gibbs sampling and synchronous updates on three Ising cells at zero field. The chromatic chain always converges to p_θ; the synchronous chain converges to π_{sync}, and its error does not decrease with run time.

Part 7: What the numbers show.

(a) *Exactness costs nothing in parallelism.* The plaquette needs two phases per sweep and the King's cells four, independent of how many cells are tiled together, and in every row the chromatic chain converges to the exact p_θ. The synchronous chain uses one phase per sweep but converges to π_{sync}. Its error is a property of its stationary law, so running it longer does not help.

(b) *The synchronous error is worst exactly where sampling is interesting.* On the bipartite plaquette it erases all nearest-neighbor correlations for both signs of J. On the King's cells it reproduces the ferromagnet reasonably only deep in the ordered regime, where both distributions sit on the two ordered states, and it gets the antiferromagnet qualitatively wrong

at every coupling, returning ferromagnetic correlations for a frustrated antiferromagnet. The divergence $D_{\mathrm{KL}}(\pi_{\mathrm{sync}} \| p_\theta)$ reaches 29.6 nats on the 3×3 antiferromagnetic cell at $K = -1$.

(c) *The chromatic gap is set by the physics.* For the bipartite plaquette, Eq. (2.103) gives $\gamma_{\mathrm{chr}} = \operatorname{sech}^2 2K$ for both signs. On the King's cells the two signs differ sharply. The frustrated antiferromagnet has many degenerate low-energy states separated by small barriers (on $\mathcal{K}_4$, leaving an $M = 0$ ground state by one flip costs only $2\,|J|$), and it mixes in a few sweeps even at $|K| = 1$. The ferromagnet has two ordered states separated by an energy barrier: to reverse the magnetization a domain wall must cross the cell, and the gap falls to 1.4×10^{-6} for the 3×3 cell at $K = 1$, requiring about 3×10^6 sweeps. This is the discrete counterpart of Kramers escape over the barrier of the double-well sampler. Its quantitative theory, $\tau_{\mathrm{mix}} \sim e^{\beta \Delta E}$, is developed in Chapter 7, its spin-glass generalization in Section 3.5, and the way Denoising Thermodynamic Models avoid it by keeping each sampling step well-conditioned in Section 16.3.

(d) *Spectral structure.* The two-color sweep has a real, non-negative spectrum with $\lambda_2 = \varrho^2$, while the four-color sweep has complex eigenvalues. Neither has the eigenvalue near -1 that the synchronous chain develops at strong coupling.

Summary

Proved:

- Equivalence of the pairwise, local, and global Markov properties for positive distributions (Proposition 2.3); the Hammersley–Clifford theorem with explicit Möbius clique potentials (Theorem 2.5).
- Single-site conditionals on factor graphs: effective fields for k-local Ising models and softmax for Potts models (Proposition 2.7, Examples 2.8 and 2.9).
- Spectral decomposition, mixing bound, and Dirichlet-form characterization of the gap for reversible chains (Theorem 2.11, Lemma 2.15, and Proposition 2.16).
- Detailed balance of Metropolis–Hastings and heat-bath kernels; block Gibbs updates as conditional-expectation projections; properties of random and systematic scans (Propositions 2.12 and 2.14 and Lemma 2.13).
- The Little–Peretto law of synchronous dynamics, its blindness to the sign of $\mathbf{J}$, and its product form on bipartite graphs; the Gaussian analogue (Theorem 2.20, Corollary 2.21, and Example 2.27).
- Exactness and L^2 contraction of chromatic block Gibbs sampling, with $\lambda_2 = \varrho^2$ for two colors (Proposition 2.24 and Theorem 2.26); exact results for the plaquette and King's cells (Worked computation 2.28).

Assumed:

- Strict positivity $p_\theta > 0$, and a finite state space for all spectral statements; continuous state spaces were handled by replacing sums with integrals, without proof.
- Symmetric pairwise couplings for the Little–Peretto theorem.
- Ideal hardware phases: each active node draws an exact sample from its conditional, using settled neighbor fields that are held fixed.

Taken up later:

- Markov kernels on continuous spaces (Chapter C); Langevin and Fokker–Planck counterparts of the gap and the Poincaré inequality (Chapters 5 and 6).
- Kramers theory of the exponentially small gaps of ordered cells (Chapter 7); frustration and mixing bottlenecks in spin glasses (Chapter 3).
- Device-level p-bits and analog synapses (Chapter 14); chromatic scheduling and energy budgets of the denoising architecture (Chapters 16 and 17).

Open questions:

- Error bounds for partially asynchronous hardware, in which some coupled neighbors update within each other's field-settling window.
- Blocks larger than color classes (for example, trees sampled exactly by message passing) that increase the gap while keeping updates local.
- Non-reversible chromatic schedules, and the stationary laws of synchronous dynamics with higher-order factors.

Bibliographic notes

The equivalence between Markov random fields and Gibbs distributions was proved by Hammersley and Clifford in an unpublished 1971 manuscript [45]; the Möbius-inversion proof given here follows the standard modern presentation. Factor graphs, Ising and Potts models as graphical models, and the statistical-physics view of conditional independence are treated in depth by Mézard and Montanari [76]. The Gibbs sampler was introduced for image restoration on lattice Markov random fields by Geman and Geman [40], building on Glauber's heat-bath dynamics for the Ising model [42], and the same stochastic update rule underlies the Boltzmann machine of Ackley et al. [1]. The pseudo-likelihood and the coding (coloring) schemes for lattice fields go back to Besag [11]. The spectral and Dirichlet-form theory of reversible chains is developed systematically by Levin et al. [65], and the identification of the two-block Gibbs rate with the squared maximal correlation is due to Liu et al. [68].

Synchronous parallel dynamics of Ising-type neural networks and its stationary distribution were studied by Little [67] and Peretto [87]. Chromatic and other parallel Gibbs schedules for large graphical models were analyzed by Gonzalez et al. [43]. Probabilistic-bit networks that implement heat-bath updates in physical hardware were introduced by Camsari et al. [18] and demonstrated at scale by Borders et al. [14]. Extropic's block-Gibbs software framework `THRML` and its mapping onto thermodynamic sampling hardware are described in Extropic Corporation [31], and the use of chromatic sampling inside Denoising Thermodynamic Models in Jelinčič et al. [56].

Chapter 3

Spin Glasses, Geometric Frustration, and the Mixing-Time Bottleneck

In Chapter 1 we treated an Energy-Based Model as a canonical ensemble $p_\theta(\mathbf{s}) = Z^{-1}e^{-\beta E_\theta(\mathbf{s})}$, and in Chapter 2 we built local Markov chains (Glauber and chromatic block Gibbs samplers) whose stationary law is p_θ. Most examples there had uniform couplings, as in the ferromagnetic Ising model and the Curie–Weiss baseline of Example 1.11. A trained EBM does not. Its couplings J_{ij} are fixed numbers produced by a learning algorithm. They take both signs, they vary from bond to bond, and they encode many competing constraints at once. The induced Hebbian couplings $\mathbf{J}_{\text{eff}} = \mathbf{W}^\top \mathbf{G}^{-1}\mathbf{W}$ found in Worked computation 1.19 are a typical example. The physics of such systems is the physics of *spin glasses*: magnets whose couplings are random but frozen in time.

Spin glasses matter for thermodynamic computing because they are the cleanest setting in which an equilibrium distribution can be written down in one line, yet cannot be sampled in any reasonable time by a local stochastic dynamics. Two ingredients cause this. The first is *frustration*: when couplings of both signs close a loop, no spin configuration satisfies every bond, and the energy landscape acquires a very large number of nearly degenerate local minima. The second is the size of the barriers between these minima. In mean-field spin glasses these barriers grow with the number of spins N, so the time a local sampler needs to move between minima grows faster than any power of N. For a hardware sampler built from p-bits, this is a limit that no improvement in device speed can remove.

The first half of the chapter develops the equilibrium theory: frustration and gauge symmetry (Section 3.1), the Thouless–Anderson–Palmer (TAP) mean-field equations (Section 3.2), the replica solution of the Sherrington–Kirkpatrick (SK) model (Section 3.3), and replica symmetry breaking and barrier scaling (Section 3.4). The second half (Section 3.5) turns the picture into a theorem: a bottleneck bound on the spectral gap of any reversible sampler, which shows that a distribution with well-separated modes cannot be sampled quickly by any reversible local dynamics. A worked computation (Worked computation 3.33) evaluates the replica results and computes trapping times in two small p-bit models.

The dynamical theory of barrier crossing (mean first-passage times, Kramers' formula, and the spectral gap of a double well) is developed in Chapter 7, which uses the barrier scaling of this chapter as input. The way around the bottleneck in hardware, Denoising Thermodynamic Models that split one hard sampling problem into a chain of well-conditioned ones, is the subject of Chapter 16.

3.1 Quenched Disorder, Frustration, and Rugged Energy Landscapes

We work throughout with N Ising spins $\mathbf{s} = (s_1, \dots, s_N) \in \{-1, +1\}^N$ and the pairwise Hamiltonian of Eq. (1.67),

$$E_{\mathbf{J}}(\mathbf{s}) = -\frac{1}{2}\sum_{i,j=1}^{N} J_{ij} s_i s_j - \sum_{i=1}^{N} h_i s_i = -\sum_{i<j} J_{ij} s_i s_j - \sum_{i=1}^{N} h_i s_i, \qquad J_{ij} = J_{ji},\ J_{ii} = 0, \tag{3.1}$$

with Boltzmann distribution $p_{\mathbf{J}}(\mathbf{s}) = Z_{\mathbf{J}}^{-1} e^{-\beta E_{\mathbf{J}}(\mathbf{s})}$. Thermal averages are written $\langle \cdot \rangle$ (or $\langle \cdot \rangle_{\mathbf{J}}$ when the dependence on couplings matters). A pair (i, j) with $J_{ij} \neq 0$ is a *bond*, and the bonds form the interaction graph $\mathcal{G} = (V, \mathcal{B})$ with vertex set $V = \{1, \dots, N\}$.

3.1.1 Quenched versus Annealed Disorder

In a spin glass the couplings $\mathbf{J} = (J_{ij})$ are drawn once from a probability distribution and then held fixed while the spins fluctuate. The physical example is a dilute magnetic alloy such as Mn in Cu, where the conduction-electron-mediated (RKKY) exchange between two Mn moments oscillates in sign with their separation and the Mn positions are frozen in at the quench from the melt. The machine-learning example is a trained EBM: the weights are fixed once training stops, and the sampler explores spins at fixed weights.

Definition 3.1 (Quenched and Annealed Free Energy)**.** Let $\mathbf{J}$ be random with law $\mathbb{P}_{\mathbf{J}}$, and write $\overline{X} := \mathbb{E}_{\mathbf{J}}[X]$ for the average over the disorder. The *quenched* free energy per spin and the *annealed* free energy per spin are

$$f_{\mathrm{q}}(\beta) := -\frac{1}{\beta N}\,\overline{\ln Z_{\mathbf{J}}}, \qquad f_{\mathrm{a}}(\beta) := -\frac{1}{\beta N}\,\ln \overline{Z_{\mathbf{J}}}. \tag{3.2}$$

The two averages describe two different physical situations. The quenched average is correct when the couplings do not equilibrate on the time scale of the experiment: each sample of the alloy, or each trained model, has its own $Z_{\mathbf{J}}$, and the measured free energy is the typical value of $\ln Z_{\mathbf{J}}$. The annealed average would be correct if the couplings were themselves thermal degrees of freedom that equilibrate together with the spins, so that the joint partition function is $\sum_{\mathbf{J}} \sum_{\mathbf{s}} (\cdots)$. Because $\ln$ is concave, Jensen's inequality gives $\overline{\ln Z_{\mathbf{J}}} \leq \ln \overline{Z_{\mathbf{J}}}$, so

$$f_{\mathrm{q}}(\beta) \geq f_{\mathrm{a}}(\beta). \tag{3.3}$$

The annealed free energy is a lower bound. It is easy to compute because the average of an exponential is often explicit. The quenched free energy is the physically relevant one and is hard to compute.

Example 3.2 (Annealed Free Energy of the Gaussian Spin Glass)**.** Let all $N(N-1)/2$ couplings be independent Gaussians $J_{ij} \sim \mathcal{N}(0, J^2/N)$ and set $h_i = 0$. Since $\overline{e^{tJ_{ij}}} = e^{t^2 J^2/(2N)}$ for a centered Gaussian of variance J^2/N, and $(s_i s_j)^2 = 1$,

$$\overline{Z_{\mathbf{J}}} = \sum_{\mathbf{s} \in \{-1,+1\}^N} \prod_{i<j} \overline{e^{\beta J_{ij} s_i s_j}} = \sum_{\mathbf{s} \in \{-1,+1\}^N} \prod_{i<j} e^{\beta^2 J^2/(2N)} = 2^N \exp\!\left(\frac{\beta^2 J^2}{2N} \cdot \frac{N(N-1)}{2} \right). \tag{3.4}$$

Hence $-\beta f_{\mathrm{a}} = \ln 2 + \beta^2 J^2/4 + \mathcal{O}(1/N)$, the annealed energy per spin is $u_{\mathrm{a}} = \partial_\beta(\beta f_{\mathrm{a}}) = -\beta J^2/2$, and the annealed entropy per spin is

$$s_{\mathrm{a}} = \beta(u_{\mathrm{a}} - f_{\mathrm{a}}) = \ln 2 - \frac{\beta^2 J^2}{4}. \tag{3.5}$$

This entropy becomes negative for $k_{\mathrm{B}}T < J/(2\sqrt{\ln 2}) \approx 0.60\,J$. An Ising system has at most 2^N states, each with nonnegative probability, so its entropy is nonnegative. The annealed calculation therefore fails at low temperature, and (as we will see) it already fails at the spin-glass transition $k_{\mathrm{B}}T_c = J$, before its entropy turns negative. A method for averaging $\ln Z_{\mathbf{J}}$ itself is needed. That method is the replica trick of Section 3.3.

A free energy that depends on a random object is useful only if it does not fluctuate much from sample to sample. The following proposition shows that the free energy per spin of the Sherrington–Kirkpatrick (SK) model, defined below, is self-averaging. The proof is a short application of the Gaussian Poincaré inequality.

Proposition 3.3 (Self-Averaging of the Quenched Free Energy)**.** *Let $J_{ij} = (J/\sqrt{N})\, g_{ij}$ for $i < j$, with g_{ij} independent standard Gaussians, and let $\mathbf{h}$ be fixed. Then*

$$\operatorname{Var}_{\mathbf{J}}\!\left(\frac{1}{N}\ln Z_{\mathbf{J}}\right) \le \frac{\beta^2 J^2}{2N}. \tag{3.6}$$

In particular, $\frac{1}{N}\ln Z_{\mathbf{J}}$ concentrates around its mean $\frac{1}{N}\overline{\ln Z_{\mathbf{J}}}$ as $N \to \infty$.

Proof. We use the Gaussian Poincaré inequality: for a standard Gaussian vector $\mathbf{g} \in \mathbb{R}^M$ and a smooth function Φ with square-integrable gradient, $\operatorname{Var}(\Phi(\mathbf{g})) \le \mathbb{E}\,\|\nabla\Phi(\mathbf{g})\|^2$. This is the Poincaré inequality of Definition 6.12 for the Ornstein–Uhlenbeck process with unit stiffness, whose spectral gap equals 1 (Worked computation 6.21). Take $\Phi(\mathbf{g}) = \ln Z_{\mathbf{J}(\mathbf{g})}$ with $M = N(N-1)/2$. Differentiating $\ln \sum_{\mathbf{s}} e^{-\beta E_{\mathbf{J}}(\mathbf{s})}$ gives

$$\frac{\partial \ln Z_{\mathbf{J}}}{\partial g_{ij}} = \frac{J}{\sqrt{N}}\,\frac{\partial \ln Z_{\mathbf{J}}}{\partial J_{ij}} = \frac{\beta J}{\sqrt{N}}\,\langle s_i s_j\rangle_{\mathbf{J}}\,. \tag{3.7}$$

Since $|\langle s_i s_j\rangle| \le 1$, we get $\|\nabla\Phi\|^2 \le \frac{\beta^2 J^2}{N}\cdot\frac{N(N-1)}{2} \le \frac{\beta^2 J^2 N}{2}$. The Gaussian Poincaré inequality gives $\operatorname{Var}(\ln Z_{\mathbf{J}}) \le \beta^2 J^2 N/2$, and dividing by N^2 yields Eq. (3.6). □

The same argument shows that any observable that is a Lipschitz function of the couplings, with Lipschitz constant $\mathcal{O}(\sqrt{N})$, has relative fluctuations of order $N^{-1/2}$. This is why a single large sample of a spin glass, or a single large trained model, has a well-defined free energy per spin.

Notation Watch: Coupling Scale and Three Kinds of Averages

In the SK model the couplings have standard deviation $J/\sqrt{N}$, not J/N. A ferromagnet with all-to-all couplings needs $J_{ij} = J_0/N$ so that the energy $-\sum_{i<j} J_{ij}s_i s_j \approx -J_0 N/2$ is extensive. A spin glass with random signs has $\sum_j J_{ij}s_j$ of order $\sqrt{N}\cdot(J/\sqrt{N}) = \mathcal{O}(1)$ for a typical configuration, so the $1/\sqrt{N}$ scaling is the one that makes local fields of order one. Some references set $J = 1$ and absorb it into β. We keep J explicit, so that the SK critical temperature reads $k_{\mathrm{B}}T_c = J$.
We use three kinds of averages: $\langle\cdot\rangle$ is the thermal (Boltzmann) average at fixed couplings; $\overline{\cdot}$

is the average over the quenched couplings; and $\mathbb{E}_z[\cdot]$ is an average over an auxiliary standard Gaussian variable $z \sim \mathcal{N}(0,1)$. The Edwards–Anderson order parameter, for example, is $q_{\mathrm{EA}} = \overline{\frac{1}{N}\sum_i \langle s_i \rangle^2}$: a thermal average, squared, then averaged over disorder.

Two standard random-coupling models will be used. The *Edwards–Anderson* (EA) model [27] places $\pm J$ or Gaussian couplings on the nearest-neighbor bonds of a finite-dimensional lattice $\mathbb{Z}^d$. The *Sherrington–Kirkpatrick* (SK) model [99] places independent Gaussian couplings $J_{ij} \sim \mathcal{N}(0, J^2/N)$ on all pairs $i < j$, with a uniform field $h_i = h$. The SK model is the mean-field limit of the EA model, in the same way that the Curie–Weiss model is the mean-field limit of the nearest-neighbor Ising ferromagnet.

3.1.2 Toulouse Frustration

Call a bond (i,j) *satisfied* in the configuration $\mathbf{s}$ if $J_{ij}s_is_j > 0$, so that it contributes $-|J_{ij}|$ to the energy. The energy satisfies $E_{\mathbf{J}}(\mathbf{s}) \geq -\sum_{(i,j)\in\mathcal{B}} |J_{ij}|$ at $\mathbf{h} = \mathbf{0}$, with equality if and only if every bond is satisfied. In a ferromagnet ($J_{ij} > 0$) the two aligned states satisfy all bonds. In a spin glass this is usually impossible, for a purely combinatorial reason identified by Toulouse [110].

Definition 3.4 (Frustrated Cycle)**.** A cycle $\mathcal{C} = (i_1, i_2, \ldots, i_\ell, i_1)$ in the interaction graph is *frustrated* if the product of its couplings is negative:

$$\Phi_{\mathcal{C}} := \prod_{(i,j)\in\mathcal{C}} J_{ij} < 0. \tag{3.8}$$

The coupling set $\mathbf{J}$ is *unfrustrated* if no cycle of $\mathcal{G}$ is frustrated.

Lemma 3.5 (A Frustrated Cycle Contains an Unsatisfied Bond)**.** *If $\mathcal{C}$ is frustrated, then for every $\mathbf{s} \in \{-1,+1\}^N$ at least one bond of $\mathcal{C}$ is unsatisfied. Consequently the ground-state energy at $\mathbf{h} = \mathbf{0}$ obeys $E_0 \geq -\sum_{\mathcal{B}} |J_{ij}| + 2\min_{(i,j)\in\mathcal{C}} |J_{ij}|$, and if $\mathcal{C}_1, \ldots, \mathcal{C}_K$ are frustrated cycles with pairwise disjoint bond sets,*

$$E_0 \geq -\sum_{(i,j)\in\mathcal{B}} |J_{ij}| + 2\sum_{k=1}^{K} \min_{(i,j)\in\mathcal{C}_k} |J_{ij}|\,. \tag{3.9}$$

Proof. Multiply the bond variables $J_{ij}s_is_j$ around the cycle. Each vertex i_r of $\mathcal{C}$ belongs to exactly two bonds of the cycle, so each spin appears twice and $s_{i_r}^2 = 1$:

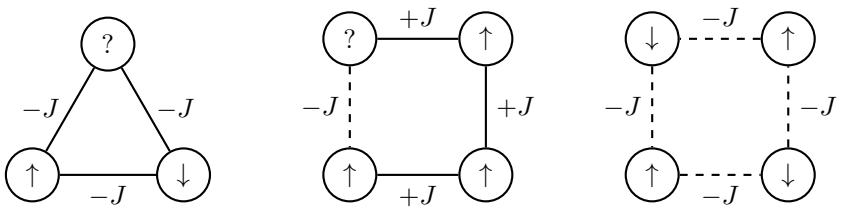
$$\prod_{(i,j)\in\mathcal{C}} (J_{ij}s_is_j) = \Phi_{\mathcal{C}} \prod_{r=1}^{\ell} s_{i_r}^2 = \Phi_{\mathcal{C}} < 0. \tag{3.10}$$

A product of real numbers is negative only if at least one factor is negative, so at least one bond has $J_{ij}s_is_j < 0$. An unsatisfied bond contributes $+|J_{ij}|$ instead of $-|J_{ij}|$, which raises the energy by $2|J_{ij}| \geq 2\min_{\mathcal{C}} |J_{ij}|$ above the all-satisfied value. For bond-disjoint cycles these penalties come from different bonds and add. □

Frustration is a property of the gauge-invariant loop products $\Phi_{\mathcal{C}}$, not of the individual coupling signs. To make this precise we introduce gauge transformations.

Definition 3.6 (Mattis Gauge Transformation)**.** For any $\boldsymbol{\varepsilon} \in \{-1,+1\}^N$, the *gauge transformation* $\mathcal{T}_{\boldsymbol{\varepsilon}}$ acts on spins and couplings by

$$s_i \mapsto s_i' = \varepsilon_i s_i, \qquad J_{ij} \mapsto J_{ij}' = \varepsilon_i\varepsilon_j J_{ij}, \qquad h_i \mapsto h_i' = \varepsilon_i h_i. \tag{3.11}$$



(a) $\Phi_{\mathcal{C}} = -J^3 < 0$ (b) $\Phi_{\mathcal{C}} = -J^4 < 0$ (c) $\Phi_{\mathcal{C}} = +J^4 > 0$

Figure 3.1: Toulouse frustration. (a) An antiferromagnetic triangle: once two spins are anti-aligned, the third cannot anti-align with both. (b) A square plaquette with one negative bond: going around the loop, the sign constraint cannot close. (c) A square with four negative bonds is *not* frustrated: the staggered configuration satisfies every bond, and the gauge transformation $\varepsilon_i = (-1)^i$ maps it to a ferromagnet.

Since $\varepsilon_i^2 = 1$, we have $J'_{ij} s'_i s'_j = \varepsilon_i^2 \varepsilon_j^2 J_{ij} s_i s_j = J_{ij} s_i s_j$ and $h'_i s'_i = h_i s_i$, so $E_{\mathbf{J}'}(\mathbf{s}') = E_{\mathbf{J}}(\mathbf{s})$. The map $\mathbf{s} \mapsto \mathbf{s}'$ is a bijection of $\{-1,+1\}^N$, so $Z_{\mathbf{J}'} = Z_{\mathbf{J}}$ and all thermodynamics is gauge invariant. Every loop product is gauge invariant as well, since each vertex of a cycle contributes $\varepsilon_i^2 = 1$: $\Phi'_{\mathcal{C}} = \Phi_{\mathcal{C}}$.

Theorem 3.7 (Frustration-Free Couplings Are Disguised Ferromagnets)**.** *Let $\mathcal{G}$ be connected and let all couplings on bonds be nonzero. The following are equivalent:*

(i) $\mathbf{J}$ *is unfrustrated: $\Phi_{\mathcal{C}} > 0$ for every cycle $\mathcal{C}$;*

(ii) there is $\boldsymbol{\xi} \in \{-1,+1\}^N$ with $J_{ij} = \xi_i \xi_j \, |J_{ij}|$ for every bond (a Mattis *coupling set [72]);*

(iii) there is a configuration $\mathbf{s}^\star$ that satisfies every bond simultaneously.

In this case the gauge transformation with $\boldsymbol{\varepsilon} = \boldsymbol{\xi}$ maps $\mathbf{J}$ to the ferromagnet $J'_{ij} = |J_{ij}|$ and the ground states to $\pm\boldsymbol{\xi}$.

Proof. (ii)⇒(iii): take $\mathbf{s}^\star = \boldsymbol{\xi}$; then $J_{ij} s_i^\star s_j^\star = \xi_i^2 \xi_j^2 \, |J_{ij}| = |J_{ij}| > 0$.

(iii)⇒(i): if every bond is satisfied by $\mathbf{s}^\star$, then by the computation in the proof of Lemma 3.5, $\Phi_{\mathcal{C}} = \prod_{\mathcal{C}} (J_{ij} s_i^\star s_j^\star) > 0$ for every cycle.

(i)⇒(ii): choose a spanning tree $\mathcal{T}$ of $\mathcal{G}$ and a root vertex r. Set $\xi_r = +1$ and define ξ along the tree: if j is the child of i in $\mathcal{T}$, put $\xi_j = \xi_i \operatorname{sgn}(J_{ij})$. Every vertex is reached exactly once because $\mathcal{T}$ is a spanning tree, so $\boldsymbol{\xi}$ is well defined, and by construction $\operatorname{sgn}(J_{ij}) = \xi_i \xi_j$ on every tree bond. Now let (k,l) be a bond that is not in $\mathcal{T}$. Adding it to $\mathcal{T}$ closes a unique cycle $\mathcal{C} = (k,l) \cup \pi_{lk}$, where π_{lk} is the tree path from l back to k. Along the tree path the signs telescope:

$$\prod_{(i,j) \in \pi_{lk}} \operatorname{sgn}(J_{ij}) = \prod_{(i,j) \in \pi_{lk}} \xi_i \xi_j = \xi_l \xi_k, \tag{3.12}$$

because every interior vertex of the path appears twice. Hypothesis (i) gives $\operatorname{sgn}(\Phi_{\mathcal{C}}) = \operatorname{sgn}(J_{kl}) \, \xi_l \xi_k = +1$, hence $\operatorname{sgn}(J_{kl}) = \xi_k \xi_l$. So $J_{ij} = \xi_i \xi_j \, |J_{ij}|$ holds on all bonds. The final statement follows from Definition 3.6 with $\boldsymbol{\varepsilon} = \boldsymbol{\xi}$, since $\varepsilon_i \varepsilon_j J_{ij} = \xi_i^2 \xi_j^2 \, |J_{ij}|$. □

Theorem 3.7 tells us that sign disorder alone is harmless. A Mattis model $J_{ij} = J \xi_i \xi_j / N$ with random $\boldsymbol{\xi}$ looks disordered, but it is the Curie–Weiss ferromagnet in disguise. It is also the single-pattern Hopfield model (an associative memory whose couplings store patterns $\boldsymbol{\xi}$ by the

Hebbian rule), the rank-one case of the Hebbian couplings Eq. (1.151). Its ground states $\pm\boldsymbol{\xi}$ are the stored pattern and its negative. With P patterns $\boldsymbol{\xi}^1, \dots, \boldsymbol{\xi}^P$ the Hebbian sum $J_{ij} = N^{-1}\sum_\mu \xi_i^\mu \xi_j^\mu$ is no longer of Mattis form, loops become frustrated, and once P is a finite fraction of N the model develops spin-glass states. What makes a spin glass is the frustration of loops.

Example 3.8 (The Frustrated Antiferromagnetic Triangle)**.** Take three spins with $J_{12} = J_{23} = J_{31} = -J$, $J > 0$, so $E(\mathbf{s}) = J(s_1s_2 + s_2s_3 + s_3s_1)$. The triangle is frustrated since $\Phi = -J^3 < 0$. Using $(s_1 + s_2 + s_3)^2 = 3 + 2(s_1s_2 + s_2s_3 + s_3s_1)$ we get $E = \frac{J}{2}\big[(s_1 + s_2 + s_3)^2 - 3\big]$. The sum $s_1 + s_2 + s_3$ takes the values ± 1 (six configurations, two spins one way and one the other) and ± 3 (two aligned configurations). Hence

$$E = -J \text{ (six-fold degenerate)}, \qquad E = +3J \text{ (two-fold)}. \tag{3.13}$$

The ground-state energy $-J$ lies above the all-satisfied value $-3J$ by exactly $2J$, as Lemma 3.5 requires. The ground-state entropy is $\ln 6$ instead of the $\ln 2$ of a ferromagnet. This is the first signature of frustration: a large number of ground states. A lattice of such triangles (the triangular Ising antiferromagnet) has an extensive ground-state entropy per spin. Degeneracy is not the same as slow dynamics, though. In Worked computation 3.33 we show that a single-spin-flip sampler on this triangle mixes in $\mathcal{O}(1)$ steps even at $T \to 0$, because the six ground states are connected to one another by zero-energy spin flips.

3.1.3 Rugged Landscapes and Metastable States

In a frustrated system with many loops, the ground-state problem is a competition among many incompatible constraints. Finding the ground state of the EA model on a nonplanar graph is NP-hard in general, and the SK ground state is a dense instance of the maximum-cut problem. Our concern is a related property: the number and separation of *local* minima.

Definition 3.9 (One-Flip Stable State)**.** A configuration $\mathbf{s}$ is *one-flip stable* for $E_{\mathbf{J}}$ if flipping any single spin does not lower the energy, i.e. if every spin is aligned with its local field:

$$s_i\, h_i^{\mathrm{loc}}(\mathbf{s}) \geq 0 \quad \forall i, \qquad h_i^{\mathrm{loc}}(\mathbf{s}) := h_i + \sum_{j \neq i} J_{ij} s_j. \tag{3.14}$$

Flipping spin i changes the energy by $\Delta_i E = E(\mathbf{s}^{(i)}) - E(\mathbf{s}) = 2 s_i h_i^{\mathrm{loc}}(\mathbf{s})$, where $\mathbf{s}^{(i)}$ is $\mathbf{s}$ with spin i reversed. Indeed, only the terms containing s_i change sign, and they sum to $-s_i h_i^{\mathrm{loc}}$. A one-flip stable state is a configuration from which a zero-temperature single-spin-flip sampler cannot move. In a ferromagnet at $\mathbf{h} = \mathbf{0}$ there are essentially two such states. In the SK model their number is exponential in N: the mean number grows as $\overline{\mathcal{N}_{1\mathrm{F}}} \approx e^{0.199N}$ [107]. Each of them is a trap for a $T = 0$ local search, and a finite-temperature local sampler must climb out of these traps by thermal activation.

The physical picture, which the rest of the chapter makes quantitative, is a *rugged landscape*. There are exponentially many valleys. They are separated by ridges whose heights are spread over a wide range, and the valleys are organized hierarchically: small valleys sit inside larger valleys, which sit inside still larger ones. A local sampler at temperature T crosses ridges of height a few $k_{\mathrm{B}}T$ quickly and is trapped behind the higher ones. The TAP equations of the next section give a finite-temperature version of the valleys. Each solution of the TAP equations is a local minimum of a free-energy function of the magnetizations, and at low temperature the SK model has exponentially many of them.

3.2 From Curie–Weiss to Thouless–Anderson–Palmer Mean-Field Theory

Example 1.11 derived the naive mean-field equations $m_i = \tanh\big(\beta(h_i + \sum_j J_{ij} m_j)\big)$ by minimizing the Gibbs–Bogoliubov–Feynman functional over product distributions. For the Curie–Weiss ferromagnet, $J_{ij} = J_0/N$, these equations become exact as $N \to \infty$. For the SK model they are wrong, and they are wrong at leading order. This section explains why and derives the correction.

3.2.1 Why Naive Mean-Field Theory Fails for Spin Glasses

Naive mean-field theory replaces the fluctuating local field $h_i^{\mathrm{loc}} = h_i + \sum_j J_{ij} s_j$ by its mean $h_i + \sum_j J_{ij} m_j$. The fluctuation of the field around its mean is controlled by

$$\mathrm{Var}\Big(\sum_j J_{ij} s_j\Big) \approx \sum_j J_{ij}^2 (1 - m_j^2), \tag{3.15}$$

where we have kept only diagonal terms $j = k$ in $\sum_{j,k} J_{ij} J_{ik} \,\mathrm{Cov}(s_j, s_k)$. For Curie–Weiss, $J_{ij}^2 = J_0^2/N^2$ and the variance is $\mathcal{O}(1/N)$: the field is sharp and naive mean-field theory is exact. For SK, $J_{ij}^2 \approx J^2/N$ and the variance is $J^2(1-q) = \mathcal{O}(1)$ with $q = N^{-1}\sum_j m_j^2$. The field fluctuates by an amount of order one, and replacing it by its mean is not justified.

A finite field variance is not by itself a problem. What makes naive mean-field theory wrong is a feedback effect. Part of the mean field $\sum_j J_{ij} m_j$ felt by spin i is caused by spin i itself. Spin i polarizes each neighbor j through the coupling J_{ji}, so m_j contains a contribution proportional to $J_{ji} m_i$, and this contribution feeds back to i through J_{ij}. The feedback is of order $\sum_j J_{ij} J_{ji} \chi_{jj} m_i = \sum_j J_{ij}^2 \chi_{jj} m_i$. Its terms all have the same sign because $J_{ij}^2 > 0$, so they add up to an $\mathcal{O}(1)$ total in SK. In a physical medium this feedback is Onsager's *reaction field*: a dipole polarizes its surroundings, and the surroundings act back on the dipole along its own direction. That part of the field does not orient the dipole, so it must be subtracted. Thouless, Anderson, and Palmer did this subtraction for the SK model [108].

3.2.2 The Cavity Derivation of the TAP Equations

We begin from an exact identity. Let $\mathbf{s}_{\setminus i}$ denote all spins other than s_i. The conditional law of s_i given the rest is the p-bit law of Example 1.6 in the local field $h_i^{\mathrm{loc}}(\mathbf{s})$:

$$p(s_i \mid \mathbf{s}_{\setminus i}) = \frac{e^{\beta s_i h_i^{\mathrm{loc}}}}{2\cosh(\beta h_i^{\mathrm{loc}})}, \qquad \Big\langle s_i \mid \mathbf{s}_{\setminus i} \Big\rangle = \tanh(\beta h_i^{\mathrm{loc}}). \tag{3.16}$$

Averaging over $\mathbf{s}_{\setminus i}$ gives the exact *Callen identity* $m_i = \Big\langle \tanh(\beta h_i^{\mathrm{loc}}) \Big\rangle$. The quantity averaged is the conditional mean of the heat-bath update that a p-bit performs (Section 2.3). Naive mean-field theory is the approximation $\Big\langle \tanh(\beta h_i^{\mathrm{loc}}) \Big\rangle \approx \tanh(\beta \Big\langle h_i^{\mathrm{loc}} \Big\rangle)$.

To do better, remove spin i from the system. The *cavity system* is the system with spin i deleted, with Hamiltonian $E^{(\setminus i)}$ containing all terms of Eq. (3.1) that do not involve s_i; write $\langle\cdot\rangle_{\setminus i}$ for its Boltzmann average. The full Hamiltonian is $E = E^{(\setminus i)} - s_i(h_i + H_i)$, where $H_i := \sum_{j \neq i} J_{ij} s_j$ is the *cavity field*. Summing the full Boltzmann weight over $s_i = \pm 1$ first gives the exact representation

$$m_i = \frac{\Big\langle e^{\beta(h_i + H_i)} - e^{-\beta(h_i + H_i)} \Big\rangle_{\setminus i}}{\Big\langle e^{\beta(h_i + H_i)} + e^{-\beta(h_i + H_i)} \Big\rangle_{\setminus i}}. \tag{3.17}$$

Everything is now a question about the distribution of H_i in the cavity system, where spin i is absent and cannot act back.

Assumption 3.10 (Cavity Hypotheses for Dense Weak Couplings)**.** In the cavity system:

(C1) the cavity field H_i is Gaussian with mean $\mu_i = \sum_j J_{ij} m_j^{(\setminus i)}$, where $m_j^{(\setminus i)} = \langle s_j \rangle_{\setminus i}$, and with some variance σ_i^2;

(C2) connected correlations between distinct spins are weak, so that in the response of m_j to a small field only the diagonal susceptibility $\chi_{jj} = \partial m_j / \partial h_j = \beta(1 - m_j^2)$ matters at the order we keep;

(C3) the shift of m_j caused by adding spin i is given by linear response.

These hypotheses hold in the SK model in its high-temperature phase. Each J_{ij} is $\mathcal{O}(N^{-1/2})$ and has a random sign, so the cavity field is a sum of many weakly correlated terms, and the off-diagonal contributions $\sum_{j \neq k} J_{ij} \chi_{jk} J_{ki}$ are sums of $\mathcal{O}(N^2)$ terms of random sign, each of size $\mathcal{O}(N^{-3/2})$, which add to $\mathcal{O}(N^{-1/2})$. They fail for short-range models on finite-dimensional lattices, where the same neighbors appear in many loops.

Theorem 3.11 (TAP Equations)**.** *Under Assumption 3.10, the magnetizations satisfy, up to corrections that vanish as $N \to \infty$,*

$$m_i = \tanh\left(\beta \Big[h_i + \sum_j J_{ij} m_j - \beta\, m_i \sum_j J_{ij}^2 (1 - m_j^2) \Big] \right), \qquad i = 1, \dots, N. \tag{3.18}$$

For the SK model, $\sum_j J_{ij}^2 (1 - m_j^2) = J^2(1 - q) + \mathcal{O}(N^{-1/2})$ with $q = N^{-1} \sum_j m_j^2$, so

$$m_i = \tanh\Big(\beta \big[h_i + \textstyle\sum_j J_{ij} m_j - \beta J^2 (1 - q)\, m_i \big] \Big). \tag{3.19}$$

Proof. Step 1: Gaussian cavity field. By (C1), for $\sigma = \pm 1$,

$$\Big\langle e^{\sigma \beta (h_i + H_i)} \Big\rangle_{\setminus i} = \exp\Big(\sigma \beta (h_i + \mu_i) + \tfrac{1}{2} \beta^2 \sigma_i^2 \Big), \tag{3.20}$$

by the Gaussian moment generating function $\mathbb{E}[e^{tX}] = e^{t\mu + t^2 \sigma^2 / 2}$, with $\sigma^2 = 1$ in the exponent. Insert this into Eq. (3.17). The common factor $e^{\beta^2 \sigma_i^2 / 2}$ cancels between numerator and denominator, leaving

$$m_i = \tanh(\beta(h_i + \mu_i)), \qquad \mu_i = \sum_j J_{ij} m_j^{(\setminus i)}. \tag{3.21}$$

The variance σ_i^2 drops out. This is special to Ising spins, for which $s_i^2 = 1$.

Step 2: Reaction correction. Equation Eq. (3.21) involves the cavity magnetizations, not the true ones. Adding spin i back applies to each j the extra field $J_{ji} s_i$. On average over the fluctuations of s_i this field is $J_{ji} m_i$. By (C2) and (C3), the response of m_j is

$$m_j = m_j^{(\setminus i)} + \chi_{jj} J_{ji} m_i + \mathcal{O}(N^{-1}) = m_j^{(\setminus i)} + \beta(1 - m_j^2) J_{ji} m_i + \mathcal{O}(N^{-1}). \tag{3.22}$$

In the susceptibility we replaced $1 - (m_j^{(\setminus i)})^2$ by $1 - m_j^2$; the two differ by $\mathcal{O}(N^{-1/2})$, which changes μ_i only at subleading order. Solving for $m_j^{(\setminus i)}$ and substituting into μ_i, with $J_{ij} = J_{ji}$:

$$\mu_i = \sum_j J_{ij} m_j - \beta m_i \sum_j J_{ij}^2 (1 - m_j^2). \tag{3.23}$$

Insert this into Eq. (3.21) to obtain Eq. (3.18).

Step 3: SK self-averaging of the reaction coefficient. For SK, $J_{ij}^2 = (J^2/N)g_{ij}^2$ with $\overline{g_{ij}^2} = 1$. The sum $\sum_j (J^2/N) g_{ij}^2 (1 - m_j^2)$ has mean $J^2 N^{-1} \sum_j (1 - m_j^2) = J^2(1 - q)$. Its fluctuation is a sum of N terms with variance $\mathcal{O}(N^{-2})$, hence $\mathcal{O}(N^{-1/2})$. This uses the fact that $m_j^{(\setminus i)}$ does not depend on the g_{ij}, which is the reason for introducing the cavity. Inserting gives Eq. (3.19). □

Heuristic Derivation: The Onsager Term as a Self-Field to Be Subtracted

Picture spin i with magnetization m_i embedded in a sea of neighbors. It pushes neighbor j with field $J_{ji} m_i$, and j responds with an extra magnetization $\delta m_j = \beta(1 - m_j^2) J_{ji} m_i$. That extra magnetization pushes back on i with field $J_{ij} \delta m_j = \beta J_{ij}^2 (1 - m_j^2) m_i$. Summing over j, the reaction field is $\beta m_i \sum_j J_{ij}^2 (1 - m_j^2)$, always parallel to m_i. It is included in the naive mean field $\sum_j J_{ij} m_j$, but it is not an external cause of the polarization of i: it would vanish if m_i vanished. The correct orienting field is the naive one minus this echo. For a ferromagnet the echo per neighbor is $\mathcal{O}(N^{-2})$ and the total is $\mathcal{O}(N^{-1})$. For a spin glass each echo is $\mathcal{O}(N^{-1})$, they do not cancel because $J_{ij}^2 > 0$, and the total over N neighbors is $\mathcal{O}(1)$.

3.2.3 The TAP Free Energy and the Plefka Expansion

The TAP equations are the stationarity conditions of an explicit function of $\mathbf{m}$:

$$\mathcal{F}_{\mathrm{TAP}}(\mathbf{m}) = -\sum_{i<j} J_{ij} m_i m_j - \sum_i h_i m_i + k_{\mathrm{B}} T \sum_{i=1}^{N} \left[\frac{1+m_i}{2} \ln \frac{1+m_i}{2} + \frac{1-m_i}{2} \ln \frac{1-m_i}{2} \right] - \frac{\beta}{4} \sum_{i,j=1}^{N} J_{ij}^2 (1 - m_i^2)(1 - m_j^2). \tag{3.24}$$

The first three terms are the naive mean-field free energy Eq. (1.73). To check stationarity, differentiate term by term. The energy terms give $-\sum_j J_{ij} m_j - h_i$. The entropy term gives $k_{\mathrm{B}} T \operatorname{artanh}(m_i)$, as computed in Example 1.11. In the last term m_i appears in the (i, j) and (j, i) summands, and $J_{ij}^2 = J_{ji}^2$, so

$$\frac{\partial}{\partial m_i} \left[-\frac{\beta}{4} \sum_{k,l} J_{kl}^2 (1 - m_k^2)(1 - m_l^2) \right] = -\frac{\beta}{4} \cdot 2 \sum_j J_{ij}^2 (-2 m_i)(1 - m_j^2) = \beta m_i \sum_j J_{ij}^2 (1 - m_j^2). \tag{3.25}$$

Setting the sum of the three pieces to zero gives $k_{\mathrm{B}} T \operatorname{artanh}(m_i) = h_i + \sum_j J_{ij} m_j - \beta m_i \sum_j J_{ij}^2 (1 - m_j^2)$, which is Eq. (3.18).

The function $\mathcal{F}_{\mathrm{TAP}}$ is not a variational upper bound on the free energy in the sense of Theorem 1.9. It is the second-order truncation of a systematic expansion due to Plefka [88]. One fixes the magnetizations $\mathbf{m}$ by Lagrange multipliers (a Legendre transform of $\ln Z$ with respect to the fields), scales the couplings by α, and expands the resulting Gibbs free energy $G(\mathbf{m}; \alpha)$ in powers of α at fixed $\mathbf{m}$. The order α^0 term is the independent-spin entropy, the order α^1 term is the mean-field energy $-\sum_{i<j} J_{ij} m_i m_j$, and the order α^2 term is the Onsager term above. For SK the higher orders vanish as $N \to \infty$ provided

$$\beta^2 J^2 \, \frac{1}{N} \sum_{i=1}^{N} (1 - m_i^2)^2 < 1. \tag{3.26}$$

Plefka identified this as the convergence condition of the expansion [88]. We will meet the same combination twice more: as the condition for a finite spin-glass susceptibility (below), and as the de Almeida–Thouless stability condition of the replica-symmetric solution (Section 3.4).

3.2.4 The Paramagnetic Instability: $T_c = J$, Not $2J$

As a first application, we locate the spin-glass transition of SK at $\mathbf{h} = \mathbf{0}$. The paramagnet $\mathbf{m} = \mathbf{0}$ always solves Eq. (3.19). It loses stability when a small nonzero $\mathbf{m}$ first becomes self-consistent. Linearizing Eq. (3.19) around $\mathbf{m} = \mathbf{0}$, with $\tanh x \approx x$ and $q = \mathcal{O}(\|\mathbf{m}\|^2)$, gives

$$m_i = \beta \sum_j J_{ij} m_j - \beta^2 J^2 m_i \quad \Longleftrightarrow \quad \mathbf{J}\mathbf{m} = \frac{1+\beta^2 J^2}{\beta}\,\mathbf{m}. \tag{3.27}$$

A nonzero solution exists only if $(1+\beta^2 J^2)/\beta$ is an eigenvalue of the coupling matrix $\mathbf{J}$. For the SK ensemble the empirical spectrum of $\mathbf{J}$ converges to Wigner's semicircle law on $[-2J, 2J]$, so the largest eigenvalue is $\lambda_{\max} \to 2J$. The paramagnet first becomes unstable when

$$\frac{1+\beta^2 J^2}{\beta} = 2J \quad \Longleftrightarrow \quad (\beta J - 1)^2 = 0 \quad \Longleftrightarrow \quad k_{\mathrm{B}} T_c = J. \tag{3.28}$$

For $T \neq T_c$, $1+\beta^2 J^2 > 2\beta J$ by the inequality of arithmetic and geometric means, so the paramagnet is linearly stable above T_c and marginal at T_c. (Below T_c the linearization again has no nonzero solution, but there $\beta^2 J^2 > 1$ violates the convergence condition Eq. (3.26) at $\mathbf{m} = \mathbf{0}$, so the TAP expansion about the paramagnet is no longer valid.)

Compare this with naive mean-field theory. Dropping the Onsager term turns Eq. (3.27) into $\mathbf{J}\mathbf{m} = \beta^{-1}\mathbf{m}$, which predicts an instability at $k_{\mathrm{B}}T = \lambda_{\max} = 2J$, too high by a factor of two. Naive mean-field theory follows the top eigenvector of $\mathbf{J}$ as if that eigenvector could be magnetized independently of the reaction of the rest of the system. The Onsager term removes the self-field and moves the instability to the correct $T_c = J$, which agrees with the replica calculation of Section 3.3.

Heuristic Derivation: Spin-Glass Susceptibility and Its Divergence

The response matrix $\chi_{ij} = \partial m_i / \partial h_j = \beta(\langle s_i s_j \rangle - m_i m_j)$ has random signs off the diagonal, so the uniform susceptibility $N^{-1}\sum_{ij}\chi_{ij}$ stays finite at the spin-glass transition. The quantity that diverges is the *spin-glass susceptibility* $\chi_{\mathrm{SG}} = N^{-1}\sum_{i,j}\chi_{ij}^2$. To estimate it, write an off-diagonal response as a sum over chains of distinct sites, $\chi_{ij} \approx \chi_{ii}J_{ij}\chi_{jj} + \sum_k \chi_{ii}J_{ik}\chi_{kk}J_{kj}\chi_{jj} + \cdots$. Backtracking chains are absent because the Onsager term has already removed them. Square, average over the independent couplings ($\overline{J_{ij}J_{kl}} = 0$ unless the bonds coincide), and keep only the pairings that match each chain with itself (*ladder* diagrams). Each extra link contributes J^2/N, and the sum over the new intermediate site contributes $N \cdot \overline{\chi_{kk}^2}$. With $\chi_{kk} = \beta(1-m_k^2)$ and $c_2 := N^{-1}\sum_k (1-m_k^2)^2$, the chains of length ℓ contribute $\beta^2 c_2 (\beta^2 J^2 c_2)^\ell$, and the geometric series gives

$$\chi_{\mathrm{SG}} \approx \frac{\beta^2 c_2}{1-\beta^2 J^2 c_2}. \tag{3.29}$$

This diverges when $\beta^2 J^2 c_2 \to 1$, which is the boundary of Plefka's condition Eq. (3.26). At $\mathbf{h} = \mathbf{0}$ above T_c we have $m_k = 0$, $c_2 = 1$, and $\chi_{\mathrm{SG}} = \beta^2/(1-\beta^2 J^2)$ diverges at $k_{\mathrm{B}}T_c = J$, in agreement with Eq. (3.28).

3.2.5 Counting TAP Solutions: Complexity

Below T_c the TAP equations have many solutions. Each locally stable solution $\mathbf{m}^\alpha$ is a local minimum of $\mathcal{F}_{\mathrm{TAP}}$, the finite-temperature analog of a one-flip stable state. Bray and Moore, and independently De Dominicis and collaborators, computed the mean number of such solutions in the SK model and found that it grows as $e^{N\Sigma(T)}$ with a *complexity* $\Sigma(T) > 0$ for all $T < T_c$, which tends at $T \to 0$ to the exponent 0.199 quoted above for one-flip stable states [77]. The Gibbs measure is then a weighted mixture of many TAP states, $p_{\mathbf{J}} \approx \sum_\alpha w_\alpha p_\alpha$, and the weights $w_\alpha \propto e^{-\beta \mathcal{F}_{\mathrm{TAP}}(\mathbf{m}^\alpha)}$ are dominated by the lowest states. The replica method of the next section computes the thermodynamics of this mixture without enumerating its components.

3.3 The Sherrington–Kirkpatrick Model and the Replica Method

The SK model [99] is the Hamiltonian Eq. (3.1) on the complete graph with

$$E_{\mathbf{J}}(\mathbf{s}) = -\sum_{1 \le i < j \le N} J_{ij} s_i s_j - h \sum_{i=1}^{N} s_i, \qquad J_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\!\left(0, \frac{J^2}{N}\right). \tag{3.30}$$

Our goal is the quenched free energy $f(\beta, h) = -\lim_{N\to\infty} (\beta N)^{-1} \overline{\ln Z_{\mathbf{J}}}$. By Proposition 3.3, this limit also describes a typical single sample.

3.3.1 The Replica Trick

The disorder average of a logarithm is hard, but the disorder average of a power of Z is a Gaussian integral. The replica trick connects the two.

Lemma 3.12 (Replica Identity)**.** *For any random variable $Z > 0$ such that $\overline{Z^n} < \infty$ for n in a neighborhood of 0,*

$$\overline{\ln Z} = \lim_{n \to 0} \frac{\overline{Z^n} - 1}{n} = \lim_{n \to 0} \frac{1}{n} \ln \overline{Z^n}. \tag{3.31}$$

Proof. For fixed Z, $Z^n = e^{n \ln Z} = 1 + n \ln Z + \frac{n^2}{2} (\ln Z)^2 e^{\theta n \ln Z}$ for some $\theta \in (0,1)$, by Taylor's theorem with Lagrange remainder. Averaging, $(\overline{Z^n} - 1)/n = \overline{\ln Z} + \mathcal{O}(n)$, where the remainder is bounded by $\frac{n}{2}\,\overline{(\ln Z)^2 (Z^{|n|} + Z^{-|n|})}$, which is finite under the moment hypothesis. This gives the first equality. For the second, $\ln \overline{Z^n} = \ln(1 + n \overline{\ln Z} + \mathcal{O}(n^2)) = n \overline{\ln Z} + \mathcal{O}(n^2)$. $\square$

The lemma is rigorous for real $n \to 0$. The difficulty is practical: we can compute $\overline{Z^n}$ only for positive integers n, where Z^n is the partition function of n independent copies (*replicas*) of the system with the same couplings. The replica method computes $\overline{Z^n}$ for integer n, writes the answer as an analytic expression in n, and continues it to $n \to 0$. That continuation is not unique in general: a function known on the positive integers is determined by those values only under growth conditions (Carlson's theorem) that are not verified here. We will also exchange the limits $n \to 0$ and $N \to \infty$. Both steps are assumptions; they are listed in the chapter summary. The theorems of Guerra and Talagrand discussed in Section 3.4 later confirmed the final answer.

3.3.2 Averaging the Replicated Partition Function

Fix an integer $n \ge 1$ and label replicas by $a, b \in \{1, \dots, n\}$. Then

$$Z_{\mathbf{J}}^n = \sum_{\{\mathbf{s}^a\}} \exp\!\Big(\beta \sum_{i<j} J_{ij} \sum_{a=1}^{n} s_i^a s_j^a + \beta h \sum_{i=1}^{N} \sum_{a=1}^{n} s_i^a\Big), \tag{3.32}$$

where the sum runs over all n-tuples of configurations $\mathbf{s}^1, \dots, \mathbf{s}^n \in \{-1,+1\}^N$. The couplings appear linearly in the exponent. Since the J_{ij} are independent, the average factorizes over pairs, and each factor is a Gaussian moment generating function $\overline{e^{tJ_{ij}}} = e^{t^2 J^2/(2N)}$ with $t = \beta \sum_a s_i^a s_j^a$:

$$\overline{Z_{\mathbf{J}}^n} = \sum_{\{\mathbf{s}^a\}} \exp\Big(\frac{\beta^2 J^2}{2N} \sum_{i<j} \Big(\sum_{a=1}^{n} s_i^a s_j^a \Big)^2 + \beta h \sum_{i,a} s_i^a \Big). \tag{3.33}$$

The replicas, which were independent at fixed $\mathbf{J}$, are now coupled. Averaging over a common random environment induces an effective attraction between copies.

Next we expand the square. Splitting the double sum over replicas into $a = b$ (where $(s_i^a)^2 (s_j^a)^2 = 1$) and $a \neq b$ (where each unordered pair appears twice),

$$\sum_{i<j} \Big(\sum_{a} s_i^a s_j^a \Big)^2 = \sum_{i<j} \sum_{a,b} s_i^a s_i^b s_j^a s_j^b = \frac{nN(N-1)}{2} + 2 \sum_{a<b} \sum_{i<j} (s_i^a s_i^b)(s_j^a s_j^b). \tag{3.34}$$

For each pair $a < b$, apply $\sum_{i<j} x_i x_j = \tfrac{1}{2}\big[(\sum_i x_i)^2 - \sum_i x_i^2 \big]$ with $x_i = s_i^a s_i^b$, $x_i^2 = 1$:

$$\sum_{i<j} (s_i^a s_i^b)(s_j^a s_j^b) = \frac{1}{2} \Big[\Big(\sum_{i=1}^{N} s_i^a s_i^b \Big)^2 - N \Big]. \tag{3.35}$$

Putting the pieces into Eq. (3.33),

$$\overline{Z_{\mathbf{J}}^n} = e^{\beta^2 J^2 [n(N-1) - n(n-1)]/4} \sum_{\{\mathbf{s}^a\}} \exp\Big(\frac{\beta^2 J^2}{2N} \sum_{a<b} \Big(\sum_{i=1}^{N} s_i^a s_i^b \Big)^2 + \beta h \sum_{i,a} s_i^a \Big). \tag{3.36}$$

Here the prefactor collects $\frac{\beta^2 J^2}{2N} \big[\frac{nN(N-1)}{2} - N \frac{n(n-1)}{2} \big]$. To leading order in N it equals $e^{N\beta^2 J^2 n/4}$; the remainder $e^{-\beta^2 J^2 n^2/4}$ is independent of N and drops out of the free energy per spin.

The sum over configurations in Eq. (3.36) depends on the spins only through the $n(n-1)/2$ *overlaps* between replicas,

$$Q_{ab}(\mathbf{s}^a, \mathbf{s}^b) := \frac{1}{N} \sum_{i=1}^{N} s_i^a s_i^b \in [-1,1], \qquad a < b. \tag{3.37}$$

In terms of the overlaps, the exponent reads $\frac{N\beta^2 J^2}{2} \sum_{a<b} Q_{ab}^2$. It is quadratic in the overlaps, which are sums over sites, so the sites are still coupled. We decouple them by a Hubbard–Stratonovich transformation.

3.3.3 Hubbard–Stratonovich Decoupling and the Saddle Point

By the one-dimensional Gaussian integral (Corollary B.3), for $A > 0$ and real X,

$$e^{X^2 B^2/(2A)} = \sqrt{\frac{A}{2\pi}} \int_{\mathbb{R}} \mathrm{d}q \, \exp\Big(-\frac{A}{2} q^2 + BXq \Big). \tag{3.38}$$

To verify, complete the square: $-\frac{A}{2}q^2 + BXq = -\frac{A}{2}\big(q - \frac{BX}{A}\big)^2 + \frac{B^2 X^2}{2A}$, and the Gaussian integral of the first term is $\sqrt{2\pi/A}$. Apply Eq. (3.38) to each pair $a < b$ with $X = \sum_i s_i^a s_i^b$, $A = N\beta^2 J^2$,

$B = \beta^2 J^2$; then $B^2 X^2/(2A) = \beta^2 J^2 X^2/(2N)$, as required. This introduces one real integration variable q_{ab} per pair:

$$\exp\Big(\frac{\beta^2 J^2}{2N} X_{ab}^2\Big) = \sqrt{\frac{N\beta^2 J^2}{2\pi}} \int \mathrm{d}q_{ab} \exp\Big(-\frac{N\beta^2 J^2}{2} q_{ab}^2 + \beta^2 J^2 q_{ab} \sum_{i=1}^{N} s_i^a s_i^b\Big). \tag{3.39}$$

Now the spin dependence is linear in the products $s_i^a s_i^b$ at each site, and the sum over $\{\mathbf{s}^a\}$ factorizes into N identical single-site sums. Let $\boldsymbol{\sigma} = (\sigma^1, \dots, \sigma^n) \in \{-1, +1\}^n$ denote the n replicas of one spin. Then

$$\sum_{\{\mathbf{s}^a\}} \prod_{i=1}^{N} \exp\Big(\beta^2 J^2 \sum_{a<b} q_{ab} s_i^a s_i^b + \beta h \sum_a s_i^a\Big) = \Big[\mathrm{Tr}_{\boldsymbol{\sigma}}\, e^{\mathcal{L}_Q(\boldsymbol{\sigma})}\Big]^N, \tag{3.40}$$

with the single-site replica Hamiltonian

$$\mathcal{L}_Q(\boldsymbol{\sigma}) := \beta^2 J^2 \sum_{a<b} q_{ab} \sigma^a \sigma^b + \beta h \sum_{a=1}^{n} \sigma^a, \qquad \mathrm{Tr}_{\boldsymbol{\sigma}} := \sum_{\boldsymbol{\sigma} \in \{-1,+1\}^n}. \tag{3.41}$$

Collecting everything, we have an exact representation for integer n.

Proposition 3.13 (Replicated Action of the SK Model)**.** *For every positive integer n and every N,*

$$\overline{Z_{\mathbf{J}}^n} = C_{n,N} \int_{\mathbb{R}^{n(n-1)/2}} \prod_{a<b} \mathrm{d}q_{ab}\; e^{N\, G_n(Q)}, \qquad G_n(Q) := \frac{\beta^2 J^2 n}{4} - \frac{\beta^2 J^2}{2} \sum_{a<b} q_{ab}^2 + \ln \mathrm{Tr}_{\boldsymbol{\sigma}}\, e^{\mathcal{L}_Q(\boldsymbol{\sigma})}, \tag{3.42}$$

where $C_{n,N} = (N\beta^2 J^2/2\pi)^{n(n-1)/4} e^{-\beta^2 J^2 n^2/4}$ grows at most polynomially in N.

The integrand is $e^{NG_n(Q)}$, so for fixed integer n and $N \to \infty$ Laplace's method gives $\frac{1}{N} \ln \overline{Z^n} \to \max_Q G_n(Q)$. Combining with Lemma 3.12, and exchanging the limits $N \to \infty$ and $n \to 0$,

$$-\beta f = \lim_{n \to 0} \frac{1}{n} G_n(Q^\star), \tag{3.43}$$

where $Q^\star$ is the appropriate stationary point of G_n. The stationarity condition follows from differentiating Eq. (3.42):

$$\frac{\partial G_n}{\partial q_{ab}} = -\beta^2 J^2 q_{ab} + \beta^2 J^2 \frac{\mathrm{Tr}_{\boldsymbol{\sigma}}\, \sigma^a \sigma^b e^{\mathcal{L}_Q}}{\mathrm{Tr}_{\boldsymbol{\sigma}}\, e^{\mathcal{L}_Q}} = 0 \quad \Longleftrightarrow \quad q_{ab} = \big\langle \sigma^a \sigma^b \big\rangle_{\mathcal{L}_Q}. \tag{3.44}$$

The saddle-point value of q_{ab} is the thermal average of $\sigma^a \sigma^b$ in the single-site replica system that q_{ab} itself defines. This is the replica version of a mean-field self-consistency condition.

Remark 3.14 (Maximum or Minimum)**.** For integer $n \geq 2$, Laplace's method selects the *maximum* of G_n. The number of variables q_{ab} is $n(n-1)/2$, which is negative for $0 < n < 1$. As a result, in the analytic continuation the physical saddle point turns out to *minimize* G_n/n with respect to the replica-symmetric order parameter q instead of maximizing it, and the stability analysis of Section 3.4 must be interpreted with care. The working rule, confirmed by the rigorous results, is that the physical solution makes the free energy f a *maximum* over the Parisi order-parameter function. We will not use this rule for any proof. We rely only on the stationarity equations and on the replicon eigenvalue, which has a direct physical meaning (Eq. (3.29)).

3.3.4 Physical Meaning of the Overlap Matrix

The replica order parameter is a measurable quantity. Take two replicas $\mathbf{s}^1, \mathbf{s}^2$ drawn independently from the same Boltzmann distribution $p_{\mathbf{J}}$, and consider their overlap $Q_{12} = N^{-1}\sum_i s_i^1 s_i^2$. Its thermal average is

$$\langle Q_{12}\rangle_{p_{\mathbf{J}}\otimes p_{\mathbf{J}}} = \frac{1}{N}\sum_{i=1}^{N}\langle s_i\rangle\,\langle s_i\rangle = \frac{1}{N}\sum_{i=1}^{N} m_i^2, \tag{3.45}$$

because the two replicas are independent at fixed $\mathbf{J}$. The disorder average of the right-hand side is the Edwards–Anderson order parameter q_{EA}. More generally, define the overlap distribution

$$P_{\mathbf{J}}(q) := \sum_{\mathbf{s}^1,\mathbf{s}^2} p_{\mathbf{J}}(\mathbf{s}^1)p_{\mathbf{J}}(\mathbf{s}^2)\,\delta\big(q - Q_{12}(\mathbf{s}^1,\mathbf{s}^2)\big), \qquad P(q) := \overline{P_{\mathbf{J}}(q)}. \tag{3.46}$$

$P(q)$ is the probability that two independent samples of the same model have overlap q. In the paramagnet, $P(q) = \delta(q)$ at $h = 0$. In a ferromagnet below T_c, $P(q) = \tfrac{1}{2}\delta(q - m^2) + \tfrac{1}{2}\delta(q + m^2)$, reflecting the two states $\pm m$. If the Gibbs measure is a mixture $\sum_\alpha w_\alpha p_\alpha$ of pure states with mutual overlaps $q_{\alpha\gamma} = N^{-1}\sum_i m_i^\alpha m_i^\gamma$, then $P_{\mathbf{J}}(q) = \sum_{\alpha,\gamma} w_\alpha w_\gamma \delta(q - q_{\alpha\gamma})$. The saddle-point matrix $Q^\star_{ab}$ encodes this distribution. In the replica-symmetric case all $Q^\star_{ab}$ are equal, which means two samples always have the same overlap, as for a single pure state. In Section 3.4 the matrix acquires several distinct values, and $P(q)$ acquires several peaks or a continuous part.

3.3.5 The Replica-Symmetric Saddle Point

The simplest ansatz treats all replicas alike.

Definition 3.15 (Replica-Symmetric Ansatz)**.** The *replica-symmetric* (RS) ansatz is $q_{ab} = q$ for all $a < b$.

Theorem 3.16 (Replica-Symmetric Free Energy of the SK Model)**.** *Under the RS ansatz and the replica assumptions above, the quenched free energy per spin is*

$$-\beta f_{\mathrm{RS}}(\beta, h) = \frac{\beta^2 J^2}{4}(1 - q)^2 + \mathbb{E}_z\Big[\ln 2\cosh \Xi(z)\Big], \qquad \Xi(z) := \beta(J\sqrt{q}\,z + h), \tag{3.47}$$

with $z \sim \mathcal{N}(0,1)$, and q solves the self-consistency equation

$$q = \mathbb{E}_z\big[\tanh^2 \Xi(z)\big]. \tag{3.48}$$

The magnetization per spin is $m = \mathbb{E}_z[\tanh \Xi(z)]$, and the internal energy per spin is

$$u_{\mathrm{RS}} = -\frac{\beta J^2}{2}(1 - q^2) - hm. \tag{3.49}$$

Proof. Step 1: Evaluate the quadratic term. With $q_{ab} = q$, $\sum_{a<b} q_{ab}^2 = \frac{n(n-1)}{2}q^2$.

Step 2: Rewrite the single-site Hamiltonian. Using $\sum_{a<b}\sigma^a\sigma^b = \tfrac{1}{2}\big[(\sum_a \sigma^a)^2 - n\big]$ (since $(\sigma^a)^2 = 1$),

$$\mathcal{L}_Q(\boldsymbol{\sigma}) = \frac{\beta^2 J^2 q}{2}\Big(\sum_{a=1}^{n}\sigma^a\Big)^2 - \frac{\beta^2 J^2 q n}{2} + \beta h\sum_{a=1}^{n}\sigma^a. \tag{3.50}$$

Step 3: Second Hubbard–Stratonovich transformation. The replicas of one site are still coupled through $(\sum_a \sigma^a)^2$. Linearize it with a Gaussian average. For $q \geq 0$ and real Y, $e^{\beta^2 J^2 q Y^2/2} =$

$\mathbb{E}_z\big[e^{\beta J\sqrt{q}\,zY}\big]$, which is the moment generating function of $z\sim\mathcal{N}(0,1)$ at $t=\beta J\sqrt{q}\,Y$. With $Y=\sum_a\sigma^a$, the replicas decouple:

$$\mathrm{Tr}_{\boldsymbol{\sigma}}\,e^{\mathcal{L}_Q}=e^{-\beta^2J^2qn/2}\,\mathbb{E}_z\Big[\prod_{a=1}^{n}\sum_{\sigma^a=\pm1}e^{\beta(J\sqrt{q}z+h)\sigma^a}\Big]=e^{-\beta^2J^2qn/2}\,\mathbb{E}_z\Big[\big(2\cosh\Xi(z)\big)^n\Big]. \tag{3.51}$$

The Gaussian variable z is a random local field. Each replica of the site is an independent p-bit in the field $J\sqrt{q}z+h$, and z is shared by all n replicas.

Step 4: Assemble G_n. Substituting Steps 1–3 into Eq. (3.42),

$$G_n^{\mathrm{RS}}(q)=\frac{\beta^2J^2n}{4}-\frac{\beta^2J^2n(n-1)}{4}q^2-\frac{\beta^2J^2qn}{2}+\ln\mathbb{E}_z\Big[\big(2\cosh\Xi\big)^n\Big]. \tag{3.52}$$

Step 5: The limit $n\to0$. Expand $(2\cosh\Xi)^n=e^{n\ln2\cosh\Xi}=1+n\ln2\cosh\Xi+\mathcal{O}(n^2)$, so $\ln\mathbb{E}_z[(2\cosh\Xi)^n]=n\mathbb{E}_z[\ln2\cosh\Xi]+\mathcal{O}(n^2)$. Dividing G_n^{RS} by n and letting $n\to0$ (so $n-1\to-1$):

$$\lim_{n\to0}\frac{G_n^{\mathrm{RS}}(q)}{n}=\frac{\beta^2J^2}{4}+\frac{\beta^2J^2}{4}q^2-\frac{\beta^2J^2}{2}q+\mathbb{E}_z\big[\ln2\cosh\Xi\big]. \tag{3.53}$$

The first three terms combine into $\frac{\beta^2J^2}{4}(1-2q+q^2)=\frac{\beta^2J^2}{4}(1-q)^2$. With Eq. (3.43), this is Eq. (3.47).

Step 6: Stationarity in q. Differentiate Eq. (3.47) with respect to q, using $\partial_q\Xi=\beta Jz/(2\sqrt{q})$:

$$\frac{\partial(-\beta f_{\mathrm{RS}})}{\partial q}=-\frac{\beta^2J^2}{2}(1-q)+\frac{\beta J}{2\sqrt{q}}\mathbb{E}_z\big[z\tanh\Xi(z)\big]. \tag{3.54}$$

By Stein's lemma (Gaussian integration by parts, Lemma B.9), $\mathbb{E}_z[zg(z)]=\mathbb{E}_z[g'(z)]$. With $g=\tanh\Xi$, $g'(z)=\beta J\sqrt{q}\,\mathrm{sech}^2\Xi$, so $\mathbb{E}_z[z\tanh\Xi]=\beta J\sqrt{q}\,\mathbb{E}_z[1-\tanh^2\Xi]$. Substituting,

$$\frac{\partial(-\beta f_{\mathrm{RS}})}{\partial q}=\frac{\beta^2J^2}{2}\Big[-(1-q)+1-\mathbb{E}_z\tanh^2\Xi\Big]=\frac{\beta^2J^2}{2}\Big[q-\mathbb{E}_z\tanh^2\Xi\Big], \tag{3.55}$$

which vanishes if and only if Eq. (3.48) holds. Equivalently, Eq. (3.48) is Eq. (3.44) at $n\to0$:

$$\big\langle\sigma^a\sigma^b\big\rangle_{\mathcal{L}_Q}=\frac{\mathbb{E}_z\big[(2\cosh\Xi)^n\tanh^2\Xi\big]}{\mathbb{E}_z\big[(2\cosh\Xi)^n\big]}\xrightarrow{n\to0}\mathbb{E}_z\big[\tanh^2\Xi\big]. \tag{3.56}$$

Step 7: Magnetization and energy. The magnetization is $m=-\partial f/\partial h$. Since q is stationary, only the explicit h dependence contributes: $\partial_h(-\beta f)=\beta\mathbb{E}_z\tanh\Xi$, so $m=\mathbb{E}_z\tanh\Xi$. For the energy, $u=\partial_\beta(\beta f)$, again at fixed q:

$$-u=\frac{\beta J^2}{2}(1-q)^2+\mathbb{E}_z\big[(J\sqrt{q}z+h)\tanh\Xi\big]=\frac{\beta J^2}{2}(1-q)^2+\beta J^2q(1-q)+hm, \tag{3.57}$$

where the z term was again integrated by parts: $J\sqrt{q}\,\mathbb{E}_z[z\tanh\Xi]=\beta J^2q\,\mathbb{E}_z[\mathrm{sech}^2\Xi]=\beta J^2q(1-q)$. Since $(1-q)^2+2q(1-q)=(1-q)(1+q)=1-q^2$, this is Eq. (3.49). □

The structure of Eq. (3.47) has a direct reading. Each spin sees a random static field $J\sqrt{q}z+h$, a Gaussian of variance J^2q produced by the frozen magnetizations of all other spins, and responds to it as an isolated p-bit. The term $\frac{\beta^2J^2}{4}(1-q)^2$ is the remaining Gaussian field fluctuation that is not frozen. Compare the cavity analysis of Section 3.2: there the cavity mean field $\mu_i=\sum_j J_{ij}m_j^{(\setminus i)}$ is a sum of N terms with independent random coefficients, hence Gaussian across sites with variance $\sum_j J_{ij}^2(m_j^{(\setminus i)})^2\approx J^2q$. Then Eq. (3.21) says $m_i=\tanh(\beta(h+J\sqrt{q}z_i))$ with z_i a standard Gaussian, and averaging m_i^2 over sites reproduces Eq. (3.48). The RS solution is therefore the site statistics of a single TAP solution. Worked computation 3.33 develops this correspondence in detail.

Corollary 3.17 (Paramagnetic Phase and the Spin-Glass Transition at $h = 0$)**.** *At $h = 0$, $q = 0$ always solves Eq. (3.48), with $-\beta f = \ln 2 + \beta^2 J^2/4$, which equals the annealed free energy of Example 3.2. A solution with $q > 0$ branches off continuously at $k_{\mathrm{B}}T_c = J$, and for $\tau := 1 - k_{\mathrm{B}}T/J \to 0^+$ it behaves as $q = \tau + \mathcal{O}(\tau^2)$.*

Proof. At $h = 0$, $\Xi = \beta J\sqrt{q}z$ and the annealed value follows from $\mathbb{E}_z \ln 2\cosh(0) = \ln 2$. For small q, expand $\tanh^2 x = x^2 - \frac{2}{3}x^4 + \mathcal{O}(x^6)$ and use $\mathbb{E}_z z^2 = 1$, $\mathbb{E}_z z^4 = 3$:

$$\mathbb{E}_z \tanh^2 \Xi = \beta^2 J^2 q - \frac{2}{3} \cdot 3\,\beta^4 J^4 q^2 + \mathcal{O}(q^3) = \beta^2 J^2 q - 2\beta^4 J^4 q^2 + \mathcal{O}(q^3). \tag{3.58}$$

Setting this equal to q and dividing by $q > 0$: $1 = \beta^2 J^2 - 2\beta^4 J^4 q + \mathcal{O}(q^2)$, so $q = (\beta^2 J^2 - 1)/(2\beta^4 J^4) + \mathcal{O}(\tau^2)$. A positive solution requires $\beta J > 1$. With $\beta J = (1-\tau)^{-1}$, $\beta^2 J^2 - 1 = 2\tau + \mathcal{O}(\tau^2)$ and $\beta^4 J^4 = 1 + \mathcal{O}(\tau)$, so $q = \tau + \mathcal{O}(\tau^2)$. □

The transition temperature agrees with the TAP result Eq. (3.28) and with the divergence of χ_{SG} in Eq. (3.29). The order parameter q grows linearly below T_c, as for a continuous transition, while the magnetization stays zero. The low-temperature phase is a frozen state with no net magnetization.

Replica symmetry cannot be the whole story. Already Sherrington and Kirkpatrick noticed that the RS entropy becomes negative at low temperature and tends to $s_{\mathrm{RS}}(T \to 0) = -1/(2\pi)$ (derived in Worked computation 3.33). For an Ising system this is impossible, so the RS solution must be wrong somewhere below T_c. The next section identifies where.

3.4 Replica Symmetry Breaking, Ultrametricity, and Barrier Scaling

The RS solution is a stationary point of G_n. To accept it we must check that it is stable against fluctuations of the $n(n-1)/2$ variables q_{ab} that break the symmetry among replicas. De Almeida and Thouless carried out this check in 1978 [23]. They found that one family of fluctuations, the *replicon* modes, becomes unstable in a region of the (h, T) plane that contains the whole $h = 0$ low-temperature phase. Parisi then found the correct saddle point, which breaks replica symmetry in a hierarchical way [83].

3.4.1 The Replica Hessian and the de Almeida–Thouless Eigenvalue

Write $q_{ab} = q + \eta_{ab}$ with small η_{ab} and expand G_n to second order: $G_n(Q) = G_n(Q_{\mathrm{RS}}) + \frac{1}{2}\sum_{(ab),(cd)} M_{(ab),(cd)}\eta_{ab}\eta_{cd} + \cdots$, where (ab) runs over unordered pairs $a < b$ and the first-order term vanishes by Eq. (3.44). Differentiating Eq. (3.44) once more,

$$M_{(ab),(cd)} = \frac{\partial^2 G_n}{\partial q_{ab}\partial q_{cd}} = -\beta^2 J^2 \delta_{(ab),(cd)} + \beta^4 J^4 \Big[\big\langle \sigma^a\sigma^b\sigma^c\sigma^d \big\rangle_{\mathcal{L}_Q} - \big\langle \sigma^a\sigma^b \big\rangle_{\mathcal{L}_Q} \big\langle \sigma^c\sigma^d \big\rangle_{\mathcal{L}_Q} \Big]. \tag{3.59}$$

The second term is the derivative of the average $\big\langle \sigma^a\sigma^b \big\rangle_{\mathcal{L}_Q}$ with respect to the coupling $\beta^2 J^2 q_{cd}$ of $\sigma^c\sigma^d$ in $\mathcal{L}_Q$, which is a connected correlation (Theorem 1.13). At the RS point every average of a product of distinct replicas is computed as in Step 3 of Theorem 3.16: each replica is an independent p-bit in the shared field, so in the limit $n \to 0$, $\langle \sigma^{a_1}\cdots\sigma^{a_k} \rangle_{\mathcal{L}_Q} \to \mathbb{E}_z[\tanh^k \Xi]$ for distinct $a_1, \ldots, a_k$. Because $(\sigma^a)^2 = 1$, a product with a repeated index reduces to a shorter product. Define

$$q = \mathbb{E}_z[\tanh^2 \Xi], \qquad r := \mathbb{E}_z[\tanh^4 \Xi]. \tag{3.60}$$

By the permutation symmetry of the RS point, M has only three distinct entries, depending on how many replica indices the pairs (ab) and (cd) share:

$$P := M_{(ab),(ab)} = -\beta^2 J^2 + \beta^4 J^4 (1 - q^2), \tag{3.61}$$

$$Q := M_{(ab),(ac)} = \beta^4 J^4 (\langle \sigma^b \sigma^c \rangle - q^2) = \beta^4 J^4 (q - q^2), \qquad b \neq c, \tag{3.62}$$

$$R := M_{(ab),(cd)} = \beta^4 J^4 (r - q^2), \qquad \{a, b\} \cap \{c, d\} = \emptyset. \tag{3.63}$$

In Eq. (3.61) we used $(\sigma^a \sigma^b)^2 = 1$, and in Eq. (3.62) we used $\sigma^a \sigma^b \sigma^a \sigma^c = \sigma^b \sigma^c$.

Proposition 3.18 (Replicon Eigenvalue)**.** *Let $\boldsymbol{\eta} = (\eta_{ab})_{a<b}$ be any vector (extended symmetrically, $\eta_{ba} = \eta_{ab}$, $\eta_{aa} = 0$) satisfying the* replicon *constraints*

$$\sum_{b \neq a} \eta_{ab} = 0 \qquad \text{for every } a = 1, \ldots, n. \tag{3.64}$$

Then $M\boldsymbol{\eta} = \lambda_{\mathrm{rep}} \boldsymbol{\eta}$ with

$$\lambda_{\mathrm{rep}} = P - 2Q + R = -\beta^2 J^2 \Big[1 - \beta^2 J^2 \, \mathbb{E}_z \big[\operatorname{sech}^4 \Xi(z) \big] \Big]. \tag{3.65}$$

For $n \geq 4$ the replicon subspace has dimension $n(n-3)/2$.

Proof. Fix a pair (ab). The component $(M\boldsymbol{\eta})_{ab}$ collects three kinds of terms: the diagonal term $P\eta_{ab}$; the pairs sharing exactly one index with (ab), which are (ac) and (bc) for $c \notin \{a, b\}$, each with weight Q; and the pairs disjoint from $\{a, b\}$, each with weight R:

$$(M\boldsymbol{\eta})_{ab} = P\eta_{ab} + Q \sum_{c \notin \{a,b\}} (\eta_{ac} + \eta_{bc}) + R \sum_{\substack{c<d \\ \{c,d\} \cap \{a,b\} = \emptyset}} \eta_{cd}. \tag{3.66}$$

By Eq. (3.64) for row a, $\sum_{c \notin \{a,b\}} \eta_{ac} = \sum_{c \neq a} \eta_{ac} - \eta_{ab} = -\eta_{ab}$; likewise for row b. So the Q term equals $-2Q\eta_{ab}$. For the R term, first compute the sum over all pairs, $\sum_{c<d} \eta_{cd} = \frac{1}{2} \sum_c \sum_{d \neq c} \eta_{cd} = 0$. The pairs that meet $\{a, b\}$ are those containing a (total $\sum_{c \neq a} \eta_{ac} = 0$) and those containing b (total 0), with the pair (ab) counted in both groups; their total is $0 + 0 - \eta_{ab} = -\eta_{ab}$. Hence the sum over disjoint pairs is $0 - (-\eta_{ab}) = \eta_{ab}$, and the R term equals $R\eta_{ab}$. Altogether $(M\boldsymbol{\eta})_{ab} = (P - 2Q + R)\eta_{ab}$.

Now evaluate, using Eqs. (3.61) to (3.63):

$$P - 2Q + R = -\beta^2 J^2 + \beta^4 J^4 [1 - q^2 - 2q + 2q^2 + r - q^2] = -\beta^2 J^2 + \beta^4 J^4 [1 - 2q + r]. \tag{3.67}$$

Finally $1 - 2q + r = \mathbb{E}_z[1 - 2\tanh^2 \Xi + \tanh^4 \Xi] = \mathbb{E}_z[(1 - \tanh^2 \Xi)^2] = \mathbb{E}_z[\operatorname{sech}^4 \Xi]$, which gives Eq. (3.65). The dimension count: the space of symmetric η with zero diagonal has dimension $n(n-1)/2$, and Eq. (3.64) imposes n linearly independent conditions when $n \geq 4$, leaving $n(n-1)/2 - n = n(n-3)/2$. □

The other eigenvectors of M (the *longitudinal* mode $\eta_{ab} = \text{const}$ and the $n - 1$ *anomalous* modes $\eta_{ab} = \xi_a + \xi_b$ with $\sum_a \xi_a = 0$) have eigenvalues that do not change sign at the AT line [23, 77]; we do not need them. The instability comes from the replicon. Up to the positive factor $\beta^2 J^2$, the relevant quantity is

$$\boxed{\lambda_{\mathrm{AT}} := 1 - \beta^2 J^2 \, \mathbb{E}_z \big[\operatorname{sech}^4 (\beta (J\sqrt{q}\, z + h)) \big]} \tag{3.68}$$

and the RS solution is stable if and only if $\lambda_{\mathrm{AT}} > 0$. Following Remark 3.14, the sign convention is fixed by physics, not by the formal Hessian: λ_{AT} is the denominator of the spin-glass susceptibility Eq. (3.29), and $\chi_{\mathrm{SG}} > 0$ requires $\lambda_{\mathrm{AT}} > 0$.

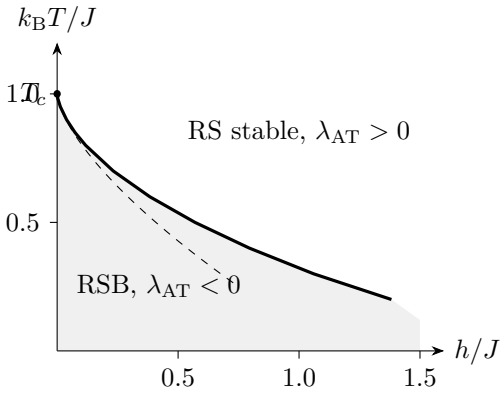


Figure 3.2: The de Almeida–Thouless line of the SK model in the (h, T) plane, obtained by solving $q = \mathbb{E}_z \tanh^2 \Xi$ and $\lambda_{\mathrm{AT}} = 0$ numerically (solid). The dashed curve is the asymptotic form Eq. (3.70), accurate near T_c. Below the line (shaded) the replica-symmetric saddle point is unstable and replica symmetry is broken.

Remark 3.19 (Reading the AT Condition)**.** In the RS solution the local magnetizations are $m_i = \tanh \Xi(z_i)$ with independent standard Gaussians z_i (Section 3.3). Since $\operatorname{sech}^2 \Xi = 1 - \tanh^2 \Xi = 1 - m^2$,

$$\mathbb{E}_z\big[\operatorname{sech}^4 \Xi\big] = \mathbb{E}_z\big[(1-m^2)^2\big] = \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} (1 - m_i^2)^2. \tag{3.69}$$

So Eq. (3.68) is the same combination as Plefka's convergence condition Eq. (3.26) and the ladder denominator Eq. (3.29). The AT condition is sometimes abbreviated $1 - \beta^2 J^2 \langle (1-q)^2 \rangle > 0$. The bracket there must be read as the site average of $(1 - m_i^2)^2$, not as $(1-q)^2$ with $q = \mathbb{E}_z[m^2]$. The two are different: by Jensen's inequality $\mathbb{E}_z[(1-m^2)^2] \geq (1 - \mathbb{E}_z[m^2])^2 = (1-q)^2$, with equality only if all $|m_i|$ are equal. Replacing the site average by $(1-q)^2$ would overestimate the stability of the RS solution. At $h = 0$ and $T \geq T_c$ we have $q = 0$, all $m_i = 0$, and both readings give the familiar $\lambda_{\mathrm{AT}} = 1 - \beta^2 J^2$, which is positive for $T > T_c$ and vanishes at $k_{\mathrm{B}} T_c = J$.

The explicit evaluation of Eq. (3.68) is carried out in Worked computation 3.33. The results are:

(a) At $h = 0$ below T_c, $\lambda_{\mathrm{AT}} = -\frac{4}{3}\tau^2 + \mathcal{O}(\tau^3)$ with $\tau = 1 - k_{\mathrm{B}}T/J$. The RS solution is unstable throughout the zero-field spin-glass phase.

(b) In a field, the RS solution is stable above the *AT line* and unstable below it. Near T_c the line is

$$\frac{h_{\mathrm{AT}}^2}{J^2} = \frac{4}{3}\Big(1 - \frac{k_{\mathrm{B}}T}{J}\Big)^3 + \mathcal{O}(\tau^4). \tag{3.70}$$

(c) At low temperature h_{AT} grows without bound (numerically, $h_{\mathrm{AT}} \approx 0.57J$ at $k_{\mathrm{B}}T = 0.5J$ and $\approx 1.38J$ at $k_{\mathrm{B}}T = 0.2J$). Every field is eventually inside the unstable region as $T \to 0$.

3.4.2 The Random Energy Model: The Simplest Broken Replica Symmetry

Before describing Parisi's general solution, we solve a model in which replica symmetry breaking can be seen without replicas. Derrida's *Random Energy Model* (REM) [26] discards all structure from the spin glass and keeps only the statistics of the energy levels.

Definition 3.20 (Random Energy Model)**.** The REM has 2^N configurations $\mathbf{s} \in \{-1,+1\}^N$ with energies $E_{\mathbf{s}}$ that are independent Gaussians, $E_{\mathbf{s}} \sim \mathcal{N}(0, NJ^2/2)$. The partition function is $Z = \sum_{\mathbf{s}} e^{-\beta E_{\mathbf{s}}}$.

The variance $NJ^2/2$ matches the SK energy variance: in SK, $\overline{E_{\mathbf{J}}(\mathbf{s})^2} = \sum_{i<j} J^2/N = J^2(N-1)/2$ at $h = 0$. The REM is also the $p \to \infty$ limit of the p-spin glass, whose Hamiltonian

$$E_{\mathbf{J}}(\mathbf{s}) = -\sum_{i_1<\cdots<i_p} J_{i_1\cdots i_p} s_{i_1}\cdots s_{i_p}$$

has independent Gaussian couplings and in which the energies of two configurations with overlap Q have correlation $\propto Q^p \to 0$ for $|Q| < 1$ [76].

Proposition 3.21 (Free Energy of the REM)**.** *Let $\beta_c := 2\sqrt{\ln 2}/J$. With probability tending to one as $N \to \infty$,*

$$-\beta f = \lim_{N\to\infty} \frac{1}{N} \ln Z = \begin{cases} \ln 2 + \dfrac{\beta^2 J^2}{4}, & \beta \le \beta_c, \\ \beta J \sqrt{\ln 2}, & \beta \ge \beta_c. \end{cases} \tag{3.71}$$

The entropy per spin $s(\beta) = \ln 2 - \beta^2 J^2/4$ decreases to zero at β_c and stays zero for $\beta > \beta_c$, where the free energy is frozen at the ground-state value $f = -J\sqrt{\ln 2}$.

Derivation. Count levels in a window. Let $\mathcal{N}(\varepsilon)$ be the number of configurations with $E_{\mathbf{s}} \in [N\varepsilon, N(\varepsilon + \mathrm{d}\varepsilon)]$. Its mean is 2^N times the Gaussian probability of the window, which to exponential accuracy is

$$\overline{\mathcal{N}(\varepsilon)} \asymp 2^N \exp\Big(-\frac{(N\varepsilon)^2}{2 \cdot NJ^2/2}\Big) = \exp\Big(N\Big[\ln 2 - \frac{\varepsilon^2}{J^2}\Big]\Big). \tag{3.72}$$

For $|\varepsilon| < \varepsilon_0 := J\sqrt{\ln 2}$ the exponent is positive and $\mathcal{N}(\varepsilon)$ is a sum of many independent indicators, so it concentrates on its mean: its variance is at most its mean, and Chebyshev's inequality gives $\mathcal{N}/\overline{\mathcal{N}} \to 1$ in probability. For $|\varepsilon| > \varepsilon_0$ the mean is exponentially small, and by Markov's inequality $\mathbb{P}(\mathcal{N}(\varepsilon) \ge 1) \le \overline{\mathcal{N}(\varepsilon)} \to 0$: there are no levels there. Hence the microcanonical entropy is $\mathfrak{s}(\varepsilon) = \ln 2 - \varepsilon^2/J^2$ on $[-\varepsilon_0, \varepsilon_0]$ and $-\infty$ outside, and the ground-state energy per spin is $-\varepsilon_0$. Laplace's method then gives

$$\frac{1}{N} \ln Z \to \max_{|\varepsilon| \le \varepsilon_0} \big[\mathfrak{s}(\varepsilon) - \beta\varepsilon\big]. \tag{3.73}$$

The unconstrained maximizer solves $\mathfrak{s}'(\varepsilon) = \beta$, i.e. $\varepsilon^\star = -\beta J^2/2$, giving $\ln 2 + \beta^2 J^2/4$. It lies inside the allowed interval as long as $\beta J^2/2 \le J\sqrt{\ln 2}$, i.e. $\beta \le \beta_c$. For $\beta > \beta_c$ the maximum sits at the edge $\varepsilon = -\varepsilon_0$, where $\mathfrak{s} = 0$, giving $\beta\varepsilon_0 = \beta J\sqrt{\ln 2}$. The entropy is $\mathfrak{s}(\varepsilon^\star) = \ln 2 - \beta^2 J^2/4$ above the transition and 0 below. □

Above T_c the REM free energy equals the annealed one: the partition function is dominated by exponentially many levels near $\varepsilon^\star$, and $\ln Z$ is self-averaging around $\ln \overline{Z}$. Below T_c the annealed calculation would put the dominant energy at $\varepsilon^\star < -\varepsilon_0$, where in a typical sample there are no levels at all. The annealed average is then dominated by rare samples, and the system freezes into the few lowest levels.

The frozen phase shows replica symmetry breaking in its simplest form. Draw two configurations independently from the Gibbs measure. Either they coincide (overlap 1) or they are different and, in the REM, uncorrelated (overlap 0). The probability of coinciding is $\overline{\sum_{\mathbf{s}} p_{\mathbf{s}}^2}$, which for $T < T_c$ equals $1 - T/T_c$ [76]. So

$$P(q) = \frac{T}{T_c}\,\delta(q) + \Big(1 - \frac{T}{T_c}\Big)\delta(q-1), \qquad T < T_c. \tag{3.74}$$

Two values of the overlap, with a temperature-dependent weight: this is the *one-step* replica symmetry breaking (1RSB) structure. In replica language the $n \times n$ matrix Q_{ab} is divided into n/x diagonal blocks of size x, with $Q_{ab} = 1$ inside blocks and 0 between blocks, and the saddle point gives $x = T/T_c$. For integer n the block size x is an integer between 1 and n. After continuation to $n \to 0$ it becomes a number in $(0, 1)$ and plays the role of a probability: it is the weight of the $\delta(q)$ peak in Eq. (3.74).

3.4.3 Parisi's Hierarchical Ansatz

Parisi's solution of the SK model [83] iterates the block construction.

Definition 3.22 (k-Step Parisi Matrix)**.** Choose overlap values $0 \le q_0 < q_1 < \cdots < q_k \le 1$ and integers $n = x_0 \ge x_1 \ge \cdots \ge x_k \ge x_{k+1} = 1$, each dividing the previous one. Partition the n replicas into groups of size x_1, each of those into groups of size x_2, and so on. For $a \neq b$, set $Q_{ab} = q_\ell$, where ℓ is the largest level at which a and b are still in the same group of size x_ℓ (with $\ell = 0$ if they are only in the same group of size $x_0 = n$).

For $k = 0$ this is the RS ansatz; for $k = 1$ it is the REM structure. The trace $\mathrm{Tr}_{\boldsymbol{\sigma}}\, e^{\mathcal{L}_Q}$ for this matrix can be computed by nested Gaussian integrations, one Hubbard–Stratonovich variable per level, generalizing Step 3 of Theorem 3.16. In the limit $n \to 0$ the inequalities between the group sizes reverse, $0 \le x_1 \le x_2 \le \cdots \le x_k \le 1$, and the data $\{q_\ell, x_\ell\}$ define a nondecreasing step function $q(x)$ on $[0, 1]$, or equivalently its inverse, $x(q)$. Parisi's solution of the SK model takes $k \to \infty$, which gives a continuous function. Its physical meaning is that $x(q)$ is the cumulative distribution of the overlap:

$$x(q) = \mathbb{P}(|Q_{12}| \le q) = \int_{-q}^{q} P(q')\, \mathrm{d}q', \qquad 0 \le q \le 1. \tag{3.75}$$

Theorem 3.23 (The Parisi Formula; Guerra 2003, Talagrand 2006)**.** *For the SK model Eq. (3.30) at inverse temperature β and field h,*

$$\lim_{N\to\infty} \frac{1}{N} \overline{\ln Z_{\mathbf{J}}} = \inf_{x}\, \mathcal{P}[x], \qquad \mathcal{P}[x] := \ln 2 + \Phi_x(0, \beta h) - \frac{\beta^2 J^2}{2} \int_0^1 x(q)\, q\, \mathrm{d}q, \tag{3.76}$$

where the infimum is over nondecreasing right-continuous $x : [0, 1] \to [0, 1]$, and $\Phi_x(q, y)$ solves the backward parabolic Parisi equation

$$\frac{\partial \Phi_x}{\partial q} = -\frac{\beta^2 J^2}{2} \left[\frac{\partial^2 \Phi_x}{\partial y^2} + x(q) \Big(\frac{\partial \Phi_x}{\partial y} \Big)^2 \right], \qquad \Phi_x(1, y) = \ln \cosh y. \tag{3.77}$$

We do not prove this theorem. Guerra proved, by an interpolation argument, that $\mathcal{P}[x]$ is an upper bound for every x [44]. Talagrand proved the matching lower bound [106]. Both proofs are beyond our scope, and we use the result as an assumption. We do check that it contains the RS answer. Take the one-step function $x(q) = 0$ for $q < q_0$ and $x(q) = 1$ for $q \ge q_0$, i.e. $P(q) = \delta(q - q_0)$.

- On $[q_0, 1]$, $x = 1$, and the substitution $\Phi = \ln \Psi$ turns Eq. (3.77) into the backward heat equation $\partial_q \Psi = -\frac{\beta^2 J^2}{2} \partial_y^2 \Psi$, since $\partial_y^2 \ln \Psi + (\partial_y \ln \Psi)^2 = \Psi^{-1} \partial_y^2 \Psi$. With $\Psi(1, y) = \cosh y$, the solution is $\Psi(q, y) = \mathbb{E}_z \cosh(y + \beta J \sqrt{1 - q}\, z) = \cosh(y)\, e^{\beta^2 J^2 (1-q)/2}$. So $\Phi(q_0, y) = \ln \cosh y + \frac{\beta^2 J^2}{2}(1 - q_0)$.
- On $[0, q_0)$, $x = 0$, and Eq. (3.77) is itself a backward heat equation, so $\Phi(0, y) = \mathbb{E}_z \Phi(q_0, y + \beta J \sqrt{q_0}\, z) = \mathbb{E}_z \ln \cosh(y + \beta J \sqrt{q_0} z) + \frac{\beta^2 J^2}{2}(1 - q_0)$.

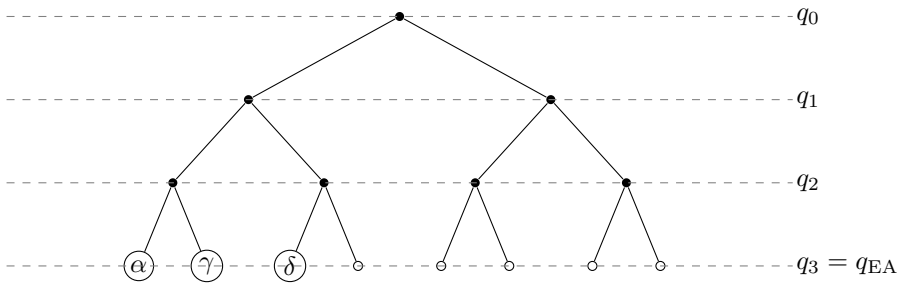


Figure 3.3: Ultrametric organization of pure states. Leaves are pure states. The overlap of two states equals the level of their lowest common ancestor: $q_{\alpha\gamma} = q_2$, while $q_{\alpha\delta} = q_{\gamma\delta} = q_1 < q_2$. Every triangle of states is isosceles, with its two largest distances equal.

- The penalty term is $\frac{\beta^2 J^2}{2} \int_{q_0}^{1} q \, \mathrm{d}q = \frac{\beta^2 J^2}{4}(1 - q_0^2)$.

Adding up, $\mathcal{P} = \mathbb{E}_z \ln 2\cosh \Xi + \frac{\beta^2 J^2}{4}\left[2(1-q_0) - (1-q_0^2)\right] = \mathbb{E}_z \ln 2\cosh \Xi + \frac{\beta^2 J^2}{4}(1-q_0)^2$, which is exactly Eq. (3.47). Above the AT line the infimum is attained at this step function. Below it, the optimal $x(q)$ has a continuous increasing part between a minimum overlap $q_{\min}$ and the Edwards–Anderson value $q_{\max} = q_{\mathrm{EA}}$. The full RSB ground-state energy is $u_0 \approx -0.7633J$, compared with the RS value $-\sqrt{2/\pi}\, J \approx -0.798J$, and the full RSB entropy is nonnegative at all temperatures [77].

3.4.4 Pure States and Ultrametricity

What does a continuous $P(q)$ mean physically? Below the AT line the Gibbs measure of a large SK sample decomposes, to a good approximation, into many *pure states* (or *valleys*) $p_{\mathbf{J}} \approx \sum_{\alpha} w_\alpha p_\alpha$. Within a pure state, connected correlations decay and the TAP equations hold with magnetizations $\mathbf{m}^\alpha$. Different pure states are separated by free-energy barriers that diverge with N, so a local dynamics started in one state stays there for a time that diverges with N. The weights w_α fluctuate from sample to sample even as $N \to \infty$, which is why $P_{\mathbf{J}}(q)$ is not self-averaging even though f is.

The overlaps between pure states are organized in a very particular way. Define the distance between two states by $d_{\alpha\gamma} = q_{\mathrm{EA}} - q_{\alpha\gamma}$.

Definition 3.24 (Ultrametricity)**.** A set of states is *ultrametric* if for every triple α, γ, δ,

$$d_{\alpha\gamma} \le \max(d_{\alpha\delta}, d_{\delta\gamma}) \quad \Longleftrightarrow \quad q_{\alpha\gamma} \ge \min(q_{\alpha\delta}, q_{\delta\gamma}). \tag{3.78}$$

The ultrametric inequality is stronger than the triangle inequality $d_{\alpha\gamma} \le d_{\alpha\delta} + d_{\delta\gamma}$. It implies that in every triangle the two largest sides are equal, so every triangle is isosceles with a short base. An equivalent description is that the states are the leaves of a rooted tree. The overlap of two leaves depends only on the depth of their lowest common ancestor, as in Fig. 3.3. This is the same hierarchy as the Parisi matrix: replicas in the same innermost group are in the same state, groups at the next level are in states with a common ancestor at overlap q_{k-1}, and so on. Ultrametricity of the SK Gibbs measure was derived from the Parisi solution by Mézard, Parisi, Sourlas, Toulouse, and Virasoro [77]. It was proved rigorously by Panchenko for a class of generic mixed p-spin models [82].

3.4.5 Free-Energy Barriers Between Pure States

For sampling, the most important property of the pure states is the height of the barriers between them. Let ΔF_N denote the typical free-energy barrier that a local dynamics must cross to move

between two pure states with a small mutual overlap in an SK sample of size N. In the mean-field limit the states are separated by barriers that diverge with N; this is what makes them pure states. The rate of divergence is not known exactly. The scaling hypothesis is

$$\Delta F_N \sim c\, N^{\psi}\, k_{\mathrm{B}}T, \qquad 0 < \psi \leq 1, \tag{3.79}$$

so that the time to cross scales as $\ln \tau \sim cN^{\psi}$. Numerical simulations of SK relaxation, starting with Mackenzie and Young [70], find relaxation times consistent with Eq. (3.79) with a fractional exponent, and estimates in the range $\psi \approx 1/4$ to $1/3$ are commonly quoted. There is no derivation of the value of ψ for SK, and the precise exponent depends on how a barrier is defined (the minimax energy along single-spin-flip paths, the free energy of the transition region, or the measured relaxation time). We treat Eq. (3.79) as a conjecture supported by numerics.

In finite-dimensional short-range spin glasses, the droplet theory of Fisher and Huse [34] predicts that a compact excitation of linear size L costs free energy $\sim L^{\theta}$, with a stiffness exponent θ, and must cross a barrier $\sim L^{\psi}$ with $\theta \leq \psi \leq d-1$. Again ψ is not known analytically.

What *is* rigorous is the consequence of a barrier, once its size is given. If any reversible local sampler must cross a set of configurations whose equilibrium probability is $e^{-\beta\Delta F}$ smaller than that of the states it separates, then its spectral gap is at most of order $e^{-\beta\Delta F}$. Proving this, with explicit constants and hypotheses, is the task of the next section. Combined with Eq. (3.79), it gives a relaxation time $\exp(\mathcal{O}(N^{\psi}))$, which grows faster than any polynomial in N.

3.5 The Fundamental Expressivity–Mixing Dilemma

The previous sections described equilibrium: frustrated couplings produce many valleys separated by high free-energy barriers. We now prove the dynamical consequence. The result is general. It does not depend on replicas, on the SK model, or on any approximation. It applies to every reversible Markov chain and every reversible Langevin diffusion, and its only input is the equilibrium probability of a separating region. This is the precise form of the statement, used throughout the book, that a monolithic Energy-Based Model cannot be both expressive and fast to sample.

3.5.1 Dirichlet Forms of Reversible Markov Chains

We use the column-stochastic convention of Section 2.3: $P(y \mid x)$ is the probability of moving from x to y in one step, and distributions evolve as $\rho \mapsto \mathbf{P}\rho$. Observables evolve by the adjoint action $(\mathbf{P}^{\top}f)(x) := \sum_y P(y \mid x) f(y)$, the expected value of f after one step from x. Let π be the stationary distribution, and write $\langle f,\, g\rangle_{\pi} = \sum_x \pi(x) f(x) g(x)$. The next two lemmas restate Proposition 2.16 (where the Dirichlet form is written $\mathcal{D}$) in the form used here; the first no longer assumes reversibility.

Lemma 3.25 (Dirichlet Form of a Markov Chain)**.** *If π is stationary for $\mathbf{P}$, then for every $f : \Omega \to \mathbb{R}$,*

$$\Big\langle f,\, (\mathbf{I} - \mathbf{P}^{\top})f \Big\rangle_{\pi} = \mathcal{E}_{\mathbf{P}}(f,f) := \frac{1}{2} \sum_{x,y \in \Omega} \pi(x) P(y \mid x) \big(f(x) - f(y)\big)^2. \tag{3.80}$$

Proof. Expand the square: $\mathcal{E}_{\mathbf{P}}(f,f) = \frac{1}{2}\sum_{x,y} \pi(x)P(y \mid x)\big[f(x)^2 - 2f(x)f(y) + f(y)^2\big]$. In the first term, $\sum_y P(y \mid x) = 1$ leaves $\frac{1}{2}\sum_x \pi(x) f(x)^2$. In the third term, stationarity $\sum_x \pi(x)P(y \mid x) = \pi(y)$ leaves $\frac{1}{2}\sum_y \pi(y) f(y)^2$. The middle term is $-\sum_x \pi(x) f(x) \sum_y P(y \mid x) f(y) = -\Big\langle f,\, \mathbf{P}^{\top} f \Big\rangle_{\pi}$. Adding, $\mathcal{E}_{\mathbf{P}}(f,f) = \langle f,\, f\rangle_{\pi} - \Big\langle f,\, \mathbf{P}^{\top} f \Big\rangle_{\pi}$. □

Lemma 3.26 (Variational Characterization of the Gap)**.** *Let* $\mathbf{P}$ *be reversible with respect to* π*, with eigenvalues* $1 = \lambda_1 \geq \lambda_2 \geq \cdots$ *as in Theorem 2.11. Then the* relaxation gap $\gamma := 1 - \lambda_2$ *satisfies*

$$\gamma = \min_{f \text{ non-constant}} \frac{\mathcal{E}_{\mathbf{P}}(f, f)}{\operatorname{Var}_\pi(f)}. \tag{3.81}$$

In particular, $\gamma \leq \mathcal{E}_{\mathbf{P}}(f, f)/\operatorname{Var}_\pi(f)$ *for every non-constant test function* f*. Since the absolute gap of Eq. (2.42) obeys* $\gamma_{\text{spec}} \leq \gamma$*, the same upper bound holds for* γ_{spec}*.*

Proof. By Theorem 2.11, the left eigenvectors ϕ_k are π-orthonormal, $\phi_1 = 1$, and $\mathbf{P}^\top \phi_k = \lambda_k \phi_k$. Expand $f = \sum_k c_k \phi_k$, with $c_1 = \mathbb{E}_\pi f$. Then $\operatorname{Var}_\pi(f) = \sum_{k \geq 2} c_k^2$ and, by Lemma 3.25, $\mathcal{E}_{\mathbf{P}}(f, f) = \sum_k (1 - \lambda_k) c_k^2 = \sum_{k \geq 2} (1 - \lambda_k) c_k^2 \geq (1 - \lambda_2) \sum_{k \geq 2} c_k^2$. Equality holds for $f = \phi_2$. □

This is the discrete analog of the Poincaré inequality Eq. (6.82): the gap is the best constant C_{P} in $\operatorname{Var}_\pi(f) \leq C_{\mathrm{P}}^{-1} \mathcal{E}(f, f)$. Upper bounds on a gap are therefore easy in principle: any test function gives one. The art lies in choosing a test function with small Dirichlet form and large variance. A function that is constant on each of two large regions and changes only in between is the natural choice.

3.5.2 The Bottleneck Theorem for Reversible Chains

Definition 3.27 (Equilibrium Flux and Bottleneck Ratio)**.** For $A \subset \Omega$ with $0 < \pi(A) < 1$, the *equilibrium flux* out of A and the *bottleneck ratio* (conductance) of A are

$$Q(A, A^c) := \sum_{x \in A} \sum_{y \notin A} \pi(x) P(y \mid x), \qquad \Phi(A) := \frac{Q(A, A^c)}{\pi(A)}. \tag{3.82}$$

$\Phi(A)$ is the probability that a chain in equilibrium, conditioned to be in A now, leaves A on the next step. It is the discrete version of Kramers' flux-over-population rate (Section 7.3).

Theorem 3.28 (Bottleneck Bound on Gap and Mixing Time)**.** *Let* $\mathbf{P}$ *be a Markov chain on a finite set* Ω *with stationary distribution* $\pi > 0$*, and let* $A \subset \Omega$ *with* $0 < \pi(A) < 1$*.*

(i) If $\mathbf{P}$ *is reversible, then*

$$\gamma = 1 - \lambda_2 \leq \frac{Q(A, A^c)}{\pi(A)\pi(A^c)} = \frac{\Phi(A)}{\pi(A^c)}. \tag{3.83}$$

(ii) Without assuming reversibility, let $\pi_A(x) = \pi(x)\mathbf{1}_A(x)/\pi(A)$ *be* π *conditioned on* A*. Then for all* $t \geq 0$*,*

$$\left\| \mathbf{P}^t \pi_A - \pi \right\|_{\mathrm{TV}} \geq \pi(A^c) - t\,\Phi(A), \tag{3.84}$$

and hence the mixing time of Eq. (2.44) obeys $\tau_{\text{mix}}(\varepsilon) \geq (\pi(A^c) - \varepsilon)/\Phi(A)$*.*

(iii) If $\pi(A) \leq \frac{1}{2}$*, then* $\gamma \leq 2\Phi(A)$ *(under reversibility) and* $\tau_{\text{mix}}(1/4) \geq 1/(4\Phi(A))$*.*

Proof. (i) Use the test function $f = \mathbf{1}_A$ in Lemma 3.26. Its variance is $\pi(A) - \pi(A)^2 = \pi(A)\pi(A^c)$. In the Dirichlet form, $(f(x) - f(y))^2 = 1$ exactly when one of x, y is in A and the other is not, so

$$\mathcal{E}_{\mathbf{P}}(\mathbf{1}_A, \mathbf{1}_A) = \frac{1}{2}\big[Q(A, A^c) + Q(A^c, A)\big]. \tag{3.85}$$

Stationarity gives $Q(A^c, A) = Q(A, A^c)$: indeed $\sum_{y\in A}\sum_{x\in\Omega}\pi(x)P(y \mid x) = \pi(A)$, and splitting the x sum into $x \in A$ and $x \in A^c$,

$$Q(A^c, A) = \pi(A) - \sum_{x\in A}\sum_{y\in A}\pi(x)P(y \mid x) = \pi(A) - \big[\pi(A) - Q(A, A^c)\big] = Q(A, A^c), \tag{3.86}$$

where we used $\sum_{y\in A} P(y \mid x) = 1 - \sum_{y\notin A} P(y \mid x)$ for $x \in A$. So $\mathcal{E}_{\mathbf{P}}(\mathbf{1}_A, \mathbf{1}_A) = Q(A, A^c)$, and Eq. (3.81) gives Eq. (3.83).

(ii) First compute one step from π_A. For $y \in A$,

$$(\mathbf{P}\pi_A)(y) = \frac{1}{\pi(A)}\sum_{x\in A} P(y \mid x)\pi(x) \le \frac{1}{\pi(A)}\sum_{x\in\Omega} P(y \mid x)\pi(x) = \frac{\pi(y)}{\pi(A)} = \pi_A(y). \tag{3.87}$$

For $y \notin A$, $\pi_A(y) = 0 \le (\mathbf{P}\pi_A)(y)$. For two probability distributions μ, ν the total variation distance can be written as

$$\|\mu - \nu\|_{\mathrm{TV}} = \sum_{y:\,\mu(y)>\nu(y)} (\mu(y) - \nu(y)), \tag{3.88}$$

because $\sum_y(\mu(y) - \nu(y)) = 0$, so the positive and negative parts of $\mu - \nu$ have equal mass. Hence

$$\|\mathbf{P}\pi_A - \pi_A\|_{\mathrm{TV}} = \sum_{y\notin A}(\mathbf{P}\pi_A)(y) = \frac{1}{\pi(A)}\sum_{y\notin A}\sum_{x\in A} P(y \mid x)\pi(x) = \Phi(A). \tag{3.89}$$

Next, $\mathbf{P}$ does not increase the ℓ^1 norm of a signed vector v: $\sum_y |\sum_x P(y \mid x)v(x)| \le \sum_x |v(x)| \sum_y P(y \mid x) = \sum_x |v(x)|$. Telescoping,

$$\left\|\mathbf{P}^t\pi_A - \pi_A\right\|_{\mathrm{TV}} \le \sum_{s=0}^{t-1}\|\mathbf{P}^s(\mathbf{P}\pi_A - \pi_A)\|_{\mathrm{TV}} \le t\,\|\mathbf{P}\pi_A - \pi_A\|_{\mathrm{TV}} = t\,\Phi(A). \tag{3.90}$$

Finally, $\|\pi_A - \pi\|_{\mathrm{TV}} = \sum_{y:\pi(y)>\pi_A(y)}(\pi(y) - \pi_A(y)) = \sum_{y\notin A}\pi(y) = \pi(A^c)$, since $\pi_A \ge \pi$ on A. The triangle inequality gives $\left\|\mathbf{P}^t\pi_A - \pi\right\|_{\mathrm{TV}} \ge \|\pi_A - \pi\|_{\mathrm{TV}} - \left\|\mathbf{P}^t\pi_A - \pi_A\right\|_{\mathrm{TV}} \ge \pi(A^c) - t\Phi(A)$. The worst-case distance $\sup_{\rho^{(0)}}\left\|\mathbf{P}^t\rho^{(0)} - \pi\right\|_{\mathrm{TV}}$ is at least the value for $\rho^{(0)} = \pi_A$, so it exceeds ε for all $t < (\pi(A^c) - \varepsilon)/\Phi(A)$.

(iii) If $\pi(A) \le \frac{1}{2}$ then $\pi(A^c) \ge \frac{1}{2}$; insert into (i) and into (ii) with $\varepsilon = \frac{1}{4}$. □

Part (ii) has a direct physical reading. A chain started in equilibrium inside A leaks out at rate at most $\Phi(A)$ per step. Until it has had time to leak out, it cannot have reached the global equilibrium, which puts weight $\pi(A^c)$ outside A. The bound does not need $\pi(A)$ to be large. It applies equally to a metastable valley that carries almost no equilibrium weight, which is the situation in the three-spin example of Worked computation 3.33.

Corollary 3.29 (Single-Spin-Flip Samplers: Free-Energy and Energy Barriers)**.** *Let $\Omega = \{-1, +1\}^N$ and let $\mathbf{P}$ be the random-scan heat-bath (Glauber) or Metropolis chain for $\pi \propto e^{-\beta E}$, which picks a site uniformly at random and updates it. For $A \subset \{-1, +1\}^N$, let $\partial_{\mathrm{in}}A$ be the set of $x \in A$ that have a single-flip neighbor outside A, and let $\mathcal{F}(S) := -k_{\mathrm{B}}T \ln \sum_{x\in S} e^{-\beta E(x)}$ be the free energy of a set S. Then*

$$\Phi(A) \le \frac{\pi(\partial_{\mathrm{in}}A)}{\pi(A)} = \exp\Big(-\beta\big[\mathcal{F}(\partial_{\mathrm{in}}A) - \mathcal{F}(A)\big]\Big). \tag{3.91}$$

Moreover, if E_∂ is the lowest value of $\max(E(x), E(y))$ over pairs $x \in A$, $y \notin A$ differing in one spin, $E_A = \min_{x\in A} E(x)$, and $|\partial A|$ is the number of such pairs, then

$$\Phi(A) \le \frac{|\partial A|}{N}\, e^{-\beta(E_\partial - E_A)}. \tag{3.92}$$

Proof. A single step changes at most one spin, so $P(y \mid x) = 0$ for $x \in A$, $y \notin A$ unless $x \in \partial_{\mathrm{in}} A$. Hence $Q(A, A^c) = \sum_{x \in \partial_{\mathrm{in}} A} \pi(x) \sum_{y \notin A} P(y \mid x) \le \pi(\partial_{\mathrm{in}} A)$, which gives the inequality in Eq. (3.91). The equality is the definition of $\mathcal{F}$, since $\pi(S) = e^{-\beta \mathcal{F}(S)}/Z$. For Eq. (3.92), note that for a single-flip pair, the heat-bath rule gives $\pi(x) P(y \mid x) = \frac{1}{N} \frac{\pi(x)\pi(y)}{\pi(x)+\pi(y)} \le \frac{1}{N} \min(\pi(x), \pi(y))$, and Metropolis gives $\pi(x) \frac{1}{N} \min(1, \pi(y)/\pi(x)) = \frac{1}{N} \min(\pi(x), \pi(y))$. Since $\min(\pi(x), \pi(y)) = e^{-\beta \max(E(x), E(y))}/Z \le e^{-\beta E_\partial}/Z$, summing over the $|\partial A|$ crossing pairs gives $Q(A, A^c) \le |\partial A|\, e^{-\beta E_\partial}/(NZ)$. Divide by $\pi(A) \ge e^{-\beta E_A}/Z$. □

The two forms of the corollary answer different questions. The energy form Eq. (3.92) is Arrhenius' law: at fixed N and $T \to 0$, the escape probability per step from A is at most $e^{-\beta \Delta E}$ times a constant, where $\Delta E = E_\partial - E_A$ is the energy barrier (the lowest possible peak along any path out of A, measured from the bottom of A). For large N at fixed T, the prefactor $|\partial A|/N$ can itself be exponential in N, because many paths cross the boundary. The free-energy form Eq. (3.91) accounts for this entropy exactly and is the form relevant to the barrier scaling $\Delta F_N \sim N^{\psi}$ of Section 3.4.

3.5.3 The Bottleneck Bound for Langevin Dynamics

The continuous version uses the Poincaré inequality of Chapter 6. Consider the overdamped Langevin dynamics Eq. (7.1),

$$\mathrm{d}\mathbf{x}_t = -\mu \nabla E(\mathbf{x}_t)\, \mathrm{d}t + \sqrt{2D}\, \mathrm{d}\mathbf{W}_t, \qquad D = \mu k_{\mathrm{B}} T, \tag{3.93}$$

on $\mathbb{R}^N$ with a confining potential E. It is reversible with respect to $p_{\mathrm{eq}} = Z^{-1} e^{-\beta E}$. By Definition 6.12, its spectral gap is

$$\lambda_1 = \inf_{f \in H^1(p_{\mathrm{eq}}),\ \mathrm{Var}(f) > 0} \frac{D \int p_{\mathrm{eq}} \, \|\nabla f\|^2 \, \mathrm{d}^N x}{\mathrm{Var}_{p_{\mathrm{eq}}}(f)}. \tag{3.94}$$

Theorem 3.30 (Langevin Bottleneck Bound)**.** *Let $A, B \subset \mathbb{R}^N$ be closed sets with $\mathrm{dist}(A, B) \ge \varepsilon > 0$ and $p_{\mathrm{eq}}(A), p_{\mathrm{eq}}(B) \ge a > 0$. Let $S := \{\mathbf{x} : 0 < \mathrm{dist}(\mathbf{x}, A) < \varepsilon\}$ be the open shell of width ε around A. Then*

$$\lambda_1 \le \frac{2D}{a\, \varepsilon^2}\, p_{\mathrm{eq}}(S), \qquad \textit{equivalently} \qquad \lambda_1^{-1} \ge \frac{a\, \varepsilon^2}{2D}\, e^{\beta[\mathcal{F}(S) - F]}, \tag{3.95}$$

where $\mathcal{F}(S) := -k_{\mathrm{B}} T \ln \int_S e^{-\beta E}\, \mathrm{d}^N x$ and $F = -k_{\mathrm{B}} T \ln Z$. With $a = \frac{1}{4}$, $\lambda_1 \le 8D\, p_{\mathrm{eq}}(S)/\varepsilon^2$.

Proof. Define the test function $g(\mathbf{x}) := \min(1, \mathrm{dist}(\mathbf{x}, A)/\varepsilon)$. The distance function is 1-Lipschitz, so g is Lipschitz with constant $1/\varepsilon$, takes values in $[0, 1]$, and belongs to $H^1(p_{\mathrm{eq}})$. By Rademacher's theorem g is differentiable almost everywhere with $\|\nabla g\| \le 1/\varepsilon$. Moreover $\nabla g = 0$ almost everywhere on the level sets $\{g = 0\} = A$ and $\{g = 1\}$, as for any Lipschitz function. Hence ∇g is supported in S up to a null set, and

$$D \int p_{\mathrm{eq}} \, \|\nabla g\|^2 \, \mathrm{d}^N x \le \frac{D}{\varepsilon^2} \int_S p_{\mathrm{eq}} \, \mathrm{d}^N x = \frac{D}{\varepsilon^2} p_{\mathrm{eq}}(S). \tag{3.96}$$

For the variance, $g = 0$ on A and $g = 1$ on B (since $\mathrm{dist}(\mathbf{x}, A) \ge \varepsilon$ for $\mathbf{x} \in B$). Thus

$$\mathrm{Var}_{p_{\mathrm{eq}}}(g) = \min_{c \in \mathbb{R}} \int p_{\mathrm{eq}} (g - c)^2 \ge \min_c \big[p_{\mathrm{eq}}(A) c^2 + p_{\mathrm{eq}}(B)(1 - c)^2 \big] = \frac{p_{\mathrm{eq}}(A)\, p_{\mathrm{eq}}(B)}{p_{\mathrm{eq}}(A) + p_{\mathrm{eq}}(B)}, \tag{3.97}$$

where the minimum is at $c = p_{\mathrm{eq}}(B)/(p_{\mathrm{eq}}(A) + p_{\mathrm{eq}}(B))$. The right-hand side equals $\big(p_{\mathrm{eq}}(A)^{-1} + p_{\mathrm{eq}}(B)^{-1}\big)^{-1} \ge (2/a)^{-1} = a/2$. Inserting both estimates into Eq. (3.94) gives $\lambda_1 \le 2D p_{\mathrm{eq}}(S)/(a\varepsilon^2)$. The second form uses $p_{\mathrm{eq}}(S) = e^{-\beta[\mathcal{F}(S) - F]}$. □

Corollary 3.31 (Arrhenius Form)**.** *In addition to the hypotheses of Theorem 3.30, suppose S is bounded, $E \geq E_{\min} + \Delta E$ on S where $E_{\min} = E(\mathbf{x}_0)$ is the global minimum, and $\nabla^2 E \preceq K\mathbf{I}$ on the ball $B_r(\mathbf{x}_0)$ of radius $r = (\beta K)^{-1/2}$. Then*

$$\lambda_1 \leq \frac{2\sqrt{e}\, D}{a\, \varepsilon^2} \frac{\mathrm{Vol}(S)}{\mathrm{Vol}(B_r)}\, e^{-\beta \Delta E}. \tag{3.98}$$

Proof. Since $\nabla E(\mathbf{x}_0) = \mathbf{0}$ and $\nabla^2 E \preceq K\mathbf{I}$ on $B_r(\mathbf{x}_0)$, Taylor's theorem gives $E(\mathbf{x}) \leq E_{\min} + \frac{K}{2}\|\mathbf{x} - \mathbf{x}_0\|^2 \leq E_{\min} + \frac{Kr^2}{2} = E_{\min} + \frac{k_{\mathrm{B}}T}{2}$ there. Hence $Z \geq \int_{B_r} e^{-\beta E} \geq \mathrm{Vol}(B_r) e^{-\beta E_{\min}} e^{-1/2}$. On the other hand $\int_S e^{-\beta E} \leq \mathrm{Vol}(S) e^{-\beta(E_{\min} + \Delta E)}$. Dividing, $p_{\mathrm{eq}}(S) \leq \sqrt{e}\, \frac{\mathrm{Vol}(S)}{\mathrm{Vol}(B_r)} e^{-\beta \Delta E}$; insert into Eq. (3.95). □

At fixed dimension the prefactor in Eq. (3.98) grows only polynomially in β (as $\mathrm{Vol}(B_r)^{-1} \propto (\beta K)^{N/2}$), so the gap is exponentially small in $\beta \Delta E$. This exponent cannot be improved. The Holley–Stroock theorem (Theorem 6.20) gives a lower bound of the form $\lambda_1 \geq c\, e^{-\beta\, \mathrm{osc}(\delta E)}$, and for a double well Kramers' formula (Theorem 7.4) shows $\lambda_1 \simeq k_{A\to B} + k_{B\to A}$ with $k_{A\to B} \propto e^{-\beta \Delta E_A}$ (Section 7.5). The bottleneck bound thus gives the correct Arrhenius exponent. It is not sharp in the prefactor, and Chapter 7 supplies the exact prefactor.

3.5.4 Statement of the Dilemma

We can now state the trade-off precisely. Fix a target distribution p_{data} on $\mathbb{R}^N$ that has two well-separated modes, and ask what happens to any Energy-Based Model that represents it accurately.

Theorem 3.32 (The Fundamental Expressivity–Mixing Dilemma)**.** *Let $A, B \subset \mathbb{R}^N$ be closed with $\mathrm{dist}(A, B) \geq \varepsilon > 0$, let S be the shell of Theorem 3.30, and suppose the target satisfies*

$$p_{\mathrm{data}}(A),\ p_{\mathrm{data}}(B) \geq a, \qquad p_{\mathrm{data}}(S) \leq \eta. \tag{3.99}$$

Let $p_\theta = Z_\theta^{-1} e^{-\beta E_\theta}$ be any EBM with $\|p_\theta - p_{\mathrm{data}}\|_{\mathrm{TV}} \leq \delta < a$, sampled by the Langevin dynamics Eq. (3.93) with energy E_θ and diffusion constant D. Then:

(i) (Barrier.) *The shell carries a free-energy barrier*

$$\mathcal{F}_\theta(S) - F_\theta \geq k_{\mathrm{B}}T \ln \frac{1}{\eta + \delta}, \tag{3.100}$$

and the uniform average $\bar{E}_S := \mathrm{Vol}(S)^{-1} \int_S E_\theta \, \mathrm{d}^N x$ of the energy over the shell (if S is bounded) satisfies $\bar{E}_S \geq F_\theta + k_{\mathrm{B}}T \ln\big(\mathrm{Vol}(S)/(\eta + \delta)\big)$.

(ii) (Slow mixing.) *The spectral gap obeys*

$$\lambda_1(\theta) \leq \frac{2D}{(a - \delta)\, \varepsilon^2}\, (\eta + \delta), \qquad \lambda_1(\theta)^{-1} \geq \frac{(a - \delta)\varepsilon^2}{2D}\, e^{\beta[\mathcal{F}_\theta(S) - F_\theta]}. \tag{3.101}$$

Proof. For any event U, $|p_\theta(U) - p_{\mathrm{data}}(U)| \leq \|p_\theta - p_{\mathrm{data}}\|_{\mathrm{TV}} \leq \delta$. Hence $p_\theta(A), p_\theta(B) \geq a - \delta > 0$ and $p_\theta(S) \leq \eta + \delta$.

(i) By definition $p_\theta(S) = e^{-\beta[\mathcal{F}_\theta(S) - F_\theta]}$, so $p_\theta(S) \leq \eta + \delta$ is Eq. (3.100). For the energy, Jensen's inequality for the convex function $e^{-\beta(\cdot)}$ and the uniform measure on S gives $e^{-\beta \bar{E}_S} \leq \mathrm{Vol}(S)^{-1} \int_S e^{-\beta E_\theta} = \mathrm{Vol}(S)^{-1} e^{-\beta \mathcal{F}_\theta(S)}$. Taking logarithms, $\bar{E}_S \geq \mathcal{F}_\theta(S) + k_{\mathrm{B}}T \ln \mathrm{Vol}(S)$, and Eq. (3.100) completes the bound.

(ii) Apply Theorem 3.30 to p_θ with a replaced by $a - \delta$ and $p_{\mathrm{eq}}(S) = p_\theta(S) \leq \eta + \delta$. □

The theorem has two halves of different character, and only one of them carries real content. The *expressivity* half (i) is close to a definition. Saying that the modes are well separated means that little probability lies between them, and for a Boltzmann distribution little probability means high free energy, by the definition of $\mathcal{F}_\theta(S)$. The *mixing* half (ii) is the substantive statement: whatever parametrization or training procedure produced E_θ, a reversible local diffusion on p_θ needs a time at least ε^2/D (the time to diffuse across the shell) multiplied by $e^{\beta\Delta\mathcal{F}}$, with $\Delta\mathcal{F} \geq k_{\mathrm{B}}T\ln(1/(\eta+\delta))$. Sharper modes (smaller η) and more accurate models (smaller δ) make sampling slower in direct proportion. For a spin glass, where $\Delta\mathcal{F}$ is believed to grow as N^ψ, the relaxation time grows as $\exp(cN^\psi)$. The discrete version follows in the same way from Theorem 3.28, which applies to any reversible Gibbs sampler of a p-bit network, including the chromatic block samplers of Section 2.5. For single-site updates Corollary 3.29 evaluates the flux through the inner boundary $\partial_{\mathrm{in}}A$; for block updates the inner boundary must be replaced by the set of states from which one block move can leave A.

Physics Insight: What the Dilemma Does and Does Not Forbid

The theorem constrains one specific procedure: a reversible, local dynamics whose stationary law is p_θ itself. Each way of escaping the dilemma changes one of these hypotheses.

- *Faster hardware* increases D (or decreases the time per Gibbs sweep). This helps by a constant factor and leaves the factor $e^{\beta\Delta\mathcal{F}}$ untouched.
- *Non-reversible dynamics* with the same stationary law (adding a drift $\mathbf{b}$ with $\nabla\cdot(\mathbf{b}\,p_{\mathrm{eq}}) = 0$, as in the circulating currents of Chapter 6) can change prefactors. If $\mathbf{b}\cdot\nabla E = 0$, the Freidlin–Wentzell quasipotential is still E (Section 7.4), so the Arrhenius exponent of the escape rate is unchanged.
- *Changing the temperature* during sampling (simulated annealing, parallel tempering) samples a family of distributions instead of p_θ alone. It can help when barriers shrink at high temperature, but in a spin glass the free-energy landscape reorganizes with temperature (*temperature chaos*), and no general polynomial guarantee is known.
- *Changing the distribution being sampled* at each step is the route taken by Denoising Thermodynamic Models. A DTM never samples p_{data} directly. It samples a chain of conditional distributions $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$, each made close to unimodal by a quadratic coupling to the previous step, so that each conditional has no shell S of small probability between large regions. Section 16.3 proves that, under a bound on the curvature of the learned energy, each step then mixes in $\mathcal{O}(1)$ time.

Hardware Realization: Tuned p-Bit Barriers versus Collective Barriers

A hardware p-bit is a bistable element whose barrier is set deliberately low, a few $k_{\mathrm{B}}T$, so that it flips spontaneously on a time scale of order the node relaxation time $\tau_{RC} \sim 1$–$10\,\mathrm{ns}$ (Chapters 13 and 14). Its single-device escape rate is governed by Kramers' formula with a small, designed $\beta\Delta E$ (Worked computation 7.16). The barriers that make a network slow are of a different kind. They are collective barriers of the Boltzmann distribution of the whole network, $\Delta\mathcal{F}$ in Theorem 3.32, and they are set by the couplings J_{ij}, not by the device physics. As a representative estimate, take a collective barrier $\beta\Delta\mathcal{F} = 40$ and an attempt time of one sweep, $\tau_0 = 10\,\mathrm{ns}$, for a chromatic Gibbs sampler updating every p-bit in parallel

(Section 2.5). Then $\tau \gtrsim \tau_0 e^{40} \approx 2.4 \times 10^9$ s, about 75 years. Parallel hardware reduces τ_0; it does not change $e^{\beta\Delta\mathcal{F}}$. This is the motivation for the denoising architecture of Jelincic et al. [56] (Chapters 16 and 17), which does not try to sample a single rugged EBM and instead composes many easy conditional sampling problems.

3.5.5 Consequences for Learning

The bottleneck also affects training. The maximum-likelihood gradient of an EBM (Section 1.4) is the difference between a data average and a model average. The model average is estimated with a sampler. When the model develops separated modes, persistent chains stay trapped in the modes they started in, and the estimated model average reflects the initial occupation of the modes instead of the equilibrium weights. The gradient then pushes down the energy of modes that the chains do not visit and fails to correct their relative weights. Training becomes biased exactly when the model starts to represent multimodal structure. This is the learning-side face of the same theorem, and it is one reason why monolithic EBMs have been difficult to train on complex data.

3.6 Worked Computation: SK Thermodynamics, the AT Line, and Trapping in p-Bit Networks

Worked computation 3.33 (SK Replica-Symmetric Thermodynamics, the de Almeida–Thouless Instability, TAP Fields, and Exponential Trapping in Frustrated and Three-Spin Landscapes)**.** We carry out five computations. Parts 1–3 evaluate the RS solution of Theorem 3.16 and its stability eigenvalue Eq. (3.68) in every regime where closed forms exist, and tabulate the numerical solution elsewhere. Part 4 shows that the RS equations are the site statistics of the TAP equations. Part 5 turns to small networks of p-bits: it contrasts a frustrated triangle (degenerate but fast) with a three-spin model (unfrustrated but exponentially slow), and computes the trapping time of the latter from Theorem 3.28. Throughout, $\tau = 1 - k_{\mathrm{B}}T/J$ and $K := \beta^2 J^2$.

Part 1: The RS solution at $h = 0$.

Above T_c the only solution is $q = 0$, with

$$-\beta f = \ln 2 + \frac{K}{4}, \qquad u = -\frac{\beta J^2}{2}, \qquad s = \ln 2 - \frac{K}{4}, \qquad \lambda_{\mathrm{AT}} = 1 - K, \tag{3.102}$$

from Eqs. (3.47), (3.49) and (3.68) with $\Xi \equiv 0$. At T_c these give $f = -(\ln 2 + \frac{1}{4})J = -0.9431J$, $u = -J/2$, and $s = \ln 2 - \frac{1}{4} = 0.4431$. Below T_c, Eq. (3.48) must be solved numerically. The solution can be found by bisection on $q \in (0, 1)$ for the function $q \mapsto \mathbb{E}_z \tanh^2(\beta J\sqrt{q}z) - q$, which is positive for small q when $K > 1$ and negative at $q = 1$. The Gaussian averages are one-dimensional integrals. The results are:

$k_{\mathrm{B}}T/J$	q	f/J	u/J	s	λ_{AT}
1.0	0	−0.9431	−0.5000	0.4431	0
0.9	0.1027	−0.9014	−0.5497	0.3908	−0.0123
0.8	0.2089	−0.8651	−0.5977	0.3343	−0.0482
0.6	0.4243	−0.8104	−0.6833	0.2117	−0.2098
0.4	0.6338	−0.7809	−0.7480	0.0824	−0.6070
0.2	0.8280	−0.7772	−0.7861	−0.0445	−1.9004
0.1	0.9171	−0.7847	−0.7951	−0.1039	−4.545

Three features stand out. The order parameter grows linearly below T_c ($q \approx \tau$, Corollary 3.17). The RS entropy crosses zero near $k_{\mathrm{B}}T \approx 0.27J$ and is negative below, which is unphysical. And $\lambda_{\mathrm{AT}} < 0$ at every temperature below T_c.

Part 2: Zero-temperature limit and the negative entropy.

As $\beta \to \infty$ at $h = 0$, $q \to 1$. Set $C := \beta(1-q)$ and $a := \beta J\sqrt{q} \to \infty$. Since $1 - q = \mathbb{E}_z[\operatorname{sech}^2(az)]$, substitute $v = az$:

$$\mathbb{E}_z[\operatorname{sech}^2(az)] = \frac{1}{a}\int_{\mathbb{R}} \frac{e^{-v^2/(2a^2)}}{\sqrt{2\pi}} \operatorname{sech}^2 v \, \mathrm{d}v = \frac{1}{a}\cdot\frac{2}{\sqrt{2\pi}} + \mathcal{O}(a^{-3}), \tag{3.103}$$

because $\int \operatorname{sech}^2 v \, \mathrm{d}v = 2$ and the Gaussian factor equals $1 - \mathcal{O}(v^2/a^2)$ on the region where $\operatorname{sech}^2 v$ is not negligible. So $1 - q = \sqrt{2/\pi}/(\beta J\sqrt{q}) + \mathcal{O}(\beta^{-3})$, i.e.

$$C = \frac{c_0}{\sqrt{q}} + \mathcal{O}(\beta^{-2}) = c_0\Big(1 + \frac{C}{2\beta}\Big) + \mathcal{O}(\beta^{-2}), \qquad c_0 := \frac{1}{J}\sqrt{\frac{2}{\pi}}, \tag{3.104}$$

using $q^{-1/2} = (1 - C/\beta)^{-1/2} = 1 + C/(2\beta) + \cdots$. Hence $C = c_0 + c_1/\beta + \mathcal{O}(\beta^{-2})$ with $c_1 = c_0^2/2$.

Energy. From Eq. (3.49), $u = -\frac{\beta J^2}{2}(1-q)(1+q) = -\frac{J^2 C}{2}(2 - C/\beta) \to -J^2 c_0 = -\sqrt{2/\pi}\, J \approx -0.798J$.

Entropy. With $s = \beta(u - f)$ and Eq. (3.47),

$$s = -\frac{K}{2}(1 - q^2) + \frac{K}{4}(1-q)^2 + \mathbb{E}_z \ln 2\cosh(az). \tag{3.105}$$

Write $\ln 2\cosh x = |x| + \ln(1 + e^{-2|x|})$. The second piece has expectation $\mathcal{O}(1/a)$ by the same rescaling as above, so $\mathbb{E}_z \ln 2\cosh(az) = a\sqrt{2/\pi} + \mathcal{O}(\beta^{-1})$. Now express everything through C, keeping terms of order one:

$$-\frac{K}{2}(1-q^2) = -\frac{\beta J^2 C}{2}\Big(2 - \frac{C}{\beta}\Big) = -\beta J^2 C + \frac{J^2 C^2}{2}, \tag{3.106}$$

$$\frac{K}{4}(1-q)^2 = \frac{J^2 C^2}{4}, \tag{3.107}$$

$$a\sqrt{2/\pi} = \beta J\sqrt{2/\pi}\Big(1 - \frac{C}{2\beta} + \mathcal{O}(\beta^{-2})\Big) = \beta J^2 c_0 - \frac{J^2 c_0 C}{2} + \mathcal{O}(\beta^{-1}). \tag{3.108}$$

The terms of order β are $\beta J^2(c_0 - C) = -J^2 c_1 + \mathcal{O}(\beta^{-1})$. Collecting, with $C \to c_0$ in the order-one terms,

$$s(T \to 0) = -J^2 c_1 + \frac{J^2 c_0^2}{2} + \frac{J^2 c_0^2}{4} - \frac{J^2 c_0^2}{2} = -\frac{J^2 c_0^2}{2} + \frac{J^2 c_0^2}{4} = -\frac{J^2 c_0^2}{4} = -\frac{1}{2\pi}. \tag{3.109}$$

Numerically, at $k_{\mathrm{B}}T = 0.01J$ the solution gives $\beta(1-q) = 0.801/J$, $u = -0.7979J$, and $s = -0.154$, approaching $c_0 J = 0.798$, $-0.798J$, and $-1/(2\pi) = -0.159$. The full Parisi solution gives $u_0 \approx -0.7633J$ and $s(0) = 0$.

Part 3: The AT eigenvalue near T_c and the AT line.

Near T_c both q and h are small. Let $x = \Xi(z) = \beta(J\sqrt{q}z + h)$. We need the Taylor series

$$\tanh^2 x = x^2 - \frac{2}{3}x^4 + \frac{17}{45}x^6 - \cdots, \qquad \operatorname{sech}^4 x = (1 - \tanh^2 x)^2 = 1 - 2x^2 + \frac{7}{3}x^4 - \cdots, \tag{3.110}$$

where the second follows from the first: $(1-x^2+\frac{2}{3}x^4)^2 = 1-2x^2+x^4+\frac{4}{3}x^4+\mathcal{O}(x^6)$. The Gaussian moments of x are $\mathbb{E}_z x^2 = Kq+\beta^2h^2$ and $\mathbb{E}_z x^4 = 3K^2q^2+6Kq\beta^2h^2+\beta^4h^4$. We anticipate $q=\mathcal{O}(\tau)$ and $h^2=\mathcal{O}(\tau^3)$ on the AT line, and keep terms through order τ^2 after dividing the q equation by q.

The q equation. Equation (3.48) gives $q = Kq+\beta^2h^2-2K^2q^2+\frac{17}{3}K^3q^3+\mathcal{O}(\tau^4)$, where the x^6 coefficient used $\mathbb{E}_z z^6 = 15$ and $\frac{17}{45}\cdot 15 = \frac{17}{3}$. Divide by q:

$$K-1 = 2K^2q-\frac{17}{3}K^3q^2-\frac{\beta^2h^2}{q}+\mathcal{O}(\tau^3). \tag{3.111}$$

The AT condition. Equation (3.68) gives

$$\lambda_{\mathrm{AT}} = 1-K\left[1-2(Kq+\beta^2h^2)+7K^2q^2\right]+\mathcal{O}(\tau^3) = 1-K+2K^2q-7K^3q^2+\mathcal{O}(\tau^3), \tag{3.112}$$

where $2K\beta^2h^2=\mathcal{O}(\tau^3)$ was dropped. Substitute $1-K$ from Eq. (3.111):

$$\lambda_{\mathrm{AT}} = -2K^2q+\frac{17}{3}K^3q^2+\frac{\beta^2h^2}{q}+2K^2q-7K^3q^2+\mathcal{O}(\tau^3) = \frac{\beta^2h^2}{q}-\frac{4}{3}K^3q^2+\mathcal{O}(\tau^3). \tag{3.113}$$

Two results follow.

- At $h=0$: $\lambda_{\mathrm{AT}} = -\frac{4}{3}K^3q^2 = -\frac{4}{3}\tau^2+\mathcal{O}(\tau^3)$, using $q=\tau+\mathcal{O}(\tau^2)$ and $K=1+\mathcal{O}(\tau)$. The RS solution is unstable immediately below T_c. Numerically, at $k_{\mathrm{B}}T=0.99J$ the exact value is $\lambda_{\mathrm{AT}} = -1.32\times10^{-4}$, against $-\frac{4}{3}\tau^2 = -1.33\times10^{-4}$.
- In a field: $\lambda_{\mathrm{AT}}=0$ when $\beta^2h^2 = \frac{4}{3}K^3q^3$. To leading order $q=\tau$ (the field changes q only at order $h^2/q=\mathcal{O}(\tau^2)$) and $\beta=1/J$, which gives the AT line Eq. (3.70), $h_{\mathrm{AT}}^2 = \frac{4}{3}J^2\tau^3$. For $h<h_{\mathrm{AT}}$ the negative term wins and RS is unstable. Numerically, $h_{\mathrm{AT}}/J = 0.0133$ at $k_{\mathrm{B}}T=0.95J$ (asymptotic formula: 0.0129) and 0.0388 at $k_{\mathrm{B}}T=0.9J$ (formula: 0.0365).

Part 4: The RS equations as the statistics of a TAP solution.

Take a solution $\mathbf{m}$ of the SK TAP equations Eq. (3.19) with uniform field h. By Step 1 of the proof of Theorem 3.11, $m_i = \tanh(\beta(h+\mu_i))$ with cavity mean field $\mu_i = \sum_j J_{ij}m_j^{(\setminus i)}$. The cavity magnetizations $m_j^{(\setminus i)}$ are computed in a system that does not contain the couplings J_{ij}, $j=1,\dots,N$, so they are independent of these couplings. Conditional on the cavity system, μ_i is therefore a sum of N independent centered Gaussians $J_{ij}m_j^{(\setminus i)}$ and is itself Gaussian with mean zero and variance

$$\sum_j \frac{J^2}{N}\left(m_j^{(\setminus i)}\right)^2 = J^2q+\mathcal{O}(N^{-1/2}). \tag{3.114}$$

Write $\mu_i = J\sqrt{q}\,z_i$ with $z_i\sim\mathcal{N}(0,1)$. Then $m_i = \tanh(\beta(J\sqrt{q}z_i+h)) = \tanh\Xi(z_i)$, and the site average gives

$$q = \frac{1}{N}\sum_i m_i^2 \;\to\; \mathbb{E}_z\tanh^2\Xi, \qquad m = \frac{1}{N}\sum_i m_i \;\to\; \mathbb{E}_z\tanh\Xi, \tag{3.115}$$

which are the RS equations of Theorem 3.16. Three further identifications follow. The Onsager coefficient $\beta J^2(1-q)$ in Eq. (3.19) is the RS quantity $\beta J^2\mathbb{E}_z\operatorname{sech}^2\Xi$. The Plefka condition Eq. (3.26) becomes $1-K\mathbb{E}_z\operatorname{sech}^4\Xi>0$, which is $\lambda_{\mathrm{AT}}>0$. And the spin-glass susceptibility Eq. (3.29) becomes $\chi_{\mathrm{SG}} = \beta^2\mathbb{E}_z[\operatorname{sech}^4\Xi]/\lambda_{\mathrm{AT}}$, which diverges on the AT line. In the RS phase the cavity picture and the replica picture describe the same state. Below the AT line, the TAP equations have exponentially

many solutions and the cavity hypothesis (C1) holds only within each of them, which is the cavity version of replica symmetry breaking [77].

Part 5a: The frustrated triangle mixes fast.

Take the triangle of Example 3.8 as three p-bits updated by the random-scan heat-bath sampler: at each step pick a site uniformly and resample it from Eq. (3.16). From the ground state $(+,+,-)$, the local fields are $h_1^{\mathrm{loc}} = -J(s_2+s_3) = 0$, $h_2^{\mathrm{loc}} = 0$, and $h_3^{\mathrm{loc}} = -J(s_1+s_2) = -2J$. Spins 1 and 2 sit in zero field and are resampled to either value with probability $\frac{1}{2}$. Spin 3 flips to $+1$ with probability $(1+e^{4\beta J})^{-1}$, which tends to 0 as $T \to 0$. So, as $T \to 0$, from each ground state the chain moves to each of two other ground states with probability $\frac{1}{3}\cdot\frac{1}{2} = \frac{1}{6}$ and stays put with probability $\frac{2}{3}$. The two neighbors of $(+,+,-)$ are $(-,+,-)$ and $(+,-,-)$, and repeating the argument shows that the six ground states form a ring. The zero-temperature chain is the lazy random walk on a 6-cycle, with eigenvalues

$$\lambda_k = \frac{2}{3} + \frac{1}{3}\cos\frac{2\pi k}{6}, \qquad k = 0,\dots,5, \tag{3.116}$$

since the walk moves to each neighbor with probability $\frac{1}{6}$ and the Fourier mode $e^{2\pi i k\ell/6}$ on the ring is an eigenvector. The second-largest eigenvalue (attained at $k=1$ and $k=5$) is $\frac{2}{3}+\frac{1}{6} = \frac{5}{6}$, so the relaxation gap is $\gamma = \frac{1}{6}$: about six single-site steps, or two sweeps, even as $T \to 0$. Frustration produced a degenerate ground manifold, but no barrier: every ground state is connected to the others through zero-energy flips. Slow mixing needs an extensive barrier, and frustration on one loop does not produce one.

Part 5b: A three-spin landscape traps exponentially.

Consider N spins with the mean-field three-spin energy

$$E(\mathbf{s}) = -\frac{J}{N^2}\sum_{i,j,k=1}^{N} s_i s_j s_k = -NJm^3, \qquad m(\mathbf{s}) := \frac{1}{N}\sum_i s_i, \tag{3.117}$$

with $J>0$ and N even. By the gauge argument of Theorem 3.7, the couplings $J_{ijk} = J\xi_i\xi_j\xi_k/N^2$ with an arbitrary sign pattern $\boldsymbol{\xi}$ define the same model up to relabeling; they look disordered but carry no frustration. This is the $p=3$ member of the family of p-spin ferromagnets. It has a first-order transition: the fully magnetized state $m \approx 1$ is the ground state, and the paramagnetic state $m \approx 0$ is a separate valley at every temperature, because three-body forces vanish to second order at $m = 0$.

Free energy of a magnetization level. The energy depends on $\mathbf{s}$ only through k, the number of up spins, with $m = 2k/N - 1$. The equilibrium probability of level k is $\pi(k) = \binom{N}{k}e^{\beta NJm^3}/Z$. We need two elementary bounds on binomial coefficients. Let $H(p) = -p\ln p - (1-p)\ln(1-p)$. First, for $p = k/N$, $1 = (p+(1-p))^N \geq \binom{N}{k}p^k(1-p)^{N-k}$, and $p^k(1-p)^{N-k} = e^{-NH(p)}$, so $\binom{N}{k} \leq e^{NH(k/N)}$. Second, $\binom{N}{N/2}$ is the largest of the $N+1$ coefficients that sum to 2^N, so $\binom{N}{N/2} \geq 2^N/(N+1)$. In terms of the binary entropy $\mathfrak{s}(m) := H\big(\frac{1+m}{2}\big)$, define the free energy per spin of a magnetization level,

$$f(m) := -Jm^3 - k_{\mathrm{B}}T\,\mathfrak{s}(m), \qquad f(0) = -k_{\mathrm{B}}T\ln 2. \tag{3.118}$$

Then $\binom{N}{k}e^{\beta NJm^3} \leq e^{-N\beta f(m)}$ and $\binom{N}{N/2} \geq e^{-N\beta f(0)}/(N+1)$.

The barrier. Since $f'(m) = -3Jm^2 + k_{\mathrm{B}}T\,\operatorname{artanh} m$ and $f''(0) = k_{\mathrm{B}}T > 0$, $m=0$ is a local minimum of f. The saddle $m^\star \in (0,1)$ is the first positive root of $3Jm^2 = k_{\mathrm{B}}T\,\operatorname{artanh} m$, and the barrier per spin is $\Delta f := f(m^\star) - f(0)$. For $k_{\mathrm{B}}T \ll J$, $\operatorname{artanh} m \approx m$ and $\mathfrak{s}(m) \approx \ln 2 - m^2/2$ give $m^\star \approx k_{\mathrm{B}}T/(3J)$ and

$$\Delta f \approx -J\Big(\frac{k_{\mathrm{B}}T}{3J}\Big)^3 + \frac{k_{\mathrm{B}}T}{2}\Big(\frac{k_{\mathrm{B}}T}{3J}\Big)^2 = \frac{(k_{\mathrm{B}}T)^3}{54J^2}, \qquad \beta\Delta f \approx \frac{(k_{\mathrm{B}}T)^2}{54J^2}. \tag{3.119}$$

Solving exactly: at $k_{\mathrm{B}}T = J$, $m^\star = 0.3478$ and $\beta\Delta f = 0.01969$ (the small-T formula gives 0.0185). At $k_{\mathrm{B}}T = 0.5J$, $m^\star = 0.1683$ and $\beta\Delta f = 0.00470$. Meanwhile the ferromagnetic minimum $m_+ \approx 1$ has $f(m_+) \approx -J$, far below $f(0) = -k_{\mathrm{B}}T\ln 2$, so the paramagnetic valley is metastable.

Bottleneck bound. Let k_b be the level whose magnetization m_b is closest to $m^\star$, and let $A = \{\mathbf{s} : k(\mathbf{s}) \le k_b\}$. A single flip changes k by ± 1, so $\partial_{\mathrm{in}}A$ is the level k_b. By Corollary 3.29 and $\pi(A) \ge \pi(N/2)$,

$$\Phi(A) \le \frac{\pi(k_b)}{\pi(N/2)} \le (N+1)\, e^{-N\beta[f(m_b)-f(0)]} \le (N+1)\, e^{\beta\kappa/(2N)}\, e^{-N\beta\Delta f}, \tag{3.120}$$

where $\kappa = \max|f''|$ near $m^\star$ and we used $|m_b - m^\star| \le 1/N$ and $f'(m^\star) = 0$, so $f(m_b) \ge f(m^\star) - \kappa/(2N^2)$. For large N, $\pi(A)$ is exponentially small because $f(m_+) < f(0)$, so $\pi(A) \le \frac{1}{2}$, and Theorem 3.28(iii) gives

$$\gamma \le 2(N+1)\, e^{\beta\kappa/(2N)}\, e^{-N\beta\Delta f}, \qquad \tau_{\mathrm{mix}}(1/4) \ge \frac{e^{-\beta\kappa/(2N)}}{4(N+1)}\, e^{N\beta\Delta f} \text{ single-site steps.} \tag{3.121}$$

The trapping time grows exponentially in N, with rate set by the free-energy barrier per spin. The barrier is collective: no single spin faces a large energy cost, but leaving the paramagnetic valley requires a coordinated change of $\mathcal{O}(N)$ spins, each step of which is uphill in free energy.

Comparison with the exact gap. Because E depends on $\mathbf{s}$ only through k, the heat-bath chain projects exactly onto a birth–death chain on $k \in \{0, \dots, N\}$, with up and down probabilities $r_\pm(k) = \frac{N \mp (2k-N)}{2N}\big(1 + e^{\beta[E(k\pm1)-E(k)]}\big)^{-1}$. Every function of k that is an eigenfunction of the projected chain lifts to an eigenfunction of the full chain with the same eigenvalue, so the projected chain's gap is an upper bound on the full chain's gap, and its slowest mode is the escape from the paramagnetic valley. Diagonalizing the $(N+1) \times (N+1)$ tridiagonal matrix numerically at $k_{\mathrm{B}}T = J$:

N	exact gap γ	$2\Phi(A)$	$e^{-N\beta\Delta f}$	$1/(4\Phi(A))$ (steps)
100	1.9×10^{-4}	4.8×10^{-3}	1.4×10^{-1}	1.0×10^{2}
200	1.3×10^{-5}	4.9×10^{-4}	2.0×10^{-2}	1.0×10^{3}
400	1.3×10^{-7}	7.0×10^{-6}	3.8×10^{-4}	7.2×10^{4}
800	2.6×10^{-11}	1.9×10^{-9}	1.4×10^{-7}	2.7×10^{8}

The exact gap always lies below $2\Phi(A)$, as Theorem 3.28 requires. The ratio of $\Phi(A)$ to $e^{-N\beta\Delta f}$ is a slowly varying power of N, from the Gaussian widths of the two levels, and the ratio of γ to $\Phi(A)$ is the analog of the Kramers transmission factor. At $N = 800$ and $k_{\mathrm{B}}T = J$, escape requires more than 2.6×10^8 single-site steps, about 3×10^5 sweeps. At $N = 10^4$ the exponential factor alone is e^{197}.

Dependence on p. Replacing m^3 by m^p gives the p-spin ferromagnet. At $k_{\mathrm{B}}T = J$ the barrier per spin grows with p: $\beta\Delta f = 0.0197, 0.0690, 0.115, 0.155, 0.263, 0.391$ for $p = 3, 4, 5, 6, 10, 20$. As $p \to \infty$, $m^p \to 0$ for $|m| < 1$, the saddle moves to $m^\star \to 1$, and $\beta\Delta f \to \ln 2$. The landscape becomes a *golf course*: a flat plateau containing almost all 2^N configurations and a single deep hole. The trapping time approaches 2^N, the cost of exhaustive search. The genuinely frustrated version, the random p-spin glass with independent Gaussian couplings $J_{i_1 \cdots i_p}$, behaves similarly but with exponentially many metastable valleys instead of one. Its dynamics freezes at a temperature T_d above the static transition, where the valleys first become separated by extensive barriers [76].

Connection to Kramers and to DTMs. For large N the magnetization moves in steps of $2/N$ with drift $\propto -f'(m)$, so it performs an overdamped diffusion in the effective potential $Nf(m)$ at

temperature T. This is the escape problem of the double-well sampler (the quartic double well whose binary limit is the p-bit of Example 1.6), now with an extensive barrier $N\Delta f$, and the Kramers formula of Theorem 7.4 gives the escape rate with the correct prefactor. Chapter 16 shows how a DTM avoids this barrier. In this mean-field model, adding a quadratic coupling $\frac{\alpha}{2}N(m-c)^2$ to a reference point c makes the conditional free-energy profile $f(m)+\frac{\alpha}{2}(m-c)^2$ convex on $[0,1)$ as soon as $\alpha > \max_m[-f''(m)] = \max_m[6Jm - k_{\mathrm{B}}T/(1-m^2)]$, which equals $2.24J$ at $k_{\mathrm{B}}T = J$. The valley disappears, and the conditional sampling problem has no bottleneck. This is the mechanism analyzed in general in Section 16.3 and computed for the double well in Worked computation 16.27.

Summary

Proved:

- Self-averaging of the SK free energy per spin (Proposition 3.3); the annealed free energy and its negative entropy at low temperature (Example 3.2).
- Frustrated cycles force unsatisfied bonds (Lemma 3.5); unfrustrated couplings are gauge-equivalent to a ferromagnet (Theorem 3.7).
- The TAP equations, given the cavity hypotheses (Theorem 3.11), and the paramagnetic instability at $k_{\mathrm{B}}T_c = J$.
- The replicated action for integer n (Proposition 3.13); the RS free energy, self-consistency equation, and energy (Theorem 3.16 and Corollary 3.17); the replicon eigenvalue and the AT line (Proposition 3.18, Worked computation 3.33).
- The REM free energy and freezing transition, at the level of a first- and second-moment argument (Proposition 3.21).
- Bottleneck bounds on the gap and mixing time of reversible chains and Langevin diffusions, and the expressivity–mixing dilemma (Theorems 3.28, 3.30 and 3.32 and Corollaries 3.29 and 3.31).
- Fast mixing of the frustrated triangle and exponential trapping in the mean-field three-spin model (Worked computation 3.33).

Assumed:

- The replica method: continuation from integer n to $n \to 0$, exchange of the limits $n \to 0$ and $N \to \infty$, and the choice of saddle point (Remark 3.14).
- The Parisi formula (Theorem 3.23), proved by Guerra [44] and Talagrand [106], and ultrametricity for generic models, proved by Panchenko [82].
- The cavity hypotheses of Assumption 3.10; the semicircle law for the SK coupling matrix; the ladder resummation behind Eq. (3.29).
- Quoted numerical values: the one-flip complexity ≈ 0.199 [107], the Parisi ground-state energy $u_0 \approx -0.7633J$, and the REM overlap weight $1 - T/T_c$.

Taken up later:

- The Kramers prefactor and Freidlin–Wentzell theory (Chapter 7); lower bounds on spectral gaps (Section 6.5, Chapter C); Gaussian identities (Chapter B).

- Device-level p-bit barriers (Chapters 12 and 14) and the denoising route around the bottleneck (Section 16.3 and Worked computation 16.27).

Open questions:

- The barrier exponent ψ for the SK model and for finite-dimensional spin glasses, known only from numerics and scaling arguments [34, 70].
- Whether three-dimensional short-range spin glasses have a replica-symmetry-broken phase or a single pair of states (droplet picture).
- Which non-local, non-reversible, or tempered samplers mix in polynomial time on glassy landscapes, and whether trained EBMs develop spin-glass-like barriers in practice.

Bibliographic notes

The Edwards–Anderson model and the order parameter q_{EA} were introduced in [27], and the mean-field SK model in [99], where the replica-symmetric solution and its negative zero-temperature entropy first appeared. Frustration as a loop property was formulated by Toulouse [110], and the gauge-equivalent Mattis model in [72]. The TAP equations are from [108]; Plefka's expansion and its convergence condition are in [88]; the count of metastable states at zero temperature is due to Tanaka and Edwards [107]. The replicon instability is from de Almeida and Thouless [23], and the hierarchical solution from Parisi [83]. The Random Energy Model is Derrida's [26]. The standard physics reference for all of this material, including ultrametricity, the cavity method, and reprints of the original papers, is Mézard, Parisi, and Virasoro [77]. Mézard and Montanari [76] give a modern treatment that connects spin glasses to inference, coding, and random constraint satisfaction, including the REM and the p-spin glass. The rigorous proofs of the Parisi formula are [44, 106], and ultrametricity was proved in [82]. Early numerical evidence for barriers growing with N in SK dynamics is in [70], and the droplet theory of finite-dimensional spin glasses is in [34].

The bottleneck (conductance) bounds of Section 3.5 are standard in the theory of Markov chain mixing; [65] gives a careful treatment of Dirichlet forms, bottleneck ratios, and mixing-time lower bounds for finite chains. For diffusions, the Poincaré-inequality viewpoint and its relation to spectral gaps is developed in [7], and the Arrhenius exponent of the gap goes back to Kramers [61]. The use of this bottleneck as a design constraint for thermodynamic samplers, and its resolution by denoising, is the subject of [56].

Part II

Continuous-Time Stochastic Calculus, Langevin Dynamics, and Fokker–Planck Theory

Chapter 4

Brownian Motion, Martingales, and Itô vs. Stratonovich Stochastic Calculus

In Part I we described a target distribution $p_\theta(\mathbf{x}) = Z(\theta,\beta)^{-1}\exp(-\beta E_\theta(\mathbf{x}))$ and built discrete-time Markov chains that sample it (Chapter 2), and we saw in Chapter 3 why such chains can become trapped behind large free-energy barriers. This chapter turns to a different question: how does a physical system, evolving continuously in time, generate random trajectories that relax to p_θ?

A digital processor suppresses thermal noise with energy barriers of order 10^4–$10^5\,k_\mathrm{B}T$ and then synthesizes pseudo-random numbers in deterministic logic. A thermodynamic computer instead lets thermal noise drive the computation. Its state variables are node voltages $\mathbf{v}(t) \in \mathbb{R}^N$ on small capacitors (in subthreshold CMOS) or magnetic fluxes in Josephson circuits. As a representative estimate, a node capacitance of 1 fF at $T = 300\,\mathrm{K}$ has root-mean-square thermal voltage $\sqrt{k_\mathrm{B}T/C_\mathrm{node}} \approx 2.03\,\mathrm{mV}$, close to a tenth of the thermal voltage $V_T = k_\mathrm{B}T/q_e \approx 25.85\,\mathrm{mV}$ that sets the scale of subthreshold logic. Such circuits are described by stochastic differential equations (SDEs), and sampling happens at the speed of analog RC relaxation.

To analyze these SDEs we need a calculus for Brownian motion $\mathbf{W}_t$, the mathematical model of white thermal noise. Its paths are continuous but nowhere differentiable, and an increment over a time $\mathrm{d}t$ has size $\sqrt{\mathrm{d}t}$ rather than $\mathrm{d}t$. The second-order term of a Taylor expansion therefore survives, $(\mathrm{d}W_t)^2 = \mathrm{d}t$, and the ordinary chain rule fails. When the noise amplitude depends on the state, as it does in subthreshold MOSFETs, the stochastic integral $\int_0^t \mathbf{B}(\mathbf{v}_s)\,\mathrm{d}\mathbf{W}_s$ also depends on where in each time step the integrand is evaluated. The left endpoint gives the Itô integral, which is a martingale and matches simple numerical schemes. The midpoint gives the Stratonovich integral, which obeys the ordinary chain rule and arises when physical noise with a short correlation time is idealized as white. Confusing the two adds a spurious drift that changes the stationary distribution of the hardware.

The chapter builds this calculus from Brownian motion up to Itô's formula, then treats multiplicative noise and the Wong–Zakai theorem (Section 4.4) and Girsanov's theorem (Section 4.5). It ends with a nonlinear RC circuit in which the choice of calculus decides whether the circuit samples the Boltzmann distribution (Worked computation 4.23).

4.1 Construction and Path Properties of Standard Brownian Motion

In 1827, the botanist Robert Brown observed through a microscope that microscopic pollen grains suspended in water execute an incessant, erratic, zig-zag motion that never damps out. In 1905 Albert Einstein (and independently Marian Smoluchowski in 1906) explained this motion by the kinetic theory of heat: each colloidal particle is struck by about 10^{21} water molecules per second. Each collision imparts a tiny impulse over a collision time $\tau_{\text{coll}} \sim 10^{-13}\,\text{s}$, and the summed effect of many independent collisions over an interval $\Delta t \gg \tau_{\text{coll}}$ is Gaussian by the central limit theorem. The same mechanism operates inside an integrated circuit (Chapter 11): the voltage $v(t)$ across a capacitor connected to a resistor or subthreshold transistor channel fluctuates erratically because discrete electrons undergo thermal scattering events (Johnson–Nyquist noise, [57, 79]) or barrier crossings (shot noise, [96]) on femtosecond to picosecond timescales.

When we observe the circuit on timescales $t \gg \tau_{\text{corr}}$, where τ_{corr} is the microscopic correlation time of the charge carriers, the net charge transferred over disjoint time intervals becomes statistically independent and Gaussian. The limit $\tau_{\text{corr}} \to 0$ yields *standard Brownian motion*, which Norbert Wiener constructed in 1923 as a probability measure on the space of continuous paths.

4.1.1 Filtered Probability Spaces and the Axioms of the Wiener Process

To track how information accumulates in time as a stochastic physical system evolves, we equip our underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a time-indexed family of σ-algebras that records the history of all observable events up to time t.

Definition 4.1 (Filtration and Adapted Process)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space. A *filtration* is a family $\{\mathcal{F}_t\}_{t \geq 0}$ of sub-σ-algebras of $\mathcal{F}$ that is non-decreasing:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \qquad \text{for all } 0 \leq s \leq t < \infty. \tag{4.1}$$

We assume throughout that $\{\mathcal{F}_t\}_{t \geq 0}$ satisfies the *usual conditions*: it is right-continuous ($\mathcal{F}_t = \bigcap_{\varepsilon > 0} \mathcal{F}_{t+\varepsilon}$ for all $t \geq 0$) and $\mathcal{F}_0$ contains all $\mathbb{P}$-null sets of $\mathcal{F}$. A stochastic process $\mathbf{X} = \{\mathbf{X}_t\}_{t \geq 0}$ taking values in $\mathbb{R}^d$ is said to be *adapted* to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$ (or *non-anticipating*) if for every $t \geq 0$, the random vector $\mathbf{X}_t : \Omega \to \mathbb{R}^d$ is $\mathcal{F}_t$-measurable.

Physically, $\mathcal{F}_t$ represents the totality of trajectories and thermal kicks that have occurred up to time t. Saying that a node voltage vector $\mathbf{v}_t$ is $\mathcal{F}_t$-adapted is simply the mathematical statement of *physical causality*: the state of an electronic circuit at time t can depend on past and present thermal fluctuations over $[0, t]$, but it cannot depend on future thermal kicks that have not yet occurred at times $u > t$.

Definition 4.2 (Standard d-Dimensional Brownian Motion / Wiener Process)**.** An $\mathbb{R}^d$-valued stochastic process $\mathbf{W} = \{\mathbf{W}_t\}_{t \geq 0} = \{(W_{1,t}, W_{2,t}, \ldots, W_{d,t})^\top\}_{t \geq 0}$ defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ is called a *standard d-dimensional $\{\mathcal{F}_t\}$-Brownian motion* (or *Wiener process*) if it satisfies the following four axioms:

(i) **Initial condition:** $\mathbf{W}_0 = \mathbf{0}$ almost surely ($\mathbb{P}(\mathbf{W}_0 = \mathbf{0}) = 1$).

(ii) **Independent increments:** For every $0 \leq s < t < \infty$, the increment $\mathbf{W}_t - \mathbf{W}_s$ is statistically independent of the past σ-algebra $\mathcal{F}_s$:

$$\mathbf{W}_t - \mathbf{W}_s \perp\!\!\!\perp \mathcal{F}_s. \tag{4.2}$$

In particular, for any partition $0 = t_0 < t_1 < t_2 < \cdots < t_n$, the increments $\mathbf{W}_{t_1} - \mathbf{W}_{t_0}$, $\mathbf{W}_{t_2} - \mathbf{W}_{t_1}$, …, $\mathbf{W}_{t_n} - \mathbf{W}_{t_{n-1}}$ are mutually independent random vectors.

(iii) **Stationary Gaussian increments:** For every $0 \le s < t < \infty$, the increment $\mathbf{W}_t - \mathbf{W}_s$ is a multivariate Gaussian random vector with mean $\mathbf{0}$ and covariance matrix $(t-s)\mathbf{I}_d$:

$$\mathbf{W}_t - \mathbf{W}_s \sim \mathcal{N}(\mathbf{0},\, (t-s)\mathbf{I}_d)\,. \tag{4.3}$$

Equivalently, the transition probability density for $\mathbf{W}_t = \mathbf{y}$ given $\mathbf{W}_s = \mathbf{x}$ is the heat kernel

$$p(\mathbf{y}, t \mid \mathbf{x}, s) = \frac{1}{[2\pi(t-s)]^{d/2}} \exp\!\left(-\frac{\|\mathbf{y}-\mathbf{x}\|^2}{2(t-s)}\right). \tag{4.4}$$

(iv) **Continuous sample paths:** For $\mathbb{P}$-almost every realization $\omega \in \Omega$, the sample trajectory $t \mapsto \mathbf{W}_t(\omega)$ is a continuous function from $[0,\infty)$ into $\mathbb{R}^d$.

It is a non-trivial theorem of measure theory, first proved by Wiener with Fourier series and later simplified by Paul Lévy (dyadic interpolation) and by Andrei Kolmogorov (his continuity theorem; see [59, Ch. 2]), that a probability measure $\mathbb{P}_W$ on the canonical space $\mathcal{C}([0,\infty);\mathbb{R}^d)$ of continuous paths satisfying Conditions (i)–(iv) exists. The two-time correlation and the scaling symmetry of $\mathbf{W}_t$ follow directly from Definition 4.2.

Proposition 4.3 (Covariance Structure and Brownian Scaling Symmetry)**.** *Let $\mathbf{W}_t = (W_{1,t}, \dots, W_{d,t})^\top$ be a standard d-dimensional Brownian motion. Then:*

(a) $\{\mathbf{W}_t\}_{t\ge 0}$ is a zero-mean Gaussian process with two-time covariance function

$$\mathbb{E}[W_{i,t} W_{j,s}] = \delta_{ij} \min(t,s) \qquad \text{for all } i,j \in \{1,\dots,d\},\ t,s \ge 0. \tag{4.5}$$

*(b) **(Diffusive scale invariance)** For any constant $c > 0$, the rescaled process $\tilde{\mathbf{W}}_t := \frac{1}{\sqrt{c}} \mathbf{W}_{ct}$ is again a standard d-dimensional Brownian motion.*

Proof. To prove part (a), assume without loss of generality that $0 \le s \le t$. We decompose $W_{i,t}$ into its value at time s plus the subsequent increment:

$$W_{i,t} = W_{i,s} + (W_{i,t} - W_{i,s}). \tag{4.6}$$

Multiplying by $W_{j,s}$ and taking the expectation yields

$$\mathbb{E}[W_{i,t} W_{j,s}] = \mathbb{E}[W_{i,s} W_{j,s}] + \mathbb{E}[(W_{i,t} - W_{i,s}) W_{j,s}]\,. \tag{4.7}$$

Since $W_{j,s}$ is $\mathcal{F}_s$-measurable and the increment $W_{i,t} - W_{i,s}$ is independent of $\mathcal{F}_s$ by Eq. (4.2), the expectation of their product factors into the product of their expectations:

$$\mathbb{E}[(W_{i,t} - W_{i,s}) W_{j,s}] = \mathbb{E}[W_{i,t} - W_{i,s}]\, \mathbb{E}[W_{j,s}] = 0 \cdot 0 = 0. \tag{4.8}$$

For the first term in Eq. (4.7), setting $s = 0$ in Eq. (4.3) gives $\mathbf{W}_s = \mathbf{W}_s - \mathbf{W}_0 \sim \mathcal{N}(\mathbf{0}, s\mathbf{I}_d)$, so $\mathbb{E}[W_{i,s} W_{j,s}] = s\delta_{ij}$. Since $s = \min(t,s)$ when $s \le t$, and the symmetric argument holds when $t < s$, we obtain $\mathbb{E}[W_{i,t} W_{j,s}] = \delta_{ij} \min(t,s)$.

To prove part (b), observe that $\tilde{\mathbf{W}}_0 = c^{-1/2}\mathbf{W}_0 = \mathbf{0}$ a.s., sample paths $t \mapsto c^{-1/2}\mathbf{W}_{ct}$ are continuous, and increments over disjoint intervals $[s,t]$ correspond to increments of $\mathbf{W}$ over disjoint intervals $[cs, ct]$, which are independent. Finally, for $0 \le s < t$,

$$\tilde{\mathbf{W}}_t - \tilde{\mathbf{W}}_s = \frac{1}{\sqrt{c}}\,(\mathbf{W}_{ct} - \mathbf{W}_{cs}) \sim \frac{1}{\sqrt{c}}\,\mathcal{N}(\mathbf{0},\, c(t-s)\mathbf{I}_d) = \mathcal{N}(\mathbf{0},\, (t-s)\mathbf{I}_d)\,, \tag{4.9}$$

verifying all four axioms of Definition 4.2. □

4.1.2 Nowhere-Differentiability and Unbounded First Variation

The diffusive scaling relation $\Delta \mathbf{W}_t \sim \sqrt{\Delta t}$ of Proposition 4.3(b) has a direct geometric consequence: if we zoom into a Brownian trajectory by a factor of c in time, we must magnify its vertical amplitude by $\sqrt{c}$ to see the same statistical roughness. Over a short time step $\Delta t \ll 1$, a typical increment has magnitude $|\Delta W_t| \approx \sqrt{\Delta t} \gg \Delta t$. Consequently, the difference quotient $\Delta W_t/\Delta t$ blows up as $(\Delta t)^{-1/2}$:

$$\mathbb{E}\left[\left(\frac{W_{t+\Delta t} - W_t}{\Delta t}\right)^2\right] = \frac{\mathbb{E}[(W_{t+\Delta t} - W_t)^2]}{(\Delta t)^2} = \frac{\Delta t}{(\Delta t)^2} = \frac{1}{\Delta t} \xrightarrow{\Delta t \to 0^+} +\infty. \tag{4.10}$$

Thus the time derivative $\dot{W}_t = \frac{\mathrm{d}W_t}{\mathrm{d}t}$ ("Gaussian white noise") does not exist as an ordinary function of time; it exists only as a generalized function (distribution) in the sense of Laurent Schwartz. More generally, using the Gaussian moments $\mathbb{E}\left[|W_t - W_s|^{2p}\right] = \frac{(2p)!}{2^p p!}\,|t-s|^p$ and Kolmogorov's continuity theorem [59, Thm. 2.2.8], one proves that for every $\alpha \in (0, 1/2)$, almost every Brownian path is locally Hölder continuous of exponent α (meaning $|W_t(\omega) - W_s(\omega)| \le C_\alpha(\omega)\,|t-s|^\alpha$), whereas by a theorem of Paley, Wiener, and Zygmund (1933), for $\mathbb{P}$-almost every $\omega \in \Omega$, the path $t \mapsto W_t(\omega)$ is *nowhere differentiable* (and not even Hölder continuous of exponent $\alpha = 1/2$ at any point).

Closely tied to nowhere-differentiability is the fact that Brownian paths have *infinite total variation* (infinite arc length) on every finite time interval $[0, T]$. Recall that for a function $g : [0,T] \to \mathbb{R}$ and a partition $\Pi_n = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ with mesh size $|\Pi_n| := \max_{0 \le k \le n-1}(t_{k+1} - t_k)$, the p-variation sum is

$$V_n^{(p)}(g; [0,T]) := \sum_{k=0}^{n-1} |g(t_{k+1}) - g(t_k)|^p\,. \tag{4.11}$$

For $p = 1$ and a uniformly spaced partition $t_k = kT/n$ ($\Delta t = T/n$), each increment $\Delta W_k := W_{t_{k+1}} - W_{t_k}$ has the distribution $\sqrt{T/n}\,Z_k$ where $Z_k \sim \mathcal{N}(0,1)$ has $\mathbb{E}[|Z_k|] = \sqrt{2/\pi}$. Hence the expected first variation grows without bound as $n \to \infty$:

$$\mathbb{E}\left[V_n^{(1)}(W; [0,T])\right] = \sum_{k=0}^{n-1} \mathbb{E}[|\Delta W_k|] = n\sqrt{\frac{2}{\pi}}\sqrt{\frac{T}{n}} = \sqrt{\frac{2Tn}{\pi}} \xrightarrow{n \to \infty} +\infty. \tag{4.12}$$

Because standard Riemann–Stieltjes integration $\int_0^T H_t\,\mathrm{d}g(t)$ requires the integrator g to have bounded first variation ($V^{(1)}(g;[0,T]) < \infty$) in order for upper and lower Riemann sums to converge to a unique limit for continuous integrands H_t, Eq. (4.12) means that $\int_0^T H_t\,\mathrm{d}W_t$ *cannot* be defined as an ordinary pathwise Riemann–Stieltjes integral.

4.1.3 Quadratic Variation and the Algebra of Stochastic Differentials

The first variation ($p = 1$) of Brownian motion diverges as $\sqrt{n}$ and the higher variations ($p > 2$) vanish as $n^{1-p/2}$, but the **quadratic variation** ($p = 2$) converges to a non-zero *deterministic* limit. This theorem underlies Itô's formula, the diffusion term of the Fokker–Planck equation, and the Itô–Stratonovich drift shift, so we prove it in full.

Theorem 4.4 (Quadratic Variation and Cross-Variation of Multi-Dimensional Brownian Motion)**.** *Let $\mathbf{W}_t = (W_{1,t}, \ldots, W_{d,t})^\top$ be a standard d-dimensional Brownian motion, and let $\Pi_n = \{0 = t_0^{(n)} < t_1^{(n)} < \cdots < t_{m_n}^{(n)} = t\}$ be any sequence of deterministic partitions of $[0,t]$ whose mesh size*

$|\Pi_n| := \max_{0\le k\le m_n-1} \Delta t_k^{(n)}$ *tends to zero as* $n\to\infty$, *where* $\Delta t_k^{(n)} := t_{k+1}^{(n)} - t_k^{(n)}$ *and* $\Delta W_{i,k} := W_{i,t_{k+1}^{(n)}} - W_{i,t_k^{(n)}}$. *Define the sampled quadratic covariation sum for components* $i,j\in\{1,\dots,d\}$ *by*

$$Q_n^{ij}(t) := \sum_{k=0}^{m_n-1} \Delta W_{i,k}\,\Delta W_{j,k} = \sum_{k=0}^{m_n-1} \left(W_{i,t_{k+1}^{(n)}} - W_{i,t_k^{(n)}}\right)\left(W_{j,t_{k+1}^{(n)}} - W_{j,t_k^{(n)}}\right). \tag{4.13}$$

Then $Q_n^{ij}(t)$ *converges in mean square* ($L^2(\Omega)$) *to the deterministic quantity* $\delta_{ij}\,t$ *as* $|\Pi_n|\to 0$*:*

$$\lim_{|\Pi_n|\to 0} \mathbb{E}\left[\left(Q_n^{ij}(t) - \delta_{ij}\,t\right)^2\right] = 0. \tag{4.14}$$

We write this limit in bracket notation as $[W_i,W_j]_t = \delta_{ij}\,t$, *and in differential shorthand as*

$$\mathrm{d}W_{i,t}\,\mathrm{d}W_{j,t} = \delta_{ij}\,\mathrm{d}t, \qquad \mathrm{d}W_{i,t}\,\mathrm{d}t = 0, \qquad (\mathrm{d}t)^2 = 0. \tag{4.15}$$

Proof. We suppress the superscript (n) on t_k and Δt_k for readability. First we compute the expectation of $Q_n^{ij}(t)$ using linearity of expectation and the covariance of Gaussian increments Eq. (4.3):

$$\mathbb{E}\Big[Q_n^{ij}(t)\Big] = \sum_{k=0}^{m_n-1} \mathbb{E}[\Delta W_{i,k}\,\Delta W_{j,k}] = \sum_{k=0}^{m_n-1} \delta_{ij}\,\Delta t_k = \delta_{ij}\sum_{k=0}^{m_n-1}(t_{k+1}-t_k) = \delta_{ij}\,t, \tag{4.16}$$

since the telescoping sum of interval lengths equals $t_{m_n} - t_0 = t$. Because $\mathbb{E}[Q_n^{ij}(t)] = \delta_{ij}\,t$, the mean-square error in Eq. (4.14) is simply the variance of $Q_n^{ij}(t)$. We rewrite the difference $Q_n^{ij}(t) - \delta_{ij}\,t$ as a sum of zero-mean random variables $Y_k^{ij} := \Delta W_{i,k}\Delta W_{j,k} - \delta_{ij}\Delta t_k$:

$$Q_n^{ij}(t) - \delta_{ij}\,t = \sum_{k=0}^{m_n-1} (\Delta W_{i,k}\,\Delta W_{j,k} - \delta_{ij}\,\Delta t_k) = \sum_{k=0}^{m_n-1} Y_k^{ij}. \tag{4.17}$$

Squaring Eq. (4.17) and taking expectations, we separate the diagonal terms ($k=\ell$) from the off-diagonal cross terms ($k\neq\ell$):

$$\mathbb{E}\left[\left(Q_n^{ij}(t) - \delta_{ij}\,t\right)^2\right] = \sum_{k=0}^{m_n-1} \mathbb{E}\left[\left(Y_k^{ij}\right)^2\right] + 2\sum_{0\le k<\ell\le m_n-1} \mathbb{E}\left[Y_k^{ij}Y_\ell^{ij}\right]. \tag{4.18}$$

For $k<\ell$, the time intervals $[t_k,t_{k+1}]$ and $[t_\ell,t_{\ell+1}]$ have disjoint interiors ($t_{k+1}\le t_\ell$). By the independent increments property (Definition 4.2(ii)), Y_k^{ij} and Y_ℓ^{ij} are statistically independent, and since $\mathbb{E}\left[Y_k^{ij}\right] = 0$, every off-diagonal term vanishes identically:

$$\mathbb{E}\left[Y_k^{ij}Y_\ell^{ij}\right] = \mathbb{E}\left[Y_k^{ij}\right]\mathbb{E}\left[Y_\ell^{ij}\right] = 0\cdot 0 = 0 \qquad (k<\ell). \tag{4.19}$$

To evaluate the diagonal terms $\mathbb{E}\left[(Y_k^{ij})^2\right]$, we consider two cases.

Case 1: Diagonal components ($i=j$). Here $Y_k^{ii} = (\Delta W_{i,k})^2 - \Delta t_k$. Expanding the square gives

$$\mathbb{E}\left[\left(Y_k^{ii}\right)^2\right] = \mathbb{E}\left[\left((\Delta W_{i,k})^2 - \Delta t_k\right)^2\right] = \mathbb{E}\left[(\Delta W_{i,k})^4\right] - 2\Delta t_k\mathbb{E}\left[(\Delta W_{i,k})^2\right] + (\Delta t_k)^2. \tag{4.20}$$

Recall that for a zero-mean Gaussian random variable $X \sim \mathcal{N}(0, \sigma^2)$, integration by parts on the Gaussian density (or differentiating the moment generating function $\mathbb{E}\left[e^{\lambda X}\right] = e^{\lambda^2\sigma^2/2}$ four times at $\lambda = 0$) yields the fourth moment $\mathbb{E}[X^4] = 3\sigma^4$. Applying this to $\Delta W_{i,k} \sim \mathcal{N}(0, \Delta t_k)$ gives $\mathbb{E}[(\Delta W_{i,k})^2] = \Delta t_k$ and $\mathbb{E}[(\Delta W_{i,k})^4] = 3(\Delta t_k)^2$. Substituting these into Eq. (4.20) yields

$$\mathbb{E}\left[\left(Y_k^{ii}\right)^2\right] = 3(\Delta t_k)^2 - 2\Delta t_k(\Delta t_k) + (\Delta t_k)^2 = 2(\Delta t_k)^2. \tag{4.21}$$

Case 2: Off-diagonal components ($i \neq j$). Here $\delta_{ij} = 0$, so $Y_k^{ij} = \Delta W_{i,k}\Delta W_{j,k}$. Because the components W_i and W_j are independent for $i \neq j$, we have

$$\mathbb{E}\left[\left(Y_k^{ij}\right)^2\right] = \mathbb{E}\left[(\Delta W_{i,k})^2(\Delta W_{j,k})^2\right] = \mathbb{E}\left[(\Delta W_{i,k})^2\right]\mathbb{E}\left[(\Delta W_{j,k})^2\right] = (\Delta t_k)(\Delta t_k) = (\Delta t_k)^2. \tag{4.22}$$

Combining Eqs. (4.21) and (4.22) into Eq. (4.18), we find for all $i, j \in \{1, \ldots, d\}$ that

$$\begin{aligned}
\mathbb{E}\left[\left(Q_n^{ij}(t) - \delta_{ij}\, t\right)^2\right] &= (1 + \delta_{ij}) \sum_{k=0}^{m_n-1} (\Delta t_k)^2 \\
&\leq (1 + \delta_{ij}) \left(\max_{0\leq k\leq m_n-1} \Delta t_k\right) \sum_{k=0}^{m_n-1} \Delta t_k \\
&= (1 + \delta_{ij})\ |\Pi_n|\ t \xrightarrow{|\Pi_n|\to 0} 0.
\end{aligned} \tag{4.23}$$

Finally, a similar calculation for $\sum_k \Delta W_{i,k}\Delta t_k$ shows $\mathbb{E}[(\sum_k \Delta W_{i,k}\Delta t_k)^2] = \sum_k (\Delta t_k)^3 \leq |\Pi_n|^2\, t \to 0$, and $\sum_k (\Delta t_k)^2 \leq |\Pi_n|\, t \to 0$, establishing the differential rules Eq. (4.15). □

Remark 4.5 (Almost-Sure Convergence along Refining Partitions)**.** If the sequence of partitions $\{\Pi_n\}$ satisfies $\sum_{n=1}^{\infty} |\Pi_n| < \infty$ (for instance, the dyadic partitions $t_k^{(n)} = kt/2^n$ with $|\Pi_n| = t2^{-n}$), then summing Eq. (4.23) over n and applying Chebyshev's inequality and the Borel–Cantelli lemma shows that $Q_n^{ij}(t) \to \delta_{ij}\, t$ *almost surely* ($\mathbb{P}$-a.s.), not merely in $L^2(\Omega)$. Thus, for almost every individual sample path of thermal noise in a circuit, the sum of squared Brownian increments over $[0, t]$ converges deterministically to the elapsed time t.

4.2 The Itô Stochastic Integral, the L^2 Isometry, and Martingales

We now face the central constructive problem of stochastic calculus: given a stochastic process $H_t(\omega)$ (such as a state-dependent noise amplitude or a force evaluated along a fluctuating trajectory), how should we define the stochastic integral $\int_0^T H_t\, \mathrm{d}W_t$?

4.2.1 Why the Discretization Point Matters: A Canonical Calculation

In ordinary Riemann–Stieltjes calculus, if g is continuously differentiable (or of bounded variation) and H is continuous, the Riemann sum $\sum_{k=0}^{n-1} H_{\tau_k}\,[g(t_{k+1}) - g(t_k)]$ converges to the same integral regardless of which evaluation point $\tau_k \in [t_k, t_{k+1}]$ we choose inside each subinterval. For Brownian motion, the non-vanishing quadratic variation Theorem 4.4 destroys this invariance.

To see this explicitly, we parameterize the evaluation point inside $[t_k, t_{k+1}]$ by a weight $\lambda \in [0, 1]$, setting $\tau_k^{(\lambda)} := (1-\lambda)t_k + \lambda t_{k+1}$ (or in value space, evaluating the integrand at $(1-\lambda)W_{t_k} + \lambda W_{t_{k+1}}$), and compute the elementary integral $\int_0^T W_t\, \mathrm{d}W_t$.

Proposition 4.6 (Discretization Sensitivity of $\int_0^T W_t \, \mathrm{d}W_t$)**.** *Let $\Pi_n = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ be a sequence of partitions of $[0, T]$ with $|\Pi_n| \to 0$. For any fixed interpolation parameter $\lambda \in [0, 1]$, define the λ-Riemann sum*

$$S_n^{(\lambda)}(T) := \sum_{k=0}^{n-1} \left[(1-\lambda)W_{t_k} + \lambda W_{t_{k+1}}\right] \left(W_{t_{k+1}} - W_{t_k}\right). \tag{4.24}$$

Then as $|\Pi_n| \to 0$, $S_n^{(\lambda)}(T)$ converges in $L^2(\Omega)$ to

$$\lim_{|\Pi_n|\to 0} S_n^{(\lambda)}(T) = \frac{1}{2}W_T^2 + \left(\lambda - \frac{1}{2}\right) T. \tag{4.25}$$

Proof. We use the algebraic identity that expresses any linear combination $(1-\lambda)a + \lambda b$ in terms of the left endpoint a and the increment $b - a$:

$$(1-\lambda)W_{t_k} + \lambda W_{t_{k+1}} = W_{t_k} + \lambda(W_{t_{k+1}} - W_{t_k}). \tag{4.26}$$

Furthermore, from the identity $(b-a)^2 = b^2 - a^2 - 2a(b-a)$, we can isolate the left-endpoint product $a(b-a) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(b-a)^2$, which for $a = W_{t_k}$ and $b = W_{t_{k+1}}$ reads

$$W_{t_k}(W_{t_{k+1}} - W_{t_k}) = \frac{1}{2}\left(W_{t_{k+1}}^2 - W_{t_k}^2\right) - \frac{1}{2}\left(W_{t_{k+1}} - W_{t_k}\right)^2. \tag{4.27}$$

Substituting Eq. (4.27) into the λ-Riemann sum Eq. (4.24) gives

$$\begin{aligned} S_n^{(\lambda)}(T) &= \sum_{k=0}^{n-1} W_{t_k}(W_{t_{k+1}} - W_{t_k}) + \lambda \sum_{k=0}^{n-1} \left(W_{t_{k+1}} - W_{t_k}\right)^2 \\ &= \frac{1}{2}\sum_{k=0}^{n-1}\left(W_{t_{k+1}}^2 - W_{t_k}^2\right) + \left(\lambda - \frac{1}{2}\right)\sum_{k=0}^{n-1}\left(W_{t_{k+1}} - W_{t_k}\right)^2. \end{aligned} \tag{4.28}$$

The first sum on the right-hand side of Eq. (4.28) is a telescoping sum that collapses exactly to $\frac{1}{2}(W_{t_n}^2 - W_{t_0}^2) = \frac{1}{2}W_T^2$ (since $W_0 = 0$). The second sum is precisely $(\lambda - \frac{1}{2})Q_n^{11}(T)$, where $Q_n^{11}(T)$ is the quadratic variation sum from Theorem 4.4. Taking the $L^2(\Omega)$ limit as $|\Pi_n| \to 0$ and invoking $Q_n^{11}(T) \to T$ yields Eq. (4.25). □

Proposition 4.6 shows that a stochastic integral is defined only once a discretization convention is chosen. Three values of λ are in common use:

1. $\lambda = 0$ **(Itô / Left-Endpoint Convention):** Evaluating the integrand at the start of the time step t_k yields

$$\int_0^T W_t \cdot \mathrm{d}W_t = \frac{1}{2}W_T^2 - \frac{1}{2}T. \tag{4.29}$$

   Taking the expectation of Eq. (4.29) gives $\mathbb{E}\left[\int_0^T W_t \cdot \mathrm{d}W_t\right] = \frac{1}{2}\mathbb{E}[W_T^2] - \frac{1}{2}T = 0$, because at each step the integrand W_{t_k} is uncorrelated with the future increment $W_{t_{k+1}} - W_{t_k}$. However, the ordinary chain rule is modified by the extra term $-\frac{1}{2}T$.

2. $\lambda = 1/2$ **(Stratonovich / Midpoint Convention):** Evaluating the integrand at the symmetric midpoint yields

$$\int_0^T W_t \circ \mathrm{d}W_t = \frac{1}{2}W_T^2, \tag{4.30}$$

   which obeys the ordinary Newton–Leibniz chain rule $\mathrm{d}(\frac{1}{2}W_t^2) = W_t \circ \mathrm{d}W_t$, at the cost of having a non-zero mean $\mathbb{E}\left[\int_0^T W_t \circ \mathrm{d}W_t\right] = \frac{1}{2}T \neq 0$ because the midpoint $\frac{1}{2}(W_{t_k} + W_{t_{k+1}})$ is correlated with the increment $W_{t_{k+1}} - W_{t_k}$.

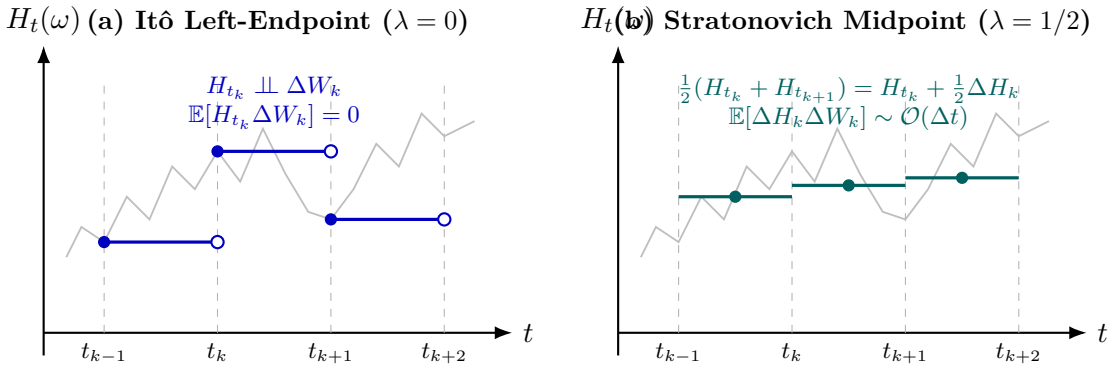


Figure 4.1: Geometric comparison of the Itô ($\lambda = 0$, left-endpoint) and Stratonovich ($\lambda = 1/2$, midpoint) discretizations of a stochastic integral $\int_0^T H_t \, \mathrm{d}W_t$. (a) In the Itô integral, the integrand is frozen at the left endpoint t_k, making H_{t_k} strictly non-anticipating and statistically independent of the future Brownian kick $\Delta W_k = W_{t_{k+1}} - W_{t_k}$. (b) In the Stratonovich integral, evaluating at the midpoint adds $\frac{1}{2}\Delta H_k \Delta W_k \sim \mathcal{O}(\Delta t)$, correlating the integrand with the noise increment and restoring the classical time-reversal-symmetric chain rule.

3. $\lambda = 1$ **(Hänggi–Klimontovich / Post-Point / Kinetic Convention):** Evaluating at the right endpoint t_{k+1} gives $\frac{1}{2}W_T^2 + \frac{1}{2}T$, which occasionally arises when deriving Fokker–Planck equations with state-dependent diffusion where one desires $\mathcal{L}_{\mathrm{FP}}\rho = \mathrm{div}(\cdots + \mathbf{D}\nabla\rho)$ with $\mathbf{D}(\mathbf{x})$ standing to the left of one gradient and to the right of the other (Section 5.5).

4.2.2 Construction of the Itô Integral and the Itô L^2 Isometry

Kiyosi Itô (1944) observed that the left-endpoint convention ($\lambda = 0$, Fig. 4.1(a)) gives the stochastic integral a Hilbert-space structure: the map from the integrand H to the integral $\int_0^T H_t \cdot \mathrm{d}W_t$ is an *isometry* in L^2. We first define the Hilbert space of square-integrable adapted processes.

Definition 4.7 (The Space $\mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$ of Square-Integrable Adapted Processes)**.** Fix a terminal time $T > 0$ and a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ carrying a standard d-dimensional Brownian motion $\mathbf{W}_t$. We denote by $\mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$ (or simply $\mathcal{M}^2([0,T])$ when $n = d = 1$) the space of all $\mathbb{R}^{n\times d}$-valued processes $\mathbf{H} = \{\mathbf{H}_t(\omega)\}_{t\in[0,T]}$ that are jointly $(\mathcal{B}([0,T]) \otimes \mathcal{F})$-measurable, $\{\mathcal{F}_t\}$-adapted, and satisfy

$$\|\mathbf{H}\|^2_{\mathcal{M}^2([0,T])} := \mathbb{E}\left[\int_0^T \|\mathbf{H}_t\|_{\mathrm{F}}^2 \, \mathrm{d}t\right] = \int_0^T \mathbb{E}\Big[\mathrm{tr}\Big(\mathbf{H}_t\mathbf{H}_t^{\top}\Big)\Big] \, \mathrm{d}t < \infty, \tag{4.31}$$

where $\|\mathbf{H}_t\|_{\mathrm{F}}^2 = \sum_{i=1}^n \sum_{\alpha=1}^d H_{i\alpha,t}^2$ is the squared Frobenius norm.

A process $\mathbf{H}^{(n)} \in \mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$ is called *elementary* (or *simple*) if there exists a deterministic partition $0 = t_0 < t_1 < \cdots < t_m = T$ and bounded $\mathcal{F}_{t_k}$-measurable random matrices $\mathbf{\Phi}_k(\omega) \in \mathbb{R}^{n\times d}$ such that

$$\mathbf{H}_t^{(n)}(\omega) = \sum_{k=0}^{m-1} \mathbf{\Phi}_k(\omega) \, \mathbf{1}_{[t_k, t_{k+1})}(t). \tag{4.32}$$

For such an elementary process, its **Itô stochastic integral** over $[0,T]$ is defined unambiguously

by the left-endpoint sum

$$I_T\Big(\mathbf{H}^{(n)}\Big) := \int_0^T \mathbf{H}_t^{(n)} \cdot \mathrm{d}\mathbf{W}_t := \sum_{k=0}^{m-1} \mathbf{\Phi}_k \left(\mathbf{W}_{t_{k+1}} - \mathbf{W}_{t_k}\right) \in \mathbb{R}^n. \tag{4.33}$$

We now prove that Eq. (4.33) preserves the L^2 norm, allowing us to extend I_T uniquely by continuity to all of $\mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$.

Theorem 4.8 (The Itô L^2 Isometry and Extension to $\mathcal{M}^2$)**.** *For every elementary process $\mathbf{H}^{(n)} \in \mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$, the Itô integral Eq. (4.33) has zero mean,*

$$\mathbb{E}\left[\int_0^T \mathbf{H}_t^{(n)} \cdot \mathrm{d}\mathbf{W}_t\right] = \mathbf{0}, \tag{4.34}$$

*and satisfies the **Itô L^2 isometry**:*

$$\mathbb{E}\left[\left\|\int_0^T \mathbf{H}_t^{(n)} \cdot \mathrm{d}\mathbf{W}_t\right\|^2\right] = \mathbb{E}\left[\int_0^T \left\|\mathbf{H}_t^{(n)}\right\|_{\mathrm{F}}^2 \,\mathrm{d}t\right]. \tag{4.35}$$

Since the subspace of elementary processes is dense in the Hilbert space $\mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$, the linear operator I_T extends uniquely to a linear isometry from $\mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$ into $L^2(\Omega,\mathcal{F}_T,\mathbb{P};\mathbb{R}^n)$ such that Eqs. (4.34) and (4.35) hold for every $\mathbf{H} \in \mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$. Moreover, for any two processes $\mathbf{H},\mathbf{G} \in \mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$, their covariance matrix is given by the inner-product polarization identity

$$\mathbb{E}\left[\left(\int_0^T \mathbf{H}_t \cdot \mathrm{d}\mathbf{W}_t\right)\left(\int_0^T \mathbf{G}_t \cdot \mathrm{d}\mathbf{W}_t\right)^\top\right] = \int_0^T \mathbb{E}\Big[\mathbf{H}_t\mathbf{G}_t^\top\Big]\,\mathrm{d}t. \tag{4.36}$$

Proof. Let $\mathbf{H}_t^{(n)} = \sum_{k=0}^{m-1} \mathbf{\Phi}_k \mathbf{1}_{[t_k,t_{k+1})}(t)$ be an elementary process, and denote $\Delta\mathbf{W}_k := \mathbf{W}_{t_{k+1}} - \mathbf{W}_{t_k}$. To prove Eq. (4.34), we use the tower property of conditional expectation ($\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_{t_k}]]$). Since $\mathbf{\Phi}_k$ is $\mathcal{F}_{t_k}$-measurable and $\Delta\mathbf{W}_k$ is independent of $\mathcal{F}_{t_k}$, we have

$$\mathbb{E}\Big[I_T\Big(\mathbf{H}^{(n)}\Big)\Big] = \sum_{k=0}^{m-1} \mathbb{E}[\mathbb{E}[\mathbf{\Phi}_k\Delta\mathbf{W}_k \mid \mathcal{F}_{t_k}]] = \sum_{k=0}^{m-1} \mathbb{E}[\mathbf{\Phi}_k\,\mathbb{E}[\Delta\mathbf{W}_k \mid \mathcal{F}_{t_k}]] = \sum_{k=0}^{m-1} \mathbb{E}[\mathbf{\Phi}_k \cdot \mathbf{0}] = \mathbf{0}. \tag{4.37}$$

Next we compute the outer-product expectation $\mathbb{E}\Big[I_T(\mathbf{H}^{(n)})I_T(\mathbf{H}^{(n)})^\top\Big]$. Expanding the double sum over $j,k \in \{0,\dots,m-1\}$ gives

$$\mathbb{E}\Big[I_T\Big(\mathbf{H}^{(n)}\Big)\, I_T\Big(\mathbf{H}^{(n)}\Big)^\top\Big] = \sum_{j=0}^{m-1}\sum_{k=0}^{m-1} \mathbb{E}\Big[\mathbf{\Phi}_j\Delta\mathbf{W}_j\Delta\mathbf{W}_k^\top\mathbf{\Phi}_k^\top\Big]. \tag{4.38}$$

Suppose first that $j < k$. Then $t_j < t_{j+1} \le t_k$, so $\mathbf{\Phi}_j$, $\Delta\mathbf{W}_j$, and $\mathbf{\Phi}_k$ are all $\mathcal{F}_{t_k}$-measurable, whereas $\Delta\mathbf{W}_k = \mathbf{W}_{t_{k+1}} - \mathbf{W}_{t_k}$ is independent of $\mathcal{F}_{t_k}$ with mean $\mathbf{0}$. Conditioning on $\mathcal{F}_{t_k}$ yields

$$\mathbb{E}\Big[\mathbf{\Phi}_j\Delta\mathbf{W}_j\Delta\mathbf{W}_k^\top\mathbf{\Phi}_k^\top\Big] = \mathbb{E}\Big[\mathbf{\Phi}_j\Delta\mathbf{W}_j\mathbb{E}\Big[\Delta\mathbf{W}_k^\top \,\Big|\, \mathcal{F}_{t_k}\Big]\mathbf{\Phi}_k^\top\Big] = \mathbf{0} \qquad (j<k). \tag{4.39}$$

By symmetry, the terms with $j > k$ also vanish. Only the diagonal terms $j = k$ survive. For $j = k$, conditioning on $\mathcal{F}_{t_k}$ and using $\mathbb{E}\Big[\Delta\mathbf{W}_k\Delta\mathbf{W}_k^\top \mid \mathcal{F}_{t_k}\Big] = \Delta t_k\mathbf{I}_d$ (from Eq. (4.3)) yields

$$\mathbb{E}\Big[\mathbf{\Phi}_k\Delta\mathbf{W}_k\Delta\mathbf{W}_k^\top\mathbf{\Phi}_k^\top\Big] = \mathbb{E}\Big[\mathbf{\Phi}_k\mathbb{E}\Big[\Delta\mathbf{W}_k\Delta\mathbf{W}_k^\top \,\Big|\, \mathcal{F}_{t_k}\Big]\mathbf{\Phi}_k^\top\Big]$$

$$= \mathbb{E}\Big[\mathbf{\Phi}_k \left(\Delta t_k \mathbf{I}_d\right) \mathbf{\Phi}_k^\top\Big] = \mathbb{E}\Big[\mathbf{\Phi}_k \mathbf{\Phi}_k^\top\Big]\, \Delta t_k. \tag{4.40}$$

Summing Eq. (4.40) over $k = 0, \ldots, m-1$ and taking the trace $\operatorname{tr}(\cdot)$ gives

$$\begin{aligned}\mathbb{E}\left[\left\|I_T\Big(\mathbf{H}^{(n)}\Big)\right\|^2\right] &= \operatorname{tr}\mathbb{E}\left[I_T\Big(\mathbf{H}^{(n)}\Big)\, I_T\Big(\mathbf{H}^{(n)}\Big)^\top\right] = \sum_{k=0}^{m-1} \mathbb{E}\Big[\operatorname{tr}\Big(\mathbf{\Phi}_k \mathbf{\Phi}_k^\top\Big)\Big]\,(t_{k+1} - t_k) \\ &= \mathbb{E}\left[\int_0^T \left\|\mathbf{H}_t^{(n)}\right\|_{\mathrm{F}}^2 \,\mathrm{d}t\right], \end{aligned} \tag{4.41}$$

which is Eq. (4.35). For a general $\mathbf{H} \in \mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$, standard L^2 approximation (truncating $\mathbf{H}$, smoothing by a backward-in-time convolution so adaptedness is preserved, and sampling on a fine partition; see [59, 80]) produces a sequence of elementary processes $\mathbf{H}^{(n)}$ with $\left\|\mathbf{H}^{(n)} - \mathbf{H}\right\|_{\mathcal{M}^2} \to 0$. By Eq. (4.35), $\{I_T(\mathbf{H}^{(n)})\}$ is a Cauchy sequence in the complete Hilbert space $L^2(\Omega;\mathbb{R}^n)$, and we define $\int_0^T \mathbf{H}_t \cdot \mathrm{d}\mathbf{W}_t := \lim_{n\to\infty} I_T(\mathbf{H}^{(n)})$ in $L^2(\Omega)$. Applying the same argument to $\mathbf{H} \pm \mathbf{G}$ and polarizing yields Eq. (4.36). □

4.2.3 The Martingale Property of Itô Integrals

When we vary the upper limit of integration $t \in [0,T]$, the Itô integral $\mathbf{I}_t := \int_0^t \mathbf{H}_s \cdot \mathrm{d}\mathbf{W}_s := \int_0^T \mathbf{H}_s \mathbf{1}_{[0,t]}(s) \cdot \mathrm{d}\mathbf{W}_s$ becomes a stochastic process in its own right. Its defining property is that it is a *martingale*: the continuous-time version of a fair game, in which the best prediction of a future value $\mathbf{I}_t$ given all history $\mathcal{F}_s$ up to time $s \le t$ is simply the present value $\mathbf{I}_s$.

Definition 4.9 (Continuous-Time Martingale)**.** An $\{\mathcal{F}_t\}$-adapted process $\mathbf{M} = \{\mathbf{M}_t\}_{t\ge 0}$ taking values in $\mathbb{R}^N$ with $\mathbb{E}[\|\mathbf{M}_t\|] < \infty$ for all $t \ge 0$ is called an $\{\mathcal{F}_t\}$-*martingale* if for all $0 \le s \le t < \infty$,

$$\mathbb{E}[\mathbf{M}_t \mid \mathcal{F}_s] = \mathbf{M}_s \qquad \mathbb{P}\text{-a.s.} \tag{4.42}$$

If $\mathbf{M}$ is scalar-valued and $\mathbb{E}[M_t \mid \mathcal{F}_s] \ge M_s$ (respectively $\le M_s$) a.s., it is called a *submartingale* (respectively *supermartingale*).

Theorem 4.10 (Martingale Property and Continuity of the Itô Integral)**.** *Let $\mathbf{H} \in \mathcal{M}^2([0,T];\mathbb{R}^{n\times d})$. Then the Itô integral process $\mathbf{I}_t = \int_0^t \mathbf{H}_s \cdot \mathrm{d}\mathbf{W}_s$ $(0 \le t \le T)$ possesses a version with continuous sample paths $t \mapsto \mathbf{I}_t(\omega)$ that is a square-integrable $\{\mathcal{F}_t\}$-martingale satisfying Doob's maximal L^2 inequality:*

$$\mathbb{E}\left[\sup_{0\le t\le T}\left\|\int_0^t \mathbf{H}_s \cdot \mathrm{d}\mathbf{W}_s\right\|^2\right] \le 4\,\mathbb{E}\left[\int_0^T \|\mathbf{H}_s\|_{\mathrm{F}}^2 \,\mathrm{d}s\right]. \tag{4.43}$$

Proof. Fix $0 \le s < t \le T$. By linearity of the stochastic integral, $\mathbf{I}_t - \mathbf{I}_s = \int_s^t \mathbf{H}_u \cdot \mathrm{d}\mathbf{W}_u$. For any elementary process $\mathbf{H}_u^{(n)} = \sum_k \mathbf{\Phi}_k \mathbf{1}_{[t_k, t_{k+1})}(u)$ whose partition includes s and t, every term in $\int_s^t \mathbf{H}_u^{(n)} \cdot \mathrm{d}\mathbf{W}_u$ has $t_k \ge s$, so $\mathcal{F}_s \subseteq \mathcal{F}_{t_k}$. By the tower property $\mathbb{E}[\mathbb{E}[\cdot \mid \mathcal{F}_{t_k}] \mid \mathcal{F}_s] = \mathbb{E}[\cdot \mid \mathcal{F}_s]$, we obtain

$$\mathbb{E}\left[\int_s^t \mathbf{H}_u^{(n)} \cdot \mathrm{d}\mathbf{W}_u \,\middle|\, \mathcal{F}_s\right] = \sum_{t_k \ge s} \mathbb{E}[\mathbf{\Phi}_k \mathbb{E}[\Delta\mathbf{W}_k \mid \mathcal{F}_{t_k}] \mid \mathcal{F}_s] = \mathbf{0}. \tag{4.44}$$

Since conditional expectation $\mathbb{E}[\cdot \mid \mathcal{F}_s]$ is a contraction on $L^2(\Omega)$, passing to the L^2 limit $n \to \infty$ gives $\mathbb{E}[\mathbf{I}_t - \mathbf{I}_s \mid \mathcal{F}_s] = \mathbf{0}$, so $\mathbb{E}[\mathbf{I}_t \mid \mathcal{F}_s] = \mathbf{I}_s$ a.s. Doob's martingale inequality then yields Eq. (4.43) for each elementary approximant $\mathbf{I}_t^{(n)}$, and choosing a fast subsequence $\left\|\mathbf{H}^{(n_k)} - \mathbf{H}\right\|_{\mathcal{M}^2} \le 2^{-k}$ implies uniform convergence on $[0,T]$ almost surely, ensuring that the limit $\mathbf{I}_t$ has continuous paths because each polygonal path $\mathbf{I}_t^{(n_k)}$ is continuous. □

Physics Insight: Why Martingales Separate Systematic Drift from Thermal Agitation

In statistical physics and stochastic thermodynamics, a stochastic differential equation is written in Itô form as $\mathrm{d}\mathbf{X}_t = \mathbf{a}(\mathbf{X}_t, t)\,\mathrm{d}t + \mathbf{B}(\mathbf{X}_t, t)\cdot\mathrm{d}\mathbf{W}_t$. Integrating from 0 to t gives the Doob–Meyer semimartingale decomposition:

$$\mathbf{X}_t = \mathbf{X}_0 + \underbrace{\int_0^t \mathbf{a}(\mathbf{X}_s, s)\,\mathrm{d}s}_{\text{bounded-variation drift}} + \underbrace{\int_0^t \mathbf{B}(\mathbf{X}_s, s)\cdot\mathrm{d}\mathbf{W}_s}_{\text{zero-mean martingale } \mathbf{M}_t}. \tag{4.45}$$

Because the Itô integral $\mathbf{M}_t$ is a martingale, $\mathbb{E}[\mathbf{M}_t - \mathbf{M}_s \mid \mathcal{F}_s] = \mathbf{0}$: it contributes *zero* conditional mean displacement over an infinitesimal step $\mathrm{d}t$. Thus, the Itô drift vector $\mathbf{a}(\mathbf{x}, t) = \lim_{\Delta t\to 0^+}(\Delta t)^{-1}\mathbb{E}[\mathbf{X}_{t+\Delta t} - \mathbf{X}_t \mid \mathbf{X}_t = \mathbf{x}]$ is the exact conditional mean velocity of the state vector. In contrast, in the Stratonovich representation, the stochastic integral $\int_0^t \mathbf{B}(\mathbf{X}_s, s)\circ\mathrm{d}\mathbf{W}_s$ is *not* a martingale when $\mathbf{B}$ depends on $\mathbf{x}$; it hides a systematic drift inside the noise integral.

4.3 Multi-Dimensional Itô's Formula

Suppose a physical state vector $\mathbf{X}_t \in \mathbb{R}^N$ (such as the N capacitor voltages $\mathbf{v}_t$ of a thermodynamic processor) evolves according to an Itô stochastic differential equation, and we want the time evolution of a smooth observable $f(\mathbf{X}_t, t)$, such as the circuit's electrostatic energy $E(\mathbf{v}_t)$, a Lyapunov function $V(\mathbf{v}_t)$, or the log-probability $\ln\rho(\mathbf{v}_t, t)$.

Definition 4.11 (Multi-Dimensional Itô Process and Diffusion Tensor)**.** An $\mathbb{R}^N$-valued $\{\mathcal{F}_t\}$-adapted continuous process $\mathbf{X} = \{\mathbf{X}_t\}_{t\in[0,T]}$ is called an *Itô process* (or *Itô semimartingale*) driven by a standard m-dimensional Brownian motion $\mathbf{W}_t = (W_{1,t}, \ldots, W_{m,t})^\top$ if there exist an $\{\mathcal{F}_t\}$-adapted drift process $\mathbf{a}_t \in \mathbb{R}^N$ with $\int_0^T \|\mathbf{a}_t\|\,\mathrm{d}t < \infty$ a.s. and a volatility (noise-coupling) matrix process $\mathbf{B}_t \in \mathcal{M}^2([0,T];\mathbb{R}^{N\times m})$ such that for all $t \in [0,T]$,

$$\mathbf{X}_t = \mathbf{X}_0 + \int_0^t \mathbf{a}_s\,\mathrm{d}s + \int_0^t \mathbf{B}_s\cdot\mathrm{d}\mathbf{W}_s, \tag{4.46}$$

which we write in differential notation (often omitting the explicit dot $\cdot$ when Itô is understood) as

$$\mathrm{d}\mathbf{X}_t = \mathbf{a}_t\,\mathrm{d}t + \mathbf{B}_t\,\mathrm{d}\mathbf{W}_t, \qquad \text{or in components} \qquad \mathrm{d}X_{i,t} = a_{i,t}\,\mathrm{d}t + \sum_{\alpha=1}^{m} B_{i\alpha,t}\,\mathrm{d}W_{\alpha,t}. \tag{4.47}$$

The symmetric positive-semidefinite matrix

$$\mathbf{D}_t := \frac{1}{2}\mathbf{B}_t\mathbf{B}_t^\top \in \mathbb{R}^{N\times N}, \qquad D_{ij,t} = \frac{1}{2}\sum_{\alpha=1}^{m} B_{i\alpha,t}B_{j\alpha,t}, \tag{4.48}$$

is called the *diffusion tensor* of the process $\mathbf{X}_t$.

Using the differential multiplication rules Eq. (4.15) ($\mathrm{d}W_{\alpha,t}\,\mathrm{d}W_{\beta,t} = \delta_{\alpha\beta}\,\mathrm{d}t$, $\mathrm{d}W_{\alpha,t}\,\mathrm{d}t = 0$, $(\mathrm{d}t)^2 = 0$), the quadratic covariation between components $X_{i,t}$ and $X_{j,t}$ is computed immediately from Eq. (4.47):

$$\mathrm{d}X_{i,t}\,\mathrm{d}X_{j,t} = \left(a_{i,t}\,\mathrm{d}t + \sum_{\alpha=1}^{m} B_{i\alpha,t}\,\mathrm{d}W_{\alpha,t}\right)\left(a_{j,t}\,\mathrm{d}t + \sum_{\beta=1}^{m} B_{j\beta,t}\,\mathrm{d}W_{\beta,t}\right)$$

$$= \sum_{\alpha=1}^{m}\sum_{\beta=1}^{m} B_{i\alpha,t}B_{j\beta,t}\,\underbrace{\mathrm{d}W_{\alpha,t}\,\mathrm{d}W_{\beta,t}}_{\delta_{\alpha\beta}\,\mathrm{d}t} = \left(\sum_{\alpha=1}^{m} B_{i\alpha,t}B_{j\alpha,t}\right)\mathrm{d}t = 2D_{ij,t}\,\mathrm{d}t. \tag{4.49}$$

We now state and prove Itô's formula, the central result of stochastic calculus.

Theorem 4.12 (Multi-Dimensional Itô's Formula)**.** *Let $\mathbf{X}_t \in \mathbb{R}^N$ be an Itô process satisfying Eq. (4.47), and let $f : \mathbb{R}^N \times [0,T] \to \mathbb{R}$ be a function of class $C^{2,1}(\mathbb{R}^N \times [0,T])$ (twice continuously differentiable in $\mathbf{x} \in \mathbb{R}^N$ and once continuously differentiable in $t \in [0,T]$). Then $Y_t := f(\mathbf{X}_t, t)$ is again a scalar Itô process, and its stochastic differential is given by*

$$\mathrm{d}f(\mathbf{X}_t,t) = \frac{\partial f}{\partial t}\,\mathrm{d}t + \sum_{i=1}^{N}\frac{\partial f}{\partial x_i}\,\mathrm{d}X_{i,t} + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{\partial^2 f}{\partial x_i \partial x_j}\,\mathrm{d}X_{i,t}\,\mathrm{d}X_{j,t} \tag{4.50}$$

$$= \left[\frac{\partial f}{\partial t} + \mathbf{a}_t \cdot \nabla f(\mathbf{X}_t,t) + \frac{1}{2}\,\mathrm{tr}\Big(\mathbf{B}_t\mathbf{B}_t^{\top}\nabla^2 f(\mathbf{X}_t,t)\Big)\right]\mathrm{d}t + \nabla f(\mathbf{X}_t,t)^{\top}\mathbf{B}_t \cdot \mathrm{d}\mathbf{W}_t, \tag{4.51}$$

where all partial derivatives of f are evaluated at $(\mathbf{X}_t, t)$ and $\mathrm{tr}(\mathbf{D}_t\nabla^2 f) = \sum_{i,j=1}^{N} D_{ij,t}\frac{\partial^2 f}{\partial x_i \partial x_j}$.

Proof. By standard localization (stopping $\mathbf{X}_t$, $\mathbf{a}_t$, and $\mathbf{B}_t$ when they exit a ball of radius R and letting $R \to \infty$), we may assume without loss of generality that f and its first and second derivatives are bounded and uniformly continuous, and that $\mathbf{a}_t$ and $\mathbf{B}_t$ are elementary processes adapted to a partition $\Pi_n = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ whose mesh $|\Pi_n| \to 0$. Over the partition Π_n, we write the total change in f as a telescoping sum:

$$f(\mathbf{X}_t,t) - f(\mathbf{X}_0,0) = \sum_{k=0}^{n-1}\left[f(\mathbf{X}_{t_{k+1}},t_{k+1}) - f(\mathbf{X}_{t_k},t_k)\right]. \tag{4.52}$$

Denoting $\Delta t_k := t_{k+1} - t_k$ and $\Delta\mathbf{X}_k := \mathbf{X}_{t_{k+1}} - \mathbf{X}_{t_k} = \mathbf{a}_{t_k}\Delta t_k + \mathbf{B}_{t_k}\Delta\mathbf{W}_k$, we apply Taylor's theorem up to first order in Δt_k and second order in $\Delta\mathbf{X}_k$ on each subinterval $[t_k, t_{k+1}]$:

$$\begin{aligned} f(\mathbf{X}_{t_{k+1}},t_{k+1}) - f(\mathbf{X}_{t_k},t_k) &= \frac{\partial f}{\partial t}(\mathbf{X}_{t_k},\theta_k)\,\Delta t_k + \sum_{i=1}^{N}\frac{\partial f}{\partial x_i}(\mathbf{X}_{t_k},t_k)\,\Delta X_{i,k} \\ &\quad + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{\partial^2 f}{\partial x_i\partial x_j}(\mathbf{X}_{t_k},t_k)\,\Delta X_{i,k}\Delta X_{j,k} + R_k, \end{aligned} \tag{4.53}$$

where $\theta_k \in [t_k, t_{k+1}]$ and the remainder R_k satisfies $|R_k| \le \omega_n(\|\Delta\mathbf{X}_k\|)\,\|\Delta\mathbf{X}_k\|^2$, with $\omega_n(\delta) := \sup_{\|\mathbf{x}-\mathbf{y}\|\le\delta}\|\nabla^2 f(\mathbf{x},t_k) - \nabla^2 f(\mathbf{y},t_k)\| \to 0$ as $\delta \to 0$ by uniform continuity of $\nabla^2 f$.

We analyze the four sums produced when Eq. (4.53) is inserted into Eq. (4.52) as $|\Pi_n| \to 0$:

Step 1: *Time derivative sum:* Because $t \mapsto \frac{\partial f}{\partial t}(\mathbf{X}_t,t)$ is continuous on $[0,t]$, the Riemann sum $\sum_{k=0}^{n-1}\frac{\partial f}{\partial t}(\mathbf{X}_{t_k},\theta_k)\Delta t_k$ converges pathwise and in $L^2(\Omega)$ to the ordinary Lebesgue integral $\int_0^t \frac{\partial f}{\partial s}(\mathbf{X}_s,s)\,\mathrm{d}s$.

Step 2: *First-order spatial increment sum:* Substituting $\Delta X_{i,k} = a_{i,t_k}\Delta t_k + \sum_{\alpha=1}^{m} B_{i\alpha,t_k}\Delta W_{\alpha,k}$ gives

$$\sum_{k=0}^{n-1}\sum_{i=1}^{N}\frac{\partial f}{\partial x_i}(\mathbf{X}_{t_k},t_k)\,\Delta X_{i,k} = \sum_{k=0}^{n-1}\mathbf{a}_{t_k}\cdot\nabla f(\mathbf{X}_{t_k},t_k)\,\Delta t_k + \sum_{k=0}^{n-1}\nabla f(\mathbf{X}_{t_k},t_k)^{\top}\mathbf{B}_{t_k}\Delta\mathbf{W}_k. \tag{4.54}$$

As $|\Pi_n| \to 0$, the first sum converges to $\int_0^t \mathbf{a}_s \cdot \nabla f(\mathbf{X}_s,s)\,\mathrm{d}s$, while by Theorem 4.8 the second sum, whose integrand is evaluated at the left endpoint t_k, converges in $L^2(\Omega)$ to the Itô integral $\int_0^t \nabla f(\mathbf{X}_s,s)^{\top}\mathbf{B}_s \cdot \mathrm{d}\mathbf{W}_s$.

Step 3: *Second-order quadratic variation sum:* Denote $H_{ij,k} := \frac{1}{2}\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{X}_{t_k}, t_k)$ and expand the product $\Delta X_{i,k}\Delta X_{j,k}$ explicitly:

$$\Delta X_{i,k}\Delta X_{j,k} = a_{i,t_k}a_{j,t_k}(\Delta t_k)^2 + \sum_{\alpha=1}^{m}\left(a_{i,t_k}B_{j\alpha,t_k} + a_{j,t_k}B_{i\alpha,t_k}\right)\Delta t_k \Delta W_{\alpha,k} + \sum_{\alpha=1}^{m}\sum_{\beta=1}^{m} B_{i\alpha,t_k}B_{j\beta,t_k}\,\Delta W_{\alpha,k}\Delta W_{\beta,k}. \tag{4.55}$$

When summed over $k = 0, \ldots, n-1$, the $(\Delta t_k)^2$ terms are bounded by $C\,|\Pi_n|\,t \to 0$, and the $\Delta t_k \Delta W_{\alpha,k}$ terms have L^2 norm bounded by $C\left(\sum_k (\Delta t_k)^3\right)^{1/2} \le C\,|\Pi_n|\,\sqrt{t} \to 0$. For the final double sum over α, β, we write $\Delta W_{\alpha,k}\Delta W_{\beta,k} = \delta_{\alpha\beta}\Delta t_k + Y_k^{\alpha\beta}$, where $Y_k^{\alpha\beta} := \Delta W_{\alpha,k}\Delta W_{\beta,k} - \delta_{\alpha\beta}\Delta t_k$ are the zero-mean increments from the proof of Theorem 4.4. Because $H_{ij,k}B_{i\alpha,t_k}B_{j\beta,t_k}$ is $\mathcal{F}_{t_k}$-measurable and bounded by some constant M, whereas $Y_k^{\alpha\beta}$ is independent of $\mathcal{F}_{t_k}$ with $\mathbb{E}\Big[(Y_k^{\alpha\beta})^2\Big] = (1+\delta_{\alpha\beta})(\Delta t_k)^2$, the error sum vanishes in $L^2(\Omega)$:

$$\mathbb{E}\left[\left(\sum_{k=0}^{n-1} H_{ij,k}B_{i\alpha,t_k}B_{j\beta,t_k}Y_k^{\alpha\beta}\right)^2\right] = \sum_{k=0}^{n-1}\mathbb{E}\Big[\left(H_{ij,k}B_{i\alpha,t_k}B_{j\beta,t_k}\right)^2\Big]\,(1+\delta_{\alpha\beta})(\Delta t_k)^2 \le 2M^2 t\;|\Pi_n| \to 0. \tag{4.56}$$

Meanwhile, the deterministic part $\delta_{\alpha\beta}\Delta t_k$ yields the Riemann sum

$$\sum_{k=0}^{n-1}\frac{1}{2}\sum_{i,j=1}^{N}\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{X}_{t_k}, t_k)\left(\sum_{\alpha=1}^{m} B_{i\alpha,t_k}B_{j\alpha,t_k}\right)\Delta t_k \xrightarrow{|\Pi_n|\to 0} \int_0^t \frac{1}{2}\,\mathrm{tr}\Big(\mathbf{B}_s\mathbf{B}_s^\top \nabla^2 f(\mathbf{X}_s, s)\Big)\,\mathrm{d}s. \tag{4.57}$$

Step 4: *Vanishing of the higher-order Taylor remainder:* Since $\mathbb{E}\Big[\|\Delta\mathbf{X}_k\|^4\Big] \le C(\Delta t_k)^2$, Cauchy–Schwarz gives $\sum_k |R_k| \to 0$ in probability as $|\Pi_n| \to 0$.

Combining Steps 1–4 completes the proof of Eqs. (4.50) and (4.51). □

Corollary 4.13 (Itô Product Rule / Stochastic Integration by Parts)**.** *Let X_t and Y_t be two scalar Itô processes. Applying Theorem 4.12 to $f(x,y) = xy$ (for which $\frac{\partial f}{\partial x} = y$, $\frac{\partial f}{\partial y} = x$, $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2} = 0$, and $\frac{\partial^2 f}{\partial x \partial y} = 1$) yields the stochastic product rule:*

$$\mathrm{d}(X_t Y_t) = X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}X_t\,\mathrm{d}Y_t, \tag{4.58}$$

or in integrated form over $[0, T]$,

$$X_T Y_T - X_0 Y_0 = \int_0^T X_t \cdot \mathrm{d}Y_t + \int_0^T Y_t \cdot \mathrm{d}X_t + \int_0^T \mathrm{d}X_t\,\mathrm{d}Y_t. \tag{4.59}$$

Example 4.14 (Electrostatic Energy Evolution in an RC Circuit)**.** Consider a single-node linear RC circuit, the one-node case of the RC sampler (the coupled RC network of Example 1.5), with capacitance C, conductance $G = 1/R$, and Johnson–Nyquist thermal current noise of spectral density $2k_{\mathrm{B}}TG$. The node voltage v_t obeys the Ornstein–Uhlenbeck SDE

$$C\,\mathrm{d}v_t = -Gv_t\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}TG}\,\mathrm{d}W_t \quad\Longleftrightarrow\quad \mathrm{d}v_t = -\frac{G}{C}v_t\,\mathrm{d}t + \frac{\sqrt{2k_{\mathrm{B}}TG}}{C}\,\mathrm{d}W_t. \tag{4.60}$$

How does the capacitor's electrostatic energy $E(v_t) = \frac{1}{2}Cv_t^2$ evolve? If we naively used ordinary calculus, we would write $\mathrm{d}E \stackrel{?}{=} Cv_t\,\mathrm{d}v_t = -Gv_t^2\,\mathrm{d}t + \sqrt{2k_\mathrm{B}TG}\,v_t\,\mathrm{d}W_t$, which upon taking expectations would predict $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[E(v_t)] \stackrel{?}{=} -G\mathbb{E}[v_t^2] \le 0$, implying that the capacitor energy decays to zero even at $T > 0$.

Applying Itô's formula (Theorem 4.12) with $f(v) = \frac{1}{2}Cv^2$, $f'(v) = Cv$, and $f''(v) = C$ instead yields

$$\begin{aligned}\mathrm{d}E(v_t) &= f'(v_t)\,\mathrm{d}v_t + \frac{1}{2}f''(v_t)\,(\mathrm{d}v_t)^2 \\ &= Cv_t\left(-\frac{G}{C}v_t\,\mathrm{d}t + \frac{\sqrt{2k_\mathrm{B}TG}}{C}\cdot\mathrm{d}W_t\right) + \frac{1}{2}C\left(\frac{2k_\mathrm{B}TG}{C^2}\,\mathrm{d}t\right) \\ &= \left(-Gv_t^2 + \frac{k_\mathrm{B}TG}{C}\right)\mathrm{d}t + \sqrt{2k_\mathrm{B}TG}\,v_t\cdot\mathrm{d}W_t. \end{aligned} \tag{4.61}$$

The quadratic variation term $\frac{1}{2}C(\mathrm{d}v_t)^2 = \frac{k_\mathrm{B}TG}{C}\,\mathrm{d}t$ is the average thermal power injected by Johnson–Nyquist fluctuations from the resistor into the capacitor. Taking the expectation of Eq. (4.61) (using the martingale property $\mathbb{E}\left[\int_0^t v_s\cdot\mathrm{d}W_s\right] = 0$) gives the exact energy relaxation ODE:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[E(v_t)] = -G\mathbb{E}\left[v_t^2\right] + \frac{k_\mathrm{B}TG}{C} = -\frac{2G}{C}\left(\mathbb{E}[E(v_t)] - \frac{1}{2}k_\mathrm{B}T\right). \tag{4.62}$$

Thus $\mathbb{E}[E(v_t)] = \frac{1}{2}C\mathbb{E}[v_t^2] \to \frac{1}{2}k_\mathrm{B}T$ exponentially at rate $2/\tau_{RC} = 2G/C$, recovering the classical equipartition theorem $\mathbb{E}[v_\infty^2] = k_\mathrm{B}T/C$ purely from Itô's second-order correction.

4.4 Multiplicative Noise, Stratonovich Calculus, and the Wong–Zakai Theorem

In Example 4.14, the noise amplitude $B = \sqrt{2k_\mathrm{B}TG}/C$ was constant (*additive noise*). Whenever the volatility matrix $\mathbf{B}$ is independent of the state $\mathbf{X}_t$, the Itô ($\lambda = 0$) and Stratonovich ($\lambda = 1/2$) integrals coincide because $\mathbf{B}(\mathbf{X}_{t_k}) = \mathbf{B}(\mathbf{X}_{(t_k+t_{k+1})/2}) = \mathbf{B}$.

In real solid-state devices, however, noise is usually *multiplicative* (state-dependent), meaning $\mathbf{B}(\mathbf{X}_t)$ varies with $\mathbf{X}_t$. For example:

- In a subthreshold MOSFET (Chapters 12 and 13), the drain current depends exponentially on the gate and drain-source voltages, $I_\mathrm{D}(\mathbf{v}) \propto e^{V_\mathrm{GS}/(nV_T)}$, and the shot/thermal current noise spectral density is proportional to the current itself, $S_I(\mathbf{v}) = 2q_e(I_+(\mathbf{v}) + I_-(\mathbf{v}))$, so the noise amplitude scales as $\mathbf{B}(\mathbf{v}) \propto \exp(V_\mathrm{GS}/(2nV_T))$.

- In a nonlinear MOS gate or p-n junction varactor, the differential capacitance $C(v) = \frac{\mathrm{d}q}{\mathrm{d}v}$ depends on the node voltage v, so dividing current noise by $C(v)$ makes the voltage noise amplitude $\mathbf{B}(v) \propto C(v)^{-1}$ state-dependent.

When we write down a physical circuit equation with state-dependent noise $\mathbf{B}(\mathbf{X}_t)\,\boldsymbol{\xi}_t$, should we interpret $\mathbf{B}(\mathbf{X}_t)\,\mathrm{d}\mathbf{W}_t$ in the sense of Itô or Stratonovich? This is the *Itô–Stratonovich dilemma* of statistical physics [39, 98, 112]. As we will see, it is settled by examining how white noise arises from the underlying physics.

4.4.1 The Stratonovich Midpoint Integral and the Exact Drift Conversion Formula

We first define the Stratonovich integral and relate it to the Itô integral.

Definition 4.15 (Stratonovich Stochastic Integral)**.** Let $\mathbf{X}_t \in \mathbb{R}^N$ be a continuous semimartingale and let $\mathbf{B} : \mathbb{R}^N \times [0,T] \to \mathbb{R}^{N\times m}$ be a $C^{1,0}$ matrix-valued function. The *Stratonovich stochastic integral* of $\mathbf{B}(\mathbf{X}_t, t)$ with respect to a standard m-dimensional Brownian motion $\mathbf{W}_t$, denoted with an open circle $\circ$, is defined as the $L^2(\Omega)$ (or convergence-in-probability) limit of the symmetric midpoint Riemann sums:

$$\int_0^T \mathbf{B}(\mathbf{X}_t, t) \circ \mathrm{d}\mathbf{W}_t := \lim_{|\Pi_n|\to 0} \sum_{k=0}^{n-1} \mathbf{B}\left(\frac{\mathbf{X}_{t_k} + \mathbf{X}_{t_{k+1}}}{2},\, t_k\right) (\mathbf{W}_{t_{k+1}} - \mathbf{W}_{t_k})\,. \tag{4.63}$$

Theorem 4.16 (Multi-Dimensional Itô–Stratonovich Drift Conversion Formula and Chain Rule)**.** *Let $\mathbf{X}_t \in \mathbb{R}^N$ satisfy the multi-dimensional Stratonovich SDE*

$$\mathrm{d}\mathbf{X}_t = \underline{\mathbf{a}}(\mathbf{X}_t, t)\ \mathrm{d}t + \mathbf{B}(\mathbf{X}_t, t) \circ \mathrm{d}\mathbf{W}_t, \qquad \text{or} \qquad \mathrm{d}X_{i,t} = \underline{a}_i(\mathbf{X}_t, t)\ \mathrm{d}t + \sum_{\alpha=1}^{m} B_{i\alpha}(\mathbf{X}_t, t) \circ \mathrm{d}W_{\alpha,t}, \tag{4.64}$$

where $\underline{\mathbf{a}} : \mathbb{R}^N \times [0,T] \to \mathbb{R}^N$ is continuous and $\mathbf{B} : \mathbb{R}^N \times [0,T] \to \mathbb{R}^{N\times m}$ is continuously differentiable in $\mathbf{x}$. Then:

(a) ***(Itô–Stratonovich Conversion)*** *The Stratonovich stochastic differential is related to the Itô stochastic differential by*

$$\mathbf{B}(\mathbf{X}_t, t) \circ \mathrm{d}\mathbf{W}_t = \mathbf{B}(\mathbf{X}_t, t) \cdot \mathrm{d}\mathbf{W}_t + \mathbf{a}_{\mathrm{sp}}(\mathbf{X}_t, t)\ \mathrm{d}t, \tag{4.65}$$

where the ***noise-induced (spurious) drift vector*** *$\mathbf{a}_{\mathrm{sp}}(\mathbf{x}, t) \in \mathbb{R}^N$ has components*

$$a_{\mathrm{sp},i}(\mathbf{x}, t) := \frac{1}{2} \sum_{j=1}^{N} \sum_{\alpha=1}^{m} B_{j\alpha}(\mathbf{x}, t)\, \frac{\partial B_{i\alpha}(\mathbf{x}, t)}{\partial x_j}, \qquad i \in \{1, \dots, N\}. \tag{4.66}$$

Consequently, the Stratonovich SDE Eq. (4.64) is pathwise equivalent to the Itô SDE

$$\mathrm{d}\mathbf{X}_t = \underbrace{[\underline{\mathbf{a}}(\mathbf{X}_t, t) + \mathbf{a}_{\mathrm{sp}}(\mathbf{X}_t, t)]}_{\mathbf{a}(\mathbf{X}_t, t)}\, \mathrm{d}t + \mathbf{B}(\mathbf{X}_t, t) \cdot \mathrm{d}\mathbf{W}_t. \tag{4.67}$$

(b) ***(Stratonovich Chain Rule)*** *For any smooth function $f \in C^{2,1}(\mathbb{R}^N \times [0,T]; \mathbb{R})$, the Stratonovich differential obeys the ordinary fundamental theorem of calculus (without any second-derivative Hessian correction):*

$$\mathrm{d}f(\mathbf{X}_t, t) = \frac{\partial f}{\partial t}(\mathbf{X}_t, t)\ \mathrm{d}t + \nabla f(\mathbf{X}_t, t) \cdot \circ\, \mathrm{d}\mathbf{X}_t = \left[\frac{\partial f}{\partial t} + \underline{\mathbf{a}} \cdot \nabla f\right] \mathrm{d}t + \nabla f^{\top} \mathbf{B} \circ \mathrm{d}\mathbf{W}_t. \tag{4.68}$$

Proof. To prove part (a), we expand the i-th component of the midpoint integrand in Eq. (4.63) around the left endpoint $\mathbf{X}_{t_k}$ using a first-order Taylor expansion in $\Delta\mathbf{X}_k := \mathbf{X}_{t_{k+1}} - \mathbf{X}_{t_k}$:

$$B_{i\alpha}\left(\mathbf{X}_{t_k} + \frac{1}{2}\Delta\mathbf{X}_k,\, t_k\right) = B_{i\alpha}(\mathbf{X}_{t_k}, t_k) + \frac{1}{2} \sum_{j=1}^{N} \frac{\partial B_{i\alpha}}{\partial x_j}(\mathbf{X}_{t_k}, t_k)\, \Delta X_{j,k} + o(\|\Delta\mathbf{X}_k\|). \tag{4.69}$$

Multiplying Eq. (4.69) by $\Delta W_{\alpha,k} := W_{\alpha,t_{k+1}} - W_{\alpha,t_k}$ and summing over $\alpha = 1, \dots, m$ and $k = 0, \dots, n-1$ gives

$$\sum_{k=0}^{n-1}\sum_{\alpha=1}^{m} B_{i\alpha}\Big(\frac{\mathbf{X}_{t_k}+\mathbf{X}_{t_{k+1}}}{2}, t_k\Big)\,\Delta W_{\alpha,k} = \sum_{k=0}^{n-1}\sum_{\alpha=1}^{m} B_{i\alpha}(\mathbf{X}_{t_k}, t_k)\,\Delta W_{\alpha,k} + \frac{1}{2}\sum_{k=0}^{n-1}\sum_{j=1}^{N}\sum_{\alpha=1}^{m}\frac{\partial B_{i\alpha}}{\partial x_j}(\mathbf{X}_{t_k}, t_k)\,\Delta X_{j,k}\,\Delta W_{\alpha,k} + \mathcal{R}_n. \quad (4.70)$$

As $|\Pi_n| \to 0$, the first sum on the right-hand side of Eq. (4.70) is the left-endpoint Riemann sum, which converges to the Itô integral $\sum_{\alpha=1}^{m}\int_0^T B_{i\alpha}(\mathbf{X}_t, t)\cdot \mathrm{d}W_{\alpha,t}$, while the remainder $\mathcal{R}_n$ is of order $\sum_k \|\Delta\mathbf{X}_k\|^2\,\|\Delta\mathbf{W}_k\| \sim \mathcal{O}(|\Pi_n|^{1/2}) \to 0$. In the second sum, we substitute $\Delta X_{j,k} = a_{j,t_k}\Delta t_k + \sum_{\beta=1}^{m} B_{j\beta}(\mathbf{X}_{t_k}, t_k)\Delta W_{\beta,k} + o(\Delta t_k)$ and apply the quadratic covariation theorem (Theorem 4.4, $\Delta W_{\beta,k}\Delta W_{\alpha,k} \to \delta_{\alpha\beta}\,\mathrm{d}t$, $\Delta t_k \Delta W_{\alpha,k} \to 0$):

$$\Delta X_{j,k}\,\Delta W_{\alpha,k} = \sum_{\beta=1}^{m} B_{j\beta}(\mathbf{X}_{t_k}, t_k)\,\underbrace{\Delta W_{\beta,k}\,\Delta W_{\alpha,k}}_{\to\,\delta_{\alpha\beta}\Delta t_k} + \mathcal{O}\Big((\Delta t_k)^{3/2}\Big) \longrightarrow B_{j\alpha}(\mathbf{X}_{t_k}, t_k)\,\mathrm{d}t. \quad (4.71)$$

Inserting Eq. (4.71) into the second sum of Eq. (4.70) produces the Riemann integral of

$$\frac{1}{2}\sum_{j=1}^{N}\sum_{\alpha=1}^{m} B_{j\alpha}\frac{\partial B_{i\alpha}}{\partial x_j} = a_{\mathrm{sp},i},$$

proving Eqs. (4.65) to (4.67).

To prove part (b), we start from the equivalent Itô SDE Eq. (4.67) with $a_i = \underline{a}_i + a_{\mathrm{sp},i}$ and apply Itô's formula (Theorem 4.12) to $f(\mathbf{X}_t, t)$:

$$\mathrm{d}f(\mathbf{X}_t, t) = \left[\frac{\partial f}{\partial t} + \sum_{i=1}^{N}(\underline{a}_i + a_{\mathrm{sp},i})\frac{\partial f}{\partial x_i} + \frac{1}{2}\sum_{i,j=1}^{N}\sum_{\alpha=1}^{m} B_{i\alpha}B_{j\alpha}\frac{\partial^2 f}{\partial x_i \partial x_j}\right]\mathrm{d}t + \sum_{i=1}^{N}\sum_{\alpha=1}^{m}\frac{\partial f}{\partial x_i}B_{i\alpha}\cdot \mathrm{d}W_{\alpha,t}. \quad (4.72)$$

On the other hand, we convert the Stratonovich differential $\sum_{i,\alpha}\left(\frac{\partial f}{\partial x_i}B_{i\alpha}\right)\circ \mathrm{d}W_{\alpha,t}$ into Itô form by applying part (a) to the composite volatility matrix $\tilde{B}_{1\alpha}(\mathbf{x}, t) := \sum_{i=1}^{N}\frac{\partial f}{\partial x_i}(\mathbf{x}, t)B_{i\alpha}(\mathbf{x}, t)$ (for the 1D process $Y_t = f(\mathbf{X}_t, t)$ whose state variations come from $\mathrm{d}X_{j,t} = \sum_\beta B_{j\beta}\,\mathrm{d}W_{\beta,t} + \dots$). Using the product rule for $\frac{\partial}{\partial x_j}\left(\frac{\partial f}{\partial x_i}B_{i\alpha}\right)$, we get

$$\begin{aligned}\sum_{i=1}^{N}\sum_{\alpha=1}^{m}\left(\frac{\partial f}{\partial x_i}B_{i\alpha}\right)\circ \mathrm{d}W_{\alpha,t} &= \sum_{i=1}^{N}\sum_{\alpha=1}^{m}\frac{\partial f}{\partial x_i}B_{i\alpha}\cdot \mathrm{d}W_{\alpha,t} + \frac{1}{2}\sum_{j=1}^{N}\sum_{\alpha=1}^{m} B_{j\alpha}\frac{\partial}{\partial x_j}\left(\sum_{i=1}^{N}\frac{\partial f}{\partial x_i}B_{i\alpha}\right)\mathrm{d}t \\ &= \sum_{i=1}^{N}\sum_{\alpha=1}^{m}\frac{\partial f}{\partial x_i}B_{i\alpha}\cdot \mathrm{d}W_{\alpha,t} + \sum_{i=1}^{N}\underbrace{\left(\frac{1}{2}\sum_{j=1}^{N}\sum_{\alpha=1}^{m} B_{j\alpha}\frac{\partial B_{i\alpha}}{\partial x_j}\right)}_{a_{\mathrm{sp},i}}\frac{\partial f}{\partial x_i}\,\mathrm{d}t \\ &\quad + \frac{1}{2}\sum_{i,j=1}^{N}\sum_{\alpha=1}^{m} B_{i\alpha}B_{j\alpha}\frac{\partial^2 f}{\partial x_i \partial x_j}\,\mathrm{d}t. \end{aligned} \quad (4.73)$$

Comparing Eq. (4.72) and Eq. (4.73), we see that both the spurious drift term $\sum_i a_{\mathrm{sp},i}\frac{\partial f}{\partial x_i}\,\mathrm{d}t$ and the Itô Hessian term $\frac{1}{2}\sum_{i,j,\alpha} B_{i\alpha}B_{j\alpha}\frac{\partial^2 f}{\partial x_i \partial x_j}\,\mathrm{d}t$ are absorbed into the Stratonovich differential $\nabla f^\top \mathbf{B}\circ \mathrm{d}\mathbf{W}_t$, leaving exactly Eq. (4.68). □

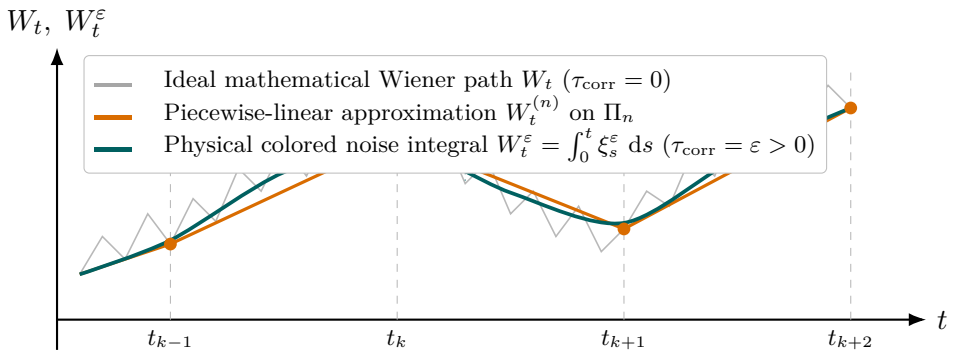


Figure 4.2: Physical motivation of the Wong–Zakai theorem. Any real physical noise source $\xi_t^\varepsilon = \dot{W}_t^\varepsilon$ has a non-zero microscopic correlation time $\tau_{\text{corr}} = \varepsilon > 0$ (teal curve), or can be approximated by a piecewise-smooth polygonal path $W_t^{(n)}$ (orange curve). For every $\varepsilon > 0$, W_t^ε is continuously differentiable and obeys ordinary calculus. In the white-noise limit $\varepsilon \to 0$, the solution X_t^ε converges to the *Stratonovich* SDE ($\lambda = 1/2$), not the Itô SDE.

Remark 4.17 (General λ-Calculus Drift Conversion)**.** Repeating the Taylor expansion Eq. (4.69) at the general evaluation point $(1-\lambda)\mathbf{X}_{t_k} + \lambda \mathbf{X}_{t_{k+1}} = \mathbf{X}_{t_k} + \lambda \Delta \mathbf{X}_k$ for any $\lambda \in [0,1]$ replaces the prefactor $\frac{1}{2}$ in Eq. (4.66) by λ:

$$\mathbf{B}(\mathbf{X}_t) \overset{(\lambda)}{\cdot} \mathrm{d}\mathbf{W}_t = \mathbf{B}(\mathbf{X}_t) \cdot \mathrm{d}\mathbf{W}_t + 2\lambda\, \mathbf{a}_{\text{sp}}(\mathbf{X}_t)\, \mathrm{d}t, \qquad (2\lambda\, \mathbf{a}_{\text{sp}}(\mathbf{x}))_i = \lambda \sum_{j=1}^{N} \sum_{\alpha=1}^{m} B_{j\alpha}(\mathbf{x}) \frac{\partial B_{i\alpha}(\mathbf{x})}{\partial x_j}. \tag{4.74}$$

In particular, the post-point / Hänggi–Klimontovich convention ($\lambda = 1$) carries twice the Stratonovich drift shift ($2\mathbf{a}_{\text{sp}}$).

4.4.2 The Wong–Zakai Theorem: From Physical Colored Noise to Stratonovich SDEs

Why does the Stratonovich midpoint ($\lambda = 1/2$) appear when physical noise is idealized as white noise? In any real physical system, the fluctuating thermal force $\boldsymbol{\xi}_t^\varepsilon$ is *not* mathematically white up to infinite frequency. Phonon baths in solids have a Debye cutoff frequency $\omega_{\text{D}} \sim 10^{13}\,\text{rad/s}$; Johnson–Nyquist electron scattering in silicon has a momentum relaxation time $\tau_m \sim 10^{-13}\,\text{s}$; and quantum thermal fluctuations have a thermal correlation time $\tau_{\text{th}} = \hbar/(k_{\text{B}}T) \approx 25\,\text{fs}$ at $T = 300\,\text{K}$. On timescales shorter than the correlation time $\varepsilon \equiv \tau_{\text{corr}} > 0$, the physical noise $\boldsymbol{\xi}_t^\varepsilon = \frac{\mathrm{d}\mathbf{W}_t^\varepsilon}{\mathrm{d}t}$ is a smooth function of time (see Fig. 4.2).

For any positive correlation time $\varepsilon > 0$, because $\mathbf{W}_t^\varepsilon(\omega)$ is a smooth C^1 path, the differential equation

$$\frac{\mathrm{d}\mathbf{X}_t^\varepsilon}{\mathrm{d}t} = \underline{\mathbf{a}}(\mathbf{X}_t^\varepsilon) + \mathbf{B}(\mathbf{X}_t^\varepsilon)\, \frac{\mathrm{d}\mathbf{W}_t^\varepsilon}{\mathrm{d}t} \tag{4.75}$$

is an *ordinary differential equation* (ODE) for each realization $\omega \in \Omega$, so there is no ambiguity between Itô and Stratonovich at $\varepsilon > 0$. Eugene Wong and Moshe Zakai proved in 1965 [119] that when $\varepsilon \to 0$, so that $\mathbf{W}_t^\varepsilon \to \mathbf{W}_t$ uniformly on $[0,T]$, the solution $\mathbf{X}_t^\varepsilon$ converges to the solution of the Stratonovich SDE.

Theorem 4.18 (Wong–Zakai Theorem)**.** *Let $\underline{a}, B : \mathbb{R} \to \mathbb{R}$ be C^2 functions with bounded first and second derivatives, and let $W_t^{(n)}$ be the piecewise-linear interpolation of a standard 1D Brownian*

motion W_t on a partition $\Pi_n = \{0 = t_0 < t_1 < \cdots < t_n = T\}$:

$$W_t^{(n)} := W_{t_k} + \frac{t - t_k}{\Delta t_k}\left(W_{t_{k+1}} - W_{t_k}\right), \qquad t \in [t_k, t_{k+1}]. \tag{4.76}$$

Let $X_t^{(n)}$ be the unique solution to the random ODE

$$\frac{\mathrm{d}X_t^{(n)}}{\mathrm{d}t} = \underline{a}\Big(X_t^{(n)}\Big) + B\Big(X_t^{(n)}\Big)\frac{\mathrm{d}W_t^{(n)}}{\mathrm{d}t}, \qquad X_0^{(n)} = x_0. \tag{4.77}$$

Then as $|\Pi_n| \to 0$, $X_t^{(n)}$ converges uniformly on $[0,T]$ in $L^2(\Omega)$ (and almost surely along refining dyadic partitions) to the unique solution X_t of the Stratonovich SDE

$$\mathrm{d}X_t = \underline{a}(X_t)\,\mathrm{d}t + B(X_t) \circ \mathrm{d}W_t = \left[\underline{a}(X_t) + \frac{1}{2}B(X_t)B'(X_t)\right]\mathrm{d}t + B(X_t)\cdot \mathrm{d}W_t. \tag{4.78}$$

Proof. We show how the drift $\frac{1}{2}B(X_t)B'(X_t)\,\mathrm{d}t$ arises on each subinterval $[t_k, t_{k+1}]$. On $[t_k, t_{k+1}]$, the derivative of the polygonal path Eq. (4.76) is the constant $\dot{W}_t^{(n)} = \Delta W_k/\Delta t_k$. Integrating Eq. (4.77) from t_k to $s \in [t_k, t_{k+1}]$ yields the Picard expansion

$$X_s^{(n)} - X_{t_k}^{(n)} = \int_{t_k}^{s} \underline{a}\Big(X_u^{(n)}\Big)\,\mathrm{d}u + \frac{\Delta W_k}{\Delta t_k}\int_{t_k}^{s} B\Big(X_u^{(n)}\Big)\,\mathrm{d}u = B\Big(X_{t_k}^{(n)}\Big)\frac{s - t_k}{\Delta t_k}\,\Delta W_k + \mathcal{O}(\Delta t_k). \tag{4.79}$$

Now expand $B(X_s^{(n)})$ around $X_{t_k}^{(n)}$ using Taylor's theorem and substitute Eq. (4.79):

$$\begin{aligned} B\Big(X_s^{(n)}\Big) &= B\Big(X_{t_k}^{(n)}\Big) + B'\Big(X_{t_k}^{(n)}\Big)\Big(X_s^{(n)} - X_{t_k}^{(n)}\Big) + \mathcal{O}\Big(\Big|X_s^{(n)} - X_{t_k}^{(n)}\Big|^2\Big) \\ &= B\Big(X_{t_k}^{(n)}\Big) + B\Big(X_{t_k}^{(n)}\Big)B'\Big(X_{t_k}^{(n)}\Big)\frac{s - t_k}{\Delta t_k}\,\Delta W_k + \mathcal{O}(\Delta t_k). \end{aligned} \tag{4.80}$$

Inserting Eq. (4.80) into the stochastic integral over $[t_k, t_{k+1}]$ gives

$$\begin{aligned} \int_{t_k}^{t_{k+1}} B\Big(X_s^{(n)}\Big)\frac{\mathrm{d}W_s^{(n)}}{\mathrm{d}s}\,\mathrm{d}s &= \frac{\Delta W_k}{\Delta t_k}\int_{t_k}^{t_{k+1}}\Big[B\Big(X_{t_k}^{(n)}\Big) + B\Big(X_{t_k}^{(n)}\Big)B'\Big(X_{t_k}^{(n)}\Big)\frac{s - t_k}{\Delta t_k}\,\Delta W_k \\ &\qquad + \mathcal{O}(\Delta t_k)\Big]\,\mathrm{d}s \\ &= B\Big(X_{t_k}^{(n)}\Big)\,\Delta W_k \\ &\quad + B\Big(X_{t_k}^{(n)}\Big)B'\Big(X_{t_k}^{(n)}\Big)\frac{(\Delta W_k)^2}{\Delta t_k}\underbrace{\int_{t_k}^{t_{k+1}}\frac{s - t_k}{\Delta t_k}\,\mathrm{d}s}_{\frac{1}{2}\Delta t_k} \\ &\quad + \mathcal{O}\Big((\Delta t_k)^{3/2}\Big) \\ &= B\Big(X_{t_k}^{(n)}\Big)\,\Delta W_k + \frac{1}{2}B\Big(X_{t_k}^{(n)}\Big)B'\Big(X_{t_k}^{(n)}\Big)(\Delta W_k)^2 + \mathcal{O}\Big((\Delta t_k)^{3/2}\Big). \end{aligned} \tag{4.81, 4.82}$$

In Eq. (4.82) the prefactor $1/2$ comes from the elementary integral $\int_0^1 u\,\mathrm{d}u = 1/2$ representing the average of the noise increment accumulated from t_k to s multiplied by the noise rate at time s. Summing Eq. (4.82) over $k = 0, \ldots, n-1$, using $(\Delta W_k)^2 \to \mathrm{d}t$ by Theorem 4.4, and bounding the difference $\left\|X_t^{(n)} - X_t\right\|$ via Grönwall's inequality establishes convergence to Eq. (4.78). □

Heuristic Derivation: Colored Ornstein–Uhlenbeck Thermal Noise and the Area Symmetry of Correlation

We can also understand the Wong–Zakai prefactor $1/2$ from a stationary physical colored-noise process ξ_t^τ with correlation time $\tau_{\text{corr}} = \tau > 0$ and exponential autocorrelation function

$$C_\tau(t-s) := \mathbb{E}[\xi_t^\tau \xi_s^\tau] = \frac{1}{2\tau} \exp\left(-\frac{|t-s|}{\tau}\right) \xrightarrow{\tau \to 0^+} \delta(t-s). \tag{4.83}$$

In the ODE $\dot{X}_t = \underline{a}(X_t) + B(X_t)\xi_t^\tau$, expanding $B(X_t) \approx B(X_{t_0}) + B'(X_{t_0}) \int_{t_0}^t B(X_{t_0})\xi_s^\tau \, \mathrm{d}s$ over a window $\Delta t \gg \tau$ yields the noise-induced drift rate

$$\lim_{\tau \to 0^+} B(x)B'(x) \int_{-\infty}^t C_\tau(t-s) \, \mathrm{d}s = B(x)B'(x) \int_0^\infty \frac{1}{2\tau} e^{-u/\tau} \, \mathrm{d}u = \frac{1}{2} B(x)B'(x). \tag{4.84}$$

Because the physical noise autocorrelation function $C_\tau(t-s) = C_\tau(s-t)$ is an *even* function of $t-s$ that integrates to 1 over $(-\infty, +\infty)$, integrating only over the causal past $s \in (-\infty, t]$ captures exactly half of the total Dirac delta weight $\int_{-\infty}^\infty \delta(u) \, \mathrm{d}u = 1$. In multi-dimensional systems where cross-correlations between different colored noise channels $\xi_\alpha^\tau, \xi_\beta^\tau$ have an antisymmetric component (*e.g.*, in a magnetic field or non-reciprocal active gyroscopic circuit), the integral $\int_0^\infty \mathbb{E}\left[\xi_\alpha^\tau(t)\xi_\beta^\tau(t-u)\right] \mathrm{d}u$ acquires an additional Lévy area correction on top of the symmetric Stratonovich midpoint term [85].

Notation Watch: Itô ($\cdot$), Stratonovich ($\circ$), and Kinetic ($\star$) Conventions Across Communities

Different scientific communities default to different stochastic integral conventions, often without warning:

- **Probability, quantitative finance, and score-based diffusion models** [59, 80, 103]: almost always use the Itô integral ($\lambda = 0$), usually omitting the dot and writing $\mathbf{B}(\mathbf{X}_t) \, \mathrm{d}\mathbf{W}_t$.
- **Physics, stochastic energetics, and differential geometry** [97, 98]: default to the Stratonovich integral ($\lambda = 1/2$, denoted $\mathbf{B}(\mathbf{X}_t) \circ \mathrm{d}\mathbf{W}_t$) when writing trajectory heat $\mathrm{d}\mathcal{Q} = -\nabla E(\mathbf{x}_t) \circ \mathrm{d}\mathbf{x}_t$ (Chapter 8) or coordinate transformations on curved manifolds, because $\circ$ obeys the ordinary chain rule Eq. (4.68).
- **This book:** whenever the noise $\mathbf{B}(\mathbf{x})$ depends on the state, $\mathbf{B}(\mathbf{x}_t) \cdot \mathrm{d}\mathbf{W}_t$ (or $\mathbf{B}(\mathbf{x}_t) \, \mathrm{d}\mathbf{W}_t$) denotes the Itô integral, and the open circle $\mathbf{B}(\mathbf{x}_t) \circ \mathrm{d}\mathbf{W}_t$ is always written explicitly for the Stratonovich integral.

Hardware Realization: Multiplicative Noise and Spurious Drift in Subthreshold CMOS Samplers

The Itô–Stratonovich distinction matters both for subthreshold CMOS sampling hardware and for the software used to simulate it, such as Extropic's `THRML` library [31]; see also [37, 38].

1. **Physical circuit versus Euler–Maruyama simulation.** In a subthreshold CMOS circuit, the thermal and shot current fluctuations of the transistors have a correlation

time set by the carrier transit time, $\tau_{\mathrm{tr}} \sim L^2/(\mu_n V_T)$, of order 0.1–1 ps for short channels. This is several orders of magnitude shorter than the node relaxation time $\tau_{RC} = C_{\mathrm{node}}/g_{\mathrm{m}}$, which is of order 1–10 ns at representative bias currents. By the Wong–Zakai theorem (Theorem 4.18), the circuit integrates these fast, smooth fluctuations in the Stratonovich sense, or more precisely according to the mesoscopic master-equation dynamics of Chapter 13. A digital simulation of the same circuit with a forward Euler–Maruyama step

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{a}(\mathbf{v}_k)\Delta t + \mathbf{B}(\mathbf{v}_k)\sqrt{\Delta t}\,\boldsymbol{\zeta}_k, \qquad \boldsymbol{\zeta}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

evaluates $\mathbf{B}(\mathbf{v}_k)$ at the start of the step and therefore implements the Itô integral ($\lambda = 0$). Unless the drift conversion term of Theorem 4.16 is added, or a midpoint (Heun) predictor-corrector step is used, the simulation converges to a stationary distribution that differs from that of the physical circuit whenever $\mathbf{B}(\mathbf{v})$ depends on $\mathbf{v}$.

2. **State-dependent mobility.** In Worked computation 4.23 and Section 5.5, we will prove that if a circuit has a state-dependent conductance/mobility $G(v)$ and samples a target energy $E(v)$ at temperature T, thermodynamic detailed balance requires the Itô drift to include the divergence of the diffusion tensor, $a(v) = -\frac{G(v)}{C^2}E'(v) + \frac{k_{\mathrm{B}}T}{C^2}G'(v)$. If the circuit is described in Stratonovich form ($\lambda = 1/2$), half of this $+\frac{k_{\mathrm{B}}T}{C^2}G'(v)$ drift is supplied by the Wong–Zakai midpoint correlation $\mathbf{a}_{\mathrm{sp}}(v) = \frac{k_{\mathrm{B}}T}{2C^2}G'(v)$, while the remaining half arises from the microscopic reversibility of bidirectional Poisson barrier transport (Chapters 11 and 13).

4.5 Strong/Weak Solutions of SDEs and Girsanov's Theorem

With the Itô and Stratonovich integrals defined, we turn to two basic questions about the stochastic differential equation

$$\mathrm{d}\mathbf{X}_t = \mathbf{a}(\mathbf{X}_t, t)\ \mathrm{d}t + \mathbf{B}(\mathbf{X}_t, t) \cdot \mathrm{d}\mathbf{W}_t, \qquad \mathbf{X}_0 = \boldsymbol{\xi}: \tag{4.85}$$

first, under what conditions on the drift $\mathbf{a}$ and volatility $\mathbf{B}$ does Eq. (4.85) possess a unique global-in-time solution $\mathbf{X}_t$? Second, how does the probability measure $\mathcal{P}[\mathbf{X}_{[0,T]}]$ on the space of continuous trajectories change when we shift the drift field $\mathbf{a}(\mathbf{x}, t) \mapsto \mathbf{a}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t)$?

4.5.1 Strong vs. Weak Well-Posedness and Lyapunov Confinement for Non-Lipschitz Potentials

In stochastic analysis, there are two distinct notions of a solution to Eq. (4.85):

- **Strong solution (pathwise causality):** The probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$, the specific Brownian motion realization $\mathbf{W}_t(\omega)$, and the initial condition $\boldsymbol{\xi}(\omega)$ are *given in advance* (for example, the actual thermal noise waveform inside a physical resistor). A strong solution is an $\{\mathcal{F}_t\}$-adapted process $\mathbf{X}_t(\omega)$ constructed as a causal functional of $\mathbf{W}_{[0,t]}(\omega)$ and $\boldsymbol{\xi}(\omega)$ that satisfies the integral equation Eq. (4.46) $\mathbb{P}$-a.s.

- **Weak solution (distributional equivalence):** Only the coefficients $\mathbf{a}(\mathbf{x}, t)$, $\mathbf{B}(\mathbf{x}, t)$, and the initial law $\mu_0 = \mathrm{Law}(\boldsymbol{\xi})$ are prescribed. A weak solution consists of a *triple*

$((\tilde{\Omega}, \tilde{\mathcal{F}}, \{\tilde{\mathcal{F}}_t\}, \tilde{\mathbb{P}}), \tilde{\mathbf{W}}_t, \tilde{\mathbf{X}}_t)$ such that $\tilde{\mathbf{W}}_t$ is a Brownian motion on $\tilde{\Omega}$, $\mathrm{Law}(\tilde{\mathbf{X}}_0) = \mu_0$, and Eq. (4.46) holds on $\tilde{\Omega}$. Weak uniqueness (uniqueness in law) means that any two solutions induce the exact same probability distribution on path space $\mathcal{C}([0,T];\mathbb{R}^N)$. Weak uniqueness is all that a probabilistic sampler requires.

Theorem 4.19 (Itô's Global Lipschitz and Khasminskii's Lyapunov Existence–Uniqueness Theorem)**.** *Consider the SDE Eq. (4.85) on $[0,T]$ with $\mathbb{E}\left[\|\boldsymbol{\xi}\|^2\right] < \infty$ and $\boldsymbol{\xi} \perp\!\!\!\perp \mathbf{W}$.*

(a) ***(Global Lipschitz & Linear Growth Case)*** *Suppose there exists a constant $L > 0$ such that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ and $t \in [0,T]$,*

$$\|\mathbf{a}(\mathbf{x},t) - \mathbf{a}(\mathbf{y},t)\| + \|\mathbf{B}(\mathbf{x},t) - \mathbf{B}(\mathbf{y},t)\|_{\mathrm{F}} \le L\,\|\mathbf{x}-\mathbf{y}\|\,, \tag{4.86}$$

$$\|\mathbf{a}(\mathbf{x},t)\| + \|\mathbf{B}(\mathbf{x},t)\|_{\mathrm{F}} \le L\,(1+\|\mathbf{x}\|)\,. \tag{4.87}$$

Then Eq. (4.85) has a pathwise unique strong solution $\mathbf{X} \in \mathcal{M}^2([0,T];\mathbb{R}^N)$.

(b) ***(Local Lipschitz + Lyapunov Coercivity Case for Physical Potentials)*** *Suppose $\mathbf{a}$ and $\mathbf{B}$ are locally Lipschitz (satisfying Eq. (4.86) on every compact ball $\|\mathbf{x}\|, \|\mathbf{y}\| \le R$ with a constant L_R), and there exists a radially unbounded Lyapunov function $V \in C^2(\mathbb{R}^N;[0,\infty))$ ($\lim_{\|\mathbf{x}\|\to\infty} V(\mathbf{x}) = \infty$) and a constant $C > 0$ such that the backward generator $\mathcal{L}V := \mathbf{a}\cdot\nabla V + \frac{1}{2}\operatorname{tr}(\mathbf{B}\mathbf{B}^\top \nabla^2 V)$ satisfies*

$$\mathcal{L}V(\mathbf{x},t) = \mathbf{a}(\mathbf{x},t)\cdot\nabla V(\mathbf{x}) + \frac{1}{2}\operatorname{tr}\Big(\mathbf{B}(\mathbf{x},t)\mathbf{B}(\mathbf{x},t)^\top \nabla^2 V(\mathbf{x})\Big) \le C\,(1+V(\mathbf{x})) \tag{4.88}$$

for all $(\mathbf{x},t) \in \mathbb{R}^N \times [0,T]$. Then Eq. (4.85) has a unique global strong solution on $[0,T]$ that never explodes in finite time.

Proof. For part (a), we define the Picard iteration $\mathbf{X}_t^{(0)} := \boldsymbol{\xi}$ and

$$\mathbf{X}_t^{(k+1)} := \boldsymbol{\xi} + \int_0^t \mathbf{a}\Big(\mathbf{X}_s^{(k)}, s\Big)\,\mathrm{d}s + \int_0^t \mathbf{B}\Big(\mathbf{X}_s^{(k)}, s\Big)\cdot \mathrm{d}\mathbf{W}_s. \tag{4.89}$$

Using $(u+v)^2 \le 2u^2 + 2v^2$, the Cauchy–Schwarz inequality on the $\mathrm{d}s$ integral, Doob's maximal inequality Eq. (4.43) on the Itô integral, and the Lipschitz bound Eq. (4.86), we estimate $D_k(t) := \mathbb{E}\Big[\sup_{0\le u\le t}\big\|\mathbf{X}_u^{(k+1)} - \mathbf{X}_u^{(k)}\big\|^2\Big]$ by

$$\begin{aligned} D_k(t) &\le 2t\int_0^t \mathbb{E}\Big[\Big\|\mathbf{a}(\mathbf{X}_s^{(k)},s) - \mathbf{a}(\mathbf{X}_s^{(k-1)},s)\Big\|^2\Big]\,\mathrm{d}s + 8\int_0^t \mathbb{E}\Big[\Big\|\mathbf{B}(\mathbf{X}_s^{(k)},s) - \mathbf{B}(\mathbf{X}_s^{(k-1)},s)\Big\|_{\mathrm{F}}^2\Big]\,\mathrm{d}s \\ &\le 2(T+4)L^2\int_0^t D_{k-1}(s)\,\mathrm{d}s \le \frac{\big[2(T+4)L^2 t\big]^k}{k!}\,D_0(T). \end{aligned} \tag{4.90}$$

Since $\sum_{k=0}^{\infty}\frac{[2(T+4)L^2T]^k}{k!} < \infty$, the Picard iterates converge uniformly on $[0,T]$ in $L^2(\Omega)$ to a strong solution $\mathbf{X}_t$, and Grönwall's lemma on $\mathbb{E}\left[\|\mathbf{X}_t - \mathbf{Y}_t\|^2\right] \le 2(T+4)L^2\int_0^t \mathbb{E}\left[\|\mathbf{X}_s - \mathbf{Y}_s\|^2\right]\mathrm{d}s$ yields $\mathbf{X}_t = \mathbf{Y}_t$ a.s.

For part (b), local Lipschitz continuity guarantees a unique strong solution up to the exit time $\tau_R := \inf\{t \ge 0 : \|\mathbf{X}_t\| \ge R\}$. Applying Itô's formula to $e^{-Ct}V(\mathbf{X}_{t\wedge\tau_R})$ and using Eq. (4.88) gives $\mathbb{E}[V(\mathbf{X}_{T\wedge\tau_R})] \le e^{CT}(\mathbb{E}[V(\boldsymbol{\xi})] + CT)$. By Markov's inequality, $\mathbb{P}(\tau_R \le T) \le \frac{\mathbb{E}[V(\mathbf{X}_{T\wedge\tau_R})]}{\inf_{\|\mathbf{x}\|\ge R} V(\mathbf{x})} \to 0$ as $R \to \infty$, proving that the explosion time $\tau_\infty = \lim_{R\to\infty}\tau_R = +\infty$ a.s. □

Example 4.20 (Non-Explosion of the Double-Well Sampler)**.** For the RC sampler, $\mathbf{C}\,\mathrm{d}\mathbf{v}_t = -\mathbf{G}(\mathbf{v}_t - \mathbf{b})\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t$, the linear drift $-\mathbf{C}^{-1}\mathbf{G}(\mathbf{v}-\mathbf{b})$ is globally Lipschitz, so Theorem 4.19(a) applies directly. The double-well sampler (Example 1.6) is different. Its potential is $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$ with $a > 0$, and its overdamped Langevin equation is

$$\mathrm{d}x_t = -\mu E'(x_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad -\mu E'(x) = -\mu\left(ax^3 - bx - h\right), \tag{4.91}$$

with mobility $\mu > 0$ and noise amplitude $\sigma = \sqrt{2\mu k_{\mathrm{B}}T}$. The drift grows like $|x|^3$ as $|x| \to \infty$ and therefore violates the global Lipschitz and linear growth conditions Eqs. (4.86) and (4.87). The trajectory nevertheless cannot explode, because the cubic term $-\mu a x^3$ of the drift points inward, toward the origin. With the Lyapunov function $V(x) = 1 + x^2$,

$$\mathcal{L}V(x) = -\mu E'(x)V'(x) + \frac{1}{2}\sigma^2 V''(x) = -2\mu x\left(ax^3 - bx - h\right) + \sigma^2 = -2\mu a x^4 + 2\mu b x^2 + 2\mu h x + \sigma^2, \tag{4.92}$$

which is bounded above on all of $\mathbb{R}$ because $-2\mu a x^4$ dominates the lower powers as $|x| \to \infty$. Hence Theorem 4.19(b) guarantees a unique, non-exploding global strong solution for all $t \geq 0$.

4.5.2 Girsanov's Theorem: Change of Measure on Path Space

In statistical mechanics of equilibrium configurations (Chapter 1), reweighting a probability density $p_0(\mathbf{x})$ by a Boltzmann factor $e^{-\beta\Delta E(\mathbf{x})}$ produces a tilted density $p_1(\mathbf{x})$ whose Radon–Nikodym derivative $\mathrm{d}p_1/\,\mathrm{d}p_0$ is well-defined on $\mathbb{R}^N$. Girsanov's theorem is the analogue of this reweighting on the path space $\mathcal{C}([0,T];\mathbb{R}^N)$.

Suppose we have two SDEs on $[0,T]$:

$$\text{Reference process } (\mathbb{P}): \qquad \mathrm{d}\mathbf{X}_t = \mathbf{a}(\mathbf{X}_t,t)\,\mathrm{d}t + \mathbf{B}(\mathbf{X}_t,t)\cdot\mathrm{d}\mathbf{W}_t, \tag{4.93}$$

$$\text{Tilted process } (\tilde{\mathbb{P}}): \qquad \mathrm{d}\mathbf{X}_t = [\mathbf{a}(\mathbf{X}_t,t) + \mathbf{B}(\mathbf{X}_t,t)\mathbf{u}_t]\,\mathrm{d}t + \mathbf{B}(\mathbf{X}_t,t)\cdot\mathrm{d}\tilde{\mathbf{W}}_t. \tag{4.94}$$

The two SDEs Eqs. (4.93) and (4.94) must have the same volatility matrix $\mathbf{B}(\mathbf{X}_t,t)$. The reason is that, by Theorem 4.4, the quadratic covariation $[\mathbf{X},\mathbf{X}]_t = \int_0^t \mathbf{B}(\mathbf{X}_s,s)\mathbf{B}(\mathbf{X}_s,s)^\top\,\mathrm{d}s$ can be read off deterministically from a single sample path $\mathbf{X}_{[0,T]}(\omega)$. If two SDEs had different diffusion tensors $\mathbf{D}_1 \neq \mathbf{D}_2$, their path measures on $\mathcal{C}([0,T];\mathbb{R}^N)$ would be *mutually singular* ($\mathbb{P}_1 \perp \mathbb{P}_2$), supported on disjoint sets of paths with different quadratic variations. When $\mathbf{B}$ is identical and the drift shift lies in the range of $\mathbf{B}$ ($\Delta\mathbf{a}_t = \mathbf{B}_t\mathbf{u}_t$), the two path measures are mutually absolutely continuous (under Novikov's condition, Eq. (4.97) below), and their Radon–Nikodym derivative is given by Girsanov's theorem. The construction rests on an exponential martingale, the Doléans-Dade stochastic exponential.

Lemma 4.21 (Doléans-Dade Stochastic Exponential and Novikov's Condition)**.** *Let $\mathbf{u} \in \mathcal{M}^2([0,T];\mathbb{R}^m)$ be an $\{\mathcal{F}_t\}$-adapted process, and define the exponential process*

$$\mathcal{E}_t(\mathbf{u}) := \exp\left(\int_0^t \mathbf{u}_s\cdot\mathrm{d}\mathbf{W}_s - \frac{1}{2}\int_0^t \|\mathbf{u}_s\|^2\,\mathrm{d}s\right), \qquad t\in[0,T]. \tag{4.95}$$

Then $\mathcal{E}_t(\mathbf{u})$ satisfies the linear Itô SDE

$$\mathrm{d}\mathcal{E}_t(\mathbf{u}) = \mathcal{E}_t(\mathbf{u})\,\mathbf{u}_t\cdot\mathrm{d}\mathbf{W}_t, \qquad \mathcal{E}_0(\mathbf{u}) = 1. \tag{4.96}$$

Moreover, if $\mathbf{u}$ satisfies ***Novikov's condition***

$$\mathbb{E}\left[\exp\left(\frac{1}{2}\int_0^T \|\mathbf{u}_s\|^2\,\mathrm{d}s\right)\right] < \infty, \tag{4.97}$$

then $\{\mathcal{E}_t(\mathbf{u})\}_{t\in[0,T]}$ is a strictly positive martingale with $\mathbb{E}[\mathcal{E}_t(\mathbf{u})] = \mathcal{E}_0(\mathbf{u}) = 1$ for all $t\in[0,T]$.

Proof. Define the scalar Itô process $Z_t := \int_0^t \mathbf{u}_s \cdot \mathrm{d}\mathbf{W}_s - \frac{1}{2}\int_0^t \|\mathbf{u}_s\|^2 \,\mathrm{d}s$, whose differential and quadratic variation are

$$\mathrm{d}Z_t = -\frac{1}{2}\|\mathbf{u}_t\|^2 \,\mathrm{d}t + \mathbf{u}_t \cdot \mathrm{d}\mathbf{W}_t, \qquad (\mathrm{d}Z_t)^2 = \sum_{\alpha=1}^{m} u_{\alpha,t}^2 \,\mathrm{d}t = \|\mathbf{u}_t\|^2 \,\mathrm{d}t. \tag{4.98}$$

Applying Itô's formula (Theorem 4.12) to $f(z) = e^z$ (for which $f'(z) = f''(z) = e^z$) gives

$$\begin{aligned}\mathrm{d}\mathcal{E}_t(\mathbf{u}) = \mathrm{d}\left(e^{Z_t}\right) &= e^{Z_t}\,\mathrm{d}Z_t + \frac{1}{2}e^{Z_t}(\mathrm{d}Z_t)^2 \\ &= \mathcal{E}_t(\mathbf{u})\left(-\frac{1}{2}\|\mathbf{u}_t\|^2\,\mathrm{d}t + \mathbf{u}_t\cdot\mathrm{d}\mathbf{W}_t + \frac{1}{2}\|\mathbf{u}_t\|^2\,\mathrm{d}t\right) = \mathcal{E}_t(\mathbf{u})\,\mathbf{u}_t\cdot\mathrm{d}\mathbf{W}_t.\end{aligned} \tag{4.99}$$

The term $-\frac{1}{2}\int_0^t \|\mathbf{u}_s\|^2\,\mathrm{d}s$ in the exponent of Eq. (4.95) is there precisely to cancel the $+\frac{1}{2}\|\mathbf{u}_t\|^2\,\mathrm{d}t$ Itô quadratic-variation term. Under Novikov's condition Eq. (4.97), uniform integrability holds (see [59]), so the local martingale $\mathcal{E}_t(\mathbf{u}) = 1 + \int_0^t \mathcal{E}_s(\mathbf{u})\mathbf{u}_s\cdot\mathrm{d}\mathbf{W}_s$ is a true martingale with $\mathbb{E}[\mathcal{E}_t(\mathbf{u})] = 1$. □

Theorem 4.22 (Girsanov's Theorem for Drift Transformations)**.** *Let $\mathbf{W}_t$ be a standard m-dimensional Brownian motion on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{P})$, and let $\mathbf{u} \in \mathcal{M}^2([0,T];\mathbb{R}^m)$ satisfy Novikov's condition Eq. (4.97). Define a new probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \mathcal{F}_T)$ via the Radon–Nikodym derivative*

$$\left.\frac{\mathrm{d}\tilde{\mathbb{P}}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} := \mathcal{E}_T(\mathbf{u}) = \exp\left(\int_0^T \mathbf{u}_s\cdot\mathrm{d}\mathbf{W}_s - \frac{1}{2}\int_0^T \|\mathbf{u}_s\|^2\,\mathrm{d}s\right). \tag{4.100}$$

Then the drift-shifted process

$$\tilde{\mathbf{W}}_t := \mathbf{W}_t - \int_0^t \mathbf{u}_s\,\mathrm{d}s, \qquad t\in[0,T], \tag{4.101}$$

is a standard m-dimensional $\{\mathcal{F}_t\}$-Brownian motion under the new measure $\tilde{\mathbb{P}}$. Consequently, if $\mathbf{X}_t$ solves the reference SDE Eq. (4.93) under $\mathbb{P}$, then under $\tilde{\mathbb{P}}$ the same trajectory $\mathbf{X}_t$ solves the drift-tilted SDE

$$\mathrm{d}\mathbf{X}_t = [\mathbf{a}(\mathbf{X}_t,t) + \mathbf{B}(\mathbf{X}_t,t)\mathbf{u}_t]\,\mathrm{d}t + \mathbf{B}(\mathbf{X}_t,t)\cdot\mathrm{d}\tilde{\mathbf{W}}_t, \tag{4.102}$$

and the KL divergence (relative entropy) between the tilted path measure $\tilde{\mathbb{P}}$ and reference path measure $\mathbb{P}$ on $[0,T]$ is given exactly by the integrated control effort

$$D_{\mathrm{KL}}\left(\tilde{\mathbb{P}}\,\middle\|\,\mathbb{P}\right) = \mathbb{E}_{\tilde{\mathbb{P}}}\left[\ln\frac{\mathrm{d}\tilde{\mathbb{P}}}{\mathrm{d}\mathbb{P}}\right] = \frac{1}{2}\,\mathbb{E}_{\tilde{\mathbb{P}}}\left[\int_0^T \|\mathbf{u}_s\|^2\,\mathrm{d}s\right]. \tag{4.103}$$

Proof. Since $\mathcal{E}_t \equiv \mathcal{E}_t(\mathbf{u})$ is an $\{\mathcal{F}_t\}$-martingale under $\mathbb{P}$, for any $\mathcal{F}_t$-measurable integrable random variable Y_t and $0 \le s \le t \le T$, the conditional expectation under $\tilde{\mathbb{P}}$ satisfies Bayes' rule for measures: $\mathbb{E}_{\tilde{\mathbb{P}}}[Y_t \mid \mathcal{F}_s] = \mathcal{E}_s^{-1}\mathbb{E}[\mathcal{E}_t Y_t \mid \mathcal{F}_s]$. To prove that $\tilde{\mathbf{W}}_t$ is a standard $\{\mathcal{F}_t\}$-Brownian motion under $\tilde{\mathbb{P}}$, it suffices by Lévy's characterization of Brownian motion (or by computing conditional characteristic functions) to show that for every deterministic vector $\boldsymbol{\theta} \in \mathbb{R}^m$, the complex exponential process

$$M_t^{\boldsymbol{\theta}} := \exp\left(i\,\boldsymbol{\theta}\cdot\tilde{\mathbf{W}}_t + \frac{1}{2}\|\boldsymbol{\theta}\|^2 t\right) \tag{4.104}$$

is an $\{\mathcal{F}_t\}$-martingale under $\tilde{\mathbb{P}}$ (which is equivalent to $\mathcal{E}_t M_t^{\boldsymbol{\theta}}$ being a martingale under $\mathbb{P}$), since $\mathbb{E}_{\tilde{\mathbb{P}}}[M_t^{\boldsymbol{\theta}} \mid \mathcal{F}_s] = M_s^{\boldsymbol{\theta}}$ rearranges immediately to $\mathbb{E}_{\tilde{\mathbb{P}}}[e^{i\boldsymbol{\theta}\cdot(\tilde{\mathbf{W}}_t - \tilde{\mathbf{W}}_s)} \mid \mathcal{F}_s] = e^{-\frac{1}{2}\|\boldsymbol{\theta}\|^2(t-s)}$, proving that $\tilde{\mathbf{W}}_t - \tilde{\mathbf{W}}_s \perp\!\!\!\perp \mathcal{F}_s$ with distribution $\mathcal{N}(\mathbf{0}, (t-s)\mathbf{I}_m)$ under $\tilde{\mathbb{P}}$.

We compute $\mathrm{d}(\mathcal{E}_t M_t^{\boldsymbol{\theta}})$ under $\mathbb{P}$ using the Itô product rule (Corollary 4.13). Since $\mathrm{d}\tilde{\mathbf{W}}_t = \mathrm{d}\mathbf{W}_t - \mathbf{u}_t\,\mathrm{d}t$ has $(\mathrm{d}\tilde{W}_{\alpha,t})(\mathrm{d}\tilde{W}_{\beta,t}) = \delta_{\alpha\beta}\,\mathrm{d}t$, applying Itô's formula to Eq. (4.104) gives

$$\begin{aligned}\mathrm{d}M_t^{\boldsymbol{\theta}} &= M_t^{\boldsymbol{\theta}}\left[i\boldsymbol{\theta}\cdot(\mathrm{d}\mathbf{W}_t - \mathbf{u}_t\,\mathrm{d}t) + \frac{1}{2}\|\boldsymbol{\theta}\|^2\,\mathrm{d}t + \frac{1}{2}(i)^2\|\boldsymbol{\theta}\|^2\,\mathrm{d}t\right]\\ &= M_t^{\boldsymbol{\theta}}\left(-i\boldsymbol{\theta}\cdot\mathbf{u}_t\,\mathrm{d}t + i\boldsymbol{\theta}\cdot\mathrm{d}\mathbf{W}_t\right).\end{aligned} \tag{4.105}$$

Recall from Eq. (4.96) that $\mathrm{d}\mathcal{E}_t = \mathcal{E}_t\,\mathbf{u}_t\cdot\mathrm{d}\mathbf{W}_t$. Therefore, the cross-variation between $\mathcal{E}_t$ and $M_t^{\boldsymbol{\theta}}$ is

$$\mathrm{d}\mathcal{E}_t\;\mathrm{d}M_t^{\boldsymbol{\theta}} = (\mathcal{E}_t\,\mathbf{u}_t\cdot\mathrm{d}\mathbf{W}_t)\left(iM_t^{\boldsymbol{\theta}}\,\boldsymbol{\theta}\cdot\mathrm{d}\mathbf{W}_t\right) = i\,\mathcal{E}_t M_t^{\boldsymbol{\theta}}\,(\boldsymbol{\theta}\cdot\mathbf{u}_t)\,\mathrm{d}t. \tag{4.106}$$

Combining Eqs. (4.105) and (4.106) in the Itô product rule $\mathrm{d}(\mathcal{E}_t M_t^{\boldsymbol{\theta}}) = \mathcal{E}_t\,\mathrm{d}M_t^{\boldsymbol{\theta}} + M_t^{\boldsymbol{\theta}}\,\mathrm{d}\mathcal{E}_t + \mathrm{d}\mathcal{E}_t\,\mathrm{d}M_t^{\boldsymbol{\theta}}$, the drift terms $\mp i\mathcal{E}_t M_t^{\boldsymbol{\theta}}(\boldsymbol{\theta}\cdot\mathbf{u}_t)\,\mathrm{d}t$ cancel identically:

$$\mathrm{d}\left(\mathcal{E}_t M_t^{\boldsymbol{\theta}}\right) = \mathcal{E}_t M_t^{\boldsymbol{\theta}}\,(\mathbf{u}_t + i\boldsymbol{\theta})\cdot\mathrm{d}\mathbf{W}_t. \tag{4.107}$$

Because the right-hand side is a pure Itô stochastic differential with zero $\mathrm{d}t$ drift, $\mathcal{E}_t M_t^{\boldsymbol{\theta}}$ is a martingale under $\mathbb{P}$, proving that $\tilde{\mathbf{W}}_t$ is a standard Brownian motion under $\tilde{\mathbb{P}}$. Substituting $\mathrm{d}\mathbf{W}_t = \mathrm{d}\tilde{\mathbf{W}}_t + \mathbf{u}_t\,\mathrm{d}t$ into Eq. (4.93) yields Eq. (4.102), and substituting $\mathrm{d}\mathbf{W}_s = \mathrm{d}\tilde{\mathbf{W}}_s + \mathbf{u}_s\,\mathrm{d}s$ into $\ln(\mathrm{d}\tilde{\mathbb{P}}/\,\mathrm{d}\mathbb{P})$ gives

$$\ln\frac{\mathrm{d}\tilde{\mathbb{P}}}{\mathrm{d}\mathbb{P}} = \int_0^T \mathbf{u}_s\cdot(\mathrm{d}\tilde{\mathbf{W}}_s + \mathbf{u}_s\,\mathrm{d}s) - \frac{1}{2}\int_0^T\|\mathbf{u}_s\|^2\,\mathrm{d}s = \int_0^T\mathbf{u}_s\cdot\mathrm{d}\tilde{\mathbf{W}}_s + \frac{1}{2}\int_0^T\|\mathbf{u}_s\|^2\,\mathrm{d}s. \tag{4.108}$$

Taking $\mathbb{E}_{\tilde{\mathbb{P}}}[\cdot]$ and using the fact that $\int_0^T\mathbf{u}_s\cdot\mathrm{d}\tilde{\mathbf{W}}_s$ is a zero-mean $\tilde{\mathbb{P}}$-martingale yields Eq. (4.103). □

4.6 Worked Computation: Multiplicative Noise in a Nonlinear *RC* Circuit

We now apply Itô's formula, the Stratonovich integral, and the Itô–Stratonovich drift conversion to a nonlinear electronic circuit, and find which form of the noisy circuit equation samples the Boltzmann distribution.

Worked computation 4.23 (Nonlinear Varactor and State-Dependent Conductance *RC* Circuit with Multiplicative Thermal Noise)**.** Consider the single-node nonlinear *RC* circuit shown in Fig. 4.3, operating in contact with a thermal reservoir at temperature T ($\beta = (k_{\mathrm{B}}T)^{-1}$). We analyze two sources of multiplicative noise that occur in subthreshold CMOS circuits (Chapters 12 and 13):

Part A: A linear capacitor C_0 coupled to a state-dependent differential conductance $G(v) > 0$ (such as a subthreshold MOSFET channel or diode-connected load) and a potential-shaping current source $-G(v)U'(v)$ designed to sample a target electrostatic energy $E(v) = C_0U(v)$ (where $U(v) = \frac{1}{2}v^2$ for a purely passive *RC* circuit).

Part B: A nonlinear voltage-dependent capacitor (varactor) with differential capacitance $C(v) := \frac{\mathrm{d}q(v)}{\mathrm{d}v} > 0$ (such as a MOS gate-to-substrate capacitance across depletion and weak inversion), discharged through a conductance that we take constant, G_0.

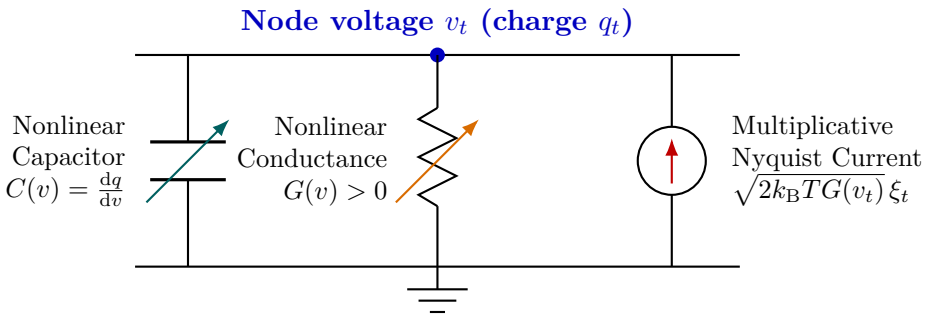


Figure 4.3: Schematic of the nonlinear RC circuit analyzed in Worked computation 4.23, featuring a voltage-dependent differential capacitance $C(v) = \frac{dq}{dv}$, a state-dependent conductance $G(v)$, and a multiplicative thermal noise current $\sqrt{2k_BTG(v_t)}\,\xi_t$ at temperature T.

Part A: State-Dependent Conductance $G(v)$ and Breakdown of Naive Itô Discretization

Suppose first that the capacitance $C_0 > 0$ is constant, and the node experiences a deterministic restoring current $I_{\text{det}}(v) = -\frac{G(v)}{C_0}E'(v)$ derived from a confining energy function $E(v)$ (with $\int_{-\infty}^{\infty} e^{-\beta E(v)}\,dv < \infty$), together with a thermal current noise of instantaneous Johnson–Nyquist amplitude $\sqrt{2k_BTG(v)}$. Writing the naive Langevin equation in the general λ-discretization ($\lambda \in [0,1]$) gives

$$dv_t = -\frac{G(v_t)}{C_0^2}E'(v_t)\,dt + \frac{\sqrt{2k_BTG(v_t)}}{C_0} \overset{(\lambda)}{\cdot} dW_t. \tag{4.109}$$

Denote the state-dependent volatility and diffusion coefficients by

$$B(v) := \frac{\sqrt{2k_BTG(v)}}{C_0}, \qquad D(v) := \frac{1}{2}B(v)^2 = \frac{k_BTG(v)}{C_0^2} = \frac{G(v)}{\beta C_0^2}. \tag{4.110}$$

The deterministic drift in Eq. (4.109) is $\underline{a}(v) = -\beta D(v)E'(v)$, and differentiating $B(v)$ with respect to v gives

$$B(v)B'(v) = \frac{1}{2}\frac{d}{dv}\left[B(v)^2\right] = \frac{dD(v)}{dv} = D'(v) = \frac{k_BTG'(v)}{C_0^2}. \tag{4.111}$$

By the general λ-to-Itô conversion formula Eq. (4.74), the λ-SDE Eq. (4.109) is equivalent to the Itô SDE

$$dv_t = \underbrace{\left[-\beta D(v_t)E'(v_t) + \lambda D'(v_t)\right]}_{a_\lambda(v_t)}dt + \sqrt{2D(v_t)}\cdot dW_t. \tag{4.112}$$

We find the stationary density $p_\lambda(v)$ of Eq. (4.112) for each $\lambda \in [0,1]$ from the stationarity condition $\mathbb{E}[\mathcal{L}f(v_\infty)] = 0$ for all smooth compactly supported test functions $f \in C_c^\infty(\mathbb{R})$. Applying Itô's formula (Theorem 4.12) to $f(v_t)$ gives

$$df(v_t) = \left[a_\lambda(v_t)f'(v_t) + D(v_t)f''(v_t)\right]dt + \sqrt{2D(v_t)}f'(v_t)\cdot dW_t. \tag{4.113}$$

In a stationary state $v_t \sim p_\lambda(v)$, $\frac{d}{dt}\mathbb{E}[f(v_t)] = 0$, so integrating by parts twice against $p_\lambda(v)$ (where boundary terms at $\pm\infty$ vanish since f has compact support) yields

$$0 = \int_{-\infty}^{\infty}\left[a_\lambda(v)f'(v) + D(v)f''(v)\right]p_\lambda(v)\,dv$$

$$= \int_{-\infty}^{\infty} f'(v) \left[a_\lambda(v) p_\lambda(v) - \frac{\mathrm{d}}{\mathrm{d}v}(D(v) p_\lambda(v)) \right] \mathrm{d}v. \tag{4.114}$$

For Eq. (4.114) to vanish for every test function f on $\mathbb{R}$ with $p_\lambda(\pm\infty) = 0$, the stationary probability current $J_\lambda(v) := a_\lambda(v)p_\lambda(v) - \frac{\mathrm{d}}{\mathrm{d}v}[D(v)p_\lambda(v)]$ must vanish identically ($J_\lambda(v) = 0$). Using the product rule $\frac{\mathrm{d}}{\mathrm{d}v}[D(v)p_\lambda(v)] = D'(v)p_\lambda(v) + D(v)p'_\lambda(v)$ and substituting $a_\lambda(v) = -\beta D(v)E'(v) + \lambda D'(v)$ gives the first-order ODE

$$D(v)p'_\lambda(v) + (1-\lambda)D'(v)p_\lambda(v) = -\beta D(v)E'(v)p_\lambda(v). \tag{4.115}$$

Dividing through by $D(v)p_\lambda(v) > 0$, we can integrate Eq. (4.115) in closed form:

$$\begin{aligned} & \frac{p'_\lambda(v)}{p_\lambda(v)} = -(1-\lambda)\frac{D'(v)}{D(v)} - \beta E'(v) \\ \Longrightarrow \quad & \frac{\mathrm{d}}{\mathrm{d}v} \ln p_\lambda(v) = \frac{\mathrm{d}}{\mathrm{d}v} \left[-(1-\lambda)\ln G(v) - \beta E(v) + \text{const} \right] \\ \Longrightarrow \quad & p_\lambda(v) = \frac{1}{Z_\lambda} \left[G(v)\right]^{-(1-\lambda)} \exp(-\beta E(v)) \,. \end{aligned} \tag{4.116}$$

Equation (4.116) has direct consequences for circuit modeling and simulation:

- **Naive Itô discretization ($\lambda = 0$).** Evaluating $\sqrt{2k_\mathrm{B}TG(v)}$ at the left endpoint without drift correction produces
$$p_\mathrm{Ito}(v) = \frac{1}{Z_0}\frac{1}{G(v)} \exp(-\beta E(v)) \,, \tag{4.117}$$
which is thermodynamically wrong. The node spends too much time in regions of low conductance $G(v)$, where the noise kicks are small and it lingers. In equilibrium statistical mechanics (Chapter 1) the canonical distribution $p_\mathrm{eq}(v) = Z^{-1}e^{-\beta E(v)}$ can depend *only* on the Hamiltonian/energy $E(v)$ and temperature T, never on a kinetic transport coefficient like the resistance $R(v) = 1/G(v)$.

- **Stratonovich without the extra drift ($\lambda = 1/2$).** Using the midpoint integral in Eq. (4.109) yields $p_{1/2}(v) \propto G(v)^{-1/2}\exp(-\beta E(v))$, which still retains half the kinetic distortion if the physical drift is written naively as $-\beta D(v)E'(v)$ without the microscopic barrier-transport drift (Section 5.5 and Chapter 13).

- **Thermodynamically consistent SDE.** For the circuit to sample the true Boltzmann distribution $p_\mathrm{eq}(v) = Z^{-1}\exp(-\beta E(v))$, the exponent $-(1-\lambda)$ on $G(v)$ in Eq. (4.116) must vanish, that is, the naive equation must be read with $\lambda = 1$. Rewritten in Itô ($\lambda = 0$) and Stratonovich ($\lambda = 1/2$) form, the consistent SDE is

$$\text{Itô form:} \quad \mathrm{d}v_t = \left[-\frac{G(v_t)}{C_0^2}E'(v_t) + \frac{k_\mathrm{B}T}{C_0^2}G'(v_t) \right] \mathrm{d}t + \frac{\sqrt{2k_\mathrm{B}TG(v_t)}}{C_0} \cdot \mathrm{d}W_t, \tag{4.118}$$

$$\text{Stratonovich form:} \quad \mathrm{d}v_t = \left[-\frac{G(v_t)}{C_0^2}E'(v_t) + \frac{k_\mathrm{B}T}{2C_0^2}G'(v_t) \right] \mathrm{d}t + \frac{\sqrt{2k_\mathrm{B}TG(v_t)}}{C_0} \circ \mathrm{d}W_t. \tag{4.119}$$

Part B: Nonlinear Varactor Capacitance $C(v)$ and Itô vs. Stratonovich Coordinate Transformation

Now suppose the capacitor itself is nonlinear, with charge-voltage relation $q = q(v)$ and strictly positive differential capacitance $C(v) := \frac{\mathrm{d}q(v)}{\mathrm{d}v} > 0$ (for example, $C(v) = C_0(1+\gamma v^2)$). Adding an

infinitesimal charge $\mathrm{d}q = C(v)\,\mathrm{d}v$ at voltage v requires electrostatic work $\mathrm{d}E = v\,\mathrm{d}q = C(v)v\,\mathrm{d}v$, so the stored energy as a function of voltage is

$$E_{\mathrm{cap}}(v) = \int_0^{q(v)} \tilde{v}(q')\,\mathrm{d}q' = \int_0^{v} C(v')\,v'\,\mathrm{d}v', \qquad \text{so that} \qquad E'_{\mathrm{cap}}(v) = C(v)\,v. \tag{4.120}$$

When $C(v)$ is not constant, $E_{\mathrm{cap}}(v)$ is *not* equal to $\frac{1}{2}C(v)v^2$.

In the charge coordinate q_t, Kirchhoff's current law states that $\frac{\mathrm{d}q_t}{\mathrm{d}t}$ equals the deterministic current $-G(v_t)v_t = -G(v(q_t))\,\frac{\partial E_{\mathrm{cap}}}{\partial q}$ plus the thermal noise current. With a constant conductance G_0, the noise in the charge coordinate q_t has constant (additive) amplitude $\sqrt{2k_{\mathrm{B}}TG_0}$, so the q-coordinate SDE is unambiguous:

$$\mathrm{d}q_t = -G_0 v(q_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}TG_0}\,\mathrm{d}W_t, \qquad (\mathrm{d}q_t)^2 = 2k_{\mathrm{B}}TG_0\,\mathrm{d}t. \tag{4.121}$$

To find the SDE obeyed by the node voltage $v_t = v(q_t)$, note that since $\frac{\mathrm{d}q}{\mathrm{d}v} = C(v)$, the inverse function $v(q)$ has first and second derivatives with respect to q given by the chain rule:

$$\frac{\mathrm{d}v}{\mathrm{d}q} = \frac{1}{C(v)}, \tag{4.122}$$

$$\frac{\mathrm{d}^2v}{\mathrm{d}q^2} = \frac{\mathrm{d}}{\mathrm{d}v}\left(\frac{1}{C(v)}\right)\frac{\mathrm{d}v}{\mathrm{d}q} = -\frac{C'(v)}{C(v)^2}\cdot\frac{1}{C(v)} = -\frac{C'(v)}{C(v)^3}. \tag{4.123}$$

We transform Eq. (4.121) from q_t to v_t in two ways:

1. **Stratonovich chain rule** (Eq. (4.68)). Dividing directly by $C(v_t)$ gives

$$\mathrm{d}v_t = \frac{\mathrm{d}v}{\mathrm{d}q}\circ\mathrm{d}q_t = -\frac{G_0 v_t}{C(v_t)}\,\mathrm{d}t + \frac{\sqrt{2k_{\mathrm{B}}TG_0}}{C(v_t)}\circ\mathrm{d}W_t. \tag{4.124}$$

   Although the noise in q_t is additive, the nonlinear capacitance makes the noise in the voltage coordinate v_t multiplicative, with $B(v) = \sqrt{2k_{\mathrm{B}}TG_0}/C(v)$.

2. **Itô's formula** (Theorem 4.12). Expanding $\mathrm{d}v(q_t) = \frac{\mathrm{d}v}{\mathrm{d}q}\,\mathrm{d}q_t + \frac{1}{2}\frac{\mathrm{d}^2v}{\mathrm{d}q^2}(\mathrm{d}q_t)^2$ using Eqs. (4.122) and (4.123) (or, equivalently, converting Eq. (4.124) to Itô form with the drift shift $\frac{1}{2}B(v)B'(v) = -k_{\mathrm{B}}TG_0C'(v)/C(v)^3$) gives the voltage SDE in Itô form:

$$\mathrm{d}v_t = \left[-\frac{G_0 v_t}{C(v_t)} - \frac{k_{\mathrm{B}}TG_0C'(v_t)}{C(v_t)^3}\right]\mathrm{d}t + \frac{\sqrt{2k_{\mathrm{B}}TG_0}}{C(v_t)}\cdot\mathrm{d}W_t. \tag{4.125}$$

We now check the stationary voltage distribution $p_{\mathrm{eq}}^{(v)}(v)$ of Eq. (4.125). In the charge coordinate q, Eq. (4.121) is an overdamped Langevin equation with constant diffusion $D_q = k_{\mathrm{B}}TG_0$ and drift $-G_0\frac{\partial E_{\mathrm{cap}}}{\partial q}$, so its stationary charge density is $p_{\mathrm{eq}}^{(q)}(q) = Z^{-1}\exp(-\beta E_{\mathrm{cap}}(q))$. Under the change of variable $q = q(v)$ with Jacobian $\frac{\mathrm{d}q}{\mathrm{d}v} = C(v)$, the equilibrium voltage density acquires the Jacobian factor $C(v)$:

$$p_{\mathrm{eq}}^{(v)}(v) = p_{\mathrm{eq}}^{(q)}(q(v))\left|\frac{\mathrm{d}q}{\mathrm{d}v}\right| = \frac{1}{Z}\,C(v)\,\exp\!\left(-\beta\int_0^v C(v')\,v'\,\mathrm{d}v'\right). \tag{4.126}$$

Indeed, substituting $D(v) = k_{\mathrm{B}}TG_0/C(v)^2$ and the Itô drift $a(v) = -G_0v/C(v) - k_{\mathrm{B}}TG_0C'(v)/C(v)^3$ from Eq. (4.125) into the zero-current condition $a(v)p_{\mathrm{eq}}^{(v)}(v) = \frac{\mathrm{d}}{\mathrm{d}v}[D(v)p_{\mathrm{eq}}^{(v)}(v)]$ yields

$$\frac{\frac{\mathrm{d}}{\mathrm{d}v}p_{\mathrm{eq}}^{(v)}(v)}{p_{\mathrm{eq}}^{(v)}(v)} = \frac{a(v) - D'(v)}{D(v)} = \frac{-\frac{G_0v}{C(v)} - \frac{k_{\mathrm{B}}TG_0C'(v)}{C(v)^3} + \frac{2k_{\mathrm{B}}TG_0C'(v)}{C(v)^3}}{\frac{k_{\mathrm{B}}TG_0}{C(v)^2}} = -\beta C(v)v + \frac{C'(v)}{C(v)}, \tag{4.127}$$

whose integral is precisely $\ln p_{\mathrm{eq}}^{(v)}(v) = \ln C(v) - \beta \int_0^v C(v')v'\,\mathrm{d}v' + \text{const}$, confirming Eq. (4.126). Had we omitted the Itô curvature drift $-k_{\mathrm{B}}TG_0C'(v)/C(v)^3$ in Eq. (4.125), the simulated circuit would have converged to $C(v)^2\exp(-\beta E_{\mathrm{cap}}(v))$ instead of Eq. (4.126).

Summary

- **Proved in this chapter:**
  - Covariance structure $\mathbb{E}[W_{i,t}W_{j,s}] = \delta_{ij}\min(t,s)$, diffusive scaling $\mathbf{W}_{ct} \overset{d}{=} \sqrt{c}\mathbf{W}_t$, unbounded first variation (Proposition 4.3, Eq. (4.12)), and the $L^2(\Omega)$ quadratic variation theorem $\mathrm{d}W_i\,\mathrm{d}W_j = \delta_{ij}\,\mathrm{d}t$ (Theorem 4.4).
  - Discretization sensitivity $\int_0^T W_t \overset{(\lambda)}{\cdot} \mathrm{d}W_t = \frac{1}{2}W_T^2 + (\lambda - \frac{1}{2})T$ (Proposition 4.6), the Itô L^2 isometry (Theorem 4.8), and the martingale property of Itô integrals (Theorem 4.10).
  - Multi-dimensional Itô's formula (Theorem 4.12), the Stratonovich chain rule, the multi-dimensional Itô–Stratonovich drift conversion $a_{\mathrm{sp},i} = \frac{1}{2}\sum_{j,\alpha} B_{j\alpha}\frac{\partial B_{i\alpha}}{\partial x_j}$ (Theorem 4.16), and the one-dimensional Wong–Zakai theorem, with the final Grönwall step sketched (Theorem 4.18).
  - Strong existence and uniqueness of SDE solutions via Picard iteration and Khasminskii Lyapunov functions (Theorem 4.19), and Girsanov's theorem for path-measure Radon–Nikodym derivatives (Lemma 4.21 and Theorem 4.22).
  - Stationary laws of the RC circuit with state-dependent conductance $G(v)$ or nonlinear capacitance $C(v)$, and the drift each needs to sample the Boltzmann distribution (Worked computation 4.23).
- **Assumed:**
  - Existence of the Wiener measure on $\mathcal{C}([0,\infty);\mathbb{R}^d)$ via Kolmogorov's extension and continuity theorems, Doob's L^2 maximal martingale inequality, and uniform integrability under Novikov's condition.
- **Taken up later:**
  - Kolmogorov's continuity theorem and the Burkholder–Davis–Gundy inequalities, which we quote from [59]; Doob's maximal inequality and the Picard existence proof are in Chapter C.
  - Microscopic derivation of why state-dependent mobility requires the full divergence drift $+k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}(\mathbf{x})$ (Section 5.5) and why mesoscopic Poisson shot noise in subthreshold CMOS generates state-dependent diffusion (Chapters 11 and 13).
  - Application of Girsanov's theorem and Stratonovich midpoint path integrals to Onsager–Machlup actions and Crooks fluctuation theorems (Chapters 8 and 9), Anderson's reverse-time diffusion (Chapter 15), and differentiable hardware calibration (Chapter 18).
- **Open questions:**
  - Multi-dimensional Wong–Zakai (Lévy-area) corrections in non-reciprocally coupled subthreshold CMOS networks, such as the NEAT-RNs of Chapter 13 [38], when different transistors have unequal noise correlation times $\tau_{\mathrm{corr},i} \neq \tau_{\mathrm{corr},j}$.

Bibliographic notes

For mathematical treatments of Brownian motion, martingales, Itô's formula, and Girsanov's theorem, the classic graduate texts are Øksendal [80] and Karatzas and Shreve [59]. Pavliotis [85] connects stochastic analysis with statistical physics. The physical perspective on the Itô–Stratonovich dilemma and colored-noise limits is developed in van Kampen [112], Gardiner [39], and Risken [90], while the original convergence theorem for smooth approximations to Brownian motion is due to Wong and Zakai [119]. The role of Stratonovich calculus in trajectory energetics was pioneered by Sekimoto [98] and reviewed by Seifert [97]. For nonlinear stochastic circuit equations, state-dependent conductances, and capacitances at the $k_{\mathrm{B}}T$ scale relevant to thermodynamic hardware, see Freitas et al. [37], Freitas et al. [38], and Extropic Corporation [30, 31].

Chapter 5

Microscopic Baths, Generalized Langevin Dynamics, and the Fluctuation-Dissipation Theorem

In Chapter 4 we learned how to integrate stochastic differential equations. We did not ask where those equations come from. Every Langevin equation in this book has the same two ingredients: a friction force that drains energy out of the system, and a random force that pumps energy back in. For the RC sampler (the coupled *RC* network of Example 1.5), whose equation is $\mathbf{C}\,\mathrm{d}\mathbf{v}_t = -\mathbf{G}(\mathbf{v}_t - \mathbf{b})\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t$, the friction is the conductance matrix $\mathbf{G}$ and the random force is the Johnson–Nyquist current noise $\mathbf{B}\,\mathrm{d}\mathbf{W}_t$. Up to now we simply asserted that the two are tied together by $\mathbf{B}\mathbf{B}^\top = 2k_\mathrm{B}T\mathbf{G}$. This chapter derives that relation from Hamiltonian mechanics and then shows what it implies for measurable quantities such as noise spectra and response functions.

The physical picture is simple. A resistor, a viscous fluid, or the phonons of a silicon lattice are all large collections of microscopic degrees of freedom. When our slow coordinate moves, it pushes on these degrees of freedom and loses energy to them; this is friction. The same degrees of freedom, being at temperature T, also push back randomly; this is noise. Because both effects come from one and the same coupling, their strengths cannot be chosen independently. The result is the *fluctuation-dissipation theorem* (FDT), which comes in two forms. The *second kind* fixes the noise correlation in terms of the friction kernel. The *first kind* fixes the linear response of the system to a weak external force in terms of its equilibrium fluctuations. The Johnson–Nyquist formula of Chapter 11, the Einstein relation $D = \mu k_\mathrm{B}T$, and the requirement that a sampler relax to the Boltzmann distribution of Chapter 1 are all special cases.

We start from the underdamped Langevin equation and its overdamped limit, then derive friction and noise from a system coupled to a bath of harmonic oscillators (Section 5.2), which yields the FDT of the second kind. Linear response theory gives the first kind (Section 5.4). We then derive the extra drift that a state-dependent mobility requires, as promised in Chapter 4 (Section 5.5), and close by solving the N-node RC circuit exactly and checking the Nyquist relation (Worked computation 5.33).

For thermodynamic computing hardware, this chapter answers a practical question. A sampling cell of the kind studied in Chapter 13 is a set of capacitor nodes connected through resistive or transconductive elements. Whether the cell samples the intended distribution depends on the balance between the deterministic currents and the noise currents. If the noise is purely thermal and the couplings are passive and reciprocal, the FDT guarantees equilibrium sampling, and it also tells us which distribution is sampled. If the noise is shot noise from biased transistors, or the

couplings are non-reciprocal, the FDT fails, and the cell settles into a non-equilibrium steady state whose covariance must be computed from the Lyapunov equation (Chapters 6 and 13). Knowing exactly where the FDT holds and where it breaks is the first step in designing either kind of circuit.

5.1 Underdamped Langevin Dynamics and Adiabatic Elimination of Momenta

We begin with the most direct generalization of Newton's equations to a system in contact with a heat bath. A particle (or, more generally, a set of N generalized coordinates) moves in a potential energy landscape $E(\mathbf{x})$. In addition to the conservative force $-\nabla E$, it feels a friction force proportional to its velocity and a rapidly fluctuating random force. We postpone the microscopic justification to Section 5.2 and first study the resulting equations.

5.1.1 The Phase-Space Langevin Equation

Let $\mathbf{x} \in \mathbb{R}^N$ be the coordinates and $\mathbf{p} \in \mathbb{R}^N$ the conjugate momenta. Let $\mathbf{M} = \mathbf{M}^\top \succ \mathbf{0}$ be a constant mass matrix, so that the Hamiltonian is

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\mathbf{p}^\top \mathbf{M}^{-1}\mathbf{p} + E(\mathbf{x}). \tag{5.1}$$

Let $\mathbf{\Gamma} = \mathbf{\Gamma}^\top \succ \mathbf{0}$ be a constant friction tensor with units of force per velocity. The friction force is $-\mathbf{\Gamma}\dot{\mathbf{x}} = -\mathbf{\Gamma}\mathbf{M}^{-1}\mathbf{p}$.

Definition 5.1 (Underdamped Langevin Equation)**.** The *underdamped* (or *kinetic*) Langevin equation at temperature T is the Itô SDE on phase space $\mathbb{R}^{2N}$

$$\mathrm{d}\mathbf{x}_t = \mathbf{M}^{-1}\mathbf{p}_t \, \mathrm{d}t, \tag{5.2}$$

$$\mathrm{d}\mathbf{p}_t = \Big[-\nabla E(\mathbf{x}_t) - \mathbf{\Gamma}\mathbf{M}^{-1}\mathbf{p}_t\Big] \, \mathrm{d}t + \sqrt{2k_{\mathrm{B}}T}\, \mathbf{\Gamma}^{1/2} \, \mathrm{d}\mathbf{W}_t, \tag{5.3}$$

where $\mathbf{W}_t$ is a standard N-dimensional Brownian motion and $\mathbf{\Gamma}^{1/2}$ is the symmetric positive square root of $\mathbf{\Gamma}$.

In one dimension with mass m and scalar friction coefficient γ_0, Eqs. (5.2) and (5.3) read $\mathrm{d}x = (p/m)\,\mathrm{d}t$ and $\mathrm{d}p = [-E'(x) - (\gamma_0/m)p]\,\mathrm{d}t + \sqrt{2\gamma_0 k_{\mathrm{B}}T}\,\mathrm{d}W_t$, which is the form used for Kramers' escape problem in Chapter 7. The subscript 0 anticipates Section 5.3, where γ_0 will turn out to be the zero-frequency value of a more general memory kernel.

Three remarks about Definition 5.1 are in order. First, the noise enters only the momentum equation and its amplitude does not depend on $\mathbf{p}$; since $\mathbf{x}_t$ is a differentiable function of time, the Itô and Stratonovich readings of Eq. (5.3) coincide even if $\mathbf{\Gamma}$ were to depend on $\mathbf{x}$. This fact will be important in Section 5.5. Second, the noise covariance is $\sqrt{2k_{\mathrm{B}}T}\mathbf{\Gamma}^{1/2}(\sqrt{2k_{\mathrm{B}}T}\mathbf{\Gamma}^{1/2})^\top = 2k_{\mathrm{B}}T\mathbf{\Gamma}$; we have built the FDT into the definition, and Theorem 5.18 will show that no other choice is consistent with thermal equilibrium. Third, the diffusion tensor on phase space is degenerate: it acts only in the $\mathbf{p}$ directions. The generator is therefore hypoelliptic rather than elliptic. Convergence to equilibrium still holds, but it relies on the Hamiltonian part of the dynamics to carry fluctuations from $\mathbf{p}$ into $\mathbf{x}$.

5.1.2 The Boltzmann–Gibbs Distribution Is Stationary

The Fokker–Planck equation for the phase-space density $\rho(\mathbf{x}, \mathbf{p}, t)$ follows from the general construction of Chapter 6. The drift vector on $\mathbb{R}^{2N}$ is $(\mathbf{M}^{-1}\mathbf{p},\, -\nabla E - \boldsymbol{\Gamma}\mathbf{M}^{-1}\mathbf{p})$ and the diffusion tensor is $k_{\mathrm{B}}T\boldsymbol{\Gamma}$ in the momentum block and zero elsewhere. Hence

$$\frac{\partial \rho}{\partial t} = \underbrace{-\nabla_{\mathbf{x}} \cdot (\mathbf{M}^{-1}\mathbf{p}\,\rho) + \nabla_{\mathbf{p}} \cdot (\nabla E(\mathbf{x})\,\rho)}_{\text{Hamiltonian (Liouville) part}} + \underbrace{\nabla_{\mathbf{p}} \cdot \Big(\boldsymbol{\Gamma}\mathbf{M}^{-1}\mathbf{p}\,\rho + k_{\mathrm{B}}T\boldsymbol{\Gamma}\nabla_{\mathbf{p}}\rho\Big)}_{\text{dissipative part}}. \tag{5.4}$$

This is the *Kramers* (or Klein–Kramers) equation.

Proposition 5.2 (Stationarity of the Phase-Space Gibbs Measure)**.** *The density*

$$\rho_{\mathrm{eq}}(\mathbf{x}, \mathbf{p}) = \frac{1}{Z} \exp\Big(-\beta\Big[\frac{1}{2}\mathbf{p}^{\top}\mathbf{M}^{-1}\mathbf{p} + E(\mathbf{x})\Big]\Big) \tag{5.5}$$

is a stationary solution of Eq. (5.4). Each of the two bracketed parts vanishes separately. The $\mathbf{x}$*-marginal of* ρ_{eq} *is* $p_{\mathrm{eq}}(\mathbf{x}) \propto e^{-\beta E(\mathbf{x})}$*, independently of* $\mathbf{M}$ *and* $\boldsymbol{\Gamma}$*.*

Proof. The gradients of ρ_{eq} are $\nabla_{\mathbf{x}}\rho_{\mathrm{eq}} = -\beta\nabla E\,\rho_{\mathrm{eq}}$ and $\nabla_{\mathbf{p}}\rho_{\mathrm{eq}} = -\beta\mathbf{M}^{-1}\mathbf{p}\,\rho_{\mathrm{eq}}$.

Hamiltonian part. Since $\mathbf{M}^{-1}\mathbf{p}$ does not depend on $\mathbf{x}$ and $\nabla E(\mathbf{x})$ does not depend on $\mathbf{p}$, the divergences act only on ρ_{eq}:

$$-\mathbf{M}^{-1}\mathbf{p} \cdot \nabla_{\mathbf{x}}\rho_{\mathrm{eq}} + \nabla E \cdot \nabla_{\mathbf{p}}\rho_{\mathrm{eq}} = \beta\,\rho_{\mathrm{eq}}\Big[(\mathbf{M}^{-1}\mathbf{p}) \cdot \nabla E - \nabla E \cdot (\mathbf{M}^{-1}\mathbf{p})\Big] = 0.$$

Dissipative part. The vector inside the $\mathbf{p}$-divergence is

$$\boldsymbol{\Gamma}\mathbf{M}^{-1}\mathbf{p}\,\rho_{\mathrm{eq}} + k_{\mathrm{B}}T\boldsymbol{\Gamma}(-\beta\mathbf{M}^{-1}\mathbf{p}\,\rho_{\mathrm{eq}}) = \boldsymbol{\Gamma}\mathbf{M}^{-1}\mathbf{p}\,\rho_{\mathrm{eq}} - \boldsymbol{\Gamma}\mathbf{M}^{-1}\mathbf{p}\,\rho_{\mathrm{eq}} = \mathbf{0},$$

using $k_{\mathrm{B}}T\beta = 1$. So its divergence vanishes as well. Finally, integrating Eq. (5.5) over $\mathbf{p}$ gives the Gaussian integral $\int \mathrm{d}^N p\, e^{-\beta\mathbf{p}^{\top}\mathbf{M}^{-1}\mathbf{p}/2} = (2\pi k_{\mathrm{B}}T)^{N/2}(\det\mathbf{M})^{1/2}$, which is independent of $\mathbf{x}$, so the $\mathbf{x}$-marginal is proportional to $e^{-\beta E(\mathbf{x})}$. □

The proof shows something sharper than stationarity: the dissipative probability current in momentum space vanishes *pointwise*, and it does so only because the noise strength $k_{\mathrm{B}}T\boldsymbol{\Gamma}$ exactly matches the friction $\boldsymbol{\Gamma}$. Had we written the noise as $\boldsymbol{\sigma}\,\mathrm{d}\mathbf{W}_t$ with $\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top} \neq 2k_{\mathrm{B}}T\boldsymbol{\Gamma}$, the Maxwellian factor $e^{-\beta\mathbf{p}^{\top}\mathbf{M}^{-1}\mathbf{p}/2}$ would not be stationary. This is the content of the FDT of the second kind in its Markovian form, which we state and prove in Section 5.3.

The same balance can be seen at the level of a single trajectory. Apply Itô's formula to the energy $H(\mathbf{x}_t, \mathbf{p}_t)$ of Eq. (5.1). The second-order term is $\frac{1}{2}\,\mathrm{d}\mathbf{p}^{\top}\mathbf{M}^{-1}\,\mathrm{d}\mathbf{p} = k_{\mathrm{B}}T\,\mathrm{tr}(\mathbf{M}^{-1}\boldsymbol{\Gamma})\,\mathrm{d}t$, and the conservative terms $\nabla E \cdot \mathrm{d}\mathbf{x}$ and $-\mathbf{M}^{-1}\mathbf{p} \cdot \nabla E\,\mathrm{d}t$ cancel, leaving, with $\dot{\mathbf{x}} = \mathbf{M}^{-1}\mathbf{p}$,

$$\mathrm{d}H = \Big[-\dot{\mathbf{x}}^{\top}\boldsymbol{\Gamma}\dot{\mathbf{x}} + k_{\mathrm{B}}T\,\mathrm{tr}\big(\mathbf{M}^{-1}\boldsymbol{\Gamma}\big)\Big]\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T}\,\dot{\mathbf{x}}^{\top}\boldsymbol{\Gamma}^{1/2}\,\mathrm{d}\mathbf{W}_t. \tag{5.6}$$

Because E does not depend on time, no work is done on the system, and every change of H is energy exchanged with the bath, that is, heat. The first term is the power removed by friction, the second the mean power injected by the noise, and the last a zero-mean fluctuating exchange. In equilibrium $\big\langle \dot{\mathbf{x}}\dot{\mathbf{x}}^{\top} \big\rangle = k_{\mathrm{B}}T\mathbf{M}^{-1}$, so $\big\langle \dot{\mathbf{x}}^{\top}\boldsymbol{\Gamma}\dot{\mathbf{x}} \big\rangle = k_{\mathrm{B}}T\,\mathrm{tr}(\boldsymbol{\Gamma}\mathbf{M}^{-1})$ and the mean heat flow vanishes. Equation (5.6) is the starting point for the trajectory energetics of Chapter 8, where the overdamped limit of this heat is taken.

5.1.3 The Electrical Analogue: A Series RLC Loop

The RC sampler is an electrical circuit, so it is useful to translate Definition 5.1 into circuit language. Consider a capacitor C in series with an inductor L and a resistor R at temperature T. Take the capacitor charge q as the coordinate and the inductor flux linkage $\lambda = LI = L\dot{q}$ as the conjugate momentum. The stored energy is $H = \lambda^2/(2L) + q^2/(2C)$. Kirchhoff's voltage law around the loop, with the Johnson–Nyquist Thévenin noise source $\delta V(t)$ of the resistor (Theorem 11.5), reads

$$\mathrm{d}q_t = \frac{\lambda_t}{L}\,\mathrm{d}t, \qquad \mathrm{d}\lambda_t = \Big[-\frac{q_t}{C} - R\,\frac{\lambda_t}{L}\Big]\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}TR}\,\mathrm{d}W_t. \tag{5.7}$$

This is Eqs. (5.2) and (5.3) with the dictionary

$$x \leftrightarrow q, \qquad p \leftrightarrow \lambda, \qquad m \leftrightarrow L, \qquad E(x) \leftrightarrow \frac{q^2}{2C}, \qquad \gamma_0 \leftrightarrow R. \tag{5.8}$$

The noise term $\sqrt{2k_{\mathrm{B}}TR}\,\mathrm{d}W_t$ corresponds to a white voltage noise with double-sided spectral density $S_V^{\mathrm{ds}} = 2k_{\mathrm{B}}TR$, which is precisely the classical Nyquist result Eq. (11.28). The two relaxation times of the loop are the inductive time $\tau_L = L/R$, which plays the role of the momentum relaxation time $\tau_p = m/\gamma_0$, and the capacitive time $\tau_{RC} = RC$, which plays the role of the slow positional relaxation time γ_0/E''.

5.1.4 Adiabatic Elimination of the Momenta

In most of the systems of interest to us the momenta relax much faster than the coordinates. A micron-sized colloidal particle in water loses its velocity memory in well under a microsecond but diffuses over its own size in a fraction of a second or more. In a CMOS circuit, the parasitic inductance of the interconnect gives an inductive time $\tau_L = L/R$ many orders of magnitude shorter than τ_{RC} (see Section 5.1.5). On time scales long compared with τ_p, the momentum is always close to its local equilibrium and can be eliminated. The following heuristic shows the mechanism in one line.

Heuristic Derivation: Dropping the Inertial Term

Rewrite Eq. (5.3) using $\boldsymbol{\Gamma}\mathbf{M}^{-1}\mathbf{p}_t\,\mathrm{d}t = \boldsymbol{\Gamma}\,\mathrm{d}\mathbf{x}_t$, which follows from Eq. (5.2):

$$\boldsymbol{\Gamma}\,\mathrm{d}\mathbf{x}_t = -\nabla E(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T}\,\boldsymbol{\Gamma}^{1/2}\,\mathrm{d}\mathbf{W}_t - \mathrm{d}\mathbf{p}_t. \tag{5.9}$$

This is an exact force balance: friction equals the sum of the conservative force, the random force, and minus the rate of change of momentum. The last term integrates to $\mathbf{p}_t - \mathbf{p}_0$. In thermal equilibrium $\mathbf{p}$ has covariance $k_{\mathrm{B}}T\mathbf{M}$, so $\mathbf{p}_t - \mathbf{p}_0$ is of order $\sqrt{mk_{\mathrm{B}}T}$, which vanishes as the mass goes to zero. Dropping it and multiplying by $\boldsymbol{\mu} := \boldsymbol{\Gamma}^{-1}$ gives

$$\mathrm{d}\mathbf{x}_t = -\boldsymbol{\mu}\nabla E(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T}\,\boldsymbol{\mu}^{1/2}\,\mathrm{d}\mathbf{W}_t, \tag{5.10}$$

because $\boldsymbol{\Gamma}^{-1}\boldsymbol{\Gamma}^{1/2} = \boldsymbol{\Gamma}^{-1/2} = \boldsymbol{\mu}^{1/2}$. The diffusion tensor of Eq. (5.10) is $\mathbf{D} = \frac{1}{2}(2k_{\mathrm{B}}T\boldsymbol{\mu}) = k_{\mathrm{B}}T\boldsymbol{\mu}$, which is the Einstein relation.

Equation (5.10) is the overdamped Langevin or *Smoluchowski* SDE. With $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$ it is the form used in Chapters 6 to 8. We now make the heuristic precise.

Assumption 5.3 (Small-Mass Regime)**.** The mass matrix is $\mathbf{M} = m\mathbf{I}$ with $m > 0$ a small parameter. The friction tensor $\mathbf{\Gamma}$ is constant, symmetric, and positive definite with smallest eigenvalue $\gamma_{\min} > 0$. The force $-\nabla E$ is bounded, $\sup_{\mathbf{x}} \|\nabla E(\mathbf{x})\| \le F_{\max}$, and globally Lipschitz with constant L_E. The initial momentum $\mathbf{p}_0$ is drawn from the Maxwell distribution $\mathcal{N}(\mathbf{0}, mk_{\mathrm{B}}T\mathbf{I})$, independently of $\mathbf{x}_0$ and of $\mathbf{W}$.

Theorem 5.4 (Smoluchowski–Kramers Limit with Constant Friction)**.** *Under Assumption 5.3, let* $(\mathbf{x}_t^m, \mathbf{p}_t^m)$ *solve Eqs. (5.2) and (5.3) and let* $\mathbf{x}_t$ *solve the overdamped SDE Eq. (5.10) with the same initial position* $\mathbf{x}_0$ *and the same Brownian motion* $\mathbf{W}$*. Then for every* $t \ge 0$*,*

$$\mathbb{E}[\|\mathbf{x}_t^m - \mathbf{x}_t\|] \le K\sqrt{m}\, e^{L_E t/\gamma_{\min}}, \qquad K := \frac{2}{\gamma_{\min}}\left(2\sqrt{Nk_{\mathrm{B}}T} + \frac{F_{\max}\sqrt{m}}{\gamma_{\min}}\right). \tag{5.11}$$

In particular $\mathbf{x}_t^m \to \mathbf{x}_t$ *in* $L^1(\Omega)$ *as* $m \to 0$*, uniformly on compact time intervals.*

Proof. Step 1: integral form. Integrating Eq. (5.9) from 0 to t and multiplying by $\boldsymbol{\mu} = \mathbf{\Gamma}^{-1}$,

$$\mathbf{x}_t^m = \mathbf{x}_0 - \int_0^t \boldsymbol{\mu}\nabla E(\mathbf{x}_s^m)\, \mathrm{d}s + \sqrt{2k_{\mathrm{B}}T}\, \boldsymbol{\mu}^{1/2}\mathbf{W}_t - \boldsymbol{\mu}(\mathbf{p}_t^m - \mathbf{p}_0). \tag{5.12}$$

The overdamped solution satisfies the same identity without the last term:

$$\mathbf{x}_t = \mathbf{x}_0 - \int_0^t \boldsymbol{\mu}\nabla E(\mathbf{x}_s)\, \mathrm{d}s + \sqrt{2k_{\mathrm{B}}T}\, \boldsymbol{\mu}^{1/2}\mathbf{W}_t. \tag{5.13}$$

The noise terms are identical and cancel in the difference $\boldsymbol{\Delta}_t := \mathbf{x}_t^m - \mathbf{x}_t$:

$$\boldsymbol{\Delta}_t = -\int_0^t \boldsymbol{\mu}[\nabla E(\mathbf{x}_s^m) - \nabla E(\mathbf{x}_s)]\, \mathrm{d}s - \boldsymbol{\mu}(\mathbf{p}_t^m - \mathbf{p}_0). \tag{5.14}$$

Step 2: bound on the momentum. With $\mathbf{M} = m\mathbf{I}$, the momentum equation is linear in $\mathbf{p}$ with a forcing term, and the variation-of-constants formula gives

$$\mathbf{p}_t^m = e^{-\mathbf{\Gamma}t/m}\mathbf{p}_0 - \int_0^t e^{-\mathbf{\Gamma}(t-s)/m}\nabla E(\mathbf{x}_s^m)\, \mathrm{d}s + \sqrt{2k_{\mathrm{B}}T}\int_0^t e^{-\mathbf{\Gamma}(t-s)/m}\mathbf{\Gamma}^{1/2}\, \mathrm{d}\mathbf{W}_s. \tag{5.15}$$

Since $\mathbf{\Gamma}$ is symmetric with eigenvalues at least $\gamma_{\min}$, the operator norm obeys $\left\|e^{-\mathbf{\Gamma}u/m}\right\| \le e^{-\gamma_{\min}u/m}$. We bound the three terms separately. The first has $\mathbb{E}\left[\left\|e^{-\mathbf{\Gamma}t/m}\mathbf{p}_0\right\|\right] \le \mathbb{E}[\|\mathbf{p}_0\|] \le (\mathbb{E}\left[\|\mathbf{p}_0\|^2\right])^{1/2} = \sqrt{Nmk_{\mathrm{B}}T}$. The second is bounded deterministically by $F_{\max}\int_0^t e^{-\gamma_{\min}(t-s)/m}\, \mathrm{d}s \le F_{\max}m/\gamma_{\min}$. For the third, the Itô isometry (Theorem 4.8) gives

$$\mathbb{E}\left[\left\|\sqrt{2k_{\mathrm{B}}T}\int_0^t e^{-\mathbf{\Gamma}(t-s)/m}\mathbf{\Gamma}^{1/2}\, \mathrm{d}\mathbf{W}_s\right\|^2\right] = 2k_{\mathrm{B}}T\int_0^t \mathrm{tr}\big(e^{-2\mathbf{\Gamma}u/m}\mathbf{\Gamma}\big)\, \mathrm{d}u = mk_{\mathrm{B}}T\ \mathrm{tr}\big(\mathbf{I} - e^{-2\mathbf{\Gamma}t/m}\big) \le Nmk_{\mathrm{B}}T,$$

where we used $\int_0^t e^{-2\mathbf{\Gamma}u/m}\mathbf{\Gamma}\, \mathrm{d}u = \frac{m}{2}(\mathbf{I} - e^{-2\mathbf{\Gamma}t/m})$, valid because $\mathbf{\Gamma}$ commutes with its own exponential. By Jensen's inequality the expected norm of the third term is at most $\sqrt{Nmk_{\mathrm{B}}T}$. Combining, and adding $\mathbb{E}[\|\mathbf{p}_0\|] \le \sqrt{Nmk_{\mathrm{B}}T}$ for the subtracted initial momentum,

$$\mathbb{E}[\|\mathbf{p}_t^m - \mathbf{p}_0\|] \le 3\sqrt{Nmk_{\mathrm{B}}T} + \frac{F_{\max}m}{\gamma_{\min}} \le 2\sqrt{m}\left(2\sqrt{Nk_{\mathrm{B}}T} + \frac{F_{\max}\sqrt{m}}{\gamma_{\min}}\right). \tag{5.16}$$

Step 3: Gronwall. Taking norms and expectations in Eq. (5.14), using $\|\boldsymbol{\mu}\| = 1/\gamma_{\min}$ and the Lipschitz bound $\|\nabla E(\mathbf{x}_s^m) - \nabla E(\mathbf{x}_s)\| \le L_E \, \|\boldsymbol{\Delta}_s\|$, the function $\delta(t) := \mathbb{E}[\|\boldsymbol{\Delta}_t\|]$ satisfies

$$\delta(t) \le \frac{L_E}{\gamma_{\min}} \int_0^t \delta(s)\, \mathrm{d}s + \frac{1}{\gamma_{\min}} \mathbb{E}[\|\mathbf{p}_t^m - \mathbf{p}_0\|] \le \frac{L_E}{\gamma_{\min}} \int_0^t \delta(s)\, \mathrm{d}s + K\sqrt{m}.$$

Gronwall's inequality for a constant inhomogeneity gives $\delta(t) \le K\sqrt{m}\, e^{L_E t/\gamma_{\min}}$, which is Eq. (5.11). □

The theorem says that the overdamped trajectory is the small-mass limit of the inertial one, with an error of order $\sqrt{m}$, that is, of order $\sqrt{\tau_p}$. The rate is set by the size of thermal momentum fluctuations, not by the mean force. Two assumptions deserve comment. The constant friction tensor was used in Step 1: when $\boldsymbol{\Gamma}$ depends on $\mathbf{x}$, the identity $\boldsymbol{\Gamma}\, \mathrm{d}\mathbf{x} = \mathrm{d}(\boldsymbol{\Gamma}\mathbf{x})$ fails, and an additional drift appears in the limit. That drift is the subject of Section 5.5. The boundedness of ∇E is technical; for confining potentials such as the quartic potential of the double-well sampler (Example 1.6), the same conclusion holds on the event that the trajectory stays in a large ball, whose probability tends to one.

5.1.5 The Crossover from Ballistic to Diffusive Motion

The elimination of momenta discards the dynamics on time scales shorter than τ_p. A free particle shows explicitly what is lost.

Example 5.5 (Ornstein–Fürth Mean-Square Displacement)**.** Set $E \equiv 0$, $N = 1$, mass m, friction γ_0, and let $u_t := p_t/m$ be the velocity. Dividing Eq. (5.3) by m gives the Ornstein–Uhlenbeck equation

$$\mathrm{d}u_t = -\frac{u_t}{\tau_p}\, \mathrm{d}t + \frac{\sqrt{2\gamma_0 k_{\mathrm{B}} T}}{m}\, \mathrm{d}W_t, \qquad \tau_p := \frac{m}{\gamma_0}. \tag{5.17}$$

Its solution is $u_t = e^{-t/\tau_p} u_0 + \frac{\sqrt{2\gamma_0 k_{\mathrm{B}} T}}{m} \int_0^t e^{-(t-s)/\tau_p}\, \mathrm{d}W_s$. If u_0 is drawn from the stationary law, the Itô isometry gives the variance $\frac{2\gamma_0 k_{\mathrm{B}} T}{m^2} \cdot \frac{\tau_p}{2} = \frac{k_{\mathrm{B}} T}{m}$, which is equipartition, and the stationary velocity autocorrelation function

$$\langle u_{t+\tau} u_t \rangle = \frac{k_{\mathrm{B}} T}{m}\, e^{-|\tau|/\tau_p}. \tag{5.18}$$

To verify Eq. (5.18) for $\tau > 0$, write $u_{t+\tau} = e^{-\tau/\tau_p} u_t + (\text{noise on } (t, t+\tau])$; the noise increment is independent of u_t and has zero mean, so $\langle u_{t+\tau} u_t \rangle = e^{-\tau/\tau_p} \langle u_t^2 \rangle$.

The displacement is $x_t - x_0 = \int_0^t u_s\, \mathrm{d}s$, so the mean-square displacement is a double integral of Eq. (5.18):

$$\begin{aligned} \Big\langle (x_t - x_0)^2 \Big\rangle &= \int_0^t \!\! \int_0^t \langle u_s u_{s'} \rangle\, \mathrm{d}s\, \mathrm{d}s' = 2 \int_0^t (t-\tau)\, \frac{k_{\mathrm{B}} T}{m} e^{-\tau/\tau_p}\, \mathrm{d}\tau \\ &= \frac{2 k_{\mathrm{B}} T}{m} \Big[t\tau_p (1 - e^{-t/\tau_p}) - \tau_p^2 + \tau_p^2 e^{-t/\tau_p} + t\tau_p e^{-t/\tau_p} \Big] \\ &= 2D \Big[t - \tau_p (1 - e^{-t/\tau_p}) \Big], \qquad D := \frac{k_{\mathrm{B}} T \tau_p}{m} = \frac{k_{\mathrm{B}} T}{\gamma_0}. \end{aligned} \tag{5.19}$$

In the first line we changed variables to $\tau = |s - s'|$; the region $\{(s, s') \in [0,t]^2 : |s - s'| \in [\tau, \tau + \mathrm{d}\tau]\}$ has area $2(t - \tau)\, \mathrm{d}\tau$. In the second line we used $\int_0^t e^{-\tau/\tau_p}\, \mathrm{d}\tau = \tau_p (1 - e^{-t/\tau_p})$ and $\int_0^t \tau e^{-\tau/\tau_p}\, \mathrm{d}\tau = \tau_p^2 - \tau_p^2 e^{-t/\tau_p} - t\tau_p e^{-t/\tau_p}$.

For $t \ll \tau_p$, expanding $1 - e^{-t/\tau_p} = t/\tau_p - t^2/(2\tau_p^2) + \mathcal{O}(t^3)$ gives $\langle (x_t - x_0)^2 \rangle \approx (k_{\mathrm{B}}T/m)t^2$: ballistic flight at the thermal speed $\sqrt{k_{\mathrm{B}}T/m}$. For $t \gg \tau_p$, $\langle (x_t - x_0)^2 \rangle \approx 2D(t - \tau_p)$: diffusion with the Einstein coefficient $D = k_{\mathrm{B}}T/\gamma_0 = \mu_0 k_{\mathrm{B}}T$, $\mu_0 = 1/\gamma_0$. The overdamped equation Eq. (5.10) reproduces only the second regime, and the offset $-2D\tau_p$ is the $\mathcal{O}(\tau_p)$ correction lost in the limit.

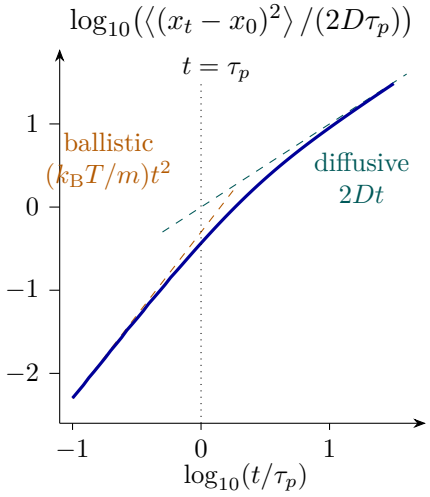


Figure 5.1: Ornstein–Fürth mean-square displacement Eq. (5.19) of a free underdamped Brownian particle, in units of $2D\tau_p$, on logarithmic axes. The curve follows the ballistic law $\frac{1}{2}(t/\tau_p)^2$ (slope 2) for $t \ll \tau_p$ and the diffusive law t/τ_p (slope 1) for $t \gg \tau_p$. The overdamped equation Eq. (5.10) retains only the second branch.

Hardware Realization: Inertial Time Scales in CMOS and Superconducting Samplers

Using the dictionary Eq. (5.8), the ratio that decides whether a circuit node is overdamped is $\tau_L/\tau_{RC} = (L/R)/(RC) = L/(R^2C)$. The following are representative orders of magnitude, not specifications of any particular chip.

- **Subthreshold CMOS nodes.** A node capacitance $C_{\mathrm{node}} \sim 1\,\mathrm{fF}$, an effective small-signal resistance $R = 1/g_{\mathrm{m}} \sim 10^6\,\Omega$ for a transistor biased at tens of nanoamperes, and an interconnect inductance $L \sim 1\,\mathrm{nH}$ give $\tau_{RC} \sim 1\,\mathrm{ns}$ and $\tau_L \sim 10^{-15}\,\mathrm{s}$. The ratio $\tau_L/\tau_{RC} \sim 10^{-6}$ is so small that inertia is irrelevant, and the overdamped equation of the RC sampler is accurate. We expect the same conclusion to hold for the subthreshold CMOS sampling cells described in [38].
- **Josephson junctions.** In the resistively and capacitively shunted junction model (Section 14.4) the phase φ plays the role of x, the junction capacitance plays the role of mass, and the shunt conductance plays the role of friction. The momentum relaxation time is the junction RC time, and the Stewart–McCumber parameter is $\beta_{\mathrm{c}} = (\omega_p RC)^2$, where ω_p is the plasma frequency of small oscillations. For $\beta_{\mathrm{c}} \gtrsim 1$ the junction is underdamped and Definition 5.1 must be used without elimination; Kramers' rate in that regime (Chapter 7) then depends on γ_0 in a non-trivial way.

5.2 The Zwanzig–Caldeira–Leggett Harmonic Bath and the Generalized Langevin Equation

The Langevin equation of Section 5.1 was postulated. We now derive it. The strategy, due to Zwanzig and to Caldeira and Leggett [17, 120], is to couple the system to a large number of harmonic oscillators, solve the oscillator equations exactly, and substitute the solution back into the equation of motion of the system. Everything is Hamiltonian and time-reversible. Friction and noise appear only after the bath has been eliminated.

Why harmonic oscillators? A macroscopic bath (the electron gas in a resistor, the phonons of a crystal, a viscous liquid) is not literally a set of oscillators. However, if each bath degree of freedom is only weakly perturbed by the system, the bath responds linearly, and any linear response can be represented by a set of normal modes. The harmonic bath is therefore the general model of a bath that is large and weakly perturbed at the level of each mode, even when the total coupling to the system is strong. In electrical engineering the same idea appears as Nyquist's representation of a resistor by a lossless transmission line (Section 11.2.1 and Section 5.2.4 below).

5.2.1 The Model Hamiltonian

For clarity we take a single system coordinate $x \in \mathbb{R}$ with mass m and potential $E(x)$; the multidimensional case is stated at the end of the section. The bath consists of K oscillators with coordinates q_k, momenta p_k, masses m_k, and frequencies $\omega_k > 0$, $k = 1, \dots, K$.

Definition 5.6 (Zwanzig–Caldeira–Leggett Hamiltonian)**.** The total Hamiltonian of system plus bath is

$$H = \frac{p^2}{2m} + E(x) + \sum_{k=1}^{K} \left[\frac{p_k^2}{2m_k} + \frac{1}{2} m_k \omega_k^2 \left(q_k - \frac{c_k}{m_k \omega_k^2}\, x \right)^2 \right], \tag{5.20}$$

where the real constants c_k (units of force per length) set the coupling strength of mode k.

Expanding the square separates three contributions:

$$H = \underbrace{\frac{p^2}{2m} + E(x)}_{H_S} + \underbrace{\sum_k \left[\frac{p_k^2}{2m_k} + \frac{1}{2} m_k \omega_k^2 q_k^2 \right]}_{H_B} \underbrace{- x \sum_k c_k q_k}_{H_{\mathrm{int}}} + \underbrace{\frac{x^2}{2} \sum_k \frac{c_k^2}{m_k \omega_k^2}}_{H_{\mathrm{ct}}}. \tag{5.21}$$

The interaction H_{int} is bilinear. The last term H_{ct} is the *counterterm*. Its role is physical, not cosmetic. Suppose the system is held at a fixed position x and the bath relaxes. The bath energy $H_B + H_{\mathrm{int}}$ is minimized at $q_k = c_k x/(m_k\omega_k^2)$, where it equals $-\frac{x^2}{2}\sum_k c_k^2/(m_k\omega_k^2)$. Without the counterterm, the bath would therefore lower the effective potential of the system by this quadratic amount, softening or even destabilizing $E(x)$. The counterterm cancels this shift exactly, so that $E(x)$ is the potential the system feels when the bath is relaxed. In the form Eq. (5.20) this is manifest: each bath oscillator is centered at the instantaneous system position, and for a static x the bath can always relax to zero energy. We will see in Section 5.3 that for an Ohmic bath the coefficient of H_{ct} diverges with the bath cutoff, so the counterterm cannot be omitted in the Markovian limit.

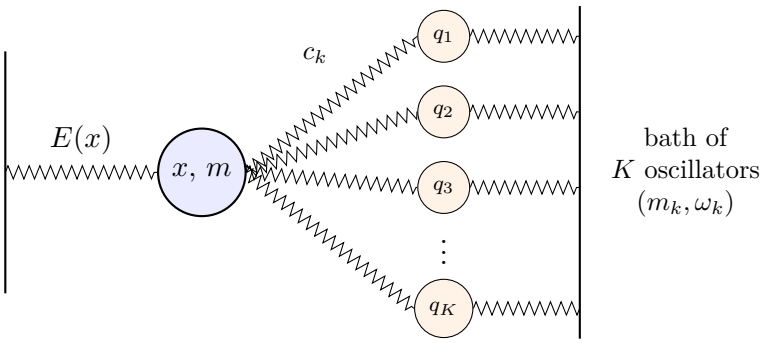


Figure 5.2: Schematic of the Zwanzig–Caldeira–Leggett model. The system coordinate x moves in the potential $E(x)$ and is coupled bilinearly, with strength c_k, to each of K harmonic bath oscillators. Eliminating the bath produces friction with memory and a random force.

5.2.2 Exact Elimination of the Bath

Hamilton's equations for Eq. (5.21) are

$$\dot{x} = \frac{p}{m}, \qquad \dot{p} = -E'(x) + \sum_k c_k q_k - x \sum_k \frac{c_k^2}{m_k \omega_k^2}, \tag{5.22}$$

$$\dot{q}_k = \frac{p_k}{m_k}, \qquad \dot{p}_k = -m_k \omega_k^2 q_k + c_k x. \tag{5.23}$$

Each bath oscillator obeys $m_k \ddot{q}_k + m_k \omega_k^2 q_k = c_k x(t)$: a harmonic oscillator driven by the system trajectory. Because this equation is linear, it can be solved for an arbitrary trajectory $x(t)$.

Lemma 5.7 (Driven Oscillator Solution)**.** *For a continuously differentiable trajectory $x(s)$, $0 \le s \le t$, the solution of Eq. (5.23) is*

$$q_k(t) = q_k(0)\cos\omega_k t + \frac{p_k(0)}{m_k\omega_k}\sin\omega_k t + \frac{c_k}{m_k\omega_k}\int_0^t \sin\big(\omega_k(t-s)\big)\, x(s)\, \mathrm{d}s, \tag{5.24}$$

and, equivalently, after integration by parts,

$$q_k(t) = \frac{c_k}{m_k\omega_k^2}x(t) + \left[q_k(0) - \frac{c_k}{m_k\omega_k^2}x(0)\right]\cos\omega_k t + \frac{p_k(0)}{m_k\omega_k}\sin\omega_k t - \frac{c_k}{m_k\omega_k^2}\int_0^t \cos\big(\omega_k(t-s)\big)\, \dot{x}(s)\, \mathrm{d}s. \tag{5.25}$$

Proof. The first two terms of Eq. (5.24) solve the free equation with the correct initial data. The third term vanishes at $t = 0$; its time derivative is $\frac{c_k}{m_k\omega_k}\big[\sin(0)x(t) + \omega_k\int_0^t \cos(\omega_k(t-s))x(s)\,\mathrm{d}s\big]$, which also vanishes at $t = 0$, and its second derivative is $\frac{c_k}{m_k}x(t) - \omega_k^2 \cdot \frac{c_k}{m_k\omega_k}\int_0^t \sin(\omega_k(t-s))x(s)\,\mathrm{d}s$. So the third term solves $\ddot{q} + \omega_k^2 q = c_k x/m_k$ with zero initial data, and Eq. (5.24) holds.

For Eq. (5.25), use $\frac{\mathrm{d}}{\mathrm{d}s}\cos(\omega_k(t-s)) = \omega_k \sin(\omega_k(t-s))$ and integrate by parts:

$$\begin{aligned}\int_0^t \sin(\omega_k(t-s))x(s)\,\mathrm{d}s &= \left[\frac{\cos(\omega_k(t-s))}{\omega_k}\,x(s)\right]_{s=0}^{s=t} - \frac{1}{\omega_k}\int_0^t \cos(\omega_k(t-s))\dot{x}(s)\,\mathrm{d}s \\ &= \frac{x(t)}{\omega_k} - \frac{\cos(\omega_k t)}{\omega_k}x(0) - \frac{1}{\omega_k}\int_0^t \cos(\omega_k(t-s))\dot{x}(s)\,\mathrm{d}s.\end{aligned}$$

Multiplying by $c_k/(m_k\omega_k)$ and collecting the $x(0)\cos\omega_k t$ term with $q_k(0)\cos\omega_k t$ gives Eq. (5.25). □

The first term in Eq. (5.25) is the adiabatic response: if x moved infinitely slowly, each oscillator would sit at its displaced equilibrium $c_k x/(m_k\omega_k^2)$. The remaining terms describe free oscillations about the initial displaced equilibrium and the lag of the bath behind a moving system.

Theorem 5.8 (Exact Generalized Langevin Equation)**.** *Define the* memory friction kernel *and the* bath force

$$\gamma(t) := \sum_{k=1}^{K} \frac{c_k^2}{m_k\omega_k^2}\cos(\omega_k t), \tag{5.26}$$

$$\xi(t) := \sum_{k=1}^{K} c_k \left\{ \left[q_k(0) - \frac{c_k}{m_k\omega_k^2}x(0) \right] \cos\omega_k t + \frac{p_k(0)}{m_k\omega_k}\sin\omega_k t \right\}. \tag{5.27}$$

Then every solution of Hamilton's equations Eqs. (5.22) and (5.23) satisfies the generalized Langevin equation *(GLE)*

$$m\ddot{x}(t) = -E'(x(t)) - \int_0^t \gamma(t-s)\,\dot{x}(s)\,\mathrm{d}s + \xi(t). \tag{5.28}$$

Proof. Multiply Eq. (5.25) by c_k and sum over k:

$$\sum_k c_k q_k(t) = x(t)\sum_k \frac{c_k^2}{m_k\omega_k^2} + \xi(t) - \int_0^t \left[\sum_k \frac{c_k^2}{m_k\omega_k^2}\cos(\omega_k(t-s))\right]\dot{x}(s)\,\mathrm{d}s.$$

The bracket is $\gamma(t-s)$. Insert this into the momentum equation Eq. (5.22):

$$m\ddot{x} = \dot{p} = -E'(x) + \underbrace{x(t)\sum_k \frac{c_k^2}{m_k\omega_k^2} - x(t)\sum_k \frac{c_k^2}{m_k\omega_k^2}}_{=0} + \xi(t) - \int_0^t \gamma(t-s)\dot{x}(s)\,\mathrm{d}s.$$

The adiabatic bath force $x(t)\sum_k c_k^2/(m_k\omega_k^2)$ cancels the counterterm force $-\partial H_{\mathrm{ct}}/\partial x$ exactly, and Eq. (5.28) follows. □

Theorem 5.8 is exact. It involves no approximation about the size of the bath, the strength of the coupling, or the form of $E(x)$. The price is that the friction term has memory: the force at time t depends on the whole velocity history through the kernel $\gamma(t-s)$. The kernel is even in its argument, $\gamma(-t) = \gamma(t)$, and its value at zero, $\gamma(0) = \sum_k c_k^2/(m_k\omega_k^2)$, is exactly twice the counterterm coefficient.

5.2.3 Statistics of the Bath Force

So far $\xi(t)$ is a deterministic function of the bath initial conditions. It becomes a random force once we specify a probability distribution for those initial conditions. The physically natural choice is that the bath starts in thermal equilibrium at temperature T in the presence of the system at its initial position $x(0) = x_0$.

Assumption 5.9 (Thermal Bath Initial Conditions)**.** Conditionally on the system initial state (x_0, p_0), the bath initial conditions $\{q_k(0), p_k(0)\}$ are distributed according to the canonical density of the *shifted* bath Hamiltonian,

$$\rho_B(\{q_k, p_k\} \mid x_0) \propto \exp\left(-\beta\sum_k \left[\frac{p_k^2}{2m_k} + \frac{1}{2}m_k\omega_k^2\left(q_k - \frac{c_k x_0}{m_k\omega_k^2}\right)^2\right]\right). \tag{5.29}$$

This is the conditional distribution of the bath given (x_0, p_0) when system and bath are jointly in the Gibbs state $e^{-\beta H}/Z$ of Eq. (5.20), because H is the sum of $H_S(x,p)$ and the shifted bath energy. It also describes a bath that has equilibrated while the system was clamped at x_0.

Proposition 5.10 (Gaussian Bath Force and the Second FDT)**.** *Under Assumption 5.9, the bath force $\xi(t)$ is a Gaussian process, independent of (x_0, p_0), with*

$$\langle \xi(t) \rangle = 0, \qquad \langle \xi(t)\, \xi(s) \rangle = k_{\mathrm{B}} T\, \gamma(t-s) \qquad \text{for all } t, s \geq 0. \tag{5.30}$$

Proof. Introduce the shifted coordinates $Q_k := q_k(0) - c_k x_0/(m_k \omega_k^2)$ and $P_k := p_k(0)$. By Eq. (5.29), the joint density of $\{Q_k, P_k\}$ is

$$\prod_k \exp\!\left(-\frac{\beta P_k^2}{2m_k}\right) \exp\!\left(-\frac{\beta m_k \omega_k^2 Q_k^2}{2}\right),$$

which does not involve (x_0, p_0). Hence the Q_k and P_k are mutually independent centered Gaussians, independent of the system initial state, with variances

$$\left\langle Q_k^2 \right\rangle = \frac{k_{\mathrm{B}} T}{m_k \omega_k^2}, \qquad \left\langle P_k^2 \right\rangle = m_k k_{\mathrm{B}} T, \qquad \langle Q_k P_l \rangle = 0, \qquad \langle Q_k Q_l \rangle = \langle P_k P_l \rangle = 0 \ (k \neq l). \tag{5.31}$$

By Eq. (5.27), $\xi(t) = \sum_k c_k \big[Q_k \cos \omega_k t + P_k \sin(\omega_k t)/(m_k \omega_k) \big]$ is a linear combination of these Gaussians, so it is a Gaussian process with zero mean. Its covariance is

$$\begin{aligned} \langle \xi(t)\xi(s) \rangle &= \sum_k c_k^2 \left[\left\langle Q_k^2 \right\rangle \cos \omega_k t \cos \omega_k s + \frac{\langle P_k^2 \rangle}{m_k^2 \omega_k^2} \sin \omega_k t \sin \omega_k s \right] \\ &= \sum_k \frac{c_k^2 k_{\mathrm{B}} T}{m_k \omega_k^2} \big[\cos \omega_k t \cos \omega_k s + \sin \omega_k t \sin \omega_k s \big] = k_{\mathrm{B}} T \sum_k \frac{c_k^2}{m_k \omega_k^2} \cos\big(\omega_k (t-s)\big), \end{aligned}$$

where cross terms vanished by independence and we used $\cos a \cos b + \sin a \sin b = \cos(a-b)$. The last sum is $k_{\mathrm{B}} T \gamma(t-s)$. □

Equation (5.30) is the **fluctuation-dissipation theorem of the second kind** in its general, non-Markovian form [62, 120]. The same function γ that describes friction in Eq. (5.28) describes the correlations of the noise, multiplied by $k_{\mathrm{B}} T$. This is not an independent physical law. It is a consequence of the fact that both terms are generated by the same coupling constants c_k and that the bath starts in thermal equilibrium. The noise is stationary, $\langle \xi(t)\xi(s) \rangle$ depending only on $t - s$, even though the GLE itself was started at a definite time $t = 0$. That stationarity is a consequence of choosing the shifted bath distribution Eq. (5.29). If the bath were instead equilibrated around $q_k = 0$ (the unshifted H_B), an extra deterministic force $-\gamma(t) x_0$ would appear, called the *initial slip*; it decays on the memory time of the kernel and does not affect long-time behavior.

Remark 5.11 (Where Irreversibility Comes From)**.** The total system Eq. (5.20) is Hamiltonian and time-reversible, and for finite K the kernel Eq. (5.26) is a quasi-periodic function that returns arbitrarily close to $\gamma(0)$ at arbitrarily late times (a recurrence). Genuine dissipation requires the limit $K \to \infty$ with the frequencies ω_k becoming dense, so that the sum in Eq. (5.26) becomes an integral over a continuous spectrum and $\gamma(t) \to 0$ as $t \to \infty$ by the Riemann–Lebesgue lemma. Energy that flows into the bath then spreads over infinitely many modes and does not come back on any finite time scale. The arrow of time enters through the initial condition, Assumption 5.9: the bath is prepared in equilibrium and the system is not.

5.2.4 Several System Coordinates

The derivation generalizes directly to $\mathbf{x} \in \mathbb{R}^N$ with mass matrix $\mathbf{M}$. Replace the coupling $c_k x$ by $\mathbf{c}_k^\top \mathbf{x} = \sum_i c_{ki} x_i$ in Eq. (5.20), so each bath mode couples to a fixed linear combination of system coordinates. Repeating the steps above line by line gives

$$\mathbf{M}\ddot{\mathbf{x}}(t) = -\nabla E(\mathbf{x}(t)) - \int_0^t \boldsymbol{\gamma}(t-s)\,\dot{\mathbf{x}}(s)\,\mathrm{d}s + \boldsymbol{\xi}(t), \qquad \gamma_{ij}(t) := \sum_k \frac{c_{ki}c_{kj}}{m_k\omega_k^2}\cos(\omega_k t), \tag{5.32}$$

with the matrix FDT

$$\langle \xi_i(t)\,\xi_j(s)\rangle = k_\mathrm{B}T\,\gamma_{ij}(t-s). \tag{5.33}$$

The kernel matrix $\boldsymbol{\gamma}(t)$ is symmetric for every t, and for every $\mathbf{u} \in \mathbb{R}^N$ the scalar kernel $\mathbf{u}^\top\boldsymbol{\gamma}(t)\mathbf{u} = \sum_k (\mathbf{c}_k^\top \mathbf{u})^2 \cos(\omega_k t)/(m_k\omega_k^2)$ is a positive combination of cosines, hence a positive-definite function of t in the sense of Bochner. This positivity is what guarantees that the friction always removes energy on average.

Remark 5.12 (Nonlinear Coupling and State-Dependent Friction)**.** Zwanzig's original treatment [120] allows the bath to couple to a nonlinear function $F(x)$ of the system coordinate: replace x by $F(x)$ inside the square in Eq. (5.20). The bath equations become $m_k\ddot{q}_k + m_k\omega_k^2 q_k = c_k F(x(t))$, and Lemma 5.7 holds with x replaced by $F(x)$ and $\dot{x}(s)$ replaced by $\frac{\mathrm{d}}{\mathrm{d}s}F(x(s)) = F'(x(s))\dot{x}(s)$. The force on the system is now $-\partial H/\partial x = -E'(x) + F'(x)\sum_k c_k(q_k - c_k F(x)/(m_k\omega_k^2))$, and the same steps give

$$m\ddot{x}(t) = -E'(x(t)) - F'(x(t))\int_0^t \gamma(t-s)\,F'(x(s))\,\dot{x}(s)\,\mathrm{d}s + F'(x(t))\,\xi(t), \tag{5.34}$$

with the same kernel γ and the same noise statistics Eq. (5.30). In the Ohmic Markovian limit of Section 5.3, this becomes $m\ddot{x} = -E'(x) - \gamma_0 F'(x)^2\dot{x} + F'(x)\xi(t)$ with $\langle\xi(t)\xi(s)\rangle = 2\gamma_0 k_\mathrm{B}T\delta(t-s)$: a state-dependent friction $\gamma(x) = \gamma_0 F'(x)^2$ accompanied by multiplicative noise whose variance $2k_\mathrm{B}T\gamma(x)$ obeys the FDT point by point. This is the microscopic origin of the inertial model Eq. (5.62) studied in Section 5.5.

Physics Insight: A Resistor Is an Ohmic Bath

Nyquist's derivation of thermal noise (Section 11.2.1) replaces a resistor R by a long lossless transmission line of characteristic impedance $Z_0 = R$. A lossless line is a chain of inductors and capacitors, so its normal modes are harmonic oscillators with a dense set of frequencies. Connecting a circuit node to the line is exactly the Caldeira–Leggett construction, with the node charge playing the role of x and the line modes playing the role of the q_k. Energy launched into the line propagates away and never returns, which is the electrical version of Remark 5.11. The thermal waves arriving from the far end of the line are the random force. In this picture, the Johnson–Nyquist formula is the FDT Eq. (5.30) applied to a bath whose spectral density is Ohmic over the whole frequency range where the line is uniform.

5.3 Spectral Densities, the Ohmic Bath, and the FDT of the Second Kind

The GLE Eq. (5.28) depends on the bath only through the kernel $\gamma(t)$, and the kernel depends on the bath only through one function of frequency. In this section we introduce that function, show

that a bath with a linear (Ohmic) spectral density produces memoryless friction and white noise, and recover the Markovian Langevin equation of Section 5.1 together with the FDT in the form $\langle \xi_i(t)\xi_j(t')\rangle = 2k_{\mathrm{B}}T\Gamma_{ij}\delta(t-t')$.

5.3.1 The Bath Spectral Density

Definition 5.13 (Spectral Density of the Bath)**.** The *spectral density* of the harmonic bath Eq. (5.20) is the function of $\omega > 0$

$$J(\omega) := \frac{\pi}{2}\sum_{k=1}^{K}\frac{c_k^2}{m_k\omega_k}\,\delta(\omega-\omega_k). \tag{5.35}$$

For a large bath with a dense set of frequencies, J is replaced by a smooth non-negative function.

The factor $\pi/2$ is conventional. With it, the memory kernel Eq. (5.26) becomes a cosine transform of $J(\omega)/\omega$:

$$\gamma(t) = \frac{2}{\pi}\int_0^\infty \frac{J(\omega)}{\omega}\cos(\omega t)\,\mathrm{d}\omega. \tag{5.36}$$

To check this, insert Eq. (5.35): $\frac{2}{\pi}\cdot\frac{\pi}{2}\sum_k \frac{c_k^2}{m_k\omega_k}\cdot\frac{\cos(\omega_k t)}{\omega_k} = \sum_k \frac{c_k^2}{m_k\omega_k^2}\cos(\omega_k t) = \gamma(t)$. The spectral density thus encodes everything about the bath that matters for the system: how many modes there are near each frequency, weighted by how strongly they couple.

It is useful to know the frequency-domain friction. Following the convention of Chapter 11, we use the Fourier factor $e^{-i\omega t}$ throughout this chapter. The one-sided transform of the kernel,

$$\hat\gamma(\omega) := \int_0^\infty \gamma(t)\,e^{-i\omega t}\,\mathrm{d}t, \tag{5.37}$$

appears whenever the GLE is solved by Fourier or Laplace methods, because the friction term in Eq. (5.28) is a causal convolution. For $\omega > 0$ its real part is

$$\operatorname{Re}\hat\gamma(\omega) = \int_0^\infty \gamma(t)\cos(\omega t)\,\mathrm{d}t = \frac{1}{2}\int_{-\infty}^{\infty}\gamma(t)\cos(\omega t)\,\mathrm{d}t = \frac{J(\omega)}{\omega}, \tag{5.38}$$

where the second equality uses that γ is even, and the third follows from Eq. (5.36) and the Fourier inversion formula for the cosine transform, $\int_0^\infty \cos(\omega t)\cos(\omega' t)\,\mathrm{d}t = \frac{\pi}{2}\delta(\omega-\omega')$ for $\omega,\omega' > 0$. Combining with Eq. (5.30), the double-sided power spectral density of the bath force is

$$S_\xi^{\mathrm{ds}}(\omega) = \int_{-\infty}^{\infty} k_{\mathrm{B}}T\,\gamma(\tau)\,e^{-i\omega\tau}\,\mathrm{d}\tau = 2k_{\mathrm{B}}T\operatorname{Re}\hat\gamma(\omega) = 2k_{\mathrm{B}}T\,\frac{J(|\omega|)}{|\omega|}. \tag{5.39}$$

Equation (5.39) is the FDT of the second kind in the frequency domain: the noise power at frequency ω equals $2k_{\mathrm{B}}T$ times the dissipative (real) part of the friction at that frequency. It has exactly the structure of the Nyquist formula $S_V^{\mathrm{ds}} = 2k_{\mathrm{B}}TR$, with $\operatorname{Re}\hat\gamma$ in place of R, and we will meet it again as $S_V^{\mathrm{ds}} = 2k_{\mathrm{B}}T\operatorname{Re}Z$ in Worked computation 5.33.

5.3.2 The Ohmic Bath and Its Markovian Limit

Definition 5.14 (Ohmic Spectral Density with Drude Cutoff)**.** A bath is *Ohmic* with friction coefficient $\gamma_0 > 0$ if $J(\omega) = \gamma_0\omega$ at low frequencies. A convenient model with a high-frequency cutoff ω_c is the *Drude* form

$$J(\omega) = \gamma_0\,\omega\,\frac{\omega_c^2}{\omega^2+\omega_c^2}. \tag{5.40}$$

The name comes from the fact that the frequency-domain friction $\operatorname{Re}\hat{\gamma}(\omega) = J(\omega)/\omega = \gamma_0$ is independent of frequency for $\omega \ll \omega_c$, like the resistance of an Ohmic conductor. The cutoff is always present physically: no real bath has modes at arbitrarily high frequency coupled with constant strength. For the resistor, ω_c is the inverse of the electron momentum relaxation time or of the parasitic RC time of the device, whichever is longer.

Proposition 5.15 (Drude Kernel and the Markovian Limit)**.** *For the Drude spectral density Eq. (5.40):*

(a) The memory kernel is exponential,

$$\gamma(t) = \gamma_0\,\omega_c\,e^{-\omega_c|t|}, \qquad \int_0^\infty \gamma(t)\,\mathrm{d}t = \gamma_0, \qquad \gamma(0) = \gamma_0\omega_c. \tag{5.41}$$

(b) For any trajectory with bounded acceleration, $\sup_s|\ddot{x}(s)| \le a_{\max}$, the friction force converges to the Markovian form,

$$\left|\int_0^t \gamma(t-s)\,\dot{x}(s)\,\mathrm{d}s - \gamma_0\dot{x}(t)\right| \le \gamma_0\,|\dot{x}(0)|\,e^{-\omega_c t} + \frac{\gamma_0 a_{\max}}{\omega_c}. \tag{5.42}$$

(c) As $\omega_c \to \infty$, $\gamma(t) \to 2\gamma_0\,\delta(t)$ in the sense of distributions on $\mathbb{R}$, and the noise covariance becomes $\langle\xi(t)\xi(s)\rangle = 2\gamma_0 k_{\mathrm{B}}T\,\delta(t-s)$.

Proof. (a) By Eq. (5.36), $\gamma(t) = \frac{2\gamma_0}{\pi}\int_0^\infty \frac{\omega_c^2}{\omega^2+\omega_c^2}\cos(\omega t)\,\mathrm{d}\omega$. The standard integral $\int_0^\infty \frac{\cos(\omega t)}{\omega^2+\omega_c^2}\,\mathrm{d}\omega = \frac{\pi}{2\omega_c}e^{-\omega_c|t|}$ (obtained by closing the contour of $\int_{-\infty}^\infty e^{i\omega|t|}/(\omega^2+\omega_c^2)\,\mathrm{d}\omega$ in the upper half plane around the pole $\omega = i\omega_c$, residue $e^{-\omega_c|t|}/(2i\omega_c)$) gives $\gamma(t) = \frac{2\gamma_0}{\pi}\omega_c^2\cdot\frac{\pi}{2\omega_c}e^{-\omega_c|t|} = \gamma_0\omega_c e^{-\omega_c|t|}$. Integrating, $\int_0^\infty \gamma_0\omega_c e^{-\omega_c t}\,\mathrm{d}t = \gamma_0$.

(b) Integrate by parts with $\frac{\mathrm{d}}{\mathrm{d}s}e^{-\omega_c(t-s)} = \omega_c e^{-\omega_c(t-s)}$:

$$\begin{aligned}\int_0^t \gamma_0\omega_c e^{-\omega_c(t-s)}\dot{x}(s)\,\mathrm{d}s &= \gamma_0\Big[e^{-\omega_c(t-s)}\dot{x}(s)\Big]_0^t - \gamma_0\int_0^t e^{-\omega_c(t-s)}\ddot{x}(s)\,\mathrm{d}s\\ &= \gamma_0\dot{x}(t) - \gamma_0 e^{-\omega_c t}\dot{x}(0) - \gamma_0\int_0^t e^{-\omega_c(t-s)}\ddot{x}(s)\,\mathrm{d}s.\end{aligned}$$

The last integral is bounded in absolute value by $a_{\max}\int_0^\infty e^{-\omega_c u}\,\mathrm{d}u = a_{\max}/\omega_c$, which gives Eq. (5.42).

(c) For a test function ϕ, $\int_{-\infty}^\infty \gamma_0\omega_c e^{-\omega_c|t|}\phi(t)\,\mathrm{d}t = \gamma_0\int_{-\infty}^\infty e^{-|u|}\phi(u/\omega_c)\,\mathrm{d}u \to \gamma_0\phi(0)\int_{-\infty}^\infty e^{-|u|}\,\mathrm{d}u = 2\gamma_0\phi(0)$ by dominated convergence. The noise statement follows from Eq. (5.30). □

Part (c) contains a factor-of-two trap. The full-line weight of the kernel is $2\gamma_0$, but the GLE convolution runs only over the past, $s \in [0,t]$, and therefore picks up only half of it, γ_0, as part (b) shows directly. This is the same half-weight phenomenon that produced the Stratonovich factor $\frac{1}{2}$ in Section 4.4.2. In fact the Drude kernel is exactly the exponentially correlated colored noise of Eq. (4.83) with correlation time $\tau = 1/\omega_c$: $\gamma(t) = 2\gamma_0\cdot\frac{\omega_c}{2}e^{-\omega_c|t|}$. The counterterm coefficient $\frac{1}{2}\gamma(0) = \frac{1}{2}\gamma_0\omega_c$ diverges in the Markovian limit, which confirms that the counterterm is needed to keep the potential $E(x)$ unchanged when the cutoff is removed.

Corollary 5.16 (Markovian Langevin Equation from the Ohmic Bath)**.** *In the limit $\omega_c \to \infty$, the GLE Eq. (5.28) with Drude kernel becomes the underdamped Langevin equation*

$$m\ddot{x} = -E'(x) - \gamma_0\dot{x} + \xi(t), \qquad \langle\xi(t)\xi(s)\rangle = 2\gamma_0 k_{\mathrm{B}}T\,\delta(t-s), \tag{5.43}$$

which in Itô form is Eqs. (5.2) and (5.3) with $\mathbf{\Gamma} = \gamma_0$ *and* $\xi(t)\,\mathrm{d}t = \sqrt{2\gamma_0 k_{\mathrm{B}}T}\,\mathrm{d}W_t$. *In the multidimensional case, a matrix Ohmic spectral density* $J_{ij}(\omega) = \Gamma_{ij}\omega$ *(with cutoff) gives* $\boldsymbol{\gamma}(t) \to 2\mathbf{\Gamma}\delta(t)$ *and*

$$\langle \xi_i(t)\,\xi_j(t')\rangle = 2k_{\mathrm{B}}T\,\Gamma_{ij}\,\delta(t-t'). \tag{5.44}$$

As noted after Definition 5.1, passing from Eq. (5.43) to the Itô SDE involves no Itô–Stratonovich ambiguity.

Example 5.17 (Markovian Embedding of the Drude Bath)**.** The exponential kernel has a useful property: the GLE with finite cutoff is itself equivalent to a Markovian SDE in an enlarged state space. Define the total bath force $z(t) := -\int_0^t \gamma(t-s)\dot{x}(s)\,\mathrm{d}s + \xi(t)$. For the Drude kernel, differentiate the convolution:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^t \gamma_0\omega_c e^{-\omega_c(t-s)}\dot{x}(s)\,\mathrm{d}s = \gamma_0\omega_c\dot{x}(t) - \omega_c\int_0^t \gamma_0\omega_c e^{-\omega_c(t-s)}\dot{x}(s)\,\mathrm{d}s.$$

A stationary Gaussian process with covariance $k_{\mathrm{B}}T\gamma_0\omega_c e^{-\omega_c|t-s|}$ is an Ornstein–Uhlenbeck process, $\mathrm{d}\xi = -\omega_c\xi\,\mathrm{d}t + \omega_c\sqrt{2\gamma_0 k_{\mathrm{B}}T}\,\mathrm{d}W_t$; indeed its stationary variance is $\omega_c^2\cdot 2\gamma_0 k_{\mathrm{B}}T/(2\omega_c) = \gamma_0\omega_c k_{\mathrm{B}}T = k_{\mathrm{B}}T\gamma(0)$. Adding the two pieces,

$$\mathrm{d}x = \frac{p}{m}\,\mathrm{d}t, \qquad \mathrm{d}p = \big[-E'(x) + z\big]\,\mathrm{d}t, \qquad \mathrm{d}z = -\omega_c\Big[z + \gamma_0\frac{p}{m}\Big]\,\mathrm{d}t + \omega_c\sqrt{2\gamma_0 k_{\mathrm{B}}T}\,\mathrm{d}W_t. \tag{5.45}$$

One checks, exactly as in Proposition 5.2, that the Gibbs density $\propto \exp\!\big(-\beta[p^2/(2m) + E(x) + z^2/(2\gamma_0\omega_c)]\big)$ is stationary for Eq. (5.45): the auxiliary variable z behaves like an extra momentum with "mass" $\gamma_0\omega_c$, coupled antisymmetrically to p. Setting $\omega_c \to \infty$ forces $z \approx -\gamma_0 p/m +$(white noise) and recovers Eq. (5.43). In circuit terms, Eq. (5.45) describes a resistor with a parasitic shunt capacitance: its impedance $R/(1 + i\omega RC_{\mathrm{p}})$ has real part $R/(1+\omega^2R^2C_{\mathrm{p}}^2)$, which is a Drude form with $\omega_c = 1/(RC_{\mathrm{p}})$.

5.3.3 The FDT of the Second Kind as a Consistency Condition

Before the general statement, a one-line energy balance shows why the noise strength is not free.

Heuristic Derivation: Power Injected by Noise Equals Power Removed by Friction

Take a free particle, $\mathrm{d}p = -(\gamma_0/m)p\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, with an arbitrary noise amplitude σ. By Itô's formula, $\mathrm{d}(p^2) = 2p\,\mathrm{d}p + (\mathrm{d}p)^2 = \big[-2(\gamma_0/m)p^2 + \sigma^2\big]\,\mathrm{d}t + 2\sigma p\,\mathrm{d}W_t$. Taking expectations, the kinetic energy $K = \langle p^2\rangle/(2m)$ obeys

$$\frac{\mathrm{d}K}{\mathrm{d}t} = \underbrace{-\frac{2\gamma_0}{m}K}_{\text{power removed by friction}} + \underbrace{\frac{\sigma^2}{2m}}_{\text{power injected by noise}}.$$

In the steady state the two balance and $K = \sigma^2/(4\gamma_0)$. Equipartition demands $K = \frac{1}{2}k_{\mathrm{B}}T$, so $\sigma^2 = 2\gamma_0 k_{\mathrm{B}}T$. A larger σ would heat the particle above the bath temperature, a smaller one would cool it below. The FDT is the statement that the bath does neither.

The harmonic bath produced the FDT as an output. The converse viewpoint is equally useful in practice: given a Langevin model with some friction and some noise, thermal equilibrium at temperature T *requires* the FDT. The following theorem makes this precise for the Markovian equations, which are the ones we simulate and build.

Theorem 5.18 (FDT of the Second Kind: Necessity and Sufficiency)**.**

(a) Underdamped. *Consider* $\mathrm{d}\mathbf{x} = \mathbf{M}^{-1}\mathbf{p}\,\mathrm{d}t$, $\mathrm{d}\mathbf{p} = [-\nabla E(\mathbf{x}) - \boldsymbol{\Gamma}\mathbf{M}^{-1}\mathbf{p}]\,\mathrm{d}t + \boldsymbol{\sigma}\,\mathrm{d}\mathbf{W}_t$ *with constant symmetric* $\boldsymbol{\Gamma}$, $\mathbf{M} \succ \mathbf{0}$, *and constant* $\boldsymbol{\sigma} \in \mathbb{R}^{N\times N}$. *The phase-space Gibbs density Eq. (5.5) is stationary if and only if* $\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top} = 2k_{\mathrm{B}}T\boldsymbol{\Gamma}$.

(b) Overdamped. *Consider* $\mathrm{d}\mathbf{x} = -\boldsymbol{\mu}\nabla E(\mathbf{x})\,\mathrm{d}t + \boldsymbol{\sigma}\,\mathrm{d}\mathbf{W}_t$ *with constant symmetric positive-definite* $\boldsymbol{\mu}$ *and constant* $\boldsymbol{\sigma}$, *and suppose the gradients* $\{\nabla E(\mathbf{x}) : \mathbf{x} \in \mathbb{R}^N\}$ *span* $\mathbb{R}^N$ *in the sense that no nonzero matrix annihilates all of them. Then* $p_{\mathrm{eq}} \propto e^{-\beta E}$ *has zero probability current if and only if* $\mathbf{D} := \tfrac{1}{2}\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top} = k_{\mathrm{B}}T\boldsymbol{\mu}$.

Proof. (a) Let $\boldsymbol{\Sigma}_\sigma := \tfrac{1}{2}\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top}$. The Fokker–Planck equation is Eq. (5.4) with $k_{\mathrm{B}}T\boldsymbol{\Gamma}\nabla_{\mathbf{p}}\rho$ replaced by $\boldsymbol{\Sigma}_\sigma\nabla_{\mathbf{p}}\rho$. The Hamiltonian part annihilates ρ_{eq} for any $\boldsymbol{\sigma}$, as shown in the proof of Proposition 5.2. For the dissipative part, $\nabla_{\mathbf{p}}\rho_{\mathrm{eq}} = -\beta\mathbf{M}^{-1}\mathbf{p}\rho_{\mathrm{eq}}$ gives

$$\nabla_{\mathbf{p}} \cdot \Big[(\boldsymbol{\Gamma} - \beta\boldsymbol{\Sigma}_\sigma)\mathbf{M}^{-1}\mathbf{p}\,\rho_{\mathrm{eq}}\Big] = \rho_{\mathrm{eq}}\Big[\mathrm{tr}(\mathbf{K}\mathbf{M}^{-1}) - \beta\,\mathbf{p}^{\top}\mathbf{M}^{-1}\mathbf{K}\mathbf{M}^{-1}\mathbf{p}\Big], \qquad \mathbf{K} := \boldsymbol{\Gamma} - \beta\boldsymbol{\Sigma}_\sigma.$$

If $\mathbf{K} = \mathbf{0}$ this vanishes. Conversely, if it vanishes for all $\mathbf{p}$, the quadratic form $\mathbf{p}^{\top}\mathbf{M}^{-1}\mathbf{K}\mathbf{M}^{-1}\mathbf{p}$ must be constant in $\mathbf{p}$, hence identically zero. Since $\mathbf{K}$ is symmetric, so is $\mathbf{M}^{-1}\mathbf{K}\mathbf{M}^{-1}$, and a symmetric matrix with vanishing quadratic form is zero. Thus $\mathbf{K} = \mathbf{0}$, that is $\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top} = 2k_{\mathrm{B}}T\boldsymbol{\Gamma}$.

(b) With constant $\mathbf{D}$, the probability current Eq. (6.19) of the density p_{eq} is $\mathbf{J} = -\boldsymbol{\mu}\nabla E\,p_{\mathrm{eq}} - \mathbf{D}\nabla p_{\mathrm{eq}} = (\beta\mathbf{D} - \boldsymbol{\mu})\nabla E\,p_{\mathrm{eq}}$. It vanishes everywhere if and only if the constant matrix $\beta\mathbf{D} - \boldsymbol{\mu}$ annihilates every $\nabla E(\mathbf{x})$, which by hypothesis means $\beta\mathbf{D} = \boldsymbol{\mu}$. □

Part (b) is the *Einstein relation* $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$ [39, 90]. For the RC circuit it is the statement that a conductance G at temperature T must carry current noise with double-sided spectral density $2k_{\mathrm{B}}TG$ (a current diffusion constant $D_I = k_{\mathrm{B}}TG$ in the notation of Section 11.1), which is Nyquist's theorem. In Section 5.4 we will see the Einstein relation reappear from a completely different direction, as a statement about linear response.

Remark 5.19 (Non-Ohmic Baths)**.** Spectral densities of the form $J(\omega) = \gamma_0\,\omega_c^{1-s}\omega^s$ (with a cutoff) are called sub-Ohmic for $0 < s < 1$ and super-Ohmic for $s > 1$. For $s > 1$ the low-frequency friction $J(\omega)/\omega \to 0$, and the long-time motion is underdamped no matter how strong the coupling. For $s < 1$ the friction diverges at low frequency and the kernel $\gamma(t)$ decays as a power law, so memory never becomes negligible; noise with a $1/f$-like spectrum, discussed for MOSFETs in Section 11.5, is of this long-memory type. Throughout the rest of the book we assume an Ohmic bath, which is an excellent approximation for resistive and transconductive elements in the frequency range where thermodynamic computing circuits operate.

Physics Insight: The Quantum FDT and the Classical Limit

Caldeira and Leggett introduced the harmonic bath to study quantum tunneling of a macroscopic variable with dissipation [17], a problem we return to for Josephson devices in Section 14.5. Quantum mechanically, each bath mode in thermal equilibrium has mean energy $\frac{\hbar\omega_k}{2}\coth\!\big(\frac{\hbar\omega_k}{2k_{\mathrm{B}}T}\big)$ instead of $k_{\mathrm{B}}T$. Repeating the proof of Proposition 5.10 with this replacement gives a symmetrized noise spectrum $\hbar\omega\coth\!\big(\frac{\hbar\omega}{2k_{\mathrm{B}}T}\big)\,\mathrm{Re}\,\hat{\gamma}(\omega)$ in place of $2k_{\mathrm{B}}T\,\mathrm{Re}\,\hat{\gamma}(\omega)$. For $\hbar\omega \ll k_{\mathrm{B}}T$, $\coth(y) \approx 1/y$ and the classical result returns. At room temperature, $k_{\mathrm{B}}T/\hbar \approx 4\times 10^{13}\,\mathrm{s}^{-1}$ (a frequency of about 6 THz), far above any frequency relevant for CMOS sampling circuits. The electrical version of this crossover is the Planck–Nyquist formula Eq. (11.30).

5.4 Linear Response, Onsager Reciprocity, and the FDT of the First Kind

The second FDT relates noise to friction. The first FDT relates a different pair of quantities: the response of a system in equilibrium to a weak external force, and the spontaneous fluctuations of the same system without the force. The physical reason for the connection is that a small fluctuation, once it has occurred, cannot be distinguished from a small displacement caused by an external push. Its subsequent decay is governed by the same dynamics in both cases. This idea, formalized by Kubo [62], is the basis of all noise-based measurements of transport coefficients, from Johnson's measurement of Boltzmann's constant with a resistor [57] to the calibration of analog sampling circuits.

5.4.1 Response Functions

Let a classical system with phase-space point $\mathbf{\Gamma}_{\mathrm{ps}}$ and Hamiltonian H_0 be in thermal equilibrium, $\rho_{\mathrm{eq}} = e^{-\beta H_0}/Z$. (In this section the system may include the bath of Section 5.2; ρ_{eq} is then the Gibbs state of system plus bath.) At times $t > t_0$ we apply a weak time-dependent field $h(t)$ that couples to an observable $B(\mathbf{\Gamma}_{\mathrm{ps}})$:

$$H(t) = H_0 - h(t)\, B. \tag{5.46}$$

For example, if $B = x$ is a position then h is an applied force; if $B = q$ is the charge on a capacitor then h is the voltage of a source in series with it.

Definition 5.20 (Linear Response Function and Susceptibility)**.** The *response function* $\chi_{AB}(t)$ of an observable A to the field h coupled to B is defined by the linear relation

$$\langle A\rangle(t) - \langle A\rangle_{\mathrm{eq}} = \int_{-\infty}^{t} \chi_{AB}(t-s)\, h(s)\, \mathrm{d}s + \mathcal{O}(h^2), \tag{5.47}$$

with $\chi_{AB}(\tau) = 0$ for $\tau < 0$ (causality). Its Fourier transform, with the convention of this chapter,

$$\chi_{AB}(\omega) := \int_{0}^{\infty} \chi_{AB}(\tau)\, e^{-i\omega\tau}\, \mathrm{d}\tau, \tag{5.48}$$

is the *dynamic susceptibility*. A field $h(t) = \mathrm{Re}(h_0 e^{i\omega t})$ then produces the response $\langle A\rangle(t) - \langle A\rangle_{\mathrm{eq}} = \mathrm{Re}(\chi_{AB}(\omega) h_0 e^{i\omega t})$, because $\int_0^\infty \chi_{AB}(\tau) e^{i\omega(t-\tau)}\,\mathrm{d}\tau = \chi_{AB}(\omega) e^{i\omega t}$. This is the phasor convention of circuit theory, in which a capacitor has impedance $1/(i\omega C)$; the alternative convention common in physics is discussed in Section 5.4.2.

We also define the equilibrium correlation function of fluctuations $\delta A = A - \langle A\rangle_{\mathrm{eq}}$,

$$C_{AB}(\tau) := \langle \delta A(\tau)\, \delta B(0)\rangle_{\mathrm{eq}}, \tag{5.49}$$

where $A(\tau) = A(\mathbf{\Phi}_\tau \mathbf{\Gamma}_{\mathrm{ps}})$ is evaluated along the unperturbed Hamiltonian flow $\mathbf{\Phi}_\tau$ of H_0, and the average is over initial conditions drawn from ρ_{eq}. Because ρ_{eq} is invariant under the flow, C_{AB} depends only on the time difference, and $C_{AB}(-\tau) = C_{BA}(\tau)$.

5.4.2 The Kubo Formula

Theorem 5.21 (Classical Kubo Formula / FDT of the First Kind)**.** *For the perturbation Eq. (5.46) of a system in canonical equilibrium,*

$$\chi_{AB}(\tau) = \beta\, \theta(\tau)\, \left\langle A(\tau)\, \dot{B}(0)\right\rangle_{\mathrm{eq}} = -\beta\, \theta(\tau)\, \frac{\mathrm{d}}{\mathrm{d}\tau} C_{AB}(\tau), \tag{5.50}$$

where θ is the Heaviside step function and $\dot{B} = \{B, H_0\}$ is the time derivative of B along the unperturbed flow.

Proof. The phase-space density obeys Liouville's equation $\partial_t \rho = \{H(t), \rho\}$, where $\{f, g\} = \sum_i (\partial_{q_i} f\, \partial_{p_i} g - \partial_{p_i} f\, \partial_{q_i} g)$ is the Poisson bracket. Write $\rho = \rho_{\mathrm{eq}} + \delta\rho$ and define the unperturbed Liouville operator $\mathcal{L}_0 \rho := \{H_0, \rho\}$. Since $\{H_0, \rho_{\mathrm{eq}}\} = 0$, to first order in h,

$$\partial_t\, \delta\rho = \mathcal{L}_0\, \delta\rho - h(t)\, \{B, \rho_{\mathrm{eq}}\}. \tag{5.51}$$

Using $\rho_{\mathrm{eq}} \propto e^{-\beta H_0}$ and the chain rule for Poisson brackets, $\{B, \rho_{\mathrm{eq}}\} = -\beta \rho_{\mathrm{eq}} \{B, H_0\} = -\beta \rho_{\mathrm{eq}} \dot{B}$. Hence

$$\partial_t\, \delta\rho = \mathcal{L}_0\, \delta\rho + \beta h(t)\, \rho_{\mathrm{eq}} \dot{B}, \qquad \delta\rho(t) = \beta \int_{-\infty}^{t} h(s)\, e^{(t-s)\mathcal{L}_0} \left[\rho_{\mathrm{eq}} \dot{B}\right] \mathrm{d}s,$$

with $\delta\rho = 0$ before the field is switched on. The operator $e^{\tau \mathcal{L}_0}$ transports a density along the flow, $(e^{\tau \mathcal{L}_0} f)(\mathbf{\Gamma}_{\mathrm{ps}}) = f(\mathbf{\Phi}_{-\tau} \mathbf{\Gamma}_{\mathrm{ps}})$, and the flow preserves phase-space volume (Liouville's theorem). Therefore

$$\int A\, e^{\tau \mathcal{L}_0} \left[\rho_{\mathrm{eq}} \dot{B}\right] \mathrm{d}\mathbf{\Gamma}_{\mathrm{ps}} = \int A(\mathbf{\Phi}_\tau \mathbf{\Gamma}_{\mathrm{ps}})\, \rho_{\mathrm{eq}}(\mathbf{\Gamma}_{\mathrm{ps}})\, \dot{B}(\mathbf{\Gamma}_{\mathrm{ps}})\, \mathrm{d}\mathbf{\Gamma}_{\mathrm{ps}} = \left\langle A(\tau) \dot{B}(0) \right\rangle_{\mathrm{eq}},$$

after the change of variables $\mathbf{\Gamma}_{\mathrm{ps}} \to \mathbf{\Phi}_\tau \mathbf{\Gamma}_{\mathrm{ps}}$. Multiplying $\delta\rho$ by A and integrating gives $\langle A \rangle(t) - \langle A \rangle_{\mathrm{eq}} = \beta \int_{-\infty}^{t} h(s) \left\langle A(t-s) \dot{B}(0) \right\rangle_{\mathrm{eq}} \mathrm{d}s$, which is the first form of Eq. (5.50). For the second form, stationarity gives $\langle A(\tau) B(s) \rangle_{\mathrm{eq}} = C_{AB}(\tau - s) + \langle A \rangle \langle B \rangle$, and differentiating in s at $s = 0$ yields $\left\langle A(\tau) \dot{B}(0) \right\rangle_{\mathrm{eq}} = -\frac{\mathrm{d}}{\mathrm{d}\tau} C_{AB}(\tau)$. □

Two features of Eq. (5.50) matter here. First, the right-hand side contains only equilibrium quantities: no knowledge of the perturbed dynamics is needed. Second, the result is exact for the full Hamiltonian system, so it applies to any subsystem, including the reduced Langevin dynamics of Sections 5.1 and 5.2. Integrating Eq. (5.50) over τ gives the static limit

$$\chi_{AB}(\omega = 0) = \int_0^\infty \chi_{AB}(\tau)\, \mathrm{d}\tau = -\beta \big[C_{AB}(\infty) - C_{AB}(0)\big] = \beta\, \mathrm{Cov}_{\mathrm{eq}}(A, B), \tag{5.52}$$

provided correlations decay, $C_{AB}(\infty) = 0$. This is the static fluctuation-response relation of Section 1.3: the static susceptibility is β times the equilibrium covariance.

Proposition 5.22 (Frequency-Domain FDT)**.** *Let $A_1, \dots, A_n$ be observables with stationary correlation matrix $C_{ij}(\tau) = \langle \delta A_i(\tau) \delta A_j(0) \rangle_{\mathrm{eq}}$ decaying to zero as $\tau \to \infty$, let $\chi_{ij}(\omega)$ be the susceptibility of A_i to a field coupled to A_j, and let $S^{\mathrm{ds}}_{ij}(\omega) = \int_{-\infty}^{\infty} C_{ij}(\tau) e^{-i\omega\tau}\, \mathrm{d}\tau$ be the double-sided cross-spectral density matrix. Then for $\omega \neq 0$,*

$$\mathbf{S}^{\mathrm{ds}}(\omega) = \frac{i k_{\mathrm{B}} T}{\omega} \Big[\boldsymbol{\chi}(\omega) - \boldsymbol{\chi}(\omega)^\dagger\Big], \qquad \textit{and for a single observable} \qquad S^{\mathrm{ds}}_{AA}(\omega) = -\frac{2 k_{\mathrm{B}} T}{\omega} \operatorname{Im} \chi_{AA}(\omega). \tag{5.53}$$

Proof. Insert Eq. (5.50) into Eq. (5.48) and integrate by parts:

$$\begin{aligned} \chi_{ij}(\omega) = -\beta \int_0^\infty \frac{\mathrm{d}C_{ij}}{\mathrm{d}\tau} e^{-i\omega\tau}\, \mathrm{d}\tau &= -\beta \Big[C_{ij}(\tau) e^{-i\omega\tau}\Big]_0^\infty - i\omega\beta \int_0^\infty C_{ij}(\tau) e^{-i\omega\tau}\, \mathrm{d}\tau \\ &= \beta C_{ij}(0) - i\omega\beta \int_0^\infty C_{ij}(\tau) e^{-i\omega\tau}\, \mathrm{d}\tau. \end{aligned}$$

The Hermitian conjugate is $(\boldsymbol{\chi}^\dagger)_{ij} = \overline{\chi_{ji}(\omega)} = \beta C_{ji}(0) + i\omega\beta \int_0^\infty C_{ji}(\tau) e^{i\omega\tau}\, \mathrm{d}\tau$. Equal-time correlations are symmetric, $C_{ji}(0) = C_{ij}(0)$, and stationarity gives $C_{ji}(\tau) = C_{ij}(-\tau)$, so the last integral equals $\int_{-\infty}^0 C_{ij}(\tau) e^{-i\omega\tau}\, \mathrm{d}\tau$. Subtracting,

$$\chi_{ij}(\omega) - \overline{\chi_{ji}(\omega)} = -i\omega\beta \int_{-\infty}^{\infty} C_{ij}(\tau) e^{-i\omega\tau}\, \mathrm{d}\tau = -i\omega\beta\, S_{ij}^{\mathrm{ds}}(\omega).$$

Solving for S_{ij}^{ds} gives the matrix statement. For one observable, $\chi - \bar{\chi} = 2i \operatorname{Im}\chi$, and $\frac{ik_{\mathrm{B}}T}{\omega} \cdot 2i \operatorname{Im}\chi = -\frac{2k_{\mathrm{B}}T}{\omega} \operatorname{Im}\chi$. □

With our $e^{-i\omega\tau}$ convention, a dissipative response has $\operatorname{Im}\chi_{AA}(\omega) < 0$ for $\omega > 0$, so the right-hand side of Eq. (5.53) is positive, as a spectral density must be. The anti-Hermitian part of the susceptibility describes absorption of energy from the field; Eq. (5.53) says that a system that absorbs at frequency ω also fluctuates at frequency ω, with a strength fixed by $k_{\mathrm{B}}T$.

Example 5.23 (The Damped Harmonic Oscillator)**.** Let $m\ddot{x} = -kx - \gamma_0\dot{x} + h(t) + \xi(t)$ with $\langle \xi(t)\xi(s)\rangle = 2\gamma_0 k_{\mathrm{B}}T\delta(t-s)$, so that h is a force coupled to $B = x$. With the convention $\mathrm{d}/\,\mathrm{d}t \to i\omega$, the averaged equation gives $(k - m\omega^2 + i\omega\gamma_0)\hat{x} = \hat{h}$, hence

$$\chi_{xx}(\omega) = \frac{1}{k - m\omega^2 + i\omega\gamma_0}, \qquad \operatorname{Im}\chi_{xx}(\omega) = \frac{-\omega\gamma_0}{(k - m\omega^2)^2 + \omega^2\gamma_0^2}.$$

Equation (5.53) predicts $S_{xx}^{\mathrm{ds}}(\omega) = -\frac{2k_{\mathrm{B}}T}{\omega}\operatorname{Im}\chi_{xx} = \frac{2\gamma_0 k_{\mathrm{B}}T}{(k-m\omega^2)^2+\omega^2\gamma_0^2}$. Directly from the Langevin equation with $h = 0$, $\hat{x} = \chi_{xx}\hat{\xi}$, so $S_{xx}^{\mathrm{ds}} = |\chi_{xx}|^2 S_\xi^{\mathrm{ds}} = 2\gamma_0 k_{\mathrm{B}}T\,|\chi_{xx}|^2$, which is the same expression. The two routes agree because the noise obeys the second FDT; the first FDT is, for linear Langevin equations, a consequence of the second. The static limit Eq. (5.52) gives $\chi_{xx}(0) = 1/k = \beta\langle x^2\rangle$, which is equipartition, $\frac{1}{2}k\langle x^2\rangle = \frac{1}{2}k_{\mathrm{B}}T$.

Notation Watch: Fourier Sign Conventions for Susceptibilities and Impedances

Physics texts following Kubo [62] usually define $\chi(\omega) = \int_0^\infty \chi(\tau) e^{+i\omega\tau}\, \mathrm{d}\tau$, which gives the familiar form $S^{\mathrm{ds}}(\omega) = \frac{2k_{\mathrm{B}}T}{\omega}\operatorname{Im}\chi(\omega)$ with $\operatorname{Im}\chi > 0$ for absorption, and the impedance of a parallel RC node $Z = (G - i\omega C)^{-1}$. Electrical engineering, Chapter 11, and this chapter use $e^{-i\omega\tau}$, which gives $S^{\mathrm{ds}}(\omega) = -\frac{2k_{\mathrm{B}}T}{\omega}\operatorname{Im}\chi(\omega)$ and $Z = (G + i\omega C)^{-1}$. The two susceptibilities are complex conjugates of each other. Power spectral densities of real stationary processes are even in ω and are the same in both conventions, as are all real parts, so $\operatorname{Re} Z(\omega)$ and the Nyquist formula do not depend on the choice.

5.4.3 The Kubo Formula for Overdamped Langevin Dynamics

Theorem 5.21 was derived for Hamiltonian dynamics. The overdamped circuits of this book are described by SDEs, and it is instructive to see that the same result holds there, and exactly which property of the dynamics it needs.

Proposition 5.24 (Kubo Formula for Reversible Diffusions)**.** *Let $\mathbf{x}_t$ obey the overdamped Itô SDE with mobility $\boldsymbol{\mu}(\mathbf{x})$, diffusion $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$, and drift $-\boldsymbol{\mu}\nabla E + k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}$ (Theorem 5.29 below; for constant $\boldsymbol{\mu}$ the divergence term vanishes), so that its generator is*

$$\mathcal{L}f = k_{\mathrm{B}}T\, e^{\beta E}\, \nabla\cdot\left(e^{-\beta E}\boldsymbol{\mu}\nabla f\right). \tag{5.54}$$

Perturb the energy as $E \to E - h(t)B(\mathbf{x})$, keeping $\boldsymbol{\mu}$ fixed. Then the response of $A(\mathbf{x})$ is given by Eq. (5.50), $\chi_{AB}(\tau) = -\beta\theta(\tau)\frac{\mathrm{d}}{\mathrm{d}\tau}C_{AB}(\tau)$, with C_{AB} the stationary correlation of the unperturbed diffusion.

Proof. The Fokker–Planck operator corresponding to Eq. (5.54) is $\mathcal{L}_{\mathrm{FP}}\rho = \nabla\cdot(\boldsymbol{\mu}[\rho\nabla E + k_{\mathrm{B}}T\nabla\rho])$ (Eq. (6.21)). The perturbation adds $-h\nabla B$ to ∇E, so to first order $\partial_t\delta\rho = \mathcal{L}_{\mathrm{FP}}\delta\rho - h(t)\nabla\cdot(\boldsymbol{\mu}\nabla B\, p_{\mathrm{eq}})$. Since $\nabla p_{\mathrm{eq}} = -\beta p_{\mathrm{eq}}\nabla E$,

$$\nabla\cdot(\boldsymbol{\mu}\nabla B\, p_{\mathrm{eq}}) = p_{\mathrm{eq}}\Big[\nabla\cdot(\boldsymbol{\mu}\nabla B) - \beta\nabla E\cdot\boldsymbol{\mu}\nabla B\Big] = \beta\, p_{\mathrm{eq}}\,\mathcal{L}B,$$

where the last equality expands Eq. (5.54). Hence $\delta\rho(t) = -\beta\int_{-\infty}^{t} h(s)\, e^{(t-s)\mathcal{L}_{\mathrm{FP}}}[p_{\mathrm{eq}}\mathcal{L}B]\,\mathrm{d}s$ and

$$\chi_{AB}(\tau) = -\beta\int A(\mathbf{x})\, e^{\tau\mathcal{L}_{\mathrm{FP}}}[p_{\mathrm{eq}}\mathcal{L}B](\mathbf{x})\,\mathrm{d}^N x = -\beta\int p_{\mathrm{eq}}(\mathbf{x})\,(\mathcal{L}B)(\mathbf{x})\,(e^{\tau\mathcal{L}}A)(\mathbf{x})\,\mathrm{d}^N x,$$

using the duality between $\mathcal{L}_{\mathrm{FP}}$ and $\mathcal{L}$ (Section 6.1). Now comes the one property we need: by Eq. (5.54) and integration by parts, $\int p_{\mathrm{eq}}\, g\,\mathcal{L}f\,\mathrm{d}^N x = -k_{\mathrm{B}}T\int p_{\mathrm{eq}}\,\nabla g\cdot\boldsymbol{\mu}\nabla f\,\mathrm{d}^N x$, which is symmetric in f and g. So $\mathcal{L}$ is self-adjoint in $L^2(p_{\mathrm{eq}})$, which is detailed balance. Moving $\mathcal{L}$ onto $e^{\tau\mathcal{L}}A$,

$$\chi_{AB}(\tau) = -\beta\int p_{\mathrm{eq}}\, B\,\mathcal{L}e^{\tau\mathcal{L}}A\,\mathrm{d}^N x = -\beta\frac{\mathrm{d}}{\mathrm{d}\tau}\int p_{\mathrm{eq}}\, B\, e^{\tau\mathcal{L}}A\,\mathrm{d}^N x = -\beta\frac{\mathrm{d}}{\mathrm{d}\tau}\,\langle A(\mathbf{x}_\tau)B(\mathbf{x}_0)\rangle_{\mathrm{eq}}\,,$$

which is Eq. (5.50) since constant shifts of A and B do not affect the derivative. □

The proof shows where the FDT of the first kind fails. In a non-equilibrium steady state the stationary density is not $e^{-\beta E}$, so the perturbation acting on p_{ss} is not $\beta p_{\mathrm{ss}}\mathcal{L}B$, and the generator is not self-adjoint in $L^2(p_{\mathrm{ss}})$. Response and fluctuations are then related by a modified formula that contains an extra term involving the probability current, and the size of the violation is tied to the rate of entropy production [97]. For the non-reciprocal NEAT-RN circuits of Chapter 13, measured noise spectra therefore do not determine the response, and vice versa.

5.4.4 Onsager Reciprocity

Theorem 5.25 (Onsager Reciprocal Relations)**.** *Suppose H_0 is invariant under the time-reversal map $\Theta(\mathbf{q},\mathbf{p}) = (\mathbf{q},-\mathbf{p})$ (no magnetic field), and let A, B have definite parities, $A\circ\Theta = \epsilon_A A$ and $B\circ\Theta = \epsilon_B B$ with $\epsilon_A,\epsilon_B\in\{+1,-1\}$. Then*

$$C_{AB}(\tau) = \epsilon_A\epsilon_B\, C_{BA}(\tau), \qquad \chi_{AB}(\tau) = \epsilon_A\epsilon_B\,\chi_{BA}(\tau). \tag{5.55}$$

Proof. Time-reversal invariance of H_0 implies that Θ preserves ρ_{eq} and phase-space volume, and that reversing the momenta and running forward for time τ is the same as running backward for time τ and then reversing: $\boldsymbol{\Phi}_\tau\circ\Theta = \Theta\circ\boldsymbol{\Phi}_{-\tau}$. Let $K_{AB}(\tau) := \langle A(\tau)B(0)\rangle_{\mathrm{eq}}$ be the correlation without subtraction of means. Changing variables $\boldsymbol{\Gamma}_{\mathrm{ps}} = \Theta\boldsymbol{\Gamma}'$,

$$\begin{aligned} K_{AB}(\tau) &= \int \rho_{\mathrm{eq}}(\Theta\boldsymbol{\Gamma}')\, A(\boldsymbol{\Phi}_\tau\Theta\boldsymbol{\Gamma}')\, B(\Theta\boldsymbol{\Gamma}')\,\mathrm{d}\boldsymbol{\Gamma}' = \int \rho_{\mathrm{eq}}(\boldsymbol{\Gamma}')\, A(\Theta\boldsymbol{\Phi}_{-\tau}\boldsymbol{\Gamma}')\,\epsilon_B B(\boldsymbol{\Gamma}')\,\mathrm{d}\boldsymbol{\Gamma}' \\ &= \epsilon_A\epsilon_B\int \rho_{\mathrm{eq}}(\boldsymbol{\Gamma}')\, A(\boldsymbol{\Phi}_{-\tau}\boldsymbol{\Gamma}')\, B(\boldsymbol{\Gamma}')\,\mathrm{d}\boldsymbol{\Gamma}' = \epsilon_A\epsilon_B\, K_{AB}(-\tau) = \epsilon_A\epsilon_B\, K_{BA}(\tau), \end{aligned}$$

where the last step is stationarity. The same change of variables shows $\langle A\rangle = \epsilon_A\langle A\rangle$, so an odd observable has zero mean, and therefore $\langle A\rangle\langle B\rangle = \epsilon_A\epsilon_B\langle A\rangle\langle B\rangle$ in every case. Subtracting the means gives $C_{AB}(\tau) = \epsilon_A\epsilon_B C_{BA}(\tau)$. The statement for χ follows from Eq. (5.50). □

In a magnetic field $\mathbf{B}_{\mathrm{mag}}$, the same argument relates the system in field $\mathbf{B}_{\mathrm{mag}}$ to the system in field $-\mathbf{B}_{\mathrm{mag}}$ (the Onsager–Casimir relations). For overdamped reversible diffusions, $C_{AB}(\tau) = C_{BA}(\tau)$ follows directly from the self-adjointness used in Proposition 5.24.

For electrical networks, Onsager reciprocity is the familiar reciprocity theorem: in a network of resistors, capacitors, and inductors, the voltage at node i produced by a current injected at node j equals the voltage at node j produced by the same current injected at node i, so the impedance matrix is symmetric, $\mathbf{Z}(\omega) = \mathbf{Z}(\omega)^{\top}$. In Worked computation 5.33 this will appear as the symmetry of $(\mathbf{G} + i\omega\mathbf{C})^{-1}$ when $\mathbf{G}$ and $\mathbf{C}$ are symmetric. Active subthreshold transconductances violate it: a transistor whose gate is node j and whose drain is node i injects current into i in response to v_j but not the other way around, so $G_{ij} \neq G_{ji}$. That broken reciprocity is the circuit-level origin of the non-equilibrium steady states analyzed in Section 6.2 and Chapter 13.

5.4.5 Green–Kubo Relations and the Einstein Relation

Transport coefficients are zero-frequency susceptibilities of currents. The Kubo formula expresses them as time integrals of current autocorrelation functions, called *Green–Kubo relations*.

Theorem 5.26 (Green–Kubo Formula for Mobility and the Einstein Relation)**.** *Let x be the coordinate of a particle in a spatially homogeneous medium in equilibrium at temperature T (for instance, a free particle coupled to a harmonic bath, or a particle in a periodic potential), with velocity $u = \dot{x}$. Assume that the stationary velocity autocorrelation function $C_{uu}(\tau) = \langle u(\tau)u(0)\rangle_{\mathrm{eq}}$ is integrable. Then:*

(a) The mobility, defined by the steady drift $\langle u\rangle = \mu_0 f$ under a weak constant force f, is

$$\mu_0 = \beta \int_0^{\infty} C_{uu}(\tau)\, \mathrm{d}\tau. \tag{5.56}$$

(b) The diffusion coefficient, defined by $\langle (x_t - x_0)^2\rangle = 2Dt + o(t)$ as $t \to \infty$, is

$$D = \int_0^{\infty} C_{uu}(\tau)\, \mathrm{d}\tau. \tag{5.57}$$

(c) Consequently $D = \mu_0 k_{\mathrm{B}} T$.

Proof. (a) A constant force is the perturbation $H = H_0 - fx$, so $B = x$ and $\dot{B} = u$. The first form of Eq. (5.50) with $A = u$ gives $\chi_{ux}(\tau) = \beta\theta(\tau)\,\langle u(\tau)u(0)\rangle_{\mathrm{eq}}$; this form uses only $\dot{B}$ and so remains meaningful even though x itself has no stationary distribution. For a force switched on at $t = 0$ and held constant, $\langle u\rangle(t) = f\int_0^t \chi_{ux}(\tau)\,\mathrm{d}\tau \to \beta f \int_0^{\infty} C_{uu}(\tau)\,\mathrm{d}\tau$.

(b) Exactly as in Example 5.5, $\langle (x_t - x_0)^2\rangle = \int_0^t\!\int_0^t C_{uu}(s - s')\,\mathrm{d}s\,\mathrm{d}s' = 2\int_0^t (t-\tau)C_{uu}(\tau)\,\mathrm{d}\tau$. Dividing by $2t$ gives $\int_0^t C_{uu}\,\mathrm{d}\tau - \frac{1}{t}\int_0^t \tau C_{uu}(\tau)\,\mathrm{d}\tau$. The first term tends to $\int_0^{\infty} C_{uu}$. The second tends to zero: split it at $\tau = \sqrt{t}$; the part over $[0,\sqrt{t}]$ is at most $t^{-1/2}\int_0^{\infty}|C_{uu}|$, and the part over $[\sqrt{t}, t]$ is at most $\int_{\sqrt{t}}^{\infty}|C_{uu}|$, both of which vanish.

(c) Compare (a) and (b). □

The Einstein relation $D = \mu_0 k_{\mathrm{B}} T$ appeared in Theorem 5.18(b) as a requirement for the Boltzmann distribution to be stationary. Here it appears as a consequence of the Kubo formula: diffusion and mobility are both integrals of the same velocity autocorrelation function.

Example 5.27 (Mobility of a Particle with Memory Friction)**.** For a free particle ($E = 0$) obeying the GLE Eq. (5.28), multiply the equation $m\dot{u}(t) = -\int_0^t \gamma(t-s)u(s)\,\mathrm{d}s + \xi(t)$ by $u(0)$ and average over the equilibrium initial state. By Proposition 5.10, $\xi(t)$ is independent of the system initial condition, so $\langle \xi(t)u(0)\rangle = 0$ and

$$m\,\frac{\mathrm{d}}{\mathrm{d}t}C_{uu}(t) = -\int_0^t \gamma(t-s)\,C_{uu}(s)\,\mathrm{d}s, \qquad C_{uu}(0) = \frac{k_{\mathrm{B}}T}{m}.$$

Take the Laplace transform $\tilde{f}(z) = \int_0^\infty f(t)e^{-zt}\,\mathrm{d}t$. The convolution becomes a product, and $m[z\tilde{C}_{uu}(z) - k_\mathrm{B}T/m] = -\tilde{\gamma}(z)\tilde{C}_{uu}(z)$, so

$$\tilde{C}_{uu}(z) = \frac{k_\mathrm{B}T}{mz + \tilde{\gamma}(z)}. \tag{5.58}$$

At $z = 0$, $\int_0^\infty C_{uu}(t)\,\mathrm{d}t = k_\mathrm{B}T/\tilde{\gamma}(0)$, and $\tilde{\gamma}(0) = \int_0^\infty \gamma(t)\,\mathrm{d}t$. For the Drude bath this is γ_0 by Eq. (5.41). By Theorem 5.26, $D = k_\mathrm{B}T/\gamma_0$ and $\mu_0 = 1/\gamma_0$, independent of the mass and of the cutoff ω_c. Long-time transport sees only the zero-frequency friction, which justifies the notation γ_0 used since Section 5.1.

Remark 5.28 (Johnson–Nyquist Noise as a Green–Kubo Relation)**.** Apply Theorem 5.26 to charge transport through a resistor held at zero voltage: the coordinate is the transferred charge Q, the velocity is the current $I = \dot{Q}$, and the force is the applied voltage V, coupled through $H' = -VQ$. The mobility is the conductance, $\langle I\rangle = GV$, so Eq. (5.56) reads $G = \beta\int_0^\infty \langle \delta I(\tau)\delta I(0)\rangle\,\mathrm{d}\tau$. By the Wiener–Khinchin theorem (Theorem 11.3), $\int_0^\infty C_{II}\,\mathrm{d}\tau = \frac{1}{2}S_I^{\mathrm{ds}}(0)$, hence $S_I^{\mathrm{ds}}(0) = 2k_\mathrm{B}TG$ and $S_I(f) = 4k_\mathrm{B}TG$ at low frequency [57, 79].

5.5 State-Dependent Mobility and the Divergence Drift

So far the friction tensor $\boldsymbol{\Gamma}$ was constant. In real devices it is not. The channel conductance of a subthreshold MOSFET depends exponentially on its gate voltage, the conductance of a diode depends on its bias, and the viscous drag on a colloid increases near a wall. When the mobility $\boldsymbol{\mu}(\mathbf{x}) = \boldsymbol{\Gamma}(\mathbf{x})^{-1}$ depends on the state, the Einstein relation still fixes the noise locally, $\mathbf{D}(\mathbf{x}) = k_\mathrm{B}T\boldsymbol{\mu}(\mathbf{x})$, and the noise becomes multiplicative. Section 4.4 showed that multiplicative noise makes the drift convention-dependent, and the worked computation Worked computation 4.23 showed that the naive Itô equation $\mathrm{d}v = -\mu(v)E'(v)\,\mathrm{d}t + \sqrt{2k_\mathrm{B}T\mu(v)}\,\mathrm{d}W_t$ samples $\mu(v)^{-1}e^{-\beta E(v)}$ instead of the Boltzmann distribution. Here we derive the correct equation from the underlying physics.

5.5.1 The Thermodynamically Consistent Overdamped Equation

We first ask which Itô drift is compatible with equilibrium, and then show that the small-mass limit of the inertial dynamics produces exactly that drift.

Theorem 5.29 (Divergence Drift for State-Dependent Mobility)**.** *Let $\boldsymbol{\mu}(\mathbf{x})$ be a smooth field of symmetric positive-definite matrices and consider the Itô SDE*

$$\mathrm{d}\mathbf{x}_t = \mathbf{a}(\mathbf{x}_t)\,\mathrm{d}t + \boldsymbol{\sigma}(\mathbf{x}_t)\,\mathrm{d}\mathbf{W}_t, \qquad \frac{1}{2}\boldsymbol{\sigma}\boldsymbol{\sigma}^\top = \mathbf{D} = k_\mathrm{B}T\boldsymbol{\mu}. \tag{5.59}$$

The Boltzmann density $p_\mathrm{eq} \propto e^{-\beta E}$ is a stationary solution with zero probability current (equilibrium detailed balance) if and only if

$$\mathbf{a}(\mathbf{x}) = -\boldsymbol{\mu}(\mathbf{x})\nabla E(\mathbf{x}) + k_\mathrm{B}T\,\nabla\cdot\boldsymbol{\mu}(\mathbf{x}), \qquad (\nabla\cdot\boldsymbol{\mu})_i := \sum_{j=1}^{N}\frac{\partial \mu_{ij}}{\partial x_j}. \tag{5.60}$$

With this drift the Fokker–Planck equation takes the form

$$\frac{\partial \rho}{\partial t} = \nabla\cdot\Big(\boldsymbol{\mu}(\mathbf{x})\big[\rho\,\nabla E + k_\mathrm{B}T\nabla\rho\big]\Big). \tag{5.61}$$

Proof. By Eq. (6.19), the probability current of an Itô diffusion is $\mathbf{J} = (\mathbf{a} - \nabla \cdot \mathbf{D})\rho - \mathbf{D}\nabla\rho$. With $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$ and $\rho = p_{\mathrm{eq}}$, $\nabla p_{\mathrm{eq}} = -\beta p_{\mathrm{eq}}\nabla E$, so

$$\mathbf{J}[p_{\mathrm{eq}}] = \Big(\mathbf{a} - k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu} + k_{\mathrm{B}}T\boldsymbol{\mu}\,\beta\nabla E\Big)p_{\mathrm{eq}} = \Big(\mathbf{a} - k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu} + \boldsymbol{\mu}\nabla E\Big)p_{\mathrm{eq}}.$$

Since $p_{\mathrm{eq}} > 0$ everywhere, $\mathbf{J}[p_{\mathrm{eq}}] \equiv \mathbf{0}$ if and only if $\mathbf{a} = -\boldsymbol{\mu}\nabla E + k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}$. Substituting this drift into the general current gives, for any ρ, $\mathbf{J} = (-\boldsymbol{\mu}\nabla E)\rho - k_{\mathrm{B}}T\boldsymbol{\mu}\nabla\rho$, and $\partial_t\rho = -\nabla\cdot\mathbf{J}$ is Eq. (5.61). □

This is the result quoted in Eq. (6.20). The extra term $k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}$ is not a force: it does not come from any energy and it disappears when $\boldsymbol{\mu}$ is constant. It is present because the Itô convention evaluates the noise amplitude at the start of each time step. A trajectory starting in a high-mobility region receives large kicks and leaves quickly, while one starting in a low-mobility region receives small kicks and lingers. Without correction this would make the process accumulate where $\boldsymbol{\mu}$ is small, which is the μ^{-1} factor found in Eq. (4.117). The divergence drift pushes the process toward higher mobility at exactly the rate needed to cancel this bias.

The Fokker–Planck form Eq. (5.61) has a transparent physical reading: the probability flux is $-\boldsymbol{\mu}(\mathbf{x})$ times the gradient of the local chemical potential $E + k_{\mathrm{B}}T\ln\rho$. This is Fick's law combined with drift, with the mobility standing outside both gradients. Diffusive transport between neighboring points is governed by the mobility at the interface between them, not by the mobility at the point of departure.

5.5.2 Microscopic Origin: The Small-Mass Limit with State-Dependent Friction

Theorem 5.29 identifies the drift required by equilibrium. It does not yet show that a physical system actually has that drift. For that we return to the inertial dynamics, which has no convention ambiguity. Consider the underdamped equation with state-dependent friction,

$$\mathrm{d}\mathbf{x} = \frac{\mathbf{p}}{m}\,\mathrm{d}t, \qquad \mathrm{d}\mathbf{p} = \Big[-\nabla E(\mathbf{x}) - \boldsymbol{\Gamma}(\mathbf{x})\frac{\mathbf{p}}{m}\Big]\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T}\,\boldsymbol{\Gamma}(\mathbf{x})^{1/2}\,\mathrm{d}\mathbf{W}_t. \tag{5.62}$$

Because $\mathbf{x}$ is differentiable in time, the noise amplitude $\boldsymbol{\Gamma}(\mathbf{x}_t)^{1/2}$ varies smoothly over a time step, and the Itô and Stratonovich interpretations of Eq. (5.62) coincide. The same argument as in Proposition 5.2, with $\boldsymbol{\Gamma}$ now depending on $\mathbf{x}$, shows that the phase-space Gibbs density remains stationary, because $\boldsymbol{\Gamma}(\mathbf{x})$ appears only inside the $\mathbf{p}$-divergence. So the inertial dynamics is unambiguously in equilibrium. The question is what survives as $m \to 0$.

Heuristic Derivation: Momentum-Moment Closure of the Kramers Equation

Integrate the Kramers equation Eq. (5.4) (with $\mathbf{M} = m\mathbf{I}$ and $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}(\mathbf{x})$) over $\mathbf{p}$. The momentum-space divergences integrate to zero, leaving the continuity equation $\partial_t\rho(\mathbf{x},t) + \nabla\cdot\mathbf{j} = 0$ for the position density $\rho = \int\rho\,\mathrm{d}^N p$ with flux $\mathbf{j} = \int(\mathbf{p}/m)\rho\,\mathrm{d}^N p$. Next multiply Eq. (5.4) by $\mathbf{p}/m$ and integrate over $\mathbf{p}$. Integrating by parts in $\mathbf{p}$,

$$\partial_t\mathbf{j} = -\nabla\cdot\int\frac{\mathbf{p}\mathbf{p}^{\top}}{m^2}\rho\,\mathrm{d}^N p - \frac{\nabla E}{m}\rho - \frac{\boldsymbol{\Gamma}(\mathbf{x})}{m}\mathbf{j}.$$

On time scales long compared with $\tau_p = m/\gamma_{\min}$, the momenta are in local Maxwellian equilibrium, $\int\mathbf{p}\mathbf{p}^{\top}\rho\,\mathrm{d}^N p \approx mk_{\mathrm{B}}T\rho(\mathbf{x})\mathbf{I}$, and the flux is slaved to the slow variables, so $\partial_t\mathbf{j}$ is negligible compared with the terms of order $1/m$. Multiplying by m and dropping $m\,\partial_t\mathbf{j}$,

$$\mathbf{0} \approx -k_{\mathrm{B}}T\nabla\rho - \rho\nabla E - \boldsymbol{\Gamma}(\mathbf{x})\mathbf{j} \qquad\Longrightarrow\qquad \mathbf{j} = -\boldsymbol{\mu}(\mathbf{x})\big[\rho\nabla E + k_{\mathrm{B}}T\nabla\rho\big].$$

This is the flux of Eq. (5.61), with the mobility outside the gradient. The Itô drift Eq. (5.60) is then read off from Theorem 5.29.

The heuristic can be made rigorous by a systematic expansion of the Kramers equation in powers of $\sqrt{m}$ [85, 90]. In one dimension there is a short pathwise derivation that makes the role of the Itô integral explicit.

Proposition 5.30 (One-Dimensional Small-Mass Limit with State-Dependent Friction)**.** *Let $N = 1$ with friction $\gamma(x) > 0$ smooth and bounded away from zero, and mobility $\mu(x) = 1/\gamma(x)$. Then the formal small-mass limit of Eq. (5.62) is the Itô SDE*

$$\mathrm{d}x_t = \Big[-\mu(x_t)E'(x_t) + k_{\mathrm{B}}T\mu'(x_t)\Big]\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\mu(x_t)}\;\mathrm{d}W_t, \tag{5.63}$$

in agreement with Eq. (5.60).

Proof. Let $\Phi(x) := \int_0^x \gamma(y)\,\mathrm{d}y$. Because $\gamma > 0$, Φ is strictly increasing with inverse Φ^{-1}. For $m > 0$ the path x_t^m is continuously differentiable, so the ordinary chain rule gives $\mathrm{d}\Phi(x_t^m) = \gamma(x_t^m)\dot{x}_t^m\,\mathrm{d}t = \gamma(x_t^m)(p_t^m/m)\,\mathrm{d}t$. The momentum equation can therefore be written as

$$\mathrm{d}p_t^m = -E'(x_t^m)\,\mathrm{d}t - \mathrm{d}\Phi(x_t^m) + \sqrt{2k_{\mathrm{B}}T\gamma(x_t^m)}\;\mathrm{d}W_t.$$

Integrating from 0 to t,

$$\Phi(x_t^m) = \Phi(x_0) - \int_0^t E'(x_s^m)\,\mathrm{d}s + \int_0^t \sqrt{2k_{\mathrm{B}}T\gamma(x_s^m)}\;\mathrm{d}W_s - (p_t^m - p_0). \tag{5.64}$$

This identity is exact for every $m > 0$. The stochastic integral is an Itô integral of the adapted integrand $\sqrt{2k_{\mathrm{B}}T\gamma(x_s^m)}$; since x^m is differentiable, it would have the same value with any other evaluation point, but its limit as $m \to 0$ is the Itô integral of the limiting process, because Itô integrals of adapted integrands converge in L^2 when the integrands do (Theorem 4.8). The momentum term is of order $\sqrt{mk_{\mathrm{B}}T}$, as in Theorem 5.4, and drops out. Hence $y_t := \Phi(x_t)$ satisfies the Itô SDE

$$\mathrm{d}y_t = -E'(x_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\gamma(x_t)}\;\mathrm{d}W_t, \qquad x_t = \Phi^{-1}(y_t).$$

Now apply Itô's formula (Theorem 4.12) to $x_t = \Phi^{-1}(y_t)$. The derivatives of the inverse function are $(\Phi^{-1})'(y) = 1/\gamma(x)$ and $(\Phi^{-1})''(y) = \frac{\mathrm{d}}{\mathrm{d}y}\frac{1}{\gamma(x)} = -\frac{\gamma'(x)}{\gamma(x)^2}\cdot\frac{1}{\gamma(x)} = -\frac{\gamma'(x)}{\gamma(x)^3}$, and the quadratic variation is $(\mathrm{d}y_t)^2 = 2k_{\mathrm{B}}T\gamma(x_t)\,\mathrm{d}t$. Therefore

$$\begin{aligned}\mathrm{d}x_t &= \frac{1}{\gamma(x_t)}\,\mathrm{d}y_t + \frac{1}{2}\left(-\frac{\gamma'(x_t)}{\gamma(x_t)^3}\right)2k_{\mathrm{B}}T\gamma(x_t)\,\mathrm{d}t \\ &= -\frac{E'(x_t)}{\gamma(x_t)}\,\mathrm{d}t + \sqrt{\frac{2k_{\mathrm{B}}T}{\gamma(x_t)}}\,\mathrm{d}W_t - k_{\mathrm{B}}T\frac{\gamma'(x_t)}{\gamma(x_t)^2}\,\mathrm{d}t.\end{aligned}$$

With $\mu = 1/\gamma$ we have $\mu' = -\gamma'/\gamma^2$, so the last term is $+k_{\mathrm{B}}T\mu'(x_t)\,\mathrm{d}t$, and Eq. (5.63) follows. □

The proof shows that the divergence drift is produced by Itô's formula itself: the coordinate in which the noise is "linear" is $\Phi(x)$, the integrated friction, and transforming back to x generates a second-order term. In several dimensions no such global change of variables exists in general, but the conclusion is the same. Hottovy, McDaniel, Volpe, and Wehr proved [53] that the small-mass limit of $m\ddot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) - \boldsymbol{\Gamma}(\mathbf{x})\dot{\mathbf{x}} + \boldsymbol{\sigma}(\mathbf{x})\boldsymbol{\eta}(t)$ is the Itô SDE with drift $\boldsymbol{\Gamma}^{-1}\mathbf{F} + \mathbf{S}$, where $S_i = \sum_{j,l}\partial_{x_l}\big[(\boldsymbol{\Gamma}^{-1})_{ij}\big]J_{jl}$ and $\mathbf{J}$ solves the Lyapunov equation $\boldsymbol{\Gamma}\mathbf{J} + \mathbf{J}\boldsymbol{\Gamma}^{\top} = \boldsymbol{\sigma}\boldsymbol{\sigma}^{\top}$. When the FDT holds, $\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top} = 2k_{\mathrm{B}}T\boldsymbol{\Gamma}$, the solution is $\mathbf{J} = k_{\mathrm{B}}T\mathbf{I}$, and $\mathbf{S} = k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}$, which is Eq. (5.60). This Lyapunov equation is the momentum-space counterpart of the one that governs the RC sampler (Worked computation 5.33): it encodes the local Maxwellian covariance $k_{\mathrm{B}}T\mathbf{I}$ of the eliminated velocities.

5.5.3 Three Equivalent Forms and the Link to Itô–Stratonovich Conversion

In one dimension, the thermodynamically consistent equation Eq. (5.63) can be written in any of the λ-conventions of Remark 4.17. With $B(x) = \sqrt{2k_{\mathrm{B}}T\mu(x)}$ we have $BB' = \frac{1}{2}(B^2)' = k_{\mathrm{B}}T\mu'$, so by Eq. (4.74) the λ-drift equals the Itô drift minus $\lambda k_{\mathrm{B}}T\mu'$:

$$\text{Itô } (\lambda = 0): \quad \mathrm{d}x = \left[-\mu E' + k_{\mathrm{B}}T\mu'\right]\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\mu}\cdot \mathrm{d}W_t, \tag{5.65}$$

$$\text{Stratonovich } (\lambda = \tfrac{1}{2}): \quad \mathrm{d}x = \left[-\mu E' + \tfrac{1}{2}k_{\mathrm{B}}T\mu'\right]\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\mu}\circ \mathrm{d}W_t, \tag{5.66}$$

$$\text{Hänggi–Klimontovich } (\lambda = 1): \quad \mathrm{d}x = -\mu E'\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\mu}\star \mathrm{d}W_t. \tag{5.67}$$

All three describe the same process. The post-point (kinetic) form Eq. (5.67) has no extra drift at all; that is the SDE counterpart of the Fokker–Planck form Eq. (5.61) with the mobility outside the gradient. The Stratonovich form, which is what the Wong–Zakai theorem (Theorem 4.18) produces from smooth colored noise, contains half of the Itô correction. This resolves the question raised in Section 4.4.2: the Wong–Zakai mechanism supplies only half of the required drift, and the other half must come from the physics of the dissipative element. In the inertial model above it comes from the momentum relaxation; in a subthreshold transistor it comes from the bidirectional barrier-crossing statistics of Section 11.4 and Chapter 13. A model that takes smooth noise $\sqrt{2k_{\mathrm{B}}T\mu(x)}\,\eta_\tau(t)$ and the bare drift $-\mu E'$ converges to Eq. (5.66) without the $\frac{1}{2}k_{\mathrm{B}}T\mu'$ term and samples $\mu^{-1/2}e^{-\beta E}$, which is wrong.

In several dimensions the Stratonovich and post-point corrections depend on the choice of square root $\boldsymbol{\sigma}(\mathbf{x})$ of $2k_{\mathrm{B}}T\boldsymbol{\mu}(\mathbf{x})$, through the term $\sum_{j,\alpha}\sigma_{j\alpha}\partial_j\sigma_{i\alpha}$ in Eq. (4.66). The Itô form Eq. (5.60) and the Fokker–Planck form Eq. (5.61) do not, which is why we use them as the reference statements throughout the book.

Corollary 5.31 (Nonlinear Conductance on a Capacitive Node)**.** *A node of constant capacitance C_0 connected to a source of voltage b through a voltage-dependent conductance $G(v)$ at temperature T has energy $E(v) = \frac{1}{2}C_0(v-b)^2$ and mobility $\mu(v) = G(v)/C_0^2$ (from $\mathrm{d}q = C_0\,\mathrm{d}v$ and the current $G(v)\cdot$(voltage drop)). Its thermodynamically consistent Itô equation is*

$$\mathrm{d}v_t = \left[-\frac{G(v_t)}{C_0}(v_t - b) + \frac{k_{\mathrm{B}}T}{C_0^2}G'(v_t)\right]\mathrm{d}t + \frac{\sqrt{2k_{\mathrm{B}}TG(v_t)}}{C_0}\,\mathrm{d}W_t, \tag{5.68}$$

and its stationary distribution is $\mathcal{N}(b, k_{\mathrm{B}}T/C_0)$ for every function $G(v) > 0$.

Proof. Insert $E' = C_0(v-b)$ and $\mu = G/C_0^2$ into Eq. (5.63): $-\mu E' = -G(v)(v-b)/C_0$, $k_{\mathrm{B}}T\mu' = k_{\mathrm{B}}TG'(v)/C_0^2$, and $\sqrt{2k_{\mathrm{B}}T\mu} = \sqrt{2k_{\mathrm{B}}TG}/C_0$. By Theorem 5.29 the stationary density is $\propto e^{-\beta C_0(v-b)^2/2}$. □

Equation (5.68) is the Itô equation Eq. (4.118) of Worked computation 4.23, now derived rather than imposed. The stationary variance $k_{\mathrm{B}}T/C_0$ does not depend on the conductance, linear or not: the conductance controls only how fast the node equilibrates.

Example 5.32 (A Voltage-Controlled Coupling Between Two Nodes)**.** Consider two nodes with equal grounded capacitances C, each tied to ground through a fixed conductance g_0, and coupled to each other through a conductance $g(v_1)$ that depends on the voltage of node 1 (for example, a pass transistor whose gate is driven by node 1). If the coupling element is in thermal equilibrium at temperature T, the charge-space mobility is the conductance matrix $\mathbf{G}(\mathbf{v}) = g_0\mathbf{I} + g(v_1)\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and the voltage-space mobility is $\boldsymbol{\mu}(\mathbf{v}) = \mathbf{G}(\mathbf{v})/C^2$. Its row divergence is

$$(\nabla\cdot\boldsymbol{\mu})_1 = \partial_{v_1}\mu_{11} + \partial_{v_2}\mu_{12} = \frac{g'(v_1)}{C^2}, \qquad (\nabla\cdot\boldsymbol{\mu})_2 = \partial_{v_1}\mu_{21} + \partial_{v_2}\mu_{22} = -\frac{g'(v_1)}{C^2},$$

since $\mu_{12} = \mu_{21} = -g(v_1)/C^2$ does not depend on v_2 and $\mu_{22} = (g_0 + g(v_1))/C^2$ does not depend on v_2. The divergence drift is $\frac{k_B T g'(v_1)}{C^2}(1, -1)^\top$: when $g' > 0$ it pushes v_1 up and v_2 down, toward larger coupling conductance. It is a purely noise-induced effect, of size $k_B T g'/C^2$, and an Euler–Maruyama simulation that omits it converges to a distorted stationary law whose density depends on the function g. With the drift included, the stationary law is the product of two independent $\mathcal{N}(0, k_B T/C)$ distributions, as Theorem 5.29 requires for the capacitive energy $\frac{C}{2}(v_1^2 + v_2^2)$.

Hardware Realization: Noise-Induced Drift in Simulated and Physical Sampling Cells

Two consequences of this section matter for thermodynamic hardware.

1. **Software models.** Any discrete-time simulator that advances a circuit with state-dependent conductances by the Euler–Maruyama rule $\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{a}(\mathbf{v}_k)\Delta t + \boldsymbol{\sigma}(\mathbf{v}_k)\sqrt{\Delta t}\,\boldsymbol{\zeta}_k$ is implementing the Itô equation, so its drift must contain the term $k_B T \nabla \cdot \boldsymbol{\mu}$ explicitly. Software used to prototype probabilistic circuits, such as `THRML` [31], must respect this whenever the noise model is multiplicative. Omitting the term is not a discretization error that shrinks with Δt: the missing drift is of the same order as the physical drift, and the simulated chain converges to the wrong stationary law, distorted by the factor μ^{-1} of Eq. (4.117) in one dimension, for every step size.

2. **Physical devices.** A physical sampling cell needs no such correction; it simply is the inertial or master-equation dynamics, and whichever drift it has is the true one. If its noise is equilibrium thermal noise, it samples $e^{-\beta E}$ automatically. If its noise is shot noise generated by biased transistors, the local Einstein relation $\mathbf{D} = k_B T \boldsymbol{\mu}$ is replaced by a bias-dependent relation (Section 11.4), and the stationary state must be computed from the mesoscopic Fokker–Planck equation of Chapter 13.

5.6 Worked Computation: The Coupled Ornstein–Uhlenbeck RC Circuit

We now apply the results of this chapter to the RC sampler. The computation below solves the N-node RC circuit exactly, computes all of its second-order statistics in the time and frequency domains, and checks the Nyquist form of the fluctuation-dissipation theorem directly from the Langevin solution. It also settles a point that is easy to get wrong: which Gaussian distribution a passive RC network actually samples.

Worked computation 5.33 (Propagator, Two-Time Covariance, Spectrum, and Nyquist Relation of the N-Node RC Circuit)**. Setup.** The circuit has N capacitive nodes with voltage vector $\mathbf{v}_t \in \mathbb{R}^N$. The capacitance matrix $\mathbf{C} = \mathbf{C}^\top \succ \mathbf{0}$ contains the grounded node capacitances on its diagonal and minus the coupling capacitances off the diagonal. The conductance matrix $\mathbf{G} \in \mathbb{R}^{N \times N}$ describes the linear (or linearized) currents between nodes and to a set of bias sources that would hold the node voltages at $\mathbf{b}$ if there were no noise. Noise currents $\mathbf{B}\,\mathrm{d}\mathbf{W}_t$ are injected at the nodes, with $\mathbf{W}_t$ a standard M-dimensional Brownian motion and $\mathbf{B} \in \mathbb{R}^{N \times M}$. Kirchhoff's current law gives the equation of the RC sampler

$$\mathbf{C}\,\mathrm{d}\mathbf{v}_t = -\mathbf{G}(\mathbf{v}_t - \mathbf{b})\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t, \tag{5.69}$$

or, dividing by $\mathbf{C}$ and writing $\mathbf{u}_t := \mathbf{v}_t - \mathbf{b}$,

$$\mathrm{d}\mathbf{u}_t = \mathbf{A}\mathbf{u}_t\,\mathrm{d}t + \mathbf{C}^{-1}\mathbf{B}\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{A} := -\mathbf{C}^{-1}\mathbf{G}, \qquad \mathbf{D} := \frac{1}{2}\mathbf{C}^{-1}\mathbf{B}\mathbf{B}^\top\mathbf{C}^{-1}. \tag{5.70}$$

We assume $\mathbf{A}$ is Hurwitz (all eigenvalues have negative real part, Definition B.14), which holds for instance whenever $\mathbf{G} + \mathbf{G}^\top \succ \mathbf{0}$. Two noise models will be compared:

- **(P) Passive thermal noise.** Every conductance carries its own Johnson–Nyquist noise at temperature T, generalized to possibly non-symmetric $\mathbf{G}$ as $\mathbf{B}\mathbf{B}^\top = k_\mathrm{B}T(\mathbf{G} + \mathbf{G}^\top)$. For a reciprocal network ($\mathbf{G} = \mathbf{G}^\top$) this is the FDT $\mathbf{B}\mathbf{B}^\top = 2k_\mathrm{B}T\mathbf{G}$ of Eq. (5.44), with the conductance playing the role of the friction tensor. The antisymmetric part of $\mathbf{G}$, which a passive network can contain only through lossless non-reciprocal elements such as gyrators, dissipates no power and therefore generates no noise.
- **(E) Engineered noise.** The conductances are effectively noiseless and dedicated sources inject noise with $\mathbf{B}\mathbf{B}^\top = 2(k_\mathrm{B}T/\tau_0)\,\mathbf{C}$, where τ_0 is a design time constant.

Step 1: Matrix-exponential propagator. Apply Itô's product rule (Corollary 4.13) to $e^{-\mathbf{A}t}\mathbf{u}_t$; since $e^{-\mathbf{A}t}$ is deterministic and differentiable, there is no Itô correction:

$$\mathrm{d}(e^{-\mathbf{A}t}\mathbf{u}_t) = -\mathbf{A}e^{-\mathbf{A}t}\mathbf{u}_t\,\mathrm{d}t + e^{-\mathbf{A}t}(\mathbf{A}\mathbf{u}_t\,\mathrm{d}t + \mathbf{C}^{-1}\mathbf{B}\,\mathrm{d}\mathbf{W}_t) = e^{-\mathbf{A}t}\mathbf{C}^{-1}\mathbf{B}\,\mathrm{d}\mathbf{W}_t.$$

Integrating from 0 to t and multiplying by $e^{\mathbf{A}t}$,

$$\mathbf{u}_t = e^{\mathbf{A}t}\mathbf{u}_0 + \int_0^t e^{\mathbf{A}(t-s)}\mathbf{C}^{-1}\mathbf{B}\,\mathrm{d}\mathbf{W}_s. \tag{5.71}$$

The stochastic integral is a linear functional of Gaussian increments, so $\mathbf{u}_t$ given $\mathbf{u}_0$ is Gaussian. Its mean is $e^{\mathbf{A}t}\mathbf{u}_0$, and by the Itô isometry (Theorem 4.8) its covariance is

$$\mathbf{\Sigma}(t) = \int_0^t e^{\mathbf{A}(t-s)}\mathbf{C}^{-1}\mathbf{B}\mathbf{B}^\top\mathbf{C}^{-1}e^{\mathbf{A}^\top(t-s)}\,\mathrm{d}s = 2\int_0^t e^{\mathbf{A}s}\mathbf{D}e^{\mathbf{A}^\top s}\,\mathrm{d}s. \tag{5.72}$$

Hence the transition density (the Green's function of the Fokker–Planck equation) is

$$P(\mathbf{v}, t \mid \mathbf{v}_0, 0) = \mathcal{N}(\mathbf{v};\ \mathbf{b} + e^{\mathbf{A}t}(\mathbf{v}_0 - \mathbf{b}),\ \mathbf{\Sigma}(t)). \tag{5.73}$$

Differentiating Eq. (5.72) gives $\dot{\mathbf{\Sigma}} = \mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top + 2\mathbf{D}$ with $\mathbf{\Sigma}(0) = \mathbf{0}$. As $t \to \infty$, $e^{\mathbf{A}t} \to \mathbf{0}$ and $\mathbf{\Sigma}(t)$ converges to the unique solution $\mathbf{\Sigma}$ of the Lyapunov equation $\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top + 2\mathbf{D} = \mathbf{0}$ (Theorem B.16).

For a reciprocal network ($\mathbf{G} = \mathbf{G}^\top \succ \mathbf{0}$) the exponential can be written in closed form. The generalized symmetric eigenproblem $\mathbf{G}\mathbf{w}_k = \lambda_k\mathbf{C}\mathbf{w}_k$ has real positive eigenvalues λ_k and eigenvectors that can be normalized so that the matrix $\mathbf{U} = [\mathbf{w}_1, \dots, \mathbf{w}_N]$ satisfies

$$\mathbf{U}^\top\mathbf{C}\mathbf{U} = \mathbf{I}, \qquad \mathbf{U}^\top\mathbf{G}\mathbf{U} = \mathbf{\Lambda} := \mathrm{diag}(\lambda_1, \dots, \lambda_N). \tag{5.74}$$

(Take $\mathbf{w}_k = \mathbf{C}^{-1/2}\mathbf{o}_k$ with $\mathbf{o}_k$ the orthonormal eigenvectors of the symmetric matrix $\mathbf{C}^{-1/2}\mathbf{G}\mathbf{C}^{-1/2}$.) From $\mathbf{G}\mathbf{U} = \mathbf{C}\mathbf{U}\mathbf{\Lambda}$ we get $\mathbf{A}\mathbf{U} = -\mathbf{U}\mathbf{\Lambda}$, and from the first relation in Eq. (5.74), $\mathbf{U}^{-1} = \mathbf{U}^\top\mathbf{C}$ and $\mathbf{C}^{-1} = \mathbf{U}\mathbf{U}^\top$. Therefore

$$e^{\mathbf{A}t} = \mathbf{U}e^{-\mathbf{\Lambda}t}\mathbf{U}^\top\mathbf{C} = \sum_{k=1}^{N} e^{-t/\tau_k}\,\mathbf{w}_k\mathbf{w}_k^\top\mathbf{C}, \qquad \tau_k := \frac{1}{\lambda_k}. \tag{5.75}$$

The τ_k are the RC time constants of the normal modes of the network.

Step 2: Stationary covariance under the two noise models.

Model (P). Here $2\mathbf{D} = k_{\mathrm{B}}T\mathbf{C}^{-1}(\mathbf{G}+\mathbf{G}^{\top})\mathbf{C}^{-1}$. Try $\boldsymbol{\Sigma} = k_{\mathrm{B}}T\mathbf{C}^{-1}$. Using $\mathbf{C}^{\top} = \mathbf{C}$, so that $\mathbf{A}^{\top} = -\mathbf{G}^{\top}\mathbf{C}^{-1}$,

$$\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^{\top} = -k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{G}\mathbf{C}^{-1} - k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{G}^{\top}\mathbf{C}^{-1} = -2\mathbf{D}.$$

By uniqueness,

$$\boldsymbol{\Sigma}_{(\mathrm{P})} = k_{\mathrm{B}}T\,\mathbf{C}^{-1}, \qquad p_{\mathrm{ss}}(\mathbf{v}) \propto \exp\Big(-\frac{\beta}{2}(\mathbf{v}-\mathbf{b})^{\top}\mathbf{C}(\mathbf{v}-\mathbf{b})\Big). \tag{5.76}$$

This is the Boltzmann distribution of the electrostatic energy stored in the capacitors, measured from the bias point. It does not depend on $\mathbf{G}$ at all. In particular, when $\mathbf{C}$ is diagonal (no coupling capacitors), the node voltages are uncorrelated at equal times no matter how strongly the resistors couple them, and each node has variance $k_{\mathrm{B}}T/C_i$, as in Proposition 11.6. This is exactly what Theorem 5.29 predicts: in charge coordinates $\mathbf{q} = \mathbf{C}\mathbf{v}$ the conductance matrix is the mobility and the capacitive energy is the energy, and the Boltzmann law depends only on the energy. The conductances set the speed of sampling, not the distribution sampled.

Model (E). Here $2\mathbf{D} = 2(k_{\mathrm{B}}T/\tau_0)\mathbf{C}^{-1}$. For symmetric $\mathbf{G}$ try $\boldsymbol{\Sigma} = (k_{\mathrm{B}}T/\tau_0)\mathbf{G}^{-1}$:

$$\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^{\top} = -\frac{k_{\mathrm{B}}T}{\tau_0}\mathbf{C}^{-1}\mathbf{G}\mathbf{G}^{-1} - \frac{k_{\mathrm{B}}T}{\tau_0}\mathbf{G}^{-1}\mathbf{G}\mathbf{C}^{-1} = -\frac{2k_{\mathrm{B}}T}{\tau_0}\mathbf{C}^{-1} = -2\mathbf{D},$$

so

$$\boldsymbol{\Sigma}_{(\mathrm{E})} = k_{\mathrm{B}}T\,(\tau_0\mathbf{G})^{-1}, \qquad p_{\mathrm{ss}}(\mathbf{v}) \propto \exp\Big(-\frac{\beta}{2}(\mathbf{v}-\mathbf{b})^{\top}\tau_0\mathbf{G}(\mathbf{v}-\mathbf{b})\Big). \tag{5.77}$$

Now the conductance matrix, multiplied by τ_0 to give units of capacitance, plays the role of the precision matrix. In the language of Section 5.5, model (E) is overdamped Langevin dynamics in voltage space with energy $\frac{\tau_0}{2}(\mathbf{v}-\mathbf{b})^{\top}\mathbf{G}(\mathbf{v}-\mathbf{b})$ and mobility $(\tau_0\mathbf{C})^{-1}$. Physically it is not a passive circuit in equilibrium with its own resistors; the noise power is supplied by the injection sources. This is the mode of operation in which the coupling matrix programs the sampled covariance, as in the NEAT-RN synthesis of Proposition 13.10, where $\mathbf{C} = C_0\mathbf{I}$ and $\mathbf{D} = d_0\mathbf{I}$ correspond to $k_{\mathrm{B}}T/\tau_0 = d_0C_0$.

Step 3: Two-time covariance. In the stationary state, write $\mathbf{u}_{t+\tau} = e^{\mathbf{A}\tau}\mathbf{u}_t + \int_t^{t+\tau} e^{\mathbf{A}(t+\tau-s)}\mathbf{C}^{-1}\mathbf{B}\,\mathrm{d}\mathbf{W}_s$ for $\tau \geq 0$. The integral involves only Brownian increments after time t, which are independent of $\mathbf{u}_t$ and have zero mean. Therefore

$$\mathbf{K}(\tau) := \big\langle \mathbf{u}_{t+\tau}\mathbf{u}_t^{\top}\big\rangle_{\mathrm{ss}} = e^{\mathbf{A}\tau}\boldsymbol{\Sigma} \quad (\tau \geq 0), \qquad \mathbf{K}(-\tau) = \mathbf{K}(\tau)^{\top} = \boldsymbol{\Sigma}e^{\mathbf{A}^{\top}\tau}, \tag{5.78}$$

in agreement with Eq. (B.88). For model (P) with symmetric $\mathbf{G}$, Eq. (5.75) and $\mathbf{C}\mathbf{C}^{-1} = \mathbf{I}$ give the mode expansion

$$\mathbf{K}(\tau) = k_{\mathrm{B}}T\,\mathbf{U}e^{-\boldsymbol{\Lambda}|\tau|}\mathbf{U}^{\top} = k_{\mathrm{B}}T\sum_{k=1}^{N} e^{-|\tau|/\tau_k}\,\mathbf{w}_k\mathbf{w}_k^{\top}. \tag{5.79}$$

This matrix is symmetric for every τ: $\mathbf{K}(\tau) = \mathbf{K}(\tau)^{\top} = \mathbf{K}(-\tau)$. That is the Onsager symmetry $C_{ij}(\tau) = C_{ji}(\tau)$ of Theorem 5.25 for the even observables v_i, and by Proposition B.17 it is equivalent to detailed balance. At $\tau = 0$ it reduces to $k_{\mathrm{B}}T\mathbf{U}\mathbf{U}^{\top} = k_{\mathrm{B}}T\mathbf{C}^{-1}$, as it must. For non-symmetric $\mathbf{G}$ in model (P), the circulation matrix of Eq. (B.85) is $\mathbf{Q} = \frac{1}{2}(\mathbf{A}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\mathbf{A}^{\top}) = -\frac{k_{\mathrm{B}}T}{2}\mathbf{C}^{-1}(\mathbf{G}-\mathbf{G}^{\top})\mathbf{C}^{-1}$, which is nonzero exactly when the network is non-reciprocal.

Step 4: Power spectral density via Wiener–Khinchin. By Theorem 11.3, the double-sided cross-spectral density matrix is the Fourier transform of $\mathbf{K}$. Split the integral at $\tau = 0$ and use Eq. (5.78):

$$\mathbf{S}^{\mathrm{ds}}(\omega) = \int_0^\infty e^{(\mathbf{A}-i\omega\mathbf{I})\tau}\boldsymbol{\Sigma}\,\mathrm{d}\tau + \int_0^\infty \boldsymbol{\Sigma}e^{(\mathbf{A}^\top+i\omega\mathbf{I})\tau}\,\mathrm{d}\tau = (i\omega\mathbf{I}-\mathbf{A})^{-1}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}(-i\omega\mathbf{I}-\mathbf{A}^\top)^{-1}.$$

Both integrals converge because $\mathbf{A}$ is Hurwitz, and we used $\int_0^\infty e^{\mathbf{M}\tau}\,\mathrm{d}\tau = -\mathbf{M}^{-1}$ for Hurwitz $\mathbf{M}$. Write $\mathbf{R}(\omega) := (i\omega\mathbf{I}-\mathbf{A})^{-1}$; since $\mathbf{A}$ is real, $\mathbf{R}(\omega)^\dagger = (-i\omega\mathbf{I}-\mathbf{A}^\top)^{-1}$. Factor $\mathbf{R}$ on the left and $\mathbf{R}^\dagger$ on the right:

$$\mathbf{S}^{\mathrm{ds}}(\omega) = \mathbf{R}\Big[\boldsymbol{\Sigma}(-i\omega\mathbf{I}-\mathbf{A}^\top) + (i\omega\mathbf{I}-\mathbf{A})\boldsymbol{\Sigma}\Big]\mathbf{R}^\dagger = \mathbf{R}\Big[-\mathbf{A}\boldsymbol{\Sigma}-\boldsymbol{\Sigma}\mathbf{A}^\top\Big]\mathbf{R}^\dagger = 2\,\mathbf{R}(\omega)\mathbf{D}\mathbf{R}(\omega)^\dagger,$$

where the $\pm i\omega\boldsymbol{\Sigma}$ terms cancelled and the Lyapunov equation was used in the last step. Now introduce the *impedance matrix* of the network, the response of node voltages to currents injected at the nodes,

$$\mathbf{Z}(\omega) := (\mathbf{G}+i\omega\mathbf{C})^{-1}. \tag{5.80}$$

Since $i\omega\mathbf{I}-\mathbf{A} = \mathbf{C}^{-1}(\mathbf{G}+i\omega\mathbf{C})$, we have $\mathbf{R} = \mathbf{Z}\mathbf{C}$, and with $2\mathbf{D} = \mathbf{C}^{-1}\mathbf{B}\mathbf{B}^\top\mathbf{C}^{-1}$,

$$\mathbf{S}_V^{\mathrm{ds}}(\omega) = \mathbf{Z}(\omega)\,\mathbf{B}\mathbf{B}^\top\,\mathbf{Z}(\omega)^\dagger. \tag{5.81}$$

This has the transparent circuit reading of Proposition 11.4: white current noise with spectral matrix $\mathbf{B}\mathbf{B}^\top$ filtered by the impedance matrix.

Step 5: Verification of the Nyquist relation. Let $\operatorname{Re}\mathbf{Z} := \frac{1}{2}(\mathbf{Z}+\mathbf{Z}^\dagger)$ denote the Hermitian part. From Eq. (5.80), $\mathbf{Z}^{-1} + (\mathbf{Z}^\dagger)^{-1} = (\mathbf{G}+i\omega\mathbf{C}) + (\mathbf{G}^\top - i\omega\mathbf{C}) = \mathbf{G}+\mathbf{G}^\top$, because $\mathbf{C}$ is real and symmetric. Multiplying on the left by $\mathbf{Z}$ and on the right by $\mathbf{Z}^\dagger$,

$$\mathbf{Z}^\dagger + \mathbf{Z} = \mathbf{Z}(\mathbf{G}+\mathbf{G}^\top)\mathbf{Z}^\dagger. \tag{5.82}$$

For model (P), $\mathbf{B}\mathbf{B}^\top = k_{\mathrm{B}}T(\mathbf{G}+\mathbf{G}^\top)$, so Eqs. (5.81) and (5.82) give

$$\boxed{\mathbf{S}_V^{\mathrm{ds}}(\omega) = k_{\mathrm{B}}T(\mathbf{Z}(\omega)+\mathbf{Z}(\omega)^\dagger) = 2k_{\mathrm{B}}T\operatorname{Re}\mathbf{Z}(\omega), \qquad \mathbf{S}_V(f) = 4k_{\mathrm{B}}T\operatorname{Re}\mathbf{Z}(2\pi f)} \tag{5.83}$$

for the double-sided angular and single-sided cyclic spectra respectively (Definition 11.2). For a reciprocal network $\mathbf{Z} = \mathbf{Z}^\top$, the Hermitian part is the entrywise real part, and the diagonal entries give the familiar statement: the voltage noise at node i is $4k_{\mathrm{B}}T$ times the real part of the driving-point impedance seen at that node [79]. For a single node, $Z = 1/(G+i\omega C)$ has $\operatorname{Re}Z = G/(G^2+\omega^2C^2) = R/(1+\omega^2R^2C^2)$, and Eq. (5.83) reproduces the Lorentzian Eq. (11.47).

As a check on normalization, integrate over frequency. For symmetric $\mathbf{G}$, Eq. (5.74) gives $\mathbf{G}+i\omega\mathbf{C} = \mathbf{U}^{-\top}(\boldsymbol{\Lambda}+i\omega\mathbf{I})\mathbf{U}^{-1}$, so $\mathbf{Z} = \mathbf{U}(\boldsymbol{\Lambda}+i\omega\mathbf{I})^{-1}\mathbf{U}^\top$ and $\operatorname{Re}\mathbf{Z} = \mathbf{U}\operatorname{diag}(\lambda_k/(\lambda_k^2+\omega^2))\mathbf{U}^\top$. Using $\int_{-\infty}^{\infty}\frac{\lambda}{\lambda^2+\omega^2}\frac{\mathrm{d}\omega}{2\pi} = \frac{1}{2}$,

$$\int_{-\infty}^{\infty}\mathbf{S}_V^{\mathrm{ds}}(\omega)\,\frac{\mathrm{d}\omega}{2\pi} = 2k_{\mathrm{B}}T\cdot\frac{1}{2}\,\mathbf{U}\mathbf{U}^\top = k_{\mathrm{B}}T\mathbf{C}^{-1} = \boldsymbol{\Sigma}_{(\mathrm{P})},$$

recovering Eq. (5.76). The resistances shape the spectrum, but the area under it is fixed by the capacitances alone.

Step 6: The impedance is the Kubo susceptibility. We now check Worked computation 5.33 against Theorem 5.21. Take model (P) with symmetric $\mathbf{G}$. Injecting a small current $\mathbf{I}(t)$ at the

nodes adds $\mathbf{I}\,\mathrm{d}t$ to the right side of Eq. (5.69). In the mobility language of Section 5.5, the charge $\mathbf{q} = \mathbf{C}\mathbf{v}$ moves with mobility $\mathbf{G}$, so the injected current is equivalent to a field $\mathbf{h} = \mathbf{G}^{-1}\mathbf{I}$ (a set of Thévenin voltages) coupled to the observable $\mathbf{q}$ through $E \to E - \mathbf{h}^{\top}\mathbf{q}$; this is the Norton–Thévenin equivalence. By Proposition 5.24, the response of $\mathbf{v}$ to $\mathbf{h}$ is

$$\boldsymbol{\chi}_{vq}(\tau) = -\beta\theta(\tau)\frac{\mathrm{d}}{\mathrm{d}\tau}\left\langle \mathbf{u}_\tau(\mathbf{C}\mathbf{u}_0)^{\top}\right\rangle = -\beta\theta(\tau)\frac{\mathrm{d}}{\mathrm{d}\tau}\left[e^{\mathbf{A}\tau}k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{C}\right] = -\theta(\tau)\,\mathbf{A}e^{\mathbf{A}\tau} = \theta(\tau)\,\mathbf{C}^{-1}\mathbf{G}e^{\mathbf{A}\tau}.$$

Its Fourier transform is $\boldsymbol{\chi}_{vq}(\omega) = \mathbf{C}^{-1}\mathbf{G}(i\omega\mathbf{I} - \mathbf{A})^{-1} = \mathbf{C}^{-1}\mathbf{G}\mathbf{Z}\mathbf{C}$. The response of $\mathbf{v}$ to the injected current is therefore $\boldsymbol{\chi}_{vq}\mathbf{G}^{-1} = \mathbf{C}^{-1}\mathbf{G}\mathbf{Z}\mathbf{C}\mathbf{G}^{-1}$. To simplify it, note the push-through identity $\mathbf{G}\mathbf{Z}\mathbf{C} = \mathbf{C}\mathbf{Z}\mathbf{G}$, which follows from

$$(\mathbf{G}\mathbf{Z}\mathbf{C})^{-1} = \mathbf{C}^{-1}(\mathbf{G} + i\omega\mathbf{C})\mathbf{G}^{-1} = \mathbf{C}^{-1} + i\omega\mathbf{G}^{-1} = \mathbf{G}^{-1}(\mathbf{G} + i\omega\mathbf{C})\mathbf{C}^{-1} = (\mathbf{C}\mathbf{Z}\mathbf{G})^{-1}.$$

Hence $\mathbf{C}^{-1}\mathbf{G}\mathbf{Z}\mathbf{C}\mathbf{G}^{-1} = \mathbf{C}^{-1}\mathbf{C}\mathbf{Z}\mathbf{G}\mathbf{G}^{-1} = \mathbf{Z}$. The Kubo formula, fed only with the equilibrium two-time covariance, reproduces the impedance matrix obtained directly from the circuit equations. Combining this with Proposition 5.22 gives a second derivation of Eq. (5.83): with $\boldsymbol{\chi}_{qq} = \mathbf{C}\mathbf{Z}\mathbf{G}$ and $\mathbf{G} = \mathbf{Z}^{-1} - i\omega\mathbf{C}$, one finds $\boldsymbol{\chi}_{qq} - \boldsymbol{\chi}_{qq}^{\dagger} = -i\omega\mathbf{C}(\mathbf{Z} + \mathbf{Z}^{\dagger})\mathbf{C}$, so $\mathbf{S}_{qq}^{\mathrm{ds}} = k_{\mathrm{B}}T\mathbf{C}(\mathbf{Z} + \mathbf{Z}^{\dagger})\mathbf{C}$ and $\mathbf{S}_{V}^{\mathrm{ds}} = \mathbf{C}^{-1}\mathbf{S}_{qq}^{\mathrm{ds}}\mathbf{C}^{-1} = k_{\mathrm{B}}T(\mathbf{Z} + \mathbf{Z}^{\dagger})$.

Step 7: When the Nyquist relation fails. Subtracting $k_{\mathrm{B}}T(\mathbf{Z} + \mathbf{Z}^{\dagger})$ from Eq. (5.81) and using Eq. (5.82) gives, for an arbitrary noise matrix,

$$\boldsymbol{\Delta}(\omega) := \mathbf{S}_{V}^{\mathrm{ds}}(\omega) - 2k_{\mathrm{B}}T\,\mathrm{Re}\,\mathbf{Z}(\omega) = \mathbf{Z}(\omega)\Big[\mathbf{B}\mathbf{B}^{\top} - k_{\mathrm{B}}T(\mathbf{G} + \mathbf{G}^{\top})\Big]\mathbf{Z}(\omega)^{\dagger}. \tag{5.84}$$

Since $\mathbf{Z}(\omega)$ is invertible, the Nyquist defect $\boldsymbol{\Delta}$ vanishes at any single frequency if and only if $\mathbf{B}\mathbf{B}^{\top} = k_{\mathrm{B}}T(\mathbf{G} + \mathbf{G}^{\top})$. Three cases illustrate the content of this statement.

- A reciprocal resistor network at uniform temperature has $\boldsymbol{\Delta} = \mathbf{0}$: this is Eq. (5.83).
- A network whose resistors sit at different temperatures has $\mathbf{B}\mathbf{B}^{\top} = \sum_r 2k_{\mathrm{B}}T_r\mathbf{G}_r$, a sum over resistors of their conductance stamps. There is no single T with $\boldsymbol{\Delta} = \mathbf{0}$, and heat flows through the circuit from hot to cold resistors.
- In model (E), $\mathbf{B}\mathbf{B}^{\top} = 2(k_{\mathrm{B}}T/\tau_0)\mathbf{C}$ is not proportional to $\mathbf{G} + \mathbf{G}^{\top}$ unless $\mathbf{G} \propto \mathbf{C}$, so $\boldsymbol{\Delta} \neq \mathbf{0}$ even though the voltage dynamics satisfies detailed balance with respect to the programmed energy. In active subthreshold circuits the noise is shot noise set by bias currents (Section 11.4) and the transconductances are non-reciprocal, so both factors in the bracket of Eq. (5.84) depart from the passive form (Chapter 13).

Step 8: A two-node example with numbers. Take two nodes with equal grounded capacitances $C = 1\,\mathrm{fF}$, each tied to a bias source through $g_0 = 1\,\mu\mathrm{S}$, and coupled to each other through $g_c = 1\,\mu\mathrm{S}$, at $T = 300\,\mathrm{K}$ ($k_{\mathrm{B}}T = 4.14 \times 10^{-21}\,\mathrm{J}$), with passive noise (P). Then

$$\mathbf{C} = C\,\mathbf{I}, \qquad \mathbf{G} = \begin{pmatrix} g_0 + g_c & -g_c \\ -g_c & g_0 + g_c \end{pmatrix}.$$

The normal modes are the common mode $\mathbf{w}_1 = (1,1)^{\top}/\sqrt{2C}$ with $\lambda_1 = g_0/C = 10^9\,\mathrm{s}^{-1}$ ($\tau_1 = 1\,\mathrm{ns}$), and the differential mode $\mathbf{w}_2 = (1,-1)^{\top}/\sqrt{2C}$ with $\lambda_2 = (g_0 + 2g_c)/C = 3 \times 10^9\,\mathrm{s}^{-1}$ ($\tau_2 \approx 0.33\,\mathrm{ns}$). By Eq. (5.79),

$$K_{11}(\tau) = \frac{k_{\mathrm{B}}T}{2C}\Big(e^{-|\tau|/\tau_1} + e^{-|\tau|/\tau_2}\Big), \qquad K_{12}(\tau) = \frac{k_{\mathrm{B}}T}{2C}\Big(e^{-|\tau|/\tau_1} - e^{-|\tau|/\tau_2}\Big).$$

At $\tau = 0$, $K_{11} = k_{\mathrm{B}}T/C = 4.14 \times 10^{-6}\,\mathrm{V}^2$ (an rms fluctuation of 2.03 mV) and $K_{12} = 0$: the resistive coupling creates no equal-time correlation. At nonzero lag the cross-correlation is positive, reaching its maximum $\frac{k_{\mathrm{B}}T}{2C} \cdot \frac{2}{3^{3/2}} \approx 0.19\, k_{\mathrm{B}}T/C$ at $|\tau| = \frac{\ln 3}{2}$ ns ≈ 0.55 ns (set the derivative of $e^{-|\tau|/\tau_1} - e^{-3|\tau|/\tau_1}$ to zero). The impedance matrix, by the same mode decomposition, has driving-point element

$$Z_{11}(\omega) = \frac{1}{2}\left[\frac{1}{g_0 + i\omega C} + \frac{1}{g_0 + 2g_c + i\omega C}\right], \qquad Z_{11}(0) = \frac{1}{2}\left(1 + \frac{1}{3}\right)\mathrm{M\Omega} \approx 0.667\,\mathrm{M\Omega},$$

which is indeed the resistance from node 1 to the bias sources (g_0 in parallel with the series combination of g_c and g_0, i.e. 1.5 μS). The low-frequency single-sided noise at node 1 is $S_V(0) = 4k_{\mathrm{B}}T \cdot 0.667\,\mathrm{M\Omega} \approx 1.1 \times 10^{-14}\,\mathrm{V}^2/\mathrm{Hz}$, or about $105\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$, and it rolls off in two Lorentzian steps at $f_k = 1/(2\pi\tau_k)$, near 160 MHz and 480 MHz.

Notation Watch: Which Gaussian Does the RC Sampler Produce?

The RC sampler equation $\mathbf{C}\,\mathrm{d}\mathbf{v} = -\mathbf{G}(\mathbf{v} - \mathbf{b})\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}$ does not by itself determine its stationary law; the noise model does.

- With passive thermal noise, $\mathbf{B}\mathbf{B}^{\top} = 2k_{\mathrm{B}}T\mathbf{G}$ for reciprocal $\mathbf{G}$, the stationary law is $\mathcal{N}(\mathbf{b}, k_{\mathrm{B}}T\mathbf{C}^{-1})$ (Eq. (5.76)). The conductances set only the relaxation rates.
- With engineered noise, $\mathbf{B}\mathbf{B}^{\top} = 2(k_{\mathrm{B}}T/\tau_0)\mathbf{C}$, the stationary law is $\mathcal{N}(\mathbf{b}, k_{\mathrm{B}}T(\tau_0\mathbf{G})^{-1})$ (Eq. (5.77)).

The shorthand $\mathcal{N}(\mathbf{b}, k_{\mathrm{B}}T\mathbf{G}^{-1})$ for the RC sampler, with energy $\frac{1}{2}(\mathbf{v} - \mathbf{b})^{\top}\mathbf{G}(\mathbf{v} - \mathbf{b})$ and mobility $\mathbf{C}^{-1}$ as in Worked computation 6.21, refers to the second case in units where $\tau_0 = 1$. It is not the equilibrium law of a passive RC network, and the two must not be mixed in dimensional estimates: $k_{\mathrm{B}}T\mathbf{G}^{-1}$ has units of $\mathrm{V}^2 \cdot \mathrm{s}$, while $k_{\mathrm{B}}T(\tau_0\mathbf{G})^{-1}$ and $k_{\mathrm{B}}T\mathbf{C}^{-1}$ have units of V^2.

Notation Watch: One-Sided and Two-Sided Nyquist Formulas

This chapter writes spectra as double-sided functions of angular frequency, $S^{\mathrm{ds}}(\omega) = \int C(\tau)e^{-i\omega\tau}\,\mathrm{d}\tau$, normalized so that $\int S^{\mathrm{ds}}\,\mathrm{d}\omega/(2\pi)$ is the variance. In this convention the FDT results read $S_\xi^{\mathrm{ds}} = 2k_{\mathrm{B}}T\,\mathrm{Re}\,\hat{\gamma}$, $S_I^{\mathrm{ds}} = 2k_{\mathrm{B}}TG$, and $S_V^{\mathrm{ds}} = 2k_{\mathrm{B}}T\,\mathrm{Re}\,Z$, and a white noise $\sqrt{2D}\,\mathrm{d}W_t$ has $S^{\mathrm{ds}} = 2D$. Measurement practice and Chapter 11 use the single-sided cyclic spectrum $S(f) = 2S^{\mathrm{ds}}(2\pi f)$ for $f \geq 0$, giving $S_I(f) = 4k_{\mathrm{B}}TG$ and $S_V(f) = 4k_{\mathrm{B}}T\,\mathrm{Re}\,Z(2\pi f)$ (Section 11.1). The factor of 2 between them is the folding of negative frequencies; neither is more correct. A third convention, the single-sided angular spectrum normalized with $\mathrm{d}\omega$ instead of $\mathrm{d}\omega/(2\pi)$, introduces a further factor of $1/\pi$ and is the usual source of errors by 2π when comparing papers.

Hardware Realization: Noise Spectra as a Diagnostic of Equilibrium Operation

Equation (5.84) turns the FDT into a measurement procedure. For a sampling cell one can measure, on the same nodes, the small-signal impedance matrix $\mathbf{Z}(\omega)$ (by injecting test currents) and the noise spectral matrix $\mathbf{S}_V(f)$ (by recording the free fluctuations). If $\mathbf{S}_V = 4k_{\mathrm{B}}T\,\mathrm{Re}\,\mathbf{Z}$ holds across the band, the cell is operating as an equilibrium sampler at temperature T and its stationary law is fixed by its stored energy. A nonzero defect $\boldsymbol{\Delta}$ quantifies how far the noise sources depart from thermal noise, and an asymmetric $\mathbf{Z}$ signals

broken reciprocity. Both departures are intentional in non-equilibrium designs such as the NEAT-RNs of Chapter 13 [38], which use them to program covariances and to speed up mixing. For such designs, the relevant design equation is the Lyapunov equation of Step 2, not the Nyquist formula.

Summary

- **Proved in this chapter:**
  - Stationarity of the phase-space Gibbs measure for the underdamped Langevin (Kramers) equation, with pointwise vanishing of the dissipative current (Proposition 5.2), and the Smoluchowski–Kramers small-mass limit with an explicit $\mathcal{O}(\sqrt{m})$ error bound for constant friction (Theorem 5.4); the Ornstein–Fürth crossover from ballistic to diffusive motion (Example 5.5).
  - Exact elimination of a harmonic bath with counterterm, yielding the generalized Langevin equation with memory kernel $\gamma(t) = \sum_k c_k^2 \cos(\omega_k t)/(m_k \omega_k^2)$ (Theorem 5.8), and the FDT of the second kind $\langle \xi(t)\xi(s)\rangle = k_{\mathrm{B}} T \gamma(t-s)$ for thermal bath initial conditions (Proposition 5.10).
  - The spectral representation $\gamma(t) = \frac{2}{\pi}\int_0^\infty J(\omega)\omega^{-1}\cos\omega t \, \mathrm{d}\omega$, the frequency-domain FDT $S_\xi^{\mathrm{ds}} = 2k_{\mathrm{B}}T \operatorname{Re}\hat{\gamma}$, the Drude kernel and its Markovian limit (Proposition 5.15 and Corollary 5.16), and necessity and sufficiency of $\boldsymbol{\sigma}\boldsymbol{\sigma}^\top = 2k_{\mathrm{B}}T\boldsymbol{\Gamma}$ and $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$ (Theorem 5.18).
  - The classical Kubo formula (Theorem 5.21), its frequency-domain form (Proposition 5.22), its validity for reversible diffusions and the role of detailed balance (Proposition 5.24), Onsager reciprocity (Theorem 5.25), and the Green–Kubo and Einstein relations (Theorem 5.26).
  - The divergence drift $+k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}$ as the unique Itô drift compatible with detailed balance (Theorem 5.29) and its microscopic derivation from the one-dimensional small-mass limit (Proposition 5.30).
  - The complete second-order statistics of the N-node RC circuit, the equilibrium covariance $k_{\mathrm{B}}T\mathbf{C}^{-1}$ under thermal noise, and the matrix Nyquist relation $\mathbf{S}_V^{\mathrm{ds}} = 2k_{\mathrm{B}}T \operatorname{Re}\mathbf{Z}$ together with the defect formula Eq. (5.84) (Worked computation 5.33).
- **Assumed:**
  - The bath starts in the canonical state of the shifted bath Hamiltonian (Assumption 5.9); a bath that is large, with a dense spectrum, so that $\gamma(t) \to 0$ (Remark 5.11); and an Ohmic spectral density with a cutoff far above all system frequencies, which is what licenses the Markovian limit.
  - Classical statistics, $\hbar\omega \ll k_{\mathrm{B}}T$ at all relevant frequencies (Section 5.3.3).
  - Bounded, Lipschitz forces and thermal initial momenta in Theorem 5.4; convergence of the Itô integral in the formal limit of Proposition 5.30.
- **Taken up later:**
  - The Fokker–Planck formalism used here (generator, duality, probability current, detailed balance) is developed in Chapter 6; the Lyapunov equation and the

circulation matrix are proved in Chapter B; analytic background on Itô integrals is in Chapter C.

- Kramers' escape rate in the underdamped and overdamped regimes (Chapter 7); trajectory heat and entropy production for Langevin dynamics (Chapter 8).
- The quantum FDT and the Planck–Nyquist spectrum (Eq. (11.30)); dissipative quantum tunneling in Josephson devices (Section 14.5).
- Shot noise and the non-equilibrium replacement of the Einstein relation in sub-threshold transistors (Section 11.4 and Chapter 13).

- **Open questions:**
  - Systematic generalizations of the FDT to non-equilibrium steady states, in which response and fluctuations differ by a term tied to the probability current, and their use for calibrating NEAT-RN cells in which both reciprocity and the Nyquist relation are deliberately broken [38, 97].
  - Rigorous small-mass limits for multidimensional state-dependent friction combined with non-thermal (shot) noise, where the noise-induced drift is no longer $k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}$.

Bibliographic notes

The harmonic-bath derivation of the generalized Langevin equation is due to Zwanzig [120], and its use with a counterterm and an Ohmic spectral density to describe macroscopic quantum tunneling with dissipation is due to Caldeira and Leggett [17]. The fluctuation-dissipation theorem in its general linear-response form, including the distinction between the first and second kinds, is reviewed by Kubo [62]. Clear textbook treatments of the Langevin and Kramers equations, the overdamped limit, and adiabatic elimination are given by Risken [90], Gardiner [39], and van Kampen [112]; Pavliotis [85] gives a mathematically careful account of the small-mass limit, of generalized Langevin equations with Markovian embeddings, and of Green–Kubo formulas. The multidimensional small-mass limit with state-dependent friction, which produces the noise-induced drift of Section 5.5, was established by Hottovy et al. [53]. Johnson's measurement of thermal noise in conductors and Nyquist's theoretical derivation appeared together [57, 79]; their connection to the electronic devices used in thermodynamic computing is taken up in Chapter 11. The stochastic energetics of Langevin dynamics, which builds directly on the equations of this chapter, is presented by Sekimoto [98] and reviewed by Seifert [97], and the circuit-level theory of thermal and shot noise in nonlinear electronic networks is developed by Freitas et al. [37].

Chapter 6

The Fokker–Planck Equation, Spectral Theory, and Functional Inequalities

In Chapters 4 and 5 we wrote down stochastic differential equations (SDEs) for individual trajectories $\mathbf{x}_t \in \mathbb{R}^N$. Whether $\mathbf{x}_t$ is the vector of node voltages of the RC sampler (the coupled linear RC circuit of Chapter 1) or the coordinate of the double-well sampler (a bistable element with energy $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$), an SDE describes one realization of the noise $\mathbf{W}_t$. A thermodynamic computer, however, is used for its distribution. Its output is a set of samples from a target law $p_\theta(\mathbf{x})$, or expectations over many thermal realizations. To understand how a circuit carries an initial voltage distribution $\rho_0(\mathbf{x})$ to its target, and how many nanoseconds that takes, we must pass from trajectories $\mathbf{x}_t$ to the evolving probability density $\rho(\mathbf{x}, t)$.

The density obeys the *Fokker–Planck equation*, known in probability theory as the *forward Kolmogorov equation*. The SDE is driven by nowhere-differentiable Brownian paths; the Fokker–Planck equation is a deterministic, linear partial differential equation (PDE) of parabolic type:

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} = \mathcal{L}_{\mathrm{FP}} \rho(\mathbf{x}, t) = -\nabla \cdot \mathbf{J}(\mathbf{x}, t).$$

Because it is linear, we can study it with spectral theory and functional analysis. We first derive it together with its probability current $\mathbf{J}$ (Section 6.1). The current separates equilibrium circuits, where detailed balance forces $\mathbf{J}_{\mathrm{eq}} \equiv \mathbf{0}$, from non-equilibrium steady states (NESS) in which currents circulate, as they do in the non-reciprocal subthreshold CMOS networks of Chapter 13 (Section 6.2). We then prove an H-theorem: the Kullback–Leibler (KL) divergence to the stationary density decreases monotonically, even in a NESS; at equilibrium this is the decrease of the variational free energy $\mathcal{F}[\rho]$ of Chapter 1 (Section 6.3). Monotonic decay says nothing about speed. To obtain rates we map the Fokker–Planck operator onto a Schrödinger operator, the Witten Hamiltonian [118], whose spectral gap λ_1 sets the relaxation time (Section 6.4). Finally, the Bakry–Émery criterion [6, 7] turns a lower bound $\nabla^2 E \succeq \kappa \mathbf{I}$ on the curvature of the energy into a dimension-free contraction rate $2\mu\kappa$ for the KL divergence (Section 6.5). This result explains why the denoising thermodynamic models of Chapter 16 add a quadratic coupling at every step: it restores the convexity that a multimodal landscape lacks.

6.1 The Forward Fokker–Planck Operator and Backward Kolmogorov Generator

Let $\mathbf{x}_t \in \mathbb{R}^N$ be a continuous-time Markov diffusion process evolving according to the general N-dimensional Itô stochastic differential equation established in Chapter 4:

$$\mathrm{d}\mathbf{x}_t = \mathbf{a}(\mathbf{x}_t, t)\ \mathrm{d}t + \mathbf{B}(\mathbf{x}_t, t)\ \mathrm{d}\mathbf{W}_t, \tag{6.1}$$

where $\mathbf{a} : \mathbb{R}^N \times [0,\infty) \to \mathbb{R}^N$ is the deterministic drift vector field, $\mathbf{B} : \mathbb{R}^N \times [0,\infty) \to \mathbb{R}^{N\times M}$ is the noise coupling (volatility) matrix, and $\mathbf{W}_t \in \mathbb{R}^M$ is a standard M-dimensional Wiener process with independent components satisfying $\mathbb{E}[\mathrm{d}W_\alpha(t)] = 0$ and $\mathrm{d}W_\alpha(t)\ \mathrm{d}W_\beta(t) = \delta_{\alpha\beta}\ \mathrm{d}t$. Throughout this chapter, unless explicitly noted otherwise, we focus on time-homogeneous autonomous systems in which $\mathbf{a}(\mathbf{x})$ and $\mathbf{B}(\mathbf{x})$ depend only on the instantaneous state $\mathbf{x} \in \mathbb{R}^N$ and are smooth (C^∞) functions with bounded derivatives or suitable confining growth at infinity so that strong solutions exist globally in time (Chapter 4).

From the noise coupling matrix $\mathbf{B}(\mathbf{x})$, we define the symmetric, positive semi-definite *diffusion tensor* $\mathbf{D}(\mathbf{x}) \in \mathbb{R}^{N\times N}$ by

$$D_{ij}(\mathbf{x}) := \frac{1}{2}\sum_{\alpha=1}^{M} B_{i\alpha}(\mathbf{x})B_{j\alpha}(\mathbf{x}), \qquad \text{or in matrix form,} \qquad \mathbf{D}(\mathbf{x}) := \frac{1}{2}\mathbf{B}(\mathbf{x})\mathbf{B}(\mathbf{x})^\top. \tag{6.2}$$

For any vector $\mathbf{u} \in \mathbb{R}^N$, we have $\mathbf{u}^\top\mathbf{D}(\mathbf{x})\mathbf{u} = \frac{1}{2}\left\|\mathbf{B}(\mathbf{x})^\top\mathbf{u}\right\|^2 \geq 0$, so $\mathbf{D}(\mathbf{x}) \succeq \mathbf{0}$. When the thermal noise sources drive all N physical degrees of freedom non-degenerately (the *uniformly elliptic* case), $\mathbf{D}(\mathbf{x}) \succ \mathbf{0}$ is strictly positive definite at every point $\mathbf{x} \in \mathbb{R}^N$.

6.1.1 The Backward Kolmogorov Generator via Itô's Formula

Suppose we prepare the circuit at time $t = 0$ in a random initial state $\mathbf{x}_0$ drawn from a smooth probability density $\rho_0(\mathbf{x})$ on $\mathbb{R}^N$. At any later time $t > 0$, the state $\mathbf{x}_t$ is distributed according to a time-dependent probability density $\rho(\mathbf{x}, t)$, which means that for any smooth, rapidly decaying test observable $f \in C_c^\infty(\mathbb{R}^N)$ (the space of infinitely differentiable functions with compact support), the ensemble expectation of $f(\mathbf{x}_t)$ is given by

$$\mathbb{E}[f(\mathbf{x}_t)] = \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}, t)\, f(\mathbf{x}). \tag{6.3}$$

How does $\mathbb{E}[f(\mathbf{x}_t)]$ change over an infinitesimal time interval $\mathrm{d}t$? We can compute this rate of change in two dual ways: first by following the stochastic trajectory $\mathbf{x}_t$ inside $f(\mathbf{x}_t)$ via Itô's formula (Chapter 4), and second by differentiating the probability density $\rho(\mathbf{x}, t)$ inside the integral Eq. (6.3). Equating these two expressions will yield both the backward generator and the forward Fokker–Planck operator.

Apply the multidimensional Itô formula (Chapter 4) to the scalar process $f(\mathbf{x}_t)$:

$$\mathrm{d}f(\mathbf{x}_t) = \sum_{i=1}^{N} \frac{\partial f}{\partial x_i}(\mathbf{x}_t)\ \mathrm{d}x_{t,i} + \frac{1}{2}\sum_{i,j=1}^{N} \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_t)\ \mathrm{d}x_{t,i}\ \mathrm{d}x_{t,j}. \tag{6.4}$$

Substituting the SDE $\mathrm{d}x_{t,i} = a_i(\mathbf{x}_t)\ \mathrm{d}t + \sum_{\alpha=1}^{M} B_{i\alpha}(\mathbf{x}_t)\ \mathrm{d}W_\alpha(t)$ and using the Itô multiplication rules $\mathrm{d}t^2 = 0$, $\mathrm{d}t\ \mathrm{d}W_\alpha = 0$, and $\mathrm{d}W_\alpha\ \mathrm{d}W_\beta = \delta_{\alpha\beta}\ \mathrm{d}t$, the quadratic variation product evaluates to

$$\mathrm{d}x_{t,i}\ \mathrm{d}x_{t,j} = \sum_{\alpha,\beta=1}^{M} B_{i\alpha}(\mathbf{x}_t)B_{j\beta}(\mathbf{x}_t)\ \mathrm{d}W_\alpha(t)\ \mathrm{d}W_\beta(t)$$

$$= \left(\sum_{\alpha=1}^{M} B_{i\alpha}(\mathbf{x}_t) B_{j\alpha}(\mathbf{x}_t) \right) \mathrm{d}t = 2D_{ij}(\mathbf{x}_t)\ \mathrm{d}t, \tag{6.5}$$

where the factor of $1/2$ in the definition Eq. (6.2) of D_{ij} cancels the Taylor factor of $1/2$ in Eq. (6.4). Inserting Eq. (6.5) into Eq. (6.4), we obtain the stochastic differential of $f(\mathbf{x}_t)$:

$$\mathrm{d}f(\mathbf{x}_t) = \left[\sum_{i=1}^{N} a_i(\mathbf{x}_t) \frac{\partial f}{\partial x_i}(\mathbf{x}_t) + \sum_{i,j=1}^{N} D_{ij}(\mathbf{x}_t) \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_t) \right] \mathrm{d}t + \sum_{i=1}^{N} \sum_{\alpha=1}^{M} \frac{\partial f}{\partial x_i}(\mathbf{x}_t) B_{i\alpha}(\mathbf{x}_t)\ \mathrm{d}W_\alpha(t). \tag{6.6}$$

The operator in the square bracket of Eq. (6.6) governs the evolution of observables, and we give it a name.

Definition 6.1 (Backward Kolmogorov Generator)**.** Let $\mathbf{x}_t \in \mathbb{R}^N$ be the Itô diffusion Eq. (6.1) with drift $\mathbf{a}(\mathbf{x})$ and diffusion tensor $\mathbf{D}(\mathbf{x}) = \frac{1}{2}\mathbf{B}(\mathbf{x})\mathbf{B}(\mathbf{x})^\top$. The *backward Kolmogorov generator* (or *infinitesimal generator*) $\mathcal{L}$ acting on twice continuously differentiable observables $f \in C^2(\mathbb{R}^N)$ is the second-order linear differential operator

$$(\mathcal{L}f)(\mathbf{x}) := \sum_{i=1}^{N} a_i(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial x_i} + \sum_{i,j=1}^{N} D_{ij}(\mathbf{x}) \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \mathbf{a}(\mathbf{x}) \cdot \nabla f(\mathbf{x}) + \mathrm{tr}\Big(\mathbf{D}(\mathbf{x}) \nabla^2 f(\mathbf{x}) \Big). \tag{6.7}$$

When we integrate Eq. (6.6) from t to $t + \Delta t$ and take the expectation $\mathbb{E}[\cdot]$, the stochastic integral $\int_t^{t+\Delta t} \nabla f(\mathbf{x}_s)^\top \mathbf{B}(\mathbf{x}_s)\ \mathrm{d}\mathbf{W}_s$ is a zero-mean martingale (since $f \in C_c^\infty(\mathbb{R}^N)$ has bounded gradient, so the integrand is square-integrable by Chapter 4). Therefore, its expectation vanishes identically, leaving

$$\mathbb{E}[f(\mathbf{x}_{t+\Delta t})] - \mathbb{E}[f(\mathbf{x}_t)] = \int_t^{t+\Delta t} \mathrm{d}s\, \mathbb{E}[(\mathcal{L}f)(\mathbf{x}_s)]\,.$$

Dividing by Δt and taking the limit $\Delta t \to 0$ yields the instantaneous rate of change of any expected observable:

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}[f(\mathbf{x}_t)] = \mathbb{E}[(\mathcal{L}f)(\mathbf{x}_t)] = \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}, t)\, (\mathcal{L}f)(\mathbf{x}). \tag{6.8}$$

If we condition on a deterministic starting point $\mathbf{x}_0 = \mathbf{x}$ and define $u(\mathbf{x}, t) := \mathbb{E}[f(\mathbf{x}_t) \mid \mathbf{x}_0 = \mathbf{x}]$, then by the Markov property Eq. (6.8) shows that $u(\mathbf{x}, t)$ solves the *backward Kolmogorov PDE*:

$$\frac{\partial u(\mathbf{x}, t)}{\partial t} = \mathcal{L}u(\mathbf{x}, t), \qquad u(\mathbf{x}, 0) = f(\mathbf{x}). \tag{6.9}$$

In Eq. (6.9), the operator $\mathcal{L}$ differentiates with respect to the *initial* state $\mathbf{x} = \mathbf{x}_0$. We use this backward equation in Chapter 7 to compute mean first-passage times out of metastable wells.

6.1.2 Adjoint Duality and the Forward Fokker–Planck Equation

To obtain the PDE governing the probability density $\rho(\mathbf{x}, t)$ itself, we now equate Eq. (6.8) with the time derivative of Eq. (6.3) and integrate by parts to move all spatial derivatives off the test observable $f(\mathbf{x})$ and onto the density $\rho(\mathbf{x}, t)$.

Theorem 6.2 (The Forward Fokker–Planck / Kolmogorov Equation)**.** *Let $\mathbf{x}_t \in \mathbb{R}^N$ obey the Itô SDE Eq. (6.1) with smooth coefficients $\mathbf{a}(\mathbf{x})$ and $\mathbf{D}(\mathbf{x}) = \frac{1}{2}\mathbf{B}(\mathbf{x})\mathbf{B}(\mathbf{x})^\top$, and suppose the law of $\mathbf{x}_t$ admits a smooth probability density $\rho(\mathbf{x}, t) \in C^{2,1}(\mathbb{R}^N \times (0, \infty))$. Then $\rho(\mathbf{x}, t)$ satisfies the* Fokker–Planck equation*:*

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} = (\mathcal{L}_{\mathrm{FP}}\rho)(\mathbf{x}, t) := -\sum_{i=1}^{N} \frac{\partial}{\partial x_i} \Big[a_i(\mathbf{x}) \rho(\mathbf{x}, t) \Big] + \sum_{i,j=1}^{N} \frac{\partial^2}{\partial x_i \partial x_j} \Big[D_{ij}(\mathbf{x}) \rho(\mathbf{x}, t) \Big], \tag{6.10}$$

where $\mathcal{L}_{\mathrm{FP}} = \mathcal{L}^\dagger$ *is the formal* $L^2(\mathbb{R}^N)$*-adjoint of the backward generator* $\mathcal{L}$*. Equivalently, Eq. (6.10) takes the local probability conservation (continuity) form*

$$\frac{\partial \rho(\mathbf{x},t)}{\partial t} + \nabla \cdot \mathbf{J}(\mathbf{x},t) = 0, \tag{6.11}$$

where the probability current vector field $\mathbf{J}(\mathbf{x},t) \in \mathbb{R}^N$ *has components*

$$J_i(\mathbf{x},t) := a_i(\mathbf{x})\rho(\mathbf{x},t) - \sum_{j=1}^{N} \frac{\partial}{\partial x_j}\Big[D_{ij}(\mathbf{x})\rho(\mathbf{x},t)\Big], \qquad i = 1,\dots,N. \tag{6.12}$$

Proof. Let $f \in C_c^\infty(\mathbb{R}^N)$ be an arbitrary smooth test function with compact support contained inside a ball $B_R = \Big\{\mathbf{x} \in \mathbb{R}^N : \|\mathbf{x}\| < R\Big\}$ of sufficiently large radius $R > 0$. Differentiating Eq. (6.3) with respect to t and equating the result to Eq. (6.8) gives

$$\int_{\mathbb{R}^N} \mathrm{d}^N x\, \frac{\partial \rho(\mathbf{x},t)}{\partial t}\, f(\mathbf{x}) = \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t) \left[\sum_{i=1}^{N} a_i(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial x_i} + \sum_{i,j=1}^{N} D_{ij}(\mathbf{x}) \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}\right]. \tag{6.13}$$

We now integrate each term on the right-hand side by parts over the ball B_R. For the first-order drift term, applying the product rule $\rho a_i \partial_{x_i} f = \partial_{x_i}(\rho a_i f) - f \partial_{x_i}(a_i \rho)$ and the divergence theorem yields

$$\int_{B_R} \mathrm{d}^N x\, \rho(\mathbf{x},t) a_i(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial x_i} = \oint_{\partial B_R} \mathrm{d}S\, \hat{n}_i\, \rho(\mathbf{x},t) a_i(\mathbf{x}) f(\mathbf{x}) - \int_{B_R} \mathrm{d}^N x\, f(\mathbf{x}) \frac{\partial}{\partial x_i}\Big[a_i(\mathbf{x})\rho(\mathbf{x},t)\Big]. \tag{6.14}$$

Because $\mathrm{supp}(f) \subset B_R$, $f(\mathbf{x}) = 0$ identically on the boundary sphere ∂B_R, so the surface integral in Eq. (6.14) vanishes identically.

For the second-order diffusion term, we perform two successive integrations by parts. In the first integration by parts (with respect to x_i), we have

$$\int_{B_R} \mathrm{d}^N x\, \rho(\mathbf{x},t) D_{ij}(\mathbf{x}) \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \oint_{\partial B_R} \mathrm{d}S\, \hat{n}_i\, \rho(\mathbf{x},t) D_{ij}(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial x_j} - \int_{B_R} \mathrm{d}^N x\, \frac{\partial f(\mathbf{x})}{\partial x_j} \frac{\partial}{\partial x_i}\Big[D_{ij}(\mathbf{x})\rho(\mathbf{x},t)\Big]. \tag{6.15}$$

Again, since f vanishes in a neighborhood of ∂B_R, its gradient $\partial_{x_j} f$ also vanishes on ∂B_R, eliminating the boundary term in Eq. (6.15). Performing a second integration by parts (now with respect to x_j) on the remaining volume integral in Eq. (6.15) gives

$$\begin{aligned} -\int_{B_R} \mathrm{d}^N x\, \frac{\partial f(\mathbf{x})}{\partial x_j} \frac{\partial}{\partial x_i}\Big[D_{ij}(\mathbf{x})\rho(\mathbf{x},t)\Big] &= -\oint_{\partial B_R} \mathrm{d}S\, \hat{n}_j\, f(\mathbf{x}) \frac{\partial}{\partial x_i}\Big[D_{ij}(\mathbf{x})\rho(\mathbf{x},t)\Big] \\ &\quad + \int_{B_R} \mathrm{d}^N x\, f(\mathbf{x}) \frac{\partial^2}{\partial x_i \partial x_j}\Big[D_{ij}(\mathbf{x})\rho(\mathbf{x},t)\Big]. \end{aligned} \tag{6.16}$$

Once more, the boundary integral vanishes because $f\big|_{\partial B_R} = 0$. Summing over $i,j = 1,\dots,N$ and substituting Eqs. (6.14) and (6.16) into Eq. (6.13), we obtain

$$\int_{\mathbb{R}^N} \mathrm{d}^N x\, f(\mathbf{x}) \left\{ \frac{\partial \rho(\mathbf{x},t)}{\partial t} + \sum_{i=1}^{N} \frac{\partial}{\partial x_i}\Big[a_i(\mathbf{x})\rho(\mathbf{x},t)\Big] - \sum_{i,j=1}^{N} \frac{\partial^2}{\partial x_i \partial x_j}\Big[D_{ij}(\mathbf{x})\rho(\mathbf{x},t)\Big] \right\} = 0. \tag{6.17}$$

Since Eq. (6.17) holds for every smooth compactly supported test function $f \in C_c^\infty(\mathbb{R}^N)$ and the quantity in curly braces is continuous, the Fundamental Lemma of the Calculus of Variations implies that the quantity in braces vanishes pointwise for all $\mathbf{x} \in \mathbb{R}^N$, establishing Eq. (6.10). Finally, factoring $-\sum_{i=1}^{N} \frac{\partial}{\partial x_i}$ out of the right-hand side of Eq. (6.10) immediately yields the continuity equation Eq. (6.11) with the current $J_i(\mathbf{x},t)$ of Eq. (6.12). □

Notation Watch: Diffusion Tensor Factor of 1/2 and Adjoint Conventions

Three distinct conventions for the diffusion coefficient and generator coexist across physics, probability theory, and machine learning:

1. **Statistical Physics Convention (Adopted in this book, following Risken [90] and Pavliotis [85]):** For the SDE $\mathrm{d}\mathbf{x}_t = \mathbf{a}\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t$, we define $\mathbf{D} := \frac{1}{2}\mathbf{B}\mathbf{B}^\top$. The backward generator is $\mathcal{L} = \mathbf{a}\cdot\nabla + \mathbf{D} : \nabla^2$ (with *no* factor of 1/2 in front of $\mathbf{D}$), and for isotropic thermal noise $\mathbf{B} = \sqrt{2D}\,\mathbf{I}$, the constant $D = \mu k_{\mathrm{B}}T$ is the physical Einstein diffusion constant.

2. **Stochastic Analysis Convention (Karatzas and Shreve [59], Øksendal [80]):** Mathematicians often define the covariance matrix $\mathbf{\Sigma}_{\mathrm{diff}} := \mathbf{B}\mathbf{B}^\top = 2\mathbf{D}$ and write $\mathcal{L} = \mathbf{a}\cdot\nabla + \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_{\mathrm{diff}}\nabla^2)$.

3. **Score-Based Diffusion Convention (Song et al. [103], used in Chapter 15):** Forward noising SDEs are written $\mathrm{d}\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{W}_t$, which corresponds to $\mathbf{B} = g(t)\mathbf{I}$ and therefore $\mathbf{D}(t) = \frac{1}{2}g(t)^2\mathbf{I}$.

Notice also the notation for the operators: in probability theory (Bakry et al. [7]), $\mathcal{L}$ denotes the backward generator acting on observables, whereas in physics (Risken [90]), the Fokker–Planck operator acting on densities is written $\mathcal{L}_{\mathrm{FP}}$. We keep both symbols, related by the $L^2(\mathbb{R}^N)$ adjoint relation $\mathcal{L}_{\mathrm{FP}} = \mathcal{L}^\dagger$.

6.1.3 Fickian Form of the Probability Current and Overdamped Smoluchowski Specialization

In the probability current expression Eq. (6.12), the derivative ∂_{x_j} acts on the product $D_{ij}(\mathbf{x})\rho(\mathbf{x},t)$. Applying the product rule $\partial_{x_j}[D_{ij}(\mathbf{x})\rho(\mathbf{x},t)] = (\partial_{x_j}D_{ij}(\mathbf{x}))\rho(\mathbf{x},t) + D_{ij}(\mathbf{x})\partial_{x_j}\rho(\mathbf{x},t)$ allows us to separate the probability current into a pure convective transport term proportional to ρ and a pure *Fickian diffusion* term proportional to the density gradient $\nabla\rho$:

$$J_i(\mathbf{x},t) = \left[a_i(\mathbf{x}) - \sum_{j=1}^{N}\frac{\partial D_{ij}(\mathbf{x})}{\partial x_j}\right]\rho(\mathbf{x},t) - \sum_{j=1}^{N} D_{ij}(\mathbf{x})\frac{\partial\rho(\mathbf{x},t)}{\partial x_j}, \tag{6.18}$$

or in compact vector-tensor notation,

$$\mathbf{J}(\mathbf{x},t) = \Big[\mathbf{a}(\mathbf{x}) - \nabla\cdot\mathbf{D}(\mathbf{x})\Big]\rho(\mathbf{x},t) - \mathbf{D}(\mathbf{x})\nabla\rho(\mathbf{x},t), \tag{6.19}$$

where $(\nabla\cdot\mathbf{D})_i := \sum_{j=1}^{N}\partial_{x_j}D_{ij}(\mathbf{x})$ is the row-divergence of the diffusion tensor.

The bracketed effective drift $\tilde{\mathbf{a}}(\mathbf{x}) := \mathbf{a}(\mathbf{x}) - \nabla\cdot\mathbf{D}(\mathbf{x})$ in Eq. (6.19) contains the divergence term $-\nabla\cdot\mathbf{D}(\mathbf{x})$ that we met in Section 5.5. For an overdamped system with state-dependent mobility tensor $\boldsymbol{\mu}(\mathbf{x}) \succ \mathbf{0}$ in a potential $E(\mathbf{x})$ at temperature T, the Einstein–Smoluchowski relation sets $\mathbf{D}(\mathbf{x}) = k_{\mathrm{B}}T\boldsymbol{\mu}(\mathbf{x})$, and the thermal divergence drift derived in Section 5.5 sets the Itô drift to

$$\mathbf{a}(\mathbf{x}) = -\boldsymbol{\mu}(\mathbf{x})\nabla E(\mathbf{x}) + k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}(\mathbf{x}) = -\beta\mathbf{D}(\mathbf{x})\nabla E(\mathbf{x}) + \nabla\cdot\mathbf{D}(\mathbf{x}). \tag{6.20}$$

When we substitute Eq. (6.20) into the general probability current Eq. (6.19), the divergence drift $+\nabla\cdot\mathbf{D}(\mathbf{x})$ in $\mathbf{a}(\mathbf{x})$ *cancels* the $-\nabla\cdot\mathbf{D}(\mathbf{x})$ term coming from the spatial derivative of the diffusion

tensor. We are left with the *Smoluchowski probability current*:

$$\begin{aligned}\mathbf{J}(\mathbf{x},t) &= -\boldsymbol{\mu}(\mathbf{x})\Big[\rho(\mathbf{x},t)\nabla E(\mathbf{x}) + k_{\mathrm{B}}T\nabla\rho(\mathbf{x},t)\Big] \\ &= -\mathbf{D}(\mathbf{x})\Big[\beta\rho(\mathbf{x},t)\nabla E(\mathbf{x}) + \nabla\rho(\mathbf{x},t)\Big]. \end{aligned} \tag{6.21}$$

Two equivalent forms of Eq. (6.21) are used repeatedly in the rest of the book. Using $\nabla\left(e^{\beta E(\mathbf{x})}\rho(\mathbf{x},t)\right) = e^{\beta E(\mathbf{x})}\left[\beta\rho\nabla E + \nabla\rho\right]$ and the Boltzmann–Gibbs equilibrium density $p_{\mathrm{eq}}(\mathbf{x}) = Z^{-1}e^{-\beta E(\mathbf{x})}$ from Chapter 1, we have

$$\mathbf{J}(\mathbf{x},t) = -\mathbf{D}(\mathbf{x})\,p_{\mathrm{eq}}(\mathbf{x})\,\nabla\left(\frac{\rho(\mathbf{x},t)}{p_{\mathrm{eq}}(\mathbf{x})}\right) = -\rho(\mathbf{x},t)\,\mathbf{D}(\mathbf{x})\,\nabla\ln\left(\frac{\rho(\mathbf{x},t)}{p_{\mathrm{eq}}(\mathbf{x})}\right). \tag{6.22}$$

Accordingly, the overdamped Fokker–Planck (Smoluchowski) equation takes the divergence-form structure

$$\frac{\partial\rho(\mathbf{x},t)}{\partial t} = \mathcal{L}_{\mathrm{FP}}\rho(\mathbf{x},t) = \nabla\cdot\left[\mathbf{D}(\mathbf{x})\,p_{\mathrm{eq}}(\mathbf{x})\,\nabla\left(\frac{\rho(\mathbf{x},t)}{p_{\mathrm{eq}}(\mathbf{x})}\right)\right] = \nabla\cdot\left[\rho(\mathbf{x},t)\,\mathbf{D}(\mathbf{x})\,\nabla\ln\left(\frac{\rho(\mathbf{x},t)}{p_{\mathrm{eq}}(\mathbf{x})}\right)\right]. \tag{6.23}$$

Equation (6.22) shows why the thermal divergence drift $+k_{\mathrm{B}}T\nabla\cdot\boldsymbol{\mu}(\mathbf{x})$ in Eq. (6.20) is needed: it ensures that when $\rho(\mathbf{x},t) = p_{\mathrm{eq}}(\mathbf{x}) \propto e^{-\beta E(\mathbf{x})}$, the ratio $\rho/p_{\mathrm{eq}} = 1$ has zero gradient everywhere, so $\mathbf{J}(\mathbf{x},t) = \mathbf{0}$ and $\partial_t p_{\mathrm{eq}} = 0$ regardless of how strongly the mobility $\boldsymbol{\mu}(\mathbf{x})$ varies with voltage or position.

6.2 Detailed Balance, Microscopic Reversibility, and Non-Equilibrium Steady States

When a time-homogeneous stochastic circuit is left to evolve under constant bias voltages and constant bath temperatures, its probability density $\rho(\mathbf{x},t)$ relaxes toward a time-independent *stationary distribution* $p_{\mathrm{ss}}(\mathbf{x})$ satisfying $\partial_t p_{\mathrm{ss}}(\mathbf{x}) = 0$. By the continuity equation Eq. (6.11), stationarity requires that the steady-state probability current $\mathbf{J}_{\mathrm{ss}}(\mathbf{x})$ be *divergence-free* (solenoidal) everywhere in state space:

$$\mathcal{L}_{\mathrm{FP}}p_{\mathrm{ss}}(\mathbf{x}) = -\nabla\cdot\mathbf{J}_{\mathrm{ss}}(\mathbf{x}) = 0, \qquad \text{where} \qquad \mathbf{J}_{\mathrm{ss}}(\mathbf{x}) := \tilde{\mathbf{a}}(\mathbf{x})p_{\mathrm{ss}}(\mathbf{x}) - \mathbf{D}(\mathbf{x})\nabla p_{\mathrm{ss}}(\mathbf{x}), \tag{6.24}$$

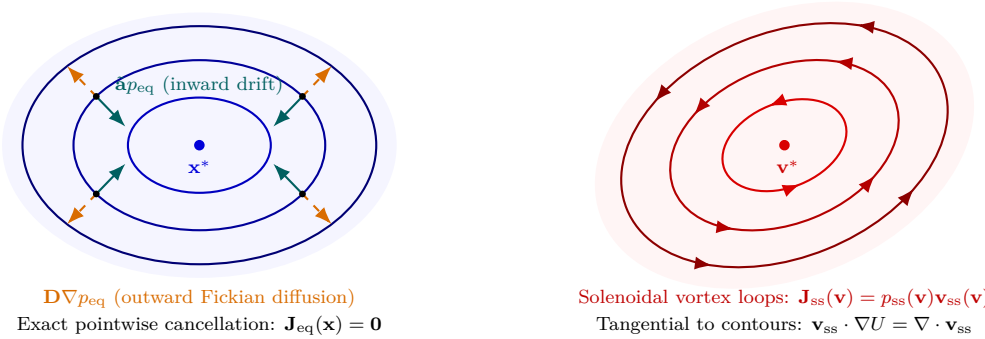
and where we have abbreviated the Fickian convective drift by

$$\tilde{\mathbf{a}}(\mathbf{x}) := \mathbf{a}(\mathbf{x}) - \nabla\cdot\mathbf{D}(\mathbf{x}). \tag{6.25}$$

The vanishing of the divergence $\nabla\cdot\mathbf{J}_{\mathrm{ss}}(\mathbf{x}) = 0$ does *not* imply that the vector field $\mathbf{J}_{\mathrm{ss}}(\mathbf{x})$ itself vanishes. Just as a steady incompressible fluid flow ($\nabla\cdot\mathbf{v} = 0$) or a static magnetic field ($\nabla\cdot\mathbf{B} = 0$) can sustain persistent closed vortex loops, a stationary stochastic process falls into one of two physically distinct classes depending on whether $\mathbf{J}_{\mathrm{ss}}(\mathbf{x})$ vanishes identically or circulates in closed loops.

Definition 6.3 (Equilibrium Detailed Balance vs. Non-Equilibrium Steady State)**.** Let $p_{\mathrm{ss}}(\mathbf{x}) > 0$ be a smooth stationary probability density of the Fokker–Planck equation $\mathcal{L}_{\mathrm{FP}}p_{\mathrm{ss}} = -\nabla\cdot\mathbf{J}_{\mathrm{ss}} = 0$ on $\mathbb{R}^N$.

(a) Equilibrium Detailed Balance ($\mathbf{J}_{\text{eq}} \equiv \mathbf{0}$) (b) Non-Equilibrium Steady State ($\nabla \cdot \mathbf{J}_{\text{ss}} = 0$, $\mathbf{J}_{\text{ss}} \neq \mathbf{0}$)



$\mathbf{D}\nabla p_{\text{eq}}$ (outward Fickian diffusion)
Exact pointwise cancellation: $\mathbf{J}_{\text{eq}}(\mathbf{x}) = \mathbf{0}$

Solenoidal vortex loops: $\mathbf{J}_{\text{ss}}(\mathbf{v}) = p_{\text{ss}}(\mathbf{v})\mathbf{v}_{\text{ss}}(\mathbf{v})$
Tangential to contours: $\mathbf{v}_{\text{ss}} \cdot \nabla U = \nabla \cdot \mathbf{v}_{\text{ss}}$

Figure 6.1: (a) An equilibrium steady state obeying detailed balance ($\mathbf{J}_{\text{eq}}(\mathbf{x}) \equiv \mathbf{0}$): the inward drift $\tilde{\mathbf{a}}(\mathbf{x})p_{\text{eq}}(\mathbf{x})$ and the outward Fickian diffusion $\mathbf{D}(\mathbf{x})\nabla p_{\text{eq}}(\mathbf{x})$ cancel at every state $\mathbf{x}$. (b) A non-equilibrium steady state, as in non-reciprocally coupled subthreshold CMOS circuits (Section 6.2.1): the stationary density $p_{\text{ss}}(\mathbf{v})$ does not change in time ($\nabla \cdot \mathbf{J}_{\text{ss}} = 0$), but a non-zero probability current $\mathbf{J}_{\text{ss}}(\mathbf{v}) \neq \mathbf{0}$ circulates around closed contours.

1. **Equilibrium Steady State (Microscopic Reversibility / Detailed Balance):** The system is in *thermodynamic equilibrium* (and satisfies continuous-state *detailed balance*) if the stationary probability current vanishes pointwise throughout state space:
$$\mathbf{J}_{\text{eq}}(\mathbf{x}) = \tilde{\mathbf{a}}(\mathbf{x})p_{\text{eq}}(\mathbf{x}) - \mathbf{D}(\mathbf{x})\nabla p_{\text{eq}}(\mathbf{x}) = \mathbf{0} \qquad \forall\, \mathbf{x} \in \mathbb{R}^N. \tag{6.26}$$
In this case we denote the stationary density by $p_{\text{eq}}(\mathbf{x})$.

2. **Non-Equilibrium Steady State (NESS):** The system is in a *non-equilibrium steady state* if $\nabla \cdot \mathbf{J}_{\text{ss}}(\mathbf{x}) = 0$ for all $\mathbf{x} \in \mathbb{R}^N$, but the steady-state probability current is non-zero:
$$\mathbf{J}_{\text{ss}}(\mathbf{x}) \neq \mathbf{0} \qquad \text{on an open subset of } \mathbb{R}^N. \tag{6.27}$$

Two questions follow. Which condition on the drift $\mathbf{a}(\mathbf{x})$ and the diffusion tensor $\mathbf{D}(\mathbf{x})$ decides whether a circuit settles into equilibrium (Fig. 6.1a) or into a circulating NESS (Fig. 6.1b)? And how does $\mathbf{J}_{\text{eq}}(\mathbf{x}) = \mathbf{0}$ relate to the statistical-mechanical definition of detailed balance, namely that a transition $\mathbf{x} \to \mathbf{x}'$ over time t is as probable in equilibrium as the reverse transition $\mathbf{x}' \to \mathbf{x}$? The following theorem answers both.

Theorem 6.4 (The Potential Condition and Equivalence of Detailed Balance Criteria)**.** *Let $\mathbf{x}_t \in \mathbb{R}^N$ be an autonomous diffusion with smooth drift $\mathbf{a}(\mathbf{x})$ and strictly positive-definite smooth diffusion tensor $\mathbf{D}(\mathbf{x}) \succ \mathbf{0}$, and let $p_{\text{eq}}(\mathbf{x}) > 0$ be a smooth normalizable probability density on $\mathbb{R}^N$. The following four statements are equivalent:*

(i) ***Zero Probability Current:*** *$p_{\text{eq}}(\mathbf{x})$ is a stationary density with vanishing probability current everywhere, $\mathbf{J}_{\text{eq}}(\mathbf{x}) \equiv \mathbf{0}$ on $\mathbb{R}^N$.*

(ii) ***Potential (Curl-Free Integrability) Condition:*** *The Fickian drift $\tilde{\mathbf{a}}(\mathbf{x}) = \mathbf{a}(\mathbf{x}) - \nabla \cdot \mathbf{D}(\mathbf{x})$ satisfies*
$$\mathbf{D}(\mathbf{x})^{-1}\tilde{\mathbf{a}}(\mathbf{x}) = -\nabla\Phi(\mathbf{x}) \tag{6.28}$$
for a smooth scalar potential $\Phi(\mathbf{x})$ with $\int_{\mathbb{R}^N} e^{-\Phi(\mathbf{x})}\, \mathrm{d}^N x < \infty$ (equivalently, the vector field $\mathbf{F}(\mathbf{x}) := \mathbf{D}(\mathbf{x})^{-1}\tilde{\mathbf{a}}(\mathbf{x})$ is curl-free on $\mathbb{R}^N$: $\frac{\partial F_j}{\partial x_i} = \frac{\partial F_i}{\partial x_j}$ for all $i, j = 1, \ldots, N$), and $p_{\text{eq}}(\mathbf{x}) = Z^{-1}e^{-\Phi(\mathbf{x})}$.

(iii) ***Self-Adjointness of $\mathcal{L}$ in $L^2(p_{\mathrm{eq}})$:*** *For all smooth compactly supported observables $f, g \in C_c^\infty(\mathbb{R}^N)$, the backward generator $\mathcal{L}$ is symmetric with respect to the p_{eq}-weighted inner product $\langle f,\, g\rangle_{p_{\mathrm{eq}}} := \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x}) f(\mathbf{x}) g(\mathbf{x})$:*

$$\langle f,\, \mathcal{L}g\rangle_{p_{\mathrm{eq}}} = \langle \mathcal{L}f,\, g\rangle_{p_{\mathrm{eq}}} = -\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \nabla f(\mathbf{x})^\top \mathbf{D}(\mathbf{x})\, \nabla g(\mathbf{x}). \tag{6.29}$$

(iv) ***Two-Time Microscopic Reversibility (Detailed Balance):*** *For every $t \geq 0$ and all $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^N$, the transition probability density $p(\mathbf{x}', t \mid \mathbf{x}, 0)$ satisfies*

$$p(\mathbf{x}', t \mid \mathbf{x}, 0)\, p_{\mathrm{eq}}(\mathbf{x}) = p(\mathbf{x}, t \mid \mathbf{x}', 0)\, p_{\mathrm{eq}}(\mathbf{x}'). \tag{6.30}$$

Proof. We prove (i) $\iff$ (ii), (i) $\iff$ (iii), and (iii) $\iff$ (iv).

Step 1: (i) $\iff$ (ii). Suppose $\mathbf{J}_{\mathrm{eq}}(\mathbf{x}) = \tilde{\mathbf{a}}(\mathbf{x}) p_{\mathrm{eq}}(\mathbf{x}) - \mathbf{D}(\mathbf{x}) \nabla p_{\mathrm{eq}}(\mathbf{x}) = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^N$. Since $p_{\mathrm{eq}}(\mathbf{x}) > 0$ and $\mathbf{D}(\mathbf{x}) \succ \mathbf{0}$ is invertible everywhere, we can divide by $p_{\mathrm{eq}}(\mathbf{x})$ and multiply on the left by $\mathbf{D}(\mathbf{x})^{-1}$ to obtain

$$\mathbf{D}(\mathbf{x})^{-1} \tilde{\mathbf{a}}(\mathbf{x}) = \frac{\nabla p_{\mathrm{eq}}(\mathbf{x})}{p_{\mathrm{eq}}(\mathbf{x})} = \nabla \ln p_{\mathrm{eq}}(\mathbf{x}). \tag{6.31}$$

Defining the dimensionless potential $\Phi(\mathbf{x}) := -\ln\left(Z p_{\mathrm{eq}}(\mathbf{x})\right)$ gives $\mathbf{D}(\mathbf{x})^{-1}\tilde{\mathbf{a}}(\mathbf{x}) = -\nabla\Phi(\mathbf{x})$, which is Eq. (6.28). By Clairaut's theorem on the symmetry of mixed partial derivatives, any gradient field $\mathbf{F} = -\nabla\Phi$ satisfies $\partial_{x_i} F_j = -\partial_{x_i}\partial_{x_j}\Phi = \partial_{x_j} F_i$. Conversely, since $\mathbb{R}^N$ is simply connected, Poincaré's lemma guarantees that if $\partial_{x_i}(\mathbf{D}^{-1}\tilde{\mathbf{a}})_j = \partial_{x_j}(\mathbf{D}^{-1}\tilde{\mathbf{a}})_i$ for all i, j, there exists a smooth scalar function $\Phi(\mathbf{x})$ (given explicitly by the line integral $\Phi(\mathbf{x}) = -\int_{\mathbf{0}}^{\mathbf{x}} \mathbf{D}(\mathbf{y})^{-1}\tilde{\mathbf{a}}(\mathbf{y})\cdot \mathrm{d}\mathbf{y}$) satisfying Eq. (6.28). Setting $p_{\mathrm{eq}}(\mathbf{x}) = Z^{-1} e^{-\Phi(\mathbf{x})}$ then yields $\nabla p_{\mathrm{eq}} = -p_{\mathrm{eq}}\nabla\Phi = p_{\mathrm{eq}}\mathbf{D}^{-1}\tilde{\mathbf{a}}$, so $\mathbf{J}_{\mathrm{eq}}(\mathbf{x}) = \tilde{\mathbf{a}} p_{\mathrm{eq}} - \mathbf{D}\nabla p_{\mathrm{eq}} = \mathbf{0}$, proving (ii) $\implies$ (i).

Step 2: (i) $\iff$ (iii). First write the backward generator $\mathcal{L}g$ in divergence form. Using the product rule $\sum_i \partial_{x_i}(D_{ij}\partial_{x_j} g) = \sum_i (\partial_{x_i} D_{ij})\partial_{x_j} g + \sum_i D_{ij}\partial_{x_i}\partial_{x_j} g$ and the symmetry $D_{ij} = D_{ji}$, we can rewrite Eq. (6.7) in terms of $\tilde{\mathbf{a}} = \mathbf{a} - \nabla\cdot\mathbf{D}$ as

$$(\mathcal{L}g)(\mathbf{x}) = \tilde{\mathbf{a}}(\mathbf{x})\cdot\nabla g(\mathbf{x}) + \nabla\cdot\Big(\mathbf{D}(\mathbf{x})\nabla g(\mathbf{x})\Big). \tag{6.32}$$

Now multiply Eq. (6.32) by $p_{\mathrm{eq}}(\mathbf{x}) f(\mathbf{x})$ and integrate over $\mathbb{R}^N$. Integrating the second term by parts (where boundary terms vanish because $f \in C_c^\infty(\mathbb{R}^N)$) gives

$$\begin{aligned}
\langle f,\, \mathcal{L}g\rangle_{p_{\mathrm{eq}}} &= \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x}) f(\mathbf{x})\, \tilde{\mathbf{a}}(\mathbf{x})\cdot\nabla g(\mathbf{x}) - \int_{\mathbb{R}^N} \mathrm{d}^N x\, \nabla\Big(p_{\mathrm{eq}}(\mathbf{x}) f(\mathbf{x})\Big)\cdot \mathbf{D}(\mathbf{x})\nabla g(\mathbf{x}) \\
&= \int_{\mathbb{R}^N} \mathrm{d}^N x\, \Big[p_{\mathrm{eq}}(\mathbf{x})\tilde{\mathbf{a}}(\mathbf{x}) - \mathbf{D}(\mathbf{x})\nabla p_{\mathrm{eq}}(\mathbf{x})\Big] f(\mathbf{x})\cdot\nabla g(\mathbf{x}) \\
&\quad - \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \nabla f(\mathbf{x})^\top \mathbf{D}(\mathbf{x})\nabla g(\mathbf{x}) \\
&= \int_{\mathbb{R}^N} \mathrm{d}^N x\, f(\mathbf{x})\, \mathbf{J}_{\mathrm{eq}}(\mathbf{x})\cdot\nabla g(\mathbf{x}) - \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \nabla f(\mathbf{x})^\top \mathbf{D}(\mathbf{x})\nabla g(\mathbf{x}).
\end{aligned} \tag{6.33}$$

In Eq. (6.33), the second integral on the right-hand side is manifestly symmetric in f and g because $\mathbf{D}(\mathbf{x}) = \mathbf{D}(\mathbf{x})^\top$. If (i) holds ($\mathbf{J}_{\mathrm{eq}} \equiv \mathbf{0}$), the first integral drops out and we obtain Eq. (6.29) immediately, proving (i) $\implies$ (iii). Conversely, subtracting the $f \leftrightarrow g$ transposed version of Eq. (6.33) from Eq. (6.33) itself yields the antisymmetric part of the generator:

$$\langle f,\, \mathcal{L}g\rangle_{p_{\mathrm{eq}}} - \langle \mathcal{L}f,\, g\rangle_{p_{\mathrm{eq}}} = \int_{\mathbb{R}^N} \mathrm{d}^N x\, \Big[f(\mathbf{x})\nabla g(\mathbf{x}) - g(\mathbf{x})\nabla f(\mathbf{x})\Big]\cdot \mathbf{J}_{\mathrm{eq}}(\mathbf{x}). \tag{6.34}$$

Integrating the second term $-\int g\nabla f\cdot\mathbf{J}_{\rm eq}\,\mathrm{d}^N x$ by parts gives $+\int f\nabla\cdot(g\mathbf{J}_{\rm eq})\,\mathrm{d}^N x=\int f\nabla g\cdot\mathbf{J}_{\rm eq}\,\mathrm{d}^N x+\int fg\nabla\cdot\mathbf{J}_{\rm eq}\,\mathrm{d}^N x$. If (iii) holds for all f,g, choosing $g\equiv 1$ on $\mathrm{supp}(f)$ gives $\int f\nabla\cdot\mathbf{J}_{\rm eq}\,\mathrm{d}^N x=0\implies\nabla\cdot\mathbf{J}_{\rm eq}=0$, and then Eq. (6.34) becomes $2\int f(\mathbf{x})\nabla g(\mathbf{x})\cdot\mathbf{J}_{\rm eq}(\mathbf{x})\,\mathrm{d}^N x=0$ for all $f,g\in C_c^\infty(\mathbb{R}^N)$, which forces $\mathbf{J}_{\rm eq}(\mathbf{x})\equiv\mathbf{0}$ everywhere, proving (iii) $\implies$ (i).

Step 3: (iii) $\iff$ (iv). The finite-time Markov propagator $P_t=e^{t\mathcal{L}}$ defined by $(P_tg)(\mathbf{x})=\int_{\mathbb{R}^N}\mathrm{d}^N x'\,p(\mathbf{x}',t\mid\mathbf{x},0)\,g(\mathbf{x}')$ has matrix elements in $L^2(p_{\rm eq})$ given by

$$\langle f,\,P_tg\rangle_{p_{\rm eq}}=\int_{\mathbb{R}^N}\mathrm{d}^N x\int_{\mathbb{R}^N}\mathrm{d}^N x'\,f(\mathbf{x})\,p_{\rm eq}(\mathbf{x})\,p(\mathbf{x}',t\mid\mathbf{x},0)\,g(\mathbf{x}'). \tag{6.35}$$

Meanwhile, $\langle P_tf,\,g\rangle_{p_{\rm eq}}=\int_{\mathbb{R}^N}\mathrm{d}^N x\int_{\mathbb{R}^N}\mathrm{d}^N x'\,f(\mathbf{x})\,p_{\rm eq}(\mathbf{x}')\,p(\mathbf{x},t\mid\mathbf{x}',0)\,g(\mathbf{x}')$. Thus P_t is self-adjoint in $L^2(p_{\rm eq})$ ($\langle f,\,P_tg\rangle_{p_{\rm eq}}=\langle P_tf,\,g\rangle_{p_{\rm eq}}$ for all f,g and all $t\geq 0$) if and only if the kernel identity Eq. (6.30) holds. Differentiating $\langle f,\,P_tg\rangle_{p_{\rm eq}}=\langle P_tf,\,g\rangle_{p_{\rm eq}}$ at $t=0$ gives $\langle f,\,\mathcal{L}g\rangle_{p_{\rm eq}}=\langle\mathcal{L}f,\,g\rangle_{p_{\rm eq}}$, and conversely self-adjointness of the generator $\mathcal{L}$ implies self-adjointness of the semigroup $P_t=e^{t\mathcal{L}}$ for all $t\geq 0$. □

6.2.1 Helmholtz–Hodge Decomposition of Non-Equilibrium Drift Fields

Now suppose the potential condition Eq. (6.28) fails ($\partial_i(\mathbf{D}^{-1}\tilde{\mathbf{a}})_j\neq\partial_j(\mathbf{D}^{-1}\tilde{\mathbf{a}})_i$), as it does in active circuits with non-reciprocal transistor couplings. No potential $E(\mathbf{x})$ then generates the drift, but any positive stationary distribution $p_{\rm ss}(\mathbf{x})>0$ defines an *effective generalized potential* (or stationary surprisal)

$$U(\mathbf{x}):=-\ln p_{\rm ss}(\mathbf{x})\qquad\Longleftrightarrow\qquad p_{\rm ss}(\mathbf{x})=e^{-U(\mathbf{x})}. \tag{6.36}$$

Using $U(\mathbf{x})$ and the steady-state probability current $\mathbf{J}_{\rm ss}(\mathbf{x})$, we can define the *stationary mean local velocity field*

$$\mathbf{v}_{\rm ss}(\mathbf{x}):=\frac{\mathbf{J}_{\rm ss}(\mathbf{x})}{p_{\rm ss}(\mathbf{x})}=\tilde{\mathbf{a}}(\mathbf{x})-\mathbf{D}(\mathbf{x})\nabla\ln p_{\rm ss}(\mathbf{x})=\tilde{\mathbf{a}}(\mathbf{x})+\mathbf{D}(\mathbf{x})\nabla U(\mathbf{x}). \tag{6.37}$$

Solving Eq. (6.37) for the Itô drift $\mathbf{a}(\mathbf{x})=\tilde{\mathbf{a}}(\mathbf{x})+\nabla\cdot\mathbf{D}(\mathbf{x})$ yields the canonical *Helmholtz–Hodge decomposition* of any stationary Fokker–Planck drift field into a reversible gradient part and an irreversible solenoidal part:

$$\mathbf{a}(\mathbf{x})=\underbrace{-\mathbf{D}(\mathbf{x})\nabla U(\mathbf{x})+\nabla\cdot\mathbf{D}(\mathbf{x})}_{\mathbf{a}_{\rm rev}(\mathbf{x})\ \text{(reversible)}}+\underbrace{\mathbf{v}_{\rm ss}(\mathbf{x})}_{\mathbf{a}_{\rm irr}(\mathbf{x})\ \text{(irreversible)}}, \tag{6.38}$$

where the stationarity equation $\nabla\cdot\mathbf{J}_{\rm ss}(\mathbf{x})=\nabla\cdot\big(e^{-U(\mathbf{x})}\mathbf{v}_{\rm ss}(\mathbf{x})\big)=0$ expands via the product rule to the orthogonality relation

$$\nabla\cdot\mathbf{v}_{\rm ss}(\mathbf{x})-\mathbf{v}_{\rm ss}(\mathbf{x})\cdot\nabla U(\mathbf{x})=0. \tag{6.39}$$

When the irreversible velocity field is itself incompressible ($\nabla\cdot\mathbf{v}_{\rm ss}=0$, which holds for every linear Ornstein–Uhlenbeck NESS), Eq. (6.39) gives $\mathbf{v}_{\rm ss}(\mathbf{x})\cdot\nabla U(\mathbf{x})=0$ pointwise. The steady-state velocity is then orthogonal to $\nabla U(\mathbf{x})$ and flows along the surfaces of constant probability $U(\mathbf{x})=\text{const}$, as depicted in Fig. 6.1(b). Under the decomposition Eq. (6.38), the backward generator splits into symmetric and skew-symmetric operators in $L^2(p_{\rm ss})$:

$$\mathcal{L}=\mathcal{L}_{\rm sym}+\mathcal{L}_{\rm skew},\qquad\text{where}\qquad\mathcal{L}_{\rm sym}f=\frac{1}{p_{\rm ss}}\nabla\cdot(p_{\rm ss}\mathbf{D}\nabla f),\quad\mathcal{L}_{\rm skew}f=\mathbf{v}_{\rm ss}\cdot\nabla f. \tag{6.40}$$

Indeed, for any $f,g\in C_c^\infty(\mathbb{R}^N)$, integrating $\langle f,\,\mathcal{L}_{\rm skew}g\rangle_{p_{\rm ss}}=\int\mathrm{d}^N x\,f(\mathbf{x})\,\mathbf{J}_{\rm ss}(\mathbf{x})\cdot\nabla g(\mathbf{x})$ by parts and using $\nabla\cdot\mathbf{J}_{\rm ss}=0$ gives $\langle f,\,\mathcal{L}_{\rm skew}g\rangle_{p_{\rm ss}}=-\langle\mathcal{L}_{\rm skew}f,\,g\rangle_{p_{\rm ss}}$. Because $\mathcal{L}_{\rm skew}$ is skew-Hermitian in

$L^2(p_{\mathrm{ss}})$, the eigenvalues of a NESS Fokker–Planck operator can acquire non-zero imaginary parts ($\lambda_n = \gamma_n \pm i\omega_n$), manifesting physically as damped oscillatory spiral trajectories and complex-conjugate poles in the circuit's correlation functions.

Example 6.5 (Circulating Currents and the Lyapunov Equation in the RC Sampler)**.** We apply this decomposition to the RC sampler in its general linear form, with possibly non-reciprocal couplings: the multidimensional Ornstein–Uhlenbeck circuit $\mathrm{d}\mathbf{v}_t = \mathbf{A}\mathbf{v}_t\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t$ on $\mathbb{R}^N$, where $\mathbf{A} \in \mathbb{R}^{N\times N}$ is a Hurwitz-stable drift matrix (all eigenvalues of $\mathbf{A}$ have strictly negative real parts) and $\mathbf{D} = \frac{1}{2}\mathbf{B}\mathbf{B}^\top \succ \mathbf{0}$ is a constant positive-definite diffusion matrix. As shown in Chapter 5, the stationary distribution is the zero-mean multivariate Gaussian

$$p_{\mathrm{ss}}(\mathbf{v}) = \frac{1}{(2\pi)^{N/2}(\det \boldsymbol{\Sigma})^{1/2}} \exp\!\left(-\frac{1}{2}\mathbf{v}^\top \boldsymbol{\Sigma}^{-1}\mathbf{v}\right), \tag{6.41}$$

whose stationary covariance matrix $\boldsymbol{\Sigma} = \mathbb{E}\left[\mathbf{v}\mathbf{v}^\top\right]_{\mathrm{ss}} \succ \mathbf{0}$ is the unique solution of the *continuous-time Lyapunov equation*

$$\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top + 2\mathbf{D} = \mathbf{0}. \tag{6.42}$$

Here $U(\mathbf{v}) = \frac{1}{2}\mathbf{v}^\top\boldsymbol{\Sigma}^{-1}\mathbf{v} + \text{const}$, so $\nabla U(\mathbf{v}) = \boldsymbol{\Sigma}^{-1}\mathbf{v}$. By Eq. (6.37), the stationary velocity field is

$$\mathbf{v}_{\mathrm{ss}}(\mathbf{v}) = \mathbf{A}\mathbf{v} + \mathbf{D}\nabla U(\mathbf{v}) = (\mathbf{A} + \mathbf{D}\boldsymbol{\Sigma}^{-1})\mathbf{v} = (\mathbf{A}\boldsymbol{\Sigma} + \mathbf{D})\boldsymbol{\Sigma}^{-1}\mathbf{v}. \tag{6.43}$$

Define the matrix $\boldsymbol{\Omega} := \mathbf{A}\boldsymbol{\Sigma} + \mathbf{D}$. Using the symmetry of $\boldsymbol{\Sigma}$ and $\mathbf{D}$ together with the Lyapunov equation Eq. (6.42), we find

$$\boldsymbol{\Omega} + \boldsymbol{\Omega}^\top = (\mathbf{A}\boldsymbol{\Sigma} + \mathbf{D}) + (\boldsymbol{\Sigma}\mathbf{A}^\top + \mathbf{D}) = \mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top + 2\mathbf{D} = \mathbf{0} \quad\Longrightarrow\quad \boldsymbol{\Omega}^\top = -\boldsymbol{\Omega}. \tag{6.44}$$

Thus $\boldsymbol{\Omega} = \frac{1}{2}(\mathbf{A}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\mathbf{A}^\top)$ is a *skew-symmetric vorticity matrix*. In terms of $\boldsymbol{\Omega}$, the drift matrix $\mathbf{A}$ splits via Eq. (6.38) into

$$\mathbf{A} = (-\mathbf{D} + \boldsymbol{\Omega})\boldsymbol{\Sigma}^{-1}, \qquad \mathbf{v}_{\mathrm{ss}}(\mathbf{v}) = \boldsymbol{\Omega}\boldsymbol{\Sigma}^{-1}\mathbf{v}, \qquad \mathbf{J}_{\mathrm{ss}}(\mathbf{v}) = p_{\mathrm{ss}}(\mathbf{v})\,\boldsymbol{\Omega}\boldsymbol{\Sigma}^{-1}\mathbf{v}. \tag{6.45}$$

Because $\boldsymbol{\Omega}$ is skew-symmetric, for every state vector $\mathbf{v} \in \mathbb{R}^N$ we have

$$\mathbf{v}_{\mathrm{ss}}(\mathbf{v}) \cdot \nabla U(\mathbf{v}) = (\boldsymbol{\Sigma}^{-1}\mathbf{v})^\top \boldsymbol{\Omega}^\top (\boldsymbol{\Sigma}^{-1}\mathbf{v}) = 0 \qquad \text{and} \qquad \nabla \cdot \mathbf{v}_{\mathrm{ss}}(\mathbf{v}) = \mathrm{tr}(\boldsymbol{\Omega}\boldsymbol{\Sigma}^{-1}) = 0.$$

Finally, when does the linear circuit satisfy detailed balance ($\mathbf{J}_{\mathrm{ss}}(\mathbf{v}) \equiv \mathbf{0}$)? Precisely when $\boldsymbol{\Omega} = \mathbf{0}$, which requires $\mathbf{A}\boldsymbol{\Sigma} = -\mathbf{D}$, or $\boldsymbol{\Sigma} = -\mathbf{A}^{-1}\mathbf{D}$. Since $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^\top$, this holds if and only if $\mathbf{A}^{-1}\mathbf{D} = \mathbf{D}(\mathbf{A}^\top)^{-1}$, which is the *Onsager symmetry condition* $\mathbf{A}\mathbf{D} = \mathbf{D}\mathbf{A}^\top$.

Hardware Realization: Broken Detailed Balance in Subthreshold CMOS NEAT-RNs

Non-equilibrium steady states with $\mathbf{J}_{\mathrm{ss}} \neq \mathbf{0}$ are not a mathematical curiosity. In passive RC networks composed solely of two-terminal resistors and capacitors at uniform temperature T, the conductance matrix is automatically symmetric ($\mathbf{G} = \mathbf{G}^\top$) by reciprocity, the thermal noise satisfies $\mathbf{D} = k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{G}(\mathbf{C}^{-1})^\top$, and Onsager symmetry $\mathbf{A}\mathbf{D} = \mathbf{D}\mathbf{A}^\top$ holds identically. Passive RC circuits therefore always satisfy detailed balance ($\boldsymbol{\Omega} = \mathbf{0}$).

Active circuits behave differently. In the subthreshold CMOS non-equilibrium analog thermodynamic random networks (NEAT-RNs) of [37, 38], whose device physics is treated in Chapters 12 and 13, capacitor nodes are coupled through three-terminal weak-inversion

MOSFET transconductance amplifiers. When node j drives the gate of a transistor connected to the drain at node i, it produces a forward transconductance current $G_{ij}v_j$ with zero reverse reaction current unless a matched feedback path is wired ($G_{ij} \neq G_{ji}$). Furthermore, biased subthreshold transistors inject non-equilibrium Poissonian shot noise (Chapter 11) whose intensity depends on the sum of forward and reverse diffusion currents rather than the symmetrized conductance. Active CMOS circuits therefore generically violate the potential condition Eq. (6.28) and operate in a NESS with circulating currents $\mathbf{J}_{\mathrm{ss}}(\mathbf{v}) = p_{\mathrm{ss}}(\mathbf{v})\boldsymbol{\Omega}\boldsymbol{\Sigma}^{-1}\mathbf{v} \neq \mathbf{0}$. Even so, Example 6.5 shows that the linearized circuit still relaxes to a Gaussian steady state $\mathcal{N}(\mathbf{v}^*, \boldsymbol{\Sigma})$, whose covariance is set by the gate biases through the Lyapunov equation Eq. (6.42). The skew-symmetric part $\boldsymbol{\Omega}\boldsymbol{\Sigma}^{-1}$ of the drift can even speed up mixing along poorly conditioned modes.

6.3 The Fokker–Planck H-Theorem and Relative Fisher Information Dissipation

In Chapter 1 (Section 1.2), we proved the static Gibbs–Bogoliubov–Feynman variational principle: for any probability density $\rho(\mathbf{x})$ on $\mathbb{R}^N$, the non-equilibrium variational free energy functional

$$\mathcal{F}[\rho] := \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}) E(\mathbf{x}) - TS[\rho] = \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}) \Big[E(\mathbf{x}) + k_{\mathrm{B}} T \ln \rho(\mathbf{x}) \Big] \tag{6.46}$$

exceeds the equilibrium Helmholtz free energy $F = -k_{\mathrm{B}}T \ln Z$ by $k_{\mathrm{B}}T$ times the KL divergence to the canonical Boltzmann–Gibbs distribution $p_{\mathrm{eq}}(\mathbf{x}) = Z^{-1} e^{-\beta E(\mathbf{x})}$:

$$\mathcal{F}[\rho] - F = k_{\mathrm{B}} T\, D_{\mathrm{KL}}(\rho \,\|\, p_{\mathrm{eq}}) = k_{\mathrm{B}} T \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}) \ln \left(\frac{\rho(\mathbf{x})}{p_{\mathrm{eq}}(\mathbf{x})} \right) \geq 0. \tag{6.47}$$

Chapter 1 asserted that a circuit in contact with a bath at temperature T performs variational inference automatically, by descending $\mathcal{F}[\rho_t]$ toward its minimum $\rho = p_{\mathrm{eq}}$. With the Fokker–Planck equation for $\rho_t(\mathbf{x}) \equiv \rho(\mathbf{x}, t)$ in hand, we can now compute $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t]$ and check this claim.

The rate of free-energy dissipation is expressed through a functional that measures the average squared gradient of the log-density ratio.

Definition 6.6 (Relative Fisher Information)**.** Let $\rho(\mathbf{x})$ and $q(\mathbf{x})$ be smooth positive probability densities on $\mathbb{R}^N$, and let $\mathbf{D}(\mathbf{x}) \succ \mathbf{0}$ be a symmetric positive-definite diffusion tensor. The $\mathbf{D}$-*weighted Relative Fisher Information* of ρ with respect to q is defined by

$$\begin{aligned} \mathcal{I}_{\mathbf{D}}(\rho \,\|\, q) &:= \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}) \left\| \nabla \ln \frac{\rho(\mathbf{x})}{q(\mathbf{x})} \right\|^2_{\mathbf{D}(\mathbf{x})} \\ &= \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}) \, \big(\nabla \ln \rho(\mathbf{x}) - \nabla \ln q(\mathbf{x}) \big)^{\top} \mathbf{D}(\mathbf{x}) \, \big(\nabla \ln \rho(\mathbf{x}) - \nabla \ln q(\mathbf{x}) \big), \end{aligned} \tag{6.48}$$

where $\|\mathbf{u}\|^2_{\mathbf{D}(\mathbf{x})} := \mathbf{u}^{\top}\mathbf{D}(\mathbf{x})\mathbf{u}$. In the unweighted isotropic case $\mathbf{D}(\mathbf{x}) = \mathbf{I}$, we write simply $\mathcal{I}(\rho \,\|\, q) := \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}) \, \|\nabla \ln(\rho/q)\|^2$.

The difference $\nabla \ln \rho(\mathbf{x}) - \nabla \ln q(\mathbf{x})$ is the mismatch between the *score function* $\nabla \ln \rho(\mathbf{x})$ of the current distribution and the score $\nabla \ln q(\mathbf{x})$ of the target, the same object that reappears in score-based diffusion models in Chapter 15. Because $\rho(\mathbf{x}) > 0$ and $\mathbf{D}(\mathbf{x}) \succ \mathbf{0}$ everywhere, the

integrand of Eq. (6.48) is non-negative pointwise, so $\mathcal{I}_{\mathbf{D}}(\rho \,\|\, q) \geq 0$, with $\mathcal{I}_{\mathbf{D}}(\rho \,\|\, q) = 0$ if and only if $\nabla \ln(\rho/q) = \mathbf{0}$ everywhere on the connected space $\mathbb{R}^N$, which implies $\rho(\mathbf{x})/q(\mathbf{x}) = \text{const} = 1$ (since both ρ and q integrate to 1).

Theorem 6.7 (The Fokker–Planck H-Theorem and Free-Energy Dissipation Identity)**.** *Let $\rho_t(\mathbf{x}) \equiv \rho(\mathbf{x}, t)$ be a smooth, positive, rapidly decaying solution of the overdamped Smoluchowski Fokker–Planck equation Eq. (6.23) with mobility $\boldsymbol{\mu}(\mathbf{x}) \succ \mathbf{0}$, diffusion tensor $\mathbf{D}(\mathbf{x}) = k_{\mathrm{B}}T\boldsymbol{\mu}(\mathbf{x})$, and equilibrium Boltzmann–Gibbs density $p_{\mathrm{eq}}(\mathbf{x}) = Z^{-1}e^{-\beta E(\mathbf{x})}$. Then at every instant $t > 0$, the variational free energy $\mathcal{F}[\rho_t]$ and KL divergence $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}})$ obey the exact dissipation identity*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t] = k_{\mathrm{B}}T\frac{\mathrm{d}}{\mathrm{d}t}D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) = -k_{\mathrm{B}}T\,\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{eq}}) \tag{6.49}$$

$$= -\int_{\mathbb{R}^N} \mathrm{d}^N x\, \frac{1}{\rho_t(\mathbf{x})}\, \mathbf{J}(\mathbf{x}, t)^{\top} \boldsymbol{\mu}(\mathbf{x})^{-1} \mathbf{J}(\mathbf{x}, t) \leq 0, \tag{6.50}$$

with equality $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t] = 0$ if and only if $\rho_t(\mathbf{x}) = p_{\mathrm{eq}}(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^N$.

More generally, if $\rho_t(\mathbf{x})$ evolves under an arbitrary (possibly irreversible NESS) Fokker–Planck operator $\partial_t\rho_t = \mathcal{L}_{\mathrm{FP}}\rho_t$ admitting a smooth positive stationary density $p_{\mathrm{ss}}(\mathbf{x}) = e^{-U(\mathbf{x})}$ with steady-state velocity field $\mathbf{v}_{\mathrm{ss}}(\mathbf{x}) = \mathbf{J}_{\mathrm{ss}}(\mathbf{x})/p_{\mathrm{ss}}(\mathbf{x})$, then the KL divergence to p_{ss} still decreases monotonically at the exact rate

$$\frac{\mathrm{d}}{\mathrm{d}t}D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}) = -\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{ss}}) = -\int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho_t(\mathbf{x}) \left\| \nabla \ln \frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})} \right\|^2_{\mathbf{D}(\mathbf{x})} \leq 0. \tag{6.51}$$

Proof. We prove the general identity Eq. (6.51) first; Eqs. (6.49) and (6.50) are its reversible special case $p_{\mathrm{ss}} = p_{\mathrm{eq}}$ ($\mathbf{v}_{\mathrm{ss}} = \mathbf{0}$).

By the Helmholtz–Hodge decomposition Eq. (6.38), for any Fokker–Planck equation with stationary density $p_{\mathrm{ss}}(\mathbf{x}) = e^{-U(\mathbf{x})}$, the Fickian drift is $\tilde{\mathbf{a}}(\mathbf{x}) = -\mathbf{D}(\mathbf{x})\nabla U(\mathbf{x}) + \mathbf{v}_{\mathrm{ss}}(\mathbf{x}) = \mathbf{D}(\mathbf{x})\nabla \ln p_{\mathrm{ss}}(\mathbf{x}) + \mathbf{v}_{\mathrm{ss}}(\mathbf{x})$. Substituting this into the time-dependent probability current Eq. (6.19) gives

$$\begin{aligned}
\mathbf{J}(\mathbf{x}, t) &= \tilde{\mathbf{a}}(\mathbf{x})\rho_t(\mathbf{x}) - \mathbf{D}(\mathbf{x})\nabla\rho_t(\mathbf{x}) \\
&= \rho_t(\mathbf{x})\mathbf{v}_{\mathrm{ss}}(\mathbf{x}) - \rho_t(\mathbf{x})\mathbf{D}(\mathbf{x})\Big(\nabla \ln \rho_t(\mathbf{x}) - \nabla \ln p_{\mathrm{ss}}(\mathbf{x})\Big) \\
&= \rho_t(\mathbf{x})\mathbf{v}_{\mathrm{ss}}(\mathbf{x}) - \rho_t(\mathbf{x})\mathbf{D}(\mathbf{x})\nabla \ln\!\left(\frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})}\right).
\end{aligned} \tag{6.52}$$

Now differentiate the KL divergence $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}) = \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho_t(\mathbf{x}) \ln(\rho_t(\mathbf{x})/p_{\mathrm{ss}}(\mathbf{x}))$ with respect to time t:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}) &= \int_{\mathbb{R}^N} \mathrm{d}^N x \left[\frac{\partial\rho_t(\mathbf{x})}{\partial t} \ln\!\left(\frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})}\right) + \rho_t(\mathbf{x})\frac{p_{\mathrm{ss}}(\mathbf{x})}{\rho_t(\mathbf{x})}\frac{1}{p_{\mathrm{ss}}(\mathbf{x})}\frac{\partial\rho_t(\mathbf{x})}{\partial t}\right] \\
&= \int_{\mathbb{R}^N} \mathrm{d}^N x\, \frac{\partial\rho_t(\mathbf{x})}{\partial t} \ln\!\left(\frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})}\right) + \frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho_t(\mathbf{x}).
\end{aligned} \tag{6.53}$$

Because total probability is conserved ($\int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho_t(\mathbf{x}) = 1$ for all t), the second integral in Eq. (6.53) vanishes identically. In the first integral, we substitute the continuity equation $\partial_t\rho_t = -\nabla \cdot \mathbf{J}(\mathbf{x}, t)$ and integrate by parts over $\mathbb{R}^N$ (where rapid decay of ρ_t and $\mathbf{J}$ at infinity eliminates the boundary surface integral):

$$\frac{\mathrm{d}}{\mathrm{d}t}D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}) = -\int_{\mathbb{R}^N} \mathrm{d}^N x \Big(\nabla \cdot \mathbf{J}(\mathbf{x}, t)\Big) \ln\!\left(\frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})}\right)$$

$$= + \int_{\mathbb{R}^N} \mathrm{d}^N x \, \mathbf{J}(\mathbf{x}, t) \cdot \nabla \ln\!\left(\frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})} \right). \tag{6.54}$$

Inserting the decomposed current Eq. (6.52) into Eq. (6.54) separates the rate of change into two integrals:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}) = & \int_{\mathbb{R}^N} \mathrm{d}^N x \, \rho_t(\mathbf{x}) \, \mathbf{v}_{\mathrm{ss}}(\mathbf{x}) \cdot \nabla \ln\!\left(\frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})} \right) \\ & - \int_{\mathbb{R}^N} \mathrm{d}^N x \, \rho_t(\mathbf{x}) \left[\nabla \ln\!\left(\frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})} \right) \right]^{\top} \mathbf{D}(\mathbf{x}) \left[\nabla \ln\!\left(\frac{\rho_t(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})} \right) \right]. \end{aligned} \tag{6.55}$$

The second integral in Eq. (6.55) is $-\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{ss}})$. To evaluate the first integral, which involves the irreversible velocity $\mathbf{v}_{\mathrm{ss}}(\mathbf{x})$, denote the density ratio by $h_t(\mathbf{x}) := \rho_t(\mathbf{x})/p_{\mathrm{ss}}(\mathbf{x})$, so that $\rho_t(\mathbf{x}) = p_{\mathrm{ss}}(\mathbf{x}) h_t(\mathbf{x})$ and $p_{\mathrm{ss}}(\mathbf{x}) \mathbf{v}_{\mathrm{ss}}(\mathbf{x}) = \mathbf{J}_{\mathrm{ss}}(\mathbf{x})$. Then using $h_t \nabla \ln h_t = \nabla h_t$, the first integral in Eq. (6.55) becomes

$$\begin{aligned} \int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{ss}}(\mathbf{x}) \mathbf{v}_{\mathrm{ss}}(\mathbf{x}) \cdot \Big(h_t(\mathbf{x}) \nabla \ln h_t(\mathbf{x}) \Big) &= \int_{\mathbb{R}^N} \mathrm{d}^N x \, \mathbf{J}_{\mathrm{ss}}(\mathbf{x}) \cdot \nabla h_t(\mathbf{x}) \\ &= - \int_{\mathbb{R}^N} \mathrm{d}^N x \, \Big(\nabla \cdot \mathbf{J}_{\mathrm{ss}}(\mathbf{x}) \Big) \, h_t(\mathbf{x}) = 0, \end{aligned} \tag{6.56}$$

where we integrated by parts and used the stationarity condition $\nabla \cdot \mathbf{J}_{\mathrm{ss}}(\mathbf{x}) = 0$. Thus the solenoidal NESS current contributes *zero* to the time derivative of $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}})$, establishing Eq. (6.51).

Finally, in the reversible equilibrium case ($p_{\mathrm{ss}} = p_{\mathrm{eq}}$, $\mathbf{v}_{\mathrm{ss}} = \mathbf{0}$, $\mathbf{D} = k_{\mathrm{B}} T \boldsymbol{\mu}$), Eq. (6.52) reduces to $\mathbf{J}(\mathbf{x}, t) = -\rho_t(\mathbf{x}) \mathbf{D}(\mathbf{x}) \nabla \ln(\rho_t / p_{\mathrm{eq}})$, or $\nabla \ln(\rho_t / p_{\mathrm{eq}}) = -\frac{1}{\rho_t} \mathbf{D}^{-1} \mathbf{J}$. Multiplying Eq. (6.51) by $k_{\mathrm{B}} T$ and using $\mathcal{F}[\rho_t] = F + k_{\mathrm{B}} T D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}})$ yields Eq. (6.49), and substituting $\nabla \ln(\rho_t / p_{\mathrm{eq}}) = -\frac{1}{\rho_t} (k_{\mathrm{B}} T \boldsymbol{\mu})^{-1} \mathbf{J}$ into $k_{\mathrm{B}} T \mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{eq}})$ yields Eq. (6.50). Since $\boldsymbol{\mu}(\mathbf{x}) \succ \mathbf{0}$ is strictly positive definite, $\mathbf{J}^{\top} \boldsymbol{\mu}^{-1} \mathbf{J} \geq 0$ vanishes if and only if $\mathbf{J}(\mathbf{x}, t) = \mathbf{0}$ everywhere, which by Eq. (6.22) occurs if and only if $\rho_t = p_{\mathrm{eq}}$. □

Physics Insight: Thermodynamic Meaning of the Free-Energy Dissipation Rate

Consider the quadratic current integral in Eq. (6.50). If we define the instantaneous mean local velocity of the probability fluid by $\mathbf{v}(\mathbf{x}, t) := \mathbf{J}(\mathbf{x}, t)/\rho_t(\mathbf{x})$, then $\boldsymbol{\mu}(\mathbf{x})^{-1} = \boldsymbol{\Gamma}(\mathbf{x})$ is the local friction tensor, and Eq. (6.50) reads

$$-\frac{\mathrm{d}}{\mathrm{d}t} \mathcal{F}[\rho_t] = T \dot{\Sigma}_{\mathrm{tot}}(t) = \int_{\mathbb{R}^N} \mathrm{d}^N x \, \rho_t(\mathbf{x}) \, \mathbf{v}(\mathbf{x}, t)^{\top} \boldsymbol{\Gamma}(\mathbf{x}) \, \mathbf{v}(\mathbf{x}, t) \geq 0. \tag{6.57}$$

In classical mechanics, $\mathbf{v}^{\top} \boldsymbol{\Gamma} \mathbf{v}$ is the Rayleigh viscous power dissipated by a particle moving at velocity $\mathbf{v}$ against friction $\boldsymbol{\Gamma}$. Equation Eq. (6.57) shows that the rate of decrease of the variational free energy $\mathcal{F}[\rho_t]$ is the ensemble-averaged viscous power dissipated by the probability current $\mathbf{J}(\mathbf{x}, t)$, which equals T times the non-negative total entropy production rate $\dot{\Sigma}_{\mathrm{tot}}(t) \geq 0$. We return to this identity in Chapter 8 (where we derive trajectory-level entropy production and the Hatano–Sasa housekeeping/excess split for NESS) and in Chapter 10 (where Eq. (6.57) becomes the metric velocity of the Jordan–Kinderlehrer–Otto Wasserstein gradient flow).

6.4 Hermitian Symmetrization and the Witten Schrödinger Hamiltonian

The Fokker–Planck H-theorem (Theorem 6.7) guarantees that $\rho(\mathbf{x},t)$ converges toward $p_{\mathrm{eq}}(\mathbf{x})$ as $t\to\infty$, but it does not tell us the characteristic relaxation time required for a circuit to equilibrate. Because the Fokker–Planck equation $\partial_t\rho=\mathcal{L}_{\mathrm{FP}}\rho$ is a linear autonomous PDE, its time evolution is governed by the eigenvalue spectrum of $\mathcal{L}_{\mathrm{FP}}$.

However, when viewed as an operator on the standard unweighted Hilbert space $L^2(\mathbb{R}^N)$ with inner product $\langle\phi|\psi\rangle=\int_{\mathbb{R}^N}\mathrm{d}^N x\,\phi(\mathbf{x})\psi(\mathbf{x})$, the Fokker–Planck operator $\mathcal{L}_{\mathrm{FP}}$ is *not* self-adjoint: its second-order diffusion part $\nabla\cdot(\mathbf{D}\nabla\cdot)$ is symmetric, whereas its first-order drift part $-\nabla\cdot(\tilde{\mathbf{a}}\cdot)$ is not. On the other hand, we proved in Theorem 6.4(iii) that when detailed balance holds ($\mathbf{J}_{\mathrm{eq}}=\mathbf{0}$), the backward generator $\mathcal{L}$ is self-adjoint in the weighted Hilbert space $L^2(p_{\mathrm{eq}})$, which means $\mathcal{L}_{\mathrm{FP}}=\mathcal{L}^\dagger$ is self-adjoint in the dual space $L^2(p_{\mathrm{eq}}^{-1})$. This suggests that we can map $\mathcal{L}_{\mathrm{FP}}$ and $\mathcal{L}$ to a single self-adjoint operator on the standard unweighted Hilbert space $L^2(\mathbb{R}^N)$ by rescaling functions by $p_{\mathrm{eq}}(\mathbf{x})^{\pm1/2}$.

6.4.1 Ground-State Similarity Transformation to the Witten Operator

To keep the algebra transparent, consider first an isotropic overdamped Langevin system on $\mathbb{R}^N$ with scalar mobility $\mu>0$, diffusion constant $D=\mu k_{\mathrm{B}}T=\mu/\beta>0$, potential $E\in C^\infty(\mathbb{R}^N)$, drift $\mathbf{a}(\mathbf{x})=-\mu\nabla E(\mathbf{x})=-D\beta\nabla E(\mathbf{x})$, and equilibrium Boltzmann density $p_{\mathrm{eq}}(\mathbf{x})=Z^{-1}e^{-\beta E(\mathbf{x})}$. (The extension to a general position-dependent diffusion tensor $\mathbf{D}(\mathbf{x})$ is direct and recorded in Remark 6.10 below.) The Fokker–Planck operator Eq. (6.23) and backward generator Eq. (6.7) take the forms

$$\mathcal{L}_{\mathrm{FP}}\rho=D\nabla\cdot\Big(\nabla\rho+\beta\rho\nabla E\Big)=D\Delta\rho+D\beta\nabla\rho\cdot\nabla E+D\beta\rho\Delta E, \tag{6.58}$$

$$\mathcal{L}f=D\Delta f-D\beta\nabla E\cdot\nabla f. \tag{6.59}$$

We now factor out the square root of the Boltzmann distribution from both the density $\rho(\mathbf{x},t)$ and the observable $f(\mathbf{x},t)$ by writing

$$\rho(\mathbf{x},t)=p_{\mathrm{eq}}(\mathbf{x})^{1/2}\,\psi(\mathbf{x},t)=\frac{1}{\sqrt{Z}}\,e^{-\beta E(\mathbf{x})/2}\,\psi(\mathbf{x},t). \tag{6.60}$$

This scaling is natural: if $\rho(\mathbf{x},t)$ approaches $p_{\mathrm{eq}}(\mathbf{x})$, then $\psi(\mathbf{x},t)=\rho(\mathbf{x},t)/\sqrt{p_{\mathrm{eq}}(\mathbf{x})}$ belongs to $L^2(\mathbb{R}^N)$ whenever the Pearson chi-squared divergence $\int_{\mathbb{R}^N}\mathrm{d}^N x\,\rho^2/p_{\mathrm{eq}}=\|\psi\|^2_{L^2(\mathbb{R}^N)}$ is finite.

Theorem 6.8 (Hermitian Symmetrization and the Witten Schrödinger Hamiltonian)**.** *Under the similarity transformation Eq. (6.60), the Fokker–Planck equation $\partial_t\rho=\mathcal{L}_{\mathrm{FP}}\rho$ is equivalent to the imaginary-time Schrödinger equation (with $\hbar=1$)*

$$\frac{\partial\psi(\mathbf{x},t)}{\partial t}=-\mathcal{H}_{\mathrm{W}}\psi(\mathbf{x},t), \tag{6.61}$$

where the Witten Schrödinger Hamiltonian $\mathcal{H}_{\mathrm{W}}$ *is defined by*

$$\mathcal{H}_{\mathrm{W}}\psi:=-p_{\mathrm{eq}}(\mathbf{x})^{-1/2}\,\mathcal{L}_{\mathrm{FP}}\Big(p_{\mathrm{eq}}(\mathbf{x})^{1/2}\psi\Big)=-p_{\mathrm{eq}}(\mathbf{x})^{1/2}\,\mathcal{L}\Big(p_{\mathrm{eq}}(\mathbf{x})^{-1/2}\psi\Big) \tag{6.62}$$

and takes the explicit Schrödinger form on $L^2(\mathbb{R}^N)$:

$$\mathcal{H}_{\mathrm{W}}=-D\Delta+V_{\mathrm{W}}(\mathbf{x}),\qquad V_{\mathrm{W}}(\mathbf{x}):=D\left(\frac{\beta^2}{4}\,\|\nabla E(\mathbf{x})\|^2-\frac{\beta}{2}\Delta E(\mathbf{x})\right). \tag{6.63}$$

Proof. We compute the spatial derivatives of $\rho(\mathbf{x},t) = Z^{-1/2}e^{-\beta E(\mathbf{x})/2}\psi(\mathbf{x},t)$. By the product rule, the gradient is

$$\nabla\left(e^{-\beta E/2}\psi\right) = e^{-\beta E/2}\left(\nabla\psi - \frac{\beta}{2}\psi\nabla E\right). \tag{6.64}$$

Before expanding the second derivative, form the inner bracket $\nabla\rho + \beta\rho\nabla E$ of the Fokker–Planck operator Eq. (6.58). Using Eq. (6.64),

$$\begin{aligned}\nabla\rho + \beta\rho\nabla E &= \frac{1}{\sqrt{Z}}e^{-\beta E/2}\left(\nabla\psi - \frac{\beta}{2}\psi\nabla E + \beta\psi\nabla E\right)\\ &= p_{\mathrm{eq}}(\mathbf{x})^{1/2}\left(\nabla\psi + \frac{\beta}{2}\psi\nabla E\right).\end{aligned} \tag{6.65}$$

Now we apply $-D\nabla\cdot$ to Eq. (6.65) using the divergence product rule

$$\nabla\cdot(p_{\mathrm{eq}}^{1/2}\mathbf{w}) = p_{\mathrm{eq}}^{1/2}\left(\nabla\cdot\mathbf{w} - \frac{\beta}{2}\nabla E\cdot\mathbf{w}\right)$$

with $\mathbf{w} = \nabla\psi + \frac{\beta}{2}\psi\nabla E$:

$$\begin{aligned}-\mathcal{L}_{\mathrm{FP}}\left(p_{\mathrm{eq}}^{1/2}\psi\right) &= -D\nabla\cdot\left[p_{\mathrm{eq}}^{1/2}\left(\nabla\psi + \frac{\beta}{2}\psi\nabla E\right)\right]\\ &= -Dp_{\mathrm{eq}}^{1/2}\left[\nabla\cdot\left(\nabla\psi + \frac{\beta}{2}\psi\nabla E\right) - \frac{\beta}{2}\nabla E\cdot\left(\nabla\psi + \frac{\beta}{2}\psi\nabla E\right)\right]\\ &= -Dp_{\mathrm{eq}}^{1/2}\left[\Delta\psi + \frac{\beta}{2}\nabla\psi\cdot\nabla E + \frac{\beta}{2}\psi\Delta E - \frac{\beta}{2}\nabla E\cdot\nabla\psi - \frac{\beta^2}{4}\psi\left\|\nabla E\right\|^2\right].\end{aligned} \tag{6.66}$$

The second and fourth terms inside the square brackets of Eq. (6.66), the first-order cross terms $+\frac{\beta}{2}\nabla\psi\cdot\nabla E$ and $-\frac{\beta}{2}\nabla E\cdot\nabla\psi$, cancel. Multiplying both sides of Eq. (6.66) by $p_{\mathrm{eq}}(\mathbf{x})^{-1/2}$ leaves

$$-p_{\mathrm{eq}}^{-1/2}\mathcal{L}_{\mathrm{FP}}\left(p_{\mathrm{eq}}^{1/2}\psi\right) = -D\Delta\psi + D\left(\frac{\beta^2}{4}\left\|\nabla E(\mathbf{x})\right\|^2 - \frac{\beta}{2}\Delta E(\mathbf{x})\right)\psi = \left(-D\Delta + V_{\mathrm{W}}(\mathbf{x})\right)\psi. \tag{6.67}$$

Similarly, to verify the second equality in Eq. (6.62) for the backward generator $\mathcal{L}$, we compute $\nabla(p_{\mathrm{eq}}^{-1/2}\psi) = p_{\mathrm{eq}}^{-1/2}(\nabla\psi + \frac{\beta}{2}\psi\nabla E)$ and $\Delta(p_{\mathrm{eq}}^{-1/2}\psi) = p_{\mathrm{eq}}^{-1/2}\left[\Delta\psi + \beta\nabla E\cdot\nabla\psi + \frac{\beta}{2}\psi\Delta E + \frac{\beta^2}{4}\psi\left\|\nabla E\right\|^2\right]$. Subtracting $\beta\nabla E\cdot\nabla(p_{\mathrm{eq}}^{-1/2}\psi) = p_{\mathrm{eq}}^{-1/2}\left[\beta\nabla E\cdot\nabla\psi + \frac{\beta^2}{2}\psi\left\|\nabla E\right\|^2\right]$ and multiplying by $-Dp_{\mathrm{eq}}^{1/2}$ cancels the $+\beta\nabla E\cdot\nabla\psi$ term and reproduces Eq. (6.67). □

6.4.2 Supersymmetric Factorization and Spectral Decomposition

The particular combination $\frac{\beta^2}{4}\left\|\nabla E\right\|^2 - \frac{\beta}{2}\Delta E$ in $V_{\mathrm{W}}(\mathbf{x})$ has a structural origin, already visible in the two-step calculation Eqs. (6.65) and (6.66): $\mathcal{H}_{\mathrm{W}}$ factorizes into a product of first-order operators. This is a *supersymmetric* (or Darboux) factorization [118].

Define the vector-valued first-order differential operator $\mathbf{Q} = (Q_1,\dots,Q_N)^\top$ and its formal $L^2(\mathbb{R}^N)$-adjoint $\mathbf{Q}^\dagger = (Q_1^\dagger,\dots,Q_N^\dagger)$ by

$$Q_i := \sqrt{D}\left(\frac{\partial}{\partial x_i} + \frac{\beta}{2}\frac{\partial E(\mathbf{x})}{\partial x_i}\right), \qquad Q_i^\dagger = \sqrt{D}\left(-\frac{\partial}{\partial x_i} + \frac{\beta}{2}\frac{\partial E(\mathbf{x})}{\partial x_i}\right), \qquad i = 1,\dots,N. \tag{6.68}$$

In terms of $p_{\mathrm{eq}}(\mathbf{x}) \propto e^{-\beta E(\mathbf{x})}$, these operators can be written as

$$Q_i\psi = \sqrt{D}\,p_{\mathrm{eq}}(\mathbf{x})^{1/2}\frac{\partial}{\partial x_i}\left(p_{\mathrm{eq}}(\mathbf{x})^{-1/2}\psi\right), \qquad Q_i^\dagger\psi = -\sqrt{D}\,p_{\mathrm{eq}}(\mathbf{x})^{-1/2}\frac{\partial}{\partial x_i}\left(p_{\mathrm{eq}}(\mathbf{x})^{1/2}\psi\right). \tag{6.69}$$

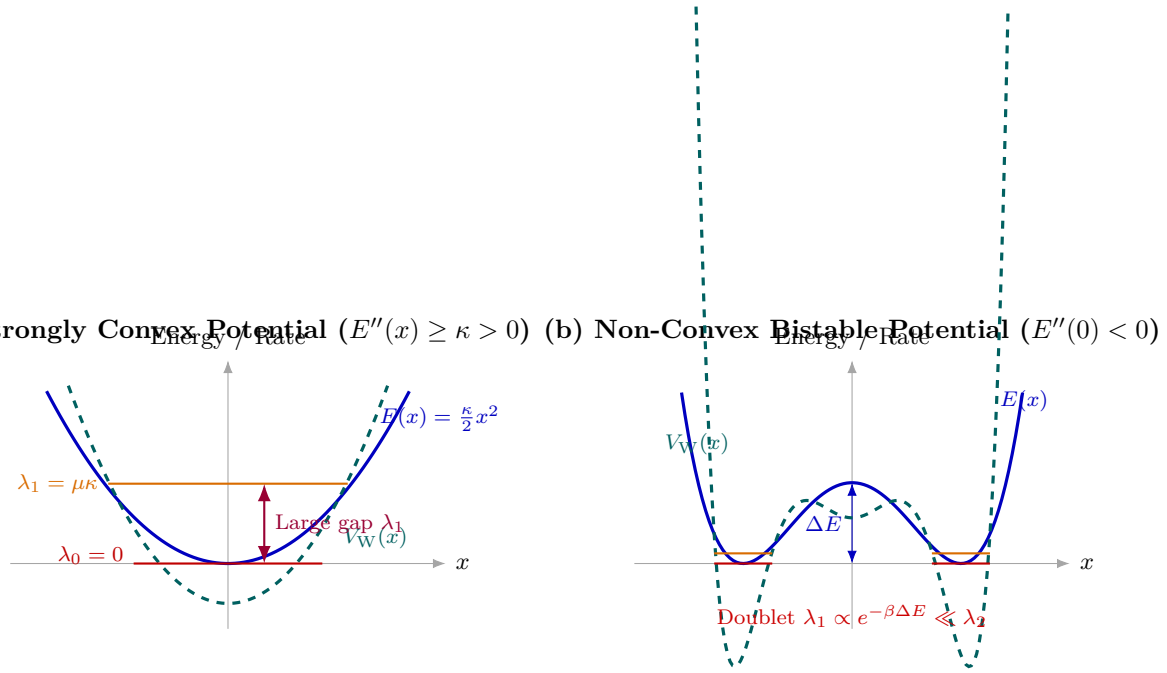


Figure 6.2: Comparison of the physical potential $E(x)$ (solid blue) and the associated Witten Schrödinger potential $V_{\mathrm{W}}(x) = D(\frac{\beta^2}{4}(E')^2 - \frac{\beta}{2}E'')$ (dashed teal). (a) When $E(x)$ is strongly convex ($E''(x) \geq \kappa > 0$), $V_{\mathrm{W}}(x)$ is a single confining well dipped below zero at the origin by $-\frac{1}{2}D\beta E''(0)$ so that its ground state sits at $\lambda_0 = 0$ and the spectral gap $\lambda_1 \geq \mu\kappa$ is large. (b) When $E(x)$ is a non-convex double well, $V_{\mathrm{W}}(x)$ dips negative at the two well minima $x = \pm x_0$ (where $E' = 0, E'' > 0$) but rises to $+\frac{1}{2}D\beta\,|E''(0)| > 0$ at the saddle $x = 0$ flanked by steep $\mathcal{O}(\beta^2)$ barriers where $(E')^2$ is large, creating a quasi-degenerate ground-state doublet $\lambda_1 \propto e^{-\beta\Delta E}$.

When we compose $\mathbf{Q}^\dagger$ with $\mathbf{Q}$, we find for any $\psi \in C^2(\mathbb{R}^N)$:

$$\begin{aligned}\mathbf{Q}^\dagger \cdot \mathbf{Q}\psi &= \sum_{i=1}^{N} Q_i^\dagger Q_i \psi = D\sum_{i=1}^{N}\left(-\frac{\partial}{\partial x_i} + \frac{\beta}{2}\frac{\partial E}{\partial x_i}\right)\left(\frac{\partial \psi}{\partial x_i} + \frac{\beta}{2}\frac{\partial E}{\partial x_i}\psi\right)\\ &= D\sum_{i=1}^{N}\left(-\frac{\partial^2\psi}{\partial x_i^2} - \frac{\beta}{2}\frac{\partial^2 E}{\partial x_i^2}\psi - \frac{\beta}{2}\frac{\partial E}{\partial x_i}\frac{\partial \psi}{\partial x_i} + \frac{\beta}{2}\frac{\partial E}{\partial x_i}\frac{\partial \psi}{\partial x_i} + \frac{\beta^2}{4}\left(\frac{\partial E}{\partial x_i}\right)^2\psi\right)\\ &= -D\Delta\psi + V_{\mathrm{W}}(\mathbf{x})\psi = \mathcal{H}_{\mathrm{W}}\psi. \end{aligned} \tag{6.70}$$

Thus the Witten Hamiltonian admits the exact positive semi-definite factorization

$$\mathcal{H}_{\mathrm{W}} = \mathbf{Q}^\dagger \cdot \mathbf{Q} = \sum_{i=1}^{N} Q_i^\dagger Q_i. \tag{6.71}$$

The spectral theory of reversible Fokker–Planck operators follows from the factorization Eq. (6.71).

Theorem 6.9 (Spectral Decomposition of Reversible Fokker–Planck Operators)**.** *Let $E \in C^\infty(\mathbb{R}^N)$ be a confining potential such that $Z = \int_{\mathbb{R}^N} e^{-\beta E(\mathbf{x})}\,\mathrm{d}^N x < \infty$ and $V_{\mathrm{W}}(\mathbf{x}) \to +\infty$ as $\|\mathbf{x}\| \to \infty$. Then:*

(i) ***Self-Adjointness and Non-Negativity:*** *$\mathcal{H}_{\mathrm{W}} = \mathbf{Q}^\dagger \cdot \mathbf{Q}$ is essentially self-adjoint and positive semi-definite on $L^2(\mathbb{R}^N)$: for every $\psi \in \mathrm{dom}(\mathcal{H}_{\mathrm{W}})$,*

$$\langle \psi | \mathcal{H}_{\mathrm{W}} | \psi \rangle = \langle \mathbf{Q}\psi | \mathbf{Q}\psi \rangle = \sum_{i=1}^{N} \|Q_i \psi\|^2_{L^2(\mathbb{R}^N)} = D \int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{eq}}(\mathbf{x}) \left\| \nabla \Big(p_{\mathrm{eq}}(\mathbf{x})^{-1/2} \psi(\mathbf{x}) \Big) \right\|^2 \geq 0. \tag{6.72}$$

(ii) ***Unique Zero-Energy Ground State:*** *$\lambda_0 = 0$ is a simple (non-degenerate) eigenvalue of $\mathcal{H}_{\mathrm{W}}$ with normalized ground-state eigenfunction*

$$\psi_0(\mathbf{x}) = p_{\mathrm{eq}}(\mathbf{x})^{1/2} = \frac{1}{\sqrt{Z}} \, e^{-\beta E(\mathbf{x})/2}, \qquad \|\psi_0\|^2_{L^2(\mathbb{R}^N)} = \int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{eq}}(\mathbf{x}) = 1. \tag{6.73}$$

(iii) ***Discrete Spectrum and Orthonormal Eigenbasis:*** *Because $V_{\mathrm{W}}(\mathbf{x}) \to +\infty$ as $\|\mathbf{x}\| \to \infty$, the resolvent $(\mathcal{H}_{\mathrm{W}} + \mathbf{I})^{-1}$ is compact on $L^2(\mathbb{R}^N)$, and $\mathcal{H}_{\mathrm{W}}$ possesses a purely discrete spectrum of real eigenvalues*

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \cdots \to +\infty \tag{6.74}$$

with a complete orthonormal basis of smooth real eigenfunctions $\{\psi_n(\mathbf{x})\}_{n=0}^{\infty}$ in $L^2(\mathbb{R}^N)$ satisfying $\mathcal{H}_{\mathrm{W}} \psi_n = \lambda_n \psi_n$ and $\langle \psi_m | \psi_n \rangle = \delta_{mn}$.

(iv) ***Eigenfunction Expansion of Densities and Observables:*** *For each $n \geq 0$, the functions*

$$\phi_n(\mathbf{x}) := p_{\mathrm{eq}}(\mathbf{x})^{1/2} \psi_n(\mathbf{x}) \qquad \text{and} \qquad f_n(\mathbf{x}) := p_{\mathrm{eq}}(\mathbf{x})^{-1/2} \psi_n(\mathbf{x}) \tag{6.75}$$

are right eigenfunctions of $\mathcal{L}_{\mathrm{FP}}$ and $\mathcal{L}$, respectively, with eigenvalue $-\lambda_n$:

$$\mathcal{L}_{\mathrm{FP}} \phi_n = -\lambda_n \phi_n, \qquad \mathcal{L} f_n = -\lambda_n f_n, \qquad \langle f_m, \, f_n \rangle_{p_{\mathrm{eq}}} = \int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{eq}}(\mathbf{x}) f_m(\mathbf{x}) f_n(\mathbf{x}) = \delta_{mn}. \tag{6.76}$$

Consequently, the transition probability density (heat kernel) $p(\mathbf{x}, t \mid \mathbf{x}_0, 0)$ and the time-dependent density $\rho(\mathbf{x}, t)$ expand as

$$p(\mathbf{x}, t \mid \mathbf{x}_0, 0) = p_{\mathrm{eq}}(\mathbf{x}) + \sum_{n=1}^{\infty} e^{-\lambda_n t} \, \phi_n(\mathbf{x}) \, f_n(\mathbf{x}_0) = p_{\mathrm{eq}}(\mathbf{x}) \left[1 + \sum_{n=1}^{\infty} e^{-\lambda_n t} \, f_n(\mathbf{x}) \, f_n(\mathbf{x}_0) \right], \tag{6.77}$$

$$\rho(\mathbf{x}, t) = p_{\mathrm{eq}}(\mathbf{x}) + \sum_{n=1}^{\infty} c_n \, e^{-\lambda_n t} \, p_{\mathrm{eq}}(\mathbf{x})^{1/2} \psi_n(\mathbf{x}), \qquad c_n = \int_{\mathbb{R}^N} \mathrm{d}^N x_0 \, \rho_0(\mathbf{x}_0) f_n(\mathbf{x}_0). \tag{6.78}$$

Proof. Part (i) follows directly by integrating $\left\langle \psi \middle| \mathbf{Q}^\dagger \cdot \mathbf{Q}\psi \right\rangle$ by parts and using Eq. (6.69). For part (ii), $\langle \psi | \mathcal{H}_{\mathrm{W}} | \psi \rangle = 0$ if and only if $Q_i \psi = 0$ for every $i = 1, \ldots, N$. By Eq. (6.69), $Q_i \psi = 0$ for all i if and only if $\nabla(p_{\mathrm{eq}}^{-1/2} \psi) = \mathbf{0}$ on the connected space $\mathbb{R}^N$, which means $p_{\mathrm{eq}}(\mathbf{x})^{-1/2} \psi(\mathbf{x}) = c$ is constant, so $\psi(\mathbf{x}) = c \, p_{\mathrm{eq}}(\mathbf{x})^{1/2}$. Choosing $c = 1$ gives the unique normalized ground state $\psi_0(\mathbf{x}) = p_{\mathrm{eq}}(\mathbf{x})^{1/2}$ in Eq. (6.73), and since $\mathbb{R}^N$ is connected, no other linearly independent state can have $\lambda = 0$, proving $\lambda_1 > 0$. Part (iii) is a standard result for Schrödinger operators with confining potentials $V_{\mathrm{W}}(\mathbf{x}) \to +\infty$ (Rellich's compactness criterion; see [85], and Section C.3 for the Gaussian case). Part (iv) follows from Eq. (6.62) by writing $e^{t\mathcal{L}_{\mathrm{FP}}} = p_{\mathrm{eq}}^{1/2} e^{-t\mathcal{H}_{\mathrm{W}}} p_{\mathrm{eq}}^{-1/2}$ and inserting the completeness relation $\sum_{n=0}^{\infty} |\psi_n\rangle \langle \psi_n| = \mathbf{I}$ in $L^2(\mathbb{R}^N)$, noting that the $n = 0$ term with $\lambda_0 = 0$, $\phi_0(\mathbf{x}) = p_{\mathrm{eq}}(\mathbf{x})$, and $f_0(\mathbf{x}_0) = 1$ separates out as the equilibrium distribution $p_{\mathrm{eq}}(\mathbf{x})$. □

Remark 6.10 (Anisotropic Position-Dependent Diffusion Tensor)**.** If the diffusion tensor $\mathbf{D}(\mathbf{x}) = k_{\mathrm{B}}T\boldsymbol{\mu}(\mathbf{x}) \succ \mathbf{0}$ depends on $\mathbf{x}$ and can be factored as $\mathbf{D}(\mathbf{x}) = \boldsymbol{\sigma}(\mathbf{x})\boldsymbol{\sigma}(\mathbf{x})^{\top}$, the exact same similarity transformation $\mathcal{H}_{\mathrm{W}} = -p_{\mathrm{eq}}^{-1/2}\mathcal{L}_{\mathrm{FP}}(p_{\mathrm{eq}}^{1/2}\,\cdot)$ yields $\mathcal{H}_{\mathrm{W}} = \mathbf{Q}^{\dagger}\cdot\mathbf{Q}$ with $Q_\alpha = \sum_{i=1}^{N}\sigma_{i\alpha}(\mathbf{x})(\partial_{x_i} + \frac{\beta}{2}\partial_{x_i}E(\mathbf{x}))$, and all four conclusions of Theorem 6.9 remain valid.

6.5 Poincaré Inequality, Logarithmic Sobolev Inequality, and the Bakry–Émery Criterion

According to the spectral expansion Eq. (6.78), the long-time convergence of $\rho(\mathbf{x},t)$ to $p_{\mathrm{eq}}(\mathbf{x})$ is governed by the *spectral gap* $\lambda_1 > 0$, which defines the asymptotic relaxation time $\tau_{\mathrm{rel}} := \lambda_1^{-1}$. In a one-dimensional system or a linear Gaussian network (Worked computation 6.21), we can often compute λ_1 explicitly. However, in an N-dimensional nonlinear thermodynamic circuit with $N \sim 10^4$–10^6 coupled nodes, solving the N-dimensional PDE $\mathcal{H}_{\mathrm{W}}\psi_1 = \lambda_1\psi_1$ directly is intractable.

We therefore need a method to bound the relaxation rate directly from the local geometry of the energy function $E(\mathbf{x})$, specifically from its Hessian matrix $\nabla^2 E(\mathbf{x})$, without solving for the eigenfunctions. Moreover, in high dimensions ($N \gg 1$), we must be careful about which divergence measure we control. This leads us to the theory of *functional inequalities* for Markov diffusion operators.

6.5.1 The Carré du Champ, Dirichlet Form, and Poincaré Inequality

To formulate functional inequalities systematically, we first introduce the bilinear differential form associated with the backward generator $\mathcal{L}$, initiated by Paul-André Meyer and developed by Dominique Bakry and Michel Émery [6, 7].

Definition 6.11 (Carré du Champ Operator and Dirichlet Form)**.** Let $\mathcal{L} = \mathbf{a}(\mathbf{x})\cdot\nabla + \mathrm{tr}(\mathbf{D}(\mathbf{x})\nabla^2)$ be the backward Kolmogorov generator of a diffusion process on $\mathbb{R}^N$.

1. The *carré du champ* ("square of the field") operator $\Gamma : C^2(\mathbb{R}^N)\times C^2(\mathbb{R}^N) \to C(\mathbb{R}^N)$ is the symmetric bilinear form measuring the failure of $\mathcal{L}$ to obey the first-order Leibniz product rule:
$$\Gamma(f,g)(\mathbf{x}) := \frac{1}{2}\Big[\mathcal{L}(fg)(\mathbf{x}) - f(\mathbf{x})\mathcal{L}g(\mathbf{x}) - g(\mathbf{x})\mathcal{L}f(\mathbf{x})\Big]. \tag{6.79}$$
Expanding $\mathcal{L}(fg) = f\mathcal{L}g + g\mathcal{L}f + 2\nabla f^{\top}\mathbf{D}\nabla g$ shows that $\Gamma(f,g)$ depends *only* on the diffusion tensor $\mathbf{D}(\mathbf{x})$ and is completely independent of the drift $\mathbf{a}(\mathbf{x})$:
$$\Gamma(f,g)(\mathbf{x}) = \nabla f(\mathbf{x})^{\top}\mathbf{D}(\mathbf{x})\nabla g(\mathbf{x}) = \sum_{i,j=1}^{N} D_{ij}(\mathbf{x})\frac{\partial f(\mathbf{x})}{\partial x_i}\frac{\partial g(\mathbf{x})}{\partial x_j}. \tag{6.80}$$
We write $\Gamma(f) \equiv \Gamma(f,f) = \|\nabla f\|_{\mathbf{D}}^2 \geq 0$.

2. For a reversible diffusion with equilibrium density $p_{\mathrm{eq}}(\mathbf{x})$, the *Dirichlet form* $\mathcal{E}(f,g)$ on $L^2(p_{\mathrm{eq}})$ is the equilibrium expectation of the carré du champ:
$$\mathcal{E}(f,g) := \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\,\Gamma(f,g)(\mathbf{x}) = -\left\langle f,\, \mathcal{L}g\right\rangle_{p_{\mathrm{eq}}} = \left\langle p_{\mathrm{eq}}^{1/2} f \middle| \mathcal{H}_{\mathrm{W}} \middle| p_{\mathrm{eq}}^{1/2} g\right\rangle. \tag{6.81}$$

Consider the Rayleigh–Ritz variational principle for the first excited eigenvalue λ_1 of $\mathcal{H}_{\mathrm{W}}$: since $\psi_0 = p_{\mathrm{eq}}^{1/2}$ is the ground state with $\lambda_0 = 0$, any function $\psi = p_{\mathrm{eq}}^{1/2} f \in L^2(\mathbb{R}^N)$ is orthogonal to ψ_0

($\langle\psi_0|\psi\rangle = \int p_{\mathrm{eq}} f \, \mathrm{d}^N x = \mathbb{E}[f]_{p_{\mathrm{eq}}} = 0$) if and only if f has zero equilibrium mean, and for any non-constant $f \in H^1(p_{\mathrm{eq}})$ we can shift $f \mapsto \tilde{f} = f - \mathbb{E}[f]_{p_{\mathrm{eq}}}$ so that $\left\|\tilde{\psi}\right\|^2_{L^2(\mathbb{R}^N)} = \int p_{\mathrm{eq}} (f - \mathbb{E}[f]_{p_{\mathrm{eq}}})^2 \, \mathrm{d}^N x = \mathrm{Var}_{p_{\mathrm{eq}}}(f)$. The variational principle $\lambda_1 = \inf_{\langle\psi|\psi_0\rangle=0} \frac{\langle\psi|\mathcal{H}_{\mathrm{W}}|\psi\rangle}{\langle\psi|\psi\rangle}$ is therefore equivalent to the *Poincaré inequality*.

Definition 6.12 (Poincaré Inequality)**.** A probability measure $p_{\mathrm{eq}}(\mathbf{x}) \, \mathrm{d}^N x$ on $\mathbb{R}^N$ with diffusion tensor $\mathbf{D}(\mathbf{x})$ satisfies a *Poincaré inequality* (or *spectral gap inequality*) with constant $C_{\mathrm{P}} > 0$ if for every smooth square-integrable observable $f \in H^1(p_{\mathrm{eq}})$,

$$\mathrm{Var}_{p_{\mathrm{eq}}}(f) \le \frac{1}{C_{\mathrm{P}}} \mathcal{E}(f,f) = \frac{1}{C_{\mathrm{P}}} \int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{eq}}(\mathbf{x}) \, \nabla f(\mathbf{x})^\top \mathbf{D}(\mathbf{x}) \nabla f(\mathbf{x}), \tag{6.82}$$

where $\mathrm{Var}_{p_{\mathrm{eq}}}(f) := \mathbb{E}[f^2]_{p_{\mathrm{eq}}} - \left(\mathbb{E}[f]_{p_{\mathrm{eq}}}\right)^2$. The largest constant C_{P} for which Eq. (6.82) holds is precisely the spectral gap λ_1 of $-\mathcal{L}$ (and of $\mathcal{H}_{\mathrm{W}}$).

Proposition 6.13 (Poincaré Inequality and Exponential Variance / χ^2 Decay)**.** *Let $\mathbf{x}_t$ be a diffusion with stationary density $p_{\mathrm{ss}}(\mathbf{x})$ (which may be either reversible p_{eq} or a NESS p_{ss}), and let $f_t(\mathbf{x}) = (P_t f)(\mathbf{x}) = \mathbb{E}[f(\mathbf{x}_t) \mid \mathbf{x}_0 = \mathbf{x}]$ solve the backward equation $\partial_t f_t = \mathcal{L} f_t$, or equivalently let $h_t(\mathbf{x}) = \rho_t(\mathbf{x})/p_{\mathrm{ss}}(\mathbf{x})$ be the density ratio. Then at every $t \ge 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathrm{Var}_{p_{\mathrm{ss}}}(f_t) = -2\mathcal{E}(f_t, f_t) = -2 \int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{ss}}(\mathbf{x}) \, \Gamma(f_t, f_t)(\mathbf{x}). \tag{6.83}$$

Consequently, the Poincaré inequality Eq. (6.82) holds with constant $C_{\mathrm{P}} > 0$ if and only if for every $f_0 \in L^2(p_{\mathrm{ss}})$ and every initial density ρ_0, the variance and the Pearson χ^2-divergence $\chi^2(\rho_t \,\|\, p_{\mathrm{ss}}) := \int_{\mathbb{R}^N} \mathrm{d}^N x \, \frac{(\rho_t - p_{\mathrm{ss}})^2}{p_{\mathrm{ss}}} = \mathrm{Var}_{p_{\mathrm{ss}}}(h_t)$ decay exponentially at rate $2C_{\mathrm{P}}$:

$$\mathrm{Var}_{p_{\mathrm{ss}}}(f_t) \le e^{-2C_{\mathrm{P}} t} \mathrm{Var}_{p_{\mathrm{ss}}}(f_0), \qquad \chi^2(\rho_t \,\|\, p_{\mathrm{ss}}) \le e^{-2C_{\mathrm{P}} t} \chi^2(\rho_0 \,\|\, p_{\mathrm{ss}}). \tag{6.84}$$

Proof. Since p_{ss} is stationary, $\mathbb{E}[f_t]_{p_{\mathrm{ss}}} = \mathbb{E}[f_0]_{p_{\mathrm{ss}}}$ is constant in time. Differentiating $\mathrm{Var}_{p_{\mathrm{ss}}}(f_t) = \int p_{\mathrm{ss}} f_t^2 \, \mathrm{d}^N x - (\mathbb{E}[f_0]_{p_{\mathrm{ss}}})^2$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathrm{Var}_{p_{\mathrm{ss}}}(f_t) = 2 \int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{ss}}(\mathbf{x}) \, f_t(\mathbf{x}) \, \mathcal{L} f_t(\mathbf{x}) = \int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{ss}}(\mathbf{x}) \Big[\mathcal{L}(f_t^2) - 2\Gamma(f_t, f_t) \Big] = -2\mathcal{E}(f_t, f_t),$$

where we used Eq. (6.79) and the stationarity identity $\int p_{\mathrm{ss}} \mathcal{L}(f_t^2) \, \mathrm{d}^N x = \int (\mathcal{L}_{\mathrm{FP}} p_{\mathrm{ss}}) f_t^2 \, \mathrm{d}^N x = 0$. If the Poincaré inequality $\mathcal{E}(f_t, f_t) \ge C_{\mathrm{P}} \mathrm{Var}_{p_{\mathrm{ss}}}(f_t)$ holds, then $\frac{\mathrm{d}}{\mathrm{d}t} \mathrm{Var}_{p_{\mathrm{ss}}}(f_t) \le -2C_{\mathrm{P}} \mathrm{Var}_{p_{\mathrm{ss}}}(f_t)$, and Grönwall's inequality yields Eq. (6.84). Conversely, differentiating Eq. (6.84) at $t = 0$ recovers $2\mathcal{E}(f_0, f_0) \ge 2C_{\mathrm{P}} \mathrm{Var}_{p_{\mathrm{ss}}}(f_0)$. □

6.5.2 Why High-Dimensional Circuits Need the Logarithmic Sobolev Inequality

The Poincaré bound $\chi^2(\rho_t \,\|\, p_{\mathrm{eq}}) \le e^{-2\lambda_1 t} \chi^2(\rho_0 \,\|\, p_{\mathrm{eq}})$ in Eq. (6.84) might seem to settle the question of mixing. In high dimensions it does not, because the initial χ^2-divergence itself can be enormous.

Consider an array of N independent RC nodes (or N uncoupled modes) at equilibrium temperature T, so $p_{\mathrm{eq}}(\mathbf{x}) = \prod_{i=1}^N p_1(x_i)$. Suppose the circuit is initialized at $t = 0$ in a product distribution $\rho_0(\mathbf{x}) = \prod_{i=1}^N q_1(x_i)$ where each individual node has a modest single-node chi-squared discrepancy $1 + \alpha = \int \frac{q_1(x_i)^2}{p_1(x_i)} \, \mathrm{d}x_i > 1$ and a modest single-node KL divergence $\delta_{\mathrm{KL}} = D_{\mathrm{KL}}(q_1 \,\|\, p_1) > 0$. The N-node divergences then scale with N as

$$\chi^2(\rho_0 \,\|\, p_{\mathrm{eq}}) = \int_{\mathbb{R}^N} \mathrm{d}^N x \, \prod_{i=1}^N \frac{q_1(x_i)^2}{p_1(x_i)} - 1 = (1+\alpha)^N - 1 \sim e^{N \ln(1+\alpha)}, \tag{6.85}$$

$$D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}}) = \sum_{i=1}^{N} D_{\mathrm{KL}}(q_1 \,\|\, p_1) = N\,\delta_{\mathrm{KL}} \sim \mathcal{O}(N). \tag{6.86}$$

The two scale very differently. For an N-node chip with $N = 10^5$, the initial χ^2-divergence $\chi^2(\rho_0 \,\|\, p_{\mathrm{eq}})$ is exponentially huge ($\sim e^{cN}$), so the Poincaré bound $\chi^2(\rho_t \,\|\, p_{\mathrm{eq}}) \le e^{cN - 2\lambda_1 t}$ requires a waiting time $t \sim \frac{cN}{2\lambda_1} = \mathcal{O}(N)$ just to bring the bound below 1, even though the N nodes are completely uncoupled and physically equilibrate in parallel in $\mathcal{O}(1)$ time. By contrast, the KL divergence $D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}}) = N\delta_{\mathrm{KL}}$ is *extensive* (additive over subsystems): an inequality that gives exponential decay of $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}})$ at rate $2\rho_{\mathrm{LS}}$ yields $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) \le N\delta_{\mathrm{KL}} e^{-2\rho_{\mathrm{LS}} t}$, which drops below any tolerance ε after a time $t \sim \frac{1}{2\rho_{\mathrm{LS}}} \ln(N\delta_{\mathrm{KL}}/\varepsilon) = \mathcal{O}(\ln N)$, only logarithmic in N.

Recall the Fokker–Planck H-theorem Eq. (6.51):

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) = -\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{eq}}).$$

Just as bounding the variance $\mathrm{Var}_{p_{\mathrm{eq}}}(f_t)$ by its rate of decrease $2\mathcal{E}(f_t, f_t)$ produced exponential decay of variance in Proposition 6.13, bounding the KL divergence $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}})$ by its rate of decrease, the Relative Fisher Information $\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{eq}})$, produces exponential decay of the KL divergence and variational free energy.

Definition 6.14 (Logarithmic Sobolev Inequality (LSI))**.** A probability measure $p_{\mathrm{eq}}(\mathbf{x})\,\mathrm{d}^N x$ on $\mathbb{R}^N$ with diffusion tensor $\mathbf{D}(\mathbf{x}) \succ \mathbf{0}$ satisfies a *Logarithmic Sobolev Inequality (LSI)* with constant $\rho_{\mathrm{LS}} > 0$ if for every smooth probability density $\rho(\mathbf{x})$ with $D_{\mathrm{KL}}(\rho \,\|\, p_{\mathrm{eq}}) < \infty$,

$$D_{\mathrm{KL}}(\rho \,\|\, p_{\mathrm{eq}}) \le \frac{1}{2\rho_{\mathrm{LS}}} \mathcal{I}_{\mathbf{D}}(\rho \,\|\, p_{\mathrm{eq}}) = \frac{1}{2\rho_{\mathrm{LS}}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}) \left\| \nabla \ln \frac{\rho(\mathbf{x})}{p_{\mathrm{eq}}(\mathbf{x})} \right\|_{\mathbf{D}(\mathbf{x})}^2 . \tag{6.87}$$

Equivalently, writing $h(\mathbf{x}) = \rho(\mathbf{x})/p_{\mathrm{eq}}(\mathbf{x}) = g(\mathbf{x})^2$ with $\int_{\mathbb{R}^N} p_{\mathrm{eq}} g^2 \,\mathrm{d}^N x = 1$ and noting that $h\,\|\nabla \ln h\|_{\mathbf{D}}^2 = \frac{1}{h}\,\|\nabla h\|_{\mathbf{D}}^2 = 4\,\|\nabla g\|_{\mathbf{D}}^2$, Eq. (6.87) takes Gross's classic $L^2(p_{\mathrm{eq}})$ form:

$$\mathrm{Ent}_{p_{\mathrm{eq}}}(g^2) := \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, g(\mathbf{x})^2 \ln\left(\frac{g(\mathbf{x})^2}{\|g\|_{L^2(p_{\mathrm{eq}})}^2} \right) \le \frac{2}{\rho_{\mathrm{LS}}}\, \mathcal{E}(g, g) = \frac{2}{\rho_{\mathrm{LS}}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \|\nabla g(\mathbf{x})\|_{\mathbf{D}(\mathbf{x})}^2 . \tag{6.88}$$

Proposition 6.15 (Consequences of the Logarithmic Sobolev Inequality)**.** *Suppose p_{eq} (or a stationary NESS p_{ss}) satisfies the Logarithmic Sobolev Inequality Eq. (6.87) with constant $\rho_{\mathrm{LS}} > 0$. Then:*

(i) ***Exponential Contraction of KL Divergence and Free Energy:*** *Along any solution ρ_t of the Fokker–Planck equation, for all $t \ge 0$,*

$$D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) \le e^{-2\rho_{\mathrm{LS}} t}\, D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}})\,, \tag{6.89}$$

$$\mathcal{F}[\rho_t] - F \le e^{-2\rho_{\mathrm{LS}} t} \Big(\mathcal{F}[\rho_0] - F \Big), \tag{6.90}$$

$$\|\rho_t - p_{\mathrm{eq}}\|_{\mathrm{TV}} \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}})}\, e^{-\rho_{\mathrm{LS}} t}. \tag{6.91}$$

(ii) ***LSI Implies the Poincaré Inequality:*** *p_{eq} also satisfies the Poincaré inequality Eq. (6.82) with spectral gap*

$$\lambda_1 = C_{\mathrm{P}} \ge \rho_{\mathrm{LS}}. \tag{6.92}$$

Proof. To prove (i), combine the H-theorem Eq. (6.51) with the LSI Eq. (6.87):

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) = -\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{eq}}) \le -2\rho_{\mathrm{LS}}\, D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) \,.$$

Applying Grönwall's differential inequality yields Eq. (6.89); multiplying by $k_{\mathrm{B}}T$ gives Eq. (6.90); and combining Eq. (6.89) with the Csiszár–Kullback–Pinsker inequality $\|\rho_t - p_{\mathrm{eq}}\|_{\mathrm{TV}}^2 \le \frac{1}{2} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}})$ [7] yields Eq. (6.91).

To prove (ii), let $f \in C_c^\infty(\mathbb{R}^N)$ be any bounded smooth observable with zero equilibrium mean, $\mathbb{E}[f]_{p_{\mathrm{eq}}} = 0$, and consider the perturbed density $\rho_\varepsilon(\mathbf{x}) := p_{\mathrm{eq}}(\mathbf{x})(1 + \varepsilon f(\mathbf{x}))$ for $|\varepsilon|$ sufficiently small that $1 + \varepsilon f(\mathbf{x}) > 0$ everywhere. Expanding the left-hand side of Eq. (6.87) in a Taylor series in ε using $(1+u)\ln(1+u) = u + \frac{1}{2}u^2 + \mathcal{O}(u^3)$ gives

$$D_{\mathrm{KL}}(\rho_\varepsilon \,\|\, p_{\mathrm{eq}}) = \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x}) \left[\varepsilon f(\mathbf{x}) + \frac{\varepsilon^2}{2} f(\mathbf{x})^2 + \mathcal{O}(\varepsilon^3) \right] = \frac{\varepsilon^2}{2} \operatorname{Var}_{p_{\mathrm{eq}}}(f) + \mathcal{O}(\varepsilon^3). \tag{6.93}$$

Meanwhile, on the right-hand side of Eq. (6.87), $\nabla \ln(\rho_\varepsilon / p_{\mathrm{eq}}) = \nabla \ln(1 + \varepsilon f) = \frac{\varepsilon \nabla f}{1 + \varepsilon f}$, so

$$\frac{1}{2\rho_{\mathrm{LS}}} \mathcal{I}_{\mathbf{D}}(\rho_\varepsilon \,\|\, p_{\mathrm{eq}}) = \frac{\varepsilon^2}{2\rho_{\mathrm{LS}}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \frac{p_{\mathrm{eq}}(\mathbf{x})}{1 + \varepsilon f(\mathbf{x})} \, \|\nabla f(\mathbf{x})\|_{\mathbf{D}(\mathbf{x})}^2 = \frac{\varepsilon^2}{2\rho_{\mathrm{LS}}} \, \mathcal{E}(f, f) + \mathcal{O}(\varepsilon^3). \tag{6.94}$$

Dividing the inequality $D_{\mathrm{KL}}(\rho_\varepsilon \,\|\, p_{\mathrm{eq}}) \le \frac{1}{2\rho_{\mathrm{LS}}} \mathcal{I}_{\mathbf{D}}(\rho_\varepsilon \,\|\, p_{\mathrm{eq}})$ by $\varepsilon^2/2$ and taking the limit $\varepsilon \to 0$ yields $\operatorname{Var}_{p_{\mathrm{eq}}}(f) \le \frac{1}{\rho_{\mathrm{LS}}} \mathcal{E}(f, f)$, which is the Poincaré inequality with $C_{\mathrm{P}} \ge \rho_{\mathrm{LS}}$. □

6.5.3 The Iterated Carré du Champ Γ_2 and the Bakry–Émery Theorem

To decide whether a given energy $E(\mathbf{x})$ satisfies the Logarithmic Sobolev Inequality Eq. (6.87), and to estimate ρ_{LS}, Bakry and Émery [6] differentiated the Relative Fisher Information $\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{eq}})$ a *second* time along the Fokker–Planck flow. This produces a new bilinear form. Just as Γ in Definition 6.11 measures how $\mathcal{L}$ fails to obey the Leibniz rule on products, the *iterated carré du champ* Γ_2 measures how $\mathcal{L}$ fails to commute with Γ.

Definition 6.16 (Iterated Carré du Champ Operator Γ_2)**.** For any Markov diffusion generator $\mathcal{L}$ with carré du champ $\Gamma(f, g)$, the *iterated carré du champ* $\Gamma_2 : C^3(\mathbb{R}^N) \times C^3(\mathbb{R}^N) \to C(\mathbb{R}^N)$ is the symmetric bilinear form defined by

$$\Gamma_2(f, g)(\mathbf{x}) := \frac{1}{2} \Big[\mathcal{L}\Gamma(f, g)(\mathbf{x}) - \Gamma(f, \mathcal{L}g)(\mathbf{x}) - \Gamma(\mathcal{L}f, g)(\mathbf{x}) \Big]. \tag{6.95}$$

On the diagonal $f = g$, we write $\Gamma_2(f) \equiv \Gamma_2(f, f) = \frac{1}{2}\mathcal{L}\Gamma(f, f) - \Gamma(f, \mathcal{L}f)$.

We compute $\Gamma_2(f, f)$ explicitly for the overdamped Langevin generator $\mathcal{L} = D\Delta - \mu \nabla E(\mathbf{x}) \cdot \nabla$ with constant scalar mobility $\mu > 0$ and diffusion constant $D = \mu k_{\mathrm{B}} T$. The resulting formula is the flat-space version of Bochner's identity from Riemannian geometry.

Lemma 6.17 (Bochner–Bakry–Émery Identity)**.** *Let $\mathcal{L} = D\Delta - \mu \nabla E(\mathbf{x}) \cdot \nabla$ on $\mathbb{R}^N$, where $D = \mu k_{\mathrm{B}} T > 0$ and $E \in C^2(\mathbb{R}^N)$. Then for every smooth function $f \in C^3(\mathbb{R}^N)$, the carré du champ is $\Gamma(f, f) = D \,\|\nabla f\|^2$, and the iterated carré du champ is given pointwise by*

$$\Gamma_2(f, f)(\mathbf{x}) = D^2 \left\| \nabla^2 f(\mathbf{x}) \right\|_{\mathrm{F}}^2 + D\mu \, \nabla f(\mathbf{x})^{\top} \nabla^2 E(\mathbf{x}) \, \nabla f(\mathbf{x}), \tag{6.96}$$

where $\|\nabla^2 f(\mathbf{x})\|_{\mathrm{F}}^2 := \sum_{i,j=1}^N \big(\frac{\partial^2 f}{\partial x_i \partial x_j}\big)^2 \ge 0$ is the squared Frobenius (Hilbert–Schmidt) norm of the Hessian of f, and $\nabla^2 E(\mathbf{x})$ is the Hessian matrix of the energy function $E(\mathbf{x})$ with entries $(\nabla^2 E)_{ij} = \frac{\partial^2 E}{\partial x_i \partial x_j}$.

Proof. We compute both terms $\frac{1}{2}\mathcal{L}\Gamma(f,f)$ and $\Gamma(f,\mathcal{L}f)$ in Eq. (6.95) explicitly in components, summing over repeated indices $i,j \in \{1,\dots,N\}$. Since $\Gamma(f,f) = D\sum_{i=1}^{N}(\partial_i f)^2$ (where $\partial_i \equiv \frac{\partial}{\partial x_i}$), we first compute its first and second derivatives:

$$\partial_j\Big(\Gamma(f,f)\Big) = 2D\sum_{i=1}^{N}(\partial_i f)(\partial_j\partial_i f), \tag{6.97}$$

$$\partial_j^2\Big(\Gamma(f,f)\Big) = 2D\sum_{i=1}^{N}\Big[(\partial_j\partial_i f)^2 + (\partial_i f)(\partial_j^2\partial_i f)\Big]. \tag{6.98}$$

Applying $\frac{1}{2}\mathcal{L} = \frac{1}{2}(D\sum_j \partial_j^2 - \mu\sum_j(\partial_j E)\partial_j)$ to $\Gamma(f,f)$ using Eqs. (6.97) and (6.98) gives

$$\frac{1}{2}\mathcal{L}\Gamma(f,f) = D^2\sum_{i,j=1}^{N}(\partial_i\partial_j f)^2 + D^2\sum_{i,j=1}^{N}(\partial_i f)(\partial_j^2\partial_i f) - D\mu\sum_{i,j=1}^{N}(\partial_j E)(\partial_i f)(\partial_j\partial_i f). \tag{6.99}$$

Next, we compute $\Gamma(f,\mathcal{L}f) = D\sum_{i=1}^{N}(\partial_i f)\,\partial_i(\mathcal{L}f)$. Differentiating $\mathcal{L}f = D\sum_j \partial_j^2 f - \mu\sum_j(\partial_j E)(\partial_j f)$ with respect to x_i via the product rule yields

$$\partial_i(\mathcal{L}f) = D\sum_{j=1}^{N}\partial_i\partial_j^2 f - \mu\sum_{j=1}^{N}(\partial_i\partial_j E)(\partial_j f) - \mu\sum_{j=1}^{N}(\partial_j E)(\partial_i\partial_j f). \tag{6.100}$$

Multiplying Eq. (6.100) by $D(\partial_i f)$ and summing over $i = 1,\dots,N$ gives

$$\Gamma(f,\mathcal{L}f) = D^2\sum_{i,j=1}^{N}(\partial_i f)(\partial_i\partial_j^2 f) - D\mu\sum_{i,j=1}^{N}(\partial_i f)(\partial_i\partial_j E)(\partial_j f) - D\mu\sum_{i,j=1}^{N}(\partial_j E)(\partial_i f)(\partial_i\partial_j f). \tag{6.101}$$

Since $f \in C^3(\mathbb{R}^N)$, mixed third partial derivatives commute ($\partial_j^2\partial_i f = \partial_i\partial_j^2 f$). Subtracting Eq. (6.101) from Eq. (6.99), the third-derivative terms $D^2\sum_{i,j}(\partial_i f)(\partial_j^2\partial_i f)$ and the first-order potential gradient terms $-D\mu\sum_{i,j}(\partial_j E)(\partial_i f)(\partial_i\partial_j f)$ cancel. What remains is

$$\Gamma_2(f,f) = \frac{1}{2}\mathcal{L}\Gamma(f,f) - \Gamma(f,\mathcal{L}f) = D^2\sum_{i,j=1}^{N}(\partial_i\partial_j f)^2 + D\mu\sum_{i,j=1}^{N}(\partial_i f)(\partial_i\partial_j E)(\partial_j f),$$

which is Eq. (6.96). □

In the Bochner–Bakry–Émery identity Eq. (6.96), the first term $D^2\left\|\nabla^2 f\right\|_{\mathrm{F}}^2$ is always non-negative. Consequently, if the energy $E(\mathbf{x})$ is *uniformly κ-strongly convex* on $\mathbb{R}^N$, meaning that the smallest eigenvalue of its Hessian matrix $\nabla^2 E(\mathbf{x})$ is bounded below by $\kappa > 0$ at every point $\mathbf{x} \in \mathbb{R}^N$:

$$\nabla^2 E(\mathbf{x}) \succeq \kappa\mathbf{I} \qquad \Longleftrightarrow \qquad \mathbf{u}^\top\nabla^2 E(\mathbf{x})\mathbf{u} \geq \kappa\left\|\mathbf{u}\right\|^2 \quad \forall\,\mathbf{x},\mathbf{u} \in \mathbb{R}^N \tag{6.102}$$

then Eq. (6.96) implies the *Bakry–Émery curvature-dimension inequality* $CD(\mu\kappa,\infty)$:

$$\Gamma_2(f,f)(\mathbf{x}) \geq D\mu\kappa\left\|\nabla f(\mathbf{x})\right\|^2 = \mu\kappa\,\Gamma(f,f)(\mathbf{x}) \qquad \forall\,\mathbf{x} \in \mathbb{R}^N. \tag{6.103}$$

The central theorem of this chapter states that the curvature inequality Eq. (6.103) implies both the Poincaré inequality and the Logarithmic Sobolev Inequality with constant $\rho_{\mathrm{LS}} = \mu\kappa$.

Theorem 6.18 (The Bakry–Émery Γ_2 Theorem [6, 7])**.** *Let $\mathcal{L} = D\Delta - \mu\nabla E(\mathbf{x})\cdot\nabla$ be the overdamped Langevin generator on $\mathbb{R}^N$ with $D = \mu k_{\mathrm{B}}T > 0$ and equilibrium density $p_{\mathrm{eq}}(\mathbf{x}) = Z^{-1}e^{-\beta E(\mathbf{x})}$. Suppose the potential $E \in C^2(\mathbb{R}^N)$ is uniformly κ-strongly convex for some $\kappa > 0$, i.e., $\nabla^2 E(\mathbf{x}) \succeq \kappa \mathbf{I}$ for all $\mathbf{x} \in \mathbb{R}^N$. Then:*

(i) ***Bakry–Émery Curvature Bound:*** *For all $f \in C^3(\mathbb{R}^N)$, $\Gamma_2(f,f) \geq \mu\kappa\,\Gamma(f,f)$.*

(ii) ***Poincaré Spectral Gap Bound:*** *p_{eq} satisfies the Poincaré inequality Eq. (6.82) with spectral gap*

$$\lambda_1 = C_{\mathrm{P}} \geq \mu\kappa = D\beta\kappa. \tag{6.104}$$

(iii) ***Logarithmic Sobolev Inequality Bound:*** *p_{eq} satisfies the Logarithmic Sobolev Inequality Eq. (6.87) with constant*

$$\rho_{\mathrm{LS}} \geq \mu\kappa = D\beta\kappa. \tag{6.105}$$

(iv) ***Dimension-Free Exponential Relaxation:*** *For every initial probability density ρ_0 on $\mathbb{R}^N$ and all $t \geq 0$, the KL divergence and variational free energy contract exponentially at rate $2\mu\kappa$:*

$$D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) \leq e^{-2\mu\kappa t}\, D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}})\,, \qquad \mathcal{F}[\rho_t] - F \leq e^{-2\mu\kappa t}\Big(\mathcal{F}[\rho_0] - F\Big), \tag{6.106}$$

and the characteristic relaxation time of the circuit is bounded above by $\tau_{\mathrm{rel}} = \lambda_1^{-1} \leq (\mu\kappa)^{-1}$, independent of the state-space dimension N.

Proof. Part (i) was proved in Lemma 6.17 and Eq. (6.103), and part (iv) follows from part (iii) by Proposition 6.15(i). Part (ii) is a corollary of part (iii) via Proposition 6.15(ii), but its direct proof is short and instructive, so we give it before the semigroup proof of (iii).

Direct Proof of the Spectral Gap Bound (ii): Let $f \in C^\infty(\mathbb{R}^N)$ be any smooth eigenfunction of $-\mathcal{L}$ with eigenvalue $\lambda > 0$ ($\mathcal{L}f = -\lambda f$). Integrate the curvature inequality $\Gamma_2(f,f) \geq \mu\kappa\,\Gamma(f,f)$ against $p_{\mathrm{eq}}(\mathbf{x})$ over $\mathbb{R}^N$:

$$\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x}) \left[\frac{1}{2}\mathcal{L}\Gamma(f,f) - \Gamma(f,\mathcal{L}f)\right] \geq \mu\kappa \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\,\Gamma(f,f) = \mu\kappa\,\mathcal{E}(f,f). \tag{6.107}$$

By stationarity of p_{eq}, the first term on the left-hand side of Eq. (6.107) vanishes: $\int p_{\mathrm{eq}}\mathcal{L}\Gamma(f,f)\ \mathrm{d}^N x = \int (\mathcal{L}_{\mathrm{FP}} p_{\mathrm{eq}})\Gamma(f,f)\ \mathrm{d}^N x = 0$. In the second term, substituting $\mathcal{L}f = -\lambda f$ yields $-\int p_{\mathrm{eq}}\Gamma(f,-\lambda f)\ \mathrm{d}^N x = \lambda\mathcal{E}(f,f)$. Thus Eq. (6.107) becomes

$$\lambda\,\mathcal{E}(f,f) \geq \mu\kappa\,\mathcal{E}(f,f).$$

Since $\lambda > 0$, f is non-constant, so $\mathcal{E}(f,f) = \lambda\,\mathrm{Var}_{p_{\mathrm{eq}}}(f) > 0$. Dividing by $\mathcal{E}(f,f)$ gives $\lambda \geq \mu\kappa$ for every non-zero eigenvalue, and in particular $\lambda_1 \geq \mu\kappa$.

Semigroup Proof of the Logarithmic Sobolev Inequality (iii): Let $h_0(\mathbf{x}) = \rho_0(\mathbf{x})/p_{\mathrm{eq}}(\mathbf{x}) > 0$ be a smooth positive density ratio, and let $h_t(\mathbf{x}) = (P_t h_0)(\mathbf{x})$ denote its evolution under the backward semigroup $P_t = e^{t\mathcal{L}}$ (which equals $\rho_t(\mathbf{x})/p_{\mathrm{eq}}(\mathbf{x})$ by symmetry of P_t in $L^2(p_{\mathrm{eq}})$). Consider the Relative Fisher Information along the flow at time t:

$$\mathcal{I}(t) := \mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{eq}}) = \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, h_t(\mathbf{x})\,\Gamma(\ln h_t)(\mathbf{x}) = \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\,\frac{\Gamma(h_t,h_t)(\mathbf{x})}{h_t(\mathbf{x})}. \tag{6.108}$$

We compute $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{I}(t)$ directly from $\partial_t h_t = \mathcal{L}h_t$. Differentiating the integrand $\Gamma(h_t)/h_t$ with respect to t gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{I}(t) = \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x}) \left[\frac{2\Gamma(h_t, \mathcal{L}h_t)}{h_t} - \frac{\Gamma(h_t, h_t)\,\mathcal{L}h_t}{h_t^2}\right]. \tag{6.109}$$

Next, for any smooth positive function $h > 0$, apply the generator $\mathcal{L}$ to the quotient $\Gamma(h)/h$. Using the quotient/product rule for $\mathcal{L}$, namely $\mathcal{L}(uv) = u\mathcal{L}v + v\mathcal{L}u + 2\Gamma(u, v)$ with $u = \Gamma(h)$ and $v = 1/h$, together with $\mathcal{L}(1/h) = -\frac{\mathcal{L}h}{h^2} + \frac{2\Gamma(h)}{h^3}$, we have

$$\begin{aligned}\mathcal{L}\left(\frac{\Gamma(h)}{h}\right) &= \frac{\mathcal{L}\Gamma(h)}{h} + \Gamma(h)\left(-\frac{\mathcal{L}h}{h^2} + \frac{2\Gamma(h)}{h^3}\right) + 2\Gamma\left(\Gamma(h), \frac{1}{h}\right) \\ &= \frac{\mathcal{L}\Gamma(h)}{h} - \frac{\Gamma(h)\mathcal{L}h}{h^2} + \frac{2\Gamma(h)^2}{h^3} - \frac{2\Gamma(\Gamma(h), h)}{h^2}.\end{aligned} \tag{6.110}$$

Because $\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \mathcal{L}(\Gamma(h_t)/h_t) = 0$ by stationarity of p_{eq}, we can subtract the integral of Eq. (6.110) from Eq. (6.109). The $-\frac{\Gamma(h_t)\mathcal{L}h_t}{h_t^2}$ terms cancel, while $\frac{2\Gamma(h_t, \mathcal{L}h_t) - \mathcal{L}\Gamma(h_t)}{h_t} = -\frac{2\Gamma_2(h_t)}{h_t}$, leaving

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{I}(t) = -2\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \frac{1}{h_t}\left[\Gamma_2(h_t) - \frac{\Gamma(\Gamma(h_t), h_t)}{h_t} + \frac{\Gamma(h_t)^2}{h_t^2}\right]. \tag{6.111}$$

To simplify the bracket in Eq. (6.111), set $u_t := \ln h_t$, so $h_t = e^{u_t}$. By the chain rule, $\partial_i h_t = h_t \partial_i u_t$ and $\partial_i\partial_j h_t = h_t(\partial_i\partial_j u_t + (\partial_i u_t)(\partial_j u_t))$. Substituting these into the Bochner identity Eq. (6.96) for $\Gamma_2(h_t)$ gives

$$\begin{aligned}\frac{\Gamma_2(h_t)}{h_t^2} &= D^2 \sum_{i,j=1}^{N} \Big(\partial_i\partial_j u_t + (\partial_i u_t)(\partial_j u_t)\Big)^2 + D\mu\, \nabla u_t^\top \nabla^2 E\, \nabla u_t \\ &= \Gamma_2(u_t) + 2D^2 \sum_{i,j=1}^{N} (\partial_i\partial_j u_t)(\partial_i u_t)(\partial_j u_t) + D^2 \left\|\nabla u_t\right\|^4,\end{aligned} \tag{6.112}$$

while $\Gamma(h_t) = h_t^2 \Gamma(u_t)$ implies

$$\frac{\Gamma(\Gamma(h_t), h_t)}{h_t^3} = 2\Gamma(u_t)^2 + \Gamma(\Gamma(u_t), u_t) = 2D^2 \left\|\nabla u_t\right\|^4 + 2D^2 \sum_{i,j} (\partial_i\partial_j u_t)(\partial_i u_t)(\partial_j u_t).$$

Subtracting these in Eq. (6.111) cancels the third- and fourth-order gradient terms in u_t identically, leaving the identity

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{I}(t) = -2\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, h_t(\mathbf{x})\, \Gamma_2(\ln h_t)(\mathbf{x}). \tag{6.113}$$

Now we apply the Bakry–Émery curvature bound (i) to $u_t = \ln h_t$: since $\Gamma_2(\ln h_t) \geq \mu\kappa\, \Gamma(\ln h_t)$, Eq. (6.113) gives the differential inequality for the Relative Fisher Information:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{I}(t) \leq -2\mu\kappa \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, h_t(\mathbf{x})\, \Gamma(\ln h_t)(\mathbf{x}) = -2\mu\kappa\, \mathcal{I}(t) \quad \Longrightarrow \quad \mathcal{I}(t) \leq e^{-2\mu\kappa t}\, \mathcal{I}(0). \tag{6.114}$$

Finally, by the H-theorem Eq. (6.51), $\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) = -\mathcal{I}(t)$. Integrating this relation from $t = 0$ to $t = \infty$ (where $D_{\mathrm{KL}}(\rho_\infty \,\|\, p_{\mathrm{eq}}) = 0$) and applying the bound Eq. (6.114) gives

$$D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}}) = \int_0^\infty \mathrm{d}t\, \mathcal{I}(t) \leq \mathcal{I}(0) \int_0^\infty \mathrm{d}t\, e^{-2\mu\kappa t} = \frac{1}{2\mu\kappa}\, \mathcal{I}_{\mathbf{D}}(\rho_0 \,\|\, p_{\mathrm{eq}}). \tag{6.115}$$

This is the Logarithmic Sobolev Inequality Eq. (6.87) with $\rho_{\mathrm{LS}} = \mu\kappa$, completing the proof. □

Remark 6.19 (Witten Supersymmetric Partner Proof of the Spectral Gap Bound)**.** An operator-theoretic argument gives $\lambda_1 \geq \mu\kappa$ directly and links Section 6.4 to Section 6.5. By Eq. (6.71), the Witten Hamiltonian acting on scalar functions (0-forms) is $\mathcal{H}_{\mathrm{W}} = \mathbf{Q}^\dagger \cdot \mathbf{Q} = \sum_{i=1}^N Q_i^\dagger Q_i$. Reversing the order of Q_i and $Q_j^\dagger$ gives the *supersymmetric partner operator* $\mathcal{H}_{\mathrm{W}}^{(1)}$, which acts on N-component vector fields (1-forms) $\boldsymbol{\xi}(\mathbf{x}) = (\xi_1(\mathbf{x}), \dots, \xi_N(\mathbf{x}))^\top \in L^2(\mathbb{R}^N; \mathbb{R}^N)$. Using the commutation relation

$$[Q_i, Q_j^\dagger] = D\left[\frac{\partial}{\partial x_i} + \frac{\beta}{2}\frac{\partial E}{\partial x_i},\, -\frac{\partial}{\partial x_j} + \frac{\beta}{2}\frac{\partial E}{\partial x_j}\right] = D\beta\frac{\partial^2 E(\mathbf{x})}{\partial x_i \partial x_j} = \mu(\nabla^2 E(\mathbf{x}))_{ij}, \tag{6.116}$$

the 1-form Witten Hamiltonian $\left(\mathcal{H}_{\mathrm{W}}^{(1)}\boldsymbol{\xi}\right)_i := \sum_{j=1}^N (\delta_{ij}\mathcal{H}_{\mathrm{W}} + [Q_i, Q_j^\dagger])\xi_j$ takes the explicit matrix form

$$\mathcal{H}_{\mathrm{W}}^{(1)} = \mathcal{H}_{\mathrm{W}} \otimes \mathbf{I}_N + \mu\, \nabla^2 E(\mathbf{x}). \tag{6.117}$$

Furthermore, $\mathbf{Q}$ intertwines $\mathcal{H}_{\mathrm{W}}$ and $\mathcal{H}_{\mathrm{W}}^{(1)}$: for any excited eigenfunction ψ_n of $\mathcal{H}_{\mathrm{W}}$ with eigenvalue $\lambda_n > 0$ ($\mathcal{H}_{\mathrm{W}}\psi_n = \lambda_n\psi_n$), the non-zero vector field $\boldsymbol{\xi}_n := \mathbf{Q}\psi_n \neq \mathbf{0}$ satisfies

$$\mathcal{H}_{\mathrm{W}}^{(1)}(\mathbf{Q}\psi_n) = \mathbf{Q}(\mathbf{Q}^\dagger \cdot \mathbf{Q}\psi_n) = \mathbf{Q}(\mathcal{H}_{\mathrm{W}}\psi_n) = \lambda_n(\mathbf{Q}\psi_n). \tag{6.118}$$

Taking the $L^2(\mathbb{R}^N; \mathbb{R}^N)$ inner product of Eq. (6.118) with $\boldsymbol{\xi}_n = \mathbf{Q}\psi_n$ and using Eq. (6.117) together with the positive semi-definiteness of $\mathcal{H}_{\mathrm{W}} \succeq 0$ gives

$$\lambda_n \left\|\boldsymbol{\xi}_n\right\|_{L^2}^2 = \langle \boldsymbol{\xi}_n | \mathcal{H}_{\mathrm{W}} \otimes \mathbf{I}_N | \boldsymbol{\xi}_n \rangle + \mu \int_{\mathbb{R}^N} \mathrm{d}^N x\, \boldsymbol{\xi}_n(\mathbf{x})^\top \nabla^2 E(\mathbf{x})\, \boldsymbol{\xi}_n(\mathbf{x}) \geq \mu\kappa \left\|\boldsymbol{\xi}_n\right\|_{L^2}^2, \tag{6.119}$$

which gives $\lambda_n \geq \mu\kappa$ for all $n \geq 1$.

6.5.4 The Holley–Stroock Bounded Perturbation Principle

Many energies are not globally strongly convex but can be written as a strongly convex reference potential $E_0(\mathbf{x})$ (with $\nabla^2 E_0(\mathbf{x}) \succeq \kappa\mathbf{I}$) plus a bounded non-convex perturbation $\delta E(\mathbf{x})$ that carves local wells. A classical theorem of Holley and Stroock quantifies how such bounded non-convexity degrades the Logarithmic Sobolev constant.

Theorem 6.20 (Holley–Stroock Perturbation Theorem)**.** *Suppose $E(\mathbf{x}) = E_0(\mathbf{x}) + \delta E(\mathbf{x})$, where the reference Boltzmann measure $p_0(\mathbf{x}) = Z_0^{-1} e^{-\beta E_0(\mathbf{x})}$ satisfies the Logarithmic Sobolev Inequality Eq. (6.88) with constant $\rho_{\mathrm{LS},0} > 0$, and the perturbation $\delta E \in C_b^1(\mathbb{R}^N)$ has finite oscillation*

$$\operatorname{osc}(\delta E) := \sup_{\mathbf{x}\in\mathbb{R}^N} \delta E(\mathbf{x}) - \inf_{\mathbf{x}\in\mathbb{R}^N} \delta E(\mathbf{x}) < \infty. \tag{6.120}$$

Then the perturbed Boltzmann distribution $p_{\mathrm{eq}}(\mathbf{x}) = Z^{-1} e^{-\beta\left(E_0(\mathbf{x}) + \delta E(\mathbf{x})\right)}$ satisfies the Logarithmic Sobolev Inequality with constant

$$\rho_{\mathrm{LS}} \geq \rho_{\mathrm{LS},0}\, \exp\Big(-\beta \operatorname{osc}(\delta E)\Big). \tag{6.121}$$

Proof. Let $\delta E_{\min} := \inf_{\mathbf{x}} \delta E(\mathbf{x})$ and $\delta E_{\max} := \sup_{\mathbf{x}} \delta E(\mathbf{x})$. We use a variational characterization of the entropy functional $\mathrm{Ent}_p(g^2) = \int p g^2 \ln(g^2/\mathbb{E}[g^2]_p)\, \mathrm{d}^N x$: for any function $g(\mathbf{x})$,

$$\mathrm{Ent}_p(g^2) = \inf_{c>0} \int_{\mathbb{R}^N} \mathrm{d}^N x\, p(\mathbf{x}) \Big[g(\mathbf{x})^2 \ln g(\mathbf{x})^2 - g(\mathbf{x})^2 \ln c - g(\mathbf{x})^2 + c \Big], \tag{6.122}$$

where the infimum is attained at $c = \mathbb{E}[g^2]_p$ and the integrand in brackets, $\phi(g^2, c) := g^2 \ln(g^2/c) - g^2 + c \geq 0$, is pointwise non-negative because $u \ln u - u + 1 \geq 0$ for all $u = g^2/c > 0$.

Now compare the density $p_{\mathrm{eq}}(\mathbf{x})$ to $p_0(\mathbf{x})$: since $Z = \int e^{-\beta(E_0+\delta E)}\, \mathrm{d}^N x \geq e^{-\beta\delta E_{\max}} Z_0$ and $Z \leq e^{-\beta\delta E_{\min}} Z_0$, we have the pointwise upper and lower bounds

$$e^{-\beta\,\mathrm{osc}(\delta E)}\, p_0(\mathbf{x}) \leq \frac{e^{-\beta\delta E(\mathbf{x})} Z_0}{Z}\, p_0(\mathbf{x}) = p_{\mathrm{eq}}(\mathbf{x}) \leq e^{+\beta\,\mathrm{osc}(\delta E)}\, p_0(\mathbf{x}). \tag{6.123}$$

Using the non-negativity of the bracketed integrand in Eq. (6.122) with $c = \mathbb{E}[g^2]_{p_0}$, the upper bound $e^{-\beta\delta E(\mathbf{x})} \leq e^{-\beta\delta E_{\min}}$ and lower bound $e^{-\beta\delta E(\mathbf{x})} \geq e^{-\beta\delta E_{\max}}$ yield

$$\begin{aligned}\mathrm{Ent}_{p_{\mathrm{eq}}}(g^2) &\leq \frac{Z_0}{Z}\, e^{-\beta\delta E_{\min}}\, \mathrm{Ent}_{p_0}(g^2) \leq \frac{Z_0}{Z}\, e^{-\beta\delta E_{\min}}\, \frac{2}{\rho_{\mathrm{LS},0}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_0(\mathbf{x})\, \|\nabla g\|_{\mathbf{D}}^2 \\ &\leq e^{\beta(\delta E_{\max} - \delta E_{\min})}\, \frac{2}{\rho_{\mathrm{LS},0}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \|\nabla g\|_{\mathbf{D}}^2\,,\end{aligned} \tag{6.124}$$

which establishes Eq. (6.121). □

The exponential penalty $e^{-\beta\,\mathrm{osc}(\delta E)}$ in Eq. (6.121) anticipates the Arrhenius–Kramers barrier factor $e^{-\beta\Delta E}$ derived in Chapter 7.

Heuristic Derivation: Why Positive Curvature Forces Exponential Score Contraction

The identity $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{I}(t) = -2\mathbb{E}[\Gamma_2(\ln h_t)]_{\rho_t}$ in Eq. (6.113) has a simple mechanism. The function $u_t(\mathbf{x}) = \ln(\rho_t(\mathbf{x})/p_{\mathrm{eq}}(\mathbf{x}))$ is the log-density error, and its gradient $\mathbf{s}_{\mathrm{err}}(\mathbf{x}, t) = \nabla u_t(\mathbf{x}) = \nabla \ln \rho_t(\mathbf{x}) + \beta \nabla E(\mathbf{x})$ is the local mismatch in the score function (or $-\mathbf{D}^{-1}\mathbf{v}(\mathbf{x}, t)$, where $\mathbf{v} = \mathbf{J}/\rho_t$ is the probability fluid velocity).

When two nearby probability fluid elements at $\mathbf{x}$ and $\mathbf{x} + \delta\mathbf{x}$ slide down a strongly convex bowl $E(\mathbf{x})$ with $\nabla^2 E(\mathbf{x}) \succeq \kappa \mathbf{I}$, the deterministic restoring drift $-\mu\nabla E(\mathbf{x})$ compresses their separation at rate $\delta\dot{\mathbf{x}} = -\mu\nabla^2 E(\mathbf{x})\, \delta\mathbf{x}$, while thermal diffusion $D\Delta$ simultaneously smooths out second-order kinks ($\|\nabla^2 u_t\|_{\mathrm{F}}^2$). Together, the confining curvature $\mu\nabla^2 E$ and the diffusive smoothing $D\Delta$ damp the squared score error $\|\nabla u_t\|^2$ at rate at least $2\mu\kappa$. Conversely, if the landscape contains a saddle barrier where $\nabla^2 E(\mathbf{x})$ has a *negative* eigenvalue $-\omega_{\mathrm{sad}}^2 < 0$, nearby trajectories on opposite sides of the ridge are pulled *apart* exponentially by the drift, destroying the global contraction bound Eq. (6.114).

Hardware Realization: Bakry–Émery Curvature Restoration in Denoising Thermodynamic Models

Theorem 6.18 explains how the denoising thermodynamic models (DTMs) proposed by Jelinčič et al. [56] sidestep the Fundamental Expressivity–Mixing Dilemma of Section 3.5.

Suppose we wish to sample a multimodal data distribution $p_{\mathrm{data}}(\mathbf{x}_0) \propto e^{-\beta E_0(\mathbf{x}_0)}$ whose energy $E_0(\mathbf{x}_0)$ has deep wells separated by saddles, where $\lambda_{\min}(\nabla^2 E_0(\mathbf{x}_0)) = -L_0 < 0$. A single Langevin circuit run on E_0 has no Bakry–Émery guarantee, because its curvature is negative in places, and it is exposed to the exponentially slow barrier crossing of Chapter 7. In a DTM (developed in Chapter 16), generation is split into a chain of T conditional sampling steps $\mathbf{x}_T \to \mathbf{x}_{T-1} \to \cdots \to \mathbf{x}_0$ (in this box T counts steps, not temperature). At step t, given the clamped state $\mathbf{x}_t$ from the previous stage, the hardware circuit relaxes in the conditional

energy landscape

$$E_{\mathrm{tot}}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \frac{\alpha_t}{2} \|\mathbf{x}_{t-1} - c_t \mathbf{x}_t\|^2 + \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t), \tag{6.125}$$

where $\frac{\alpha_t}{2} \|\mathbf{x}_{t-1} - c_t \mathbf{x}_t\|^2$ is a physical quadratic coupling to the clamped state $\mathbf{x}_t$ with stiffness $\alpha_t > 0$, and $\mathcal{E}_{\theta,t-1}$ is a learned non-convex interaction energy whose Hessian is bounded below by $\nabla^2_{\mathbf{x}_{t-1}} \mathcal{E}_{\theta,t-1} \succeq -L_{\mathcal{E}} \mathbf{I}$. Taking the Hessian of Eq. (6.125) with respect to the active variable $\mathbf{x}_{t-1}$ gives

$$\nabla^2_{\mathbf{x}_{t-1}} E_{\mathrm{tot}}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \alpha_t \mathbf{I} + \nabla^2_{\mathbf{x}_{t-1}} \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) \succeq (\alpha_t - L_{\mathcal{E}}) \mathbf{I}. \tag{6.126}$$

Whenever the step stiffness α_t exceeds the non-convexity scale $L_{\mathcal{E}}$, the curvature $\kappa_t := \alpha_t - L_{\mathcal{E}}$ is positive. By Theorem 6.18, the conditional distribution $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ then satisfies the Logarithmic Sobolev Inequality with $\rho_{\mathrm{LS},t} \geq \mu(\alpha_t - L_{\mathcal{E}})$, and the circuit relaxes at each denoising step on a time scale of order $[\mu(\alpha_t - L_{\mathcal{E}})]^{-1}$, independent of the number of nodes.

6.6 Worked Computation: Witten Spectrum of the Ornstein–Uhlenbeck Sampler and Curvature Breakdown

The following computation applies the tools of this chapter to both running examples. We first solve the Witten problem for the RC sampler exactly with ladder operators. We then follow a one-dimensional quartic potential as its quadratic confinement is weakened until it becomes the double-well sampler, and locate the point where Bakry–Émery curvature fails.

Worked computation 6.21 (Witten Spectrum of the Ornstein–Uhlenbeck Sampler and Curvature Breakdown at the Bifurcation Threshold)**.** Consider first the RC sampler at equilibrium, in the engineered-noise convention of Section 5.6 with $\tau_0 = 1$: N nodes with uniform node capacitance $C > 0$, symmetric positive-definite coupling matrix $\mathbf{G} = \mathbf{G}^\top \succ \mathbf{0}$, mobility $\mu = 1/C$, diffusion coefficient $D = k_{\mathrm{B}}T/C = (\beta C)^{-1}$, and quadratic electrostatic energy

$$E(\mathbf{v}) = \frac{1}{2} \mathbf{v}^\top \mathbf{G} \mathbf{v}, \qquad \mathbf{v} \in \mathbb{R}^N. \tag{6.127}$$

In the second part of this computation, we specialize to a single mode $x \in \mathbb{R}$ and augment the potential with a stabilizing quartic term and a tunable quadratic coefficient $\alpha - b$:

$$E(x; \alpha) = \frac{a}{4} x^4 + \frac{\alpha - b}{2} x^2 \qquad (a > 0,\ b > 0,\ \alpha \in \mathbb{R}), \tag{6.128}$$

a one-dimensional caricature of a single denoising stage (Section 6.5.4) in which the confinement stiffness α is swept across the critical value $\alpha_c = b$. At $\alpha = 0$ this is the untilted double-well sampler.

Part 1: Exact Witten Hamiltonian and Ladder Operators for the Multidimensional OU Sampler. For the quadratic energy Eq. (6.127), we have $\nabla E(\mathbf{v}) = \mathbf{G}\mathbf{v}$, $\nabla^2 E(\mathbf{v}) = \mathbf{G}$, and $\Delta E(\mathbf{v}) = \mathrm{tr}(\mathbf{G})$. Since $\mathbf{G} = \mathbf{G}^\top \succ \mathbf{0}$, there exists an orthogonal matrix $\mathbf{O} \in \mathrm{O}(N)$ that diagonalizes $\mathbf{G}$ into its positive eigenconductances $g_1, g_2, \ldots, g_N > 0$, ordered such that $0 < g_1 \leq g_2 \leq \cdots \leq g_N$:

$$\mathbf{O}^\top \mathbf{G} \mathbf{O} = \mathrm{diag}(g_1, g_2, \ldots, g_N), \qquad \mathbf{u} := \mathbf{O}^\top \mathbf{v} = (u_1, u_2, \ldots, u_N)^\top. \tag{6.129}$$

In the normal-mode voltage coordinates $\mathbf{u}$, the energy separates into N independent harmonic oscillators $E(\mathbf{u}) = \sum_{k=1}^{N} \frac{1}{2} g_k u_k^2$, and the characteristic RC relaxation rate of mode k is

$$\gamma_k := \frac{g_k}{C} = \mu g_k = D\beta g_k > 0, \qquad k = 1, \dots, N. \tag{6.130}$$

Substituting $\|\nabla E(\mathbf{u})\|^2 = \sum_{k=1}^{N} g_k^2 u_k^2$ and $\Delta E(\mathbf{u}) = \sum_{k=1}^{N} g_k$ into Eq. (6.63), we obtain the exact Witten potential $V_{\mathrm{W}}(\mathbf{u})$ and Witten Schrödinger Hamiltonian $\mathcal{H}_{\mathrm{W}}$:

$$V_{\mathrm{W}}(\mathbf{u}) = \sum_{k=1}^{N} \left(\frac{D\beta^2 g_k^2}{4} u_k^2 - \frac{D\beta g_k}{2} \right) = \sum_{k=1}^{N} \left(\frac{\gamma_k^2}{4D} u_k^2 - \frac{\gamma_k}{2} \right), \tag{6.131}$$

$$\mathcal{H}_{\mathrm{W}} = \sum_{k=1}^{N} \mathcal{H}_{\mathrm{W},k} = \sum_{k=1}^{N} \left(-D \frac{\partial^2}{\partial u_k^2} + \frac{\gamma_k^2}{4D} u_k^2 - \frac{\gamma_k}{2} \right). \tag{6.132}$$

Each normal-mode Hamiltonian $\mathcal{H}_{\mathrm{W},k}$ in Eq. (6.132) is a quantum harmonic oscillator with mass $m_{\mathrm{eff}} = 1/(2D)$ and oscillator frequency γ_k, shifted downward in energy by its exact zero-point energy $\frac{1}{2}\gamma_k$.

To diagonalize $\mathcal{H}_{\mathrm{W},k}$ algebraically, we rescale the supersymmetric operators $Q_k, Q_k^\dagger$ from Eq. (6.68) by $1/\sqrt{\gamma_k}$ to define the dimensionless *bosonic annihilation and creation (ladder) operators*:

$$\hat{a}_k := \frac{1}{\sqrt{\gamma_k}} Q_k = \sqrt{\frac{D}{\gamma_k}} \frac{\partial}{\partial u_k} + \frac{1}{2} \sqrt{\frac{\gamma_k}{D}}\, u_k, \qquad \hat{a}_k^\dagger := \frac{1}{\sqrt{\gamma_k}} Q_k^\dagger = -\sqrt{\frac{D}{\gamma_k}} \frac{\partial}{\partial u_k} + \frac{1}{2} \sqrt{\frac{\gamma_k}{D}}\, u_k. \tag{6.133}$$

Their commutator, acting on a smooth function $\psi(\mathbf{u})$, is

$$\begin{aligned}
[\hat{a}_k, \hat{a}_l^\dagger]\psi &= \left(\sqrt{\frac{D}{\gamma_k}} \frac{\partial}{\partial u_k} + \frac{1}{2}\sqrt{\frac{\gamma_k}{D}}\, u_k \right) \left(-\sqrt{\frac{D}{\gamma_l}} \frac{\partial \psi}{\partial u_l} + \frac{1}{2}\sqrt{\frac{\gamma_l}{D}}\, u_l \psi \right) \\
&\quad - \left(-\sqrt{\frac{D}{\gamma_l}} \frac{\partial}{\partial u_l} + \frac{1}{2}\sqrt{\frac{\gamma_l}{D}}\, u_l \right) \left(\sqrt{\frac{D}{\gamma_k}} \frac{\partial \psi}{\partial u_k} + \frac{1}{2}\sqrt{\frac{\gamma_k}{D}}\, u_k \psi \right) \\
&= \frac{1}{2}\sqrt{\frac{\gamma_l}{\gamma_k}} \frac{\partial (u_l \psi)}{\partial u_k} - \frac{1}{2}\sqrt{\frac{\gamma_k}{\gamma_l}} u_k \frac{\partial \psi}{\partial u_l} + \frac{1}{2}\sqrt{\frac{\gamma_k}{\gamma_l}} \frac{\partial (u_k \psi)}{\partial u_l} - \frac{1}{2}\sqrt{\frac{\gamma_l}{\gamma_k}} u_l \frac{\partial \psi}{\partial u_k} \\
&= \delta_{kl}\, \psi,
\end{aligned} \tag{6.134}$$

so $[\hat{a}_k, \hat{a}_l^\dagger] = \delta_{kl}$ and $[\hat{a}_k, \hat{a}_l] = [\hat{a}_k^\dagger, \hat{a}_l^\dagger] = 0$. In terms of $\hat{a}_k$ and $\hat{a}_k^\dagger$, the Witten Hamiltonian Eq. (6.132) is simply

$$\mathcal{H}_{\mathrm{W}} = \sum_{k=1}^{N} \gamma_k\, \hat{a}_k^\dagger \hat{a}_k. \tag{6.135}$$

Part 2: Multi-Index Spectrum, Hermite Eigenbasis, and Tightness of the Bakry–Émery Bound. From Eq. (6.135) and the canonical commutation relations Eq. (6.134), the complete set of orthonormal eigenfunctions of $\mathcal{H}_{\mathrm{W}}$ in $L^2(\mathbb{R}^N)$ is labeled by multi-indices $\mathbf{n} = (n_1, n_2, \dots, n_N) \in \mathbb{N}_0^N$ ($n_k \in \{0, 1, 2, \dots\}$):

$$\psi_{\mathbf{n}}(\mathbf{u}) = \left(\prod_{k=1}^{N} \frac{(\hat{a}_k^\dagger)^{n_k}}{\sqrt{n_k!}} \right) \psi_{\mathbf{0}}(\mathbf{u}), \qquad \mathcal{H}_{\mathrm{W}} \psi_{\mathbf{n}} = \lambda_{\mathbf{n}} \psi_{\mathbf{n}}, \qquad \lambda_{\mathbf{n}} = \sum_{k=1}^{N} n_k \gamma_k. \tag{6.136}$$

Here the vacuum state $\psi_{\mathbf{0}}(\mathbf{u})$ solved by $\hat{a}_k\psi_{\mathbf{0}} = 0$ is the square root of the equilibrium Gaussian distribution:

$$\psi_{\mathbf{0}}(\mathbf{u}) = p_{\mathrm{eq}}(\mathbf{u})^{1/2} = \prod_{k=1}^{N}\left(\frac{\gamma_k}{2\pi D}\right)^{1/4}\exp\!\left(-\frac{\gamma_k u_k^2}{4D}\right). \tag{6.137}$$

To express the backward-generator eigenfunctions $f_{\mathbf{n}}(\mathbf{u}) = p_{\mathrm{eq}}(\mathbf{u})^{-1/2}\psi_{\mathbf{n}}(\mathbf{u})$ explicitly as polynomials in the voltages u_k, we introduce the dimensionless standardized normal-mode coordinates $z_k := \sqrt{\frac{\gamma_k}{D}}\,u_k = \sqrt{\beta g_k}\,u_k$ (which have unit equilibrium variance $\mathbb{E}[z_k^2]_{p_{\mathrm{eq}}} = 1$). Using the second relation in Eq. (6.69), the action of $\hat{a}_k^\dagger$ on $\psi = p_{\mathrm{eq}}^{1/2} f$ translates to

$$p_{\mathrm{eq}}(\mathbf{u})^{-1/2}\hat{a}_k^\dagger\Big(p_{\mathrm{eq}}(\mathbf{u})^{1/2}f(\mathbf{z})\Big) = -\sqrt{\frac{D}{\gamma_k}}\,e^{z_k^2/2}\frac{\partial}{\partial u_k}\Big(e^{-z_k^2/2}f(\mathbf{z})\Big) = -e^{z_k^2/2}\frac{\partial}{\partial z_k}\Big(e^{-z_k^2/2}f(\mathbf{z})\Big). \tag{6.138}$$

Applying Eq. (6.138) n_k times to $f_{\mathbf{0}}(\mathbf{z}) = 1$ produces Rodrigues' formula for the *probabilists' Hermite polynomials* $\mathrm{He}_n(z) := (-1)^n e^{z^2/2}\frac{\mathrm{d}^n}{\mathrm{d}z^n}e^{-z^2/2}$ (with $\mathrm{He}_0(z) = 1$, $\mathrm{He}_1(z) = z$, $\mathrm{He}_2(z) = z^2 - 1$, $\mathrm{He}_3(z) = z^3 - 3z$):

$$f_{\mathbf{n}}(\mathbf{u}) = \prod_{k=1}^{N}\frac{1}{\sqrt{n_k!}}\,\mathrm{He}_{n_k}\!\left(\sqrt{\beta g_k}\,u_k\right), \qquad \phi_{\mathbf{n}}(\mathbf{u}) = p_{\mathrm{eq}}(\mathbf{u})\,f_{\mathbf{n}}(\mathbf{u}). \tag{6.139}$$

The spectral gap is the lowest non-zero eigenvalue in Eq. (6.136), obtained by exciting a single quantum $n_1 = 1$ in the slowest normal mode $k = 1$ (with $n_2 = \cdots = n_N = 0$), yielding

$$\lambda_1 = \gamma_1 = \frac{g_{\min}}{C} = \mu\lambda_{\min}(\mathbf{G}). \tag{6.140}$$

Compare this exact eigenvalue with the Bakry–Émery theorem (Theorem 6.18). For $E(\mathbf{v}) = \frac{1}{2}\mathbf{v}^\top\mathbf{G}\mathbf{v}$, the Hessian is constant everywhere: $\nabla^2 E(\mathbf{v}) = \mathbf{G} \succeq g_{\min}\mathbf{I}$, so the strong convexity constant is $\kappa = g_{\min} = \lambda_{\min}(\mathbf{G})$. Theorem 6.18 gives $\lambda_1 \geq \mu\kappa = \mu g_{\min}$ and $\rho_{\mathrm{LS}} \geq \mu g_{\min}$. Comparing with Eq. (6.140) and Proposition 6.15(ii) ($\lambda_1 \geq \rho_{\mathrm{LS}} \geq \mu g_{\min} = \lambda_1$), we see that for any Gaussian Ornstein–Uhlenbeck sampler, the Bakry–Émery Log-Sobolev and Poincaré bounds are *tight*:

$$\rho_{\mathrm{LS}} = C_{\mathrm{P}} = \lambda_1 = \frac{g_{\min}}{C} = \frac{1}{\tau_{RC}^{\max}}, \tag{6.141}$$

where $\tau_{RC}^{\max} = C/g_{\min}$ is the slowest RC time constant of the network.

Part 3: Tracking the Bifurcation from Convex Single Well to Bistable Double Well (α Across b). Now consider the 1D quartic potential Eq. (6.128), $E(x;\alpha) = \frac{a}{4}x^4 + \frac{\alpha-b}{2}x^2$. Its first and second derivatives are

$$E'(x;\alpha) = ax^3 + (\alpha - b)x, \qquad E''(x;\alpha) = 3ax^2 + (\alpha - b). \tag{6.142}$$

By Eq. (6.63), the Witten potential $V_{\mathrm{W}}(x;\alpha)$ is a sixth-degree polynomial in x:

$$V_{\mathrm{W}}(x;\alpha) = D\left[\frac{\beta^2}{4}x^2(ax^2 + \alpha - b)^2 - \frac{\beta}{2}(3ax^2 + \alpha - b)\right]. \tag{6.143}$$

There are three regimes as the confinement stiffness α is lowered across the threshold $\alpha_c = b$:

1. **Strongly Convex Regime ($\alpha > b$):** Here $\kappa(\alpha) := \inf_{x\in\mathbb{R}} E''(x;\alpha) = E''(0;\alpha) = \alpha - b > 0$. The potential $E(x;\alpha)$ has a unique global minimum at $x = 0$, and Theorem 6.18 guarantees a strictly positive Log-Sobolev constant and spectral gap bounded below by

$$\lambda_1(\alpha) \geq \rho_{\mathrm{LS}}(\alpha) \geq \mu(\alpha - b) > 0, \qquad \tau_{\mathrm{rel}}(\alpha) \leq \frac{1}{\mu(\alpha - b)}. \tag{6.144}$$

   In fact, because the positive quartic term $+3ax^2$ in $E''(x;\alpha)$ adds extra confining curvature away from $x = 0$, the true spectral gap $\lambda_1(\alpha)$ is strictly larger than $\mu(\alpha - b)$.

2. **Critical Pitchfork Bifurcation Point ($\alpha = b$):** At $\alpha = b$, the quadratic term vanishes: $E(x;b) = \frac{a}{4}x^4$ and $E''(x;b) = 3ax^2$. Because $E''(0;b) = 0$, uniform strong convexity fails at the single point $x = 0$ ($\kappa(b) = 0$). However, the Witten potential Eq. (6.143) at $\alpha = b$ is

$$V_{\mathrm{W}}(x;b) = D\left(\frac{\beta^2 a^2}{4}x^6 - \frac{3\beta a}{2}x^2\right).$$

   By rescaling $x = (\beta a)^{-1/4}y$ in the eigenvalue equation $D(-\partial_x^2 + \frac{\beta^2a^2}{4}x^6 - \frac{3\beta a}{2}x^2)\psi_1 = \lambda_1\psi_1$, all dependence on β and a factors out in front of the dimensionless operator $-\partial_y^2 + \frac{1}{4}y^6 - \frac{3}{2}y^2$, which gives the scaling law at the critical point:

$$\lambda_1(\alpha = b) = c_1\, D(\beta a)^{1/2} = c_1\, \mu\sqrt{k_{\mathrm{B}}Ta} > 0, \tag{6.145}$$

   where $c_1 \approx 1.369$ is the first excited eigenvalue of the dimensionless sextic oscillator. At criticality the gap is therefore still positive and scales as a power of temperature, $\lambda_1 \propto T^{1/2}$.

3. **Non-Convex Bistable Regime ($\alpha < b$):** Once $\alpha < b$, the curvature at the origin becomes strictly negative: $E''(0;\alpha) = -(b - \alpha) < 0$. The Bakry–Émery curvature condition fails on the entire spinodal interval

$$\mathcal{I}_{\mathrm{neg}} = \{x \in \mathbb{R} : E''(x;\alpha) < 0\} = \left(-\sqrt{\frac{b-\alpha}{3a}}, +\sqrt{\frac{b-\alpha}{3a}}\right). \tag{6.146}$$

   Simultaneously, $E(x;\alpha)$ splits into two symmetric minima at $x_\pm = \pm\sqrt{(b-\alpha)/a}$ with positive curvature $E''(x_\pm;\alpha) = 2(b - \alpha) > 0$, separated by a central saddle barrier at $x = 0$ of height

$$\Delta E(\alpha) = E(0;\alpha) - E(x_\pm;\alpha) = \frac{(b-\alpha)^2}{4a}. \tag{6.147}$$

   For $\alpha < b$ the Witten potential $V_{\mathrm{W}}(x;\alpha)$ of Eq. (6.143) has three features:

   - At the two well bottoms $x = x_\pm$, $E'(x_\pm) = 0$ and $E''(x_\pm) = 2(b - \alpha) > 0$, so $V_{\mathrm{W}}(x_\pm) = -D\beta(b - \alpha) < 0$.
   - At the saddle $x = 0$, $E'(0) = 0$ and $E''(0) = -(b - \alpha) < 0$, so $V_{\mathrm{W}}(0) = +\frac{1}{2}D\beta(b - \alpha) > 0$.
   - At the intermediate points $x^2 = \frac{b-\alpha}{3a}$ inside the barrier flanks, $(E')^2 = \frac{4(b-\alpha)^3}{27a}$ produces two tall $\mathcal{O}(\beta^2)$ potential barriers of height $V_{\mathrm{W}} \approx \frac{D\beta^2(b-\alpha)^3}{27a}$ separating the two negative wells at $x_\pm$.

As Chapter 7 shows (Worked computation 7.16), tunneling of the eigenfunction $\psi_1(x)$ through this $\mathcal{O}(\beta^2)$ Witten barrier makes the spectral gap collapse exponentially with the barrier height Eq. (6.147):

$$\lambda_1(\alpha) \simeq \frac{\sqrt{2}\,\mu(b-\alpha)}{\pi} \exp\left(-\frac{\beta(b-\alpha)^2}{4a}\right) \qquad (\alpha < b,\ \beta\Delta E(\alpha) \gg 1). \tag{6.148}$$

Equations Eqs. (6.144), (6.145) and (6.148) together describe the transition from rapid $\mathcal{O}(\tau_{RC})$ Bakry–Émery relaxation ($\alpha > b$) to exponential Kramers metastability ($\alpha < b$).

Summary

Proved:

- The Fokker–Planck equation as the $L^2(\mathbb{R}^N)$ adjoint of the backward generator, $\mathcal{L}_{\mathrm{FP}} = \mathcal{L}^\dagger$, with the probability current $\mathbf{J}$ (Theorem 6.2).
- Four equivalent forms of detailed balance: zero current, the potential condition, self-adjointness of $\mathcal{L}$ in $L^2(p_{\mathrm{eq}})$, and two-time reversibility (Theorem 6.4); the Helmholtz–Hodge split of a non-equilibrium drift, worked out for the linear Gaussian NESS (Example 6.5).
- The H-theorem: $\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}) = -\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{ss}}) \le 0$ for any steady state, and $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t] = -k_{\mathrm{B}}T\mathcal{I}_{\mathbf{D}}(\rho_t \,\|\, p_{\mathrm{eq}})$ at equilibrium (Theorem 6.7).
- The Witten Hamiltonian $\mathcal{H}_{\mathrm{W}} = \mathbf{Q}^\dagger \cdot \mathbf{Q}$ and the spectral expansion of reversible Fokker–Planck dynamics (Theorems 6.8 and 6.9).
- The Bakry–Émery theorem: $\nabla^2 E \succeq \kappa \mathbf{I}$ implies $\lambda_1 \ge \mu\kappa$ and $\rho_{\mathrm{LS}} \ge \mu\kappa$ (Theorem 6.18).
- The exact spectrum of the RC sampler, for which the Bakry–Émery bound is tight (Worked computation 6.21).

Assumed:

- Smooth, confining potentials, so that boundary terms vanish in every integration by parts.
- Uniform ellipticity, $\mathbf{D}(\mathbf{x}) \succ \mathbf{0}$.
- Discreteness of the spectrum for confining V_{W} (Rellich's criterion), cited rather than proved.

Taken up later:

- The exponentially small gap $\lambda_1 \propto e^{-\beta\Delta E}$ of the double well (Chapter 7).
- The Fokker–Planck equation as a Wasserstein gradient flow of $\mathcal{F}[\rho]$ (Chapter 10).
- The non-reciprocal drift and shot-noise diffusion of subthreshold CMOS networks (Chapter 13), and denoising thermodynamic models (Chapter 16).

Open:

- How much a designed skew-symmetric drift $\mathbf{\Omega}$ can speed up mixing in non-equilibrium CMOS circuits, and at what cost in housekeeping heat $\dot{\mathcal{Q}}_{\mathrm{hk}}$.

Bibliographic notes

Standard physics treatments of the Fokker–Planck equation, probability currents, and eigenfunction expansions are given by Risken [90], Gardiner [39], and van Kampen [112]. Pavliotis [85] provides an accessible mathematical exposition bridging stochastic differential equations, the Helmholtz–Hodge decomposition of reversible vs. non-equilibrium diffusions, and spectral theory. The supersymmetric factorization of the symmetrized Fokker–Planck operator and its connection to Morse theory were introduced by Witten [118]. The Γ_2 curvature-dimension calculus and its application to Logarithmic Sobolev and Poincaré inequalities originate with Bakry and Émery [6]; the standard modern monograph is Bakry et al. [7]. The role of non-equilibrium steady-state probability currents and Lyapunov covariance equations in subthreshold CMOS circuits is developed in Freitas et al. [37] and Freitas et al. [38], while the use of quadratic forward couplings to restore positive Bakry–Émery curvature in denoising thermodynamic models is proposed in Jelinčič et al. [56].

Chapter 7

Metastability, Large Deviations, and Kramers' Escape Rate Theory

In Chapter 6, we proved via the Bakry–Émery Γ_2 criterion (Theorem 6.18) that whenever an energy landscape $E(\mathbf{x})$ is uniformly κ-strongly convex ($\nabla^2 E(\mathbf{x}) \succeq \kappa \mathbf{I}$ with $\kappa > 0$), the probability density $\rho(\mathbf{x}, t)$ of an overdamped Langevin circuit converges exponentially to its Boltzmann equilibrium $p_{\mathrm{eq}}(\mathbf{x}) = Z^{-1} e^{-\beta E(\mathbf{x})}$ at rate $2\mu\kappa$, independent of the state-space dimension N. However, as we saw in Chapters 1 and 3, any distribution capable of representing multiple distinct hypotheses, discrete logic states, or separated modes of a complex dataset must possess multiple local minima separated by energy barriers. A smooth function $E(\mathbf{x})$ with two or more local minima cannot be convex. Along any path from one minimum $\mathbf{x}_a$ to another $\mathbf{x}_b$ the energy must rise and fall again, and the lowest such pass is, generically, a saddle point $\mathbf{x}_{\mathrm{sad}}$ at which the Hessian $\nabla^2 E(\mathbf{x}_{\mathrm{sad}})$ has a negative eigenvalue.

What happens to the relaxation dynamics of a stochastic circuit when $\nabla^2 E(\mathbf{x})$ loses positive definiteness and a potential barrier of height $\Delta E := E(\mathbf{x}_{\mathrm{sad}}) - E(\mathbf{x}_a) > 0$ separates two basins of attraction? When the thermal energy $k_{\mathrm{B}}T = \beta^{-1}$ is small compared to the barrier height ($\beta \Delta E \gg 1$), the dynamics splits into two widely separated timescales. On a fast intra-well timescale $\tau_{\mathrm{intra}} \sim (\mu \omega_a^2)^{-1} \sim \tau_{RC}$ governed by the positive local curvature $\omega_a^2 = E''(x_a) > 0$ at the bottom of the well, the probability distribution relaxes to a quasi-stationary local Gaussian equilibrium inside its initial basin. On a much longer inter-well timescale $\tau_{\mathrm{escape}} \sim \tau_{\mathrm{intra}}\, e^{+\beta \Delta E}$, rare sequences of constructively aligned thermal impulses push the state up the potential slope against the restoring drift $-\mu \nabla E(\mathbf{x})$ and across the saddle $\mathbf{x}_{\mathrm{sad}}$ into the neighboring basin. This combination of rapid local thermalization and exponentially slow global equilibration across barriers is called *metastability*.

This chapter computes escape rates from three complementary viewpoints. First-passage theory expresses the mean escape time as the solution of a backward boundary-value problem; in the low-noise limit it yields Kramers' escape rate [61] and Langer's multidimensional generalization [64] (Sections 7.2 and 7.3). Large-deviation theory [36] gives the exponent of the rate and the most probable escape path, even for non-equilibrium circuits that have no potential (Section 7.4). Spectral theory identifies the sum of the forward and backward rates with the spectral gap of the Witten Hamiltonian of Chapter 6 (Section 7.5). The running example is the double-well sampler, with energy $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$. The worked computation at the end shows why barriers of a few $k_{\mathrm{B}}T$ make fast hardware p-bits (Chapter 14), and why large barriers motivate the denoising thermodynamic models of Chapter 16.

7.1 Bistable Energy Landscapes and Timescale Separation

Consider an overdamped Langevin system on $\mathbb{R}^N$ evolving according to

$$\mathrm{d}\mathbf{x}_t = -\mu \nabla E(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}\mathbf{W}_t, \qquad D = \mu k_{\mathrm{B}} T = \frac{\mu}{\beta}, \tag{7.1}$$

where $\mu = \gamma^{-1} > 0$ is the scalar mobility (inverse friction coefficient) and $E \in C^\infty(\mathbb{R}^N)$ is a confining potential ($E(\mathbf{x}) \to +\infty$ as $\|\mathbf{x}\| \to \infty$). Instead of assuming that $E(\mathbf{x})$ is convex, we now assume that $E(\mathbf{x})$ is a *bistable landscape* possessing two local minima $\mathbf{x}_a$ and $\mathbf{x}_b$.

7.1.1 Critical Points, Basins of Attraction, and Barrier Heights

To analyze the geometry of $E(\mathbf{x})$ precisely, we classify its stationary points $\left\{\mathbf{x}^* \in \mathbb{R}^N : \nabla E(\mathbf{x}^*) = \mathbf{0}\right\}$ by the eigenvalue spectrum of the symmetric Hessian matrix $\nabla^2 E(\mathbf{x}^*)$ (the *Morse index* of $\mathbf{x}^*$, defined as the number of strictly negative eigenvalues of $\nabla^2 E(\mathbf{x}^*)$):

1. **Local Minima (Morse Index 0):** At $\mathbf{x}_a$ and $\mathbf{x}_b$, $\nabla E(\mathbf{x}_a) = \nabla E(\mathbf{x}_b) = \mathbf{0}$ and the Hessian matrices are strictly positive definite:

$$\nabla^2 E(\mathbf{x}_a) \succ \mathbf{0}, \qquad \nabla^2 E(\mathbf{x}_b) \succ \mathbf{0}. \tag{7.2}$$

   In one dimension ($N = 1$), this simply means $E'(x_a) = E'(x_b) = 0$ with positive curvatures $\omega_a^2 := E''(x_a) > 0$ and $\omega_b^2 := E''(x_b) > 0$.

2. **Transition State / Index-1 Saddle Point (Morse Index 1):** Separating the two minima is a mountain pass or *index-1 saddle point* $\mathbf{x}_{\mathrm{sad}}$ where $\nabla E(\mathbf{x}_{\mathrm{sad}}) = \mathbf{0}$ and $\nabla^2 E(\mathbf{x}_{\mathrm{sad}})$ has *exactly one strictly negative eigenvalue* $-\lambda_{\mathrm{u}} < 0$ (along the unstable reaction coordinate $\hat{\mathbf{e}}_1$) and $N-1$ strictly positive eigenvalues $\lambda_2^{(s)}, \dots, \lambda_N^{(s)} > 0$ (along the transverse stable directions orthogonal to $\hat{\mathbf{e}}_1$). In one dimension ($N = 1$), $x_{\mathrm{sad}} \in (x_a, x_b)$ is a local maximum of $E(x)$ with negative curvature

$$E''(x_{\mathrm{sad}}) = -\omega_{\mathrm{sad}}^2 = -\left|E''(x_{\mathrm{sad}})\right| < 0. \tag{7.3}$$

In the noiseless limit $D \to 0$ ($T \to 0$), the deterministic gradient flow $\dot{\mathbf{x}}_t = -\mu \nabla E(\mathbf{x}_t)$ decomposes $\mathbb{R}^N$ into two disjoint *basins of attraction*:

$$\mathcal{A} := \left\{\mathbf{x}_0 \in \mathbb{R}^N : \lim_{t\to\infty} \mathbf{x}_t = \mathbf{x}_a\right\}, \qquad \mathcal{B} := \left\{\mathbf{x}_0 \in \mathbb{R}^N : \lim_{t\to\infty} \mathbf{x}_t = \mathbf{x}_b\right\}, \tag{7.4}$$

separated by an $(N-1)$-dimensional hypersurface $\Sigma_{\mathrm{sep}} = \partial\mathcal{A} = \partial\mathcal{B}$ called the *separatrix* (or watershed manifold), which is the stable manifold of the saddle point $\mathbf{x}_{\mathrm{sad}}$. Any deterministic trajectory starting on Σ_{sep} flows into $\mathbf{x}_{\mathrm{sad}}$ as $t \to \infty$; any trajectory starting in $\mathcal{A}$ flows into $\mathbf{x}_a$; and any trajectory starting in $\mathcal{B}$ flows into $\mathbf{x}_b$.

The minimum energy required to climb from the bottom of well $\mathcal{A}$ (or well $\mathcal{B}$) to the transition state $\mathbf{x}_{\mathrm{sad}}$ defines the forward and backward *activation barrier heights* (Fig. 7.1):

$$\Delta E_A := E(\mathbf{x}_{\mathrm{sad}}) - E(\mathbf{x}_a) > 0, \qquad \Delta E_B := E(\mathbf{x}_{\mathrm{sad}}) - E(\mathbf{x}_b) > 0. \tag{7.5}$$

The difference between the two barrier heights equals the energy difference (tilt) between the two bottoms:

$$\Delta E_B - \Delta E_A = E(\mathbf{x}_a) - E(\mathbf{x}_b). \tag{7.6}$$

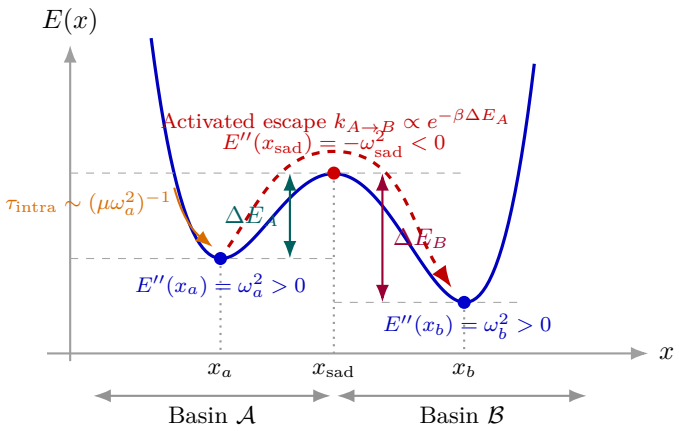


Figure 7.1: Geometry and timescale separation of a bistable potential $E(x)$, such as that of the double-well sampler. Local minima at $x_a \in \mathcal{A}$ and $x_b \in \mathcal{B}$ with positive curvatures $E''(x_a) = \omega_a^2 > 0$ and $E''(x_b) = \omega_b^2 > 0$ are separated by a saddle barrier at $x_{\rm sad}$ with negative curvature $E''(x_{\rm sad}) = -\omega_{\rm sad}^2 < 0$. When $\beta\Delta E_{A,B} \gg 1$, trajectories equilibrate locally inside basin $\mathcal{A}$ on the fast RC timescale $\tau_{\rm intra} \sim (\mu\omega_a^2)^{-1}$ before making a rare thermally activated escape over $x_{\rm sad}$ at rate $k_{A\to B} \propto e^{-\beta\Delta E_A}$.

7.1.2 Intra-Well Equilibration vs. Inter-Well Escape and Coarse-Grained Master Equation

Suppose we initialize the system at $t = 0$ near $\mathbf{x}_a$ inside basin $\mathcal{A}$, and suppose that both dimensionless barrier heights are large compared to thermal energy:

$$\beta\Delta E_A = \frac{\Delta E_A}{k_{\rm B}T} \gg 1 \qquad \text{and} \qquad \beta\Delta E_B = \frac{\Delta E_B}{k_{\rm B}T} \gg 1. \tag{7.7}$$

Inside a neighborhood of $\mathbf{x}_a$, we can expand $E(\mathbf{x})$ to quadratic order as $E(\mathbf{x}) \approx E(\mathbf{x}_a) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_a)^\top \nabla^2 E(\mathbf{x}_a)(\mathbf{x} - \mathbf{x}_a)$. Locally within basin $\mathcal{A}$, the potential is strongly convex with minimum curvature $\kappa_a = \lambda_{\min}(\nabla^2 E(\mathbf{x}_a)) > 0$ (or $\omega_a^2 = E''(x_a)$ in 1D). By the Ornstein–Uhlenbeck analysis of Worked computation 6.21, any initial density $\rho_0(\mathbf{x})$ supported inside $\mathcal{A}$ relaxes locally to the intra-well Gaussian distribution

$$\rho_{\mathcal{A}}^{\rm loc}(\mathbf{x}) \approx \frac{\mathbf{1}_{\mathcal{A}}(\mathbf{x})}{Z_A}\, e^{-\beta E(\mathbf{x})} \approx \frac{(\beta)^{N/2}\sqrt{\det \nabla^2 E(\mathbf{x}_a)}}{(2\pi)^{N/2}} \exp\!\left(-\frac{\beta}{2}(\mathbf{x} - \mathbf{x}_a)^\top \nabla^2 E(\mathbf{x}_a)(\mathbf{x} - \mathbf{x}_a)\right) \tag{7.8}$$

on the fast *intra-well relaxation time*

$$\tau_{\rm intra}^{(A)} := \frac{1}{\mu\lambda_{\min}(\nabla^2 E(\mathbf{x}_a))} \quad \left(= \frac{1}{\mu E''(x_a)} = \frac{\gamma}{\omega_a^2} \text{ in 1D}\right). \tag{7.9}$$

Here $Z_A := \int_{\mathcal{A}} \mathrm{d}^N x\, e^{-\beta E(\mathbf{x})}$ and $Z_B := \int_{\mathcal{B}} \mathrm{d}^N x\, e^{-\beta E(\mathbf{x})}$ are the restricted partition functions of basins $\mathcal{A}$ and $\mathcal{B}$, with $Z = Z_A + Z_B$, defining the local basin free energies $F_A := -k_{\rm B}T \ln Z_A$ and $F_B := -k_{\rm B}T \ln Z_B$.

Once local equilibrium Eq. (7.8) is established on timescale $t \gg \tau_{\rm intra}$, the probability density inside each basin maintains the fixed spatial profile $\rho_{\mathcal{A}}^{\rm loc}(\mathbf{x})$ and $\rho_{\mathcal{B}}^{\rm loc}(\mathbf{x})$, while only the total coarse-grained basin probabilities

$$P_A(t) := \int_{\mathcal{A}} \mathrm{d}^N x\, \rho(\mathbf{x}, t), \qquad P_B(t) := \int_{\mathcal{B}} \mathrm{d}^N x\, \rho(\mathbf{x}, t) = 1 - P_A(t) \tag{7.10}$$

evolve slowly in time:

$$\rho(\mathbf{x},t) \approx P_A(t)\,\rho_{\mathcal{A}}^{\mathrm{loc}}(\mathbf{x}) + P_B(t)\,\rho_{\mathcal{B}}^{\mathrm{loc}}(\mathbf{x}) \qquad (t \gg \tau_{\mathrm{intra}}). \tag{7.11}$$

Because memory of where the trajectory started inside $\mathcal{A}$ is erased on the fast timescale $\tau_{\mathrm{intra}} \ll \tau_{\mathrm{escape}}$, the subsequent escape across Σ_{sep} into $\mathcal{B}$ is a memoryless Poisson process with a constant hazard rate $k_{A\to B}$ (the *forward escape rate*), and similarly escape from $\mathcal{B}$ to $\mathcal{A}$ occurs at rate $k_{B\to A}$. Thus, on timescales $t \gg \tau_{\mathrm{intra}}$, the continuous Fokker–Planck PDE reduces to the two-state continuous-time Markov chain (or *p-bit master equation*):

$$\frac{\mathrm{d}P_A(t)}{\mathrm{d}t} = -k_{A\to B}P_A(t) + k_{B\to A}P_B(t), \qquad \frac{\mathrm{d}P_B(t)}{\mathrm{d}t} = +k_{A\to B}P_A(t) - k_{B\to A}P_B(t). \tag{7.12}$$

Using $P_B(t) = 1 - P_A(t)$, the first equation in Eq. (7.12) becomes $\frac{\mathrm{d}P_A}{\mathrm{d}t} = -(k_{A\to B}+k_{B\to A})P_A + k_{B\to A}$, which integrates immediately to

$$P_A(t) = P_A^{\mathrm{eq}} + \left(P_A(0) - P_A^{\mathrm{eq}}\right) e^{-\lambda_{\mathrm{macro}}t}, \tag{7.13}$$

where the stationary basin probabilities and the macroscopic inter-well relaxation rate λ_{macro} are given by

$$\frac{P_A^{\mathrm{eq}}}{P_B^{\mathrm{eq}}} = \frac{k_{B\to A}}{k_{A\to B}} = \frac{Z_A}{Z_B} = e^{-\beta(F_A - F_B)}, \qquad \lambda_{\mathrm{macro}} = k_{A\to B} + k_{B\to A}. \tag{7.14}$$

Our task in the next two sections is to calculate the escape rates $k_{A\to B}$ and $k_{B\to A}$ from first principles.

7.2 Mean First-Passage Time and Pontryagin's Backward Equation

To compute the escape rate $k_{A\to B}$ out of basin $\mathcal{A}$, we formulate barrier crossing as a *first-passage problem*. Let $\Omega \subset \mathbb{R}^N$ be a connected open domain with smooth boundary $\partial\Omega$, and suppose the stochastic process $\mathbf{x}_t$ starts at a deterministic initial point $\mathbf{x}_0 = \mathbf{x} \in \Omega$.

Definition 7.1 (First-Passage Stopping Time and Mean First-Passage Time)**.** For an open domain $\Omega \subset \mathbb{R}^N$ and initial state $\mathbf{x}_0 = \mathbf{x} \in \Omega$, the *first-passage time* (or *first exit time*) from Ω is the random stopping time

$$\tau_\Omega(\mathbf{x}) := \inf\left\{t \geq 0 : \mathbf{x}_t \notin \Omega \mid \mathbf{x}_0 = \mathbf{x}\right\}. \tag{7.15}$$

The *Mean First-Passage Time (MFPT)* $\tau(\mathbf{x})$ is the expected time required to reach the boundary $\partial\Omega$ starting from $\mathbf{x}$:

$$\tau(\mathbf{x}) := \mathbb{E}[\tau_\Omega(\mathbf{x})] = \mathbb{E}[\tau_\Omega \mid \mathbf{x}_0 = \mathbf{x}]. \tag{7.16}$$

More generally, for each integer $n \geq 1$, the n-th first-passage moment is $\tau_n(\mathbf{x}) := \mathbb{E}[\tau_\Omega(\mathbf{x})^n]$, with $\tau_0(\mathbf{x}) \equiv 1$ and $\tau_1(\mathbf{x}) = \tau(\mathbf{x})$.

If $\mathbf{x} \in \partial\Omega$ already lies on the boundary at $t = 0$, the trajectory exits immediately, so $\tau(\mathbf{x}) = 0$ for all $\mathbf{x} \in \partial\Omega$. In the interior, $\tau(\mathbf{x})$ satisfies a linear elliptic PDE built from the *backward Kolmogorov generator* $\mathcal{L}$ of Definition 6.1, first derived by Pontryagin, Andronov, and Vitt in 1933.

Theorem 7.2 (Pontryagin–Andronov–Vitt–Dynkin Backward Equation for the MFPT)**.** *Let $\mathbf{x}_t \in \mathbb{R}^N$ be a time-homogeneous diffusion with backward generator $\mathcal{L} = \mathbf{a}(\mathbf{x})\cdot\nabla + \mathrm{tr}\big(\mathbf{D}(\mathbf{x})\nabla^2\big)$ where*

$\mathbf{D}(\mathbf{x}) \succ \mathbf{0}$, *and let* $\Omega \subset \mathbb{R}^N$ *be a domain such that* $\tau(\mathbf{x}) = \mathbb{E}[\tau_\Omega(\mathbf{x})] < \infty$ *for all* $\mathbf{x} \in \Omega$. *Then* $\tau \in C^2(\Omega) \cap C(\overline{\Omega})$ *solves the boundary-value problem*

$$(\mathcal{L}\tau)(\mathbf{x}) = \sum_{i=1}^{N} a_i(\mathbf{x}) \frac{\partial \tau(\mathbf{x})}{\partial x_i} + \sum_{i,j=1}^{N} D_{ij}(\mathbf{x}) \frac{\partial^2 \tau(\mathbf{x})}{\partial x_i \partial x_j} = -1 \quad (\mathbf{x} \in \Omega), \qquad \tau(\mathbf{x}) = 0 \quad (\mathbf{x} \in \partial\Omega). \tag{7.17}$$

More generally, the n*-th moment* $\tau_n(\mathbf{x}) = \mathbb{E}[\tau_\Omega(\mathbf{x})^n]$ *(*$n \geq 1$*) solves the recursive hierarchy*

$$(\mathcal{L}\tau_n)(\mathbf{x}) = -n\, \tau_{n-1}(\mathbf{x}) \quad (\mathbf{x} \in \Omega), \qquad \tau_n(\mathbf{x}) = 0 \quad (\mathbf{x} \in \partial\Omega). \tag{7.18}$$

Proof. We give two proofs: a stochastic-calculus proof via Itô's formula (Dynkin's formula), and a PDE proof via the survival probability.

Proof 1 (Via Itô's Formula and Stopped Martingales): Apply Itô's formula Eq. (6.6) to $\tau(\mathbf{x}_t)$ from $t = 0$ (where $\mathbf{x}_0 = \mathbf{x} \in \Omega$) up to the stopped time $T \wedge \tau_\Omega := \min(T, \tau_\Omega)$, where $T > 0$ is any fixed deterministic time horizon:

$$\tau(\mathbf{x}_{T\wedge\tau_\Omega}) - \tau(\mathbf{x}) = \int_0^{T\wedge\tau_\Omega} (\mathcal{L}\tau)(\mathbf{x}_s)\, \mathrm{d}s + \int_0^{T\wedge\tau_\Omega} \nabla\tau(\mathbf{x}_s)^\top \mathbf{B}(\mathbf{x}_s)\, \mathrm{d}\mathbf{W}_s. \tag{7.19}$$

Take the conditional expectation $\mathbb{E}[\,\cdot \mid \mathbf{x}_0 = \mathbf{x}]$. The Itô integral is a martingale (Chapter 4), and $T \wedge \tau_\Omega$ is a bounded stopping time, so by the optional stopping theorem the stopped stochastic integral has expectation zero. This leaves

$$\mathbb{E}[\tau(\mathbf{x}_{T\wedge\tau_\Omega}) \mid \mathbf{x}_0 = \mathbf{x}] - \tau(\mathbf{x}) = \mathbb{E}\left[\int_0^{T\wedge\tau_\Omega} (\mathcal{L}\tau)(\mathbf{x}_s)\, \mathrm{d}s \;\middle|\; \mathbf{x}_0 = \mathbf{x}\right]. \tag{7.20}$$

Now let $T \to \infty$. Since $\mathbb{E}[\tau_\Omega(\mathbf{x})] < \infty$, $\tau_\Omega < \infty$ almost surely, so $T \wedge \tau_\Omega \to \tau_\Omega$ and $\mathbf{x}_{T\wedge\tau_\Omega} \to \mathbf{x}_{\tau_\Omega} \in \partial\Omega$. Because $\tau(\mathbf{y}) = 0$ for all $\mathbf{y} \in \partial\Omega$, we have $\tau(\mathbf{x}_{\tau_\Omega}) = 0$ almost surely. Thus in the limit $T \to \infty$, Eq. (7.20) becomes

$$-\tau(\mathbf{x}) = \mathbb{E}\left[\int_0^{\tau_\Omega} (\mathcal{L}\tau)(\mathbf{x}_s)\, \mathrm{d}s \;\middle|\; \mathbf{x}_0 = \mathbf{x}\right]. \tag{7.21}$$

On the other hand, by definition of $\tau(\mathbf{x}) = \mathbb{E}[\tau_\Omega \mid \mathbf{x}_0 = \mathbf{x}]$, we can write $-\tau(\mathbf{x})$ as the expected integral of -1 from 0 to τ_Ω:

$$-\tau(\mathbf{x}) = -\mathbb{E}[\tau_\Omega \mid \mathbf{x}_0 = \mathbf{x}] = \mathbb{E}\left[\int_0^{\tau_\Omega} (-1)\, \mathrm{d}s \;\middle|\; \mathbf{x}_0 = \mathbf{x}\right]. \tag{7.22}$$

Equating Eqs. (7.21) and (7.22) shows that $\mathcal{L}\tau + 1$ integrates to zero along trajectories from every starting point. To obtain the pointwise statement, use Eq. (7.20) with a short horizon $T = \Delta t$ together with the Markov property $\tau(\mathbf{x}) = \mathbb{E}[\Delta t \wedge \tau_\Omega] + \mathbb{E}[\tau(\mathbf{x}_{\Delta t\wedge\tau_\Omega})]$; dividing by Δt and letting $\Delta t \to 0$ gives $(\mathcal{L}\tau)(\mathbf{x}) = -1$ for all $\mathbf{x} \in \Omega$.

Proof 2 (Via Survival Probability and the Backward Kolmogorov PDE): Let $p_\Omega(\mathbf{y}, t \mid \mathbf{x}, 0)$ denote the transition probability density of $\mathbf{x}_t$ in Ω with an *absorbing boundary condition* $p_\Omega(\mathbf{y}, t \mid \mathbf{x}, 0) = 0$ for $\mathbf{x} \in \partial\Omega$ or $\mathbf{y} \in \partial\Omega$. The *survival probability* $S(\mathbf{x}, t)$, the probability that a trajectory starting at $\mathbf{x} \in \Omega$ has not yet touched $\partial\Omega$ by time t, is

$$S(\mathbf{x}, t) := \mathbb{P}(\tau_\Omega(\mathbf{x}) > t) = \int_\Omega \mathrm{d}^N y\, p_\Omega(\mathbf{y}, t \mid \mathbf{x}, 0). \tag{7.23}$$

At $t = 0$, $p_\Omega(\mathbf{y}, 0 \mid \mathbf{x}, 0) = \delta(\mathbf{y} - \mathbf{x})$ for $\mathbf{x} \in \Omega$, so $S(\mathbf{x}, 0) = 1$; as $t \to \infty$, all trajectories eventually exit Ω, so $\lim_{t\to\infty} S(\mathbf{x}, t) = 0$. Furthermore, since $\mathbb{P}(\tau_\Omega(\mathbf{x}) \in [t, t + \mathrm{d}t]) = -\frac{\partial S(\mathbf{x},t)}{\partial t}\,\mathrm{d}t$, integrating by parts in t expresses every moment $\tau_n(\mathbf{x})$ directly in terms of $S(\mathbf{x}, t)$:

$$\tau_n(\mathbf{x}) = \int_0^\infty \mathrm{d}t\, t^n \left(-\frac{\partial S(\mathbf{x}, t)}{\partial t}\right) = n \int_0^\infty \mathrm{d}t\, t^{n-1} S(\mathbf{x}, t) \qquad (n \geq 1), \tag{7.24}$$

and in particular for $n = 1$,

$$\tau(\mathbf{x}) = \int_0^\infty \mathrm{d}t\, S(\mathbf{x}, t). \tag{7.25}$$

Now, as a function of the initial position $\mathbf{x}$, $p_\Omega(\mathbf{y}, t \mid \mathbf{x}, 0)$ satisfies the backward Kolmogorov equation Eq. (6.9): $\frac{\partial}{\partial t} p_\Omega(\mathbf{y}, t \mid \mathbf{x}, 0) = \mathcal{L}_\mathbf{x} p_\Omega(\mathbf{y}, t \mid \mathbf{x}, 0)$. Integrating over $\mathbf{y} \in \Omega$ shows that $S(\mathbf{x}, t)$ itself satisfies the backward Kolmogorov PDE:

$$\frac{\partial S(\mathbf{x}, t)}{\partial t} = \mathcal{L} S(\mathbf{x}, t) \quad (\mathbf{x} \in \Omega), \qquad S(\mathbf{x}, 0) = 1. \tag{7.26}$$

Applying $\mathcal{L}$ to Eq. (7.25) and substituting Eq. (7.26), the Fundamental Theorem of Calculus in t gives

$$(\mathcal{L}\tau)(\mathbf{x}) = \int_0^\infty \mathrm{d}t\, \mathcal{L} S(\mathbf{x}, t) = \int_0^\infty \mathrm{d}t\, \frac{\partial S(\mathbf{x}, t)}{\partial t} = \lim_{t\to\infty} S(\mathbf{x}, t) - S(\mathbf{x}, 0) = 0 - 1 = -1. \tag{7.27}$$

Similarly, applying $\mathcal{L}$ to Eq. (7.24) and integrating by parts in t gives $(\mathcal{L}\tau_n)(\mathbf{x}) = n \int_0^\infty \mathrm{d}t\, t^{n-1} \partial_t S(\mathbf{x}, t) = -n\, \tau_{n-1}(\mathbf{x})$, proving Eq. (7.18). □

Physics Insight: Why First-Passage Times Obey the Backward Generator $\mathcal{L}$ Rather Than $\mathcal{L}_{\mathrm{FP}}$

Why does the mean first-passage time obey $\mathcal{L}\tau(\mathbf{x}) = -1$ rather than $\mathcal{L}_{\mathrm{FP}}\tau(\mathbf{x}) = -1$? The answer lies in which variable is being varied. In the definition $\tau(\mathbf{x}) = \mathbb{E}[\tau_\Omega \mid \mathbf{x}_0 = \mathbf{x}]$, the final absorption surface $\partial\Omega$ is held fixed while we vary the *initial starting coordinate* $\mathbf{x} = \mathbf{x}_0$. Over an infinitesimal first step $\mathrm{d}t$ starting from $\mathbf{x}$, the particle moves to $\mathbf{x} + \mathrm{d}\mathbf{x}_t$, and by the Markov property the remaining time to exit is $\tau(\mathbf{x} + \mathrm{d}\mathbf{x}_t)$, so $\tau(\mathbf{x}) = \mathrm{d}t + \mathbb{E}[\tau(\mathbf{x} + \mathrm{d}\mathbf{x}_t)]$. Expanding $\mathbb{E}[\tau(\mathbf{x} + \mathrm{d}\mathbf{x}_t)] = \tau(\mathbf{x}) + (\mathcal{L}\tau)(\mathbf{x})\,\mathrm{d}t$ immediately yields $0 = 1 + (\mathcal{L}\tau)(\mathbf{x})$, or $\mathcal{L}\tau = -1$.

7.2.1 Exact Double-Integral Quadrature of the 1D MFPT Equation

In one spatial dimension ($N = 1$), Pontryagin's PDE Eq. (7.17) is a second-order ordinary differential equation that is first-order in the derivative $\tau'(x)$, so it can be solved in closed form by two successive quadratures for *any* potential $E(x)$.

Proposition 7.3 (Exact 1D Mean First-Passage Time Formula)**.** *Consider a 1D overdamped Langevin particle* $\mathrm{d}x_t = -\mu E'(x_t)\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}W_t$ *($D = \mu k_\mathrm{B} T = \mu/\beta$) on the interval $\Omega = (r, b)$ with a reflecting boundary at $x = r$ (where $\tau'(r) = 0$) and an absorbing boundary at $x = b$ (where $\tau(b) = 0$). Then for any initial position $x_0 \in [r, b]$, the Mean First-Passage Time to reach b is given exactly by*

$$\tau(x_0 \to b) = \frac{1}{D} \int_{x_0}^b \mathrm{d}y\, e^{\beta E(y)} \int_r^y \mathrm{d}z\, e^{-\beta E(z)}. \tag{7.28}$$

In particular, if the left boundary is pushed to $r \to -\infty$ in a confining potential ($\lim_{x\to-\infty} E(x) = +\infty$), the exact MFPT from $x_0 < b$ to b is

$$\tau(x_0 \to b) = \frac{1}{D}\int_{x_0}^{b} \mathrm{d}y\, e^{\beta E(y)} \int_{-\infty}^{y} \mathrm{d}z\, e^{-\beta E(z)} = \frac{1}{D}\int_{x_0}^{b} \frac{\mathrm{d}y}{p_{\mathrm{eq}}(y)} \int_{-\infty}^{y} \mathrm{d}z\, p_{\mathrm{eq}}(z). \tag{7.29}$$

Proof. In 1D, the backward generator Eq. (6.59) is $\mathcal{L}\tau = D\tau''(x) - \mu E'(x)\tau'(x) = D[\tau''(x) - \beta E'(x)\tau'(x)]$. Thus Pontryagin's equation $(\mathcal{L}\tau)(x) = -1$ reads

$$\tau''(x) - \beta E'(x)\tau'(x) = -\frac{1}{D}. \tag{7.30}$$

The left-hand side of Eq. (7.30) becomes an exact derivative when multiplied by the Boltzmann integrating factor $e^{-\beta E(x)}$:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[e^{-\beta E(x)}\tau'(x)\right] = e^{-\beta E(x)}\left[\tau''(x) - \beta E'(x)\tau'(x)\right] = -\frac{1}{D}\, e^{-\beta E(x)}. \tag{7.31}$$

Integrating Eq. (7.31) with respect to x (relabeled as dummy variable z) from the reflecting boundary $z = r$ to an interior point $z = y \in (r, b)$ gives

$$e^{-\beta E(y)}\tau'(y) - e^{-\beta E(r)}\tau'(r) = -\frac{1}{D}\int_{r}^{y} \mathrm{d}z\, e^{-\beta E(z)}. \tag{7.32}$$

Because $x = r$ is a reflecting boundary ($\tau'(r) = 0$, or in the limit $r \to -\infty$, $e^{-\beta E(r)} \to 0$), the boundary term $e^{-\beta E(r)}\tau'(r)$ vanishes. Multiplying Eq. (7.32) by $e^{+\beta E(y)}$ yields the first derivative of the MFPT at any point y:

$$\tau'(y) = -\frac{1}{D}\, e^{\beta E(y)} \int_{r}^{y} \mathrm{d}z\, e^{-\beta E(z)} < 0. \tag{7.33}$$

Finally, we integrate Eq. (7.33) with respect to y from $y = x_0$ to the absorbing boundary $y = b$, where $\tau(b) = 0$:

$$\tau(b) - \tau(x_0) = \int_{x_0}^{b} \mathrm{d}y\, \tau'(y) = -\frac{1}{D}\int_{x_0}^{b} \mathrm{d}y\, e^{\beta E(y)} \int_{r}^{y} \mathrm{d}z\, e^{-\beta E(z)}. \tag{7.34}$$

Setting $\tau(b) = 0$ and multiplying by -1 gives Eq. (7.28), and taking $r \to -\infty$ gives Eq. (7.29). □

7.3 Kramers' Overdamped Escape Rate and Langer's Multidimensional Saddle Formula

Consider the exact 1D mean first-passage time Eq. (7.29) for a particle starting at the bottom of well $\mathcal{A}$ ($x_0 = x_a$) and escaping across the saddle barrier x_{sad} to the bottom of well $\mathcal{B}$ ($b = x_b$, as in Fig. 7.1):

$$\tau(x_a \to x_b) = \frac{1}{D}\int_{x_a}^{x_b} \mathrm{d}y\, e^{+\beta E(y)} \int_{-\infty}^{y} \mathrm{d}z\, e^{-\beta E(z)}. \tag{7.35}$$

In the high-barrier regime $\beta\Delta E_A \gg 1$ this integral simplifies for two reasons.

1. In the outer integral over $y \in [x_a, x_b]$, the integrand contains $e^{+\beta E(y)}$ with a *plus* sign in the exponent. Because $E(y)$ achieves its strict maximum on $[x_a, x_b]$ at the barrier top $y = x_{\mathrm{sad}}$, the factor $e^{+\beta E(y)}$ is sharply peaked in a narrow Gaussian window of width $\delta y \sim (\beta\,|E''(x_{\mathrm{sad}})|)^{-1/2}$ around $y = x_{\mathrm{sad}}$, and is exponentially negligible everywhere else.

2. For y in this narrow window around x_{sad}, the upper limit of the inner integral $\int_{-\infty}^{y} \mathrm{d}z\, e^{-\beta E(z)}$ can be replaced by $y \approx x_{\text{sad}}$. In this inner integral over $z \in (-\infty, x_{\text{sad}})$, the integrand contains $e^{-\beta E(z)}$ with a *minus* sign in the exponent. Because $E(z)$ achieves its strict minimum on $(-\infty, x_{\text{sad}})$ at the bottom of well $\mathcal{A}$ ($z = x_a$), the factor $e^{-\beta E(z)}$ is sharply peaked in a narrow Gaussian window of width $\delta z \sim (\beta E''(x_a))^{-1/2}$ around $z = x_a$.

Thus the two integrals in Eq. (7.35) decouple into a product of two independent Gaussian integrals, one around the well minimum x_a and one around the barrier maximum x_{sad}. Carrying out this double Laplace approximation yields Kramers' escape rate formula.

Theorem 7.4 (Kramers' Overdamped Escape Rate Formula [61])**.** *Let $E \in C^3(\mathbb{R})$ be a 1D bistable potential with local minima at $x_a < x_b$ ($E''(x_a) > 0$, $E''(x_b) > 0$) separated by a local maximum at $x_{\text{sad}} \in (x_a, x_b)$ ($E''(x_{\text{sad}}) = -\,|E''(x_{\text{sad}})| < 0$), and barrier height $\Delta E_A = E(x_{\text{sad}}) - E(x_a) > 0$. In the low-temperature / high-barrier limit $\beta \Delta E_A \to \infty$ (with $D = \mu/\beta = (\gamma\beta)^{-1}$), the Mean First-Passage Time from x_a to x_b and its inverse, the* Kramers escape rate *$k_{A\to B} := \tau(x_a \to x_b)^{-1}$, are given asymptotically by*

$$\tau(x_a \to x_b) = \frac{2\pi\gamma}{\sqrt{E''(x_a)\,|E''(x_{\text{sad}})|}}\, \exp(+\beta \Delta E_A)\left[1 + \mathcal{O}\Big(\beta^{-1}\Big)\right], \tag{7.36}$$

$$\begin{aligned} k_{A\to B} &= \frac{\sqrt{E''(x_a)\,|E''(x_{\text{sad}})|}}{2\pi\gamma}\, \exp(-\beta \Delta E_A)\left[1 + \mathcal{O}\Big(\beta^{-1}\Big)\right] \\ &= \frac{\mu\sqrt{E''(x_a)\,|E''(x_{\text{sad}})|}}{2\pi}\, \exp\!\left(-\frac{\Delta E_A}{k_{\mathrm{B}}T}\right)\left[1 + \mathcal{O}(k_{\mathrm{B}}T)\right]. \end{aligned} \tag{7.37}$$

Moreover, if the absorbing boundary is placed directly at the saddle point $b = x_{\text{sad}}$ instead of at x_b, then $\tau(x_a \to x_{\text{sad}}) = \frac{1}{2}\tau(x_a \to x_b)$ because half of the Gaussian peak at $y = x_{\text{sad}}$ lies in $(-\infty, x_{\text{sad}}]$, reflecting the physical fact that a trajectory reaching the top of the barrier x_{sad} has probability $1/2$ of committing forward to x_b and probability $1/2$ of falling back into x_a.

Proof. We evaluate the inner and outer integrals of Eq. (7.35) step by step via Laplace's method.

Step 1: Laplace Evaluation of the Inner Integral around $z = x_a$. For any y in a neighborhood of x_{sad}, the unique minimum of $E(z)$ on $(-\infty, y]$ is at $z = x_a$, where $E'(x_a) = 0$ and $E''(x_a) > 0$. Expanding $E(z)$ in a Taylor series around $z = x_a$ gives

$$E(z) = E(x_a) + \frac{1}{2}E''(x_a)(z - x_a)^2 + \frac{1}{6}E'''(x_a)(z - x_a)^3 + \mathcal{O}\Big((z - x_a)^4\Big). \tag{7.38}$$

Substituting Eq. (7.38) into the inner integral of Eq. (7.35) and changing variables to $u = \sqrt{\beta E''(x_a)}(z - x_a)$ shows that for any y within $\mathcal{O}(\beta^{-1/2})$ of x_{sad}, the upper limit $(y - x_a)\sqrt{\beta E''(x_a)} \sim \mathcal{O}(\beta^{+1/2}) \to +\infty$ up to exponentially small errors $\mathcal{O}(e^{-\beta \Delta E_A})$. Completing the Gaussian integral over $z \in (-\infty, \infty)$ gives

$$\begin{aligned} \int_{-\infty}^{y} \mathrm{d}z\, e^{-\beta E(z)} &= e^{-\beta E(x_a)} \int_{-\infty}^{\infty} \mathrm{d}z\, \exp\!\left(-\frac{1}{2}\beta E''(x_a)(z - x_a)^2\right)\left[1 + \mathcal{O}(\beta^{-1})\right] \\ &= \sqrt{\frac{2\pi}{\beta E''(x_a)}}\, e^{-\beta E(x_a)}\left[1 + \mathcal{O}(\beta^{-1})\right], \end{aligned} \tag{7.39}$$

where the odd cubic term $\frac{1}{6}\beta E'''(x_a)(z - x_a)^3$ integrates to zero by symmetry and the quartic and squared-cubic corrections contribute at order $\mathcal{O}(\beta^{-1})$.

Step 2: Laplace Evaluation of the Outer Integral around $y = x_{\rm sad}$. Next, we expand $E(y)$ in a Taylor series around its maximum $y = x_{\rm sad} \in (x_a, x_b)$, where $E'(x_{\rm sad}) = 0$ and $E''(x_{\rm sad}) = -\,|E''(x_{\rm sad})| < 0$:

$$E(y) = E(x_{\rm sad}) - \frac{1}{2}\,|E''(x_{\rm sad})|\,(y - x_{\rm sad})^2 + \frac{1}{6}E'''(x_{\rm sad})(y - x_{\rm sad})^3 + \mathcal{O}\Big((y - x_{\rm sad})^4\Big)\,. \tag{7.40}$$

Because both endpoints $x_a < x_{\rm sad} < x_b$ lie at distance $\mathcal{O}(1)$ from $x_{\rm sad}$, extending the limits of integration from $[x_a, x_b]$ to $(-\infty, \infty)$ introduces only an exponentially negligible error, and the outer Gaussian integral evaluates to

$$\begin{aligned}\int_{x_a}^{x_b} \mathrm{d}y\, e^{+\beta E(y)} &= e^{+\beta E(x_{\rm sad})} \int_{-\infty}^{\infty} \mathrm{d}y\, \exp\!\Big(-\frac{1}{2}\beta\,|E''(x_{\rm sad})|\,(y - x_{\rm sad})^2\Big)\,\Big[1 + \mathcal{O}(\beta^{-1})\Big] \\ &= \sqrt{\frac{2\pi}{\beta\,|E''(x_{\rm sad})|}}\, e^{+\beta E(x_{\rm sad})}\,\Big[1 + \mathcal{O}(\beta^{-1})\Big]\,. \end{aligned} \tag{7.41}$$

(If the upper limit is set at $b = x_{\rm sad}$ instead of x_b, the Gaussian integral runs over $(-\infty, x_{\rm sad}]$, which is exactly half of the full line $(-\infty, \infty)$, yielding $\frac{1}{2}\sqrt{\frac{2\pi}{\beta|E''(x_{\rm sad})|}}\, e^{+\beta E(x_{\rm sad})}$.)

Step 3: Combining the Two Saddle-Point Integrals. Substituting Eqs. (7.39) and (7.41) into Eq. (7.35) and using $D = (\gamma\beta)^{-1}$ and $E(x_{\rm sad}) - E(x_a) = \Delta E_A$, the factors of $\beta^{-1/2}$ from the two Gaussian widths multiply to β^{-1}, which cancels the prefactor $\frac{1}{D} = \gamma\beta$. We obtain

$$\begin{aligned}\tau(x_a \to x_b) &= \gamma\beta \left(\sqrt{\frac{2\pi}{\beta E''(x_a)}}\, e^{-\beta E(x_a)}\right)\left(\sqrt{\frac{2\pi}{\beta\,|E''(x_{\rm sad})|}}\, e^{+\beta E(x_{\rm sad})}\right)\Big[1 + \mathcal{O}(\beta^{-1})\Big] \\ &= \frac{2\pi\gamma}{\sqrt{E''(x_a)\;|E''(x_{\rm sad})|}}\, \exp\!\Big(\beta\big[E(x_{\rm sad}) - E(x_a)\big]\Big)\,\Big[1 + \mathcal{O}(\beta^{-1})\Big]\,, \end{aligned} \tag{7.42}$$

which is Eq. (7.36), and inverting gives Eq. (7.37). Finally, consider the higher-moment hierarchy Eq. (7.18), $\mathcal{L}\tau_n = -n\tau_{n-1}$. Because $\tau_{n-1}(z)$ is essentially constant and equal to $\tau_{n-1}(x_a)$ throughout basin $\mathcal{A}$, repeating the exact same double integration yields $\tau_n(x_a) = n\,\tau(x_a)\,\tau_{n-1}(x_a)[1 + \mathcal{O}(e^{-\beta\Delta E_A})]$, or by induction $\mathbb{E}[\tau_\Omega^n] \simeq n!\,\tau(x_a)^n$. A random variable whose n-th moment is $n!\,\tau^n$ has the exponential distribution $\mathbb{P}(\tau_\Omega > t) = e^{-t/\tau} = e^{-k_{A\to B}t}$, confirming that barrier crossing is a Poisson process with rate $k_{A\to B} = \tau(x_a \to x_b)^{-1}$. □

7.3.1 Kramers' Flux-Over-Population Method and Langer's Multidimensional Formula

In N dimensions the exact double integral Eq. (7.35) is no longer available. Kramers' own derivation of Eq. (7.37) [61] used a different method, the *flux-over-population* (quasi-stationary current) method, which does generalize; Langer extended it to multidimensional saddle points in 1969 [64].

Imagine maintaining a steady-state probability density $\rho(\mathbf{x})$ in which particles are gently replenished near the bottom of well $\mathcal{A}$ and absorbed once they cross the saddle into well $\mathcal{B}$. Outside the immediate source/sink neighborhoods, $\rho(\mathbf{x})$ is stationary ($\partial_t\rho = -\nabla\cdot\mathbf{J}(\mathbf{x}) = 0$), and the escape rate $k_{A\to B}$ is the ratio of the total steady-state probability flux $I_{A\to B}$ crossing the separatrix $\Sigma_{\rm sep}$ to the total probability population N_A residing in well $\mathcal{A}$:

$$k_{A\to B} = \frac{I_{A\to B}}{N_A} = \frac{\int_{\Sigma_{\rm sep}} \mathrm{d}^{N-1}S\;\hat{\mathbf{n}}\cdot\mathbf{J}(\mathbf{x})}{\int_{\mathcal{A}} \mathrm{d}^N x\,\rho(\mathbf{x})}. \tag{7.43}$$

In 1D, Eq. (7.43) reproduces Kramers' rate in a few lines:

- In 1D, $\partial_x J(x) = 0$ implies that the probability current $J(x) = -De^{-\beta E(x)}\frac{\mathrm{d}}{\mathrm{d}x}\big(e^{\beta E(x)}\rho(x)\big) = I_0$ is spatially constant across the barrier region. Dividing by $-De^{-\beta E(x)}$ and integrating from x_a (where $e^{\beta E(x)}\rho(x) \approx e^{\beta E(x_a)}\rho(x_a)$ is in local Boltzmann equilibrium) to x_b (where the absorbing sink forces $\rho(x_b) \approx 0$) gives

$$e^{\beta E(x_a)}\rho(x_a) = \frac{I_0}{D}\int_{x_a}^{x_b} \mathrm{d}y\, e^{\beta E(y)} \simeq \frac{I_0}{D}\sqrt{\frac{2\pi}{\beta\,|E''(x_{\mathrm{sad}})|}}\, e^{\beta E(x_{\mathrm{sad}})}.$$

- Meanwhile, integrating the local Boltzmann density $\rho(x) \approx \rho(x_a)e^{-\beta[E(x)-E(x_a)]}$ around x_a gives the well population $N_A = \int_{\mathcal{A}} \rho(x)\,\mathrm{d}x \simeq \rho(x_a)\sqrt{\frac{2\pi}{\beta E''(x_a)}}$. Taking the ratio $k_{A\to B} = I_0/N_A$ cancels $\rho(x_a)$ and reproduces Eq. (7.37).

We now extend the flux-over-population construction to an N-dimensional system $\mathrm{d}\mathbf{x}_t = -\boldsymbol{\mu}\nabla E(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T}\,\boldsymbol{\mu}^{1/2}\,\mathrm{d}\mathbf{W}_t$ crossing an index-1 saddle point $\mathbf{x}_{\mathrm{sad}} \in \mathbb{R}^N$.

Theorem 7.5 (Langer's Multidimensional Saddle-Point Escape Rate [64])**.** *Let $E \in C^3(\mathbb{R}^N)$ be an N-dimensional bistable energy landscape with a local minimum at $\mathbf{x}_a \in \mathcal{A}$ ($\nabla^2 E(\mathbf{x}_a) \succ \mathbf{0}$) and an index-1 saddle point at $\mathbf{x}_{\mathrm{sad}} \in \Sigma_{\mathrm{sep}}$ whose Hessian $\nabla^2 E(\mathbf{x}_{\mathrm{sad}})$ has one negative eigenvalue $-\lambda_{\mathrm{u}} < 0$ and $N-1$ positive eigenvalues $\lambda_2^{(s)}, \dots, \lambda_N^{(s)} > 0$. Let $\boldsymbol{\mu} \succ \mathbf{0}$ be the mobility tensor at the saddle, and let $\kappa_{\mathrm{sad}} > 0$ be the unique positive eigenvalue of the linearized deterministic drift matrix $-\boldsymbol{\mu}\nabla^2 E(\mathbf{x}_{\mathrm{sad}})$ at the saddle (so that $\kappa_{\mathrm{sad}} = \mu\lambda_{\mathrm{u}}$ in the isotropic case $\boldsymbol{\mu} = \mu\mathbf{I}$). Then in the low-noise limit $\beta\Delta E_A = \beta\big[E(\mathbf{x}_{\mathrm{sad}}) - E(\mathbf{x}_a)\big] \to \infty$, the multidimensional escape rate across $\mathbf{x}_{\mathrm{sad}}$ is*

$$k_{A\to B} = \frac{\kappa_{\mathrm{sad}}}{2\pi}\sqrt{\frac{\det\nabla^2 E(\mathbf{x}_a)}{|\det\nabla^2 E(\mathbf{x}_{\mathrm{sad}})|}}\,\exp(-\beta\Delta E_A)\left[1 + \mathcal{O}(\beta^{-1})\right]. \tag{7.44}$$

In the isotropic overdamped case $\boldsymbol{\mu} = \mu\mathbf{I}$, where $\det\nabla^2 E(\mathbf{x}_a) = \prod_{k=1}^N \lambda_k^{(a)}$ and $|\det\nabla^2 E(\mathbf{x}_{\mathrm{sad}})| = \lambda_{\mathrm{u}}\prod_{k=2}^N \lambda_k^{(s)}$, Eq. (7.44) takes the factored form

$$k_{A\to B} = \underbrace{\frac{\mu\sqrt{\lambda_1^{(a)}\lambda_{\mathrm{u}}}}{2\pi}}_{\text{1D Kramers prefactor}} \times \underbrace{\prod_{k=2}^{N}\sqrt{\frac{\lambda_k^{(a)}}{\lambda_k^{(s)}}}}_{\text{entropic channel ratio}} \times \exp(-\beta\Delta E_A). \tag{7.45}$$

Proof. We compute the denominator N_A and numerator $I_{A\to B}$ of Eq. (7.43) in turn.

Step 1: Population N_A of Basin $\mathcal{A}$. Deep inside basin $\mathcal{A}$, the quasi-stationary density is in local Boltzmann equilibrium: $\rho(\mathbf{x}) \simeq c_0\, e^{-\beta E(\mathbf{x})}$. Expanding $E(\mathbf{x}) = E(\mathbf{x}_a) + \frac{1}{2}(\mathbf{x}-\mathbf{x}_a)^\top\nabla^2 E(\mathbf{x}_a)(\mathbf{x}-\mathbf{x}_a) + \dots$ and performing the N-dimensional Gaussian integral (Chapter B) gives

$$N_A = \int_{\mathcal{A}} \mathrm{d}^N x\,\rho(\mathbf{x}) \simeq c_0\, e^{-\beta E(\mathbf{x}_a)}\,\frac{(2\pi)^{N/2}}{\beta^{N/2}\sqrt{\det\nabla^2 E(\mathbf{x}_a)}}. \tag{7.46}$$

Step 2: Boundary-Layer Solution and Total Flux $I_{A\to B}$ Across the Saddle $\mathbf{x}_{\mathrm{sad}}$. Write $\rho(\mathbf{x}) = c_0\, e^{-\beta E(\mathbf{x})}\,\eta(\mathbf{x})$, where the crossover function $\eta(\mathbf{x})$ satisfies $\eta(\mathbf{x}) \to 1$ inside $\mathcal{A}$, $\eta(\mathbf{x}) \to 0$ inside $\mathcal{B}$, and solves the stationary backward equation $\nabla\cdot\big(e^{-\beta E(\mathbf{x})}\boldsymbol{\mu}\nabla\eta(\mathbf{x})\big) = 0$. In a neighborhood of the saddle $\mathbf{x}_{\mathrm{sad}}$, where $E(\mathbf{x}) \approx E(\mathbf{x}_{\mathrm{sad}}) + \frac{1}{2}\delta\mathbf{x}^\top\mathbf{H}_s\,\delta\mathbf{x}$ with $\delta\mathbf{x} = \mathbf{x} - \mathbf{x}_{\mathrm{sad}}$ and $\mathbf{H}_s := \nabla^2 E(\mathbf{x}_{\mathrm{sad}})$, the equation for $\eta(\delta\mathbf{x})$ becomes

$$\nabla\cdot(\boldsymbol{\mu}\nabla\eta) - \beta\,\delta\mathbf{x}^\top\mathbf{H}_s\boldsymbol{\mu}\nabla\eta = 0. \tag{7.47}$$

Look for a boundary-layer solution $\eta(\delta\mathbf{x}) = f(u)$ that depends on $\delta\mathbf{x}$ through a single linear combination $u := \mathbf{q}^\top \delta\mathbf{x}$, where $\mathbf{q} \in \mathbb{R}^N$ is an eigenvector of $\mathbf{H}_s\boldsymbol{\mu}$ with negative eigenvalue $-\kappa_{\mathrm{sad}} < 0$, that is, a left eigenvector of the linearized drift matrix $-\boldsymbol{\mu}\mathbf{H}_s$ with positive eigenvalue:

$$\mathbf{H}_s\boldsymbol{\mu}\mathbf{q} = -\kappa_{\mathrm{sad}}\mathbf{q} \qquad \Longleftrightarrow \qquad (-\boldsymbol{\mu}\mathbf{H}_s)^\top \mathbf{q} = +\kappa_{\mathrm{sad}}\mathbf{q} \quad (\kappa_{\mathrm{sad}} > 0). \tag{7.48}$$

The matrix $-\boldsymbol{\mu}\mathbf{H}_s$ has exactly one positive eigenvalue. Since $\boldsymbol{\mu} \succ \mathbf{0}$ has a symmetric positive-definite square root, $-\boldsymbol{\mu}\mathbf{H}_s$ is similar to $-\boldsymbol{\mu}^{1/2}\mathbf{H}_s\boldsymbol{\mu}^{1/2}$, and by Sylvester's law of inertia this congruent matrix has the same signature as $-\mathbf{H}_s$: one positive eigenvalue $+\kappa_{\mathrm{sad}}$ and $N-1$ negative eigenvalues. Normalizing $\mathbf{q}$ so that $\mathbf{q}^\top\boldsymbol{\mu}\mathbf{q} = \kappa_{\mathrm{sad}}$, we have $\nabla\eta = \mathbf{q}f'(u)$ and $\nabla\cdot(\boldsymbol{\mu}\nabla\eta) = (\mathbf{q}^\top\boldsymbol{\mu}\mathbf{q})f''(u) = \kappa_{\mathrm{sad}}f''(u)$, while $\delta\mathbf{x}^\top\mathbf{H}_s\boldsymbol{\mu}\mathbf{q} = -\kappa_{\mathrm{sad}}\delta\mathbf{x}^\top\mathbf{q} = -\kappa_{\mathrm{sad}}u$. Thus Eq. (7.47) reduces to the 1D ordinary differential equation

$$f''(u) + \beta u f'(u) = 0, \qquad f(-\infty) = 1, \quad f(+\infty) = 0, \tag{7.49}$$

whose unique solution is the complementary error function profile

$$f'(u) = -\sqrt{\frac{\beta}{2\pi}}\,\exp\!\left(-\frac{1}{2}\beta u^2\right), \qquad \eta(\delta\mathbf{x}) = \sqrt{\frac{\beta}{2\pi}}\int_{\mathbf{q}^\top\delta\mathbf{x}}^{\infty} \mathrm{d}s\, e^{-\beta s^2/2}. \tag{7.50}$$

Substituting Eq. (7.50) into the probability current $\mathbf{J}(\mathbf{x}) = -k_{\mathrm{B}}Tc_0 e^{-\beta E(\mathbf{x})}\boldsymbol{\mu}\nabla\eta(\mathbf{x})$ near the saddle gives

$$\mathbf{J}(\delta\mathbf{x}) = k_{\mathrm{B}}Tc_0\, e^{-\beta E(\mathbf{x}_{\mathrm{sad}})}\sqrt{\frac{\beta}{2\pi}}\,\boldsymbol{\mu}\mathbf{q}\,\exp\!\left(-\frac{\beta}{2}\delta\mathbf{x}^\top(\mathbf{H}_s + \mathbf{q}\mathbf{q}^\top)\delta\mathbf{x}\right). \tag{7.51}$$

We integrate the flux over the $(N-1)$-dimensional hyperplane $\Pi := \left\{\delta\mathbf{x} : u = \mathbf{q}^\top\delta\mathbf{x} = 0\right\}$ with unit normal $\hat{\mathbf{n}} = \mathbf{q}/\left\|\mathbf{q}\right\|$. On Π the added term $(\mathbf{q}^\top\delta\mathbf{x})^2 = u^2$ in the exponent of Eq. (7.51) vanishes, so the integrand is $\exp(-\frac{\beta}{2}\delta\mathbf{x}^\top\mathbf{H}_s\delta\mathbf{x})$ restricted to Π, and the normal component of the current carries the factor $\hat{\mathbf{n}}\cdot\boldsymbol{\mu}\mathbf{q} = \mathbf{q}^\top\boldsymbol{\mu}\mathbf{q}/\left\|\mathbf{q}\right\| = \kappa_{\mathrm{sad}}/\left\|\mathbf{q}\right\|$. The restriction $\mathbf{H}_s|_\Pi$ is positive definite: if $\mathbf{y}^\top\mathbf{H}_s\mathbf{y} \le 0$ for some nonzero $\mathbf{y} \in \Pi$, then $\mathbf{H}_s$ would be negative semi-definite on the two-dimensional span of $\mathbf{y}$ and $\boldsymbol{\mu}\mathbf{q}$ (because $(\boldsymbol{\mu}\mathbf{q})^\top\mathbf{H}_s(\boldsymbol{\mu}\mathbf{q}) = -\kappa_{\mathrm{sad}}\mathbf{q}^\top\boldsymbol{\mu}\mathbf{q} = -\kappa_{\mathrm{sad}}^2 < 0$ and $\mathbf{y}^\top\mathbf{H}_s\boldsymbol{\mu}\mathbf{q} = -\kappa_{\mathrm{sad}}\mathbf{y}^\top\mathbf{q} = 0$ by Eq. (7.48)), which by the Courant–Fischer min-max principle would force a second non-positive eigenvalue of $\mathbf{H}_s$. Its determinant follows from the Schur-complement identity $\det(\mathbf{H}_s|_\Pi) = \det\mathbf{H}_s \cdot \mathbf{q}^\top\mathbf{H}_s^{-1}\mathbf{q}/\left\|\mathbf{q}\right\|^2$, together with $\mathbf{q}^\top\mathbf{H}_s^{-1}\mathbf{q} = -\mathbf{q}^\top\boldsymbol{\mu}\mathbf{q}/\kappa_{\mathrm{sad}} = -1$ (multiply $\mathbf{H}_s\boldsymbol{\mu}\mathbf{q} = -\kappa_{\mathrm{sad}}\mathbf{q}$ by $\mathbf{q}^\top\mathbf{H}_s^{-1}$) and $\det\mathbf{H}_s < 0$, giving $\det(\mathbf{H}_s|_\Pi) = |\det\mathbf{H}_s|\,/\left\|\mathbf{q}\right\|^2$. The $(N-1)$-dimensional Gaussian integral over Π is therefore $(2\pi/\beta)^{(N-1)/2}\left\|\mathbf{q}\right\|/\sqrt{|\det\mathbf{H}_s|}$; the factor $\left\|\mathbf{q}\right\|$ cancels against $1/\left\|\mathbf{q}\right\|$ from the normal current, and using $k_{\mathrm{B}}T\sqrt{\beta/2\pi}\,(2\pi/\beta)^{(N-1)/2} = (2\pi)^{-1}(2\pi/\beta)^{N/2}$ we obtain

$$I_{A\to B} = \int_{u=0} \mathrm{d}^{N-1}S\,\hat{\mathbf{n}}\cdot\mathbf{J}(\delta\mathbf{x}) = c_0\, e^{-\beta E(\mathbf{x}_{\mathrm{sad}})}\,\frac{\kappa_{\mathrm{sad}}}{2\pi}\,\frac{(2\pi)^{N/2}}{\beta^{N/2}\sqrt{|\det\nabla^2 E(\mathbf{x}_{\mathrm{sad}})|}}. \tag{7.52}$$

Dividing $I_{A\to B}$ in Eq. (7.52) by N_A in Eq. (7.46) cancels $c_0(2\pi/\beta)^{N/2}$ and yields Langer's formula Eq. (7.44). □

Remark 7.6 (Entropic Channel Width and Underdamped Kramers Turnover)**.** Two physical features of Langer's formula Eqs. (7.44) and (7.45) deserve emphasis:

1. **Transverse Entropic Barrier:** In Eq. (7.45), the transverse eigenvalue product $\prod_{k=2}^{N}\sqrt{\lambda_k^{(a)}/\lambda_k^{(s)}}$ is the ratio of the thermal width of the saddle channel ($\propto (\lambda_k^{(s)})^{-1/2}$) to the thermal width

of the initial well ($\propto (\lambda_k^{(a)})^{-1/2}$). If the mountain pass is much narrower in the transverse directions than the valley ($\lambda_k^{(s)} \gg \lambda_k^{(a)}$), this factor suppresses $k_{A\to B}$ at a given energetic barrier ΔE_A. This is an *entropic bottleneck*.

2. **Underdamped Phase-Space Escape:** Langer's flux-over-population construction also applies when the linearized drift at the saddle is not symmetric and the noise is degenerate, as happens when the drift contains a conservative Hamiltonian part; κ_{sad} is then the unique positive eigenvalue of the full linearized drift. Applied to the 2D underdamped phase-space Langevin equation $\dot{x} = p/m$, $\dot{p} = -E'(x) - (\gamma/m)p + \sqrt{2\gamma k_{\text{B}}T}\,\xi(t)$ of Chapter 5, it gives Kramers' moderate-to-high damping escape rate [46, 61]:

$$k_{A\to B}^{\text{mod-high}} = \frac{\omega_a}{2\pi\omega_{\text{sad}}}\left(\sqrt{\frac{\gamma^2}{4m^2} + \omega_{\text{sad}}^2} - \frac{\gamma}{2m}\right)\exp(-\beta\Delta E_A)\,, \tag{7.53}$$

where $\omega_a = \sqrt{E''(x_a)/m}$ and $\omega_{\text{sad}} = \sqrt{|E''(x_{\text{sad}})|/m}$ are the mechanical angular frequencies, and $\kappa_{\text{sad}} = \sqrt{\frac{\gamma^2}{4m^2} + \omega_{\text{sad}}^2} - \frac{\gamma}{2m}$ is the positive root of the linearized phase-space saddle characteristic equation $\kappa^2 + \frac{\gamma}{m}\kappa - \omega_{\text{sad}}^2 = 0$. In the overdamped limit $\frac{\gamma}{m\omega_{\text{sad}}} \gg 1$, expanding $\sqrt{1 + \frac{4m^2\omega_{\text{sad}}^2}{\gamma^2}} \approx 1 + \frac{2m^2\omega_{\text{sad}}^2}{\gamma^2}$ in Eq. (7.53) recovers $\frac{m\omega_a\omega_{\text{sad}}}{2\pi\gamma}e^{-\beta\Delta E_A} = \frac{\sqrt{E''(x_a)|E''(x_{\text{sad}})|}}{2\pi\gamma}e^{-\beta\Delta E_A}$, which is Eq. (7.37). At very weak damping a different bottleneck, diffusion in energy, takes over and the rate becomes proportional to γ. Section 7.3.2 treats this regime and the resulting *Kramers turnover* [46].

Hardware Realization: Kramers Switching Bandwidth of Hardware p-Bits vs. Digital Memory Retention

Kramers' law $k_{A\to B} = f_0\exp(-\Delta E/k_{\text{B}}T)$, with f_0 the attempt frequency, shows that a *single physical bistable circuit* spans the range from a fast probabilistic bit (p-bit) to a non-volatile memory cell as the dimensionless barrier $\beta\Delta E = \Delta E/k_{\text{B}}T$ is tuned [14, 18, 31]. The attempt frequency depends on the technology. In a subthreshold CMOS latch biased at nanoamperes it is set by $g_{\text{m}}/C_{\text{node}}$ and is of order $f_0 \sim 10^8\,\text{s}^{-1}$; in a cryogenic Josephson parametron it is of order $10^{10}\,\text{s}^{-1}$ (Chapter 14). Taking the CMOS value $f_0 = 10^8\,\text{s}^{-1}$ as a representative estimate:

- **Stochastic p-bit regime ($\Delta E \sim 0$–$3\,k_{\text{B}}T$):** With $\Delta E = 1\,k_{\text{B}}T$, the switching rate is $k_{A\to B} = f_0 e^{-1} \approx 3.7\times10^7$ flips/s (mean dwell time ≈ 27 ns). The circuit produces tens of millions of thermal random bits per second.
- **Metastable regime ($\Delta E = 15\,k_{\text{B}}T$):** The switching rate drops by six orders of magnitude to $k_{A\to B} = 10^8 e^{-15} \approx 31$ flips/s (dwell time ≈ 33 ms).
- **Non-volatile memory regime ($\Delta E = 45\,k_{\text{B}}T \approx 1.16$ eV at 300 K):** The mean escape time reaches $(10^8\,\text{s}^{-1})^{-1}e^{45} \approx 3.5\times10^{11}$ s, about 10^4 years.

The same exponential sensitivity has two consequences. It allows fast hardware p-bits when ΔE is a few $k_{\text{B}}T$. It also means that a large energy-based model whose barriers grow with system size, $\Delta E \sim cN^{\psi}\,k_{\text{B}}T$ (Chapter 3), is effectively frozen once N is large. The denoising thermodynamic models of Chapter 16 are designed to avoid such barriers.

7.3.2 Kramers' Turnover: Energy Diffusion versus Spatial Diffusion

The overdamped formula Eq. (7.37) assumes that the slowest step in an escape event is diffusion of the position x across the barrier top. When inertia matters, a second bottleneck appears. A weakly damped particle in well $\mathcal{A}$ oscillates many times at nearly constant total energy before friction and noise change that energy appreciably. If the coupling to the bath is weak enough, the particle simply cannot acquire the energy $E(x_{\mathrm{sad}})$ fast enough, and the rate becomes limited by slow diffusion in *energy* rather than in position. Kramers worked out both limits in his 1940 paper [61]; the review [46] collects the later refinements. We now state the three regimes with their conditions.

The underdamped (Klein–Kramers) Langevin equation for a particle of mass m with friction coefficient γ is

$$\mathrm{d}x_t = \frac{p_t}{m}\,\mathrm{d}t, \qquad \mathrm{d}p_t = -E'(x_t)\,\mathrm{d}t - \frac{\gamma}{m}p_t\,\mathrm{d}t + \sqrt{2\gamma k_{\mathrm{B}}T}\,\mathrm{d}W_t, \tag{7.54}$$

whose stationary density is the phase-space Boltzmann distribution $p_{\mathrm{eq}}(x,p) \propto e^{-\beta H(x,p)}$ with $H(x,p) = p^2/2m + E(x)$. We write $\Gamma := \gamma/m$ for the velocity damping rate, $\omega_a := \sqrt{E''(x_a)/m}$ and $\omega_{\mathrm{sad}} := \sqrt{|E''(x_{\mathrm{sad}})|/m}$ for the mechanical frequencies, and

$$I(\mathcal{E}) := \oint_{H(x,p)=\mathcal{E}} p\,\mathrm{d}x = 2\int_{x_-(\mathcal{E})}^{x_+(\mathcal{E})} \mathrm{d}x\,\sqrt{2m(\mathcal{E}-E(x))} \tag{7.55}$$

for the action of the closed orbit of energy $\mathcal{E}$ in well $\mathcal{A}$ (with turning points $x_\pm(\mathcal{E})$). Its derivative is the orbital period, $I'(\mathcal{E}) = \oint \mathrm{d}x\,(m/p) = T(\mathcal{E})$, which tends to $2\pi/\omega_a$ at the bottom of the well. We abbreviate $I_{\mathrm{sad}} := I(E(x_{\mathrm{sad}}))$.

Notation Watch: Three Damping Symbols

In this chapter γ is the friction coefficient (force per unit velocity), $\mu = \gamma^{-1}$ is the overdamped mobility used in Eq. (7.1), and $\Gamma = \gamma/m$ is the velocity damping rate (inverse seconds). Overdamped formulas contain only the combination $E''/\gamma = \mu E''$, which is a relaxation rate. Underdamped formulas compare Γ with the mechanical frequencies ω_a, ω_{sad} and with the energy-exchange scale $k_{\mathrm{B}}T/I_{\mathrm{sad}}$. In the RCSJ model of a Josephson junction (Chapter 14) the same roles are played by the capacitance (mass), the shunt conductance (friction), and $\Gamma = 1/RC$.

Regime 1: transition-state theory (the upper bound). Suppose the well population is in full Boltzmann equilibrium in phase space and every trajectory that crosses $x = x_{\mathrm{sad}}$ with $p > 0$ is counted as a completed escape. The rate is then the equilibrium one-way flux through the dividing line divided by the well population:

$$k_{A\to B}^{\mathrm{TST}} = \frac{\displaystyle\int_0^\infty \mathrm{d}p\,\frac{p}{m}\,e^{-\beta[p^2/2m+E(x_{\mathrm{sad}})]}}{\displaystyle\int_{\mathcal{A}}\mathrm{d}x\int_{-\infty}^{\infty}\mathrm{d}p\,e^{-\beta[p^2/2m+E(x)]}} = \frac{k_{\mathrm{B}}T\,e^{-\beta E(x_{\mathrm{sad}})}}{\sqrt{2\pi m k_{\mathrm{B}}T}\,\sqrt{2\pi k_{\mathrm{B}}T/(m\omega_a^2)}\;e^{-\beta E(x_a)}}. \tag{7.56}$$

In the numerator we used $\int_0^\infty \mathrm{d}p\,(p/m)e^{-\beta p^2/2m} = k_{\mathrm{B}}T$; in the denominator we did the Gaussian momentum integral and the Laplace integral around x_a with $E''(x_a) = m\omega_a^2$. The factors of $k_{\mathrm{B}}T$ and m cancel, leaving

$$k_{A\to B}^{\mathrm{TST}} = \frac{\omega_a}{2\pi}\,e^{-\beta\Delta E_A}. \tag{7.57}$$

This is an upper bound on the true rate because it ignores two effects: trajectories that cross $x_{\rm sad}$ and are then pushed back by the noise (recrossings), and depletion of the high-energy part of the well population by the escape itself.

Regime 2: spatial diffusion (moderate to strong damping). When Γ is large enough that the well stays thermalized up to the barrier energy, only recrossings correct Eq. (7.57). Linearizing Eq. (7.54) at the saddle, $\delta\dot{x} = \delta p/m$, $\delta\dot{p} = m\omega_{\rm sad}^2\delta x - \Gamma\delta p$, gives a drift matrix whose eigenvalues κ solve $\kappa^2 + \Gamma\kappa - \omega_{\rm sad}^2 = 0$. Its unique positive root $\kappa_{\rm sad}$ plays the role of $\kappa_{\rm sad}$ in Langer's formula Eq. (7.44). The phase-space Hessian of H is $\mathrm{diag}(m\omega_a^2, 1/m)$ at the well and $\mathrm{diag}(-m\omega_{\rm sad}^2, 1/m)$ at the saddle, so the determinant ratio is $\omega_a^2/\omega_{\rm sad}^2$. Langer's formula gives

$$k_{A\to B}^{\rm sd} = \frac{\kappa_{\rm sad}}{\omega_{\rm sad}}\,\frac{\omega_a}{2\pi}\,e^{-\beta\Delta E_A} = \frac{\kappa_{\rm sad}}{\omega_{\rm sad}}\,k_{A\to B}^{\rm TST}, \qquad \kappa_{\rm sad} = \sqrt{\frac{\Gamma^2}{4} + \omega_{\rm sad}^2} - \frac{\Gamma}{2}, \tag{7.58}$$

which is Eq. (7.53). The *transmission factor* $\kappa_{\rm sad}/\omega_{\rm sad} \in (0,1)$ is the fraction of TST crossings that are not undone by recrossing. It tends to 1 as $\Gamma/\omega_{\rm sad} \to 0$ and to $\omega_{\rm sad}/\Gamma$ as $\Gamma/\omega_{\rm sad} \to \infty$; the second limit gives back the overdamped result Eq. (7.37). The condition for Eq. (7.58) to hold is that the energy dissipated during one oscillation near the barrier energy is large compared with $k_{\rm B}T$:

$$\Gamma\, I_{\rm sad} \gg k_{\rm B}T, \qquad \beta\Delta E_A \gg 1. \tag{7.59}$$

To see why ΓI is the energy loss per cycle, average the frictional power $\Gamma p^2/m$ over one period: $\oint \mathrm{d}t\,\Gamma p^2/m = \Gamma \oint p\,\mathrm{d}x = \Gamma I(\mathcal{E})$.

Regime 3: energy diffusion (weak damping). When $\Gamma I_{\rm sad} \ll k_{\rm B}T$, the energy $\mathcal{E}_t = H(x_t, p_t)$ changes by much less than $k_{\rm B}T$ per orbit, and it becomes the slow variable. By Itô's formula applied to H along Eq. (7.54),

$$\mathrm{d}\mathcal{E}_t = \frac{p_t}{m}\,\mathrm{d}p_t + E'(x_t)\,\mathrm{d}x_t + \frac{1}{2m}(\mathrm{d}p_t)^2 = \Big(-\Gamma\frac{p_t^2}{m} + \Gamma k_{\rm B}T\Big)\,\mathrm{d}t + \frac{p_t}{m}\sqrt{2\gamma k_{\rm B}T}\,\mathrm{d}W_t. \tag{7.60}$$

Averaging the drift and the squared noise amplitude over one unperturbed orbit, and using $\langle p^2/m\rangle_{\rm orbit} = T(\mathcal{E})^{-1}\oint \mathrm{d}t\, p^2/m = I(\mathcal{E})/T(\mathcal{E})$, gives a one-dimensional diffusion in energy with drift $A(\mathcal{E}) = \Gamma[k_{\rm B}T - I/T]$ and diffusion coefficient $B(\mathcal{E}) = \tfrac{1}{2}\langle 2\gamma k_{\rm B}T p^2/m^2\rangle = \Gamma k_{\rm B}T\, I/T$. The density $P(\mathcal{E},t)$ obeys $\partial_t P = -\partial_{\mathcal{E}} J_{\mathcal{E}}$ with current

$$J_{\mathcal{E}} = AP - \partial_{\mathcal{E}}(BP) = -\Gamma k_{\rm B}T\, I(\mathcal{E})\, e^{-\beta\mathcal{E}}\,\frac{\mathrm{d}}{\mathrm{d}\mathcal{E}}\left[\frac{e^{\beta\mathcal{E}}P(\mathcal{E})}{T(\mathcal{E})}\right]. \tag{7.61}$$

The second form is checked by expanding the derivative and using $I' = T$: $-\Gamma k_{\rm B}T I e^{-\beta\mathcal{E}}\partial_{\mathcal{E}}(e^{\beta\mathcal{E}}P/T) = -\Gamma I P/T - \Gamma k_{\rm B}T I\,\partial_{\mathcal{E}}(P/T)$, which equals $AP - \partial_{\mathcal{E}}(BP) = \Gamma k_{\rm B}T P - \Gamma I P/T - \Gamma k_{\rm B}T\,\partial_{\mathcal{E}}(IP/T)$ because $\partial_{\mathcal{E}}(IP/T) = P + I\,\partial_{\mathcal{E}}(P/T)$. The current vanishes for $P_{\rm eq} \propto T(\mathcal{E})e^{-\beta\mathcal{E}}$, the microcanonical density of states times the Boltzmann factor, as it must.

Now apply the flux-over-population method in energy. Orbits with $\mathcal{E} > E(x_{\rm sad})$ leave the well, so we impose $P(E(x_{\rm sad})) = 0$ and look for a steady state with constant current $J_{\mathcal{E}} = J_0$. Deep in the well P is close to the equilibrium profile, $e^{\beta\mathcal{E}}P/T \approx N_A\,\beta\omega_a e^{\beta E(x_a)}/(2\pi)$, where we normalized with $\int \mathrm{d}\mathcal{E}\,T e^{-\beta\mathcal{E}} \approx (2\pi/\omega_a)k_{\rm B}T e^{-\beta E(x_a)}$. Integrating Eq. (7.61) from a low energy up to $E(x_{\rm sad})$ gives

$$N_A\,\frac{\beta\omega_a}{2\pi}\,e^{\beta E(x_a)} = \frac{J_0}{\Gamma k_{\rm B}T}\int^{E(x_{\rm sad})}\mathrm{d}\mathcal{E}\,\frac{e^{\beta\mathcal{E}}}{I(\mathcal{E})} \simeq \frac{J_0}{\Gamma k_{\rm B}T}\,\frac{e^{\beta E(x_{\rm sad})}}{\beta I_{\rm sad}}, \tag{7.62}$$

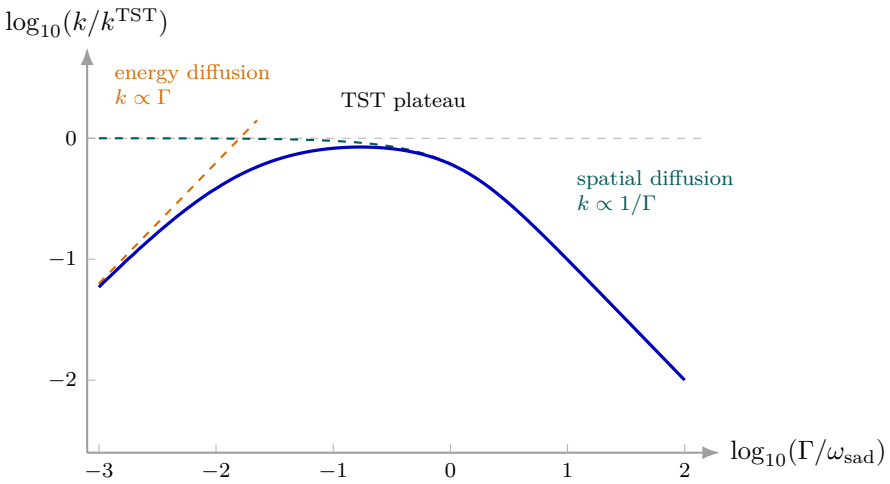


Figure 7.2: Kramers' turnover for $\beta\Delta E_A = 10$ and $\omega_a = \omega_{\mathrm{sad}}$, using the harmonic estimate $I_{\mathrm{sad}} = 2\pi\Delta E_A/\omega_a$. Dashed orange: energy-diffusion rate Eq. (7.63). Dashed teal: spatial-diffusion rate Eq. (7.58). Solid blue: the interpolation $[1/k^{\mathrm{ed}} + 1/k^{\mathrm{sd}}]^{-1}$, which peaks at $\Gamma_{\max} \approx 0.17\,\omega_{\mathrm{sad}}$ with $k \approx 0.85\,k^{\mathrm{TST}}$. Overdamped circuits, such as CMOS realizations of the double-well sampler, live at the far right of this plot, where $k \propto 1/\Gamma$.

because the integrand is dominated by its upper end within $\delta\mathcal{E} \sim k_{\mathrm{B}}T$. Solving for $k = J_0/N_A$ gives Kramers' energy-diffusion rate

$$k^{\mathrm{ed}}_{A\to B} = \Gamma\,\beta I_{\mathrm{sad}}\,\frac{\omega_a}{2\pi}\,e^{-\beta\Delta E_A} = (\beta\Gamma I_{\mathrm{sad}})\,k^{\mathrm{TST}}_{A\to B}, \qquad \text{valid for } \Gamma I_{\mathrm{sad}} \ll k_{\mathrm{B}}T,\ \beta\Delta E_A \gg 1. \tag{7.63}$$

For a nearly harmonic well, $I(\mathcal{E}) \approx 2\pi(\mathcal{E} - E(x_a))/\omega_a$, so $k^{\mathrm{ed}}_{A\to B} \approx \Gamma\,\beta\Delta E_A\,e^{-\beta\Delta E_A}$: the rate is proportional to the friction, the opposite of the overdamped $1/\gamma$ law.

The turnover. The three regimes join as follows. For $\Gamma \ll k_{\mathrm{B}}T/I_{\mathrm{sad}}$ the rate grows linearly with Γ Eq. (7.63); for $k_{\mathrm{B}}T/I_{\mathrm{sad}} \ll \Gamma \ll \omega_{\mathrm{sad}}$ it sits on the TST plateau Eq. (7.57); for $\Gamma \gg \omega_{\mathrm{sad}}$ it falls as $\omega_{\mathrm{sad}}/\Gamma$ Eq. (7.58). Since $I_{\mathrm{sad}} \approx 2\pi\Delta E_A/\omega_a$, the plateau is wide when $\beta\Delta E_A$ is large. At its left edge, $\Gamma = k_{\mathrm{B}}T/I_{\mathrm{sad}} \approx \omega_a/(2\pi\beta\Delta E_A)$, the energy-diffusion rate equals k^{TST} and the true rate is still only about half of it. The maximum lies inside the plateau: minimizing $1/k^{\mathrm{ed}} + 1/k^{\mathrm{sd}} \approx (k^{\mathrm{TST}})^{-1}[k_{\mathrm{B}}T/(\Gamma I_{\mathrm{sad}}) + 1 + \Gamma/(2\omega_{\mathrm{sad}})]$ over Γ (for $\Gamma \ll \omega_{\mathrm{sad}}$) gives $\Gamma_{\max} \approx \sqrt{2\omega_{\mathrm{sad}}k_{\mathrm{B}}T/I_{\mathrm{sad}}}$, the geometric mean of the two edges up to a factor $\sqrt{2}$, still far below ω_{sad} when $\beta\Delta E_A \gg 1$. A simple interpolation $k \approx [1/k^{\mathrm{ed}} + 1/k^{\mathrm{sd}}]^{-1}$ captures the shape to within a factor of order one; the uniform theory of Mel'nikov and Meshkov, reviewed in [46], is accurate across the whole range. Figure 7.2 plots the three curves.

For the rest of this book, the relevant regime is clear. A subthreshold CMOS node has no inertia worth mentioning: its dynamics is first order, $C_{\mathrm{node}}\,\mathrm{d}v = I(v)\,\mathrm{d}t + \text{noise}$, which is the overdamped limit with $\gamma \leftrightarrow C_{\mathrm{node}}$ playing the role of friction and no mass at all. All CMOS p-bits and NEAT-RN cells are therefore described by Eq. (7.37) and Eq. (7.44). The underdamped and energy-diffusion regimes matter for superconducting circuits with large Stewart–McCumber parameter β_{c}, where the junction capacitance supplies inertia (Chapter 14).

7.4 Freidlin–Wentzell Large Deviations, Quasipotentials, and Instantons

Kramers' and Langer's formulas rest on two facts: a scalar potential $E(\mathbf{x})$ exists, and the equilibrium density $p_{\mathrm{eq}} \propto e^{-\beta E}$ is known in closed form. Neither holds for the non-reciprocal circuits of Section 6.2, whose drift $\mathbf{a}(\mathbf{x})$ is not a gradient and whose stationary density p_{ss} is usually unknown. We still want the leading exponential behavior of rare transitions in such circuits: how unlikely a given fluctuation is, which path a transition follows, and how the mean escape time scales with temperature. Freidlin–Wentzell theory [36] answers all three questions using only the drift and the noise amplitude. The price is that it gives exponents, not prefactors.

Throughout this section we consider

$$\mathrm{d}\mathbf{x}_t = \mathbf{a}(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T}\,\boldsymbol{\mu}^{1/2}\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{D} = k_{\mathrm{B}}T\,\boldsymbol{\mu}, \tag{7.64}$$

with a constant symmetric positive-definite mobility tensor $\boldsymbol{\mu}$ and a temperature-independent, locally Lipschitz drift $\mathbf{a}(\mathbf{x})$. The gradient case is $\mathbf{a} = -\boldsymbol{\mu}\nabla E$. The small parameter is $k_{\mathrm{B}}T$, measured against the energy scales built into $\mathbf{a}$ and $\boldsymbol{\mu}$.

7.4.1 The Action Functional

Fix a time step Δt and consider the Euler–Maruyama transition density of Eq. (7.64) from $\mathbf{x}_k$ to $\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta\mathbf{x}_k$:

$$p(\mathbf{x}_{k+1} \mid \mathbf{x}_k) = \frac{1}{\left[(4\pi\Delta t)^N \det \mathbf{D}\right]^{1/2}} \exp\!\left(-\frac{\Delta t}{4k_{\mathrm{B}}T}\left(\frac{\Delta\mathbf{x}_k}{\Delta t} - \mathbf{a}(\mathbf{x}_k)\right)^{\!\top} \boldsymbol{\mu}^{-1}\left(\frac{\Delta\mathbf{x}_k}{\Delta t} - \mathbf{a}(\mathbf{x}_k)\right)\right). \tag{7.65}$$

Multiplying K such factors along a discretized path $\boldsymbol{\varphi}$ and letting $\Delta t \to 0$ gives a path weight $\exp(-\mathcal{S}_T[\boldsymbol{\varphi}]/k_{\mathrm{B}}T)$, where $\mathcal{S}_T$ is defined below. This is the Onsager–Machlup action Eq. (8.57) of Chapter 8 with $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$, except that the term $\frac{1}{2}\int \mathrm{d}t\,\nabla\cdot\mathbf{a}$ has been dropped. That term does not scale with $1/k_{\mathrm{B}}T$, so it changes path probabilities by factors of order one and does not affect the exponent as $k_{\mathrm{B}}T \to 0$.

Definition 7.7 (Freidlin–Wentzell Action Functional)**.** For an absolutely continuous path $\boldsymbol{\varphi} : [0,T] \to \mathbb{R}^N$, the *Freidlin–Wentzell action* of Eq. (7.64) is

$$\mathcal{S}_T[\boldsymbol{\varphi}] := \frac{1}{4}\int_0^T \mathrm{d}t\,\big(\dot{\boldsymbol{\varphi}}_t - \mathbf{a}(\boldsymbol{\varphi}_t)\big)^{\top}\boldsymbol{\mu}^{-1}\big(\dot{\boldsymbol{\varphi}}_t - \mathbf{a}(\boldsymbol{\varphi}_t)\big), \tag{7.66}$$

and $\mathcal{S}_T[\boldsymbol{\varphi}] = +\infty$ for paths that are not absolutely continuous. It has units of energy, it is non-negative, and it vanishes exactly on solutions of the noiseless flow $\dot{\boldsymbol{\varphi}} = \mathbf{a}(\boldsymbol{\varphi})$.

Notation Watch: Freidlin–Wentzell Action versus Onsager–Machlup Action

The Onsager–Machlup action $\mathcal{S}_{\mathrm{OM}}$ of Chapter 8 is dimensionless and includes the divergence term $\frac{1}{2}\int \nabla\cdot\mathbf{a}$. The Freidlin–Wentzell action $\mathcal{S}_T$ used here is measured in energy units, omits the divergence term, and satisfies $\mathcal{S}_{\mathrm{OM}} = \mathcal{S}_T/k_{\mathrm{B}}T + \mathcal{O}(1)$. Mathematical texts write the noise as $\sqrt{\varepsilon}\,\boldsymbol{\sigma}\,\mathrm{d}\mathbf{W}$ and use the action $\frac{1}{2}\int |\dot{\boldsymbol{\varphi}} - \mathbf{a}|^2_{(\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top})^{-1}}$; the translation is $\varepsilon \leftrightarrow 2k_{\mathrm{B}}T$, $\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top} \leftrightarrow \boldsymbol{\mu}$, which turns their $\frac{1}{2}$ into our $\frac{1}{4}$ and their ε into our $k_{\mathrm{B}}T$ up to that factor of two.

Theorem 7.8 (Freidlin–Wentzell Large Deviation Principle [36])**.** *Let* $\mathbf{x}_t$ *solve Eq. (7.64) with* $\mathbf{x}_0 = \mathbf{x}$ *and suppose* $\mathbf{a}$ *is globally Lipschitz. For every* $T > 0$ *and every set* $\mathcal{G}$ *of continuous paths on* $[0, T]$ *starting at* $\mathbf{x}$*, with interior* $\mathcal{G}^\circ$ *and closure* $\overline{\mathcal{G}}$ *in the uniform topology,*

$$-\inf_{\boldsymbol{\varphi}\in\mathcal{G}^\circ} \mathcal{S}_T[\boldsymbol{\varphi}] \le \liminf_{k_\mathrm{B}T\to 0} k_\mathrm{B}T \ln \mathbb{P}(\mathbf{x}_{[0,T]} \in \mathcal{G}) \le \limsup_{k_\mathrm{B}T\to 0} k_\mathrm{B}T \ln \mathbb{P}(\mathbf{x}_{[0,T]} \in \mathcal{G}) \le -\inf_{\boldsymbol{\varphi}\in\overline{\mathcal{G}}} \mathcal{S}_T[\boldsymbol{\varphi}]. \quad (7.67)$$

Informally, $\mathbb{P}(\mathbf{x}_{[0,T]} \approx \boldsymbol{\varphi}) \asymp \exp(-\mathcal{S}_T[\boldsymbol{\varphi}]/k_\mathrm{B}T)$*, where* $\asymp$ *means equality of the logarithms to leading order in* $1/k_\mathrm{B}T$*.*

Sketch of proof. We sketch the lower bound for a small tube $\mathcal{G}_\delta = \{\mathbf{y} : \sup_t \|\mathbf{y}_t - \boldsymbol{\varphi}_t\| < \delta\}$ around a smooth path $\boldsymbol{\varphi}$. Let $\tilde{\mathbb{P}}$ be the law of the process with drift $\dot{\boldsymbol{\varphi}}_t + \mathbf{a}(\mathbf{x}_t) - \mathbf{a}(\boldsymbol{\varphi}_t)$, which follows $\boldsymbol{\varphi}$ on average and stays in $\mathcal{G}_\delta$ with $\tilde{\mathbb{P}}$-probability tending to one as $k_\mathrm{B}T \to 0$. The two drifts differ by $\mathbf{u}_t = \dot{\boldsymbol{\varphi}}_t - \mathbf{a}(\boldsymbol{\varphi}_t)$ (plus a term of order δ inside the tube). By Girsanov's theorem (Theorem 4.22) the Radon–Nikodym derivative is

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}} = \exp\left(-\frac{1}{\sqrt{2k_\mathrm{B}T}} \int_0^T (\boldsymbol{\mu}^{-1/2}\mathbf{u}_t)^\top \,\mathrm{d}\tilde{\mathbf{W}}_t - \frac{1}{4k_\mathrm{B}T} \int_0^T \mathbf{u}_t^\top \boldsymbol{\mu}^{-1} \mathbf{u}_t \,\mathrm{d}t \right).$$

The deterministic second term equals $-\mathcal{S}_T[\boldsymbol{\varphi}]/k_\mathrm{B}T$. The stochastic first term has standard deviation of order $(k_\mathrm{B}T)^{-1/2}$, which is negligible against $1/k_\mathrm{B}T$. Hence $\mathbb{P}(\mathcal{G}_\delta) = \tilde{\mathbb{E}}[\mathbf{1}_{\mathcal{G}_\delta}\, \mathrm{d}\mathbb{P}/\,\mathrm{d}\tilde{\mathbb{P}}] \ge \exp(-[\mathcal{S}_T[\boldsymbol{\varphi}] + o(1)]/k_\mathrm{B}T)$, where $o(1) \to 0$ as $\delta \to 0$. The matching upper bound uses exponential tightness of the Brownian paths (Schilder's theorem) and the continuity of the map from noise to path; see [36, Ch. 3–4]. □

7.4.2 The Quasipotential and its Hamilton–Jacobi Equation

The theorem converts questions about rare events into minimization problems. The cheapest way to reach a state $\mathbf{x}$ from the attractor $\mathbf{x}_a$, allowing any amount of time, defines the central object of the theory.

Definition 7.9 (Quasipotential)**.** Let $\mathbf{x}_a$ be a stable fixed point of $\dot{\mathbf{x}} = \mathbf{a}(\mathbf{x})$. The *quasipotential* relative to $\mathbf{x}_a$ is

$$\mathcal{U}(\mathbf{x}) := \inf_{T>0} \inf\{\mathcal{S}_T[\boldsymbol{\varphi}] : \boldsymbol{\varphi}_0 = \mathbf{x}_a,\ \boldsymbol{\varphi}_T = \mathbf{x}\}. \quad (7.68)$$

A path that attains (or approaches) the infimum is called a *minimum-action path* or *instanton*.

If $\mathbf{x}_a$ is the only attractor and its basin is all of $\mathbb{R}^N$, the stationary density obeys the WKB form

$$p_\mathrm{ss}(\mathbf{x}) \asymp \exp\left(-\frac{\mathcal{U}(\mathbf{x})}{k_\mathrm{B}T}\right), \qquad \text{i.e.} \qquad \lim_{k_\mathrm{B}T\to 0} k_\mathrm{B}T\, U(\mathbf{x}) = \mathcal{U}(\mathbf{x}) + \text{const}, \quad (7.69)$$

where $U = -\ln p_\mathrm{ss}$ is the generalized potential of Eq. (6.36). The quasipotential is thus the low-temperature limit of the non-equilibrium free-energy landscape. (With several attractors, the global p_ss is assembled from the local quasipotentials by Freidlin and Wentzell's graph construction [36, Ch. 6]; the two-well case reduces to the master equation Eq. (7.12).)

Proposition 7.10 (Hamilton–Jacobi Equation for the Quasipotential)**.** *Wherever* $\mathcal{U}$ *is differentiable, it satisfies the static Hamilton–Jacobi equation*

$$\mathcal{H}(\mathbf{x}, \nabla\mathcal{U}(\mathbf{x})) = \mathbf{a}(\mathbf{x}) \cdot \nabla\mathcal{U}(\mathbf{x}) + \nabla\mathcal{U}(\mathbf{x})^\top \boldsymbol{\mu}\, \nabla\mathcal{U}(\mathbf{x}) = 0, \quad (7.70)$$

with the Freidlin–Wentzell Hamiltonian $\mathcal{H}(\mathbf{x},\mathbf{p}) = \mathbf{p}\cdot\mathbf{a}(\mathbf{x}) + \mathbf{p}^{\top}\boldsymbol{\mu}\mathbf{p}$. *Instantons solve Hamilton's equations*

$$\dot{\boldsymbol{\varphi}} = \frac{\partial \mathcal{H}}{\partial \mathbf{p}} = \mathbf{a}(\boldsymbol{\varphi}) + 2\boldsymbol{\mu}\mathbf{p}, \qquad \dot{\mathbf{p}} = -\frac{\partial \mathcal{H}}{\partial \boldsymbol{\varphi}} = -\big(\nabla \mathbf{a}(\boldsymbol{\varphi})\big)^{\top}\mathbf{p}, \tag{7.71}$$

on the zero-energy surface $\mathcal{H} = 0$, *with* $\mathbf{p} = \nabla \mathcal{U}(\boldsymbol{\varphi})$ *along the path.*

Proof. We give two derivations.

Derivation 1 (dynamic programming). Write the action as $\mathcal{S}_T = \int_0^T \mathcal{L}(\boldsymbol{\varphi}, \dot{\boldsymbol{\varphi}})\, \mathrm{d}t$ with Lagrangian $\mathcal{L}(\mathbf{x},\mathbf{v}) = \frac{1}{4}(\mathbf{v}-\mathbf{a})^{\top}\boldsymbol{\mu}^{-1}(\mathbf{v}-\mathbf{a})$. From $\mathbf{x}$, one may move with any velocity $\mathbf{v}$ for a short time $\mathrm{d}t$. Since $\mathcal{U}$ is the least cost of reaching a point, the cost of reaching $\mathbf{x}+\mathbf{v}\,\mathrm{d}t$ through $\mathbf{x}$ can only exceed the optimum:

$$\mathcal{U}(\mathbf{x}+\mathbf{v}\,\mathrm{d}t) \le \mathcal{U}(\mathbf{x}) + \mathcal{L}(\mathbf{x},\mathbf{v})\,\mathrm{d}t \quad\Longrightarrow\quad \nabla\mathcal{U}\cdot\mathbf{v} - \mathcal{L}(\mathbf{x},\mathbf{v}) \le 0 \quad \text{for all } \mathbf{v}, \tag{7.72}$$

with equality for the velocity of the optimal path through $\mathbf{x}$. Therefore $\sup_{\mathbf{v}}[\nabla\mathcal{U}\cdot\mathbf{v} - \mathcal{L}(\mathbf{x},\mathbf{v})] = 0$. The supremum is the Legendre transform of $\mathcal{L}$. Setting the $\mathbf{v}$-gradient to zero gives $\mathbf{p} = \partial\mathcal{L}/\partial\mathbf{v} = \frac{1}{2}\boldsymbol{\mu}^{-1}(\mathbf{v}-\mathbf{a})$, i.e. $\mathbf{v} = \mathbf{a} + 2\boldsymbol{\mu}\mathbf{p}$, and substituting back,

$$\mathbf{p}\cdot\mathbf{v} - \mathcal{L} = \mathbf{p}\cdot\mathbf{a} + 2\mathbf{p}^{\top}\boldsymbol{\mu}\mathbf{p} - \frac{1}{4}(2\boldsymbol{\mu}\mathbf{p})^{\top}\boldsymbol{\mu}^{-1}(2\boldsymbol{\mu}\mathbf{p}) = \mathbf{p}\cdot\mathbf{a} + \mathbf{p}^{\top}\boldsymbol{\mu}\mathbf{p} = \mathcal{H}(\mathbf{x},\mathbf{p}). \tag{7.73}$$

With $\mathbf{p} = \nabla\mathcal{U}$ this is Eq. (7.70). Hamilton's equations Eq. (7.71) are the Euler–Lagrange equations of $\mathcal{L}$ written in canonical form, and $\mathcal{H} = 0$ because the final time is free: the optimal path has no incentive to speed up or slow down, which is the condition that the conserved "energy" $\mathcal{H}$ vanish.

Derivation 2 (WKB expansion of the stationary Fokker–Planck equation). Insert $p_{\mathrm{ss}} = \exp(-\mathcal{U}/k_{\mathrm{B}}T)$ into $0 = -\nabla\cdot(\mathbf{a}p_{\mathrm{ss}}) + k_{\mathrm{B}}T\,\nabla\cdot(\boldsymbol{\mu}\nabla p_{\mathrm{ss}})$. Using $\nabla p_{\mathrm{ss}} = -(\nabla\mathcal{U}/k_{\mathrm{B}}T)p_{\mathrm{ss}}$,

$$\begin{aligned} -\nabla\cdot(\mathbf{a}p_{\mathrm{ss}}) &= \Big[-\nabla\cdot\mathbf{a} + \frac{1}{k_{\mathrm{B}}T}\mathbf{a}\cdot\nabla\mathcal{U}\Big]p_{\mathrm{ss}}, \\ k_{\mathrm{B}}T\,\nabla\cdot(\boldsymbol{\mu}\nabla p_{\mathrm{ss}}) &= \Big[\frac{1}{k_{\mathrm{B}}T}\nabla\mathcal{U}^{\top}\boldsymbol{\mu}\nabla\mathcal{U} - \nabla\cdot(\boldsymbol{\mu}\nabla\mathcal{U})\Big]p_{\mathrm{ss}}. \end{aligned} \tag{7.74}$$

The sum must vanish. The terms of order $1/k_{\mathrm{B}}T$ give $\mathbf{a}\cdot\nabla\mathcal{U} + \nabla\mathcal{U}^{\top}\boldsymbol{\mu}\nabla\mathcal{U} = 0$, which is Eq. (7.70). The order-one terms fix the prefactor of p_{ss}, which the large deviation principle does not see. □

7.4.3 Gradient Systems: The Instanton Is the Time-Reversed Relaxation Path

For an equilibrium circuit the variational problem can be solved exactly, and the answer has a clear physical meaning.

Theorem 7.11 (Quasipotential and Instanton of a Gradient System)**.** *Let* $\mathbf{a} = -\boldsymbol{\mu}\nabla E$ *and let* $\mathbf{x}$ *lie in the closure of the basin* $\mathcal{A}$ *of the minimum* $\mathbf{x}_a$. *Then*

$$\mathcal{U}(\mathbf{x}) = E(\mathbf{x}) - E(\mathbf{x}_a), \tag{7.75}$$

and the infimum in Eq. (7.68) is approached, as $T\to\infty$, *by the path obeying the time-reversed gradient flow*

$$\dot{\boldsymbol{\varphi}}_t = +\boldsymbol{\mu}\nabla E(\boldsymbol{\varphi}_t). \tag{7.76}$$

In particular, the cheapest path from $\mathbf{x}_a$ *to the saddle* $\mathbf{x}_{\mathrm{sad}}$ *costs exactly* $\mathcal{U}(\mathbf{x}_{\mathrm{sad}}) = \Delta E_A$, *and it runs along the unstable manifold of the saddle, traversed backwards.*

Proof. **Lower bound.** For any path from $\mathbf{x}_a$ to $\mathbf{x}$, expand the integrand of Eq. (7.66) with $\mathbf{a} = -\boldsymbol{\mu}\nabla E$:

$$\begin{aligned}(\dot{\boldsymbol{\varphi}} + \boldsymbol{\mu}\nabla E)^{\top}\boldsymbol{\mu}^{-1}(\dot{\boldsymbol{\varphi}} + \boldsymbol{\mu}\nabla E) &= \dot{\boldsymbol{\varphi}}^{\top}\boldsymbol{\mu}^{-1}\dot{\boldsymbol{\varphi}} + 2\dot{\boldsymbol{\varphi}}\cdot\nabla E + \nabla E^{\top}\boldsymbol{\mu}\nabla E \\ &= (\dot{\boldsymbol{\varphi}} - \boldsymbol{\mu}\nabla E)^{\top}\boldsymbol{\mu}^{-1}(\dot{\boldsymbol{\varphi}} - \boldsymbol{\mu}\nabla E) + 4\,\dot{\boldsymbol{\varphi}}\cdot\nabla E. \end{aligned} \tag{7.77}$$

The last term is a total derivative, $\dot{\boldsymbol{\varphi}}\cdot\nabla E(\boldsymbol{\varphi}) = \frac{\mathrm{d}}{\mathrm{d}t}E(\boldsymbol{\varphi}_t)$. Integrating over $[0,T]$ and dividing by four,

$$\mathcal{S}_T[\boldsymbol{\varphi}] = \frac{1}{4}\int_0^T \mathrm{d}t\,(\dot{\boldsymbol{\varphi}} - \boldsymbol{\mu}\nabla E)^{\top}\boldsymbol{\mu}^{-1}(\dot{\boldsymbol{\varphi}} - \boldsymbol{\mu}\nabla E) + E(\mathbf{x}) - E(\mathbf{x}_a) \;\geq\; E(\mathbf{x}) - E(\mathbf{x}_a). \tag{7.78}$$

Attainability. Let $\boldsymbol{\psi}_s$ be the downhill relaxation path $\dot{\boldsymbol{\psi}} = -\boldsymbol{\mu}\nabla E(\boldsymbol{\psi})$ started at $\boldsymbol{\psi}_0 = \mathbf{x}$. Because $\mathbf{x}$ is in the closure of $\mathcal{A}$, $\boldsymbol{\psi}_s \to \mathbf{x}_a$ as $s \to \infty$ (if $\mathbf{x}$ is on the separatrix, choose $\mathbf{x}$ on the unstable manifold of the saddle and approach the separatrix as a limit). Define $\boldsymbol{\varphi}_t = \boldsymbol{\psi}_{T-t}$ on $[0,T]$; it obeys Eq. (7.76), so the first term of Eq. (7.78) vanishes identically and the action is $E(\mathbf{x}) - E(\boldsymbol{\psi}_T)$. Prepending a short straight segment from $\mathbf{x}_a$ to $\boldsymbol{\psi}_T$, whose cost tends to zero as $\boldsymbol{\psi}_T \to \mathbf{x}_a$, gives paths from $\mathbf{x}_a$ to $\mathbf{x}$ with action tending to $E(\mathbf{x}) - E(\mathbf{x}_a)$ as $T \to \infty$. □

The theorem says that, in an equilibrium system, the most probable way to climb out of a well is to run the movie of relaxation backward. Every fluctuation path costs at least the energy it gains, and the excess cost is the "distance" between the path and the reversed relaxation path. This is Onsager's principle of microscopic reversibility at the level of trajectories, and it is the trajectory counterpart of detailed balance (Theorem 6.4). Once the path passes the saddle, it can follow the forward relaxation flow into $\mathcal{B}$ at zero cost, so the complete transition path is the backward-run unstable manifold from $\mathbf{x}_a$ to $\mathbf{x}_{\mathrm{sad}}$ followed by the forward-run unstable manifold from $\mathbf{x}_{\mathrm{sad}}$ to $\mathbf{x}_b$. The total action of the transition is ΔE_A.

Physics Insight: Why the Instanton Takes Infinite Time

Both endpoints of the instanton are fixed points of the drift, so the reversed flow Eq. (7.76) leaves $\mathbf{x}_a$ and arrives at $\mathbf{x}_{\mathrm{sad}}$ only asymptotically, as $t \to \mp\infty$. Physically, the escaping trajectory spends a long time fluctuating near the well bottom, then makes a comparatively quick excursion up the instanton. The duration of the climb is of order $(\mu\lambda)^{-1}\ln(\Delta E_A/k_{\mathrm{B}}T)$, where $\mu\lambda$ is a typical relaxation rate: logarithmic in $1/k_{\mathrm{B}}T$ and much shorter than the waiting time $\sim e^{\beta\Delta E_A}$. Recorded escape events in bistable circuits therefore look like sudden jumps along a reproducible path, which is what single-trajectory recordings of bistable systems show.

Theorem 7.12 (Freidlin–Wentzell Exit Law [36])**.** *Let Ω be a bounded domain with smooth boundary containing the stable fixed point $\mathbf{x}_a$, contained in its basin, with $\mathbf{a}\cdot\hat{\mathbf{n}} < 0$ on $\partial\Omega$. Let $\mathcal{U}^* := \min_{\mathbf{y}\in\partial\Omega}\mathcal{U}(\mathbf{y})$. Then for every $\mathbf{x} \in \Omega$,*

$$\lim_{k_{\mathrm{B}}T\to 0} k_{\mathrm{B}}T\ln\mathbb{E}[\tau_\Omega(\mathbf{x})] = \mathcal{U}^*, \tag{7.79}$$

and if the minimum of $\mathcal{U}$ on $\partial\Omega$ is attained at a unique point $\mathbf{y}^*$, the exit position $\mathbf{x}_{\tau_\Omega}$ converges in probability to $\mathbf{y}^*$.

The proof compares two estimates. From any starting point, the process returns to a small ball around $\mathbf{x}_a$ quickly; from that ball, by Theorem 7.8, each attempt of bounded duration reaches $\partial\Omega$

with probability between $e^{-(\mathcal{U}^*+\delta)/k_{\mathrm{B}}T}$ and $e^{-(\mathcal{U}^*-\delta)/k_{\mathrm{B}}T}$. The number of attempts before success is geometrically distributed, which gives Eq. (7.79) and, in addition, the exponential law of the exit time found in the proof of Theorem 7.4. Letting Ω grow to the basin $\mathcal{A}$ of a gradient system, $\mathcal{U}^* \to \min_{\Sigma_{\mathrm{sep}}}[E - E(\mathbf{x}_a)] = \Delta E_A$, because the lowest point of the separatrix is the saddle. We recover the Arrhenius law

$$\lim_{k_{\mathrm{B}}T\to 0} k_{\mathrm{B}}T \ln \tau(\mathbf{x}_a \to \mathcal{B}) = \Delta E_A \qquad \Longleftrightarrow \qquad k_{A\to B} = \exp\!\left(-\frac{\Delta E_A + o(1)}{k_{\mathrm{B}}T}\right), \tag{7.80}$$

which agrees with the exponent of Kramers' and Langer's formulas. The prefactor is invisible at this level of approximation; it comes from Gaussian fluctuations around the instanton, which is exactly what the Laplace integrals of Section 7.3 computed. A rigorous proof of the full Eyring–Kramers formula, prefactor included, for reversible diffusions was given by Bovier, Eckhoff, Gayrard, and Klein using potential theory [15, 16].

7.4.4 Non-Gradient Circuits: Currents Bend the Escape Path

Now let the circuit violate detailed balance. Suppose the drift can be split as

$$\mathbf{a}(\mathbf{x}) = -\boldsymbol{\mu}\nabla\mathcal{U}(\mathbf{x}) + \boldsymbol{\ell}(\mathbf{x}), \qquad \boldsymbol{\ell}(\mathbf{x}) \cdot \nabla\mathcal{U}(\mathbf{x}) = 0. \tag{7.81}$$

Substituting into Eq. (7.70) gives $-\nabla\mathcal{U}^\top\boldsymbol{\mu}\nabla\mathcal{U} + \boldsymbol{\ell}\cdot\nabla\mathcal{U} + \nabla\mathcal{U}^\top\boldsymbol{\mu}\nabla\mathcal{U} = 0$, so $\mathcal{U}$ is indeed the quasipotential. Conversely, any solution of Eq. (7.70) produces such a split by setting $\boldsymbol{\ell} := \mathbf{a} + \boldsymbol{\mu}\nabla\mathcal{U}$. This is the low-temperature form of the Helmholtz–Hodge decomposition Eq. (6.38): as $k_{\mathrm{B}}T \to 0$, $k_{\mathrm{B}}TU \to \mathcal{U}$ and the stationary velocity $\mathbf{v}_{\mathrm{ss}} = \mathbf{a} + k_{\mathrm{B}}T\boldsymbol{\mu}\nabla U \to \boldsymbol{\ell}$, while the orthogonality relation Eq. (6.39), multiplied by $k_{\mathrm{B}}T$, reduces at leading order to $\boldsymbol{\ell}\cdot\nabla\mathcal{U} = 0$.

The instanton follows from Eq. (7.71) with $\mathbf{p} = \nabla\mathcal{U}$:

$$\text{instanton (uphill):}\quad \dot{\boldsymbol{\varphi}} = \mathbf{a} + 2\boldsymbol{\mu}\nabla\mathcal{U} = +\boldsymbol{\mu}\nabla\mathcal{U} + \boldsymbol{\ell}, \qquad \text{relaxation (downhill):}\quad \dot{\boldsymbol{\varphi}} = -\boldsymbol{\mu}\nabla\mathcal{U} + \boldsymbol{\ell}. \tag{7.82}$$

The gradient part reverses, but the circulating part $\boldsymbol{\ell}$ does not. The uphill path is therefore not the reversed downhill path; it is the reversed relaxation path of the *adjoint* dynamics with drift $-\boldsymbol{\mu}\nabla\mathcal{U} - \boldsymbol{\ell}$, the dynamics that generates the time-reversed stationary process. Along the instanton, $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{U} = \nabla\mathcal{U}\cdot\dot{\boldsymbol{\varphi}} = \nabla\mathcal{U}^\top\boldsymbol{\mu}\nabla\mathcal{U}$, so the action accumulated per unit time, $\mathcal{L} = \frac{1}{4}(2\boldsymbol{\mu}\nabla\mathcal{U})^\top\boldsymbol{\mu}^{-1}(2\boldsymbol{\mu}\nabla\mathcal{U}) = \nabla\mathcal{U}^\top\boldsymbol{\mu}\nabla\mathcal{U}$, equals the rate of climb in $\mathcal{U}$, consistent with $\mathcal{S} = \Delta\mathcal{U}$.

Example 7.13 (Instantons of the Non-Reciprocal RC Sampler)**.** Take the RC sampler (the coupled RC circuit of Chapter 1) in the linear Ornstein–Uhlenbeck form of Example 6.5, with drift $\mathbf{a}(\mathbf{v}) = \mathbf{A}\mathbf{v}$ and $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$. Its stationary covariance is $\boldsymbol{\Sigma} = k_{\mathrm{B}}T\hat{\boldsymbol{\Sigma}}$, where $\hat{\boldsymbol{\Sigma}}$ solves the temperature-free Lyapunov equation $\mathbf{A}\hat{\boldsymbol{\Sigma}} + \hat{\boldsymbol{\Sigma}}\mathbf{A}^\top + 2\boldsymbol{\mu} = \mathbf{0}$ (divide Eq. (6.42) by $k_{\mathrm{B}}T$). Try $\mathcal{U}(\mathbf{v}) = \frac{1}{2}\mathbf{v}^\top\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{v}$. Then $\nabla\mathcal{U} = \hat{\boldsymbol{\Sigma}}^{-1}\mathbf{v}$ and

$$\mathbf{a}\cdot\nabla\mathcal{U} + \nabla\mathcal{U}^\top\boldsymbol{\mu}\nabla\mathcal{U} = \mathbf{v}^\top\Big[\tfrac{1}{2}(\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{A} + \mathbf{A}^\top\hat{\boldsymbol{\Sigma}}^{-1}) + \hat{\boldsymbol{\Sigma}}^{-1}\boldsymbol{\mu}\hat{\boldsymbol{\Sigma}}^{-1}\Big]\mathbf{v} = \tfrac{1}{2}\mathbf{v}^\top\hat{\boldsymbol{\Sigma}}^{-1}\Big[\mathbf{A}\hat{\boldsymbol{\Sigma}} + \hat{\boldsymbol{\Sigma}}\mathbf{A}^\top + 2\boldsymbol{\mu}\Big]\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{v} = 0,$$

so $\mathcal{U}$ solves Eq. (7.70), and $\exp(-\mathcal{U}/k_{\mathrm{B}}T) = \exp(-\frac{1}{2}\mathbf{v}^\top\boldsymbol{\Sigma}^{-1}\mathbf{v})$ is the exact Gaussian p_{ss}, as it should be. With $\hat{\boldsymbol{\Omega}} := \mathbf{A}\hat{\boldsymbol{\Sigma}} + \boldsymbol{\mu}$ (skew-symmetric by the Lyapunov equation, as in Eq. (6.44)), the drift is $\mathbf{A}\mathbf{v} = (-\boldsymbol{\mu} + \hat{\boldsymbol{\Omega}})\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{v}$, and Eq. (7.82) gives

$$\text{relaxation: } \dot{\mathbf{v}} = (-\boldsymbol{\mu} + \hat{\boldsymbol{\Omega}})\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{v}, \qquad \text{instanton: } \dot{\mathbf{v}} = (+\boldsymbol{\mu} + \hat{\boldsymbol{\Omega}})\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{v} = -\hat{\boldsymbol{\Sigma}}\mathbf{A}^\top\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{v}. \tag{7.83}$$

For a concrete case, let $N = 2$, $\boldsymbol{\mu} = \mu\mathbf{I}$ and $\mathbf{A} = \mu\begin{pmatrix} -g & w \\ -w & -g \end{pmatrix}$ with $g > 0$, a pair of identical RC nodes with antisymmetric (non-reciprocal) coupling w. Then $\mathbf{A} + \mathbf{A}^\top = -2\mu g\mathbf{I}$, so $\hat{\boldsymbol{\Sigma}} = g^{-1}\mathbf{I}$ and $\mathcal{U} = \frac{g}{2}\|\mathbf{v}\|^2$, independent of w. In polar coordinates the relaxation path spirals inward, $\dot{r} = -\mu g r$, $\dot{\theta} = -\mu w$, while the instanton spirals outward, $\dot{r} = +\mu g r$, $\dot{\theta} = -\mu w$, rotating in the *same* sense. A large fluctuation of the node voltages is reached by a path that winds around the origin in the direction of the circulating current, not by retracing the relaxation spiral.

Hardware Realization: Rare-Event Statistics in Non-Reciprocal Analog Cells

The non-reciprocal couplings of subthreshold CMOS NEAT-RN cells (Section 6.2.1, Chapter 13) make the drift non-gradient, so Kramers' and Langer's formulas cannot be applied directly. Equations (7.69) and (7.79) still give the exponents. Two practical consequences follow. First, the frequency of rare large excursions, such as a node voltage crossing a comparator threshold, is set by the quasipotential $\mathcal{U}$ at the threshold, which can be computed from the drift alone by solving Eq. (7.70) or, for linear cells, the Lyapunov equation. Second, because circulation bends the instanton, the exit point on a threshold surface is shifted relative to the equilibrium prediction; a designer who reads off the "weakest point" of a decision boundary from the potential part alone can be misled.

7.5 The Witten Ground-State Doublet and the Spectral Gap

We now connect escape rates to the spectral theory of Section 6.4. For a reversible circuit, every relaxation process is a superposition of the decaying modes $e^{-\lambda_n t}$ of the Witten Hamiltonian $\mathcal{H}_{\mathrm{W}} = -D\Delta + V_{\mathrm{W}}$ (Theorem 6.9), and the slowest non-trivial mode decays at the spectral gap λ_1, which is also the optimal Poincaré constant (Definition 6.12). For a strongly convex landscape, Bakry–Émery gives $\lambda_1 \geq \mu\kappa$. For the double well of Fig. 7.1, the two-state master equation Eq. (7.12) predicts that $P_A(t)$ relaxes at $\lambda_{\mathrm{macro}} = k_{A\to B} + k_{B\to A}$, an exponentially small number. This section shows that λ_{macro} is the spectral gap: to leading order, $\lambda_1 = k_{A\to B} + k_{B\to A}$, and the eigenfunction of λ_1 is the committor.

7.5.1 Harmonic Analysis of the Low-Lying Spectrum

To see why $\mathcal{H}_{\mathrm{W}}$ has exactly one exponentially small non-zero eigenvalue in a double well, look at V_{W} near each critical point $\mathbf{x}_c$ of E. In one dimension, write $y = x - x_c$ and $E''(x_c) = h_c$. Then $E'(x) \approx h_c y$ and

$$\mathcal{H}_{\mathrm{W}} \approx -D\frac{\mathrm{d}^2}{\mathrm{d}y^2} + \frac{D\beta^2 h_c^2}{4}\,y^2 - \frac{D\beta h_c}{2}. \tag{7.84}$$

The operator $-D\,\mathrm{d}^2/\mathrm{d}y^2 + Ky^2$ has eigenvalues $(n+\frac{1}{2})\,2\sqrt{DK}$. With $K = D\beta^2 h_c^2/4$ this is $(n+\frac{1}{2})D\beta\,|h_c| = (n+\frac{1}{2})\mu\,|h_c|$, using $D\beta = \mu$. Adding the constant $-\frac{1}{2}\mu h_c$ gives the local levels

$$\lambda_n^{(c)} \approx \mu\Big[n\,|h_c| + \tfrac{1}{2}\big(|h_c| - h_c\big)\Big] = \begin{cases} n\,\mu h_c, & h_c > 0 \text{ (minimum)}, \\ (n+1)\,\mu\,|h_c|\,, & h_c < 0 \text{ (saddle or maximum)}, \end{cases} \qquad n = 0, 1, 2, \ldots \tag{7.85}$$

In N dimensions the same computation in the Hessian eigenbasis gives $\mu\sum_k[n_k\,|h_k| + \frac{1}{2}(|h_k| - h_k)]$, which vanishes only if all $h_k > 0$ and all $n_k = 0$. Each local minimum therefore contributes one

level at zero, while saddles and higher excitations contribute levels of order $\mu\,|h|$. For the double well, the harmonic approximation predicts two zero levels (one per well) and a third level at

$$\lambda_2 \approx \mu \min\big(E''(x_a),\, E''(x_b),\, |E''(x_{\mathrm{sad}})|\big). \tag{7.86}$$

The harmonic picture is exact only as $\beta \to \infty$. At finite β, the two well-localized zero modes overlap through the $\mathcal{O}(\beta^2)$ Witten barrier of Fig. 6.2(b). One combination stays exactly at $\lambda_0 = 0$, since the ground state $\psi_0 = p_{\mathrm{eq}}^{1/2}$ is known exactly (Eq. (6.73)). The other is pushed up to an exponentially small λ_1. This is the *ground-state doublet*. Counting exponentially small eigenvalues by counting local minima is Witten's observation [118]; more generally, the low-lying levels of the Witten Laplacian on k-forms count the critical points of Morse index k, which is his analytic proof of the Morse inequalities.

7.5.2 The Spectral Gap Equals the Sum of the Escape Rates

We work in one dimension with the bistable potential of Theorem 7.4. Define the restricted partition functions $Z_A = \int_{-\infty}^{x_{\mathrm{sad}}} e^{-\beta E}$, $Z_B = \int_{x_{\mathrm{sad}}}^{\infty} e^{-\beta E}$, the basin weights $P_A = Z_A/Z$, $P_B = Z_B/Z$, and the barrier integral

$$\mathcal{K} := \int_{x_a}^{x_b} \mathrm{d}y\, e^{\beta E(y)}. \tag{7.87}$$

The *committor* is the probability that the process started at x reaches x_b before x_a. It solves $\mathcal{L}q = 0$ on (x_a, x_b) with $q(x_a) = 0$, $q(x_b) = 1$. Since $\mathcal{L}q = De^{\beta E}(e^{-\beta E}q')'$, the solution is

$$q(x) = \frac{1}{\mathcal{K}} \int_{x_a}^{x} \mathrm{d}y\, e^{\beta E(y)} \quad (x_a \le x \le x_b), \qquad q(x) = 0 \ (x < x_a), \qquad q(x) = 1 \ (x > x_b). \tag{7.88}$$

Because $e^{\beta E}$ is sharply peaked at the barrier top, q is close to 0 throughout well $\mathcal{A}$, close to 1 throughout well $\mathcal{B}$, and switches over a layer of width $(\beta\,|E''(x_{\mathrm{sad}})|)^{-1/2}$ around x_{sad}. Near the saddle it is the error-function profile of Eq. (7.50).

Theorem 7.14 (Spectral Gap of a Bistable Landscape)**.** *Under the hypotheses of Theorem 7.4 for both wells ($\beta\Delta E_A \gg 1$ and $\beta\Delta E_B \gg 1$), the spectral gap of $\mathcal{H}_{\mathrm{W}}$ (equivalently of $-\mathcal{L}$ in $L^2(p_{\mathrm{eq}})$) satisfies:*

(i) the exact variational upper bound

$$\lambda_1 \le \frac{\mathcal{E}(q,q)}{\mathrm{Var}_{p_{\mathrm{eq}}}(q)} = \frac{D}{Z\,\mathcal{K}\,\mathrm{Var}_{p_{\mathrm{eq}}}(q)}; \tag{7.89}$$

(ii) the asymptotic evaluation

$$\lambda_1 = \Big(k_{A\to B} + k_{B\to A}\Big)\big[1 + o(1)\big], \tag{7.90}$$

$$k_{A\to B} = \frac{\mu\sqrt{E''(x_a)\,|E''(x_{\mathrm{sad}})|}}{2\pi} e^{-\beta\Delta E_A}, \qquad k_{B\to A} = \frac{\mu\sqrt{E''(x_b)\,|E''(x_{\mathrm{sad}})|}}{2\pi} e^{-\beta\Delta E_B},$$

with the next eigenvalue λ_2 bounded away from zero as in Eq. (7.86);

(iii) the corresponding normalized eigenfunctions are, to leading order,

$$f_1 \approx \frac{q - P_B}{\sqrt{P_A P_B}}, \qquad \psi_0 = \sqrt{P_A}\,\phi_A + \sqrt{P_B}\,\phi_B, \qquad \psi_1 \approx -\sqrt{P_B}\,\phi_A + \sqrt{P_A}\,\phi_B, \tag{7.91}$$

where $\phi_A = p_{\mathrm{eq}}^{1/2}\mathbf{1}_{\mathcal{A}}/\sqrt{P_A}$ and $\phi_B = p_{\mathrm{eq}}^{1/2}\mathbf{1}_{\mathcal{B}}/\sqrt{P_B}$ are the normalized local ground states of the two wells.

Proof. **(i) Upper bound.** By the Rayleigh–Ritz principle of Definition 6.12, $\lambda_1 \le \mathcal{E}(f,f)/\operatorname{Var}_{p_{\mathrm{eq}}}(f)$ for every non-constant trial function f. Take $f = q$. Its derivative is $q' = e^{\beta E}/\mathcal{K}$ on (x_a, x_b) and zero elsewhere, so with $p_{\mathrm{eq}} = e^{-\beta E}/Z$,

$$\mathcal{E}(q,q) = D\int \mathrm{d}x\, p_{\mathrm{eq}}(x)\, q'(x)^2 = \frac{D}{Z\,\mathcal{K}^2}\int_{x_a}^{x_b} \mathrm{d}x\, e^{-\beta E(x)} e^{2\beta E(x)} = \frac{D}{Z\,\mathcal{K}}. \tag{7.92}$$

This is Eq. (7.89).

(ii) Evaluation of the bound. The equilibrium density lives in the two wells, where q equals 0 or 1 up to corrections of relative size $e^{-\beta(E(x_{\mathrm{sad}})-E(x))}$. Hence $\mathbb{E}[q]_{p_{\mathrm{eq}}} = P_B[1+o(1)]$, $\mathbb{E}[q^2]_{p_{\mathrm{eq}}} = P_B[1+o(1)]$, and

$$\operatorname{Var}_{p_{\mathrm{eq}}}(q) = P_B - P_B^2 + o(P_A P_B) = P_A P_B\big[1+o(1)\big] = \frac{Z_A Z_B}{Z^2}\big[1+o(1)\big]. \tag{7.93}$$

Inserting Eq. (7.93) into Eq. (7.89) and using $Z = Z_A + Z_B$,

$$\frac{\mathcal{E}(q,q)}{\operatorname{Var}_{p_{\mathrm{eq}}}(q)} = \frac{D\,Z}{\mathcal{K}\,Z_A Z_B}\big[1+o(1)\big] = \left(\frac{D}{\mathcal{K}\,Z_A} + \frac{D}{\mathcal{K}\,Z_B}\right)\big[1+o(1)\big]. \tag{7.94}$$

Each term is an inverse mean first-passage time. In Eq. (7.35) the inner integral equals $Z_A[1+o(1)]$ for all y in the peak of $e^{\beta E(y)}$, so $\tau(x_a \to x_b) = \mathcal{K} Z_A[1+o(1)]/D$ and $D/(\mathcal{K}Z_A) = k_{A\to B}[1+o(1)]$. The same argument from the right gives $D/(\mathcal{K}Z_B) = k_{B\to A}[1+o(1)]$. The Laplace evaluations Eqs. (7.39) and (7.41) of Z_A, Z_B, and $\mathcal{K}$ then give the explicit rates in Eq. (7.90). This proves $\lambda_1 \le (k_{A\to B} + k_{B\to A})[1+o(1)]$.

To see that the bound is saturated, use the eigenvalue equation itself. In one dimension $p_{\mathrm{eq}}\mathcal{L}f = D(p_{\mathrm{eq}}f')'$, so integrating $\mathcal{L}f_1 = -\lambda_1 f_1$ against p_{eq} over $(-\infty, x_{\mathrm{sad}})$ gives the exact flux identity

$$D\,p_{\mathrm{eq}}(x_{\mathrm{sad}})\,f_1'(x_{\mathrm{sad}}) = -\lambda_1 \int_{-\infty}^{x_{\mathrm{sad}}} \mathrm{d}x\, p_{\mathrm{eq}}(x) f_1(x). \tag{7.95}$$

Inside each well the local generator has a gap of order $\mu E''(x_{a,b})$, much larger than λ_1, so f_1 is flat there to relative accuracy $\lambda_1/(\mu E'')$: $f_1 \approx c_A$ in $\mathcal{A}$ and $f_1 \approx c_B$ in $\mathcal{B}$. Across the barrier, $\mathcal{L}f_1 = -\lambda_1 f_1$ is an exponentially small source, so to leading order $\mathcal{L}f_1 = 0$ and $f_1 = c_A + (c_B - c_A)q$. Orthogonality to the constant mode, $\mathbb{E}[f_1]_{p_{\mathrm{eq}}} = 0$, gives $c_A P_A + c_B P_B = 0$, i.e. $c_B - c_A = -c_A/P_B$. Substituting $f_1'(x_{\mathrm{sad}}) = (c_B - c_A)e^{\beta E(x_{\mathrm{sad}})}/\mathcal{K}$, $p_{\mathrm{eq}}(x_{\mathrm{sad}}) = e^{-\beta E(x_{\mathrm{sad}})}/Z$, and $\int_{-\infty}^{x_{\mathrm{sad}}} p_{\mathrm{eq}} f_1 \approx c_A P_A$ into Eq. (7.95),

$$\lambda_1 \approx -\frac{D\,(c_B - c_A)}{Z\,\mathcal{K}\,c_A P_A} = \frac{D}{Z\,\mathcal{K}\,P_A P_B}, \tag{7.96}$$

which coincides with the leading term of Eq. (7.94). This matched-asymptotics argument is the standard physicist's proof; a rigorous lower bound of the same order uses either semiclassical analysis of $\mathcal{H}_{\mathrm{W}}$ or the potential-theoretic method mentioned in Section 7.4.

(iii) Eigenfunctions. Normalizing $f_1 = c_A + (c_B - c_A)q$ with $c_A = -\sqrt{P_B/P_A}$ and $c_B = \sqrt{P_A/P_B}$ gives $\mathbb{E}[f_1]_{p_{\mathrm{eq}}} = -\sqrt{P_A P_B} + \sqrt{P_A P_B} = 0$ and $\mathbb{E}[f_1^2]_{p_{\mathrm{eq}}} = P_B + P_A = 1$, which is the first formula in Eq. (7.91). Multiplying by $p_{\mathrm{eq}}^{1/2} = \sqrt{P_A}\phi_A + \sqrt{P_B}\phi_B$ (exact up to the thin barrier layer) gives $\psi_1 = p_{\mathrm{eq}}^{1/2} f_1$. □

The proof shows where the error in Kramers' formula comes from. In the worked computation of Section 7.6 below, the committor quotient Eq. (7.89), evaluated by numerical quadrature, agrees with the exact λ_1 to about 0.1% at $\beta\Delta E = 5$, while the Laplace-evaluated rates Eq. (7.90) are about 9% too large. The two-state reduction is very accurate; the Gaussian evaluation of the integrals is what costs accuracy at moderate barriers.

7.5.3 The Two-Level Witten Hamiltonian and Tunneling

Restricted to the doublet, $\mathcal{H}_{\mathrm{W}}$ becomes a 2×2 matrix. The cleanest way to obtain it is to symmetrize the master equation. Write Eq. (7.12) as $\dot{\mathbf{P}} = \mathbf{W}\mathbf{P}$ with $\mathbf{W} = \begin{pmatrix} -k_{A\to B} & k_{B\to A} \\ k_{A\to B} & -k_{B\to A} \end{pmatrix}$, and change variables to $c_i = P_i/\sqrt{P_i^{\mathrm{eq}}}$, the discrete analogue of $\psi = \rho/p_{\mathrm{eq}}^{1/2}$ in Eq. (6.60). Then $\dot{\mathbf{c}} = -\mathbf{H}_{\mathrm{eff}}\mathbf{c}$ with

$$\mathbf{H}_{\mathrm{eff}} = \begin{pmatrix} k_{A\to B} & -t_{AB} \\ -t_{AB} & k_{B\to A} \end{pmatrix}, \qquad t_{AB} := k_{B\to A}\sqrt{\frac{P_B^{\mathrm{eq}}}{P_A^{\mathrm{eq}}}} = \sqrt{k_{A\to B}\,k_{B\to A}}, \tag{7.97}$$

where the last equality uses detailed balance $P_A^{\mathrm{eq}}k_{A\to B} = P_B^{\mathrm{eq}}k_{B\to A}$. The matrix is symmetric, as a Hamiltonian should be. Its trace is $k_{A\to B} + k_{B\to A}$ and its determinant is $k_{A\to B}k_{B\to A} - t_{AB}^2 = 0$, so its eigenvalues are exactly 0 and $k_{A\to B} + k_{B\to A}$, with zero-mode $(\sqrt{P_A^{\mathrm{eq}}}, \sqrt{P_B^{\mathrm{eq}}})$, matching Eq. (7.91).

Equation (7.97) reads like the tight-binding Hamiltonian of a particle in a quantum double well. The off-diagonal element t_{AB} is a tunneling amplitude between the local ground states ϕ_A and ϕ_B; the diagonal elements are the leakage rates out of each well. The exponent of t_{AB} can be checked against WKB tunneling through the Witten barrier. In the barrier region $V_{\mathrm{W}} \approx D\beta^2(E')^2/4$, so the WKB decay constant of a zero-energy state is $\sqrt{V_{\mathrm{W}}/D} \approx \frac{\beta}{2}\,|E'(x)|$, and the tunneling exponent from x_a to x_b is

$$\int_{x_a}^{x_b} \mathrm{d}x\, \frac{\beta}{2}\,|E'(x)| = \frac{\beta}{2}\Big[(E(x_{\mathrm{sad}}) - E(x_a)) + (E(x_{\mathrm{sad}}) - E(x_b))\Big] = \frac{\beta}{2}(\Delta E_A + \Delta E_B), \tag{7.98}$$

which matches $t_{AB} = \sqrt{k_{A\to B}k_{B\to A}} \propto e^{-\beta(\Delta E_A + \Delta E_B)/2}$. For a symmetric double well ($\Delta E_A = \Delta E_B$), $\lambda_1 = 2t_{AB}$ is the familiar tunnel splitting. For an asymmetric well, however, the gap is set by the diagonal elements, not by t_{AB}:

$$\lambda_1 = k_{A\to B} + k_{B\to A} \propto e^{-\beta\min(\Delta E_A,\,\Delta E_B)} \gg t_{AB}. \tag{7.99}$$

Relaxation is limited by escape from the shallower well, which is faster than the geometric mean of the two rates. A purely symmetric-well tunneling estimate would underestimate the gap of a tilted p-bit by the factor $e^{\beta|\Delta E_A - \Delta E_B|/2}$.

Remark 7.15 (Higher Dimensions and Many Wells)**.** The argument extends to N dimensions with Langer's rates Eq. (7.44) in place of the 1D Kramers rates: the trial function is Langer's boundary-layer profile η of Eq. (7.50), and the Dirichlet form is evaluated by the same Gaussian integral over the saddle region that produced the flux Eq. (7.52). For n local minima, $\mathcal{H}_{\mathrm{W}}$ has n exponentially small eigenvalues $0 = \lambda_0 < \lambda_1 \le \cdots \le \lambda_{n-1}$, which are the eigenvalues of the n-state master equation built from the Langer rates between neighboring wells. Above them sits a gap of order $\mu \min|h|$. This is the spectral content of the statement that a multimodal energy-based model (Chapter 3) reduces at low temperature to a Markov chain on its modes.

Hardware Realization: The Spectral Gap Is the Sample Rate of a p-Bit

Read out a bistable circuit through its sign, $s(t) = \operatorname{sgn} x(t) \in \{-1, +1\}$. On timescales longer than τ_{intra}, $s(t)$ is a two-state Markov chain with rates $k_{A\to B}$ and $k_{B\to A}$. Its mean is $m = P_B - P_A$, and its autocorrelation follows from Eq. (7.13):

$$\langle s(t)s(0)\rangle - m^2 = (1 - m^2)\,e^{-\lambda_1|t|}, \qquad \lambda_1 = k_{A\to B} + k_{B\to A}, \tag{7.100}$$

with the Lorentzian spectrum $S_s(\omega) = 2(1-m^2)\lambda_1/(\lambda_1^2+\omega^2)$. This is random telegraph noise, the same process that describes oxide traps in Section 11.5. A p-bit therefore delivers roughly one statistically independent sample per correlation time $1/\lambda_1$. Since λ_1 is dominated by the faster of the two rates, Eq. (7.99), a strongly biased p-bit still flips at a rate set by its shallower well. For a representative subthreshold CMOS attempt frequency $f_0 \sim 10^8\,\mathrm{s}^{-1}$ and $\Delta E \approx 2$–$3\,k_\mathrm{B}T$, this gives independent-sample rates of order 10^6–$10^7\,\mathrm{s}^{-1}$ per cell (an order-of-magnitude estimate, since Kramers' formula is not accurate at such low barriers; see Worked computation 7.16); the exact numbers depend on the circuit and are discussed in Chapter 14.

7.6 Worked Computation: Kramers Rates, Magnetization, and Spectral Gap of the Tilted Quartic

We now apply the results of this chapter to the double-well sampler. The computation produces closed-form expressions, valid for small tilt, for the forward and backward escape rates, the stationary magnetization of the resulting p-bit, and the spectral gap, and it closes the loop opened in Worked computation 6.21, where the exponential collapse of λ_1 was announced but not derived.

Worked computation 7.16 (Escape Rates, Magnetization, and Spectral Gap of the Tilted Quartic Double Well)**. Setup.** Consider the overdamped dynamics Eq. (7.1) in one dimension with

$$E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx, \qquad a>0,\ b>0, \tag{7.101}$$

where the tilt h is a bias field (a bias current or voltage in a circuit realization). Its derivatives are

$$E'(x) = ax^3 - bx - h, \qquad E''(x) = 3ax^2 - b, \qquad E'''(x) = 6ax, \qquad E''''(x) = 6a. \tag{7.102}$$

At $h=0$ the critical points are $x=0$ and $x=\pm x_0$ with

$$x_0 := \sqrt{\frac{b}{a}}, \qquad \Delta E_0 := E(0) - E(\pm x_0) = \frac{b^2}{4a}, \qquad E''(\pm x_0) = 2b, \qquad E''(0) = -b. \tag{7.103}$$

Two identities are used repeatedly below: $ax_0 = b/x_0$ and $bx_0^2 = b^2/a = 4\Delta E_0$. The double-well structure survives as long as $E'(x)=0$ has three real roots. The cubic $ax^3 - bx$ has a local maximum $\frac{2b}{3}\sqrt{b/(3a)}$ at $x = -\sqrt{b/(3a)}$, so three roots exist for

$$|h| < h_\mathrm{c} := \frac{2}{3\sqrt{3}}\,\frac{b^{3/2}}{a^{1/2}} = \frac{2}{3\sqrt{3}}\,b\,x_0 \approx 0.385\,b\,x_0. \tag{7.104}$$

At $|h| = h_\mathrm{c}$ one minimum merges with the saddle (a saddle-node or spinodal point) and the barrier vanishes. We work in the regime $|h| \ll h_\mathrm{c}$, expand in the small dimensionless ratio $h/(bx_0)$, and label the left well x_a (well $\mathcal{A}$) and the right well x_b (well $\mathcal{B}$), so that $h>0$ favors $\mathcal{B}$, as in Fig. 7.1.

Step 1: Critical points. Write a critical point as $x^* + \delta$, where $x^* \in \{-x_0, 0, x_0\}$ is an unperturbed critical point, and expand $E'(x^*+\delta) = 0$ in δ. Since $E'(x^*) = -h$ at the unperturbed point (the term $-h$ is the only h-dependence),

$$E''(x^*)\,\delta + \frac{1}{2}E'''(x^*)\,\delta^2 + \frac{1}{6}E''''\,\delta^3 = h, \tag{7.105}$$

where E'', E''', E'''' are now the h-independent derivatives of $\frac{a}{4}x^4 - \frac{b}{2}x^2$. Solve order by order with $\delta = \delta_1 + \delta_2 + \delta_3 + \cdots$, $\delta_k = \mathcal{O}(h^k)$:

$$\delta_1 = \frac{h}{E''(x^*)}, \qquad \delta_2 = -\frac{E'''(x^*)\,\delta_1^2}{2E''(x^*)}, \qquad \delta_3 = -\frac{E''''\,\delta_1^3}{6E''(x^*)} - \frac{E'''(x^*)\,\delta_1\delta_2}{E''(x^*)}. \tag{7.106}$$

Right well ($x^* = x_0$, $E'' = 2b$, $E''' = 6ax_0 = 6b/x_0$): $\delta_1 = h/(2b)$ and $\delta_2 = -\frac{6b}{x_0}\frac{h^2}{4b^2}\frac{1}{4b} = -\frac{3h^2}{8b^2x_0}$. *Left well* ($x^* = -x_0$, $E''' = -6b/x_0$): $\delta_1 = h/(2b)$ and $\delta_2 = +\frac{3h^2}{8b^2x_0}$. *Saddle* ($x^* = 0$, $E'' = -b$, $E''' = 0$): $\delta_1 = -h/b$, $\delta_2 = 0$, and $\delta_3 = -\frac{6a}{6}\frac{(-h/b)^3}{(-b)} = -\frac{ah^3}{b^4}$. Collecting,

$$x_b = x_0 + \frac{h}{2b} - \frac{3h^2}{8b^2x_0} + \mathcal{O}(h^3), \qquad x_a = -x_0 + \frac{h}{2b} + \frac{3h^2}{8b^2x_0} + \mathcal{O}(h^3), \qquad x_{\mathrm{sad}} = -\frac{h}{b} - \frac{ah^3}{b^4} + \mathcal{O}(h^5). \tag{7.107}$$

Both wells shift toward positive x by $h/(2b)$, half as far as the saddle shifts toward negative x, because the saddle curvature $|E''(0)| = b$ is half the well curvature $2b$. The barrier top moves toward the disfavored well, which is what lowers ΔE_A.

Step 2: Energies and barrier heights. Rather than substituting Eq. (7.107) into E, use the envelope theorem. Let $E^*(h) = E(x^*(h); h)$ be the energy at a critical point. Because $\partial_x E = 0$ there,

$$\frac{\mathrm{d}E^*}{\mathrm{d}h} = \frac{\partial E}{\partial x}\Big|_{x^*}\frac{\mathrm{d}x^*}{\mathrm{d}h} + \frac{\partial E}{\partial h}\Big|_{x^*} = 0 - x^*(h). \tag{7.108}$$

Integrating with $x^*(h) = x^*(0) + h/E''(x^*(0)) + \mathcal{O}(h^2)$ gives $E^*(h) = E^*(0) - x^*(0)\,h - h^2/(2E''(x^*(0))) + \mathcal{O}(h^3)$. Therefore

$$E(x_b) = -\Delta E_0 - x_0 h - \frac{h^2}{4b} + \mathcal{O}(h^3), \qquad E(x_a) = -\Delta E_0 + x_0 h - \frac{h^2}{4b} + \mathcal{O}(h^3),$$
$$E(x_{\mathrm{sad}}) = +\frac{h^2}{2b} + \mathcal{O}(h^4), \tag{7.109}$$

and the forward and backward barriers are

$$\Delta E_A = \Delta E_0 - x_0 h + \frac{3h^2}{4b} + \mathcal{O}(h^3), \qquad \Delta E_B = \Delta E_0 + x_0 h + \frac{3h^2}{4b} + \mathcal{O}(h^3). \tag{7.110}$$

As a check, $\Delta E_B - \Delta E_A = 2x_0 h = E(x_a) - E(x_b)$, in agreement with Eq. (7.6). The linear term $\mp x_0 h$ is the work done by the field on a dipole of length x_0. The quadratic term $+3h^2/(4b)$ is the same for both barriers: the field polarizes the wells (lowering them by $h^2/4b$) and raises the saddle (by $h^2/2b$), so it slightly stiffens both barriers.

Step 3: Curvatures. Squaring Eq. (7.107), $x_{a,b}^2 = x_0^2 \mp x_0 h/b + \mathcal{O}(h^2)$ (upper sign for x_a). With $3ax_0 h/b = 3h/x_0$,

$$E''(x_a) = 2b - \frac{3h}{x_0} + \mathcal{O}(h^2) = 2b(1-\eta), \qquad E''(x_b) = 2b(1+\eta), \qquad \eta := \frac{3h}{2bx_0},$$
$$E''(x_{\mathrm{sad}}) = -b + \frac{3ah^2}{b^2} + \mathcal{O}(h^4). \tag{7.111}$$

The favored well is stiffer and the disfavored one softer. To first order in h the saddle curvature is unchanged, $|E''(x_{\mathrm{sad}})| = b$.

Step 4: Forward and backward Kramers rates. Insert Eqs. (7.110) and (7.111) into Eq. (7.37). With the symmetric rate

$$k_0 := \frac{\mu\sqrt{2b\cdot b}}{2\pi}\, e^{-\beta\Delta E_0} = \frac{\sqrt{2}\,\mu b}{2\pi}\, e^{-\beta b^2/4a}, \tag{7.112}$$

the rates to first order in h in the prefactor and second order in the exponent are

$$k_{A\to B} = k_0\sqrt{1-\eta}\,\exp\!\Big(+\beta x_0 h - \frac{3\beta h^2}{4b}\Big), \qquad k_{B\to A} = k_0\sqrt{1+\eta}\,\exp\!\Big(-\beta x_0 h - \frac{3\beta h^2}{4b}\Big). \tag{7.113}$$

The exponential factors $e^{\pm\beta x_0 h}$ carry essentially all of the bias dependence when $\beta x_0 h = \mathcal{O}(1)$: the prefactor corrections $\sqrt{1\mp\eta}$ are of relative size $h/(bx_0)$, which is smaller than $\beta x_0 h$ by the factor $1/(\beta b x_0^2) = 1/(4\beta\Delta E_0)$. Their ratio satisfies detailed balance with the Laplace-evaluated basin weights: $k_{A\to B}/k_{B\to A} = \sqrt{E''(x_a)/E''(x_b)}\, e^{\beta[E(x_a)-E(x_b)]} = Z_B/Z_A$, as Eq. (7.14) requires.

Step 5: Stationary magnetization. Define the p-bit output $s = \mathrm{sgn}(x - x_{\mathrm{sad}})$ and its mean $m = P_B - P_A$. From the master equation, $P_B/P_A = k_{A\to B}/k_{B\to A}$, so

$$m = \frac{k_{A\to B} - k_{B\to A}}{k_{A\to B} + k_{B\to A}} = \tanh\!\left(\frac{1}{2}\ln\frac{k_{A\to B}}{k_{B\to A}}\right) = \tanh\!\left(\beta x_0 h - \frac{1}{2}\,\mathrm{artanh}\,\eta\right), \tag{7.114}$$

where we used $\frac{1}{2}\ln\frac{1-\eta}{1+\eta} = -\,\mathrm{artanh}\,\eta$ and the fact that the h^2 terms in the two exponents cancel. To first order in η, and using $bx_0^2 = 4\Delta E_0$,

$$m \approx \tanh\!\left(\beta x_0 h - \frac{3h}{4bx_0}\right) = \tanh\!\left(\beta h x_0\Big[1 - \frac{3}{16\,\beta\Delta E_0}\Big]\right). \tag{7.115}$$

The leading behavior $m = \tanh(\beta h x_0)$ is the response of an Ising spin $s = \pm 1$ with moment x_0 in the field h: this is the two-state (Ising) limit in which the continuous circuit behaves as the discrete p-bit of Chapter 1, with effective field $h_{\mathrm{eff}} = x_0 h$. The correction factor $1 - 3/(16\beta\Delta E_0)$ is an entropic effect. The favored well is stiffer by Eq. (7.111), hence narrower, and holds less thermal phase space than its depth alone would suggest. The correction vanishes as $\beta\Delta E_0 \to \infty$. The same factor appears in the mean position: adding the field shift $h/2b$ of the well centers from Eq. (7.107) and the cubic-anharmonic thermal shift $\langle x - x_b\rangle \approx -E'''(x_0)k_{\mathrm{B}}T/(2E''(x_0)^2) = -3k_{\mathrm{B}}T/(4bx_0)$ (and the opposite sign in well $\mathcal{A}$) gives

$$\langle x\rangle \approx m\, x_0\Big[1 - \frac{3}{16\,\beta\Delta E_0}\Big] + \frac{h}{2b}. \tag{7.116}$$

Step 6: Spectral gap. By Theorem 7.14, $\lambda_1 = k_{A\to B} + k_{B\to A}$. Expanding $\sqrt{1\mp\eta} \approx 1 \mp \eta/2$ in Eq. (7.113),

$$\lambda_1 \approx 2k_0\, e^{-3\beta h^2/4b}\Big[\cosh(\beta x_0 h) - \frac{3h}{4bx_0}\sinh(\beta x_0 h)\Big]. \tag{7.117}$$

At zero tilt this is

$$\lambda_1(h=0) = 2k_0 = \frac{\sqrt{2}\,\mu b}{\pi}\exp\!\left(-\frac{\beta b^2}{4a}\right), \tag{7.118}$$

which is exactly the formula Eq. (6.148) announced in Chapter 6, with $b - \alpha \to b$ (no denoising confinement, $\alpha = 0$). The tilt increases the gap through the factor $\cosh(\beta x_0 h)$, because escape from the raised well $\mathcal{A}$ speeds up exponentially while escape from $\mathcal{B}$ slows down by the same factor, and the

sum is dominated by the faster rate. The next eigenvalue is, by Eq. (7.86), $\lambda_2 \approx \mu \min(2b, b) = \mu b$, set by the saddle curvature. The ratio

$$\frac{\lambda_1}{\lambda_2} \approx \frac{\sqrt{2}}{\pi}\, e^{-\beta \Delta E_0} \cosh(\beta x_0 h) \tag{7.119}$$

measures the timescale separation that makes the two-state description valid.

Step 7: Numerical example. Measure energies in units of $k_{\mathrm{B}}T$ ($\beta = 1$), positions in a reference length ℓ (for a circuit, a reference voltage), and rates in units of $\mu k_{\mathrm{B}}T/\ell^2$. Take $b = 2$ and two values of a:

- $a = 0.2$: $x_0 = \sqrt{10} \approx 3.162$, $\Delta E_0 = 5\, k_{\mathrm{B}}T$, $h_{\mathrm{c}} \approx 2.43$ (a fast p-bit);
- $a = 0.05$: $x_0 = \sqrt{40} \approx 6.325$, $\Delta E_0 = 20\, k_{\mathrm{B}}T$, $h_{\mathrm{c}} \approx 4.87$ (a slow, nearly frozen bit).

For $a = 0.2$, $h = 0.1$, Step 1 gives $x_a = -3.1370$, $x_{\mathrm{sad}} = -0.0500$, $x_b = 3.1870$, in agreement with the numerical roots of $E'(x) = 0$ to four decimals. Step 2 gives $\Delta E_A = 5 - 0.3162 + 0.0038 = 4.6875\, k_{\mathrm{B}}T$ and $\Delta E_B = 5.3200\, k_{\mathrm{B}}T$; Step 3 gives $E''(x_a) = 3.905$, $E''(x_b) = 4.095$; and Eq. (7.113) gives $k_{A\to B} = 4.09 \times 10^{-3}$, $k_{B\to A} = 2.23 \times 10^{-3}$, $\lambda_1 = 6.32 \times 10^{-3}$. Tables 7.1 and 7.2 compare these asymptotic results with exact values: the exact mean first-passage time from Eq. (7.29) by numerical quadrature, and the exact λ_1 from inverse iteration of the generator.

$\beta\Delta E_0$	h	$\beta\Delta E_A$	$\beta\Delta E_B$	$k_{A\to B}$ (Kramers)	$1/\tau(x_a \to x_b)$ (exact)	λ_1 (Kramers)	λ_1 (exact)
5	0	5.000	5.000	3.03×10^{-3}	2.74×10^{-3}	6.07×10^{-3}	5.55×10^{-3}
5	0.1	4.688	5.320	4.09×10^{-3}	3.68×10^{-3}	6.32×10^{-3}	5.78×10^{-3}
5	0.3	4.085	5.982	7.26×10^{-3}	6.46×10^{-3}	8.43×10^{-3}	7.66×10^{-3}
20	0	20.000	20.000	9.28×10^{-10}	9.10×10^{-10}	1.86×10^{-9}	1.82×10^{-9}
20	0.05	19.685	20.317	1.27×10^{-9}	1.24×10^{-9}	1.95×10^{-9}	1.91×10^{-9}
20	0.1	19.371	20.636	1.73×10^{-9}	1.69×10^{-9}	2.22×10^{-9}	2.18×10^{-9}

Table 7.1: Escape rates and spectral gap of the tilted quartic Eq. (7.101) with $b = 2$, in units $k_{\mathrm{B}}T = 1$ and rates in units of $\mu k_{\mathrm{B}}T/\ell^2$. The exact λ_2 is 1.64 for $\Delta E_0 = 5\, k_{\mathrm{B}}T$ and 1.92 for $\Delta E_0 = 20\, k_{\mathrm{B}}T$, compared with the harmonic estimate $\mu b = 2$.

$\beta\Delta E_0$	h	$\beta x_0 h$	$\tanh(\beta x_0 h)$ (Ising)	Eq. (7.115)	exact $\langle s \rangle$
5	0.1	0.316	0.306	0.295	0.292
5	0.3	0.949	0.739	0.723	0.716
20	0.05	0.316	0.306	0.303	0.303
20	0.1	0.632	0.560	0.556	0.555

Table 7.2: Stationary magnetization of the tilted quartic: Ising limit, entropy-corrected two-state formula, and exact equilibrium average.

Three lessons follow from the tables.

1. *Accuracy of Kramers' formula.* The asymptotic rates overestimate the exact ones by about 10% at $\beta\Delta E_0 = 5$ and by about 2% at $\beta\Delta E_0 = 20$. The error scales as $1/(\beta\Delta E)$, as expected from the $\mathcal{O}(\beta^{-1})$ corrections in Eq. (7.36). As noted after Theorem 7.14, the committor Rayleigh quotient Eq. (7.89) evaluated by quadrature is far more accurate (5.559×10^{-3} versus

the exact 5.552×10^{-3} at $h = 0$, $\Delta E_0 = 5\, k_{\mathrm{B}}T$). For barriers of 1–3 $k_{\mathrm{B}}T$, which is where fast hardware p-bits operate, the Laplace evaluation is not reliable and one should use the exact quadrature Eq. (7.29) or the committor quotient directly.

2. *The two-state picture is accurate even at* $5\, k_{\mathrm{B}}T$. The gap ratio $\lambda_1/\lambda_2 \approx 3 \times 10^{-3}$ at $\Delta E_0 = 5\, k_{\mathrm{B}}T$, so the slow and fast timescales are well separated, and the entropy-corrected magnetization Eq. (7.115) is within 0.01 of the exact value. The naive Ising law $\tanh(\beta x_0 h)$ is off by a few percent at $5\, k_{\mathrm{B}}T$ and by about one percent at $20\, k_{\mathrm{B}}T$.

3. *Exponential cost of barriers.* Raising the barrier from 5 to $20\, k_{\mathrm{B}}T$ reduces λ_1 by a factor of 3×10^6. If the saddle relaxation time $(\mu b)^{-1}$ is a representative 1 ns, then $\mu k_{\mathrm{B}}T/\ell^2 = 0.5\,\mathrm{ns}^{-1}$ in these units, and the zero-tilt gap is $\lambda_1 \approx 2.8 \times 10^6\,\mathrm{s}^{-1}$ at $5\, k_{\mathrm{B}}T$ but only $\approx 0.9\,\mathrm{s}^{-1}$ at $20\, k_{\mathrm{B}}T$. The first is a useful random-bit source; the second is a sampler that produces about one independent sample per second.

Step 8: What the computation says about hardware design. For a single p-bit, the barrier is a design parameter: choosing $\beta\Delta E_0$ of a few units gives fast, nearly unbiased telegraph switching whose bias is controlled through $\tanh(\beta x_0 h)$ (Chapter 14). For a large energy-based model, the barriers between modes are not chosen by the designer; they are set by the learned landscape and typically grow with system size (Chapter 3). The gap then behaves as in Eq. (7.118), $\lambda_1 \propto e^{-\beta\Delta E}$, and Bakry–Émery's guarantee is lost. In Chapter 6 we saw that adding a quadratic confinement $\frac{\alpha}{2}x^2$ shifts $b \to b - \alpha$; the barrier $\Delta E = (b-\alpha)^2/4a$ then shrinks and disappears for $\alpha \geq b$. This is the mechanism behind the Denoising Thermodynamic Models of Chapter 16: each denoising step solves an easy problem whose landscape has no barrier, instead of one hard problem with $\lambda_1 \propto e^{-\beta\Delta E}$.

Summary

Proved:

- Reduction of bistable Langevin dynamics to the two-state master equation, with stationary ratio $P_A/P_B = Z_A/Z_B$ and relaxation rate $k_{A\to B} + k_{B\to A}$ (Section 7.1).
- Pontryagin's backward equation $\mathcal{L}\tau = -1$ and the moment hierarchy $\mathcal{L}\tau_n = -n\tau_{n-1}$ (Theorem 7.2), and the exact double-integral MFPT in one dimension (Proposition 7.3).
- Kramers' overdamped rate by Laplace evaluation of the exact MFPT, with the exponential law of the escape time (Theorem 7.4); Langer's multidimensional formula by the flux-over-population method (Theorem 7.5).
- Transition-state, spatial-diffusion, and energy-diffusion rates of the underdamped problem with their validity conditions, and the location of the Kramers turnover (Section 7.3.2).
- The Hamilton–Jacobi equation for the Freidlin–Wentzell quasipotential (Proposition 7.10); the identity $\mathcal{U} = E - E(\mathbf{x}_a)$ and the time-reversed-relaxation form of the instanton for gradient systems (Theorem 7.11); the bending of instantons by circulating currents, solved exactly for the linear non-reciprocal circuit (Example 7.13).
- The exact committor upper bound on the spectral gap and its asymptotic evaluation $\lambda_1 \simeq k_{A\to B} + k_{B\to A}$, with the doublet eigenfunctions and the two-level Witten Hamiltonian (Theorem 7.14, Eq. (7.97)).

- Closed-form critical points, barriers, curvatures, rates, magnetization, and spectral gap of the tilted quartic, including the hand-off formula Eq. (6.148) of Chapter 6 (Worked computation 7.16).

Assumed:

- Smooth confining potentials with non-degenerate critical points; high barriers $\beta\Delta E \gg 1$ for all asymptotic statements; a single index-1 saddle dominating each transition.
- Constant (or slowly varying) mobility; Markovian white-noise baths. Colored noise and memory friction, which modify the rate through the Grote–Hynes frequency-dependent friction, are not treated.
- For the energy-diffusion rate, orbit averaging at weak damping (a separation between the orbital period and the energy-relaxation time).
- For the non-gradient Freidlin–Wentzell results, existence of a differentiable quasipotential and of the transverse decomposition Eq. (7.81).

Taken up later:

- The full proof of the Freidlin–Wentzell large deviation principle and exit law (upper bound via exponential tightness), which we only sketched; see [36].
- A rigorous lower bound matching the committor upper bound on λ_1 (semiclassical analysis of $\mathcal{H}_{\mathrm{W}}$ or potential theory), and the spectral facts about Schrödinger operators used in Section 7.5 (Chapter C).
- The physical realization of the tilted quartic in CMOS latches and Josephson parametrons, and the calibration of a, b, h from device parameters (Chapters 12 and 14); the use of barrier-free denoising steps to avoid $\lambda_1 \propto e^{-\beta\Delta E}$ (Chapter 16).
- The heat and work associated with driven switching of the double well (Chapters 8 and 9).

Open:

- Prefactors for escape in non-gradient systems when the quasipotential is not smooth (caustics, switching lines), and their measurement in non-reciprocal analog circuits.
- Escape in high-dimensional glassy landscapes where many saddles of similar height compete and the single-saddle Langer picture fails; the relation of the resulting mixing times to the barrier scaling $\Delta E \sim N^{\psi}$ of Chapter 3.
- Optimal shaping of barriers and bias in arrays of coupled p-bits to maximize independent samples per joule.

Bibliographic notes

The theory of thermally activated escape starts with Kramers [61], who derived both the spatial-diffusion and the energy-diffusion rates from the Klein–Kramers equation and identified the turnover between them. Hänggi et al. [46] is the standard review of the fifty years that followed: transition-state theory, the flux-over-population method, multidimensional rates, the turnover problem, memory friction, and quantum corrections. Langer's multidimensional saddle-point formula, originally developed for the decay of metastable states in field theory and nucleation, is in Langer [64]. Textbook treatments of first-passage times and the backward equation, with many worked examples, are given by Gardiner [39] and Risken [90], and Pavliotis [85] covers the same material with more mathematical detail, including the connection between the spectral gap and escape rates. The large-deviation theory of small random perturbations of dynamical systems, including the action functional, the quasipotential, the exit problem, and the graph construction for several attractors, is developed in Freidlin and Wentzell [36]. The path-integral weight from which the action functional descends is due to Onsager and Machlup [81]. The relation between low-lying eigenvalues of the supersymmetric Witten Laplacian, critical points, and tunneling is in Witten [118]. The rigorous potential-theoretic proof of the Eyring–Kramers formula for reversible diffusions, and of the asymptotic equality between small eigenvalues and inverse mean exit times, is due to Bovier, Eckhoff, Gayrard, and Klein [15, 16]. For the use of tunable-barrier bistable devices as probabilistic bits see Camsari et al. [18] and Borders et al. [14]; for the Denoising Thermodynamic Models that avoid high barriers altogether see Jelinčič et al. [56].

Part III

Stochastic Thermodynamics and the Physical Limits of Computation

Chapter 8

Sekimoto's Trajectory Energetics, Path Integrals, and Entropy Production

In Part II (Chapters 4 to 7) we described a noisy circuit at two levels. A single realization obeys a Langevin equation, interpreted with the Itô or Stratonovich calculus of Chapter 4 and derived from a microscopic bath in Chapter 5. Its state $\mathbf{x}_t \in \mathbb{R}^N$ might be the vector of capacitor node voltages $\mathbf{v}_t$ in an analog RC network or the phases of a Josephson junction array, kicked by thermal fluctuations from an Ohmic bath at temperature T. An ensemble of realizations is described by the Fokker–Planck equation of Chapter 6. Its density $\rho(\mathbf{x}, t)$ relaxes toward the Boltzmann–Gibbs distribution $p_{\mathrm{eq}}(\mathbf{x}) \propto \exp(-\beta E(\mathbf{x}))$, and the Fokker–Planck H-theorem states that along the way the variational free energy $\mathcal{F}[\rho_t]$ decreases at a rate set by the relative Fisher information.

Classical thermodynamics defines internal energy U, work $\mathcal{W}$, heat $\mathcal{Q}$, and entropy S for macroscopic systems with $N \sim 10^{23}$ degrees of freedom, whose relative fluctuations, of order $N^{-1/2} \sim 10^{-11}$, are negligible. A thermodynamic computer works in the opposite regime. In subthreshold CMOS sampling circuits, such as the non-equilibrium analog thermodynamic random networks (NEAT-RNs) studied in Chapter 13 [31, 37, 38], a computational node has a capacitance of order a femtofarad (we take $C_{\mathrm{node}} \sim 0.5$–$2\,\mathrm{fF}$ as representative) and its voltage swings are comparable to the thermal voltage $V_T = k_{\mathrm{B}}T/q_e \approx 25.85\,\mathrm{mV}$ at $T = 300\,\mathrm{K}$. The electrostatic energy stored on such a node is only tens to hundreds of $k_{\mathrm{B}}T$. During a sampling or denoising cycle the circuit is also driven out of equilibrium, either in finite time τ by time-dependent control parameters $\boldsymbol{\lambda}_t$ (programmable gate biases or clock signals) or permanently, into a non-equilibrium steady state (NESS), by non-reciprocal transistor transconductances ($G_{ij} \neq G_{ji}$).

In this regime one run of a randomized algorithm is a single stochastic trajectory $\mathbf{x}_{[0,\tau]} \equiv \{\mathbf{x}_t\}_{0 \le t \le \tau}$. Thermal fluctuations along it are not small corrections to a deterministic mean; they drive the state transitions. To find the energy cost of generating a sample, clamping a variable, or holding a circuit in a NESS, we need the First and Second Laws along one fluctuating trajectory, before any ensemble average is taken.

This chapter develops that theory, stochastic thermodynamics [22, 48, 97, 98]. Reading the Langevin equation as a force balance gives a First Law on every trajectory. Adding Seifert's trajectory entropy $s_{\mathrm{sys}} = -k_{\mathrm{B}} \ln \rho(\mathbf{x}_t, t)$ gives an entropy production whose average cannot be negative. The Onsager–Machlup path integral then assigns a probability to a whole trajectory, and the central result of the chapter (Section 8.4) is that the log-ratio of the probabilities of a trajectory and of its time reverse equals the entropy the trajectory produces. That identity is the starting point for the fluctuation theorems of Chapter 9. We close by separating the entropy produced in a non-equilibrium circuit into a housekeeping part, paid to maintain steady currents, and an excess

part, paid for changing the distribution, and we compute the housekeeping heat of a non-reciprocal version of the RC sampler (the coupled linear RC circuit of Chapter 1).

8.1 Sekimoto's Stochastic Energetics and the Trajectory First Law

To define work and heat along a single fluctuating path, we must distinguish three kinds of variables: the degrees of freedom of the computational system, the external parameters set by the hardware clock or the user, and the microscopic degrees of freedom of the thermal environment.

8.1.1 Mechanical Force Balance and Control Protocols in Overdamped Langevin Systems

Consider an N-dimensional continuous state vector $\mathbf{x}_t = (x_{1,t}, \dots, x_{N,t})^\top \in \mathbb{R}^N$ in contact with a memoryless Ohmic bath at temperature T (inverse temperature $\beta = (k_{\mathrm{B}}T)^{-1}$), of the kind derived from the Caldeira–Leggett oscillator model in Chapter 5. Suppose the system is governed by a smooth potential energy function $E(\mathbf{x}, \boldsymbol{\lambda}_t)$, where

$$\boldsymbol{\lambda}_t \;=\; \big(\lambda_1(t), \lambda_2(t), \dots, \lambda_M(t)\big)^\top \;\in\; \mathbb{R}^M$$

is a deterministic, externally prescribed **control protocol** over the time interval $t \in [0, \tau]$. In a thermodynamic circuit, $\mathbf{x}_t$ represents the vector of capacitor node voltages $\mathbf{v}_t$ (or node charges $\mathbf{q}_t$), while $\boldsymbol{\lambda}_t$ represents programmable bias voltages, variable coupling conductances, or flux-bias currents injected by digital-to-analog converters (DACs) and timing circuitry.

In addition to the conservative restoring force $-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t)$, an active electronic circuit may be subjected to a non-conservative force field $\mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t) \in \mathbb{R}^N$ that cannot be written as the gradient of any single-valued scalar potential (that is, $\partial_{x_i} f_{\mathrm{nc},j} \neq \partial_{x_j} f_{\mathrm{nc},i}$). As we saw in Chapter 6 and will analyze at the transistor level in Chapter 13, such non-conservative forces arise naturally whenever active subthreshold CMOS transconductors couple node voltages asymmetrically ($G_{ij} \neq G_{ji}$) or when external voltage sources drive persistent currents around closed circuit loops. The total mechanical or electrostatic force acting on the system coordinates $\mathbf{x}_t$ is therefore

$$\mathbf{F}_{\mathrm{tot}}(\mathbf{x}_t, \boldsymbol{\lambda}_t, t) \;\equiv\; -\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t). \tag{8.1}$$

Let $\boldsymbol{\Gamma} \in \mathbb{R}^{N \times N}$ be a constant, symmetric, positive-definite friction tensor ($\boldsymbol{\Gamma} = \boldsymbol{\Gamma}^\top \succ \mathbf{0}$), with mobility tensor $\boldsymbol{\mu} \equiv \boldsymbol{\Gamma}^{-1}$ and diffusion tensor $\mathbf{D} \equiv k_{\mathrm{B}}T\boldsymbol{\mu} = k_{\mathrm{B}}T\boldsymbol{\Gamma}^{-1}$. This Einstein relation is the Markovian form of the fluctuation–dissipation theorem of the second kind (Chapter 5). In the overdamped limit where inertial momentum relaxation occurs on a timescale much shorter than the diffusive RC relaxation time τ_{RC}, the state vector $\mathbf{x}_t$ obeys the overdamped Langevin stochastic differential equation (SDE):

$$\boldsymbol{\Gamma}\, \mathrm{d}\mathbf{x}_t \;=\; \big[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\big]\, \mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\boldsymbol{\Gamma}}\, \mathrm{d}\mathbf{W}_t, \tag{8.2}$$

where $\mathbf{W}_t \in \mathbb{R}^N$ is a standard N-dimensional Wiener process with independent components satisfying $\mathbb{E}[\mathrm{d}W_{i,t}] = 0$ and $\mathrm{d}W_{i,t}\, \mathrm{d}W_{j,t} = \delta_{ij}\, \mathrm{d}t$. Multiplying Eq. (8.2) on the left by $\boldsymbol{\mu} = \boldsymbol{\Gamma}^{-1}$ yields the kinematic drift-diffusion form

$$\mathrm{d}\mathbf{x}_t \;=\; \mathbf{a}(\mathbf{x}_t, t)\, \mathrm{d}t + \mathbf{B}\, \mathrm{d}\mathbf{W}_t, \tag{8.3}$$

where the total deterministic drift vector field $\mathbf{a}(\mathbf{x}, t)$ and noise amplitude matrix $\mathbf{B}$ are given by

$$\mathbf{a}(\mathbf{x}, t) \;\equiv\; \boldsymbol{\mu}\big[-\nabla_{\mathbf{x}} E(\mathbf{x}, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}, t)\big], \qquad \mathbf{B} \;\equiv\; \sqrt{2k_{\mathrm{B}}T\boldsymbol{\mu}}, \qquad \frac{1}{2}\mathbf{B}\mathbf{B}^\top \;=\; k_{\mathrm{B}}T\boldsymbol{\mu} \;=\; \mathbf{D}. \tag{8.4}$$

Because $\mathbf{B}$ is independent of $\mathbf{x}_t$ in this baseline setting (we treat state-dependent mobility $\boldsymbol{\mu}(\mathbf{x})$ separately in Section 8.3), the Itô and Stratonovich interpretations of the noise term $\mathbf{B}\,\mathrm{d}\mathbf{W}_t$ in Eq. (8.3) coincide for $\mathbf{x}_t$ itself. However, as soon as we evaluate nonlinear functions of $\mathbf{x}_t$, above all the energy $E(\mathbf{x}_t, \boldsymbol{\lambda}_t)$ and the system entropy $s_{\mathrm{sys}}(t) = -k_{\mathrm{B}} \ln \rho(\mathbf{x}_t, t)$ defined in Section 8.2, the choice between Itô $(\cdot)$ and Stratonovich $(\circ)$ products matters.

8.1.2 Trajectory Work, Trajectory Heat, and the First Law

Sekimoto's idea [98] is to read the Langevin equation Eq. (8.2) as a balance of forces (Newton's third law, or Kirchhoff's current law for a circuit) at the interface between the system and the bath. Rearranging Eq. (8.2), we isolate the total force exerted *by the thermal bath on the system*:

$$\mathbf{F}_{\mathrm{bath}}(t) \;\equiv\; -\boldsymbol{\Gamma}\,\dot{\mathbf{x}}_t + \sqrt{2k_{\mathrm{B}}T\boldsymbol{\Gamma}}\,\boldsymbol{\xi}_t \;=\; \nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) - \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t), \tag{8.5}$$

where $\boldsymbol{\xi}_t = \mathrm{d}\mathbf{W}_t/\,\mathrm{d}t$ is the formal white noise of Chapter 4; products involving $\dot{\mathbf{x}}_t$ and $\boldsymbol{\xi}_t$ are understood as stochastic integrals against $\mathrm{d}\mathbf{x}_t$ and $\mathrm{d}\mathbf{W}_t$. The two terms in $\mathbf{F}_{\mathrm{bath}}(t)$ have a direct physical meaning. The first, $-\boldsymbol{\Gamma}\,\dot{\mathbf{x}}_t$, is the systematic viscous drag (or Ohmic resistive voltage drop) opposing the motion of the system, while the second, $+\sqrt{2k_{\mathrm{B}}T\boldsymbol{\Gamma}}\,\boldsymbol{\xi}_t$, is the random Johnson–Nyquist or Brownian kick delivered by the microscopic bath degrees of freedom. Both come from the same reservoir, as the Caldeira–Leggett model of Chapter 5 shows, and $\mathbf{F}_{\mathrm{bath}}\,\mathrm{d}t$ is the net impulse exchanged with the heat bath during the interval $[t, t+\mathrm{d}t]$.

By action–reaction symmetry, the force exerted *by the system on the thermal bath* is $-\mathbf{F}_{\mathrm{bath}}(t)$. In classical mechanics, the mechanical energy transferred from one body to another during an infinitesimal displacement $\mathrm{d}\mathbf{x}_t$ is the inner product of the applied force with the displacement. The microscopic thermal noise has a small but finite correlation time $\tau_{\mathrm{corr}} > 0$ before the white-noise limit $\tau_{\mathrm{corr}} \to 0$ is taken. The Wong–Zakai theorem (Chapter 4) therefore requires the work done by the system on the bath, which is the heat $\mathrm{d}\mathcal{Q}$ dissipated into the bath, to be evaluated with the Stratonovich midpoint product $(\circ)$:

$$\mathrm{d}\mathcal{Q}(t) \;\equiv\; \big(-\mathbf{F}_{\mathrm{bath}}(t)\big) \circ \mathrm{d}\mathbf{x}_t \;=\; \big(\boldsymbol{\Gamma}\,\dot{\mathbf{x}}_t - \sqrt{2k_{\mathrm{B}}T\boldsymbol{\Gamma}}\,\boldsymbol{\xi}_t\big) \circ \mathrm{d}\mathbf{x}_t. \tag{8.6}$$

Substituting the force balance Eq. (8.5) into Eq. (8.6) allows us to express the dissipated heat directly in terms of the system's potential gradient and non-conservative driving force:

$$\mathrm{d}\mathcal{Q}(t) \;=\; \big[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\big] \circ \mathrm{d}\mathbf{x}_t. \tag{8.7}$$

Similarly, work is performed *on* the system in two distinct ways during $[t, t+\mathrm{d}t]$: first, by the external hardware controller varying the protocol parameters $\boldsymbol{\lambda}_t \to \boldsymbol{\lambda}_t + \mathrm{d}\boldsymbol{\lambda}_t$, which shifts the potential energy surface $E(\mathbf{x}_t, \boldsymbol{\lambda}_t)$ at the instantaneous state $\mathbf{x}_t$; and second, by the non-conservative force $\mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)$ pushing the coordinate through the displacement $\mathrm{d}\mathbf{x}_t$. This motivates the following definitions.

Definition 8.1 (Sekimoto Trajectory Energetics)**.** Let $\mathbf{x}_{[0,\tau]} = \{\mathbf{x}_t\}_{0 \le t \le \tau}$ be a continuous trajectory of the overdamped Langevin SDE Eq. (8.2) under a smooth deterministic control protocol $\boldsymbol{\lambda}_t \in C^1([0,\tau]; \mathbb{R}^M)$ and non-conservative force $\mathbf{f}_{\mathrm{nc}} \in C^1(\mathbb{R}^N \times [0,\tau]; \mathbb{R}^N)$. Along this single trajectory:

1. The **trajectory internal (potential) energy** at time t is the random variable

$$E_t \;\equiv\; E(\mathbf{x}_t, \boldsymbol{\lambda}_t). \tag{8.8}$$

2. The **infinitesimal trajectory work** $\mathrm{d}\mathcal{W}(t)$ performed *on* the system during $[t, t+\mathrm{d}t]$ is the sum of the protocol work $\mathrm{d}\mathcal{W}_{\mathrm{prot}}(t)$ and the non-conservative driving work $\mathrm{d}\mathcal{W}_{\mathrm{nc}}(t)$:

$$\mathrm{d}\mathcal{W}(t) \;\equiv\; \mathrm{d}\mathcal{W}_{\mathrm{prot}}(t) + \mathrm{d}\mathcal{W}_{\mathrm{nc}}(t) \;=\; \frac{\partial E(\mathbf{x}_t, \boldsymbol{\lambda}_t)}{\partial \boldsymbol{\lambda}_t} \cdot \mathrm{d}\boldsymbol{\lambda}_t + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t) \circ \mathrm{d}\mathbf{x}_t, \tag{8.9}$$

where $\frac{\partial E}{\partial \boldsymbol{\lambda}_t} \cdot \mathrm{d}\boldsymbol{\lambda}_t = \sum_{\alpha=1}^{M} \frac{\partial E}{\partial \lambda_\alpha} \dot{\lambda}_\alpha(t)\, \mathrm{d}t$.

3. The **infinitesimal trajectory heat** $\mathrm{d}\mathcal{Q}(t)$ dissipated *by* the system *into* the thermal bath at temperature T during $[t, t+\mathrm{d}t]$ is

$$\mathrm{d}\mathcal{Q}(t) \;\equiv\; \left[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\right] \circ \mathrm{d}\mathbf{x}_t. \tag{8.10}$$

4. Integrating over $t \in [0, \tau]$, the **total trajectory work** $\mathcal{W}[\mathbf{x}_{[0,\tau]}]$ and **total dissipated trajectory heat** $\mathcal{Q}[\mathbf{x}_{[0,\tau]}]$ are path functionals given by

$$\mathcal{W}[\mathbf{x}_{[0,\tau]}] \;\equiv\; \int_0^\tau \frac{\partial E(\mathbf{x}_t, \boldsymbol{\lambda}_t)}{\partial \boldsymbol{\lambda}_t} \cdot \dot{\boldsymbol{\lambda}}_t \, \mathrm{d}t + \int_0^\tau \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t) \circ \mathrm{d}\mathbf{x}_t, \tag{8.11}$$

$$\mathcal{Q}[\mathbf{x}_{[0,\tau]}] \;\equiv\; \int_0^\tau \left[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\right] \circ \mathrm{d}\mathbf{x}_t. \tag{8.12}$$

For purely conservative forces, $\mathbf{f}_{\mathrm{nc}} \equiv \mathbf{0}$, these reduce to Sekimoto's canonical expressions $\mathrm{d}\mathcal{W} = \frac{\partial E}{\partial \boldsymbol{\lambda}_t} \cdot \mathrm{d}\boldsymbol{\lambda}_t$ and $\mathrm{d}\mathcal{Q} = -\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) \circ \mathrm{d}\mathbf{x}_t$.

Notation Watch: Sign Convention for Heat $\mathrm{d}\mathcal{Q}$

Across physics, chemistry, and electrical engineering, two opposite sign conventions for heat coexist:

- **Fluid/Chemical Convention ($\mathrm{d}E = \mathrm{d}\mathcal{W} + \mathrm{d}\mathcal{Q}_{\mathrm{in}}$):** Heat $\mathrm{d}\mathcal{Q}_{\mathrm{in}} > 0$ is defined as energy *absorbed by the system from the bath.*
- **Computing/Circuit Dissipation Convention ($\mathrm{d}E = \mathrm{d}\mathcal{W} - \mathrm{d}\mathcal{Q}$):** Heat $\mathrm{d}\mathcal{Q} > 0$ is defined as energy *dissipated by the system into the ambient bath* (e.g., Joule heating $I^2 R\, \mathrm{d}t > 0$ transferred from a silicon die into its heat sink).

Throughout this book we use the dissipation convention: $\mathrm{d}\mathcal{W} > 0$ is work done *on* the circuit by external bias/clock sources, and $\mathrm{d}\mathcal{Q} > 0$ is heat *dissipated into* the thermal bath at temperature T. Consequently, the First Law reads $\mathrm{d}E = \mathrm{d}\mathcal{W} - \mathrm{d}\mathcal{Q}$, and the medium entropy produced in the bath is $\mathrm{d}s_{\mathrm{med}} = +\mathrm{d}\mathcal{Q}/T$.

With these definitions energy is conserved along every trajectory. The next theorem proves this and gives the Itô and ensemble-averaged forms of the energy balance.

Theorem 8.2 (Trajectory First Law of Stochastic Thermodynamics)**.** *Let $\mathbf{x}_t \in \mathbb{R}^N$ evolve according to the overdamped Langevin SDE Eq. (8.2) with $E \in C^{2,1}(\mathbb{R}^N \times [0,\tau]; \mathbb{R})$. Then:*

1. ***Trajectory First Law:*** *Along every individual realization $\mathbf{x}_{[0,\tau]}$, the differential and integrated energy balances hold identically:*

$$\mathrm{d}E(\mathbf{x}_t, \boldsymbol{\lambda}_t) \;=\; \mathrm{d}\mathcal{W}(t) - \mathrm{d}\mathcal{Q}(t), \qquad \Delta E \;\equiv\; E(\mathbf{x}_\tau, \boldsymbol{\lambda}_\tau) - E(\mathbf{x}_0, \boldsymbol{\lambda}_0) \;=\; \mathcal{W}[\mathbf{x}_{[0,\tau]}] - \mathcal{Q}[\mathbf{x}_{[0,\tau]}]. \tag{8.13}$$

2. ***Itô Decomposition of Dissipated Heat:*** *In terms of the Itô stochastic integral $(\cdot)$, the infinitesimal dissipated heat $\mathrm{d}\mathcal{Q}(t)$ decomposes into a deterministic drift rate plus a zero-mean martingale fluctuation:*

$$\begin{aligned}\mathrm{d}\mathcal{Q}(t) \;&=\; \Big[(-\nabla_{\mathbf{x}}E+\mathbf{f}_{\mathrm{nc}})^{\top}\boldsymbol{\mu}(-\nabla_{\mathbf{x}}E+\mathbf{f}_{\mathrm{nc}})+k_{\mathrm{B}}T\,\nabla_{\mathbf{x}}\cdot\big(\boldsymbol{\mu}(-\nabla_{\mathbf{x}}E+\mathbf{f}_{\mathrm{nc}})\big)\Big]\,\mathrm{d}t\\ &\quad+(-\nabla_{\mathbf{x}}E+\mathbf{f}_{\mathrm{nc}})^{\top}\sqrt{2k_{\mathrm{B}}T\boldsymbol{\mu}}\cdot\mathrm{d}\mathbf{W}_t. \end{aligned}\tag{8.14}$$

3. ***Ensemble-Averaged First Law:*** *If $\rho(\mathbf{x},t)$ is the smooth solution of the associated Fokker–Planck equation $\partial_t\rho+\nabla_{\mathbf{x}}\cdot\mathbf{J}(\mathbf{x},t)=0$ with probability current*

$$\mathbf{J}(\mathbf{x},t) \;=\; \boldsymbol{\mu}\big[-\nabla_{\mathbf{x}}E(\mathbf{x},\boldsymbol{\lambda}_t)+\mathbf{f}_{\mathrm{nc}}(\mathbf{x},t)\big]\rho(\mathbf{x},t)-k_{\mathrm{B}}T\boldsymbol{\mu}\nabla_{\mathbf{x}}\rho(\mathbf{x},t),\tag{8.15}$$

and $\rho(\mathbf{x},t)$ decays rapidly as $\|\mathbf{x}\|\to\infty$, then the ensemble-averaged work and heat rates satisfy $\frac{\mathrm{d}}{\mathrm{d}t}\langle E(\mathbf{x}_t,\boldsymbol{\lambda}_t)\rangle=\big\langle\dot{\mathcal{W}}(t)\big\rangle-\big\langle\dot{\mathcal{Q}}(t)\big\rangle$ with

$$\Big\langle\dot{\mathcal{W}}(t)\Big\rangle \;=\; \int_{\mathbb{R}^N}\mathrm{d}^N x\,\rho(\mathbf{x},t)\,\frac{\partial E(\mathbf{x},\boldsymbol{\lambda}_t)}{\partial\boldsymbol{\lambda}_t}\cdot\dot{\boldsymbol{\lambda}}_t+\int_{\mathbb{R}^N}\mathrm{d}^N x\,\mathbf{f}_{\mathrm{nc}}(\mathbf{x},t)\cdot\mathbf{J}(\mathbf{x},t),\tag{8.16}$$

$$\Big\langle\dot{\mathcal{Q}}(t)\Big\rangle \;=\; \int_{\mathbb{R}^N}\mathrm{d}^N x\,\big[-\nabla_{\mathbf{x}}E(\mathbf{x},\boldsymbol{\lambda}_t)+\mathbf{f}_{\mathrm{nc}}(\mathbf{x},t)\big]\cdot\mathbf{J}(\mathbf{x},t).\tag{8.17}$$

Proof. We prove the three statements in turn.

Part 1 (Trajectory First Law): Because the Stratonovich integral obeys the ordinary chain rule (Chapter 4), applying the Stratonovich chain rule to the smooth scalar function $E(\mathbf{x}_t,\boldsymbol{\lambda}_t)$ along a realization $\mathbf{x}_t$ yields

$$\mathrm{d}E(\mathbf{x}_t,\boldsymbol{\lambda}_t) \;=\; \frac{\partial E(\mathbf{x}_t,\boldsymbol{\lambda}_t)}{\partial\boldsymbol{\lambda}_t}\cdot\mathrm{d}\boldsymbol{\lambda}_t+\nabla_{\mathbf{x}}E(\mathbf{x}_t,\boldsymbol{\lambda}_t)\circ\mathrm{d}\mathbf{x}_t.\tag{8.18}$$

Now add and subtract the non-conservative work differential $\mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t,t)\circ\mathrm{d}\mathbf{x}_t$ on the right-hand side of Eq. (8.18):

$$\begin{aligned}\mathrm{d}E(\mathbf{x}_t,\boldsymbol{\lambda}_t) \;&=\; \underbrace{\Big[\frac{\partial E(\mathbf{x}_t,\boldsymbol{\lambda}_t)}{\partial\boldsymbol{\lambda}_t}\cdot\mathrm{d}\boldsymbol{\lambda}_t+\mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t,t)\circ\mathrm{d}\mathbf{x}_t\Big]}_{\mathrm{d}\mathcal{W}(t)}-\underbrace{\big[-\nabla_{\mathbf{x}}E(\mathbf{x}_t,\boldsymbol{\lambda}_t)+\mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t,t)\big]\circ\mathrm{d}\mathbf{x}_t}_{\mathrm{d}\mathcal{Q}(t)}\\ &=\; \mathrm{d}\mathcal{W}(t)-\mathrm{d}\mathcal{Q}(t).\end{aligned}\tag{8.19}$$

Integrating both sides from $t=0$ to $t=\tau$ yields $\Delta E=\mathcal{W}[\mathbf{x}_{[0,\tau]}]-\mathcal{Q}[\mathbf{x}_{[0,\tau]}]$, establishing Eq. (8.13).

Part 2 (Itô Decomposition of Dissipated Heat): Let $\mathbf{F}(\mathbf{x}_t,t)\equiv-\nabla_{\mathbf{x}}E(\mathbf{x}_t,\boldsymbol{\lambda}_t)+\mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t,t)$ denote the total systematic force vector, so that $\mathrm{d}\mathcal{Q}(t)=\mathbf{F}(\mathbf{x}_t,t)\circ\mathrm{d}\mathbf{x}_t$. Recall from Chapter 4 the general conversion rule between the Stratonovich midpoint product and the Itô product for any smooth vector field $\mathbf{F}(\mathbf{x}_t,t)$ driven by $\mathrm{d}\mathbf{x}_t=\mathbf{a}(\mathbf{x}_t,t)\,\mathrm{d}t+\mathbf{B}\,\mathrm{d}\mathbf{W}_t$:

$$\mathbf{F}(\mathbf{x}_t,t)\circ\mathrm{d}\mathbf{x}_t \;=\; \mathbf{F}(\mathbf{x}_t,t)\cdot\mathrm{d}\mathbf{x}_t+\frac{1}{2}\sum_{i,j=1}^{N}\frac{\partial F_i(\mathbf{x}_t,t)}{\partial x_j}\,\mathrm{d}x_{i,t}\,\mathrm{d}x_{j,t}.\tag{8.20}$$

Using the quadratic variation table $\mathrm{d}W_{k,t}\,\mathrm{d}W_{l,t} = \delta_{kl}\,\mathrm{d}t$ and $\mathrm{d}t\,\mathrm{d}W_{k,t} = (\mathrm{d}t)^2 = 0$, the quadratic variation of $\mathrm{d}\mathbf{x}_t$ is

$$\mathrm{d}x_{i,t}\,\mathrm{d}x_{j,t} \;=\; \sum_{k,l=1}^{N} B_{ik}B_{jl}\,\mathrm{d}W_{k,t}\,\mathrm{d}W_{l,t} \;=\; \left(\mathbf{B}\mathbf{B}^{\top}\right)_{ij}\mathrm{d}t \;=\; 2k_{\mathrm{B}}T\,\mu_{ij}\,\mathrm{d}t.$$

Substituting this quadratic variation and $\mathrm{d}\mathbf{x}_t = \boldsymbol{\mu}\mathbf{F}(\mathbf{x}_t,t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\boldsymbol{\mu}}\,\mathrm{d}\mathbf{W}_t$ into Eq. (8.20), we obtain

$$\begin{aligned}\mathrm{d}\mathcal{Q}(t) \;&=\; \mathbf{F}(\mathbf{x}_t,t)^{\top}\boldsymbol{\mu}\mathbf{F}(\mathbf{x}_t,t)\,\mathrm{d}t + \mathbf{F}(\mathbf{x}_t,t)^{\top}\sqrt{2k_{\mathrm{B}}T\boldsymbol{\mu}}\cdot\mathrm{d}\mathbf{W}_t + \frac{1}{2}\sum_{i,j=1}^{N}\frac{\partial F_i(\mathbf{x}_t,t)}{\partial x_j}\left(2k_{\mathrm{B}}T\,\mu_{ij}\right)\mathrm{d}t \\ &=\; \Big[\mathbf{F}(\mathbf{x}_t,t)^{\top}\boldsymbol{\mu}\mathbf{F}(\mathbf{x}_t,t) + k_{\mathrm{B}}T\,\nabla_{\mathbf{x}}\cdot\left(\boldsymbol{\mu}\mathbf{F}(\mathbf{x}_t,t)\right)\Big]\mathrm{d}t + \mathbf{F}(\mathbf{x}_t,t)^{\top}\sqrt{2k_{\mathrm{B}}T\boldsymbol{\mu}}\cdot\mathrm{d}\mathbf{W}_t, \qquad (8.21)\end{aligned}$$

where we used the symmetry $\mu_{ij} = \mu_{ji}$ and constancy of $\boldsymbol{\mu}$ to write $\sum_{i,j}\mu_{ij}\partial_{x_j}F_i = \nabla_{\mathbf{x}}\cdot(\boldsymbol{\mu}\mathbf{F})$. Substituting $\mathbf{F} = -\nabla_{\mathbf{x}}E + \mathbf{f}_{\mathrm{nc}}$ gives Eq. (8.14).

Part 3 (Ensemble-Averaged First Law): Because the Itô stochastic integral in Eq. (8.21) is a martingale with zero expectation ($\mathbb{E}\Big[\mathbf{F}(\mathbf{x}_t,t)^{\top}\sqrt{2k_{\mathrm{B}}T\boldsymbol{\mu}}\cdot\mathrm{d}\mathbf{W}_t\Big] = 0$), taking the expectation over the instantaneous density $\rho(\mathbf{x},t)$ and dividing by $\mathrm{d}t$ yields the ensemble heat dissipation rate:

$$\left\langle\dot{\mathcal{Q}}(t)\right\rangle \;=\; \int_{\mathbb{R}^N}\mathrm{d}^N x\,\rho(\mathbf{x},t)\,\mathbf{F}(\mathbf{x},t)^{\top}\boldsymbol{\mu}\mathbf{F}(\mathbf{x},t) + k_{\mathrm{B}}T\int_{\mathbb{R}^N}\mathrm{d}^N x\,\rho(\mathbf{x},t)\,\nabla_{\mathbf{x}}\cdot\left(\boldsymbol{\mu}\mathbf{F}(\mathbf{x},t)\right). \qquad (8.22)$$

We now integrate the second integral in Eq. (8.22) by parts over $\mathbb{R}^N$. Because $\rho(\mathbf{x},t)\to 0$ rapidly as $\|\mathbf{x}\|\to\infty$, the boundary surface integral at infinity vanishes identically:

$$k_{\mathrm{B}}T\int_{\mathbb{R}^N}\mathrm{d}^N x\,\rho(\mathbf{x},t)\,\nabla_{\mathbf{x}}\cdot\left(\boldsymbol{\mu}\mathbf{F}(\mathbf{x},t)\right) \;=\; \underbrace{k_{\mathrm{B}}T\oint_{\partial\mathbb{R}^N}\mathrm{d}\mathbf{S}\cdot\left(\rho\,\boldsymbol{\mu}\mathbf{F}\right)}_{=\,0} - k_{\mathrm{B}}T\int_{\mathbb{R}^N}\mathrm{d}^N x\,\left(\nabla_{\mathbf{x}}\rho(\mathbf{x},t)\right)^{\top}\boldsymbol{\mu}\mathbf{F}(\mathbf{x},t).$$

Combining this with the first term in Eq. (8.22) and using the symmetry $\boldsymbol{\mu} = \boldsymbol{\mu}^{\top}$, we factor out $\mathbf{F}(\mathbf{x},t)$:

$$\begin{aligned}\left\langle\dot{\mathcal{Q}}(t)\right\rangle \;&=\; \int_{\mathbb{R}^N}\mathrm{d}^N x\,\mathbf{F}(\mathbf{x},t)\cdot\Big[\boldsymbol{\mu}\mathbf{F}(\mathbf{x},t)\rho(\mathbf{x},t) - k_{\mathrm{B}}T\boldsymbol{\mu}\nabla_{\mathbf{x}}\rho(\mathbf{x},t)\Big] \\ &=\; \int_{\mathbb{R}^N}\mathrm{d}^N x\,\left[-\nabla_{\mathbf{x}}E(\mathbf{x},\boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x},t)\right]\cdot\mathbf{J}(\mathbf{x},t), \qquad (8.23)\end{aligned}$$

which is precisely Eq. (8.17). Exactly the same Itô-to-Stratonovich conversion and integration by parts applied to $\mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t,t)\circ\mathrm{d}\mathbf{x}_t$ yields Eq. (8.16). Finally, subtracting Eq. (8.17) from Eq. (8.16), the $\mathbf{f}_{\mathrm{nc}}\cdot\mathbf{J}$ terms cancel, leaving

$$\begin{aligned}\left\langle\dot{\mathcal{W}}(t)\right\rangle - \left\langle\dot{\mathcal{Q}}(t)\right\rangle \;&=\; \int_{\mathbb{R}^N}\mathrm{d}^N x\,\rho(\mathbf{x},t)\,\frac{\partial E(\mathbf{x},\boldsymbol{\lambda}_t)}{\partial\boldsymbol{\lambda}_t}\cdot\dot{\boldsymbol{\lambda}}_t + \int_{\mathbb{R}^N}\mathrm{d}^N x\,\nabla_{\mathbf{x}}E(\mathbf{x},\boldsymbol{\lambda}_t)\cdot\mathbf{J}(\mathbf{x},t) \\ &=\; \int_{\mathbb{R}^N}\mathrm{d}^N x\,\left[\rho(\mathbf{x},t)\,\frac{\partial E}{\partial\boldsymbol{\lambda}_t}\cdot\dot{\boldsymbol{\lambda}}_t - E(\mathbf{x},\boldsymbol{\lambda}_t)\,\nabla_{\mathbf{x}}\cdot\mathbf{J}(\mathbf{x},t)\right] \\ &=\; \int_{\mathbb{R}^N}\mathrm{d}^N x\,\left[\rho(\mathbf{x},t)\,\frac{\partial E}{\partial t} + E(\mathbf{x},\boldsymbol{\lambda}_t)\,\frac{\partial\rho(\mathbf{x},t)}{\partial t}\right] \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\left\langle E(\mathbf{x}_t,\boldsymbol{\lambda}_t)\right\rangle, \qquad (8.24)\end{aligned}$$

where we integrated by parts in the second line and used the Fokker–Planck continuity equation $-\nabla_{\mathbf{x}}\cdot\mathbf{J} = \partial_t\rho$ in the third line. □

Physics Insight: Why Energy Conservation Requires the Stratonovich Product

Theorem 8.2 shows that the choice of stochastic calculus in energetics is not a matter of taste. Suppose one naively defined the heat dissipated during $[t, t+\mathrm{d}t]$ using the Itô left-endpoint product, $\mathrm{d}\mathcal{Q}_{\mathrm{Itô}} \equiv -\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) \cdot \mathrm{d}\mathbf{x}_t$ (with $\mathbf{f}_{\mathrm{nc}} = \mathbf{0}$). By Itô's Lemma (Chapter 4), the true differential of the energy $E(\mathbf{x}_t, \boldsymbol{\lambda}_t)$ is

$$\mathrm{d}E(\mathbf{x}_t, \boldsymbol{\lambda}_t) \;=\; \frac{\partial E}{\partial \boldsymbol{\lambda}_t} \cdot \mathrm{d}\boldsymbol{\lambda}_t + \nabla_{\mathbf{x}} E \cdot \mathrm{d}\mathbf{x}_t + k_{\mathrm{B}}T \;\mathrm{tr}\big(\boldsymbol{\mu} \nabla_{\mathbf{x}}^2 E(\mathbf{x}_t, \boldsymbol{\lambda}_t)\big)\; \mathrm{d}t.$$

Even in a static harmonic trap $E(x) = \frac{1}{2}kx^2$ sitting in thermal equilibrium ($\dot{\lambda} = 0$, $\mathbf{J} = \mathbf{J}_{\mathrm{eq}} = 0$), the naive Itô heat rate would have a non-zero ensemble average:

$$\left\langle \dot{\mathcal{Q}}_{\mathrm{Itô}} \right\rangle \;=\; -\left\langle kx_t \cdot \left(-\mu k x_t + \sqrt{2k_{\mathrm{B}}T\mu}\, \frac{\mathrm{d}W_t}{\mathrm{d}t} \right) \right\rangle \;=\; +\mu k^2 \left\langle x^2 \right\rangle_{\mathrm{eq}} \;=\; +\mu k k_{\mathrm{B}}T \;>\; 0.$$

The naive Itô definition would have an RC circuit in equilibrium, with no source attached and no work done on it, steadily dissipating heat into its bath. That violates energy conservation. The error is one of timing. In Itô's product $F(x_t) \cdot (x_{t+\mathrm{d}t} - x_t)$, the force $F(x_t) = -kx_t$ is evaluated *before* the thermal kick $\mathrm{d}W_t$ arrives, so it is statistically uncorrelated with $\mathrm{d}W_t$. It therefore captures the viscous cooling $\mu k^2 x_t^2\, \mathrm{d}t$ as the particle slides down the potential well, but misses the systematic heating $-k_{\mathrm{B}}T\mu k\, \mathrm{d}t$ caused by the correlation between the thermal kick $\mathrm{d}W_t$ and the change in the restoring force across the step $[t, t+\mathrm{d}t]$. The Stratonovich midpoint evaluation $F(\frac{x_t + x_{t+\mathrm{d}t}}{2}) \cdot (x_{t+\mathrm{d}t} - x_t)$ captures both the viscous cooling and the thermal heating on an equal footing: in thermal equilibrium, the two terms in Eq. (8.14) cancel exactly on average ($\mu k^2 \langle x^2 \rangle_{\mathrm{eq}} - k_{\mathrm{B}}T\mu k = 0$), yielding $\left\langle \dot{\mathcal{Q}} \right\rangle = 0$, as energy conservation requires.

8.1.3 Consistency with Underdamped Phase-Space Energetics

Before leaving the First Law, we check that the overdamped heat formula Eq. (8.10) is not an ad hoc postulate. It follows from the underdamped phase-space dynamics of Chapter 5.

In full phase space $(\mathbf{x}_t, \mathbf{p}_t) \in \mathbb{R}^N \times \mathbb{R}^N$ with diagonal mass (or inductance) matrix $\mathbf{M} = \mathbf{M}^\top \succ \mathbf{0}$, the total mechanical energy of the system is the Hamiltonian

$$H(\mathbf{x}_t, \mathbf{p}_t, \boldsymbol{\lambda}_t) \;=\; \frac{1}{2}\mathbf{p}_t^\top \mathbf{M}^{-1} \mathbf{p}_t + E(\mathbf{x}_t, \boldsymbol{\lambda}_t), \tag{8.25}$$

and the underdamped Langevin equations of motion are

$$\mathrm{d}\mathbf{x}_t \;=\; \mathbf{M}^{-1}\mathbf{p}_t\, \mathrm{d}t, \tag{8.26}$$

$$\mathrm{d}\mathbf{p}_t \;=\; \left[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t) - \boldsymbol{\Gamma}\mathbf{M}^{-1}\mathbf{p}_t \right] \mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\boldsymbol{\Gamma}}\; \mathrm{d}\mathbf{W}_t. \tag{8.27}$$

Because $\mathrm{d}\mathbf{x}_t = \mathbf{v}_t\, \mathrm{d}t$ (where $\mathbf{v}_t \equiv \mathbf{M}^{-1}\mathbf{p}_t$) has zero quadratic variation ($\mathrm{d}x_{i,t}\, \mathrm{d}x_{j,t} = 0$), all position differentials are ordinary smooth differentials. Only the momentum $\mathbf{p}_t$ couples directly to the white noise $\mathrm{d}\mathbf{W}_t$. Applying the Stratonovich chain rule to the total Hamiltonian Eq. (8.25) gives

$$\mathrm{d}H(\mathbf{x}_t, \mathbf{p}_t, \boldsymbol{\lambda}_t) \;=\; \frac{\partial E(\mathbf{x}_t, \boldsymbol{\lambda}_t)}{\partial \boldsymbol{\lambda}_t} \cdot \mathrm{d}\boldsymbol{\lambda}_t + \nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) \cdot \mathrm{d}\mathbf{x}_t + (\mathbf{M}^{-1}\mathbf{p}_t) \circ \mathrm{d}\mathbf{p}_t. \tag{8.28}$$

Using $\mathrm{d}\mathbf{x}_t = \mathbf{M}^{-1}\mathbf{p}_t\,\mathrm{d}t$, we can substitute $\nabla_{\mathbf{x}}E\cdot\mathrm{d}\mathbf{x}_t = (\nabla_{\mathbf{x}}E)\cdot(\mathbf{M}^{-1}\mathbf{p}_t)\,\mathrm{d}t$ and insert $\mathrm{d}\mathbf{p}_t$ from Eq. (8.27) into Eq. (8.28). The conservative force terms $-\nabla_{\mathbf{x}}E\cdot(\mathbf{M}^{-1}\mathbf{p}_t)\,\mathrm{d}t$ and $+\nabla_{\mathbf{x}}E\cdot\mathrm{d}\mathbf{x}_t$ cancel identically along every trajectory, leaving

$$\mathrm{d}H(\mathbf{x}_t,\mathbf{p}_t,\boldsymbol{\lambda}_t) \;=\; \underbrace{\left[\frac{\partial E}{\partial\boldsymbol{\lambda}_t}\cdot\mathrm{d}\boldsymbol{\lambda}_t + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t,t)\cdot\mathrm{d}\mathbf{x}_t\right]}_{\mathrm{d}\mathcal{W}(t)} - \underbrace{(\boldsymbol{\Gamma}\mathbf{v}_t\,\mathrm{d}t - \sqrt{2k_{\mathrm{B}}T\boldsymbol{\Gamma}}\,\mathrm{d}\mathbf{W}_t)\circ\mathbf{v}_t}_{\mathrm{d}\mathcal{Q}_{\mathrm{ud}}(t)}. \tag{8.29}$$

Thus, in the underdamped theory, the heat dissipated into the bath is again $\mathrm{d}\mathcal{Q}_{\mathrm{ud}}(t) = (-\mathbf{F}_{\mathrm{bath}})\circ\mathrm{d}\mathbf{x}_t$, now with the underdamped bath force $\mathbf{F}_{\mathrm{bath}} = -\boldsymbol{\Gamma}\mathbf{v}_t + \sqrt{2k_{\mathrm{B}}T\boldsymbol{\Gamma}}\,\boldsymbol{\xi}_t$, which by Eq. (8.27) equals $[-\nabla_{\mathbf{x}}E + \mathbf{f}_{\mathrm{nc}}]\cdot\mathrm{d}\mathbf{x}_t - \mathrm{d}(\frac{1}{2}\mathbf{p}_t^{\top}\mathbf{M}^{-1}\mathbf{p}_t)$. When the momentum relaxation time $\tau_p = \|\mathbf{M}\|\,\|\boldsymbol{\Gamma}^{-1}\|$ is much shorter than the observation window, the kinetic energy stays thermalized at $\frac{1}{2}\left\langle\mathbf{p}_t^{\top}\mathbf{M}^{-1}\mathbf{p}_t\right\rangle = \frac{N}{2}k_{\mathrm{B}}T$, its change ΔE_{kin} stays of order $k_{\mathrm{B}}T$ rather than growing with time, and the work and heat reduce to Definition 8.1.

Example 8.3 (Trajectory Energetics of a Driven RC Node)**.** Consider a single-node linear RC circuit, the $N = 1$ building block of the RC sampler (the coupled linear RC circuit of Chapter 1), consisting of a capacitor C connected through a resistor of conductance $G = 1/R$ to a programmable voltage source $b(t) \equiv \lambda(t)$ at temperature T. The state coordinate is the node voltage $v_t \in \mathbb{R}$, and the electrostatic free energy of the node relative to the source bias $b(t)$ is

$$E(v_t, b_t) \;=\; \frac{1}{2}C(v_t - b_t)^2. \tag{8.30}$$

In our canonical Langevin notation Eq. (8.2), the restoring force is $-\partial_v E(v_t,b_t) = -C(v_t-b_t)$. Because Kirchhoff's current law reads $C\,\mathrm{d}v_t = -G(v_t-b_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}TG}\,\mathrm{d}W_t$, multiplying by C/G puts the equation into the force-balance form Eq. (8.2) with effective friction $\Gamma = C^2/G$ (so $\mu = G/C^2$ and $\tau_{RC} = C/G = \Gamma/C$):

$$\frac{C^2}{G}\,\mathrm{d}v_t \;=\; -C(v_t-b_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\,\frac{C^2}{G}}\,\mathrm{d}W_t.$$

By Definition 8.1, the trajectory work performed by the external voltage source b_t during $[t, t+\mathrm{d}t]$ is

$$\mathrm{d}\mathcal{W}(t) \;=\; \frac{\partial E(v_t,b_t)}{\partial b_t}\,\mathrm{d}b_t \;=\; -C(v_t-b_t)\,\dot{b}_t\,\mathrm{d}t, \tag{8.31}$$

while the trajectory heat dissipated as Joule heating in the resistor $R = 1/G$ is

$$\mathrm{d}\mathcal{Q}(t) \;=\; -\frac{\partial E(v_t,b_t)}{\partial v_t}\circ\mathrm{d}v_t \;=\; -C(v_t-b_t)\circ\mathrm{d}v_t. \tag{8.32}$$

Applying the Itô decomposition Eq. (8.14) with $\mu = G/C^2$ and $-\partial_v E = -C(v_t-b_t)$, we find the explicit Itô expression for the instantaneous Joule heat in the resistor:

$$\mathrm{d}\mathcal{Q}(t) \;=\; \left[G(v_t-b_t)^2 - \frac{k_{\mathrm{B}}T}{\tau_{RC}}\right]\mathrm{d}t - (v_t-b_t)\sqrt{2k_{\mathrm{B}}TG}\cdot\mathrm{d}W_t. \tag{8.33}$$

The two drift terms have a direct interpretation: $G(v_t-b_t)^2\,\mathrm{d}t = \frac{(v_t-b_t)^2}{R}\,\mathrm{d}t$ is the classical macroscopic Joule dissipation from the voltage drop across R, while $-\frac{k_{\mathrm{B}}T}{\tau_{RC}}\,\mathrm{d}t = -\frac{G}{C}k_{\mathrm{B}}T\,\mathrm{d}t$ is the steady thermal power injected *from* the Johnson–Nyquist resistor noise *into* the capacitor. At thermal equilibrium ($\dot{b}_t = 0$), the equipartition theorem gives $\langle(v_t-b)^2\rangle_{\mathrm{eq}} = k_{\mathrm{B}}T/C$, so the mean Joule dissipation $Gk_{\mathrm{B}}T/C$ and the mean noise injection $-k_{\mathrm{B}}TG/C$ cancel, yielding $\left\langle\dot{\mathcal{Q}}\right\rangle_{\mathrm{eq}} = 0$.

8.2 Stochastic System, Medium, and Total Entropy Production

Having established the trajectory First Law $\mathrm{d}E = \mathrm{d}\mathcal{W} - \mathrm{d}\mathcal{Q}$, we now turn to the Second Law. In macroscopic equilibrium thermodynamics, the total entropy change of the universe is the sum of the entropy change of the system, ΔS_{sys}, and the entropy change of the heat bath (or surrounding medium), $\Delta S_{\mathrm{med}} = \mathcal{Q}/T$. The question is how to define the entropy of the system along a *single* trajectory $\mathbf{x}_t$.

8.2.1 Seifert's Trajectory System Entropy and Total Entropy Production

At first glance, assigning an entropy to a single microstate $\mathbf{x}_t$ seems paradoxical, because the Gibbs–Shannon entropy $S[\rho_t] = -k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t) \ln \rho(\mathbf{x},t)$ is a functional of the entire probability density $\rho(\mathbf{x},t)$. Seifert resolved this (see his review [97]) by observing that once the initial distribution $\rho(\mathbf{x},0) = \rho_0(\mathbf{x})$ and the driving protocol $\boldsymbol{\lambda}_{[0,\tau]}$ are specified, the Fokker–Planck equation $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$ uniquely determines the scalar density field $\rho(\mathbf{x},t)$ at every point in space and time. One can therefore evaluate the surprise (or self-information) $-k_{\mathrm{B}} \ln \rho(\mathbf{x},t)$ *along the stochastic trajectory* $\mathbf{x}_t$ *itself.*

Definition 8.4 (Trajectory System, Medium, and Total Entropy)**.** Let $\rho(\mathbf{x},t) > 0$ be the instantaneous probability density solving the Fokker–Planck equation $\partial_t \rho(\mathbf{x},t) = -\nabla_{\mathbf{x}} \cdot \mathbf{J}(\mathbf{x},t)$ associated with the Langevin SDE Eq. (8.2). Along an individual trajectory $\mathbf{x}_{[0,\tau]}$:

1. The **stochastic system entropy** at time t is the state-dependent random variable

$$s_{\mathrm{sys}}(t) \;\equiv\; -k_{\mathrm{B}} \ln \rho(\mathbf{x}_t, t), \tag{8.34}$$

   whose ensemble average is the macroscopic Gibbs–Shannon differential entropy:

$$S(t) \;\equiv\; \langle s_{\mathrm{sys}}(t) \rangle \;=\; -k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t) \ln \rho(\mathbf{x},t). \tag{8.35}$$

2. The **infinitesimal medium entropy production** $\mathrm{d}s_{\mathrm{med}}(t)$ in the reversible thermal bath at temperature T is the heat dissipated into the bath divided by T:

$$\mathrm{d}s_{\mathrm{med}}(t) \;\equiv\; \frac{\mathrm{d}\mathcal{Q}(t)}{T} \;=\; \frac{1}{T}\big[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\big] \circ \mathrm{d}\mathbf{x}_t. \tag{8.36}$$

3. The **infinitesimal total trajectory entropy production** $\mathrm{d}s_{\mathrm{tot}}(t)$ is the sum of the system entropy change and the medium entropy production:

$$\mathrm{d}s_{\mathrm{tot}}(t) \;\equiv\; \mathrm{d}s_{\mathrm{sys}}(t) + \mathrm{d}s_{\mathrm{med}}(t) \;=\; -k_{\mathrm{B}}\, \mathrm{d}\ln \rho(\mathbf{x}_t, t) + \frac{\mathrm{d}\mathcal{Q}(t)}{T}, \tag{8.37}$$

   with integrated total trajectory entropy production $\Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}] \equiv \Delta s_{\mathrm{sys}} + \Delta s_{\mathrm{med}}$ over $[0,\tau]$, and ensemble-averaged total entropy production

$$\Sigma_{\mathrm{tot}}(\tau) \;\equiv\; \Big\langle \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}] \Big\rangle \;=\; \Delta S + \frac{\langle \mathcal{Q} \rangle}{T}. \tag{8.38}$$

Notation Watch: Lowercase Trajectory Entropies vs. Uppercase Ensemble Entropies

We maintain a strict typographical distinction between fluctuating trajectory quantities and deterministic ensemble expectations:

- $s_{\mathrm{sys}}(t)$, $\Delta s_{\mathrm{med}}[\mathbf{x}_{[0,\tau]}]$, and $\Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]$ (lowercase s) denote random variables along a single trajectory. In particular, $\Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]$ fluctuates in sign from run to run and can be strictly negative on rare trajectories (Chapter 9).
- $S(t) = \langle s_{\mathrm{sys}}(t)\rangle$ and $\Sigma_{\mathrm{tot}}(\tau) = \left\langle \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]\right\rangle$ (uppercase S and Σ) denote ensemble averages, which are deterministic numbers obeying the macroscopic Second Law $\Sigma_{\mathrm{tot}}(\tau) \geq 0$.

Both carry physical dimensions of Boltzmann's constant k_{B} (J/K); dividing by k_{B} yields dimensionless entropy in nats.

8.2.2 The Local Mean Velocity and the Trajectory Entropy SDE

To simplify $\mathrm{d}s_{\mathrm{tot}}(t)$, we introduce the **local mean (hydrodynamic) velocity field** $\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t) \in \mathbb{R}^N$ of the probability fluid:

$$\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t) \;\equiv\; \frac{\mathbf{J}(\mathbf{x},t)}{\rho(\mathbf{x},t)} \;=\; \boldsymbol{\mu}\big[-\nabla_{\mathbf{x}}E(\mathbf{x},\boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x},t)\big] - \mathbf{D}\nabla_{\mathbf{x}}\ln\rho(\mathbf{x},t), \tag{8.39}$$

where we used $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$ and $\frac{\nabla_{\mathbf{x}}\rho}{\rho} = \nabla_{\mathbf{x}}\ln\rho$. In terms of $\mathbf{v}_{\mathrm{loc}}$ the continuity equation $\partial_t\rho + \nabla_{\mathbf{x}}\cdot\mathbf{J} = 0$ takes the advection form $\partial_t\rho + \nabla_{\mathbf{x}}\cdot(\rho\,\mathbf{v}_{\mathrm{loc}}) = 0$. Solving Eq. (8.39) for the total force divided by T gives an identity connecting the medium force to the local probability velocity and the score function $\nabla_{\mathbf{x}}\ln\rho$:

$$\frac{-\nabla_{\mathbf{x}}E(\mathbf{x},\boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x},t)}{T} \;=\; k_{\mathrm{B}}\mathbf{D}^{-1}\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t) + k_{\mathrm{B}}\nabla_{\mathbf{x}}\ln\rho(\mathbf{x},t). \tag{8.40}$$

With it we can compute the differential of $s_{\mathrm{tot}}(t)$ along a trajectory.

Proposition 8.5 (Trajectory Entropy Production Differential)**.** *Along any trajectory $\mathbf{x}_t$ of the overdamped Langevin SDE Eq. (8.2), the Stratonovich differentials of the system entropy $s_{\mathrm{sys}}(t)$ and total entropy production $s_{\mathrm{tot}}(t)$ are given by*

$$\mathrm{d}s_{\mathrm{sys}}(t) \;=\; -k_{\mathrm{B}}\,\frac{\partial_t\rho(\mathbf{x}_t,t)}{\rho(\mathbf{x}_t,t)}\,\mathrm{d}t - k_{\mathrm{B}}\,\nabla_{\mathbf{x}}\ln\rho(\mathbf{x}_t,t)\circ\mathrm{d}\mathbf{x}_t, \tag{8.41}$$

$$\mathrm{d}s_{\mathrm{tot}}(t) \;=\; -k_{\mathrm{B}}\,\frac{\partial_t\rho(\mathbf{x}_t,t)}{\rho(\mathbf{x}_t,t)}\,\mathrm{d}t + k_{\mathrm{B}}(\mathbf{D}^{-1}\mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t,t))\circ\mathrm{d}\mathbf{x}_t. \tag{8.42}$$

Equivalently, written in Itô form ($\cdot$), the total trajectory entropy production obeys the SDE

$$\begin{aligned}\mathrm{d}s_{\mathrm{tot}}(t) \;=\; k_{\mathrm{B}}\Big[&\mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t,t)^{\top}\mathbf{D}^{-1}\mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t,t) \\ &+ \frac{1}{\rho(\mathbf{x}_t,t)}\nabla_{\mathbf{x}}\cdot(\rho(\mathbf{x}_t,t)\mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t,t)) \\ &+ \mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t,t)\cdot\nabla_{\mathbf{x}}\ln\rho(\mathbf{x}_t,t) + \nabla_{\mathbf{x}}\cdot\mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t,t)\Big]\,\mathrm{d}t \\ &+ k_{\mathrm{B}}\,\mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t,t)^{\top}\mathbf{D}^{-1}\mathbf{B}\cdot\mathrm{d}\mathbf{W}_t.\end{aligned} \tag{8.43, 8.44}$$

Proof. Applying the Stratonovich chain rule to $s_{\mathrm{sys}}(t) = -k_{\mathrm{B}} \ln \rho(\mathbf{x}_t, t)$ gives Eq. (8.41):

$$\mathrm{d}s_{\mathrm{sys}}(t) \;=\; -k_{\mathrm{B}} \left[\frac{\partial \ln \rho(\mathbf{x}_t, t)}{\partial t}\, \mathrm{d}t + \nabla_{\mathbf{x}} \ln \rho(\mathbf{x}_t, t) \circ \mathrm{d}\mathbf{x}_t \right] \;=\; -k_{\mathrm{B}}\, \frac{\partial_t \rho(\mathbf{x}_t, t)}{\rho(\mathbf{x}_t, t)}\, \mathrm{d}t - k_{\mathrm{B}}\, \nabla_{\mathbf{x}} \ln \rho(\mathbf{x}_t, t) \circ \mathrm{d}\mathbf{x}_t.$$

Next, we substitute the force decomposition Eq. (8.40) into the medium entropy production Eq. (8.36):

$$\mathrm{d}s_{\mathrm{med}}(t) \;=\; \frac{-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)}{T} \circ \mathrm{d}\mathbf{x}_t \;=\; k_{\mathrm{B}} (\mathbf{D}^{-1} \mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t, t)) \circ \mathrm{d}\mathbf{x}_t + k_{\mathrm{B}}\, \nabla_{\mathbf{x}} \ln \rho(\mathbf{x}_t, t) \circ \mathrm{d}\mathbf{x}_t. \tag{8.45}$$

Adding Eq. (8.41) and Eq. (8.45), the spatial score terms $\mp k_{\mathrm{B}} \nabla_{\mathbf{x}} \ln \rho(\mathbf{x}_t, t) \circ \mathrm{d}\mathbf{x}_t$ cancel identically along every trajectory, yielding Eq. (8.42).

To convert Eq. (8.42) into the Itô form Eq. (8.43), note first from the Fokker–Planck continuity equation that $-\partial_t \rho = \nabla_{\mathbf{x}} \cdot \mathbf{J} = \nabla_{\mathbf{x}} \cdot (\rho \mathbf{v}_{\mathrm{loc}})$. Second, from Eqs. (8.3) and (8.39), the Langevin drift is $\mathbf{a}(\mathbf{x}, t) = \mathbf{v}_{\mathrm{loc}}(\mathbf{x}, t) + \mathbf{D} \nabla_{\mathbf{x}} \ln \rho(\mathbf{x}, t)$. Using the Stratonovich-to-Itô conversion formula Eq. (8.20) with $\frac{1}{2}\mathbf{B}\mathbf{B}^{\top} = \mathbf{D}$, we expand the second term in Eq. (8.42):

$$\begin{aligned} (\mathbf{D}^{-1} \mathbf{v}_{\mathrm{loc}}(\mathbf{x}_t, t)) \circ \mathrm{d}\mathbf{x}_t \;&=\; (\mathbf{D}^{-1} \mathbf{v}_{\mathrm{loc}}) \cdot \Big[(\mathbf{v}_{\mathrm{loc}} + \mathbf{D} \nabla_{\mathbf{x}} \ln \rho)\, \mathrm{d}t + \mathbf{B}\, \mathrm{d}\mathbf{W}_t \Big] + \sum_{i,j=1}^{N} D_{ij}\, \frac{\partial}{\partial x_j} (\mathbf{D}^{-1} \mathbf{v}_{\mathrm{loc}})_i\, \mathrm{d}t \\ &=\; \Big[\mathbf{v}_{\mathrm{loc}}^{\top} \mathbf{D}^{-1} \mathbf{v}_{\mathrm{loc}} + \mathbf{v}_{\mathrm{loc}} \cdot \nabla_{\mathbf{x}} \ln \rho + \nabla_{\mathbf{x}} \cdot \mathbf{v}_{\mathrm{loc}} \Big]\, \mathrm{d}t + \mathbf{v}_{\mathrm{loc}}^{\top} \mathbf{D}^{-1} \mathbf{B} \cdot \mathrm{d}\mathbf{W}_t. \end{aligned} \tag{8.46}$$

Combining this with $-\frac{\partial_t \rho}{\rho} = \frac{1}{\rho} \nabla_{\mathbf{x}} \cdot (\rho \mathbf{v}_{\mathrm{loc}})$ completes the proof of Eq. (8.43). □

8.2.3 Ensemble Second Law and Recovery of the Fokker–Planck H-Theorem

On taking the ensemble average of Proposition 8.5, the boundary terms vanish by conservation of probability, and a simple expression for the rate of total entropy production remains.

Theorem 8.6 (Ensemble Second Law and Local Velocity Quadratic Form)**.** *Under the hypotheses of Theorem 8.2, the ensemble-averaged rate of total entropy production* $\dot{\Sigma}_{\mathrm{tot}}(t) \equiv \frac{\mathrm{d}}{\mathrm{d}t} \langle s_{\mathrm{tot}}(t) \rangle$ *satisfies*

$$\dot{\Sigma}_{\mathrm{tot}}(t) \;=\; \left\langle \frac{\mathrm{d}s_{\mathrm{tot}}}{\mathrm{d}t} \right\rangle \;=\; k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \frac{\mathbf{J}(\mathbf{x}, t)^{\top} \mathbf{D}^{-1} \mathbf{J}(\mathbf{x}, t)}{\rho(\mathbf{x}, t)} \;=\; k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}, t)\, \|\mathbf{v}_{\mathrm{loc}}(\mathbf{x}, t)\|^2_{\mathbf{D}^{-1}} \;\geq\; 0, \tag{8.47}$$

where $\|\mathbf{u}\|^2_{\mathbf{D}^{-1}} \equiv \mathbf{u}^{\top} \mathbf{D}^{-1} \mathbf{u}$. *Equality* $\dot{\Sigma}_{\mathrm{tot}}(t) = 0$ *holds at time* t *if and only if the probability current vanishes identically everywhere (*$\mathbf{J}(\mathbf{x}, t) = \mathbf{0}$ *for all* $\mathbf{x} \in \mathbb{R}^N$*), which requires that the non-conservative force vanish (*$\mathbf{f}_{\mathrm{nc}} = \mathbf{0}$*) and that the density coincide at that instant with the Boltzmann–Gibbs density for the current parameters,* $\rho(\mathbf{x}, t) = p_{\mathrm{eq}}(\mathbf{x}; \boldsymbol{\lambda}_t)$*. For this to hold over a time interval rather than at a single instant, the protocol must also be frozen,* $\dot{\boldsymbol{\lambda}}_t = \mathbf{0}$*.*

Moreover, whenever $\mathbf{f}_{\mathrm{nc}} \equiv \mathbf{0}$*, the total entropy production rate is related to the Gibbs–Bogoliubov–Feynman variational free energy* $\mathcal{F}[\rho_t; \boldsymbol{\lambda}_t] \equiv \langle E(\mathbf{x}_t, \boldsymbol{\lambda}_t) \rangle - T S[\rho_t]$ *by the non-equilibrium free-energy balance:*

$$T \dot{\Sigma}_{\mathrm{tot}}(t) \;=\; \left\langle \dot{\mathcal{W}}(t) \right\rangle - \frac{\mathrm{d}}{\mathrm{d}t} \mathcal{F}[\rho_t; \boldsymbol{\lambda}_t]. \tag{8.48}$$

Proof. First, take the expectation of $-k_{\mathrm{B}} \frac{\partial_t \rho(\mathbf{x}_t, t)}{\rho(\mathbf{x}_t, t)}$ over $\rho(\mathbf{x}, t)$:

$$\left\langle -k_{\mathrm{B}}\, \frac{\partial_t \rho(\mathbf{x}_t, t)}{\rho(\mathbf{x}_t, t)} \right\rangle \;=\; -k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}, t)\, \frac{\partial_t \rho(\mathbf{x}, t)}{\rho(\mathbf{x}, t)} \;=\; -k_{\mathrm{B}}\, \frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x}, t) \;=\; -k_{\mathrm{B}}\, \frac{\mathrm{d}}{\mathrm{d}t}(1) \;=\; 0, \tag{8.49}$$

by normalization of probability. Similarly, in the Itô drift of Eq. (8.43), note the vector calculus identity

$$\begin{aligned}\rho(\mathbf{x},t)\Big[\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t)\cdot\nabla_{\mathbf{x}}\ln\rho(\mathbf{x},t)+\nabla_{\mathbf{x}}\cdot\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t)\Big] &= \mathbf{v}_{\mathrm{loc}}\cdot\nabla_{\mathbf{x}}\rho+\rho\,\nabla_{\mathbf{x}}\cdot\mathbf{v}_{\mathrm{loc}}\\ &= \nabla_{\mathbf{x}}\cdot\big(\rho(\mathbf{x},t)\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t)\big) \;=\; \nabla_{\mathbf{x}}\cdot\mathbf{J}(\mathbf{x},t).\end{aligned}$$

Integrating this divergence over $\mathbb{R}^N$ gives zero by the divergence theorem since $\mathbf{J}(\mathbf{x},t)\to\mathbf{0}$ as $\|\mathbf{x}\|\to\infty$. And because the Itô stochastic integral in Eq. (8.43) has zero expectation, only the first quadratic term in Eq. (8.43) survives the ensemble average:

$$\dot{\Sigma}_{\mathrm{tot}}(t) \;=\; k_{\mathrm{B}}\int_{\mathbb{R}^N}\mathrm{d}^N x\,\rho(\mathbf{x},t)\,\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t)^{\top}\mathbf{D}^{-1}\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t) \;=\; k_{\mathrm{B}}\int_{\mathbb{R}^N}\mathrm{d}^N x\,\frac{\mathbf{J}(\mathbf{x},t)^{\top}\mathbf{D}^{-1}\mathbf{J}(\mathbf{x},t)}{\rho(\mathbf{x},t)} \;\geq\; 0,$$

where non-negativity follows because $\mathbf{D}^{-1}=(k_{\mathrm{B}}T\boldsymbol{\mu})^{-1}\succ\mathbf{0}$ is strictly positive definite and $\rho(\mathbf{x},t)>0$.

Finally, when $\mathbf{f}_{\mathrm{nc}}\equiv\mathbf{0}$, multiplying $\dot{\Sigma}_{\mathrm{tot}}(t)=\dot{S}(t)+\big\langle\dot{\mathcal{Q}}(t)\big\rangle/T$ by T and substituting $\big\langle\dot{\mathcal{Q}}(t)\big\rangle=\big\langle\dot{\mathcal{W}}(t)\big\rangle-\frac{\mathrm{d}}{\mathrm{d}t}\langle E\rangle$ from Theorem 8.2 yields

$$\begin{aligned}T\dot{\Sigma}_{\mathrm{tot}}(t) &= T\dot{S}(t)+\big\langle\dot{\mathcal{W}}(t)\big\rangle-\frac{\mathrm{d}}{\mathrm{d}t}\langle E(\mathbf{x}_t,\boldsymbol{\lambda}_t)\rangle\\ &= \big\langle\dot{\mathcal{W}}(t)\big\rangle-\frac{\mathrm{d}}{\mathrm{d}t}\Big(\langle E(\mathbf{x}_t,\boldsymbol{\lambda}_t)\rangle-TS[\rho_t]\Big)\\ &= \big\langle\dot{\mathcal{W}}(t)\big\rangle-\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t;\boldsymbol{\lambda}_t].\end{aligned}$$

In the special case of a static protocol ($\dot{\boldsymbol{\lambda}}_t=\mathbf{0}$), $\big\langle\dot{\mathcal{W}}(t)\big\rangle=0$ and Eq. (8.48) reduces to $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t]=-T\dot{\Sigma}_{\mathrm{tot}}(t)\leq 0$, recovering the Fokker–Planck H-theorem proved in Chapter 6. □

8.3 Onsager–Machlup and MSRJD Path Integrals

So far we have used differential stochastic calculus. To prove fluctuation theorems (Chapter 9), to describe most probable escape paths such as the instantons of Chapter 7, or to compare the probability of a whole forward trajectory $\mathbf{x}_{[0,\tau]}$ with that of its time-reversed twin $\tilde{\mathbf{x}}_{[0,\tau]}$, we need an explicit probability density functional $\mathcal{P}[\mathbf{x}_{[0,\tau]}\mid\mathbf{x}_0]$ on path space. It is given by the Onsager–Machlup path integral [81] and by its field-theoretic counterpart, the Martin–Siggia–Rose–Janssen–De Dominicis (MSRJD) response-field formalism [71].

8.3.1 Time-Slicing the Overdamped Langevin Equation and the Discretization Parameter α

Consider the N-dimensional overdamped Langevin equation with general smooth drift $\mathbf{a}(\mathbf{x}_t,t)\in\mathbb{R}^N$ and constant positive-definite diffusion tensor $\mathbf{D}=\frac{1}{2}\mathbf{B}\mathbf{B}^{\top}\succ\mathbf{0}$:

$$\mathrm{d}\mathbf{x}_t \;=\; \mathbf{a}(\mathbf{x}_t,t)\,\mathrm{d}t+\mathbf{B}\,\mathrm{d}\mathbf{W}_t,\qquad t\in[0,\tau], \tag{8.50}$$

starting from a prescribed initial state $\mathbf{x}_0$ at $t=0$. Because $\mathbf{B}$ is constant, the SDE Eq. (8.50) has no multiplicative-noise ambiguity in the noise term itself. Nevertheless, when we construct a path integral by slicing the time interval $[0,\tau]$ into K small subintervals of duration $\Delta t=\tau/K$ with grid

points $t_k = k\Delta t$ $(k = 0, 1, \ldots, K)$ and states $\mathbf{x}_k \equiv \mathbf{x}_{t_k}$, we must choose *at which point inside the k-th step* $[\mathbf{x}_{k-1}, \mathbf{x}_k]$ *to evaluate the nonlinear drift field* $\mathbf{a}(\mathbf{x}, t)$.

Let $\alpha \in [0, 1]$ be a fixed discretization parameter, and define the α-interpolated coordinate and time on the k-th slice $[t_{k-1}, t_k]$ by

$$\bar{\mathbf{x}}_k^{(\alpha)} \;\equiv\; (1-\alpha)\mathbf{x}_{k-1} + \alpha\mathbf{x}_k, \qquad \bar{t}_k^{(\alpha)} \;\equiv\; (1-\alpha)t_{k-1} + \alpha t_k. \tag{8.51}$$

The three standard choices of α are:

- $\alpha = 0$ (**Itô / pre-point discretization**): $\bar{\mathbf{x}}_k^{(0)} = \mathbf{x}_{k-1}$, where the drift is evaluated exclusively at the start of the time step;
- $\alpha = 1/2$ (**Stratonovich / midpoint discretization**): $\bar{\mathbf{x}}_k^{(1/2)} = \frac{1}{2}(\mathbf{x}_{k-1} + \mathbf{x}_k)$, which preserves time-reversal symmetry between $\mathbf{x}_{k-1}$ and $\mathbf{x}_k$;
- $\alpha = 1$ (**Anti-Itô / post-point discretization**): $\bar{\mathbf{x}}_k^{(1)} = \mathbf{x}_k$.

In the α-discretization scheme, the discrete evolution equation over the k-th slice is

$$\Delta\mathbf{x}_k \;\equiv\; \mathbf{x}_k - \mathbf{x}_{k-1} \;=\; \mathbf{a}\big(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)}\big)\,\Delta t + \mathbf{B}\,\Delta\mathbf{W}_k, \qquad k = 1, 2, \ldots, K, \tag{8.52}$$

where the Wiener increments $\Delta\mathbf{W}_k \equiv \mathbf{W}_{t_k} - \mathbf{W}_{t_{k-1}} \in \mathbb{R}^N$ are mutually independent Gaussian random vectors distributed as $\mathcal{N}(\mathbf{0}, \Delta t\,\mathbf{I}_N)$. Consequently, the joint probability density of the K noise increments $(\Delta\mathbf{W}_1, \ldots, \Delta\mathbf{W}_K)$ is the exact product of K Gaussian densities:

$$P_W(\Delta\mathbf{W}_1, \ldots, \Delta\mathbf{W}_K) \prod_{k=1}^{K} \mathrm{d}^N(\Delta\mathbf{W}_k) \;=\; \prod_{k=1}^{K} \left[\frac{1}{(2\pi\Delta t)^{N/2}} \exp\!\left(-\frac{\|\Delta\mathbf{W}_k\|^2}{2\Delta t} \right) \mathrm{d}^N(\Delta\mathbf{W}_k) \right]. \tag{8.53}$$

Heuristic Derivation: Why Changing Variables from Noise $\Delta\mathbf{W}_k$ to Coordinates $\mathbf{x}_k$ Produces a Jacobian

A common pitfall in heuristic path-integral derivations is to solve Eq. (8.52) for $\mathbf{B}\Delta\mathbf{W}_k = \Delta\mathbf{x}_k - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})\,\Delta t$, plug this directly into the Gaussian exponent Eq. (8.53), and replace $\dot{\mathbf{x}}_t = \Delta\mathbf{x}_k/\Delta t$ without computing the change-of-variables Jacobian determinant $|\det(\partial\Delta\mathbf{W}_k/\partial\mathbf{x}_j)|$.

This fails for $\alpha > 0$. In Eq. (8.52) with $\alpha > 0$, the target coordinate $\mathbf{x}_k$ at the end of the step appears *inside* the argument of the drift $\mathbf{a}((1-\alpha)\mathbf{x}_{k-1} + \alpha\mathbf{x}_k, \bar{t}_k^{(\alpha)})$. Differentiating $\Delta\mathbf{W}_k = \mathbf{B}^{-1}[\mathbf{x}_k - \mathbf{x}_{k-1} - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)})\Delta t]$ with respect to $\mathbf{x}_k$ produces

$$\frac{\partial\Delta\mathbf{W}_k}{\partial\mathbf{x}_k} \;=\; \mathbf{B}^{-1}\Big[\mathbf{I}_N - \alpha\,\Delta t\,\nabla_{\mathbf{x}}\mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)})\Big].$$

Taking the determinant over all K time slices multiplies $K = \tau/\Delta t$ factors of the form $1 - \alpha\,\Delta t\,\nabla_{\mathbf{x}}\cdot\mathbf{a}$. In the continuum limit $K \to \infty$, $(1 - \alpha\,\frac{\tau}{K}\nabla\cdot\mathbf{a})^K$ does *not* converge to 1; it converges to the exponential $\exp(-\alpha\int_0^\tau \mathrm{d}t\,\nabla_{\mathbf{x}}\cdot\mathbf{a}(\mathbf{x}_t, t))$. Conversely, $\alpha = 0$ (Itô) makes this Jacobian equal to 1, but then expanding the squared exponent $\|\Delta\mathbf{x}_k - \mathbf{a}\Delta t\|^2$ produces the Itô cross-term $-\frac{1}{2}\int \mathbf{a}^\top\mathbf{D}^{-1}\cdot\mathrm{d}\mathbf{x}_t$, which equals the time-symmetric Stratonovich cross-term $-\frac{1}{2}\int \mathbf{a}^\top\mathbf{D}^{-1}\circ\mathrm{d}\mathbf{x}_t$ plus $\frac{1}{2}\int \mathrm{d}t\,\nabla\cdot\mathbf{a}$. As the theorem below shows, the continuum path measure does not depend on α, provided the Jacobian and the cross-term are treated consistently.

8.3.2 Derivation of the Midpoint Jacobian and the Onsager–Machlup Action

We now change variables from the noise increments $(\Delta\mathbf{W}_1, \dots, \Delta\mathbf{W}_K)$ to the trajectory coordinates $(\mathbf{x}_1, \dots, \mathbf{x}_K)$, conditional on $\mathbf{x}_0$.

Theorem 8.7 (Onsager–Machlup Path-Integral Measure)**.** *Let $\mathbf{x}_t \in \mathbb{R}^N$ obey the overdamped Langevin SDE $\mathrm{d}\mathbf{x}_t = \mathbf{a}(\mathbf{x}_t, t)\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t$ with $\mathbf{a} \in C^{2,1}(\mathbb{R}^N \times [0,\tau]; \mathbb{R}^N)$ and constant nonsingular $\mathbf{B} \in \mathbb{R}^{N\times N}$ ($\mathbf{D} = \frac{1}{2}\mathbf{B}\mathbf{B}^\top \succ \mathbf{0}$). Then:*

1. *For any discretization parameter $\alpha \in [0,1]$, the time-sliced conditional path probability density for $(\mathbf{x}_1, \dots, \mathbf{x}_K)$ given $\mathbf{x}_0$ is*

$$P_K(\mathbf{x}_1, \dots, \mathbf{x}_K \mid \mathbf{x}_0) \;=\; \frac{1}{\left[(4\pi\Delta t)^N \det \mathbf{D}\right]^{K/2}} \exp\Big(-\mathcal{S}_{\mathrm{OM}}^{(K,\alpha)}[\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_K]\Big), \tag{8.54}$$

   where the discrete α-discretized Onsager–Machlup action up to errors vanishing as $K \to \infty$ is

$$\mathcal{S}_{\mathrm{OM}}^{(K,\alpha)} \;=\; \sum_{k=1}^{K} \Delta t \left[\frac{1}{4}\left(\frac{\Delta\mathbf{x}_k}{\Delta t} - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})\right)^\top \mathbf{D}^{-1} \left(\frac{\Delta\mathbf{x}_k}{\Delta t} - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})\right) + \alpha\, \nabla_{\mathbf{x}} \cdot \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})\right]. \tag{8.55}$$

2. *In the continuum limit $K \to \infty$ ($\Delta t \to 0$), written in the Stratonovich midpoint convention ($\alpha = 1/2$, so that ordinary chain-rule and time-reversal symmetries hold for the velocity $\dot{\mathbf{x}}_t$), the formal path-space measure is*

$$\mathcal{D}\mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0] \;=\; \mathcal{D}\mathbf{x}\, \exp\Big(-\mathcal{S}_{\mathrm{OM}}[\mathbf{x}_{[0,\tau]}]\Big), \tag{8.56}$$

   *where $\mathcal{D}\mathbf{x} \equiv \lim_{K\to\infty} \prod_{k=1}^{K} \frac{\mathrm{d}^N\mathbf{x}_k}{[(4\pi\Delta t)^N \det \mathbf{D}]^{1/2}}$ and the **continuum Onsager–Machlup action functional** $\mathcal{S}_{\mathrm{OM}}[\mathbf{x}_{[0,\tau]}]$ is*

$$\mathcal{S}_{\mathrm{OM}}[\mathbf{x}_{[0,\tau]}] \;=\; \int_0^\tau \mathrm{d}t \left[\frac{1}{4}(\dot{\mathbf{x}}_t - \mathbf{a}(\mathbf{x}_t,t))^\top \mathbf{D}^{-1} (\dot{\mathbf{x}}_t - \mathbf{a}(\mathbf{x}_t,t)) + \frac{1}{2}\nabla_{\mathbf{x}} \cdot \mathbf{a}(\mathbf{x}_t,t)\right], \tag{8.57}$$

   in which the cross-term $-\frac{1}{2}\int_0^\tau \mathbf{a}(\mathbf{x}_t,t)^\top \mathbf{D}^{-1} \circ \mathrm{d}\mathbf{x}_t$ is a Stratonovich stochastic integral.

Proof. From the discrete evolution equation Eq. (8.52), we express the k-th Wiener increment $\Delta\mathbf{W}_k$ as a function of the trajectory coordinates $(\mathbf{x}_1, \dots, \mathbf{x}_K)$:

$$\Delta\mathbf{W}_k(\mathbf{x}_1, \dots, \mathbf{x}_K) \;=\; \mathbf{B}^{-1}\Big[\mathbf{x}_k - \mathbf{x}_{k-1} - \mathbf{a}((1-\alpha)\mathbf{x}_{k-1} + \alpha\mathbf{x}_k, \bar{t}_k^{(\alpha)})\,\Delta t\Big]. \tag{8.58}$$

The increment $\Delta\mathbf{W}_k$ depends only on $\mathbf{x}_k$ and $\mathbf{x}_{k-1}$, not on $\mathbf{x}_j$ for $j > k$. Therefore the $NK \times NK$ Jacobian matrix $\mathcal{J}$ with $N \times N$ blocks $\mathcal{J}_{kj} \equiv \frac{\partial\Delta\mathbf{W}_k}{\partial\mathbf{x}_j}$ $(k, j \in \{1, \dots, K\})$ is block lower-triangular:

$$\mathcal{J}_{kj} \;=\; \frac{\partial\Delta\mathbf{W}_k}{\partial\mathbf{x}_j} \;=\; \mathbf{0} \qquad \text{for all } j > k.$$

The determinant of a block lower-triangular matrix is simply the product of the determinants of its diagonal blocks $\mathcal{J}_{kk}$:

$$\det \mathcal{J} \;=\; \prod_{k=1}^{K} \det\left(\frac{\partial\Delta\mathbf{W}_k}{\partial\mathbf{x}_k}\right). \tag{8.59}$$

The k-th diagonal block $\mathcal{J}_{kk} = \frac{\partial \Delta \mathbf{W}_k}{\partial \mathbf{x}_k}$ follows from differentiating Eq. (8.58) with respect to $x_{k,m}$:

$$\frac{\partial \Delta \mathbf{W}_k}{\partial \mathbf{x}_k} \;=\; \mathbf{B}^{-1}\Big[\mathbf{I}_N - \alpha\,\Delta t\,\mathbf{M}_k\Big], \qquad \text{where} \qquad (\mathbf{M}_k)_{lm} \;\equiv\; \frac{\partial a_l(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})}{\partial x_m}. \tag{8.60}$$

Taking the determinant of Eq. (8.60) and using $\det(\mathbf{B}^{-1}) = (\det \mathbf{B})^{-1} = (\det(2\mathbf{D}))^{-1/2} = (2^N \det \mathbf{D})^{-1/2}$, we have

$$\left|\det\left(\frac{\partial \Delta \mathbf{W}_k}{\partial \mathbf{x}_k}\right)\right| \;=\; \frac{1}{(2^N \det \mathbf{D})^{1/2}}\,\left|\det(\mathbf{I}_N - \alpha\,\Delta t\,\mathbf{M}_k)\right|. \tag{8.61}$$

For sufficiently small Δt, all eigenvalues of $\alpha\,\Delta t\,\mathbf{M}_k$ have magnitude strictly less than 1. We can therefore apply the matrix identity $\det(\mathbf{I}_N - \epsilon\mathbf{M}_k) = \exp(\operatorname{tr}\ln(\mathbf{I}_N - \epsilon\mathbf{M}_k))$ and expand the matrix logarithm in a Taylor series in Δt:

$$\begin{aligned}\det(\mathbf{I}_N - \alpha\,\Delta t\,\mathbf{M}_k) \;&=\; \exp\Big(\operatorname{tr}\ln(\mathbf{I}_N - \alpha\,\Delta t\,\mathbf{M}_k)\Big) \\ &=\; \exp\left(-\alpha\,\Delta t\;\operatorname{tr}(\mathbf{M}_k) - \frac{1}{2}\alpha^2(\Delta t)^2\;\operatorname{tr}(\mathbf{M}_k^2) + \mathcal{O}((\Delta t)^3)\right).\end{aligned} \tag{8.62}$$

The trace of $\mathbf{M}_k$ is the divergence of the drift:

$$\operatorname{tr}(\mathbf{M}_k) \;=\; \sum_{l=1}^{N} \frac{\partial a_l(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})}{\partial x_l} \;=\; \nabla_{\mathbf{x}} \cdot \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)}).$$

When we take the product over $k = 1, \ldots, K$ in Eq. (8.59), the $\mathcal{O}(\Delta t)$ terms sum to a Riemann sum of order $K \cdot \mathcal{O}(\Delta t) = \mathcal{O}(\tau)$, whereas the $\mathcal{O}((\Delta t)^2)$ terms sum to $K \cdot \mathcal{O}((\Delta t)^2) = \mathcal{O}(\tau\,\Delta t) \to 0$ as $K \to \infty$. Thus, up to multiplicative factors of $1 + \mathcal{O}(\Delta t)$ that vanish in the continuum limit, the total Jacobian determinant is

$$|\det \mathcal{J}| \;=\; \frac{1}{\left(2^N \det \mathbf{D}\right)^{K/2}} \exp\left(-\alpha \sum_{k=1}^{K} \Delta t\, \nabla_{\mathbf{x}} \cdot \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)}) + \mathcal{O}(\Delta t)\right). \tag{8.63}$$

Next, we substitute $\Delta \mathbf{W}_k$ from Eq. (8.58) into the Gaussian exponent of Eq. (8.53). Using $(\mathbf{B}^{-1})^\top \mathbf{B}^{-1} = (\mathbf{B}\mathbf{B}^\top)^{-1} = \frac{1}{2}\mathbf{D}^{-1}$, we find

$$\begin{aligned}\frac{\|\Delta \mathbf{W}_k\|^2}{2\Delta t} \;&=\; \frac{1}{2\Delta t}(\Delta \mathbf{x}_k - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})\,\Delta t)^\top (\mathbf{B}\mathbf{B}^\top)^{-1}(\Delta \mathbf{x}_k - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})\,\Delta t) \\ &=\; \frac{\Delta t}{4}\left(\frac{\Delta \mathbf{x}_k}{\Delta t} - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})\right)^\top \mathbf{D}^{-1}\left(\frac{\Delta \mathbf{x}_k}{\Delta t} - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})\right).\end{aligned} \tag{8.64}$$

Multiplying the Gaussian prefactor $(2\pi\Delta t)^{-NK/2}$ from Eq. (8.53) by the Jacobian prefactor $(2^N \det \mathbf{D})^{-K/2}$ from Eq. (8.63) gives $[(4\pi\Delta t)^N \det \mathbf{D}]^{-K/2}$, and combining the exponents Eqs. (8.63) and (8.64) yields Eqs. (8.54) and (8.55). Setting $\alpha = 1/2$ and taking $K \to \infty$ immediately gives Eqs. (8.56) and (8.57).

Finally, we verify that the continuum measure Eq. (8.54) is independent of the choice of $\alpha \in [0, 1]$. Expanding the quadratic form in Eq. (8.55), the only α-dependent terms are the cross-term and the Jacobian term:

$$-\frac{1}{2}\sum_{k=1}^{K} \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})^\top \mathbf{D}^{-1} \Delta \mathbf{x}_k + \alpha \sum_{k=1}^{K} \Delta t\, \nabla_{\mathbf{x}} \cdot \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)}). \tag{8.65}$$

Taylor-expanding $\mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}) = \mathbf{a}(\mathbf{x}_{k-1} + \alpha\Delta\mathbf{x}_k)$ around the midpoint $\bar{\mathbf{x}}_k^{(1/2)} = \mathbf{x}_{k-1} + \frac{1}{2}\Delta\mathbf{x}_k$ gives

$$\mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}) \;=\; \mathbf{a}(\bar{\mathbf{x}}_k^{(1/2)}) + \left(\alpha - \frac{1}{2}\right)(\Delta\mathbf{x}_k \cdot \nabla_{\mathbf{x}})\mathbf{a}(\bar{\mathbf{x}}_k^{(1/2)}) + \mathcal{O}(\|\Delta\mathbf{x}_k\|^2).$$

Substituting this into the first sum of Eq. (8.65) and replacing $\Delta x_{k,i}\Delta x_{k,j}$ by its leading quadratic variation $2D_{ij}\Delta t$, we obtain

$$\begin{aligned} -\frac{1}{2}\sum_{k=1}^{K}\mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)})^{\top}\mathbf{D}^{-1}\Delta\mathbf{x}_k \;&=\; -\frac{1}{2}\sum_{k=1}^{K}\mathbf{a}(\bar{\mathbf{x}}_k^{(1/2)})^{\top}\mathbf{D}^{-1}\Delta\mathbf{x}_k - \frac{1}{2}\left(\alpha-\frac{1}{2}\right)\sum_{k=1}^{K}\sum_{i,j,l=1}^{N}\frac{\partial a_i}{\partial x_l}(D^{-1})_{ij}\underbrace{\Delta x_{k,l}\Delta x_{k,j}}_{\to 2D_{lj}\Delta t} \\ &=\; -\frac{1}{2}\int_0^{\tau}\mathbf{a}(\mathbf{x}_t,t)^{\top}\mathbf{D}^{-1}\circ\mathrm{d}\mathbf{x}_t - \left(\alpha-\frac{1}{2}\right)\int_0^{\tau}\mathrm{d}t\,\nabla_{\mathbf{x}}\cdot\mathbf{a}(\mathbf{x}_t,t). \end{aligned} \tag{8.66}$$

Adding $+\alpha\int_0^{\tau}\mathrm{d}t\,\nabla_{\mathbf{x}}\cdot\mathbf{a}(\mathbf{x}_t,t)$ from the Jacobian in Eq. (8.65), the α dependence cancels identically: $-\left(\alpha-\frac{1}{2}\right)+\alpha = +\frac{1}{2}$, leaving the α-independent Onsager–Machlup action Eq. (8.57). □

8.3.3 The MSRJD Response-Field Path Integral

While the Onsager–Machlup action Eq. (8.57) is quadratic in the velocity $\dot{\mathbf{x}}_t$, it is often convenient in perturbation theory (for spin-glass dynamics, say, or the weakly nonlinear CMOS circuits of Chapter 13) to make the dependence on $\dot{\mathbf{x}}_t - \mathbf{a}(\mathbf{x}_t,t)$ linear by introducing an auxiliary **response field** $\hat{\mathbf{x}}_t \in \mathbb{R}^N$. This yields the MSRJD path integral [71].

To construct the MSRJD representation from the discrete α-sliced action Eq. (8.55), we apply the standard N-dimensional Gaussian Hubbard–Stratonovich (Fourier) identity at each time slice $k = 1,\dots,K$:

$$\exp\!\left(-\frac{\Delta t}{4}\,\mathbf{u}_k^{\top}\mathbf{D}^{-1}\mathbf{u}_k\right) \;=\; \left[(4\pi\Delta t)^N\det\mathbf{D}\right]^{1/2}\int_{\mathbb{R}^N}\frac{\mathrm{d}^N\hat{\mathbf{x}}_k}{(2\pi)^N}\,\exp\!\Big(-i\Delta t\,\hat{\mathbf{x}}_k^{\top}\mathbf{u}_k - \Delta t\,\hat{\mathbf{x}}_k^{\top}\mathbf{D}\hat{\mathbf{x}}_k\Big), \tag{8.67}$$

where $\mathbf{u}_k \equiv \frac{\Delta\mathbf{x}_k}{\Delta t} - \mathbf{a}(\bar{\mathbf{x}}_k^{(\alpha)}, \bar{t}_k^{(\alpha)})$. The prefactor $[(4\pi\Delta t)^N\det\mathbf{D}]^{1/2}$ in Eq. (8.67) cancels the divergent normalization in Eq. (8.54) slice by slice. Taking the continuum limit $K\to\infty$ gives the MSRJD generating functional:

$$\mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0] \;=\; \int\mathcal{D}\hat{\mathbf{x}}\,\exp\!\Big(-\mathcal{S}^{(\alpha)}_{\mathrm{MSRJD}}[\mathbf{x},\hat{\mathbf{x}}]\Big), \tag{8.68}$$

where $\mathcal{D}\hat{\mathbf{x}} \equiv \lim_{K\to\infty}\prod_{k=1}^{K}\frac{\mathrm{d}^N\hat{\mathbf{x}}_k}{(2\pi)^N}$ and the **MSRJD action** is

$$\mathcal{S}^{(\alpha)}_{\mathrm{MSRJD}}[\mathbf{x},\hat{\mathbf{x}}] \;=\; \int_0^{\tau}\mathrm{d}t\Big[i\hat{\mathbf{x}}_t^{\top}\left(\dot{\mathbf{x}}_t - \mathbf{a}(\mathbf{x}_t,t)\right) + \hat{\mathbf{x}}_t^{\top}\mathbf{D}\hat{\mathbf{x}}_t + \alpha\,\nabla_{\mathbf{x}}\cdot\mathbf{a}(\mathbf{x}_t,t)\Big]. \tag{8.69}$$

In the Itô convention ($\alpha = 0$), the divergence term $\alpha\nabla\cdot\mathbf{a}$ vanishes completely (eliminating determinant loops in Feynman diagrams), while in the Stratonovich convention ($\alpha = 1/2$), the midpoint Jacobian $+\frac{1}{2}\nabla\cdot\mathbf{a}$ is retained explicitly so that ordinary integration by parts applies to $\dot{\mathbf{x}}_t$.

8.3.4 State-Dependent Mobility and Multiplicative Thermal Noise

In nonlinear analog circuits, such as subthreshold MOSFET networks with voltage-dependent drain conductances $g_{\mathrm{ds}}(\mathbf{v})$ or varactor diodes with state-dependent capacitances (Chapters 4 and 5), the mobility tensor $\boldsymbol{\mu}(\mathbf{x})$ and the resulting diffusion tensor $\mathbf{D}(\mathbf{x}) = k_{\mathrm{B}}T\boldsymbol{\mu}(\mathbf{x}) = \frac{1}{2}\mathbf{B}(\mathbf{x})\mathbf{B}(\mathbf{x})^{\top}$ depend explicitly on the state coordinate $\mathbf{x}$. As shown in Chapter 5, to ensure that the Fokker–Planck

equation relaxes to the Boltzmann–Gibbs distribution $p_{\mathrm{eq}}(\mathbf{x}) \propto \exp(-\beta E(\mathbf{x}))$ in equilibrium, the Itô Langevin equation must include the thermal divergence drift $+\nabla_{\mathbf{x}} \cdot \mathbf{D}(\mathbf{x})$:

$$\mathrm{d}\mathbf{x}_t \;=\; \Big[\boldsymbol{\mu}(\mathbf{x}_t)\big(-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\big) + \nabla_{\mathbf{x}} \cdot \mathbf{D}(\mathbf{x}_t)\Big]\,\mathrm{d}t + \mathbf{B}(\mathbf{x}_t) \cdot \mathrm{d}\mathbf{W}_t, \tag{8.70}$$

where $(\nabla_{\mathbf{x}} \cdot \mathbf{D}(\mathbf{x}))_i \equiv \sum_{j=1}^{N} \partial_{x_j} D_{ij}(\mathbf{x})$. When constructing the time-sliced path integral for Eq. (8.70), the slice normalization factor in Eq. (8.54) becomes state-dependent, replacing $(\det \mathbf{D})^{K/2}$ with $\prod_{k=1}^{K} [\det \mathbf{D}(\bar{\mathbf{x}}_k^{(\alpha)})]^{1/2}$ inside the functional integration measure $\mathcal{D}\mathbf{x}$. Meanwhile, the local hydrodynamic velocity remains $\mathbf{v}_{\mathrm{loc}}(\mathbf{x}, t) = \mathbf{J}(\mathbf{x}, t)/\rho(\mathbf{x}, t)$, where

$$\mathbf{J}(\mathbf{x}, t) \;=\; \boldsymbol{\mu}(\mathbf{x})\big[-\nabla_{\mathbf{x}} E(\mathbf{x}, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}, t)\big]\rho(\mathbf{x}, t) - \mathbf{D}(\mathbf{x})\nabla_{\mathbf{x}}\rho(\mathbf{x}, t). \tag{8.71}$$

Because the divergence drift $+\nabla_{\mathbf{x}} \cdot \mathbf{D}(\mathbf{x})$ is precisely what converts the second-order Fokker–Planck diffusion term $\sum_{i,j} \partial_{x_i}\partial_{x_j}(D_{ij}\rho)$ into the divergence of the Fickian flux $-\nabla_{\mathbf{x}} \cdot (\mathbf{D}(\mathbf{x})\nabla_{\mathbf{x}}\rho)$, both the trajectory First Law (Theorem 8.2) and the ensemble Second Law (Theorem 8.6) with $\mathbf{D}(\mathbf{x})^{-1}$ inside the integral Eq. (8.47) remain valid without modification.

8.4 Microscopic Irreversibility and the Crooks Path-Measure Ratio

We now connect the thermodynamic entropy production $\Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]$ defined in Section 8.2 to the path-integral action $\mathcal{S}_{\mathrm{OM}}[\mathbf{x}_{[0,\tau]}]$ derived in Section 8.3. Specifically, we ask: if a movie of a stochastic trajectory $\mathbf{x}_{[0,\tau]}$ is played backward in time while the external control protocol $\boldsymbol{\lambda}_t$ is also reversed, how much more (or less) probable is the forward trajectory than its time-reversed counterpart?

8.4.1 Forward vs. Time-Reversed Driving Protocols and Conjugate Trajectories

To compare forward and backward evolutions precisely, we define a pair of conjugate physical experiments over the time interval $t \in [0, \tau]$:

1. **The Forward Process (F):** The system starts at $t = 0$ from an initial probability density $\rho_0(\mathbf{x}_0)$ and evolves from $t = 0$ to $t = \tau$ under the forward control protocol

$$\boldsymbol{\lambda}_t^F \;\equiv\; \boldsymbol{\lambda}_t, \qquad \mathbf{a}_F(\mathbf{x}, t) \;=\; \boldsymbol{\mu}\big[-\nabla_{\mathbf{x}} E(\mathbf{x}, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}, t)\big], \tag{8.72}$$

   generating a forward trajectory $\mathbf{x}_{[0,\tau]} = \{\mathbf{x}_t\}_{0 \le t \le \tau}$ that terminates at $\mathbf{x}_\tau$, where the ensemble probability density at time $t = \tau$ is $\rho_\tau(\mathbf{x}_\tau)$.

2. **The Time-Reversed Process (R):** In the conjugate reverse experiment, the external hardware controller executes the **time-reversed protocol**

$$\tilde{\boldsymbol{\lambda}}_t \;\equiv\; \boldsymbol{\lambda}_{\tau - t}, \qquad t \in [0, \tau], \tag{8.73}$$

   so that $\tilde{\boldsymbol{\lambda}}_0 = \boldsymbol{\lambda}_\tau$ and $\tilde{\boldsymbol{\lambda}}_\tau = \boldsymbol{\lambda}_0$. Assuming the non-conservative force $\mathbf{f}_{\mathrm{nc}}$ is even under time reversal (such as an electrostatic transconductance force; for magnetic or Lorentz forces that are odd under time reversal, one also flips the magnetic field $\mathbf{B}_{\mathrm{mag}} \to -\mathbf{B}_{\mathrm{mag}}$), the drift field at time t in the reverse experiment is

$$\tilde{\mathbf{a}}(\mathbf{x}, t) \;\equiv\; \mathbf{a}_F(\mathbf{x}, \tau - t) \;=\; \boldsymbol{\mu}\big[-\nabla_{\mathbf{x}} E(\mathbf{x}, \boldsymbol{\lambda}_{\tau - t}) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}, \tau - t)\big]. \tag{8.74}$$

   For any given forward trajectory $\mathbf{x}_{[0,\tau]}$, we define its **conjugate time-reversed trajectory** $\tilde{\mathbf{x}}_{[0,\tau]} = \{\tilde{\mathbf{x}}_t\}_{0 \le t \le \tau}$ by

$$\tilde{\mathbf{x}}_t \;\equiv\; \mathbf{x}_{\tau - t}, \qquad t \in [0, \tau]. \tag{8.75}$$

The reverse trajectory starts at $\tilde{\mathbf{x}}_0 = \mathbf{x}_\tau$, ends at $\tilde{\mathbf{x}}_\tau = \mathbf{x}_0$, and has instantaneous velocity opposite to that of the forward trajectory at the corresponding instant $u = \tau - t$:

$$\dot{\tilde{\mathbf{x}}}_t \;=\; \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{x}_{\tau-t}) \;=\; -\dot{\mathbf{x}}_{\tau-t}\big|_{u=\tau-t}. \tag{8.76}$$

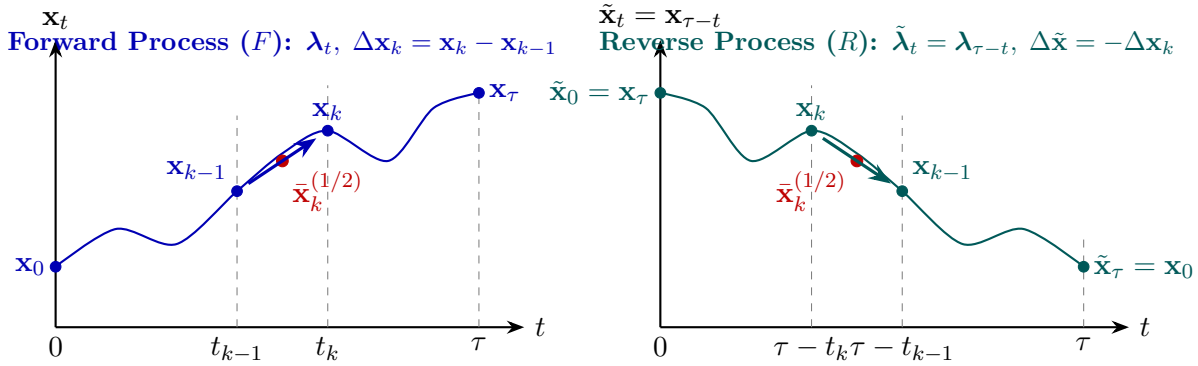


Figure 8.1: Geometric symmetry of the Stratonovich midpoint discretization ($\alpha = 1/2$) under time reversal. On the k-th forward time slice $[t_{k-1}, t_k]$ (left), the midpoint state $\bar{\mathbf{x}}_k^{(1/2)} = \frac{1}{2}(\mathbf{x}_{k-1} + \mathbf{x}_k)$ coincides identically with the midpoint state on the conjugate reverse time slice $[\tau - t_k, \tau - t_{k-1}]$ (right), while the coordinate increment reverses sign ($\Delta\tilde{\mathbf{x}}_{K-k+1} = \mathbf{x}_{k-1} - \mathbf{x}_k = -\Delta\mathbf{x}_k$).

8.4.2 Proof of the Crooks Path-Measure Time-Reversal Identity

Figure 8.1 shows why the midpoint discretization ($\alpha = 1/2$) of Theorem 8.7 is the natural one for comparing forward and time-reversed trajectories. The midpoint $\bar{\mathbf{x}}_k^{(1/2)} = \frac{1}{2}(\mathbf{x}_{k-1} + \mathbf{x}_k)$ and the midpoint time $\tau - \bar{t}_{K-k+1}^{(1/2)} = \bar{t}_k^{(1/2)}$ are invariant under time reversal, whereas a pre-point ($\alpha = 0$) discretization of the forward path becomes a post-point ($\alpha = 1$) discretization of the reversed path. Using this symmetry, we now prove Crooks's relation [22] between microscopic irreversibility and entropy production.

Theorem 8.8 (Crooks Path-Measure Ratio and Microscopic Irreversibility)**.** *Let $\mathbf{x}_{[0,\tau]}$ be a trajectory of the overdamped Langevin equation Eq. (8.2) under the forward protocol $\boldsymbol{\lambda}_t$ starting from $\mathbf{x}_0 \sim \rho_0(\mathbf{x}_0)$, and let $\tilde{\mathbf{x}}_{[0,\tau]}$ be the conjugate time-reversed trajectory Eq. (8.75) under the time-reversed protocol $\tilde{\boldsymbol{\lambda}}_t = \boldsymbol{\lambda}_{\tau-t}$.*

1. ***Conditional Path-Measure Ratio (Medium Entropy Production):*** *The ratio of the conditional forward path probability density $\mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0]$ to the conditional reverse path probability density $\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]} \mid \tilde{\mathbf{x}}_0]$ satisfies*

$$k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0]}{\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]} \mid \tilde{\mathbf{x}}_0]} \;=\; k_{\mathrm{B}}\Big(\tilde{\mathcal{S}}_{\mathrm{OM}}[\tilde{\mathbf{x}}_{[0,\tau]}] - \mathcal{S}_{\mathrm{OM}}[\mathbf{x}_{[0,\tau]}]\Big) \;=\; \frac{\mathcal{Q}[\mathbf{x}_{[0,\tau]}]}{T} \;=\; \Delta s_{\mathrm{med}}[\mathbf{x}_{[0,\tau]}]. \tag{8.77}$$

2. ***Unconditional Path-Measure Ratio (Total Entropy Production):*** *Let $\mathcal{P}[\mathbf{x}_{[0,\tau]}] \equiv \rho_0(\mathbf{x}_0)\,\mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0]$ be the full path measure of the forward trajectory, and let $\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]}] \equiv$*

$\rho_\tau(\tilde{\mathbf{x}}_0)\,\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]} \mid \tilde{\mathbf{x}}_0]$ be the full path measure of the reverse trajectory initialized from the final forward ensemble density $\rho_\tau(\mathbf{x}_\tau)$. Then

$$k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]}]}{\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]}]} \;=\; k_{\mathrm{B}} \ln \frac{\rho_0(\mathbf{x}_0)}{\rho_\tau(\mathbf{x}_\tau)} + \frac{\mathcal{Q}[\mathbf{x}_{[0,\tau]}]}{T} \;=\; \Delta s_{\mathrm{sys}}[\mathbf{x}_{[0,\tau]}] + \Delta s_{\mathrm{med}}[\mathbf{x}_{[0,\tau]}] \;=\; \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]. \tag{8.78}$$

Proof. We work first on the time-sliced grid (K slices, $\alpha = 1/2$) and then take the continuum limit $K \to \infty$.

Step 1 (Discrete Midpoint Action Comparison): On the K-slice grid with $\alpha = 1/2$, the forward trajectory visits the sequence of states $(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_K)$ with midpoint states $\bar{\mathbf{x}}_k = \frac{1}{2}(\mathbf{x}_{k-1} + \mathbf{x}_k)$, midpoint times $\bar{t}_k = (k - \frac{1}{2})\Delta t$, and increments $\Delta\mathbf{x}_k = \mathbf{x}_k - \mathbf{x}_{k-1}$. The reverse trajectory visits the states $\tilde{\mathbf{x}}_j = \mathbf{x}_{K-j}$ $(j = 0, 1, \dots, K)$. Under the re-indexing $k = K - j + 1$, the j-th reverse time slice corresponds to the k-th forward time slice traversed in reverse:

$$\bar{\tilde{\mathbf{x}}}_j \;=\; \frac{1}{2}(\tilde{\mathbf{x}}_{j-1} + \tilde{\mathbf{x}}_j) \;=\; \frac{1}{2}(\mathbf{x}_k + \mathbf{x}_{k-1}) \;=\; \bar{\mathbf{x}}_k, \tag{8.79}$$

$$\tilde{\mathbf{a}}(\bar{\tilde{\mathbf{x}}}_j, \bar{t}_j) \;=\; \mathbf{a}_F(\bar{\mathbf{x}}_k, \tau - \bar{t}_j) \;=\; \mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k), \tag{8.80}$$

$$\Delta\tilde{\mathbf{x}}_j \;=\; \tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_{j-1} \;=\; \mathbf{x}_{k-1} - \mathbf{x}_k \;=\; -\Delta\mathbf{x}_k. \tag{8.81}$$

The slice normalization $[(4\pi\Delta t)^N \det \mathbf{D}]^{-K/2}$ in Eq. (8.54) is the same for the forward and reverse path densities and cancels in the ratio P_K^F / P_K^R. Substituting $\Delta\tilde{\mathbf{x}}_j = -\Delta\mathbf{x}_k$ into the discrete midpoint action Eq. (8.55) with $\alpha = 1/2$, we compare the forward action $\mathcal{S}_{\mathrm{OM}}^{(K,1/2)}$ and the reverse action $\tilde{\mathcal{S}}_{\mathrm{OM}}^{(K,1/2)}$ term by term:

$$\begin{aligned}\mathcal{S}_{\mathrm{OM}}^{(K,1/2)}[\mathbf{x}] \;=\; \sum_{k=1}^{K} \Delta t \Bigg[&\frac{1}{4}\left(\frac{\Delta\mathbf{x}_k}{\Delta t}\right)^{\!\top} \mathbf{D}^{-1} \left(\frac{\Delta\mathbf{x}_k}{\Delta t}\right) - \frac{1}{2}\mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k)^\top \mathbf{D}^{-1} \left(\frac{\Delta\mathbf{x}_k}{\Delta t}\right) \\ &+ \frac{1}{4}\mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k)^\top \mathbf{D}^{-1} \mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k) + \frac{1}{2}\nabla_{\mathbf{x}} \cdot \mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k) \Bigg],\end{aligned} \tag{8.82}$$

$$\begin{aligned}\tilde{\mathcal{S}}_{\mathrm{OM}}^{(K,1/2)}[\tilde{\mathbf{x}}] \;=\; \sum_{k=1}^{K} \Delta t \Bigg[&\frac{1}{4}\left(\frac{-\Delta\mathbf{x}_k}{\Delta t}\right)^{\!\top} \mathbf{D}^{-1} \left(\frac{-\Delta\mathbf{x}_k}{\Delta t}\right) - \frac{1}{2}\mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k)^\top \mathbf{D}^{-1} \left(\frac{-\Delta\mathbf{x}_k}{\Delta t}\right) \\ &+ \frac{1}{4}\mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k)^\top \mathbf{D}^{-1} \mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k) + \frac{1}{2}\nabla_{\mathbf{x}} \cdot \mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k) \Bigg].\end{aligned} \tag{8.83}$$

Consider the four terms in the brackets of Eqs. (8.82) and (8.83):

- The kinetic term $\frac{1}{4}(\frac{\Delta\mathbf{x}_k}{\Delta t})^\top \mathbf{D}^{-1}(\frac{\Delta\mathbf{x}_k}{\Delta t})$ is quadratic in $\Delta\mathbf{x}_k$, so it is even under $\Delta\mathbf{x}_k \to -\Delta\mathbf{x}_k$ and cancels in $\tilde{\mathcal{S}}_{\mathrm{OM}} - \mathcal{S}_{\mathrm{OM}}$.
- The squared-drift term $\frac{1}{4}\mathbf{a}_F^\top \mathbf{D}^{-1}\mathbf{a}_F$ depends only on the midpoint $\bar{\mathbf{x}}_k$, so it is even under time reversal and cancels in $\tilde{\mathcal{S}}_{\mathrm{OM}} - \mathcal{S}_{\mathrm{OM}}$.
- The midpoint Jacobian divergence term $+\frac{1}{2}\nabla_{\mathbf{x}} \cdot \mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k)$ also depends only on the midpoint $\bar{\mathbf{x}}_k$, so it is even under time reversal and cancels in $\tilde{\mathcal{S}}_{\mathrm{OM}} - \mathcal{S}_{\mathrm{OM}}$.
- Only the linear velocity-drift cross-term $-\frac{1}{2}\mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k)^\top \mathbf{D}^{-1}\Delta\mathbf{x}_k$ is odd under $\Delta\mathbf{x}_k \to -\Delta\mathbf{x}_k$, so subtracting Eq. (8.82) from Eq. (8.83) doubles this term:

$$\tilde{\mathcal{S}}_{\mathrm{OM}}^{(K,1/2)}[\tilde{\mathbf{x}}] - \mathcal{S}_{\mathrm{OM}}^{(K,1/2)}[\mathbf{x}] \;=\; \sum_{k=1}^{K} \mathbf{a}_F(\bar{\mathbf{x}}_k, \bar{t}_k)^\top \mathbf{D}^{-1}\,\Delta\mathbf{x}_k. \tag{8.84}$$

Step 2 (Continuum Limit and Identification with Dissipated Heat): Because $\bar{\mathbf{x}}_k = \frac{1}{2}(\mathbf{x}_{k-1} + \mathbf{x}_k)$ is the midpoint of the k-th slice, the $K \to \infty$ limit of the Riemann–Stieltjes sum in Eq. (8.84) is, by definition, the Stratonovich stochastic integral:

$$\ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0]}{\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]} \mid \tilde{\mathbf{x}}_0]} \;=\; \tilde{\mathcal{S}}_{\mathrm{OM}}[\tilde{\mathbf{x}}_{[0,\tau]}] - \mathcal{S}_{\mathrm{OM}}[\mathbf{x}_{[0,\tau]}] \;=\; \int_0^\tau \mathbf{a}_F(\mathbf{x}_t, t)^\top \mathbf{D}^{-1} \circ \mathrm{d}\mathbf{x}_t. \tag{8.85}$$

The Einstein relation $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$ of Eq. (8.4) implies $\mathbf{D}^{-1} = (k_{\mathrm{B}}T)^{-1}\boldsymbol{\mu}^{-1}$, while the forward drift is $\mathbf{a}_F(\mathbf{x}_t, t) = \boldsymbol{\mu}[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)]$. Using the symmetry of $\boldsymbol{\mu}$, the integrand in Eq. (8.85) simplifies to

$$\begin{aligned} \mathbf{a}_F(\mathbf{x}_t, t)^\top \mathbf{D}^{-1} \circ \mathrm{d}\mathbf{x}_t \;&=\; \frac{1}{k_{\mathrm{B}}T}\big[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\big]^\top \boldsymbol{\mu}^\top \boldsymbol{\mu}^{-1} \circ \mathrm{d}\mathbf{x}_t \\ &=\; \frac{1}{k_{\mathrm{B}}T}\big[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\big] \circ \mathrm{d}\mathbf{x}_t \;=\; \frac{\mathrm{d}\mathcal{Q}(t)}{k_{\mathrm{B}}T}. \end{aligned} \tag{8.86}$$

Integrating Eq. (8.86) from $t = 0$ to $t = \tau$ and multiplying by k_{B} yields Eq. (8.77).

Step 3 (Boundary Densities and Total Entropy Production): To obtain the unconditional path ratio Eq. (8.78), we multiply the conditional path measures by their respective initial state distributions $\rho_0(\mathbf{x}_0)$ and $\rho_\tau(\tilde{\mathbf{x}}_0) = \rho_\tau(\mathbf{x}_\tau)$:

$$\begin{aligned} k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]}]}{\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]}]} \;&=\; k_{\mathrm{B}} \ln \frac{\rho_0(\mathbf{x}_0)}{\rho_\tau(\mathbf{x}_\tau)} + k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0]}{\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]} \mid \tilde{\mathbf{x}}_0]} \\ &=\; \Big[\big(-k_{\mathrm{B}} \ln \rho_\tau(\mathbf{x}_\tau)\big) - \big(-k_{\mathrm{B}} \ln \rho_0(\mathbf{x}_0)\big)\Big] + \frac{\mathcal{Q}[\mathbf{x}_{[0,\tau]}]}{T} \\ &=\; [s_{\mathrm{sys}}(\tau) - s_{\mathrm{sys}}(0)] + \Delta s_{\mathrm{med}}[\mathbf{x}_{[0,\tau]}] \;=\; \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}], \end{aligned} \tag{8.87}$$

completing the proof. □

Theorem 8.8 provides an information-theoretic interpretation of the ensemble Second Law: total entropy production is the Kullback–Leibler divergence on path space between forward trajectories and their time-reversed twins.

Corollary 8.9 (Path-Space Kullback–Leibler Divergence Identity)**.** *The ensemble-averaged total entropy production* $\Sigma_{\mathrm{tot}}(\tau) = \big\langle \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}] \big\rangle$ *over* $[0, \tau]$ *is proportional to the relative entropy (Kullback–Leibler divergence) between the forward path measure* $\mathcal{P}$ *and the time-reversed path measure* $\tilde{\mathcal{P}}$*:*

$$\Sigma_{\mathrm{tot}}(\tau) \;=\; \int \mathcal{D}\mathbf{x}\, \mathcal{P}[\mathbf{x}_{[0,\tau]}]\, k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]}]}{\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]}]} \;=\; k_{\mathrm{B}}\, D_{\mathrm{KL}}\Big(\mathcal{P}[\mathbf{x}_{[0,\tau]}] \,\Big\|\, \tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]}]\Big) \;\geq\; 0. \tag{8.88}$$

Proof. Taking the expectation of Eq. (8.78) with respect to the forward path measure $\mathcal{D}\mathcal{P}[\mathbf{x}_{[0,\tau]}] = \mathcal{D}\mathbf{x}\, \mathcal{P}[\mathbf{x}_{[0,\tau]}]$ immediately yields Eq. (8.88). Because the Kullback–Leibler divergence $D_{\mathrm{KL}}(P \,\|\, Q)$ between any two normalized probability measures is non-negative and vanishes if and only if $P = Q$ almost everywhere, we have $\Sigma_{\mathrm{tot}}(\tau) \geq 0$, with $\Sigma_{\mathrm{tot}}(\tau) = 0$ if and only if the forward and time-reversed path ensembles are statistically indistinguishable ($\mathcal{P}[\mathbf{x}_{[0,\tau]}] = \tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]}]$). □

8.5 Housekeeping vs. Excess Entropy Production in Non-Equilibrium Circuits

In a passive physical system whose forces derive solely from a conservative potential ($\mathbf{f}_{\text{nc}} \equiv \mathbf{0}$), holding the control parameters fixed ($\dot{\boldsymbol{\lambda}}_t = \mathbf{0}$) allows the system to relax to the Boltzmann–Gibbs equilibrium distribution $p_{\text{eq}}(\mathbf{x};\boldsymbol{\lambda}) = Z(\boldsymbol{\lambda})^{-1}\exp(-\beta E(\mathbf{x},\boldsymbol{\lambda}))$. Once in equilibrium, the probability current vanishes identically ($\mathbf{J}_{\text{eq}}(\mathbf{x}) = \mathbf{0}$), and by Theorem 8.6, the entropy production rate drops to zero ($\dot{\Sigma}_{\text{tot}} = 0$).

In active electronic circuits detailed balance is generically broken. As Chapter 13 shows for subthreshold CMOS NEAT-RNs [37, 38], a transistor is a three-terminal active device: the gate voltage of transistor j controls the drain current flowing into node i without a reciprocal gate-to-drain coupling from node i back to node j. Consequently, the effective conductance/transconductance matrix $\mathbf{G}$ is asymmetric ($\mathbf{G} \neq \mathbf{G}^{\top}$), generating a non-conservative rotational force field $\mathbf{f}_{\text{nc}}(\mathbf{x}) \neq \mathbf{0}$. Even when all external control biases $\boldsymbol{\lambda}$ are held completely static, such a circuit does not relax to a current-free Boltzmann equilibrium; instead, it relaxes to a NESS characterized by a stationary density $p_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda})$ and a persistent, divergence-free circulating probability current $\mathbf{J}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) \neq \mathbf{0}$.

For systems whose reference steady states are already out of equilibrium, Hatano and Sasa [48] showed that the total entropy production splits into two contributions:

1. **Adiabatic (Housekeeping) Entropy Production (Σ_{hk}):** The background entropy continuously produced to maintain the circulating steady-state currents $\mathbf{J}_{\text{ss}} \neq \mathbf{0}$ against viscous/Ohmic dissipation, even when the control parameters $\boldsymbol{\lambda}$ are held fixed;

2. **Non-Adiabatic (Excess) Entropy Production (Σ_{ex}):** The additional entropy produced during a finite-time transient relaxation or protocol shift as the instantaneous density $\rho(\mathbf{x},t)$ lags behind the instantaneous stationary distribution $p_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)$.

8.5.1 The Non-Equilibrium Potential and Velocity Decomposition

For any fixed value of the control parameter vector $\boldsymbol{\lambda}$, suppose the Fokker–Planck operator $\mathcal{L}_{\text{FP}}^{(\boldsymbol{\lambda})}$ admits a unique, smooth, strictly positive stationary probability density $p_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) > 0$ satisfying

$$\mathcal{L}_{\text{FP}}^{(\boldsymbol{\lambda})} p_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) \;=\; -\nabla_{\mathbf{x}} \cdot \mathbf{J}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) \;=\; 0, \tag{8.89}$$

where the steady-state probability current $\mathbf{J}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda})$ and the steady-state local hydrodynamic velocity $\mathbf{v}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda})$ are given by

$$\mathbf{v}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) \;\equiv\; \frac{\mathbf{J}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda})}{p_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda})} \;=\; \mathbf{a}(\mathbf{x};\boldsymbol{\lambda}) - \mathbf{D}\nabla_{\mathbf{x}} \ln p_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}). \tag{8.90}$$

Following Hatano and Sasa [48], we define the dimensionless **non-equilibrium potential** (or steady-state self-information) $\phi(\mathbf{x};\boldsymbol{\lambda})$ by

$$\phi(\mathbf{x};\boldsymbol{\lambda}) \;\equiv\; -\ln p_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) \qquad \Longleftrightarrow \qquad p_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) \;=\; \exp\!\big(-\phi(\mathbf{x};\boldsymbol{\lambda})\big). \tag{8.91}$$

In an equilibrium system ($\mathbf{f}_{\text{nc}} = \mathbf{0}$), $\phi(\mathbf{x};\boldsymbol{\lambda}) = \beta[E(\mathbf{x},\boldsymbol{\lambda}) - F(\boldsymbol{\lambda})]$ and $\mathbf{v}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) = \mathbf{0}$. In a NESS ($\mathbf{f}_{\text{nc}} \neq \mathbf{0}$), $\mathbf{v}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}) \neq \mathbf{0}$, and we can rewrite Eq. (8.90) as a decomposition of the total drift $\mathbf{a}(\mathbf{x};\boldsymbol{\lambda}_t)$ into a steady-state circulating velocity plus a gradient of the non-equilibrium potential:

$$\mathbf{a}(\mathbf{x};\boldsymbol{\lambda}_t) \;=\; \mathbf{v}_{\text{ss}}(\mathbf{x};\boldsymbol{\lambda}_t) - \mathbf{D}\nabla_{\mathbf{x}}\phi(\mathbf{x};\boldsymbol{\lambda}_t). \tag{8.92}$$

Now compare Eq. (8.92) with the instantaneous local velocity $\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t) = \mathbf{a}(\mathbf{x};\boldsymbol{\lambda}_t) - \mathbf{D}\nabla_{\mathbf{x}} \ln \rho(\mathbf{x},t)$ at any non-stationary density $\rho(\mathbf{x},t)$. Subtracting the two equations eliminates $\mathbf{a}(\mathbf{x};\boldsymbol{\lambda}_t)$ and decomposes the local velocity $\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t)$ into its **housekeeping component** $\mathbf{v}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)$ and its **excess component** $\mathbf{v}_{\mathrm{ex}}(\mathbf{x},t)$:

$$\mathbf{v}_{\mathrm{loc}}(\mathbf{x},t) \;=\; \mathbf{v}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t) + \mathbf{v}_{\mathrm{ex}}(\mathbf{x},t), \qquad \text{where} \qquad \mathbf{v}_{\mathrm{ex}}(\mathbf{x},t) \;\equiv\; -\mathbf{D}\nabla_{\mathbf{x}} \ln \frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)}. \tag{8.93}$$

8.5.2 The Hatano–Sasa Decomposition and Orthogonality Theorem

The velocity decomposition Eq. (8.93) induces a natural splitting of the dissipated heat and entropy production along every trajectory.

Definition 8.10 (Housekeeping and Excess Heat and Entropy Production)**.** Along a trajectory $\mathbf{x}_{[0,\tau]}$ of the Langevin equation Eq. (8.2) with instantaneous steady state $p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t) = \exp(-\phi(\mathbf{x};\boldsymbol{\lambda}_t))$:

1. The **infinitesimal trajectory housekeeping heat** $\mathrm{d}\mathcal{Q}_{\mathrm{hk}}(t)$ and **housekeeping (adiabatic) entropy production** $\mathrm{d}s_{\mathrm{hk}}(t)$ are

$$\mathrm{d}\mathcal{Q}_{\mathrm{hk}}(t) \;\equiv\; k_{\mathrm{B}}T(\mathbf{D}^{-1}\mathbf{v}_{\mathrm{ss}}(\mathbf{x}_t;\boldsymbol{\lambda}_t)) \circ \mathrm{d}\mathbf{x}_t, \qquad \mathrm{d}s_{\mathrm{hk}}(t) \;\equiv\; \frac{\mathrm{d}\mathcal{Q}_{\mathrm{hk}}(t)}{T} \;=\; k_{\mathrm{B}}(\mathbf{D}^{-1}\mathbf{v}_{\mathrm{ss}}(\mathbf{x}_t;\boldsymbol{\lambda}_t)) \circ \mathrm{d}\mathbf{x}_t. \tag{8.94}$$

2. The **infinitesimal trajectory excess heat** $\mathrm{d}\mathcal{Q}_{\mathrm{ex}}(t) \equiv \mathrm{d}\mathcal{Q}(t) - \mathrm{d}\mathcal{Q}_{\mathrm{hk}}(t)$ is

$$\mathrm{d}\mathcal{Q}_{\mathrm{ex}}(t) \;=\; -k_{\mathrm{B}}T\,\nabla_{\mathbf{x}}\phi(\mathbf{x}_t;\boldsymbol{\lambda}_t) \circ \mathrm{d}\mathbf{x}_t \;=\; k_{\mathrm{B}}T\,\nabla_{\mathbf{x}} \ln p_{\mathrm{ss}}(\mathbf{x}_t;\boldsymbol{\lambda}_t) \circ \mathrm{d}\mathbf{x}_t. \tag{8.95}$$

3. The **infinitesimal excess (non-adiabatic) trajectory entropy production** $\mathrm{d}s_{\mathrm{ex}}(t) \equiv \mathrm{d}s_{\mathrm{sys}}(t) + \mathrm{d}\mathcal{Q}_{\mathrm{ex}}(t)/T$ is

$$\mathrm{d}s_{\mathrm{ex}}(t) \;=\; -k_{\mathrm{B}}\,\mathrm{d}\ln\rho(\mathbf{x}_t,t) - k_{\mathrm{B}}\,\nabla_{\mathbf{x}}\phi(\mathbf{x}_t;\boldsymbol{\lambda}_t) \circ \mathrm{d}\mathbf{x}_t, \tag{8.96}$$

so that $\mathrm{d}s_{\mathrm{tot}}(t) = \mathrm{d}s_{\mathrm{hk}}(t) + \mathrm{d}s_{\mathrm{ex}}(t)$ holds identically along every trajectory.

At the ensemble level the housekeeping velocity $\mathbf{v}_{\mathrm{ss}}$ and the excess velocity $\mathbf{v}_{\mathrm{ex}}$ are orthogonal in the weighted inner product $\int \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \mathbf{u}^{\top}\mathbf{D}^{-1}\mathbf{w}$. As a result, the housekeeping and excess entropy production rates are separately non-negative.

Theorem 8.11 (Orthogonality of Housekeeping and Excess Entropy Production)**.** *Let $\rho(\mathbf{x},t)$ be any smooth probability density solving the Fokker–Planck equation $\partial_t\rho + \nabla_{\mathbf{x}}\cdot(\rho\mathbf{v}_{\mathrm{loc}}) = 0$, and let $p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t) = \exp(-\phi(\mathbf{x};\boldsymbol{\lambda}_t))$ be the instantaneous stationary distribution satisfying $\nabla_{\mathbf{x}}\cdot(p_{\mathrm{ss}}\mathbf{v}_{\mathrm{ss}}) = 0$. Then:*

1. ***Orthogonal Decomposition of Total Entropy Production Rate:*** *The ensemble total entropy production rate $\dot{\Sigma}_{\mathrm{tot}}(t)$ splits without cross-terms into the sum of the housekeeping rate $\dot{\Sigma}_{\mathrm{hk}}(t) \equiv \langle \dot{s}_{\mathrm{hk}}(t)\rangle$ and the excess rate $\dot{\Sigma}_{\mathrm{ex}}(t) \equiv \langle \dot{s}_{\mathrm{ex}}(t)\rangle$:*

$$\dot{\Sigma}_{\mathrm{tot}}(t) \;=\; \dot{\Sigma}_{\mathrm{hk}}(t) + \dot{\Sigma}_{\mathrm{ex}}(t), \tag{8.97}$$

where each component is separately non-negative:

$$\dot{\Sigma}_{\mathrm{hk}}(t) \;=\; k_{\mathrm{B}}\int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \|\mathbf{v}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)\|^2_{\mathbf{D}^{-1}} \;\geq\; 0, \tag{8.98}$$

$$\dot{\Sigma}_{\mathrm{ex}}(t) \;=\; k_{\mathrm{B}}\int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \|\mathbf{v}_{\mathrm{ex}}(\mathbf{x},t)\|^2_{\mathbf{D}^{-1}} \;=\; k_{\mathrm{B}}\int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \left\|\nabla_{\mathbf{x}} \ln \frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)}\right\|^2_{\mathbf{D}} \;\geq\; 0. \tag{8.99}$$

2. ***Non-Equilibrium Relative Entropy Identity (Generalized H-Theorem):*** *The excess entropy production rate is related to the time derivative of the Kullback–Leibler divergence* $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}(\cdot;\boldsymbol{\lambda}_t))$ *by*

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}(\cdot;\boldsymbol{\lambda}_t)) \;=\; -\frac{1}{k_{\mathrm{B}}}\dot{\Sigma}_{\mathrm{ex}}(t) + \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \frac{\partial \phi(\mathbf{x};\boldsymbol{\lambda}_t)}{\partial \boldsymbol{\lambda}_t} \cdot \dot{\boldsymbol{\lambda}}_t. \tag{8.100}$$

In particular, for any fixed control parameter ($\dot{\boldsymbol{\lambda}}_t = \mathbf{0}$), $\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}) = -\frac{1}{k_{\mathrm{B}}}\dot{\Sigma}_{\mathrm{ex}}(t) \le 0$, *so* $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}})$ *is a Lyapunov functional for relaxation to a non-equilibrium steady state, as already found by a different route in Section 6.3.*

Proof. We begin by substituting the velocity decomposition $\mathbf{v}_{\mathrm{loc}} = \mathbf{v}_{\mathrm{ss}} + \mathbf{v}_{\mathrm{ex}}$ from Eq. (8.93) into the quadratic form Eq. (8.47) for $\dot{\Sigma}_{\mathrm{tot}}(t)$:

$$\begin{aligned}
\dot{\Sigma}_{\mathrm{tot}}(t) \;&=\; k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, (\mathbf{v}_{\mathrm{ss}} + \mathbf{v}_{\mathrm{ex}})^{\top} \mathbf{D}^{-1} (\mathbf{v}_{\mathrm{ss}} + \mathbf{v}_{\mathrm{ex}}) \\
&=\; k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\; \|\mathbf{v}_{\mathrm{ss}}\|^2_{\mathbf{D}^{-1}} \qquad (8.101) \\
&\quad + k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\; \|\mathbf{v}_{\mathrm{ex}}\|^2_{\mathbf{D}^{-1}} \\
&\quad + 2k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \mathbf{v}_{\mathrm{ss}}^{\top} \mathbf{D}^{-1} \mathbf{v}_{\mathrm{ex}}. \qquad (8.102)
\end{aligned}$$

Consider the cross-term integral $\mathcal{I}_{\mathrm{cross}} \equiv \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \mathbf{v}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)^{\top} \mathbf{D}^{-1} \mathbf{v}_{\mathrm{ex}}(\mathbf{x},t)$. Substituting $\mathbf{v}_{\mathrm{ex}}(\mathbf{x},t) = -\mathbf{D}\nabla_{\mathbf{x}} \ln \frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)}$ from Eq. (8.93), the factor $\mathbf{D}^{-1}\mathbf{D} = \mathbf{I}_N$ cancels:

$$\mathcal{I}_{\mathrm{cross}} \;=\; -\int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \mathbf{v}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t) \cdot \nabla_{\mathbf{x}} \left(\ln \frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)} \right). \tag{8.103}$$

Now write $\rho(\mathbf{x},t)\mathbf{v}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t) = \frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)}\, p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)\mathbf{v}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t) = \frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)}\, \mathbf{J}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)$, and use the identity $\frac{\rho}{p_{\mathrm{ss}}} \nabla_{\mathbf{x}} \ln \frac{\rho}{p_{\mathrm{ss}}} = \nabla_{\mathbf{x}} \left(\frac{\rho}{p_{\mathrm{ss}}} \right)$:

$$\mathcal{I}_{\mathrm{cross}} \;=\; -\int_{\mathbb{R}^N} \mathrm{d}^N x\, \mathbf{J}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t) \cdot \nabla_{\mathbf{x}} \left(\frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)} \right). \tag{8.104}$$

Integrating Eq. (8.104) by parts over $\mathbb{R}^N$ (with vanishing boundary flux at infinity), we obtain

$$\mathcal{I}_{\mathrm{cross}} \;=\; +\int_{\mathbb{R}^N} \mathrm{d}^N x\, \left(\frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)} \right) \underbrace{\nabla_{\mathbf{x}} \cdot \mathbf{J}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)}_{=\,0} \;=\; 0, \tag{8.105}$$

because the steady-state current $\mathbf{J}_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)$ is divergence-free at every instant t by Eq. (8.89). Exactly the same integration by parts shows that the ensemble average of $\dot{s}_{\mathrm{hk}}(t) = k_{\mathrm{B}}(\mathbf{D}^{-1}\mathbf{v}_{\mathrm{ss}}) \circ \dot{\mathbf{x}}_t$ equals $k_{\mathrm{B}} \int \mathrm{d}^N x\, \mathbf{v}_{\mathrm{ss}}^{\top} \mathbf{D}^{-1} \mathbf{J} = k_{\mathrm{B}} \int \mathrm{d}^N x\, \rho \mathbf{v}_{\mathrm{ss}}^{\top} \mathbf{D}^{-1} (\mathbf{v}_{\mathrm{ss}} + \mathbf{v}_{\mathrm{ex}}) = k_{\mathrm{B}} \int \mathrm{d}^N x\, \rho\, \|\mathbf{v}_{\mathrm{ss}}\|^2_{\mathbf{D}^{-1}}$, establishing Eqs. (8.97) to (8.99).

To prove Part 2, we differentiate $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}(\cdot;\boldsymbol{\lambda}_t)) = \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t) \big[\ln \rho(\mathbf{x},t) + \phi(\mathbf{x};\boldsymbol{\lambda}_t) \big]$ with respect to t:

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{ss}}(\cdot;\boldsymbol{\lambda}_t)) \;=\; \int_{\mathbb{R}^N} \mathrm{d}^N x\, \partial_t \rho(\mathbf{x},t)\, \ln \frac{\rho(\mathbf{x},t)}{p_{\mathrm{ss}}(\mathbf{x};\boldsymbol{\lambda}_t)}$$

$$+\underbrace{\int_{\mathbb{R}^N} \mathrm{d}^N x\, \partial_t \rho(\mathbf{x},t)}_{=0} + \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \frac{\partial \phi(\mathbf{x};\boldsymbol{\lambda}_t)}{\partial \boldsymbol{\lambda}_t} \cdot \dot{\boldsymbol{\lambda}}_t. \tag{8.106}$$

Substituting $\partial_t \rho = -\nabla_{\mathbf{x}} \cdot (\rho \mathbf{v}_{\mathrm{loc}}) = -\nabla_{\mathbf{x}} \cdot (\rho \mathbf{v}_{\mathrm{ss}} + \rho \mathbf{v}_{\mathrm{ex}})$ into the first integral and integrating by parts gives

$$\begin{aligned} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \partial_t \rho\, \ln \frac{\rho}{p_{\mathrm{ss}}} &= \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, (\mathbf{v}_{\mathrm{ss}} + \mathbf{v}_{\mathrm{ex}}) \cdot \nabla_{\mathbf{x}} \ln \frac{\rho}{p_{\mathrm{ss}}} \\ &= -\mathcal{I}_{\mathrm{cross}} - \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho(\mathbf{x},t)\, \mathbf{v}_{\mathrm{ex}}^{\top} \mathbf{D}^{-1} \mathbf{v}_{\mathrm{ex}} \;=\; -\frac{1}{k_{\mathrm{B}}} \dot{\Sigma}_{\mathrm{ex}}(t), \end{aligned} \tag{8.107}$$

which proves Eq. (8.100). □

Hardware Realization: Housekeeping Power and Non-Reciprocal Samplers

Theorem 8.11 clarifies a trade-off in the design of analog thermodynamic processors [31, 37, 38]:

- **Passive / Reciprocal Equilibrium Samplers ($\mathbf{G} = \mathbf{G}^{\top}$, $\mathbf{f}_{\mathrm{nc}} = \mathbf{0}$):** When capacitor nodes or probabilistic bits (p-bits, Chapter 1) are coupled purely through passive resistors, capacitors, or symmetric reciprocal transmission gates, the steady state is a true Boltzmann equilibrium ($\mathbf{J}_{\mathrm{ss}} = \mathbf{0}$, $\mathbf{v}_{\mathrm{ss}} = \mathbf{0}$). Once the circuit relaxes after a parameter update ($\rho_t \to p_{\mathrm{eq}}$), the excess entropy production rate $\dot{\Sigma}_{\mathrm{ex}}(t)$ decays exponentially to zero, and the housekeeping entropy production rate is identically zero ($\dot{\Sigma}_{\mathrm{hk}} = 0$). Thermodynamic work is dissipated only during transient parameter switching ($\dot{\boldsymbol{\lambda}}_t \neq \mathbf{0}$).
- **Active / Non-Reciprocal Subthreshold CMOS NEAT-RNs ($\mathbf{G} \neq \mathbf{G}^{\top}$, $\mathbf{f}_{\mathrm{nc}} \neq \mathbf{0}$):** In the subthreshold CMOS NEAT-RNs of Chapter 13, active transconductance amplifiers allow unilateral coupling ($G_{ij} \neq G_{ji}$) and impedance buffering without loading the upstream node. This breaks detailed balance and drives a steady-state circulating current $\mathbf{J}_{\mathrm{ss}}(\mathbf{v}) \neq \mathbf{0}$ even when all gate biases are static. As a result, the circuit continuously dissipates a positive **housekeeping heat rate**

$$\dot{\mathcal{Q}}_{\mathrm{hk}} \;=\; T\dot{\Sigma}_{\mathrm{hk}} \;=\; k_{\mathrm{B}}T \int \mathrm{d}^N v\, p_{\mathrm{ss}}(\mathbf{v})\, \|\mathbf{v}_{\mathrm{ss}}(\mathbf{v})\|^2_{\mathbf{D}^{-1}} \;>\; 0,$$

  drawn from the DC bias supply of the transistors (a small part of their total bias power $\sum_k I_{D,k} V_{\mathrm{DD}}$). The stationary covariance $\boldsymbol{\Sigma}$ of a non-reciprocal drift matrix $\mathbf{A} = -\mathbf{C}^{-1}\mathbf{G}$ is still fixed by the Lyapunov equation $\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^{\top} + 2\mathbf{D} = \mathbf{0}$, so the circuit can still be programmed to sample a target Gaussian. What the housekeeping power buys is speed: the circulating currents $\mathbf{J}_{\mathrm{ss}}$ carry probability along the level sets of p_{ss} and can shorten the relaxation time relative to a reversible circuit with the same stationary distribution (Chapter 13).

8.6 Worked Computation: Vorticity and Housekeeping Heat in a Non-Reciprocal 2-Node RC Circuit

The following computation applies the tools of this chapter (Sekimoto's energetics, the Onsager–Machlup action, the Crooks path ratio, and the housekeeping–excess decomposition) to a non-equilibrium version of the RC sampler: two RC nodes coupled by an antisymmetric transconductance.

Worked computation 8.12 (Onsager–Machlup Action, Vorticity Current, and Housekeeping Heat in a Non-Reciprocal 2-Node RC Sampler)**.** Consider a two-node continuous-voltage circuit with state vector $\mathbf{v}_t = (v_{1,t}, v_{2,t})^\top \in \mathbb{R}^2$ and matched node capacitances $\mathbf{C} = C\,\mathbf{I}_2$ ($C > 0$). Each node $i \in \{1, 2\}$ is connected to ground through a passive Ohmic conductance $g > 0$ at temperature T, and the two nodes are cross-coupled by an antisymmetric, non-reciprocal active transconductance $\eta \in \mathbb{R}$ (where node 2 drives node 1 with transconductance $+\eta$, while node 1 drives node 2 with transconductance $-\eta$, realized by a pair of subthreshold CMOS transconductors as analyzed in Chapter 13). The total conductance matrix $\mathbf{G}$ and current noise spectral matrix $\mathbf{S}_I = 2k_\mathrm{B}Tg\,\mathbf{I}_2$ yield the coupled Langevin SDE:

$$C\,\mathrm{d}\begin{pmatrix} v_{1,t} \\ v_{2,t} \end{pmatrix} \;=\; -\underbrace{\begin{pmatrix} g & -\eta \\ +\eta & g \end{pmatrix}}_{\mathbf{G}} \begin{pmatrix} v_{1,t} \\ v_{2,t} \end{pmatrix} \mathrm{d}t + \sqrt{2k_\mathrm{B}Tg}\;\mathrm{d}\begin{pmatrix} W_{1,t} \\ W_{2,t} \end{pmatrix}. \tag{8.108}$$

We will carry out a six-step derivation to determine:

1. The decomposition of the drift field $\mathbf{a}(\mathbf{v})$ into conservative RC relaxation and chiral non-conservative rotation;
2. The stationary Gaussian distribution $p_\mathrm{ss}(\mathbf{v})$, from the continuous-time Lyapunov equation;
3. The steady-state probability current $\mathbf{J}_\mathrm{ss}(\mathbf{v})$, local velocity $\mathbf{v}_\mathrm{ss}(\mathbf{v})$, and phase-space vorticity;
4. The Onsager–Machlup action $\mathcal{S}_\mathrm{OM}[\mathbf{v}_{[0,\tau]}]$, including the midpoint Jacobian;
5. The Crooks path-measure ratio $\Delta s_\mathrm{tot}[\mathbf{v}_{[0,\tau]}]$ in terms of the stochastic area swept in the (v_1, v_2) plane;
6. The steady-state housekeeping heat rate $\dot{\mathcal{Q}}_\mathrm{hk}$ and its size for representative circuit values.

Step 1: Drift Decomposition and Canonical Force Balance. Dividing Eq. (8.108) by C, the kinematic drift-diffusion SDE Eq. (8.3) takes the form

$$\mathrm{d}\mathbf{v}_t \;=\; \mathbf{A}\mathbf{v}_t\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}\mathbf{W}_t, \tag{8.109}$$

where the drift matrix $\mathbf{A} \equiv -\mathbf{C}^{-1}\mathbf{G}$ and the scalar diffusion coefficient D ($\mathbf{D} = D\,\mathbf{I}_2$) are

$$\mathbf{A} \;=\; -\frac{1}{\tau_{RC}}\mathbf{I}_2 - \omega\boldsymbol{\varepsilon} \;=\; \begin{pmatrix} -1/\tau_{RC} & +\omega \\ -\omega & -1/\tau_{RC} \end{pmatrix}, \qquad \tau_{RC} \;\equiv\; \frac{C}{g}, \qquad \omega \;\equiv\; \frac{\eta}{C}, \qquad D \;\equiv\; \frac{k_\mathrm{B}Tg}{C^2} \;=\; \frac{k_\mathrm{B}T}{\tau_{RC}C}, \tag{8.110}$$

and $\boldsymbol{\varepsilon}$ denotes the two-dimensional antisymmetric Levi-Civita rotation generator:

$$\boldsymbol{\varepsilon} \;\equiv\; \begin{pmatrix} 0 & -1 \\ +1 & 0 \end{pmatrix}, \qquad \boldsymbol{\varepsilon}^\top \;=\; -\boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon}^2 \;=\; -\mathbf{I}_2. \tag{8.111}$$

To match our canonical Sekimoto force balance Eq. (8.2) with mobility $\boldsymbol{\mu} = \frac{D}{k_\mathrm{B}T}\mathbf{I}_2 = \frac{g}{C^2}\mathbf{I}_2$ (and friction $\boldsymbol{\Gamma} = \boldsymbol{\mu}^{-1} = \frac{C^2}{g}\mathbf{I}_2$), we decompose the total electrostatic force $\mathbf{F}_\mathrm{tot}(\mathbf{v}) = \boldsymbol{\Gamma}\mathbf{A}\mathbf{v}$ into

a conservative gradient of the capacitive electrostatic energy $E(\mathbf{v})$ and a divergence-free non-conservative rotational force $\mathbf{f}_{\rm nc}(\mathbf{v})$:

$$E(\mathbf{v}) \;=\; \frac{1}{2}C(v_1^2+v_2^2), \qquad -\nabla_{\mathbf{v}}E(\mathbf{v}) \;=\; -C\mathbf{v}, \qquad \mathbf{f}_{\rm nc}(\mathbf{v}) \;=\; -\frac{C\eta}{g}\boldsymbol{\varepsilon}\mathbf{v} \;=\; \frac{C\eta}{g}\begin{pmatrix} v_2 \\ -v_1 \end{pmatrix}. \quad (8.112)$$

The curl of $\mathbf{f}_{\rm nc}(\mathbf{v})$ in the (v_1, v_2) plane is non-zero whenever $\eta \neq 0$:

$$\frac{\partial f_{\rm nc,2}}{\partial v_1} - \frac{\partial f_{\rm nc,1}}{\partial v_2} \;=\; -\frac{C\eta}{g} - \frac{C\eta}{g} \;=\; -\frac{2C\eta}{g} \;\neq\; 0,$$

confirming that $\mathbf{f}_{\rm nc}(\mathbf{v})$ cannot be derived from any scalar potential.

Step 2: Stationary Distribution from the Lyapunov Equation. Because Eq. (8.109) is a linear SDE with eigenvalues $\lambda_\pm(\mathbf{A}) = -\frac{1}{\tau_{RC}} \pm i\omega$ having strictly negative real parts ($-\frac{1}{\tau_{RC}} < 0$), the system relaxes to a unique zero-mean Gaussian stationary distribution $p_{\rm ss}(\mathbf{v}) = \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, where the stationary covariance matrix $\boldsymbol{\Sigma} = \left\langle \mathbf{v}\mathbf{v}^\top \right\rangle_{\rm ss}$ solves the continuous-time Lyapunov equation (Chapters 5 and 13):

$$\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top + 2\mathbf{D} \;=\; \mathbf{0}. \quad (8.113)$$

Substituting $\mathbf{A} = -\frac{1}{\tau_{RC}}\mathbf{I}_2 - \omega\boldsymbol{\varepsilon}$, $\mathbf{A}^\top = -\frac{1}{\tau_{RC}}\mathbf{I}_2 + \omega\boldsymbol{\varepsilon}$, and $\mathbf{D} = \frac{k_{\rm B}T}{\tau_{RC}C}\mathbf{I}_2$ into Eq. (8.113) gives

$$-\frac{2}{\tau_{RC}}\boldsymbol{\Sigma} - \omega(\boldsymbol{\varepsilon}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\boldsymbol{\varepsilon}) + \frac{2k_{\rm B}T}{\tau_{RC}C}\mathbf{I}_2 \;=\; \mathbf{0}. \quad (8.114)$$

Testing the isotropic ansatz $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}_2$, we see that the commutator $[\boldsymbol{\varepsilon}, \sigma^2\mathbf{I}_2] = \boldsymbol{\varepsilon}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\boldsymbol{\varepsilon} = \mathbf{0}$ vanishes identically. Equation Eq. (8.114) then reduces to $-\frac{2}{\tau_{RC}}\sigma^2 + \frac{2k_{\rm B}T}{\tau_{RC}C} = 0$, which yields the unique solution

$$\boldsymbol{\Sigma} \;=\; \frac{k_{\rm B}T}{C}\mathbf{I}_2, \qquad p_{\rm ss}(\mathbf{v}) \;=\; \frac{C}{2\pi k_{\rm B}T}\exp\!\left(-\frac{C}{2k_{\rm B}T}(v_1^2+v_2^2)\right),$$
$$\phi(\mathbf{v}) \;=\; -\ln p_{\rm ss}(\mathbf{v}) \;=\; \frac{C}{2k_{\rm B}T}(v_1^2+v_2^2) + \ln\!\left(\frac{2\pi k_{\rm B}T}{C}\right). \quad (8.115)$$

The antisymmetric coupling η leaves the stationary distribution $p_{\rm ss}(\mathbf{v})$ unchanged: it is the passive equipartition distribution $p_{\rm eq}(\mathbf{v}) \propto \exp(-\beta E(\mathbf{v}))$. A snapshot of the node voltages at a single instant t cannot distinguish $\eta \neq 0$ from $\eta = 0$.

Step 3: Steady-State Probability Current $\mathbf{J}_{\rm ss}(\mathbf{v})$ and Local Velocity $\mathbf{v}_{\rm ss}(\mathbf{v})$. Although the stationary density $p_{\rm ss}(\mathbf{v})$ is independent of η, the two-time dynamics and the steady-state probability current $\mathbf{J}_{\rm ss}(\mathbf{v})$ depend on it. Substituting $\mathbf{a}(\mathbf{v}) = -\frac{1}{\tau_{RC}}\mathbf{v} - \omega\boldsymbol{\varepsilon}\mathbf{v}$ and $\nabla_{\mathbf{v}}\ln p_{\rm ss}(\mathbf{v}) = -\frac{C}{k_{\rm B}T}\mathbf{v}$ into Eq. (8.90) with $D = \frac{k_{\rm B}T}{\tau_{RC}C}$, the radial conservative drift $-\frac{1}{\tau_{RC}}\mathbf{v}$ cancels the diffusive Fickian flux $-D\nabla_{\mathbf{v}}\ln p_{\rm ss} = +\frac{1}{\tau_{RC}}\mathbf{v}$, leaving a purely rotational steady-state velocity and current:

$$\mathbf{v}_{\rm ss}(\mathbf{v}) \;=\; \left(-\frac{1}{\tau_{RC}}\mathbf{v} - \omega\boldsymbol{\varepsilon}\mathbf{v}\right) - \frac{k_{\rm B}T}{\tau_{RC}C}\left(-\frac{C}{k_{\rm B}T}\mathbf{v}\right) \;=\; -\omega\boldsymbol{\varepsilon}\mathbf{v} \;=\; \omega\begin{pmatrix} +v_2 \\ -v_1 \end{pmatrix} \;=\; \frac{\eta}{C}\begin{pmatrix} +v_2 \\ -v_1 \end{pmatrix}, \quad (8.116)$$

$$\mathbf{J}_{\rm ss}(\mathbf{v}) \;=\; p_{\rm ss}(\mathbf{v})\,\mathbf{v}_{\rm ss}(\mathbf{v}) \;=\; \frac{\eta}{2\pi k_{\rm B}T}\begin{pmatrix} +v_2 \\ -v_1 \end{pmatrix}\exp\!\left(-\frac{C}{2k_{\rm B}T}(v_1^2+v_2^2)\right). \quad (8.117)$$

The current has two characteristic properties:

1. *Zero divergence (incompressibility):*

$$\begin{aligned}\nabla_{\mathbf{v}} \cdot \mathbf{J}_{\mathrm{ss}}(\mathbf{v}) &= \frac{\partial}{\partial v_1}(\omega v_2\, p_{\mathrm{ss}}(v_1, v_2)) + \frac{\partial}{\partial v_2}(-\omega v_1\, p_{\mathrm{ss}}(v_1, v_2)) \\ &= \omega v_2 \left(-\frac{C v_1}{k_{\mathrm{B}} T} p_{\mathrm{ss}}\right) - \omega v_1 \left(-\frac{C v_2}{k_{\mathrm{B}} T} p_{\mathrm{ss}}\right) = 0.\end{aligned}$$

2. *Orthogonality to potential gradients (circulation along equipotential contours):*

$$\nabla_{\mathbf{v}} E(\mathbf{v}) \cdot \mathbf{J}_{\mathrm{ss}}(\mathbf{v}) = C(v_1, v_2) \begin{pmatrix} \omega v_2 \\ -\omega v_1 \end{pmatrix} p_{\mathrm{ss}}(\mathbf{v}) = C\omega(v_1 v_2 - v_2 v_1) p_{\mathrm{ss}}(\mathbf{v}) = 0.$$

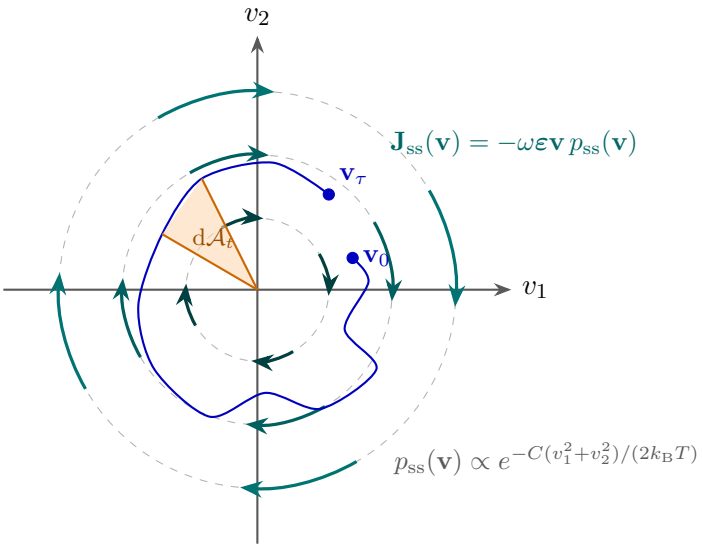


Figure 8.2: Phase-space portrait of the non-reciprocal two-node RC circuit (Worked computation 8.12) for $\eta > 0$ ($\omega = \eta/C > 0$). Dashed circles are contours of constant energy and constant stationary density $p_{\mathrm{ss}}(\mathbf{v})$, while teal arrows depict the divergence-free clockwise steady-state probability current $\mathbf{J}_{\mathrm{ss}}(\mathbf{v}) = (\omega v_2, -\omega v_1)^\top p_{\mathrm{ss}}(\mathbf{v})$. Along a stochastic trajectory $\mathbf{v}_t$ (blue curve), the irreversible medium entropy production is proportional to the signed area $\mathcal{A}[\mathbf{v}_{[0,\tau]}] = \frac{1}{2}\int_0^\tau (v_{1,t} \circ \mathrm{d}v_{2,t} - v_{2,t} \circ \mathrm{d}v_{1,t})$ swept around the origin.

Step 4: Onsager–Machlup Action $\mathcal{S}_{\mathrm{OM}}[\mathbf{v}_{[0,\tau]}]$. The midpoint Jacobian term $\frac{1}{2}\nabla_{\mathbf{v}} \cdot \mathbf{a}(\mathbf{v})$ for the drift field $\mathbf{a}(\mathbf{v}) = (-\frac{1}{\tau_{RC}} v_1 + \omega v_2,\ -\omega v_1 - \frac{1}{\tau_{RC}} v_2)^\top$:

$$\frac{1}{2}\nabla_{\mathbf{v}} \cdot \mathbf{a}(\mathbf{v}) = \frac{1}{2}\left(\frac{\partial a_1}{\partial v_1} + \frac{\partial a_2}{\partial v_2}\right) = \frac{1}{2}\left(-\frac{1}{\tau_{RC}} - \frac{1}{\tau_{RC}}\right) = -\frac{1}{\tau_{RC}}. \tag{8.118}$$

The rotation $-\omega \boldsymbol{\varepsilon} \mathbf{v}$ is divergence-free and does not contribute; the RC relaxation contributes the constant $-1/\tau_{RC} = -g/C$. Substituting $\mathbf{D}^{-1} = D^{-1}\mathbf{I}_2 = \frac{\tau_{RC} C}{k_{\mathrm{B}} T}\mathbf{I}_2$ and Eq. (8.118) into Theorem 8.7, the continuum Onsager–Machlup action Eq. (8.57) for any voltage trajectory $\mathbf{v}_{[0,\tau]}$ is

$$\mathcal{S}_{\mathrm{OM}}[\mathbf{v}_{[0,\tau]}] = \frac{\tau_{RC} C}{4 k_{\mathrm{B}} T} \int_0^\tau \mathrm{d}t \left[\left(\dot{v}_{1,t} + \frac{v_{1,t}}{\tau_{RC}} - \omega v_{2,t}\right)^2 + \left(\dot{v}_{2,t} + \frac{v_{2,t}}{\tau_{RC}} + \omega v_{1,t}\right)^2\right] - \frac{\tau}{\tau_{RC}}. \tag{8.119}$$

Step 5: Crooks Path-Measure Ratio and Stochastic Area (Vorticity). Suppose the circuit is operating in its stationary state $p_{\rm ss}(\mathbf{v})$ with fixed parameters ($\dot{\boldsymbol{\lambda}} = \mathbf{0}$). If we compare the probability of observing a forward trajectory $\mathbf{v}_{[0,\tau]}$ against the probability of observing the time-reversed trajectory $\tilde{\mathbf{v}}_t = \mathbf{v}_{\tau-t}$ in the *same* circuit (so $\tilde{\mathcal{P}} = \mathcal{P}$), Theorem 8.8 gives

$$\Delta s_{\rm tot}[\mathbf{v}_{[0,\tau]}] \;=\; k_{\rm B}\ln\frac{\mathcal{P}[\mathbf{v}_{[0,\tau]}]}{\mathcal{P}[\tilde{\mathbf{v}}_{[0,\tau]}]} \;=\; k_{\rm B}\ln\frac{p_{\rm ss}(\mathbf{v}_0)}{p_{\rm ss}(\mathbf{v}_\tau)} + \frac{1}{T}\int_0^\tau \left[-\nabla_{\mathbf{v}}E(\mathbf{v}_t) + \mathbf{f}_{\rm nc}(\mathbf{v}_t)\right]\circ \mathrm{d}\mathbf{v}_t. \tag{8.120}$$

Now evaluate the two terms in Eq. (8.120) explicitly:

1. Because $p_{\rm ss}(\mathbf{v}) \propto \exp(-E(\mathbf{v})/k_{\rm B}T)$ where $E(\mathbf{v}) = \frac{1}{2}C(v_1^2+v_2^2)$, the boundary system entropy change is

$$\Delta s_{\rm sys} \;=\; k_{\rm B}\ln\frac{p_{\rm ss}(\mathbf{v}_0)}{p_{\rm ss}(\mathbf{v}_\tau)} \;=\; \frac{E(\mathbf{v}_\tau)-E(\mathbf{v}_0)}{T} \;=\; \frac{1}{T}\int_0^\tau \nabla_{\mathbf{v}}E(\mathbf{v}_t)\circ\mathrm{d}\mathbf{v}_t,$$

   which is $-1/T$ times the conservative excess heat $\mathcal{Q}_{\rm ex}[\mathbf{v}_{[0,\tau]}] = -\Delta E$.

2. When we add $\Delta s_{\rm sys}$ to $\Delta s_{\rm med} = \mathcal{Q}/T$ in Eq. (8.120), the conservative potential terms $\pm\frac{1}{T}\int_0^\tau \nabla_{\mathbf{v}}E(\mathbf{v}_t)\circ\mathrm{d}\mathbf{v}_t$ cancel identically along every trajectory, leaving only the non-conservative housekeeping contribution:

$$\begin{aligned}\Delta s_{\rm tot}[\mathbf{v}_{[0,\tau]}] \;=\; \Delta s_{\rm hk}[\mathbf{v}_{[0,\tau]}] \;&=\; \frac{1}{T}\int_0^\tau \mathbf{f}_{\rm nc}(\mathbf{v}_t)\circ\mathrm{d}\mathbf{v}_t \\ &=\; \frac{C\eta}{gT}\int_0^\tau (v_{2,t}\circ\mathrm{d}v_{1,t} - v_{1,t}\circ\mathrm{d}v_{2,t}) \;=\; -\frac{2C\eta}{gT}\,\mathcal{A}[\mathbf{v}_{[0,\tau]}],\end{aligned} \tag{8.121}$$

   where $\mathcal{A}[\mathbf{v}_{[0,\tau]}] \equiv \frac{1}{2}\int_0^\tau (v_{1,t}\circ\mathrm{d}v_{2,t} - v_{2,t}\circ\mathrm{d}v_{1,t})$ is the signed geometric area swept by the voltage vector $\mathbf{v}_t$ around the origin in the (v_1, v_2) plane (Fig. 8.2).

Equation Eq. (8.121) shows how broken detailed balance appears at the level of trajectories: when $\eta > 0$, clockwise trajectories ($\mathcal{A} < 0$) have $\Delta s_{\rm tot} > 0$ and are exponentially more probable than their counterclockwise time-reversed twins ($\mathcal{A} > 0$).

Step 6: Steady-State Housekeeping Heat Rate $\dot{\mathcal{Q}}_{\rm hk}$. We write the housekeeping heat differential $\mathrm{d}\mathcal{Q}_{\rm hk}(t) = \mathbf{f}_{\rm nc}(\mathbf{v}_t)\circ\mathrm{d}\mathbf{v}_t$ in Itô form and average it in the steady state to obtain $\dot{\mathcal{Q}}_{\rm hk} = T\dot{\Sigma}_{\rm hk}$. Because $\mathbf{f}_{\rm nc}(\mathbf{v}) = \frac{C\eta}{g}(v_2, -v_1)^\top$ is divergence-free ($\nabla_{\mathbf{v}}\cdot\mathbf{f}_{\rm nc} = \partial_{v_1}(\frac{C\eta}{g}v_2) + \partial_{v_2}(-\frac{C\eta}{g}v_1) = 0$), the Stratonovich-to-Itô midpoint correction $k_{\rm B}T\mu\,\nabla_{\mathbf{v}}\cdot\mathbf{f}_{\rm nc}\,\mathrm{d}t$ vanishes identically. Substituting $\mathrm{d}\mathbf{v}_t = \mathbf{A}\mathbf{v}_t\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}\mathbf{W}_t$ into $\mathrm{d}\mathcal{Q}_{\rm hk}(t) = \mathbf{f}_{\rm nc}(\mathbf{v}_t)\cdot\mathrm{d}\mathbf{v}_t$, we obtain:

$$\begin{aligned}\mathrm{d}\mathcal{Q}_{\rm hk}(t) \;&=\; \frac{C\eta}{g}\begin{pmatrix} v_{2,t} & -v_{1,t}\end{pmatrix}\left[\begin{pmatrix} -\frac{1}{\tau_{RC}}v_{1,t} + \omega v_{2,t} \\ -\omega v_{1,t} - \frac{1}{\tau_{RC}}v_{2,t}\end{pmatrix}\mathrm{d}t + \sqrt{\frac{2k_{\rm B}Tg}{C^2}}\cdot\mathrm{d}\begin{pmatrix} W_{1,t}\\ W_{2,t}\end{pmatrix}\right] \\ &=\; \frac{\eta^2}{g}(v_{1,t}^2+v_{2,t}^2)\,\mathrm{d}t + \eta\sqrt{\frac{2k_{\rm B}T}{g}}(v_{2,t}\cdot\mathrm{d}W_{1,t} - v_{1,t}\cdot\mathrm{d}W_{2,t}),\end{aligned} \tag{8.122}$$

where the radial terms $-\frac{1}{\tau_{RC}}(v_2v_1 - v_1v_2)$ canceled algebraically and we used $\frac{C\eta}{g}\omega = \frac{\eta^2}{g}$. Taking the expectation over $p_{\rm ss}(\mathbf{v})$ using $\langle v_1^2\rangle_{\rm ss} = \langle v_2^2\rangle_{\rm ss} = \frac{k_{\rm B}T}{C}$ (or equivalently evaluating Eq. (8.98) directly), the steady-state housekeeping power and entropy production rate are

$$\dot{\mathcal{Q}}_{\rm hk} \;=\; T\dot{\Sigma}_{\rm hk} \;=\; \frac{\eta^2}{g}\Big(\big\langle v_1^2\big\rangle_{\rm ss} + \big\langle v_2^2\big\rangle_{\rm ss}\Big) \;=\; \frac{2\eta^2}{g}\,\frac{k_{\rm B}T}{C} \;=\; 2\left(\frac{\eta}{g}\right)^2\frac{k_{\rm B}T}{\tau_{RC}}. \tag{8.123}$$

For a sense of scale, take representative values for a subthreshold CMOS sampling cell, chosen here for illustration rather than taken from a specific chip: $T = 300\,\mathrm{K}$ ($k_\mathrm{B}T \approx 4.142 \times 10^{-21}\,\mathrm{J}$), node capacitance $C = 1\,\mathrm{fF}$, passive conductance $g = 10\,\mathrm{nS}$ (so $\tau_{RC} = C/g = 100\,\mathrm{ns}$), and coupling $\eta/g = 0.5$. The housekeeping heat rate of the voltage fluctuations is then

$$\dot{\mathcal{Q}}_\mathrm{hk} \;=\; 2(0.5)^2\,\frac{k_\mathrm{B}T}{100\,\mathrm{ns}} \;=\; 0.5\times 10^{7}\,k_\mathrm{B}T/\mathrm{s} \;\approx\; 2.07\times 10^{-14}\,\mathrm{W} \;=\; 20.7\,\mathrm{fW},$$

or $0.5\,k_\mathrm{B}T$ per relaxation time τ_{RC}.

Summary

- **Proved:**
  1. The trajectory First Law $\mathrm{d}E = \mathrm{d}\mathcal{W} - \mathrm{d}\mathcal{Q}$ on every realization, with its Itô form and ensemble average (Theorem 8.2); consistency with underdamped energetics.
  2. The trajectory entropy differentials (Proposition 8.5) and the ensemble Second Law $\dot{\Sigma}_\mathrm{tot} = k_\mathrm{B}\int \mathrm{d}^N x\,\rho\,\|\mathbf{v}_\mathrm{loc}\|^2_{\mathbf{D}^{-1}} \geq 0$ (Theorem 8.6).
  3. The Onsager–Machlup action and the MSRJD action for any discretization α, including the midpoint Jacobian $+\frac{1}{2}\int_0^\tau \mathrm{d}t\,\nabla\cdot\mathbf{a}$ and the α-independence of the continuum measure (Theorem 8.7).
  4. The Crooks path ratio $k_\mathrm{B}\ln(\mathcal{P}/\tilde{\mathcal{P}}) = \Delta s_\mathrm{tot}$ and the path-space Kullback–Leibler form of Σ_tot (Theorem 8.8 and Corollary 8.9).
  5. The housekeeping–excess decomposition $\dot{\Sigma}_\mathrm{tot} = \dot{\Sigma}_\mathrm{hk} + \dot{\Sigma}_\mathrm{ex}$ with both parts non-negative, and the housekeeping heat rate $\dot{\mathcal{Q}}_\mathrm{hk} = 2\eta^2 k_\mathrm{B}T/(gC)$ of the non-reciprocal two-node RC sampler (Theorem 8.11 and Worked computation 8.12).
- **Assumed:**
  1. A Markovian white-noise bath at a single temperature T, with the Einstein relation $\mathbf{D} = k_\mathrm{B}T\boldsymbol{\mu}$ (Chapter 5).
  2. Potentials smooth and confining enough that $\rho > 0$ and that ρ and $\mathbf{J}$ decay at infinity, so integrations by parts have no boundary terms.
- **Taken up later:**
  1. The measure-theoretic form of path-measure ratios, via Girsanov's theorem, is in Chapter 4.
  2. Entropy production when the discreteness of charge q_e is resolved, through the shot-noise master equation, is treated in Chapters 11 and 13.
- **Open:**
  1. How to choose non-reciprocal couplings $\mathbf{G} \neq \mathbf{G}^\top$ in large subthreshold CMOS networks to maximize the speed-up of mixing per unit of housekeeping power is an open design problem [38].

Bibliographic notes

Stochastic energetics along individual Langevin trajectories was introduced by Sekimoto in 1997 and is presented in his monograph *Stochastic Energetics* [98]. The trajectory system entropy $s_{\mathrm{sys}}(t) = -k_{\mathrm{B}} \ln \rho(\mathbf{x}_t, t)$ and the total entropy production Δs_{tot} are due to Seifert, whose review [97] remains the standard reference for stochastic thermodynamics, fluctuation theorems, and molecular machines. The path-integral representation of diffusion processes originates with Onsager and Machlup [81], and the response-field formulation was developed by Martin, Siggia, and Rose [71] (and independently, in functional form, by Janssen and by De Dominicis). Treatments of the discretization parameter α and of the related Itô–Stratonovich conversions can be found in Risken [90], Gardiner [39], and Pavliotis [85]. The identification of the log-ratio of forward and time-reversed path probabilities with medium entropy production is due to Crooks [22]. The steady-state thermodynamics of Langevin systems and the splitting of entropy production into housekeeping and excess parts were introduced by Hatano and Sasa [48]. The stochastic thermodynamics of nonlinear electronic circuits was developed by Freitas, Delvenne, and Esposito [37], and its application to subthreshold CMOS NEAT-RNs by Freitas and collaborators at Extropic [38].

Chapter 9

Fluctuation Theorems, Landauer's Principle, and the Thermodynamics of Information

Chapter 8 ended with an exact statement about single trajectories. For an overdamped Langevin system driven by a protocol $\boldsymbol{\lambda}_t$, the log-ratio of the probability of a forward path $\mathbf{x}_{[0,\tau]}$ to the probability of its time-reversed twin $\tilde{\mathbf{x}}_{[0,\tau]}$ is the total entropy produced along that path (Theorem 8.8):

$$k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]}]}{\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]}]} \;=\; \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}].$$

Averaging this identity over the forward ensemble gave the Second Law $\Sigma_{\mathrm{tot}}(\tau) = k_{\mathrm{B}}\, D_{\mathrm{KL}}\!\left(\mathcal{P} \,\middle\|\, \tilde{\mathcal{P}}\right) \geq 0$ (Corollary 8.9). An average discards most of the information in a distribution. This chapter extracts the rest. The same path-measure ratio fixes the entire probability distribution of the entropy production, of the work, and of the heat, in a form that holds arbitrarily far from equilibrium and for systems with only a handful of degrees of freedom.

The physical question behind the chapter is the energy cost of probabilistic computation. A thermodynamic sampling unit turns thermal noise into samples. A digital memory turns samples into stored bits and later erases them. Both operations change the Shannon entropy of a physical distribution, and both must obey the Second Law. We will find that the Second Law, applied to non-equilibrium distributions, takes the form $\langle \mathcal{W} \rangle \geq \Delta \mathcal{F}$, where $\mathcal{F}[\rho] = \langle E \rangle - TS[\rho]$ is the variational free energy of Definition 1.7. Erasing one bit lowers S by $k_{\mathrm{B}} \ln 2$ and therefore costs at least $k_{\mathrm{B}}T \ln 2$ of work, all of which leaves as heat. Randomizing a bit raises S by $k_{\mathrm{B}} \ln 2$ and can, in principle, return up to $k_{\mathrm{B}}T \ln 2$ of work, but only once and only if the randomization is carried out quasistatically. A free-running sampler at fixed bias produces samples at no thermodynamic cost at all. The cost appears when those samples are read out, stored, and reset.

We first generalize the path-measure ratio of Chapter 8 and derive from it the integral and detailed fluctuation theorems; choosing Boltzmann reference densities then gives the Crooks and Jarzynski relations for work. The non-equilibrium Landauer bound (Section 9.3) and its extension to measurement and feedback follow. The last result is the thermodynamic uncertainty relation (Section 9.5), which bounds the precision of any steady-state current by the entropy it produces. A closing worked computation tests these results on two exactly solvable systems: an RC node driven by a ramped source, whose work distribution is Gaussian, and a bistable p-bit that is reset by tilting its double-well potential.

For thermodynamic sampling hardware, the results of this chapter set the floor under the energy estimates of Chapter 17. A p-bit that flips under thermal noise at fixed bias dissipates nothing on average; a p-bit whose state is copied into a register and then reset dissipates at least $k_{\mathrm{B}}T\ln 2$ per reset; and a non-equilibrium circuit that produces a precise current, such as a clock or a counter of p-bit flips, must dissipate an amount of heat that grows as the square of the desired precision. These are the reference points against which the $\sim 10^4$–$10^9\, k_{\mathrm{B}}T$ per-operation costs of present-day digital and analog hardware should be read.

9.1 The Integral and Detailed Fluctuation Theorems

Throughout this chapter the system obeys the overdamped Langevin equation Eq. (8.2),

$$\boldsymbol{\Gamma}\, \mathrm{d}\mathbf{x}_t \;=\; \left[-\nabla_{\mathbf{x}} E(\mathbf{x}_t, \boldsymbol{\lambda}_t) + \mathbf{f}_{\mathrm{nc}}(\mathbf{x}_t, t)\right] \mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\boldsymbol{\Gamma}}\; \mathrm{d}\mathbf{W}_t, \qquad \mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\Gamma}^{-1}, \tag{9.1}$$

with the sign conventions of Chapter 8: $\mathcal{W}$ is work done *on* the system, $\mathcal{Q}$ is heat dissipated *into* the bath, the First Law reads $\Delta E = \mathcal{W} - \mathcal{Q}$, and the medium entropy change is $\Delta s_{\mathrm{med}} = \mathcal{Q}/T$. When we speak of a discrete system (a two-state p-bit or a random walk on a ring), the same quantities are defined from the master equation, as in Eq. (7.12).

9.1.1 A Path-Measure Ratio with an Arbitrary Reference Density

Theorem 8.8 compared the forward process started from ρ_0 with a reverse process started from the final forward density ρ_τ. Every fluctuation theorem in this chapter uses the same comparison with a different choice of starting density for the reverse process. We therefore prove the comparison once, for an arbitrary choice.

Lemma 9.1 (Generalized Path-Measure Ratio)**.** *Assume the hypotheses of Theorem 8.8. Let $\rho_0(\mathbf{x}) > 0$ be the initial density of the forward process and let $q(\mathbf{x}) > 0$ be any normalized probability density on $\mathbb{R}^N$. Define the reverse path measure started from q,*

$$\tilde{\mathcal{P}}_q[\tilde{\mathbf{x}}_{[0,\tau]}] \;\equiv\; q(\tilde{\mathbf{x}}_0)\, \tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]} \mid \tilde{\mathbf{x}}_0], \tag{9.2}$$

where $\tilde{\mathcal{P}}[\,\cdot \mid \tilde{\mathbf{x}}_0]$ is the conditional path density of the time-reversed protocol $\tilde{\boldsymbol{\lambda}}_t = \boldsymbol{\lambda}_{\tau - t}$. Define the path functional

$$\mathcal{R}_q[\mathbf{x}_{[0,\tau]}] \;\equiv\; k_{\mathrm{B}} \ln \frac{\rho_0(\mathbf{x}_0)}{q(\mathbf{x}_\tau)} + \frac{\mathcal{Q}[\mathbf{x}_{[0,\tau]}]}{T}. \tag{9.3}$$

Then, for every forward trajectory,

$$k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]}]}{\tilde{\mathcal{P}}_q[\tilde{\mathbf{x}}_{[0,\tau]}]} \;=\; \mathcal{R}_q[\mathbf{x}_{[0,\tau]}], \tag{9.4}$$

and the forward average of $e^{-\mathcal{R}_q/k_{\mathrm{B}}}$ is exactly one:

$$\left\langle e^{-\mathcal{R}_q[\mathbf{x}_{[0,\tau]}]/k_{\mathrm{B}}} \right\rangle_F \;=\; 1. \tag{9.5}$$

Proof. The forward path density factorizes as $\mathcal{P}[\mathbf{x}_{[0,\tau]}] = \rho_0(\mathbf{x}_0)\, \mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0]$. The reverse trajectory starts at $\tilde{\mathbf{x}}_0 = \mathbf{x}_\tau$, so by Eq. (9.2),

$$k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]}]}{\tilde{\mathcal{P}}_q[\tilde{\mathbf{x}}_{[0,\tau]}]} \;=\; k_{\mathrm{B}} \ln \frac{\rho_0(\mathbf{x}_0)}{q(\mathbf{x}_\tau)} + k_{\mathrm{B}} \ln \frac{\mathcal{P}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0]}{\tilde{\mathcal{P}}[\tilde{\mathbf{x}}_{[0,\tau]} \mid \tilde{\mathbf{x}}_0]}.$$

The second term equals $\mathcal{Q}[\mathbf{x}_{[0,\tau]}]/T$ by the conditional ratio Eq. (8.77). This proves Eq. (9.4).

For Eq. (9.5), work on the time-sliced grid $t_k = k\tau/K$, $k = 0, \dots, K$, used in Section 8.3, where a path is the finite list $(\mathbf{x}_0, \dots, \mathbf{x}_K)$ and $\mathcal{D}\mathbf{x} = \prod_{k=0}^{K} \mathrm{d}^N x_k$. The reversed path is the list $(\tilde{\mathbf{x}}_0, \dots, \tilde{\mathbf{x}}_K) = (\mathbf{x}_K, \dots, \mathbf{x}_0)$. This map only relabels the integration variables, so it is a bijection of $\mathbb{R}^{N(K+1)}$ onto itself with Jacobian determinant of absolute value one, and $\mathcal{D}\tilde{\mathbf{x}} = \mathcal{D}\mathbf{x}$. Using Eq. (9.4),

$$\left\langle e^{-\mathcal{R}_q/k_{\mathrm{B}}} \right\rangle_F \;=\; \int \mathcal{D}\mathbf{x}\, \mathcal{P}[\mathbf{x}_{[0,\tau]}]\, \frac{\tilde{\mathcal{P}}_q[\tilde{\mathbf{x}}_{[0,\tau]}]}{\mathcal{P}[\mathbf{x}_{[0,\tau]}]} \;=\; \int \mathcal{D}\mathbf{x}\, \tilde{\mathcal{P}}_q[\tilde{\mathbf{x}}_{[0,\tau]}] \;=\; \int \mathcal{D}\tilde{\mathbf{x}}\, \tilde{\mathcal{P}}_q[\tilde{\mathbf{x}}_{[0,\tau]}] \;=\; 1.$$

The last equality is the normalization of the reverse process: integrating its conditional path density over $\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_K$ gives one for every $\tilde{\mathbf{x}}_0$, and q is normalized. The cancellation in the first step requires $\mathcal{P}[\mathbf{x}_{[0,\tau]}] > 0$, which holds because $\rho_0 > 0$ and the Gaussian transition kernels of Theorem 8.7 are strictly positive. The identity survives the continuum limit $K \to \infty$ because it holds exactly at every K. □

If ρ_0 vanishes on a set of positive measure, the cancellation fails on paths that start there. The same computation then gives $\left\langle e^{-\mathcal{R}_q/k_{\mathrm{B}}} \right\rangle_F = \int_{\rho_0(\mathbf{x}_0)>0} \mathcal{D}\tilde{\mathbf{x}}\, \tilde{\mathcal{P}}_q \le 1$. This one-sided version is still enough for all the inequalities below, but the equalities require a strictly positive initial density. Discrete-state analogues of Lemma 9.1 hold with sums over jump sequences in place of path integrals.

9.1.2 The Integral Fluctuation Theorem and the Second Law

Choosing the reference density $q = \rho_\tau$, the final density of the forward process, turns $\mathcal{R}_q$ into the total entropy production.

Theorem 9.2 (Integral Fluctuation Theorem)**.** *Under the hypotheses of Theorem 8.8, with $\rho_0 > 0$, the total trajectory entropy production $\Delta s_{\mathrm{tot}} = \Delta s_{\mathrm{sys}} + \Delta s_{\mathrm{med}}$ satisfies, for every protocol and every duration τ,*

$$\left\langle e^{-\Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]/k_{\mathrm{B}}} \right\rangle \;=\; 1. \tag{9.6}$$

Proof. Set $q = \rho_\tau$ in Eq. (9.3). Then

$$\mathcal{R}_{\rho_\tau}[\mathbf{x}_{[0,\tau]}] \;=\; \left[-k_{\mathrm{B}} \ln \rho_\tau(\mathbf{x}_\tau)\right] - \left[-k_{\mathrm{B}} \ln \rho_0(\mathbf{x}_0)\right] + \frac{\mathcal{Q}}{T} \;=\; s_{\mathrm{sys}}(\tau) - s_{\mathrm{sys}}(0) + \Delta s_{\mathrm{med}} \;=\; \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}],$$

by Definition 8.4. Equation Eq. (9.5) becomes Eq. (9.6). □

The Integral Fluctuation Theorem (IFT) is an equality, not an inequality. It constrains the whole distribution of Δs_{tot}: positive and negative values must balance so that the exponential average is exactly one. The ordinary Second Law follows from it in one line.

Corollary 9.3 (Second Law from the IFT)**.** $\Sigma_{\mathrm{tot}}(\tau) = \langle \Delta s_{\mathrm{tot}} \rangle \ge 0$*, with equality if and only if $\Delta s_{\mathrm{tot}} = 0$ on almost every trajectory.*

Proof. The function $u \mapsto e^{-u}$ is strictly convex. Jensen's inequality gives $e^{-\langle \Delta s_{\mathrm{tot}} \rangle / k_{\mathrm{B}}} \le \left\langle e^{-\Delta s_{\mathrm{tot}}/k_{\mathrm{B}}} \right\rangle = 1$. Taking logarithms, $-\langle \Delta s_{\mathrm{tot}} \rangle / k_{\mathrm{B}} \le 0$. Strict convexity makes Jensen's inequality an equality only when the random variable is almost surely constant; the constant must then be zero by Eq. (9.6). □

Corollary 8.9 already gave $\Sigma_{\mathrm{tot}} \ge 0$ as a Kullback–Leibler divergence. The two proofs are the same fact seen from two sides: $\langle \Delta s_{\mathrm{tot}} \rangle = k_{\mathrm{B}}\, D_{\mathrm{KL}}\left(\mathcal{P} \,\middle\|\, \tilde{\mathcal{P}}_{\rho_\tau}\right)$ is the mean of the log-likelihood ratio, while Eq. (9.6) is the normalization of the likelihood ratio itself.

9.1.3 Negative Entropy Production Is Exponentially Rare

The IFT requires some trajectories to have $\Delta s_{\mathrm{tot}} < 0$, unless Δs_{tot} vanishes identically. If every trajectory had $\Delta s_{\mathrm{tot}} \geq 0$ with some strictly positive, the average of $e^{-\Delta s_{\mathrm{tot}}/k_{\mathrm{B}}}$ would be strictly less than one. Trajectories that reduce the total entropy are therefore not an anomaly. They are a necessary part of the distribution. The IFT also bounds how often they occur.

Proposition 9.4 (Exponential Bound on Negative Entropy Production)**.** *For every $a > 0$,*

$$\mathbb{P}(\Delta s_{\mathrm{tot}} \leq -a\, k_{\mathrm{B}}) \;\leq\; e^{-a}. \tag{9.7}$$

Proof. The event $\{\Delta s_{\mathrm{tot}} \leq -ak_{\mathrm{B}}\}$ is the event $\left\{e^{-\Delta s_{\mathrm{tot}}/k_{\mathrm{B}}} \geq e^{a}\right\}$. Markov's inequality for the non-negative random variable $Y = e^{-\Delta s_{\mathrm{tot}}/k_{\mathrm{B}}}$ states $\mathbb{P}(Y \geq c) \leq \langle Y \rangle / c$ for $c > 0$. With $c = e^{a}$ and $\langle Y \rangle = 1$,

$$\mathbb{P}(\Delta s_{\mathrm{tot}} \leq -ak_{\mathrm{B}}) \;=\; \mathbb{P}(Y \geq e^{a}) \;\leq\; e^{-a}. \qquad \square$$

The bound contains no system-specific constant. It does not depend on N, on the protocol, or on how far from equilibrium the system is driven. For a macroscopic process that produces $\Sigma_{\mathrm{tot}} \sim 10^{20}\, k_{\mathrm{B}}$, a violation of order Σ_{tot} has probability below $e^{-10^{20}}$, which is why the Second Law looks deterministic in the laboratory. For a single p-bit or a single femtofarad capacitor node, typical entropy productions per operation are of order k_{B}, and trajectories with $\Delta s_{\mathrm{tot}} \approx -k_{\mathrm{B}}$ occur with probability of order e^{-1}.

Physics Insight: What Negative-Entropy-Production Trajectories Are and Are Not

A trajectory with $\Delta s_{\mathrm{tot}} < 0$ is one along which the bath gives up heat to the system ($\mathcal{Q} < 0$) by more than the system entropy grows. Negative medium entropy production alone is commonplace: when a p-bit at fixed bias flips from its low-energy state to its high-energy state, the bath supplies the energy difference, so $\Delta s_{\mathrm{med}} < 0$ on that trajectory. In equilibrium this is exactly offset by the system entropy, $\Delta s_{\mathrm{sys}} = k_{\mathrm{B}} \ln[p_{\mathrm{eq}}(\mathrm{low})/p_{\mathrm{eq}}(\mathrm{high})] = -\Delta s_{\mathrm{med}}$, so $\Delta s_{\mathrm{tot}} = 0$. A strictly negative Δs_{tot} needs a density that is out of equilibrium, for example a p-bit whose bias has just been switched and which then flips against the new bias. There is nothing unusual about such trajectories. They occur with probability bounded by Eq. (9.7), they are balanced on average by trajectories with $\Delta s_{\mathrm{tot}} > 0$, and no device without measurement can accumulate them. A ratchet that tried to rectify them with a fixed mechanism would itself be part of the system, and the IFT would apply to the enlarged system.

A sampler does use them, in the following limited sense. Drawing a sample from $p_{\mathrm{eq}} \propto e^{-\beta E}$ requires the system to visit high-energy states with the correct Boltzmann frequency, and every such visit is an uphill excursion paid for by the bath. A thermodynamic sampler obtains these excursions from the bath at no average cost, while a digital sampler must synthesize them in arithmetic. Extracting net work from individual fluctuations requires information about which fluctuation occurred. Section 9.4 shows that such information is itself a thermodynamic resource whose acquisition and erasure close the books.

Example 9.5 (Gaussian Entropy Production Must Have Variance $2k_{\mathrm{B}}\Sigma_{\mathrm{tot}}$)**.** Suppose Δs_{tot} is Gaussian with mean Σ_{tot} and variance s^2. The Gaussian moment-generating function $\left\langle e^{uX} \right\rangle =$

$e^{um+u^2s^2/2}$ for $X \sim \mathcal{N}(m, s^2)$, evaluated at $u = -1/k_{\mathrm{B}}$, gives

$$\left\langle e^{-\Delta s_{\mathrm{tot}}/k_{\mathrm{B}}} \right\rangle \;=\; \exp\!\left(-\frac{\Sigma_{\mathrm{tot}}}{k_{\mathrm{B}}} + \frac{s^2}{2k_{\mathrm{B}}^2}\right).$$

The IFT forces the exponent to vanish, so $s^2 = 2k_{\mathrm{B}}\Sigma_{\mathrm{tot}}$. A Gaussian entropy production is therefore completely fixed by its mean, and the probability of a negative value is

$$\mathbb{P}(\Delta s_{\mathrm{tot}} < 0) \;=\; \Phi\!\left(-\frac{\Sigma_{\mathrm{tot}}}{\sqrt{2k_{\mathrm{B}}\Sigma_{\mathrm{tot}}}}\right) \;=\; \frac{1}{2}\,\mathrm{erfc}\!\left(\frac{1}{2}\sqrt{\frac{\Sigma_{\mathrm{tot}}}{k_{\mathrm{B}}}}\right),$$

where Φ is the standard normal cumulative distribution function and we used $\Phi(-z) = \frac{1}{2}\,\mathrm{erfc}(z/\sqrt{2})$. For $\Sigma_{\mathrm{tot}} = k_{\mathrm{B}}$ this is 0.240; for $\Sigma_{\mathrm{tot}} = 100\,k_{\mathrm{B}}$ it is 7.7×10^{-13}. The relation $s^2 = 2k_{\mathrm{B}}\Sigma_{\mathrm{tot}}$ will reappear twice: as $\langle \mathcal{W}_{\mathrm{diss}} \rangle = \frac{\beta}{2}\,\mathrm{Var}(\mathcal{W})$ in Section 9.2, and as the equality case of the thermodynamic uncertainty relation in Section 9.5.

9.1.4 The Detailed Fluctuation Theorem

The IFT is a statement about one average. A detailed fluctuation theorem (DFT) is a statement about the probability density of Δs_{tot} point by point. Its correct general form compares two different experiments.

Theorem 9.6 (Detailed Fluctuation Theorem)**.** *Assume the hypotheses of Theorem 9.2.*

1. ***Transient form.*** *Let $P_F(\sigma)$ be the probability density of $\Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]$ in the forward process started from ρ_0. In the reverse process started from ρ_τ, define for each reverse path $\mathbf{y}_{[0,\tau]}$ the functional $r[\mathbf{y}_{[0,\tau]}] \equiv -\Delta s_{\mathrm{tot}}[\tilde{\mathbf{y}}_{[0,\tau]}]$, where $\tilde{\mathbf{y}}_t = \mathbf{y}_{\tau-t}$ and Δs_{tot} is evaluated with the forward densities ρ_0, ρ_τ and forward protocol. Let $P_R(\sigma)$ be the probability density of r in the reverse process. Then*

$$\frac{P_F(\sigma)}{P_R(-\sigma)} \;=\; e^{\sigma/k_{\mathrm{B}}}. \tag{9.8}$$

2. ***Steady-state form.*** *Suppose in addition that the protocol is constant ($\boldsymbol{\lambda}_t \equiv \boldsymbol{\lambda}$), that $\mathbf{f}_{\mathrm{nc}}$ is time-independent and even under time reversal, and that $\rho_0 = p_{\mathrm{ss}}$ is the stationary density. Then the reverse process coincides with the forward process, $\Delta s_{\mathrm{tot}}[\tilde{\mathbf{x}}] = -\Delta s_{\mathrm{tot}}[\mathbf{x}]$, and the single distribution $P(\sigma)$ of Δs_{tot} obeys, for every $\tau > 0$,*

$$\frac{P(-\sigma)}{P(+\sigma)} \;=\; e^{-\sigma/k_{\mathrm{B}}}. \tag{9.9}$$

Proof. (1) Write the density of Δs_{tot} as a path integral with a delta function and use Eq. (9.4) with $q = \rho_\tau$:

$$\begin{aligned} P_F(\sigma) \;&=\; \int \mathcal{D}\mathbf{x}\; \mathcal{P}[\mathbf{x}_{[0,\tau]}]\, \delta\big(\sigma - \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]\big) \;=\; \int \mathcal{D}\mathbf{x}\; \tilde{\mathcal{P}}_{\rho_\tau}[\tilde{\mathbf{x}}_{[0,\tau]}]\, e^{\Delta s_{\mathrm{tot}}[\mathbf{x}]/k_{\mathrm{B}}}\, \delta\big(\sigma - \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]\big) \\ &=\; e^{\sigma/k_{\mathrm{B}}} \int \mathcal{D}\mathbf{x}\; \tilde{\mathcal{P}}_{\rho_\tau}[\tilde{\mathbf{x}}_{[0,\tau]}]\, \delta\big(\sigma - \Delta s_{\mathrm{tot}}[\mathbf{x}_{[0,\tau]}]\big). \end{aligned}$$

In the last integral, change variables to $\mathbf{y} = \tilde{\mathbf{x}}$ (unit Jacobian, as in the proof of Lemma 9.1). Then $\mathbf{x} = \tilde{\mathbf{y}}$ and $\Delta s_{\mathrm{tot}}[\mathbf{x}] = \Delta s_{\mathrm{tot}}[\tilde{\mathbf{y}}] = -r[\mathbf{y}]$, so the integral is $\int \mathcal{D}\mathbf{y}\, \tilde{\mathcal{P}}_{\rho_\tau}[\mathbf{y}]\, \delta(\sigma + r[\mathbf{y}]) = P_R(-\sigma)$.

(2) With a constant protocol and an even, time-independent $\mathbf{f}_{\mathrm{nc}}$, the reverse drift Eq. (8.74) is $\tilde{\mathbf{a}}(\mathbf{x}, t) = \mathbf{a}(\mathbf{x})$, the same as the forward drift. Since $\rho_0 = p_{\mathrm{ss}}$ is stationary, $\rho_\tau = p_{\mathrm{ss}}$ too, so the reverse process starts from the same density and evolves under the same dynamics: $\tilde{\mathcal{P}}_{\rho_\tau} = \mathcal{P}$ as measures on paths. The ratio Eq. (9.4) then reads $k_{\mathrm{B}} \ln(\mathcal{P}[\mathbf{x}]/\mathcal{P}[\tilde{\mathbf{x}}]) = \Delta s_{\mathrm{tot}}[\mathbf{x}]$ for every path $\mathbf{x}$. Applying the same identity to the path $\tilde{\mathbf{x}}$, whose reversal is $\mathbf{x}$, gives $\Delta s_{\mathrm{tot}}[\tilde{\mathbf{x}}] = k_{\mathrm{B}} \ln(\mathcal{P}[\tilde{\mathbf{x}}]/\mathcal{P}[\mathbf{x}]) = -\Delta s_{\mathrm{tot}}[\mathbf{x}]$. Hence $r[\mathbf{y}] = -\Delta s_{\mathrm{tot}}[\tilde{\mathbf{y}}] = \Delta s_{\mathrm{tot}}[\mathbf{y}]$, and $P_R = P_F \equiv P$. Equation Eq. (9.8) becomes $P(\sigma) = e^{\sigma/k_{\mathrm{B}}} P(-\sigma)$, which is Eq. (9.9). □

Notation Watch: Which Distribution Appears in the Denominator

The single-distribution symmetry Eq. (9.9) is often quoted without hypotheses. In a transient process (for example, a bit reset started from a uniform distribution) the forward and reverse experiments are different, and the density of Δs_{tot} in the forward experiment alone need not satisfy Eq. (9.9). The safe forms are the IFT Eq. (9.6), which always holds, and the two-experiment relation Eq. (9.8). The steady-state form holds for any τ in a stationary state with time-independent driving, which is the situation of a non-equilibrium circuit such as the non-reciprocal RC sampler of Worked computation 8.12. It also holds for a time-symmetric protocol ($\boldsymbol{\lambda}_{\tau-t} = \boldsymbol{\lambda}_t$) that starts and ends in the same equilibrium state, since the reverse experiment is again identical to the forward one.

Integrating Eq. (9.9) against $e^{-\sigma/k_{\mathrm{B}}}$ recovers the IFT, and the steady-state DFT sharpens Proposition 9.4: in a steady state, $\mathbb{P}(\Delta s_{\mathrm{tot}} \in [-\sigma - \mathrm{d}\sigma, -\sigma]) = e^{-\sigma/k_{\mathrm{B}}} \mathbb{P}(\Delta s_{\mathrm{tot}} \in [\sigma, \sigma + \mathrm{d}\sigma])$, so negative values are suppressed exponentially relative to their positive mirror images.

Example 9.7 (Exact Steady-State Fluctuation Theorem for a Biased Ring)**.** A particle hops around a ring of $L \geq 3$ sites, clockwise at rate k_+ and counterclockwise at rate k_-, at every site. This is the simplest non-equilibrium steady state: the stationary distribution is uniform, but a net current circulates. We return to it in Section 9.5. Clockwise and counterclockwise jumps form independent Poisson processes of rates k_+ and k_-, and under local detailed balance each clockwise jump dissipates heat $k_{\mathrm{B}}T\mathcal{A}$ with affinity $\mathcal{A} = \ln(k_+/k_-)$. The system entropy is constant, so the total entropy production over $[0, \tau]$ is $\Delta s_{\mathrm{tot}} = k_{\mathrm{B}}\mathcal{A}J_\tau$, where $J_\tau = N_+ - N_-$ is the net number of clockwise jumps.

The distribution of the difference of two independent Poisson counts follows from summing over the number m of counterclockwise jumps:

$$\mathbb{P}(J_\tau = n) \;=\; \sum_{m \geq \max(0,-n)} \frac{(k_+\tau)^{n+m} e^{-k_+\tau}}{(n+m)!} \, \frac{(k_-\tau)^m e^{-k_-\tau}}{m!} \;=\; e^{-(k_+ + k_-)\tau} \Big(\frac{k_+}{k_-}\Big)^{n/2} \sum_m \frac{\big(\tau\sqrt{k_+k_-}\big)^{n+2m}}{m!\,(n+m)!},$$

where we wrote $(k_+\tau)^{n+m}(k_-\tau)^m = (k_+/k_-)^{n/2}(\tau\sqrt{k_+k_-})^{n+2m}$. The remaining sum is the series of the modified Bessel function, $I_n(z) = \sum_m (z/2)^{n+2m}/[m!\,(n+m)!]$, at $z = 2\tau\sqrt{k_+k_-}$. Hence

$$\mathbb{P}(J_\tau = n) \;=\; e^{-(k_+ + k_-)\tau}\, e^{n\mathcal{A}/2}\, I_n\big(2\tau\sqrt{k_+k_-}\big),$$

the Skellam distribution. Because $I_{-n} = I_n$ for integer n,

$$\frac{\mathbb{P}(J_\tau = -n)}{\mathbb{P}(J_\tau = +n)} \;=\; e^{-n\mathcal{A}} \;=\; e^{-\Delta s_{\mathrm{tot}}/k_{\mathrm{B}}}\Big|_{J_\tau = n},$$

for every $\tau > 0$, which is the steady-state DFT Eq. (9.9) in exact form. For a weakly driven ring with $\mathcal{A} = 0.1$, a backward excursion of ten net steps is only e^{-1} times as likely as the corresponding forward excursion; for $\mathcal{A} = 5$ it is e^{-50} times as likely.

9.2 The Crooks Work Theorem and the Jarzynski Equality

Entropy production involves the density $\rho_\tau(\mathbf{x})$, which is rarely known in practice. Work is different: the work done on a system along one trajectory is a mechanical quantity that can be measured from the protocol and the state alone. The Crooks and Jarzynski relations are fluctuation theorems for work, obtained by choosing the two reference densities in Lemma 9.1 to be Boltzmann distributions.

9.2.1 Setting and the Time-Reversal Parity of Work

Assumption 9.8 (Equilibrium Endpoints)**.** In this section $\mathbf{f}_{\mathrm{nc}} \equiv \mathbf{0}$, so the dynamics at each fixed $\boldsymbol{\lambda}$ satisfies detailed balance with respect to $p_{\mathrm{eq}}(\mathbf{x};\boldsymbol{\lambda}) = Z(\boldsymbol{\lambda})^{-1}e^{-\beta E(\mathbf{x},\boldsymbol{\lambda})}$, and $F(\boldsymbol{\lambda}) = -k_{\mathrm{B}}T\ln Z(\boldsymbol{\lambda})$ is the equilibrium free energy. The forward process starts in equilibrium, $\rho_0 = p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_0)$. The reverse process, with protocol $\tilde{\boldsymbol{\lambda}}_t = \boldsymbol{\lambda}_{\tau-t}$, starts in equilibrium at its own initial parameter, $q = p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_\tau)$. We write $\Delta F = F(\boldsymbol{\lambda}_\tau) - F(\boldsymbol{\lambda}_0)$.

Neither process needs to be in equilibrium at its final time. The forward process ends in some density ρ_τ that in general differs from $p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_\tau)$, and the reverse process ends in a density that in general differs from $p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_0)$.

Lemma 9.9 (Work Is Odd Under Time Reversal)**.** *Let $\mathcal{W}[\mathbf{x}_{[0,\tau]}] = \int_0^\tau \frac{\partial E(\mathbf{x}_t,\boldsymbol{\lambda}_t)}{\partial\boldsymbol{\lambda}}\cdot\dot{\boldsymbol{\lambda}}_t\,\mathrm{d}t$ be the forward work Eq. (8.11), and let $\tilde{\mathcal{W}}[\tilde{\mathbf{x}}_{[0,\tau]}] = \int_0^\tau \frac{\partial E(\tilde{\mathbf{x}}_t,\tilde{\boldsymbol{\lambda}}_t)}{\partial\boldsymbol{\lambda}}\cdot\dot{\tilde{\boldsymbol{\lambda}}}_t\,\mathrm{d}t$ be the work done on the reversed path in the reverse process. Then $\tilde{\mathcal{W}}[\tilde{\mathbf{x}}_{[0,\tau]}] = -\mathcal{W}[\mathbf{x}_{[0,\tau]}]$.*

Proof. Since $\tilde{\boldsymbol{\lambda}}_t = \boldsymbol{\lambda}_{\tau-t}$, the chain rule gives $\dot{\tilde{\boldsymbol{\lambda}}}_t = -\dot{\boldsymbol{\lambda}}_{\tau-t}$. With $\tilde{\mathbf{x}}_t = \mathbf{x}_{\tau-t}$,

$$\tilde{\mathcal{W}}[\tilde{\mathbf{x}}] \;=\; -\int_0^\tau \frac{\partial E(\mathbf{x}_{\tau-t},\boldsymbol{\lambda}_{\tau-t})}{\partial\boldsymbol{\lambda}}\cdot\dot{\boldsymbol{\lambda}}_{\tau-t}\,\mathrm{d}t \;=\; -\int_0^\tau \frac{\partial E(\mathbf{x}_u,\boldsymbol{\lambda}_u)}{\partial\boldsymbol{\lambda}}\cdot\dot{\boldsymbol{\lambda}}_u\,\mathrm{d}u \;=\; -\mathcal{W}[\mathbf{x}],$$

where we substituted $u = \tau - t$, $\mathrm{d}t = -\,\mathrm{d}u$, and reversed the limits of integration. □

9.2.2 The Crooks Work Fluctuation Theorem

Theorem 9.10 (Crooks Work Fluctuation Theorem)**.** *Under Assumption 9.8 and the hypotheses of Theorem 8.8:*

1. *For every forward trajectory,*

$$\frac{\mathcal{P}_F[\mathbf{x}_{[0,\tau]}]}{\mathcal{P}_R[\tilde{\mathbf{x}}_{[0,\tau]}]} \;=\; e^{\beta(\mathcal{W}[\mathbf{x}_{[0,\tau]}]-\Delta F)}, \tag{9.10}$$

   where $\mathcal{P}_R \equiv \tilde{\mathcal{P}}_{p_{\mathrm{eq}}(\cdot;\boldsymbol{\lambda}_\tau)}$ is the reverse path measure started in equilibrium.

2. *Let $P_F(\mathcal{W})$ be the probability density of the work in the forward process and $P_R(\mathcal{W})$ the probability density of the work in the reverse process. Then*

$$\frac{P_F(+\mathcal{W})}{P_R(-\mathcal{W})} \;=\; e^{\beta(\mathcal{W}-\Delta F)}. \tag{9.11}$$

Proof. (1) Apply Lemma 9.1 with $\rho_0 = p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_0)$ and $q = p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_\tau)$. The boundary term is

$$k_{\mathrm{B}}\ln\frac{p_{\mathrm{eq}}(\mathbf{x}_0;\boldsymbol{\lambda}_0)}{p_{\mathrm{eq}}(\mathbf{x}_\tau;\boldsymbol{\lambda}_\tau)} \;=\; k_{\mathrm{B}}\Big[-\beta E(\mathbf{x}_0,\boldsymbol{\lambda}_0) - \ln Z(\boldsymbol{\lambda}_0) + \beta E(\mathbf{x}_\tau,\boldsymbol{\lambda}_\tau) + \ln Z(\boldsymbol{\lambda}_\tau)\Big] \;=\; \frac{1}{T}(\Delta E - \Delta F),$$

where $\Delta E = E(\mathbf{x}_\tau, \boldsymbol{\lambda}_\tau) - E(\mathbf{x}_0, \boldsymbol{\lambda}_0)$ and we used $k_\mathrm{B}\beta = 1/T$ and $-k_\mathrm{B}\ln Z = F/T$. Adding $\mathcal{Q}/T$ and using the trajectory First Law $\Delta E + \mathcal{Q} = \mathcal{W}$ (Theorem 8.2),

$$\mathcal{R}_q \;=\; \frac{1}{T}(\Delta E + \mathcal{Q} - \Delta F) \;=\; \frac{1}{T}(\mathcal{W} - \Delta F).$$

Dividing Eq. (9.4) by k_B and exponentiating gives Eq. (9.10).

(2) Insert a delta function and use Eq. (9.10):

$$\begin{aligned} P_F(\mathcal{W}) \;&=\; \int \mathcal{D}\mathbf{x}\, \mathcal{P}_F[\mathbf{x}]\, \delta(\mathcal{W} - \mathcal{W}[\mathbf{x}]) \;=\; \int \mathcal{D}\mathbf{x}\, \mathcal{P}_R[\tilde{\mathbf{x}}]\, e^{\beta(\mathcal{W}[\mathbf{x}]-\Delta F)}\, \delta(\mathcal{W} - \mathcal{W}[\mathbf{x}]) \\ &=\; e^{\beta(\mathcal{W}-\Delta F)} \int \mathcal{D}\tilde{\mathbf{x}}\, \mathcal{P}_R[\tilde{\mathbf{x}}]\, \delta(\mathcal{W} + \tilde{\mathcal{W}}[\tilde{\mathbf{x}}]) \;=\; e^{\beta(\mathcal{W}-\Delta F)}\, P_R(-\mathcal{W}). \end{aligned}$$

In the third step we changed variables from $\mathbf{x}$ to $\tilde{\mathbf{x}}$ with unit Jacobian and used $\mathcal{W}[\mathbf{x}] = -\tilde{\mathcal{W}}[\tilde{\mathbf{x}}]$ from Lemma 9.9. □

Unlike the transient DFT Eq. (9.8), the Crooks theorem compares two physically natural experiments: run the protocol forward from equilibrium, and run it backward from equilibrium. Both are directly realizable on hardware.

Two consequences can be read off Eq. (9.11) directly (Fig. 9.1). First, the forward and reverse work densities $P_F(\mathcal{W})$ and $P_R(-\mathcal{W})$ cross at exactly one point, $\mathcal{W} = \Delta F$, where the exponential equals one. This crossing is used in single-molecule experiments to measure free-energy differences without knowing the dissipation. Second, the forward process is exponentially unlikely to produce work much below ΔF:

$$\mathbb{P}_F(\mathcal{W} \le \Delta F - \zeta) \;=\; \int_{-\infty}^{\Delta F - \zeta} \mathrm{d}\mathcal{W}\; P_R(-\mathcal{W})\, e^{\beta(\mathcal{W}-\Delta F)} \;\le\; e^{-\beta\zeta} \int_{-\infty}^{\infty} \mathrm{d}\mathcal{W}\; P_R(-\mathcal{W}) \;=\; e^{-\beta\zeta}, \tag{9.12}$$

for every $\zeta > 0$, since $\mathcal{W} - \Delta F \le -\zeta$ on the domain of integration.

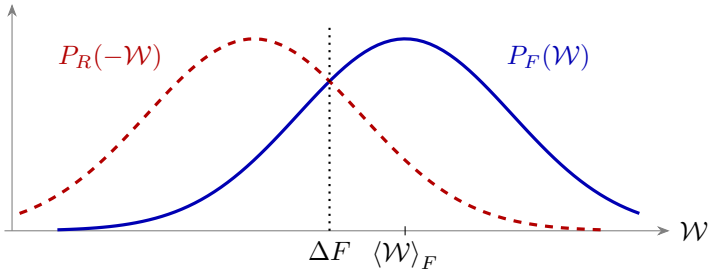


Figure 9.1: Crooks theorem for a Gaussian work distribution with $\langle \mathcal{W}_\mathrm{diss} \rangle = k_\mathrm{B}T$. The forward density $P_F(\mathcal{W})$ (solid) and the mirrored reverse density $P_R(-\mathcal{W})$ (dashed) have the same variance $2k_\mathrm{B}T\,\langle \mathcal{W}_\mathrm{diss} \rangle$ and cross exactly at $\mathcal{W} = \Delta F$. To the left of the crossing the forward density is suppressed by the factor $e^{\beta(\mathcal{W}-\Delta F)} < 1$.

9.2.3 The Jarzynski Equality

Theorem 9.11 (Jarzynski Equality)**.** *Under Assumption 9.8, for every protocol $\boldsymbol{\lambda}_{[0,\tau]}$ and every duration τ,*

$$\left\langle e^{-\beta\mathcal{W}} \right\rangle_F \;=\; e^{-\beta\Delta F}. \tag{9.13}$$

Consequently $\langle \mathcal{W} \rangle_F \ge \Delta F$.

Proof from the Crooks theorem. Write Eq. (9.11) as $P_F(\mathcal{W}) = P_R(-\mathcal{W})\, e^{\beta(\mathcal{W}-\Delta F)}$, multiply both sides by $e^{-\beta\mathcal{W}}$, and integrate over $\mathcal{W}$:

$$\int \mathrm{d}\mathcal{W}\; P_F(\mathcal{W})\, e^{-\beta\mathcal{W}} \;=\; e^{-\beta\Delta F} \int \mathrm{d}\mathcal{W}\; P_R(-\mathcal{W}) \;=\; e^{-\beta\Delta F},$$

by normalization of P_R. The inequality follows from Jensen's inequality exactly as in Corollary 9.3: $e^{-\beta\langle\mathcal{W}\rangle} \le \left\langle e^{-\beta\mathcal{W}} \right\rangle = e^{-\beta\Delta F}$. □

The Jarzynski equality does not mention any path integral, so it should have a proof that does not use one. Jarzynski's own argument uses the Feynman–Kac formula, and we give it here because it isolates the one physical input: at each fixed $\boldsymbol{\lambda}$, the Boltzmann distribution is stationary.

Second proof, by the Feynman–Kac formula. Define the work accumulated up to time t, $\mathcal{W}_t = \int_0^t \partial_{\boldsymbol{\lambda}} E(\mathbf{x}_s, \boldsymbol{\lambda}_s) \cdot \dot{\boldsymbol{\lambda}}_s \, \mathrm{d}s$, and the weighted density

$$g(\mathbf{x}, t) \;\equiv\; \left\langle \delta(\mathbf{x} - \mathbf{x}_t)\, e^{-\beta\mathcal{W}_t} \right\rangle_F , \tag{9.14}$$

so that $\int \mathrm{d}^N x\, g(\mathbf{x}, t) = \left\langle e^{-\beta\mathcal{W}_t} \right\rangle_F$. Write $w(\mathbf{x}, t) = \partial_{\boldsymbol{\lambda}} E(\mathbf{x}, \boldsymbol{\lambda}_t) \cdot \dot{\boldsymbol{\lambda}}_t$ for the instantaneous power, so $\mathrm{d}\mathcal{W}_t = w(\mathbf{x}_t, t)\, \mathrm{d}t$ with no stochastic term. For a smooth, compactly supported test function $\phi(\mathbf{x})$, Itô's lemma (Section 4.3) applied to $\phi(\mathbf{x}_t) e^{-\beta\mathcal{W}_t}$ gives

$$\mathrm{d}\big[\phi(\mathbf{x}_t) e^{-\beta\mathcal{W}_t}\big] \;=\; e^{-\beta\mathcal{W}_t} \Big[\mathcal{L}_t \phi(\mathbf{x}_t) - \beta w(\mathbf{x}_t, t) \phi(\mathbf{x}_t)\Big]\, \mathrm{d}t + (\text{Itô martingale increment}),$$

where $\mathcal{L}_t$ is the backward generator at parameter $\boldsymbol{\lambda}_t$ (Definition 6.1). The product rule has no Itô correction because $\mathcal{W}_t$ has finite variation. Taking expectations and moving $\mathcal{L}_t$ onto g by integration by parts (its adjoint is the Fokker–Planck operator $\mathcal{L}_{\mathrm{FP}}^{(t)}$ of Theorem 6.2),

$$\frac{\mathrm{d}}{\mathrm{d}t} \int \mathrm{d}^N x\; \phi(\mathbf{x})\, g(\mathbf{x}, t) \;=\; \int \mathrm{d}^N x\; \phi(\mathbf{x}) \Big[\mathcal{L}_{\mathrm{FP}}^{(t)} g(\mathbf{x}, t) - \beta w(\mathbf{x}, t)\, g(\mathbf{x}, t)\Big].$$

Since ϕ is arbitrary, g solves the Feynman–Kac equation

$$\partial_t g(\mathbf{x}, t) \;=\; \mathcal{L}_{\mathrm{FP}}^{(t)} g(\mathbf{x}, t) - \beta\, \partial_{\boldsymbol{\lambda}} E(\mathbf{x}, \boldsymbol{\lambda}_t) \cdot \dot{\boldsymbol{\lambda}}_t\, g(\mathbf{x}, t), \qquad g(\mathbf{x}, 0) = p_{\mathrm{eq}}(\mathbf{x}; \boldsymbol{\lambda}_0). \tag{9.15}$$

We now check that the unnormalized instantaneous Boltzmann factor

$$g_\star(\mathbf{x}, t) \;\equiv\; \frac{e^{-\beta E(\mathbf{x}, \boldsymbol{\lambda}_t)}}{Z(\boldsymbol{\lambda}_0)}$$

solves Eq. (9.15). At $t = 0$ it equals $p_{\mathrm{eq}}(\mathbf{x}; \boldsymbol{\lambda}_0)$. Its time derivative is $\partial_t g_\star = -\beta\, \partial_{\boldsymbol{\lambda}} E \cdot \dot{\boldsymbol{\lambda}}_t\, g_\star$, which matches the last term of Eq. (9.15). The remaining term vanishes: $g_\star$ is proportional to $p_{\mathrm{eq}}(\cdot\,; \boldsymbol{\lambda}_t)$, and $\mathcal{L}_{\mathrm{FP}}^{(t)} p_{\mathrm{eq}}(\cdot\,; \boldsymbol{\lambda}_t) = 0$ because, with $\mathbf{f}_{\mathrm{nc}} = \mathbf{0}$, the current $-\boldsymbol{\mu}\nabla E\, p_{\mathrm{eq}} - \mathbf{D}\nabla p_{\mathrm{eq}} = -\boldsymbol{\mu}(\nabla E - k_{\mathrm{B}} T \beta \nabla E) p_{\mathrm{eq}}$ is identically zero. Equation Eq. (9.15) is linear with a unique solution for given initial data, so $g = g_\star$. Integrating over $\mathbf{x}$,

$$\left\langle e^{-\beta\mathcal{W}_t} \right\rangle_F \;=\; \frac{1}{Z(\boldsymbol{\lambda}_0)} \int \mathrm{d}^N x\; e^{-\beta E(\mathbf{x}, \boldsymbol{\lambda}_t)} \;=\; \frac{Z(\boldsymbol{\lambda}_t)}{Z(\boldsymbol{\lambda}_0)} \;=\; e^{-\beta[F(\boldsymbol{\lambda}_t) - F(\boldsymbol{\lambda}_0)]}.$$

At $t = \tau$ this is Eq. (9.13). □

The second proof shows what the equality does *not* say. The weighted density $g(\mathbf{x}, t)$ is always the instantaneous Boltzmann factor, but the actual density $\rho(\mathbf{x}, t) = \langle \delta(\mathbf{x} - \mathbf{x}_t) \rangle$ is not. The exponential weight $e^{-\beta \mathcal{W}_t}$ reweights each trajectory so that the lag of ρ behind $p_{\mathrm{eq}}(\cdot\,; \boldsymbol{\lambda}_t)$ is exactly compensated. The same reweighting idea, applied to a sequence of Markov-chain transitions instead of continuous Langevin dynamics, is annealed importance sampling in machine learning [78]: the importance weights are $e^{-\beta \mathcal{W}}$ and their average estimates the partition-function ratio $Z(\boldsymbol{\lambda}_\tau)/Z(\boldsymbol{\lambda}_0)$.

Remark 9.12 (Non-Conservative Forces)**.** If $\mathbf{f}_{\mathrm{nc}} \neq \mathbf{0}$ but is even under time reversal, the Crooks and Jarzynski relations still hold, with $\mathcal{W}$ the total work Eq. (8.9) including the non-conservative contribution, and with $F(\boldsymbol{\lambda}) = -k_{\mathrm{B}}T \ln \int e^{-\beta E(\mathbf{x}, \boldsymbol{\lambda})}\, \mathrm{d}^N x$ defined from E alone. The path-measure proof goes through unchanged, because Eq. (9.4) and the First Law hold for any $\mathbf{f}_{\mathrm{nc}}$; the non-conservative work is odd under reversal by the same substitution as in Lemma 9.9, since the Stratonovich increment $\mathrm{d}\mathbf{x}_t$ changes sign. The Feynman–Kac proof does not carry over directly: $p_{\mathrm{eq}}(\cdot\,; \boldsymbol{\lambda}_t)$ is no longer annihilated by $\mathcal{L}_{\mathrm{FP}}^{(t)}$, and the Stratonovich work increment $\mathbf{f}_{\mathrm{nc}} \circ \mathrm{d}\mathbf{x}_t$ turns Eq. (9.15) into an equation with a tilted generator; we rely on the path-measure proof in that case. In a non-equilibrium steady state the appropriate generalization replaces $\mathcal{W}$ by the excess work of Section 8.5 (the Hatano–Sasa relation [48]).

9.2.4 Cumulant Expansion and Dissipated Work

The **dissipated work** along a trajectory is $\mathcal{W}_{\mathrm{diss}} \equiv \mathcal{W} - \Delta F$. The Jarzynski equality reads $\left\langle e^{-\beta \mathcal{W}_{\mathrm{diss}}} \right\rangle = 1$, and by Eq. (9.10) together with Lemma 9.1, $\langle \mathcal{W}_{\mathrm{diss}} \rangle = k_{\mathrm{B}}T\, D_{\mathrm{KL}}(\mathcal{P}_F \,\|\, \mathcal{P}_R) \geq 0$. The cumulants of $\mathcal{W}$ relate the mean dissipation to the fluctuations.

Proposition 9.13 (Cumulant Expansion of the Dissipated Work)**.** *Let κ_n denote the n-th cumulant of $\mathcal{W}$ in the forward process, so $\kappa_1 = \langle \mathcal{W} \rangle$, $\kappa_2 = \mathrm{Var}(\mathcal{W})$, and $\kappa_3 = \langle (\mathcal{W} - \langle \mathcal{W} \rangle)^3 \rangle$. If the cumulant generating function $\ln \left\langle e^{u\mathcal{W}} \right\rangle$ is analytic at $u = -\beta$ with a convergent cumulant series, then*

$$\langle \mathcal{W}_{\mathrm{diss}} \rangle \;=\; \langle \mathcal{W} \rangle - \Delta F \;=\; \sum_{n=2}^{\infty} \frac{(-1)^n \beta^{n-1}}{n!}\, \kappa_n \;=\; \frac{\beta}{2}\, \mathrm{Var}(\mathcal{W}) - \frac{\beta^2}{6}\kappa_3 + \frac{\beta^3}{24}\kappa_4 - \cdots . \tag{9.16}$$

If $\mathcal{W}$ is Gaussian, all cumulants beyond the second vanish and $\langle \mathcal{W}_{\mathrm{diss}} \rangle = \frac{\beta}{2}\, \mathrm{Var}(\mathcal{W})$ exactly.

Proof. By definition of the cumulants, $\ln \left\langle e^{u\mathcal{W}} \right\rangle = \sum_{n \geq 1} \kappa_n u^n / n!$. At $u = -\beta$ the Jarzynski equality gives $\ln \left\langle e^{-\beta \mathcal{W}} \right\rangle = -\beta \Delta F$. Hence

$$-\beta \Delta F \;=\; -\beta \kappa_1 + \frac{\beta^2}{2}\kappa_2 - \frac{\beta^3}{6}\kappa_3 + \frac{\beta^4}{24}\kappa_4 - \cdots .$$

Adding $\beta \kappa_1$ to both sides and dividing by β gives $\kappa_1 - \Delta F = \frac{\beta}{2}\kappa_2 - \frac{\beta^2}{6}\kappa_3 + \cdots$, which is Eq. (9.16). The Gaussian case is the truncation after $n = 2$. □

The Gaussian identity $\langle \mathcal{W}_{\mathrm{diss}} \rangle = \frac{\beta}{2}\, \mathrm{Var}(\mathcal{W})$ is a fluctuation–dissipation relation for work. It holds exactly for linear systems driven by protocols that enter linearly, as we will verify for the RC node in Worked computation 9.30, and approximately for any system driven slowly, because the work is then a sum of many weakly correlated increments and the central limit theorem makes it close to Gaussian. Far from equilibrium the higher cumulants matter, and a Gaussian fit to a skewed work distribution misestimates ΔF.

9.2.5 Slow Driving and Thermodynamic Friction

When the protocol changes slowly compared with the relaxation time of the system, the dissipated work can be computed from equilibrium fluctuations alone. Consider a single control parameter λ_t and its conjugate force $X(\mathbf{x};\lambda) = \partial_\lambda E(\mathbf{x},\lambda)$, so that $\mathrm{d}\mathcal{W} = X\,\dot\lambda\,\mathrm{d}t$. Let $C_X(t;\lambda) = \langle \delta X(t)\,\delta X(0)\rangle_{\mathrm{eq},\lambda}$ be its equilibrium autocorrelation function at fixed λ.

Heuristic Derivation: Dissipated Power at Slow Driving

Step response. Suppose λ jumps by a small amount $\Delta\lambda$ at $t = 0$ after the system has equilibrated at λ. Immediately after the jump the distribution is still $p_{\mathrm{eq}}(\cdot\,;\lambda)$, so $\langle X\rangle_{0^+} = \langle X\rangle_{\mathrm{eq},\lambda} + \langle \partial_\lambda X\rangle_{\mathrm{eq}}\,\Delta\lambda$. The new equilibrium value is $\langle X\rangle_{\mathrm{eq},\lambda+\Delta\lambda} = \langle X\rangle_{\mathrm{eq},\lambda} + [\langle \partial_\lambda X\rangle_{\mathrm{eq}} - \beta C_X(0)]\Delta\lambda$, where the last term comes from differentiating the Boltzmann weight, $\partial_\lambda p_{\mathrm{eq}} = -\beta(X - \langle X\rangle)p_{\mathrm{eq}}$. The initial excess over the new equilibrium is therefore $\beta C_X(0)\Delta\lambda$. By the fluctuation–dissipation theorem for relaxation from a perturbed equilibrium (Section 5.4), this excess decays in proportion to $C_X(t)$:

$$\langle X\rangle_t - \langle X\rangle_{\mathrm{eq},\lambda+\Delta\lambda} \;\approx\; \beta\,\Delta\lambda\,C_X(t).$$

Superposition. A slow ramp is a sequence of small steps $\dot\lambda_s\,\mathrm{d}s$. To first order in $\dot\lambda$, the excesses add:

$$\langle X\rangle_t - \langle X\rangle_{\mathrm{eq},\lambda_t} \;\approx\; \beta\int_{-\infty}^{t} \dot\lambda_s\,C_X(t-s)\,\mathrm{d}s \;\approx\; \beta\,\dot\lambda_t\int_0^\infty C_X(t')\,\mathrm{d}t' \;\equiv\; \zeta(\lambda_t)\,\dot\lambda_t,$$

where the second step uses that $\dot\lambda$ hardly changes over the correlation time, and $\zeta(\lambda) = \beta\int_0^\infty C_X(t')\,\mathrm{d}t'$ is the **thermodynamic friction coefficient**.
Dissipation. Since $\int \langle X\rangle_{\mathrm{eq},\lambda_t}\,\dot\lambda_t\,\mathrm{d}t = \Delta F$ (because $\partial_\lambda F = \langle X\rangle_{\mathrm{eq}}$), the dissipated work is

$$\langle \mathcal{W}_{\mathrm{diss}}\rangle \;\approx\; \int_0^\tau \zeta(\lambda_t)\,\dot\lambda_t^2\,\mathrm{d}t \;\geq\; 0.$$

For a constant ramp over a time τ long compared with the correlation time, the work $\mathcal{W} = \int X\dot\lambda\,\mathrm{d}t$ is a time integral of a weakly correlated stationary signal, so it is nearly Gaussian with $\mathrm{Var}(\mathcal{W}) \approx 2\dot\lambda^2\tau\int_0^\infty C_X\,\mathrm{d}t' = 2k_{\mathrm{B}}T\,\langle\mathcal{W}_{\mathrm{diss}}\rangle$, which is the Gaussian identity of Proposition 9.13.

Both systems of Worked computation 9.30 fit this formula. For the RC node, $\lambda = b$, $X = -C(v-b)$, $C_X(t) = Ck_{\mathrm{B}}Te^{-t/\tau_{RC}}$, so $\zeta = C\tau_{RC}$ and $\langle\mathcal{W}_{\mathrm{diss}}\rangle \approx C\tau_{RC}u^2\tau$, the long-ramp limit of the exact result. For the two-state p-bit, $\lambda = \varepsilon$, $X = -s$, $C_X(t) = \mathrm{sech}^2(\beta\varepsilon)e^{-\lambda_{\mathrm{G}}t}$ with Glauber relaxation rate λ_{G}, so $\zeta = \beta\,\mathrm{sech}^2(\beta\varepsilon)/\lambda_{\mathrm{G}}$. The quadratic form $\int\zeta\dot\lambda^2\,\mathrm{d}t$ is a length in parameter space with metric ζ, and minimizing it over protocols with fixed endpoints is a geodesic problem; Chapter 10 develops this geometry and its exact finite-time counterpart.

Heuristic Derivation: Why Jarzynski Estimators Need Many Samples

To estimate ΔF from M measured work values one forms $\widehat{\Delta F} = -k_{\mathrm{B}}T\ln\!\big(M^{-1}\sum_{m=1}^{M} e^{-\beta\mathcal{W}_m}\big)$. The average is dominated by the rare trajectories with $\mathcal{W}$ near or below ΔF, which by Eq. (9.12) are exponentially rare when $\langle\mathcal{W}_{\mathrm{diss}}\rangle \gg k_{\mathrm{B}}T$. For a Gaussian work distribution

with mean $\Delta F + \langle \mathcal{W}_{\mathrm{diss}} \rangle$ and variance $2k_{\mathrm{B}}T \langle \mathcal{W}_{\mathrm{diss}} \rangle$, the integrand $P_F(\mathcal{W})e^{-\beta \mathcal{W}}$ is a Gaussian of the same variance centered at $\Delta F - \langle \mathcal{W}_{\mathrm{diss}} \rangle$, obtained by completing the square:

$$-\frac{(\mathcal{W} - \Delta F - \langle \mathcal{W}_{\mathrm{diss}} \rangle)^2}{4k_{\mathrm{B}}T \langle \mathcal{W}_{\mathrm{diss}} \rangle} - \beta \mathcal{W} \;=\; -\frac{(\mathcal{W} - \Delta F + \langle \mathcal{W}_{\mathrm{diss}} \rangle)^2}{4k_{\mathrm{B}}T \langle \mathcal{W}_{\mathrm{diss}} \rangle} - \beta \Delta F.$$

The dominant trajectories sit $2\langle \mathcal{W}_{\mathrm{diss}} \rangle$ below the typical ones, which is $\sqrt{2\beta \langle \mathcal{W}_{\mathrm{diss}} \rangle}$ standard deviations away. The Gaussian tail beyond z standard deviations is of order $e^{-z^2/2}$, so the fraction of samples that land there is of order $e^{-\beta \langle \mathcal{W}_{\mathrm{diss}} \rangle}$, and a reliable estimate needs $M \gtrsim e^{\beta \langle \mathcal{W}_{\mathrm{diss}} \rangle}$ samples, more for heavy-tailed distributions. The equality is exact for any τ, but it is useful as an estimator only when the protocol dissipates at most a few $k_{\mathrm{B}}T$. This is the same weight-degeneracy problem that limits annealed importance sampling, and it is the reason the optimal low-dissipation protocols of Chapter 10 matter in practice.

9.3 The Non-Equilibrium Landauer Principle

The Jarzynski equality assumes that the process starts in equilibrium. A computation rarely does. A memory cell that holds a definite bit, a p-bit register that holds the last sample of a Markov chain, and a denoising sampler midway through its schedule are all described by distributions that are not Boltzmann distributions of the current energy function. This section states the Second Law for arbitrary initial and final distributions and applies it to the erasure and randomization of a bit.

9.3.1 Work, Variational Free Energy, and Entropy Production

For a density ρ and a parameter value $\boldsymbol{\lambda}$, the variational (non-equilibrium) free energy of Definition 1.7 is

$$\mathcal{F}[\rho; \boldsymbol{\lambda}] \;\equiv\; \int_{\mathbb{R}^N} \mathrm{d}^N x \; \rho(\mathbf{x}) E(\mathbf{x}, \boldsymbol{\lambda}) - T S[\rho], \qquad S[\rho] = -k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x \; \rho(\mathbf{x}) \ln \rho(\mathbf{x}). \tag{9.17}$$

By Eq. (1.57), $\mathcal{F}[\rho; \boldsymbol{\lambda}] = F(\boldsymbol{\lambda}) + k_{\mathrm{B}}T \, D_{\mathrm{KL}}(\rho \,\|\, p_{\mathrm{eq}}(\cdot\,; \boldsymbol{\lambda})) \geq F(\boldsymbol{\lambda})$, with equality only at equilibrium. The excess $k_{\mathrm{B}}T \, D_{\mathrm{KL}}$ is the work that an ideal machine could extract while the distribution relaxes to equilibrium, and the work that is lost as heat if it relaxes spontaneously.

Theorem 9.14 (Non-Equilibrium Second Law and Landauer Bound)**.** *Let $\mathbf{x}_t$ obey Eq. (9.1) with $\mathbf{f}_{\mathrm{nc}} \equiv \mathbf{0}$, from an arbitrary initial density ρ_0, under a protocol $\boldsymbol{\lambda}_{[0,\tau]}$. Write $\mathcal{F}_0 = \mathcal{F}[\rho_0; \boldsymbol{\lambda}_0]$ and $\mathcal{F}_\tau = \mathcal{F}[\rho_\tau; \boldsymbol{\lambda}_\tau]$. Then:*

1. *The mean work and the change in variational free energy satisfy*

$$\langle \mathcal{W} \rangle - \Delta \mathcal{F} \;=\; T \, \Sigma_{\mathrm{tot}}(\tau) \;\geq\; 0, \qquad \Delta \mathcal{F} = \mathcal{F}_\tau - \mathcal{F}_0. \tag{9.18}$$

2. *Equivalently, in terms of equilibrium free energies,*

$$\langle \mathcal{W} \rangle \;=\; \Delta F + T \Sigma_{\mathrm{tot}}(\tau) + k_{\mathrm{B}}T \Big[D_{\mathrm{KL}}(\rho_\tau \,\|\, p_{\mathrm{eq}}(\cdot\,; \boldsymbol{\lambda}_\tau)) - D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}}(\cdot\,; \boldsymbol{\lambda}_0)) \Big]. \tag{9.19}$$

3. *The mean heat dissipated into the bath is bounded below by the decrease of the system's Shannon entropy:*

$$\langle \mathcal{Q} \rangle \;\geq\; -T \, \Delta S, \qquad \Delta S = S[\rho_\tau] - S[\rho_0]. \tag{9.20}$$

4. At the trajectory level, define the stochastic free energy $\varphi(\mathbf{x},t) \equiv E(\mathbf{x},\boldsymbol{\lambda}_t) + k_{\mathrm{B}}T\ln\rho(\mathbf{x},t)$*, whose mean is* $\mathcal{F}[\rho_t;\boldsymbol{\lambda}_t]$*. Then* $T\Delta s_{\mathrm{tot}} = \mathcal{W} - \Delta\varphi$ *on every trajectory, and*

$$\left\langle e^{-\beta(\mathcal{W}-\Delta\varphi)} \right\rangle \;=\; 1. \tag{9.21}$$

Proof. (4) On a single trajectory, $T\Delta s_{\mathrm{sys}} = -k_{\mathrm{B}}T[\ln\rho(\mathbf{x}_\tau,\tau) - \ln\rho(\mathbf{x}_0,0)]$ and $T\Delta s_{\mathrm{med}} = \mathcal{Q} = \mathcal{W} - \Delta E$ by the First Law. Adding,

$$T\Delta s_{\mathrm{tot}} \;=\; \mathcal{W} - \Big[E(\mathbf{x}_\tau,\boldsymbol{\lambda}_\tau) + k_{\mathrm{B}}T\ln\rho(\mathbf{x}_\tau,\tau)\Big] + \Big[E(\mathbf{x}_0,\boldsymbol{\lambda}_0) + k_{\mathrm{B}}T\ln\rho(\mathbf{x}_0,0)\Big] \;=\; \mathcal{W} - \Delta\varphi.$$

Substituting into the IFT Eq. (9.6) gives Eq. (9.21).

(1) Averaging the trajectory identity, $\langle\varphi(\mathbf{x}_t,t)\rangle = \langle E\rangle + k_{\mathrm{B}}T\int\rho_t\ln\rho_t = \langle E\rangle - TS[\rho_t] = \mathcal{F}[\rho_t;\boldsymbol{\lambda}_t]$, so $T\Sigma_{\mathrm{tot}} = \langle\mathcal{W}\rangle - \Delta\mathcal{F}$. Non-negativity is Corollary 9.3; if ρ_0 vanishes on part of the state space, the one-sided form $\left\langle e^{-\Delta s_{\mathrm{tot}}/k_{\mathrm{B}}}\right\rangle \le 1$ noted after Lemma 9.1 gives the same conclusion by Jensen's inequality, and Eq. (9.21) becomes ≤ 1. The same result follows by integrating the free-energy balance Eq. (8.48) from 0 to τ.

(2) Substitute $\mathcal{F}[\rho;\boldsymbol{\lambda}] = F(\boldsymbol{\lambda}) + k_{\mathrm{B}}T\,D_{\mathrm{KL}}(\rho\,\|\,p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}))$ at $t=0$ and $t=\tau$ into Eq. (9.18).

(3) By Eq. (8.38), $\Sigma_{\mathrm{tot}} = \Delta S + \langle\mathcal{Q}\rangle/T \ge 0$. Multiply by T. □

Equation Eq. (9.19) contains the Jarzynski inequality $\langle\mathcal{W}\rangle \ge \Delta F$ as the special case $\rho_0 = p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_0)$, since then the initial divergence vanishes and the final one is non-negative. It also shows how a non-equilibrium initial state is a resource: if ρ_0 is far from equilibrium, $\langle\mathcal{W}\rangle$ can be smaller than ΔF by up to $k_{\mathrm{B}}T\,D_{\mathrm{KL}}(\rho_0\,\|\,p_{\mathrm{eq}})$. The heat form Eq. (9.20) is the version Landauer originally stated: an operation that reduces the entropy of the system must export at least $-T\Delta S$ of heat to the environment [63].

Physics Insight: The Bound Concerns Distributions, Not Individual States

The variational free energy is a property of the distribution ρ, not of the microstate. A bit that is known to be 1 and a bit that is equally likely to be 0 or 1 can occupy exactly the same physical configuration on a given run. They differ in what the controller knows, and that knowledge is encoded in ρ. The same operation on the same hardware therefore has a different thermodynamic cost depending on the distribution of its inputs. Erasing a memory whose contents have Shannon entropy H (in nats) costs at least $k_{\mathrm{B}}TH$, not $k_{\mathrm{B}}T\ln 2$ per stored bit. A register holding a highly predictable sample is cheap to reset; a register holding a fair coin flip is not.

9.3.2 Coarse-Grained Bits and the Information-Theoretic Free Energy

To apply Theorem 9.14 to a memory, we need the variational free energy of a distribution over a coarse-grained logical state. Let the state space be partitioned into disjoint regions, $\mathbb{R}^N = \bigcup_i \Omega_i$, one for each logical value i. For the double-well sampler (the quartic double well $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$ of Chapter 1) these are the basins $\mathcal{A}$ and $\mathcal{B}$ of Section 7.1. For each region define the restricted partition function, local free energy, and local equilibrium density

$$Z_i(\boldsymbol{\lambda}) = \int_{\Omega_i} \mathrm{d}^N x\; e^{-\beta E(\mathbf{x},\boldsymbol{\lambda})}, \qquad F_i = -k_{\mathrm{B}}T\ln Z_i, \qquad \rho_i^{\mathrm{loc}}(\mathbf{x}) = \frac{\mathbf{1}_{\Omega_i}(\mathbf{x})}{Z_i}\, e^{-\beta E(\mathbf{x},\boldsymbol{\lambda})}, \tag{9.22}$$

as in Eq. (7.8). A density ρ assigns probability $p_i = \int_{\Omega_i}\rho$ to logical state i, and its normalized restriction to Ω_i is $\rho_{|i} = \mathbf{1}_{\Omega_i}\rho/p_i$ for $p_i > 0$.

Lemma 9.15 (Free Energy of a Coarse-Grained Memory)**.** *For any density ρ with occupations p_i (terms with $p_i = 0$ are omitted from the sums),*

$$\mathcal{F}[\rho;\boldsymbol{\lambda}] \;=\; \sum_i p_i F_i(\boldsymbol{\lambda}) + k_{\mathrm{B}}T\sum_i p_i \ln p_i + k_{\mathrm{B}}T\sum_i p_i\, D_{\mathrm{KL}}\Big(\rho_{|i}\,\Big\|\,\rho_i^{\mathrm{loc}}\Big)\,. \tag{9.23}$$

In particular, among all densities with given occupations $\{p_i\}$, the local-equilibrium density $\rho = \sum_i p_i \rho_i^{\mathrm{loc}}$ has the smallest variational free energy, $\mathcal{F} = \sum_i p_i F_i - T S_{\mathrm{info}}(p)$, where $S_{\mathrm{info}}(p) = -k_{\mathrm{B}}\sum_i p_i \ln p_i$ is the Shannon entropy of the logical state in units of k_{B}.

Proof. Split the integrals in Eq. (9.17) over the regions and write $\rho = p_i \rho_{|i}$ on Ω_i:

$$\begin{aligned}\mathcal{F}[\rho;\boldsymbol{\lambda}] \;&=\; \sum_i \int_{\Omega_i} \mathrm{d}^N x\; p_i\rho_{|i}\,[E + k_{\mathrm{B}}T\ln(p_i\rho_{|i})] \\ &=\; \sum_i p_i \int_{\Omega_i} \mathrm{d}^N x\; \rho_{|i}\,[E + k_{\mathrm{B}}T\ln\rho_{|i}] + k_{\mathrm{B}}T\sum_i p_i\ln p_i \int_{\Omega_i}\rho_{|i}.\end{aligned}$$

The last integral is one. The first bracket is the variational free energy of $\rho_{|i}$ for the energy E restricted to Ω_i, which by Eq. (1.57) applied on Ω_i equals $F_i + k_{\mathrm{B}}T\,D_{\mathrm{KL}}\big(\rho_{|i}\,\big\|\,\rho_i^{\mathrm{loc}}\big)$. This gives Eq. (9.23). The divergence terms are non-negative and vanish if and only if $\rho_{|i} = \rho_i^{\mathrm{loc}}$ for every i. □

The variational free energy of a memory thus splits into a physical part $\sum_i p_i F_i$, set by the wells, and an informational part $-TS_{\mathrm{info}}$, set by the uncertainty about which well is occupied. The Second Law Eq. (9.18) treats both on the same footing.

9.3.3 Erasure Costs Work; Randomization Can Return It

A **symmetric bit** is a two-region memory, $i \in \{0,1\}$, whose potential at the start and end of the operation satisfies $F_0 = F_1$ and $\langle E\rangle_{\rho_0^{\mathrm{loc}}} = \langle E\rangle_{\rho_1^{\mathrm{loc}}}$, as for the double-well sampler with $h = 0$, $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2$. Denote by $H(p) = -p\ln p - (1-p)\ln(1-p)$ the binary entropy in nats.

Corollary 9.16 (Landauer's Principle for Erasure and Randomization)**.** *Consider a symmetric bit whose potential is the same at $t = 0$ and $t = \tau$, initially in local equilibrium with occupation $p_1(0)$, and finally holding occupation $p_1(\tau)$. Then:*

1. ***Erasure (reset).*** *If $p_1(0) = \frac{1}{2}$ and $p_1(\tau) = \delta$, then*

$$\langle \mathcal{W}\rangle \;\geq\; k_{\mathrm{B}}T\big[\ln 2 - H(\delta)\big] \;\xrightarrow{\;\delta\to 0\;}\; k_{\mathrm{B}}T\ln 2. \tag{9.24}$$

   If the bit is then left at the fixed final potential until it reaches local equilibrium, the total mean heat dissipated over the whole operation equals the total mean work and obeys the same bound.

2. ***Randomization.*** *If $p_1(0) = 0$ and $p_1(\tau) = \frac{1}{2}$, then $\langle\mathcal{W}\rangle \geq -k_{\mathrm{B}}T\ln 2$: at most $k_{\mathrm{B}}T\ln 2$ of work can be extracted.*

3. ***Spontaneous randomization.*** *If the randomization in (2) is carried out with $\mathcal{W} = 0$ on every trajectory (for example, by lowering the barrier symmetrically and letting the bit relax), the total entropy produced is $\Sigma_{\mathrm{tot}} = k_{\mathrm{B}}\ln 2$.*

Proof. By Lemma 9.15, the initial state has $\mathcal{F}_0 = F_0 + k_\mathrm{B}T[p\ln p + (1-p)\ln(1-p)] = F_0 - k_\mathrm{B}TH(p)$ with $p = p_1(0)$, and every final state with occupation δ has $\mathcal{F}_\tau \geq F_0 - k_\mathrm{B}TH(\delta)$, using $F_0 = F_1$.

(1) $\Delta\mathcal{F} \geq -k_\mathrm{B}TH(\delta) + k_\mathrm{B}TH(\frac{1}{2}) = k_\mathrm{B}T[\ln 2 - H(\delta)]$, and Eq. (9.18) gives Eq. (9.24). During the final relaxation at fixed potential no work is done, so the total work is unchanged. At the end the bit is in local equilibrium in the same symmetric potential it started in, with $\langle E\rangle_{\rho_0^{\mathrm{loc}}} = \langle E\rangle_{\rho_1^{\mathrm{loc}}}$, so $\Delta\langle E\rangle = 0$ and the averaged First Law gives $\langle\mathcal{Q}\rangle = \langle\mathcal{W}\rangle$.

(2) Here $\mathcal{F}_0 = F_0 - k_\mathrm{B}TH(0) = F_0$ and, if the final state is in local equilibrium, $\mathcal{F}_\tau = F_0 - k_\mathrm{B}T\ln 2$, so $\Delta\mathcal{F} = -k_\mathrm{B}T\ln 2$ and $\langle\mathcal{W}\rangle \geq -k_\mathrm{B}T\ln 2$. If the final state is not in local equilibrium, $\mathcal{F}_\tau$ is larger and the bound is stronger.

(3) With $\mathcal{W} = 0$, Eq. (9.18) gives $T\Sigma_{\mathrm{tot}} = -\Delta\mathcal{F} = k_\mathrm{B}T\ln 2$. □

The corollary separates three statements that are often run together.

- The cost $k_\mathrm{B}T\ln 2$ is attached to a reduction of the Shannon entropy of the logical state. It is a bound on work and, for a cyclic operation on a symmetric memory, on heat. It is achieved only in the quasistatic limit. The worked computation Worked computation 9.30 gives an explicit protocol that reaches $k_\mathrm{B}T[\ln 2 - H(\delta)]$ and computes the extra cost of finishing in finite time.

- Randomization runs the other way. Turning a known bit into a fair coin raises the entropy by $k_\mathrm{B}\ln 2$ and is allowed to return up to $k_\mathrm{B}T\ln 2$ of work. This is the precise sense in which drawing a random sample is thermodynamically downhill: the bath supplies the randomness, and the free energy of the ordered state can in principle be harvested. The return is available only once per known bit, and only with a quasistatic, work-extracting protocol.

- A sampler that is randomized by simply letting thermal noise act, with no work-extracting protocol, forfeits that return and produces $k_\mathrm{B}\ln 2$ of entropy per bit. A full cycle of spontaneous randomization followed by quasistatic reset then costs $k_\mathrm{B}T\ln 2$ per bit. With quasistatic work extraction on the randomization step, the cycle costs zero, consistent with the Second Law for a cyclic process.

Example 9.17 (Resetting the Output Register of a Sampler)**.** Suppose a p-bit with bias h has produced a sample s with $p(s) = e^{\beta hs}/(2\cosh\beta h)$, and the value has been copied into a symmetric one-bit register. Resetting the register to a standard value with residual error δ has, by Lemma 9.15 applied with initial occupation $p_+ = \frac{1}{2}(1+\tanh\beta h)$, the minimum cost

$$\langle\mathcal{W}\rangle \;\geq\; k_\mathrm{B}T\Big[H\big(\tfrac{1}{2}(1+\tanh\beta h)\big) - H(\delta)\Big].$$

For an unbiased p-bit this is $k_\mathrm{B}T\ln 2$. For a strongly biased one, $\beta h = 3$, the sample is $+1$ with probability 0.9975 and the bound drops to $0.017\,k_\mathrm{B}T$. The same argument applied to a register of N bits holding a sample of an energy-based model p_θ over $\{-1,+1\}^N$ (regions $\Omega_{\mathbf{s}}$, one per configuration) gives the minimum reset cost $TS[p_\theta] = \langle E_\theta\rangle_{p_\theta} - F$, the canonical entropy of Definition 1.3. The thermodynamic cost of recording and forgetting samples is set by the entropy of the model, not by the number of bits.

Example 9.18 (Compressing the Thermal Distribution of an RC Node)**.** Landauer's bound is not specific to discrete memories. Consider a single node of the RC sampler (the coupled linear RC circuit of Chapter 1) with energy $E(v;k) = \frac{1}{2}kv^2$, where the stiffness k (an effective capacitance) is programmable, in equilibrium at stiffness k_0. Its equilibrium distribution is $\mathcal{N}(0, k_\mathrm{B}T/k)$, with

$$F(k) = \frac{k_\mathrm{B}T}{2}\ln\frac{k}{2\pi k_\mathrm{B}T}, \qquad S(k) = \frac{k_\mathrm{B}}{2}\ln\frac{2\pi e\,k_\mathrm{B}T}{k}, \qquad \langle E\rangle = \frac{k_\mathrm{B}T}{2},$$

where the entropy is the Gaussian differential entropy $\frac{k_B}{2}\ln(2\pi e\sigma^2)$ with $\sigma^2 = k_BT/k$. Raising the stiffness from k_0 to $k_1 > k_0$ narrows the distribution and lowers the entropy by $\frac{k_B}{2}\ln(k_1/k_0)$, while the mean energy stays $k_BT/2$ by equipartition. By Theorem 9.14 with equilibrium endpoints, the minimum work and the minimum heat are both $\frac{k_BT}{2}\ln(k_1/k_0) = -T\Delta S$, reached by a quasistatic stiffening. Halving the standard deviation ($k_1 = 4k_0$) costs at least $k_BT\ln 2$: compressing the thermal distribution of an analog node by a factor of two costs the same as erasing one bit. A sudden stiffening followed by relaxation instead costs $\langle \mathcal{W} \rangle = \frac{1}{2}(k_1 - k_0)\langle v^2 \rangle = \frac{k_BT}{2}(k_1/k_0 - 1)$, which exceeds $\frac{k_BT}{2}\ln(k_1/k_0)$ because $r - 1 > \ln r$ for $r > 1$; for $k_1 = 4k_0$ it is $1.5\,k_BT$.

Bennett pointed out that measurement and logically reversible computation have no minimum thermodynamic cost; only the logically irreversible step of discarding information, that is, erasure, carries the Landauer cost [10]. A computation that keeps a copy of every intermediate result can in principle run with arbitrarily small dissipation, at the price of a growing memory. The cost reappears when that memory is cleared.

Hardware Realization: Where the Landauer Cost Sits in a Thermodynamic Sampler

A p-bit held at fixed bias and fixed barrier is a system in thermal equilibrium. Its state flips at the Kramers rate of Section 7.3, and each flip changes the energy by an amount supplied or absorbed by the bath, but the average heat flow and the average entropy production are zero. Generating samples in this way costs nothing thermodynamically. The energy bill of a real sampler comes from three other places: (i) the work done to change biases and couplings, which is bounded below by $\Delta\mathcal{F}$ and exceeds it by $T\Sigma_{\text{tot}}$ at finite speed (Theorem 9.14 and Chapter 10); (ii) the logically irreversible reset of any register that records a sample, bounded below by k_BT times the entropy of the recorded value; and (iii) the static and dynamic power of the readout, clocking, and bias circuitry, which in present CMOS technology is set by device physics far above any thermodynamic bound.

For scale, $k_BT\ln 2 \approx 2.9 \times 10^{-21}$ J at 300 K. The preface estimates that charging a 1 fF digital gate to 0.7 V dissipates about $6 \times 10^4\,k_BT$, and a 16-bit floating-point multiply-accumulate on a GPU about 10^8–$10^9\,k_BT$. These are order-of-magnitude figures. They show that present hardware, digital or thermodynamic, operates several orders of magnitude above the Landauer floor, and that the advantage of thermodynamic sampling comes from replacing the synthesis of randomness by its direct physical generation, not from approaching $k_BT\ln 2$. The detailed accounting is the subject of Section 17.4.

9.4 Measurement, Feedback, and Mutual Information

Maxwell imagined a being that watches individual molecules and opens a trapdoor only for fast ones, sorting a gas into hot and cold halves without doing work. In the language of this chapter, the demon uses the information it gathers to choose a protocol that harvests the negative-entropy-production trajectories of Proposition 9.4. The fluctuation theorems of the previous sections assume a protocol fixed in advance. We now allow the protocol to depend on the outcome of a measurement and find how much the Second Law changes. The answer, due to Sagawa and Ueda [92], is that the change is exactly the mutual information between the system and the measurement record.

9.4.1 Measurement Model and Stochastic Mutual Information

Assumption 9.19 (Single Measurement Followed by Feedback)**.** 1. At $t = 0$ the system is in equilibrium, $\mathbf{x}_0 \sim p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_0)$, and $\mathbf{f}_{\mathrm{nc}} \equiv \mathbf{0}$.

2. At $t = 0$ a classical measurement produces an outcome y in a finite set $\mathcal{Y}$ with likelihood $p(y \mid \mathbf{x}_0)$. The measurement does not disturb $\mathbf{x}_0$.

3. For each outcome y, the controller then applies a protocol $\boldsymbol{\lambda}^{(y)}_{[0,\tau]}$ with $\boldsymbol{\lambda}^{(y)}_0 = \boldsymbol{\lambda}_0$. The work along the trajectory is $\mathcal{W} = \int_0^\tau \partial_{\boldsymbol{\lambda}} E(\mathbf{x}_t, \boldsymbol{\lambda}^{(y)}_t) \cdot \dot{\boldsymbol{\lambda}}^{(y)}_t \, \mathrm{d}t$ and the free-energy change is $\Delta F_y = F(\boldsymbol{\lambda}^{(y)}_\tau) - F(\boldsymbol{\lambda}_0)$.

The marginal probability of outcome y is $p(y) = \int \mathrm{d}^N x \, p_{\mathrm{eq}}(\mathbf{x};\boldsymbol{\lambda}_0)\, p(y \mid \mathbf{x})$. The **stochastic mutual information** between the initial state and the outcome is the random variable

$$I(\mathbf{x}_0; y) \;\equiv\; \ln \frac{p(y \mid \mathbf{x}_0)}{p(y)} \;=\; \ln \frac{p(\mathbf{x}_0, y)}{p_{\mathrm{eq}}(\mathbf{x}_0;\boldsymbol{\lambda}_0)\, p(y)}, \tag{9.25}$$

measured in nats. Its average is the mutual information

$$\mathcal{I}(X_0; Y) \;=\; \langle I \rangle \;=\; \sum_y \int \mathrm{d}^N x \; p(\mathbf{x}, y) \ln \frac{p(\mathbf{x}, y)}{p_{\mathrm{eq}}(\mathbf{x};\boldsymbol{\lambda}_0) p(y)} \;=\; D_{\mathrm{KL}}\big(p(\mathbf{x}, y) \,\|\, p_{\mathrm{eq}}(\mathbf{x};\boldsymbol{\lambda}_0) p(y)\big) \;\geq\; 0, \tag{9.26}$$

which vanishes if and only if the outcome is statistically independent of the state. For an error-free measurement of a discrete logical state, $\mathcal{I}$ equals the Shannon entropy of that state.

Notation Watch: Units of Information and Signs of Work

We measure I and $\mathcal{I}$ in nats, so the corresponding energy is $k_{\mathrm{B}}T\mathcal{I}$; one bit of information is $\mathcal{I} = \ln 2$. We keep $\mathcal{W}$ as work done *on* the system and write $\mathcal{W}_{\mathrm{ext}} = -\mathcal{W}$ for work extracted by the controller. A statement that an engine extracts $k_{\mathrm{B}}T \ln 2$ means $\langle \mathcal{W} \rangle = -k_{\mathrm{B}}T \ln 2$.

9.4.2 The Sagawa–Ueda Fluctuation Theorem

Theorem 9.20 (Fluctuation Theorem with Feedback)**.** *Under Assumption 9.19 and the hypotheses of Theorem 8.8,*

$$\left\langle e^{-\beta(\mathcal{W} - \Delta F_y) - I(\mathbf{x}_0; y)} \right\rangle \;=\; 1, \tag{9.27}$$

where the average is over $\mathbf{x}_0$, y, and the trajectory. Consequently

$$\langle \mathcal{W} \rangle \;\geq\; \langle \Delta F_y \rangle - k_{\mathrm{B}}T\, \mathcal{I}(X_0; Y), \qquad \textit{equivalently} \qquad \langle \mathcal{W}_{\mathrm{ext}} \rangle \;\leq\; -\langle \Delta F_y \rangle + k_{\mathrm{B}}T\, \mathcal{I}(X_0; Y). \tag{9.28}$$

Proof. The joint probability of $(\mathbf{x}_0, y)$ and the subsequent path is $p_{\mathrm{eq}}(\mathbf{x}_0;\boldsymbol{\lambda}_0)\, p(y \mid \mathbf{x}_0)\, \mathcal{P}^{(y)}[\mathbf{x}_{[0,\tau]} \mid \mathbf{x}_0]$, where $\mathcal{P}^{(y)}$ is the conditional path density under protocol $\boldsymbol{\lambda}^{(y)}$. Since $e^{-I} = p(y)/p(y \mid \mathbf{x}_0)$, the likelihood factor cancels:

$$\begin{aligned}
\left\langle e^{-\beta(\mathcal{W} - \Delta F_y) - I} \right\rangle \;&=\; \sum_y \int \mathrm{d}^N x_0 \; p_{\mathrm{eq}}(\mathbf{x}_0;\boldsymbol{\lambda}_0)\, p(y \mid \mathbf{x}_0)\, \frac{p(y)}{p(y \mid \mathbf{x}_0)} \int \mathcal{D}\mathbf{x}\, \mathcal{P}^{(y)}[\mathbf{x} \mid \mathbf{x}_0]\, e^{-\beta(\mathcal{W}[\mathbf{x}] - \Delta F_y)} \\
&=\; \sum_y p(y) \int \mathrm{d}^N x_0 \; p_{\mathrm{eq}}(\mathbf{x}_0;\boldsymbol{\lambda}_0) \int \mathcal{D}\mathbf{x}\, \mathcal{P}^{(y)}[\mathbf{x} \mid \mathbf{x}_0]\, e^{-\beta(\mathcal{W}[\mathbf{x}] - \Delta F_y)}.
\end{aligned}$$

For fixed y, the double integral is the average of $e^{-\beta(\mathcal{W}-\Delta F_y)}$ in an ordinary, measurement-free process that starts in $p_{\mathrm{eq}}(\cdot\,;\boldsymbol{\lambda}_0)$ and follows the fixed protocol $\boldsymbol{\lambda}^{(y)}$. By the Jarzynski equality Eq. (9.13) it equals one. Hence the whole expression is $\sum_y p(y) = 1$. The cancellation requires $p(y \mid \mathbf{x}_0) > 0$ whenever the pair occurs, which holds by definition on the support of the joint distribution.

For the inequality, Jensen's inequality gives $\exp(-\beta\,\langle \mathcal{W} - \Delta F_y\rangle - \langle I\rangle) \le 1$, so $\beta\,\langle \mathcal{W} - \Delta F_y\rangle + \mathcal{I} \ge 0$. Multiply by $k_{\mathrm{B}}T$. □

Information therefore enters the Second Law on the same footing as free energy. A controller holding $\mathcal{I}$ nats about the state can extract up to $k_{\mathrm{B}}T\mathcal{I}$ more work than any measurement-free protocol with the same endpoints. The theorem extends to measurements at intermediate times and to repeated measurements, with I replaced by the accumulated stochastic mutual information between the trajectory and the record; we do not prove this generalization here. The trajectory-level form Eq. (9.27) also shows that a protocol may extract more than $-\Delta F_y + k_{\mathrm{B}}TI$ on individual runs; only the average is bounded.

9.4.3 The Szilard Engine as a p-Bit Protocol

Szilard reduced Maxwell's demon to a single particle in a box: insert a partition, observe which side the particle is on, and let the particle push the partition quasistatically toward the empty side, extracting $k_{\mathrm{B}}T\ln 2$ from the bath. The p-bit limit of the double-well sampler gives the same engine in the variables of this book. Consider a symmetric bistable p-bit with logical state $s \in \{-1, +1\}$ and two-state energy $E(s;\varepsilon) = -\varepsilon s$, where the bias ε is under the controller's command. (For the quartic well $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$ with a high barrier, $s = \mathrm{sgn}(x)$ and $\varepsilon \approx hx_\star$ with $x_\star = \sqrt{b/a}$; see Worked computation 9.30.) At bias ε the equilibrium occupations are $p(s) = e^{\beta\varepsilon s}/(2\cosh\beta\varepsilon)$, the magnetization is $m = \tanh(\beta\varepsilon)$, and the free energy is $F(\varepsilon) = -k_{\mathrm{B}}T\ln(2\cosh\beta\varepsilon)$, as in Example 1.6.

Example 9.21 (Szilard Engine with a Noisy Measurement)**.** Start at $\varepsilon = 0$ in equilibrium, so $p(s = \pm 1) = \frac{1}{2}$, with the barrier raised so that s is frozen. Measure s through a binary symmetric channel with error probability $\delta \in [0, \frac{1}{2}]$: the outcome is $y = s$ with probability $1 - \delta$ and $y = -s$ with probability δ.

Information. By symmetry $p(y = \pm 1) = \frac{1}{2}$. The stochastic mutual information is $I = \ln[(1-\delta)/\frac{1}{2}] = \ln 2(1-\delta)$ when the measurement is correct and $I = \ln 2\delta$ when it is wrong. Its average is

$$\mathcal{I} \;=\; (1-\delta)\ln 2(1-\delta) + \delta\ln 2\delta \;=\; \ln 2 + (1-\delta)\ln(1-\delta) + \delta\ln\delta \;=\; \ln 2 - H(\delta).$$

For an error-free measurement, $\mathcal{I} = \ln 2$; for a useless one ($\delta = \frac{1}{2}$), $\mathcal{I} = 0$.

Feedback protocol. Given outcome y, the posterior is $p(s = y \mid y) = 1 - \delta$. Choose $\varepsilon_\delta > 0$ such that this posterior is the Boltzmann distribution at bias $y\varepsilon_\delta$, that is, $\tanh(\beta\varepsilon_\delta) = 1 - 2\delta$. The protocol has two steps.

1. With the barrier still raised, switch the bias suddenly from 0 to $y\varepsilon_\delta$. The state cannot move, so the work is the energy change, $\mathcal{W}_1 = E(s; y\varepsilon_\delta) - E(s;0) = -\varepsilon_\delta\, sy$. Its average is $\langle \mathcal{W}_1\rangle = -\varepsilon_\delta[(1-\delta) - \delta] = -\varepsilon_\delta\tanh(\beta\varepsilon_\delta)$.

2. Lower the barrier and return the bias quasistatically from $y\varepsilon_\delta$ to 0. The system starts this step in equilibrium at bias $y\varepsilon_\delta$ by construction of ε_δ, and a quasistatic protocol keeps it in equilibrium, so $\mathcal{W}_2 = F(0) - F(y\varepsilon_\delta) = k_{\mathrm{B}}T\ln\cosh(\beta\varepsilon_\delta)$ on average.

The total mean work is

$$\langle \mathcal{W} \rangle \;=\; -k_{\mathrm{B}}T\Big[\beta\varepsilon_\delta \tanh(\beta\varepsilon_\delta) - \ln\cosh(\beta\varepsilon_\delta)\Big].$$

To evaluate the bracket, let $z = \beta\varepsilon_\delta$ and $t = \tanh z = 1 - 2\delta$. Then $1 \pm t$ equal $2(1-\delta)$ and 2δ, so $z = \operatorname{artanh} t = \frac{1}{2}\ln\frac{1-\delta}{\delta}$ and $\cosh z = (1-t^2)^{-1/2} = [4\delta(1-\delta)]^{-1/2}$. Substituting,

$$z\tanh z - \ln\cosh z \;=\; \frac{1-2\delta}{2}\ln\frac{1-\delta}{\delta} + \frac{1}{2}\ln 4\delta(1-\delta) \;=\; \ln 2 + (1-\delta)\ln(1-\delta) + \delta\ln\delta \;=\; \ln 2 - H(\delta),$$

where we collected the coefficients of $\ln(1-\delta)$, namely $\frac{1-2\delta}{2} + \frac{1}{2} = 1-\delta$, and of $\ln\delta$, namely $-\frac{1-2\delta}{2} + \frac{1}{2} = \delta$. Therefore

$$\langle \mathcal{W}_{\mathrm{ext}} \rangle \;=\; k_{\mathrm{B}}T\big[\ln 2 - H(\delta)\big] \;=\; k_{\mathrm{B}}T\,\mathcal{I}, \tag{9.29}$$

and since the bias starts and ends at zero, $\Delta F_y = 0$. The protocol saturates the Sagawa–Ueda bound Eq. (9.28). With an error-free measurement ($\delta \to 0$, $\varepsilon_\delta \to \infty$) it extracts $k_{\mathrm{B}}T\ln 2$, Szilard's value.

The protocol of Example 9.21 satisfies the Sagawa–Ueda equality Eq. (9.27) in an unusually strong way. In the quasistatic limit, stage 2 contributes the deterministic work $k_{\mathrm{B}}T\ln\cosh(\beta\varepsilon_\delta)$ on every run, so $\mathcal{W} = -\varepsilon_\delta sy + k_{\mathrm{B}}T\ln\cosh(\beta\varepsilon_\delta)$. On a run with a correct outcome ($sy = +1$, probability $1-\delta$), $I = \ln 2(1-\delta)$ and $2(1-\delta) = e^{\beta\varepsilon_\delta}/\cosh(\beta\varepsilon_\delta)$, so

$$e^{-\beta\mathcal{W}-I} \;=\; \frac{e^{\beta\varepsilon_\delta}}{\cosh(\beta\varepsilon_\delta)} \cdot \frac{\cosh(\beta\varepsilon_\delta)}{e^{\beta\varepsilon_\delta}} \;=\; 1.$$

On a run with a wrong outcome ($sy = -1$, probability δ), $I = \ln 2\delta$ and $2\delta = e^{-\beta\varepsilon_\delta}/\cosh(\beta\varepsilon_\delta)$, and again $e^{-\beta\mathcal{W}-I} = 1$. The random variable $\beta\mathcal{W} + I$ is therefore zero on every trajectory, not only on average. This is the feedback analogue of a reversible process: an optimal feedback protocol leaves no fluctuation in the quantity whose exponential average the theorem fixes, so Jensen's inequality holds with equality and the bound is saturated.

The identity $z\tanh z - \ln\cosh z = \ln 2 - H\big(\frac{1-\tanh z}{2}\big)$ derived in the example will be used again for the bit reset in Worked computation 9.30. It is not a coincidence: the Szilard protocol run backward is a reset protocol, and Eq. (9.29) is the Landauer bound Eq. (9.24) read in reverse.

9.4.4 Closing the Cycle: the Cost of the Record

The engine of Example 9.21 ends with the system back in its initial state, but the controller now holds the outcome y, a fair coin. To run the engine again, the record must be reset to a standard value. By Corollary 9.16 applied to the memory that holds y, resetting it costs at least $k_{\mathrm{B}}TH(Y) = k_{\mathrm{B}}T\ln 2 \geq k_{\mathrm{B}}T\mathcal{I}$. More generally, Sagawa and Ueda showed that the work spent on measurement and on erasure of the record together obeys $\langle \mathcal{W}_{\mathrm{meas}} \rangle + \langle \mathcal{W}_{\mathrm{erase}} \rangle \geq k_{\mathrm{B}}T\mathcal{I}$, although either term alone can be smaller [84]. Adding this to Eq. (9.28), the net work extracted over a complete cycle of measurement, feedback, and erasure is at most zero. This is Bennett's resolution of Maxwell's demon [10]: the demon is not defeated by the cost of looking, which can be made small, but by the cost of forgetting.

Example 9.22 (Measurement by a Coupled Sensor p-Bit)**.** A measurement can itself be a physical process inside the sampler. Let a sensor p-bit $y \in \{-1, +1\}$ interact with the system p-bit s through the coupling energy $-Jsy$. Starting from $J = 0$ with both bits in equilibrium and unbiased, raise J

quasistatically to $J_0 > 0$, freeze both bits by raising their barriers, and switch the coupling back to zero with the states frozen.

The joint distribution after the first step is $p(s,y) = e^{\beta J_0 sy}/(4\cosh\beta J_0)$, so the marginal of s is uniform and $p(y \mid s) = e^{\beta J_0 sy}/(2\cosh\beta J_0)$. This is a binary symmetric channel with error probability $\delta = (1+e^{2\beta J_0})^{-1}$, that is, $\tanh(\beta J_0) = 1-2\delta$, and $\mathcal{I} = \ln 2 - H(\delta)$. The quasistatic coupling step costs $\Delta F = -k_{\mathrm{B}}T\ln\cosh(\beta J_0)$, from $Z(J) = \sum_{s,y} e^{\beta J sy} = 4\cosh\beta J$. The frozen decoupling step costs $\mathcal{W} = \int \partial_J(-Jsy)\,\mathrm{d}J = J_0 sy$, with mean $J_0\tanh(\beta J_0)$. The measurement work is therefore

$$\langle \mathcal{W}_{\mathrm{meas}} \rangle \;=\; k_{\mathrm{B}}T\big[\beta J_0\tanh(\beta J_0) - \ln\cosh(\beta J_0)\big] \;=\; k_{\mathrm{B}}T\big[\ln 2 - H(\delta)\big] \;=\; k_{\mathrm{B}}T\,\mathcal{I},$$

by the identity of Example 9.21. In this implementation the measurement costs exactly what the subsequent feedback can extract. The general bound only requires that measurement plus erasure cost at least $k_{\mathrm{B}}T\mathcal{I}$; this protocol places the whole cost on the measurement.

Hardware Realization: Conditional Clamping as Feedback

In a hardware sampler used for inference, observed variables are clamped: the bias applied to a latent p-bit is set by the value of an evidence variable, and the latent bit is then sampled from its conditional distribution. When the evidence is itself the state of another physical register, as in Example 9.22, clamping is a measurement-and-feedback operation in the sense of Assumption 9.19. The Sagawa–Ueda bound then states that the conditional protocol can save at most $k_{\mathrm{B}}T\mathcal{I}$ of work relative to an unconditional protocol with the same free-energy change, where $\mathcal{I}$ is the mutual information between the clamped register and the latent state. Conversely, a bias that is copied from one register into another creates a correlation that must eventually be erased at a cost of order $k_{\mathrm{B}}T$ per nat. For present devices these costs are far below the readout and clocking energies discussed in Section 17.4; they mark the floor that no clamping scheme can beat.

9.5 Thermodynamic Uncertainty Relations

The fluctuation theorems constrain the distribution of entropy production. A different kind of constraint links entropy production to the fluctuations of *other* observables. In a non-equilibrium steady state a circuit carries persistent currents: charge flowing through a biased transistor, net rotations of the probability current in the non-reciprocal RC sampler of Worked computation 8.12, net flips of a driven ring oscillator, ticks of a clock. Each such current fluctuates. The thermodynamic uncertainty relation (TUR), found by Barato and Seifert [8] and proved in general by Gingrich, Horowitz, Perunov, and England [41], states that the relative fluctuations of any current can be made small only by producing entropy.

9.5.1 Currents and Statement of the Relation

We work with a continuous-time Markov jump process on a finite state space $\{1,\dots,M\}$, the natural description of coupled p-bits (Eq. (7.12)) and of charge transport through mesoscopic conductors (Chapters 11 and 13). The rate from state i to state $j \neq i$ is $k_{ij} \geq 0$. We assume that $k_{ij} > 0$ if and only if $k_{ji} > 0$, that the process is irreducible, and that it starts in its unique stationary distribution $p_{\mathrm{ss}} = (p_1,\dots,p_M)$ with all $p_i > 0$. The stationary edge current is

$$j_{ij} \;\equiv\; p_i k_{ij} - p_j k_{ji} \;=\; -j_{ji}, \qquad \sum_{j\neq i} j_{ij} \;=\; 0 \text{ for every } i, \tag{9.30}$$

where the second relation is the stationary master equation. Let $N_{ij}(\tau)$ be the number of jumps from i to j during $[0,\tau]$.

Definition 9.23 (Time-Integrated Current)**.** A **current** is a random variable of the form $J_\tau = \sum_{i\neq j} d_{ij} N_{ij}(\tau)$ with antisymmetric increments $d_{ij} = -d_{ji}$. In a stationary state its mean is $\langle J_\tau \rangle = \tau \sum_{i\neq j} d_{ij} p_i k_{ij} = \tau \sum_{i<j} d_{ij} j_{ij}$.

Examples are the net number of clockwise jumps around a cycle, the net charge transferred across a junction, and the net number of $0 \to 1$ flips of a particular p-bit. Under local detailed balance, each jump $i \to j$ deposits heat $k_{\mathrm{B}} T \ln(k_{ij}/k_{ji})$ in the bath. Together with the system-entropy change $k_{\mathrm{B}} \ln(p_i/p_j)$, the total entropy produced by the jump is $k_{\mathrm{B}} \ln(p_i k_{ij}/p_j k_{ji})$, and the mean total entropy production over $[0,\tau]$ is Schnakenberg's expression [97]

$$\Sigma_{\mathrm{tot}}(\tau) \;=\; \tau k_{\mathrm{B}} \sum_{i<j} j_{ij} \ln \frac{p_i k_{ij}}{p_j k_{ji}} \;\geq\; 0. \tag{9.31}$$

Each term is non-negative, because j_{ij} and $\ln(p_i k_{ij}/p_j k_{ji})$ have the same sign. Entropy production is itself a current, with $d_{ij} = k_{\mathrm{B}} \ln(p_i k_{ij}/p_j k_{ji})$.

Theorem 9.24 (Thermodynamic Uncertainty Relation)**.** *Under the hypotheses above, every current J_τ with $\langle J_\tau \rangle \neq 0$ satisfies, for every $\tau > 0$,*

$$\frac{\mathrm{Var}(J_\tau)}{\langle J_\tau \rangle^2} \;\geq\; \frac{2k_{\mathrm{B}}}{\Sigma_{\mathrm{tot}}(\tau)}. \tag{9.32}$$

The relation is a trade-off between precision and cost. The squared relative uncertainty $\epsilon^2 = \mathrm{Var}(J_\tau)/\langle J_\tau \rangle^2$ of any current, multiplied by the entropy produced, cannot fall below $2k_{\mathrm{B}}$. We first prove it for the simplest case, where every quantity is explicit, and then prove the general case.

9.5.2 Complete Proof for the Biased Ring

Proposition 9.25 (TUR for a Biased Random Walk on a Ring)**.** *A particle hops on a ring of $L \geq 3$ sites, clockwise with rate k_+ and counterclockwise with rate k_-, independent of position. Let $J_\tau = N_+(\tau) - N_-(\tau)$ be the net number of clockwise jumps. Under local detailed balance, each clockwise jump dissipates heat $k_{\mathrm{B}} T \mathcal{A}$ with affinity $\mathcal{A} = \ln(k_+/k_-)$. Then*

$$\langle J_\tau \rangle = (k_+ - k_-)\tau, \qquad \mathrm{Var}(J_\tau) = (k_+ + k_-)\tau, \qquad \Sigma_{\mathrm{tot}}(\tau) = k_{\mathrm{B}} \mathcal{A} (k_+ - k_-)\tau, \tag{9.33}$$

and

$$\frac{\mathrm{Var}(J_\tau)}{\langle J_\tau \rangle^2} \cdot \frac{\Sigma_{\mathrm{tot}}(\tau)}{k_{\mathrm{B}}} \;=\; \mathcal{A} \coth \frac{\mathcal{A}}{2} \;\geq\; 2, \tag{9.34}$$

with equality approached only in the linear-response limit $\mathcal{A} \to 0$.

Proof. Because the rates do not depend on position, clockwise jumps occur as a Poisson process of rate k_+ and counterclockwise jumps as an independent Poisson process of rate k_-. (Equivalently, jumps of either kind occur at total rate $k_+ + k_-$, and each jump is independently clockwise with probability $k_+/(k_+ + k_-)$; thinning a Poisson process produces independent Poisson processes.) A Poisson count of mean $k\tau$ has variance $k\tau$, so

$$\langle J_\tau \rangle = k_+\tau - k_-\tau, \qquad \mathrm{Var}(J_\tau) = \mathrm{Var}(N_+) + \mathrm{Var}(N_-) = k_+\tau + k_-\tau,$$

using independence for the variance of the difference. The stationary distribution is uniform, $p_i = 1/L$, so the system entropy $-k_{\mathrm{B}} \ln p_{x_t}$ is constant and only the medium contributes. Each clockwise jump produces $k_{\mathrm{B}}\mathcal{A}$ and each counterclockwise jump produces $-k_{\mathrm{B}}\mathcal{A}$, so $\Delta s_{\mathrm{tot}} = k_{\mathrm{B}}\mathcal{A} J_\tau$ and $\Sigma_{\mathrm{tot}} = k_{\mathrm{B}}\mathcal{A} \langle J_\tau \rangle$. This proves Eq. (9.33); it also agrees with Eq. (9.31), since each of the L edges carries current $j = (k_+ - k_-)/L$.

Multiplying,

$$\frac{\mathrm{Var}(J_\tau)}{\langle J_\tau \rangle^2} \cdot \frac{\Sigma_{\mathrm{tot}}}{k_{\mathrm{B}}} \;=\; \frac{(k_+ + k_-)\tau}{(k_+ - k_-)^2 \tau^2}\, \mathcal{A}(k_+ - k_-)\tau \;=\; \mathcal{A}\, \frac{k_+ + k_-}{k_+ - k_-}.$$

With $k_+/k_- = e^{\mathcal{A}}$, $\frac{k_+ + k_-}{k_+ - k_-} = \frac{e^{\mathcal{A}}+1}{e^{\mathcal{A}}-1} = \coth \frac{\mathcal{A}}{2}$, which gives the equality in Eq. (9.34). For the inequality, set $y = \mathcal{A}/2 \neq 0$. Then $\mathcal{A} \coth(\mathcal{A}/2) = 2y \coth y$, and $y \coth y = y/\tanh y \geq 1$ because $|\tanh y| \leq |y|$ with the same sign as y. The bound $|\tanh y| \leq |y|$ follows from $\tanh 0 = 0$ and $0 < \frac{\mathrm{d}}{\mathrm{d}y} \tanh y = \operatorname{sech}^2 y \leq 1$. As $y \to 0$, $y \coth y = 1 + y^2/3 + O(y^4) \to 1$. □

Far from equilibrium, $\mathcal{A} \to \infty$ and the left side of Eq. (9.34) grows like $\mathcal{A}$: the counting statistics become those of a single Poisson process, $\mathrm{Var}(J_\tau)/\langle J_\tau \rangle \to 1$, while the entropy cost per jump grows without limit. A strongly driven clock spends far more entropy than the TUR requires. Near equilibrium the bound is tight.

The same two regimes apply to a counter of net transitions around a cycle of p-bit states held in a steady state by a constant non-reciprocal bias, which is the ring of Proposition 9.25 with the sites read as joint configurations. With a large affinity per cycle the count is nearly Poissonian, so the relative uncertainty after n net cycles is $\epsilon \approx n^{-1/2}$, while the heat grows as $n k_{\mathrm{B}} T \mathcal{A}$; the TUR would allow the same precision for $2n k_{\mathrm{B}} T$. Driving the counter harder makes it faster but not more precise per unit of dissipated heat.

Example 9.26 (A Drift–Diffusion Current Saturates the Bound)**.** Let x_t live on a circle of circumference ℓ and obey $\mathrm{d}x_t = \mu f \, \mathrm{d}t + \sqrt{2D}\, \mathrm{d}W_t$ with a constant non-conservative force f and $D = \mu k_{\mathrm{B}} T$. Take J_τ to be the net displacement $\int_0^\tau \mathrm{d}x_t$ on the universal cover (the winding distance). Then $\langle J_\tau \rangle = \mu f \tau$ and $\mathrm{Var}(J_\tau) = 2D\tau$. The stationary density is uniform, so Σ_{tot} is the mean heat divided by T; the heat is the work done by f, so $\Sigma_{\mathrm{tot}} = f \langle J_\tau \rangle / T = \mu f^2 \tau / T$. Then

$$\frac{\mathrm{Var}(J_\tau)}{\langle J_\tau \rangle^2} \cdot \frac{\Sigma_{\mathrm{tot}}}{k_{\mathrm{B}}} \;=\; \frac{2D\tau}{\mu^2 f^2 \tau^2} \cdot \frac{\mu f^2 \tau}{k_{\mathrm{B}} T} \;=\; \frac{2D}{\mu k_{\mathrm{B}} T} \;=\; 2,$$

for every f and every τ. The overdamped Langevin limit of the ring, in which jumps become infinitesimal, saturates the TUR exactly. This is the continuous counterpart of the Gaussian identity in Example 9.5: here J_τ is proportional to Δs_{tot}, and a Gaussian entropy production has $\mathrm{Var} = 2k_{\mathrm{B}}\Sigma_{\mathrm{tot}}$.

9.5.3 Proof of the General Relation

The general proof uses a statistical idea. If a small change of a parameter θ shifts the mean of J_τ, then J_τ can be used to estimate θ, and the Cramér–Rao inequality bounds the variance of any estimator by the inverse Fisher information. We choose a perturbation that scales every stationary current by $1 + \theta$ and show that its Fisher information is at most $\Sigma_{\mathrm{tot}}/(2k_{\mathrm{B}})$.

Lemma 9.27 (Cramér–Rao Inequality on Path Space)**.** *Let $\mathcal{P}_\theta$, $\theta \in (-\theta_0, \theta_0)$, be a family of path measures with common support, differentiable in θ, and let $\mathfrak{s} = \partial_\theta \ln \mathcal{P}_\theta\big|_{\theta=0}$ be the score with*

Fisher information $\mathfrak{F} = \langle \mathfrak{s}^2 \rangle_0 < \infty$. *For any observable* J *with finite variance under* $\mathcal{P}_0$ *for which differentiation under the integral sign is permitted,*

$$\left(\partial_\theta \langle J \rangle_\theta \big|_{\theta=0}\right)^2 \;\le\; \mathrm{Var}_0(J)\, \mathfrak{F}. \tag{9.35}$$

Proof. Differentiating the normalization $\int \mathcal{D}\mathbf{x}\, \mathcal{P}_\theta = 1$ gives $\langle \mathfrak{s} \rangle_0 = \int \mathcal{D}\mathbf{x}\, \mathcal{P}_0\, \partial_\theta \ln \mathcal{P}_\theta = \partial_\theta \int \mathcal{D}\mathbf{x}\, \mathcal{P}_\theta = 0$. Differentiating $\langle J \rangle_\theta = \int \mathcal{D}\mathbf{x}\, \mathcal{P}_\theta J$ gives $\partial_\theta \langle J \rangle_\theta \big|_0 = \langle J\mathfrak{s} \rangle_0 = \langle (J - \langle J \rangle_0)\mathfrak{s} \rangle_0$, where we subtracted $\langle J \rangle_0 \langle \mathfrak{s} \rangle_0 = 0$. The Cauchy–Schwarz inequality gives $\langle (J - \langle J \rangle_0)\mathfrak{s} \rangle_0^2 \le \langle (J - \langle J \rangle_0)^2 \rangle_0 \langle \mathfrak{s}^2 \rangle_0$. □

Lemma 9.28 (Fisher Information of a Stationary Jump Process)**.** *Let the rates depend smoothly on* θ, $k_{ij}(\theta) = k_{ij}(1 + \theta c_{ij}) + O(\theta^2)$, *with the initial distribution independent of* θ, *and let the process at* $\theta = 0$ *start in* p_{ss}. *Then the Fisher information of the path measure over* $[0, \tau]$ *is*

$$\mathfrak{F} \;=\; \tau \sum_{i \neq j} p_i k_{ij} c_{ij}^2. \tag{9.36}$$

Derivation. A path with jumps $i_0 \to i_1 \to \cdots \to i_n$ at times $t_1 < \cdots < t_n$ has density, relative to a θ-independent reference measure,

$$\ln \mathcal{P}_\theta \;=\; \ln p_{i_0} + \sum_{m=1}^{n} \ln k_{i_{m-1} i_m}(\theta) - \int_0^\tau \mathrm{d}t \sum_{j \neq x_t} k_{x_t j}(\theta),$$

the product of the jump rates times the survival probabilities $e^{-\int (\text{escape rate})\, \mathrm{d}t}$ between jumps. Differentiating at $\theta = 0$, using $\partial_\theta \ln k_{ij}(\theta)\big|_0 = c_{ij}$ and $\partial_\theta k_{ij}(\theta)\big|_0 = k_{ij} c_{ij}$,

$$\mathfrak{s} \;=\; \sum_{i \neq j} c_{ij} \Big[N_{ij}(\tau) - \int_0^\tau \mathrm{d}t\; k_{ij} \mathbf{1}\{x_t = i\} \Big] \;\equiv\; \sum_{i \neq j} c_{ij} \mathcal{M}_{ij}(\tau).$$

Each $\mathcal{M}_{ij}$ is the jump count of edge $i \to j$ minus its compensator. We use two standard properties of these compensated counting processes, which we take as given: (i) each $\mathcal{M}_{ij}$ is a zero-mean martingale with $\langle \mathcal{M}_{ij}(\tau)^2 \rangle = \langle \int_0^\tau k_{ij} \mathbf{1}\{x_t = i\}\, \mathrm{d}t \rangle$, because on a short interval $\mathrm{d}t$ the increment $\mathrm{d}N_{ij}$ is a Bernoulli variable with success probability $k_{ij} \mathbf{1}\{x_t = i\}\, \mathrm{d}t$; (ii) for distinct edges, $\langle \mathcal{M}_{ij}(\tau) \mathcal{M}_{kl}(\tau) \rangle = 0$, because two different jumps never occur at the same instant, so the increments are orthogonal. In the stationary state $\langle \mathbf{1}\{x_t = i\} \rangle = p_i$, so $\langle \mathfrak{s}^2 \rangle = \sum_{i \neq j} c_{ij}^2 \tau p_i k_{ij}$. □

Proof of Theorem 9.24. Step 1: the perturbation. Define, for each edge with $k_{ij} > 0$,

$$k_{ij}(\theta) \;=\; k_{ij}(1 + \theta c_{ij}), \qquad c_{ij} \;\equiv\; \frac{j_{ij}}{p_i k_{ij} + p_j k_{ji}} \;=\; -c_{ji}. \tag{9.37}$$

For $|\theta| < 1$ these rates are positive, since $|c_{ij}| \le 1$. The perturbed stationary edge currents, evaluated at the *unperturbed* distribution p_{ss}, are

$$p_i k_{ij}(\theta) - p_j k_{ji}(\theta) \;=\; j_{ij} + \theta(c_{ij} p_i k_{ij} - c_{ji} p_j k_{ji}) \;=\; j_{ij} + \theta c_{ij}(p_i k_{ij} + p_j k_{ji}) \;=\; (1 + \theta)\, j_{ij}.$$

By Eq. (9.30), $\sum_{j \neq i} (1 + \theta) j_{ij} = 0$ for every i, so p_{ss} remains stationary for every θ, and every edge current is multiplied by $1 + \theta$. Starting the perturbed process in p_{ss}, the mean of any current is $\langle J_\tau \rangle_\theta = \tau \sum_{i<j} d_{ij}(1 + \theta) j_{ij} = (1 + \theta) \langle J_\tau \rangle_0$, hence $\partial_\theta \langle J_\tau \rangle_\theta = \langle J_\tau \rangle_0$.

Step 2: the Fisher information. By Lemma 9.28, grouping the terms (i, j) and (j, i) and using $c_{ji}^2 = c_{ij}^2$,

$$\mathfrak{F} \;=\; \tau \sum_{i<j} c_{ij}^2 (p_i k_{ij} + p_j k_{ji}) \;=\; \tau \sum_{i<j} \frac{j_{ij}^2}{p_i k_{ij} + p_j k_{ji}}.$$

Step 3: comparison with entropy production. For $a, b > 0$ we claim

$$\frac{(a-b)^2}{a+b} \;\le\; \frac{1}{2}(a-b)\ln\frac{a}{b}. \tag{9.38}$$

Write $r = a/b$ and divide by $b > 0$; the claim becomes $(r-1)\big[\ln r - 2(r-1)/(r+1)\big] \ge 0$. The function $g(r) = \ln r - 2(r-1)/(r+1)$ has $g(1) = 0$ and $g'(r) = \frac{1}{r} - \frac{4}{(r+1)^2} = \frac{(r-1)^2}{r(r+1)^2} \ge 0$, so $g(r)$ has the sign of $r-1$, and the product is non-negative. Applying Eq. (9.38) with $a = p_i k_{ij}$, $b = p_j k_{ji}$ to each edge and comparing with Eq. (9.31),

$$\mathfrak{F} \;\le\; \frac{\tau}{2}\sum_{i<j} j_{ij}\ln\frac{p_i k_{ij}}{p_j k_{ji}} \;=\; \frac{\Sigma_{\mathrm{tot}}(\tau)}{2k_{\mathrm{B}}}.$$

Step 4: Cramér–Rao. By Lemma 9.27 and Steps 1–3,

$$\langle J_\tau\rangle^2 \;=\; \Big(\partial_\theta \langle J_\tau\rangle_\theta\Big)^2 \;\le\; \mathrm{Var}(J_\tau)\,\mathfrak{F} \;\le\; \mathrm{Var}(J_\tau)\,\frac{\Sigma_{\mathrm{tot}}(\tau)}{2k_{\mathrm{B}}}.$$

Dividing by $\langle J_\tau\rangle^2\,\Sigma_{\mathrm{tot}}(\tau)/(2k_{\mathrm{B}}) > 0$ gives Eq. (9.32). □

The proof shows where the bound can be tight. Inequality Eq. (9.38) is an equality only when $a = b$, so on a lattice of discrete jumps the TUR is saturated only in the linear-response limit, as in Proposition 9.25. For overdamped Langevin dynamics the same argument works with the virtual drift $\mathbf{a}_\theta = \mathbf{a} + \theta\,\mathbf{v}_{\mathrm{ss}}$, where $\mathbf{v}_{\mathrm{ss}} = \mathbf{J}_{\mathrm{ss}}/p_{\mathrm{ss}}$ is the stationary local velocity of Section 8.5. The perturbed current is $(1+\theta)\mathbf{J}_{\mathrm{ss}}$, still divergence-free, so p_{ss} stays stationary; by Girsanov's theorem (Section 4.5) the Fisher information is $\mathfrak{F} = \frac{\tau}{2}\int \mathrm{d}^N x\, p_{\mathrm{ss}}\,\mathbf{v}_{\mathrm{ss}}^{\top}\mathbf{D}^{-1}\mathbf{v}_{\mathrm{ss}} = \Sigma_{\mathrm{tot}}/(2k_{\mathrm{B}})$ by Eq. (8.47), with equality instead of inequality Eq. (9.38). The only inequality left is Cauchy–Schwarz, which is why the drift–diffusion current of Example 9.26 saturates the bound.

Remark 9.29 (The Large-Deviation Route and the Limits of the TUR)**.** Gingrich, Horowitz, Perunov, and England proved the long-time version of Eq. (9.32) by a different route [41]. For $\tau \to \infty$, the empirical current J_τ/τ obeys a large-deviation principle $\mathbb{P}(J_\tau/\tau \approx \jmath) \asymp e^{-\tau\mathcal{I}(\jmath)}$. Using the level-2.5 large-deviation functional for empirical densities and edge currents, they constructed an explicit quadratic upper bound $\mathcal{I}(\jmath) \le \frac{(\jmath-\bar\jmath)^2}{4\bar\jmath^2}\,\frac{\sigma}{k_{\mathrm{B}}}$, where $\bar\jmath = \langle J_\tau\rangle/\tau$ and $\sigma = \Sigma_{\mathrm{tot}}/\tau$. Since the curvature of $\mathcal{I}$ at its minimum is the inverse of the asymptotic variance rate, $\mathcal{I}''(\bar\jmath) = \lim_{\tau\to\infty}\tau/\mathrm{Var}(J_\tau)$, comparing curvatures of the two sides yields Eq. (9.32). The quadratic bound is stronger than the TUR, because it also bounds the probability of large current fluctuations. We do not reproduce the level-2.5 construction.

The hypotheses matter. The TUR in the form Eq. (9.32) can fail for underdamped dynamics with momentum-odd variables, for systems in magnetic fields, for time-dependent (periodically driven) protocols, and for processes that do not start in the stationary state. Modified bounds exist in each case.

9.5.4 Precision Costs Entropy: Clocks and Counters

Rearranged, the TUR is a lower bound on dissipation:

$$\Sigma_{\mathrm{tot}}(\tau) \;\ge\; \frac{2k_{\mathrm{B}}}{\epsilon^2}, \qquad \epsilon^2 \;\equiv\; \frac{\mathrm{Var}(J_\tau)}{\langle J_\tau\rangle^2}. \tag{9.39}$$

A clock is a non-equilibrium system whose ticks form a current. Over a window containing $\langle J_\tau \rangle = n$ ticks, write the Fano factor as $\mathfrak{f} = \mathrm{Var}(J_\tau)/\langle J_\tau \rangle$, so $\epsilon^2 = \mathfrak{f}/n$. Then Eq. (9.39) gives a minimum heat per tick,

$$\frac{T\Sigma_{\mathrm{tot}}(\tau)}{n} \geq \frac{2k_{\mathrm{B}}T}{\mathfrak{f}}. \tag{9.40}$$

A clock with Poissonian ticks ($\mathfrak{f} = 1$) must dissipate at least $2k_{\mathrm{B}}T$ per tick, and a clock ten times more regular in variance ($\mathfrak{f} = 0.1$) at least $20k_{\mathrm{B}}T$ per tick. To measure a time interval to relative precision $\epsilon = 10^{-2}$ requires $\Sigma_{\mathrm{tot}} \geq 2 \times 10^4\, k_{\mathrm{B}}$, a minimum heat of $2 \times 10^4\, k_{\mathrm{B}}T \approx 8 \times 10^{-17}\,\mathrm{J}$ at 300 K, however long the interval.

Hardware Realization: Precision, Throughput, and Thermodynamic Inference in Non-Equilibrium Circuits

Two uses of the TUR are relevant to stochastic hardware. First, any on-chip process that must deliver a precise count, such as a ring oscillator used as a timing reference, a counter of accepted samples, or a charge pump delivering a fixed charge, is a current in a non-equilibrium steady state and pays at least Eq. (9.40). The bound is far below present device energies, but it sets the scaling: halving the relative timing jitter ϵ quadruples the minimum dissipation. Second, the TUR can be read in reverse as a lower bound on dissipation from observed statistics. If a circuit is run in a steady state and the mean and variance of any current are measured, for example the net rotation of the voltage vector in the non-reciprocal RC sampler of Worked computation 8.12, then $\Sigma_{\mathrm{tot}} \geq 2k_{\mathrm{B}} \langle J_\tau \rangle^2 / \mathrm{Var}(J_\tau)$ without any model of the internal dynamics. This kind of thermodynamic inference gives a model-independent check on the housekeeping heat computed in Chapter 8 and can be applied to calibration data of the kind discussed in Chapter 18.

9.6 Worked Computation: Work Statistics of a Driven RC Node and a Reset p-Bit

We now test every result of the chapter on two systems that can be solved exactly. The first is a single node of the RC sampler whose bias source is ramped linearly; it has a Gaussian work distribution and serves as the control case. The second is the p-bit limit of the double-well sampler, reduced to two states, which we tilt suddenly, reset quasistatically, and reset in finite time.

Worked computation 9.30 (Work Distributions, Jarzynski Checks, and Landauer Heat for a Dragged RC Node and a Reset p-Bit)**. Part 1: the driven RC node.** Take the single-node circuit of Example 8.3: a capacitor C connected through a conductance G to a programmable source b_t, with energy $E(v, b) = \frac{1}{2}C(v - b)^2$ and Langevin equation $C\,\mathrm{d}v_t = -G(v_t - b_t)\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}TG}\,\mathrm{d}W_t$. The node starts in equilibrium at b_0, so $v_0 - b_0 \sim \mathcal{N}(0, k_{\mathrm{B}}T/C)$, and the source is ramped at constant rate $\dot{b}_t = u$ for $0 \leq t \leq \tau$. The relaxation time is $\tau_{RC} = C/G$.

Equation for the lag. Let $y_t = v_t - b_t$ be the voltage across the resistor. Dividing the Langevin equation by C and subtracting $\mathrm{d}b_t = u\,\mathrm{d}t$,

$$\mathrm{d}y_t = -\frac{y_t}{\tau_{RC}}\,\mathrm{d}t - u\,\mathrm{d}t + \frac{\sqrt{2k_{\mathrm{B}}TG}}{C}\,\mathrm{d}W_t. \tag{9.41}$$

The work done by the source is, by Eq. (8.31), $\mathrm{d}\mathcal{W} = \partial_b E\,\mathrm{d}b = -Cy_t u\,\mathrm{d}t$, so

$$\mathcal{W} = -Cu\int_0^\tau y_t\,\mathrm{d}t. \tag{9.42}$$

Equation Eq. (9.41) is linear with Gaussian noise and a Gaussian initial condition, so y_t is a Gaussian process, and $\mathcal{W}$, a linear functional of it, is exactly Gaussian. We need its mean and variance.

Mean. Taking expectations, $\frac{\mathrm{d}}{\mathrm{d}t}\langle y_t\rangle = -\langle y_t\rangle/\tau_{RC} - u$ with $\langle y_0\rangle = 0$, whose solution is

$$\langle y_t\rangle \;=\; -u\tau_{RC}\bigl(1 - e^{-t/\tau_{RC}}\bigr). \tag{9.43}$$

The node voltage trails the source by $u\tau_{RC}$ once the transient has died out. Substituting into Eq. (9.42),

$$\langle \mathcal{W}\rangle \;=\; Cu^2\tau_{RC}\int_0^\tau \bigl(1 - e^{-t/\tau_{RC}}\bigr)\,\mathrm{d}t \;=\; Cu^2\tau_{RC}\Bigl[\tau - \tau_{RC}\bigl(1 - e^{-\tau/\tau_{RC}}\bigr)\Bigr]. \tag{9.44}$$

Variance. The fluctuation $\delta y_t = y_t - \langle y_t\rangle$ obeys $\mathrm{d}\,\delta y_t = -\delta y_t\,\mathrm{d}t/\tau_{RC} + (\sqrt{2k_{\mathrm{B}}TG}/C)\,\mathrm{d}W_t$, the Ornstein–Uhlenbeck equation without the drive. Its stationary variance is $\frac{2k_{\mathrm{B}}TG}{C^2}\cdot\frac{\tau_{RC}}{2} = \frac{k_{\mathrm{B}}T}{C}$, which is also its initial variance, so δy_t is stationary from $t = 0$ with covariance $\langle \delta y_t\,\delta y_s\rangle = \frac{k_{\mathrm{B}}T}{C}e^{-|t-s|/\tau_{RC}}$ (Worked computation 5.33). Hence

$$\mathrm{Var}(\mathcal{W}) \;=\; C^2u^2\int_0^\tau\!\!\int_0^\tau \langle \delta y_t\,\delta y_s\rangle\,\mathrm{d}s\,\mathrm{d}t \;=\; Ck_{\mathrm{B}}Tu^2\int_0^\tau\!\!\int_0^\tau e^{-|t-s|/\tau_{RC}}\,\mathrm{d}s\,\mathrm{d}t.$$

The integrand is symmetric under $t \leftrightarrow s$, so the double integral is twice the integral over $s < t$:

$$2\int_0^\tau \mathrm{d}t\int_0^t e^{-(t-s)/\tau_{RC}}\,\mathrm{d}s \;=\; 2\int_0^\tau \tau_{RC}\bigl(1 - e^{-t/\tau_{RC}}\bigr)\,\mathrm{d}t \;=\; 2\tau_{RC}\Bigl[\tau - \tau_{RC}\bigl(1 - e^{-\tau/\tau_{RC}}\bigr)\Bigr].$$

Comparing with Eq. (9.44),

$$\mathrm{Var}(\mathcal{W}) \;=\; 2k_{\mathrm{B}}T\,\langle \mathcal{W}\rangle\,, \tag{9.45}$$

exactly, for every u, τ, and τ_{RC}.

Fluctuation theorems. The partition function $Z(b) = \int \mathrm{d}v\, e^{-\beta C(v-b)^2/2} = \sqrt{2\pi k_{\mathrm{B}}T/C}$ does not depend on b, so $\Delta F = 0$. For the Gaussian work, the Jarzynski average is

$$\bigl\langle e^{-\beta\mathcal{W}}\bigr\rangle \;=\; \exp\Bigl(-\beta\,\langle \mathcal{W}\rangle + \frac{\beta^2}{2}\,\mathrm{Var}(\mathcal{W})\Bigr) \;=\; \exp\bigl(-\beta\,\langle \mathcal{W}\rangle + \beta\,\langle \mathcal{W}\rangle\bigr) \;=\; 1 \;=\; e^{-\beta\Delta F},$$

and the cumulant expansion Eq. (9.16) terminates: $\langle \mathcal{W}_{\mathrm{diss}}\rangle = \langle \mathcal{W}\rangle = \frac{\beta}{2}\,\mathrm{Var}(\mathcal{W})$. The reverse protocol ramps the source down at rate $-u$; by the reflection $y \to -y$ it has the same work distribution, $P_R = P_F = \mathcal{N}(\langle \mathcal{W}\rangle, 2k_{\mathrm{B}}T\,\langle \mathcal{W}\rangle)$. The Crooks ratio is then

$$\frac{P_F(\mathcal{W})}{P_R(-\mathcal{W})} \;=\; \exp\left(\frac{-(\mathcal{W} - \langle \mathcal{W}\rangle)^2 + (\mathcal{W} + \langle \mathcal{W}\rangle)^2}{4k_{\mathrm{B}}T\,\langle \mathcal{W}\rangle}\right) \;=\; \exp\left(\frac{4\mathcal{W}\,\langle \mathcal{W}\rangle}{4k_{\mathrm{B}}T\,\langle \mathcal{W}\rangle}\right) \;=\; e^{\beta\mathcal{W}},$$

in agreement with Eq. (9.11) for $\Delta F = 0$. The fraction of runs in which the source receives energy from the node ($\mathcal{W} < 0$) is, by Example 9.5 with Σ_{tot} replaced by $\langle \mathcal{W}\rangle/T$, $\mathbb{P}(\mathcal{W} < 0) = \frac{1}{2}\,\mathrm{erfc}\bigl(\frac{1}{2}\sqrt{\beta\,\langle \mathcal{W}\rangle}\bigr)$.

Limits. For a fast ramp, $\tau \ll \tau_{RC}$, expanding $1 - e^{-\tau/\tau_{RC}} = \tau/\tau_{RC} - \tau^2/(2\tau_{RC}^2) + O(\tau^3)$ gives $\langle \mathcal{W}\rangle \approx \frac{1}{2}C(u\tau)^2$: the source jumps by $\Delta b = u\tau$ before the node can respond, and the work is the electrostatic energy of the sudden offset. For a slow ramp, $\tau \gg \tau_{RC}$, $\langle \mathcal{W}\rangle \approx Cu^2\tau_{RC}\tau = (Cu)^2\tau/G$, the Joule heat $I^2R\tau$ of the charging current $I = Cu$. This is the minimum entropy production for translating a Gaussian by Δb in time τ derived in Section 10.5: with $\mathcal{W}_2 = u\tau$ and diffusion coefficient $D = k_{\mathrm{B}}TG/C^2$ for the voltage, $k_{\mathrm{B}}T\mathcal{W}_2^2/(D\tau) = C^2u^2\tau/G$.

Work versus entropy production. Here $\langle \mathcal{W} \rangle - \Delta F$ is not equal to $T\Sigma_{\mathrm{tot}}$. At $t = \tau$ the density is $\mathcal{N}(b_\tau + \langle y_\tau \rangle, k_{\mathrm{B}}T/C)$, a shifted copy of $p_{\mathrm{eq}}(\cdot\,; b_\tau)$. The divergence between two Gaussians of equal variance $k_{\mathrm{B}}T/C$ whose means differ by $\langle y_\tau \rangle$ is $C\langle y_\tau \rangle^2/(2k_{\mathrm{B}}T)$, so by Eq. (9.19),

$$T\Sigma_{\mathrm{tot}}(\tau) \;=\; \langle \mathcal{W} \rangle - \frac{C}{2}\langle y_\tau \rangle^2 \;=\; \langle \mathcal{W} \rangle - \frac{C}{2}u^2\tau_{RC}^2(1 - e^{-\tau/\tau_{RC}})^2.$$

The subtracted term is the free energy stored in the final lag. It is dissipated later, as the node catches up with the source, unless some later protocol recovers it. For the fast ramp the two terms cancel to leading order: a sudden switch produces almost no entropy during the switch itself, and the whole $\frac{1}{2}C(u\tau)^2$ is dissipated in the subsequent relaxation.

Numbers. Take $C = 1\,\mathrm{fF}$ and $G = 10\,\mathrm{nS}$ ($\tau_{RC} = 100\,\mathrm{ns}$) at $T = 300\,\mathrm{K}$, where the thermal voltage noise is $\sqrt{k_{\mathrm{B}}T/C} = 2.0\,\mathrm{mV}$ rms. Ramp the source by $\Delta b = 10\,\mathrm{mV}$ in $\tau = 1\,\mu\mathrm{s} = 10\,\tau_{RC}$ ($u = 10^4\,\mathrm{V/s}$). Then Eq. (9.44) gives $\langle \mathcal{W} \rangle = 9.0 \times 10^{-21}\,\mathrm{J} = 2.17\,k_{\mathrm{B}}T$, with standard deviation $\sqrt{2 \times 2.17}\,k_{\mathrm{B}}T = 2.08\,k_{\mathrm{B}}T$. The probability of negative work is $\frac{1}{2}\operatorname{erfc}(\frac{1}{2}\sqrt{2.17}) = 0.15$. The lag free energy is $0.12\,k_{\mathrm{B}}T$, so $T\Sigma_{\mathrm{tot}}(\tau) = 2.05\,k_{\mathrm{B}}T$. On one run in seven the source gains energy from the thermal fluctuations of the node; on average it pays about $2\,k_{\mathrm{B}}T$.

Part 2: two-state reduction of the bistable p-bit. The quartic double well $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$ has, at $h = 0$, minima at $\pm x_\star$ with $x_\star = \sqrt{b/a}$, curvature $E''(\pm x_\star) = 2b$, and barrier $\Delta E = b^2/(4a)$. For a small tilt the minima shift by $h/(2b)$ and their energies become $-\frac{b^2}{4a} \mp hx_\star - \frac{h^2}{4b} + O(h^3)$. The first-order term distinguishes the wells; the second-order term is common to both and drops out of all differences. When $\beta\Delta E \gg 1$, the coarse-graining of Section 7.1 reduces the dynamics to a two-state process with logical variable $s = \mathrm{sgn}(x) \in \{-1, +1\}$ and effective energy

$$E(s; \varepsilon) \;=\; -\varepsilon s, \qquad \varepsilon \;\equiv\; hx_\star, \tag{9.46}$$

up to an s-independent constant. The work increment of the continuous model, $\mathrm{d}\mathcal{W} = \partial_h E\,\mathrm{d}h = -x\,\mathrm{d}h$, becomes $\mathrm{d}\mathcal{W} = -x_\star s\,\mathrm{d}h = -s\,\mathrm{d}\varepsilon$, consistent with Eq. (9.46); we neglect the intra-well spread of x around $\pm x_\star$. The reduction requires the tilt to leave both wells intact, $h < h_c = \frac{2}{3\sqrt{3}}\,b^{3/2}a^{-1/2}$, that is $\varepsilon < \frac{8}{3\sqrt{3}}\Delta E \approx 1.54\,\Delta E$.

For the inter-well dynamics we take Glauber rates

$$k_{-\to+} \;=\; \frac{\lambda}{2}(1 + \tanh\beta\varepsilon), \qquad k_{+\to-} \;=\; \frac{\lambda}{2}(1 - \tanh\beta\varepsilon), \tag{9.47}$$

which obey detailed balance, $k_{-\to+}/k_{+\to-} = e^{2\beta\varepsilon}$, with a total rate $\lambda = k_{-\to+} + k_{+\to-}$ that we treat as a separate control knob: $\lambda > 0$ when the barrier is lowered, $\lambda \to 0$ when it is raised and the bit is frozen. (Kramers rates from Section 7.3 split the tilt between forward and backward barriers, which makes λ depend on ε; this changes the finite-time numbers below but not the quasistatic ones.) The magnetization $m = p_+ - p_-$ then obeys, from Eq. (7.12),

$$\dot{m} \;=\; 2\dot{p}_+ \;=\; k_{-\to+}(1 - m) - k_{+\to-}(1 + m) \;=\; -\lambda\big[m - \tanh(\beta\varepsilon)\big].$$

We also assume that raising and lowering the barrier at fixed ε does no net work on the logical degree of freedom, because it moves both wells identically.

Part 3: sudden tilt and the exact two-point work distribution. Start in equilibrium at $\varepsilon = 0$, so $s = \pm 1$ with probability $\frac{1}{2}$, and switch the bias instantaneously to $\varepsilon_1 > 0$. The state cannot change during the switch, so $\mathcal{W} = E(s; \varepsilon_1) - E(s; 0) = -\varepsilon_1 s$, and

$$P_F(\mathcal{W}) \;=\; \tfrac{1}{2}\,\delta(\mathcal{W} + \varepsilon_1) + \tfrac{1}{2}\,\delta(\mathcal{W} - \varepsilon_1).$$

The free-energy change is $\Delta F = -k_{\mathrm{B}}T\ln(2\cosh\beta\varepsilon_1) + k_{\mathrm{B}}T\ln 2 = -k_{\mathrm{B}}T\ln\cosh\beta\varepsilon_1$. The Jarzynski check is immediate: $\left\langle e^{-\beta\mathcal{W}}\right\rangle = \frac{1}{2}(e^{\beta\varepsilon_1} + e^{-\beta\varepsilon_1}) = \cosh\beta\varepsilon_1 = e^{-\beta\Delta F}$.

For the Crooks check, the reverse process starts in equilibrium at ε_1, where $p(s) = e^{\beta\varepsilon_1 s}/(2\cosh\beta\varepsilon_1)$, and switches suddenly to 0, with work $\tilde{\mathcal{W}} = 0 - (-\varepsilon_1 s) = \varepsilon_1 s$. At $\mathcal{W} = -\varepsilon_1$ (forward runs with $s = +1$), $P_R(-\mathcal{W}) = P_R(\tilde{\mathcal{W}} = \varepsilon_1) = e^{\beta\varepsilon_1}/(2\cosh\beta\varepsilon_1)$, so

$$\frac{P_F(-\varepsilon_1)}{P_R(+\varepsilon_1)} \;=\; \frac{1/2}{e^{\beta\varepsilon_1}/(2\cosh\beta\varepsilon_1)} \;=\; e^{-\beta\varepsilon_1}\cosh\beta\varepsilon_1 \;=\; e^{\beta(-\varepsilon_1-\Delta F)},$$

as Eq. (9.11) requires; the case $\mathcal{W} = +\varepsilon_1$ is identical with $\varepsilon_1 \to -\varepsilon_1$ in the exponentials.

The dissipated work is $\langle\mathcal{W}_{\mathrm{diss}}\rangle = 0 - \Delta F = k_{\mathrm{B}}T\ln\cosh\beta\varepsilon_1$. The cumulants of the symmetric two-point distribution are $\kappa_2 = \varepsilon_1^2$, $\kappa_3 = 0$, and $\kappa_4 = \langle\mathcal{W}^4\rangle - 3\langle\mathcal{W}^2\rangle^2 = -2\varepsilon_1^4$, so the first two terms of Eq. (9.16) are $\frac{\beta}{2}\varepsilon_1^2 + \frac{\beta^3}{24}(-2\varepsilon_1^4) = k_{\mathrm{B}}T\left[\frac{x^2}{2} - \frac{x^4}{12}\right]$ with $x = \beta\varepsilon_1$. This is the Taylor series $\ln\cosh x = \frac{x^2}{2} - \frac{x^4}{12} + O(x^6)$, as it must be. The distribution is far from Gaussian, and the Gaussian estimate $\frac{\beta}{2}\mathrm{Var}(\mathcal{W})$ overestimates the dissipation once $x \gtrsim 1$. Half of all runs have $\mathcal{W} = -\varepsilon_1 < \Delta F$, which is consistent with the tail bound Eq. (9.12): for $\zeta = \varepsilon_1 + \Delta F$ the bound is $e^{-\beta\zeta} = e^{-x}\cosh x = \frac{1}{2}(1 + e^{-2x}) \geq \frac{1}{2}$.

Part 4: quasistatic reset to $s = +1$. The reset protocol has four stages (Fig. 9.2):

(i) lower the barrier ($\lambda > 0$) at $\varepsilon = 0$;

(ii) ramp the bias from 0 to $\varepsilon_{\max}$ quasistatically;

(iii) raise the barrier ($\lambda \to 0$), freezing s;

(iv) ramp the bias from $\varepsilon_{\max}$ back to 0 with s frozen.

Stages (i) and (iii) cost no work by assumption. In stage (ii) the bit stays in equilibrium, and its many flips make the work self-averaging, so on every run $\mathcal{W}_{\mathrm{ii}} \to F(\varepsilon_{\max}) - F(0) = -k_{\mathrm{B}}T\ln\cosh\beta\varepsilon_{\max}$ as the ramp time tends to infinity. In stage (iv), $\mathcal{W}_{\mathrm{iv}} = -\int_{\varepsilon_{\max}}^{0} s\,\mathrm{d}\varepsilon = \varepsilon_{\max}s$, where s is the frozen state, equal to $+1$ with probability $1-\delta$ and to -1 with probability δ, with

$$\delta \;=\; \frac{1 - \tanh\beta\varepsilon_{\max}}{2} \;=\; \frac{1}{1 + e^{2\beta\varepsilon_{\max}}}.$$

The exact work distribution of the quasistatic reset is therefore two-point,

$$P(\mathcal{W}) \;=\; (1-\delta)\,\delta(\mathcal{W} - \mathcal{W}_+) + \delta\,\delta(\mathcal{W} - \mathcal{W}_-), \qquad \mathcal{W}_\pm \;=\; \pm\varepsilon_{\max} - k_{\mathrm{B}}T\ln\cosh\beta\varepsilon_{\max}. \tag{9.48}$$

Its mean is

$$\langle\mathcal{W}\rangle \;=\; \varepsilon_{\max}\tanh(\beta\varepsilon_{\max}) - k_{\mathrm{B}}T\ln\cosh(\beta\varepsilon_{\max}) \;=\; k_{\mathrm{B}}T\big[\ln 2 - H(\delta)\big],$$

by the identity derived in Example 9.21 with $z = \beta\varepsilon_{\max}$. This equals the lower bound Eq. (9.24): the quasistatic protocol saturates Landauer's bound for every error probability δ. The energy function is the same at the start and the end, so the averaged First Law gives $\langle\mathcal{Q}\rangle = \langle\mathcal{W}\rangle$: the work leaves as heat.

The Jarzynski equality also holds for this cycle, since it starts in equilibrium at $\varepsilon = 0$ and ends with the same energy function, so $\Delta F = 0$. Using $1 - \delta = e^{\beta\varepsilon_{\max}}/(2\cosh\beta\varepsilon_{\max})$ and $\delta = e^{-\beta\varepsilon_{\max}}/(2\cosh\beta\varepsilon_{\max})$,

$$\left\langle e^{-\beta\mathcal{W}}\right\rangle \;=\; \frac{e^{\beta\varepsilon_{\max}}}{2\cosh\beta\varepsilon_{\max}}\,e^{-\beta\varepsilon_{\max}}\cosh\beta\varepsilon_{\max} + \frac{e^{-\beta\varepsilon_{\max}}}{2\cosh\beta\varepsilon_{\max}}\,e^{+\beta\varepsilon_{\max}}\cosh\beta\varepsilon_{\max} \;=\; \frac{1}{2} + \frac{1}{2} \;=\; 1.$$

Half of the exponential average comes from the runs that ended in the wrong state, which occur with probability δ and return work $|\mathcal{W}_-| \approx 2\varepsilon_{\max}$. For a reset with $\beta\varepsilon_{\max} = 7$ ($\delta = 8.3 \times 10^{-7}$), a direct Monte Carlo estimate of $\left\langle e^{-\beta\mathcal{W}} \right\rangle$ from 2×10^5 runs is expected to contain only 0.17 failed runs, so with probability about 0.84 it contains none and returns 0.50, not 1. This is the sampling problem of the heuristic in Section 9.2 in its sharpest form: the equality is exact, but its estimator is controlled by events of probability δ.

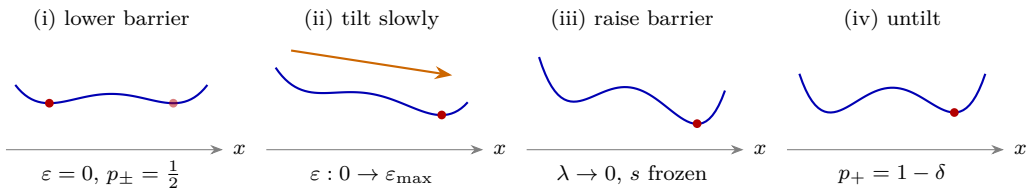


Figure 9.2: The four-stage reset of a bistable p-bit in the quartic double well $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$. In stage (ii) the bit follows the tilt in equilibrium and work $F(\varepsilon_{\max}) - F(0) < 0$ is extracted; in stage (iv) the frozen bit is lifted back and work $\varepsilon_{\max}\tanh(\beta\varepsilon_{\max})$ is done on it on average. The net cost $k_{\mathrm{B}}T[\ln 2 - H(\delta)]$ equals the Landauer bound.

Part 5: finite-time reset. Now perform stage (ii) as a linear ramp $\varepsilon_t = \varepsilon_{\max}t/\tau$ of duration τ, starting from $m(0) = 0$. The mean work in stage (ii) is $\langle\mathcal{W}_{\mathrm{ii}}\rangle = -\int_0^\tau m(t)\,\dot{\varepsilon}_t\,\mathrm{d}t$, and its excess over the quasistatic value $-\int_0^\tau m_{\mathrm{eq}}(\varepsilon_t)\dot{\varepsilon}_t\,\mathrm{d}t$, where $m_{\mathrm{eq}}(\varepsilon) = \tanh\beta\varepsilon$, is

$$\langle\mathcal{W}_{\mathrm{ii}}\rangle - \Delta F_{\mathrm{ii}} \;=\; -\int_0^\tau \big[m(t) - m_{\mathrm{eq}}(\varepsilon_t)\big]\dot{\varepsilon}_t\,\mathrm{d}t.$$

When $\lambda\tau \gg \beta\varepsilon_{\max}$ the magnetization stays close to equilibrium and we can solve $\dot{m} = -\lambda(m - m_{\mathrm{eq}})$ iteratively: $m = m_{\mathrm{eq}} - \lambda^{-1}\dot{m} = m_{\mathrm{eq}} - \lambda^{-1}\frac{\mathrm{d}m_{\mathrm{eq}}}{\mathrm{d}t} + O(\lambda^{-2})$. With $\frac{\mathrm{d}m_{\mathrm{eq}}}{\mathrm{d}t} = \beta\,\mathrm{sech}^2(\beta\varepsilon_t)\,\dot{\varepsilon}_t$,

$$\langle\mathcal{W}_{\mathrm{ii}}\rangle - \Delta F_{\mathrm{ii}} \;\approx\; \frac{\beta}{\lambda}\int_0^\tau \mathrm{sech}^2(\beta\varepsilon_t)\,\dot{\varepsilon}_t^2\,\mathrm{d}t \;=\; \frac{\beta\varepsilon_{\max}}{\lambda\tau}\int_0^{\varepsilon_{\max}} \mathrm{sech}^2(\beta\varepsilon)\,\mathrm{d}\varepsilon \;=\; \frac{\varepsilon_{\max}}{\lambda\tau}\tanh(\beta\varepsilon_{\max}), \tag{9.49}$$

where we changed variables $\mathrm{d}\varepsilon = \dot{\varepsilon}\,\mathrm{d}t$ with $\dot{\varepsilon} = \varepsilon_{\max}/\tau$ constant. The integrand $\beta\,\mathrm{sech}^2(\beta\varepsilon)/\lambda$ is the variance of the conjugate force $\partial_\varepsilon E = -s$ times its correlation time, divided by $k_{\mathrm{B}}T$: it is the thermodynamic friction coefficient $\zeta(\varepsilon)$ of Section 9.2, the same quantity that gives $\zeta = C\tau_{RC}$ for the RC node of Part 1, and the starting point of the optimal-protocol theory of Chapter 10. The excess dissipation decays as $1/\tau$.

Table 9.1 compares Eq. (9.49) with the exact solution of the magnetization equation, obtained by numerical integration, for $\beta\varepsilon_{\max} = 7$ (quasistatic error $\delta = 8.3 \times 10^{-7}$, quasistatic cost $0.69313\,k_{\mathrm{B}}T$). The slow-driving formula is accurate to better than 1% once $\lambda\tau \geq 10$. A faster ramp also leaves the bit lagging at the moment the barrier is raised, so the final error probability exceeds its quasistatic value.

Combining Eq. (9.49) with the quasistatic cost and with $\varepsilon_{\max} = \frac{k_{\mathrm{B}}T}{2}\ln\frac{1-\delta}{\delta}$ gives, for $\delta \ll 1$ and $\lambda\tau \gg \ln(1/\delta)$,

$$\langle\mathcal{Q}_{\mathrm{reset}}\rangle \;=\; \langle\mathcal{W}_{\mathrm{reset}}\rangle \;\approx\; k_{\mathrm{B}}T\ln 2 + \frac{k_{\mathrm{B}}T}{2\lambda\tau}\ln\frac{1}{\delta}. \tag{9.50}$$

Resetting a p-bit with error δ in a time τ costs the Landauer value plus an excess that grows logarithmically with the required accuracy and falls inversely with the available time measured in units of the flip time $1/\lambda$. For a representative p-bit with $\lambda \sim 10^8\,\mathrm{s}^{-1}$, a reset in 300 ns ($\lambda\tau = 30$) to $\delta \approx 10^{-6}$ costs about $0.93\,k_{\mathrm{B}}T$ in this model, about 4×10^{-21} J at room temperature.

Table 9.1: Finite-time reset of the two-state p-bit with a linear ramp to $\beta\varepsilon_{\max} = 7$ and Glauber rates of total rate λ. Energies in units of $k_{\mathrm{B}}T$. The total mean work is $\langle \mathcal{W}_{\mathrm{ii}} \rangle + \varepsilon_{\max}\, m(\tau)$.

$\lambda\tau$	excess work, exact	excess work, Eq. (9.49)	final error δ	total mean work
3	2.171	2.333	3.5×10^{-2}	2.378
10	0.6998	0.7000	1.2×10^{-4}	1.391
30	0.2333	0.2333	1.6×10^{-6}	0.926
100	0.0700	0.0700	9.7×10^{-7}	0.763
∞	0	0	8.3×10^{-7}	0.693

As a check on the fluctuation theorem in finite time, we simulated the full four-stage reset with $\beta\varepsilon_{\max} = 3$ and $\lambda\tau = 10$, using 2×10^6 trajectories of the two-state jump process (split-step propagation over 10^3 steps per ramp: a bias increment followed by exact two-state relaxation at the new bias). The simulation gives $\langle \mathcal{W} \rangle = 0.951\, k_{\mathrm{B}}T$, in agreement with the value $0.9514\, k_{\mathrm{B}}T$ from the magnetization equation; it gives $\mathbb{P}(\mathcal{W} < 0) = 0.17$; and it gives $\left\langle e^{-\beta \mathcal{W}} \right\rangle = 0.99 \pm 0.01$, consistent with the Jarzynski value 1. With $\beta\varepsilon_{\max} = 3$ the failed runs have probability 6×10^{-3} and are well sampled, which is why the estimator converges here and not for $\beta\varepsilon_{\max} = 7$.

Summary. The linear RC node has Gaussian work with $\mathrm{Var}(\mathcal{W}) = 2k_{\mathrm{B}}T\,\langle \mathcal{W} \rangle$, satisfies the Jarzynski and Crooks relations by a two-line Gaussian computation, and dissipates the Joule heat that the finite-time bound of Chapter 10 identifies as optimal for slow ramps. The bistable p-bit has strongly non-Gaussian, two-point work distributions. Its quasistatic reset costs exactly $k_{\mathrm{B}}T[\ln 2 - H(\delta)]$, all of it released as heat, and a finite-time reset adds an excess $\approx k_{\mathrm{B}}T \ln(1/\delta)/(2\lambda\tau)$. In both systems a sizable fraction of runs have work below ΔF, as the fluctuation theorems require.

Summary

- **Proved:**
  1. The generalized path-measure ratio $k_{\mathrm{B}} \ln(\mathcal{P}/\tilde{\mathcal{P}}_q) = k_{\mathrm{B}} \ln[\rho_0(\mathbf{x}_0)/q(\mathbf{x}_\tau)] + \mathcal{Q}/T$ for an arbitrary reference density q, and the normalization identity $\left\langle e^{-\mathcal{R}_q/k_{\mathrm{B}}} \right\rangle = 1$ (Lemma 9.1), from the Crooks ratio of Theorem 8.8.
  2. The Integral Fluctuation Theorem $\left\langle e^{-\Delta s_{\mathrm{tot}}/k_{\mathrm{B}}} \right\rangle = 1$, the Second Law by Jensen's inequality, and the exponential bound $\mathbb{P}(\Delta s_{\mathrm{tot}} \le -a k_{\mathrm{B}}) \le e^{-a}$ (Theorem 9.2, Corollary 9.3, and Proposition 9.4).
  3. The Detailed Fluctuation Theorem in its two-experiment form, and its single-distribution form for non-equilibrium steady states with time-independent, time-reversal-even driving (Theorem 9.6).
  4. The Crooks work fluctuation theorem $P_F(\mathcal{W})/P_R(-\mathcal{W}) = e^{\beta(\mathcal{W} - \Delta F)}$, the Jarzynski equality by two independent routes (path measure and Feynman–Kac), the tail bound $\mathbb{P}(\mathcal{W} \le \Delta F - \zeta) \le e^{-\beta\zeta}$, and the cumulant expansion $\langle \mathcal{W}_{\mathrm{diss}} \rangle = \frac{\beta}{2}\mathrm{Var}(\mathcal{W}) - \frac{\beta^2}{6}\kappa_3 + \cdots$ (Theorems 9.10 and 9.11 and Proposition 9.13).
  5. The non-equilibrium Landauer bound $\langle \mathcal{W} \rangle - \Delta\mathcal{F} = T\Sigma_{\mathrm{tot}} \ge 0$ with its trajectory form, the coarse-grained free energy of a memory, and the erasure and randomization bounds $\langle \mathcal{W} \rangle \ge k_{\mathrm{B}}T[\ln 2 - H(\delta)]$ and $\langle \mathcal{W} \rangle \ge -k_{\mathrm{B}}T \ln 2$ (Theorem 9.14, Lemma 9.15, and Corollary 9.16).

6. The Sagawa–Ueda fluctuation theorem $\left\langle e^{-\beta(\mathcal{W}-\Delta F_y)-I}\right\rangle = 1$ and the bound $\langle \mathcal{W}_{\mathrm{ext}}\rangle \geq -\langle \Delta F_y\rangle + k_{\mathrm{B}}T\mathcal{I}$ for a single error-prone measurement at $t = 0$, and saturation of the bound by a Szilard protocol on a p-bit (Theorem 9.20 and Example 9.21).
7. The thermodynamic uncertainty relation for the biased ring in closed form, and for general finite-state stationary Markov jump processes via the Cramér–Rao inequality and a current-scaling virtual perturbation (Proposition 9.25 and Theorem 9.24); saturation by drift–diffusion currents (Example 9.26).
8. Exact Gaussian work statistics $\mathrm{Var}(\mathcal{W}) = 2k_{\mathrm{B}}T\,\langle \mathcal{W}\rangle$ for the ramped RC node, exact two-point work distributions for the sudden tilt and the quasistatic reset of a two-state p-bit, and saturation of the Landauer bound by the quasistatic reset (Worked computation 9.30).

- **Assumed:**

1. All hypotheses of Theorem 8.8: Markovian bath at a single temperature, Stratonovich (midpoint) energetics, smooth confining potentials, and the existence of the continuum limit of the time-sliced path integral. Equalities need a strictly positive initial density; otherwise only the inequalities survive.
2. The martingale and orthogonality properties of compensated jump counts used in the Fisher-information formula Eq. (9.36), and local detailed balance relating jump rates to dissipated heat in Eq. (9.31).
3. In the worked computation: the two-state reduction of the quartic well with $\beta\Delta E \gg 1$ and $\varepsilon < 1.54\,\Delta E$; Glauber rates with a total rate λ controlled independently of ε; and zero net work for raising and lowering the barrier. The slow-driving formula Eq. (9.49) is a first-order expansion in $(\lambda\tau)^{-1}$, checked numerically in Table 9.1.
4. Classical, non-invasive measurement in Assumption 9.19.

- **Taken up later:**

1. The Girsanov-theorem derivation of the Fisher information for the Langevin version of the TUR, sketched after the proof of Theorem 9.24, relies on Section 4.5.
2. The large-deviation proof of the TUR by Gingrich et al. through the level-2.5 rate functional, and the resulting quadratic bound on the current rate function, are described but not proved (Remark 9.29).
3. The Sagawa–Ueda relation for repeated and intermediate-time measurements, and the general bound $\langle \mathcal{W}_{\mathrm{meas}}\rangle + \langle \mathcal{W}_{\mathrm{erase}}\rangle \geq k_{\mathrm{B}}T\mathcal{I}$, are stated with references only.
4. Optimal finite-time protocols that minimize the excess work of Eq. (9.49) and the Wasserstein lower bound on Σ_{tot} are developed in Chapter 10. The device-level energy budget of readout and reset is in Section 17.4.

- **Open:**

1. Tight speed–accuracy–energy trade-offs for resetting and reading out large arrays of coupled p-bits, beyond the single-bit estimate Eq. (9.50), are an active research topic.

2. Whether thermodynamic inference from current fluctuations via the TUR can serve as a practical on-chip monitor of dissipation in non-reciprocal analog samplers has not been established experimentally.

Bibliographic notes

The work fluctuation theorem for non-equilibrium driving starting from equilibrium was found by Jarzynski [55], whose original argument is the Feynman–Kac proof given here. Crooks related it to microscopic reversibility and proved the work fluctuation theorem that bears his name [22]. Seifert introduced the trajectory entropy $s_{\mathrm{sys}} = -k_{\mathrm{B}} \ln \rho(\mathbf{x}_t, t)$, proved the integral fluctuation theorem for total entropy production, and organized the field in his review [97], which also treats the Schnakenberg entropy production of master equations and the steady-state detailed fluctuation theorem. The Hatano–Sasa relation for transitions between non-equilibrium steady states is in [48]. The minimum heat of erasure is due to Landauer [63], and the analysis of reversible computation and of Maxwell's demon in terms of erasure is due to Bennett [10]. The generalized Jarzynski equality with feedback control is due to Sagawa and Ueda [92]; the review by Parrondo, Horowitz, and Sagawa [84] covers measurement and erasure costs, Szilard engines, and the non-equilibrium free energy $\mathcal{F}[\rho]$ used in Section 9.3. The thermodynamic uncertainty relation was proposed by Barato and Seifert [8] and proved for Markov jump processes in the long-time limit by Gingrich, Horowitz, Perunov, and England [41]. The Cramér–Rao route with a current-scaling virtual perturbation, used in our proof of Theorem 9.24, follows later work by Dechant and Sasa [24] and by Hasegawa and Van Vu [47]. Annealed importance sampling, the machine-learning counterpart of the Jarzynski estimator, is due to Neal [78]. Sekimoto's monograph [98] discusses the energetics of the two-state and Brownian models used in the worked computation, and the energy scales of p-bit hardware are taken from [18, 31].

Chapter 10

Finite-Time Thermodynamics and Optimal Transport

A thermodynamic computer does not sit in equilibrium. Each time it produces a sample, it changes its control parameters (bias voltages, coupling conductances, clamped inputs) and the probability density of its node voltages must move from one distribution ρ_0 to another distribution ρ_τ. The Second Law of Chapter 8 tells us that the total entropy production of such a process is non-negative, $\Sigma_{\mathrm{tot}} \geq 0$, and that it can be made arbitrarily small only in the quasistatic limit. A sampler that waits forever produces no samples. The physical question of this chapter is therefore quantitative: if the distribution must be moved from ρ_0 to ρ_τ in a fixed time τ, what is the smallest entropy production that physics allows, and which control protocol achieves it?

The answer is geometric. We will show that for overdamped Langevin dynamics with mobility μ and diffusion constant $D = \mu k_{\mathrm{B}} T$ (the convention of Chapters 6 and 8),

$$\Sigma_{\mathrm{tot}}(\tau) \;\geq\; \frac{k_{\mathrm{B}}}{D\tau}\,\mathcal{W}_2^2(\rho_0, \rho_\tau), \qquad \text{equivalently} \qquad \langle \mathcal{W} \rangle - \Delta\mathcal{F} \;\geq\; \frac{\mathcal{W}_2^2(\rho_0, \rho_\tau)}{\mu\tau}, \tag{10.1}$$

where $\mathcal{W}_2$ is the L^2-Wasserstein distance of optimal transport. The distance $\mathcal{W}_2(\rho_0, \rho_\tau)$ measures, in units of length, how far the probability mass has to travel. The bound says that dissipation is the product of a squared distance and a rate $1/\tau$, exactly like the work done against viscous drag when a bead is pulled through a fluid. The bound is tight: there is a protocol $E(\mathbf{x}, t)$ that attains it.

To reach Eq. (10.1) we build the necessary optimal transport theory from first principles. A dragged harmonic trap first exhibits the $1/\tau$ law. We then define $\mathcal{W}_2$ and show, following Benamou and Brenier [9], that $\mathcal{W}_2^2$ is the minimum kinetic action of a probability fluid obeying the continuity equation. The same fluid picture makes the Fokker–Planck equation of Chapter 6 a gradient flow of the variational free energy $\mathcal{F}[\rho]$ of Chapter 1 [58]. Combining Benamou–Brenier with the entropy production formula of Chapter 8 then proves Eq. (10.1) and yields the optimal protocol (Section 10.5) [5, 25, 113].

The chapter ends with a worked computation on the RC sampler, the coupled linear RC circuit of Example 1.5. For Gaussian distributions the Wasserstein distance has a closed form (the Bures–Wasserstein distance, Chapter B), the optimal transport is linear, and the optimal protocol is a time-dependent quadratic energy. We compute the optimal conductance schedule $\mathbf{G}(t)$ and bias schedule $\mathbf{b}(t)$ for a linear RC sampler that must move its stationary Gaussian from one mean and covariance to another in time τ, verify that the dissipated work equals $\mathcal{W}_2^2/(\mu\tau)$, and evaluate it for a single node. These results set the thermodynamic floor that Chapter 17 compares with the energy budget of the denoising thermodynamic computer architecture proposed by Jelincic et al.

[56], and the transport viewpoint reappears in the probability-flow formulation of diffusion models in Chapter 15.

10.1 The Finite-Time Dilemma

We begin with the facts we import from Chapters 6 and 8. A state vector $\mathbf{x}_t \in \mathbb{R}^N$ obeys the overdamped Langevin equation with a time-dependent potential $E(\mathbf{x}, t)$, scalar mobility $\mu > 0$, and temperature T:

$$\mathrm{d}\mathbf{x}_t = -\mu \nabla E(\mathbf{x}_t, t)\, \mathrm{d}t + \sqrt{2D}\, \mathrm{d}\mathbf{W}_t, \qquad D = \mu k_{\mathrm{B}} T. \tag{10.2}$$

The time dependence of E is the control protocol, written $E(\mathbf{x}, \boldsymbol{\lambda}_t)$ in Chapter 8; here we suppress the parameters and let the whole function change in time. The density $\rho_t(\mathbf{x}) \equiv \rho(\mathbf{x}, t)$ obeys the Smoluchowski equation Eq. (6.23),

$$\frac{\partial \rho_t}{\partial t} + \nabla \cdot (\rho_t \mathbf{v}_t) = 0, \qquad \mathbf{v}_t(\mathbf{x}) := \frac{\mathbf{J}(\mathbf{x}, t)}{\rho_t(\mathbf{x})} = -\mu \nabla E(\mathbf{x}, t) - D \nabla \ln \rho_t(\mathbf{x}), \tag{10.3}$$

where $\mathbf{v}_t$ is the local mean velocity $\mathbf{v}_{\mathrm{loc}}$ of Eq. (8.39). We drop the subscript loc in this chapter because $\mathbf{v}_t$ will be compared with general velocity fields of fluid mechanics. The ensemble entropy production rate is Eq. (8.47) with $\mathbf{D} = D\,\mathbf{I}$:

$$\dot{\Sigma}_{\mathrm{tot}}(t) = \frac{k_{\mathrm{B}}}{D} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho_t(\mathbf{x})\, \|\mathbf{v}_t(\mathbf{x})\|^2 = \frac{1}{\mu T} \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho_t(\mathbf{x})\, \|\mathbf{v}_t(\mathbf{x})\|^2 . \tag{10.4}$$

The second form uses $k_{\mathrm{B}}/D = k_{\mathrm{B}}/(\mu k_{\mathrm{B}} T) = 1/(\mu T)$. Finally, the non-equilibrium free-energy balance Eq. (8.48) integrated over $[0, \tau]$ gives

$$T\Sigma_{\mathrm{tot}}(\tau) = \langle \mathcal{W} \rangle - \Delta \mathcal{F}, \qquad \Delta \mathcal{F} := \mathcal{F}[\rho_\tau; E(\cdot, \tau)] - \mathcal{F}[\rho_0; E(\cdot, 0)], \tag{10.5}$$

with $\mathcal{F}[\rho; E] = \int \mathrm{d}^N x\, \rho\, (E + k_{\mathrm{B}} T \ln \rho)$ the variational free energy of Eq. (6.46). If ρ_0 and ρ_τ are the Boltzmann distributions of the initial and final potentials, $\Delta \mathcal{F}$ is the equilibrium free-energy difference ΔF and $T\Sigma_{\mathrm{tot}}$ is the dissipated work $\mathcal{W}_{\mathrm{diss}} = \langle \mathcal{W} \rangle - \Delta F$. In general we define

$$\mathcal{W}_{\mathrm{diss}} := \langle \mathcal{W} \rangle - \Delta \mathcal{F} = T\Sigma_{\mathrm{tot}}(\tau) \;\geq\; 0. \tag{10.6}$$

10.1.1 An Exactly Solvable Finite-Time Process: The Dragged Trap

The simplest finite-time process is a harmonic trap whose center is moved at constant speed. In circuit language it is the driven RC node of Example 8.3: a node voltage pulled by a programmable source $b(t)$. We use mechanical notation here and return to circuit notation in Section 10.6.

Take $N = 1$, the potential $E(x, t) = \frac{k}{2}(x - \lambda_t)^2$ with stiffness $k > 0$, and the linear protocol $\lambda_t = Lt/\tau$ for $0 \leq t \leq \tau$, so that the trap moves a distance L at speed $u = L/\tau$. The system starts in equilibrium, $\rho_0 = \mathcal{N}(0, k_{\mathrm{B}} T/k)$. The Langevin equation Eq. (10.2) is linear,

$$\mathrm{d}x_t = -\mu k (x_t - \lambda_t)\, \mathrm{d}t + \sqrt{2D}\, \mathrm{d}W_t,$$

so ρ_t stays Gaussian. Taking expectations, the mean $m_t = \langle x_t \rangle$ obeys $\dot{m}_t = -\mu k (m_t - \lambda_t)$. Applying Itô's formula to $(x_t - m_t)^2$ and averaging, the variance $s_t = \langle (x_t - m_t)^2 \rangle$ obeys $\dot{s}_t = -2\mu k s_t + 2D$, whose solution with $s_0 = k_{\mathrm{B}} T/k$ is $s_t = D/(\mu k) = k_{\mathrm{B}} T/k$ for all t: moving the trap does not change the width of the distribution. Define the relaxation time and the lag

$$\tau_r := \frac{1}{\mu k}, \qquad \delta_t := \lambda_t - m_t. \tag{10.7}$$

The lag obeys $\dot{\delta}_t = \dot{\lambda}_t - \dot{m}_t = u - \delta_t/\tau_r$ with $\delta_0 = 0$, so

$$\delta_t = u\tau_r(1 - e^{-t/\tau_r}). \tag{10.8}$$

After a transient of duration τ_r the distribution trails the trap center by the constant distance $u\tau_r$, at which the restoring force $k\,u\tau_r = u/\mu$ balances the viscous drag on the mean.

The average work rate is $\left\langle \dot{\mathcal{W}} \right\rangle = \langle \partial_t E \rangle = \langle -k(x_t - \lambda_t) \rangle\, \dot{\lambda}_t = k\delta_t u$ (Eq. (8.16)). Integrating,

$$\begin{aligned} \langle \mathcal{W} \rangle &= ku \int_0^\tau \mathrm{d}t\, u\tau_r(1 - e^{-t/\tau_r}) = ku^2\tau_r\Big[\tau - \tau_r(1 - e^{-\tau/\tau_r})\Big] \\ &= \frac{L^2}{\mu\tau}\left[1 - \frac{\tau_r}{\tau}\Big(1 - e^{-\tau/\tau_r}\Big)\right], \end{aligned} \tag{10.9}$$

where we used $k\tau_r = 1/\mu$ and $u = L/\tau$. The equilibrium free energy of a harmonic trap does not depend on its center, so $\Delta F = 0$ and all of this work is eventually dissipated as heat.

Two limits are instructive. For a sudden jump, $\tau \ll \tau_r$, expand $1 - e^{-a} = a - a^2/2 + \mathcal{O}(a^3)$ with $a = \tau/\tau_r$: the bracket in Eq. (10.9) becomes $a/2 + \mathcal{O}(a^2)$ and $\langle \mathcal{W} \rangle \to L^2/(2\mu\tau_r) = \frac{1}{2}kL^2$, the energy of a particle suddenly displaced by L from the trap center. For slow driving, $\tau \gg \tau_r$, the bracket tends to 1 and

$$\langle \mathcal{W} \rangle \simeq \frac{L^2}{\mu\tau} \qquad (\tau \gg \tau_r). \tag{10.10}$$

The dissipated work decays as $1/\tau$ and vanishes only in the quasistatic limit $\tau \to \infty$. Equation Eq. (10.10) is also the work done on a bead dragged through a viscous fluid with friction coefficient μ^{-1} at speed L/τ over a distance L: force $L/(\mu\tau)$ times distance L. The thermal fluctuations do not enter, because the width of the distribution is unchanged and only its center is transported.

At $t = \tau$ the distribution $\rho_\tau = \mathcal{N}(L - \delta_\tau, k_\mathrm{B}T/k)$ lags behind the trap. Its variational free energy exceeds the equilibrium value by $k_\mathrm{B}T\, D_\mathrm{KL}(\rho_\tau \,\|\, p_\mathrm{eq}(\cdot; L)) = \frac{k}{2}\delta_\tau^2$ (Eq. (6.47) and Eq. (B.21)). The entropy produced during $[0, \tau]$ is therefore $T\Sigma_\mathrm{tot}(\tau) = \langle \mathcal{W} \rangle - \frac{k}{2}\delta_\tau^2$, and the remaining $\frac{k}{2}\delta_\tau^2$ is dissipated after the protocol stops, while the distribution relaxes into the trap.

10.1.2 Why Zero Dissipation Requires Infinite Time

The $1/\tau$ law is not a property of harmonic traps. It follows from the structure of Eq. (10.4).

Heuristic Derivation: The $1/\tau$ scaling of dissipation

Suppose a protocol must move a distribution so that a typical element of probability mass travels a distance ℓ in time τ. The local mean velocity is then of order ℓ/τ, and by Eq. (10.4) the entropy production rate is of order $k_\mathrm{B}\ell^2/(D\tau^2)$. Integrating over the duration τ gives

$$\Sigma_\mathrm{tot}(\tau) \sim \frac{k_\mathrm{B}}{D} \cdot \frac{\ell^2}{\tau^2} \cdot \tau = \frac{k_\mathrm{B}\ell^2}{D\tau}.$$

A process can have small Σ_tot only if the velocities are small, and it can transport probability over a finite distance with small velocities only if it takes a long time. The quantity that replaces ℓ^2 in a rigorous statement must be a distance between probability distributions that measures how far mass has to move, not how much the densities overlap. The Kullback–Leibler divergence and the total variation distance do not have this property: two narrow Gaussians separated by 1 mV and by 1 V are both nearly orthogonal, yet the second requires a thousand times more transport. The Wasserstein distance of Section 10.2 has exactly this property.

The power dissipated by a sampler that repeats such a transport every τ seconds is

$$P \sim \frac{\mathcal{W}_{\mathrm{diss}}}{\tau} \sim \frac{\ell^2}{\mu \tau^2}. \tag{10.11}$$

Doubling the sampling rate quadruples the power. This is the finite-time dilemma of thermodynamic computing: the Landauer-type bounds of Chapter 9 concern the free-energy change $\Delta\mathcal{F}$, which is independent of speed, while the cost of speed sits entirely in $\mathcal{W}_{\mathrm{diss}}$. To design a fast sampler we need the exact minimum of $\mathcal{W}_{\mathrm{diss}}$ at fixed τ, and the protocol that achieves it.

Hardware Realization: Throughput and dissipation in an RC sampler node

For a single passive RC node (Example 8.3) with capacitance C and conductance G, the thermal voltage spread is $\sigma_V = \sqrt{k_{\mathrm{B}}T/C}$, the relaxation time is $\tau_{RC} = C/G$, and the effective mobility is $\mu = G/C^2$, so that $D = \mu k_{\mathrm{B}}T = \sigma_V^2/\tau_{RC}$. The slow-driving cost Eq. (10.10) of shifting the node distribution by $L = n\sigma_V$ in time τ is

$$\mathcal{W}_{\mathrm{diss}} \simeq \frac{L^2}{\mu\tau} = k_{\mathrm{B}}T\,\frac{L^2}{\sigma_V^2}\,\frac{\tau_{RC}}{\tau} = n^2\,\frac{\tau_{RC}}{\tau}\,k_{\mathrm{B}}T.$$

As a representative estimate take $C = 1\,\mathrm{fF}$ at $300\,\mathrm{K}$, so $\sigma_V \approx 2\,\mathrm{mV}$, and $\tau_{RC} = 100\,\mathrm{ns}$ as in Worked computation 8.12. Shifting by one thermal width ($n = 1$) in $\tau = 1\,\mu\mathrm{s}$ costs about $0.1\,k_{\mathrm{B}}T \approx 4 \times 10^{-22}\,\mathrm{J}$ of dissipated work per node. At $\tau = \tau_{RC} = 100\,\mathrm{ns}$ the slow-driving formula would give $1\,k_{\mathrm{B}}T$, but the protocol is no longer slow: the exact work Eq. (10.9) is $e^{-1}k_{\mathrm{B}}T \approx 0.37\,k_{\mathrm{B}}T$, and the mean of the distribution covers only $L - \delta_\tau = e^{-1}L \approx 0.37\,L$ of the distance to the new bias by $t = \tau$. Of that work, $\frac{1}{2}C\delta_\tau^2 \approx 0.20\,k_{\mathrm{B}}T$ is dissipated after the ramp ends, while the lagging distribution catches up with the source. A cheap fast protocol is one that does not finish the job. These numbers are far below the transistor-level costs computed in Chapter 17, which shows that the transport floor is not what limits present hardware. It is the floor that remains once the circuit overheads are removed.

10.2 The Monge–Kantorovich Problem and the Wasserstein Distance

Optimal transport asks for the cheapest way to rearrange one distribution of mass into another when moving a unit of mass from $\mathbf{x}$ to $\mathbf{y}$ costs $\|\mathbf{x}-\mathbf{y}\|^2$. The quadratic cost is the one that matters for overdamped dynamics, because viscous dissipation is quadratic in velocity. Throughout this section, $\mathcal{P}_2(\mathbb{R}^N)$ denotes the set of probability measures on $\mathbb{R}^N$ with finite second moment, $\int \|\mathbf{x}\|^2\,\rho(\mathrm{d}\mathbf{x}) < \infty$. When a measure has a density we use the same symbol for both.

10.2.1 Monge's Problem: Transport Maps

Definition 10.1 (Push-forward and transport map)**.** Let $\rho_0 \in \mathcal{P}_2(\mathbb{R}^N)$ and let $\mathbf{T} : \mathbb{R}^N \to \mathbb{R}^N$ be measurable. The **push-forward** $\mathbf{T}_{\#}\rho_0$ is the law of $\mathbf{T}(\mathbf{X})$ when $\mathbf{X} \sim \rho_0$; equivalently, for every bounded continuous test function f,

$$\int_{\mathbb{R}^N} f(\mathbf{y})\,(\mathbf{T}_{\#}\rho_0)(\mathrm{d}\mathbf{y}) = \int_{\mathbb{R}^N} f\big(\mathbf{T}(\mathbf{x})\big)\,\rho_0(\mathbf{x})\,\mathrm{d}^N x. \tag{10.12}$$

A map with $\mathbf{T}_{\#}\rho_0 = \rho_1$ is called a **transport map** from ρ_0 to ρ_1.

If $\mathbf{T}$ is a smooth diffeomorphism and $\rho_1 = \mathbf{T}_{\#}\rho_0$ has a density, then substituting $\mathbf{y} = \mathbf{T}(\mathbf{x})$ in the left side of Eq. (10.12) gives $\int f(\mathbf{T}(\mathbf{x}))\rho_1(\mathbf{T}(\mathbf{x}))\,|\det \nabla \mathbf{T}(\mathbf{x})|\,\mathrm{d}^N x$. Since this equals the right side for every f,

$$\rho_1(\mathbf{T}(\mathbf{x}))\;|\det \nabla \mathbf{T}(\mathbf{x})| = \rho_0(\mathbf{x}), \tag{10.13}$$

which is conservation of mass: the mass $\rho_0\,\mathrm{d}^N x$ in a small cell is carried into the image cell, whose volume is $|\det \nabla \mathbf{T}|\,\mathrm{d}^N x$.

Monge's problem is to minimize the transport cost $\int \|\mathbf{x} - \mathbf{T}(\mathbf{x})\|^2 \rho_0(\mathbf{x})\,\mathrm{d}^N x$ over all transport maps $\mathbf{T}$ from ρ_0 to ρ_1. The problem can fail to have any admissible map. If $\rho_0 = \delta_{\mathbf{0}}$ and $\rho_1 = \frac{1}{2}\delta_{\mathbf{a}} + \frac{1}{2}\delta_{-\mathbf{a}}$, every map sends the single point $\mathbf{0}$ to a single point, so no map can split the mass. Physically, a deterministic map cannot represent a process in which one microstate fans out into several. Kantorovich's relaxation removes this restriction.

10.2.2 Kantorovich's Problem: Couplings

Definition 10.2 (Couplings and the L^2-Wasserstein distance)**.** A **coupling** of $\rho_0, \rho_1 \in \mathcal{P}_2(\mathbb{R}^N)$ is a probability measure π on $\mathbb{R}^N \times \mathbb{R}^N$ whose first and second marginals are ρ_0 and ρ_1. Equivalently, it is the joint law of a pair of random vectors $(\mathbf{X}, \mathbf{Y})$ with $\mathbf{X} \sim \rho_0$ and $\mathbf{Y} \sim \rho_1$. The set of couplings is denoted $\Pi(\rho_0, \rho_1)$. The **L^2-Wasserstein distance** is

$$\mathcal{W}_2(\rho_0, \rho_1) := \left(\inf_{\pi \in \Pi(\rho_0,\rho_1)} \int_{\mathbb{R}^N \times \mathbb{R}^N} \|\mathbf{x} - \mathbf{y}\|^2 \,\pi(\mathrm{d}\mathbf{x}, \mathrm{d}\mathbf{y}) \right)^{1/2} = \left(\inf_{(\mathbf{X},\mathbf{Y})} \mathbb{E}\Big[\|\mathbf{X} - \mathbf{Y}\|^2\Big] \right)^{1/2}. \tag{10.14}$$

A coupling that attains the infimum is called an **optimal coupling** or optimal plan.

Every transport map defines a coupling, the law of $(\mathbf{X}, \mathbf{T}(\mathbf{X}))$, so $\mathcal{W}_2^2$ is at most the Monge infimum. The set $\Pi(\rho_0, \rho_1)$ is never empty, because the independent coupling $\rho_0 \otimes \rho_1$ belongs to it, and the elementary inequality $\|\mathbf{x} - \mathbf{y}\|^2 \le 2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2$ shows that $\mathcal{W}_2$ is finite on $\mathcal{P}_2(\mathbb{R}^N)$. In the splitting example above, the only coupling of $\delta_{\mathbf{0}}$ and $\frac{1}{2}(\delta_{\mathbf{a}} + \delta_{-\mathbf{a}})$ is the product, and $\mathcal{W}_2^2 = \|\mathbf{a}\|^2$.

Remark 10.3 (Wasserstein, Kantorovich, and earth mover's distances)**.** The machine-learning literature often uses the L^1 cost $\|\mathbf{x} - \mathbf{y}\|$, which defines $\mathcal{W}_1$ (the earth mover's distance of Wasserstein GANs). Its dynamical interpretation is different and it does not control entropy production. In this book $\mathcal{W}_2$ always means the quadratic cost Eq. (10.14), measured in the units of $\mathbf{x}$ (volts for node voltages), so $\mathcal{W}_2^2$ has units of V^2 and $\mathcal{W}_2^2/\mu$ has units of energy times time.

Proposition 10.4 (Elementary properties of $\mathcal{W}_2$)**.** *Let $\rho_0, \rho_1, \rho_2 \in \mathcal{P}_2(\mathbb{R}^N)$ have means $\mathbf{m}_0, \mathbf{m}_1, \mathbf{m}_2$.*

(a) (Point masses.) $\mathcal{W}_2(\delta_{\mathbf{a}}, \delta_{\mathbf{b}}) = \|\mathbf{a} - \mathbf{b}\|$.

(b) (Separation of the mean.) Let $\bar{\rho}_i$ be ρ_i shifted to zero mean. Then

$$\mathcal{W}_2^2(\rho_0, \rho_1) = \|\mathbf{m}_0 - \mathbf{m}_1\|^2 + \mathcal{W}_2^2(\bar{\rho}_0, \bar{\rho}_1) \;\ge\; \|\mathbf{m}_0 - \mathbf{m}_1\|^2. \tag{10.15}$$

In particular a pure translation, $\rho_1(\mathbf{x}) = \rho_0(\mathbf{x} - \mathbf{a})$, has $\mathcal{W}_2(\rho_0, \rho_1) = \|\mathbf{a}\|$.

(c) (Metric.) $\mathcal{W}_2$ is symmetric, satisfies $\mathcal{W}_2(\rho_0, \rho_1) = 0$ if and only if $\rho_0 = \rho_1$, and obeys the triangle inequality $\mathcal{W}_2(\rho_0, \rho_2) \le \mathcal{W}_2(\rho_0, \rho_1) + \mathcal{W}_2(\rho_1, \rho_2)$.

Proof. (a) The only coupling of two point masses is $\delta_{(\mathbf{a},\mathbf{b})}$.

(b) Given any coupling $(\mathbf{X},\mathbf{Y})$ of ρ_0,ρ_1, write $\mathbf{X}=\mathbf{m}_0+\bar{\mathbf{X}}$ and $\mathbf{Y}=\mathbf{m}_1+\bar{\mathbf{Y}}$, so that $(\bar{\mathbf{X}},\bar{\mathbf{Y}})$ is a coupling of $\bar\rho_0,\bar\rho_1$. Conversely every coupling of the centered measures arises this way. Expanding the square,

$$\mathbb{E}\Big[\|\mathbf{X}-\mathbf{Y}\|^2\Big]=\|\mathbf{m}_0-\mathbf{m}_1\|^2+2(\mathbf{m}_0-\mathbf{m}_1)\cdot\mathbb{E}\Big[\bar{\mathbf{X}}-\bar{\mathbf{Y}}\Big]+\mathbb{E}\Big[\Big\|\bar{\mathbf{X}}-\bar{\mathbf{Y}}\Big\|^2\Big],$$

and the cross term vanishes because $\mathbb{E}\big[\bar{\mathbf{X}}\big]=\mathbb{E}\big[\bar{\mathbf{Y}}\big]=\mathbf{0}$. Taking the infimum over the (bijectively related) couplings gives Eq. (10.15). For a translation, $\bar\rho_0=\bar\rho_1$, and the coupling $\bar{\mathbf{Y}}=\bar{\mathbf{X}}$ gives $\mathcal{W}_2(\bar\rho_0,\bar\rho_1)=0$.

(c) Symmetry is clear because swapping $\mathbf{X}$ and $\mathbf{Y}$ maps $\Pi(\rho_0,\rho_1)$ onto $\Pi(\rho_1,\rho_0)$. If $\rho_0=\rho_1$, the diagonal coupling $\mathbf{Y}=\mathbf{X}$ gives zero. Conversely, if $\mathcal{W}_2(\rho_0,\rho_1)=0$, an optimal coupling (whose existence we assume) has $\mathbf{X}=\mathbf{Y}$ almost surely, so the laws agree. For the triangle inequality we prove the version we need, in which ρ_1 has a smooth positive density. Let $\pi_{01}(\mathbf{x},\mathbf{y})$ and $\pi_{12}(\mathbf{y},\mathbf{z})$ be optimal couplings, written as densities for notational ease (the argument is identical for measures). Define the glued three-point law

$$\pi_{012}(\mathbf{x},\mathbf{y},\mathbf{z}):=\frac{\pi_{01}(\mathbf{x},\mathbf{y})\,\pi_{12}(\mathbf{y},\mathbf{z})}{\rho_1(\mathbf{y})},\tag{10.16}$$

which makes $\mathbf{X}$ and $\mathbf{Z}$ conditionally independent given $\mathbf{Y}$. Integrating out $\mathbf{z}$ and using $\int\pi_{12}(\mathbf{y},\mathbf{z})\,\mathrm{d}^N z=\rho_1(\mathbf{y})$ returns $\pi_{01}(\mathbf{x},\mathbf{y})$, and integrating out $\mathbf{x}$ gives π_{12}. The pair $(\mathbf{X},\mathbf{Z})$ is therefore a coupling of ρ_0 and ρ_2, and by Minkowski's inequality in L^2,

$$\mathcal{W}_2(\rho_0,\rho_2)\le\mathbb{E}\Big[\|\mathbf{X}-\mathbf{Z}\|^2\Big]^{1/2}\le\mathbb{E}\Big[\|\mathbf{X}-\mathbf{Y}\|^2\Big]^{1/2}+\mathbb{E}\Big[\|\mathbf{Y}-\mathbf{Z}\|^2\Big]^{1/2}=\mathcal{W}_2(\rho_0,\rho_1)+\mathcal{W}_2(\rho_1,\rho_2).\ \square$$

The lower bound in Eq. (10.15) already contains the physics of the dragged trap: no protocol can move the mean by L without transporting mass over a distance of at least L.

10.2.3 The One-Dimensional Problem: Monotone Rearrangement

In one dimension the optimal transport problem can be solved in closed form. Let $F_i(x)=\int_{-\infty}^{x}\rho_i(\mathrm{d}x')$ be the cumulative distribution functions and $F_i^{-1}(u):=\inf\{x:F_i(x)\ge u\}$ the quantile functions, $u\in(0,1)$. The quantile function satisfies $F_i^{-1}(u)\le x\iff u\le F_i(x)$, and if U is uniform on $(0,1)$, then $F_i^{-1}(U)\sim\rho_i$ because $\mathbb{P}(F_i^{-1}(U)\le x)=\mathbb{P}(U\le F_i(x))=F_i(x)$.

Theorem 10.5 (Optimal transport on the line)**.** *For $\rho_0,\rho_1\in\mathcal{P}_2(\mathbb{R})$, the coupling $X=F_0^{-1}(U)$, $Y=F_1^{-1}(U)$, with U uniform on $(0,1)$, is optimal, and*

$$\mathcal{W}_2^2(\rho_0,\rho_1)=\int_0^1\Big|F_0^{-1}(u)-F_1^{-1}(u)\Big|^2\,\mathrm{d}u.\tag{10.17}$$

If ρ_0 has a positive density, this optimal coupling is given by the non-decreasing transport map $T=F_1^{-1}\circ F_0$.

Proof. For any coupling, $\mathbb{E}[(X-Y)^2]=\mathbb{E}[X^2]+\mathbb{E}[Y^2]-2\mathbb{E}[X]\,\mathbb{E}[Y]-2\,\mathrm{Cov}(X,Y)$. The first three terms are fixed by the marginals, so we must show that the covariance is largest for the quantile coupling. Let $H(x,y)=\mathbb{P}(X\le x,Y\le y)$ be the joint distribution function. We first prove Hoeffding's identity

$$\mathrm{Cov}(X,Y)=\int_{\mathbb{R}}\int_{\mathbb{R}}\Big[H(x,y)-F_0(x)F_1(y)\Big]\,\mathrm{d}x\,\mathrm{d}y.\tag{10.18}$$

Let (X_1, Y_1) and (X_2, Y_2) be two independent copies of (X, Y). Then $\mathbb{E}[(X_1 - X_2)(Y_1 - Y_2)] = 2\operatorname{Cov}(X, Y)$. For any reals a, b we have $a - b = \int_{\mathbb{R}} [\mathbf{1}\{b \le x\} - \mathbf{1}\{a \le x\}]\, dx$, since the integrand equals 1 exactly for $b \le x < a$ (and -1 for $a \le x < b$). Hence

$$2\operatorname{Cov}(X, Y) = \iint \mathbb{E}\big[(\mathbf{1}\{X_2 \le x\} - \mathbf{1}\{X_1 \le x\})(\mathbf{1}\{Y_2 \le y\} - \mathbf{1}\{Y_1 \le y\})\big]\, dx\, dy,$$

where the exchange of expectation and integrals is allowed by Fubini's theorem because the second moments are finite. Expanding the product, the two diagonal terms each give $H(x, y)$ and the two cross terms each give $-F_0(x)F_1(y)$, by independence of the copies. This proves Eq. (10.18).

Next, for any coupling, $H(x, y) \le \mathbb{P}(X \le x) = F_0(x)$ and $H(x, y) \le F_1(y)$, so $H(x, y) \le \min(F_0(x), F_1(y))$ pointwise (the Fréchet upper bound). For the quantile coupling,

$$H(x, y) = \mathbb{P}(F_0^{-1}(U) \le x,\, F_1^{-1}(U) \le y) = \mathbb{P}(U \le F_0(x),\, U \le F_1(y)) = \min(F_0(x), F_1(y)),$$

which attains the bound at every (x, y). By Eq. (10.18) it maximizes the covariance and therefore minimizes $\mathbb{E}[(X - Y)^2]$, which gives Eq. (10.17). Finally, if ρ_0 has a positive density, F_0 is continuous and strictly increasing, $U = F_0(X)$ is uniform, and $Y = F_1^{-1}(F_0(X))$. □

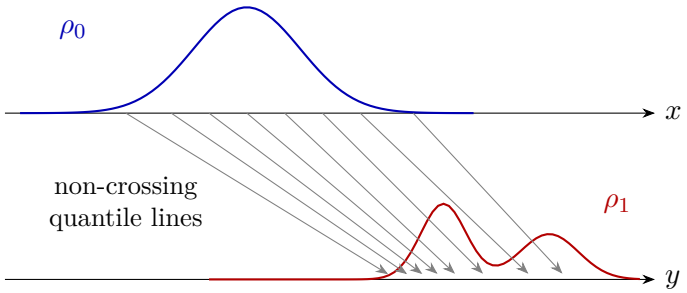


Figure 10.1: Monotone rearrangement on the line. The u-quantile of ρ_0 is sent to the u-quantile of ρ_1, so the transport lines never cross. A crossing pair of lines can always be uncrossed at lower quadratic cost, as shown in Eq. (10.19).

The theorem has a simple mechanical reading. Suppose a transport plan sends $x_1 \mapsto y_1$ and $x_2 \mapsto y_2$ with $x_1 < x_2$ but $y_1 > y_2$, so the two paths cross. Swapping the destinations changes the cost by

$$[(x_1-y_2)^2+(x_2-y_1)^2]-[(x_1-y_1)^2+(x_2-y_2)^2] = 2(x_1y_1+x_2y_2-x_1y_2-x_2y_1) = -2(x_2-x_1)(y_1-y_2) < 0. \tag{10.19}$$

Crossing paths are never optimal. In higher dimensions the same swap argument shows that an optimal plan is supported on a set where $(\mathbf{x}_1 - \mathbf{x}_2) \cdot (\mathbf{y}_1 - \mathbf{y}_2) \ge 0$ for all pairs of points; this monotonicity is what Brenier's theorem below turns into the statement that the optimal map is the gradient of a convex function.

Example 10.6 (One-dimensional Gaussians)**.** For $\rho_i = \mathcal{N}(m_i, \sigma_i^2)$ we have $F_i^{-1}(u) = m_i + \sigma_i \Phi^{-1}(u)$, where Φ is the standard normal distribution function. Since $\int_0^1 \Phi^{-1}(u)\, du = \mathbb{E}[Z] = 0$ and $\int_0^1 \Phi^{-1}(u)^2\, du = \mathbb{E}[Z^2] = 1$ for $Z \sim \mathcal{N}(0, 1)$, Eq. (10.17) gives

$$\mathcal{W}_2^2(\rho_0, \rho_1) = \int_0^1 \big[(m_0 - m_1) + (\sigma_0 - \sigma_1)\Phi^{-1}(u)\big]^2\, du = (m_0 - m_1)^2 + (\sigma_0 - \sigma_1)^2. \tag{10.20}$$

The optimal map is the affine map $T(x) = m_1 + (\sigma_1/\sigma_0)(x - m_0)$. In the space of one-dimensional Gaussians, $\mathcal{W}_2$ is the Euclidean distance in the half-plane of coordinates (m, σ). Compressing a node distribution from σ_0 to $\sigma_0/2$ is a transport over distance $\sigma_0/2$, equivalent in cost to translating it by $\sigma_0/2$.

10.2.4 Contraction Under Lipschitz Maps and Synchronous Coupling

Couplings are also a practical tool for bounding how stochastic dynamics moves distributions. The next two results are used in Chapters 15 and 17.

Lemma 10.7 (Lipschitz maps and kernels contract $\mathcal{W}_2$)**.** *(a) If the map $\mathbf{T} : \mathbb{R}^N \to \mathbb{R}^N$ is L-Lipschitz, that is, $\|\mathbf{T}(\mathbf{x}) - \mathbf{T}(\mathbf{y})\| \le L\,\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y}$, then $\mathcal{W}_2(\mathbf{T}_\#\rho, \mathbf{T}_\#\sigma) \le L\,\mathcal{W}_2(\rho, \sigma)$.*

(b) Let $K(\cdot\,|\,\mathbf{x})$ be a Markov transition kernel such that $\mathcal{W}_2(K(\cdot\,|\,\mathbf{x}), K(\cdot\,|\,\mathbf{y})) \le L\,\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y}$. Then the one-step laws $\rho K = \int K(\cdot\,|\,\mathbf{x})\rho(\mathrm{d}\mathbf{x})$ obey $\mathcal{W}_2(\rho K, \sigma K) \le L\,\mathcal{W}_2(\rho, \sigma)$.

Proof. (a) Let $(\mathbf{X}, \mathbf{Y})$ be an optimal coupling of ρ and σ. Then $(\mathbf{T}(\mathbf{X}), \mathbf{T}(\mathbf{Y}))$ is a coupling of the push-forwards and $\mathbb{E}\big[\|\mathbf{T}(\mathbf{X}) - \mathbf{T}(\mathbf{Y})\|^2\big] \le L^2\mathbb{E}\big[\|\mathbf{X} - \mathbf{Y}\|^2\big] = L^2\mathcal{W}_2^2(\rho, \sigma)$. (b) Draw $(\mathbf{X}, \mathbf{Y})$ from an optimal coupling of ρ, σ. Given $(\mathbf{X}, \mathbf{Y}) = (\mathbf{x}, \mathbf{y})$, draw $(\mathbf{X}', \mathbf{Y}')$ from an optimal coupling of $K(\cdot\,|\,\mathbf{x})$ and $K(\cdot\,|\,\mathbf{y})$ (that this choice can be made measurably in $(\mathbf{x}, \mathbf{y})$ is a standard selection result we assume). Then $\mathbf{X}' \sim \rho K$, $\mathbf{Y}' \sim \sigma K$, and

$$\mathcal{W}_2^2(\rho K, \sigma K) \le \mathbb{E}\big[\|\mathbf{X}' - \mathbf{Y}'\|^2\big] = \mathbb{E}\big[\mathcal{W}_2^2(K(\cdot\,|\,\mathbf{X}), K(\cdot\,|\,\mathbf{Y}))\big] \le L^2\mathbb{E}\big[\|\mathbf{X} - \mathbf{Y}\|^2\big] = L^2\mathcal{W}_2^2(\rho, \sigma). \qquad \square$$

Proposition 10.8 (Synchronous coupling and Wasserstein contraction)**.** *Let $E(\mathbf{x}, t)$ be uniformly κ-convex in $\mathbf{x}$, $\nabla^2 E(\mathbf{x}, t) \succeq \kappa\mathbf{I}$ with $\kappa > 0$, and let ρ_t and σ_t be the laws at time t of two solutions of Eq. (10.2) with the same potential and initial laws ρ_0, σ_0. Then*

$$\mathcal{W}_2(\rho_t, \sigma_t) \le e^{-\mu\kappa t}\,\mathcal{W}_2(\rho_0, \sigma_0). \tag{10.21}$$

Proof. Let $(\mathbf{X}_0, \mathbf{Y}_0)$ be an optimal coupling of ρ_0, σ_0, and drive both copies with the *same* Wiener process:

$$\mathrm{d}\mathbf{X}_t = -\mu\nabla E(\mathbf{X}_t, t)\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}\mathbf{W}_t, \qquad \mathrm{d}\mathbf{Y}_t = -\mu\nabla E(\mathbf{Y}_t, t)\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}\mathbf{W}_t.$$

The noise cancels in the difference $\boldsymbol{\Delta}_t = \mathbf{X}_t - \mathbf{Y}_t$, which therefore obeys the ordinary differential equation $\dot{\boldsymbol{\Delta}}_t = -\mu[\nabla E(\mathbf{X}_t, t) - \nabla E(\mathbf{Y}_t, t)]$ path by path. By the fundamental theorem of calculus along the segment from $\mathbf{Y}_t$ to $\mathbf{X}_t$,

$$\nabla E(\mathbf{X}_t, t) - \nabla E(\mathbf{Y}_t, t) = \left[\int_0^1 \nabla^2 E\big(\mathbf{Y}_t + s\boldsymbol{\Delta}_t, t\big)\,\mathrm{d}s\right]\boldsymbol{\Delta}_t,$$

and the bracket is $\succeq \kappa\mathbf{I}$. Hence $\frac{\mathrm{d}}{\mathrm{d}t}\|\boldsymbol{\Delta}_t\|^2 = 2\boldsymbol{\Delta}_t \cdot \dot{\boldsymbol{\Delta}}_t \le -2\mu\kappa\,\|\boldsymbol{\Delta}_t\|^2$, and Grönwall's inequality gives $\|\boldsymbol{\Delta}_t\|^2 \le e^{-2\mu\kappa t}\,\|\boldsymbol{\Delta}_0\|^2$. Since $(\mathbf{X}_t, \mathbf{Y}_t)$ couples ρ_t and σ_t,

$$\mathcal{W}_2^2(\rho_t, \sigma_t) \le \mathbb{E}\big[\|\boldsymbol{\Delta}_t\|^2\big] \le e^{-2\mu\kappa t}\mathbb{E}\big[\|\boldsymbol{\Delta}_0\|^2\big] = e^{-2\mu\kappa t}\mathcal{W}_2^2(\rho_0, \sigma_0). \qquad \square$$

Taking $\sigma_0 = p_{\mathrm{eq}}$ for a time-independent E gives $\mathcal{W}_2(\rho_t, p_{\mathrm{eq}}) \le e^{-\mu\kappa t}\mathcal{W}_2(\rho_0, p_{\mathrm{eq}})$, the Wasserstein counterpart of the Bakry–Émery decay of Section 6.5. The rate $\mu\kappa$ for $\mathcal{W}_2$ corresponds to the rate $2\mu\kappa$ for $\mathcal{W}_2^2$ and for the KL divergence. The proof needs no functional inequality, only convexity, but it gives nothing for the non-convex potential of the double-well sampler (the tilted quartic of Worked computation 7.16), where two synchronously coupled copies on opposite sides of the barrier do not approach each other.

10.2.5 Kantorovich Duality and Brenier's Theorem

The optimal plan has a rigid structure for the quadratic cost. To see it, we pass to the dual problem.

Proposition 10.9 (Kantorovich duality)**.** *For $\rho_0, \rho_1 \in \mathcal{P}_2(\mathbb{R}^N)$,*

$$\mathcal{W}_2^2(\rho_0,\rho_1) = \sup\left\{ \int f \,\mathrm{d}\rho_0 + \int g \,\mathrm{d}\rho_1 \;:\; f(\mathbf{x}) + g(\mathbf{y}) \le \|\mathbf{x}-\mathbf{y}\|^2 \;\text{ for all } \mathbf{x},\mathbf{y} \right\}, \tag{10.22}$$

the supremum being over integrable functions f, g.

Proof of the inequality $\sup \le \mathcal{W}_2^2$*; equality assumed.* For any admissible (f,g) and any coupling $(\mathbf{X},\mathbf{Y})$, $\mathbb{E}[f(\mathbf{X})] + \mathbb{E}[g(\mathbf{Y})] = \mathbb{E}[f(\mathbf{X})+g(\mathbf{Y})] \le \mathbb{E}\left[\|\mathbf{X}-\mathbf{Y}\|^2\right]$. Taking the supremum on the left and the infimum on the right gives $\sup \le \mathcal{W}_2^2$. The absence of a duality gap is a minimax theorem whose proof we do not reproduce. □

The dual problem has an economic reading that suits a physicist. A shipping company offers to carry the mass: it charges $f(\mathbf{x})$ to pick up a unit of mass at $\mathbf{x}$ and $g(\mathbf{y})$ to deliver it at $\mathbf{y}$ (negative prices are allowed). Its prices are competitive only if no pair of charges exceeds the cost of moving the mass oneself, $f(\mathbf{x}) + g(\mathbf{y}) \le \|\mathbf{x}-\mathbf{y}\|^2$. The company maximizes its revenue $\int f \,\mathrm{d}\rho_0 + \int g \,\mathrm{d}\rho_1$, and duality says that the maximal competitive revenue equals the optimal transport cost.

Suppose now that an optimal plan π and an optimal dual pair (f,g) exist, with f differentiable. Comparing Eq. (10.14) with Eq. (10.22), $\int [\|\mathbf{x}-\mathbf{y}\|^2 - f(\mathbf{x}) - g(\mathbf{y})]\,\mathrm{d}\pi = 0$, and the integrand is non-negative, so $f(\mathbf{x})+g(\mathbf{y}) = \|\mathbf{x}-\mathbf{y}\|^2$ on the support of π. Fix $(\mathbf{x},\mathbf{y})$ in the support. The function $\mathbf{x}' \mapsto \|\mathbf{x}'-\mathbf{y}\|^2 - f(\mathbf{x}')$ is bounded below by $g(\mathbf{y})$ everywhere (by the dual constraint) and equals $g(\mathbf{y})$ at $\mathbf{x}' = \mathbf{x}$, so it has a minimum at $\mathbf{x}$ and its gradient vanishes there: $2(\mathbf{x}-\mathbf{y}) - \nabla f(\mathbf{x}) = \mathbf{0}$. Hence

$$\mathbf{y} = \mathbf{x} - \tfrac{1}{2}\nabla f(\mathbf{x}) = \nabla\varphi(\mathbf{x}), \qquad \varphi(\mathbf{x}) := \tfrac{1}{2}\|\mathbf{x}\|^2 - \tfrac{1}{2} f(\mathbf{x}). \tag{10.23}$$

The destination is a deterministic function of the origin. Moreover the optimal f can be taken of the form $f(\mathbf{x}) = \inf_{\mathbf{y}}[\|\mathbf{x}-\mathbf{y}\|^2 - g(\mathbf{y})]$ (this is the largest f compatible with the constraint for the given g, so replacing f by it can only increase the dual objective), and then

$$\varphi(\mathbf{x}) = \sup_{\mathbf{y}} \left[\mathbf{x}\cdot\mathbf{y} - \tfrac{1}{2}\|\mathbf{y}\|^2 + \tfrac{1}{2} g(\mathbf{y}) \right]$$

is a supremum of affine functions of $\mathbf{x}$, hence convex. This heuristic is made rigorous by the following theorem.

Theorem 10.10 (Brenier)**.** *Let $\rho_0, \rho_1 \in \mathcal{P}_2(\mathbb{R}^N)$ and suppose ρ_0 has a density. Then there is a convex function $\varphi : \mathbb{R}^N \to \mathbb{R}$, unique up to an additive constant on the support of ρ_0, such that $\mathbf{T} = \nabla\varphi$ pushes ρ_0 forward to ρ_1. The coupling $(\mathbf{X}, \nabla\varphi(\mathbf{X}))$ is the unique optimal coupling, and*

$$\mathcal{W}_2^2(\rho_0,\rho_1) = \int_{\mathbb{R}^N} \|\mathbf{x} - \nabla\varphi(\mathbf{x})\|^2 \,\rho_0(\mathbf{x})\,\mathrm{d}^N x. \tag{10.24}$$

Conversely, if φ is convex and $(\nabla\varphi)_{\#}\rho_0 = \rho_1$, then $\nabla\varphi$ is the optimal map.

We use this theorem without proof; see [114]. Two consequences matter for us. First, the optimal map is monotone: $(\nabla\varphi(\mathbf{x}_1) - \nabla\varphi(\mathbf{x}_2))\cdot(\mathbf{x}_1-\mathbf{x}_2) \ge 0$, the multidimensional version of the non-crossing property Eq. (10.19). Second, if φ is smooth and strictly convex, then $\nabla\mathbf{T} = \nabla^2\varphi$ is symmetric positive definite and the Jacobian equation Eq. (10.13) becomes the **Monge–Ampère equation**

$$\rho_1(\nabla\varphi(\mathbf{x}))\,\det\nabla^2\varphi(\mathbf{x}) = \rho_0(\mathbf{x}). \tag{10.25}$$

For Gaussian ρ_0, ρ_1 this equation is solved by a quadratic φ, which we construct in Section 10.6. In one dimension every non-decreasing map is the derivative of a convex function, and Theorem 10.10 reduces to Theorem 10.5.

10.3 The Benamou–Brenier Dynamical Formulation

The definition Eq. (10.14) is static: it pairs initial and final positions and says nothing about the path in between. Thermodynamics needs a dynamical statement, because entropy production Eq. (10.4) is an integral over time of the kinetic energy of the probability fluid. Benamou and Brenier showed that $\mathcal{W}_2^2$ is the minimal kinetic action of a fluid that carries ρ_0 into ρ_1 in unit time [9]. This section proves that statement for smooth positive densities.

10.3.1 Probability as a Fluid

Equation Eq. (10.3) already wrote the Fokker–Planck equation as the continuity equation of a compressible fluid with density ρ_t and velocity $\mathbf{v}_t = -\mu\nabla E - D\nabla \ln \rho_t$. The diffusive part of the motion, which at the level of single trajectories is the nowhere-differentiable Brownian increment, appears at the level of the density as the smooth osmotic velocity $-D\nabla \ln \rho_t$ pointing down the density gradient. The same rewriting works for the general Itô diffusion of Eq. (6.1): with the current Eq. (6.19), $\mathbf{v} = \tilde{\mathbf{a}} - \mathbf{D}\nabla \ln \rho$. At the density level, every diffusion process is a deterministic flow. We now study such flows for an arbitrary velocity field.

Definition 10.11 (Admissible transport paths)**.** Let ρ_0, ρ_1 be smooth positive probability densities on $\mathbb{R}^N$ with finite second moments. An **admissible path** on $[0,1]$ is a pair $(\rho_t, \mathbf{v}_t)_{t\in[0,1]}$ of smooth positive densities and smooth velocity fields such that

(i) $\mathbf{v}_t$ is Lipschitz in $\mathbf{x}$ with a constant independent of t;

(ii) the continuity equation $\partial_t \rho_t + \nabla\cdot(\rho_t \mathbf{v}_t) = 0$ holds on $(0,1)\times\mathbb{R}^N$;

(iii) $\rho_t \to \rho_0$ as $t \to 0$ and $\rho_t \to \rho_1$ as $t \to 1$.

Its **kinetic action** is

$$\mathcal{A}[\rho, \mathbf{v}] := \int_0^1 \mathrm{d}t \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho_t(\mathbf{x})\, \|\mathbf{v}_t(\mathbf{x})\|^2 . \tag{10.26}$$

The action is twice the time-integrated kinetic energy of a fluid of unit total mass. The Lipschitz condition (i) guarantees that the ordinary differential equation of fluid particles has unique global solutions; it is a technical convenience that the sharp theory relaxes.

Lemma 10.12 (Lagrangian representation of the continuity equation)**.** *Let $\mathbf{v}_t$ be smooth and Lipschitz uniformly in t, and let $\mathbf{X}_t(\mathbf{x})$ be its flow map,*

$$\partial_t \mathbf{X}_t(\mathbf{x}) = \mathbf{v}_t(\mathbf{X}_t(\mathbf{x})), \qquad \mathbf{X}_0(\mathbf{x}) = \mathbf{x}. \tag{10.27}$$

If ρ_t is a smooth solution of $\partial_t\rho_t + \nabla\cdot(\rho_t\mathbf{v}_t) = 0$, then $\rho_t = (\mathbf{X}_t)_{\#}\rho_0$ for every t. Conversely, $(\mathbf{X}_t)_{\#}\rho_0$ solves the continuity equation in the weak sense.

Proof. Each $\mathbf{X}_t$ is a diffeomorphism of $\mathbb{R}^N$ (its inverse is obtained by running the flow backwards). Let $J_t(\mathbf{x}) = \det \nabla \mathbf{X}_t(\mathbf{x})$, with $J_0 = 1$. Differentiating Eq. (10.27) in $\mathbf{x}$ gives $\partial_t \nabla \mathbf{X}_t = \nabla\mathbf{v}_t(\mathbf{X}_t)\,\nabla\mathbf{X}_t$, and Jacobi's formula $\partial_t \det \mathbf{M} = \det\mathbf{M}\,\operatorname{tr}(\mathbf{M}^{-1}\partial_t\mathbf{M})$ yields Liouville's equation

$$\partial_t J_t(\mathbf{x}) = \big(\nabla\cdot\mathbf{v}_t\big)\big(\mathbf{X}_t(\mathbf{x})\big)\, J_t(\mathbf{x}),$$

so $J_t > 0$. Expanding the continuity equation as $\partial_t \rho_t + \mathbf{v}_t \cdot \nabla \rho_t = -\rho_t \nabla \cdot \mathbf{v}_t$, the chain rule along a fluid particle gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[\rho_t(\mathbf{X}_t(\mathbf{x}))\, J_t(\mathbf{x})\Big] = \Big[\partial_t \rho_t + \mathbf{v}_t \cdot \nabla \rho_t\Big]_{\mathbf{X}_t(\mathbf{x})} J_t + \rho_t(\mathbf{X}_t)\,(\nabla \cdot \mathbf{v}_t)(\mathbf{X}_t)\, J_t = 0.$$

Hence $\rho_t(\mathbf{X}_t(\mathbf{x}))\det \nabla \mathbf{X}_t(\mathbf{x}) = \rho_0(\mathbf{x})$, which is the Jacobian equation Eq. (10.13) for the map $\mathbf{X}_t$, so $\rho_t = (\mathbf{X}_t)_{\#}\rho_0$. For the converse, take a smooth compactly supported test function f and differentiate the push-forward identity Eq. (10.12):

$$\frac{\mathrm{d}}{\mathrm{d}t}\int f \,\mathrm{d}\big((\mathbf{X}_t)_{\#}\rho_0\big) = \int \nabla f(\mathbf{X}_t(\mathbf{x}))\cdot \mathbf{v}_t(\mathbf{X}_t(\mathbf{x}))\,\rho_0(\mathbf{x})\,\mathrm{d}^N x = \int \nabla f \cdot \mathbf{v}_t \,\mathrm{d}\big((\mathbf{X}_t)_{\#}\rho_0\big),$$

which is the weak form of the continuity equation (integrate the right side by parts to see $-\int f\,\nabla\cdot(\rho\mathbf{v})$). □

The lemma says that a density evolving by the continuity equation is nothing but the initial density carried along by fluid particles. The kinetic action can then be computed particle by particle.

10.3.2 The Benamou–Brenier Theorem

Theorem 10.13 (Benamou–Brenier)**.** *Let ρ_0, ρ_1 be smooth positive probability densities on $\mathbb{R}^N$ with finite second moments, and assume the Brenier potential φ of Theorem 10.10 is smooth with $\nabla^2\varphi \succeq c\,\mathbf{I}$ for some $c > 0$ and $\nabla\varphi - \mathbf{x}$ globally Lipschitz (these regularity properties hold, for example, for the Gaussian case of Section 10.6). Then*

$$\mathcal{W}_2^2(\rho_0,\rho_1) = \min_{(\rho_t,\mathbf{v}_t)} \int_0^1 \mathrm{d}t \int_{\mathbb{R}^N} \mathrm{d}^N x\, \rho_t(\mathbf{x})\, \|\mathbf{v}_t(\mathbf{x})\|^2, \tag{10.28}$$

*where the minimum is over admissible paths (Definition 10.11). It is attained by the **displacement interpolation***

$$\rho_t = (\mathbf{T}_t)_{\#}\rho_0, \qquad \mathbf{T}_t(\mathbf{x}) := (1-t)\,\mathbf{x} + t\,\nabla\varphi(\mathbf{x}), \tag{10.29}$$

with velocity field $\mathbf{v}_t = (\nabla\varphi - \mathrm{id})\circ \mathbf{T}_t^{-1}$.

Proof. Step 1: every admissible path has action at least $\mathcal{W}_2^2$. Let $(\rho_t, \mathbf{v}_t)$ be admissible and $\mathbf{X}_t$ the flow map of $\mathbf{v}_t$. By Lemma 10.12, $\rho_t = (\mathbf{X}_t)_{\#}\rho_0$. Using the push-forward identity Eq. (10.12) with $f = \|\mathbf{v}_t\|^2$ at each fixed t, and then exchanging the order of integration (Tonelli's theorem, as the integrand is non-negative),

$$\mathcal{A}[\rho,\mathbf{v}] = \int_0^1 \mathrm{d}t \int \rho_0(\mathbf{x})\, \|\mathbf{v}_t(\mathbf{X}_t(\mathbf{x}))\|^2\, \mathrm{d}^N x = \int \rho_0(\mathbf{x}) \left[\int_0^1 \|\partial_t \mathbf{X}_t(\mathbf{x})\|^2\, \mathrm{d}t\right] \mathrm{d}^N x.$$

For each starting point $\mathbf{x}$, the Cauchy–Schwarz inequality in time gives

$$\|\mathbf{X}_1(\mathbf{x}) - \mathbf{x}\|^2 = \left\| \int_0^1 \partial_t \mathbf{X}_t(\mathbf{x})\, \mathrm{d}t \right\|^2 \le \int_0^1 \mathrm{d}t \cdot \int_0^1 \|\partial_t \mathbf{X}_t(\mathbf{x})\|^2\, \mathrm{d}t = \int_0^1 \|\partial_t \mathbf{X}_t(\mathbf{x})\|^2\, \mathrm{d}t, \tag{10.30}$$

with equality if and only if the particle moves at constant velocity on a straight line. Therefore

$$\mathcal{A}[\rho,\mathbf{v}] \ge \int \rho_0(\mathbf{x})\, \|\mathbf{X}_1(\mathbf{x}) - \mathbf{x}\|^2\, \mathrm{d}^N x \ge \mathcal{W}_2^2(\rho_0,\rho_1),$$

where the last inequality holds because $(\mathbf{X}, \mathbf{X}_1(\mathbf{X}))$ with $\mathbf{X} \sim \rho_0$ is a coupling of ρ_0 and $(\mathbf{X}_1)_{\#}\rho_0 = \rho_1$.

Step 2: the displacement interpolation is admissible and has action $\mathcal{W}_2^2$. Write $\mathbf{T} = \nabla\varphi$. The map $\mathbf{T}_t$ of Eq. (10.29) is the gradient of $\varphi_t(\mathbf{x}) = \frac{1-t}{2}\|\mathbf{x}\|^2 + t\varphi(\mathbf{x})$, whose Hessian $(1-t)\mathbf{I} + t\nabla^2\varphi$ is positive definite for $t \in [0,1]$. The gradient of a strictly convex function is injective, because for $\mathbf{x}_1 \neq \mathbf{x}_2$,

$$(\nabla\varphi_t(\mathbf{x}_1) - \nabla\varphi_t(\mathbf{x}_2)) \cdot (\mathbf{x}_1 - \mathbf{x}_2) = \int_0^1 (\mathbf{x}_1 - \mathbf{x}_2)^\top \nabla^2\varphi_t(\mathbf{x}_2 + s(\mathbf{x}_1 - \mathbf{x}_2))(\mathbf{x}_1 - \mathbf{x}_2)\, \mathrm{d}s > 0.$$

So $\mathbf{T}_t$ is a diffeomorphism onto its image, and $\mathbf{v}_t = (\mathbf{T} - \mathrm{id}) \circ \mathbf{T}_t^{-1}$ is well defined on the support of ρ_t. Along the curve $t \mapsto \mathbf{T}_t(\mathbf{x})$,

$$\partial_t \mathbf{T}_t(\mathbf{x}) = \mathbf{T}(\mathbf{x}) - \mathbf{x} = \mathbf{v}_t(\mathbf{T}_t(\mathbf{x})),$$

so $\mathbf{T}_t$ is the flow map of $\mathbf{v}_t$ and, by Lemma 10.12, $\rho_t = (\mathbf{T}_t)_\# \rho_0$ solves the continuity equation, with $\rho_t = \rho_0$ at $t = 0$ and $\rho_t = \mathbf{T}_\# \rho_0 = \rho_1$ at $t = 1$. Every particle moves on a straight line at constant velocity, so Eq. (10.30) holds with equality, and

$$\mathcal{A}[\rho, \mathbf{v}] = \int \rho_0(\mathbf{x}) \|\mathbf{T}(\mathbf{x}) - \mathbf{x}\|^2 \, \mathrm{d}^N x = \mathcal{W}_2^2(\rho_0, \rho_1)$$

by Eq. (10.24). Combining the two steps proves Eq. (10.28) and shows that the minimum is attained. □

The proof shows where each half of the theorem comes from. The lower bound is purely kinematic: whatever the fluid does, each particle travels from $\mathbf{x}$ to $\mathbf{X}_1(\mathbf{x})$, and the cheapest way to cover a displacement in unit time is to move straight at constant speed. The upper bound needs Brenier's theorem to guarantee that the straight-line motions $\mathbf{x} \mapsto \nabla\varphi(\mathbf{x})$ do not collide, so that they define a velocity *field*.

Remark 10.14 (Displacement interpolation is not a mixture)**.** The linear mixture $(1-t)\rho_0 + t\rho_1$ also connects ρ_0 to ρ_1, and it is the path one obtains by switching the circuit abruptly between two settings with probability t. For separated Gaussians the mixture is bimodal at $t = \frac{1}{2}$, and generating it requires mass to jump from one mode to the other, which a continuous diffusion can do only through intermediate states. Displacement interpolation instead slides the mass continuously, as in Fig. 10.2. This difference is why optimal finite-time protocols move the bias and the stiffness of a trap continuously rather than toggling between endpoints.

10.3.3 Structure of the Optimal Flow: Gradient Velocities and Hamilton–Jacobi

Corollary 10.15 (Optimal velocity fields)**.** *Under the hypotheses of Theorem 10.13, the optimal velocity field $\mathbf{v}_t$ satisfies:*

(a) (Pressureless Euler / inviscid Burgers.) $\partial_t \mathbf{v}_t + (\mathbf{v}_t \cdot \nabla)\mathbf{v}_t = \mathbf{0}$: the velocity is constant along each particle path.

(b) (Potential flow.) $\mathbf{v}_t = \nabla\psi_t$ for a scalar potential ψ_t that solves the Hamilton–Jacobi equation

$$\partial_t \psi_t + \tfrac{1}{2}\|\nabla\psi_t\|^2 = 0, \qquad \psi_0 = \varphi - \tfrac{1}{2}\|\mathbf{x}\|^2. \tag{10.31}$$

Proof. (a) From the proof of Theorem 10.13, $\mathbf{v}_t(\mathbf{T}_t(\mathbf{x})) = \mathbf{T}(\mathbf{x}) - \mathbf{x}$ does not depend on t. Differentiating in t with the chain rule gives $\partial_t \mathbf{v}_t + (\nabla \mathbf{v}_t)\, \partial_t \mathbf{T}_t = \mathbf{0}$ at $\mathbf{T}_t(\mathbf{x})$, and $\partial_t \mathbf{T}_t = \mathbf{v}_t(\mathbf{T}_t)$.

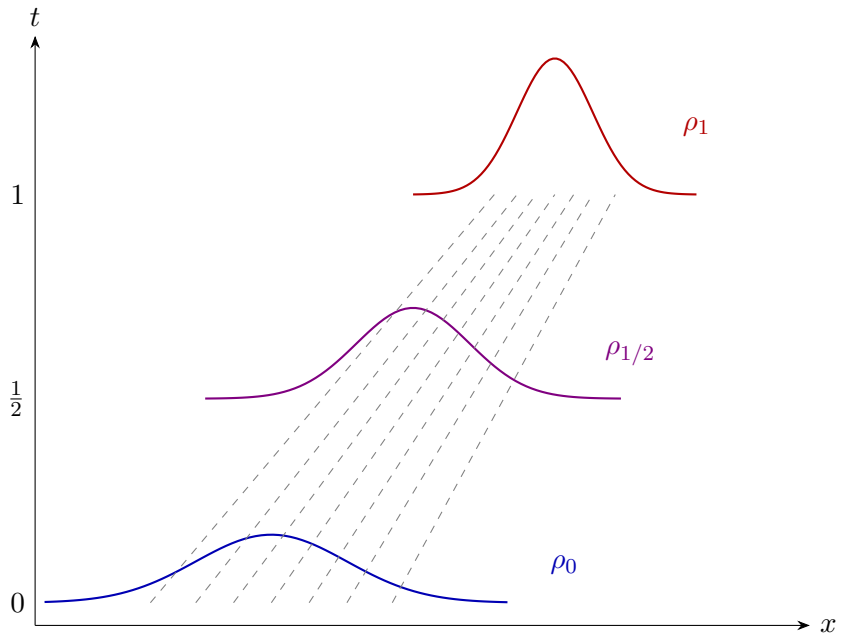


Figure 10.2: Displacement interpolation between $\rho_0 = \mathcal{N}(-1.5, 0.8^2)$ and $\rho_1 = \mathcal{N}(1.5, 0.4^2)$. Dashed lines are fluid-particle trajectories $t \mapsto \mathbf{T}_t(\mathbf{x})$: straight, at constant speed, and non-crossing. The intermediate density $\rho_{1/2} = \mathcal{N}(0, 0.6^2)$ has the averaged mean and the averaged standard deviation, not the averaged variance. The density is drawn with its peak height scaled by $1/\sigma$; the vertical offsets mark time.

(b) Differentiating $\mathbf{v}_t(\mathbf{T}_t(\mathbf{x})) = \nabla\varphi(\mathbf{x}) - \mathbf{x}$ in $\mathbf{x}$ gives

$$\nabla\mathbf{v}_t(\mathbf{T}_t(\mathbf{x})) = \big(\nabla^2\varphi(\mathbf{x}) - \mathbf{I}\big)\big((1-t)\mathbf{I} + t\nabla^2\varphi(\mathbf{x})\big)^{-1}.$$

Both factors are functions of the same symmetric matrix $\nabla^2\varphi(\mathbf{x})$, so they commute and their product is symmetric. A smooth vector field on $\mathbb{R}^N$ with symmetric Jacobian is curl-free and hence a gradient, $\mathbf{v}_t = \nabla\psi_t$. For a gradient field, $(\mathbf{v}\cdot\nabla)\mathbf{v} = \nabla(\frac{1}{2}\|\mathbf{v}\|^2)$, so (a) reads $\nabla\big(\partial_t\psi_t + \frac{1}{2}\|\nabla\psi_t\|^2\big) = \mathbf{0}$. The bracket is a function of t alone, which we absorb into ψ_t, giving Eq. (10.31). At $t = 0$, $\mathbf{v}_0 = \nabla\varphi - \mathbf{x} = \nabla(\varphi - \frac{1}{2}\|\mathbf{x}\|^2)$. □

Physics Insight: Optimal transport as free-particle mechanics

The Hamilton–Jacobi equation Eq. (10.31) is that of a free particle of unit mass, with ψ_t playing the role of Hamilton's principal function divided by mass. The same structure follows from a formal variational calculation. Adjoin the continuity constraint to $\frac{1}{2}\mathcal{A}$ with a Lagrange multiplier $\psi(\mathbf{x}, t)$ and integrate by parts (boundary terms at $t = 0, 1$ do not affect variations with fixed endpoints):

$$\mathcal{L} = \int_0^1 \mathrm{d}t \int \mathrm{d}^N x \left[\tfrac{1}{2}\rho\|\mathbf{v}\|^2 + \psi(\partial_t\rho + \nabla\cdot(\rho\mathbf{v}))\right] \simeq \int_0^1 \mathrm{d}t \int \mathrm{d}^N x \left[\tfrac{1}{2}\rho\|\mathbf{v}\|^2 - \rho\,\partial_t\psi - \rho\,\mathbf{v}\cdot\nabla\psi\right].$$

Varying $\mathbf{v}$ gives $\rho(\mathbf{v} - \nabla\psi) = \mathbf{0}$, so $\mathbf{v} = \nabla\psi$. Varying ρ gives $\frac{1}{2}\|\mathbf{v}\|^2 - \partial_t\psi - \mathbf{v}\cdot\nabla\psi = 0$, which with $\mathbf{v} = \nabla\psi$ is $\partial_t\psi + \frac{1}{2}\|\nabla\psi\|^2 = 0$. Benamou and Brenier observed that in the variables $(\rho, \mathbf{m}) = (\rho, \rho\mathbf{v})$ the constraint $\partial_t\rho + \nabla\cdot\mathbf{m} = 0$ is linear and the integrand $\|\mathbf{m}\|^2/\rho$ is jointly

convex, because it is a supremum of linear functions: $\|\mathbf{m}\|^2/\rho = \sup_{\mathbf{b}}(2\mathbf{b}\cdot\mathbf{m} - \|\mathbf{b}\|^2\rho)$ for $\rho > 0$, the supremum being attained at $\mathbf{b} = \mathbf{m}/\rho$. A convex problem with linear constraints has no spurious local minima, so a critical point of $\mathcal{L}$ is a global minimizer. This convexity is also what makes Eq. (10.28) a practical numerical method [9].

10.3.4 Length, Speed, and the Projection onto Gradient Fields

The action Eq. (10.26) is an energy. The corresponding length is often more natural, and the time-rescaled form is what thermodynamics uses.

Definition 10.16 (Wasserstein speed)**.** Let (ρ_t) be a smooth curve of positive densities. A velocity field $\mathbf{v}_t$ *generates* the curve at time t if $\partial_t\rho_t + \nabla\cdot(\rho_t\mathbf{v}_t) = 0$. The **Wasserstein speed** of the curve is

$$\|\partial_t\rho_t\|_{\rho_t} := \inf\left\{\|\mathbf{v}\|_{L^2(\rho_t)} : \partial_t\rho_t + \nabla\cdot(\rho_t\mathbf{v}) = 0\right\}, \qquad \|\mathbf{v}\|^2_{L^2(\rho)} := \int \rho\,\|\mathbf{v}\|^2\,\mathrm{d}^N x. \tag{10.32}$$

Proposition 10.17 (Helmholtz projection)**.** *Let $\rho > 0$ be smooth and let $\mathbf{v}$ generate $\partial_t\rho$. Suppose the weighted Poisson equation $\nabla\cdot(\rho\nabla\psi) = \nabla\cdot(\rho\mathbf{v})$ has a smooth solution ψ with $\rho\psi\nabla\psi$ and $\rho\psi\mathbf{v}$ decaying at infinity. Then $\mathbf{w} := \mathbf{v} - \nabla\psi$ satisfies $\nabla\cdot(\rho\mathbf{w}) = 0$, and*

$$\int \rho\,\|\mathbf{v}\|^2\,\mathrm{d}^N x = \int \rho\,\|\nabla\psi\|^2\,\mathrm{d}^N x + \int \rho\,\|\mathbf{w}\|^2\,\mathrm{d}^N x. \tag{10.33}$$

Hence the gradient field $\nabla\psi$ is the unique generator of minimal kinetic energy, and $\|\partial_t\rho\|_\rho = \|\nabla\psi\|_{L^2(\rho)}$.

Proof. By construction $\nabla\cdot(\rho\mathbf{w}) = \nabla\cdot(\rho\mathbf{v}) - \nabla\cdot(\rho\nabla\psi) = 0$. Expanding $\|\nabla\psi + \mathbf{w}\|^2$, the cross term is

$$2\int \rho\,\nabla\psi\cdot\mathbf{w}\;\mathrm{d}^N x = -2\int \psi\,\nabla\cdot(\rho\mathbf{w})\;\mathrm{d}^N x = 0,$$

by the divergence theorem (the surface term vanishes by the decay assumption). This gives Eq. (10.33). Any other generator differs from $\nabla\psi$ by a field $\mathbf{w}'$ with $\nabla\cdot(\rho\mathbf{w}') = 0$ and so has kinetic energy larger by $\int\rho\,\|\mathbf{w}'\|^2$. □

The solenoidal part $\mathbf{w}$ moves probability around closed loops without changing the density. In the language of Chapters 6 and 8 it is the circulating current $\rho\mathbf{w}$ of a non-equilibrium steady state. Such a current costs kinetic energy, and hence dissipation, but does no transport. We return to this point in Section 10.5.

Corollary 10.18 (Length and time-rescaled forms)**.** *Under the hypotheses of Theorem 10.13:*

(a) $\mathcal{W}_2(\rho_0,\rho_1) = \min\int_0^1 \|\mathbf{v}_t\|_{L^2(\rho_t)}\,\mathrm{d}t$ *over admissible paths, and for any smooth curve* $\mathcal{W}_2(\rho_0,\rho_1) \le \int_0^1 \|\partial_t\rho_t\|_{\rho_t}\,\mathrm{d}t$.

(b) For any admissible path on a time interval $[0,\tau]$ (defined as in Definition 10.11 with $[0,1]$ replaced by $[0,\tau]$),

$$\int_0^\tau \mathrm{d}t\int_{\mathbb{R}^N}\mathrm{d}^N x\,\rho_t\,\|\mathbf{v}_t\|^2 \;\ge\; \frac{1}{\tau}\left(\int_0^\tau \|\mathbf{v}_t\|_{L^2(\rho_t)}\,\mathrm{d}t\right)^2 \;\ge\; \frac{\mathcal{W}_2^2(\rho_0,\rho_\tau)}{\tau}, \tag{10.34}$$

with equality throughout for the displacement interpolation run at constant speed, $\rho_t = (\mathbf{T}_{t/\tau})_\#\rho_0$.

Proof. (a) Let $\ell_t = \|\mathbf{v}_t\|_{L^2(\rho_t)}$. By Cauchy–Schwarz, $(\int_0^1 \ell_t \,\mathrm{d}t)^2 \le \int_0^1 \ell_t^2 \,\mathrm{d}t = \mathcal{A}$, and for the displacement interpolation ℓ_t is constant, $\ell_t^2 = \mathcal{W}_2^2$, so the minimal length is at most $\mathcal{W}_2$. Conversely, given any admissible path with $\ell_t > 0$ and total length $\Lambda = \int_0^1 \ell_t \,\mathrm{d}t$, reparametrize time by $s(t) = \Lambda^{-1} \int_0^t \ell_{t'} \,\mathrm{d}t'$. The path $\tilde{\rho}_s = \rho_{t(s)}$ is generated by $\tilde{\mathbf{v}}_s = \mathbf{v}_{t(s)} \,\mathrm{d}t/\,\mathrm{d}s$, whose $L^2(\tilde{\rho}_s)$ norm is $\ell_t \,(\Lambda/\ell_t) = \Lambda$. Its action is Λ^2, which by Theorem 10.13 is at least $\mathcal{W}_2^2$. Hence $\Lambda \ge \mathcal{W}_2$. Applying this with the minimal generators $\nabla\psi_t$ of Proposition 10.17 gives the second statement.

(b) The first inequality is Cauchy–Schwarz on $[0,\tau]$: $(\int_0^\tau \ell_t \,\mathrm{d}t)^2 \le \tau \int_0^\tau \ell_t^2 \,\mathrm{d}t$. For the second, rescale time, $s = t/\tau$: the path $\hat{\rho}_s = \rho_{\tau s}$ on $[0,1]$ is generated by $\hat{\mathbf{v}}_s = \tau\, \mathbf{v}_{\tau s}$, and its length $\int_0^1 \|\hat{\mathbf{v}}_s\|_{L^2(\hat{\rho}_s)} \,\mathrm{d}s = \int_0^\tau \ell_t \,\mathrm{d}t$ is at least $\mathcal{W}_2(\rho_0, \rho_\tau)$ by (a). Equality in both steps requires constant speed along a minimizing path. □

10.4 Otto Calculus and the Jordan–Kinderlehrer–Otto Scheme

The Benamou–Brenier formula says that $\mathcal{W}_2$ is the geodesic distance of a length structure on densities whose squared speed is the kinetic energy $\int \rho \|\nabla\psi\|^2$. Otto observed that this length structure comes from a formal Riemannian metric, and that in this metric the Fokker–Planck equation is a gradient flow [58]. In Chapter 1 we said that a thermal circuit performs variational inference by descending the variational free energy $\mathcal{F}[\rho]$. In Chapter 6 we proved that $\mathcal{F}[\rho_t]$ decreases (the H-theorem). Here we identify the precise sense in which the descent is steepest. We present Otto's calculus as a formal computation, as is customary: the manifold of densities is infinite-dimensional and we do not verify the functional-analytic conditions under which each step is rigorous. The JKO scheme in Section 10.4.3 then gives a construction that can be made rigorous, and whose Euler–Lagrange equation we derive explicitly.

10.4.1 The Otto Metric

Let $\mathcal{M}$ denote the set of smooth positive probability densities on $\mathbb{R}^N$ with finite second moment. A tangent vector at $\rho \in \mathcal{M}$ is the time derivative $s = \partial_t \rho_t$ of a curve through ρ; since mass is conserved, $\int s \,\mathrm{d}^N x = 0$. Define the weighted Laplacian

$$\mathcal{K}_\rho \psi := -\nabla \cdot (\rho \nabla \psi). \tag{10.35}$$

It is symmetric and non-negative on $L^2(\mathrm{d}^N x)$: integrating by parts twice, $\int \psi_1 \,\mathcal{K}_\rho \psi_2 \,\mathrm{d}^N x = \int \rho \,\nabla\psi_1 \cdot \nabla\psi_2 \,\mathrm{d}^N x = \int \psi_2 \,\mathcal{K}_\rho \psi_1 \,\mathrm{d}^N x$. Its kernel consists of constants. By Proposition 10.17, the minimal-energy generator of a tangent vector s is $\nabla\psi$ with $\mathcal{K}_\rho \psi = s$, and we *assume* this equation is solvable (formally, $\psi = \mathcal{K}_\rho^{-1} s$, unique up to a constant).

Definition 10.19 (Otto metric (formal))**.** For tangent vectors s_1, s_2 at ρ, with potentials $\psi_i = \mathcal{K}_\rho^{-1} s_i$, the **Otto metric** is

$$g_\rho(s_1, s_2) := \int_{\mathbb{R}^N} \rho \,\nabla\psi_1 \cdot \nabla\psi_2 \,\mathrm{d}^N x = \int_{\mathbb{R}^N} s_1 \,\mathcal{K}_\rho^{-1} s_2 \,\mathrm{d}^N x. \tag{10.36}$$

The second form shows that the metric tensor is the operator $\mathcal{K}_\rho^{-1}$, the inverse of the weighted Laplacian, and the inverse metric tensor is $\mathcal{K}_\rho$ itself. The metric depends on the base point ρ: moving mass where ρ is small is expensive, because a given change s there requires a large velocity. By Definition 10.16 and Proposition 10.17, $g_\rho(\partial_t\rho, \partial_t\rho) = \|\partial_t\rho\|_\rho^2$, and by Corollary 10.18 the length of a curve in this metric is bounded below by $\mathcal{W}_2$ of its endpoints, with equality for displacement interpolation. So $\mathcal{W}_2$ is the geodesic distance of g, and displacement interpolations are its constant-speed geodesics.

Remark 10.20 (Three different metrics)**.** The space of densities carries several natural metrics. The Otto metric Eq. (10.36) is built from transport and knows about distances in $\mathbb{R}^N$. The Fisher–Rao metric $\int s_1 s_2/\rho \, \mathrm{d}^N x$ is built from information and is blind to distances in $\mathbb{R}^N$; it is the infinitesimal form of the KL divergence and defines the statistical (Fisher) length of a family of distributions. Finally, the friction tensor $\zeta_{ij}(\boldsymbol{\lambda})$ of linear-response thermodynamics is a metric on the space of control parameters. In Section 10.5 we show that for overdamped dynamics the friction tensor is the pull-back of the Otto metric, divided by μ, to the manifold of equilibrium states.

10.4.2 Wasserstein Gradients and the Fokker–Planck Equation

Let $\mathcal{F} : \mathcal{M} \to \mathbb{R}$ be a smooth functional with functional derivative $\frac{\delta \mathcal{F}}{\delta \rho}$, so that along any curve

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t] = \int_{\mathbb{R}^N} \frac{\delta \mathcal{F}}{\delta \rho}(\mathbf{x}) \, \partial_t \rho_t(\mathbf{x}) \, \mathrm{d}^N x. \tag{10.37}$$

The **Wasserstein gradient** $\mathrm{grad}_{\mathcal{W}_2}\mathcal{F}$ at ρ is the tangent vector defined by $g_\rho(\mathrm{grad}_{\mathcal{W}_2}\mathcal{F}, s) = \int \frac{\delta \mathcal{F}}{\delta \rho}\, s \, \mathrm{d}^N x$ for every tangent vector s.

Proposition 10.21 (Wasserstein gradient (formal))**.**

$$\mathrm{grad}_{\mathcal{W}_2}\mathcal{F} = \mathcal{K}_\rho \frac{\delta \mathcal{F}}{\delta \rho} = -\nabla \cdot \left(\rho \, \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right), \qquad g_\rho(\mathrm{grad}_{\mathcal{W}_2}\mathcal{F}, \mathrm{grad}_{\mathcal{W}_2}\mathcal{F}) = \int \rho \left\| \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right\|^2 \mathrm{d}^N x. \tag{10.38}$$

Proof. Let $u = \frac{\delta \mathcal{F}}{\delta \rho}$ and s a tangent vector with $\psi_s = \mathcal{K}_\rho^{-1} s$. The candidate $\mathcal{K}_\rho u$ has potential u, so by Eq. (10.36), $g_\rho(\mathcal{K}_\rho u, s) = \int \rho \, \nabla u \cdot \nabla \psi_s = \int u \, \mathcal{K}_\rho \psi_s = \int u \, s$, as required. The norm follows by taking $s = \mathcal{K}_\rho u$. □

A gradient flow with mobility μ is the evolution $\partial_t \rho_t = -\mu \, \mathrm{grad}_{\mathcal{W}_2}\mathcal{F}[\rho_t]$. By Eq. (10.38), it is the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0, \qquad \mathbf{v}_t = -\mu \, \nabla \frac{\delta \mathcal{F}}{\delta \rho}[\rho_t], \tag{10.39}$$

in which each element of the probability fluid moves with velocity equal to mobility times the force $-\nabla \frac{\delta \mathcal{F}}{\delta \rho}$.

Theorem 10.22 (Fokker–Planck as a Wasserstein gradient flow (Otto))**.** *Let $\mathcal{F}[\rho] = \int \rho \, E \, \mathrm{d}^N x + k_\mathrm{B} T \int \rho \ln \rho \, \mathrm{d}^N x$ be the variational free energy Eq. (6.46). Then the gradient flow Eq. (10.39) is the Smoluchowski equation*

$$\partial_t \rho_t = \nabla \cdot \Big(\mu \rho_t \nabla E + D \nabla \rho_t \Big), \qquad D = \mu k_\mathrm{B} T, \tag{10.40}$$

and along it

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t] = -\frac{1}{\mu} \, g_{\rho_t}(\partial_t \rho_t, \partial_t \rho_t) = -\mu \, g_{\rho_t}(\mathrm{grad}_{\mathcal{W}_2}\mathcal{F}, \mathrm{grad}_{\mathcal{W}_2}\mathcal{F}) = -T \dot{\Sigma}_{\mathrm{tot}}(t). \tag{10.41}$$

Proof. The functional derivative is $\frac{\delta \mathcal{F}}{\delta \rho} = E + k_\mathrm{B} T (\ln \rho + 1)$, whose gradient is $\nabla E + k_\mathrm{B} T \nabla \rho / \rho$. Substituting into Eq. (10.39),

$$\rho_t \mathbf{v}_t = -\mu \rho_t \nabla E - \mu k_\mathrm{B} T \nabla \rho_t = -\mu \rho_t \nabla E - D \nabla \rho_t,$$

which is the Smoluchowski current Eq. (6.21) with $\mathbf{D} = D\mathbf{I}$, and Eq. (10.40) follows. The velocity $\mathbf{v}_t = -\mu\nabla\frac{\delta\mathcal{F}}{\delta\rho}$ is a gradient, so it is the minimal generator of $\partial_t\rho_t$ and $g_{\rho_t}(\partial_t\rho_t, \partial_t\rho_t) = \int \rho_t \|\mathbf{v}_t\|^2$. By Eq. (10.37) and the continuity equation, integrating by parts,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[\rho_t] = -\int \frac{\delta\mathcal{F}}{\delta\rho}\,\nabla\cdot(\rho_t\mathbf{v}_t) = \int \rho_t\,\mathbf{v}_t\cdot\nabla\frac{\delta\mathcal{F}}{\delta\rho} = -\frac{1}{\mu}\int \rho_t\,\|\mathbf{v}_t\|^2 = -\mu\int \rho_t\left\|\nabla\frac{\delta\mathcal{F}}{\delta\rho}\right\|^2.$$

The last expression is $-\mu\, g(\operatorname{grad}\mathcal{F}, \operatorname{grad}\mathcal{F})$ by Eq. (10.38), and $\frac{1}{\mu}\int \rho_t\|\mathbf{v}_t\|^2 = T\dot{\Sigma}_{\mathrm{tot}}$ by Eq. (10.4). □

Using $\nabla\frac{\delta\mathcal{F}}{\delta\rho} = k_{\mathrm{B}}T\nabla\ln(\rho/p_{\mathrm{eq}})$, the last form of the dissipation is $\mu(k_{\mathrm{B}}T)^2\int\rho\,\|\nabla\ln(\rho/p_{\mathrm{eq}})\|^2 = k_{\mathrm{B}}T\,\mathcal{I}_{D\mathbf{I}}(\rho\,\|\,p_{\mathrm{eq}})$, which is the H-theorem Eq. (6.49). The new information in Theorem 10.22 is not that $\mathcal{F}$ decreases but *how*: among all ways of changing ρ that conserve mass locally and have the same kinetic cost, the Fokker–Planck flow decreases $\mathcal{F}$ fastest. Since the two expressions in Eq. (10.41) are equal, the dissipation rate can be split symmetrically,

$$\mathcal{F}[\rho_\tau] + \int_0^\tau \left[\frac{1}{2\mu}\,\|\partial_t\rho_t\|_{\rho_t}^2 + \frac{\mu}{2}\,\|\operatorname{grad}_{\mathcal{W}_2}\mathcal{F}[\rho_t]\|_{\rho_t}^2\right]\mathrm{d}t = \mathcal{F}[\rho_0]. \tag{10.42}$$

This energy-dissipation equality characterizes the gradient flow. For an arbitrary curve the left side is at least the right side, by the inequality $\frac{a^2}{2\mu} + \frac{\mu b^2}{2} \ge ab \ge -\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}$, with equality only if the velocity is exactly $-\mu$ times the gradient.

Physics Insight: Onsager's principle of least dissipation

Theorem 10.22 is a continuum version of Onsager's variational principle for overdamped dynamics. Define the Rayleighian of a trial velocity field $\mathbf{v}$ as half the dissipation rate plus the rate of change of free energy it would cause:

$$\mathcal{R}[\mathbf{v}] = \frac{1}{2\mu}\int \rho\,\|\mathbf{v}\|^2\,\mathrm{d}^N x + \int \rho\,\mathbf{v}\cdot\nabla\frac{\delta\mathcal{F}}{\delta\rho}\,\mathrm{d}^N x.$$

Completing the square, $\mathcal{R}[\mathbf{v}] = \frac{1}{2\mu}\int\rho\left\|\mathbf{v} + \mu\nabla\frac{\delta\mathcal{F}}{\delta\rho}\right\|^2 - \frac{\mu}{2}\int\rho\left\|\nabla\frac{\delta\mathcal{F}}{\delta\rho}\right\|^2$, so $\mathcal{R}$ is minimized by $\mathbf{v} = -\mu\nabla\frac{\delta\mathcal{F}}{\delta\rho}$, the Fokker–Planck velocity. The physical system chooses the motion that balances viscous dissipation against the release of free energy. The factor $\frac{1}{2}$ in front of the dissipation is essential, and it reappears in the JKO scheme below.

10.4.3 The JKO Minimizing-Movement Scheme

The formal calculus can be replaced by a time discretization that uses only the distance $\mathcal{W}_2$, not the metric tensor. For a Euclidean gradient flow $\dot{\mathbf{x}} = -\mu\nabla f(\mathbf{x})$, the implicit Euler step $\mathbf{x}^{k+1} = \mathbf{x}^k - h\mu\nabla f(\mathbf{x}^{k+1})$ is the Euler–Lagrange equation of the minimization of $\left\|\mathbf{x} - \mathbf{x}^k\right\|^2/(2\mu h) + f(\mathbf{x})$. Jordan, Kinderlehrer, and Otto replaced the Euclidean distance by $\mathcal{W}_2$ [58].

Definition 10.23 (JKO scheme)**.** Fix a time step $h > 0$ and an initial density ρ^0. Define recursively

$$\rho^{k+1} \in \operatorname*{arg\,min}_{\rho\in\mathcal{P}_2(\mathbb{R}^N)}\left\{\frac{\mathcal{W}_2^2(\rho,\rho^k)}{2\mu h} + \mathcal{F}[\rho]\right\}, \qquad \mathcal{F}[\rho] = \int \rho E\,\mathrm{d}^N x + k_{\mathrm{B}}T\int\rho\ln\rho\,\mathrm{d}^N x. \tag{10.43}$$

Each step is a trade-off. Lowering $\mathcal{F}$ requires moving mass, and moving it a Wasserstein distance $\mathcal{W}_2$ in time h is penalized by $\mathcal{W}_2^2/(2\mu h)$, half the minimal dissipated work of Eq. (10.1). Jordan, Kinderlehrer, and Otto proved that for smooth $E \geq 0$ with $\|\nabla E\| \leq c\,(E+1)$ for some constant c, the minimizers exist and are unique, and the piecewise-constant interpolation $\rho^{(h)}(t) = \rho^k$ for $t \in [kh, (k+1)h)$ converges, as $h \to 0$, to the weak solution of the Fokker–Planck equation Eq. (10.40) with initial datum ρ^0. We take existence, uniqueness, and convergence as given, and verify the essential point: that the optimality condition of one step is a discretized Fokker–Planck equation.

Proposition 10.24 (Euler–Lagrange equation of a JKO step)**.** *Assume the minimizer ρ^{k+1} of Eq. (10.43) is smooth and positive, and let $\mathbf{T}$ be the Brenier map from ρ^{k+1} to ρ^k. Then*

$$\mathbf{T}(\mathbf{x}) - \mathbf{x} = \mu h\, \nabla\Big(E(\mathbf{x}) + k_{\mathrm{B}}T \ln \rho^{k+1}(\mathbf{x})\Big) \qquad \text{for } \rho^{k+1}\text{-almost every } \mathbf{x}, \tag{10.44}$$

and for every smooth compactly supported test function f,

$$\int f\, \frac{\rho^{k+1} - \rho^k}{h}\, \mathrm{d}^N x = -\int \nabla f \cdot \Big(\mu \rho^{k+1} \nabla E + D \nabla \rho^{k+1}\Big)\, \mathrm{d}^N x + \mathcal{O}\left(\frac{\|\nabla^2 f\|_\infty\, \mathcal{W}_2^2(\rho^{k+1}, \rho^k)}{h}\right), \tag{10.45}$$

where the error term is $\mathcal{O}(h)$ because Eq. (10.44) gives $\mathcal{W}_2^2(\rho^{k+1}, \rho^k) = \mu^2 h^2 \int \rho^{k+1} \left\|\nabla(E + k_{\mathrm{B}}T \ln \rho^{k+1})\right\|^2$.

Proof. Write $\rho = \rho^{k+1}$ and let $J[\cdot]$ be the functional minimized in Eq. (10.43). We perturb ρ by transporting it along a smooth compactly supported vector field $\boldsymbol{\xi}$: $\rho_\varepsilon := (\mathrm{id} + \varepsilon\boldsymbol{\xi})_{\#}\rho$. For $|\varepsilon|$ small, $\mathrm{id} + \varepsilon\boldsymbol{\xi}$ is a diffeomorphism. We compute the first variation of each term at $\varepsilon = 0$.

Potential energy. By Eq. (10.12), $\int E\, \rho_\varepsilon = \int E(\mathbf{x} + \varepsilon\boldsymbol{\xi}(\mathbf{x}))\rho(\mathbf{x})\, \mathrm{d}^N x$, whose ε-derivative at 0 is $\int \nabla E \cdot \boldsymbol{\xi}\, \rho$.

Entropy. The Jacobian equation Eq. (10.13) gives $\rho_\varepsilon(\mathbf{x} + \varepsilon\boldsymbol{\xi}(\mathbf{x}))\det(\mathbf{I} + \varepsilon\nabla\boldsymbol{\xi}(\mathbf{x})) = \rho(\mathbf{x})$. Changing variables $\mathbf{y} = \mathbf{x} + \varepsilon\boldsymbol{\xi}(\mathbf{x})$,

$$\int \rho_\varepsilon(\mathbf{y}) \ln \rho_\varepsilon(\mathbf{y})\, \mathrm{d}^N y = \int \rho(\mathbf{x}) \Big[\ln \rho(\mathbf{x}) - \ln\det(\mathbf{I} + \varepsilon\nabla\boldsymbol{\xi}(\mathbf{x}))\Big]\, \mathrm{d}^N x.$$

Since $\ln\det(\mathbf{I} + \varepsilon\mathbf{M}) = \varepsilon\,\mathrm{tr}\,\mathbf{M} + \mathcal{O}(\varepsilon^2)$, the derivative at 0 is $-\int \rho\, \nabla\cdot\boldsymbol{\xi} = \int \nabla\rho\cdot\boldsymbol{\xi}$, after an integration by parts.

Transport term. The pair $(\mathbf{X} + \varepsilon\boldsymbol{\xi}(\mathbf{X}), \mathbf{T}(\mathbf{X}))$ with $\mathbf{X} \sim \rho$ is a coupling of ρ_ε and ρ^k, so

$$\mathcal{W}_2^2(\rho_\varepsilon, \rho^k) \leq \int \|\mathbf{x} + \varepsilon\boldsymbol{\xi}(\mathbf{x}) - \mathbf{T}(\mathbf{x})\|^2\, \rho(\mathbf{x})\, \mathrm{d}^N x,$$

with equality at $\varepsilon = 0$ because $\mathbf{T}$ is optimal. Define

$$\Phi(\varepsilon) := \frac{1}{2\mu h} \int \|\mathbf{x} + \varepsilon\boldsymbol{\xi} - \mathbf{T}\|^2\, \rho\, \mathrm{d}^N x + \mathcal{F}[\rho_\varepsilon] \;\geq\; J[\rho_\varepsilon] \;\geq\; J[\rho] = \Phi(0).$$

The smooth function Φ therefore has a minimum at $\varepsilon = 0$, and $\Phi'(0) = 0$:

$$\frac{1}{\mu h} \int (\mathbf{x} - \mathbf{T}(\mathbf{x}))\cdot\boldsymbol{\xi}(\mathbf{x})\, \rho(\mathbf{x})\, \mathrm{d}^N x + \int \Big(\rho\nabla E + k_{\mathrm{B}}T\nabla\rho\Big)\cdot\boldsymbol{\xi}\, \mathrm{d}^N x = 0.$$

Since $\boldsymbol{\xi}$ is arbitrary, $\rho(\mathbf{x} - \mathbf{T})/(\mu h) = -\rho\nabla E - k_{\mathrm{B}}T\nabla\rho$ pointwise; dividing by $\rho > 0$ gives Eq. (10.44).

For Eq. (10.45), use $\mathbf{T}_{\#}\rho^{k+1} = \rho^k$ to write

$$\int f\, (\rho^{k+1} - \rho^k)\, \mathrm{d}^N x = \int \Big[f(\mathbf{x}) - f(\mathbf{T}(\mathbf{x}))\Big]\rho^{k+1}(\mathbf{x})\, \mathrm{d}^N x.$$

Taylor's theorem gives $f(\mathbf{T}(\mathbf{x})) = f(\mathbf{x}) + \nabla f(\mathbf{x})\cdot(\mathbf{T}(\mathbf{x}) - \mathbf{x}) + R(\mathbf{x})$ with $|R| \leq \frac{1}{2}\|\nabla^2 f\|_\infty \|\mathbf{T}(\mathbf{x}) - \mathbf{x}\|^2$. Inserting Eq. (10.44) for $\mathbf{T} - \mathrm{id}$ and dividing by h gives the main term of Eq. (10.45), while the remainder integrates to at most $\frac{1}{2}\|\nabla^2 f\|_\infty\, \mathcal{W}_2^2(\rho^{k+1}, \rho^k)/h$. □

Equation Eq. (10.44) has a direct particle reading. The map $\mathbf{T}$ sends each new position $\mathbf{x}$ back to where its mass came from, so $\mathbf{x} = \mathbf{T}(\mathbf{x}) + h\,\mathbf{v}(\mathbf{x})$ with $\mathbf{v} = -\mu\nabla(E + k_{\mathrm{B}}T\ln\rho^{k+1})$: each particle takes an implicit Euler step with the Fokker–Planck velocity Eq. (10.3) evaluated at the new density. The diffusion does not enter as a random kick but as the deterministic osmotic force $-k_{\mathrm{B}}T\nabla\ln\rho$ that pushes mass out of crowded regions. The JKO scheme thus builds the Fokker–Planck equation without ever mentioning Brownian motion: only free energy and transport cost appear.

10.4.4 Displacement Convexity

Gradient flows of convex functions contract. The relevant notion of convexity on $(\mathcal{M}, \mathcal{W}_2)$ is convexity along displacement interpolations.

Proposition 10.25 (Displacement convexity of the free energy)**.** *Let $\nabla^2 E \succeq \kappa\mathbf{I}$ with $\kappa \in \mathbb{R}$, and let $\rho_s = (\mathbf{T}_s)_{\#}\rho_0$, $s \in [0,1]$, be the displacement interpolation of Theorem 10.13 with smooth Brenier potential φ. Then*

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2}\mathcal{F}[\rho_s] \;\geq\; \kappa\,\mathcal{W}_2^2(\rho_0,\rho_1). \tag{10.46}$$

Proof. Potential energy. By Eq. (10.12), $\int E\,\rho_s = \int E(\mathbf{x}+s\mathbf{u}(\mathbf{x}))\rho_0(\mathbf{x})\,\mathrm{d}^N x$ with $\mathbf{u} = \nabla\varphi - \mathrm{id}$. Differentiating twice under the integral, the second derivative is $\int \mathbf{u}^{\top}\nabla^2 E(\mathbf{x}+s\mathbf{u})\,\mathbf{u}\,\rho_0 \geq \kappa\int\|\mathbf{u}\|^2\rho_0 = \kappa\mathcal{W}_2^2$.

Entropy. As in the proof of Proposition 10.24, $\int\rho_s\ln\rho_s = \int\rho_0[\ln\rho_0 - \ln\det\mathbf{A}_s]$ with $\mathbf{A}_s = \nabla\mathbf{T}_s = \mathbf{I} + s\mathbf{B}$ and $\mathbf{B} = \nabla^2\varphi - \mathbf{I}$ symmetric. Using $\frac{\mathrm{d}}{\mathrm{d}s}\ln\det\mathbf{A}_s = \mathrm{tr}(\mathbf{A}_s^{-1}\mathbf{B})$ and $\frac{\mathrm{d}}{\mathrm{d}s}\mathbf{A}_s^{-1} = -\mathbf{A}_s^{-1}\mathbf{B}\mathbf{A}_s^{-1}$,

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2}(-\ln\det\mathbf{A}_s) = \mathrm{tr}(\mathbf{A}_s^{-1}\mathbf{B}\mathbf{A}_s^{-1}\mathbf{B}) = \mathrm{tr}(\mathbf{M}_s^2) \geq 0, \qquad \mathbf{M}_s := \mathbf{A}_s^{-1/2}\mathbf{B}\mathbf{A}_s^{-1/2},$$

where we used cyclicity of the trace and the fact that $\mathbf{M}_s$ is symmetric. So the entropic term $k_{\mathrm{B}}T\int\rho_s\ln\rho_s$ is convex in s, and adding the two contributions proves Eq. (10.46). □

The entropy part is displacement convex for every ρ_0: along a straight-line interpolation, $\int\rho_s\ln\rho_s$ is a convex function of s. When $\kappa > 0$ the free energy is uniformly convex along geodesics, and the gradient flow contracts $\mathcal{W}_2$ at rate $\mu\kappa$. We proved this contraction directly, by synchronous coupling, in Proposition 10.8. In the Bakry–Émery language of Section 6.5, the same hypothesis $\nabla^2 E \succeq \kappa\mathbf{I}$ appears as a lower bound on the Γ_2 curvature. For the double-well sampler, $\kappa < 0$ near the barrier, the free energy is not displacement convex, and the gradient flow can stall in a metastable state, which is the Kramers problem of Chapter 7.

Remark 10.26 (Other dynamics, other metrics)**.** The Wasserstein structure is tied to dynamics in which probability flows continuously through space, as in the Langevin equation of a node voltage. Jump processes, such as the stochastic flips of a p-bit (the discrete limit of the double-well sampler), move probability between discrete states without passing through intermediate positions. For such Markov jump processes an analogous gradient-flow structure and an analogous finite-time bound hold with a discrete transport distance defined on the graph of allowed transitions [113]. The geometric lesson carries over: the dissipation cost of a finite-time process is set by a transport distance adapted to the dynamics.

10.5 The Geometric Second Law and Optimal Protocols

We now return to the finite-time dilemma of Section 10.1 with the tools of Sections 10.3 and 10.4. The link is that the entropy production rate Eq. (10.4) is, up to the factor k_{B}/D, the kinetic energy of the probability fluid, and Benamou–Brenier bounds the time integral of that kinetic energy by $\mathcal{W}_2^2/\tau$. This connection between minimal entropy production and optimal transport was made in the stochastic-thermodynamics setting by Aurell, Mejía-Monasterio, and Muratore-Ginanneschi [5]. Closely related bounds of currents by entropy production in Langevin systems were derived by Dechant and Sasa [25], and Van Vu and Hasegawa formulated the geometric bound for Markov jump processes [113]. We call the result the geometric second law.

10.5.1 The Finite-Time Bound

Theorem 10.27 (Geometric second law)**.** *Let $\mathbf{x}_t$ obey the overdamped Langevin equation Eq. (10.2) on $[0,\tau]$ with a smooth time-dependent potential $E(\mathbf{x},t)$, scalar mobility μ, and $D=\mu k_{\mathrm{B}}T$. Assume the density ρ_t is smooth and positive and the local mean velocity $\mathbf{v}_t$ of Eq. (10.3) is Lipschitz in $\mathbf{x}$ uniformly in t. Then:*

(a) The entropy production rate equals the squared Wasserstein speed of the density:

$$\dot{\Sigma}_{\mathrm{tot}}(t)=\frac{k_{\mathrm{B}}}{D}\,\|\partial_t\rho_t\|_{\rho_t}^2\,. \tag{10.47}$$

(b) The total entropy production is bounded below by the squared Wasserstein distance between the initial and final densities:

$$\Sigma_{\mathrm{tot}}(\tau)\;\geq\;\frac{k_{\mathrm{B}}}{D\tau}\left(\int_0^\tau \|\partial_t\rho_t\|_{\rho_t}\,\mathrm{d}t\right)^2\;\geq\;\frac{k_{\mathrm{B}}}{D\tau}\,\mathcal{W}_2^2(\rho_0,\rho_\tau). \tag{10.48}$$

(c) Equivalently, in terms of the average work and the variational free energy,

$$\mathcal{W}_{\mathrm{diss}}=\langle\mathcal{W}\rangle-\Delta\mathcal{F}=T\Sigma_{\mathrm{tot}}(\tau)\;\geq\;\frac{\mathcal{W}_2^2(\rho_0,\rho_\tau)}{\mu\tau}. \tag{10.49}$$

Proof. (a) The local mean velocity $\mathbf{v}_t=-\nabla(\mu E+D\ln\rho_t)$ is the gradient of a scalar. A gradient field generating $\partial_t\rho_t$ is the minimal generator (Proposition 10.17), so $\int\rho_t\,\|\mathbf{v}_t\|^2=\|\partial_t\rho_t\|_{\rho_t}^2$. Inserting this in Eq. (10.4) gives Eq. (10.47).

(b) Integrate Eq. (10.47) over $[0,\tau]$ and apply Corollary 10.18(b) to the admissible path $(\rho_t,\mathbf{v}_t)_{t\in[0,\tau]}$:

$$\Sigma_{\mathrm{tot}}(\tau)=\frac{k_{\mathrm{B}}}{D}\int_0^\tau\mathrm{d}t\int\rho_t\,\|\mathbf{v}_t\|^2\,\mathrm{d}^Nx\geq\frac{k_{\mathrm{B}}}{D}\cdot\frac{1}{\tau}\left(\int_0^\tau\|\mathbf{v}_t\|_{L^2(\rho_t)}\,\mathrm{d}t\right)^2\geq\frac{k_{\mathrm{B}}}{D}\cdot\frac{\mathcal{W}_2^2(\rho_0,\rho_\tau)}{\tau}.$$

The first inequality is the Cauchy–Schwarz inequality in time; the second is the Benamou–Brenier length bound.

(c) Multiply (b) by T, use $Tk_{\mathrm{B}}/D=Tk_{\mathrm{B}}/(\mu k_{\mathrm{B}}T)=1/\mu$, and use the integrated free-energy balance Eq. (10.5). □

The two inequalities in Eq. (10.48) have separate physical content. The second is a statement about *where* the distribution goes: a process that moves mass a Wasserstein distance $\mathcal{W}_2$ must travel at least that far. The first is a statement about *how evenly in time* it goes: for a fixed path

length, entropy production is minimized by moving at constant Wasserstein speed. A protocol that idles and then rushes pays a penalty, exactly as a car that covers a fixed distance in a fixed time spends least energy against quadratic drag when it drives at constant speed.

Corollary 10.28 (Thermodynamic speed limit)**.** *Under the hypotheses of Theorem 10.27, any process that transforms ρ_0 into ρ_τ while dissipating at most $\mathcal{W}_{\mathrm{diss}}$ of work requires a time*

$$\tau \;\geq\; \frac{\mathcal{W}_2^2(\rho_0,\rho_\tau)}{\mu\,\mathcal{W}_{\mathrm{diss}}} = \frac{k_{\mathrm{B}}\,\mathcal{W}_2^2(\rho_0,\rho_\tau)}{D\,\Sigma_{\mathrm{tot}}(\tau)}. \tag{10.50}$$

For the RC node of the hardware box in Section 10.1, $\mu = D/k_{\mathrm{B}}T = \sigma_V^2/(k_{\mathrm{B}}T\,\tau_{RC})$, and Eq. (10.50) reads $\tau/\tau_{RC} \geq (\mathcal{W}_2/\sigma_V)^2\,(k_{\mathrm{B}}T/\mathcal{W}_{\mathrm{diss}})$: a node cannot move its distribution by n thermal widths in less than $n^2\tau_{RC}$ while dissipating less than one $k_{\mathrm{B}}T$.

Remark 10.29 (Anisotropic diffusion and non-conservative forces)**.** If the diffusion tensor is a constant positive-definite matrix $\mathbf{D} = k_{\mathrm{B}}T\boldsymbol{\mu}$, the change of variables $\mathbf{y} = \mathbf{D}^{-1/2}\mathbf{x}$ turns Eq. (8.47) into the isotropic form with $D = 1$, and Eq. (10.48) holds with $\mathcal{W}_2^2/D$ replaced by the Wasserstein distance for the cost $(\mathbf{x}-\mathbf{y})^{\top}\mathbf{D}^{-1}(\mathbf{x}-\mathbf{y})$. If a non-conservative force $\mathbf{f}_{\mathrm{nc}}$ acts, as in the non-reciprocal circuits of Worked computation 8.12, the local mean velocity acquires a component that is not a gradient. By Proposition 10.17, $\dot{\Sigma}_{\mathrm{tot}} = \frac{k_{\mathrm{B}}}{D}\left(\|\partial_t\rho_t\|_{\rho_t}^2 + \int \rho_t\,\|\mathbf{w}_t\|^2\right)$, where $\mathbf{w}_t$ is the solenoidal part of $\mathbf{v}_t$. The bound Eq. (10.48) still holds, but it can no longer be saturated: the circulating part of the current dissipates without transporting. This gives a geometric split of entropy production into a transport (excess-like) part and a circulation (housekeeping-like) part, which differs in general from the Hatano–Sasa split of Section 8.5.

10.5.2 The Optimal Protocol

The bound Eq. (10.48) is attained when both inequalities are equalities: the density must follow the Wasserstein geodesic from ρ_0 to ρ_τ at constant speed. The geodesic is the displacement interpolation, so the optimal density path is

$$\rho_t^* = (\mathbf{T}_{t/\tau})_{\#}\rho_0, \qquad \mathbf{T}_s(\mathbf{x}) = (1-s)\,\mathbf{x} + s\,\nabla\varphi(\mathbf{x}), \tag{10.51}$$

with $\nabla\varphi$ the Brenier map from ρ_0 to ρ_τ. Every fluid particle moves on the straight line from $\mathbf{x}$ to $\nabla\varphi(\mathbf{x})$ at the constant velocity $(\nabla\varphi(\mathbf{x})-\mathbf{x})/\tau$. By Corollary 10.15, rescaled to the interval $[0,\tau]$, the Eulerian velocity field is a gradient, $\mathbf{v}_t^* = \nabla\psi_t^*$, with

$$\partial_t\psi_t^* + \tfrac{1}{2}\,\|\nabla\psi_t^*\|^2 = 0, \qquad \psi_0^*(\mathbf{x}) = \frac{1}{\tau}\left(\varphi(\mathbf{x}) - \tfrac{1}{2}\,\|\mathbf{x}\|^2\right). \tag{10.52}$$

The remaining question is which potential $E(\mathbf{x},t)$ makes the Langevin dynamics produce this velocity field.

Theorem 10.30 (Minimum-dissipation protocol)**.** *Under the hypotheses of Theorem 10.13 for the pair ρ_0,ρ_τ, define for $0<t<\tau$*

$$E^*(\mathbf{x},t) := -k_{\mathrm{B}}T\ln\rho_t^*(\mathbf{x}) - \frac{1}{\mu}\,\psi_t^*(\mathbf{x}). \tag{10.53}$$

If the Langevin equation Eq. (10.2) is run with $E = E^*$ starting from ρ_0, then $\rho_t = \rho_t^*$ for all $t\in[0,\tau]$, and the entropy production attains the bound,

$$\Sigma_{\mathrm{tot}}(\tau) = \frac{k_{\mathrm{B}}}{D\tau}\mathcal{W}_2^2(\rho_0,\rho_\tau), \qquad \mathcal{W}_{\mathrm{diss}} = \frac{\mathcal{W}_2^2(\rho_0,\rho_\tau)}{\mu\tau}. \tag{10.54}$$

The potential E^* is unique up to an additive function of t alone among protocols attaining the bound.

Proof. The Fokker–Planck velocity Eq. (10.3) generated by E^*, evaluated on the density ρ_t^*, is

$$-\mu\nabla E^* - D\nabla\ln\rho_t^* = \mu k_{\mathrm{B}}T\nabla\ln\rho_t^* + \nabla\psi_t^* - D\nabla\ln\rho_t^* = \nabla\psi_t^* = \mathbf{v}_t^*,$$

because $\mu k_{\mathrm{B}}T = D$. Since ρ_t^* solves $\partial_t\rho_t^* + \nabla\cdot(\rho_t^*\mathbf{v}_t^*) = 0$ (Theorem 10.13), it solves the Fokker–Planck equation with potential E^* and initial condition ρ_0. That equation is a linear parabolic equation whose solution is unique, so $\rho_t = \rho_t^*$. Its entropy production is $\frac{k_{\mathrm{B}}}{D}\int_0^\tau\int\rho_t^*\,\|\mathbf{v}_t^*\|^2 = \frac{k_{\mathrm{B}}}{D}\cdot\frac{\mathcal{W}_2^2}{\tau}$, the equality case of Corollary 10.18(b). Conversely, a protocol attaining the bound must have $\rho_t = \rho_t^*$ (equality in both steps of Eq. (10.48) forces the constant-speed geodesic, which is unique because the optimal coupling is unique by Theorem 10.10), and then $\mu\nabla E = -\mathbf{v}_t^* - D\nabla\ln\rho_t^*$ determines ∇E and hence E up to a function of t. □

The structure of Eq. (10.53) is easy to read. The first term, $-k_{\mathrm{B}}T\ln\rho_t^*$, is the potential whose Boltzmann distribution is the current density: on its own it would hold the distribution still. The second term, $-\psi_t^*/\mu$, adds the force $\nabla\psi_t^*/\mu = \mathbf{v}_t^*/\mu$ that pushes each fluid element with exactly the velocity it needs. The optimal protocol keeps the system in a moving, slightly tilted local equilibrium, with a tilt proportional to $1/\tau$.

Remark 10.31 (Jumps at the endpoints)**.** Suppose the circuit starts in equilibrium in a given potential E_0, so that $\rho_0 \propto e^{-\beta E_0}$, and must end in a prescribed potential E_1 with $\rho_\tau \propto e^{-\beta E_1}$. Then $E^*(\mathbf{x},0^+) = E_0(\mathbf{x}) - \psi_0^*(\mathbf{x})/\mu + \text{const}$ and $E^*(\mathbf{x},\tau^-) = E_1(\mathbf{x}) - \psi_\tau^*(\mathbf{x})/\mu + \text{const}$. Unless $\rho_0 = \rho_\tau$, the potential ψ^* is not constant, and the optimal protocol *jumps* discontinuously at $t = 0$ and at $t = \tau$, by amounts of order $1/(\mu\tau)$. A jump of the potential is instantaneous, so the density does not change during it and no entropy is produced; the work done at a jump equals the change of $\langle E\rangle$, which equals the change of $\mathcal{F}$. The jumps therefore cost no dissipation. Their role is to set up, at $t = 0^+$, the force that starts the mass moving at full speed, and to remove it at $t = \tau$. Smoothing the jumps over a short interval ϵ costs extra dissipation that vanishes as $\epsilon \to 0$.

10.5.3 Entropy-Optimal Versus Work-Optimal: The Dragged Trap Revisited

Return to the trap $E(x,t) = \frac{k}{2}(x-\lambda_t)^2$ of Section 10.1, with $\lambda_0 = 0$, $\tau_r = 1/(\mu k)$, and the trap center to be moved to L. Two different optimization problems are natural, and they must be kept apart.

Problem I (minimum entropy production with prescribed final distribution). The final density must be the equilibrium of the final trap, $\rho_\tau = \mathcal{N}(L, k_{\mathrm{B}}T/k)$. By Proposition 10.4(b), $\mathcal{W}_2(\rho_0,\rho_\tau) = L$, so Theorem 10.30 gives $\mathcal{W}_{\mathrm{diss}}^{\mathrm{I}} = L^2/(\mu\tau)$. The optimal path translates the Gaussian at constant speed, $\rho_t^* = \mathcal{N}(Lt/\tau, k_{\mathrm{B}}T/k)$, with $\mathbf{v}^* = L/\tau$ and $\psi_t^* = Lx/\tau$. Equation Eq. (10.53) gives, after completing the square,

$$E^*(x,t) = \frac{k}{2}\Big(x - \frac{Lt}{\tau}\Big)^2 - \frac{L}{\mu\tau}\,x + \text{const} = \frac{k}{2}\Big(x - \lambda_t^{\mathrm{I}}\Big)^2 + \text{const}, \qquad \lambda_t^{\mathrm{I}} = L\,\frac{t+\tau_r}{\tau} \quad (0<t<\tau).$$

The trap leads the distribution by the constant distance $L\tau_r/\tau$, which supplies the drag force $kL\tau_r/\tau = L/(\mu\tau)$. The trap jumps from 0 to $L\tau_r/\tau$ at $t = 0$ and from $L(1+\tau_r/\tau)$ back to L at $t = \tau$, in accordance with Remark 10.31. Compare the linear protocol of Section 10.1: it starts with zero lead, so the distribution falls behind and never catches up during $[0,\tau]$.

Problem II (minimum work with prescribed final potential, Schmiedl–Seifert). Now only the final trap position $\lambda_{\tau^+} = L$ is prescribed, and the final distribution is free. This is the problem posed and solved by Schmiedl and Seifert [95]. Because the width of the distribution never changes, the state is described by the mean m_t, which is continuous in time and obeys $\dot m_t = \mu k(\lambda_t - m_t)$ for $0 < t < \tau$, so

$\lambda_t = m_t + \dot m_t/(\mu k)$. Allow jumps of λ at $t=0$ and $t=\tau$. Since $\langle E\rangle = \frac12 k_{\mathrm B}T + \frac{k}{2}(m_t-\lambda_t)^2$, the work at the initial jump is $\frac{k}{2}\lambda_{0^+}^2 = \dot m_{0^+}^2/(2\mu^2 k)$, and the work at the final jump is $\frac{k}{2}(L-m_\tau)^2 - \dot m_{\tau^-}^2/(2\mu^2 k)$. In between, the work rate is

$$\langle \partial_t E\rangle = k(\lambda_t - m_t)\dot\lambda_t = \frac{\dot m_t}{\mu}\left(\dot m_t + \frac{\ddot m_t}{\mu k}\right) = \frac{\dot m_t^2}{\mu} + \frac{1}{2\mu^2 k}\frac{\mathrm d}{\mathrm dt}\dot m_t^2.$$

Integrating over $(0,\tau)$, the total derivative produces $(\dot m_{\tau^-}^2 - \dot m_{0^+}^2)/(2\mu^2 k)$, which cancels exactly against the jump contributions. The total work is

$$\langle \mathcal W\rangle = \frac{1}{\mu}\int_0^\tau \dot m_t^2\ \mathrm dt + \frac{k}{2}(L-m_\tau)^2. \tag{10.55}$$

The first term is the transport cost $\mathcal W_2^2/(\mu\tau)$ if m_t moves at constant speed. The second term is the free energy left over in the lagging distribution, $k_{\mathrm B}T\,D_{\mathrm{KL}}(\rho_\tau\,\|\,p_L)$, where p_L is the equilibrium density of the final trap. For fixed m_τ, Cauchy–Schwarz gives $\int \dot m^2 \ge m_\tau^2/\tau$ with equality at constant speed, so it remains to minimize $m_\tau^2/(\mu\tau) + \frac{k}{2}(L-m_\tau)^2$ over m_τ. Setting the derivative to zero, $2m_\tau/(\mu\tau) = k(L-m_\tau)$, so

$$m_\tau = \frac{L\,\tau}{\tau+2\tau_r}, \qquad \langle\mathcal W\rangle^{\mathrm{II}} = \frac{L^2}{\mu(\tau+2\tau_r)}, \qquad \lambda_t^{\mathrm{II}} = L\,\frac{t+\tau_r}{\tau+2\tau_r}\quad (0<t<\tau). \tag{10.56}$$

The protocol jumps by $L\tau_r/(\tau+2\tau_r)$ at both ends: from 0 up to $\lambda_{0^+}^{\mathrm{II}}$, and from $\lambda_{\tau^-}^{\mathrm{II}} = L(\tau+\tau_r)/(\tau+2\tau_r)$ up to L. These symmetric jumps are the classic Schmiedl–Seifert result.

The two problems give different answers, and the difference is physical. In Problem II the work $L^2/(\mu(\tau+2\tau_r))$ is smaller than $L^2/(\mu\tau)$, but the system is not in equilibrium at $t=\tau$: the free energy $\frac{k}{2}(L-m_\tau)^2$ is dissipated afterwards, during a relaxation of duration $\sim\tau_r$ that is not counted in τ. Problem II effectively borrows time from the subsequent relaxation. In general, Eq. (10.5) and Eq. (6.47) give, for a process ending in a prescribed potential E_1 with equilibrium density p_1,

$$\langle\mathcal W\rangle - \Delta F = T\Sigma_{\mathrm{tot}}(\tau) + k_{\mathrm B}T\,D_{\mathrm{KL}}(\rho_\tau\,\|\,p_1)\ \ge\ \frac{\mathcal W_2^2(\rho_0,\rho_\tau)}{\mu\tau} + k_{\mathrm B}T\,D_{\mathrm{KL}}(\rho_\tau\,\|\,p_1)\,, \tag{10.57}$$

and the work-optimal protocol minimizes the right side over the free endpoint ρ_τ [5]. For a sampler, whose output is the distribution at time τ, Problem I is the relevant one: a distribution that has not reached its target is a sampling error, not a saving.

10.5.4 Slow Driving, Thermodynamic Length, and the Comparison with Quenching

Most practical protocols vary a few control parameters $\boldsymbol\lambda$ along a fixed curve, $\boldsymbol\lambda_t = \boldsymbol\Lambda(t/\tau)$, and the question is how their cost compares with the optimum.

Proposition 10.32 (Slow driving and the Wasserstein friction tensor)**.** *Let $E(\mathbf x,\boldsymbol\lambda)$ depend smoothly on parameters $\boldsymbol\lambda\in\mathbb R^M$, let $p_{\mathrm{eq}}(\cdot;\boldsymbol\lambda)$ be its Boltzmann density, and drive the system along $\boldsymbol\lambda_t=\boldsymbol\Lambda(t/\tau)$ from equilibrium. Assume that the density follows the instantaneous equilibrium with a deviation that is $\mathcal O(1/\tau)$ together with its time derivative, $\rho_t = p_{\mathrm{eq}}(\cdot;\boldsymbol\lambda_t)+\mathcal O(\tau^{-1})$. Then*

$$\mathcal W_{\mathrm{diss}} = \frac{1}{\tau}\int_0^1 \boldsymbol\Lambda'(s)^{\top}\boldsymbol\zeta(\boldsymbol\Lambda(s))\boldsymbol\Lambda'(s)\ \mathrm ds + \mathcal O(\tau^{-2}), \qquad \zeta_{ij}(\boldsymbol\lambda) := \frac{1}{\mu}\,g_{p_{\mathrm{eq}}}\!\left(\frac{\partial p_{\mathrm{eq}}}{\partial\lambda_i},\frac{\partial p_{\mathrm{eq}}}{\partial\lambda_j}\right). \tag{10.58}$$

Derivation. By Eq. (10.47), $T\dot{\Sigma}_{\mathrm{tot}} = \frac{1}{\mu} g_{\rho_t}(\partial_t \rho_t, \partial_t \rho_t)$ exactly. Under the assumption, $\partial_t \rho_t = \sum_i \partial_{\lambda_i} p_{\mathrm{eq}}\, \dot{\lambda}_i + \mathcal{O}(\tau^{-2})$ with $\dot{\boldsymbol{\lambda}} = \boldsymbol{\Lambda}'/\tau$, and the metric evaluated at ρ_t differs from that at p_{eq} by $\mathcal{O}(1/\tau)$. Hence $T\dot{\Sigma}_{\mathrm{tot}} = \tau^{-2} \boldsymbol{\Lambda}'^{\top} \boldsymbol{\zeta} \boldsymbol{\Lambda}' + \mathcal{O}(\tau^{-3})$, and integrating over $t \in [0, \tau]$ (with $\mathrm{d}t = \tau\, \mathrm{d}s$) gives Eq. (10.58). □

For overdamped dynamics, the linear-response friction tensor of slowly driven systems is therefore the Otto metric pulled back to the equilibrium manifold, divided by μ. Two checks on the harmonic trap: for the trap center, $\partial_\lambda p_{\mathrm{eq}} = -\partial_x p_{\mathrm{eq}}$ is generated by the uniform velocity 1, so $\zeta_{\lambda\lambda} = 1/\mu$ and Eq. (10.58) reproduces Eq. (10.10). For the stiffness k at fixed center, $\sigma = \sqrt{k_{\mathrm{B}}T/k}$ and, by Example 10.6, the Wasserstein speed is $|\dot{\sigma}| = \frac{1}{2}\sqrt{k_{\mathrm{B}}T}\, k^{-3/2} \left|\dot{k}\right|$, so $\zeta_{kk} = k_{\mathrm{B}}T/(4\mu k^3)$. The same value follows from the Green–Kubo formula $\zeta_{kk} = \beta \int_0^\infty \langle \delta X(t) \delta X(0) \rangle\, \mathrm{d}t$ for the conjugate force $X = \partial_k E = \frac{1}{2}x^2$, whose variance $\sigma^4/2$ decays at rate $2\mu k$: $\beta \cdot \frac{\sigma^4}{2} \cdot \frac{1}{2\mu k} = \frac{k_{\mathrm{B}}T}{4\mu k^3}$.

By Cauchy–Schwarz, $\mathcal{W}_{\mathrm{diss}} \geq \mathcal{L}^2/\tau$ to leading order, where $\mathcal{L} = \int_0^1 (\boldsymbol{\Lambda}'^{\top} \boldsymbol{\zeta} \boldsymbol{\Lambda}')^{1/2}\, \mathrm{d}s$ is the thermodynamic length of the parameter curve. Since $\mu\, \boldsymbol{\Lambda}'^{\top} \boldsymbol{\zeta} \boldsymbol{\Lambda}'$ is the squared Wasserstein speed of a curve of densities, Corollary 10.18 gives $\mathcal{L} \geq \mathcal{W}_2(\rho_0, \rho_\tau)/\sqrt{\mu}$. The slow-driving optimum within a parameter family coincides with the exact optimum Eq. (10.54) when the Wasserstein geodesic stays inside the family of equilibrium densities. This is the case for Gaussians with a quadratic energy, which is why the RC sampler admits an exact finite-time optimum within the space of programmable quadratic potentials.

Remark 10.33 (Optimal transport versus quench-and-wait)**.** The opposite extreme to a slow protocol is to switch the potential from E_0 to E_1 instantaneously and let the system relax. During the switch the density is ρ_0, so the work is $\int \rho_0 (E_1 - E_0)$. Writing $E_0 = F_0 - k_{\mathrm{B}}T \ln \rho_0$ and $E_1 = F_1 - k_{\mathrm{B}}T \ln p_1$ (the Boltzmann form of the two equilibria),

$$\mathcal{W}_{\mathrm{diss}}^{\mathrm{quench}} = \int \rho_0 (E_1 - E_0)\, \mathrm{d}^N x - (F_1 - F_0) = k_{\mathrm{B}}T\, D_{\mathrm{KL}}(\rho_0 \,\|\, p_1)\,,$$

independent of the waiting time, which must in any case be several relaxation times. If $\nabla^2 E_1 \succeq \kappa \mathbf{I}$, the Talagrand transport inequality [7], which we assume, states $\mathcal{W}_2^2(\rho, p_1) \leq (2k_{\mathrm{B}}T/\kappa)\, D_{\mathrm{KL}}(\rho \,\|\, p_1)$ for all ρ. Combining with Eq. (10.54),

$$\frac{\mathcal{W}_{\mathrm{diss}}^*(\tau)}{\mathcal{W}_{\mathrm{diss}}^{\mathrm{quench}}} = \frac{\mathcal{W}_2^2(\rho_0, p_1)}{\mu\tau\, k_{\mathrm{B}}T\, D_{\mathrm{KL}}(\rho_0 \,\|\, p_1)} \leq \frac{2}{\mu\kappa\tau} = \frac{2\tau_\kappa}{\tau}, \qquad \tau_\kappa := \frac{1}{\mu\kappa}.$$

An optimally controlled transport of duration τ dissipates less than a quench whenever $\tau > 2\tau_\kappa$, and by a factor that grows linearly with τ. For a translated Gaussian Talagrand's inequality is an equality, since $\mathcal{W}_2^2 = L^2$ and $k_{\mathrm{B}}T\, D_{\mathrm{KL}}(\rho_0 \,\|\, p_1) = \frac{k}{2}L^2$.

Example 10.34 (Finite-time erasure in the double-well sampler)**.** Let $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2$ (the double-well sampler at zero bias) with deep wells at $\pm x_0$, $x_0 = \sqrt{b/a}$, of thermal width $\sigma \ll x_0$. Erasing a bit means transforming the symmetric equilibrium $\rho_0 \approx \frac{1}{2}n(x + x_0) + \frac{1}{2}n(x - x_0)$, where n is a narrow normalized peak of width σ, into the reset state $\rho_\tau \approx n(x - x_0)$, at the end of a cycle that restores the original potential. By Theorem 10.5, monotone rearrangement sends the left half of the mass (quantiles $u < \frac{1}{2}$) across the barrier to the left half of the right well, a displacement of about $2x_0$, while the right half of the mass moves by $\mathcal{O}(\sigma)$. Hence $\mathcal{W}_2^2(\rho_0, \rho_\tau) = \frac{1}{2}(2x_0)^2 + \mathcal{O}(x_0\sigma) \approx 2x_0^2 = 2b/a$. The free-energy change is $\Delta\mathcal{F} = k_{\mathrm{B}}T \ln 2$ (same average energy, entropy lowered by $k_{\mathrm{B}} \ln 2$), and Eq. (10.49) gives the finite-time Landauer bound

$$\langle \mathcal{W} \rangle \;\geq\; k_{\mathrm{B}}T \ln 2 + \frac{2x_0^2}{\mu\tau}. \tag{10.59}$$

The correction to Landauer's $k_{\mathrm{B}}T\ln 2$ (Section 9.3) decays as $1/\tau$ and grows with the square of the well separation. Achieving it requires a protocol that reshapes the potential so as to slide half the mass continuously across the barrier region, as in Eq. (10.53); the bound says nothing about how hard that is to build.

Hardware Realization: A transport floor for denoising samplers

Each step of a denoising thermodynamic sampler (Chapters 15 to 17) changes the conditional distribution held on a block of nodes, and must do so in a fixed clock period τ_{step}. Theorem 10.27 gives a floor of $\mathcal{W}_2^2(\rho_{\mathrm{before}},\rho_{\mathrm{after}})/(\mu\tau_{\mathrm{step}})$ for each step, independent of how the step is implemented. With $D=\sigma_V^2/\tau_{RC}$ for an RC node, this floor is $k_{\mathrm{B}}T\,(\mathcal{W}_2/\sigma_V)^2\,(\tau_{RC}/\tau_{\mathrm{step}})$ per node: of order $k_{\mathrm{B}}T$ when each step moves each node by about one thermal width and lasts about one relaxation time. Section 17.4.3 compares this floor with the transistor-level energy budget of the hardware. The bound also suggests a design rule: since dissipation scales as the square of the distance moved per step, splitting a large transport into K equal steps of duration τ/K each, along a geodesic, leaves the total at $\mathcal{W}_2^2/(\mu\tau)$, while taking large uneven steps costs more.

10.6 Worked Computation: Optimal Transport of the Gaussian RC Sampler

For linear dynamics everything in this chapter can be computed in closed form. The RC sampler holds a Gaussian distribution of node voltages, and reprogramming it (a new bias vector, a new coupling matrix) asks the circuit to transport one Gaussian into another. We compute the exact Wasserstein distance between the two Gaussians, the geodesic, the optimal schedules of conductance matrix and bias, and the minimal dissipated work.

Notation Watch: The conductance matrix in this computation

We follow the normalization of Worked computation 6.21: the node voltages $\mathbf{v}\in\mathbb{R}^N$ obey an overdamped Langevin equation with scalar mobility μ and the quadratic energy $E(\mathbf{v})=\frac{1}{2}(\mathbf{v}-\mathbf{b})^{\top}\mathbf{G}(\mathbf{v}-\mathbf{b})$, so that the deterministic relaxation rate matrix is $\mu\mathbf{G}$ and the stationary law is $\mathcal{N}(\mathbf{b},k_{\mathrm{B}}T\mathbf{G}^{-1})$. In this normalization $\mathbf{G}$ is a stiffness matrix (energy per volt squared). For a single passive node the stiffness is the capacitance and the conductance sets the mobility (Example 8.3); in the programmable networks of Chapter 13, active transconductance elements set the effective stiffness matrix, which is the matrix we call $\mathbf{G}$. The relaxation time of the initial setting, $\tau_0:=1/(\mu g_0)$ for a node of stiffness g_0, plays the role of τ_{RC}.

Worked computation 10.35 (Bures–Wasserstein geodesic and optimal conductance schedule for the RC sampler)**. Setup.** The sampler obeys

$$\mathrm{d}\mathbf{v}_t=-\mu\,\mathbf{G}(t)\,(\mathbf{v}_t-\mathbf{b}(t))\,\mathrm{d}t+\sqrt{2D}\,\mathrm{d}\mathbf{W}_t,\qquad D=\mu k_{\mathrm{B}}T, \tag{10.60}$$

with a symmetric matrix $\mathbf{G}(t)$ and bias $\mathbf{b}(t)$ that we may program as functions of time. Initially the circuit is in equilibrium with $\mathbf{G}_0=k_{\mathrm{B}}T\boldsymbol{\Sigma}_0^{-1}$ and $\mathbf{b}_0=\mathbf{m}_0$, so $\rho_0=\mathcal{N}(\mathbf{m}_0,\boldsymbol{\Sigma}_0)$. At time τ it must hold $\rho_\tau=\mathcal{N}(\mathbf{m}_1,\boldsymbol{\Sigma}_1)$, the equilibrium of $\mathbf{G}_1=k_{\mathrm{B}}T\boldsymbol{\Sigma}_1^{-1}$, $\mathbf{b}_1=\mathbf{m}_1$. Both covariances are positive definite.

Since Eq. (10.60) is linear, a Gaussian initial law stays Gaussian. Taking the expectation of Eq. (10.60) and applying Itô's formula to $(\mathbf{v}_t-\mathbf{m}_t)(\mathbf{v}_t-\mathbf{m}_t)^\top$ gives the moment equations

$$\dot{\mathbf{m}}_t=-\mu\mathbf{G}(t)(\mathbf{m}_t-\mathbf{b}(t)),\qquad \dot{\mathbf{\Sigma}}_t=-\mu(\mathbf{G}(t)\mathbf{\Sigma}_t+\mathbf{\Sigma}_t\mathbf{G}(t))+2D\,\mathbf{I}, \tag{10.61}$$

the time-dependent version of the Lyapunov equation of Chapter B (Eq. (B.76)).

Part 1: The Bures–Wasserstein distance. By Proposition 10.4(b), $\mathcal{W}_2^2(\rho_0,\rho_\tau)=\|\mathbf{m}_0-\mathbf{m}_1\|^2+\mathcal{W}_2^2(\bar\rho_0,\bar\rho_1)$, where $\bar\rho_i=\mathcal{N}(\mathbf{0},\mathbf{\Sigma}_i)$. Let $(\mathbf{X},\mathbf{Y})$ be any coupling of $\bar\rho_0$ and $\bar\rho_1$, with cross-covariance $\mathbf{C}=\mathbb{E}\big[\mathbf{X}\mathbf{Y}^\top\big]$. Then

$$\mathbb{E}\Big[\|\mathbf{X}-\mathbf{Y}\|^2\Big]=\operatorname{tr}\mathbf{\Sigma}_0+\operatorname{tr}\mathbf{\Sigma}_1-2\operatorname{tr}\mathbf{C}. \tag{10.62}$$

The cost depends on the coupling only through $\mathbf{C}$, so we must find the largest $\operatorname{tr}\mathbf{C}$ compatible with the marginals. The joint covariance $\left(\begin{smallmatrix}\mathbf{\Sigma}_0 & \mathbf{C}\\ \mathbf{C}^\top & \mathbf{\Sigma}_1\end{smallmatrix}\right)$ must be positive semidefinite. Since $\mathbf{\Sigma}_0\succ\mathbf{0}$, this holds if and only if the Schur complement (Definition B.5) satisfies $\mathbf{\Sigma}_1-\mathbf{C}^\top\mathbf{\Sigma}_0^{-1}\mathbf{C}\succeq\mathbf{0}$. Writing $\mathbf{C}=\mathbf{\Sigma}_0^{1/2}\mathbf{K}\mathbf{\Sigma}_1^{1/2}$, the condition becomes $\mathbf{\Sigma}_1^{1/2}(\mathbf{I}-\mathbf{K}^\top\mathbf{K})\mathbf{\Sigma}_1^{1/2}\succeq\mathbf{0}$, that is, $\mathbf{K}^\top\mathbf{K}\preceq\mathbf{I}$: every singular value of $\mathbf{K}$ is at most 1, and so are the absolute values of the entries of $\mathbf{V}^\top\mathbf{K}\mathbf{U}$ for any orthogonal $\mathbf{U},\mathbf{V}$.

Now let $\mathbf{A}:=\mathbf{\Sigma}_1^{1/2}\mathbf{\Sigma}_0^{1/2}$ have singular value decomposition $\mathbf{A}=\mathbf{U}\mathbf{S}\mathbf{V}^\top$ with $\mathbf{S}=\operatorname{diag}(s_1,\dots,s_N)$, $s_i\ge0$. By cyclicity of the trace,

$$\operatorname{tr}\mathbf{C}=\operatorname{tr}(\mathbf{\Sigma}_0^{1/2}\mathbf{K}\mathbf{\Sigma}_1^{1/2})=\operatorname{tr}(\mathbf{K}\mathbf{A})=\operatorname{tr}(\mathbf{V}^\top\mathbf{K}\mathbf{U}\,\mathbf{S})=\sum_{i=1}^{N}s_i\,(\mathbf{V}^\top\mathbf{K}\mathbf{U})_{ii}\le\sum_{i=1}^{N}s_i.$$

The squares s_i^2 are the eigenvalues of $\mathbf{A}^\top\mathbf{A}=\mathbf{\Sigma}_0^{1/2}\mathbf{\Sigma}_1\mathbf{\Sigma}_0^{1/2}:=\mathbf{M}$, so $\sum_i s_i=\operatorname{tr}\mathbf{M}^{1/2}$. Hence $\operatorname{tr}\mathbf{C}\le\operatorname{tr}\mathbf{M}^{1/2}$ for every coupling.

The bound is attained by the linear map

$$\mathbf{T}:=\mathbf{\Sigma}_0^{-1/2}\,\mathbf{M}^{1/2}\,\mathbf{\Sigma}_0^{-1/2},\qquad \mathbf{M}=\mathbf{\Sigma}_0^{1/2}\mathbf{\Sigma}_1\mathbf{\Sigma}_0^{1/2}. \tag{10.63}$$

It is symmetric and positive definite, and it pushes $\bar\rho_0$ to $\bar\rho_1$ because the covariance of $\mathbf{T}\mathbf{X}$ is

$$\mathbf{T}\mathbf{\Sigma}_0\mathbf{T}=\mathbf{\Sigma}_0^{-1/2}\mathbf{M}^{1/2}\mathbf{\Sigma}_0^{-1/2}\,\mathbf{\Sigma}_0\,\mathbf{\Sigma}_0^{-1/2}\mathbf{M}^{1/2}\mathbf{\Sigma}_0^{-1/2}=\mathbf{\Sigma}_0^{-1/2}\mathbf{M}\mathbf{\Sigma}_0^{-1/2}=\mathbf{\Sigma}_1.$$

For the coupling $\mathbf{Y}=\mathbf{T}\mathbf{X}$, $\mathbf{C}=\mathbb{E}\big[\mathbf{X}\mathbf{X}^\top\big]\,\mathbf{T}=\mathbf{\Sigma}_0\mathbf{T}$, and $\operatorname{tr}(\mathbf{\Sigma}_0\mathbf{T})=\operatorname{tr}(\mathbf{\Sigma}_0^{1/2}\mathbf{M}^{1/2}\mathbf{\Sigma}_0^{-1/2})=\operatorname{tr}\mathbf{M}^{1/2}$. Including the means, the optimal map is $\mathbf{v}\mapsto\mathbf{m}_1+\mathbf{T}(\mathbf{v}-\mathbf{m}_0)$, the gradient of the convex quadratic $\varphi(\mathbf{v})=\mathbf{m}_1\cdot\mathbf{v}+\frac12(\mathbf{v}-\mathbf{m}_0)^\top\mathbf{T}(\mathbf{v}-\mathbf{m}_0)$, in agreement with Brenier's theorem. Inserting $\operatorname{tr}\mathbf{C}=\operatorname{tr}\mathbf{M}^{1/2}$ in Eq. (10.62),

$$\mathcal{W}_2^2\big(\mathcal{N}(\mathbf{m}_0,\mathbf{\Sigma}_0),\mathcal{N}(\mathbf{m}_1,\mathbf{\Sigma}_1)\big)=\|\mathbf{m}_0-\mathbf{m}_1\|^2+\operatorname{tr}\Big(\mathbf{\Sigma}_0+\mathbf{\Sigma}_1-2(\mathbf{\Sigma}_0^{1/2}\mathbf{\Sigma}_1\mathbf{\Sigma}_0^{1/2})^{1/2}\Big), \tag{10.64}$$

which is the Bures–Wasserstein formula Eq. (B.22). When $\mathbf{\Sigma}_0$ and $\mathbf{\Sigma}_1$ commute, $\mathbf{M}^{1/2}=\mathbf{\Sigma}_0^{1/2}\mathbf{\Sigma}_1^{1/2}$ and the covariance part is $\big\|\mathbf{\Sigma}_0^{1/2}-\mathbf{\Sigma}_1^{1/2}\big\|_{\mathrm F}^2=\sum_i(\sigma_{0,i}-\sigma_{1,i})^2$ in the common eigenbasis: each normal mode contributes the one-dimensional result Eq. (10.20).

Part 2: The geodesic and its velocity field. With $s=t/\tau$, the displacement interpolation Eq. (10.51) sends $\mathbf{v}\mapsto(1-s)\mathbf{v}+s[\mathbf{m}_1+\mathbf{T}(\mathbf{v}-\mathbf{m}_0)]$. Writing $\mathbf{v}=\mathbf{m}_0+\bar{\mathbf{v}}$, the image is $\mathbf{m}_s+\mathbf{A}_s\bar{\mathbf{v}}$ with

$$\mathbf{m}_s=(1-s)\mathbf{m}_0+s\mathbf{m}_1,\qquad \mathbf{A}_s=(1-s)\mathbf{I}+s\mathbf{T},\qquad \rho_t^*=\mathcal{N}\big(\mathbf{m}_{t/\tau},\mathbf{\Sigma}_t^*\big),\quad \mathbf{\Sigma}_t^*=\mathbf{A}_{t/\tau}\mathbf{\Sigma}_0\mathbf{A}_{t/\tau}. \tag{10.65}$$

The geodesic stays in the Gaussian family: the mean moves on a straight line and the covariance follows the curve $\mathbf{A}_s\boldsymbol{\Sigma}_0\mathbf{A}_s$, which is not the linear interpolation of covariances. A particle at $\mathbf{m}_s + \mathbf{A}_s\bar{\mathbf{v}}$ moves with velocity $\frac{1}{\tau}[\Delta\mathbf{m} + (\mathbf{T}-\mathbf{I})\bar{\mathbf{v}}]$, where $\Delta\mathbf{m} := \mathbf{m}_1 - \mathbf{m}_0$. Expressing $\bar{\mathbf{v}}$ through the current position, the Eulerian velocity field is

$$\mathbf{v}_t^*(\mathbf{v}) = \frac{1}{\tau}\Big[\Delta\mathbf{m} + \mathbf{S}_t(\mathbf{v} - \mathbf{m}_{t/\tau})\Big], \qquad \mathbf{S}_t := (\mathbf{T}-\mathbf{I})\,\mathbf{A}_{t/\tau}^{-1}. \tag{10.66}$$

Since $\mathbf{A}_s$ is a polynomial in $\mathbf{T}$, it commutes with $\mathbf{T}$, and $\mathbf{S}_t$ is symmetric. So $\mathbf{v}_t^* = \nabla\psi_t^*$ with $\psi_t^*(\mathbf{v}) = \frac{1}{\tau}[\Delta\mathbf{m}\cdot(\mathbf{v}-\mathbf{m}_{t/\tau}) + \frac{1}{2}(\mathbf{v}-\mathbf{m}_{t/\tau})^\top\mathbf{S}_t(\mathbf{v}-\mathbf{m}_{t/\tau})]$, as Corollary 10.15 requires. As a check, the kinetic energy is constant in time:

$$\int \rho_t^*\,\|\mathbf{v}_t^*\|^2 = \frac{1}{\tau^2}\Big[\|\Delta\mathbf{m}\|^2 + \operatorname{tr}(\mathbf{S}_t\boldsymbol{\Sigma}_t^*\mathbf{S}_t)\Big], \qquad \mathbf{S}_t\boldsymbol{\Sigma}_t^*\mathbf{S}_t = (\mathbf{T}-\mathbf{I})\boldsymbol{\Sigma}_0(\mathbf{T}-\mathbf{I}),$$

using $\mathbf{S}_t\mathbf{A}_{t/\tau} = \mathbf{T}-\mathbf{I}$ and the symmetry of $\mathbf{A}$. The trace is $\operatorname{tr}\boldsymbol{\Sigma}_0 - 2\operatorname{tr}(\mathbf{T}\boldsymbol{\Sigma}_0) + \operatorname{tr}(\mathbf{T}\boldsymbol{\Sigma}_0\mathbf{T}) = \operatorname{tr}\boldsymbol{\Sigma}_0 - 2\operatorname{tr}\mathbf{M}^{1/2} + \operatorname{tr}\boldsymbol{\Sigma}_1$, so $\int_0^\tau \mathrm{d}t\int\rho_t^*\,\|\mathbf{v}_t^*\|^2 = \mathcal{W}_2^2/\tau$, the equality case of Eq. (10.34).

Part 3: The optimal conductance and bias schedules. By Theorem 10.30, the optimal potential is $E^* = -k_\mathrm{B}T\ln\rho_t^* - \psi_t^*/\mu$. For the Gaussian ρ_t^*, $-k_\mathrm{B}T\ln\rho_t^* = \frac{k_\mathrm{B}T}{2}(\mathbf{v}-\mathbf{m}_{t/\tau})^\top(\boldsymbol{\Sigma}_t^*)^{-1}(\mathbf{v}-\mathbf{m}_{t/\tau}) + \text{const}$. Collecting the quadratic and linear terms in $\mathbf{u} = \mathbf{v} - \mathbf{m}_{t/\tau}$,

$$E^* = \frac{1}{2}\mathbf{u}^\top\mathbf{G}^*(t)\,\mathbf{u} - \frac{1}{\mu\tau}\Delta\mathbf{m}\cdot\mathbf{u} + \text{const}, \qquad \mathbf{G}^*(t) := k_\mathrm{B}T\big(\boldsymbol{\Sigma}_t^*\big)^{-1} - \frac{1}{\mu\tau}\mathbf{S}_t.$$

Completing the square, $\frac{1}{2}(\mathbf{u}-\boldsymbol{\delta})^\top\mathbf{G}^*(\mathbf{u}-\boldsymbol{\delta}) = \frac{1}{2}\mathbf{u}^\top\mathbf{G}^*\mathbf{u} - \boldsymbol{\delta}^\top\mathbf{G}^*\mathbf{u} + \text{const}$, which matches the linear term when $\mathbf{G}^*\boldsymbol{\delta} = \Delta\mathbf{m}/(\mu\tau)$. The optimal protocol is therefore again a quadratic energy of the RC-sampler form, $E^*(\mathbf{v},t) = \frac{1}{2}(\mathbf{v}-\mathbf{b}^*(t))^\top\mathbf{G}^*(t)(\mathbf{v}-\mathbf{b}^*(t)) + \text{const}$, with

$$\boxed{\mathbf{G}^*(t) = k_\mathrm{B}T\Big(\mathbf{A}_{t/\tau}\boldsymbol{\Sigma}_0\mathbf{A}_{t/\tau}\Big)^{-1} - \frac{1}{\mu\tau}(\mathbf{T}-\mathbf{I})\mathbf{A}_{t/\tau}^{-1}, \qquad \mathbf{b}^*(t) = \mathbf{m}_{t/\tau} + \frac{1}{\mu\tau}\,\mathbf{G}^*(t)^{-1}\Delta\mathbf{m},} \tag{10.67}$$

for $0 < t < \tau$ (assuming $\mathbf{G}^*(t)$ is invertible). The first term of $\mathbf{G}^*$ is the stiffness whose equilibrium is the current covariance; the second, of order $1/\tau$, softens the stiffness in directions that must expand ($\mathbf{T} \succ \mathbf{I}$) and stiffens it in directions that must contract. The bias leads the mean by $(\mu\tau)^{-1}\mathbf{G}^{*-1}\Delta\mathbf{m}$, exactly the lead of the dragged trap in Problem I.

We verify Eq. (10.67) independently with the moment equations Eq. (10.61). For the mean, $-\mu\mathbf{G}^*(\mathbf{m}_{t/\tau} - \mathbf{b}^*) = \mu\mathbf{G}^*\,(\mu\tau)^{-1}\mathbf{G}^{*-1}\Delta\mathbf{m} = \Delta\mathbf{m}/\tau = \dot{\mathbf{m}}_{t/\tau}$. For the covariance, write $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_t^*$ and $\mathbf{S} = \mathbf{S}_t$. Then

$$-\mu(\mathbf{G}^*\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{G}^*) + 2\mu k_\mathrm{B}T\mathbf{I} = -\mu\Big(2k_\mathrm{B}T\mathbf{I} - \frac{\mathbf{S}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{S}}{\mu\tau}\Big) + 2\mu k_\mathrm{B}T\mathbf{I} = \frac{1}{\tau}(\mathbf{S}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{S}),$$

while differentiating $\boldsymbol{\Sigma}_t^* = \mathbf{A}_s\boldsymbol{\Sigma}_0\mathbf{A}_s$ with $\dot{\mathbf{A}}_s = (\mathbf{T}-\mathbf{I})/\tau = \mathbf{S}\mathbf{A}_s/\tau = \mathbf{A}_s\mathbf{S}/\tau$ gives

$$\dot{\boldsymbol{\Sigma}}_t^* = \frac{1}{\tau}\Big(\mathbf{S}\mathbf{A}_s\boldsymbol{\Sigma}_0\mathbf{A}_s + \mathbf{A}_s\boldsymbol{\Sigma}_0\mathbf{A}_s\mathbf{S}\Big) = \frac{1}{\tau}(\mathbf{S}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{S}).$$

The two agree, so the schedule Eq. (10.67) drives the sampler exactly along the Bures–Wasserstein geodesic.

Endpoint jumps. At $t = 0^+$, $\mathbf{A}_0 = \mathbf{I}$ and $\mathbf{G}^*(0^+) = \mathbf{G}_0 - (\mathbf{T}-\mathbf{I})/(\mu\tau)$; at $t = \tau^-$, $\mathbf{A}_1 = \mathbf{T}$ and $\mathbf{G}^*(\tau^-) = \mathbf{G}_1 - (\mathbf{I}-\mathbf{T}^{-1})/(\mu\tau)$. The conductance matrix jumps by $-(\mathbf{T}-\mathbf{I})/(\mu\tau)$ when the protocol starts and by $+(\mathbf{I}-\mathbf{T}^{-1})/(\mu\tau)$ when it ends, and the bias jumps from $\mathbf{m}_0$ to $\mathbf{m}_0 + (\mu\tau)^{-1}\mathbf{G}^*(0^+)^{-1}\Delta\mathbf{m}$

and from $\mathbf{m}_1 + (\mu\tau)^{-1}\mathbf{G}^*(\tau^-)^{-1}\Delta\mathbf{m}$ to $\mathbf{m}_1$. All jumps vanish as $1/\tau$, and none of them produces entropy (Remark 10.31).

Positivity. A passive network requires $\mathbf{G}^*(t) \succ \mathbf{0}$. In a normal mode with standard deviations $\sigma_0 \to \sigma_1$ (commuting case), $\mathbf{T}$ has eigenvalue σ_1/σ_0, the mode's standard deviation along the geodesic is $\sigma_t = \sigma_0 + (\sigma_1 - \sigma_0)t/\tau$, and the mode stiffness is

$$g^*(t) = \frac{k_{\mathrm{B}}T}{\sigma_t^2} - \frac{\sigma_1 - \sigma_0}{\mu\tau\,\sigma_t}. \tag{10.68}$$

Contraction ($\sigma_1 < \sigma_0$) always gives $g^* > 0$. Expansion requires $\mu\tau k_{\mathrm{B}}T > (\sigma_1 - \sigma_0)\sigma_t$ for all t, the strongest condition being at $t = \tau$: with $r = \sigma_1/\sigma_0$ and $\tau_0 = \sigma_0^2/(\mu k_{\mathrm{B}}T)$ the initial relaxation time, $\tau > r(r-1)\tau_0$. Expanding a distribution faster than this requires a transiently repulsive (negative-stiffness) potential, which a passive circuit cannot supply.

Part 4: Minimum work. The initial and final states are equilibria, so $\Delta\mathcal{F} = \Delta F$. With $F = -k_{\mathrm{B}}T \ln Z$ and $Z \propto \det(2\pi k_{\mathrm{B}}T\mathbf{G}^{-1})^{1/2} = \det(2\pi\boldsymbol{\Sigma})^{1/2}$ (Theorem B.2), $\Delta F = \frac{k_{\mathrm{B}}T}{2}\ln(\det\boldsymbol{\Sigma}_0/\det\boldsymbol{\Sigma}_1)$. By Eq. (10.54), the minimum average work over all protocols of duration τ is

$$\langle\mathcal{W}\rangle^*(\tau) = \frac{k_{\mathrm{B}}T}{2}\ln\frac{\det\boldsymbol{\Sigma}_0}{\det\boldsymbol{\Sigma}_1} + \frac{1}{\mu\tau}\left[\|\Delta\mathbf{m}\|^2 + \mathrm{tr}\Big(\boldsymbol{\Sigma}_0 + \boldsymbol{\Sigma}_1 - 2\big(\boldsymbol{\Sigma}_0^{1/2}\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_0^{1/2}\big)^{1/2}\Big)\right]. \tag{10.69}$$

The first term is the reversible free-energy cost, paid even quasistatically: compressing the distribution (lowering $\det\boldsymbol{\Sigma}$) costs free energy and expanding it releases free energy, as in the isothermal compression of an ideal gas. The second term is the irreducible price of finishing in time τ, and it decays exactly as $1/\tau$ for every τ, not only asymptotically. No protocol, Gaussian or not, can do better, since Theorem 10.27 bounds all protocols.

Part 5: A one-node example. Take $N = 1$ and measure voltages in units of the initial thermal width $\sigma_0 = \sqrt{k_{\mathrm{B}}T/g_0}$ and times in units of $\tau_0 = 1/(\mu g_0)$, so that $\sigma_0^2/\mu = k_{\mathrm{B}}T\,\tau_0$. The task is to shift the mean by two thermal widths and halve the width: $m_0 = 0$, $m_1 = 2\sigma_0$, $\sigma_1 = \sigma_0/2$, so that $g_1 = 4g_0$ and $b_1 = 2\sigma_0$. By Eq. (10.20),

$$\mathcal{W}_2^2 = (2\sigma_0)^2 + (\sigma_0/2)^2 = 4.25\,\sigma_0^2, \qquad \mathcal{W}_{\mathrm{diss}}^* = \frac{4.25\,\sigma_0^2}{\mu\tau} = 4.25\,k_{\mathrm{B}}T\,\frac{\tau_0}{\tau}, \qquad \Delta F = k_{\mathrm{B}}T\ln\frac{\sigma_0}{\sigma_1} = k_{\mathrm{B}}T\ln 2.$$

Most of the transport cost (4 of the 4.25) comes from the shift of the mean. On the geodesic, $\sigma_t = \sigma_0(1 - u/2)$ with $u = t/\tau$, and Eqs. (10.67) and (10.68) give

$$\frac{g^*(t)}{g_0} = \frac{1}{(1-u/2)^2} + \frac{\tau_0}{2\tau}\,\frac{1}{1-u/2}, \qquad \frac{b^*(t)}{\sigma_0} = 2u + \frac{2\tau_0}{\tau}\,\frac{g_0}{g^*(t)}. \tag{10.70}$$

For $\tau = 10\,\tau_0$ the schedule is:

$u = t/\tau$	0^+	0.25	0.5	0.75	1^-
g^*/g_0	1.050	1.363	1.844	2.640	4.100
b^*/σ_0	0.190	0.647	1.108	1.576	2.049

The stiffness jumps from 1 to $1.05\,g_0$ at the start and from 4.1 to $4\,g_0$ at the end, and the bias jumps from 0 to $0.19\,\sigma_0$ and from $2.05\,\sigma_0$ to $2\,\sigma_0$. The minimum dissipated work is $0.425\,k_{\mathrm{B}}T$, and the total work is $k_{\mathrm{B}}T\ln 2 + 0.425\,k_{\mathrm{B}}T \approx 1.118\,k_{\mathrm{B}}T$. At $\tau = \tau_0$ the jumps are ten times larger (g^* runs from $1.5\,g_0$ to $5\,g_0$, b^* from $1.33\,\sigma_0$ to $2.4\,\sigma_0$) and $\mathcal{W}_{\mathrm{diss}}^* = 4.25\,k_{\mathrm{B}}T$. We integrated the moment equations Eq. (10.61) numerically with this schedule, including the work done at the jumps; the

computed dissipated work agrees with $4.25\,k_{\mathrm{B}}T\,\tau_0/\tau$ to better than 10^{-8} and the final state equals the target.

Comparison with simpler protocols. (i) A quench, switching instantly to (g_1, b_1) and waiting, dissipates $k_{\mathrm{B}}T\,D_{\mathrm{KL}}(\rho_0 \,\|\, p_1) = \frac{k_{\mathrm{B}}T}{2}[\sigma_0^2/\sigma_1^2 + (m_1-m_0)^2/\sigma_1^2 - 1 + \ln(\sigma_1^2/\sigma_0^2)] = \frac{k_{\mathrm{B}}T}{2}(4+16-1-\ln 4) \approx 8.81\,k_{\mathrm{B}}T$ (Eq. (B.21)), twenty times the optimum at $\tau = 10\,\tau_0$. (ii) A linear ramp of both g and b from their initial to final values over the same $\tau = 10\,\tau_0$ dissipates $0.429\,k_{\mathrm{B}}T$ during the ramp and leaves the node slightly short of the target, with $D_{\mathrm{KL}}(\rho_\tau \,\|\, p_1) = 0.0052$; including the subsequent relaxation it dissipates $0.434\,k_{\mathrm{B}}T$, about 2% above the optimum. Its large-τ coefficient follows from Proposition 10.32: the bias ramp contributes $\Delta b^2/\mu = 4\,k_{\mathrm{B}}T\tau_0$, the same as the optimum, while the linear stiffness ramp $g = g_0(1+3u)$ makes the width follow $\sigma(u) = \sigma_0(1+3u)^{-1/2}$, whose squared Wasserstein length is

$$\int_0^1 \left(\frac{\mathrm{d}\sigma}{\mathrm{d}u}\right)^2 \mathrm{d}u = \frac{9\sigma_0^2}{4}\int_0^1 (1+3u)^{-3}\,\mathrm{d}u = \frac{9\sigma_0^2}{4}\cdot\frac{1}{6}\left(1-\frac{1}{16}\right) \approx 0.352\,\sigma_0^2,$$

compared with $(\sigma_0-\sigma_1)^2 = 0.25\,\sigma_0^2$ along the geodesic, where σ changes linearly in time. So the linear ramp costs $4.352\,k_{\mathrm{B}}T\tau_0/\tau$ at large τ, against $4.25\,k_{\mathrm{B}}T\tau_0/\tau$. In this example most of the benefit of optimal control comes from finishing on time with the bias leading the mean, and only a few percent from the shape of the stiffness schedule. The shape of the stiffness schedule matters more for strong compressions. Repeating the computation for a compression ratio $r = \sigma_0/\sigma_1$, with g ramped linearly from g_0 to $r^2 g_0$, the width contribution of the linear ramp exceeds the geodesic value $(\sigma_0-\sigma_1)^2$ by a factor 1.41 at $r = 2$, 3.3 at $r = 4$, and 15 at $r = 10$: a linear stiffness ramp does almost all of its compressing in the first instants of the protocol, at high speed.

Hardware scale. With the representative values $\tau_0 = \tau_{RC} = 100\,\mathrm{ns}$ and $T = 300\,\mathrm{K}$, reprogramming the node in $\tau = 1\,\mu\mathrm{s}$ has a transport floor of $0.425\,k_{\mathrm{B}}T \approx 1.8\times10^{-21}\,\mathrm{J}$, or an average power of about 2 fW per node if repeated every microsecond. Halving τ doubles the energy per operation and quadruples the power, as anticipated in Eq. (10.11).

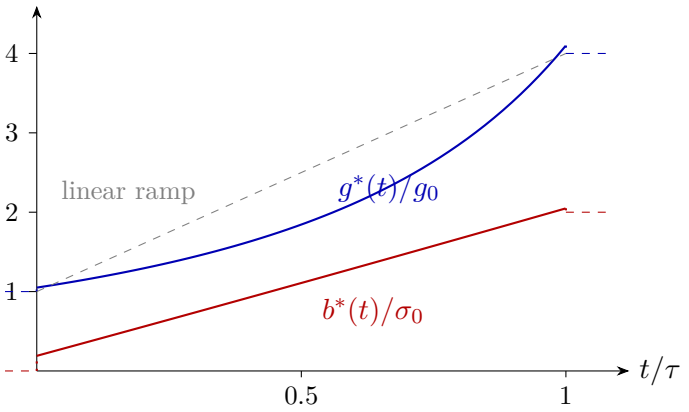


Figure 10.3: Optimal stiffness (blue) and bias (red) schedules Eq. (10.70) for shifting a node distribution by $2\sigma_0$ while halving its width, in time $\tau = 10\,\tau_0$. Short dashes mark the initial and final settings $(g_0, 0)$ and $(4g_0, 2\sigma_0)$; the small vertical segments at $t = 0$ and $t = \tau$ are the endpoint jumps of order τ_0/τ. The gray dashed line is the linear stiffness ramp, which compresses the distribution too early.

Summary

- **Proved:**
  1. The exact work and lag of a harmonic trap dragged at constant speed, with $\mathcal{W}_{\mathrm{diss}} \simeq L^2/(\mu\tau)$ for slow driving Eqs. (10.9) and (10.10).
  2. Elementary properties of $\mathcal{W}_2$: point masses, separation of the mean, and the metric axioms (triangle inequality via explicit gluing for smooth densities) (Proposition 10.4); the complete solution of one-dimensional optimal transport by monotone rearrangement, via Hoeffding's identity and the Fréchet bound (Theorem 10.5); contraction of $\mathcal{W}_2$ under Lipschitz maps and kernels, and exponential contraction under κ-convex Langevin dynamics by synchronous coupling (Lemma 10.7 and Proposition 10.8).
  3. Weak Kantorovich duality, and the derivation of the gradient structure $\mathbf{y} = \nabla\varphi(\mathbf{x})$ of optimal plans from complementary slackness (Proposition 10.9).
  4. The Lagrangian representation of the continuity equation and the Benamou–Brenier theorem for smooth positive densities with a regular Brenier potential, in both directions (Lemma 10.12 and Theorem 10.13); potential-flow and Hamilton–Jacobi structure of optimal velocities; the Helmholtz projection; and the length and time-rescaled forms Eq. (10.34).
  5. The Euler–Lagrange equation of a JKO step and its consistency with the Fokker–Planck equation to $\mathcal{O}(h)$ (Proposition 10.24); displacement convexity of $\mathcal{F}$ for κ-convex E (Proposition 10.25).
  6. The geometric second law $\Sigma_{\mathrm{tot}} \geq k_{\mathrm{B}}\mathcal{W}_2^2/(D\tau)$, equivalently $\mathcal{W}_{\mathrm{diss}} \geq \mathcal{W}_2^2/(\mu\tau)$, for overdamped dynamics with $D = \mu k_{\mathrm{B}}T$, its saturation by the explicit protocol $E^* = -k_{\mathrm{B}}T \ln \rho_t^* - \psi_t^*/\mu$, and the absence of dissipation at its endpoint jumps (Theorems 10.27 and 10.30).
  7. The distinction between entropy-optimal and work-optimal protocols for the dragged trap, including the Schmiedl–Seifert solution Eq. (10.56); the quench cost $k_{\mathrm{B}}T\, D_{\mathrm{KL}}(\rho_0 \,\|\, p_1)$.
  8. For the RC sampler: the Bures–Wasserstein formula with a self-contained optimality proof, the Gaussian geodesic, the exact optimal conductance and bias schedules Eq. (10.67) verified against the moment equations, and the minimum work Eq. (10.69).
- **Assumed:**
  1. Existence of optimal couplings, the absence of a duality gap in Eq. (10.22), and Brenier's theorem (Theorem 10.10) together with the regularity of the Brenier potential used in Theorem 10.13 (smoothness and $\nabla^2\varphi \succeq c\mathbf{I}$). These are proved in the optimal transport literature [114].
  2. Measurable selection of optimal couplings in Lemma 10.7(b).
  3. Solvability of the weighted Poisson equation $\mathcal{K}_\rho\psi = s$; Otto's calculus (Definition 10.19, Proposition 10.21, and Theorem 10.22) is formal.
  4. Existence and uniqueness of JKO minimizers and convergence of the scheme as $h \to 0$ [58].

5. The adiabatic-following hypothesis in Proposition 10.32, and the Talagrand transport inequality in Remark 10.33 [7].
6. Smooth positive densities with sufficient decay to discard boundary terms, as in Chapters 6 and 8.

- **Taken up later:**

1. Gaussian identities (Schur complements, Gaussian KL divergence, Lyapunov equations) are collected in Chapter B; stochastic-calculus details of the moment equations are in Chapter 4 and Chapter C.
2. The non-equilibrium Landauer principle and fluctuation theorems that complement the finite-time bound are developed in Chapter 9; the comparison of the transport floor with transistor-level energy budgets is in Chapter 17.
3. Transport and probability-flow formulations of diffusion models are developed in Chapter 15.

- **Open / Frontier:**

1. Optimal protocols under realistic hardware constraints (bounded or positive stiffness, finite programming bandwidth that forbids jumps, a restricted set of tunable couplings) are not given by Theorem 10.30 and are generally not known in closed form.
2. Tight finite-time bounds for non-equilibrium circuits with circulating currents (Remark 10.29), where the housekeeping part of the dissipation cannot be removed by any potential.
3. Finite-time optimal transport for multimodal distributions such as those of the double-well sampler, where the Wasserstein geodesic must pass mass through low-probability barrier regions that the native dynamics crosses only by rare Kramers events.

Bibliographic notes

The transport problem goes back to Monge (1781), and its relaxation to couplings and the duality theory to Kantorovich (1942). Brenier's theorem on the gradient structure of optimal maps appeared in 1991; the textbook by Villani [114] gives complete proofs of the results we assumed, including existence of optimal plans, duality, and Brenier's theorem. The dynamical formulation proved in Section 10.3 is due to Benamou and Brenier [9], who used the convexity in the variables $(\rho, \rho\mathbf{v})$ as the basis of a numerical method. The interpretation of the Fokker–Planck equation as a gradient flow of the free energy in the Wasserstein metric, and the minimizing-movement scheme, are due to Jordan, Kinderlehrer, and Otto [58]; Otto's formal Riemannian calculus was developed shortly afterwards, and displacement convexity is due to McCann. Functional inequalities linking transport, entropy, and curvature, including Talagrand's inequality, are treated by Bakry, Gentil, and Ledoux [7].

On the thermodynamic side, optimal finite-time protocols for overdamped systems, including the endpoint jumps, were found by Schmiedl and Seifert [95]; the review by Seifert [97] places them in the framework of stochastic thermodynamics, and Sekimoto's energetics [98] supplies the

definitions of work and heat used here. The connection of minimal entropy production to optimal transport was made by Aurell, Mejía-Monasterio, and Muratore-Ginanneschi [5]. Dechant and Sasa [25] bounded generalized currents in Langevin systems by the entropy production, a closely related family of finite-time inequalities, and Van Vu and Hasegawa [113] extended the geometric speed limit to Markov jump processes with a discrete transport distance. The Bures–Wasserstein formula for Gaussians, used in Worked computation 10.35, is classical in both optimal transport and quantum information, where it appears as the Bures distance between density matrices.

Part IV

Solid-State, Mesoscopic, and Superconducting Physics of Extropic Hardware

Chapter 11

Electronic Noise in Mesoscopic Conductors: Thermal, Shot, and $1/f$ Fluctuations

Up to this point the noise in our Langevin equations has been a mathematical primitive: a white Gaussian force $\sqrt{2\mathbf{D}}\,\mathrm{d}\mathbf{W}_t$ whose strength is fixed by the fluctuation-dissipation theorem (FDT) of Chapter 5. A circuit on a chip has no such generator. Its fluctuations come from discrete charge carriers, electrons and holes of charge $q_e \approx 1.602 \times 10^{-19}\,\mathrm{C}$, that scatter off phonons, cross potential barriers, and tunnel into and out of defects in the gate oxide. This chapter derives the noise of such conductors from that microscopic picture, so that the Langevin models of Part II can be connected to the devices of Part IV.

The character of the noise depends on how far the conductor is from equilibrium, measured by the bias voltage V in units of the thermal voltage $V_T := k_{\mathrm{B}}T/q_e \approx 25.85\,\mathrm{mV}$ at $T = 300\,\mathrm{K}$. Near equilibrium the current shows Johnson–Nyquist thermal noise, $S_I(f) = 4k_{\mathrm{B}}TG$, which does not involve q_e. Far from equilibrium it shows Schottky shot noise, $S_I(f) = 2q_e\,\langle I\rangle$, which does not involve T. The central result of the chapter (Section 11.4) is that both are limits of a single formula, $S_I(f) = 2q_eI\coth(q_eV/2k_{\mathrm{B}}T)$, obtained by treating transport across a barrier as two opposing Poisson streams of electrons linked by local detailed balance [37, 93]. Thermal noise is then the sum of two equal and opposite shot noises whose mean currents cancel. We also treat the low-frequency random telegraph and $1/f$ noise of oxide traps, which sets a lower limit on useful sampling frequencies, and close with the full counting statistics of a biased barrier, which reproduce the fluctuation theorems of Chapter 9 at the level of single electron jumps.

11.1 Stationary Stochastic Currents and the Wiener–Khinchin Theorem

Consider a two-terminal mesoscopic conductor (a metallic resistor, a semiconductor channel, or a tunnel junction) held at constant temperature T and constant terminal voltage V. An ideal ammeter of unlimited bandwidth attached to the leads would show a current $I(t)$ that fluctuates about its ensemble average $\langle I\rangle$:

$$I(t) = \langle I\rangle + \delta I(t), \qquad \langle \delta I(t)\rangle = 0. \tag{11.1}$$

Because T and V are held fixed, the process $I(t)$ quickly relaxes to a time-translation-invariant steady state: thermal equilibrium p_{eq} when $V = 0$, or a non-equilibrium steady state p_{ss} when

$V \neq 0$, in the sense of Chapters 6 and 8.

To describe the frequency content of $\delta I(t)$ we meet an immediate obstacle. A stationary signal with constant non-zero variance $\langle \delta I(t)^2 \rangle = \sigma_I^2 > 0$ does not decay as $t \to \pm\infty$, so its sample paths belong neither to $L^1(\mathbb{R})$ nor to $L^2(\mathbb{R})$, and the Fourier integral $\int_{-\infty}^{\infty} \delta I(t) e^{-i\omega t}\, \mathrm{d}t$ diverges almost surely. We therefore define spectral densities by truncating the signal to a finite time window and then taking an ensemble average.

Definition 11.1 (Wide-sense stationary current)**.** A real-valued continuous-time current process $\{I(t) : t \in \mathbb{R}\}$ with finite second moments ($\mathbb{E}[I(t)^2] < \infty$) is *wide-sense stationary* (WSS) if:

(i) Its mean current is independent of time:

$$\langle I(t) \rangle = \bar{I} \equiv \langle I \rangle \qquad \forall\, t \in \mathbb{R}. \tag{11.2}$$

(ii) Its two-time autocovariance, which we call the autocorrelation function, depends only on the time lag $\tau = t - t'$:

$$C_{II}(\tau) := \mathrm{Cov}\left(I(t+\tau),\, I(t)\right) = \langle \delta I(t+\tau)\, \delta I(t) \rangle \qquad \forall\, t, \tau \in \mathbb{R}, \tag{11.3}$$

where $\delta I(t) := I(t) - \langle I \rangle$.

Three properties of $C_{II}(\tau)$ follow directly from Definition 11.1:

(a) *Zero-lag variance:* At $\tau = 0$, the autocorrelation equals the instantaneous current variance,

$$C_{II}(0) = \left\langle \delta I(t)^2 \right\rangle = \mathrm{Var}(I) \geq 0. \tag{11.4}$$

(b) *Even parity:* Because $\delta I(t)$ is a real-valued scalar process and the ensemble average is invariant under shifting the time origin by $-\tau$, we have

$$C_{II}(-\tau) = \langle \delta I(t-\tau)\, \delta I(t) \rangle = \langle \delta I(t')\, \delta I(t'+\tau) \rangle = C_{II}(\tau). \tag{11.5}$$

(c) *Cauchy–Schwarz bound:* For every lag $\tau \in \mathbb{R}$, the Cauchy–Schwarz inequality on the $L^2(\Omega, \mathbb{P})$ probability space gives

$$|C_{II}(\tau)| = |\langle \delta I(t+\tau)\, \delta I(t) \rangle| \leq \sqrt{\langle \delta I(t+\tau)^2 \rangle\, \langle \delta I(t)^2 \rangle} = C_{II}(0). \tag{11.6}$$

To pass to the frequency domain we restrict the fluctuation $\delta I(t)$ to a symmetric observation window $t \in [-\mathcal{T}/2, \mathcal{T}/2]$ of duration $\mathcal{T} > 0$:

$$\delta I_{\mathcal{T}}(t) := \begin{cases} \delta I(t), & \text{if } |t| \leq \mathcal{T}/2, \\ 0, & \text{if } |t| > \mathcal{T}/2. \end{cases} \tag{11.7}$$

Because $\delta I_{\mathcal{T}}(t)$ has compact support and finite second moment, its Fourier transform

$$\delta \hat{I}_{\mathcal{T}}(\omega) := \int_{-\infty}^{\infty} \delta I_{\mathcal{T}}(t)\, e^{-i\omega t}\, \mathrm{d}t = \int_{-\mathcal{T}/2}^{\mathcal{T}/2} \delta I(t)\, e^{-i\omega t}\, \mathrm{d}t \tag{11.8}$$

is well-defined in mean square for every angular frequency $\omega \in \mathbb{R}$. By Parseval's theorem, the total integrated squared fluctuation in the window $[-\mathcal{T}/2, \mathcal{T}/2]$ is

$$\int_{-\mathcal{T}/2}^{\mathcal{T}/2} |\delta I(t)|^2\, \mathrm{d}t = \int_{-\infty}^{\infty} |\delta I_{\mathcal{T}}(t)|^2\, \mathrm{d}t = \int_{-\infty}^{\infty} \left|\delta \hat{I}_{\mathcal{T}}(\omega)\right|^2 \frac{\mathrm{d}\omega}{2\pi}. \tag{11.9}$$

Dividing Eq. (11.9) by the window duration $\mathcal{T}$ and taking the ensemble average gives the mean power per unit time, which motivates the following definition.

Definition 11.2 (Double-sided and single-sided power spectral densities)**.** Let $\{I(t) : t \in \mathbb{R}\}$ be a wide-sense stationary current process.

(i) The *double-sided power spectral density* (PSD) with respect to angular frequency $\omega \in (-\infty, \infty)$ is the long-time limit

$$S_I^{\mathrm{ds}}(\omega) := \lim_{\mathcal{T}\to\infty} \frac{1}{\mathcal{T}} \left\langle \left| \delta\hat{I}_{\mathcal{T}}(\omega) \right|^2 \right\rangle, \qquad \omega \in \mathbb{R}. \tag{11.10}$$

(ii) Because $\delta I(t)$ is real-valued, $\delta\hat{I}_{\mathcal{T}}(-\omega) = \delta\hat{I}_{\mathcal{T}}(\omega)^*$, so $S_I^{\mathrm{ds}}(-\omega) = S_I^{\mathrm{ds}}(\omega)$ is an even, non-negative function of ω. Experimentalists and circuit engineers measure power only at positive cyclic frequencies $f = \omega/(2\pi) \in [0, \infty)$ and fold the negative-frequency power onto the positive axis. The *single-sided power spectral density* $S_I(f)$ (with units of A^2/Hz) is defined for $f \geq 0$ by

$$S_I(f) := 2\, S_I^{\mathrm{ds}}(2\pi f) = \lim_{\mathcal{T}\to\infty} \frac{2}{\mathcal{T}} \left\langle \left| \delta\hat{I}_{\mathcal{T}}(2\pi f) \right|^2 \right\rangle, \qquad f \geq 0. \tag{11.11}$$

Notation Watch: Double-sided $S_I^{\mathrm{ds}}(\omega)$, single-sided $S_I(f)$, and the Langevin diffusion coefficient

The distinction between $S_I^{\mathrm{ds}}(\omega)$ and $S_I(f)$ is a common source of factor-of-2 and factor-of-2π errors when moving between the physics, stochastic-calculus, and engineering literatures. The two are related as follows.

1. Integration over all frequencies recovers the variance $\mathrm{Var}(I)$:

$$\begin{aligned} \mathrm{Var}(I) = \left\langle \delta I(t)^2 \right\rangle &= \int_{-\infty}^{\infty} S_I^{\mathrm{ds}}(\omega)\, \frac{\mathrm{d}\omega}{2\pi} = \int_{-\infty}^{\infty} S_I^{\mathrm{ds}}(2\pi f)\, \mathrm{d}f \\ &= \int_0^{\infty} 2\, S_I^{\mathrm{ds}}(2\pi f)\, \mathrm{d}f = \int_0^{\infty} S_I(f)\, \mathrm{d}f. \end{aligned} \tag{11.12}$$

2. For white noise, the PSD is fixed by the Langevin diffusion constant. Suppose a white-noise current $\delta I(t) = \sigma_I \xi(t)$ has delta-correlated autocorrelation $C_{II}(\tau) = \langle \delta I(t+\tau)\delta I(t) \rangle = 2D_I\delta(\tau)$, corresponding in Itô stochastic calculus (Chapters 4 and 5) to a charge increment $\mathrm{d}Q_t = \langle I \rangle\, \mathrm{d}t + \sqrt{2D_I}\, \mathrm{d}W_t$ with $\mathrm{d}W_t^2 = \mathrm{d}t$. Then:

$$S_I^{\mathrm{ds}}(\omega) = 2D_I, \qquad S_I(f) = 2\, S_I^{\mathrm{ds}}(2\pi f) = 4D_I. \tag{11.13}$$

For an equilibrium resistor of conductance G, the FDT of Chapter 5 gives $D_I = k_{\mathrm{B}}TG$, which translates via Eq. (11.13) to $S_I^{\mathrm{ds}}(\omega) = 2k_{\mathrm{B}}TG$ and $S_I(f) = 4k_{\mathrm{B}}TG$. Similarly, for Poissonian shot noise with $D_I = \frac{1}{2}q_e \langle I \rangle$, we have $S_I^{\mathrm{ds}}(\omega) = q_e \langle I \rangle$ and $S_I(f) = 2q_e \langle I \rangle$. Throughout Part IV, $S_I(f)$ (or simply S_I) without superscript denotes the single-sided PSD, while $S_I^{\mathrm{ds}}(\omega)$ explicitly marks the double-sided angular spectrum.

The next theorem connects the autocorrelation function to the power spectral density.

Theorem 11.3 (Wiener–Khinchin)**.** *Let $\{I(t) : t \in \mathbb{R}\}$ be a wide-sense stationary current process whose autocorrelation function $C_{II}(\tau) = \langle \delta I(t+\tau)\delta I(t) \rangle$ is continuous and satisfies the integrability condition*

$$\int_{-\infty}^{\infty} (1 + |\tau|)\, |C_{II}(\tau)|\, \mathrm{d}\tau < \infty. \tag{11.14}$$

Then the limit in Eq. (11.10) exists for every $\omega \in \mathbb{R}$ and equals the Fourier transform of $C_{II}(\tau)$:

$$S_I^{\mathrm{ds}}(\omega) = \int_{-\infty}^{\infty} C_{II}(\tau)\, e^{-i\omega\tau}\, \mathrm{d}\tau = 2\int_0^{\infty} C_{II}(\tau)\, \cos(\omega\tau)\, \mathrm{d}\tau. \tag{11.15}$$

Equivalently, the single-sided PSD for $f \geq 0$ is given by

$$S_I(f) = 2\int_{-\infty}^{\infty} C_{II}(\tau)\, e^{-i2\pi f\tau}\, \mathrm{d}\tau = 4\int_0^{\infty} C_{II}(\tau)\, \cos(2\pi f\tau)\, \mathrm{d}\tau, \tag{11.16}$$

and the autocorrelation function is recovered by the inverse Fourier transform:

$$C_{II}(\tau) = \int_{-\infty}^{\infty} S_I^{\mathrm{ds}}(\omega)\, e^{i\omega\tau}\, \frac{\mathrm{d}\omega}{2\pi} = \int_0^{\infty} S_I(f)\, \cos(2\pi f\tau)\, \mathrm{d}f. \tag{11.17}$$

Proof. We write the squared modulus of the windowed Fourier transform Eq. (11.8) as a double integral over the square $[-\mathcal{T}/2, \mathcal{T}/2]^2$:

$$\begin{aligned}\frac{1}{\mathcal{T}}\left\langle \left|\delta\hat{I}_{\mathcal{T}}(\omega)\right|^2 \right\rangle &= \frac{1}{\mathcal{T}}\left\langle \left(\int_{-\mathcal{T}/2}^{\mathcal{T}/2} \delta I(t)\, e^{-i\omega t}\, \mathrm{d}t\right)\left(\int_{-\mathcal{T}/2}^{\mathcal{T}/2} \delta I(t')\, e^{+i\omega t'}\, \mathrm{d}t'\right)\right\rangle \\ &= \frac{1}{\mathcal{T}}\int_{-\mathcal{T}/2}^{\mathcal{T}/2}\mathrm{d}t \int_{-\mathcal{T}/2}^{\mathcal{T}/2}\mathrm{d}t'\, \left\langle \delta I(t)\, \delta I(t')\right\rangle\, e^{-i\omega(t-t')}. \end{aligned} \tag{11.18}$$

where we used Fubini's theorem (justified because $\mathbb{E}[\delta I(t)^2] = C_{II}(0) < \infty$ on the compact square $[-\mathcal{T}/2, \mathcal{T}/2]^2$) to bring the expectation inside the double integral. By wide-sense stationarity (Definition 11.1), $\langle \delta I(t)\, \delta I(t')\rangle = C_{II}(t-t')$.

We now perform a linear change of coordinates from (t, t') to the relative lag τ and midpoint time v:

$$\tau := t - t', \qquad v := \frac{t+t'}{2}, \qquad \Longleftrightarrow \qquad t = v + \frac{\tau}{2}, \qquad t' = v - \frac{\tau}{2}. \tag{11.19}$$

The Jacobian determinant of this transformation is

$$\left|\det \frac{\partial(t,t')}{\partial(\tau,v)}\right| = \left|\det \begin{pmatrix} 1/2 & 1 \\ -1/2 & 1 \end{pmatrix}\right| = \left|\frac{1}{2} - \left(-\frac{1}{2}\right)\right| = 1. \tag{11.20}$$

Next we find the domain of integration in the (τ, v)-plane that corresponds to $-\mathcal{T}/2 \leq t \leq \mathcal{T}/2$ and $-\mathcal{T}/2 \leq t' \leq \mathcal{T}/2$. The difference $\tau = t - t'$ ranges over $[-\mathcal{T}, \mathcal{T}]$. For a fixed value of $\tau \in [-\mathcal{T}, \mathcal{T}]$, the midpoint v must simultaneously satisfy

$$-\frac{\mathcal{T}}{2} - \frac{\tau}{2} \leq v \leq \frac{\mathcal{T}}{2} - \frac{\tau}{2} \qquad \text{and} \qquad -\frac{\mathcal{T}}{2} + \frac{\tau}{2} \leq v \leq \frac{\mathcal{T}}{2} + \frac{\tau}{2}. \tag{11.21}$$

Combining the two inequalities in Eq. (11.21) yields the exact interval for v at fixed τ:

$$-\frac{\mathcal{T} - |\tau|}{2} \leq v \leq +\frac{\mathcal{T} - |\tau|}{2}. \tag{11.22}$$

The integrand $C_{II}(\tau)\, e^{-i\omega\tau}$ in Eq. (11.18) does not depend on the midpoint coordinate v, so the integration over v multiplies the integrand by the length $(\mathcal{T} - |\tau|)$ of the interval Eq. (11.22):

$$\frac{1}{\mathcal{T}}\left\langle \left|\delta\hat{I}_{\mathcal{T}}(\omega)\right|^2 \right\rangle = \frac{1}{\mathcal{T}}\int_{-\mathcal{T}}^{\mathcal{T}} \mathrm{d}\tau\, C_{II}(\tau)\, e^{-i\omega\tau} \int_{-(\mathcal{T}-|\tau|)/2}^{+(\mathcal{T}-|\tau|)/2} \mathrm{d}v$$

$$= \int_{-\mathcal{T}}^{\mathcal{T}} \left(1 - \frac{|\tau|}{\mathcal{T}}\right) C_{II}(\tau)\, e^{-i\omega\tau}\, \mathrm{d}\tau. \tag{11.23}$$

The triangular factor is the Bartlett kernel $K_{\mathcal{T}}(\tau) := (1 - |\tau|/\mathcal{T})\mathbf{1}_{|\tau|\le\mathcal{T}}$. To take the limit $\mathcal{T} \to \infty$, we bound the difference between the finite-time integral Eq. (11.23) and the infinite-time Fourier integral:

$$\begin{aligned}\left| \int_{-\mathcal{T}}^{\mathcal{T}} \left(1 - \frac{|\tau|}{\mathcal{T}}\right) C_{II}(\tau)\, e^{-i\omega\tau}\, \mathrm{d}\tau - \int_{-\infty}^{\infty} C_{II}(\tau)\, e^{-i\omega\tau}\, \mathrm{d}\tau \right| &\le \frac{1}{\mathcal{T}} \int_{-\mathcal{T}}^{\mathcal{T}} |\tau|\, |C_{II}(\tau)|\, \mathrm{d}\tau + \int_{|\tau|>\mathcal{T}} |C_{II}(\tau)|\, \mathrm{d}\tau \\ &\le \frac{1}{\mathcal{T}} \int_{-\infty}^{\infty} |\tau|\, |C_{II}(\tau)|\, \mathrm{d}\tau + \int_{|\tau|>\mathcal{T}} |C_{II}(\tau)|\, \mathrm{d}\tau.\end{aligned} \tag{11.24}$$

By hypothesis Eq. (11.14), $\int_{-\infty}^{\infty} |\tau|\, |C_{II}(\tau)|\, \mathrm{d}\tau < \infty$ and $\int_{-\infty}^{\infty} |C_{II}(\tau)|\, \mathrm{d}\tau < \infty$. As $\mathcal{T} \to \infty$, both terms on the right-hand side of Eq. (11.24) vanish uniformly in ω, proving the first equality in Eq. (11.15). Using the even parity $C_{II}(-\tau) = C_{II}(\tau)$ from Eq. (11.5) and Euler's formula $e^{-i\omega\tau} = \cos(\omega\tau) - i\sin(\omega\tau)$, the odd sine integral vanishes on $(-\infty, \infty)$, yielding the second equality in Eq. (11.15). Multiplying by 2 and setting $\omega = 2\pi f$ yields Eq. (11.16), and standard Fourier inversion on $L^1(\mathbb{R})$ yields Eq. (11.17). □

In a circuit, a fluctuating current source usually drives a linear time-invariant (LTI) impedance $Z(\omega)$, such as a parallel RC combination, and produces a fluctuating node voltage $V(t)$. The impedance reshapes the spectrum as follows.

Proposition 11.4 (Spectral filtering by a linear impedance)**.** *Suppose a wide-sense stationary current fluctuation $\delta I(t)$ with single-sided PSD $S_I(f)$ drives a stable linear time-invariant two-terminal network with impulse response $z(t)$ ($z(t) = 0$ for $t < 0$, $\int_0^\infty |z(t)|\, \mathrm{d}t < \infty$) and complex impedance $Z(\omega) = \int_0^\infty z(t) e^{-i\omega t}\, \mathrm{d}t$, so that $\delta V(t) = \int_{-\infty}^{\infty} z(t-s)\, \delta I(s)\, \mathrm{d}s$. Then $\delta V(t)$ is wide-sense stationary with single-sided voltage PSD*

$$S_V(f) = |Z(2\pi f)|^2\, S_I(f). \tag{11.25}$$

Proof. We compute the voltage autocorrelation function $C_{VV}(\tau) = \langle \delta V(t+\tau)\, \delta V(t) \rangle$ directly from the convolution representation:

$$\begin{aligned} C_{VV}(\tau) &= \left\langle \left(\int_{-\infty}^{\infty} z(u)\, \delta I(t+\tau-u)\, \mathrm{d}u \right) \left(\int_{-\infty}^{\infty} z(u')\, \delta I(t-u')\, \mathrm{d}u' \right) \right\rangle \\ &= \int_{-\infty}^{\infty} \mathrm{d}u \int_{-\infty}^{\infty} \mathrm{d}u'\, z(u)\, z(u')\, C_{II}(\tau - u + u'). \end{aligned} \tag{11.26}$$

Taking the Fourier transform of Eq. (11.26) with respect to τ and changing the inner variable to $s = \tau - u + u'$ ($\mathrm{d}\tau = \mathrm{d}s$ at fixed u, u') factors the triple integral into three single integrals:

$$\begin{aligned} S_V^{\mathrm{ds}}(\omega) &= \int_{-\infty}^{\infty} C_{VV}(\tau)\, e^{-i\omega\tau}\, \mathrm{d}\tau \\ &= \int_{-\infty}^{\infty} \mathrm{d}u\, z(u) e^{-i\omega u} \int_{-\infty}^{\infty} \mathrm{d}u'\, z(u') e^{+i\omega u'} \int_{-\infty}^{\infty} \mathrm{d}s\, C_{II}(s) e^{-i\omega s} \\ &= Z(\omega)\, Z(\omega)^*\, S_I^{\mathrm{ds}}(\omega) = |Z(\omega)|^2\, S_I^{\mathrm{ds}}(\omega). \end{aligned} \tag{11.27}$$

Multiplying both sides of Eq. (11.27) by 2 and setting $\omega = 2\pi f$ gives Eq. (11.25). □

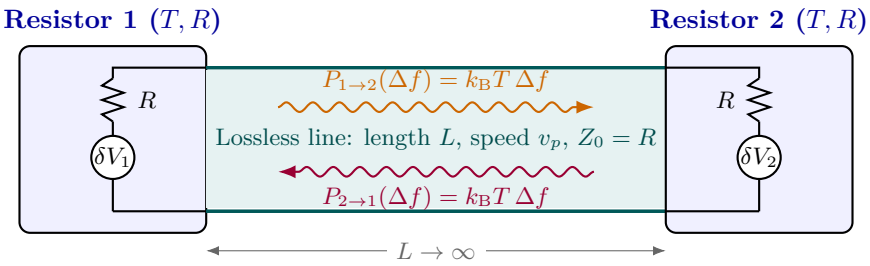


Figure 11.1: Nyquist's thought experiment [79]. Two resistors of resistance R at temperature T terminate a lossless transmission line of characteristic impedance $Z_0 = R$ and length L. Because $Z_0 = R$, each resistor absorbs all incident waves without reflection. Detailed balance in each frequency band $[f, f + \Delta f]$ requires $P_{1\to2}(\Delta f) = P_{2\to1}(\Delta f) = k_B T\,\Delta f$.

11.2 Equilibrium Johnson–Nyquist Thermal Noise

In 1928 John B. Johnson at Bell Telephone Laboratories showed that every conductor in thermal equilibrium at temperature T has spontaneous voltage fluctuations across its open terminals [57]. Their spectral density depended only on the resistance R and the temperature T, not on the material (copper, nichrome, carbon, or electrolyte), the carrier density, the geometry, or the scattering mechanism. In the next article of the same issue of *Physical Review*, Harry Nyquist explained this universality with a thought experiment that combines the Second Law with transmission-line theory [79].

The universality has a thermodynamic origin. Any conductor can be coupled to any other system at the same temperature, including a lossless transmission line, and the Second Law forbids a net flow of heat between them. We give two derivations. Nyquist's transmission-line argument explains *why* the result cannot depend on material details. The Landauer–Büttiker scattering derivation shows *how* the same law arises from the Fermi–Dirac statistics of electrons in a mesoscopic channel.

11.2.1 Nyquist's Transmission-Line Derivation

Consider two resistors of equal resistance $R_1 = R_2 = R$, both in a thermal bath at temperature T, connected by a lossless transmission line of length L, wave propagation speed v_p, and characteristic impedance $Z_0 = \sqrt{\mathcal{L}_0/\mathcal{C}_0} = R$, as depicted in Fig. 11.1.

Theorem 11.5 (Johnson–Nyquist thermal noise)**.** *Let a linear two-terminal conductor of resistance R (and conductance $G = 1/R$) be maintained in thermal equilibrium at temperature T.*

(i) In the classical regime $hf \ll k_B T$ (where $h \approx 6.626 \times 10^{-34}\,\mathrm{J\cdot s}$ is Planck's constant), the open-circuit voltage fluctuation $\delta V(t)$ can be modeled as a Thévenin voltage source in series with R having white single-sided power spectral density

$$S_V(f) = 4k_B T R, \qquad S_V^{\mathrm{ds}}(\omega) = 2k_B T R. \tag{11.28}$$

(ii) Equivalently, by Norton's theorem ($\delta I(t) = \delta V(t)/R$), the short-circuit current fluctuation can be modeled as a parallel Norton current source across $G = 1/R$ having white single-sided power spectral density

$$S_I(f) = \frac{S_V(f)}{R^2} = 4k_B T G, \qquad S_I^{\mathrm{ds}}(\omega) = 2k_B T G. \tag{11.29}$$

(iii) At all frequencies $f \in [0,\infty)$, replacing classical equipartition by the Bose–Einstein occupation of the line modes gives the Planck–Nyquist formula

$$S_V^{\mathrm{Q}}(f) = 4R\,\frac{hf}{\exp(hf/k_{\mathrm{B}}T)-1}. \tag{11.30}$$

Proof. We follow Nyquist's three-step argument [79].

Step 1: Power transfer from resistor 1 to resistor 2. Model each noisy resistor $j \in \{1,2\}$ by its noiseless resistance R in series with a fluctuating open-circuit Thévenin voltage source $\delta V_j(t)$ having single-sided PSD $S_V(f)$. Because the line has characteristic impedance $Z_0 = R$, the input impedance seen by resistor 1 at the left port is $Z_{\mathrm{in}} = Z_0 = R$, and any wave traveling to the right is absorbed by resistor 2 without reflection ($\Gamma_{\mathrm{refl}} = (R - Z_0)/(R + Z_0) = 0$).

In a narrow frequency band $[f, f+\Delta f]$, the mean-square open-circuit voltage of resistor 1 is $\langle \delta V_1^2 \rangle_{\Delta f} = S_V(f)\,\Delta f$. Because resistor 1 (internal resistance R) drives a matched load $Z_{\mathrm{in}} = R$, half of the Thévenin voltage appears across the input port:

$$\delta V_{\mathrm{port},1}(t) = \frac{Z_0}{R+Z_0}\,\delta V_1(t) = \frac{R}{2R}\,\delta V_1(t) = \frac{1}{2}\delta V_1(t), \qquad \delta I_{\mathrm{port},1}(t) = \frac{\delta V_1(t)}{2R}. \tag{11.31}$$

The average power that resistor 1 emits into the right-moving modes of the line within the bandwidth $[f, f+\Delta f]$ is

$$P_{1\to 2}(\Delta f) = \langle \delta V_{\mathrm{port},1}\,\delta I_{\mathrm{port},1} \rangle_{\Delta f} = \frac{\langle \delta V_1^2 \rangle_{\Delta f}}{4R} = \frac{S_V(f)\,\Delta f}{4R}. \tag{11.32}$$

By symmetry, resistor 2 emits the same average power $P_{2\to 1}(\Delta f) = \frac{S_V(f)\,\Delta f}{4R}$ into the left-moving modes, all of which is absorbed by resistor 1.

Step 2: Counting the normal modes of the line. Suppose that at $t = 0$, after the resistors and the line have reached thermal equilibrium at temperature T, we short-circuit both ends of the line ($x = 0$ and $x = L$) with ideal switches. The energy that was in transit along the line is now trapped inside a one-dimensional cavity of length L with boundary conditions $V(0,t) = V(L,t) = 0$. The standing-wave normal modes have wavelengths $\lambda_m = 2L/m$ and cyclic frequencies

$$f_m = \frac{v_p}{\lambda_m} = \frac{m v_p}{2L}, \qquad m = 1, 2, 3, \ldots \tag{11.33}$$

Taking the long-line limit $L \to \infty$ so that the mode spacing $\delta f = v_p/(2L) \ll \Delta f$, the number of standing-wave modes $\Delta N_{\mathrm{modes}}$ in the frequency interval $[f, f+\Delta f]$ is

$$\Delta N_{\mathrm{modes}} = \frac{\Delta f}{\delta f} = \frac{2L}{v_p}\,\Delta f. \tag{11.34}$$

Step 3: Mode energy and transit time. Each normal mode m is a harmonic oscillator with electric energy ($\propto \mathcal{C}_0 V_m^2/2$) and magnetic energy ($\propto \mathcal{L}_0 I_m^2/2$), where $\mathcal{C}_0$ and $\mathcal{L}_0$ are the capacitance and inductance per unit length. In the classical regime $hf \ll k_{\mathrm{B}}T$, equipartition (Eq. (1.39)) assigns each mode a mean energy $\frac{1}{2}k_{\mathrm{B}}T + \frac{1}{2}k_{\mathrm{B}}T = k_{\mathrm{B}}T$. The total energy stored on the line of length L within the frequency band $[f, f+\Delta f]$ is

$$\Delta E_{\mathrm{line}}(\Delta f) = k_{\mathrm{B}}T\,\Delta N_{\mathrm{modes}} = \frac{2L}{v_p}\,k_{\mathrm{B}}T\,\Delta f. \tag{11.35}$$

Half of this energy belongs to right-moving waves (emitted by resistor 1) and half to left-moving waves (emitted by resistor 2):

$$\Delta E_{1\to 2}(\Delta f) = \frac{1}{2}\,\Delta E_{\mathrm{line}}(\Delta f) = \frac{L}{v_p}\,k_{\mathrm{B}}T\,\Delta f. \tag{11.36}$$

Right-moving waves cross the line at speed v_p, so all right-moving energy on the line was emitted by resistor 1 during the transit time $\tau_{\text{line}} = L/v_p$:

$$\Delta E_{1\to 2}(\Delta f) = P_{1\to 2}(\Delta f)\,\tau_{\text{line}} = P_{1\to 2}(\Delta f)\,\frac{L}{v_p}. \tag{11.37}$$

Equating Eq. (11.36) and Eq. (11.37) eliminates the line length L and the speed v_p, so the available noise power per unit bandwidth is universal:

$$P_{1\to 2}(\Delta f) = k_{\text{B}} T\,\Delta f. \tag{11.38}$$

Substituting Eq. (11.32) into Eq. (11.38) yields

$$\frac{S_V(f)\,\Delta f}{4R} = k_{\text{B}} T\,\Delta f \qquad \Longrightarrow \qquad S_V(f) = 4k_{\text{B}} T R, \tag{11.39}$$

which establishes Eq. (11.28) and Eq. (11.29). Finally, replacing the classical energy $k_{\text{B}}T$ per mode in Eq. (11.35) by the mean thermal energy $\bar{\varepsilon}(f) = \frac{hf}{e^{hf/k_{\text{B}}T}-1}$ of a quantum oscillator of frequency f (excluding zero-point energy), the same steps give the Planck–Nyquist formula Eq. (11.30). □

The Planck–Nyquist formula Eq. (11.30) removes the ultraviolet divergence of the classical result. The white spectrum $S_V(f) = 4k_{\text{B}}TR$ would give infinite power if integrated to $f = \infty$, whereas the quantum factor $\frac{hf}{e^{hf/k_{\text{B}}T}-1}$ cuts off exponentially above the thermal quantum frequency

$$f_{\text{th}} := \frac{k_{\text{B}}T}{h} \approx \begin{cases} 6.25 \times 10^{12}\,\text{Hz} \;\; (6.25\,\text{THz}), & \text{at } T = 300\,\text{K}, \\ 8.75 \times 10^{10}\,\text{Hz} \;\; (87.5\,\text{GHz}), & \text{at } T = 4.2\,\text{K}. \end{cases} \tag{11.40}$$

CMOS circuit nodes operate at frequencies $f \lesssim 10^{10}\,\text{Hz} \ll 6.25\,\text{THz}$, so at room temperature the classical white-noise formula $S_I(f) = 4k_{\text{B}}TG$ is accurate to better than 0.1% over the whole operating band.

11.2.2 Landauer–Büttiker Scattering Derivation

Nyquist's argument shows *that* $S_I(f) = 4k_{\text{B}}TG$ must hold in equilibrium. The scattering picture shows *how* the electrons produce it. In the Landauer–Büttiker formalism [13], a mesoscopic conductor is a phase-coherent scattering region connecting a left electron reservoir (L) and a right electron reservoir (R) through M transverse channels (subbands).

In each channel $n \in \{1, \ldots, M\}$ (per spin orientation; spin enters below as a factor of 2) and at energy E, an electron incident from the left reservoir is transmitted into the right reservoir with probability $\mathcal{T}_n(E) \in [0, 1]$ and reflected back into the left reservoir with probability $\mathcal{R}_n(E) = 1 - \mathcal{T}_n(E)$. In thermal equilibrium at chemical potential $\mu_L = \mu_R = \mu_0$ and temperature T, the occupation number $\hat{n}_{\alpha,n}(E) \in \{0, 1\}$ of incoming states from reservoir $\alpha \in \{L, R\}$ has the Fermi–Dirac mean

$$\langle \hat{n}_{\alpha,n}(E) \rangle = f_0(E) := \frac{1}{\exp\!\big((E-\mu_0)/k_{\text{B}}T\big) + 1}. \tag{11.41}$$

Because $\hat{n}_{\alpha,n}(E)$ is a fermionic occupation number ($\hat{n}_{\alpha,n}^2 = \hat{n}_{\alpha,n}$), its variance is

$$\text{Var}\left(\hat{n}_{\alpha,n}(E)\right) = \left\langle \hat{n}_{\alpha,n}(E)^2 \right\rangle - \langle \hat{n}_{\alpha,n}(E) \rangle^2 = f_0(E)\big[1 - f_0(E)\big]. \tag{11.42}$$

Differentiating Eq. (11.41) with respect to E gives the identity

$$-k_{\mathrm{B}}T\frac{\partial f_0(E)}{\partial E} = k_{\mathrm{B}}T\frac{\frac{1}{k_{\mathrm{B}}T}e^{(E-\mu_0)/k_{\mathrm{B}}T}}{\left(e^{(E-\mu_0)/k_{\mathrm{B}}T}+1\right)^2} = f_0(E)\big[1-f_0(E)\big]. \tag{11.43}$$

When a small bias voltage V shifts $\mu_L - \mu_R = q_e V$, the mean current is given by the Landauer formula (summing over the two spin orientations):

$$\begin{aligned}\langle I\rangle &= \frac{2q_e}{h}\sum_{n=1}^{M}\int_{-\infty}^{\infty}\mathcal{T}_n(E)\,\big[f_L(E)-f_R(E)\big]\,\mathrm{d}E \\ &\xrightarrow[V\to 0]{} \left(\frac{2q_e^2}{h}\sum_{n=1}^{M}\int_{-\infty}^{\infty}\mathcal{T}_n(E)\left(-\frac{\partial f_0(E)}{\partial E}\right)\mathrm{d}E\right)V = G\,V.\end{aligned} \tag{11.44}$$

In the scattering theory of current fluctuations [13], the zero-frequency single-sided current PSD at equilibrium ($f_L = f_R = f_0$) is set by the occupation fluctuations Eq. (11.42) of the incident beams; the factor 2 after $4q_e^2/h$ counts spin:

$$S_I(0) = \frac{4q_e^2}{h}\cdot 2\sum_{n=1}^{M}\int_{-\infty}^{\infty}\mathcal{T}_n(E)\,f_0(E)\big[1-f_0(E)\big]\,\mathrm{d}E. \tag{11.45}$$

Substituting the Fermi–Dirac derivative identity Eq. (11.43) into Eq. (11.45) and comparing with the Landauer conductance Eq. (11.44), we obtain

$$S_I(0) = 4k_{\mathrm{B}}T\left(\frac{2q_e^2}{h}\sum_{n=1}^{M}\int_{-\infty}^{\infty}\mathcal{T}_n(E)\left(-\frac{\partial f_0(E)}{\partial E}\right)\mathrm{d}E\right) = 4k_{\mathrm{B}}TG. \tag{11.46}$$

If $\mathcal{T}_n(E) \approx \mathcal{T}_n(\mu_0)$ is constant across the thermal window $|E-\mu_0| \sim k_{\mathrm{B}}T$, the integral $\int_{-\infty}^{\infty}(-\frac{\partial f_0}{\partial E})\,\mathrm{d}E = f_0(-\infty) - f_0(+\infty) = 1$, giving $G = \frac{2q_e^2}{h}\sum_n \mathcal{T}_n$ and $S_I(0) = 4k_{\mathrm{B}}TG$ for *any* set of transmission eigenvalues $\{\mathcal{T}_n\}_{n=1}^{M}$.

11.2.3 Thermal Noise on a Capacitive Node

In a real circuit, and in particular in the RC sampler (the coupled RC circuit of Example 1.5, whose fluctuation-dissipation relation is derived in Chapter 5), a resistive or transconductive channel of conductance $G = 1/R$ sits in parallel with a node capacitance C. The current noise $S_I(f) = 4k_{\mathrm{B}}TG$ of the conductor is white up to terahertz frequencies, but the capacitor shunts high-frequency current to ground and acts as a first-order low-pass filter. Increasing R raises the low-frequency voltage noise $S_V(0) = 4k_{\mathrm{B}}TR$, which suggests that a larger resistor gives larger voltage fluctuations. It does not: the growth of the noise amplitude with R is cancelled exactly by the narrowing of the filter bandwidth as $1/R$.

Proposition 11.6 (The $k_{\mathrm{B}}T/C$ variance of an RC node)**.** *Consider a parallel combination of a noisy equilibrium resistor $R = 1/G$ at temperature T and a lossless capacitor C. The node voltage fluctuation $\delta V(t)$ across the capacitor has:*

(i) A Lorentzian single-sided power spectral density

$$S_V(f) = \frac{4k_{\mathrm{B}}TR}{1+(2\pi f RC)^2} = \frac{4k_{\mathrm{B}}T/G}{1+(2\pi f\tau_{RC})^2}, \qquad \tau_{RC} := RC = \frac{C}{G}; \tag{11.47}$$

(ii) An exponential time-domain autocorrelation function

$$C_{VV}(\tau) = \langle \delta V(t+\tau)\, \delta V(t) \rangle = \frac{k_{\mathrm{B}} T}{C}\, \exp\!\left(-\frac{|\tau|}{\tau_{RC}}\right); \tag{11.48}$$

(iii) The voltage variance

$$\mathrm{Var}(V) = \left\langle \delta V(t)^2 \right\rangle = \int_0^\infty S_V(f)\, \mathrm{d}f = \frac{k_{\mathrm{B}} T}{C}, \tag{11.49}$$

which depends only on T and C and not on R.

Proof. Represent the noisy resistor by its Norton equivalent: a noiseless conductance $G = 1/R$ in parallel with a thermal current source $\delta I(t)$ of single-sided PSD $S_I(f) = 4k_{\mathrm{B}}TG$. The parallel combination of conductance G and capacitance C has complex admittance $Y(\omega) = G + i\omega C$ and complex impedance

$$Z(\omega) = \frac{1}{G + i\omega C} = \frac{R}{1 + i\omega RC}. \tag{11.50}$$

Taking the squared modulus of Eq. (11.50) gives $|Z(2\pi f)|^2 = \frac{R^2}{1+(2\pi f RC)^2}$. By Proposition 11.4 (Eq. (11.25)), the single-sided voltage PSD is

$$S_V(f) = |Z(2\pi f)|^2\, S_I(f) = \frac{R^2}{1 + (2\pi f RC)^2} \left(\frac{4k_{\mathrm{B}}T}{R}\right) = \frac{4k_{\mathrm{B}}TR}{1 + (2\pi f RC)^2}, \tag{11.51}$$

proving Eq. (11.47).

To prove Eq. (11.49), integrate $S_V(f)$ over $f \in [0, \infty)$ using Eq. (11.12):

$$\left\langle \delta V(t)^2 \right\rangle = \int_0^\infty S_V(f)\, \mathrm{d}f = \int_0^\infty \frac{4k_{\mathrm{B}}TR}{1 + (2\pi f RC)^2}\, \mathrm{d}f. \tag{11.52}$$

Substitute $u := 2\pi f RC$, so that $\mathrm{d}f = \mathrm{d}u/(2\pi RC)$. The factor R in the numerator of Eq. (11.52) cancels the R in $\mathrm{d}f$:

$$\begin{aligned} \left\langle \delta V(t)^2 \right\rangle &= 4k_{\mathrm{B}}TR \left(\frac{1}{2\pi RC}\right) \int_0^\infty \frac{\mathrm{d}u}{1+u^2} \\ &= \frac{2k_{\mathrm{B}}T}{\pi C} \Big[\arctan(u)\Big]_{u=0}^{u\to\infty} = \frac{2k_{\mathrm{B}}T}{\pi C}\left(\frac{\pi}{2} - 0\right) = \frac{k_{\mathrm{B}}T}{C}, \end{aligned} \tag{11.53}$$

proving Eq. (11.49). This agrees with equipartition applied to the electrostatic energy $E(V) = \frac{1}{2}CV^2$: $\left\langle \frac{1}{2} C\, \delta V^2 \right\rangle = \frac{1}{2}k_{\mathrm{B}}T$ gives $\langle \delta V^2 \rangle = k_{\mathrm{B}}T/C$. Finally, the autocorrelation function Eq. (11.48) follows from the inverse cosine integral Eq. (11.17) via the residue theorem on the complex u-plane (closing the contour in the upper half-plane for $\tau > 0$ around the simple pole at $u = +i$):

$$C_{VV}(\tau) = \frac{k_{\mathrm{B}}T}{\pi C} \int_{-\infty}^{\infty} \frac{e^{iu|\tau|/\tau_{RC}}}{1+u^2}\, \mathrm{d}u = \frac{k_{\mathrm{B}}T}{\pi C}\, (2\pi i)\, \frac{e^{-|\tau|/\tau_{RC}}}{2i} = \frac{k_{\mathrm{B}}T}{C}\, \exp\!\left(-\frac{|\tau|}{\tau_{RC}}\right). \tag{11.54}$$

□

Example 11.7 (Thermal fluctuations on a femtofarad node)**.** As a representative estimate, take a state node in an advanced CMOS process with total gate-plus-parasitic capacitance $C_{\mathrm{node}} = 1.0\,\mathrm{fF} = 1.0 \times 10^{-15}\,\mathrm{F}$ at room temperature $T = 300\,\mathrm{K}$ ($k_{\mathrm{B}}T \approx 4.142 \times 10^{-21}\,\mathrm{J}$, $V_T = k_{\mathrm{B}}T/q_e \approx 25.85\,\mathrm{mV}$). By Proposition 11.6:

(a) The RMS thermal voltage on the capacitor is

$$\sigma_V = \sqrt{\frac{k_{\mathrm{B}}T}{C_{\mathrm{node}}}} = \sqrt{\frac{4.142\times10^{-21}\,\mathrm{J}}{1.0\times10^{-15}\,\mathrm{F}}} \approx 2.035\times10^{-3}\,\mathrm{V} = 2.035\,\mathrm{mV}. \tag{11.55}$$

(b) The RMS charge fluctuation $\sigma_Q = C_{\mathrm{node}}\sigma_V = \sqrt{k_{\mathrm{B}}TC_{\mathrm{node}}}$, in units of q_e, is

$$\frac{\sigma_Q}{q_e} = \frac{\sqrt{k_{\mathrm{B}}TC_{\mathrm{node}}}}{q_e} = \sqrt{\frac{C_{\mathrm{node}}V_T}{q_e}} \equiv \sqrt{\Omega}, \qquad \Omega := \frac{C_{\mathrm{node}}V_T}{q_e} \approx \frac{(10^{-15}\,\mathrm{F})(0.02585\,\mathrm{V})}{1.602\times10^{-19}\,\mathrm{C}} \approx 161.4, \tag{11.56}$$

so $\sigma_Q \approx 12.7\,q_e$. A voltage change of V_T corresponds to only $\Omega \approx 161$ electrons on a 1 fF capacitor, so thermal fluctuations are a sizable fraction, $\sigma_V/V_T = 1/\sqrt{\Omega} \approx 7.9\%$, of the subthreshold voltage scale V_T. At the same time $\Omega \gg 1$, which justifies the continuum (Kramers–Moyal) expansion of Section 13.2.

11.3 Shot Noise

Neither the Johnson–Nyquist formula $S_I(f) = 4k_{\mathrm{B}}TG$ (Theorem 11.5) nor the equipartition result $\langle \delta V^2 \rangle = k_{\mathrm{B}}T/C$ (Proposition 11.6) contains the electron charge q_e. Even if electricity were a continuous fluid, a resistor of conductance G in equilibrium would still show $S_I(f) = 4k_{\mathrm{B}}TG$.

Ten years before Johnson and Nyquist, Walter Schottky found a different kind of fluctuation in vacuum diodes [96]: **shot noise** (*Schroteffekt*, after the patter of lead shot falling on a metal plate). When electrons cross a high potential barrier, such as the cathode barrier of a vacuum tube, a *pn* junction, a tunnel junction, or the source-channel barrier of a subthreshold MOSFET (Chapter 12), each electron crosses independently and carries a discrete charge q_e.

11.3.1 Campbell's Theorem for Poisson Pulse Trains

Suppose electrons cross a barrier at random times $\cdots < t_{-1} < t_0 < t_1 < t_2 < \cdots$ that form a homogeneous Poisson process of rate λ (electrons per second). An electron crossing at time t_k induces a current pulse $i_0(t - t_k)$ in the external circuit. By the Shockley–Ramo theorem [89, 100], the current induced in the electrodes flows only while the charge is moving, so the pulse lasts for the transit time τ_{tr} of the barrier; and since the charge moves all the way from one electrode to the other, the pulse delivers one elementary charge:

$$\int_{-\infty}^{\infty} i_0(u)\,\mathrm{d}u = q_e. \tag{11.57}$$

The total current is the superposition of all pulses:

$$I(t) = \sum_{k} i_0(t - t_k). \tag{11.58}$$

Theorem 11.8 (Campbell's and Carson's theorems)**.** *Let $I(t) = \sum_k i_0(t - t_k)$ be a stationary Poisson pulse train of arrival rate $\lambda > 0$, where $i_0 \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ satisfies Eq. (11.57) and has Fourier transform $\hat{i}_0(\omega) = \int_{-\infty}^{\infty} i_0(u)e^{-i\omega u}\,\mathrm{d}u$. Then:*

(i) (Mean current.)

$$\langle I(t) \rangle = \lambda \int_{-\infty}^{\infty} i_0(u)\,\mathrm{d}u = q_e\lambda. \tag{11.59}$$

(ii) (Autocorrelation and variance.)

$$C_{II}(\tau) = \langle \delta I(t+\tau)\, \delta I(t) \rangle = \lambda \int_{-\infty}^{\infty} i_0(u)\, i_0(u+\tau)\, \mathrm{d}u, \qquad \mathrm{Var}(I) = \lambda \int_{-\infty}^{\infty} i_0(u)^2\, \mathrm{d}u. \tag{11.60}$$

(iii) (Carson's theorem.) The double-sided and single-sided current PSDs are

$$S_I^{\mathrm{ds}}(\omega) = \lambda \left| \hat{i}_0(\omega) \right|^2, \qquad S_I(f) = 2\lambda \left| \hat{i}_0(2\pi f) \right|^2. \tag{11.61}$$

Proof. Consider a large observation interval $[-\mathcal{T}/2, \mathcal{T}/2]$ of duration $\mathcal{T}$. For a homogeneous Poisson process of rate λ, the total number of arrivals K inside $[-\mathcal{T}/2, \mathcal{T}/2]$ obeys the Poisson distribution

$$\mathbb{P}(K = k) = \frac{(\lambda \mathcal{T})^k}{k!}\, e^{-\lambda \mathcal{T}}, \qquad k = 0, 1, 2, \ldots, \tag{11.62}$$

with $\mathbb{E}[K] = \lambda\mathcal{T}$ and $\mathbb{E}[K(K-1)] = (\lambda\mathcal{T})^2$. Conditioned on $\{K = k\}$, the k arrival times $(t_1, \ldots, t_k)$ are i.i.d. uniform on $[-\mathcal{T}/2, \mathcal{T}/2]$, with density $1/\mathcal{T}$.

Proof of (i). By the law of total expectation, the mean of the truncated pulse train $I_{\mathcal{T}}(t) = \sum_{j=1}^{K} i_0(t - t_j)$ is

$$\begin{aligned} \langle I_{\mathcal{T}}(t) \rangle &= \sum_{k=0}^{\infty} \mathbb{P}(K = k)\, \mathbb{E}\left[\sum_{j=1}^{k} i_0(t - t_j) \,\middle|\, K = k \right] \\ &= \sum_{k=0}^{\infty} \mathbb{P}(K = k)\, k \left(\frac{1}{\mathcal{T}} \int_{-\mathcal{T}/2}^{\mathcal{T}/2} i_0(t-u)\, \mathrm{d}u \right) = \frac{\mathbb{E}[K]}{\mathcal{T}} \int_{t-\mathcal{T}/2}^{t+\mathcal{T}/2} i_0(s)\, \mathrm{d}s. \end{aligned} \tag{11.63}$$

Substituting $\mathbb{E}[K] = \lambda\mathcal{T}$ and taking $\mathcal{T} \to \infty$ gives $\langle I(t) \rangle = \lambda \int_{-\infty}^{\infty} i_0(s)\, \mathrm{d}s = q_e \lambda$.

Proof of (ii) and (iii). Split the double sum $\sum_{j=1}^{K} \sum_{\ell=1}^{K}$ in $\langle I_{\mathcal{T}}(t+\tau)\, I_{\mathcal{T}}(t) \rangle$ into the K diagonal terms ($j = \ell$, same electron) and the $K(K-1)$ off-diagonal terms ($j \neq \ell$, distinct independent electrons):

$$\begin{aligned} \langle I_{\mathcal{T}}(t+\tau)\, I_{\mathcal{T}}(t) \rangle = \sum_{k=0}^{\infty} \mathbb{P}(K = k) \Big[& k\, \mathbb{E}[i_0(t+\tau-t_1)\, i_0(t-t_1)] \\ & + k(k-1)\, \mathbb{E}[i_0(t+\tau-t_1)]\, \mathbb{E}[i_0(t-t_2)] \Big] \\ = \frac{\mathbb{E}[K]}{\mathcal{T}} & \int_{-\mathcal{T}/2}^{\mathcal{T}/2} i_0(t+\tau-t_1)\, i_0(t-t_1)\, \mathrm{d}t_1 \\ + \frac{\mathbb{E}[K(K-1)]}{\mathcal{T}^2} & \left(\int_{-\mathcal{T}/2}^{\mathcal{T}/2} i_0(t+\tau-t_1)\, \mathrm{d}t_1 \right) \left(\int_{-\mathcal{T}/2}^{\mathcal{T}/2} i_0(t-t_2)\, \mathrm{d}t_2 \right). \end{aligned} \tag{11.64}$$

Now use the factorial moments of the Poisson distribution, $\mathbb{E}[K] = \lambda\mathcal{T}$ and $\mathbb{E}[K(K-1)] = \mathrm{Var}(K) + \mathbb{E}[K]^2 - \mathbb{E}[K] = \lambda\mathcal{T} + (\lambda\mathcal{T})^2 - \lambda\mathcal{T} = (\lambda\mathcal{T})^2$. Substituting these into Eq. (11.64), taking $\mathcal{T} \to \infty$, and setting $u = t - t_1$ in the first integral yields

$$\langle I(t+\tau)\, I(t) \rangle = \lambda \int_{-\infty}^{\infty} i_0(u+\tau)\, i_0(u)\, \mathrm{d}u + \langle I(t+\tau) \rangle \langle I(t) \rangle. \tag{11.65}$$

Subtracting $\langle I \rangle^2$ from both sides proves Eq. (11.60). Finally, Fourier transforming $C_{II}(\tau)$ by the Wiener–Khinchin theorem (Theorem 11.3) and changing variables to $s = u + \tau$ ($\mathrm{d}\tau = \mathrm{d}s$) gives

$$S_I^{\mathrm{ds}}(\omega) = \lambda \int_{-\infty}^{\infty} \mathrm{d}u\, i_0(u)\, e^{+i\omega u} \int_{-\infty}^{\infty} \mathrm{d}s\, i_0(s)\, e^{-i\omega s} = \lambda\, \hat{i}_0(\omega)^*\, \hat{i}_0(\omega) = \lambda \left| \hat{i}_0(\omega) \right|^2, \tag{11.66}$$

and $S_I(f) = 2\, S_I^{\mathrm{ds}}(2\pi f) = 2\lambda \left| \hat{i}_0(2\pi f) \right|^2$, completing the proof. □

11.3.2 Schottky's Formula in the Broadband Limit

In a nanoscale barrier or a subthreshold MOSFET channel of length $L \sim 20$–$100\,\mathrm{nm}$, the transit time is of order a picosecond or less ($\tau_{\mathrm{tr}} \sim 10^{-13}$–$10^{-12}\,\mathrm{s}$). For circuit frequencies well below $1/(2\pi\tau_{\mathrm{tr}})$, which is 100 GHz or more, the phase factor $e^{-i2\pi fu}$ is essentially 1 over the support of $i_0(u)$:

$$\hat{i}_0(2\pi f) = \int_{-\infty}^{\infty} i_0(u)\, e^{-i2\pi fu}\, \mathrm{d}u \underset{2\pi f\tau_{\mathrm{tr}} \ll 1}{\approx} \int_{-\infty}^{\infty} i_0(u)\, \mathrm{d}u = q_e, \tag{11.67}$$

which amounts to treating each crossing as an impulse $i_0(t - t_k) = q_e\, \delta(t - t_k)$. Substituting Eq. (11.67) and $\lambda = \langle I \rangle / q_e$ from Eq. (11.59) into Carson's theorem Eq. (11.61) gives Schottky's formula.

Corollary 11.9 (Schottky's formula [96])**.** *For unidirectional Poisson transport of charges q_e at mean current $\langle I \rangle = q_e\lambda$, in the broadband regime $2\pi f\tau_{\mathrm{tr}} \ll 1$, the current autocorrelation function and PSDs are*

$$C_{II}(\tau) = q_e \langle I \rangle\, \delta(\tau), \qquad S_I^{\mathrm{ds}}(\omega) = q_e \langle I \rangle, \qquad S_I(f) = 2q_e \langle I \rangle. \tag{11.68}$$

Physics Insight: Thermal noise versus shot noise

Johnson–Nyquist noise (Theorem 11.5) and Schottky shot noise (Corollary 11.9) depend on different constants:

$$S_I^{\mathrm{JN}}(f) = 4k_{\mathrm{B}}TG \quad \text{(equilibrium, no } q_e\text{)}, \qquad S_I^{\mathrm{shot}}(f) = 2q_e \langle I \rangle \quad \text{(unidirectional, no } T\text{)}. \tag{11.69}$$

The charge q_e enters S_I^{shot} because in a Poisson process of rate $\lambda = \langle I \rangle / q_e$, the number of events $N(\mathcal{T})$ in time $\mathcal{T}$ has variance equal to its mean: $\mathrm{Var}(N(\mathcal{T})) = \langle N(\mathcal{T}) \rangle = \lambda\mathcal{T}$. Since the total transferred charge is $Q(\mathcal{T}) = q_e N(\mathcal{T})$, its variance scales with q_e^2:

$$\mathrm{Var}\left(Q(\mathcal{T})\right) = q_e^2\, \mathrm{Var}\left(N(\mathcal{T})\right) = q_e^2(\lambda\mathcal{T}) = q_e \langle I \rangle\, \mathcal{T}. \tag{11.70}$$

If we subdivided the electron into m fractional charges $q_e' = q_e/m$ while keeping the macroscopic current $\langle I \rangle = q_e'\lambda'$ fixed ($\lambda' = m\lambda$), the shot noise $S_I^{\mathrm{shot}}(f) = 2q_e' \langle I \rangle = \frac{2q_e}{m} \langle I \rangle$ would vanish in the continuum limit $m \to \infty$. Shot noise is therefore a macroscopic measurement of charge quantization. Yet, as Section 11.4 shows, for a barrier in thermal equilibrium the same Poisson formula $2q_eI_{\pm}$ applied to the forward and reverse currents, combined with the Boltzmann factor $e^{-q_eV/k_{\mathrm{B}}T}$, reproduces $4k_{\mathrm{B}}TG$, with q_e cancelling at zero bias.

11.4 Bidirectional Barrier Transport

How does a conductor pass from Johnson–Nyquist noise $S_I(f) = 4k_{\mathrm{B}}TG$ at $V = 0$ to Schottky shot noise $S_I(f) = 2q_e \langle I \rangle$ at $q_eV \gg k_{\mathrm{B}}T$? A barrier crossed in both directions answers the question.

Consider two electron reservoirs, a source (S, left) and a drain (D, right), separated by a potential-energy barrier of height $E_{\mathrm{b}} \gg k_{\mathrm{B}}T$ (Fig. 11.2). As Chapter 12 shows, the same barrier picture governs transport in a subthreshold (weak-inversion) MOSFET, and it also describes thermionic diodes and biased pn junctions [29, 37, 93].

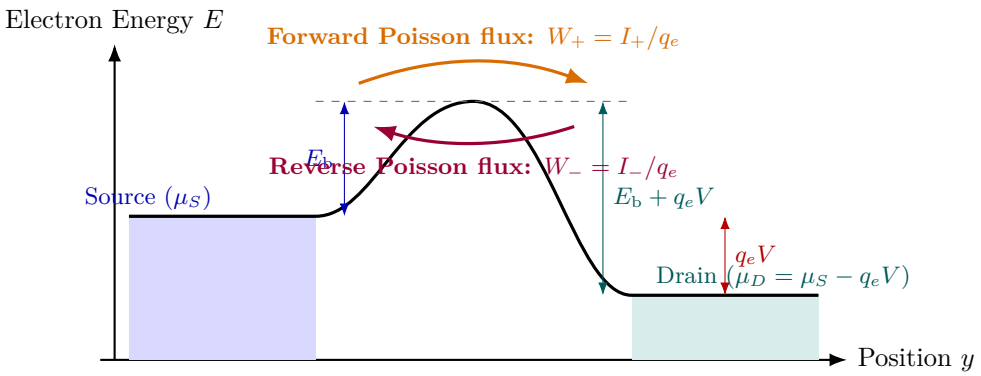


Figure 11.2: Bidirectional Poisson transport across a potential barrier separating a source reservoir at chemical potential μ_S from a drain reservoir at $\mu_D = \mu_S - q_eV$. Electrons thermally excited above the barrier from the source cross to the drain as a Poisson process of rate $W_+ = I_+/q_e \propto e^{-E_b/k_BT}$, while electrons from the drain cross back to the source as an independent Poisson process of rate $W_- = I_-/q_e \propto e^{-(E_b+q_eV)/k_BT}$.

11.4.1 The Master Equation

Because E_b is several times k_BT, only electrons in the high-energy Maxwell–Boltzmann tail of each reservoir's Fermi–Dirac distribution can cross. In this dilute tail the occupation of any single state above the barrier is small ($f(E) \ll 1$), so Pauli blocking ($1 - f(E) \approx 1$) is negligible and successive crossings are independent.

Transport across the barrier therefore consists of two independent, oppositely directed Poisson processes:

(i) a *forward* process $\{N_+(t) : t \geq 0\}$ of rate W_+ (electrons per second crossing from left to right), with mean forward current $I_+ := q_eW_+$;

(ii) a *reverse* process $\{N_-(t) : t \geq 0\}$ of rate W_- (electrons per second crossing from right to left), with mean reverse current $I_- := q_eW_-$.

Let $n(t) := N_+(t) - N_-(t) \in \mathbb{Z}$ be the net number of electrons transferred from left to right during $[0, t]$, with $n(0) = 0$. The net transferred charge is $Q(t) = q_en(t)$, and the current is $I(t) = \frac{dQ(t)}{dt}$.

Let $P_n(t) := \mathbb{P}(n(t) = n)$. During $[t, t + dt]$, state n is entered from $n - 1$ by a forward jump (probability $W_+ dt$) or from $n + 1$ by a reverse jump (probability $W_- dt$), and it is left by either jump (probability $(W_+ + W_-) dt$). This gives the master equation of bidirectional barrier transport:

$$\frac{dP_n(t)}{dt} = W_+P_{n-1}(t) + W_-P_{n+1}(t) - (W_+ + W_-)\,P_n(t), \qquad n \in \mathbb{Z}, \quad P_n(0) = \delta_{n,0}. \tag{11.71}$$

The mean current is proportional to the *difference* of the two rates, while the current noise is proportional to their *sum*.

Theorem 11.10 (Mean current and noise of bidirectional transport)**.** *For a barrier governed by the master equation Eq. (11.71) with independent forward and reverse Poisson jump rates $W_+ = I_+/q_e$ and $W_- = I_-/q_e$, in the broadband regime $2\pi f\tau_{tr} \ll 1$:*

(i) The mean current is the difference of the forward and reverse currents:

$$I \equiv \langle I(t)\rangle = q_e \frac{d\,\langle n(t)\rangle}{dt} = q_e(W_+ - W_-) = I_+ - I_-. \tag{11.72}$$

(ii) The variance of the transferred charge $Q(t) = q_e n(t)$ grows linearly in time at a rate set by the sum of the rates:

$$\operatorname{Var}\big(Q(t)\big) = q_e^2 \operatorname{Var}\big(n(t)\big) = q_e^2 (W_+ + W_-)\, t = q_e (I_+ + I_-)\, t. \tag{11.73}$$

(iii) The current PSDs are the sums of the forward and reverse shot-noise spectra:

$$S_I(f) = 2q_e^2 (W_+ + W_-) = 2q_e (I_+ + I_-), \qquad S_I^{\mathrm{ds}}(\omega) = q_e (I_+ + I_-). \tag{11.74}$$

Proof. We give two derivations, one from the master equation Eq. (11.71) and one from the superposition of the two pulse trains.

Method 1: Moments of the master equation. Multiply Eq. (11.71) by n and sum over $n \in \mathbb{Z}$ to obtain the evolution of the first moment $\langle n(t)\rangle = \sum_{n=-\infty}^{\infty} n P_n(t)$:

$$\frac{\mathrm{d}\,\langle n(t)\rangle}{\mathrm{d}t} = W_+ \sum_{n=-\infty}^{\infty} n P_{n-1}(t) + W_- \sum_{n=-\infty}^{\infty} n P_{n+1}(t) - (W_+ + W_-) \sum_{n=-\infty}^{\infty} n P_n(t). \tag{11.75}$$

Shifting the index to $m = n-1$ in the first sum and to $m = n+1$ in the second, and using $\sum_m P_m(t) = 1$, we find

$$\begin{aligned} \frac{\mathrm{d}\,\langle n(t)\rangle}{\mathrm{d}t} &= W_+ \sum_{m=-\infty}^{\infty} (m+1) P_m(t) + W_- \sum_{m=-\infty}^{\infty} (m-1) P_m(t) - (W_+ + W_-)\,\langle n(t)\rangle \\ &= W_+\big(\langle n(t)\rangle + 1\big) + W_-\big(\langle n(t)\rangle - 1\big) - (W_+ + W_-)\,\langle n(t)\rangle \\ &= W_+ - W_-. \end{aligned} \tag{11.76}$$

Multiplying by q_e gives $\langle I\rangle = q_e \frac{\mathrm{d}\langle n(t)\rangle}{\mathrm{d}t} = q_e(W_+ - W_-) = I_+ - I_-$, proving Eq. (11.72).

Next, multiply Eq. (11.71) by n^2 and sum with the same index shifts:

$$\begin{aligned} \frac{\mathrm{d}\,\langle n(t)^2\rangle}{\mathrm{d}t} &= W_+ \sum_{m=-\infty}^{\infty} (m+1)^2 P_m(t) + W_- \sum_{m=-\infty}^{\infty} (m-1)^2 P_m(t) - (W_+ + W_-)\left\langle n(t)^2\right\rangle \\ &= W_+\big(\left\langle n(t)^2\right\rangle + 2\,\langle n(t)\rangle + 1\big) + W_-\big(\left\langle n(t)^2\right\rangle - 2\,\langle n(t)\rangle + 1\big) - (W_+ + W_-)\left\langle n(t)^2\right\rangle \\ &= 2(W_+ - W_-)\,\langle n(t)\rangle + (W_+ + W_-). \end{aligned} \tag{11.77}$$

Using $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{Var}(n(t)) = \frac{\mathrm{d}}{\mathrm{d}t}\big(\langle n(t)^2\rangle - \langle n(t)\rangle^2\big) = \frac{\mathrm{d}\langle n(t)^2\rangle}{\mathrm{d}t} - 2\,\langle n(t)\rangle \frac{\mathrm{d}\langle n(t)\rangle}{\mathrm{d}t}$ and substituting Eq. (11.76) and Eq. (11.77), the $2(W_+ - W_-)\,\langle n(t)\rangle$ terms cancel:

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{Var}\big(n(t)\big) = 2(W_+ - W_-)\,\langle n(t)\rangle + (W_+ + W_-) - 2(W_+ - W_-)\,\langle n(t)\rangle = W_+ + W_-. \tag{11.78}$$

Integrating Eq. (11.78) from 0 to t with $\operatorname{Var}(n(0)) = 0$ and multiplying by q_e^2 proves Eq. (11.73).

Method 2: Autocorrelation of the pulse trains. Write the current as $I(t) = I_+(t) - I_-(t)$, where $I_+(t) = \sum_k q_e \delta(t - t_k^+)$ and $I_-(t) = \sum_\ell q_e \delta(t - t_\ell^-)$ are independent Poisson impulse trains of rates W_+ and W_-. By independence, $\langle \delta I_+(t+\tau)\, \delta I_-(t)\rangle = 0$, so the autocorrelation of the difference is the *sum* of the individual autocorrelations:

$$\begin{aligned} C_{II}(\tau) &= \big\langle \big(\delta I_+(t+\tau) - \delta I_-(t+\tau)\big)\big(\delta I_+(t) - \delta I_-(t)\big)\big\rangle \\ &= \langle \delta I_+(t+\tau)\, \delta I_+(t)\rangle + \langle \delta I_-(t+\tau)\, \delta I_-(t)\rangle \\ &= (q_e I_+ + q_e I_-)\, \delta(\tau) = q_e^2 (W_+ + W_-)\, \delta(\tau), \end{aligned} \tag{11.79}$$

where we used Eq. (11.68) for each stream. Fourier transforming with Theorem 11.3 gives $S_I^{\mathrm{ds}}(\omega) = q_e(I_+ + I_-)$ and $S_I(f) = 2q_e(I_+ + I_-)$, proving Eq. (11.74). □

11.4.2 Local Detailed Balance and the coth Crossover Formula

To relate I_+ and I_- to the voltage V and temperature T, consider the energetics of crossing the barrier of Fig. 11.2. An electron in the source must have energy at least E_b above the source Fermi level μ_S to pass the top of the barrier, so the forward rate carries an Arrhenius factor:

$$W_+ = \nu_0 \exp\!\left(-\frac{E_\mathrm{b}}{k_\mathrm{B}T}\right) \qquad \Longrightarrow \qquad I_+ = q_e W_+ = I_0, \tag{11.80}$$

where ν_0 is an attempt frequency set by the density of states and thermal velocity in the channel, and $I_0 := q_e \nu_0 e^{-E_\mathrm{b}/k_\mathrm{B}T}$ is the equilibrium exchange current. A positive voltage $V > 0$ lowers the drain Fermi level to $\mu_D = \mu_S - q_e V$, so an electron in the drain faces a barrier $E_\mathrm{b} + q_e V$:

$$W_- = \nu_0 \exp\!\left(-\frac{E_\mathrm{b} + q_e V}{k_\mathrm{B}T}\right) \qquad \Longrightarrow \qquad I_- = q_e W_- = I_0 \exp\!\left(-\frac{q_e V}{k_\mathrm{B}T}\right). \tag{11.81}$$

The ratio of Eq. (11.80) and Eq. (11.81) is the local detailed balance condition used for jump processes in Chapter 9:

$$\frac{W_+}{W_-} = \frac{I_+}{I_-} = \exp\!\left(\frac{q_e V}{k_\mathrm{B}T}\right) = \exp\!\left(\frac{V}{V_T}\right), \tag{11.82}$$

The exponent $q_e V / k_\mathrm{B}T$ equals $\Delta s_\mathrm{med}/k_\mathrm{B}$, where $\Delta s_\mathrm{med} = q_e V / T$ is the medium entropy produced in the reservoirs when one electron falls through the potential difference V.

Combining Theorem 11.10 with Eq. (11.82) gives a single formula that covers thermal noise, shot noise, and the crossover between them.

Theorem 11.11 (Thermal-to-shot noise crossover [13, 37, 93])**.** *Let a barrier conductor obey the bidirectional Poisson master equation Eq. (11.71) with forward and reverse currents satisfying the local detailed balance relation $I_+/I_- = \exp(q_e V/k_\mathrm{B}T)$. Then:*

(i) For any bias $V \neq 0$, the single-sided current PSD is related to the net current $I = I_+ - I_-$ by

$$S_I(f) = 2 q_e I \coth\!\left(\frac{q_e V}{2 k_\mathrm{B}T}\right) = 2 q_e I \coth\!\left(\frac{V}{2V_T}\right). \tag{11.83}$$

(ii) (Equilibrium limit, $|V| \ll V_T$.) Define the zero-bias conductance $G_0 := \left.\frac{\mathrm{d}I}{\mathrm{d}V}\right|_{V=0} = \frac{q_e I_0}{k_\mathrm{B}T} = \frac{I_0}{V_T}$. As $V \to 0$, Eq. (11.83) reduces to the Johnson–Nyquist formula:

$$\lim_{V \to 0} S_I(f) = 4 k_\mathrm{B} T G_0 = 4 q_e I_0. \tag{11.84}$$

(iii) (Shot-noise limit, $|V| \gg V_T$.) For $q_e |V| \gg 2 k_\mathrm{B}T$, the reverse flux is exponentially suppressed ($I_-/I_+ = e^{-q_e V/k_\mathrm{B}T} \ll 1$) and Eq. (11.83) reduces to Schottky's shot-noise formula:

$$S_I(f) = 2 q_e |I| \left(1 + 2e^{-q_e|V|/k_\mathrm{B}T} + \mathcal{O}\!\left(e^{-2q_e|V|/k_\mathrm{B}T}\right)\right) \underset{|V| \gg V_T}{\longrightarrow} 2 q_e |I|. \tag{11.85}$$

Proof. By Eq. (11.82), $I_- = I_+ e^{-q_e V/k_\mathrm{B}T}$. Using Theorem 11.10, express I and $S_I(f)$ in terms of I_+ and $x := \frac{q_e V}{2 k_\mathrm{B}T} = \frac{V}{2V_T}$:

$$I = I_+ - I_- = I_+(1 - e^{-2x}) = I_+ e^{-x}(e^x - e^{-x}) = 2 I_+ e^{-x} \sinh(x), \tag{11.86}$$

$$S_I(f) = 2 q_e (I_+ + I_-) = 2 q_e I_+ (1 + e^{-2x}) = 2 q_e I_+ e^{-x}(e^x + e^{-x}) = 4 q_e I_+ e^{-x} \cosh(x). \tag{11.87}$$

Dividing Eq. (11.87) by Eq. (11.86) eliminates the prefactor I_+e^{-x}:

$$\frac{S_I(f)}{I} = \frac{4q_eI_+e^{-x}\cosh(x)}{2I_+e^{-x}\sinh(x)} = 2q_e\frac{\cosh(x)}{\sinh(x)} = 2q_e\coth(x) = 2q_e\coth\left(\frac{q_eV}{2k_BT}\right), \tag{11.88}$$

which proves Eq. (11.83).

For the equilibrium limit Eq. (11.84), first compute G_0 from $I(V) = I_0(1 - e^{-q_eV/k_BT})$:

$$G_0 = \left.\frac{dI}{dV}\right|_{V=0} = I_0\left(\frac{q_e}{k_BT}\right)e^{-q_eV/k_BT}\Big|_{V=0} = \frac{q_eI_0}{k_BT}. \tag{11.89}$$

For $|x| = \left|\frac{q_eV}{2k_BT}\right| \ll 1$, we expand $I(V) = G_0V + \mathcal{O}(V^2)$ and use the Laurent expansion $\coth(x) = \frac{1}{x} + \frac{x}{3} - \frac{x^3}{45} + \mathcal{O}(x^5)$:

$$\begin{aligned} S_I(f) &= 2q_e(G_0V + \mathcal{O}(V^2))\left(\frac{2k_BT}{q_eV} + \frac{q_eV}{6k_BT} + \mathcal{O}(V^3)\right) \\ &= 4k_BTG_0\left(1 + \mathcal{O}\left(\frac{q_eV}{k_BT}\right)\right) \xrightarrow[V\to 0]{} 4k_BTG_0 = 4q_eI_0, \end{aligned} \tag{11.90}$$

where the last equality uses Eq. (11.89). The charge q_e in the shot-noise prefactor $2q_e$ cancels the $1/q_e$ in $2k_BT/(q_eV)$, leaving $4k_BTG_0$.

Finally, for $x = \frac{q_eV}{2k_BT} \gg 1$, we write $\coth(x) = \frac{1+e^{-2x}}{1-e^{-2x}} = (1 + e^{-2x})(1 + e^{-2x} + e^{-4x} + \cdots) = 1 + 2e^{-2x} + \mathcal{O}(e^{-4x})$, which yields Eq. (11.85). □

Heuristic Derivation: Thermal noise as two shot noises

Theorem 11.11 gives a simple picture of thermal noise in barrier devices [93]. At $V = 0$ the conductor is not quiet: a forward current $I_+ = I_0$ of thermally excited electrons crosses the barrier from left to right, and an equal reverse current $I_- = I_0$ crosses from right to left.

- For the *mean current*, the two opposing streams cancel: $I = I_+ - I_- = I_0 - I_0 = 0$.
- For the *fluctuations*, forward and reverse jumps are independent Poisson events, so their variances *add*:

$$S_I(f) = 2q_eI_+ + 2q_eI_- = 2q_eI_0 + 2q_eI_0 = 4q_eI_0. \tag{11.91}$$

A small probe voltage δV reduces the reverse current to $I_0e^{-q_e\delta V/k_BT} \approx I_0(1 - q_e\delta V/k_BT)$, giving a net current $\delta I = (q_eI_0/k_BT)\,\delta V$. The conductance is therefore $G_0 = q_eI_0/k_BT$, that is, $I_0 = k_BTG_0/q_e$. Substituting into Eq. (11.91) gives $S_I(f) = 4k_BTG_0$.

Johnson–Nyquist noise $4k_BTG_0$ is thus the sum of the shot noises $2q_eI_0 + 2q_eI_0$ of the two equilibrium exchange currents.

Hardware Realization: Bidirectional shot noise in subthreshold CMOS

Thermodynamic sampling circuits built from CMOS, such as the non-equilibrium analog thermodynamic random networks (NEAT-RNs) of Chapter 13 [38] and the denoising architecture of Chapter 17 [56], bias their transistors in the subthreshold (weak-inversion) regime. Representative drain currents are of order 0.1 to tens of nanoamperes. Section 12.5 shows

that the channel noise of such a transistor is the bidirectional shot noise of Theorem 11.10. A designer therefore has two operating modes:

1. **Resistive mode** ($V_{\mathrm{DS}} \ll V_T \approx 26\,\mathrm{mV}$). Forward and reverse currents are both active ($I_+ \approx I_-$). The device is a reciprocal conductance $G_0 = I_0/V_T$, tuned exponentially by its gate voltage, and its noise obeys the equilibrium Johnson–Nyquist relation $S_I(f) = 4k_{\mathrm{B}}TG_0$, consistent with detailed balance (Chapter 6).

2. **Saturation mode** ($V_{\mathrm{DS}} \gtrsim 4V_T \approx 100\,\mathrm{mV}$). The reverse current is suppressed by $e^{-V_{\mathrm{DS}}/V_T} < 0.02$, so $I_- \approx 0$ and $I_{\mathrm{D}} \approx I_+$. The transistor is a unidirectional current source with shot noise $S_I(f) = 2q_eI_{\mathrm{D}}$, and its gate transconductance is $g_{\mathrm{m}} = \frac{\partial I_{\mathrm{D}}}{\partial V_{\mathrm{GS}}} = q_eI_{\mathrm{D}}/(nk_{\mathrm{B}}T)$ (Chapter 12). The ratio of noise to transconductance is then independent of the bias current:
$$\frac{S_I(f)}{g_{\mathrm{m}}} = \frac{2q_eI_{\mathrm{D}}}{q_eI_{\mathrm{D}}/(nk_{\mathrm{B}}T)} = 2nk_{\mathrm{B}}T, \tag{11.92}$$
where $n \approx 1.2$–1.5 is the subthreshold slope factor. Because of Eq. (11.92), a NEAT-RN node whose conductance is set by g_{m} has an effective noise temperature proportional to nT that does not change as the bias current is varied over several decades (Chapter 13).

11.5 Random Telegraph Noise and $1/f$ Flicker Noise

Thermal noise (Section 11.2) and shot noise (Sections 11.3 and 11.4) both have white spectra, flat from $f = 0$ up to the inverse transit time $1/(2\pi\tau_{\mathrm{tr}}) \gtrsim 100\,\mathrm{GHz}$. Real devices also show excess noise at low frequencies, with a spectrum $S_I^{\mathrm{flicker}}(f) \propto 1/f^{\alpha}$ and $\alpha \approx 1$.

This low-frequency noise matters for thermodynamic computing because the Markovian Langevin and Fokker–Planck dynamics of Chapters 4 to 6 assume memoryless white noise. Low-frequency noise instead has long-lived correlations and acts as a slow, non-Markovian drift of the circuit's energy parameters $\theta(t)$. We first derive the Lorentzian spectrum of a single oxide trap, which produces **random telegraph noise** (RTN), and then show how traps spread through the depth of the gate oxide add up to $1/f$ **(flicker) noise** through McWhorter's tunneling mechanism.

11.5.1 A Single Oxide Trap: The Lorentzian RTN Spectrum

Near the interface between a crystalline silicon channel and an amorphous gate dielectric (SiO_2 or HfO_2), defects and dangling bonds create localized trap states inside the oxide, at distance z from the interface. Consider a single trap with occupation $s(t) \in \{0, 1\}$, where $s = 0$ means empty and $s = 1$ means occupied by an electron captured from the channel.

When the trap captures an electron ($0 \to 1$), its negative charge screens a small patch of the channel and shifts the local threshold voltage V_{th} by ΔV_{th}, so the drain current steps down by ΔI. When the trap emits the electron ($1 \to 0$), the current steps back up. The trap contributes $I_{\mathrm{trap}}(t) = -\Delta I\, s(t)$.

Let $\gamma_c := 1/\tau_c$ be the capture rate ($0 \to 1$; τ_c is the mean empty time) and $\gamma_e := 1/\tau_e$ the emission rate ($1 \to 0$; τ_e is the mean occupied time). The probabilities $P_1(t) = \mathbb{P}(s(t) = 1)$ and $P_0(t) = 1 - P_1(t)$ obey the two-state master equation
$$\frac{\mathrm{d}P_1(t)}{\mathrm{d}t} = \gamma_cP_0(t) - \gamma_eP_1(t) = \frac{1}{\tau_c}(1 - P_1(t)) - \frac{1}{\tau_e}P_1(t) = \frac{1}{\tau_c} - \left(\frac{1}{\tau_c} + \frac{1}{\tau_e}\right)P_1(t). \tag{11.93}$$

Proposition 11.12 (Spectrum of a single trap [69])**.** *Let $s(t) \in \{0,1\}$ be a stationary two-state Markov process governed by Eq. (11.93) with mean capture time τ_c, mean emission time τ_e, and current step amplitude ΔI. Define the relaxation time τ_{eff} by*

$$\frac{1}{\tau_{\mathrm{eff}}} := \frac{1}{\tau_c} + \frac{1}{\tau_e} \qquad \Longleftrightarrow \qquad \tau_{\mathrm{eff}} = \frac{\tau_c \tau_e}{\tau_c + \tau_e}. \tag{11.94}$$

Then:

(i) The stationary occupation probabilities and current variance are

$$\begin{aligned} p_1 := \lim_{t\to\infty} P_1(t) = \frac{\tau_e}{\tau_c+\tau_e}, \qquad p_0 = 1 - p_1 = \frac{\tau_c}{\tau_c+\tau_e}, \\ \mathrm{Var}(I_{\mathrm{trap}}) = (\Delta I)^2 p_1 p_0 = (\Delta I)^2 \frac{\tau_c \tau_e}{(\tau_c+\tau_e)^2}. \end{aligned} \tag{11.95}$$

(ii) The current autocorrelation function decays exponentially with time constant τ_{eff}:

$$C_{\mathrm{RTN}}(\tau) = \langle \delta I_{\mathrm{trap}}(t+\tau)\, \delta I_{\mathrm{trap}}(t) \rangle = (\Delta I)^2 \frac{\tau_c \tau_e}{(\tau_c+\tau_e)^2} \exp\!\left(-\frac{|\tau|}{\tau_{\mathrm{eff}}}\right). \tag{11.96}$$

(iii) The single-sided power spectral density is a Lorentzian with corner frequency $f_{\mathrm{RTN}} = 1/(2\pi\tau_{\mathrm{eff}})$:

$$S_I^{\mathrm{RTN}}(f) = 4(\Delta I)^2 \frac{\tau_c \tau_e}{(\tau_c+\tau_e)^2} \frac{\tau_{\mathrm{eff}}}{1 + (2\pi f \tau_{\mathrm{eff}})^2} = \frac{4(\Delta I)^2}{(\tau_c+\tau_e)\left[\tau_{\mathrm{eff}}^{-2} + (2\pi f)^2\right]}. \tag{11.97}$$

Proof. Setting $\frac{\mathrm{d}P_1}{\mathrm{d}t} = 0$ in Eq. (11.93) gives $\frac{1}{\tau_c} = \frac{1}{\tau_{\mathrm{eff}}} p_1$, so $p_1 = \frac{\tau_{\mathrm{eff}}}{\tau_c} = \frac{\tau_e}{\tau_c+\tau_e}$ and $p_0 = 1 - p_1 = \frac{\tau_c}{\tau_c+\tau_e}$. Because $s(t)^2 = s(t)$, its variance is $\mathrm{Var}(s) = \langle s^2 \rangle - \langle s \rangle^2 = p_1 - p_1^2 = p_1(1-p_1) = p_1 p_0$, which multiplies $(\Delta I)^2$ to give Eq. (11.95).

For $\tau \geq 0$, the master equation Eq. (11.93) can be written in terms of p_1 and τ_{eff} as

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\big(P_1(\tau) - p_1\big) = -\frac{1}{\tau_{\mathrm{eff}}}\big(P_1(\tau) - p_1\big). \tag{11.98}$$

Solving Eq. (11.98) with the conditional initial condition $P_{1|1}(0) = \mathbb{P}(s(t)=1 \mid s(t)=1) = 1$ gives the conditional transition probability

$$P_{1|1}(\tau) := \mathbb{P}\big(s(t+\tau)=1 \,\big|\, s(t)=1\big) = p_1 + (1-p_1)\, e^{-\tau/\tau_{\mathrm{eff}}} = p_1 + p_0\, e^{-\tau/\tau_{\mathrm{eff}}}, \qquad \tau \geq 0. \tag{11.99}$$

The product $s(t+\tau)s(t)$ is non-zero only if $s(t)=1$ and $s(t+\tau)=1$. Hence, for any $\tau \in \mathbb{R}$ (using even parity for $\tau < 0$),

$$\langle s(t+\tau)\, s(t) \rangle = \mathbb{P}\big(s(t)=1\big)\, P_{1|1}(|\tau|) = p_1\big(p_1 + p_0\, e^{-|\tau|/\tau_{\mathrm{eff}}}\big) = p_1^2 + p_1 p_0\, e^{-|\tau|/\tau_{\mathrm{eff}}}. \tag{11.100}$$

Subtracting $\langle s \rangle^2 = p_1^2$ and multiplying by $(\Delta I)^2$ yields the autocorrelation function Eq. (11.96).

Finally, apply the single-sided Wiener–Khinchin formula Eq. (11.16) to Eq. (11.96):

$$\begin{aligned} S_I^{\mathrm{RTN}}(f) &= 4(\Delta I)^2 p_1 p_0 \int_0^\infty e^{-\tau/\tau_{\mathrm{eff}}} \cos(2\pi f \tau)\, \mathrm{d}\tau \\ &= 4(\Delta I)^2 p_1 p_0\, \mathrm{Re}\!\left[\int_0^\infty \exp\!\left(-\left(\frac{1}{\tau_{\mathrm{eff}}} - i2\pi f\right)\tau\right) \mathrm{d}\tau\right] \\ &= 4(\Delta I)^2 p_1 p_0\, \mathrm{Re}\!\left[\frac{1}{\tau_{\mathrm{eff}}^{-1} - i2\pi f}\right] = 4(\Delta I)^2 p_1 p_0 \frac{\tau_{\mathrm{eff}}}{1 + (2\pi f \tau_{\mathrm{eff}})^2}, \end{aligned} \tag{11.101}$$

which establishes Eq. (11.97). □

11.5.2 McWhorter's Model of $1/f$ Noise

A single trap gives a Lorentzian Eq. (11.97) that is flat for $f \ll f_{\mathrm{RTN}}$ and falls as $1/f^2$ for $f \gg f_{\mathrm{RTN}}$. Yet transistors show a $1/f$ spectrum over six to ten decades of frequency.

McWhorter's explanation [74] is that channel electrons reach the traps by tunneling, and the traps are spread over the depth $z \in [0, t_{\mathrm{ox}}]$ of the gate oxide of thickness t_{ox}. The wave function of a channel electron decays exponentially into the oxide band gap with decay constant $\kappa_{\mathrm{ox}} = \sqrt{2m^*_{\mathrm{ox}}\Phi_{\mathrm{B}}}/\hbar \approx 6.4\,\mathrm{nm}^{-1}$ (where $\Phi_{\mathrm{B}} \approx 3.1\,\mathrm{eV}$ is the $\mathrm{Si/SiO_2}$ barrier height and we take an oxide effective mass $m^*_{\mathrm{ox}} \approx 0.5\,m_e$). In the WKB approximation the tunneling probability to a trap at depth z is proportional to $\exp(-2\kappa_{\mathrm{ox}} z)$, so the trap relaxation time grows exponentially with depth:

$$\tau(z) = \tau_0 \exp(2\kappa_{\mathrm{ox}} z), \qquad z \in [z_{\min}, z_{\max}], \tag{11.102}$$

where $\tau_0 \sim 10^{-10}\,\mathrm{s}$ is the capture time at the interface, $z = 0$. This map is very sensitive. As z varies over just $3\,\mathrm{nm}$ (from $z_{\min} = 0$ to $z_{\max} = 3\,\mathrm{nm}$), the exponent $2\kappa_{\mathrm{ox}} z$ runs from 0 to about 38, and $\tau(z)$ spans a factor $e^{38} \approx 3 \times 10^{16}$, about sixteen decades, from $10^{-10}\,\mathrm{s}$ to roughly $10^{6.5}\,\mathrm{s}$.

Suppose the traps are distributed uniformly in depth over $[z_{\min}, z_{\max}]$, with density $p(z) = 1/(z_{\max} - z_{\min})$. Changing variables through the monotonic map Eq. (11.102), the density $g(\tau)$ of relaxation times $\tau \in [\tau_{\min}, \tau_{\max}]$ (where $\tau_{\min} = \tau_0 e^{2\kappa_{\mathrm{ox}} z_{\min}}$ and $\tau_{\max} = \tau_0 e^{2\kappa_{\mathrm{ox}} z_{\max}}$) is

$$g(\tau) = p(z(\tau)) \left|\frac{\mathrm{d}z}{\mathrm{d}\tau}\right| = \frac{1}{z_{\max} - z_{\min}} \left(\frac{1}{2\kappa_{\mathrm{ox}}\tau}\right) = \frac{1}{\ln(\tau_{\max}/\tau_{\min})}\,\frac{1}{\tau}, \qquad \tau \in [\tau_{\min}, \tau_{\max}]. \tag{11.103}$$

Theorem 11.13 (McWhorter–Surdin $1/f$ spectrum [74, 105])**.** *Suppose a conductor contains N_{tr} independent oxide traps whose relaxation times $\tau \in [\tau_{\min}, \tau_{\max}]$ are distributed according to the log-uniform law $g(\tau) = \frac{1}{\ln(\tau_{\max}/\tau_{\min})}\frac{1}{\tau}$ of Eq. (11.103), each contributing a Lorentzian PSD Eq. (11.97) with mean-square current fluctuation $\overline{\sigma^2_{\mathrm{trap}}} = (\Delta I)^2 \overline{p_1 p_0}$. Then the total PSD is*

$$S_I^{1/f}(f) = \frac{2N_{\mathrm{tr}}\,\overline{\sigma^2_{\mathrm{trap}}}}{\pi \ln(\tau_{\max}/\tau_{\min})}\,\frac{\arctan(2\pi f\tau_{\max}) - \arctan(2\pi f\tau_{\min})}{f}. \tag{11.104}$$

In the band $\frac{1}{2\pi\tau_{\max}} \ll f \ll \frac{1}{2\pi\tau_{\min}}$, which spans many decades, Eq. (11.104) reduces to a $1/f$ spectrum:

$$S_I^{1/f}(f) \approx \frac{N_{\mathrm{tr}}\,\overline{\sigma^2_{\mathrm{trap}}}}{\ln(\tau_{\max}/\tau_{\min})}\,\frac{1}{f}. \tag{11.105}$$

Proof. The N_{tr} traps fluctuate independently, so their PSDs add. Averaging the single-trap Lorentzian Eq. (11.97) over the distribution $g(\tau)$ from Eq. (11.103) gives

$$\begin{aligned} S_I^{1/f}(f) &= N_{\mathrm{tr}} \int_{\tau_{\min}}^{\tau_{\max}} g(\tau) \left(\frac{4\overline{\sigma^2_{\mathrm{trap}}}\,\tau}{1 + (2\pi f\tau)^2}\right) \mathrm{d}\tau \\ &= \frac{4N_{\mathrm{tr}}\,\overline{\sigma^2_{\mathrm{trap}}}}{\ln(\tau_{\max}/\tau_{\min})} \int_{\tau_{\min}}^{\tau_{\max}} \frac{1}{\tau}\,\frac{\tau}{1 + (2\pi f\tau)^2}\,\mathrm{d}\tau. \end{aligned} \tag{11.106}$$

The factor $1/\tau$ from $g(\tau)$ cancels the τ in the numerator of the Lorentzian. The substitution $u = 2\pi f\tau$ then pulls out a factor $1/(2\pi f)$:

$$S_I^{1/f}(f) = \frac{4N_{\mathrm{tr}}\,\overline{\sigma^2_{\mathrm{trap}}}}{\ln(\tau_{\max}/\tau_{\min})} \left(\frac{1}{2\pi f}\right) \int_{2\pi f\tau_{\min}}^{2\pi f\tau_{\max}} \frac{\mathrm{d}u}{1 + u^2}$$

$$= \frac{2N_{\mathrm{tr}}\,\overline{\sigma^2_{\mathrm{trap}}}}{\pi \ln(\tau_{\max}/\tau_{\min})} \frac{\arctan(2\pi f\tau_{\max}) - \arctan(2\pi f\tau_{\min})}{f}, \tag{11.107}$$

proving Eq. (11.104). When $\frac{1}{2\pi\tau_{\max}} \ll f \ll \frac{1}{2\pi\tau_{\min}}$, we have $2\pi f\tau_{\max} \gg 1$ so $\arctan(2\pi f\tau_{\max}) \approx \frac{\pi}{2}$, and $2\pi f\tau_{\min} \ll 1$ so $\arctan(2\pi f\tau_{\min}) \approx 0$. Substituting $\pi/2$ for the bracket in Eq. (11.107) gives Eq. (11.105). □

11.5.3 The Corner Frequency and Timescale Separation

In a MOSFET of gate width W and channel length L, the number of active oxide traps scales with the gate area WL, while the threshold shift produced by one trapped electron, $\Delta V_{\mathrm{th}} = q_e/(WLC_{\mathrm{ox}})$, scales inversely with the total gate capacitance WLC_{ox} (here C_{ox} is the oxide capacitance per unit area). The summed trap fluctuations act as a gate-referred voltage noise whose PSD scales as $(WL)(\Delta V_{\mathrm{th}})^2 \propto 1/(WLC_{\mathrm{ox}}^2)$; it is conventional to absorb one factor of C_{ox} and the trap density into a process constant and write

$$S_V^{1/f}(f) = \frac{K_g}{C_{\mathrm{ox}} WL\, f}, \tag{11.108}$$

where K_g (units $\mathrm{V^2\,F}$) is of order $10^{-25}\,\mathrm{V^2\,F}$ for n-channel devices. The gate voltage noise reaches the drain through the transconductance $g_{\mathrm{m}} = I_{\mathrm{D}}/(nV_T)$ of a weakly inverted device (Chapter 12). Adding this $1/f$ contribution to the white spectrum Eq. (11.83) gives the total drain-current PSD

$$S_I^{\mathrm{tot}}(f) = S_I^{\mathrm{white}} + \frac{g_{\mathrm{m}}^2 K_g}{C_{\mathrm{ox}} WL\, f} = S_I^{\mathrm{white}} + \frac{K_g\, I_{\mathrm{D}}^2}{n^2 V_T^2\, C_{\mathrm{ox}} WL\, f} = S_I^{\mathrm{white}} \left(1 + \frac{f_c}{f}\right), \tag{11.109}$$

where the **corner frequency** f_c is the frequency at which the $1/f$ part equals the white part. Using $S_I^{\mathrm{white}} = 2q_e I_{\mathrm{D}} \coth(V_{\mathrm{DS}}/2V_T)$,

$$f_c := \frac{K_g\, I_{\mathrm{D}}^2}{n^2 V_T^2\, C_{\mathrm{ox}} WL\, S_I^{\mathrm{white}}} = \frac{K_g\, I_{\mathrm{D}}}{2q_e n^2 V_T^2\, C_{\mathrm{ox}} WL \coth(V_{\mathrm{DS}}/2V_T)}. \tag{11.110}$$

In weak inversion $S_I^{\mathrm{white}} \propto I_{\mathrm{D}}$, so the corner frequency is proportional to the bias current I_{D} and inversely proportional to the gate area. For $K_g = 10^{-25}\,\mathrm{V^2\,F}$, $n = 1.3$, and saturation ($\coth \to 1$), a $1\,\mu\mathrm{m}^2$ device ($C_{\mathrm{ox}}WL \approx 10\,\mathrm{fF}$) at $I_{\mathrm{D}} = 1\,\mathrm{nA}$ has $f_c \approx 30\,\mathrm{Hz}$, while a minimum-size $0.01\,\mu\mathrm{m}^2$ device at $10\,\mathrm{nA}$ has $f_c \approx 30\,\mathrm{kHz}$. The relaxation bandwidth of a femtofarad RC node is $f_{RC} = 1/(2\pi\tau_{RC})$, in the megahertz range or above for these currents. Because $f_{RC}/f_c \gtrsim 10^3$, almost all of the fluctuation power within a sampling window is white, Markovian noise. The slow $1/f$ drift below f_c acts like a slowly varying parameter offset, which the hardware-in-the-loop calibration of Section 18.5 can remove if it is repeated periodically.

11.6 Worked Computation: Full Counting Statistics and Fano Factor of a Biased Barrier

Section 11.4 gave the first two moments of the charge transferred across a biased barrier, the mean current I and the noise S_I. A jump process is not Gaussian, and its full distribution is described by all of its cumulants, known together as the **full counting statistics** [13, 37]. We now compute the cumulant generating function of a biased barrier, all of its cumulants, the Fano factor, and the fluctuation symmetry of the distribution.

Worked computation 11.14 (Full counting statistics and Fano factor of a biased barrier)**.** Consider a barrier (for example a subthreshold MOSFET channel) governed by the bidirectional Poisson master equation Eq. (11.71):

$$\frac{\mathrm{d}P_n(t)}{\mathrm{d}t} = W_+ P_{n-1}(t) + W_- P_{n+1}(t) - (W_+ + W_-)\, P_n(t), \qquad n \in \mathbb{Z}, \quad P_n(0) = \delta_{n,0}, \tag{11.111}$$

where the rates satisfy local detailed balance, $W_+/W_- = \exp(q_e V/k_{\mathrm{B}}T) = \exp(V/V_T)$, with $W_+ = I_0/q_e$ and $W_- = (I_0/q_e)e^{-V/V_T}$.

Step 1: The generating function. Introduce a counting field $\chi \in [-\pi, \pi]$ conjugate to the count $n(t)$, and define the characteristic function $M(\chi, t)$ and the cumulant generating function (CGF) $K(\chi, t)$:

$$M(\chi, t) := \left\langle e^{i\chi n(t)} \right\rangle = \sum_{n=-\infty}^{\infty} e^{i\chi n}\, P_n(t), \qquad K(\chi, t) := \ln M(\chi, t). \tag{11.112}$$

Differentiating $M(\chi, t)$ with respect to t and substituting the master equation Eq. (11.111) gives

$$\begin{aligned} \frac{\partial M(\chi,t)}{\partial t} &= \sum_{n=-\infty}^{\infty} e^{i\chi n}\Big[W_+ P_{n-1}(t) + W_- P_{n+1}(t) - (W_+ + W_-)\, P_n(t)\Big] \\ &= W_+ e^{+i\chi} \sum_{n=-\infty}^{\infty} e^{i\chi(n-1)} P_{n-1}(t) + W_- e^{-i\chi} \sum_{n=-\infty}^{\infty} e^{i\chi(n+1)} P_{n+1}(t) - (W_+ + W_-) M(\chi,t) \\ &= \Big[W_+(e^{i\chi} - 1) + W_-(e^{-i\chi} - 1)\Big]\, M(\chi, t). \end{aligned} \tag{11.113}$$

Since $n(0) = 0$, $M(\chi, 0) = 1$, and the linear equation Eq. (11.113) has the solution

$$M(\chi, t) = \exp\big(t\, \Lambda(\chi)\big), \qquad \Lambda(\chi) := W_+(e^{i\chi} - 1) + W_-(e^{-i\chi} - 1). \tag{11.114}$$

The CGF $K(\chi, t) = t\, \Lambda(\chi)$ is linear in t, because the difference $n(t) = N_+(t) - N_-(t)$ of two Poisson processes (a Skellam process) has stationary independent increments. $\Lambda(\chi)$ is the CGF per unit time.

Step 2: Cumulants of all orders. The m-th cumulant $\kappa_m(t)$ of the integer electron count $n(t)$ (and the corresponding charge cumulant $\kappa_m^{(Q)}(t) = q_e^m \kappa_m(t)$ of $Q(t) = q_e n(t)$) is obtained by differentiating $K(\chi, t) = t\, \Lambda(\chi)$ m times with respect to $i\chi$ at $\chi = 0$:

$$\kappa_m(t) := \left.\frac{\partial^m K(\chi,t)}{\partial (i\chi)^m}\right|_{\chi=0} = t\, \left.\frac{\mathrm{d}^m}{\mathrm{d}(i\chi)^m}\Big[W_+(e^{i\chi} - 1) + W_-(e^{-i\chi} - 1)\Big]\right|_{\chi=0}. \tag{11.115}$$

Because $\frac{\mathrm{d}^m}{\mathrm{d}(i\chi)^m} e^{i\chi} = e^{i\chi}$ and $\frac{\mathrm{d}^m}{\mathrm{d}(i\chi)^m} e^{-i\chi} = (-1)^m e^{-i\chi}$, evaluating Eq. (11.115) at $\chi = 0$ gives every cumulant $m \geq 1$:

$$\kappa_m(t) = \Big(W_+ + (-1)^m W_-\Big)\, t = \begin{cases} (W_+ - W_-)\, t = \dfrac{I}{q_e}\, t, & \text{if } m \text{ is odd } (m = 1, 3, 5, \dots), \\[2ex] (W_+ + W_-)\, t = \dfrac{S_I(0)}{2q_e^2}\, t, & \text{if } m \text{ is even } (m = 2, 4, 6, \dots). \end{cases} \tag{11.116}$$

The first four charge cumulants $\kappa_m^{(Q)}(t) = q_e^m \kappa_m(t)$ are

$$\text{Mean charge } (m = 1): \qquad \langle Q(t) \rangle = \kappa_1^{(Q)}(t) = q_e(W_+ - W_-)\, t = (I_+ - I_-)\, t = I\, t, \tag{11.117}$$

$$\text{Charge variance } (m=2): \qquad \mathrm{Var}\,(Q(t)) = \kappa_2^{(Q)}(t) = q_e^2(W_+ + W_-)\,t = q_e(I_+ + I_-)\,t = \frac{1}{2}S_I(0)\,t, \tag{11.118}$$

$$\text{Third cumulant } (m=3): \qquad \Big\langle \big(Q(t) - \langle Q(t)\rangle\big)^3 \Big\rangle = \kappa_3^{(Q)}(t) = q_e^3(W_+ - W_-)\,t = q_e^2 I\,t, \tag{11.119}$$

$$\text{Fourth cumulant } (m=4): \qquad \kappa_4^{(Q)}(t) = \Big\langle \delta Q^4 \Big\rangle - 3\big(\Big\langle \delta Q^2 \Big\rangle\big)^2 = q_e^4(W_+ + W_-)\,t = \frac{1}{2}q_e^2 S_I(0)\,t. \tag{11.120}$$

The third and fourth cumulants Eqs. (11.119) and (11.120) measure the departure from Gaussian statistics:

(a) At equilibrium ($V = 0$, so $W_+ = W_- := W_0$), all odd cumulants vanish and $P_n(t)$ is symmetric, but all even cumulants equal $2W_0 t > 0$, giving a positive excess kurtosis $\gamma_2(t) := \frac{\kappa_4(t)}{\kappa_2(t)^2} = \frac{1}{2W_0 t}$.

(b) Out of equilibrium ($V > 0$, so $W_+ > W_-$), the third cumulant $\kappa_3^{(Q)}(t) = q_e^2 I t$ is positive, and the skewness is

$$\gamma_1(t) := \frac{\kappa_3(t)}{\kappa_2(t)^{3/2}} = \frac{W_+ - W_-}{(W_+ + W_-)^{3/2}\,t^{1/2}} = \frac{\tanh(V/2V_T)}{\sqrt{(W_+ + W_-)\,t}}. \tag{11.121}$$

When the observation time t is long compared with the mean time between jumps, $(W_+ + W_-)^{-1}$, the number of jumps $N_{\text{jumps}} \sim (W_+ + W_-)t$ is large, and both the skewness $\gamma_1 \propto N_{\text{jumps}}^{-1/2}$ and the excess kurtosis $\gamma_2 \propto N_{\text{jumps}}^{-1}$ tend to zero. The central limit theorem then recovers the Gaussian Langevin description. For a circuit node the relevant time is the relaxation time τ_{RC}, and $(W_+ + W_-)\tau_{RC}$ is of the order of the charge scale $\Omega = C_{\text{node}}V_T/q_e$ of Example 11.7.

Step 3: The Fano factor. The **Fano factor** F is the ratio of the count variance $\kappa_2(t) = \mathrm{Var}(n(t))$ to the absolute mean count $|\kappa_1(t)| = |\langle n(t)\rangle|$:

$$F\left(\frac{V}{V_T}\right) := \frac{\mathrm{Var}\,(n(t))}{|\langle n(t)\rangle|} = \frac{S_I(0)}{2q_e\,|I|}. \tag{11.122}$$

For a unidirectional Poisson process ($W_- = 0$), $\mathrm{Var}(n(t)) = \langle n(t)\rangle = W_+ t$, so $F = 1$ is the reference value for uncorrelated shot noise. Substituting Eq. (11.116) and $W_+/W_- = e^{V/V_T}$ into Eq. (11.122) gives the Fano factor of the barrier:

$$F\left(\frac{V}{V_T}\right) = \frac{W_+ + W_-}{|W_+ - W_-|} = \frac{e^{V/(2V_T)} + e^{-V/(2V_T)}}{\left|e^{V/(2V_T)} - e^{-V/(2V_T)}\right|} = \coth\!\left(\frac{|V|}{2V_T}\right). \tag{11.123}$$

Its behavior across bias regimes:

- *Near equilibrium* ($|V| \ll V_T$). As $V \to 0$, $F(V/V_T) = \frac{2V_T}{|V|} + \frac{|V|}{6V_T} + \mathcal{O}((V/V_T)^3) \to \infty$ because the mean current $|I| \propto |V|$ vanishes while the Johnson–Nyquist noise $S_I(0) = 4k_{\mathrm{B}}TG_0$ stays finite.
- *Intermediate bias.* At $V = V_T \ln 3 \approx 28.4\,\mathrm{mV}$ ($W_+ = 3W_-$), the Fano factor is $F = \frac{3+1}{3-1} = 2$. At $V = 2V_T \approx 51.7\,\mathrm{mV}$, $F = \coth(1) \approx 1.313$.
- *Shot-noise regime* ($|V| \geq 4V_T \approx 103\,\mathrm{mV}$). At $V = 4V_T$, $F = \coth(2) \approx 1.037$, within 3.7% of the unidirectional Poisson value $F = 1$.

Step 4: The fluctuation symmetry. Finally, we check that $\Lambda(\chi)$ in Eq. (11.114) has the symmetry behind the detailed fluctuation theorem of Chapter 9. Let $\mathcal{A} := \ln(W_+/W_-) = \beta q_e V$ be the affinity, the entropy produced per forward jump in units of k_{B}. Using $W_+ e^{-\mathcal{A}} = W_-$ and $W_- e^{\mathcal{A}} = W_+$, evaluate Λ at the shifted complex argument $-\chi + i\mathcal{A}$:

$$\begin{aligned}
\Lambda(-\chi + i\mathcal{A}) &= W_+\Big(e^{i(-\chi+i\mathcal{A})} - 1\Big) + W_-\Big(e^{-i(-\chi+i\mathcal{A})} - 1\Big) \\
&= W_+\Big(e^{-i\chi}e^{-\mathcal{A}} - 1\Big) + W_-\Big(e^{+i\chi}e^{+\mathcal{A}} - 1\Big) \\
&= (W_+e^{-\mathcal{A}})e^{-i\chi} - W_+ + (W_-e^{+\mathcal{A}})e^{+i\chi} - W_- \\
&= W_-e^{-i\chi} - W_+ + W_+e^{i\chi} - W_- \\
&= W_+(e^{i\chi} - 1) + W_-(e^{-i\chi} - 1) = \Lambda(\chi).
\end{aligned} \tag{11.124}$$

The identity $\Lambda(\chi) = \Lambda(-\chi + i\beta q_e V)$ is known as the Gallavotti–Cohen (or Lebowitz–Spohn) symmetry of the CGF. To see what it implies for the probabilities of transferring $+n$ and $-n$ charges in time t, invert the Fourier series $M(\chi, t) = \exp(t\Lambda(\chi))$:

$$P_{-n}(t) = \int_{-\pi}^{\pi} \frac{\mathrm{d}\chi}{2\pi}\, e^{i\chi n}\, M(\chi, t) = \int_{-\pi}^{\pi} \frac{\mathrm{d}\chi}{2\pi}\, e^{i\chi n}\, M(-\chi + i\mathcal{A}, t). \tag{11.125}$$

Changing the integration variable to $\chi' = -\chi + i\mathcal{A}$ (and shifting the periodic horizontal contour back to the real axis $[-\pi, \pi]$ by Cauchy's theorem, since $M(\chi', t)$ is an entire function of χ' and 2π-periodic in $\mathrm{Re}\,\chi'$) gives

$$P_{-n}(t) = \int_{-\pi}^{\pi} \frac{\mathrm{d}\chi'}{2\pi}\, e^{i(-\chi'+i\mathcal{A})n}\, M(\chi', t) = e^{-\mathcal{A}n} \int_{-\pi}^{\pi} \frac{\mathrm{d}\chi'}{2\pi}\, e^{-i\chi' n}\, M(\chi', t) = e^{-\mathcal{A}n}\, P_{+n}(t). \tag{11.126}$$

Rearranging Eq. (11.126) and using $\mathcal{A}n = nq_eV/k_{\mathrm{B}}T = QV/k_{\mathrm{B}}T = \Delta s_{\mathrm{med}}/k_{\mathrm{B}}$ gives the detailed fluctuation theorem for charge transport:

$$\frac{P_{+n}(t)}{P_{-n}(t)} = \exp\!\left(\frac{nq_eV}{k_{\mathrm{B}}T}\right) = \exp\!\left(\frac{\Delta s_{\mathrm{med}}(n)}{k_{\mathrm{B}}}\right) \qquad \forall\, n \in \mathbb{Z},\ t > 0. \tag{11.127}$$

Physically, Eq. (11.127) says that even when a positive voltage drives a net current from source to drain, there is a non-zero probability $P_{-n}(t) = P_{+n}(t)e^{-nq_eV/k_{\mathrm{B}}T}$ that n electrons are transferred *against* the voltage, absorbing heat nq_eV from the lattice. Setting $\chi = i\mathcal{A}$ in Eq. (11.124) gives $\Lambda(i\mathcal{A}) = \Lambda(0) = 0$, which is the integral fluctuation theorem of Chapter 9:

$$\left\langle \exp\!\left(-\frac{\Delta s_{\mathrm{med}}(t)}{k_{\mathrm{B}}}\right)\right\rangle = \left\langle e^{-\mathcal{A}n(t)}\right\rangle = M(i\mathcal{A}, t) = e^{t\,\Lambda(i\mathcal{A})} = e^0 = 1. \tag{11.128}$$

The fluctuation theorems of Part III thus hold exactly at the level of individual electron jumps across a biased barrier.

Summary

Proved:

- The Wiener–Khinchin theorem relating the double-sided and single-sided PSDs to the autocorrelation function, and the filtering law $S_V(f) = |Z(2\pi f)|^2\, S_I(f)$ (Theorem 11.3 and Proposition 11.4).

- The Johnson–Nyquist formula $S_I(f) = 4k_{\mathrm{B}}TG$, by Nyquist's transmission-line argument and by Landauer–Büttiker scattering theory, and the R-independent variance $k_{\mathrm{B}}T/C$ of an RC node (Theorem 11.5 and Proposition 11.6).
- Campbell's and Carson's theorems for Poisson pulse trains and Schottky's formula $S_I(f) = 2q_e \langle I \rangle$ (Theorem 11.8 and Corollary 11.9).
- For bidirectional barrier transport, $I = I_+ - I_-$ and $S_I(f) = 2q_e(I_+ + I_-)$; with local detailed balance, the crossover formula $S_I(f) = 2q_e I \coth(q_e V/2k_{\mathrm{B}}T)$ (Theorems 11.10 and 11.11).
- The Lorentzian spectrum of a single trap and the $1/f$ spectrum of a log-uniform distribution of trap times (Proposition 11.12 and Theorem 11.13).
- The full counting statistics of a biased barrier, its Fano factor $F = \coth(|V|/2V_T)$, and the detailed fluctuation theorem $P_{+n}(t)/P_{-n}(t) = e^{nq_eV/k_{\mathrm{B}}T}$ (Worked computation 11.14).

Assumed:

- Fermi–Dirac statistics in the reservoirs; independent (Poisson) barrier crossings when $E_{\mathrm{b}} \gg k_{\mathrm{B}}T$; transit times τ_{tr} much shorter than circuit time constants.
- The scattering-theory expression Eq. (11.45) for equilibrium noise, and the WKB tunneling law Eq. (11.102) for trap times.

Taken up later:

- How the gate controls the barrier height of a MOSFET, giving the EKV subthreshold current and transconductance (Chapter 12, Chapter D).
- How networks of subthreshold transistors with bidirectional shot noise produce programmable Gaussian distributions, through the expansion in $\Omega = C_{\mathrm{node}}V_T/q_e$ (Chapter 13).
- Calibration of threshold offsets and slow drift in hardware (Chapter 18).

Open:

- Whether the non-Gaussian cumulants of shot noise in very small nodes ($\Omega \sim 1$–10) can be used directly to sample non-Gaussian distributions.

Bibliographic notes

The experimental discovery and thermodynamic explanation of equilibrium thermal noise were published back-to-back by Johnson [57] and Nyquist [79]; their generalization to arbitrary linear response functions is the fluctuation-dissipation theorem of Kubo [62]. Unidirectional shot noise was first formulated by Schottky [96] for vacuum tubes. Blanter and Büttiker [13] review the scattering theory of noise in mesoscopic conductors, including Landauer–Büttiker formulas, Fano factors, and full counting statistics.

The view of thermal noise in barrier devices as the sum of two opposing shot-noise currents was

emphasized in analog VLSI by Mead [75] and Sarpeshkar et al. [93], and is developed in the MOSFET monographs of Enz and Vittoz [29] and Tsividis and McAndrew [111]. A stochastic-thermodynamic treatment of nonlinear electronic circuits as jump processes, including the fluctuation symmetry of the circuit master equation and its Kramers–Moyal expansion, was given by Freitas et al. [37] and applied to subthreshold CMOS random networks by Freitas et al. [38]. Standard references on stochastic point processes, Campbell's theorem, and Lorentzian telegraph noise include van Kampen [112] and Gardiner [39].

Chapter 12

Physics of the Subthreshold (Weak-Inversion) MOSFET

Chapter 11 treated a barrier conductor as an abstract object: electrons hop across a barrier in both directions as two independent Poisson streams with currents I_+ and I_-, the net current is $I = I_+ - I_-$, and the single-sided current noise is $S_I = 2q_e(I_+ + I_-)$ (Section 11.4). That chapter left open where the barrier comes from, how a gate electrode sets its height, and why the forward and reverse currents should depend on the source and drain voltages in the specific exponential way that local detailed balance requires. This chapter answers those questions for the device from which CMOS sampling circuits are built: the metal-oxide-semiconductor field-effect transistor (MOSFET) biased in weak inversion, also called the subthreshold regime.

The physical picture is simple. In weak inversion the gate voltage is too low to form a dense conducting electron layer at the silicon surface. The few electrons that do sit at the surface are minority carriers in thermal equilibrium with the source and drain reservoirs, and their density follows the Boltzmann factor of the surface potential. Because there are so few of them, they barely screen the gate, so the surface potential is nearly the same everywhere along the channel. Electrons therefore move from source to drain mainly by diffusion down a density gradient, exactly as minority carriers cross the base of a bipolar transistor. The result is a current that is exponential in the gate voltage, exponential in the source and drain voltages, and naturally split into a forward part set by the source and a reverse part set by the drain. These two parts are the I_+ and I_- of Chapter 11.

The gate acts on the surface potential through a capacitive divider, which introduces the slope factor n. Integrating the diffusion current along the channel then gives the Enz-Krummenacher-Vittoz (EKV) law (Theorem 12.13), and integrating the local thermal noise of the channel shows that its current noise is exactly the bidirectional shot noise $2q_e(I_F + I_R)$ of Chapter 11 (Theorem 12.19). The chapter ends with the five-transistor subthreshold transconductance amplifier (Worked computation 12.24). Its tanh transfer characteristic reappears in Chapter 14 as the activation function of CMOS probabilistic bits, and the voltage variance $2nk_BT/C$ of a capacitor it holds in feedback is the noise temperature used in Chapter 13.

Throughout, we use the thermal voltage $V_T = k_BT/q_e \approx 25.85\,\mathrm{mV}$ at $T = 300\,\mathrm{K}$, measure all terminal voltages of an n-channel device relative to its bulk (substrate) terminal unless stated otherwise, and follow the single-sided noise convention $S_I(f)$ of Definition 11.2. Device numbers quoted in examples are representative order-of-magnitude values for a generic bulk CMOS process, not specifications of any particular chip.

12.1 Semiconductor Electrostatics of the MOS Structure

A MOSFET is a MOS capacitor with two extra electrodes. The capacitor consists of a gate electrode, a thin insulating layer of silicon dioxide of thickness t_{ox}, and a lightly doped silicon substrate. For an n-channel device the substrate is p-type, doped with acceptors of density N_A. Two heavily doped n^+ regions, the source and the drain, are implanted at either end of the gate. They act as reservoirs of electrons. Current flows only if electrons from these reservoirs can enter the region under the gate. Whether they can is decided by the electrostatic potential at the silicon surface, which is in turn set by the gate voltage. This section computes that potential and the charges it produces, first in a one-dimensional slice perpendicular to the surface.

12.1.1 Coordinates, Carrier Statistics, and the Fermi Potential

Let $x \geq 0$ be the depth into the silicon measured from the oxide-silicon interface, and let $y \in [0, L]$ be the position along the channel from the source ($y = 0$) to the drain ($y = L$). The channel has width W in the third direction. Let $\psi(x, y)$ be the electrostatic potential in the silicon measured relative to the neutral bulk far from the surface, so $\psi \to 0$ as $x \to \infty$. We write $\psi_s(y) := \psi(0, y)$ for the *surface potential*. The quantity $q_e\psi$ is the downward bending of the energy bands relative to the bulk.

In the neutral bulk, holes are the majority carriers with density $p_0 = N_A$ (complete ionization), and electrons are minority carriers with density $n_0 = n_i^2/N_A$, where $n_i \approx 10^{10}\,\mathrm{cm}^{-3}$ is the intrinsic carrier density of silicon at 300 K. The distance between the bulk Fermi level and the intrinsic level, expressed as a potential, is the *Fermi potential*

$$\phi_F := V_T \ln\!\left(\frac{N_A}{n_i}\right), \qquad \text{so that} \qquad n_0 = \frac{n_i^2}{N_A} = N_A\, e^{-2\phi_F/V_T}. \tag{12.1}$$

Holes are in equilibrium with the bulk contact. Electrons near the surface exchange particles with the n^+ source and drain, which may sit at a potential V above the bulk. In that case the electron quasi-Fermi level near the surface lies q_eV below the bulk Fermi level. We call V the *channel voltage*; it equals V_S at the source end and V_D at the drain end. For nondegenerate carriers, Boltzmann statistics then gives

$$p(x) = N_A\, e^{-\psi(x)/V_T}, \qquad n(x) = \frac{n_i^2}{N_A}\, e^{(\psi(x)-V)/V_T} = N_A\, e^{(\psi(x)-2\phi_F-V)/V_T}. \tag{12.2}$$

The second form in Eq. (12.2) shows that the surface electron density equals the bulk hole density N_A when $\psi_s = 2\phi_F + V$. This is the conventional onset of strong inversion. The surface regimes for V fixed are:

- *accumulation*, $\psi_s < 0$: holes pile up at the surface;
- *depletion*, $0 < \psi_s < \phi_F + V$: holes are pushed away, leaving ionized acceptors, and electrons are still rarer than in intrinsic silicon;
- *weak inversion*, $\phi_F + V < \psi_s < 2\phi_F + V$: the surface is n-type ($n_s > p_s$), but the electron charge is still small compared with the depletion charge;
- *strong inversion*, $\psi_s > 2\phi_F + V$ (in practice, more than a few V_T above it): the electron layer dominates the surface charge and screens further changes in gate voltage.

The boundary between weak and strong inversion is not sharp. Section 12.4 replaces it by a smooth interpolation (moderate inversion).

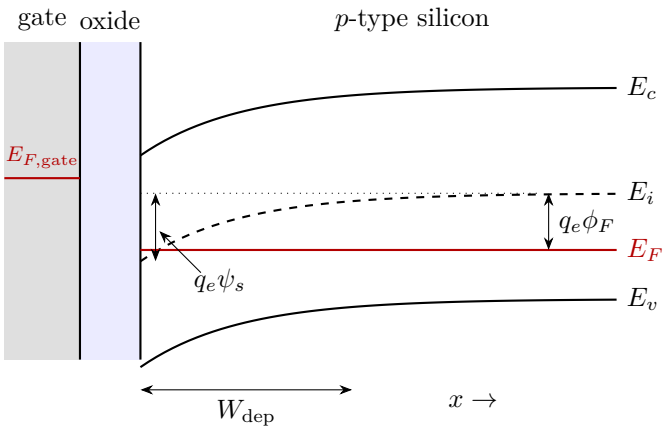


Figure 12.1: Energy band diagram of an n-channel MOS structure in weak inversion, perpendicular to the surface. A positive gate voltage bends the bands down by $q_e\psi_s$ at the surface. Once ψ_s exceeds ϕ_F, the intrinsic level E_i crosses below the Fermi level E_F near the surface and the surface becomes n-type. The bending extends over the depletion width W_{dep}. In weak inversion $\phi_F < \psi_s < 2\phi_F$ (here drawn for channel voltage $V = 0$).

12.1.2 The Poisson-Boltzmann Equation and Its First Integral

The potential $\psi(x)$ obeys Poisson's equation $\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = -\rho(x)/\varepsilon_{\text{si}}$, where $\varepsilon_{\text{si}} \approx 11.7\,\varepsilon_0$ is the permittivity of silicon. The charge density comes from holes, electrons, and ionized acceptors. We neglect the tiny equilibrium electron density n_0 in the bulk neutrality condition, so that $\rho \to 0$ in the bulk requires $p_0 = N_A$. Inserting Eq. (12.2) gives the *Poisson-Boltzmann equation*

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = -\frac{q_e}{\varepsilon_{\text{si}}}\left[N_A e^{-\psi/V_T} - N_A - n_0\, e^{(\psi - V)/V_T}\right]. \tag{12.3}$$

This is a one-dimensional problem of a particle in a potential, with x playing the role of time, and it has a first integral.

Proposition 12.1 (Surface Field from the Poisson-Boltzmann First Integral)**.** *Let $\psi(x)$ solve Eq. (12.3) with $\psi \to 0$ and $\frac{\mathrm{d}\psi}{\mathrm{d}x} \to 0$ as $x \to \infty$, and let $\psi_s = \psi(0) > 0$. Then the surface electric field $\mathcal{E}_s := -\frac{\mathrm{d}\psi}{\mathrm{d}x}\big|_{x=0} > 0$ is*

$$\mathcal{E}_s^2 = \frac{2q_eN_A}{\varepsilon_{\text{si}}}\left[\psi_s + V_T\big(e^{-\psi_s/V_T} - 1\big) + V_T\, e^{-2\phi_F/V_T}\big(e^{(\psi_s - V)/V_T} - e^{-V/V_T}\big)\right], \tag{12.4}$$

and by Gauss's law the total semiconductor charge per unit area is $Q_s = -\varepsilon_{\text{si}}\mathcal{E}_s$.

Proof. Multiply both sides of Eq. (12.3) by $\frac{\mathrm{d}\psi}{\mathrm{d}x}$. The left side becomes $\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}\big(\frac{\mathrm{d}\psi}{\mathrm{d}x}\big)^2$. The right side is $-\rho(\psi)\,\frac{\mathrm{d}\psi}{\mathrm{d}x}/\varepsilon_{\text{si}}$, a total derivative with respect to x of $-\frac{1}{\varepsilon_{\text{si}}}\int_0^{\psi(x)}\rho(u)\,\mathrm{d}u$. Integrating from $x = \infty$ (where $\psi = 0$ and $\frac{\mathrm{d}\psi}{\mathrm{d}x} = 0$) to x gives

$$\frac{1}{2}\left(\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)^2 = -\frac{1}{\varepsilon_{\text{si}}}\int_0^{\psi}\rho(u)\,\mathrm{d}u. \tag{12.5}$$

We evaluate the integral term by term:

$$-\frac{1}{q_e}\int_0^{\psi}\rho(u)\,\mathrm{d}u = -\int_0^{\psi}\left[N_A e^{-u/V_T} - N_A - n_0 e^{(u-V)/V_T}\right]\mathrm{d}u$$

$$= -\Big[- N_A V_T e^{-u/V_T} - N_A u - n_0 V_T e^{(u-V)/V_T} \Big]_0^{\psi}$$
$$= N_A V_T (e^{-\psi/V_T} - 1) + N_A \psi + n_0 V_T (e^{(\psi - V)/V_T} - e^{-V/V_T}). \tag{12.6}$$

Setting $x = 0$, using $n_0 = N_A e^{-2\phi_F/V_T}$ from Eq. (12.1), and multiplying by $2q_e/\varepsilon_{\mathrm{si}}$ gives Eq. (12.4). For $\psi_s > 0$ the potential decreases into the bulk, so $\frac{\mathrm{d}\psi}{\mathrm{d}x} < 0$ at the surface and $\mathcal{E}_s > 0$. Gauss's law across the surface, with zero field deep in the bulk, gives $Q_s = -\varepsilon_{\mathrm{si}} \mathcal{E}_s$. □

The three terms inside the bracket of Eq. (12.4) have direct meanings. The term ψ_s is the ionized acceptor charge of the depletion layer. The term $V_T(e^{-\psi_s/V_T} - 1)$ is the hole correction, which equals $-V_T$ once ψ_s exceeds a few V_T. The last term is the electron (inversion) charge, which grows as e^{ψ_s/V_T}.

12.1.3 Depletion Charge, Inversion Charge, and the Depletion Capacitance

In the subthreshold regime the bracket in Eq. (12.4) is dominated by ψ_s, which is several hundred millivolts, while the electron term is at most of order V_T. We drop the hole correction (an error of relative size $V_T/\psi_s \approx 3\%$, which can be absorbed into a redefinition of ψ_s) and the term $e^{-V/V_T} e^{-2\phi_F/V_T}$, which is smaller than the electron term by $e^{-\psi_s/V_T}$. This leaves

$$Q_s = -\sqrt{2 q_e \varepsilon_{\mathrm{si}} N_A}\, \sqrt{\psi_s + V_T\, e^{(\psi_s - 2\phi_F - V)/V_T}}. \tag{12.7}$$

Definition 12.2 (Depletion Charge and Depletion Capacitance)**.** The *depletion charge* per unit area $Q_{\mathrm{dep}}(\psi_s)$ and the *depletion capacitance* per unit area $C_{\mathrm{dep}}(\psi_s)$ are

$$Q_{\mathrm{dep}}(\psi_s) := -\sqrt{2 q_e \varepsilon_{\mathrm{si}} N_A \psi_s}, \qquad C_{\mathrm{dep}}(\psi_s) := -\frac{\mathrm{d}Q_{\mathrm{dep}}}{\mathrm{d}\psi_s} = \sqrt{\frac{q_e \varepsilon_{\mathrm{si}} N_A}{2\psi_s}} = \frac{\varepsilon_{\mathrm{si}}}{W_{\mathrm{dep}}}, \tag{12.8}$$

where $W_{\mathrm{dep}} := \sqrt{2\varepsilon_{\mathrm{si}} \psi_s / (q_e N_A)}$ is the depletion width. The *inversion charge* per unit area is $Q_{\mathrm{inv}} := Q_s - Q_{\mathrm{dep}}$.

The last equality in Eq. (12.8) follows from $\varepsilon_{\mathrm{si}}/W_{\mathrm{dep}} = \varepsilon_{\mathrm{si}} \sqrt{q_e N_A/(2\varepsilon_{\mathrm{si}}\psi_s)} = \sqrt{q_e \varepsilon_{\mathrm{si}} N_A/(2\psi_s)}$. So C_{dep} is the capacitance of a parallel-plate capacitor of thickness W_{dep} filled with silicon: changing ψ_s moves the edge of the depletion layer and uncovers or covers ionized acceptors there.

Proposition 12.3 (Inversion Charge in Weak Inversion)**.** *Suppose $V_T\, e^{(\psi_s - 2\phi_F - V)/V_T} \ll \psi_s$. Then to first order in this small ratio*

$$Q_{\mathrm{inv}}(\psi_s, V) = -C_{\mathrm{dep}}(\psi_s)\, V_T\, e^{(\psi_s - 2\phi_F - V)/V_T}. \tag{12.9}$$

Equivalently, $|Q_{\mathrm{inv}}| = q_e\, n_s\, \ell_{\mathrm{inv}}$, where $n_s = N_A e^{(\psi_s - 2\phi_F - V)/V_T}$ is the surface electron density and $\ell_{\mathrm{inv}} := V_T/\mathcal{E}_s$ is the thickness of the electron layer.

Proof. Write $a = \psi_s$ and $b = V_T e^{(\psi_s - 2\phi_F - V)/V_T}$, so that $Q_s = -\sqrt{2 q_e \varepsilon_{\mathrm{si}} N_A}\sqrt{a + b}$. For $b \ll a$, the Taylor expansion $\sqrt{a+b} = \sqrt{a} + \frac{b}{2\sqrt{a}} + \mathcal{O}(b^2/a^{3/2})$ gives

$$Q_s = -\sqrt{2 q_e \varepsilon_{\mathrm{si}} N_A \psi_s} - \frac{\sqrt{2 q_e \varepsilon_{\mathrm{si}} N_A}}{2\sqrt{\psi_s}}\, V_T\, e^{(\psi_s - 2\phi_F - V)/V_T} + \cdots. \tag{12.10}$$

The first term is Q_{dep}. In the second, $\sqrt{2 q_e \varepsilon_{\mathrm{si}} N_A}/(2\sqrt{\psi_s}) = \sqrt{q_e \varepsilon_{\mathrm{si}} N_A/(2\psi_s)} = C_{\mathrm{dep}}$, which proves Eq. (12.9). For the second statement, the leading-order field is $\mathcal{E}_s = \sqrt{2 q_e N_A \psi_s/\varepsilon_{\mathrm{si}}}$, so

$$q_e\, n_s\, \frac{V_T}{\mathcal{E}_s} = q_e N_A e^{(\psi_s - 2\phi_F - V)/V_T}\, V_T \sqrt{\frac{\varepsilon_{\mathrm{si}}}{2 q_e N_A \psi_s}} = V_T \sqrt{\frac{q_e \varepsilon_{\mathrm{si}} N_A}{2\psi_s}}\, e^{(\psi_s - 2\phi_F - V)/V_T}, \tag{12.11}$$

which equals $C_{\mathrm{dep}} V_T e^{(\psi_s - 2\phi_F - V)/V_T} = |Q_{\mathrm{inv}}|$. □

The second form has a direct physical reading. Near the surface the potential falls linearly, $\psi(x) \approx \psi_s - \mathcal{E}_s x$. The Boltzmann factor $e^{\psi(x)/V_T}$ then makes the electron density decay as $n_s e^{-x\mathcal{E}_s/V_T}$, an exponential with decay length $V_T/\mathcal{E}_s$. The electrons are a thin, thermally broadened sheet held against the oxide by the depletion field. For the numbers of Example 12.5 below, $\mathcal{E}_s \approx 3.5 \times 10^7\,\mathrm{V/m}$ and $\ell_{\mathrm{inv}} \approx 0.7\,\mathrm{nm}$. This justifies treating the inversion layer as a sheet of charge, the *charge-sheet approximation*, in the transport calculation of Section 12.3.

Equation (12.9) is the single most important equation of this section. It says that, at fixed surface potential, the inversion charge depends on the local channel voltage only through the Boltzmann factor e^{-V/V_T}. Raising the channel voltage by V_T lowers the electron density by a factor e, because the electrons at that point are in equilibrium with a reservoir whose Fermi level is lower by $q_e V_T$.

12.1.4 The Gate Voltage Equation and the Threshold Voltage

The gate, oxide, and semiconductor form a series circuit. The oxide has no charge (ideally), so the field in it is uniform, and charge neutrality of the whole stack requires the gate charge per unit area to be $Q_G = -Q_s$ (ignoring oxide and interface charges, which we fold into the flat-band voltage). The voltage across the oxide is Q_G/C_{ox}, where $C_{\mathrm{ox}} = \varepsilon_{\mathrm{ox}}/t_{\mathrm{ox}}$ is the oxide capacitance per unit area. Going from the bulk contact to the gate contact, the voltage drops are the surface potential ψ_s, the oxide voltage $-Q_s/C_{\mathrm{ox}}$, and the work-function difference between gate and substrate, together with fixed oxide charge, collected into the *flat-band voltage* V_{FB}. Therefore

$$V_G = V_{\mathrm{FB}} + \psi_s - \frac{Q_s(\psi_s, V)}{C_{\mathrm{ox}}}. \tag{12.12}$$

In weak inversion $Q_s \approx Q_{\mathrm{dep}}$, and Eq. (12.12) becomes

$$V_G = V_{\mathrm{FB}} + \psi_s + \gamma\sqrt{\psi_s}, \qquad \gamma := \frac{\sqrt{2 q_e \varepsilon_{\mathrm{si}} N_A}}{C_{\mathrm{ox}}}, \tag{12.13}$$

where γ is the *body-effect coefficient* (units $\mathrm{V}^{1/2}$). Equation (12.13) is a monotonic relation between V_G and ψ_s that does not involve the channel voltage V. This is the reason the surface potential is nearly uniform along a weakly inverted channel, a point we make precise in Section 12.3.

Definition 12.4 (Bulk-Referred Threshold Voltage)**.** The (bulk-referred) *threshold voltage* V_{th} is the gate voltage at which Eq. (12.13) gives $\psi_s = 2\phi_F$:

$$V_{\mathrm{th}} := V_{\mathrm{FB}} + 2\phi_F + \gamma\sqrt{2\phi_F}. \tag{12.14}$$

At $V_G = V_{\mathrm{th}}$ and channel voltage $V = 0$, Eq. (12.2) gives a surface electron density equal to N_A. With this definition the subthreshold region is $V_G < V_{\mathrm{th}}$ for a channel at $V = 0$. Other conventions exist. The EKV model, for example, defines the threshold where the weak- and strong-inversion asymptotes of the inversion charge meet, which lies a few nV_T higher. A change of threshold convention only rescales the current prefactor I_0 in Section 12.3, as discussed in Section 12.3.4.

Example 12.5 (Representative MOS Numbers)**.** Take $N_A = 5 \times 10^{17}\,\mathrm{cm^{-3}}$, $n_i = 10^{10}\,\mathrm{cm^{-3}}$, $t_{\mathrm{ox}} = 4\,\mathrm{nm}$ ($\varepsilon_{\mathrm{ox}} = 3.9\varepsilon_0$), and $T = 300\,\mathrm{K}$. Then:

- Fermi potential: $\phi_F = 25.85\,\mathrm{mV} \times \ln(5 \times 10^7) = 25.85\,\mathrm{mV} \times 17.73 = 0.458\,\mathrm{V}$, so $2\phi_F = 0.917\,\mathrm{V}$.
- Oxide capacitance: $C_{\mathrm{ox}} = 3.9 \times 8.854 \times 10^{-12}\,\mathrm{F/m}/(4 \times 10^{-9}\,\mathrm{m}) = 8.63 \times 10^{-3}\,\mathrm{F/m^2} = 8.63\,\mathrm{fF}/\mu\mathrm{m}^2$.

- Depletion capacitance at $\psi_s = 0.8\,\mathrm{V}$ (a weak-inversion value): $q_e\varepsilon_{\mathrm{si}}N_A = 1.602\times10^{-19}\times1.036\times10^{-10}\times5\times10^{23} = 8.30\times10^{-6}$ in SI units, so $C_{\mathrm{dep}} = \sqrt{8.30\times10^{-6}/1.6} = 2.28\times10^{-3}\,\mathrm{F/m^2} = 2.28\,\mathrm{fF}/\mu\mathrm{m}^2$, and $W_{\mathrm{dep}} = \varepsilon_{\mathrm{si}}/C_{\mathrm{dep}} = 45\,\mathrm{nm}$.
- Body-effect coefficient: $\gamma = \sqrt{2\times8.30\times10^{-6}}/(8.63\times10^{-3}) = 0.47\,\mathrm{V}^{1/2}$.

The ratio $C_{\mathrm{dep}}/C_{\mathrm{ox}} \approx 0.26$ will reappear as $n-1$ in Section 12.2. A $1\,\mu\mathrm{m}\times1\,\mu\mathrm{m}$ gate in this process has a total oxide capacitance of 8.6 fF; a minimum-size device in a modern node has a gate capacitance well below 1 fF.

12.2 The Capacitive Divider and the Subthreshold Slope Factor

The gate does not set the surface potential directly. Between the gate and the bulk there are two capacitances in series: the oxide capacitance C_{ox} between the gate and the silicon surface, and the depletion capacitance C_{dep} between the surface and the neutral bulk. A small change δV_G in gate voltage divides between them, and only the part across C_{dep} changes the surface potential. This section turns that statement into the slope factor n and shows how it fixes the steepness of the subthreshold current.

12.2.1 Small-Signal Divider

Proposition 12.6 (Slope Factor as a Capacitive Divider)**.** *In weak inversion, at fixed channel voltage V, the surface potential responds to the gate voltage as*

$$\frac{\mathrm{d}\psi_s}{\mathrm{d}V_G} = \frac{C_{\mathrm{ox}}}{C_{\mathrm{ox}}+C_{\mathrm{dep}}(\psi_s)} = \frac{1}{n(\psi_s)}, \qquad n(\psi_s) := 1+\frac{C_{\mathrm{dep}}(\psi_s)}{C_{\mathrm{ox}}} = 1+\frac{\gamma}{2\sqrt{\psi_s}}. \tag{12.15}$$

More generally, keeping the inversion charge in Eq. (12.12),

$$\left.\frac{\partial\psi_s}{\partial V_G}\right|_V = \frac{C_{\mathrm{ox}}}{C_{\mathrm{ox}}+C_{\mathrm{dep}}+C_{\mathrm{inv}}}, \qquad C_{\mathrm{inv}} := -\left.\frac{\partial Q_{\mathrm{inv}}}{\partial\psi_s}\right|_V \geq 0. \tag{12.16}$$

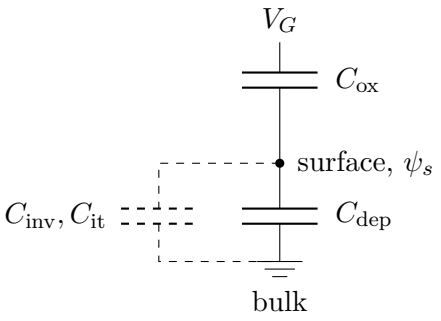
Proof. Differentiate Eq. (12.13) with respect to ψ_s:

$$\frac{\mathrm{d}V_G}{\mathrm{d}\psi_s} = 1+\frac{\gamma}{2\sqrt{\psi_s}} = 1+\frac{\sqrt{2q_e\varepsilon_{\mathrm{si}}N_A}}{2C_{\mathrm{ox}}\sqrt{\psi_s}} = 1+\frac{1}{C_{\mathrm{ox}}}\sqrt{\frac{q_e\varepsilon_{\mathrm{si}}N_A}{2\psi_s}} = 1+\frac{C_{\mathrm{dep}}}{C_{\mathrm{ox}}}. \tag{12.17}$$

Since $\frac{\mathrm{d}V_G}{\mathrm{d}\psi_s} > 0$, the inverse function theorem gives $\frac{\mathrm{d}\psi_s}{\mathrm{d}V_G} = 1/(1+C_{\mathrm{dep}}/C_{\mathrm{ox}}) = C_{\mathrm{ox}}/(C_{\mathrm{ox}}+C_{\mathrm{dep}})$. For the general case, differentiate Eq. (12.12) at fixed V with $Q_s = Q_{\mathrm{dep}}+Q_{\mathrm{inv}}$: $\partial V_G/\partial\psi_s = 1+(C_{\mathrm{dep}}+C_{\mathrm{inv}})/C_{\mathrm{ox}}$, and invert. $C_{\mathrm{inv}} \geq 0$ because $|Q_{\mathrm{inv}}|$ increases with ψ_s (Eq. (12.9), and Lemma 12.16 below for the general case). □

The divider picture (Fig. 12.2) also tells us how other charges enter. Interface traps at the oxide-silicon boundary change their occupation when ψ_s moves, which adds a capacitance $C_{\mathrm{it}} = q_e^2 D_{\mathrm{it}}$ per unit area in parallel with C_{dep}, where D_{it} is the trap density per unit energy and area. The slope factor then becomes $n = 1+(C_{\mathrm{dep}}+C_{\mathrm{it}})/C_{\mathrm{ox}}$. In strong inversion the inversion capacitance C_{inv} grows exponentially and shorts out the lower branch, so ψ_s becomes pinned and stops following V_G. In weak inversion, $C_{\mathrm{inv}} = |Q_{\mathrm{inv}}|/V_T$ (from Eq. (12.9)) is much smaller than C_{dep}, and the divider ratio is $1/n$.

For the numbers of Example 12.5, $n = 1+2.28/8.63 = 1.26$ at $\psi_s = 0.8\,\mathrm{V}$. Typical bulk CMOS values are $n \approx 1.2$–1.5. In circuit literature the reciprocal $\kappa := 1/n$ is often used and called the gate coupling coefficient; $\kappa \approx 0.65$–0.85 [75].



$$\delta\psi_s = \frac{C_{\mathrm{ox}}}{C_{\mathrm{ox}} + C_{\mathrm{dep}}}\,\delta V_G = \frac{\delta V_G}{n}$$

Figure 12.2: Small-signal capacitive divider between gate and bulk. The oxide capacitance C_{ox} and the depletion capacitance C_{dep} are in series; the surface potential is the midpoint. The inversion-layer capacitance C_{inv} and the interface-trap capacitance C_{it} (dashed) appear in parallel with C_{dep}. In weak inversion C_{inv} is negligible.

12.2.2 Linearizing the Surface Potential

The slope factor depends weakly on ψ_s through $C_{\mathrm{dep}} \propto \psi_s^{-1/2}$. Over the weak-inversion range, ψ_s changes by a few hundred millivolts at most, so we linearize Eq. (12.13) about the threshold point $\psi_s = 2\phi_F$, $V_G = V_{\mathrm{th}}$:

$$\psi_s(V_G) = 2\phi_F + \frac{V_G - V_{\mathrm{th}}}{n} + \mathcal{O}\!\left((V_G - V_{\mathrm{th}})^2\right), \tag{12.18}$$

where n is now a constant evaluated at a representative point. The size of the neglected term follows from the second derivative:

$$\frac{\mathrm{d}^2 V_G}{\mathrm{d}\psi_s^2} = -\frac{\gamma}{4\psi_s^{3/2}}, \qquad \text{so} \qquad \frac{\delta n}{n} = \frac{1}{n}\frac{\mathrm{d}^2 V_G}{\mathrm{d}\psi_s^2}\,\delta\psi_s = -\frac{\gamma\,\delta\psi_s}{4n\psi_s^{3/2}}. \tag{12.19}$$

With $\gamma = 0.47\,\mathrm{V}^{1/2}$, $\psi_s = 0.8\,\mathrm{V}$, $n = 1.26$, and a swing $\delta\psi_s = 0.1\,\mathrm{V}$, this gives $|\delta n|/n \approx 1.3\%$. A constant n is therefore accurate to about one percent over the subthreshold decades used in practice. We keep n constant from here on.

12.2.3 The Subthreshold Swing and the Boltzmann Limit

We will show in Section 12.3 that the subthreshold current is proportional to the surface electron density, which by Eq. (12.2) is proportional to e^{ψ_s/V_T}. Granting this for the moment, Eq. (12.18) gives $I_{\mathrm{D}} \propto e^{V_G/(nV_T)}$, and the gate voltage needed to change the current by a factor of ten is the *subthreshold swing*

$$\mathcal{S} := \left(\frac{\mathrm{d}\log_{10} I_{\mathrm{D}}}{\mathrm{d}V_G}\right)^{-1} = nV_T \ln 10 = n\,\frac{k_{\mathrm{B}}T}{q_e}\ln 10. \tag{12.20}$$

At $T = 300\,\mathrm{K}$, $V_T \ln 10 = 25.85\,\mathrm{mV} \times 2.303 = 59.5\,\mathrm{mV}$, so $\mathcal{S} \approx 60n\,\mathrm{mV/decade}$. For $n = 1.26$, $\mathcal{S} \approx 75\,\mathrm{mV/decade}$.

Physics Insight: Why 60 mV/decade Is a Thermal Limit

The factor $V_T \ln 10$ in Eq. (12.20) does not depend on any material parameter. It comes from the Boltzmann tail of the electron energy distribution in the source: lowering a barrier by $q_e\,\delta\psi$ raises the number of electrons with enough energy to cross it by $e^{\delta\psi/V_T}$, and no more. The best a gate can do is move the barrier one-for-one with its own voltage, which is the limit

$n \to 1$ ($C_{\mathrm{ox}} \gg C_{\mathrm{dep}}$). A transistor that injects carriers thermally over a barrier therefore cannot switch faster than 60 mV/decade at room temperature. Devices that inject carriers by band-to-band tunneling, or that place a negative capacitance in series with the gate, aim to beat this limit by changing the injection mechanism or the divider itself. For thermodynamic computing the limit is not a defect: it is the statement that the device's current is set by the same Boltzmann factor that sets its noise, which is the link between g_{m} and S_I we exploit in Section 12.5.

Example 12.7 (How Many Decades Does Weak Inversion Span?)**.** Weak inversion runs from $\psi_s = \phi_F$ to $\psi_s = 2\phi_F$ (at $V = 0$). Since the current is proportional to e^{ψ_s/V_T}, this range covers

$$\frac{\phi_F}{V_T \ln 10} = \frac{0.458\,\mathrm{V}}{59.5\,\mathrm{mV}} \approx 7.7 \text{ decades} \tag{12.21}$$

of current for the doping of Example 12.5. The corresponding gate-voltage range follows from Eq. (12.13):

$$V_G(2\phi_F) - V_G(\phi_F) = \phi_F + \gamma\left(\sqrt{2\phi_F} - \sqrt{\phi_F}\right) = 0.458\,\mathrm{V} + 0.47 \times (0.957 - 0.677)\,\mathrm{V} \approx 0.59\,\mathrm{V}. \tag{12.22}$$

The ratio $0.59\,\mathrm{V}/0.458\,\mathrm{V} = 1.29$ is the slope factor averaged over the range, slightly above the value 1.26 at $\psi_s = 0.8\,\mathrm{V}$ because C_{dep} is larger at smaller ψ_s. In practice the lowest usable currents are set by junction leakage, not by the end of weak inversion, but several decades of exponential behavior are always available.

The swing also tells us how the current depends on temperature at fixed bias. Since $I_{\mathrm{D}} \propto e^{(V_G - V_{\mathrm{th}})/(nV_T)}$ and $V_T \propto T$, a subthreshold circuit with fixed gate voltages is extremely temperature sensitive. Practical subthreshold circuits therefore set currents with current mirrors referenced to a bias current, rather than by fixed gate voltages, so that the exponential dependence cancels between matched devices.

12.2.4 Source-Referred Form and the Body Effect

Circuit designers often measure voltages from the source rather than the bulk. The weak-inversion current will turn out to depend on the bulk-referred gate and source voltages through the combination $e^{V_G/(nV_T)} e^{-V_S/V_T}$. Writing $V_G = V_{\mathrm{GS}} + V_S$ with $V_S = V_{\mathrm{SB}}$ (the source-to-bulk voltage),

$$e^{V_G/(nV_T)}\, e^{-V_S/V_T} = e^{V_{\mathrm{GS}}/(nV_T)}\, e^{-(1-1/n)V_{\mathrm{SB}}/V_T}. \tag{12.23}$$

At fixed V_{GS}, raising the source above the bulk reduces the current by $e^{-(1-\kappa)V_{\mathrm{SB}}/V_T}$. This is the subthreshold *body effect*: the bulk acts as a second, weaker gate with coupling $1 - \kappa = (n-1)/n = C_{\mathrm{dep}}/(C_{\mathrm{ox}} + C_{\mathrm{dep}})$, the other half of the capacitive divider. In bulk-referred variables the body effect is invisible, and gate, source, and drain enter symmetrically. As an example, with $n = 1.26$ the body coupling is $1 - 1/n = 0.21$, and raising the source 100 mV above the bulk at fixed V_{GS} multiplies the current by $e^{-0.21 \times 100/25.85} = e^{-0.80} = 0.45$. This is why the EKV model, and this chapter, refer everything to the bulk.

12.3 Drift-Diffusion Transport in Weak Inversion and the EKV Law

We now let the source and drain sit at different voltages and compute the current. The approach is the classical gradual-channel, charge-sheet model. We assume a long channel, so that the potential

varies slowly along y compared with its variation along x. Each slice y of the channel can then be treated as the one-dimensional MOS structure of Section 12.1, with its own channel voltage $V(y)$ running from $V(0) = V_S$ to $V(L) = V_D$.

Assumption 12.8 (Long-Channel Charge-Sheet Model)**.** (i) (Gradual channel) The potential varies slowly along the channel, in the sense that $|\partial^2\psi/\partial y^2| \ll |\partial^2\psi/\partial x^2|$ in the depletion region, so the vertical electrostatics of each slice is that of Section 12.1 with the local channel voltage $V(y)$.

(ii) (Charge sheet) The inversion layer is thin compared with the depletion width ($\ell_{\text{inv}} \ll W_{\text{dep}}$), so the electron quasi-Fermi potential V and the potential ψ_s can be taken constant across it.

(iii) (Nondegenerate, local equilibrium) Electrons obey Boltzmann statistics with the local quasi-Fermi level, and the electron mobility μ_n is constant along the channel. The diffusion coefficient obeys the Einstein relation $D_n = \mu_n V_T$.

(iv) (Stationarity) The current is time-independent, and there is no recombination or generation in the channel, so the total current is the same through every slice.

(v) (Constant slope factor) C_{dep}, and hence n, is evaluated at one representative surface potential, as in Section 12.2.2.

12.3.1 Drift and Diffusion Currents in the Channel

The electron current density along the channel is the sum of a drift term and a diffusion term,

$$J_y = q_e \mu_n n\, \mathcal{E}_y + q_e D_n \frac{\partial n}{\partial y}, \qquad \mathcal{E}_y = -\frac{\partial \psi}{\partial y}. \tag{12.24}$$

Here J_y is the conventional current density in the $+y$ direction. Electrons drift opposite to the field, so the drift term carries a positive sign when written as conventional current. Integrating over the cross section of the channel, and defining the positive inversion charge magnitude

$$Q_{\text{i}}(y) := -Q_{\text{inv}}(y) = q_e \int_0^\infty n(x,y)\, \mathrm{d}x \;\geq 0, \tag{12.25}$$

the total conventional current flowing in the $+y$ direction is

$$I_y = W \int_0^\infty J_y \, \mathrm{d}x = W\mu_n \left[-Q_{\text{i}} \frac{\mathrm{d}\psi_s}{\mathrm{d}y} + V_T \frac{\mathrm{d}Q_{\text{i}}}{\mathrm{d}y} \right], \tag{12.26}$$

where we used the charge-sheet assumption to replace $\partial\psi/\partial y$ by $\frac{\mathrm{d}\psi_s}{\mathrm{d}y}$ inside the inversion layer, and the Einstein relation $D_n = \mu_n V_T$. The drain current I_{D} is defined as the conventional current entering the drain terminal and leaving through the source, that is, flowing in the $-y$ direction through the channel. Hence

$$I_{\text{D}} = -I_y = \underbrace{W\mu_n Q_{\text{i}} \frac{\mathrm{d}\psi_s}{\mathrm{d}y}}_{\text{drift}} \underbrace{- W\mu_n V_T \frac{\mathrm{d}Q_{\text{i}}}{\mathrm{d}y}}_{\text{diffusion}}. \tag{12.27}$$

Both terms are positive in normal operation: electrons drift toward the drain if the surface potential rises toward the drain, and they diffuse toward the drain if their density falls toward the drain. In terms of the signed inversion charge, the diffusion part reads $I_y^{\text{diff}} = -W\mu_n V_T \frac{\mathrm{d}Q_{\text{inv}}}{\mathrm{d}y}$.

There is an equivalent form in terms of the quasi-Fermi potential. From Eq. (12.2), $n = N_A e^{(\psi - 2\phi_F - V)/V_T}$, so $V_T \frac{\partial n}{\partial y} = n(\frac{\partial \psi}{\partial y} - \frac{\mathrm{d}V}{\mathrm{d}y})$. Substituting into Eq. (12.24),

$$J_y = -q_e \mu_n n \frac{\partial \psi}{\partial y} + q_e \mu_n n \left(\frac{\partial \psi}{\partial y} - \frac{\mathrm{d}V}{\mathrm{d}y} \right) = -q_e \mu_n n \frac{\mathrm{d}V}{\mathrm{d}y}, \tag{12.28}$$

and integrating over x,

$$I_{\mathrm{D}} = W \mu_n Q_{\mathrm{i}} \frac{\mathrm{d}V}{\mathrm{d}y}. \tag{12.29}$$

The quasi-Fermi form Eq. (12.29) contains drift and diffusion together. It says that the channel behaves locally like a resistor whose sheet conductance is $\mu_n Q_{\mathrm{i}}$, driven by the gradient of the electrochemical potential. We will use this form in Section 12.5.

12.3.2 Why the Lateral Field Is Small in Weak Inversion

Equation (12.27) has a drift and a diffusion part. Which one dominates? The lateral field is $\mathcal{E}_y = -\frac{\mathrm{d}\psi_s}{\mathrm{d}y}$, and ψ_s is tied to the local charges by the gate voltage equation. The answer is that the field is not exactly zero, but in weak inversion it is smaller than the diffusion driving force by the ratio of the inversion charge to $nC_{\mathrm{ox}}V_T$.

Proposition 12.9 (Drift-to-Diffusion Ratio)**.** *Under Assumption 12.8, with the gate voltage uniform along the channel,*

$$\frac{\mathrm{d}\psi_s}{\mathrm{d}y} = -\frac{1}{nC_{\mathrm{ox}}} \frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}, \tag{12.30}$$

and therefore the ratio of the drift current to the diffusion current at every point of the channel is

$$\frac{I_{\mathrm{drift}}}{I_{\mathrm{diff}}} = \frac{Q_{\mathrm{i}}(y)}{nC_{\mathrm{ox}}V_T}. \tag{12.31}$$

Proof. Write the gate voltage equation Eq. (12.12) at position y, with $Q_s = Q_{\mathrm{dep}}(\psi_s) - Q_{\mathrm{i}}$:

$$C_{\mathrm{ox}}(V_G - V_{\mathrm{FB}} - \psi_s(y)) = |Q_{\mathrm{dep}}(\psi_s(y))| + Q_{\mathrm{i}}(y). \tag{12.32}$$

The left side is the gate charge per unit area above slice y. Since V_G does not depend on y, differentiating with respect to y gives

$$-C_{\mathrm{ox}} \frac{\mathrm{d}\psi_s}{\mathrm{d}y} = C_{\mathrm{dep}} \frac{\mathrm{d}\psi_s}{\mathrm{d}y} + \frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}, \qquad \text{so} \qquad (C_{\mathrm{ox}} + C_{\mathrm{dep}}) \frac{\mathrm{d}\psi_s}{\mathrm{d}y} = -\frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}, \tag{12.33}$$

using $\frac{\mathrm{d}|Q_{\mathrm{dep}}|}{\mathrm{d}\psi_s} = C_{\mathrm{dep}}$. Dividing by $C_{\mathrm{ox}} + C_{\mathrm{dep}} = nC_{\mathrm{ox}}$ proves Eq. (12.30). Inserting it into the drift term of Eq. (12.27),

$$I_{\mathrm{drift}} = W \mu_n Q_{\mathrm{i}} \frac{\mathrm{d}\psi_s}{\mathrm{d}y} = -\frac{W \mu_n Q_{\mathrm{i}}}{nC_{\mathrm{ox}}} \frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}, \qquad I_{\mathrm{diff}} = -W \mu_n V_T \frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}. \tag{12.34}$$

Dividing the two gives Eq. (12.31). □

The physical content is a screening argument. The gate charge above each slice is fixed by $V_G - \psi_s$. If the inversion charge varies along the channel, the depletion charge and the surface potential must adjust to keep the slice neutral. A change δQ_{i} is absorbed by a change $\delta\psi_s = -\delta Q_{\mathrm{i}}/(nC_{\mathrm{ox}})$ in surface potential. When the inversion charge is tiny, so is the variation of ψ_s, and the lateral field is weak. The diffusion force, in contrast, depends on the *relative* variation of Q_{i}, which is of order one across the channel no matter how small Q_{i} is.

Definition 12.10 (Weak Inversion)**.** A channel point is in *weak inversion* when $Q_{\mathrm{i}} \ll nC_{\mathrm{ox}}V_T$. A transistor is in weak inversion when this holds at the source end, where Q_{i} is largest.

Take the numbers of Example 12.5. At the threshold point $\psi_s = 2\phi_F + V$, Eq. (12.9) gives $Q_{\mathrm{i}} = C_{\mathrm{dep}}V_T$, and the ratio Eq. (12.31) is $C_{\mathrm{dep}}/(nC_{\mathrm{ox}}) = (n-1)/n \approx 0.2$. So at the classical threshold the drift current is still about a fifth of the diffusion current. Each $nV_T \ln 10 \approx 75\,\mathrm{mV}$ of gate voltage below threshold reduces Q_{i} and the ratio by a factor of ten. At 150 mV below threshold the drift current is about 0.2% of the total. The statement "the lateral field vanishes in weak inversion" means exactly this: $\mathcal{E}_y$ is nonzero, but its contribution to the current is suppressed by $Q_{\mathrm{i}}/(nC_{\mathrm{ox}}V_T)$, which is exponentially small in $(V_{\mathrm{th}} - V_G)/(nV_T)$.

Example 12.11 (How Flat Is the Surface Potential?)**.** Integrating Eq. (12.30) from source to drain gives the total change of surface potential along the channel,

$$\psi_s(L) - \psi_s(0) = \frac{Q_{\mathrm{i}S} - Q_{\mathrm{i}D}}{nC_{\mathrm{ox}}}. \tag{12.35}$$

In saturation $Q_{\mathrm{i}D} \approx 0$. At the 1 nA bias point of Example 12.14 below, $Q_{\mathrm{i}S} = C_{\mathrm{dep}}V_T/60$, so

$$\psi_s(L) - \psi_s(0) = \frac{C_{\mathrm{dep}}}{nC_{\mathrm{ox}}} \cdot \frac{V_T}{60} = \frac{0.26}{1.26} \times \frac{25.85\,\mathrm{mV}}{60} \approx 0.09\,\mathrm{mV}. \tag{12.36}$$

The quasi-Fermi potential, by contrast, drops by the full $V_{\mathrm{DS}} \geq 100\,\mathrm{mV}$. Almost all of the electrochemical potential drop is a drop in carrier density, not in electrostatic potential. (Near the drain junction, where the channel ends, the two-dimensional field of the drain does bend ψ_s strongly; the gradual-channel model applies to the channel interior, and short channels need the correction of Section 12.4.4.)

Remark 12.12 (The Charge-Based Current Without Neglecting Drift)**.** Proposition 12.9 lets us integrate Eq. (12.27) exactly, without dropping drift. Adding the two parts of Eq. (12.34),

$$I_{\mathrm{D}} = -W\mu_n \left(\frac{Q_{\mathrm{i}}}{nC_{\mathrm{ox}}} + V_T \right) \frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}. \tag{12.37}$$

Since I_{D} does not depend on y, integrate from $y = 0$ to $y = L$:

$$I_{\mathrm{D}} = \frac{W\mu_n}{L} \left[\frac{Q_{\mathrm{i}S}^2 - Q_{\mathrm{i}D}^2}{2nC_{\mathrm{ox}}} + V_T(Q_{\mathrm{i}S} - Q_{\mathrm{i}D}) \right], \tag{12.38}$$

where $Q_{\mathrm{i}S} = Q_{\mathrm{i}}(0)$ and $Q_{\mathrm{i}D} = Q_{\mathrm{i}}(L)$. This is the charge-based form of the EKV model [29]. The quadratic term is drift, which dominates in strong inversion; the linear term is diffusion, which dominates in weak inversion. The ratio of the two terms is $(Q_{\mathrm{i}S} + Q_{\mathrm{i}D})/(2nC_{\mathrm{ox}}V_T)$, which is the local ratio Eq. (12.31) averaged over the two ends of the channel.

12.3.3 Integrating the Diffusion Current

In weak inversion we keep only the diffusion term:

$$I_{\mathrm{D}} = -W\mu_n V_T \frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}. \tag{12.39}$$

This is Fick's law for a sheet of electrons. Since I_{D} is the same in every slice, $\frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}$ is constant, and the inversion charge falls linearly from source to drain:

$$Q_{\mathrm{i}}(y) = Q_{\mathrm{i}S} - (Q_{\mathrm{i}S} - Q_{\mathrm{i}D})\frac{y}{L}. \tag{12.40}$$

This is the same linear minority-carrier profile found in the base of a bipolar transistor. Integrating Eq. (12.39) from 0 to L:

$$I_{\mathrm{D}}\,L = -W\mu_n V_T \int_0^L \frac{\mathrm{d}Q_{\mathrm{i}}}{\mathrm{d}y}\,\mathrm{d}y = W\mu_n V_T (Q_{\mathrm{i}S} - Q_{\mathrm{i}D}). \tag{12.41}$$

The current depends only on the inversion charges at the two ends. To leading order in weak inversion, the surface potential is the same at both ends (it differs by $\Delta\psi_s = (Q_{\mathrm{i}S} - Q_{\mathrm{i}D})/(nC_{\mathrm{ox}})$ by Eq. (12.30), which is negligible in the same approximation). By Proposition 12.3, the end charges are

$$Q_{\mathrm{i}S} = C_{\mathrm{dep}} V_T\, e^{(\psi_s - 2\phi_F)/V_T}\, e^{-V_S/V_T}, \qquad Q_{\mathrm{i}D} = C_{\mathrm{dep}} V_T\, e^{(\psi_s - 2\phi_F)/V_T}\, e^{-V_D/V_T}. \tag{12.42}$$

Substituting Eq. (12.42) into Eq. (12.41),

$$I_{\mathrm{D}} = \frac{W}{L}\,\mu_n C_{\mathrm{dep}} V_T^2\, e^{(\psi_s - 2\phi_F)/V_T}\left[e^{-V_S/V_T} - e^{-V_D/V_T}\right]. \tag{12.43}$$

Finally, insert the linearized surface potential Eq. (12.18), $\psi_s - 2\phi_F = (V_G - V_{\mathrm{th}})/n$. This gives the main result of the chapter.

Theorem 12.13 (Bulk-Referred EKV Law in Weak Inversion [29, 116])**.** *Under Assumption 12.8, in weak inversion (Definition 12.10) at both ends of the channel, the drain current of an n-channel MOSFET with bulk-referred terminal voltages V_G, V_S, V_D is*

$$I_{\mathrm{D}} = I_0 \exp\!\left(\frac{V_G - V_{\mathrm{th}}}{nV_T}\right)\left[\exp\!\left(-\frac{V_S}{V_T}\right) - \exp\!\left(-\frac{V_D}{V_T}\right)\right], \tag{12.44}$$

with the current scale

$$I_0 := \frac{W}{L}\,\mu_n C_{\mathrm{dep}} V_T^2 = \frac{W}{L}\,\mu_n (n-1) C_{\mathrm{ox}} V_T^2. \tag{12.45}$$

The current splits as $I_{\mathrm{D}} = I_F - I_R$ into a forward current *that depends only on the source and a* reverse current *that depends only on the drain:*

$$I_F := I_0\, e^{(V_G - V_{\mathrm{th}})/(nV_T)}\, e^{-V_S/V_T}, \qquad I_R := I_0\, e^{(V_G - V_{\mathrm{th}})/(nV_T)}\, e^{-V_D/V_T}. \tag{12.46}$$

Proof. Combine Eq. (12.43) with Eq. (12.18) and use $C_{\mathrm{dep}} = (n-1)C_{\mathrm{ox}}$ from Eq. (12.15). □

Figure 12.3 shows the physics of Theorem 12.13. The surface conduction band forms a flat barrier between the n^+ source and drain. Its height above the source Fermi level is $E_{b,S} = q_e(\mathrm{const} - \psi_s + V_S)$. The density of source electrons at the barrier, and hence I_F, is proportional to $e^{-E_{b,S}/k_{\mathrm{B}}T} \propto e^{\psi_s/V_T} e^{-V_S/V_T}$. The gate lowers the barrier through $\psi_s = \mathrm{const} + V_G/n$, giving the factor $e^{V_G/(nV_T)}$. The source voltage enters with full strength e^{-V_S/V_T} because it shifts the source Fermi level one-for-one. The drain does the same for I_R. This is exactly the bidirectional barrier of Fig. 11.2, with the gate controlling a common prefactor.

12.3.4 Saturation and Local Detailed Balance

Factor Eq. (12.44) as

$$I_{\mathrm{D}} = I_F(1 - e^{-V_{\mathrm{DS}}/V_T}), \qquad V_{\mathrm{DS}} = V_D - V_S. \tag{12.47}$$

For $V_{\mathrm{DS}} \gtrsim 4V_T \approx 100\,\mathrm{mV}$, the reverse current is less than $e^{-4} \approx 2\%$ of the forward current, and the drain current saturates at $I_{\mathrm{D}} \approx I_F$, independent of the drain voltage. This saturation voltage, about

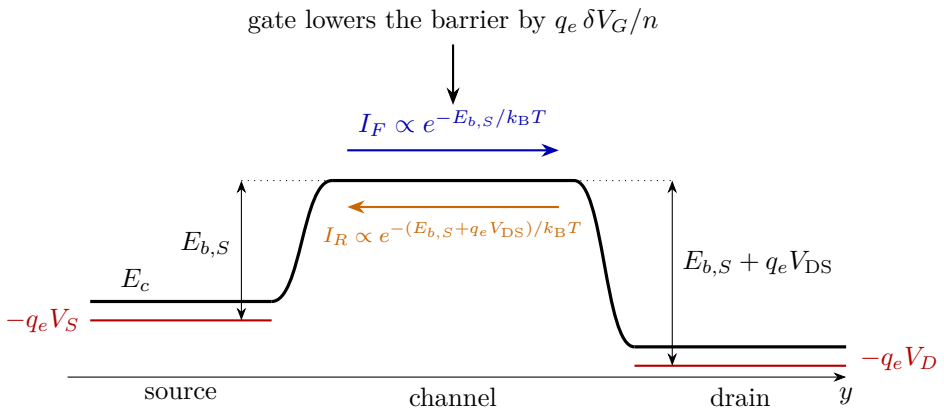


Figure 12.3: Conduction band edge at the silicon surface along the channel in weak inversion (electron energies increase upward). The surface potential is nearly flat along the channel, forming a barrier between source and drain. Electrons from the source that reach the barrier top diffuse to the drain (forward current I_F); electrons from the drain do the same in reverse (I_R). The gate moves the barrier top by $q_e\,\delta V_G/n$. Raising the drain voltage lowers the drain Fermi level and suppresses only I_R.

100 mV, is far smaller than in strong inversion, where saturation requires V_{DS} to exceed the gate overdrive. It is what allows subthreshold circuits to run from supplies of a few hundred millivolts.

The ratio of forward to reverse current is

$$\frac{I_F}{I_R} = \exp\!\left(\frac{V_D - V_S}{V_T}\right) = \exp\!\left(\frac{q_eV_{DS}}{k_BT}\right), \tag{12.48}$$

independent of the gate voltage and of all device parameters. This is the local detailed balance condition Eq. (11.82) with $V = V_{DS}$: one electron carried from source to drain releases free energy q_eV_{DS} into the reservoirs. Chapter 13 uses Eq. (12.48) to write the circuit master equation for networks of transistors (see Eq. (13.16)).

Physics Insight: Dissipation per Electron in a Subthreshold Channel

Local detailed balance turns the EKV law into a statement about entropy production. Treat the channel as two Poisson streams with rates $w_+ = I_F/q_e$ and $w_- = I_R/q_e$, as in Chapter 11. The mean entropy production rate of such a pair of jump processes is

$$\dot{\Sigma} = k_B\,(w_+ - w_-)\ln\frac{w_+}{w_-} = k_B\,\frac{I_D}{q_e}\cdot\frac{V_{DS}}{V_T} = \frac{I_DV_{DS}}{T}, \tag{12.49}$$

where we used $\ln(w_+/w_-) = V_{DS}/V_T$ from Eq. (12.48). The result is just the Joule heat I_DV_{DS} divided by T, as it must be. Per transported electron, the dissipated heat is $q_eV_{DS} = (V_{DS}/V_T)\,k_BT$. A saturated subthreshold transistor at $V_{DS} = 100$ mV dissipates about $3.9\,k_BT$ per electron. Raising V_{DS} further increases the dissipation per electron linearly, while the current and its noise, $2q_eI_D$, stay the same. In a sampling circuit, drain-source voltage beyond a few V_T buys nothing but heat, which is why subthreshold cells are run at the lowest supply that keeps every device saturated.

Heuristic Derivation: The Subthreshold MOSFET as a Gated Bipolar Transistor

In a bipolar transistor, the emitter injects minority carriers into a thin base with density $\propto e^{V_{BE}/V_T}$ at the emitter edge, the carriers diffuse across with a linear profile, and the collector removes them. The subthreshold MOSFET is the same device turned on its side: the source is the emitter, the weakly inverted surface is the base, and the drain is the collector. Two things differ. The base potential is set capacitively by the gate through the divider $1/n$, which is why the gate exponent is $1/(nV_T)$ instead of $1/V_T$. And the device is symmetric, so the drain also injects, producing I_R. With these two changes, Eq. (12.44) is the Ebers-Moll equation of a symmetric lateral bipolar transistor.

Example 12.14 (Size of the Current Scale)**.** Take $\mu_n = 0.04\,\mathrm{m}^2/(\mathrm{V\,s})$ (an effective surface mobility of $400\,\mathrm{cm}^2/(\mathrm{V\,s})$), $W/L = 1$, and $C_{\mathrm{dep}} = 2.3 \times 10^{-3}\,\mathrm{F/m}^2$ from Example 12.5. Then

$$I_0 = 0.04 \times 2.3 \times 10^{-3} \times (0.02585)^2\,\mathrm{A} = 6.1 \times 10^{-8}\,\mathrm{A} \approx 60\,\mathrm{nA}. \tag{12.50}$$

With $V_S = 0$ and $V_{\mathrm{DS}} \geq 4V_T$, a current of 1 nA requires $e^{(V_G - V_{\mathrm{th}})/(nV_T)} = 1/60$, that is, $V_G = V_{\mathrm{th}} - nV_T \ln 60 \approx V_{\mathrm{th}} - 133\,\mathrm{mV}$ for $n = 1.26$. The current density at that point is 1 nA per square, and the source-end ratio Eq. (12.31) is $(n-1)/(60n) \approx 0.3\%$, comfortably in weak inversion.

Notation Watch: Bulk vs. Source Reference, n vs. κ, and the Prefactor I_0

Several conventions for Eq. (12.44) coexist in the literature, and mixing them causes errors of factors n and e^{const}.

1. *Reference terminal.* This book refers V_G, V_S, V_D to the bulk, as in the EKV model [29]. Source-referred forms use V_{GS} and V_{SB} and carry the body-effect factor of Eq. (12.23). Gate, source, and drain appear symmetrically only in the bulk-referred form.

2. *Slope factor.* We use $n = 1 + C_{\mathrm{dep}}/C_{\mathrm{ox}} \geq 1$. Neuromorphic literature [75] writes $\kappa = 1/n$, so that $e^{(V_G - V_{\mathrm{th}})/(nV_T)} = e^{\kappa(V_G - V_{\mathrm{th}})/V_T}$.

3. *Prefactor.* We define $I_0 = (W/L)\mu_n C_{\mathrm{dep}} V_T^2$ with V_{th} the gate voltage at which $\psi_s = 2\phi_F$ (Definition 12.4). EKV instead uses the specific current $I_{\mathrm{spec}} = 2n\mu_n C_{\mathrm{ox}}(W/L)V_T^2$ together with a threshold at which the weak- and strong-inversion asymptotes meet. The two forms agree when the thresholds differ by $nV_T \ln[2n/(n-1)]$, about 70 mV for $n = 1.3$. Any measured device fixes only the product $I_0 e^{-V_{\mathrm{th}}/(nV_T)}$, so the split between I_0 and V_{th} is a convention.

4. *Direction labels.* Here "forward" current I_F means electrons injected by the source and collected by the drain, which is conventional current from drain to source. Chapter 11 calls this I_+, and Chapter 13 writes the corresponding jump rate as $w_+ = I_F/q_e$ (Eq. (13.15)).

5. *Polarity.* For a p-channel device all voltages are measured downward from the n-well, and the same formulas hold with $V \to -V$ and holes in place of electrons.

12.4 Small-Signal Conductances and the Transconductance Efficiency

A circuit uses a transistor by moving its terminal voltages by small amounts around a bias point and reading the resulting change in current. This section computes the three small-signal conductances of the weak-inversion MOSFET, proves that its transconductance per unit current is the largest that any operating point of a barrier-controlled channel can reach, connects weak and strong inversion through the EKV interpolation, and introduces the main short-channel correction, drain-induced barrier lowering.

12.4.1 Gate, Source, and Drain Conductances

Because the bulk-referred EKV law Eq. (12.44) depends on the three terminal voltages separately, a small change of each gives

$$\delta I_{\mathrm{D}} = g_{\mathrm{m}}\,\delta V_G - g_s\,\delta V_S + g_d\,\delta V_D, \tag{12.51}$$

with the gate transconductance and the source and drain conductances

$$g_{\mathrm{m}} := \frac{\partial I_{\mathrm{D}}}{\partial V_G} = \frac{I_F - I_R}{nV_T} = \frac{I_{\mathrm{D}}}{nV_T}, \qquad g_s := -\frac{\partial I_{\mathrm{D}}}{\partial V_S} = \frac{I_F}{V_T}, \qquad g_d := \frac{\partial I_{\mathrm{D}}}{\partial V_D} = \frac{I_R}{V_T}. \tag{12.52}$$

Each follows by differentiating one exponential in Eq. (12.44): $\partial I_F/\partial V_G = I_F/(nV_T)$, $\partial I_F/\partial V_S = -I_F/V_T$, and $\partial I_R/\partial V_D = -I_R/V_T$, with $I_{\mathrm{D}} = I_F - I_R$. Two exact relations follow:

$$g_s - g_d = \frac{I_F - I_R}{V_T} = n\,g_{\mathrm{m}}, \qquad \frac{g_d}{g_s} = \frac{I_R}{I_F} = e^{-V_{\mathrm{DS}}/V_T}. \tag{12.53}$$

The first identity is the small-signal form of the capacitive divider: moving the gate by δV has the same effect as moving both source and drain by $-\delta V/n$. In saturation ($V_{\mathrm{DS}} \gtrsim 4V_T$), $I_R \to 0$, so

$$g_{\mathrm{m}} = \frac{I_{\mathrm{D}}}{nV_T}, \qquad g_s = \frac{I_{\mathrm{D}}}{V_T} = n\,g_{\mathrm{m}}, \qquad g_d \approx 0. \tag{12.54}$$

The transconductance depends only on the bias current, not on the device geometry, mobility, or oxide thickness. Two transistors carrying the same current have the same g_{m}, whatever their size. In the triode (linear) region near $V_{\mathrm{DS}} = 0$, on the other hand, $I_F \approx I_R$ and the channel acts as a resistor with conductance

$$g_{\mathrm{ds0}} := \left.\frac{\partial I_{\mathrm{D}}}{\partial V_D}\right|_{V_D = V_S} = \left.\frac{I_R}{V_T}\right|_{V_D = V_S} = \frac{I_F}{V_T} = \frac{I_0}{V_T}\,e^{(V_G - V_{\mathrm{th}})/(nV_T)}\,e^{-V_S/V_T}. \tag{12.55}$$

This conductance is controlled exponentially by the gate. A subthreshold transistor at small V_{DS} is therefore a voltage-controlled resistor that can be tuned over many decades, which is how the sampling circuits of Chapter 13 realize programmable conductances.

Example 12.15 (A Gate-Tuned Resistor)**.** With $I_0 = 60\,\mathrm{nA}$ (Example 12.14), $V_S = V_D = 0$, and $n = 1.26$, the zero-bias conductance Eq. (12.55) is

$$g_{\mathrm{ds0}} = \frac{60\,\mathrm{nA}}{25.85\,\mathrm{mV}}\,e^{(V_G - V_{\mathrm{th}})/(nV_T)} = 2.3\,\mu\mathrm{S} \times e^{(V_G - V_{\mathrm{th}})/(32.6\,\mathrm{mV})}. \tag{12.56}$$

At $V_G = V_{\mathrm{th}} - 150\,\mathrm{mV}$ the exponential is $e^{-4.6} = 0.010$, so $g_{\mathrm{ds0}} = 23\,\mathrm{nS}$ ($R = 43\,\mathrm{M\Omega}$). Each further $75\,\mathrm{mV}$ of gate voltage changes R by a factor of ten. The device remains linear only for $|V_{\mathrm{DS}}| \ll V_T$; for larger drops the current saturates at I_F by Eq. (12.47).

12.4.2 The Boltzmann Bound on $g_{\mathrm{m}}/I_{\mathrm{D}}$

The ratio $g_{\mathrm{m}}/I_{\mathrm{D}}$, the transconductance efficiency, measures how much control a circuit obtains per unit of bias current. Since bias current times supply voltage is power, $g_{\mathrm{m}}/I_{\mathrm{D}}$ is also transconductance per unit power. In weak inversion it equals $1/(nV_T) \approx 31\,\mathrm{V}^{-1}$ for $n = 1.26$. We now prove that no operating point of the long-channel MOSFET does better. The argument needs one fact about the inversion charge that holds well beyond weak inversion.

Lemma 12.16 (Boltzmann Bound on the Inversion Charge)**.** *Let $Q_{\mathrm{i}}(\psi_s, V) = -(Q_s - Q_{\mathrm{dep}})$ with Q_s given by Eq. (12.7), which assumes only $\psi_s \gg V_T$ and not weak inversion. Then for all $\psi_s \geq V_T/2$,*

$$0 \leq C_{\mathrm{inv}} = \left.\frac{\partial Q_{\mathrm{i}}}{\partial \psi_s}\right|_V \leq \frac{Q_{\mathrm{i}}}{V_T}, \tag{12.57}$$

with $V_T C_{\mathrm{inv}}/Q_{\mathrm{i}} \to 1 - V_T/(2\psi_s) \approx 1$ in weak inversion.

Proof. Let $\Gamma = \sqrt{2q_e\varepsilon_{\mathrm{si}}N_A}$, $b = V_T e^{(\psi_s - 2\phi_F - V)/V_T}$, $r = \sqrt{\psi_s}$, and $s = \sqrt{\psi_s + b}$, so $s > r > 0$, $Q_{\mathrm{i}} = \Gamma(s - r)$, and $\frac{\partial b}{\partial \psi_s} = b/V_T$. Then

$$\frac{C_{\mathrm{inv}}}{\Gamma} = \frac{1 + b/V_T}{2s} - \frac{1}{2r} = -\frac{s - r}{2rs} + \frac{b}{2V_T s} = (s - r)\left[-\frac{1}{2rs} + \frac{s + r}{2V_T s}\right], \tag{12.58}$$

where we wrote $\frac{1}{2s} - \frac{1}{2r} = -\frac{s-r}{2rs}$ and $b = s^2 - r^2 = (s - r)(s + r)$. Dividing by $Q_{\mathrm{i}}/\Gamma = s - r > 0$,

$$\frac{V_T\, C_{\mathrm{inv}}}{Q_{\mathrm{i}}} = \frac{s + r}{2s} - \frac{V_T}{2rs}. \tag{12.59}$$

The first term is at most 1 because $r < s$, and the second is negative, so $V_T C_{\mathrm{inv}}/Q_{\mathrm{i}} < 1$. For the lower bound, the first form in Eq. (12.58) shows that $C_{\mathrm{inv}} \geq 0$ is equivalent to $(1 + b/V_T)\, r \geq s$, which holds since $(1 + b/V_T)^2 r^2 \geq (1 + 2b/V_T)\psi_s \geq \psi_s + b = s^2$ whenever $\psi_s \geq V_T/2$. In weak inversion $b \ll \psi_s$, so $s \to r$, and Eq. (12.59) tends to $1 - V_T/(2\psi_s) \approx 1$ for $\psi_s \gg V_T$. □

The lemma says that the inversion charge can never grow faster with surface potential than the Boltzmann factor e^{ψ_s/V_T}. It grows exactly that fast when the electrons are a dilute, unscreened gas, and more slowly once they screen the field that holds them.

Proposition 12.17 (Maximum Transconductance Efficiency)**.** *Under Assumption 12.8 items (i)–(iv), at any bias with $V_D \geq V_S$ and any level of inversion, the drain current and transconductance satisfy*

$$\frac{g_{\mathrm{m}}}{I_{\mathrm{D}}} \leq \frac{1}{n_{\min} V_T}, \tag{12.60}$$

where $n_{\min}$ is the smallest value of $1 + C_{\mathrm{dep}}(\psi_s)/C_{\mathrm{ox}}$ along the channel. Equality holds in the limit of weak inversion with constant n.

Proof. From the quasi-Fermi form Eq. (12.29), $I_{\mathrm{D}}\, \mathrm{d}y = W\mu_n Q_{\mathrm{i}}\, \mathrm{d}V$. In the gradual-channel picture, Q_{i} at a point depends only on V_G and the local channel voltage V. Integrating from source to drain,

$$I_{\mathrm{D}} = \frac{W\mu_n}{L} \int_{V_S}^{V_D} Q_{\mathrm{i}}(V_G, V)\, \mathrm{d}V. \tag{12.61}$$

Differentiating under the integral sign, $g_{\mathrm{m}} = \frac{W\mu_n}{L} \int_{V_S}^{V_D} \left.\frac{\partial Q_{\mathrm{i}}}{\partial V_G}\right|_V \mathrm{d}V$. At fixed V, the chain rule and Eq. (12.16) give

$$\left.\frac{\partial Q_{\mathrm{i}}}{\partial V_G}\right|_V = C_{\mathrm{inv}} \left.\frac{\partial \psi_s}{\partial V_G}\right|_V = \frac{C_{\mathrm{inv}}\, C_{\mathrm{ox}}}{C_{\mathrm{ox}} + C_{\mathrm{dep}} + C_{\mathrm{inv}}} \leq \frac{C_{\mathrm{inv}}}{n(\psi_s)} \leq \frac{Q_{\mathrm{i}}}{n(\psi_s) V_T} \leq \frac{Q_{\mathrm{i}}}{n_{\min} V_T}, \tag{12.62}$$

where the first inequality drops $C_{\text{inv}} \geq 0$ from the denominator and the second uses Lemma 12.16. Integrating Eq. (12.62) over $V \in [V_S, V_D]$ and comparing with Eq. (12.61) gives $g_{\text{m}} \leq I_{\text{D}}/(n_{\min} V_T)$. In weak inversion with constant n, $C_{\text{inv}} \ll C_{\text{dep}}$ and $V_T C_{\text{inv}}/Q_{\text{i}} \to 1$ (up to the relative correction $V_T/(2\psi_s)$ from the ψ_s-dependence of C_{dep}, which assumption (v) neglects), all three inequalities in Eq. (12.62) become equalities, and $g_{\text{m}} = I_{\text{D}}/(nV_T)$ as in Eq. (12.52). □

The first two inequalities in the proof are two separate physical losses. The first is the screening loss: once there is enough inversion charge, C_{inv} appears in the divider of Fig. 12.2 and the surface potential stops following the gate. The second is the self-screening of the electron sheet: once the electrons carry a noticeable part of the surface charge, they weaken the field that confines them, and their number grows more slowly than the Boltzmann factor, as in the lemma. Neither loss is present in weak inversion. The remaining factor $1/n$ is the irreducible loss of the divider, which only a thinner oxide (larger C_{ox}) or a lighter-doped body (smaller C_{dep}) can reduce.

12.4.3 Moderate Inversion and the EKV Interpolation

To see how fast $g_{\text{m}}/I_{\text{D}}$ falls as the device leaves weak inversion, we use the EKV charge linearization [29]. At fixed V_G, the gate balance gives $(C_{\text{ox}} + C_{\text{dep}})\,\mathrm{d}\psi_s = -\,\mathrm{d}Q_{\text{i}}$, exactly as in Eq. (12.33). Extending the Boltzmann relation of Eq. (12.9) to all inversion levels, $\mathrm{d}Q_{\text{i}}/Q_{\text{i}} = (\mathrm{d}\psi_s - \mathrm{d}V)/V_T$. Eliminating $\mathrm{d}\psi_s$:

$$\mathrm{d}V = -\left(\frac{1}{nC_{\text{ox}}} + \frac{V_T}{Q_{\text{i}}}\right)\mathrm{d}Q_{\text{i}}. \tag{12.63}$$

Introduce the normalized charge $q := Q_{\text{i}}/(2nC_{\text{ox}}V_T)$. Then Eq. (12.63) reads $\mathrm{d}V = -V_T\,(2+1/q)\,\mathrm{d}q$, which integrates to the EKV charge-voltage relation

$$V_P - V = V_T(2q + \ln q), \tag{12.64}$$

where the integration constant V_P is the *pinch-off voltage*. In weak inversion ($q \ll 1$), $q \approx e^{(V_P - V)/V_T}$, which reproduces the Boltzmann dependence of Eq. (12.9). In strong inversion ($q \gg 1$), $Q_{\text{i}} \approx nC_{\text{ox}}(V_P - V)$, the familiar linear charge of the square-law model. In both limits V_P depends linearly on the gate with slope $1/n$, and the EKV model takes $V_P = (V_G - V_{\text{th}}^{\text{EKV}})/n$ throughout.

Written in terms of q, the charge-based current Eq. (12.38) becomes

$$I_{\text{D}} = I_{\text{spec}}\Big[(q_S^2 + q_S) - (q_D^2 + q_D)\Big], \qquad I_{\text{spec}} := 2n\mu_n C_{\text{ox}}\frac{W}{L}V_T^2, \tag{12.65}$$

since $Q_{\text{i}}^2/(2nC_{\text{ox}}) = 2nC_{\text{ox}}V_T^2 q^2$ and $V_T Q_{\text{i}} = 2nC_{\text{ox}}V_T^2 q$. In saturation $q_D \to 0$ and the *inversion coefficient* is $\text{IC} := I_{\text{D}}/I_{\text{spec}} = q_S^2 + q_S$. Differentiating Eq. (12.64) at the source with $\partial V_P/\partial V_G = 1/n$ gives $\partial q_S/\partial V_G = q_S/(nV_T(2q_S+1))$, so

$$g_{\text{m}} = I_{\text{spec}}(2q_S+1)\frac{\partial q_S}{\partial V_G} = \frac{I_{\text{spec}}\,q_S}{nV_T}, \qquad \frac{g_{\text{m}}}{I_{\text{D}}} = \frac{1}{nV_T}\,\frac{1}{1+q_S} = \frac{1}{nV_T}\,\frac{2}{1+\sqrt{1+4\,\text{IC}}}, \tag{12.66}$$

where the last step solves $q_S^2 + q_S = \text{IC}$ for $q_S = (\sqrt{1+4\,\text{IC}} - 1)/2$. The efficiency is 0.92 of its maximum at $\text{IC} = 0.1$, 0.62 at $\text{IC} = 1$ (moderate inversion), and 0.27 at $\text{IC} = 10$. In strong inversion it approaches $1/(nV_T\sqrt{\text{IC}})$, which is the square-law result $g_{\text{m}}/I_{\text{D}} = 2/(n(V_P - V_S))$.

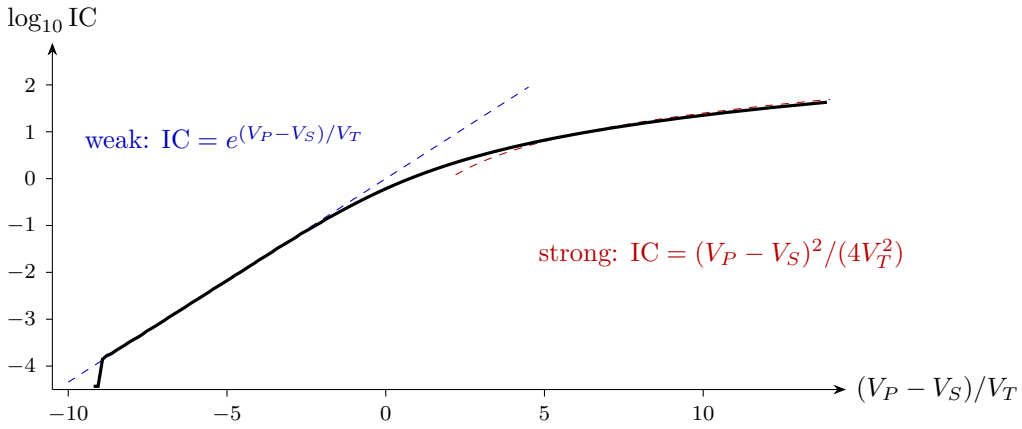


Figure 12.4: Saturation current in units of I_{spec} (the inversion coefficient) against the pinch-off voltage, from the EKV charge relations Eqs. (12.64) and (12.65) with $q_D = 0$ (solid). The dashed lines are the weak-inversion exponential and the strong-inversion square law. Since $V_P = (V_G - V_{\text{th}}^{\text{EKV}})/n$, the slope of the left branch is one decade per $nV_T \ln 10$ of gate voltage, the subthreshold swing Eq. (12.20). Moderate inversion is the region $0.1 \lesssim \text{IC} \lesssim 10$.

Physics Insight: Efficiency Against Speed

Weak inversion maximizes $g_{\text{m}}/I_{\text{D}}$ but at low current density. For a fixed current, a device in weak inversion must be wide, so its gate capacitance is large, and its intrinsic speed $g_{\text{m}}/C_{\text{gate}}$ is low. Strong inversion gives the opposite trade. For thermodynamic sampling circuits the relevant capacitance is usually not the transistor's own gate but the node capacitance C_{node} it drives, and the relevant figure is the relaxation time $C_{\text{node}}/g_{\text{m}} = nV_T C_{\text{node}}/I_{\text{D}}$. At fixed C_{node} and fixed relaxation time, weak inversion uses the least current, which is why the CMOS samplers of Chapters 13 and 17 are biased there.

12.4.4 Drain-Induced Barrier Lowering and Output Conductance

In a long channel, the drain affects the current only through I_R, and in saturation the drain conductance $g_d = I_R/V_T$ vanishes exponentially. In a short channel the drain is close enough to the source that its field reaches the barrier top in Fig. 12.3 and lowers it. This is *drain-induced barrier lowering* (DIBL). It is a two-dimensional electrostatic effect, a failure of the gradual-channel assumption (i), and we model it phenomenologically by letting the effective threshold depend on the drain voltage:

$$V_{\text{th}} \;\to\; V_{\text{th}} - \eta\, V_D, \qquad \eta = -\frac{\partial V_{\text{th}}}{\partial V_D} > 0, \tag{12.67}$$

where the dimensionless DIBL coefficient η is typically quoted in mV/V. Representative values are a few mV/V for long analog devices and tens of mV/V for minimum-length devices in advanced nodes. Because the barrier is common to both directions, DIBL multiplies both I_F and I_R by the same factor $e^{\eta V_D/(nV_T)}$, and the local detailed balance ratio Eq. (12.48) is unchanged. This must be so: the ratio of forward to reverse rates is fixed by thermodynamics, not by the barrier shape. In saturation,

$$g_d = \frac{\partial I_{\text{D}}}{\partial V_D} = \frac{\eta\, I_{\text{D}}}{nV_T} = \eta\, g_{\text{m}}, \qquad A_{\text{i}} := \frac{g_{\text{m}}}{g_d} = \frac{1}{\eta}. \tag{12.68}$$

The intrinsic voltage gain A_{i} is independent of the bias current, a well-known property of weak inversion. For $\eta = 0.02$ (that is, 20 mV/V) the intrinsic gain is 50. Channel-length modulation adds a second contribution $g_d \approx I_{\mathrm{D}}/V_E$ with an Early voltage V_E proportional to L; both contributions are proportional to I_{D}, so the gain remains bias-independent. Analog subthreshold circuits use channel lengths several times the minimum to keep η small.

12.5 Intrinsic Current Noise of the Subthreshold MOSFET

The EKV law describes the mean current. The channel, however, carries a finite number of electrons, each of which moves by a random walk, so the current fluctuates. This section computes the spectrum of these fluctuations. The result connects Theorem 12.13 to the bidirectional Poisson model of Section 11.4.

12.5.1 Two Pictures of the Same Noise

Theorem 12.13 splits the drain current into a forward current I_F, set by the source, and a reverse current I_R, set by the drain, with ratio $I_F/I_R = e^{V_{\mathrm{DS}}/V_T}$. If we identify $I_+ = I_F$ and $I_- = I_R$ and treat each as an independent Poisson stream of electrons crossing the channel, Theorem 11.10 immediately gives the single-sided current noise

$$S_I(f) = 2q_e(I_F + I_R). \tag{12.69}$$

This is the picture of Sarpeshkar et al. [93]. It is not obviously correct. Electrons in the channel do not hop from source to drain in one step; they diffuse with many collisions, and an electron that enters the channel from the source may return to the source without ever reaching the drain. To check Eq. (12.69), we compute the noise a second way, starting from the local thermal noise of the channel material and integrating it along the channel, as in the classical treatment of long-channel MOSFET noise [111].

Assumption 12.18 (Local Thermal Noise of the Channel)**.** In addition to Assumption 12.8:

(vi) Divide the channel into segments of length Δy, short compared with L but long compared with the electron mean free path. Each segment is in local equilibrium at temperature T, with conductance $g_{\mathrm{seg}}(y) = W\mu_n Q_{\mathrm{i}}(y)/\Delta y$ from Eq. (12.29). It carries a Norton noise current $\delta i(y)$ in parallel with g_{seg}, with the Johnson-Nyquist single-sided PSD $4k_{\mathrm{B}}Tg_{\mathrm{seg}}(y)$ of Theorem 11.5.

(vii) Noise sources in different segments are uncorrelated.

(viii) The frequency f is small compared with the inverse transit time of the channel, so the channel responds quasi-statically.

(ix) The source, drain, gate, and bulk terminals are held at fixed voltages (a short circuit for the noise current).

Item (vi) applies the equilibrium fluctuation-dissipation relation to a small piece of a channel that, as a whole, may be far from equilibrium. This is justified when each segment is thin enough that the drop in quasi-Fermi level across it is much smaller than V_T, which is always possible by taking Δy small.

12.5.2 The Channel Noise Theorem

Theorem 12.19 (Subthreshold Channel Noise [93])**.** *Under Assumptions 12.8 and 12.18, the single-sided PSD of the drain current fluctuations is*

$$S_I(f) = \frac{4k_{\mathrm{B}}T\,W\mu_n}{L^2}\int_0^L Q_{\mathrm{i}}(y)\,\mathrm{d}y = \frac{4k_{\mathrm{B}}T\,W^2\mu_n^2}{L^2\,I_{\mathrm{D}}}\int_{V_S}^{V_D} Q_{\mathrm{i}}(V)^2\,\mathrm{d}V, \tag{12.70}$$

valid at any level of inversion. In weak inversion it reduces to

$$S_I(f) = 2q_e(I_F + I_R) = 2q_e I_{\mathrm{D}}\coth\!\left(\frac{V_{\mathrm{DS}}}{2V_T}\right), \tag{12.71}$$

with I_F and I_R the forward and reverse currents of Eq. (12.46).

Proof. Step 1: response of the terminal current to a local voltage jump. Insert a small voltage source δv in series with the channel at position y_0, so that the quasi-Fermi potential jumps from V_1 to $V_1+\delta v$ across it. The terminal voltages are fixed, and the current $I_{\mathrm{D}}+\delta I$ is the same on both sides of y_0. Applying $I_{\mathrm{D}}\,\mathrm{d}y = W\mu_n Q_{\mathrm{i}}\,\mathrm{d}V$ to each side and integrating,

$$(I_{\mathrm{D}}+\delta I)\,y_0 = W\mu_n\int_{V_S}^{V_1} Q_{\mathrm{i}}\,\mathrm{d}V, \qquad (I_{\mathrm{D}}+\delta I)(L-y_0) = W\mu_n\int_{V_1+\delta v}^{V_D} Q_{\mathrm{i}}\,\mathrm{d}V. \tag{12.72}$$

Adding the two equations,

$$(I_{\mathrm{D}}+\delta I)\,L = W\mu_n\left[\int_{V_S}^{V_D} Q_{\mathrm{i}}\,\mathrm{d}V - \int_{V_1}^{V_1+\delta v} Q_{\mathrm{i}}\,\mathrm{d}V\right] = I_{\mathrm{D}}L - W\mu_n Q_{\mathrm{i}}(y_0)\,\delta v + \mathcal{O}(\delta v^2), \tag{12.73}$$

where we used Eq. (12.61) for the unperturbed current. Hence

$$\delta I = -\frac{W\mu_n Q_{\mathrm{i}}(y_0)}{L}\,\delta v. \tag{12.74}$$

Step 2: from noise current to terminal current. A Norton noise current δi in parallel with the segment conductance g_{seg} is equivalent to a Thévenin voltage $\delta v = \delta i/g_{\mathrm{seg}} = \delta i\,\Delta y/(W\mu_n Q_{\mathrm{i}})$ in series with it. Substituting into Eq. (12.74),

$$\delta I = -\frac{W\mu_n Q_{\mathrm{i}}(y_0)}{L}\cdot\frac{\Delta y}{W\mu_n Q_{\mathrm{i}}(y_0)}\,\delta i = -\frac{\Delta y}{L}\,\delta i. \tag{12.75}$$

Each segment's noise reaches the terminals reduced by the fraction $\Delta y/L$ of the channel it occupies, independent of where it sits and of the local charge.

Step 3: summing independent segments. By item (vii) the contributions add in power. Using the segment PSD $4k_{\mathrm{B}}Tg_{\mathrm{seg}}$,

$$\begin{aligned} S_I(f) &= \sum_{\text{segments}}\left(\frac{\Delta y}{L}\right)^2 4k_{\mathrm{B}}T\,\frac{W\mu_n Q_{\mathrm{i}}(y)}{\Delta y} = \frac{4k_{\mathrm{B}}T\,W\mu_n}{L^2}\sum_{\text{segments}} Q_{\mathrm{i}}(y)\,\Delta y \\ &\longrightarrow \frac{4k_{\mathrm{B}}T\,W\mu_n}{L^2}\int_0^L Q_{\mathrm{i}}(y)\,\mathrm{d}y. \end{aligned} \tag{12.76}$$

This is the first form of Eq. (12.70). The second follows from the change of variables $\mathrm{d}y = W\mu_n Q_{\mathrm{i}}\,\mathrm{d}V/I_{\mathrm{D}}$.

Step 4: weak inversion. In weak inversion $Q_{\mathrm{i}}(y)$ is linear in y (Eq. (12.40)), so

$$\int_0^L Q_{\mathrm{i}}(y)\,\mathrm{d}y = \frac{L}{2}(Q_{\mathrm{i}S} + Q_{\mathrm{i}D}). \tag{12.77}$$

From Eq. (12.41) and Eq. (12.46), the forward and reverse currents are $I_F = (W\mu_n/L)V_T Q_{\mathrm{i}S}$ and $I_R = (W\mu_n/L)V_T Q_{\mathrm{i}D}$. Therefore

$$S_I(f) = \frac{4k_{\mathrm{B}}T\,W\mu_n}{L^2}\cdot\frac{L}{2}(Q_{\mathrm{i}S}+Q_{\mathrm{i}D}) = \frac{2k_{\mathrm{B}}T}{V_T}\cdot\frac{W\mu_n V_T}{L}(Q_{\mathrm{i}S}+Q_{\mathrm{i}D}) = 2q_e(I_F+I_R), \tag{12.78}$$

using $k_{\mathrm{B}}T/V_T = q_e$. The coth form follows from $I_R = I_F e^{-V_{\mathrm{DS}}/V_T}$ exactly as in the proof of Theorem 11.11. □

The theorem shows that the two pictures agree. Integrating the equilibrium thermal noise of every small piece of a diffusive channel gives exactly the shot noise of two independent streams of electrons crossing the whole channel. The Poisson model of Chapter 11 is therefore a valid coarse-grained description of the subthreshold channel at frequencies below the inverse transit time, even though no electron crosses the channel in one jump. The agreement depends on weak inversion: the linear profile Eq. (12.40) is what turns the integral of Q_{i} into the sum of the end charges.

Remark 12.20 (Equilibrium Check at Any Inversion Level)**.** The general formula Eq. (12.70) passes the fluctuation-dissipation test at every inversion level, not only in weak inversion. At $V_D = V_S$ the channel is uniform, $Q_{\mathrm{i}}(y) = Q_{\mathrm{i}}$, so $\int_0^L Q_{\mathrm{i}}\,\mathrm{d}y = LQ_{\mathrm{i}}$ and $S_I = 4k_{\mathrm{B}}T\,W\mu_n Q_{\mathrm{i}}/L$. From Eq. (12.61), the zero-bias conductance is $g_{\mathrm{ds0}} = \partial I_{\mathrm{D}}/\partial V_D|_{V_D=V_S} = W\mu_n Q_{\mathrm{i}}/L$. Hence $S_I = 4k_{\mathrm{B}}T g_{\mathrm{ds0}}$ exactly. This is a check of item (vi) of Assumption 12.18: assigning Johnson-Nyquist noise to each segment is consistent with Johnson-Nyquist noise for the whole channel.

Remark 12.21 (Strong Inversion for Comparison)**.** In strong-inversion saturation, $Q_{\mathrm{i}}(V) = nC_{\mathrm{ox}}(V_P - V)$ for $V_S \le V \le V_P$ (Section 12.4.3), and the channel pinches off at $V = V_P$. Then

$$\int_{V_S}^{V_P} Q_{\mathrm{i}}^2\,\mathrm{d}V = \frac{n^2C_{\mathrm{ox}}^2(V_P-V_S)^3}{3}, \qquad I_{\mathrm{D}} = \frac{W\mu_n}{L}\cdot\frac{nC_{\mathrm{ox}}(V_P-V_S)^2}{2}, \qquad g_{\mathrm{m}} = \frac{W\mu_n}{L}C_{\mathrm{ox}}(V_P-V_S). \tag{12.79}$$

Inserting these into Eq. (12.70),

$$S_I(f) = \frac{4k_{\mathrm{B}}TW^2\mu_n^2}{L^2}\cdot\frac{n^2C_{\mathrm{ox}}^2(V_P-V_S)^3/3}{(W\mu_n/L)\,nC_{\mathrm{ox}}(V_P-V_S)^2/2} = 4k_{\mathrm{B}}T\cdot\frac{2}{3}\,n\,g_{\mathrm{m}}. \tag{12.80}$$

In weak inversion, Eq. (12.71) in saturation gives $S_I = 2q_eI_{\mathrm{D}} = 2q_enV_Tg_{\mathrm{m}} = 4k_{\mathrm{B}}T\cdot\frac{1}{2}\,n\,g_{\mathrm{m}}$. Writing $S_I = 4k_{\mathrm{B}}T\,\gamma_{\mathrm{n}}\,n\,g_{\mathrm{m}}$, the channel noise factor γ_{n} moves from $1/2$ in weak inversion to $2/3$ in strong inversion [29, 111].

12.5.3 Equilibrium and Saturation Limits

Corollary 12.22 (Johnson-Nyquist and Shot-Noise Limits of Channel Noise)**.** *Under the hypotheses of Theorem 12.19, in weak inversion:*

(i) at $V_{\mathrm{DS}} = 0$ the channel noise is the Johnson-Nyquist noise of the channel conductance,

$$S_I(f) = 4q_eI_F = 4k_{\mathrm{B}}T\,g_{\mathrm{ds0}}; \tag{12.81}$$

(ii) in saturation ($V_{\mathrm{DS}} \gg V_T$) the channel noise is full shot noise of the drain current,

$$S_I(f) = 2q_e I_{\mathrm{D}}(1 + 2e^{-V_{\mathrm{DS}}/V_T} + \cdots) \longrightarrow 2q_e I_{\mathrm{D}}. \tag{12.82}$$

Proof. (i) At $V_D = V_S$, $I_R = I_F$, so Eq. (12.71) gives $S_I = 2q_e(2I_F) = 4q_e I_F$. By Eq. (12.55), $g_{\mathrm{ds0}} = I_F/V_T$, so $4k_{\mathrm{B}}Tg_{\mathrm{ds0}} = 4k_{\mathrm{B}}TI_F/V_T = 4q_e I_F$. (ii) Write $I_F + I_R = I_{\mathrm{D}}(I_F + I_R)/(I_F - I_R) = I_{\mathrm{D}}(1 + e^{-V_{\mathrm{DS}}/V_T})/(1 - e^{-V_{\mathrm{DS}}/V_T})$ and expand $1/(1-u) = 1 + u + \cdots$ for $u = e^{-V_{\mathrm{DS}}/V_T}$: $(1+u)(1+u+\cdots) = 1 + 2u + \mathcal{O}(u^2)$. □

Part (i) is the consistency check that equilibrium statistical mechanics demands: with no applied bias, the transistor is a passive resistor in thermal equilibrium, and its noise must obey the fluctuation-dissipation theorem whatever the microscopic transport mechanism. Part (ii) is the regime used for transconductance. At $V_{\mathrm{DS}} = 4V_T$ the correction factor is $1 + 2e^{-4} = 1.037$, so the saturation noise is within 4% of $2q_e I_{\mathrm{D}}$. Between the two limits the noise follows the coth crossover of Theorem 11.11. In saturation the transistor is far from equilibrium: its noise $2q_e I_{\mathrm{D}}$ is set by the standing current, while its output conductance g_d is nearly zero. A saturated transistor is a current source with shot noise, not a resistor with Johnson noise. This is the microscopic origin of the second detailed-balance-breaking mechanism of Theorem 13.6.

Heuristic Derivation: Fano Factor and the Precision-Dissipation Trade-Off

The Fano factor of the channel, the noise in units of full shot noise, follows from Eq. (12.71):

$$F := \frac{S_I}{2q_e I_{\mathrm{D}}} = \coth\!\left(\frac{V_{\mathrm{DS}}}{2V_T}\right) \geq 1, \tag{12.83}$$

as in Worked computation 11.14. A weakly inverted channel is never quieter than Poisson: the reverse stream only adds noise while it subtracts current. This fits the thermodynamic uncertainty relation [8]. Over a long time t, the transported charge Q_t has mean $I_{\mathrm{D}}t$ and variance $\frac{1}{2}S_I t = q_e(I_F + I_R)t$, and the total entropy produced is $\Sigma_t = I_{\mathrm{D}}V_{\mathrm{DS}}t/T$ by Eq. (12.49). With $x = V_{\mathrm{DS}}/(2V_T)$,

$$\frac{\mathrm{Var}(Q_t)}{\langle Q_t\rangle^2} = \frac{q_e \coth x}{I_{\mathrm{D}}t}, \qquad \frac{2k_{\mathrm{B}}}{\Sigma_t} = \frac{2k_{\mathrm{B}}T}{I_{\mathrm{D}}V_{\mathrm{DS}}t} = \frac{q_e}{I_{\mathrm{D}}t\,x}, \tag{12.84}$$

so the uncertainty relation $\mathrm{Var}(Q_t)/\langle Q_t\rangle^2 \geq 2k_{\mathrm{B}}/\Sigma_t$ reduces to $x \coth x \geq 1$, which holds for all x and becomes an equality as $x \to 0$. A precise current needs either many electrons or a large dissipation per electron; in saturation the transistor uses the first route.

Notation Watch: One-Sided Spectra and Diffusion Coefficients for Channel Noise

Following Section 11.1, $S_I(f)$ always denotes the single-sided PSD, with $\mathrm{Var}(I) = \int_0^\infty S_I(f)\,\mathrm{d}f$. The double-sided spectrum is half as large, $S_I^{\mathrm{ds}}(\omega) = q_e(I_F + I_R)$, and the charge diffusion coefficient in the Itô form $\mathrm{d}Q = I_{\mathrm{D}}\,\mathrm{d}t + \sqrt{2D_I}\,\mathrm{d}W_t$ is $D_I = \frac{1}{2}S_I^{\mathrm{ds}} = \frac{1}{2}q_e(I_F + I_R)$. Thus:

- saturated channel: $S_I = 2q_e I_{\mathrm{D}}$, $S_I^{\mathrm{ds}} = q_e I_{\mathrm{D}}$, $D_I = \frac{1}{2}q_e I_{\mathrm{D}}$;
- channel at $V_{\mathrm{DS}} = 0$: $S_I = 4k_{\mathrm{B}}Tg_{\mathrm{ds0}}$, $S_I^{\mathrm{ds}} = 2k_{\mathrm{B}}Tg_{\mathrm{ds0}}$, $D_I = k_{\mathrm{B}}Tg_{\mathrm{ds0}}$.

Chapter 13 assigns each transistor channel ρ the double-sided strength $S_\rho = q_e(I_{\mathrm{D}})_\rho$ in saturation and sums $D_q = \frac{1}{2}\sum_\rho S_\rho$ (Eq. (13.66)); this is the same convention. Input-referred

spectra $S_V(f)$ below are also single-sided.

12.5.4 Input-Referred Gate Noise

A circuit designer usually compares noise with signals at the gate. Dividing the drain current noise by g_{m}^2 gives the gate voltage noise that would produce the same output fluctuation in a noiseless device:

$$S_{V_G}(f) := \frac{S_I(f)}{g_{\mathrm{m}}^2} = \frac{2q_e(I_F + I_R)\, n^2 V_T^2}{(I_F - I_R)^2}. \tag{12.85}$$

In saturation, with $g_{\mathrm{m}} = I_{\mathrm{D}}/(nV_T)$,

$$S_{V_G}(f) = \frac{2q_e I_{\mathrm{D}}}{g_{\mathrm{m}}^2} = \frac{2q_e\, n^2 V_T^2}{I_{\mathrm{D}}} = \frac{2nk_{\mathrm{B}}T}{g_{\mathrm{m}}}. \tag{12.86}$$

The last form follows from $2q_e n^2 V_T^2/I_{\mathrm{D}} = 2q_e nV_T \cdot (nV_T/I_{\mathrm{D}}) = 2nk_{\mathrm{B}}T/g_{\mathrm{m}}$. Comparing with the Johnson-Nyquist voltage noise $4k_{\mathrm{B}}TR$ of a resistor, the transistor's gate noise equals that of a resistor $R_{\mathrm{n}} = n/(2g_{\mathrm{m}})$. The ratio

$$\frac{S_I}{g_{\mathrm{m}}} = 2nk_{\mathrm{B}}T \tag{12.87}$$

is fixed by temperature and the slope factor alone (Eq. (11.92)). It is the effective-temperature relation that Chapter 13 uses: a saturated subthreshold transconductor that acts as a conductance g_{m} injects noise as if it were a resistor at temperature $nT/2$ in the Norton sense $S_I = 4k_{\mathrm{B}}(nT/2)g_{\mathrm{m}}$.

Example 12.23 (Gate Noise at Nanoampere Currents)**.** Take $n = 1.3$ and $T = 300\,\mathrm{K}$. At $I_{\mathrm{D}} = 1\,\mathrm{nA}$, $g_{\mathrm{m}} = I_{\mathrm{D}}/(nV_T) = 29.8\,\mathrm{nS}$, and

$$S_{V_G} = \frac{2 \times 1.602 \times 10^{-19} \times 1.69 \times (0.02585)^2}{10^{-9}}\,\mathrm{V^2/Hz} = 3.6 \times 10^{-13}\,\mathrm{V^2/Hz}, \tag{12.88}$$

that is, $0.60\,\mu\mathrm{V}/\sqrt{\mathrm{Hz}}$. At $I_{\mathrm{D}} = 10\,\mathrm{nA}$ the noise power is ten times smaller, $0.19\,\mu\mathrm{V}/\sqrt{\mathrm{Hz}}$. Integrated over a 10 MHz bandwidth, the 1 nA device has an rms gate noise of $0.60\,\mu\mathrm{V} \times \sqrt{10^7} = 1.9\,\mathrm{mV}$, comparable to the $\sqrt{k_{\mathrm{B}}T/C}$ voltage on a 1 fF node (2.0 mV, Example 11.7). In subthreshold circuits at femtofarad scales, transistor noise is not a small correction; it is of the same size as the thermal fluctuations the circuit is meant to sample.

The white spectrum extends up to frequencies of order the inverse diffusion transit time $\tau_{\mathrm{tr}} \approx L^2/(2D_n) = L^2/(2\mu_n V_T)$. For $L = 1\,\mu\mathrm{m}$ and $\mu_n = 0.04\,\mathrm{m^2/(V\,s)}$, $\tau_{\mathrm{tr}} \approx 0.5\,\mathrm{ns}$, so assumption (viii) holds for all node bandwidths below roughly 100 MHz. Shorter channels push this limit higher.

12.5.5 Flicker Noise and Mismatch

At low frequencies the channel noise is dominated by $1/f$ flicker noise from trapping and release of channel electrons by oxide traps, treated in Section 11.5. Each trapped electron shifts the local threshold voltage, so flicker noise is most naturally written as a gate voltage noise,

$$S_{V_G}^{1/f}(f) = \frac{K_g}{C_{\mathrm{ox}} W L\, f}, \tag{12.89}$$

where $C_{\text{ox}}WL$ is the total gate capacitance and K_g (units $\text{V}^2\,\text{F}$) is a process constant, of order $10^{-25}\,\text{V}^2\,\text{F}$ for n-channel devices. The inverse area dependence is the McWhorter result of Section 11.5.2: more traps average to a smaller relative fluctuation. Equating Eq. (12.89) with the white gate noise Eq. (12.86) gives the corner frequency

$$f_c = \frac{K_g\, I_{\text{D}}}{2q_e n^2 V_T^2\, C_{\text{ox}} WL}, \tag{12.90}$$

which grows linearly with bias current, as in Eq. (11.110). For $K_g = 10^{-25}\,\text{V}^2\,\text{F}$, $C_{\text{ox}}WL = 8.6\,\text{fF}$ ($1\,\mu\text{m}^2$ in Example 12.5), and $I_{\text{D}} = 1\,\text{nA}$, $f_c \approx 30\,\text{Hz}$; for a $0.01\,\mu\text{m}^2$ device at 10 nA it is about 30 kHz. In either case it lies far below the MHz to 100 MHz relaxation bandwidths of femtofarad sampling nodes.

The zero-frequency limit of the same physics is device mismatch. Two nominally identical transistors differ in threshold by a random amount with standard deviation $\sigma(\Delta V_{\text{th}}) = A_V/\sqrt{WL}$ [86], where A_V is a few $\text{mV}\,\mu\text{m}$ in modern processes. In weak inversion a threshold offset becomes a current ratio, $\Delta I_{\text{D}}/I_{\text{D}} = \Delta V_{\text{th}}/(nV_T)$, so a 3 mV offset gives a 9% current error at $n = 1.3$. Mismatch is static, so it biases the sampled distribution rather than adding noise to it. Chapter 18 treats its calibration.

12.6 Worked Computation: The Subthreshold Differential Transconductance Amplifier

The basic analog building block of subthreshold sampling circuits is the five-transistor operational transconductance amplifier (OTA) shown in Fig. 12.5. It converts a differential input voltage into an output current. Wired as a follower driving a capacitor, it acts as an active conductance G_m with a programmable value, and the result is a one-node version of the RC sampler (the coupled linear RC circuit of Example 1.5), and the basic node of the CMOS samplers of Chapter 13. Its large-signal hyperbolic-tangent characteristic is also the activation function used by the CMOS p-bits of Chapter 14, which realize the p-bit limit of the double-well sampler. We now derive, from Theorems 12.13 and 12.19, everything needed to use it as a stochastic circuit element.

Worked computation 12.24 (Electrostatics, Transfer Characteristic, and Noise of the Subthreshold OTA)**.** All n-channel devices share a grounded bulk; voltages are bulk-referred. All transistors are matched, have slope factor n, and are in weak-inversion saturation ($V_{\text{DS}} \geq 4V_T$), so each draws I_F from Eq. (12.46) and has the channel noise $2q_e I_{\text{D}}$ of Corollary 12.22.

Step 1: the common source node. The two input transistors carry

$$I_k = I_0\, e^{(V_k - V_{\text{th}})/(nV_T)}\, e^{-V_C/V_T}, \qquad k = 1, 2. \tag{12.91}$$

The tail transistor fixes their sum, $I_1 + I_2 = I_b$, independent of V_C as long as M_5 stays saturated. Adding the two equations Eq. (12.91) and solving for e^{-V_C/V_T},

$$e^{-V_C/V_T} = \frac{I_b\, e^{V_{\text{th}}/(nV_T)}}{I_0\big(e^{V_1/(nV_T)} + e^{V_2/(nV_T)}\big)}, \tag{12.92}$$

so that

$$V_C = V_T \ln\frac{I_0}{I_b} - \frac{V_{\text{th}}}{n} + V_T \ln\Big(e^{V_1/(nV_T)} + e^{V_2/(nV_T)}\Big). \tag{12.93}$$

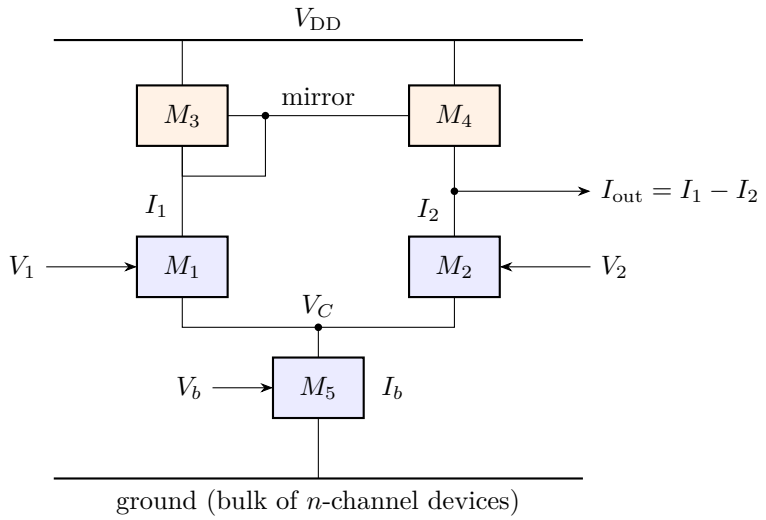


Figure 12.5: Five-transistor subthreshold OTA. The n-channel differential pair M_1, M_2 shares a common source node V_C fed by the tail transistor M_5, which sinks the bias current I_b set by V_b. The p-channel current mirror M_3 (diode-connected), M_4 copies I_1 to the output branch, so the output current is $I_{\mathrm{out}} = I_1 - I_2$. All transistors operate in weak-inversion saturation.

At balance, $V_1 = V_2 = V_{\mathrm{cm}}$, this is

$$V_C = \frac{V_{\mathrm{cm}} - V_{\mathrm{th}}}{n} + V_T \ln \frac{2I_0}{I_b}. \tag{12.94}$$

The common source follows the input common-mode voltage with gain $1/n = \kappa$. This is the capacitive divider of Section 12.2 acting at the circuit level: with constant current, $e^{V_G/(nV_T)} e^{-V_S/V_T}$ must stay fixed, so V_S moves by $\delta V_G/n$.

Step 2: headroom. The tail stays saturated if $V_C \geq 4V_T$. With Eq. (12.94) this sets a minimum input common-mode voltage,

$$V_{\mathrm{cm}} \geq V_{\mathrm{th}} + nV_T\Big(4 - \ln \frac{2I_0}{I_b}\Big). \tag{12.95}$$

For $I_0 = 60\,\mathrm{nA}$ (Example 12.14) and $I_b = 1\,\mathrm{nA}$, $\ln(120) = 4.79$, so $V_{\mathrm{cm}} \geq V_{\mathrm{th}} - 0.79\,nV_T \approx V_{\mathrm{th}} - 27\,\mathrm{mV}$ for $n = 1.3$. The input devices are saturated if their drains sit at least $4V_T$ above V_C, and the mirror needs one p-channel gate-source drop. The whole stack therefore fits in a supply of a few hundred millivolts, because each saturated subthreshold device needs only about 100 mV.

Step 3: the hyperbolic-tangent transfer characteristic. Dividing Eq. (12.91) for $k = 1$ by the sum $I_1 + I_2 = I_b$, the unknowns I_0, V_{th}, and V_C cancel:

$$I_1 = I_b \frac{e^{V_1/(nV_T)}}{e^{V_1/(nV_T)} + e^{V_2/(nV_T)}} = \frac{I_b}{1 + e^{-\Delta V/(nV_T)}}, \qquad \Delta V := V_1 - V_2. \tag{12.96}$$

With the identity $1/(1 + e^{-2x}) = \tfrac{1}{2}(1 + \tanh x)$ and $x = \Delta V/(2nV_T)$,

$$I_1 = \frac{I_b}{2}\left[1 + \tanh\Big(\frac{\Delta V}{2nV_T}\Big)\right], \qquad I_2 = \frac{I_b}{2}\left[1 - \tanh\Big(\frac{\Delta V}{2nV_T}\Big)\right]. \tag{12.97}$$

The mirror copies I_1 into M_4, so with the output held at a fixed voltage the output current is

$$I_{\text{out}} = I_1 - I_2 = I_b \tanh\!\left(\frac{\Delta V}{2nV_T}\right) = I_b \tanh\!\left(\frac{\kappa\,\Delta V}{2V_T}\right). \tag{12.98}$$

The characteristic depends on the process only through n. It is independent of the threshold voltage, of I_0, and of the common-mode voltage, because the pair compares two Boltzmann factors that share all of those. It saturates at $\pm I_b$: for $|\Delta V| \geq 4nV_T \approx 134\,\text{mV}$ ($n = 1.3$), $|I_{\text{out}}| \geq \tanh(2)\, I_b = 0.96\, I_b$. The Taylor series $\tanh x = x - x^3/3 + \cdots$ shows that the cubic error reaches 1% of the linear term at $x^2/3 = 0.01$, that is, at $|\Delta V| = 0.35\, nV_T \approx 12\,\text{mV}$.

Step 4: small-signal parameters. The transconductance at balance is

$$G_m := \left.\frac{\partial I_{\text{out}}}{\partial \Delta V}\right|_{\Delta V = 0} = \frac{I_b}{2nV_T}, \tag{12.99}$$

which equals the transconductance $g_{\text{m}} = (I_b/2)/(nV_T)$ of each input transistor. It is set by the tail current alone and can be programmed over several decades through V_b. The output conductance is the sum of the drain conductances of M_2 and M_4. With DIBL coefficients η_n, η_p and Eq. (12.68), $g_{\text{out}} \approx (\eta_n + \eta_p)\, I_b/(2nV_T)$ (taking the same n for both polarities), so the open-circuit voltage gain is

$$A_v = \frac{G_m}{g_{\text{out}}} \approx \frac{1}{\eta_n + \eta_p}, \tag{12.100}$$

about 25 for $\eta_n = \eta_p = 0.02$, and independent of I_b.

Step 5: output current noise at arbitrary imbalance. Hold both gates and the output at fixed voltages and let each transistor carry its own channel noise current i_k from drain to source, with single-sided PSDs

$$S_1 = 2q_e I_1, \qquad S_2 = 2q_e I_2, \qquad S_5 = 2q_e I_b, \qquad S_3 = S_4 = 2q_e I_1, \tag{12.101}$$

since M_3 and M_4 both carry I_1. The noise sources are independent. We need the response of the internal node V_C. By Eq. (12.51) with fixed gates and saturated drains, the drain current fluctuations of the pair are $\delta I_k = -g_{s,k}\, v_c + i_k$, with $g_{s,k} = I_k/V_T$ from Eq. (12.54). The saturated tail passes only its own noise, $\delta I_5 = i_5$. Kirchhoff's current law at the common source node, $\delta I_1 + \delta I_2 = \delta I_5$, gives

$$v_c = \frac{i_1 + i_2 - i_5}{g_{s,1} + g_{s,2}} = \frac{V_T}{I_b}(i_1 + i_2 - i_5), \tag{12.102}$$

and therefore

$$\delta I_1 = i_1 - \frac{I_1}{I_b}(i_1 + i_2 - i_5) = \frac{I_2\, i_1 - I_1\, i_2 + I_1\, i_5}{I_b}, \tag{12.103}$$

$$\delta I_2 = i_2 - \frac{I_2}{I_b}(i_1 + i_2 - i_5) = \frac{I_1\, i_2 - I_2\, i_1 + I_2\, i_5}{I_b}. \tag{12.104}$$

The diode-connected M_3 must carry δI_1. Its current is $g_{m3}u + i_3$, where u is the small-signal source-gate voltage of the mirror, so $u = (\delta I_1 - i_3)/g_{m3}$. The output device M_4 has the same gate voltage and the same transconductance, so it carries $g_{m4}u + i_4 = \delta I_1 - i_3 + i_4$. The output current fluctuation is the difference between the currents of M_4 and M_2:

$$\delta I_{\text{out}} = \delta I_1 - \delta I_2 - i_3 + i_4 = \frac{2I_2\, i_1 - 2I_1\, i_2 + (I_1 - I_2)\, i_5}{I_b} - i_3 + i_4. \tag{12.105}$$

Adding the independent contributions in power with Eq. (12.101),

$$\begin{aligned} S_{\text{out}} &= \frac{4I_2^2\,(2q_eI_1) + 4I_1^2\,(2q_eI_2) + (I_1-I_2)^2\,(2q_eI_b)}{I_b^2} + 2q_eI_1 + 2q_eI_1 \\ &= \frac{2q_e}{I_b}\Big[4I_1I_2 + (I_1-I_2)^2\Big] + 4q_eI_1 = \frac{2q_e(I_1+I_2)^2}{I_b} + 4q_eI_1 \\ &= 2q_eI_b + 4q_eI_1, \end{aligned} \tag{12.106}$$

where the first line used $I_1 + I_2 = I_b$ to write $8q_eI_1I_2(I_1+I_2)/I_b^2 = 8q_eI_1I_2/I_b$. With Eq. (12.97),

$$S_{\text{out}}(\Delta V) = 2q_eI_b\left[2 + \tanh\!\left(\frac{\Delta V}{2nV_T}\right)\right]. \tag{12.107}$$

The output noise therefore ranges between the two limits of full imbalance,

$$S_{\text{out}} \to 6q_eI_b \quad (\Delta V \gg 2nV_T,\ I_1 = I_b), \qquad S_{\text{out}} \to 2q_eI_b \quad (\Delta V \ll -2nV_T,\ I_1 = 0). \tag{12.108}$$

In the first limit the tail noise and both mirror devices pass to the output; in the second the mirror carries no current and only the tail noise, carried by M_2, remains.

Two features stand out. First, the differential pair and the tail together always contribute exactly $2q_eI_b$, the shot noise of the full tail current, whatever the imbalance. At balance the tail noise is a common-mode current that splits equally and cancels at the output, while each input device contributes $2q_e(I_b/2)$. At full imbalance the whole tail current flows through one device, and its shot noise passes straight to the output. Second, the mirror contribution $4q_eI_1$ is not symmetric in ΔV, because the mirror devices carry I_1 only. The output noise of this single-ended circuit is therefore not an even function of ΔV. Fully differential cells with symmetric loads remove this asymmetry, which is what Chapter 13 assumes for its balanced architectures. At balance,

$$S_{\text{out}}(0) = 4q_eI_b. \tag{12.109}$$

Step 6: input-referred noise. Dividing Eq. (12.109) by G_m^2 from Eq. (12.99),

$$S_{V,\text{in}} = \frac{4q_eI_b}{G_m^2} = 4q_eI_b \cdot \frac{4n^2V_T^2}{I_b^2} = \frac{16\,q_en^2V_T^2}{I_b}. \tag{12.110}$$

This is four times the gate noise Eq. (12.86) of a single device at $I_{\text{D}} = I_b/2$: two input devices and two mirror devices, each contributing $2q_e(I_b/2)$ at the output with unit gain. For $n = 1.3$:

$$\begin{aligned} I_b = 1\,\text{nA}: &\qquad S_{V,\text{in}} = 2.9\times10^{-12}\,\text{V}^2/\text{Hz} \quad (1.7\,\mu\text{V}/\sqrt{\text{Hz}}), \\ I_b = 10\,\text{nA}: &\qquad S_{V,\text{in}} = 2.9\times10^{-13}\,\text{V}^2/\text{Hz} \quad (0.54\,\mu\text{V}/\sqrt{\text{Hz}}). \end{aligned} \tag{12.111}$$

Step 7: the OTA-C follower as a sampling node. Connect the output to the inverting input V_2 and to a grounded capacitor C, and hold $V_1 = v_{\text{in}}$. The internal nodes (V_C and the mirror gate) have small capacitances and settle much faster than the output, so the OTA acts on C as a Norton source with current Eq. (12.98) and noise Eq. (12.107). Writing the noise in the Itô form of Section 12.5.3, with $D_q = \frac{1}{4}S_{\text{out}}$,

$$C\,\mathrm{d}v_t = I_b\tanh\!\left(\frac{v_{\text{in}}-v_t}{2nV_T}\right)\mathrm{d}t + \sqrt{2D_q(v_t)}\,\mathrm{d}W_t, \qquad D_q(v) = \frac{q_eI_b}{2}\left[2 + \tanh\!\left(\frac{v_{\text{in}}-v}{2nV_T}\right)\right]. \tag{12.112}$$

For small excursions around $v = v_{\text{in}}$, tanh is linear and $D_q = q_e I_b$, so

$$C\,\mathrm{d}v_t = -G_m\,(v_t - v_{\text{in}})\,\mathrm{d}t + \sqrt{2q_e I_b}\,\mathrm{d}W_t. \tag{12.113}$$

This is the one-node version of the RC-sampler equation $\mathbf{C}\,\mathrm{d}\mathbf{v}_t = -\mathbf{G}(\mathbf{v}_t - \mathbf{b})\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t$, with $\mathbf{G} = G_m$, $\mathbf{b} = v_{\text{in}}$, and $\mathbf{B}\mathbf{B}^\top = 2q_e I_b$. Dividing by C gives an Ornstein-Uhlenbeck process with relaxation rate G_m/C and noise intensity $2q_e I_b/C^2$, whose stationary variance is

$$\mathrm{Var}(v) = \frac{2q_e I_b/C^2}{2G_m/C} = \frac{q_e I_b}{G_m C} = \frac{q_e I_b \cdot 2nV_T}{I_b\,C} = \frac{2nk_{\mathrm{B}}T}{C}. \tag{12.114}$$

The same result follows from the spectrum. The voltage PSD is $S_{\text{out}}/(G_m^2+(2\pi fC)^2)$, and the integral $\int_0^\infty \mathrm{d}f/(G_m^2 + (2\pi fC)^2) = 1/(4G_mC)$ evaluated in Eq. (11.53) gives $\mathrm{Var}(v) = 4q_eI_b/(4G_mC) = 2nk_{\mathrm{B}}T/C$. Compared with a passive resistor on the same capacitor ($k_{\mathrm{B}}T/C$), the OTA node is hotter by the factor

$$\frac{T_{\text{eff}}}{T} = \frac{S_{\text{out}}}{4k_{\mathrm{B}}T\,G_m} = \frac{4q_eI_b \cdot 2nV_T}{4k_{\mathrm{B}}T\,I_b} = 2n, \tag{12.115}$$

in agreement with Eq. (13.67). The factor $2n$ is a product of two effects. Each saturated channel carries noise $2q_e(I_b/2) = 2nk_{\mathrm{B}}T\,G_m$, which is $n/2$ times the Johnson-Nyquist value $4k_{\mathrm{B}}TG_m$ of a resistor with the same conductance; the factor n is the transconductance lost in the capacitive divider. And four such channels, rather than one, feed the output with unit gain. Together, $4 \times n/2 = 2n$.

Step 8: numbers at femtofarad scale. Take $C = 1\,\mathrm{fF}$, $n = 1.3$, $T = 300\,\mathrm{K}$.

- Stationary rms voltage: $\sqrt{2nk_{\mathrm{B}}T/C} = \sqrt{2.6 \times 4.14 \times 10^{-21}/10^{-15}}\,\mathrm{V} = 3.3\,\mathrm{mV}$, compared with $\sqrt{k_{\mathrm{B}}T/C} = 2.0\,\mathrm{mV}$ for a passive node.
- Relaxation time $\tau = C/G_m = 2nV_TC/I_b$: $\tau = 67\,\mathrm{ns}$ at $I_b = 1\,\mathrm{nA}$ and $6.7\,\mathrm{ns}$ at $I_b = 10\,\mathrm{nA}$, corresponding to bandwidths $1/(2\pi\tau)$ of $2.4\,\mathrm{MHz}$ and $24\,\mathrm{MHz}$. Both lie far above the flicker corners of Section 12.5.5 and far below the inverse transit time of Section 12.5.4.
- Linearity: the rms excursion is $3.3\,\mathrm{mV}/(2nV_T) = 0.05$ in units of the tanh scale $2nV_T = 67\,\mathrm{mV}$. The cubic term of tanh and the v-dependence of D_q in Eq. (12.112) are therefore small, and they enter the stationary statistics only through powers of this ratio. Section 13.5 develops the systematic expansion.
- Energy per relaxation time: the cell draws I_b from the supply, so with $V_{\mathrm{DD}} = 0.5\,\mathrm{V}$,

$$E_\tau = I_bV_{\mathrm{DD}}\,\tau = 2nV_TC\,V_{\mathrm{DD}} = 2n\,\Omega\,\frac{V_{\mathrm{DD}}}{V_T}\,k_{\mathrm{B}}T, \qquad \Omega := \frac{CV_T}{q_e}, \tag{12.116}$$

  where $\Omega = 161$ is the number of thermal charges on the node used in Chapter 13. Numerically $E_\tau = 2 \times 1.3 \times 0.02585 \times 10^{-15} \times 0.5\,\mathrm{J} = 3.4 \times 10^{-17}\,\mathrm{J} \approx 8 \times 10^3\,k_{\mathrm{B}}T$. This is independent of I_b: a faster node costs proportionally more power, not more energy per relaxation. The scaling with Ω and V_{DD}/V_T is the one used in the energy budgets of Chapter 17.

Step 9: toward p-bits. Without feedback, Eq. (12.98) maps an input voltage to a bounded, odd, sigmoidal output. If the input carries noise and the output is compared against zero, the probability of a positive output is a sigmoidal function of the mean input, with a width set by the input-referred noise Eq. (12.110) and the scale $2nV_T$ of Eq. (12.98). Section 14.1 builds CMOS p-bits from elements of this kind and derives their activation function.

Hardware Realization: Subthreshold Sampling Cells in CMOS Thermodynamic Hardware

The CMOS thermodynamic hardware described by Extropic and collaborators builds its stochastic elements from transistors biased in weak inversion [31, 38, 56]. The results of this chapter supply the device-level relations behind such elements. The numbers below are this book's representative order-of-magnitude estimates, not chip specifications.

- *NEAT-RN nodes (Chapter 13).* Each continuous node is a small capacitor, of order 1 fF, coupled to other nodes through subthreshold transconductors. By Eq. (12.99) each coupling strength is a tail current divided by $2nV_T$, so it is programmed by a bias current over several decades. By Eq. (12.107) each coupling also injects shot noise set by its standing current, so the noise is not tied to the conductance by the equilibrium fluctuation-dissipation relation. This is the origin of the non-equilibrium steady states studied in Chapter 13.
- *Programmable conductances.* A single transistor at small V_{DS} is a resistor $g_{\mathrm{ds0}} \propto e^{V_G/(nV_T)}$ (Eq. (12.55)) with exact Johnson-Nyquist noise (Corollary 12.22). It provides reciprocal, detailed-balance-respecting couplings when they are wanted.
- *p-bits (Chapter 14).* The differential pair's tanh characteristic Eq. (12.98) and the gate noise Eq. (12.86) set the activation slope and the noise level of subthreshold p-bit cells.
- *Representative scales.* Bias currents of 0.1–50 nA per device, relaxation times of nanoseconds to about 100 ns on femtofarad nodes, $\Omega \sim 10^2$–10^3 thermal charges per node, and of order $10^4\, k_{\mathrm{B}}T$ of supply energy per relaxation time for a single OTA-C cell at a 0.5 V supply.

Summary

Proved:

- *MOS electrostatics (Propositions 12.1 and 12.3).* The first integral of the Poisson-Boltzmann equation, and the weak-inversion charge partition $Q_{\mathrm{inv}} = -C_{\mathrm{dep}}V_T e^{(\psi_s - 2\phi_F - V)/V_T}$, interpreted as a Boltzmann sheet of thickness $V_T/\mathcal{E}_s$.
- *Slope factor (Proposition 12.6).* $\frac{\mathrm{d}\psi_s}{\mathrm{d}V_G} = 1/n$ with $n = 1 + C_{\mathrm{dep}}/C_{\mathrm{ox}}$, the subthreshold swing $nV_T \ln 10 \approx 60n$ mV/decade at 300 K, and the body effect as the complementary half of the divider.
- *Drift versus diffusion (Proposition 12.9).* The lateral field is not zero, but the drift current is smaller than the diffusion current by $Q_{\mathrm{i}}/(nC_{\mathrm{ox}}V_T)$, which is exponentially small in weak inversion.
- *EKV law (Theorem 12.13).* $I_{\mathrm{D}} = I_0 e^{(V_G - V_{\mathrm{th}})/(nV_T)}\left[e^{-V_S/V_T} - e^{-V_D/V_T}\right]$, the split $I_{\mathrm{D}} = I_F - I_R$, and local detailed balance $I_F/I_R = e^{V_{\mathrm{DS}}/V_T}$ independent of the gate and of DIBL.
- *Transconductance bound (Lemma 12.16 and Proposition 12.17).* $g_{\mathrm{m}}/I_{\mathrm{D}} \le 1/(n_{\min}V_T)$ at every level of inversion in the long-channel charge-sheet model, with equality in weak inversion; the EKV interpolation $g_{\mathrm{m}}nV_T/I_{\mathrm{D}} = 2/(1 + \sqrt{1 + 4\,\mathrm{IC}})$.

- *Channel noise (Theorem 12.19 and Corollary 12.22).* Integrating local Johnson-Nyquist noise along a weakly inverted channel gives exactly $S_I = 2q_e(I_F + I_R)$: $4k_{\mathrm{B}}Tg_{\mathrm{ds0}}$ at $V_{\mathrm{DS}} = 0$ and $2q_eI_{\mathrm{D}}$ in saturation; input-referred gate noise $2nk_{\mathrm{B}}T/g_{\mathrm{m}}$.
- *Subthreshold OTA (Worked computation 12.24).* $I_{\mathrm{out}} = I_b \tanh(\Delta V/2nV_T)$, $G_m = I_b/(2nV_T)$, output noise $2q_eI_b[2 + \tanh(\Delta V/2nV_T)]$, input noise $16q_en^2V_T^2/I_b$, and the follower variance $2nk_{\mathrm{B}}T/C$.

Assumed:

- Nondegenerate Boltzmann statistics with local quasi-Fermi levels, complete dopant ionization, and the Einstein relation $D_n = \mu_nV_T$ with constant mobility.
- The gradual-channel and charge-sheet approximations (Assumption 12.8), a constant slope factor, and no oxide or interface charge beyond what is absorbed in V_{FB} and C_{it}.
- Local equilibrium of channel segments with uncorrelated Johnson-Nyquist noise (Assumption 12.18), valid below the inverse transit time.
- DIBL and channel-length modulation treated phenomenologically; flicker-noise and mismatch constants taken as process inputs.

Taken up later:

- The complete Poisson-Boltzmann integral without the $\psi_s \gg V_T$ simplifications, the full EKV interpolation of the inversion charge, and short-channel corrections (Chapter D).
- The stochastic dynamics of networks of these devices: the circuit master equation, its system-size expansion in $\Omega = C_{\mathrm{node}}V_T/q_e$, broken detailed balance, and Lyapunov covariances of NEAT-RNs (Chapter 13).
- CMOS p-bits built from differential pairs and latches (Chapter 14), energy budgets of the Denoising Thermodynamic Computer Architecture (Chapter 17), and calibration of mismatch and $1/f$ drift (Chapter 18).

Open questions:

- How far does the bidirectional Poisson description of the channel hold in very short channels, where transport becomes partly ballistic and the transit time approaches circuit timescales?
- Can steep-slope devices with $\mathcal{S} < 60\,\mathrm{mV/decade}$ (band-to-band tunneling or negative-capacitance gates) lower the energy per sample of thermodynamic samplers, given that their noise is no longer tied to their transconductance by $S_I/g_{\mathrm{m}} = 2nk_{\mathrm{B}}T$?
- What is the minimum supply energy per independent sample of an active sampling node, given that the OTA-C node of Eq. (12.116) costs $2n\Omega(V_{\mathrm{DD}}/V_T)k_{\mathrm{B}}T$ per relaxation time?

Bibliographic notes

The exponential subthreshold characteristic and its use for micropower analog circuits were established by Vittoz and Fellrath [116], who analyzed weak-inversion MOSFETs as lateral bipolar devices and built the first circuits that exploit them. The bulk-referred, source-drain symmetric formulation used throughout this chapter, the charge-based description of all inversion levels, the inversion coefficient, and the $g_{\mathrm{m}}/I_{\mathrm{D}}$ interpolation are developed in detail in the EKV monograph of Enz and Vittoz [29]. Tsividis and McAndrew [111] gives the most careful textbook treatment of MOS electrostatics, the Poisson-Boltzmann first integral, the charge-sheet model, and the integration of local channel noise along the channel that underlies Theorem 12.19.

The use of subthreshold transistors as the native elements of analog computation, the gate coupling coefficient $\kappa = 1/n$, and the differential-pair transconductance amplifier with its tanh characteristic were popularized by Mead [75]. The identification of subthreshold channel noise with the shot noise of independent forward and reverse diffusion currents, and its reduction to Johnson-Nyquist noise at zero bias, is due to Sarpeshkar et al. [93]. Threshold-voltage mismatch and its inverse-area scaling are from Pelgrom et al. [86]. The stochastic-thermodynamic treatment of circuits built from such devices, which the next chapter develops, is in Freitas et al. [37] and, for the Gaussian samplers of Chapter 13, Freitas et al. [38].

Chapter 13

Non-Equilibrium Gaussian Samplers in Subthreshold CMOS

Chapter 11 showed that charge crossing a barrier in both directions produces the current noise $S_I = 2q_e(I_+ + I_-)$, which interpolates between Johnson–Nyquist noise in equilibrium and shot noise far from it. Chapter 12 showed that a MOSFET in weak inversion is such a barrier device: its current is exponential in the terminal voltages, and its transconductance per unit current, $g_\mathrm{m}/I_\mathrm{D} = 1/(nV_T)$, is the largest that Boltzmann statistics allows. This chapter asks what happens when many such transistors and capacitors are wired into a network, and how such a network can sample from a programmable Gaussian distribution.

A passive network of capacitors and resistors cannot do this job. Its stationary distribution is the Boltzmann distribution of its electrostatic energy $U(\mathbf{v}) = \frac{1}{2}\mathbf{v}^\top \mathbf{C}\mathbf{v}$, so its voltage covariance is $k_\mathrm{B}T\mathbf{C}^{-1}$ whatever the resistor values are. The resistors set only how fast the network relaxes. We call this limitation the *passive equipartition lock*; the name is an informal one used in this book, not a standard term. On-chip capacitances cannot be negative and cannot be reprogrammed over a wide range at reasonable cost, so a passive network cannot be set to an arbitrary precision matrix $\mathbf{\Lambda} = \mathbf{\Sigma}^{-1}$.

Transistors powered from a supply rail V_DD remove the lock in two ways. The gate of a transistor controls its drain current without drawing current itself, so a node that drives a gate is not acted on in return by the node at the drain, and the conductance matrix is no longer symmetric. And a saturated transistor injects shot noise set by its standing bias current rather than by its small-signal conductance, so the noise no longer obeys the equilibrium fluctuation–dissipation relation. An active network therefore settles into a non-equilibrium steady state (NESS) with circulating probability currents and a steady housekeeping heat flow (Chapters 6 and 8).

The stochastic thermodynamics of nonlinear circuits was developed by Freitas et al. [37], and Freitas et al. [38] applied it to subthreshold Gaussian samplers, which we call Non-Equilibrium Analog Thermodynamic Random Networks (NEAT-RNs). We follow the same route. Starting from a master equation for discrete charges (Section 13.1), we expand in the inverse number of thermal charges per node to obtain a Langevin equation for the node voltages (Section 13.2). We then show how active devices break detailed balance (Section 13.3) and how the linearized network samples a Gaussian whose covariance solves a Lyapunov equation (Section 13.4). The chapter ends with a two-node cell solved in closed form. It is the hardware form of the RC sampler (the coupled linear RC circuit of Example 1.5).

13.1 Stochastic Thermodynamics of Nonlinear Electronic Circuits

At femtofarad scales a single electron changes a node voltage by a small but finite fraction of V_T. To know when a continuous description is valid, we start from the dynamics of discrete charges. Following Freitas et al. [37], we treat a circuit as an open system coupled to a heat bath at temperature T (inverse temperature $\beta = (k_{\mathrm{B}}T)^{-1}$) and to ideal voltage sources that supply energy and hold the circuit away from equilibrium.

13.1.1 Network Topology, Capacitive Electrostatics, and Charging Energy

Consider an electronic circuit composed of N internal dynamical nodes, indexed by $i \in \{1, \dots, N\}$, and M external voltage nodes (the power supply V_{DD}, ground, and gate biases set by programmable digital-to-analog converters, or DACs), indexed by $\alpha \in \{1, \dots, M\}$. Let $\mathbf{v} = (v_1, \dots, v_N)^\top \in \mathbb{R}^N$ denote the column vector of electrostatic potentials at the internal nodes, and let $\mathbf{v}_{\mathrm{ext}} = (V_{\mathrm{ext},1}, \dots, V_{\mathrm{ext},M})^\top \in \mathbb{R}^M$ denote the fixed potentials of the external voltage sources.

Every node in an integrated circuit is coupled electrostatically to ground, to the external rails, and to neighboring internal nodes through gate-oxide, depletion, junction, and interconnect capacitances. Let $c_{i0} \geq 0$ denote the capacitance between internal node i and the ground reference, $c_{i\alpha} \geq 0$ the capacitance between internal node i and external source α, and $C_{ij} = C_{ji} \geq 0$ $(i \neq j)$ the mutual coupling capacitance between internal nodes i and j. By Gauss's law, the net charge q_i on the plates connected to internal node i is

$$q_i = c_{i0} v_i + \sum_{\alpha=1}^{M} c_{i\alpha}(v_i - V_{\mathrm{ext},\alpha}) + \sum_{j \neq i} C_{ij}(v_i - v_j), \qquad i = 1, \dots, N. \tag{13.1}$$

Collecting terms proportional to v_i and v_j, we define the $N \times N$ internal Maxwell capacitance matrix $\mathbf{C}$ and the $N \times M$ external cross-capacitance matrix $\mathbf{C}_{\mathrm{ext}}$ with entries

$$[\mathbf{C}]_{ii} := c_{i0} + \sum_{\alpha=1}^{M} c_{i\alpha} + \sum_{j \neq i} C_{ij}, \qquad [\mathbf{C}]_{ij} := -C_{ij} \quad (i \neq j), \qquad [\mathbf{C}_{\mathrm{ext}}]_{i\alpha} := c_{i\alpha}. \tag{13.2}$$

The matrix $\mathbf{C}$ is symmetric with positive diagonal entries and is diagonally dominant. If every connected component of the internal capacitive network has at least one nonzero capacitance to ground or to an external rail, the dominance is strict in at least one row of each component, and such a matrix is positive definite ($\mathbf{C} \succ 0$), hence invertible. Defining the induced offset charge vector $\mathbf{q}_0(\mathbf{v}_{\mathrm{ext}}) := -\mathbf{C}_{\mathrm{ext}}\mathbf{v}_{\mathrm{ext}}$, Eq. (13.1) takes the compact vector form

$$\mathbf{q} = \mathbf{C}\mathbf{v} + \mathbf{q}_0(\mathbf{v}_{\mathrm{ext}}) \quad \Longleftrightarrow \quad \mathbf{v}(\mathbf{q}) = \mathbf{C}^{-1}(\mathbf{q} - \mathbf{q}_0(\mathbf{v}_{\mathrm{ext}}))\,. \tag{13.3}$$

Because charge is quantized in units of $q_e \approx 1.602 \times 10^{-19}\,\mathrm{C}$, the state space of the internal nodes is the charge lattice

$$\Gamma_q := (q_e\mathbb{Z})^N = \left\{ \mathbf{q} = q_e\mathbf{n} \;\middle|\; \mathbf{n} = (n_1, \dots, n_N)^\top \in \mathbb{Z}^N \right\}. \tag{13.4}$$

When the external voltages $\mathbf{v}_{\mathrm{ext}}$ are held constant, the relevant energy is the Legendre transform of the total capacitor energy with respect to $\mathbf{v}_{\mathrm{ext}}$ [37]:

$$U(\mathbf{q}) := \frac{1}{2}\,(\mathbf{q} - \mathbf{q}_0)^\top\,\mathbf{C}^{-1}\,(\mathbf{q} - \mathbf{q}_0) = \frac{1}{2}\mathbf{v}(\mathbf{q})^\top \mathbf{C}\,\mathbf{v}(\mathbf{q}). \tag{13.5}$$

Its gradient with respect to the charges is the vector of node voltages:

$$\nabla_{\mathbf{q}} U(\mathbf{q}) = \mathbf{C}^{-1}(\mathbf{q} - \mathbf{q}_0) = \mathbf{v}(\mathbf{q}). \tag{13.6}$$

13.1.2 Elementary Conductive Channels and Local Detailed Balance

Charge changes only when electrons cross the conducting devices of the circuit: MOSFET channels, p-n junctions, tunnel barriers, or resistors. We index these conducting channels (branches) by $\rho \in \{1, \dots, R\}$. Each branch carries charges q_e either between two internal nodes $i_\rho \to j_\rho$, or between an external voltage source α_ρ and an internal node i_ρ.

For each branch ρ we define the *incidence vector* $\mathbf{\Delta}_\rho \in \{-1, 0, +1\}^N$ and the source voltage $\mathcal{E}_\rho$:

(i) *Internal branch ($i_\rho \to j_\rho$).* When one elementary charge $+q_e$ moves from internal node i_ρ to internal node j_ρ, node i_ρ loses charge q_e and node j_ρ gains charge q_e. Thus

$$(\mathbf{\Delta}_\rho)_k = \delta_{k,j_\rho} - \delta_{k,i_\rho}, \qquad \mathcal{E}_\rho = 0. \tag{13.7}$$

(ii) *Source branch ($\alpha_\rho \to i_\rho$).* When one elementary charge $+q_e$ enters internal node i_ρ from external voltage source α_ρ at potential $V_{\mathrm{ext},\alpha_\rho}$, the external source performs work $q_e V_{\mathrm{ext},\alpha_\rho}$ while internal node i_ρ gains charge $+q_e$. Thus

$$(\mathbf{\Delta}_\rho)_k = +\delta_{k,i_\rho}, \qquad \mathcal{E}_\rho = V_{\mathrm{ext},\alpha_\rho}. \tag{13.8}$$

(If the branch is oriented from internal node i_ρ to source α_ρ, then $(\mathbf{\Delta}_\rho)_k = -\delta_{k,i_\rho}$ and $\mathcal{E}_\rho = -V_{\mathrm{ext},\alpha_\rho}$.)

Collecting the vectors $\mathbf{\Delta}_\rho$ as the columns of the $N \times R$ incidence matrix $\mathbf{\Delta} := [\mathbf{\Delta}_1, \dots, \mathbf{\Delta}_R]$, a forward elementary transition across channel ρ maps the circuit state according to $\mathbf{q} \to \mathbf{q} + q_e \mathbf{\Delta}_\rho$, while the reverse transition maps $\mathbf{q} \to \mathbf{q} - q_e \mathbf{\Delta}_\rho$.

The change in electrostatic energy in a single forward jump across branch ρ is $\Delta U_\rho(\mathbf{q}) := U(\mathbf{q} + q_e \mathbf{\Delta}_\rho) - U(\mathbf{q})$. Expanding the quadratic form Eq. (13.5),

$$\begin{aligned}
\Delta U_\rho(\mathbf{q}) &= \frac{1}{2} \left(\mathbf{q} - \mathbf{q}_0 + q_e \mathbf{\Delta}_\rho\right)^\top \mathbf{C}^{-1} \left(\mathbf{q} - \mathbf{q}_0 + q_e \mathbf{\Delta}_\rho\right) - \frac{1}{2} \left(\mathbf{q} - \mathbf{q}_0\right)^\top \mathbf{C}^{-1} \left(\mathbf{q} - \mathbf{q}_0\right) \\
&= q_e \mathbf{\Delta}_\rho^\top \mathbf{C}^{-1} (\mathbf{q} - \mathbf{q}_0) + \frac{q_e^2}{2} \mathbf{\Delta}_\rho^\top \mathbf{C}^{-1} \mathbf{\Delta}_\rho \\
&= q_e \mathbf{\Delta}_\rho^\top \mathbf{v}(\mathbf{q}) + E_{\mathrm{C}}^{(\rho)},
\end{aligned} \tag{13.9}$$

where the *single-electron charging energy* of branch ρ is

$$E_{\mathrm{C}}^{(\rho)} := \frac{q_e^2}{2} \mathbf{\Delta}_\rho^\top \mathbf{C}^{-1} \mathbf{\Delta}_\rho > 0. \tag{13.10}$$

The node voltages halfway through the jump are

$$\mathbf{v}\left(\mathbf{q} + \frac{1}{2} q_e \mathbf{\Delta}_\rho\right) = \mathbf{v}(\mathbf{q}) + \frac{q_e}{2} \mathbf{C}^{-1} \mathbf{\Delta}_\rho, \tag{13.11}$$

so that $\Delta U_\rho(\mathbf{q})$ can be written symmetrically as $\Delta U_\rho(\mathbf{q}) = q_e \mathbf{\Delta}_\rho^\top \mathbf{v}(\mathbf{q} + \frac{1}{2} q_e \mathbf{\Delta}_\rho)$.

By the first law of stochastic energetics (Chapter 8), when a charge $+q_e$ crosses branch ρ, the external source does work $q_e \mathcal{E}_\rho$, the stored energy changes by $\Delta U_\rho(\mathbf{q})$, and the heat released into the bath is

$$\mathcal{Q}_\rho(\mathbf{q}) = q_e \mathcal{E}_\rho - \Delta U_\rho(\mathbf{q}) = q_e \left[\mathcal{E}_\rho - \mathbf{\Delta}_\rho^\top \mathbf{v}(\mathbf{q})\right] - E_{\mathrm{C}}^{(\rho)}. \tag{13.12}$$

Here $\Delta V_\rho(\mathbf{q}) := \mathcal{E}_\rho - \mathbf{\Delta}_\rho^\top \mathbf{v}(\mathbf{q})$ is the voltage drop across branch ρ in the forward direction (e.g., $v_{i_\rho} - v_{j_\rho}$ for an internal branch, or $V_{\mathrm{ext},\alpha_\rho} - v_{i_\rho}$ for a source branch). The entropy delivered to the bath in the jump is $\Delta s_{\mathrm{med}}^{(\rho)}(\mathbf{q}) = \mathcal{Q}_\rho(\mathbf{q})/T$.

We assume that the electrons inside each device thermalize with the bath much faster than the nodes charge. Then the forward rate $W_{+\rho}(\mathbf{q})$ (for $\mathbf{q} \to \mathbf{q}+q_e\mathbf{\Delta}_\rho$) and the reverse rate $W_{-\rho}(\mathbf{q}+q_e\mathbf{\Delta}_\rho)$ obey *local detailed balance* [37, 97]:

$$\frac{W_{+\rho}(\mathbf{q})}{W_{-\rho}(\mathbf{q}+q_e\mathbf{\Delta}_\rho)} = \exp\!\left(\frac{\Delta s^{(\rho)}_{\mathrm{med}}(\mathbf{q})}{k_{\mathrm{B}}}\right) = \exp\!\left(-\frac{U(\mathbf{q}+q_e\mathbf{\Delta}_\rho) - U(\mathbf{q}) - q_e\mathcal{E}_\rho}{k_{\mathrm{B}}T}\right). \tag{13.13}$$

With Eq. (13.11) and $V_T = k_{\mathrm{B}}T/q_e$, this can be written in terms of the midpoint voltages $\bar{\mathbf{v}}_\rho := \mathbf{v}(\mathbf{q}+\frac{1}{2}q_e\mathbf{\Delta}_\rho)$:

$$\frac{W_{+\rho}(\mathbf{q})}{W_{-\rho}(\mathbf{q}+q_e\mathbf{\Delta}_\rho)} = \exp\!\left(\frac{\mathcal{E}_\rho - \mathbf{\Delta}_\rho^\top \bar{\mathbf{v}}_\rho}{V_T}\right) = \exp\!\left(\frac{\Delta V_\rho(\bar{\mathbf{v}}_\rho)}{V_T}\right). \tag{13.14}$$

The subthreshold MOSFET satisfies this condition. For an n-channel device in weak inversion with bulk-referred gate, source, and drain voltages V_G, V_S, V_D, Section 11.4 and Section 12.3 showed that transport across the channel is a bidirectional Poisson process. Its forward rate (conventional current from drain to source) and reverse rate, evaluated at the midpoint voltages, are

$$w_{+\rho}(\bar{\mathbf{v}}) = \frac{I_0}{q_e}\exp\!\left(\frac{V_G - V_{\mathrm{th}}}{nV_T}\right)\exp\!\left(-\frac{V_S}{V_T}\right), \qquad w_{-\rho}(\bar{\mathbf{v}}) = \frac{I_0}{q_e}\exp\!\left(\frac{V_G - V_{\mathrm{th}}}{nV_T}\right)\exp\!\left(-\frac{V_D}{V_T}\right). \tag{13.15}$$

In the ratio, the gate-dependent prefactor $\exp[(V_G - V_{\mathrm{th}})/(nV_T)]$ cancels:

$$\frac{w_{+\rho}(\bar{\mathbf{v}})}{w_{-\rho}(\bar{\mathbf{v}})} = \exp\!\left(\frac{V_D - V_S}{V_T}\right) = \exp\!\left(\frac{\Delta V_{DS}}{V_T}\right), \tag{13.16}$$

which is Eq. (13.14). The gate sets the height of the channel barrier and so scales both rates by the same factor, but each channel still obeys local detailed balance with respect to its own drain-source voltage.

Definition 13.1 (Chemical Master Equation of a Nonlinear Electronic Circuit)**.** Let $P(\mathbf{q},t)$ be the probability that the internal nodes carry charge $\mathbf{q} \in \Gamma_q$ at time t. Under Poisson jumps through the R branches, $P(\mathbf{q},t)$ obeys the *chemical master equation* (CME) [37]

$$\frac{\mathrm{d}}{\mathrm{d}t}P(\mathbf{q},t) = \sum_{\rho=1}^{R}\sum_{\sigma\in\{+,-\}}\left[W_{\sigma\rho}(\mathbf{q}-\sigma q_e\mathbf{\Delta}_\rho)\,P(\mathbf{q}-\sigma q_e\mathbf{\Delta}_\rho,t) - W_{\sigma\rho}(\mathbf{q})\,P(\mathbf{q},t)\right]. \tag{13.17}$$

With the net probability flux across branch ρ from $\mathbf{q}$ to $\mathbf{q}+q_e\mathbf{\Delta}_\rho$,

$$J_\rho(\mathbf{q},t) := W_{+\rho}(\mathbf{q})\,P(\mathbf{q},t) - W_{-\rho}(\mathbf{q}+q_e\mathbf{\Delta}_\rho)\,P(\mathbf{q}+q_e\mathbf{\Delta}_\rho,t), \tag{13.18}$$

the master equation takes the discrete continuity form

$$\frac{\mathrm{d}}{\mathrm{d}t}P(\mathbf{q},t) = -\sum_{\rho=1}^{R}\left[J_\rho(\mathbf{q},t) - J_\rho(\mathbf{q}-q_e\mathbf{\Delta}_\rho,t)\right]. \tag{13.19}$$

13.1.3 First and Second Laws of Thermodynamics at the Discrete-Charge Level

The master equation and local detailed balance give the first and second laws for any nonlinear circuit, without a continuum approximation.

Proposition 13.2 (Discrete-Charge Energy Balance and Schnakenberg Entropy Production)**.** *For any $P(\mathbf{q},t)$ evolving under Eq. (13.17) at constant external voltages $\mathbf{v}_{\mathrm{ext}}$:*

(i) (First law.) The rate of change of the mean electrostatic energy $\langle U(\mathbf{q})\rangle_t := \sum_{\mathbf{q}\in\Gamma_q} U(\mathbf{q})P(\mathbf{q},t)$ *equals the electrical power* $\dot{\mathcal{W}}_{\mathrm{ext}}(t)$ *supplied by the external voltage sources minus the mean heat dissipation rate* $\dot{\mathcal{Q}}(t)$ *into the thermal bath:*

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle U(\mathbf{q})\rangle_t = \dot{\mathcal{W}}_{\mathrm{ext}}(t) - \dot{\mathcal{Q}}(t), \tag{13.20}$$

where

$$\dot{\mathcal{W}}_{\mathrm{ext}}(t) := \sum_{\rho=1}^{R} q_e \mathcal{E}_\rho \sum_{\mathbf{q}\in\Gamma_q} J_\rho(\mathbf{q},t), \qquad \dot{\mathcal{Q}}(t) := \sum_{\rho=1}^{R}\sum_{\mathbf{q}\in\Gamma_q} J_\rho(\mathbf{q},t)\,\mathcal{Q}_\rho(\mathbf{q}). \tag{13.21}$$

(ii) (Second law.) Let

$$S_{\mathrm{sys}}(t) := -k_{\mathrm{B}} \sum_{\mathbf{q}\in\Gamma_q} P(\mathbf{q},t)\ln P(\mathbf{q},t)$$

denote the Gibbs–Shannon entropy of the discrete charge distribution. The total entropy production rate

$$\dot{\Sigma}_{\mathrm{tot}}(t) := \frac{\mathrm{d}}{\mathrm{d}t}S_{\mathrm{sys}}(t) + \frac{1}{T}\dot{\mathcal{Q}}(t)$$

is given by the Schnakenberg form

$$\dot{\Sigma}_{\mathrm{tot}}(t) = k_{\mathrm{B}}\sum_{\rho=1}^{R}\sum_{\mathbf{q}\in\Gamma_q} J_\rho(\mathbf{q},t)\,\ln\!\left(\frac{W_{+\rho}(\mathbf{q})\,P(\mathbf{q},t)}{W_{-\rho}(\mathbf{q}+q_e\boldsymbol{\Delta}_\rho)\,P(\mathbf{q}+q_e\boldsymbol{\Delta}_\rho,t)}\right) \geq 0, \tag{13.22}$$

which vanishes if and only if every branch flux satisfies $J_\rho(\mathbf{q},t) = 0$ *for all* $\mathbf{q}\in\Gamma_q$ *and* $\rho\in\{1,\dots,R\}$.

Proof. For part (i), multiply the continuity equation (Eq. (13.19)) by $U(\mathbf{q})$ and sum over the lattice $\mathbf{q}\in\Gamma_q$. Shifting the summation index $\mathbf{q}\to\mathbf{q}+q_e\boldsymbol{\Delta}_\rho$ in the second term gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\sum_{\mathbf{q}\in\Gamma_q} U(\mathbf{q})P(\mathbf{q},t) &= -\sum_{\rho=1}^{R}\sum_{\mathbf{q}\in\Gamma_q} U(\mathbf{q})J_\rho(\mathbf{q},t) + \sum_{\rho=1}^{R}\sum_{\mathbf{q}\in\Gamma_q} U(\mathbf{q}+q_e\boldsymbol{\Delta}_\rho)J_\rho(\mathbf{q},t)\\ &= \sum_{\rho=1}^{R}\sum_{\mathbf{q}\in\Gamma_q}\Delta U_\rho(\mathbf{q})\,J_\rho(\mathbf{q},t).\end{aligned} \tag{13.23}$$

Substituting $\Delta U_\rho(\mathbf{q}) = q_e\mathcal{E}_\rho - \mathcal{Q}_\rho(\mathbf{q})$ from Eq. (13.12) yields Eq. (13.20)–Eq. (13.21).

For part (ii), differentiate $S_{\mathrm{sys}}(t) = -k_{\mathrm{B}}\sum_{\mathbf{q}} P(\mathbf{q},t)\ln P(\mathbf{q},t)$ with respect to time using $\sum_{\mathbf{q}}\frac{\mathrm{d}P(\mathbf{q},t)}{\mathrm{d}t} = 0$ and the index shift $\mathbf{q}\to\mathbf{q}+q_e\boldsymbol{\Delta}_\rho$:

$$\begin{aligned}\frac{\mathrm{d}S_{\mathrm{sys}}(t)}{\mathrm{d}t} &= -k_{\mathrm{B}}\sum_{\mathbf{q}\in\Gamma_q}\frac{\mathrm{d}P(\mathbf{q},t)}{\mathrm{d}t}\ln P(\mathbf{q},t) = k_{\mathrm{B}}\sum_{\rho=1}^{R}\sum_{\mathbf{q}\in\Gamma_q}\left[J_\rho(\mathbf{q},t) - J_\rho(\mathbf{q}-q_e\boldsymbol{\Delta}_\rho,t)\right]\ln P(\mathbf{q},t)\\ &= k_{\mathrm{B}}\sum_{\rho=1}^{R}\sum_{\mathbf{q}\in\Gamma_q} J_\rho(\mathbf{q},t)\,\ln\!\left(\frac{P(\mathbf{q},t)}{P(\mathbf{q}+q_e\boldsymbol{\Delta}_\rho,t)}\right).\end{aligned} \tag{13.24}$$

Meanwhile, from local detailed balance (Eq. (13.13)), the heat per jump satisfies $\frac{\mathcal{Q}_\rho(\mathbf{q})}{T} = k_{\mathrm{B}}\ln\frac{W_{+\rho}(\mathbf{q})}{W_{-\rho}(\mathbf{q}+q_e\boldsymbol{\Delta}_\rho)}$, so the medium entropy production rate is

$$\frac{\dot{\mathcal{Q}}(t)}{T} = k_{\mathrm{B}}\sum_{\rho=1}^{R}\sum_{\mathbf{q}\in\Gamma_q} J_\rho(\mathbf{q},t)\,\ln\!\left(\frac{W_{+\rho}(\mathbf{q})}{W_{-\rho}(\mathbf{q}+q_e\boldsymbol{\Delta}_\rho)}\right). \tag{13.25}$$

Adding Eq. (13.24) and Eq. (13.25) produces Eq. (13.22). Since $(x-y)\ln(x/y) \geq 0$ for all $x, y > 0$ with equality if and only if $x = y$, every summand in Eq. (13.22) is non-negative, completing the proof. □

Example 13.3 (Master Equation of a Subthreshold CMOS Inverter)**.** Consider a single subthreshold CMOS inverter driving an output node with capacitance C to ground and discrete charge $q = mq_e$ ($m \in \mathbb{Z}$), corresponding to voltage $v_m = mq_e/C$. The input gate voltage is held at a constant bias $V_{\text{in}} \in [0, V_{\text{DD}}]$. The circuit has $R = 2$ conductive branches connected to node 1:

1. *PMOS pull-up branch ($\rho = 1$, to V_{DD}).* It carries positive charge from V_{DD} onto the output capacitor ($\Delta_1 = +1$, $\mathcal{E}_1 = V_{\text{DD}}$). From the EKV law for a p-channel device with source at V_{DD} and drain at v, its charging ($m \to m+1$) and discharging ($m+1 \to m$) rates at the midpoint voltage $\bar{v}_{m+1/2} = (m + \frac{1}{2})q_e/C$ are

$$W_{+1}(m) = \frac{I_{0,p}}{q_e} \exp\left(\frac{V_{\text{DD}} - V_{\text{in}} - |V_{\text{th},p}|}{n_p V_T}\right), \qquad W_{-1}(m+1) = W_{+1}(m) \exp\left(-\frac{V_{\text{DD}} - \bar{v}_{m+1/2}}{V_T}\right). \tag{13.26}$$

2. *NMOS pull-down branch ($\rho = 2$, to ground).* It carries charge between ground ($V_S = 0$) and the output capacitor ($\Delta_2 = +1$, $\mathcal{E}_2 = 0$). Its charging ($m \to m+1$) and discharging ($m+1 \to m$) rates are

$$W_{-2}(m+1) = \frac{I_{0,n}}{q_e} \exp\left(\frac{V_{\text{in}} - V_{\text{th},n}}{n_n V_T}\right), \qquad W_{+2}(m) = W_{-2}(m+1) \exp\left(-\frac{\bar{v}_{m+1/2}}{V_T}\right). \tag{13.27}$$

This is a birth-death chain on $\mathbb{Z}$ with upward rate $W_+(m) = W_{+1}(m) + W_{+2}(m)$ and downward rate $W_-(m+1) = W_{-1}(m+1) + W_{-2}(m+1)$. In the stationary state the net flux $J(m) = W_+(m)P_{\text{ss}}(m) - W_-(m+1)P_{\text{ss}}(m+1)$ must vanish, which gives the recurrence

$$\frac{P_{\text{ss}}(m+1)}{P_{\text{ss}}(m)} = \frac{W_{+1}(m) + W_{+2}(m)}{W_{-1}(m+1) + W_{-2}(m+1)}. \tag{13.28}$$

The summed flux $J_1(m) + J_2(m)$ vanishes, but the branch fluxes $J_1(m) = -J_2(m) > 0$ do not. Charge flows steadily from V_{DD} through the PMOS device and out to ground through the NMOS device, and the leakage current I_{leak} dissipates housekeeping heat $\dot{\mathcal{Q}}_{\text{hk}} = I_{\text{leak}} V_{\text{DD}} > 0$. The stationary state is not an equilibrium state even though the chain is one-dimensional.

Physics Insight: Coulomb Blockade vs. Mesoscopic Subthreshold CMOS

The charging energy $E_{\text{C}}^{(\rho)} \sim q_e^2/(2C_{\text{node}})$ in Eq. (13.9) matters when it is comparable to $k_{\text{B}}T$. The ratio is

$$\frac{E_{\text{C}}}{k_{\text{B}}T} = \frac{q_e^2}{2C_{\text{node}}k_{\text{B}}T} = \frac{q_e}{2C_{\text{node}}V_T} = \frac{1}{2\Omega},$$

where $\Omega := C_{\text{node}}V_T/q_e$ is the characteristic number of thermal elementary charges on node capacitance C_{node}.

- In single-electron transistors and small quantum dots at cryogenic temperatures ($T \lesssim 1\,\text{K}$, $C_{\text{node}} \sim 10\,\text{aF}$), $\Omega \ll 1$ and $E_{\text{C}} \gg k_{\text{B}}T$. One extra electron on the island raises its energy by much more than $k_{\text{B}}T$, which blocks transport unless the gate tunes the island to a charge-degeneracy point (*Coulomb blockade*).

- In room-temperature subthreshold CMOS ($T = 300\,\mathrm{K}$, $V_T \approx 25.85\,\mathrm{mV}$) with $C_{\mathrm{node}} \sim 0.5$–$5\,\mathrm{fF}$, the single-electron voltage step is $\Delta V_e = q_e/C_{\mathrm{node}} \approx 32$–$320\,\mu\mathrm{V}$, yielding $\Omega = V_T/\Delta V_e \approx 80$–$800 \gg 1$ and $E_{\mathrm{C}}/k_{\mathrm{B}}T = 1/(2\Omega) \sim 6\times10^{-4}$–$6\times10^{-3} \ll 1$.

With $\Omega \sim 10^2$–10^3, subthreshold CMOS is in an intermediate regime. Ω is large enough that there is no Coulomb blockade and the voltages can be described by a diffusion equation expanded in $1/\Omega$. It is small enough that the rms thermal voltage $\sigma_V \sim V_T/\sqrt{\Omega} = \sqrt{k_{\mathrm{B}}T/C_{\mathrm{node}}} \sim 1$–$3\,\mathrm{mV}$ is 4%–12% of V_T, large enough to use for sampling.

13.2 The Mesoscopic System-Size Expansion: From Discrete Charges to Circuit SDEs

The master equation is exact but hard to use. Circuit design works with continuous voltages, Fokker–Planck equations, and Langevin equations. Instead of adding white-noise sources by hand to the Kirchhoff equations, we derive them by expanding the master equation in Ω^{-1}, the inverse number of thermal charges per node [37, 38, 112]. The expansion gives the drift and the state-dependent diffusion matrix, and it also fixes where the diffusion matrix sits relative to the derivatives in the Fokker–Planck equation, a choice that matters when the noise depends on the state.

13.2.1 The Mesoscopic Scale Ω

Let C_0 denote a characteristic node capacitance of the circuit (for instance, $C_0 = \frac{1}{N}\operatorname{tr}\mathbf{C} \sim C_{\mathrm{node}}$), and define the dimensionless scale parameter

$$\Omega := \frac{C_0 V_T}{q_e} = \frac{C_0 k_{\mathrm{B}} T}{q_e^2} \gg 1. \tag{13.29}$$

It is the number of elementary charges needed to charge C_0 by one thermal voltage. We introduce the dimensionless capacitance matrix $\tilde{\mathbf{C}} := \mathbf{C}/C_0 = \mathcal{O}(1)$ so that $\mathbf{C} = C_0\tilde{\mathbf{C}} = (q_e\Omega/V_T)\tilde{\mathbf{C}}$.

For the expansion we imagine scaling the devices up together: capacitances and current prefactors $I_0 \propto W/L$ grow in proportion to Ω, while node voltages stay of order V_T. By Eq. (13.3), a single electron jump $\mathbf{q} \to \mathbf{q} + q_e\boldsymbol{\Delta}_\rho$ shifts the internal node voltage vector $\mathbf{v}$ by the small increment

$$\delta\mathbf{v}_\rho := q_e\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho = \frac{V_T}{\Omega}\tilde{\mathbf{C}}^{-1}\boldsymbol{\Delta}_\rho = \mathcal{O}(\Omega^{-1}). \tag{13.30}$$

The transition rates $W_{\pm\rho}(\mathbf{q})$ depend on the midpoint voltage $\bar{\mathbf{v}}_\rho = \mathbf{v}(\mathbf{q} \pm \frac{1}{2}q_e\boldsymbol{\Delta}_\rho) = \mathbf{v} \pm \frac{1}{2}\delta\mathbf{v}_\rho$:

$$W_{\pm\rho}(\mathbf{q}) = w_{\pm\rho}\!\left(\mathbf{v}(\mathbf{q}) \pm \frac{q_e}{2}\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho\right), \tag{13.31}$$

where $w_{\pm\rho}(\mathbf{v}) \propto \Omega$ are the jump rates as functions of the continuous voltage (for instance, Eq. (13.15)). Note the sign: for a forward jump $\mathbf{q} \to \mathbf{q} + q_e\boldsymbol{\Delta}_\rho$, the midpoint is $\mathbf{v} + \frac{1}{2}\delta\mathbf{v}_\rho$, whereas for a reverse jump starting at $\mathbf{q}$ ($\mathbf{q} \to \mathbf{q} - q_e\boldsymbol{\Delta}_\rho$), the midpoint between $\mathbf{q}$ and $\mathbf{q} - q_e\boldsymbol{\Delta}_\rho$ is $\mathbf{v} - \frac{1}{2}\delta\mathbf{v}_\rho$.

13.2.2 Kramers–Moyal Expansion of the Circuit Master Equation

We replace $P(\mathbf{q}, t)$ on the charge lattice by a smooth density $\rho(\mathbf{v}, t)$ over voltages, with $P(\mathbf{q}, t) = (q_e^N \det\mathbf{C})\,\rho(\mathbf{v}(\mathbf{q}), t)$. Since $\mathbf{v}(\mathbf{q} - \sigma q_e\boldsymbol{\Delta}_\rho) = \mathbf{v} - \sigma\delta\mathbf{v}_\rho$, a shift by one jump acts on a smooth function

$f(\mathbf{v})$ through its Taylor series:

$$f(\mathbf{v}-\sigma\delta\mathbf{v}_\rho) = \exp\Big(-\sigma\,\delta\mathbf{v}_\rho^\top\nabla_{\mathbf{v}}\Big)\,f(\mathbf{v}) \\ = f(\mathbf{v}) - \sigma\sum_{i=1}^{N}(\delta\mathbf{v}_\rho)_i\frac{\partial f}{\partial v_i} + \frac{1}{2}\sum_{i,j=1}^{N}(\delta\mathbf{v}_\rho)_i(\delta\mathbf{v}_\rho)_j\frac{\partial^2 f}{\partial v_i\partial v_j} + \mathcal{O}(\Omega^{-3}). \tag{13.32}$$

Theorem 13.4 (Mesoscopic Circuit Fokker–Planck and Langevin Equations)**.** *Consider a circuit obeying the master equation Definition 13.1 with transition rates $W_{\sigma\rho}(\mathbf{q}) = w_{\sigma\rho}(\mathbf{v}+\frac{\sigma}{2}\delta\mathbf{v}_\rho)$ satisfying local detailed balance (Eq. (13.14)). Define the mean branch current $I_\rho(\mathbf{v})$ and the branch noise intensity $S_\rho(\mathbf{v})$ by*

$$I_\rho(\mathbf{v}) := q_e\left[w_{+\rho}(\mathbf{v}) - w_{-\rho}(\mathbf{v})\right], \tag{13.33}$$

$$S_\rho(\mathbf{v}) := q_e^2\left[w_{+\rho}(\mathbf{v}) + w_{-\rho}(\mathbf{v})\right] = q_e I_\rho(\mathbf{v})\coth\!\left(\frac{\Delta V_\rho(\mathbf{v})}{2V_T}\right), \tag{13.34}$$

and the net nodal current vector $\mathbf{I}_{\mathrm{net}}(\mathbf{v}) \in \mathbb{R}^N$ and the symmetric positive-semidefinite charge-diffusion matrix $\mathbf{D}_q(\mathbf{v}) \in \mathbb{R}^{N\times N}$ by

$$\mathbf{I}_{\mathrm{net}}(\mathbf{v}) := \sum_{\rho=1}^{R}\boldsymbol{\Delta}_\rho\, I_\rho(\mathbf{v}), \qquad \mathbf{D}_q(\mathbf{v}) := \frac{1}{2}\sum_{\rho=1}^{R}\boldsymbol{\Delta}_\rho\boldsymbol{\Delta}_\rho^\top S_\rho(\mathbf{v}). \tag{13.35}$$

Then, retaining all terms up to second order in $\delta\mathbf{v}_\rho = q_e\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho = \mathcal{O}(\Omega^{-1})$, the probability density $\rho(\mathbf{v},t)$ evolves according to the nonlinear circuit Fokker–Planck equation:

$$\frac{\partial\rho(\mathbf{v},t)}{\partial t} = -\nabla_{\mathbf{v}}\cdot\Big[\mathbf{C}^{-1}\mathbf{I}_{\mathrm{net}}(\mathbf{v})\,\rho(\mathbf{v},t)\Big] \\ + \sum_{i,j=1}^{N}\frac{\partial}{\partial v_i}\left\{\Big[\mathbf{C}^{-1}\mathbf{D}_q(\mathbf{v})\mathbf{C}^{-1}\Big]_{ij}\frac{\partial\rho(\mathbf{v},t)}{\partial v_j}\right\}, \tag{13.36}$$

or, written out from the midpoint expansion Eq. (13.31),

$$\frac{\partial\rho(\mathbf{v},t)}{\partial t} = -\nabla_{\mathbf{v}}\cdot\Big\{\mathbf{C}^{-1}\mathbf{I}_{\mathrm{net}}(\mathbf{v})\,\rho(\mathbf{v},t) \\ -\frac{1}{2}\sum_{\rho=1}^{R}\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho\Big[\nabla_{\mathbf{v}}\cdot\Big(\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho S_\rho(\mathbf{v})\,\rho(\mathbf{v},t)\Big) - \Big(\nabla_{\mathbf{v}}S_\rho(\mathbf{v})\cdot\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho\Big)\,\rho(\mathbf{v},t)\Big]\Big\}. \tag{13.37}$$

At the trajectory level, this diffusion process is generated by the circuit Langevin equation

$$\mathbf{C}\,\mathrm{d}\mathbf{v}_t = \left[\mathbf{I}_{\mathrm{net}}(\mathbf{v}_t) + \mathbf{I}_{\mathrm{sp}}(\mathbf{v}_t)\right]\mathrm{d}t + \sum_{\rho=1}^{R}\boldsymbol{\Delta}_\rho\sqrt{S_\rho(\mathbf{v}_t)}\cdot\mathrm{d}W_{\rho,t}, \tag{13.38}$$

where $W_{1,t},\ldots,W_{R,t}$ are independent standard Wiener processes, $\cdot$ denotes the Itô stochastic integral (Chapter 4), and $\mathbf{I}_{\mathrm{sp}}(\mathbf{v}) := \frac{1}{2}\sum_{\rho=1}^{R}\boldsymbol{\Delta}_\rho\left(\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho\cdot\nabla_{\mathbf{v}}S_\rho(\mathbf{v})\right) = \mathcal{O}(\Omega^{-1}\mathbf{I}_{\mathrm{net}})$ is the Itô drift correction induced by the midpoint rates.

Proof. For each branch ρ and jump direction $\sigma\in\{+1,-1\}$, define $F_{\sigma\rho}(\mathbf{v}) := W_{\sigma\rho}(\mathbf{q})\,\rho(\mathbf{v},t)$, where $W_{\sigma\rho}(\mathbf{q}) = w_{\sigma\rho}(\mathbf{v}+\frac{\sigma}{2}\delta\mathbf{v}_\rho)$. The master equation (Eq. (13.17)) reads

$$\frac{\partial\rho(\mathbf{v},t)}{\partial t} = \sum_{\rho=1}^{R}\sum_{\sigma\in\{+,-\}}\left[F_{\sigma\rho}(\mathbf{v}-\sigma\delta\mathbf{v}_\rho) - F_{\sigma\rho}(\mathbf{v})\right]. \tag{13.39}$$

Applying the second-order Taylor expansion (Eq. (13.32)) to $F_{\sigma\rho}(\mathbf{v}-\sigma\delta\mathbf{v}_\rho)$ and noting that $\sigma^2=1$, we obtain

$$\frac{\partial\rho(\mathbf{v},t)}{\partial t}=\sum_{\rho=1}^{R}\sum_{\sigma\in\{+,-\}}\left[-\sigma(\delta\mathbf{v}_\rho\cdot\nabla_{\mathbf{v}})F_{\sigma\rho}(\mathbf{v})+\frac{1}{2}(\delta\mathbf{v}_\rho\cdot\nabla_{\mathbf{v}})^2F_{\sigma\rho}(\mathbf{v})+\mathcal{O}(\delta\mathbf{v}_\rho^3)\right]. \tag{13.40}$$

To keep all terms of order $\delta\mathbf{v}_\rho^2$ in Eq. (13.40), we also expand the rates to first order:

$$w_{\sigma\rho}\left(\mathbf{v}+\frac{\sigma}{2}\delta\mathbf{v}_\rho\right)=w_{\sigma\rho}(\mathbf{v})+\frac{\sigma}{2}(\delta\mathbf{v}_\rho\cdot\nabla_{\mathbf{v}})w_{\sigma\rho}(\mathbf{v})+\mathcal{O}(\delta\mathbf{v}_\rho^2). \tag{13.41}$$

Multiplying by $\rho(\mathbf{v},t)$ and forming the antisymmetric and symmetric sums over σ,

$$\begin{aligned}\sum_{\sigma\in\{+,-\}}\sigma F_{\sigma\rho}(\mathbf{v})&=\left[w_{+\rho}(\mathbf{v})-w_{-\rho}(\mathbf{v})\right]\rho(\mathbf{v},t)+\frac{1}{2}\left[(\delta\mathbf{v}_\rho\cdot\nabla_{\mathbf{v}})\left(w_{+\rho}(\mathbf{v})+w_{-\rho}(\mathbf{v})\right)\right]\rho(\mathbf{v},t)+\mathcal{O}(\delta\mathbf{v}_\rho^2)\\&=\frac{1}{q_e}I_\rho(\mathbf{v})\,\rho(\mathbf{v},t)+\frac{1}{2q_e^2}\left[(\delta\mathbf{v}_\rho\cdot\nabla_{\mathbf{v}})S_\rho(\mathbf{v})\right]\rho(\mathbf{v},t)+\mathcal{O}(\delta\mathbf{v}_\rho^2),\end{aligned} \tag{13.42}$$

and, to leading order needed for the second-derivative term in Eq. (13.40),

$$\sum_{\sigma\in\{+,-\}}F_{\sigma\rho}(\mathbf{v})=\left[w_{+\rho}(\mathbf{v})+w_{-\rho}(\mathbf{v})\right]\rho(\mathbf{v},t)+\mathcal{O}(\delta\mathbf{v}_\rho)=\frac{1}{q_e^2}S_\rho(\mathbf{v})\,\rho(\mathbf{v},t)+\mathcal{O}(\delta\mathbf{v}_\rho). \tag{13.43}$$

Substituting these into Eq. (13.40), with $\delta\mathbf{v}_\rho=q_e\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho$ a constant vector, the factors of q_e cancel:

$$\begin{aligned}\frac{\partial\rho(\mathbf{v},t)}{\partial t}=&-\sum_{\rho=1}^{R}(\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho\cdot\nabla_{\mathbf{v}})\left\{\left[I_\rho(\mathbf{v})+\frac{1}{2}(\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho\cdot\nabla_{\mathbf{v}})S_\rho(\mathbf{v})\right]\rho(\mathbf{v},t)\right\}\\&+\frac{1}{2}\sum_{\rho=1}^{R}(\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho\cdot\nabla_{\mathbf{v}})^2\left[S_\rho(\mathbf{v})\,\rho(\mathbf{v},t)\right].\end{aligned} \tag{13.44}$$

The drift correction combines with the second-order term. With the constant vector $\mathbf{u}_\rho:=\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho$, the product rule gives

$$\frac{1}{2}(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}})^2(S_\rho\rho)-\frac{1}{2}(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}})\left[\rho(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}}S_\rho)\right]=\frac{1}{2}(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}})\left[S_\rho(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}}\rho)\right]. \tag{13.45}$$

(Expanding the first term gives $\frac{1}{2}(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}})\left[\rho(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}}S_\rho)+S_\rho(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}}\rho)\right]$, and subtracting $\frac{1}{2}(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}})\left[\rho(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}}S_\rho)\right]$ leaves $\frac{1}{2}(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}})\left[S_\rho(\mathbf{u}_\rho\cdot\nabla_{\mathbf{v}}\rho)\right]$.) Summing over $\rho=1,\dots,R$ and recalling from Eq. (13.35) that $\sum_\rho\mathbf{u}_\rho I_\rho(\mathbf{v})=\mathbf{C}^{-1}\mathbf{I}_{\text{net}}(\mathbf{v})$ and $\frac{1}{2}\sum_\rho\mathbf{u}_\rho\mathbf{u}_\rho^\top S_\rho(\mathbf{v})=\mathbf{C}^{-1}\mathbf{D}_q(\mathbf{v})\mathbf{C}^{-1}$ (using $\mathbf{C}^{-\top}=\mathbf{C}^{-1}$), Eq. (13.44) becomes the Hänggi–Klimontovich (kinetic) form of the Fokker–Planck equation:

$$\frac{\partial\rho(\mathbf{v},t)}{\partial t}=-\nabla_{\mathbf{v}}\cdot\left[\mathbf{C}^{-1}\mathbf{I}_{\text{net}}(\mathbf{v})\,\rho(\mathbf{v},t)\right]+\nabla_{\mathbf{v}}\cdot\left[\mathbf{C}^{-1}\mathbf{D}_q(\mathbf{v})\mathbf{C}^{-1}\nabla_{\mathbf{v}}\rho(\mathbf{v},t)\right]. \tag{13.46}$$

To write the equivalent Itô SDE (Eq. (13.38)), recall from Section 4.4 and Section 6.1 that an Itô SDE $\mathrm{d}\mathbf{v}_t=\mathbf{a}_{\text{Ito}}(\mathbf{v}_t)\,\mathrm{d}t+\sum_\rho\mathbf{b}_\rho(\mathbf{v}_t)\cdot\mathrm{d}W_{\rho,t}$ with $\mathbf{b}_\rho(\mathbf{v})=\mathbf{C}^{-1}\boldsymbol{\Delta}_\rho\sqrt{S_\rho(\mathbf{v})}$ generates the Fokker–Planck equation $\partial_t\rho=-\nabla\cdot(\mathbf{a}_{\text{Ito}}\rho)+\frac{1}{2}\sum_{i,j}\partial_i\partial_j[(\sum_\rho\mathbf{b}_\rho\mathbf{b}_\rho^\top)_{ij}\rho]$. Here $\frac{1}{2}\sum_\rho\mathbf{b}_\rho\mathbf{b}_\rho^\top=\frac{1}{2}\sum_\rho\mathbf{u}_\rho\mathbf{u}_\rho^\top S_\rho=\mathbf{D}$, the same diffusion matrix as in Eq. (13.46). Writing $\partial_i(D_{ij}\partial_j\rho)=\partial_i\partial_j(D_{ij}\rho)-\partial_i((\partial_jD_{ij})\rho)$

turns Eq. (13.46) into the Itô form with drift $\mathbf{a}_{\mathrm{Ito}} = \mathbf{C}^{-1}\mathbf{I}_{\mathrm{net}} + \nabla_{\mathbf{v}} \cdot \mathbf{D}$, where $(\nabla_{\mathbf{v}} \cdot \mathbf{D})_i = \sum_j \partial_j D_{ij} = \frac{1}{2}\sum_\rho (\mathbf{u}_\rho)_i \, (\mathbf{u}_\rho \cdot \nabla_{\mathbf{v}} S_\rho)$. Multiplying by $\mathbf{C}$ gives $\mathbf{I}_{\mathrm{net}} + \mathbf{I}_{\mathrm{sp}}$, which confirms Eq. (13.38). For comparison, the Stratonovich form of the same process has the smaller correction $\frac{1}{2}\nabla_{\mathbf{v}} \cdot \mathbf{D}$, because the Stratonovich integral $\mathbf{b}_\rho \circ \mathrm{d}W_{\rho,t}$ already contains the drift $\frac{1}{2}(\mathbf{b}_\rho \cdot \nabla)\mathbf{b}_\rho = \frac{1}{4}\mathbf{u}_\rho(\mathbf{u}_\rho \cdot \nabla S_\rho)$. Finally, the coth relation in Eq. (13.34), the circuit form of Theorem 11.11, follows from local detailed balance $w_{+\rho}/w_{-\rho} = e^{\Delta V_\rho / V_T}$:

$$\frac{S_\rho(\mathbf{v})}{q_e I_\rho(\mathbf{v})} = \frac{w_{+\rho} + w_{-\rho}}{w_{+\rho} - w_{-\rho}} = \frac{e^{\Delta V_\rho/V_T} + 1}{e^{\Delta V_\rho/V_T} - 1} = \frac{e^{\Delta V_\rho/(2V_T)} + e^{-\Delta V_\rho/(2V_T)}}{e^{\Delta V_\rho/(2V_T)} - e^{-\Delta V_\rho/(2V_T)}} = \coth\!\left(\frac{\Delta V_\rho(\mathbf{v})}{2V_T}\right), \tag{13.47}$$

completing the proof. □

Notation Watch: Itô, Stratonovich, and Hänggi–Klimontovich Ordering in Circuit Noise

Theorem 13.4 settles an old puzzle about noise in nonlinear circuits, related to Brillouin's paradox. Take a nonlinear resistor or diode with state-dependent conductance $G(v)$ and add an Itô noise term $\sqrt{2k_{\mathrm{B}}TG(v)} \cdot \mathrm{d}W_t$ without the midpoint shift $\bar{\mathbf{v}}_\rho = \mathbf{v} + \frac{\sigma}{2}\delta\mathbf{v}_\rho$. The resulting Fokker–Planck equation does not relax to the Boltzmann distribution $p_{\mathrm{eq}}(v) \propto \exp(-\frac{Cv^2}{2k_{\mathrm{B}}T})$; it predicts a spurious rectified voltage in equilibrium (Worked computation 4.23).
The derivation Eq. (13.45)–Eq. (13.46) shows that the midpoint rates of local detailed balance (Eq. (13.14)) place the diffusion matrix $\mathbf{D}(\mathbf{v}) := \mathbf{C}^{-1}\mathbf{D}_q(\mathbf{v})\mathbf{C}^{-1}$ *between* the two derivatives:

$$\frac{\partial \rho}{\partial t} = -\nabla_{\mathbf{v}} \cdot \left[\mathbf{C}^{-1}\mathbf{I}_{\mathrm{net}}(\mathbf{v})\,\rho - \mathbf{D}(\mathbf{v})\nabla_{\mathbf{v}}\rho\right].$$

This placement is the *Hänggi–Klimontovich* (kinetic, or isothermal) convention. With it, a passive nonlinear circuit obeying local detailed balance relaxes to the Boltzmann distribution $p_{\mathrm{eq}}(\mathbf{v}) \propto \exp(-\beta U(\mathbf{v}))$ at this order in Ω^{-1}. When we linearize about an operating point $\mathbf{v}^*$ in Section 13.4, $\mathbf{D}$ is evaluated at $\mathbf{v}^*$ and is constant, so the three conventions coincide at leading order.

13.2.3 Van Kampen's System-Size Expansion and Higher-Order Terms

We truncated the Taylor series Eq. (13.32) at second order. Pawula's theorem [90] states that truncating a Kramers–Moyal expansion at any finite order $m \geq 3$ produces an equation that does not preserve positivity, so keeping more terms is not the way to improve the approximation. Van Kampen's *system-size expansion* [112] justifies stopping at second order. It splits the node voltage into a deterministic trajectory $\bar{\mathbf{v}}(t)$ and a fluctuation of size $\Omega^{-1/2}V_T$:

$$\mathbf{v}(t) = \bar{\mathbf{v}}(t) + \Omega^{-1/2}V_T\,\boldsymbol{\xi}(t). \tag{13.48}$$

We substitute Eq. (13.48) into the full Kramers–Moyal expansion. Under the change of variables $\boldsymbol{\xi} = \frac{\Omega^{1/2}}{V_T}(\mathbf{v} - \bar{\mathbf{v}}(t))$, the density becomes $\Pi(\boldsymbol{\xi}, t) = (V_T\Omega^{-1/2})^N \rho(\bar{\mathbf{v}}(t) + \Omega^{-1/2}V_T\boldsymbol{\xi}, t)$, and the derivatives transform as

$$\frac{\partial \rho}{\partial t}\Big|_{\mathbf{v}} = \left(\frac{\Omega^{1/2}}{V_T}\right)^N \left[\frac{\partial \Pi}{\partial t}\Big|_{\boldsymbol{\xi}} - \frac{\Omega^{1/2}}{V_T}\frac{\mathrm{d}\bar{\mathbf{v}}}{\mathrm{d}t} \cdot \nabla_{\boldsymbol{\xi}}\Pi\right], \qquad \nabla_{\mathbf{v}} = \frac{\Omega^{1/2}}{V_T}\nabla_{\boldsymbol{\xi}}. \tag{13.49}$$

Since $\delta\mathbf{v}_\rho = \frac{V_T}{\Omega}\tilde{\mathbf{C}}^{-1}\boldsymbol{\Delta}_\rho$, the jump operator in the $\boldsymbol{\xi}$ coordinates is

$$\delta\mathbf{v}_\rho \cdot \nabla_{\mathbf{v}} = \Omega^{-1/2}\left(\tilde{\mathbf{C}}^{-1}\boldsymbol{\Delta}_\rho \cdot \nabla_{\boldsymbol{\xi}}\right). \tag{13.50}$$

The rates scale as $w_{\pm\rho}(\mathbf{v}) = \Omega\,\tilde{w}_{\pm\rho}(\mathbf{v}/V_T)$ with $\tilde{w}_{\pm\rho} = \mathcal{O}(\Omega^0)$. The m-th Kramers–Moyal term in the $\boldsymbol{\xi}$ variables therefore scales as

$$\frac{1}{m!}\sum_{\rho,\sigma}(-\sigma\delta\mathbf{v}_\rho\cdot\nabla_{\mathbf{v}})^m\,[W_{\sigma\rho}(\mathbf{q})\,\Pi(\boldsymbol{\xi},t)] = \Omega\cdot\left(\Omega^{-1/2}\right)^m\mathcal{O}(\Omega^0) = \mathcal{O}\left(\Omega^{1-m/2}\right). \tag{13.51}$$

Collecting powers of $\Omega^{-1/2}$:

1. *Order $\Omega^{1/2}$ ($m=1$).* The terms proportional to $\nabla_{\boldsymbol{\xi}}\Pi$ in Eq. (13.49) and in the $m=1$ term cancel if and only if $\bar{\mathbf{v}}(t)$ obeys the deterministic Kirchhoff equation

$$\mathbf{C}\frac{\mathrm{d}\bar{\mathbf{v}}(t)}{\mathrm{d}t} = \mathbf{I}_{\mathrm{net}}(\bar{\mathbf{v}}(t))\,. \tag{13.52}$$

2. *Order Ω^0 ($m=1$ and $m=2$).* Expanding $\mathbf{I}_{\mathrm{net}}(\bar{\mathbf{v}}+\Omega^{-1/2}V_T\boldsymbol{\xi}) = \mathbf{I}_{\mathrm{net}}(\bar{\mathbf{v}}) - \Omega^{-1/2}V_T\mathbf{G}(\bar{\mathbf{v}})\boldsymbol{\xi} + \mathcal{O}(\Omega^{-1})$ in the $m=1$ term and keeping the leading $m=2$ term gives a linear Fokker–Planck equation for $\Pi(\boldsymbol{\xi},t)$:

$$\frac{\partial\Pi(\boldsymbol{\xi},t)}{\partial t} = -\nabla_{\boldsymbol{\xi}}\cdot[\mathbf{A}(\bar{\mathbf{v}}(t))\,\boldsymbol{\xi}\,\Pi(\boldsymbol{\xi},t)] + \frac{\Omega}{V_T^2}\nabla_{\boldsymbol{\xi}}\cdot[\mathbf{D}(\bar{\mathbf{v}}(t))\,\nabla_{\boldsymbol{\xi}}\Pi(\boldsymbol{\xi},t)]\,. \tag{13.53}$$

3. *Order $\Omega^{-1/2}$ and beyond ($m\geq 3$).* The third-order term, proportional to the net current $I_\rho(\mathbf{v})$, is suppressed by $\Omega^{-1/2}$ and vanishes for a branch carrying no net current. The fourth-order term, proportional to $S_\rho(\mathbf{v})$, is suppressed by Ω^{-1}.

13.3 Why Active Subthreshold CMOS Breaks Detailed Balance

With the circuit Fokker–Planck equation Eq. (13.46) we can compare passive networks with active subthreshold networks. The comparison shows why a passive RC circuit cannot be programmed to a given covariance, and how active transistors make this possible by breaking detailed balance.

13.3.1 The Passive Equipartition Lock in Equilibrium RC Networks

Consider first a passive circuit in which all external voltage sources are grounded ($\mathbf{v}_{\mathrm{ext}} = \mathbf{0}$, so $\mathbf{q}_0 = \mathbf{0}$ and $\mathcal{E}_\rho = 0$ for every branch ρ). Every branch ρ is then a two-terminal passive device with voltage drop $\Delta V_\rho(\mathbf{v}) = -\boldsymbol{\Delta}_\rho^\top\mathbf{v} = v_{i_\rho} - v_{j_\rho}$.

Proposition 13.5 (The Passive Equipartition Lock)**.** *In any passive electronic circuit with $\mathcal{E}_\rho = 0$ and two-terminal branches obeying local detailed balance (Eq. (13.14)) around $\mathbf{v}^* = \mathbf{0}$:*

(i) The linearized conductance matrix $\mathbf{G} := -\frac{\partial\mathbf{I}_{\mathrm{net}}}{\partial\mathbf{v}}\big|_{\mathbf{v}=\mathbf{0}}$ is symmetric positive-semidefinite ($\mathbf{G} = \mathbf{G}^\top \succeq 0$), and the charge-diffusion matrix obeys the Johnson–Nyquist (fluctuation–dissipation) relation

$$\mathbf{D}_q(\mathbf{0}) = k_{\mathrm{B}}T\,\mathbf{G}. \tag{13.54}$$

(ii) Whatever the branch conductances $g_\rho > 0$ (provided the conducting branches connect every node to ground, so that $\mathbf{G}\succ 0$), the stationary distribution of the linearized Fokker–Planck equation is the Boltzmann distribution

$$p_{\mathrm{eq}}(\mathbf{v}) = \frac{1}{Z}\exp\left(-\frac{U(\mathbf{v})}{k_{\mathrm{B}}T}\right) = \sqrt{\frac{\det\mathbf{C}}{(2\pi k_{\mathrm{B}}T)^N}}\exp\left(-\frac{1}{2k_{\mathrm{B}}T}\mathbf{v}^\top\mathbf{C}\mathbf{v}\right), \tag{13.55}$$

with zero probability current $\mathbf{J}_{\mathrm{eq}}(\mathbf{v}) = \mathbf{0}$ *and covariance*

$$\boldsymbol{\Sigma}_{\mathrm{eq}} = \mathrm{Cov}_{p_{\mathrm{eq}}}(\mathbf{v}) = k_{\mathrm{B}}T\,\mathbf{C}^{-1}. \tag{13.56}$$

Proof. For each passive two-terminal branch ρ with $\Delta V_\rho(\mathbf{v}) = -\boldsymbol{\Delta}_\rho^\top \mathbf{v}$, local detailed balance $w_{+\rho}/w_{-\rho} = \exp(\Delta V_\rho/V_T)$ implies that at $\mathbf{v} = \mathbf{0}$ ($\Delta V_\rho = 0$), $w_{+\rho}(\mathbf{0}) = w_{-\rho}(\mathbf{0}) := w_\rho^{(0)}$, so $I_\rho(\mathbf{0}) = 0$. Differentiating $I_\rho(\Delta V_\rho) = q_e w_{-\rho}(\Delta V_\rho)(e^{\Delta V_\rho/V_T} - 1)$ at $\Delta V_\rho = 0$ gives the branch conductance

$$g_\rho := \frac{\mathrm{d}I_\rho}{\mathrm{d}\Delta V_\rho}\Big|_{\Delta V_\rho=0} = \frac{q_e w_\rho^{(0)}}{V_T} = \frac{q_e^2 w_\rho^{(0)}}{k_{\mathrm{B}}T} \geq 0. \tag{13.57}$$

By the chain rule, $\frac{\partial I_\rho}{\partial \mathbf{v}}\big|_{\mathbf{0}} = -g_\rho \boldsymbol{\Delta}_\rho^\top$, so the network conductance matrix is

$$\mathbf{G} = -\frac{\partial \mathbf{I}_{\mathrm{net}}}{\partial \mathbf{v}}\Big|_{\mathbf{0}} = \sum_{\rho=1}^{R} g_\rho\, \boldsymbol{\Delta}_\rho \boldsymbol{\Delta}_\rho^\top = \mathbf{G}^\top \succeq 0. \tag{13.58}$$

Meanwhile, evaluating $S_\rho(\mathbf{0}) = q_e^2[w_{+\rho}(\mathbf{0}) + w_{-\rho}(\mathbf{0})] = 2q_e^2 w_\rho^{(0)} = 2k_{\mathrm{B}}T g_\rho$ and substituting into Eq. (13.35) gives

$$\mathbf{D}_q(\mathbf{0}) = \frac{1}{2}\sum_{\rho=1}^{R} \boldsymbol{\Delta}_\rho \boldsymbol{\Delta}_\rho^\top (2k_{\mathrm{B}}T g_\rho) = k_{\mathrm{B}}T \sum_{\rho=1}^{R} g_\rho\, \boldsymbol{\Delta}_\rho \boldsymbol{\Delta}_\rho^\top = k_{\mathrm{B}}T\,\mathbf{G}, \tag{13.59}$$

proving part (i). For part (ii), the linearized circuit has $\mathbf{I}_{\mathrm{net}}(\mathbf{v}) = -\mathbf{G}\mathbf{v}$ and $\mathbf{D} = \mathbf{C}^{-1}\mathbf{D}_q\mathbf{C}^{-1} = k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{G}\mathbf{C}^{-1}$. We evaluate the probability current of Eq. (13.46) on the Boltzmann distribution $p_{\mathrm{eq}}(\mathbf{v}) \propto \exp(-\frac{1}{2k_{\mathrm{B}}T}\mathbf{v}^\top\mathbf{C}\mathbf{v})$. Since $\nabla_{\mathbf{v}} p_{\mathrm{eq}}(\mathbf{v}) = -\frac{1}{k_{\mathrm{B}}T}\mathbf{C}\mathbf{v}\, p_{\mathrm{eq}}(\mathbf{v})$, we have

$$\begin{aligned}
\mathbf{J}_{\mathrm{eq}}(\mathbf{v}) &= \mathbf{C}^{-1}\mathbf{I}_{\mathrm{net}}(\mathbf{v})\, p_{\mathrm{eq}}(\mathbf{v}) - \mathbf{D}\,\nabla_{\mathbf{v}} p_{\mathrm{eq}}(\mathbf{v}) \\
&= -\mathbf{C}^{-1}\mathbf{G}\mathbf{v}\, p_{\mathrm{eq}}(\mathbf{v}) - \left(k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{G}\mathbf{C}^{-1}\right)\left(-\frac{1}{k_{\mathrm{B}}T}\mathbf{C}\mathbf{v}\, p_{\mathrm{eq}}(\mathbf{v})\right) \\
&= -\mathbf{C}^{-1}\mathbf{G}\mathbf{v}\, p_{\mathrm{eq}}(\mathbf{v}) + \mathbf{C}^{-1}\mathbf{G}\mathbf{v}\, p_{\mathrm{eq}}(\mathbf{v}) = \mathbf{0}.
\end{aligned} \tag{13.60}$$

Thus $p_{\mathrm{eq}}(\mathbf{v})$ satisfies detailed balance ($\mathbf{J}_{\mathrm{eq}}(\mathbf{v}) \equiv \mathbf{0}$) and has covariance $\boldsymbol{\Sigma}_{\mathrm{eq}} = k_{\mathrm{B}}T\mathbf{C}^{-1}$, independent of $\mathbf{G}$. □

The proof shows where the lock comes from. In Eq. (13.60) the same matrix $\mathbf{G}$ appears in the restoring drift $-\mathbf{C}^{-1}\mathbf{G}\mathbf{v}$ and in the diffusion matrix $k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{G}\mathbf{C}^{-1}$, so it cancels from the stationary distribution. To program the covariance electronically, with couplings of either sign, a circuit must break the tie $\mathbf{D}_q = k_{\mathrm{B}}T\mathbf{G}$ between noise and drift.

13.3.2 Two Ways Active Subthreshold CMOS Breaks Detailed Balance

In an active network powered from V_{DD} and ground, transistors act as transconductance elements with three or four terminals. This changes both the linearized conductance matrix $\mathbf{G} = -\frac{\partial \mathbf{I}_{\mathrm{net}}}{\partial \mathbf{v}}\big|_{\mathbf{v}^*}$ and the charge-diffusion matrix $\mathbf{D}_q(\mathbf{v}^*)$ at an operating point $\mathbf{v}^*$ where $\mathbf{I}_{\mathrm{net}}(\mathbf{v}^*) = \mathbf{0}$.

Theorem 13.6 (Two Mechanisms of Detailed-Balance Violation in Active Subthreshold Networks)**.** *Consider an active subthreshold network at an operating point* $\mathbf{v}^*$ *with* $\mathbf{I}_{\mathrm{net}}(\mathbf{v}^*) = \mathbf{0}$*, small-signal*

conductance matrix $G_{ij} := -\frac{\partial I_{\mathrm{net},i}}{\partial v_j}\big|_{\mathbf{v}^*}$, *and small-signal charge-diffusion matrix* $\mathbf{D}_q := \mathbf{D}_q(\mathbf{v}^*)$. *The linearized dynamics around* $\mathbf{v}^*$,

$$\mathbf{C}\,\mathrm{d}(\delta\mathbf{v}_t) = -\mathbf{G}\,\delta\mathbf{v}_t\,\mathrm{d}t + \sqrt{2\mathbf{D}_q}\,\mathrm{d}\mathbf{W}_t, \qquad \delta\mathbf{v}_t := \mathbf{v}_t - \mathbf{v}^*, \tag{13.61}$$

can violate detailed balance ($\mathbf{J}_{\mathrm{ss}}(\delta\mathbf{v}) \neq \mathbf{0}$) *through two distinct mechanisms:*

(i) Non-reciprocal transconductance ($\mathbf{G} \neq \mathbf{G}^\top$). *When the gate of a subthreshold MOSFET, or the input of an operational transconductance amplifier (OTA, Worked computation 12.24), is connected to node j and its drain or output to node i, the current into node i depends on v_j with transconductance $g_{\mathrm{m}} = I_{\mathrm{D}}/(nV_T) > 0$. The insulated gate draws no current, so node i does not act back on node j:*

$$G_{ij} = -\frac{\partial I_{\mathrm{net},i}}{\partial v_j}\big|_{\mathbf{v}^*} \neq G_{ji} = -\frac{\partial I_{\mathrm{net},j}}{\partial v_i}\big|_{\mathbf{v}^*}, \tag{13.62}$$

so the skew-symmetric part $\mathbf{G}_{\mathrm{skew}} := \frac{1}{2}(\mathbf{G} - \mathbf{G}^\top) \neq \mathbf{0}$.

(ii) Bias-dependent shot noise ($\mathbf{D}_q \neq k_{\mathrm{B}}T_{\mathrm{eff}}\mathbf{G}_{\mathrm{sym}}$). *Even when transconductance cells are paired so that* $\mathbf{G} = \mathbf{G}^\top$, *each saturated transistor ($|V_{\mathrm{DS}}| \gg V_T$) carries a standing current I_{bias} from V_{DD} to ground and produces shot noise $2q_eI_{\mathrm{bias}}$ in the single-sided convention of Section 12.5, that is, $S_\rho = q_eI_{\mathrm{bias}}$ in the convention of Eq. (13.34). Its transconductance is $g_{\mathrm{m}} = I_{\mathrm{bias}}/(nV_T)$, while its output conductance in saturation is only $g_{\mathrm{ds}} \approx I_{\mathrm{bias}}/V_E \ll g_{\mathrm{m}}$, with V_E the Early voltage. The noise matrix $\mathbf{D}_q$ is therefore set by the standing bias currents, not by $k_{\mathrm{B}}T\mathbf{G}_{\mathrm{sym}}$.*

More generally, with the drift matrix $\mathbf{A} := -\mathbf{C}^{-1}\mathbf{G}$ *and the voltage diffusion matrix* $\mathbf{D} := \mathbf{C}^{-1}\mathbf{D}_q\mathbf{C}^{-1}$, *the linearized circuit Eq. (13.61) satisfies detailed balance* ($\mathbf{J}_{\mathrm{ss}} \equiv \mathbf{0}$) *if and only if the Onsager symmetry condition holds:*

$$\mathbf{A}\mathbf{D} = \mathbf{D}\mathbf{A}^\top \quad\Longleftrightarrow\quad \mathbf{G}\mathbf{C}^{-1}\mathbf{D}_q = \mathbf{D}_q\mathbf{C}^{-1}\mathbf{G}^\top. \tag{13.63}$$

Proof. Statements (i) and (ii) follow from the EKV current and noise laws of Chapter 12 and from Theorem 13.4. For a transistor channel ρ connecting node i to ground or V_{DD} (Eq. (13.8)) whose gate is driven by node j, the incidence vector $\boldsymbol{\Delta}_\rho$ has a non-zero entry only at component i, so $\frac{\partial I_{\mathrm{net},i}}{\partial v_j} = \pm g_{\mathrm{m}}^{(\rho)}$ while $\frac{\partial I_{\mathrm{net},j}}{\partial v_i} = 0$, establishing Eq. (13.62). In saturation ($\Delta V_{DS} \geq 4V_T$, so $\coth(\Delta V_{DS}/(2V_T)) \approx 1$), Eq. (13.34) gives $S_\rho = q_eI_{\mathrm{bias},\rho}$, so the diagonal entry $[\mathbf{D}_q]_{ii} = \frac{1}{2}\sum_{\rho\ni i} q_eI_{\mathrm{bias},\rho}$ depends on the total standing bias current flowing through node i, which can be varied independently of the cross-transconductances G_{ij} ($i \neq j$).

For the condition Eq. (13.63), suppose $\mathbf{A}$ is Hurwitz stable and let $p_{\mathrm{ss}}(\delta\mathbf{v}) = \frac{1}{Z}\exp(-\frac{1}{2}\delta\mathbf{v}^\top\boldsymbol{\Sigma}^{-1}\delta\mathbf{v})$ be the stationary Gaussian distribution of the Fokker–Planck equation $\partial_t\rho = -\nabla\cdot(\mathbf{A}\delta\mathbf{v}\,\rho - \mathbf{D}\nabla\rho)$. Detailed balance requires that the stationary probability current vanish identically for all $\delta\mathbf{v} \in \mathbb{R}^N$:

$$\mathbf{J}_{\mathrm{ss}}(\delta\mathbf{v}) = \Big(\mathbf{A}\delta\mathbf{v} + \mathbf{D}\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\Big)\,p_{\mathrm{ss}}(\delta\mathbf{v}) = \mathbf{0} \quad \forall\,\delta\mathbf{v} \in \mathbb{R}^N \quad\Longleftrightarrow\quad \mathbf{A} = -\mathbf{D}\boldsymbol{\Sigma}^{-1} \quad\Longleftrightarrow\quad \boldsymbol{\Sigma} = -\mathbf{A}^{-1}\mathbf{D}. \tag{13.64}$$

Since the covariance matrix $\boldsymbol{\Sigma}$ is necessarily symmetric ($\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^\top$), Eq. (13.64) holds if and only if $\mathbf{A}^{-1}\mathbf{D} = (\mathbf{A}^{-1}\mathbf{D})^\top = \mathbf{D}\mathbf{A}^{-\top}$. Multiplying on the left by $\mathbf{A}$ and on the right by $\mathbf{A}^\top$ yields $\mathbf{D}\mathbf{A}^\top = \mathbf{A}\mathbf{D}$. Substituting $\mathbf{A} = -\mathbf{C}^{-1}\mathbf{G}$ and $\mathbf{D} = \mathbf{C}^{-1}\mathbf{D}_q\mathbf{C}^{-1}$ and multiplying on both sides by $\mathbf{C}$ gives $\mathbf{G}\mathbf{C}^{-1}\mathbf{D}_q = \mathbf{D}_q\mathbf{C}^{-1}\mathbf{G}^\top$, proving Eq. (13.63). □

Example 13.7 (Effective Noise Temperature of a Subthreshold Transconductor)**.** The simplest case of mechanism (ii) is a capacitor C driven by a subthreshold OTA in unity-gain negative feedback, which acts as an active restoring conductance (a g_m-C resistor emulator). By Worked computation 12.24, with tail current I_{bias} and all devices saturated, the output current for small excursions δv about the balance point $v^* = 0$ is

$$I_{\text{net}}(\delta v) = -I_{\text{bias}} \tanh\!\left(\frac{\delta v}{2nV_T}\right) = -g_{\text{m}}\,\delta v + \mathcal{O}(\delta v^3), \qquad g_{\text{m}} := \frac{I_{\text{bias}}}{2nV_T} = \frac{q_e I_{\text{bias}}}{2nk_{\text{B}}T}. \tag{13.65}$$

Here g_{m} denotes the OTA transconductance G_m of Eq. (12.99), half the transconductance of a single device at the full bias current. At balance, the two input transistors and the two mirror transistors each carry $I_{\text{bias}}/2$, while the tail noise cancels at the output (Worked computation 12.24, Step 5). Each saturated channel contributes $S_\rho = q_e(I_{\text{D}})_\rho = \frac{1}{2}q_e I_{\text{bias}}$, so the four channels give

$$D_q = \frac{1}{2}\sum_{\rho=1}^{4} S_\rho = \frac{1}{2}\cdot 4 \cdot \left(\frac{1}{2}q_e I_{\text{bias}}\right) = q_e I_{\text{bias}}. \tag{13.66}$$

Writing $I_{\text{bias}} = 2nk_{\text{B}}T g_{\text{m}}/q_e$ and comparing with the Johnson–Nyquist value $D_q^{\text{eq}} = k_{\text{B}}T g_{\text{m}}$ of a resistor with the same conductance (Eq. (13.54)),

$$D_q = 2nk_{\text{B}}T\, g_{\text{m}} := k_{\text{B}}T_{\text{eff}}\, g_{\text{m}}, \qquad T_{\text{eff}} = 2n\,T. \tag{13.67}$$

The stationary voltage variance on the capacitor is

$$\Sigma = \frac{D_q}{C g_{\text{m}}} = \frac{2nk_{\text{B}}T}{C} = \frac{k_{\text{B}}T_{\text{eff}}}{C} > \frac{k_{\text{B}}T}{C}. \tag{13.68}$$

Now add at the output node a pair of current sources (one sourcing, one sinking) with standing current I_{aux} and gates held at fixed bias. They add no transconductance but raise the shot noise to $D_q = q_e(I_{\text{bias}} + I_{\text{aux}})$, so the variance becomes $\Sigma = \frac{2nk_{\text{B}}T}{C}\left(1 + \frac{I_{\text{aux}}}{I_{\text{bias}}}\right)$. The effective noise temperature of each node can thus be set electronically, over orders of magnitude, without changing the die temperature T.

13.4 Non-Equilibrium Analog Thermodynamic Random Networks (NEAT-RNs)

A NEAT-RN, in the sense of Freitas et al. [38], uses subthreshold transconductance amplifiers with programmable bias currents to set the drift matrix $\mathbf{A} = -\mathbf{C}^{-1}\mathbf{G}$, and standing bias currents to set the diffusion matrix $\mathbf{D} = \mathbf{C}^{-1}\mathbf{D}_q\mathbf{C}^{-1}$. Its non-equilibrium steady state is then a programmable multivariate Gaussian.

13.4.1 The Lyapunov Equation and the Stationary Current

Let $\mathbf{v}^* \in \mathbb{R}^N$ be a fixed point of the network, $\mathbf{I}_{\text{net}}(\mathbf{v}^*) = \mathbf{0}$. Linearizing the Langevin equation Eq. (13.38) in $\delta\mathbf{v}_t := \mathbf{v}_t - \mathbf{v}^*$ and multiplying by $\mathbf{C}^{-1}$ gives the multivariate Ornstein–Uhlenbeck process

$$\mathrm{d}(\delta\mathbf{v}_t) = \mathbf{A}\,\delta\mathbf{v}_t\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t, \tag{13.69}$$

with drift matrix $\mathbf{A}$, $N \times R$ noise matrix $\mathbf{B}$, and diffusion matrix $\mathbf{D}$ given by

$$\mathbf{A} := -\mathbf{C}^{-1}\mathbf{G}, \qquad \mathbf{B} := \mathbf{C}^{-1}\boldsymbol{\Delta}\,\operatorname{diag}\!\left(\sqrt{S_1(\mathbf{v}^*)}, \ldots, \sqrt{S_R(\mathbf{v}^*)}\right), \qquad \mathbf{D} := \frac{1}{2}\mathbf{B}\mathbf{B}^\top = \mathbf{C}^{-1}\mathbf{D}_q(\mathbf{v}^*)\mathbf{C}^{-1}. \tag{13.70}$$

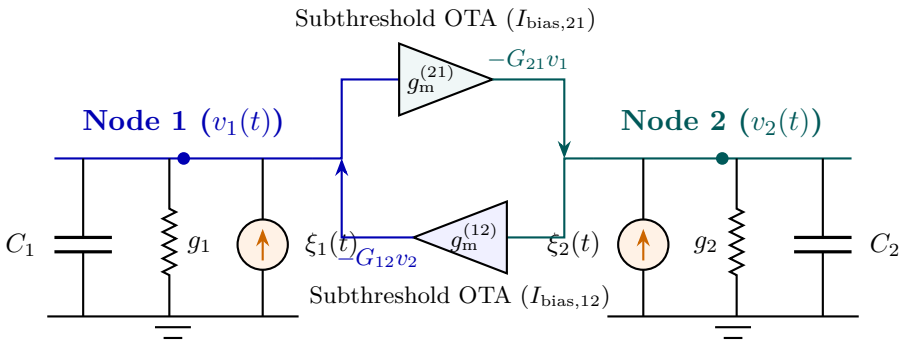


Figure 13.1: A two-node NEAT-RN cell. Each node i has capacitance C_i, self-conductance g_i, and a noise current ξ_i. Two OTAs with tail currents $I_{\text{bias},12}$ and $I_{\text{bias},21}$ each sense one node without drawing current and drive current into the other, with separately programmable transconductances $-G_{12} = g_{\text{m}}^{(12)}$ and $-G_{21} = g_{\text{m}}^{(21)}$. When $G_{12}D_2 \neq G_{21}D_1$ the cell breaks detailed balance and settles into a Gaussian NESS with circulating probability current $\mathbf{J}_{\text{ss}} \neq \mathbf{0}$.

The corresponding Fokker–Planck equation is

$$\frac{\partial \rho(\delta\mathbf{v}, t)}{\partial t} = \mathcal{L}_{\text{FP}}\rho(\delta\mathbf{v}, t) = -\nabla_{\delta\mathbf{v}} \cdot \mathbf{J}(\delta\mathbf{v}, t), \qquad \mathbf{J}(\delta\mathbf{v}, t) := \mathbf{A}\delta\mathbf{v}\,\rho(\delta\mathbf{v}, t) - \mathbf{D}\nabla_{\delta\mathbf{v}}\rho(\delta\mathbf{v}, t). \tag{13.71}$$

Theorem 13.8 (Lyapunov Equation and Gaussian NESS of a NEAT-RN)**.** *Suppose* $\mathbf{A} = -\mathbf{C}^{-1}\mathbf{G}$ *is Hurwitz stable (all its eigenvalues have negative real parts) and* $\mathbf{D} \succ 0$*. Then:*

(i) For any initial distribution ρ_0*, the solution* $\rho(\delta\mathbf{v}, t)$ *of Eq. (13.71) converges exponentially to the unique stationary Gaussian*

$$p_{\text{ss}}(\delta\mathbf{v}) = \frac{1}{\sqrt{(2\pi)^N \det \boldsymbol{\Sigma}}} \exp\left(-\frac{1}{2}\delta\mathbf{v}^\top \boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\right), \tag{13.72}$$

whose covariance $\boldsymbol{\Sigma} := \mathbb{E}\left[\delta\mathbf{v}\,\delta\mathbf{v}^\top\right]_{p_{\text{ss}}} \succ 0$ *is*

$$\boldsymbol{\Sigma} = 2\int_0^\infty e^{\mathbf{A}s}\,\mathbf{D}\,e^{\mathbf{A}^\top s}\,\mathrm{d}s, \tag{13.73}$$

and is the unique symmetric positive-definite solution of the continuous-time Lyapunov equation *[37, 38]*

$$\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top + 2\mathbf{D} = \mathbf{0}. \tag{13.74}$$

In terms of the circuit matrices $(\mathbf{C}, \mathbf{G}, \mathbf{D}_q)$ *it reads*

$$\mathbf{G}\boldsymbol{\Sigma}\mathbf{C} + \mathbf{C}\boldsymbol{\Sigma}\mathbf{G}^\top = 2\mathbf{D}_q(\mathbf{v}^*). \tag{13.75}$$

(ii) Define the skew-symmetric vorticity matrix $\mathbf{Q} = -\mathbf{Q}^\top \in \mathbb{R}^{N\times N}$ *by*

$$\mathbf{Q} := \frac{1}{2}\left(\mathbf{A}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\mathbf{A}^\top\right) = \mathbf{A}\boldsymbol{\Sigma} + \mathbf{D}. \tag{13.76}$$

Then the stationary probability current is

$$\mathbf{J}_{\text{ss}}(\delta\mathbf{v}) = \mathbf{Q}\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\,p_{\text{ss}}(\delta\mathbf{v}), \tag{13.77}$$

which is divergence-free, and the steady-state entropy production rate of the voltage dynamics is

$$\dot{\Sigma}_{\mathrm{tot}} = k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N(\delta v)\, \frac{\mathbf{J}_{\mathrm{ss}}(\delta\mathbf{v})^\top \mathbf{D}^{-1} \mathbf{J}_{\mathrm{ss}}(\delta\mathbf{v})}{p_{\mathrm{ss}}(\delta\mathbf{v})} = k_{\mathrm{B}}\, \mathrm{tr}\Big(\mathbf{Q}^\top \mathbf{D}^{-1} \mathbf{Q} \boldsymbol{\Sigma}^{-1}\Big) \geq 0, \tag{13.78}$$

with equality $\dot{\Sigma}_{\mathrm{tot}} = 0$ *if and only if* $\mathbf{Q} = \mathbf{0}$ *(detailed balance).*

Proof. Step 1: the Lyapunov equation. Let $\boldsymbol{\Sigma}(t) := \mathbb{E}\Big[\delta\mathbf{v}_t\, \delta\mathbf{v}_t^\top\Big]$. Applying the multidimensional Itô product rule (Section 4.3) to the outer product $\delta\mathbf{v}_t\, \delta\mathbf{v}_t^\top$ along trajectories of Eq. (13.69) and using the quadratic variation $\mathrm{d}\mathbf{W}_t\, \mathrm{d}\mathbf{W}_t^\top = \mathbf{I}\, \mathrm{d}t$, we obtain

$$\begin{aligned} \mathrm{d}(\delta\mathbf{v}_t\, \delta\mathbf{v}_t^\top) &= (\mathrm{d}\delta\mathbf{v}_t)\, \delta\mathbf{v}_t^\top + \delta\mathbf{v}_t\, (\mathrm{d}\delta\mathbf{v}_t)^\top + (\mathrm{d}\delta\mathbf{v}_t)(\mathrm{d}\delta\mathbf{v}_t)^\top \\ &= \Big(\mathbf{A}\delta\mathbf{v}_t\delta\mathbf{v}_t^\top + \delta\mathbf{v}_t\delta\mathbf{v}_t^\top\mathbf{A}^\top + \mathbf{B}\mathbf{B}^\top\Big)\, \mathrm{d}t + \mathbf{B}\, \mathrm{d}\mathbf{W}_t\, \delta\mathbf{v}_t^\top + \delta\mathbf{v}_t\, \mathrm{d}\mathbf{W}_t^\top \mathbf{B}^\top. \end{aligned} \tag{13.79}$$

Taking the expectation (the Itô integrals have zero mean) and using $\mathbf{B}\mathbf{B}^\top = 2\mathbf{D}$ yields the differential Lyapunov equation

$$\frac{\mathrm{d}\boldsymbol{\Sigma}(t)}{\mathrm{d}t} = \mathbf{A}\boldsymbol{\Sigma}(t) + \boldsymbol{\Sigma}(t)\mathbf{A}^\top + 2\mathbf{D}, \tag{13.80}$$

whose explicit solution is $\boldsymbol{\Sigma}(t) = e^{\mathbf{A}t}\boldsymbol{\Sigma}(0)e^{\mathbf{A}^\top t} + 2\int_0^t e^{\mathbf{A}s}\mathbf{D}e^{\mathbf{A}^\top s}\, \mathrm{d}s$. Because $\mathbf{A}$ is Hurwitz stable, $\|e^{\mathbf{A}t}\| \leq Me^{-\kappa t}$ for some $\kappa > 0$, so as $t \to \infty$ the initial condition decays exponentially and $\boldsymbol{\Sigma}(t) \to \boldsymbol{\Sigma} = 2\int_0^\infty e^{\mathbf{A}s}\mathbf{D}e^{\mathbf{A}^\top s}\, \mathrm{d}s$. Setting $\frac{\mathrm{d}\boldsymbol{\Sigma}}{\mathrm{d}t} = \mathbf{0}$ in Eq. (13.80) (or integrating $\frac{\mathrm{d}}{\mathrm{d}s}(e^{\mathbf{A}s}\mathbf{D}e^{\mathbf{A}^\top s}) = \mathbf{A}(e^{\mathbf{A}s}\mathbf{D}e^{\mathbf{A}^\top s}) + (e^{\mathbf{A}s}\mathbf{D}e^{\mathbf{A}^\top s})\mathbf{A}^\top$ from $s = 0$ to $s = \infty$) yields $\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top = -2\mathbf{D}$, which is Eq. (13.74). Multiplying on the left and right by $\mathbf{C}$ gives Eq. (13.75). The process is linear with additive Gaussian noise, so its law stays Gaussian, and its mean obeys $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\delta\mathbf{v}_t] = \mathbf{A}\mathbb{E}[\delta\mathbf{v}_t]$ and decays to zero. The limit is therefore the Gaussian Eq. (13.72); Step 2 checks directly that it is stationary.

Step 2: the stationary current. Split $\mathbf{A}\boldsymbol{\Sigma}$ into symmetric and skew-symmetric parts:

$$\mathbf{A}\boldsymbol{\Sigma} = \underbrace{\frac{1}{2}\Big(\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top\Big)}_{=-\mathbf{D}\text{ by Eq. (13.74)}} + \underbrace{\frac{1}{2}\Big(\mathbf{A}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\mathbf{A}^\top\Big)}_{=\mathbf{Q}\text{ by Eq. (13.76)}} = -\mathbf{D} + \mathbf{Q}. \tag{13.81}$$

Multiplying on the right by $\boldsymbol{\Sigma}^{-1}$ gives the drift decomposition

$$\mathbf{A} = -\mathbf{D}\boldsymbol{\Sigma}^{-1} + \mathbf{Q}\boldsymbol{\Sigma}^{-1}. \tag{13.82}$$

Evaluate the probability current $\mathbf{J}(\delta\mathbf{v})$ in Eq. (13.71) on the Gaussian distribution $p_{\mathrm{ss}}(\delta\mathbf{v})$ in Eq. (13.72). Since $\nabla_{\delta\mathbf{v}} p_{\mathrm{ss}}(\delta\mathbf{v}) = -\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\, p_{\mathrm{ss}}(\delta\mathbf{v})$, we find

$$\begin{aligned} \mathbf{J}_{\mathrm{ss}}(\delta\mathbf{v}) &= \left(\mathbf{A}\delta\mathbf{v} - \mathbf{D}\nabla_{\delta\mathbf{v}} \ln p_{\mathrm{ss}}(\delta\mathbf{v})\right) p_{\mathrm{ss}}(\delta\mathbf{v}) \\ &= \Big[\Big(-\mathbf{D}\boldsymbol{\Sigma}^{-1} + \mathbf{Q}\boldsymbol{\Sigma}^{-1}\Big)\, \delta\mathbf{v} + \mathbf{D}\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\Big]\, p_{\mathrm{ss}}(\delta\mathbf{v}) = \mathbf{Q}\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\, p_{\mathrm{ss}}(\delta\mathbf{v}), \end{aligned} \tag{13.83}$$

which is Eq. (13.77). Its divergence is

$$\begin{aligned} \nabla_{\delta\mathbf{v}} \cdot \mathbf{J}_{\mathrm{ss}}(\delta\mathbf{v}) &= \Big[\nabla_{\delta\mathbf{v}} \cdot \Big(\mathbf{Q}\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\Big)\Big]\, p_{\mathrm{ss}}(\delta\mathbf{v}) + \Big(\mathbf{Q}\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\Big) \cdot \nabla_{\delta\mathbf{v}} p_{\mathrm{ss}}(\delta\mathbf{v}) \\ &= \Big[\mathrm{tr}\Big(\mathbf{Q}\boldsymbol{\Sigma}^{-1}\Big) - \delta\mathbf{v}^\top\boldsymbol{\Sigma}^{-1}\mathbf{Q}^\top\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\Big]\, p_{\mathrm{ss}}(\delta\mathbf{v}). \end{aligned} \tag{13.84}$$

Because $\mathbf{Q}^\top = -\mathbf{Q}$ is skew-symmetric and $\boldsymbol{\Sigma}^{-1}$ is symmetric, the trace of their product vanishes:

$$\mathrm{tr}(\mathbf{Q}\boldsymbol{\Sigma}^{-1}) = \mathrm{tr}((\mathbf{Q}\boldsymbol{\Sigma}^{-1})^\top) = \mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{Q}^\top) = -\,\mathrm{tr}(\mathbf{Q}\boldsymbol{\Sigma}^{-1}) \quad \Longrightarrow \quad \mathrm{tr}(\mathbf{Q}\boldsymbol{\Sigma}^{-1}) = 0. \tag{13.85}$$

Likewise, the matrix $\mathbf{M} := \mathbf{\Sigma}^{-1}\mathbf{Q}^\top\mathbf{\Sigma}^{-1}$ is skew-symmetric ($\mathbf{M}^\top = \mathbf{\Sigma}^{-1}\mathbf{Q}\mathbf{\Sigma}^{-1} = -\mathbf{M}$), so the quadratic form $\delta\mathbf{v}^\top\mathbf{M}\delta\mathbf{v} = (\delta\mathbf{v}^\top\mathbf{M}\delta\mathbf{v})^\top = -\delta\mathbf{v}^\top\mathbf{M}\delta\mathbf{v} = 0$ vanishes for every $\delta\mathbf{v} \in \mathbb{R}^N$. Thus $\nabla_{\delta\mathbf{v}} \cdot \mathbf{J}_{\rm ss} = 0$, and $p_{\rm ss}$ is stationary.

Step 3: entropy production. Substituting $\mathbf{J}_{\rm ss}(\delta\mathbf{v}) = \mathbf{Q}\mathbf{\Sigma}^{-1}\delta\mathbf{v}\, p_{\rm ss}(\delta\mathbf{v})$ into the Fokker–Planck entropy production integral (Chapter 8),

$$\begin{aligned}\dot{\Sigma}_{\rm tot} &= k_{\rm B}\int_{\mathbb{R}^N} \mathrm{d}^N(\delta v)\, p_{\rm ss}(\delta\mathbf{v})\, \delta\mathbf{v}^\top\mathbf{\Sigma}^{-1}\mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{Q}\mathbf{\Sigma}^{-1}\delta\mathbf{v} \\ &= k_{\rm B}\,\mathrm{tr}\Big(\mathbf{\Sigma}^{-1}\mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{Q}\mathbf{\Sigma}^{-1}\mathbb{E}\Big[\delta\mathbf{v}\,\delta\mathbf{v}^\top\Big]_{p_{\rm ss}}\Big) = k_{\rm B}\,\mathrm{tr}\Big(\mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{Q}\mathbf{\Sigma}^{-1}\Big)\,. \end{aligned} \tag{13.86}$$

Writing $\mathbf{F} := \mathbf{D}^{-1/2}\mathbf{Q}\mathbf{\Sigma}^{-1/2}$, we see that $\mathrm{tr}(\mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{Q}\mathbf{\Sigma}^{-1}) = \mathrm{tr}(\mathbf{F}^\top\mathbf{F}) = \|\mathbf{F}\|_{\rm F}^2 \geq 0$, which vanishes if and only if $\mathbf{F} = \mathbf{0} \iff \mathbf{Q} = \mathbf{0}$, completing the proof. □

Heuristic Derivation: How Active Transconductance Breaks the $k_{\rm B}T/C$ Lock

Take two nodes with grounded capacitors $C_1 = C_2 = C$ and no capacitor between them. In a passive circuit, a resistor between the nodes injects correlated thermal noise into both and also adds damping to both. By Proposition 13.5 the two effects cancel exactly, and $\mathrm{Cov}(v_1, v_2)$ stays zero.
In a NEAT-RN (Fig. 13.1) the nodes are coupled instead by transconductors whose inputs sense v_1 and v_2 and whose outputs drive current into the opposite node. Then:

1. The coupling G_{12} is proportional to the tail current $I_{\rm bias,12} = I_0 e^{(V_{b,12}-V_{\rm th})/(nV_T)}$, which a DAC sets exponentially through the bias voltage $V_{b,12}$.
2. The sign of G_{12} is chosen by which output of the differential pair is used; no negative capacitance is needed.
3. The shot noise $\mathbf{D}_q$ is diagonal, because each OTA injects noise only into its own output node; it is not proportional to $\mathbf{G}$.

With $\mathbf{D}_q$ diagonal and $\mathbf{G}$ having programmable off-diagonal entries, the Lyapunov equation $\mathbf{G}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{G}^\top = \frac{2}{C}\mathbf{D}_q$ forces off-diagonal covariances $\Sigma_{ij} \neq 0$ that are set by the bias currents.

13.4.2 Two-Time Correlations and the Measurable Signature of Vorticity

The stationary distribution $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ depends only on $\mathbf{\Sigma}$, so equal-time measurements cannot tell an equilibrium state ($\mathbf{Q} = \mathbf{0}$) from a NESS with the same covariance. Two-time correlations can. Their asymmetry under time reversal, or equivalently the imaginary part of the cross-spectrum, measures $\mathbf{Q}$ directly.

Proposition 13.9 (Time-Reversal Asymmetry and Imaginary Cross-Spectrum of a NEAT-RN)**.** *In the stationary state of Eq. (13.69), define the two-time covariance for lag $\tau \in \mathbb{R}$, $\mathbf{C}_{vv}(\tau) := \mathbb{E}\Big[\delta\mathbf{v}(t+\tau)\,\delta\mathbf{v}(t)^\top\Big]_{p_{\rm ss}}$. Then:*

(i) For all $\tau \geq 0$, $\mathbf{C}_{vv}(\tau) = e^{\mathbf{A}\tau}\mathbf{\Sigma}$ and $\mathbf{C}_{vv}(-\tau) = \mathbf{C}_{vv}(\tau)^\top = \mathbf{\Sigma}e^{\mathbf{A}^\top\tau}$. The short-time asymmetry is the vorticity matrix:

$$\lim_{\tau\to 0^+} \frac{\mathbf{C}_{vv}(\tau) - \mathbf{C}_{vv}(-\tau)}{2\tau} = \frac{1}{2}\left(\mathbf{A}\mathbf{\Sigma} - \mathbf{\Sigma}\mathbf{A}^\top\right) = \mathbf{Q}. \tag{13.87}$$

(ii) By the Wiener–Khinchin theorem (Section 11.1), the double-sided cross-spectral density $\mathbf{S}_{vv}(\omega) := \int_{-\infty}^{\infty} \mathbf{C}_{vv}(\tau)\, e^{-i\omega\tau}\, \mathrm{d}\tau$ *is*

$$\mathbf{S}_{vv}(\omega) = 2\,(\mathbf{A} - i\omega\mathbf{I})^{-1}\,\mathbf{D}\left(\mathbf{A}^\top + i\omega\mathbf{I}\right)^{-1} = (i\omega\mathbf{I} - \mathbf{A})^{-1}\,\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\left(-i\omega\mathbf{I} - \mathbf{A}^\top\right)^{-1}. \quad (13.88)$$

Its imaginary part $\operatorname{Im}\mathbf{S}_{vv}(\omega) = \frac{1}{2i}\left[\mathbf{S}_{vv}(\omega) - \mathbf{S}_{vv}(-\omega)\right]$ *vanishes for all* $\omega \in \mathbb{R}$ *if and only if* $\mathbf{Q} = \mathbf{0}$ *(detailed balance).*

Proof. For $\tau \geq 0$, integrating the linear SDE (Eq. (13.69)) from t to $t+\tau$ gives $\delta\mathbf{v}(t+\tau) = e^{\mathbf{A}\tau}\delta\mathbf{v}(t) + \int_t^{t+\tau} e^{\mathbf{A}(t+\tau-s)}\mathbf{B}\,\mathrm{d}\mathbf{W}_s$. Because the future Wiener increments $\mathrm{d}\mathbf{W}_s$ $(s \geq t)$ are independent of $\delta\mathbf{v}(t)$, multiplying on the right by $\delta\mathbf{v}(t)^\top$ and taking the stationary expectation yields $\mathbf{C}_{vv}(\tau) = e^{\mathbf{A}\tau}\mathbb{E}\left[\delta\mathbf{v}(t)\delta\mathbf{v}(t)^\top\right]_{p_{\mathrm{ss}}} = e^{\mathbf{A}\tau}\boldsymbol{\Sigma}$. By time-translation invariance of the stationary process, $\mathbf{C}_{vv}(-\tau) = \mathbb{E}\left[\delta\mathbf{v}(t)\delta\mathbf{v}(t+\tau)^\top\right]_{p_{\mathrm{ss}}} = \mathbf{C}_{vv}(\tau)^\top = \boldsymbol{\Sigma}e^{\mathbf{A}^\top\tau}$. Expanding $e^{\mathbf{A}\tau} = \mathbf{I} + \mathbf{A}\tau + \mathcal{O}(\tau^2)$ gives Eq. (13.87).

For part (ii), split the Fourier integral at $\tau = 0$:

$$\begin{aligned}\mathbf{S}_{vv}(\omega) &= \int_0^{\infty} e^{(\mathbf{A}-i\omega\mathbf{I})\tau}\boldsymbol{\Sigma}\,\mathrm{d}\tau + \int_{-\infty}^{0} \boldsymbol{\Sigma}e^{-(\mathbf{A}^\top+i\omega\mathbf{I})\tau}\,\mathrm{d}\tau \\ &= (i\omega\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}(-i\omega\mathbf{I} - \mathbf{A}^\top)^{-1}. \end{aligned} \quad (13.89)$$

Multiplying the first term on the right by $(-i\omega\mathbf{I} - \mathbf{A}^\top)(-i\omega\mathbf{I} - \mathbf{A}^\top)^{-1}$ and the second term on the left by $(i\omega\mathbf{I} - \mathbf{A})^{-1}(i\omega\mathbf{I} - \mathbf{A})$, and using the Lyapunov equation $-\mathbf{A}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\mathbf{A}^\top = 2\mathbf{D}$, we obtain

$$\begin{aligned}\mathbf{S}_{vv}(\omega) &= (i\omega\mathbf{I} - \mathbf{A})^{-1}\left[\boldsymbol{\Sigma}(-i\omega\mathbf{I} - \mathbf{A}^\top) + (i\omega\mathbf{I} - \mathbf{A})\boldsymbol{\Sigma}\right](-i\omega\mathbf{I} - \mathbf{A}^\top)^{-1} \\ &= 2(i\omega\mathbf{I} - \mathbf{A})^{-1}\mathbf{D}(-i\omega\mathbf{I} - \mathbf{A}^\top)^{-1}. \end{aligned} \quad (13.90)$$

Finally, since $\mathbf{C}_{vv}(-\tau) = \mathbf{C}_{vv}(\tau)^\top$, the imaginary part $\operatorname{Im}\mathbf{S}_{vv}(\omega) = -\int_0^{\infty}[\mathbf{C}_{vv}(\tau) - \mathbf{C}_{vv}(\tau)^\top]\sin(\omega\tau)\,\mathrm{d}\tau$ is the sine transform of $e^{\mathbf{A}\tau}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}e^{\mathbf{A}^\top\tau}$. By uniqueness of the Fourier sine transform and the identity $e^{\mathbf{A}\tau}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}e^{\mathbf{A}^\top\tau} = 2\int_0^{\tau} e^{\mathbf{A}s}\mathbf{Q}e^{\mathbf{A}^\top(\tau-s)}\,\mathrm{d}s$, $\operatorname{Im}\mathbf{S}_{vv}(\omega) \equiv \mathbf{0}$ if and only if $\mathbf{Q} = \mathbf{0}$. □

13.5 Synthesizing Programmable Multivariate Gaussians and Weakly Nonlinear Corrections

This section inverts the Lyapunov equation Eq. (13.74): given a target mean and precision matrix, which conductances produce them? The answer leaves a skew-symmetric matrix free, and that freedom can be used to speed up relaxation. We then estimate how the nonlinearity of the subthreshold characteristic distorts the Gaussian at finite Ω.

13.5.1 Precision-Matrix Synthesis and Faster Mixing

Suppose the goal is to generate samples $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$ from a multivariate Gaussian distribution with specified mean $\boldsymbol{\mu} \in \mathbb{R}^N$ and symmetric positive-definite precision matrix $\boldsymbol{\Lambda} \in \mathbb{R}^{N\times N}$, or to solve the linear system $\boldsymbol{\Lambda}\mathbf{x} = \mathbf{b}$ and evaluate conditional distributions $\mathbf{x}_A \mid \mathbf{x}_B$ (Chapter 5, [3, 38]).

Suppose every node has the same capacitance C_0 to ground, with negligible capacitance between nodes, so $\mathbf{C} = C_0\mathbf{I}$, and the same shot-noise intensity $D_{q,ii} = q_e I_{\mathrm{noise}}$, produced by a dedicated noise current source at each node. Then the diffusion matrix is isotropic:

$$\mathbf{D} = \mathbf{C}^{-1}\mathbf{D}_q\mathbf{C}^{-1} = d_0\,\mathbf{I}, \qquad d_0 := \frac{q_e I_{\mathrm{noise}}}{C_0^2} > 0. \quad (13.91)$$

Injecting a programmable DC current vector $\mathbf{I}_{\text{bias}}^{\text{DC}}$ moves the fixed point to $\mathbf{v}^* = \mathbf{G}^{-1}\mathbf{I}_{\text{bias}}^{\text{DC}}$, which encodes the target mean $\boldsymbol{\mu}$.

The covariance should be $\boldsymbol{\Sigma} = \sigma_0^2\boldsymbol{\Lambda}^{-1}$, where σ_0 converts the dimensionless variables $\mathbf{x} = \delta\mathbf{v}/\sigma_0$ to volts. The drift decomposition Eq. (13.82), $\mathbf{A} = (-\mathbf{D} + \mathbf{Q})\boldsymbol{\Sigma}^{-1}$, gives every conductance matrix $\mathbf{G} = -C_0\mathbf{A}$ that does this.

Proposition 13.10 (Parametrization of Precision-Matrix Synthesis)**.** *Let $\mathbf{C} = C_0\mathbf{I}$ and $\mathbf{D} = d_0\mathbf{I}$ with $d_0 = q_e I_{\text{noise}}/C_0^2$. A Hurwitz-stable drift matrix $\mathbf{A} = -C_0^{-1}\mathbf{G}$ produces the target stationary covariance $\boldsymbol{\Sigma} = \sigma_0^2\boldsymbol{\Lambda}^{-1}$ if and only if the transconductance matrix $\mathbf{G}$ has the form*

$$\mathbf{G} = \frac{C_0}{\sigma_0^2}\,(d_0\,\mathbf{I} - \mathbf{Q})\,\boldsymbol{\Lambda}, \tag{13.92}$$

where $\mathbf{Q} = -\mathbf{Q}^\top$ is an arbitrary skew-symmetric matrix, equal to the vorticity matrix of Eq. (13.76). In particular:

(i) Symmetric synthesis ($\mathbf{Q} = \mathbf{0}$). *Symmetric conductances give*

$$\mathbf{G} = g_{\text{ref}}\,\boldsymbol{\Lambda}, \qquad g_{\text{ref}} := \frac{C_0 d_0}{\sigma_0^2} = \frac{q_e I_{\text{noise}}}{C_0\sigma_0^2}, \tag{13.93}$$

so each element Λ_{ij} is set by the bias current $I_{\text{bias},ij}$ of the transconductor coupling nodes i and j (for a single transistor; an OTA has half this transconductance, Eq. (12.99)):

$$G_{ij} = g_{\text{m}}^{(ij)} = \frac{I_{\text{bias},ij}}{nV_T} = \frac{I_0}{nV_T}\exp\!\left(\frac{V_{\text{DAC},ij} - V_{\text{th}}}{nV_T}\right). \tag{13.94}$$

(ii) Non-reciprocal synthesis ($\mathbf{Q} \neq \mathbf{0}$). *A nonzero $\mathbf{Q}$ in Eq. (13.92) leaves the stationary distribution $\mathcal{N}(\mathbf{0}, \sigma_0^2\boldsymbol{\Lambda}^{-1})$ unchanged and never decreases the decay rate of the slowest mode:*

$$\gamma_{\text{mix}}(\mathbf{Q}) := \min_k \operatorname{Re}\lambda_k(-\mathbf{A}) \geq \gamma_{\text{mix}}(\mathbf{0}) = \frac{d_0}{\sigma_0^2}\lambda_{\min}(\boldsymbol{\Lambda}). \tag{13.95}$$

Proof. Substituting $\mathbf{A} = -C_0^{-1}\mathbf{G}$ and $\boldsymbol{\Sigma} = \sigma_0^2\boldsymbol{\Lambda}^{-1}$ into Eq. (13.74) with $\mathbf{D} = d_0\mathbf{I}$ gives

$$-\frac{\sigma_0^2}{C_0}\left(\mathbf{G}\boldsymbol{\Lambda}^{-1} + \boldsymbol{\Lambda}^{-1}\mathbf{G}^\top\right) + 2d_0\mathbf{I} = \mathbf{0} \quad\Longleftrightarrow\quad \frac{1}{2}\left(\mathbf{G}\boldsymbol{\Lambda}^{-1} + \boldsymbol{\Lambda}^{-1}\mathbf{G}^\top\right) = \frac{C_0 d_0}{\sigma_0^2}\mathbf{I}. \tag{13.96}$$

So the symmetric part of $\mathbf{M} := \mathbf{G}\boldsymbol{\Lambda}^{-1}$ must equal $(C_0 d_0/\sigma_0^2)\mathbf{I}$, while its skew-symmetric part $\frac{1}{2}(\mathbf{G}\boldsymbol{\Lambda}^{-1} - \boldsymbol{\Lambda}^{-1}\mathbf{G}^\top) := -(C_0/\sigma_0^2)\mathbf{Q}$ can be any skew-symmetric matrix. Multiplying $\mathbf{G}\boldsymbol{\Lambda}^{-1} = \frac{C_0}{\sigma_0^2}(d_0\mathbf{I} - \mathbf{Q})$ on the right by $\boldsymbol{\Lambda}$ gives Eq. (13.92). For the bound, let $(\lambda, \mathbf{u})$, with $\mathbf{u} \in \mathbb{C}^N$, be an eigenpair of $-\mathbf{A} = \sigma_0^{-2}(d_0\mathbf{I} - \mathbf{Q})\boldsymbol{\Lambda}$, so $\sigma_0^{-2}(d_0\mathbf{I} - \mathbf{Q})\boldsymbol{\Lambda}\mathbf{u} = \lambda\mathbf{u}$. Multiplying on the left by $\mathbf{u}^*\boldsymbol{\Lambda}$ gives

$$\lambda\,(\mathbf{u}^*\boldsymbol{\Lambda}\mathbf{u}) = \frac{d_0}{\sigma_0^2}\mathbf{u}^*\boldsymbol{\Lambda}^2\mathbf{u} - \frac{1}{\sigma_0^2}\mathbf{u}^*\boldsymbol{\Lambda}\mathbf{Q}\boldsymbol{\Lambda}\mathbf{u}. \tag{13.97}$$

Because $\mathbf{Q}^\top = -\mathbf{Q}$ is real skew-symmetric, the matrix $\boldsymbol{\Lambda}\mathbf{Q}\boldsymbol{\Lambda}$ is also real skew-symmetric, so for any complex vector $\mathbf{w} = \boldsymbol{\Lambda}\mathbf{u}$ the quadratic form $\mathbf{w}^*\mathbf{Q}\mathbf{w}$ is purely imaginary: $\operatorname{Re}(\mathbf{u}^*\boldsymbol{\Lambda}\mathbf{Q}\boldsymbol{\Lambda}\mathbf{u}) = 0$. Taking the real part of Eq. (13.97) and using the Cauchy–Schwarz inequality $(\mathbf{u}^*\boldsymbol{\Lambda}\mathbf{u})^2 \leq (\mathbf{u}^*\mathbf{u})(\mathbf{u}^*\boldsymbol{\Lambda}^2\mathbf{u})$ together with $\mathbf{u}^*\boldsymbol{\Lambda}\mathbf{u} \geq \lambda_{\min}(\boldsymbol{\Lambda})\mathbf{u}^*\mathbf{u}$, we obtain

$$\operatorname{Re}\lambda = \frac{d_0}{\sigma_0^2}\frac{\mathbf{u}^*\boldsymbol{\Lambda}^2\mathbf{u}}{\mathbf{u}^*\boldsymbol{\Lambda}\mathbf{u}} \geq \frac{d_0}{\sigma_0^2}\frac{\mathbf{u}^*\boldsymbol{\Lambda}\mathbf{u}}{\mathbf{u}^*\mathbf{u}} \geq \frac{d_0}{\sigma_0^2}\lambda_{\min}(\boldsymbol{\Lambda}) = \gamma_{\text{mix}}(\mathbf{0}), \tag{13.98}$$

proving Eq. (13.95). □

Part (ii) is why non-reciprocal couplings are useful rather than a defect. When $\mathbf{\Lambda}$ is ill-conditioned ($\lambda_{\max} \gg \lambda_{\min}$), a symmetric network relaxes slowly along the soft eigenvectors of $\mathbf{\Lambda}$, at a rate proportional to $\lambda_{\min}(\mathbf{\Lambda})$. A skew-symmetric part $\mathbf{Q}$ drives a circulating current $\mathbf{J}_{\rm ss} = \mathbf{Q}\mathbf{\Sigma}^{-1}\delta\mathbf{v}\, p_{\rm ss}$ that mixes slow directions with fast ones. Since $\mathrm{tr}(\mathbf{Q}\mathbf{\Lambda}) = 0$, the sum of the decay rates, $\mathrm{tr}(-\mathbf{A}) = d_0\,\mathrm{tr}(\mathbf{\Lambda})/\sigma_0^2$, does not change; a well-chosen $\mathbf{Q}$ moves the real parts of the eigenvalues toward their mean $\mathrm{tr}(-\mathbf{A})/N$, while the stationary distribution stays the same.

Hardware Realization: Gaussian Conditioning by Clamping Nodes

One proposed use of such Gaussian samplers is conditional sampling and matrix inversion [3, 38]. Split the N nodes into a free subset A of size N_A and a clamped subset B of size $N_B = N - N_A$:

$$\delta\mathbf{v} = \begin{pmatrix}\delta\mathbf{v}_A\\ \delta\mathbf{v}_B\end{pmatrix}, \qquad \mathbf{\Lambda} = \begin{pmatrix}\mathbf{\Lambda}_{AA} & \mathbf{\Lambda}_{AB}\\ \mathbf{\Lambda}_{BA} & \mathbf{\Lambda}_{BB}\end{pmatrix}, \qquad \mathbf{\Sigma} = \mathbf{\Lambda}^{-1} = \begin{pmatrix}\mathbf{\Sigma}_{AA} & \mathbf{\Sigma}_{AB}\\ \mathbf{\Sigma}_{BA} & \mathbf{\Sigma}_{BB}\end{pmatrix}.$$

On a digital processor, sampling from $p(\delta\mathbf{v}_A \mid \delta\mathbf{v}_B = \mathbf{u}_B)$ requires the Schur complement $\mathbf{\Sigma}_{A|B} = \mathbf{\Lambda}_{AA}^{-1} = \mathbf{\Sigma}_{AA} - \mathbf{\Sigma}_{AB}\mathbf{\Sigma}_{BB}^{-1}\mathbf{\Sigma}_{BA}$, typically computed by a Cholesky factorization in $\mathcal{O}(N_A^3)$ operations.

In a network programmed with $\mathbf{G} = g_{\rm ref}\mathbf{\Lambda}$, one conditions by connecting the nodes in B to low-impedance buffers held at $\mathbf{v}_B^* + \mathbf{u}_B$. The free nodes then obey

$$C_0\,\mathrm{d}(\delta\mathbf{v}_{A,t}) = -g_{\rm ref}\,(\mathbf{\Lambda}_{AA}\delta\mathbf{v}_{A,t} + \mathbf{\Lambda}_{AB}\mathbf{u}_B)\,\mathrm{d}t + \sqrt{2q_e I_{\rm noise}}\;\mathrm{d}\mathbf{W}_{A,t}.$$

After a few relaxation times $\tau_{RC} = C_0/(g_{\rm ref}\lambda_{\min}(\mathbf{\Lambda}_{AA}))$, of order 1–10 ns for representative parameters, the free voltages are distributed as the conditional Gaussian

$$\delta\mathbf{v}_A \mid (\delta\mathbf{v}_B = \mathbf{u}_B) \;\sim\; \mathcal{N}\Big(-\mathbf{\Lambda}_{AA}^{-1}\mathbf{\Lambda}_{AB}\mathbf{u}_B,\; \sigma_0^2\mathbf{\Lambda}_{AA}^{-1}\Big).$$

The time-averaged voltage $\langle\delta\mathbf{v}_A\rangle$ solves the linear system $\mathbf{\Lambda}_{AA}\mathbf{x}_A = -\mathbf{\Lambda}_{AB}\mathbf{u}_B$, and the fluctuations about that mean have covariance proportional to $\mathbf{\Lambda}_{AA}^{-1}$.

13.5.2 Weakly Nonlinear Corrections

So far we have linearized the device characteristics about $\mathbf{v}^*$, which is exact only as $\Omega \to \infty$, where the fluctuations $\delta\mathbf{v} = \mathcal{O}(\Omega^{-1/2}V_T)$ vanish compared with V_T. With $\Omega \sim 10^2$–10^3 the rms fluctuation is $\sigma_V/V_T \sim 0.03$–0.10, and over this range the exponential characteristic of a single transistor and the tanh characteristic of an OTA (Worked computation 12.24) are visibly curved.

Following Freitas et al. [38], we expand in this curvature to find the leading non-Gaussian cumulants and the leading finite-Ω shift of $\mathbf{\Sigma}$. Expanding the Itô drift of Eq. (13.38), $\mathbf{a}(\delta\mathbf{v}) := \mathbf{C}^{-1}[\mathbf{I}_{\rm net} + \mathbf{I}_{\rm sp}](\mathbf{v}^* + \delta\mathbf{v})$, and the voltage diffusion tensor $\mathbf{D}(\delta\mathbf{v}) := \mathbf{C}^{-1}\mathbf{D}_q(\mathbf{v}^* + \delta\mathbf{v})\mathbf{C}^{-1}$ in Taylor series around $\delta\mathbf{v} = \mathbf{0}$ up to third order in $\delta\mathbf{v}$ gives, in index notation (with Einstein summation over repeated indices $j, k, l \in \{1, \ldots, N\}$):

$$a_i(\delta\mathbf{v}) = A_{ij}\delta v_j + \frac{1}{2}H_{ijk}\delta v_j\delta v_k + \frac{1}{6}T_{ijkl}\delta v_j\delta v_k\delta v_l + \mathcal{O}(\delta\mathbf{v}^4), \tag{13.99}$$

$$D_{ij}(\delta\mathbf{v}) = D_{ij}^{(0)} + D_{ij,k}^{(1)}\delta v_k + \frac{1}{2}D_{ij,kl}^{(2)}\delta v_k\delta v_l + \mathcal{O}(\delta\mathbf{v}^3), \tag{13.100}$$

where the expansion tensors are defined at the fixed point $\mathbf{v}^*$ by

$$A_{ij} = \frac{\partial a_i}{\partial v_j}\big|_{\mathbf{v}^*}, \quad H_{ijk} = \frac{\partial^2 a_i}{\partial v_j \partial v_k}\big|_{\mathbf{v}^*}, \quad T_{ijkl} = \frac{\partial^3 a_i}{\partial v_j \partial v_k \partial v_l}\big|_{\mathbf{v}^*}, \quad D^{(1)}_{ij,k} = \frac{\partial D_{ij}}{\partial v_k}\big|_{\mathbf{v}^*}, \quad D^{(2)}_{ij,kl} = \frac{\partial^2 D_{ij}}{\partial v_k \partial v_l}\big|_{\mathbf{v}^*}. \tag{13.101}$$

The correction $\mathbf{I}_{\mathrm{sp}}$ is smaller than $\mathbf{I}_{\mathrm{net}}$ by a factor of order Ω^{-1}, so it shifts $\mathbf{A}$ only slightly. That shift changes $\mathbf{\Sigma}$ at order Ω^{-2}, the same order as the corrections computed below, which is why it must be kept in $\mathbf{a}$.

In balanced differential architectures, each coupling current has the odd form $I_{ij}(\Delta v) = I_{\mathrm{bias},ij}\tanh\!\left(\frac{\Delta v}{2nV_T}\right)$, and the shot noise of each differential pair is set by its constant tail current. The symmetry $\delta\mathbf{v} \to -\delta\mathbf{v}$ then removes the even-order drift derivatives and the odd-order diffusion derivatives:

$$H_{ijk} = 0, \qquad D^{(1)}_{ij,k} = 0. \tag{13.102}$$

From $\tanh(x) = x - \frac{1}{3}x^3 + \mathcal{O}(x^5)$ with $x = \Delta v/(2nV_T)$, the leading nonlinearity is the cubic tensor T_{ijkl}:

$$I_{\mathrm{bias}}\tanh\!\left(\frac{\Delta v}{2nV_T}\right) = g_{\mathrm{m}}\,\Delta v - \frac{g_{\mathrm{m}}}{12(nV_T)^2}(\Delta v)^3 + \mathcal{O}(\Delta v^5), \qquad g_{\mathrm{m}} = \frac{I_{\mathrm{bias}}}{2nV_T}. \tag{13.103}$$

Proposition 13.11 (Covariance Shift and Fourth Cumulant in Balanced NEAT-RNs)**.** *Consider a balanced NEAT-RN obeying Eq. (13.99)–Eq. (13.102) with cubic drift tensor T_{ijkl}, quadratic diffusion tensor $D^{(2)}_{ij,kl}$, and unperturbed Gaussian covariance $\mathbf{\Sigma}^{(0)} = \mathcal{O}(\Omega^{-1})$ solving the linear Lyapunov equation $\mathbf{A}\mathbf{\Sigma}^{(0)} + \mathbf{\Sigma}^{(0)}\mathbf{A}^\top + 2\mathbf{D}^{(0)} = \mathbf{0}$. Then:*

(i) All odd moments vanish by symmetry ($\mathbb{E}[\delta v_i] = 0$, $\kappa^{(3)}_{ijk} = \mathbb{E}[\delta v_i \delta v_j \delta v_k] = 0$).

(ii) The covariance $\mathbf{\Sigma} = \mathbf{\Sigma}^{(0)} + \delta\mathbf{\Sigma} + \mathcal{O}(\Omega^{-3})$ has a shift $\delta\mathbf{\Sigma} = \mathcal{O}(\Omega^{-2})$ that solves the inhomogeneous Lyapunov equation

$$A_{im}\,\delta\Sigma_{mj} + \delta\Sigma_{im}\,A_{jm} + 2\,\widetilde{D}_{ij} = 0, \tag{13.104}$$

where

$$2\,\widetilde{D}_{ij} := D^{(2)}_{ij,kl}\Sigma^{(0)}_{kl} + \frac{1}{2}\left(T_{iklm}\Sigma^{(0)}_{kl}\Sigma^{(0)}_{mj} + T_{jklm}\Sigma^{(0)}_{kl}\Sigma^{(0)}_{mi}\right). \tag{13.105}$$

(iii) The fourth cumulant

$$\kappa^{(4)}_{ijkl} := \mathbb{E}[\delta v_i \delta v_j \delta v_k \delta v_l] - \Sigma_{ij}\Sigma_{kl} - \Sigma_{ik}\Sigma_{jl} - \Sigma_{il}\Sigma_{jk} = \mathcal{O}(\Omega^{-3}) \tag{13.106}$$

satisfies

$$A_{im}\kappa^{(4)}_{mjkl} + A_{jm}\kappa^{(4)}_{imkl} + A_{km}\kappa^{(4)}_{ijml} + A_{lm}\kappa^{(4)}_{ijkm} + \mathcal{S}_{ijkl}\left\{T_{imnp}\Sigma^{(0)}_{mj}\Sigma^{(0)}_{nk}\Sigma^{(0)}_{pl} + 2D^{(2)}_{ij,mn}\Sigma^{(0)}_{mk}\Sigma^{(0)}_{nl}\right\} = 0, \tag{13.107}$$

where $\mathcal{S}_{ijkl}\{\cdot\}$ denotes symmetrization over the indices (i,j,k,l) (summing the 4 cyclic/pair permutations for T and the 6 distinct pair partitions for $D^{(2)}$).

Proof. For the Fokker–Planck equation $\partial_t\rho = -\partial_i[a_i(\delta\mathbf{v})\rho - \partial_j(D_{ij}(\delta\mathbf{v})\rho)]$, the backward generator (Section 6.1) is

$$(\mathcal{L}f)(\delta\mathbf{v}) = a_m(\delta\mathbf{v})\frac{\partial f}{\partial \delta v_m} + D_{mn}(\delta\mathbf{v})\frac{\partial^2 f}{\partial \delta v_m \partial \delta v_n}, \tag{13.108}$$

and every polynomial f satisfies $\mathbb{E}[\mathcal{L}f]_{p_{\mathrm{ss}}} = 0$ in the stationary state.

For part (ii), take $f = \delta v_i \delta v_j$. With $\frac{\partial f}{\partial \delta v_m} = \delta_{im}\delta v_j + \delta_{jm}\delta v_i$ and $\frac{\partial^2 f}{\partial \delta v_m \partial \delta v_n} = \delta_{im}\delta_{jn} + \delta_{in}\delta_{jm}$, stationarity gives

$$\mathbb{E}[a_i(\delta\mathbf{v})\delta v_j + \delta v_i a_j(\delta\mathbf{v}) + 2D_{ij}(\delta\mathbf{v})]_{p_{\rm ss}} = 0. \tag{13.109}$$

Substituting the expansions Eqs. (13.99) and (13.100) with $H_{ijk} = 0$ and $D^{(1)}_{ij,k} = 0$,

$$A_{im}\Sigma_{mj} + \Sigma_{im}A_{jm} + \frac{1}{6}T_{iklm}\mathbb{E}[\delta v_k \delta v_l \delta v_m \delta v_j] + \frac{1}{6}T_{jklm}\mathbb{E}[\delta v_k \delta v_l \delta v_m \delta v_i] + 2D^{(0)}_{ij} + D^{(2)}_{ij,kl}\Sigma_{kl} = \mathcal{O}(\Omega^{-3}). \tag{13.110}$$

In the fourth moment we use Wick's (Isserlis's) theorem (Chapter B) with $\mathbf{\Sigma}^{(0)}$; the correction $\kappa^{(4)} = \mathcal{O}(\Omega^{-3})$ is one order smaller than the products $\Sigma^{(0)}\Sigma^{(0)} = \mathcal{O}(\Omega^{-2})$:

$$\mathbb{E}[\delta v_k \delta v_l \delta v_m \delta v_j] = \Sigma^{(0)}_{kl}\Sigma^{(0)}_{mj} + \Sigma^{(0)}_{km}\Sigma^{(0)}_{lj} + \Sigma^{(0)}_{kj}\Sigma^{(0)}_{lm} + \mathcal{O}(\Omega^{-3}). \tag{13.111}$$

Since T_{iklm} is symmetric in (k, l, m), the three contractions contribute equally:

$$\frac{1}{6}T_{iklm}\left(\Sigma^{(0)}_{kl}\Sigma^{(0)}_{mj} + \Sigma^{(0)}_{km}\Sigma^{(0)}_{lj} + \Sigma^{(0)}_{kj}\Sigma^{(0)}_{lm}\right) = \frac{3}{6}T_{iklm}\Sigma^{(0)}_{kl}\Sigma^{(0)}_{mj} = \frac{1}{2}T_{iklm}\Sigma^{(0)}_{kl}\Sigma^{(0)}_{mj}. \tag{13.112}$$

Writing $\Sigma_{ij} = \Sigma^{(0)}_{ij} + \delta\Sigma_{ij}$ in Eq. (13.110) and subtracting the zeroth-order Lyapunov equation $A_{im}\Sigma^{(0)}_{mj} + \Sigma^{(0)}_{im}A_{jm} + 2D^{(0)}_{ij} = 0$ gives Eq. (13.104)–Eq. (13.105).

For part (iii), apply the stationarity identity $\mathbb{E}[\mathcal{L}(\delta v_i \delta v_j \delta v_k \delta v_l)]_{p_{\rm ss}} = 0$ to the fourth-order monomial, substitute $\mathbb{E}[\delta v_i \delta v_j \delta v_k \delta v_l] = \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk} + \kappa^{(4)}_{ijkl}$, and use Eq. (13.110) to cancel all disconnected terms of order $\mathcal{O}(\Omega^{-2})$ and $\mathcal{O}(\Omega^{-3})$. The remaining connected contractions of $T_{imnp}\delta v_m \delta v_n \delta v_p \delta v_j \delta v_k \delta v_l$ (specifically the $3! = 6$ contractions that connect (m, n, p) separately to (j, k, l), weighted by $\frac{1}{6} \times 6 = 1$) and of $D^{(2)}_{ij,mn}\delta v_m \delta v_n \delta v_k \delta v_l$ yield Eq. (13.107). □

The physical content is simple. The cubic coefficient of tanh is negative, so for a restoring coupling the cubic drift term opposes the linear one ($T_{iiii} > 0$ while $A_{ii} < 0$). Saturation of the differential pair weakens the restoring force at large excursions. This raises the variance by $\delta\Sigma \sim \Sigma^{(0)} \cdot \Sigma^{(0)}/(nV_T)^2 = \mathcal{O}(\Omega^{-1}\Sigma^{(0)})$ and gives a small positive fourth cumulant (heavier tails than a Gaussian). For $\Omega \geq 200$ the variance shift is below 0.5% and can be absorbed by calibration (Chapter 18).

Example 13.12 (Kurtosis and Variance Shift of a Single OTA Node)**.** We evaluate Eqs. (13.104) and (13.107) for the single-node OTA circuit of Example 13.7, $C\,\mathrm{d}v_t = -I_{\rm bias}\tanh\left(\frac{v_t}{2nV_T}\right)\mathrm{d}t + \sqrt{2q_eI_{\rm bias}}\,\mathrm{d}W_t$. From Eqs. (13.66) and (13.103), the drift derivatives and the constant diffusion coefficient at $v^* = 0$ are

$$A_{11} = -\frac{g_{\rm m}}{C} = -\frac{I_{\rm bias}}{2nV_TC}, \qquad T_{1111} = +\frac{g_{\rm m}}{2(nV_T)^2C} > 0, \qquad D^{(0)}_{11} = \frac{q_eI_{\rm bias}}{C^2}, \qquad D^{(2)}_{11,11} = 0. \tag{13.113}$$

(Here $T_{1111} = \left.\frac{\mathrm{d}^3a}{\mathrm{d}v^3}\right|_0 = 6 \cdot \frac{g_{\rm m}}{12(nV_T)^2C} = +\frac{g_{\rm m}}{2(nV_T)^2C}$ because the restoring drift is $a(v) = -\frac{g_{\rm m}}{C}v + \frac{g_{\rm m}}{12(nV_T)^2C}v^3 + \mathcal{O}(v^5)$.)

The zeroth-order Lyapunov equation $2A_{11}\Sigma^{(0)} + 2D^{(0)}_{11} = 0$ gives the unperturbed variance (Eq. (13.68)):

$$\Sigma^{(0)} = -\frac{D^{(0)}_{11}}{A_{11}} = \frac{2nq_eV_T}{C} = \frac{2nV_T^2}{\Omega}, \tag{13.114}$$

where $\Omega = CV_T/q_e$. From Eq. (13.105), $2\widetilde{D}_{11} = T_{1111}(\Sigma^{(0)})^2$, so Eq. (13.104) reads $2A_{11}\delta\Sigma + T_{1111}(\Sigma^{(0)})^2 = 0$ and gives the variance shift

$$\delta\Sigma = -\frac{T_{1111}}{2A_{11}}\left(\Sigma^{(0)}\right)^2 = \frac{1}{4(nV_T)^2}\left(\Sigma^{(0)}\right)^2 = \frac{\Sigma^{(0)}}{2n\Omega} > 0. \tag{13.115}$$

For $N = 1$ the symmetrization $\mathcal{S}_{1111}\{T_{1111}(\Sigma^{(0)})^3\}$ in Eq. (13.107) consists of 4 equal terms adding to $4T_{1111}(\Sigma^{(0)})^3$, so $4A_{11}\kappa^{(4)} + 4T_{1111}(\Sigma^{(0)})^3 = 0$, which gives the fourth cumulant and the excess kurtosis

$$\begin{aligned} \kappa^{(4)} &= -\frac{T_{1111}}{A_{11}}\left(\Sigma^{(0)}\right)^3 = \frac{1}{2(nV_T)^2}\left(\Sigma^{(0)}\right)^3, \\ \gamma_2 &:= \frac{\kappa^{(4)}}{\left(\Sigma^{(0)}\right)^2} = \frac{\Sigma^{(0)}}{2(nV_T)^2} = \frac{1}{n\Omega}. \end{aligned} \tag{13.116}$$

These results can be checked against the exact stationary density. The drift derives from the potential

$$\begin{aligned} U_{\text{eff}}(v) &= \int_0^v I_{\text{bias}} \tanh\!\left(\frac{u}{2nV_T}\right) \mathrm{d}u = 2nV_T I_{\text{bias}} \ln\cosh\!\left(\frac{v}{2nV_T}\right) \\ &= \frac{D_q}{C}\left[\frac{v^2}{2\Sigma^{(0)}} - \frac{v^4}{48(nV_T)^2\Sigma^{(0)}} + \mathcal{O}(v^6)\right], \end{aligned}$$

so the stationary density is proportional to $\exp(-CU_{\text{eff}}/D_q)$. Expanding the moments

$$\int v^{2k} \exp\!\left(-\frac{v^2}{2\Sigma^{(0)}} + \frac{v^4}{48(nV_T)^2\Sigma^{(0)}}\right) \mathrm{d}v$$

to first order in the quartic term reproduces $\delta\Sigma = \frac{(\Sigma^{(0)})^2}{4(nV_T)^2}$ and $\kappa^{(4)} = \frac{(\Sigma^{(0)})^3}{2(nV_T)^2}$. For $n = 1.3$ and $\Omega = 200$, the relative variance shift is $\delta\Sigma/\Sigma^{(0)} = 1/(2n\Omega) \approx 0.19\%$ and the excess kurtosis is $\gamma_2 = 1/(n\Omega) \approx 0.38\%$.

13.6 Worked Computation: The Two-Node NEAT-RN Cell

The two-node cell of Fig. 13.1 is small enough to solve in closed form and shows every effect of this chapter: stability, programmable correlation, circulating current, and entropy production. It is the subthreshold CMOS version of the RC sampler studied in Worked computations 5.33, 6.21, 8.12 and 10.35.

Worked computation 13.13 (The Two-Node Non-Reciprocal NEAT-RN Cell)**.** Take the cell of Fig. 13.1 with node voltages $\mathbf{v} = (v_1, v_2)^\top$ measured from the operating point $\mathbf{v}^* = (0,0)^\top$, which sits at mid-rail, $V_{\text{CM}} = V_{\text{DD}}/2$. The nodes have equal capacitances $C_1 = C_2 = C$ and equal self-conductances $g_1 = g_2 = g$, implemented by diode-connected transistors carrying current I_g, so $g = I_g/(nV_T)$. The cross-coupling OTAs have a tunable non-reciprocity parameter $\eta \in (-1, \infty)$:

$$\mathbf{C} = C\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{G} = \begin{pmatrix} g & G_{12} \\ G_{21} & g \end{pmatrix} = \begin{pmatrix} g & -g_c(1+\eta) \\ -g_c(1-\eta) & g \end{pmatrix}, \tag{13.117}$$

where $g_c = I_c/(nV_T) > 0$ is the baseline coupling and η sets the asymmetry between $g_{\text{m}}^{(12)} = -G_{12} = g_c(1+\eta)$ and $g_{\text{m}}^{(21)} = -G_{21} = g_c(1-\eta)$:

- $\eta = 0$: symmetric coupling, $G_{12} = G_{21} = -g_c$.
- $0 < \eta < 1$: same sign in both directions, unequal magnitudes.
- $\eta = 1$: one-way coupling. Node 2 drives node 1 ($G_{12} = -2g_c$), and node 1 does not act on node 2 ($G_{21} = 0$).
- $\eta > 1$: opposite signs ($G_{12} < 0$, $G_{21} > 0$), an excitatory-inhibitory loop like a predator-prey system.

The bias currents at the two nodes produce independent shot noise, $\mathbf{D}_q = \mathrm{diag}(D_{q,1}, D_{q,2})$, so

$$\mathbf{A} = -\frac{1}{C}\begin{pmatrix} g & G_{12} \\ G_{21} & g \end{pmatrix} = \begin{pmatrix} -\gamma & \kappa(1+\eta) \\ \kappa(1-\eta) & -\gamma \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix}, \tag{13.118}$$

with the self-damping rate $\gamma := g/C$, the coupling rate $\kappa := g_c/C$, and $D_i := D_{q,i}/C^2$. We find the stability region, the covariance, the circulating current, and the entropy production.

Part 1: stability and eigenvalues. The characteristic polynomial of the 2×2 drift matrix $\mathbf{A}$ in Eq. (13.118) is

$$P(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A}) = \lambda^2 - (\operatorname{tr} \mathbf{A})\lambda + \det \mathbf{A} = 0, \tag{13.119}$$

where the trace and determinant are

$$\operatorname{tr} \mathbf{A} = -2\gamma = -\frac{2g}{C} < 0, \qquad \det \mathbf{A} = \frac{g^2 - G_{12}G_{21}}{C^2} = \gamma^2 - \kappa^2(1-\eta^2). \tag{13.120}$$

For a 2×2 matrix, both eigenvalues have negative real parts if and only if $\operatorname{tr} \mathbf{A} < 0$ and $\det \mathbf{A} > 0$. Since $\gamma > 0$, the cell is stable if and only if

$$\det \mathbf{A} > 0 \quad \Longleftrightarrow \quad \kappa^2(1-\eta^2) < \gamma^2 \quad \Longleftrightarrow \quad g_c^2(1-\eta^2) < g^2. \tag{13.121}$$

The eigenvalues are

$$\lambda_\pm = \frac{\operatorname{tr} \mathbf{A} \pm \sqrt{(\operatorname{tr} \mathbf{A})^2 - 4 \det \mathbf{A}}}{2} = -\gamma \pm \kappa\sqrt{1-\eta^2}. \tag{13.122}$$

Their character depends on η:

- For $|\eta| < 1$ (same-sign couplings), the eigenvalues are real: $\lambda_- = -\gamma - \kappa\sqrt{1-\eta^2} < \lambda_+ = -\gamma + \kappa\sqrt{1-\eta^2} < 0$ when Eq. (13.121) holds. As $|\eta|$ grows from 0 to 1, the slowest rate $|\lambda_+| = \gamma - \kappa\sqrt{1-\eta^2}$ rises from $\gamma - \kappa$ to γ. This is in the spirit of Proposition 13.10(ii), although here the covariance also changes with η.
- At $|\eta| = 1$ (one-way coupling), $\lambda_+ = \lambda_- = -\gamma$, and $\mathbf{A}$ is a Jordan block.
- For $|\eta| > 1$ (opposite-sign loop), the cell is stable for every coupling strength, since $\det \mathbf{A} = \gamma^2 + \kappa^2(\eta^2 - 1) > \gamma^2$. The eigenvalues form a complex pair,

$$\lambda_\pm = -\gamma \pm i\omega_{\mathrm{osc}}, \qquad \omega_{\mathrm{osc}} := \kappa\sqrt{\eta^2 - 1} = \frac{g_c}{C}\sqrt{\eta^2 - 1}, \tag{13.123}$$

  so fluctuations spiral about the origin at frequency ω_{osc} while decaying at rate γ.

Part 2: the Lyapunov equation. Write $\mathbf{\Sigma} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{12} & \Sigma_{22} \end{pmatrix}$ and solve $\mathbf{A\Sigma} + \mathbf{\Sigma A}^\top = -2\mathbf{D}$ with $A_{11} = A_{22} = -\gamma$, $A_{12} = -G_{12}/C = \kappa(1+\eta)$, and $A_{21} = -G_{21}/C = \kappa(1-\eta)$. The product is

$$\mathbf{A\Sigma} = \begin{pmatrix} -\gamma & A_{12} \\ A_{21} & -\gamma \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{12} & \Sigma_{22} \end{pmatrix} = \begin{pmatrix} -\gamma\Sigma_{11} + A_{12}\Sigma_{12} & -\gamma\Sigma_{12} + A_{12}\Sigma_{22} \\ A_{21}\Sigma_{11} - \gamma\Sigma_{12} & A_{21}\Sigma_{12} - \gamma\Sigma_{22} \end{pmatrix}. \tag{13.124}$$

Adding $\mathbf{\Sigma A}^\top = (\mathbf{A\Sigma})^\top$ and equating the $(1,1)$, $(2,2)$, and $(1,2)$ entries to $-2\mathbf{D} = \begin{pmatrix} -2D_1 & 0 \\ 0 & -2D_2 \end{pmatrix}$ gives three linear equations for $(\Sigma_{11}, \Sigma_{22}, \Sigma_{12})$:

$$\text{(1,1) entry:} \qquad -2\gamma\Sigma_{11} + 2A_{12}\Sigma_{12} = -2D_1 \quad\Longrightarrow\quad \Sigma_{11} = \frac{D_1 + A_{12}\Sigma_{12}}{\gamma}, \tag{13.125}$$

$$\text{(2,2) entry:} \qquad -2\gamma\Sigma_{22} + 2A_{21}\Sigma_{12} = -2D_2 \quad\Longrightarrow\quad \Sigma_{22} = \frac{D_2 + A_{21}\Sigma_{12}}{\gamma}, \tag{13.126}$$

$$\text{(1,2) entry:} \qquad -2\gamma\Sigma_{12} + A_{12}\Sigma_{22} + A_{21}\Sigma_{11} = 0. \tag{13.127}$$

Substituting the expressions for Σ_{11} (Eq. (13.125)) and Σ_{22} (Eq. (13.126)) into Eq. (13.127) and multiplying by γ,

$$\begin{aligned} -2\gamma^2\Sigma_{12} + A_{12}\left(D_2 + A_{21}\Sigma_{12}\right) + A_{21}\left(D_1 + A_{12}\Sigma_{12}\right) &= 0 \\ \Longleftrightarrow \quad 2\left(\gamma^2 - A_{12}A_{21}\right)\Sigma_{12} &= A_{21}D_1 + A_{12}D_2. \end{aligned} \tag{13.128}$$

Since $\gamma^2 - A_{12}A_{21} = \det\mathbf{A} > 0$ by Eq. (13.120), the cross-covariance is

$$\Sigma_{12} = \frac{A_{21}D_1 + A_{12}D_2}{2\det\mathbf{A}} = \frac{\kappa\left[(1-\eta)D_1 + (1+\eta)D_2\right]}{2\left[\gamma^2 - \kappa^2(1-\eta^2)\right]}. \tag{13.129}$$

Substituting back into Eqs. (13.125) and (13.126) gives the variances

$$\begin{aligned} \Sigma_{11} &= \frac{1}{\gamma}\left[D_1 + \frac{A_{12}(A_{21}D_1 + A_{12}D_2)}{2(\gamma^2 - A_{12}A_{21})}\right] \\ &= \frac{\left(2\gamma^2 - A_{12}A_{21}\right)D_1 + A_{12}^2 D_2}{2\gamma\det\mathbf{A}} \\ &= \frac{\left[2\gamma^2 - \kappa^2(1-\eta^2)\right]D_1 + \kappa^2(1+\eta)^2 D_2}{2\gamma\left[\gamma^2 - \kappa^2(1-\eta^2)\right]}, \end{aligned} \tag{13.130}$$

$$\begin{aligned} \Sigma_{22} &= \frac{\left(2\gamma^2 - A_{12}A_{21}\right)D_2 + A_{21}^2 D_1}{2\gamma\det\mathbf{A}} \\ &= \frac{\left[2\gamma^2 - \kappa^2(1-\eta^2)\right]D_2 + \kappa^2(1-\eta)^2 D_1}{2\gamma\left[\gamma^2 - \kappa^2(1-\eta^2)\right]}. \end{aligned} \tag{13.131}$$

To confirm $\mathbf{\Sigma} \succ 0$, compute $\det\mathbf{\Sigma} = \Sigma_{11}\Sigma_{22} - \Sigma_{12}^2$:

$$\begin{aligned} 4\gamma^2(\det\mathbf{A})^2\det\mathbf{\Sigma} &= \left[(2\gamma^2 - A_{12}A_{21})D_1 + A_{12}^2 D_2\right]\left[(2\gamma^2 - A_{12}A_{21})D_2 + A_{21}^2 D_1\right] \\ &\quad - \gamma^2\left(A_{21}D_1 + A_{12}D_2\right)^2 \\ &= 4\gamma^2\left(\gamma^2 - A_{12}A_{21}\right)D_1D_2 + \left(\gamma^2 - A_{12}A_{21}\right)\left(A_{21}D_1 - A_{12}D_2\right)^2, \end{aligned} \tag{13.132}$$

where we used the algebraic identity $(2\gamma^2 - p)^2 + p^2 - 2\gamma^2 p = 4\gamma^4 - 6\gamma^2 p + 2p^2 = 2(\gamma^2 - p)(2\gamma^2 - p)$ with $p = A_{12}A_{21}$. Dividing by $4\gamma^2(\det \mathbf{A})^2$,

$$\det \boldsymbol{\Sigma} = \frac{4\gamma^2 D_1 D_2 + (A_{21}D_1 - A_{12}D_2)^2}{4\gamma^2 \det \mathbf{A}} > 0. \tag{13.133}$$

For equal noise intensities $D_1 = D_2 = D_0$,

$$\boldsymbol{\Sigma} = \frac{D_0}{\gamma\,[\gamma^2 - \kappa^2(1-\eta^2)]} \begin{pmatrix} \gamma^2 + \kappa^2\eta(1+\eta) & \gamma\kappa \\ \gamma\kappa & \gamma^2 - \kappa^2\eta(1-\eta) \end{pmatrix}. \tag{13.134}$$

Two cases are worth noting:

(a) When $\eta = 0$ (symmetric transconductance coupling), $\boldsymbol{\Sigma} = \frac{D_0}{\gamma^2-\kappa^2}\begin{pmatrix} \gamma & \kappa \\ \kappa & \gamma \end{pmatrix}$ and $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1} = \frac{1}{D_0}\begin{pmatrix} \gamma & -\kappa \\ -\kappa & \gamma \end{pmatrix}$, with correlation coefficient $\rho_{12} = \Sigma_{12}/\sqrt{\Sigma_{11}\Sigma_{22}} = \kappa/\gamma = g_c/g$. Stability requires $g_c < g$, so the bias current I_c tunes ρ_{12} over $[0, 1)$; reversing the sign of the coupling (Section 13.4) gives negative correlations.

(b) With one-way coupling, $\eta = 1$ ($A_{21} = 0$, $A_{12} = 2\kappa$), the nodes are still correlated, $\Sigma_{12} = \kappa D_0/\gamma^2 > 0$. Node 2 keeps its uncoupled variance $\Sigma_{22} = D_0/\gamma$, and node 1 has the larger variance $\Sigma_{11} = \frac{D_0}{\gamma}\left(1 + \frac{2\kappa^2}{\gamma^2}\right)$.

Part 3: vorticity and circulating current. By Eq. (13.76), $\mathbf{Q} = \mathbf{A}\boldsymbol{\Sigma} + \mathbf{D}$. Its diagonal entries vanish by Eqs. (13.125) and (13.126) (for example $-\gamma\Sigma_{11} + A_{12}\Sigma_{12} + D_1 = 0$), and the $(1, 2)$ entry is

$$\begin{aligned} Q_{12} &= -\gamma\Sigma_{12} + A_{12}\Sigma_{22} = -\gamma\Sigma_{12} + \frac{A_{12}}{\gamma}\,(D_2 + A_{21}\Sigma_{12}) \\ &= \frac{A_{12}D_2 - (\gamma^2 - A_{12}A_{21})\Sigma_{12}}{\gamma}. \end{aligned} \tag{13.135}$$

Substituting $(\gamma^2 - A_{12}A_{21})\,\Sigma_{12} = \frac{1}{2}(A_{21}D_1 + A_{12}D_2)$ from Eq. (13.128) gives

$$\mathbf{Q} = \begin{pmatrix} 0 & \omega_Q \\ -\omega_Q & 0 \end{pmatrix}, \qquad \omega_Q := Q_{12} = \frac{A_{12}D_2 - A_{21}D_1}{2\gamma} = \frac{\kappa\,[(1+\eta)D_2 - (1-\eta)D_1]}{2\gamma}. \tag{13.136}$$

So detailed balance holds ($\mathbf{Q} = \mathbf{0}$, $\mathbf{J}_{\mathrm{ss}} \equiv \mathbf{0}$) if and only if

$$A_{12}D_2 = A_{21}D_1 \quad\Longleftrightarrow\quad G_{12}D_{q,2} = G_{21}D_{q,1} \quad\Longleftrightarrow\quad (1+\eta)D_2 = (1-\eta)D_1. \tag{13.137}$$

Both mechanisms of Theorem 13.6 appear here. With equal noise ($D_1 = D_2$), detailed balance fails whenever $\eta \neq 0$, that is, whenever $G_{12} \neq G_{21}$. With symmetric couplings ($\eta = 0$), it fails whenever $D_1 \neq D_2$, which is like coupling the two nodes to baths at different temperatures.

With $\boldsymbol{\Sigma}^{-1} = \frac{1}{\det \boldsymbol{\Sigma}}\begin{pmatrix} \Sigma_{22} & -\Sigma_{12} \\ -\Sigma_{12} & \Sigma_{11} \end{pmatrix}$ and Eq. (13.77), the stationary current is

$$\mathbf{J}_{\mathrm{ss}}(\mathbf{v}) = \mathbf{Q}\boldsymbol{\Sigma}^{-1}\mathbf{v}\,p_{\mathrm{ss}}(\mathbf{v}) = \frac{\omega_Q}{\det \boldsymbol{\Sigma}}\begin{pmatrix} -\Sigma_{12}v_1 + \Sigma_{11}v_2 \\ -\Sigma_{22}v_1 + \Sigma_{12}v_2 \end{pmatrix} p_{\mathrm{ss}}(\mathbf{v}). \tag{13.138}$$

The current is tangent to every ellipse of constant density, since $\mathbf{v}^\top \boldsymbol{\Sigma}^{-1}\mathbf{J}_{\mathrm{ss}}(\mathbf{v}) = \mathbf{v}^\top \boldsymbol{\Sigma}^{-1}\mathbf{Q}\boldsymbol{\Sigma}^{-1}\mathbf{v}\, p_{\mathrm{ss}}(\mathbf{v}) = 0$. Probability circulates counterclockwise in the (v_1, v_2) plane if $\omega_Q < 0$ and clockwise if $\omega_Q > 0$, with mean angular velocity $\mathbb{E}[v_1\dot{v}_2 - v_2\dot{v}_1]_{p_{\mathrm{ss}}} = -2\omega_Q$ (computed from the current).

Part 4: entropy production and energy. We evaluate $\dot{\Sigma}_{\mathrm{tot}} = k_{\mathrm{B}}\,\mathrm{tr}(\mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{Q}\boldsymbol{\Sigma}^{-1})$ from Eq. (13.78):

$$\begin{aligned}\mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{Q}\boldsymbol{\Sigma}^{-1} &= \begin{pmatrix} 0 & -\omega_Q \\ \omega_Q & 0 \end{pmatrix}\begin{pmatrix} D_1^{-1} & 0 \\ 0 & D_2^{-1} \end{pmatrix}\begin{pmatrix} 0 & \omega_Q \\ -\omega_Q & 0 \end{pmatrix}\frac{1}{\det\boldsymbol{\Sigma}}\begin{pmatrix} \Sigma_{22} & -\Sigma_{12} \\ -\Sigma_{12} & \Sigma_{11} \end{pmatrix} \\ &= \frac{\omega_Q^2}{\det\boldsymbol{\Sigma}}\begin{pmatrix} D_2^{-1} & 0 \\ 0 & D_1^{-1} \end{pmatrix}\begin{pmatrix} \Sigma_{22} & -\Sigma_{12} \\ -\Sigma_{12} & \Sigma_{11} \end{pmatrix}. \end{aligned} \tag{13.139}$$

Taking the trace gives

$$\dot{\Sigma}_{\mathrm{tot}} = k_{\mathrm{B}}\frac{\omega_Q^2}{\det\boldsymbol{\Sigma}}\left(\frac{\Sigma_{22}}{D_2} + \frac{\Sigma_{11}}{D_1}\right) = k_{\mathrm{B}}\frac{\omega_Q^2\,(D_1\Sigma_{22} + D_2\Sigma_{11})}{D_1 D_2 \det\boldsymbol{\Sigma}}. \tag{13.140}$$

The numerator simplifies with Eqs. (13.130) and (13.131):

$$\begin{aligned} D_1\Sigma_{22} + D_2\Sigma_{11} &= \frac{2(2\gamma^2 - A_{12}A_{21})D_1D_2 + A_{21}^2D_1^2 + A_{12}^2D_2^2}{2\gamma\det\mathbf{A}} \\ &= \frac{4\gamma^2 D_1D_2 + (A_{21}D_1 - A_{12}D_2)^2}{2\gamma\det\mathbf{A}} = 2\gamma\det\boldsymbol{\Sigma}, \end{aligned} \tag{13.141}$$

where the last step uses Eq. (13.133). Substituting Eq. (13.141) and $\omega_Q = (A_{12}D_2 - A_{21}D_1)/(2\gamma)$ into Eq. (13.140), $\det\boldsymbol{\Sigma}$ cancels:

$$\dot{\Sigma}_{\mathrm{tot}} = k_{\mathrm{B}}\frac{2\gamma\,\omega_Q^2}{D_1D_2} = k_{\mathrm{B}}\frac{(A_{12}D_2 - A_{21}D_1)^2}{2\gamma D_1D_2} = \frac{k_{\mathrm{B}}}{2gC}\frac{(G_{12}D_{q,2} - G_{21}D_{q,1})^2}{D_{q,1}D_{q,2}}. \tag{13.142}$$

For equal noise intensities $D_1 = D_2 = D_0$,

$$\dot{\Sigma}_{\mathrm{tot}} = k_{\mathrm{B}}\frac{(A_{12} - A_{21})^2}{2\gamma} = \frac{2k_{\mathrm{B}}\kappa^2\eta^2}{\gamma} = \frac{2k_{\mathrm{B}}g_c^2\eta^2}{gC}. \tag{13.143}$$

For numbers we take representative values chosen for this example, not the specifications of any chip: $T = 300\,\mathrm{K}$, $V_T = 25.85\,\mathrm{mV}$, $n = 1.3$, and

- node capacitance $C = 1.0\,\mathrm{fF}$, so $\Omega = CV_T/q_e \approx 161$;
- bias currents $I_g = 10\,\mathrm{nA}$ and $I_c = 8\,\mathrm{nA}$ at $V_{\mathrm{DD}} = 0.3\,\mathrm{V}$, which give $g = \frac{I_g}{nV_T} \approx 297.6\,\mathrm{nS}$, coupling transconductance $g_c = \frac{I_c}{nV_T} \approx 238.1\,\mathrm{nS}$, self-damping rate $\gamma = g/C \approx 2.98 \times 10^8\,\mathrm{s}^{-1}$ ($\tau_{RC} = \gamma^{-1} \approx 3.36\,\mathrm{ns}$), and coupling rate $\kappa = g_c/C \approx 2.38 \times 10^8\,\mathrm{s}^{-1}$.
- At $\eta = 0.5$, the entropy production of the voltage dynamics, expressed as a power, is $T\dot{\Sigma}_{\mathrm{tot}} = 2k_{\mathrm{B}}Tg_c^2\eta^2/(gC) \approx 3.95 \times 10^{-13}\,\mathrm{W}$. The DC power drawn by the bias currents is $P_{\mathrm{DC}} = 2(I_g + I_c)V_{\mathrm{DD}} = 2(18\,\mathrm{nA})(0.3\,\mathrm{V}) = 10.8\,\mathrm{nW}$, about 3×10^4 times larger. Almost all of the supply power is spent keeping the transistors biased, not on the circulation itself.

- Over one damping time $\tau_{RC} \approx 3.36\,\mathrm{ns}$, the cell uses

$$E_{\mathrm{sample}} = P_{\mathrm{DC}}\,\tau_{RC} = (10.8\,\mathrm{nW})(3.36\,\mathrm{ns}) \approx 36.3\,\mathrm{aJ} \approx 8.77 \times 10^3\,k_{\mathrm{B}}T,$$

  This is a rough estimate of the energy per sample; the slowest mode relaxes at $|\lambda_+| = \gamma - \kappa\sqrt{1-\eta^2} \approx 0.9 \times 10^8\,\mathrm{s}^{-1}$, so independent samples take about three times longer. For comparison, if generating and correlating two Gaussian variates on a digital processor costs of order $10\,\mathrm{pJ}$ ($\approx 2.4 \times 10^9\,k_{\mathrm{B}}T$, a rough figure assumed here), the analog cell is more than four orders of magnitude cheaper.

Summary

Proved:

- The single-jump energy change $\Delta U_\rho = q_e \boldsymbol{\Delta}_\rho^\top \mathbf{v} + E_{\mathrm{C}}^{(\rho)}$, local detailed balance at the branch level, and the first and second laws for the discrete-charge master equation (Proposition 13.2).
- The second-order expansion in Ω^{-1}, giving the circuit Fokker–Planck equation in Hänggi–Klimontovich form (Theorem 13.4).
- The passive equipartition lock $\boldsymbol{\Sigma}_{\mathrm{eq}} = k_{\mathrm{B}}T\mathbf{C}^{-1}$ (Proposition 13.5) and the two ways active devices break detailed balance (Theorem 13.6).
- The Lyapunov equation, the stationary current $\mathbf{J}_{\mathrm{ss}} = \mathbf{Q}\boldsymbol{\Sigma}^{-1}\delta\mathbf{v}\,p_{\mathrm{ss}}$, and the entropy production $k_{\mathrm{B}}\,\mathrm{tr}(\mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{Q}\boldsymbol{\Sigma}^{-1})$ (Theorem 13.8).
- The parametrization of all conductance matrices that realize a given precision matrix, and the fact that non-reciprocity never slows the slowest mode (Proposition 13.10); the leading nonlinear corrections (Proposition 13.11); the two-node cell in closed form (Worked computation 13.13).

Assumed:

- Carriers in each device thermalize much faster ($\sim 0.1\,\mathrm{ps}$) than the nodes relax ($\sim$ 1–10 ns), so jump rates depend only on the instantaneous node voltages.
- $1/f$ noise is negligible at the frequencies of interest (Section 11.5) or removed by calibration (Chapter 18).

Taken up later:

- Vectorized forms of the Lyapunov equation and Wick combinatorics (Chapter B).
- Coupling Gaussian networks to p-bits to obtain multimodal samplers (Chapters 14 and 16).
- Compensation of threshold-voltage mismatch by hardware-in-the-loop learning (Chapter 18).

Open:

- How to choose a sparse skew-symmetric $\mathbf{Q}$ on a two-dimensional chip layout to maximize the mixing rate per unit bias current.
- The behavior of strongly nonlinear networks at $\Omega \lesssim 10$, where the higher Kramers–Moyal terms are no longer small.

Bibliographic notes

The discrete-charge stochastic thermodynamics of nonlinear circuits, including the electrostatic energy $U(\mathbf{q})$, local detailed balance for device channels, and the expansion in $\Omega = C_0 V_T/q_e$, is due to Freitas et al. [37]. The treatment of subthreshold Gaussian samplers, including precision-matrix synthesis through the Lyapunov equation, non-reciprocal couplings, and the expansion for weak nonlinearities, follows Freitas et al. [38]; for the broader hardware program see Extropic Corporation [30] and Jelinčič et al. [56].

For the classical theory of the Kramers–Moyal and system-size expansions in stochastic physics, see van Kampen [112], Risken [90], and Gardiner [39]. Thermodynamic linear algebra, continuous-variable Ornstein–Uhlenbeck sampling for matrix inversion, and Schur-complement Gaussian conditioning in analog hardware are analyzed in Aifer et al. [3] and Coles et al. [20]. The acceleration of convergence by divergence-free probability currents ($\mathbf{Q} \neq \mathbf{0}$) is treated in Pavliotis [85]. Subthreshold circuit design and weak-inversion noise are covered in Mead [75], Enz and Vittoz [29], and Tsividis and McAndrew [111], as well as Vittoz and Fellrath [116] and Sarpeshkar et al. [93].

Chapter 14

Probabilistic Bits, Hybrid Samplers, and Cryogenic Josephson Parametrons

Up to now the double-well sampler, built on the quartic potential $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$ and its binary p-bit limit, has been a mathematical object. In Chapter 1 we saw that marginalizing binary spins out of a Gaussian model carves a $-\ln\cosh$ multi-well free-energy landscape, and that each spin, conditioned on its neighbors, is a Bernoulli variable with mean $\tanh(\beta h)$. In Chapter 2 we showed that a k-colorable factor graph can update all spins of one color in parallel without breaking detailed balance. In Chapter 7 we reduced the continuous Langevin dynamics in this potential to a two-state master equation with Kramers rates $k_{A\to B} \propto e^{-\beta\Delta E_A}$. This chapter builds that object in hardware twice: once from room-temperature subthreshold transistors, and once from superconducting Josephson junctions at liquid-helium temperature.

The central physical problem is the following. A Gibbs sampler for an Ising-type energy needs, at every site, a device whose binary output $s \in \{-1, +1\}$ is random, whose mean responds to an analog input as $\langle s\rangle = \tanh(\beta_{\text{eff}} I)$, and whose successive outputs decorrelate quickly. Such a device is called a probabilistic bit, or p-bit [14, 18]. We will ask three questions of every candidate circuit. Where does its randomness come from, and is the resulting activation function exactly tanh, or only approximately so? How fast does it produce independent samples? How much energy does one sample cost, measured in units of $k_{\text{B}}T$ and in joules?

We import the following results. From Chapter 7 we use the two-state reduction Eq. (7.12), its stationary ratio Eq. (7.14), the overdamped Kramers rate Eq. (7.37), and the moderately damped rate Eq. (7.53). From Chapter 11 we use Johnson–Nyquist noise, the $k_{\text{B}}T/C$ variance of a capacitive node (Proposition 11.6), and the Lorentzian spectrum of random telegraph signals (Section 11.5.1). From Chapters 12 and 13 we use the subthreshold transconductance $g_{\text{m}} = I_{\text{D}}/(nV_T)$, the shot-noise intensity $2q_e I_{\text{D}}$, and the mesoscopic scale $\Omega = C_{\text{node}}V_T/q_e$. From Chapter 1 we use the Gaussian–Bernoulli energy of Worked computation 1.19. The Caldeira–Leggett bath of Chapter 5 reappears in its original setting: the dissipative quantum mechanics of a Josephson junction.

The first half of the chapter (Sections 14.1 and 14.2) builds p-bits and their couplings from subthreshold CMOS, including the hybrid cell in which p-bits and continuous node voltages sample together. The second half (Sections 14.3 to 14.5) develops the Josephson physics needed to show that a superconducting loop, the quantum flux parametron, reduces near its bifurcation to the same quartic double well, and asks when quantum tunneling would spoil its Boltzmann statistics. The worked computation, Worked computation 14.20, compares a 300 K CMOS p-bit with a 4.2 K parametron, including the cost of refrigeration.

Extropic reports that its hardware, the X0 and XTR-0 platforms and the Z1 chip, is built

from room-temperature CMOS [32, 56]. Extropic's earlier litepaper also discussed superconducting circuits built from Josephson junctions as a route to thermodynamic sampling [30]. We treat both platforms with the same physics, because they are two realizations of one mathematical object: a noisy bistable coordinate whose barrier and tilt are under electrical control. Device numbers quoted in this chapter are representative order-of-magnitude estimates, not specifications of any product.

14.1 All-Transistor Subthreshold CMOS p-Bits

A p-bit is a circuit that turns physical noise into a binary random variable with a programmable bias. We first state what we require of it as a stochastic process, then derive its activation function from two different physical pictures, and finally describe three transistor-level architectures.

14.1.1 The p-Bit as a Random Telegraph Process

Definition 14.1 (Probabilistic bit)**.** A **probabilistic bit** (p-bit) is a physical device with an analog input I (a current or voltage) and a binary output $s(t) \in \{-1, +1\}$ such that, for each fixed input I:

(i) $s(t)$ is a stationary two-state Markov process with up-rate $k_\uparrow(I)$ (for $-1 \to +1$) and down-rate $k_\downarrow(I)$ (for $+1 \to -1$);

(ii) the stationary mean $m(I) := \langle s \rangle$ is a smooth, odd, strictly increasing function of I with $m(\pm\infty) = \pm 1$, called the **activation function**;

(iii) the activation has the logistic form $m(I) = \tanh(\beta_{\text{eff}} I)$ for a constant β_{eff} with units of inverse input.

A device that satisfies (i) and (ii) but not (iii) is an **approximate p-bit**.

The constant β_{eff} is not the physical inverse temperature $1/k_{\text{B}}T$. It is a circuit gain: it converts the input current into the dimensionless field βh that appears in the Boltzmann factor of the model being sampled. When a network of p-bits implements an Ising energy with local fields h_i, the input current is set so that $\beta_{\text{eff}} I_i = \beta h_i$.

The two-state master equation of Eq. (7.12) determines all statistics of $s(t)$. Write $P_\pm(t) = \mathbb{P}[s(t) = \pm 1]$ and $\lambda := k_\uparrow + k_\downarrow$. The master equation is $\dot{P}_+ = k_\uparrow P_- - k_\downarrow P_+$. Since $\langle s \rangle_t = P_+ - P_- = 2P_+ - 1$, we have

$$\frac{\mathrm{d}\,\langle s \rangle_t}{\mathrm{d}t} = 2\dot{P}_+ = 2k_\uparrow(1 - P_+) - 2k_\downarrow P_+ = 2k_\uparrow - \lambda(1 + \langle s \rangle_t) = (k_\uparrow - k_\downarrow) - \lambda \langle s \rangle_t . \tag{14.1}$$

The fixed point gives the stationary mean, and the solution relaxes to it exponentially:

$$m = \frac{k_\uparrow - k_\downarrow}{k_\uparrow + k_\downarrow}, \qquad \mathbb{E}[s(t) \mid s(0)] = m + \big(s(0) - m\big) e^{-\lambda t}. \tag{14.2}$$

Proposition 14.2 (Autocorrelation and spectrum of a p-bit)**.** *In the stationary state, the connected autocorrelation and the two-sided power spectral density of $s(t)$ are*

$$\mathrm{Cov}\big(s(t), s(0)\big) = (1 - m^2)\, e^{-\lambda |t|}, \qquad S_s(\omega) = \frac{2(1 - m^2)\lambda}{\lambda^2 + \omega^2}. \tag{14.3}$$

Proof. For $t \geq 0$, condition on $s(0)$ and use Eq. (14.2):

$$\mathbb{E}[s(t)s(0)] = \mathbb{E}[s(0)\,\mathbb{E}[s(t) \mid s(0)]] = \mathbb{E}\Big[s(0)m + s(0)^2 e^{-\lambda t} - m s(0) e^{-\lambda t}\Big].$$

Since $s(0)^2 = 1$ and $\mathbb{E}[s(0)] = m$, this equals $m^2 + (1-m^2)e^{-\lambda t}$. Subtracting m^2 gives the covariance, and stationarity extends it to $t < 0$ with $|t|$. The Fourier transform is

$$\int_{-\infty}^{\infty} (1-m^2) e^{-\lambda|t|} e^{-i\omega t}\, \mathrm{d}t = (1-m^2)\Big[\frac{1}{\lambda + i\omega} + \frac{1}{\lambda - i\omega}\Big] = \frac{2(1-m^2)\lambda}{\lambda^2 + \omega^2}.$$

□

This is the random telegraph spectrum of Section 11.5.1; a p-bit is a deliberately engineered telegraph noise source. The rate λ sets how often the p-bit forgets its past, and therefore how often it can be read to obtain a fresh sample.

Corollary 14.3 (Effective sample rate)**.** *Suppose $s(t)$ is read every Δt seconds and the readings are averaged to estimate m. For a long record the variance of the average equals that of $r_{\mathrm{eff}}\, t_{\mathrm{rec}}$ independent readings, where*

$$r_{\mathrm{eff}} = \frac{1}{\Delta t}\,\frac{1 - e^{-\lambda \Delta t}}{1 + e^{-\lambda \Delta t}} = \frac{\tanh(\lambda \Delta t/2)}{\Delta t} \;\leq\; \frac{\lambda}{2}. \tag{14.4}$$

Proof. With $\rho = e^{-\lambda \Delta t}$, the readings form a stationary sequence with lag-k correlation ρ^k. For n readings, $\mathrm{Var}(\frac{1}{n}\sum_j s_j) = \frac{1-m^2}{n^2}\sum_{j,l}\rho^{|j-l|} \to \frac{1-m^2}{n}(1 + 2\sum_{k\geq 1}\rho^k) = \frac{1-m^2}{n}\,\frac{1+\rho}{1-\rho}$ as $n \to \infty$. The equivalent number of independent readings is $n(1-\rho)/(1+\rho)$; dividing by the record length $n\Delta t$ gives Eq. (14.4). The identity $(1 - e^{-x})/(1 + e^{-x}) = \tanh(x/2)$ and $\tanh(y) \leq y$ give the bound, which is reached as $\Delta t \to 0$. □

A free-running p-bit therefore delivers at most $\lambda/2$ independent samples per second, no matter how fast it is read. Reading faster than λ wastes readout energy on correlated copies.

14.1.2 From a Tilted Double Well to the Hyperbolic-Tangent Activation

The most direct physical p-bit is a bistable coordinate x (a node voltage, a magnetization, or a superconducting phase) obeying the overdamped Langevin equation of Eq. (7.1) in a tilted potential, with output $s = \mathrm{sgn}(x - x_{\mathrm{sad}})$. The input I enters as the tilt. We prove the activation function in two steps: an exact equilibrium identity, then its explicit evaluation for the quartic double well.

Proposition 14.4 (Exact two-basin identity)**.** *Let x have stationary density $p_{\mathrm{eq}}(x) = Z^{-1}e^{-\beta E(x)}$ on $\mathbb{R}$, and let the readout threshold x_{th} (in practice the barrier top) split the line into basins $\mathcal{B}_- = (-\infty, x_{\mathrm{th}})$ and $\mathcal{B}_+ = (x_{\mathrm{th}}, \infty)$. Define the restricted partition functions $Z_\pm = \int_{\mathcal{B}_\pm} e^{-\beta E(x)}\,\mathrm{d}x$ and basin free energies $F_\pm = -k_{\mathrm{B}}T \ln Z_\pm$. Then for $s = \mathrm{sgn}(x - x_{\mathrm{th}})$, with no assumption on the barrier height,*

$$\langle s \rangle = \tanh(\beta h_{\mathrm{eff}}), \qquad h_{\mathrm{eff}} := \tfrac{1}{2}(F_- - F_+). \tag{14.5}$$

Proof. $\langle s \rangle = (Z_+ - Z_-)/(Z_+ + Z_-)$. Write $Z_\pm = e^{-\beta F_\pm}$ and multiply numerator and denominator by $e^{\beta(F_+ + F_-)/2}$: $\langle s \rangle = (e^{\beta(F_- - F_+)/2} - e^{-\beta(F_- - F_+)/2})/(e^{\beta(F_- - F_+)/2} + e^{-\beta(F_- - F_+)/2}) = \tanh(\beta h_{\mathrm{eff}})$. □

Proposition 14.4 is exact but not yet useful: it says the activation is a tanh of *something*. The physics is in how h_{eff} depends on the input. For the p-bit to sample an Ising model, h_{eff} must be linear in I.

Proposition 14.5 (Effective field of the tilted quartic)**.** *Let $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$ with $a, b > 0$, zero-tilt minima $\pm x_0 = \pm\sqrt{b/a}$, zero-tilt barrier $\Delta E_0 = b^2/(4a)$, and readout threshold $x_{\rm th} = 0$. Then*

$$h_{\rm eff}(h) = \langle x\rangle_+^{(0)}\, h + \mathcal{O}(h^3), \qquad \langle x\rangle_+^{(0)} = x_0\Big[1 - \frac{3}{16\,\beta\Delta E_0} + \mathcal{O}\big((\beta\Delta E_0)^{-2}\big)\Big], \tag{14.6}$$

where $\langle x\rangle_+^{(0)}$ is the mean of x restricted to $\mathcal{B}_+$ at $h = 0$. Consequently the activation is

$$\langle s\rangle = \tanh\Big(\beta x_0\Big[1 - \frac{3}{16\,\beta\Delta E_0}\Big]h\Big) + \mathcal{O}(h^3). \tag{14.7}$$

Proof. Differentiate $F_+ = -k_{\rm B}T \ln \int_{\mathcal{B}_+} e^{-\beta E}\,{\rm d}x$ with respect to h. Since $\partial E/\partial h = -x$, we get $\partial F_+/\partial h = -\langle x\rangle_+$, the mean of x in the restricted ensemble. At $h = 0$ the potential is even and $x_{\rm sad} = 0$, so $\langle x\rangle_-^{(0)} = -\langle x\rangle_+^{(0)}$ and $\partial F_-/\partial h = +\langle x\rangle_+^{(0)}$. Hence $\partial h_{\rm eff}/\partial h\big|_{h=0} = \frac{1}{2}(\langle x\rangle_+^{(0)} + \langle x\rangle_+^{(0)}) = \langle x\rangle_+^{(0)}$. The function $h_{\rm eff}(h)$ is odd in h by the symmetry $x \to -x$, $h \to -h$, so the next term is $\mathcal{O}(h^3)$. (The basins are fixed because the threshold is fixed at $x_{\rm th} = 0$, so no boundary term arises when we differentiate. The true barrier top moves to $x_{\rm sad} \approx -h/b$; using it as the threshold instead would change $h_{\rm eff}$ only by terms of relative size $e^{-\beta\Delta E_0}$, the weight of the barrier region.)

To evaluate $\langle x\rangle_+^{(0)}$, expand around x_0 with $x = x_0 + y$. The derivatives at x_0 are $E''(x_0) = 3ax_0^2 - b = 2b$, $E'''(x_0) = 6ax_0$, and $E'''' = 6a$, so

$$E(x_0 + y) - E(x_0) = b\,y^2 + ax_0\,y^3 + \tfrac{a}{4}y^4. \tag{14.8}$$

For $\beta\Delta E_0 \gg 1$ the weight $e^{-\beta by^2}$ confines y to a width $(2\beta b)^{-1/2} \ll x_0$, and the cubic term is a small perturbation. To first order, $e^{-\beta(by^2 + ax_0y^3)} \approx e^{-\beta by^2}(1 - \beta ax_0y^3)$, and the Gaussian moments $\langle y^2\rangle_0 = (2\beta b)^{-1}$, $\langle y^4\rangle_0 = 3(2\beta b)^{-2}$ give

$$\langle y\rangle \approx \frac{\langle y(1 - \beta ax_0y^3)\rangle_0}{\langle 1 - \beta ax_0y^3\rangle_0} = -\beta ax_0\left\langle y^4\right\rangle_0 = -\frac{3ax_0}{4\beta b^2}. \tag{14.9}$$

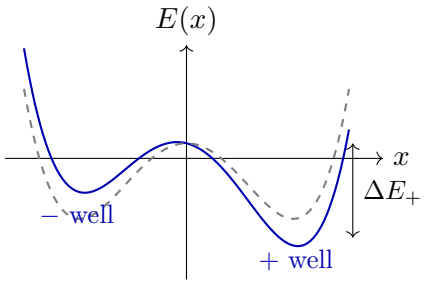
Using $b^2/a = 4\Delta E_0$ gives $\langle y\rangle/x_0 = -3/(16\beta\Delta E_0)$, which is Eq. (14.6). Inserting into Eq. (14.5) gives Eq. (14.7). □

The correction $-3/(16\beta\Delta E_0)$ has a simple meaning. Because $E'''(x_0) > 0$, the well is steeper on its outer side than on the side facing the barrier, so the thermal cloud leans toward the barrier and its mean sits slightly inside x_0. The tilt couples to x, so it couples to the thermal mean position rather than the bare minimum. The correction renormalizes $\beta_{\rm eff}$ but does not change the tanh form at linear order. The cubic remainder $\mathcal{O}(h^3)$ is the true departure from an ideal p-bit; it matters only when the tilt becomes comparable to the barrier, $hx_0 \sim \Delta E_0$.

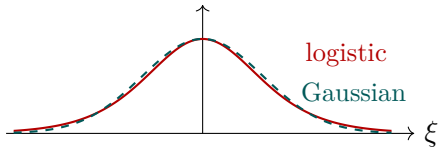
The same result follows from the kinetic picture of Chapter 7. For $\beta\Delta E_0 \gg 1$ and small tilt, the energies of the two well bottoms shift by $E(\pm x_0) \to E(\pm x_0) \mp hx_0$, while the barrier top shifts only at order h^2. The barrier seen from the + well therefore rises by hx_0 and the barrier seen from the − well falls by hx_0. With a common Kramers prefactor k_0 (Eq. (7.37)) we obtain

$$k_\uparrow = k_0\,e^{+\beta hx_0}, \qquad k_\downarrow = k_0\,e^{-\beta hx_0}, \qquad \frac{k_\uparrow}{k_\downarrow} = e^{2\beta hx_0}. \tag{14.10}$$

Inserting into Eq. (14.2) gives $m = \sinh(\beta hx_0)/\cosh(\beta hx_0) = \tanh(\beta hx_0)$ and $\lambda = 2k_0\cosh(\beta hx_0)$. If the input current I produces the tilt $h = \kappa_{\rm in}I$ through a transduction constant $\kappa_{\rm in}$, then $\beta_{\rm eff} = \beta\kappa_{\rm in}x_0$. The rate ratio $e^{2\beta_{\rm eff}I}$ is the detailed-balance condition Eq. (7.14) at leading Laplace



(a) tilt $h > 0$ lowers the + well



(b) comparator noise densities

Figure 14.1: (a) The quartic double well $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$ with zero tilt (dashed) and positive tilt (solid). The tilt raises the barrier ΔE_+ seen from the + well and lowers the barrier seen from the − well, which produces the rate ratio Eq. (14.10). (b) A comparator that thresholds an input against noise ξ produces $\langle s \rangle = 2F_\xi(u) - 1$. Logistic noise (solid) gives an exact tanh; Gaussian noise (dashed, slope-matched at the origin) gives an error function.

order. The kinetic derivation drops the change of the Kramers prefactor with h; Proposition 14.5 shows that including it only renormalizes x_0 by $1 - 3/(16\beta\Delta E_0)$.

The kinetic picture adds one fact the equilibrium identity cannot: the relaxation rate $\lambda = 2k_0 \cosh(\beta_{\text{eff}} I)$ grows with the input. A strongly biased p-bit flips back to its preferred state faster, so its correlation time shortens as $|m| \to 1$.

14.1.3 Comparator p-Bits: Exact tanh Versus the Error Function

The second way to make a p-bit is to compare the input against a random threshold. This is how most demonstrated p-bits work, whether the randomness comes from a stochastic magnetic tunnel junction [14, 18], a digital pseudorandom generator, or amplified transistor noise.

Lemma 14.6 (Comparator on noise)**.** *Let u be a deterministic input, ξ a random variable with continuous cumulative distribution function F_ξ symmetric about zero ($F_\xi(-u) = 1 - F_\xi(u)$), and $s = \operatorname{sgn}(u + \xi)$. Then*

$$\mathbb{P}[s = +1] = F_\xi(u), \qquad \langle s \rangle = 2F_\xi(u) - 1. \tag{14.11}$$

In particular:

(a) If ξ is logistic with density $\frac{\beta_{\text{eff}}}{2}\operatorname{sech}^2(\beta_{\text{eff}}\xi)$, then $\langle s \rangle = \tanh(\beta_{\text{eff}} u)$ exactly.

(b) If $\xi \sim \mathcal{N}(0, \sigma^2)$, then $\langle s \rangle = \operatorname{erf}\big(u/(\sqrt{2}\sigma)\big)$.

(c) If r is uniform on $[-1, 1]$ and $s = \operatorname{sgn}\big(\tanh(\beta_{\text{eff}} u) - r\big)$, then $\langle s \rangle = \tanh(\beta_{\text{eff}} u)$ exactly.

Proof. $\mathbb{P}[s = +1] = \mathbb{P}[\xi > -u] = 1 - F_\xi(-u) = F_\xi(u)$ by symmetry, and $\langle s \rangle = 2\mathbb{P}[s = +1] - 1$. (a) The logistic CDF is $F_\xi(u) = \int_{-\infty}^{u} \frac{\beta_{\text{eff}}}{2}\operatorname{sech}^2(\beta_{\text{eff}}\xi)\,\mathrm{d}\xi = \frac{1}{2}[1 + \tanh(\beta_{\text{eff}} u)]$, so $2F_\xi - 1 = \tanh(\beta_{\text{eff}} u)$. (b) $F_\xi(u) = \frac{1}{2}[1 + \operatorname{erf}(u/\sqrt{2}\sigma)]$. (c) $\mathbb{P}[r < \tanh(\beta_{\text{eff}} u)] = \frac{1}{2}[1 + \tanh(\beta_{\text{eff}} u)]$ because r is uniform on $[-1, 1]$. □

Case (c) is the update rule of Camsari et al. [18]: a deterministic tanh stage followed by a comparison against a uniform random number. It is exact by construction, but it moves the problem elsewhere. One needs a deterministic tanh transfer function, which the subthreshold differential pair supplies naturally (its output current is $I_{\text{tail}} \tanh(V_{\text{d}}/2nV_T)$; see Worked computation 12.24),

and one needs a source of uniform random numbers. A digital pseudorandom generator costs far more energy than the analog stage it feeds.

Case (b) is what one gets for free from amplified thermal or shot noise, because the central limit theorem makes the sum of many independent electron transits Gaussian. The activation is then a probit (error function), not a tanh. To measure the difference, match the slopes at the origin. Since $\frac{\mathrm{d}}{\mathrm{d}u}\operatorname{erf}(u/\sqrt{2}\sigma)\big|_0 = \frac{2}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{2}\sigma}$, this fixes

$$\beta_{\text{eff}} = \sqrt{\frac{2}{\pi}}\,\frac{1}{\sigma} \approx \frac{0.798}{\sigma}. \tag{14.12}$$

A direct numerical comparison gives $\max_u \left|\operatorname{erf}(u/\sqrt{2}\sigma) - \tanh(0.798\,u/\sigma)\right| \approx 0.035$, attained near $|u| \approx 1.7\sigma$. If instead β_{eff} is chosen to minimize the maximum deviation, the optimum is $\beta_{\text{eff}} \approx 0.851/\sigma$ and the maximum deviation drops to ≈ 0.019. In the mean, a Gaussian comparator is a tanh p-bit to within two to four percent, depending on how β_{eff} is chosen.

The tails tell a different story. The log-odds of a tanh p-bit are linear in the input, $\ln(P_+/P_-) = 2\beta_{\text{eff}}u$. For the probit, with $z = u/\sigma$, the standard Mills-ratio asymptotics give $\ln(P_+/P_-) = \ln[\Phi(z)/\Phi(-z)] = \frac{z^2}{2} + \ln(z\sqrt{2\pi}) + o(1)$ as $z \to \infty$, which grows quadratically. A probit p-bit is therefore far more deterministic than a tanh p-bit when it is strongly driven. In a Gibbs sampler this matters most for strongly coupled, low-temperature models, where rare minority configurations carry the physics.

Remark 14.7 (What a probit Gibbs sampler actually samples)**.** Suppose every site of a network uses the probit conditional $\mathbb{P}[s_i = +1 \mid \mathbf{s}_{-i}] = \Phi(h_i/\sigma)$ with $h_i = \sum_j J_{ij}s_j + b_i$. For a joint distribution to have these conditionals, the ratio $\mathbb{P}[s_i = +1 \mid \mathbf{s}_{-i}]/\mathbb{P}[s_i = -1 \mid \mathbf{s}_{-i}]$ must factor consistently across sites (Brook's lemma, the compatibility condition behind the Hammersley–Clifford theorem of Theorem 2.5). With pairwise linear fields this holds for the logistic conditional and fails in general for the probit. The chromatic Gibbs chain still has a unique stationary distribution, but it is not exactly the Boltzmann distribution of $(\mathbf{J}, \mathbf{b})$. In practice the discrepancy is absorbed by training on the hardware itself: contrastive learning adjusts $(\mathbf{J}, \mathbf{b})$ using statistics measured from the physical sampler, so the learned parameters compensate the activation shape along with mismatch and offsets (Section 18.5).

14.1.4 Three Transistor Architectures

We now describe three all-transistor p-bits that run in subthreshold CMOS. All three reduce to either the tilted double well of Section 14.1.2 or the comparator of Lemma 14.6.

(1) Noise-amplifying comparator. A transistor biased in weak inversion carries shot noise of single-sided spectral density $S_I = 2q_eI_{\mathrm{D}}$ (Section 12.5). A chain of high-gain amplifiers raises this noise to logic levels, and a clocked comparator thresholds the sum of the input signal $u = R_{\text{in}}I$ and the amplified noise. For a single-pole amplifier with input node capacitance C and transconductance g_{m}, the input-referred noise integrated over the amplifier bandwidth is

$$\sigma^2 = \int_0^\infty \frac{S_I(f)}{g_{\mathrm{m}}^2}\,\frac{\mathrm{d}f}{1 + (2\pi fC/g_{\mathrm{m}})^2} = \frac{S_I}{g_{\mathrm{m}}^2}\cdot\frac{g_{\mathrm{m}}}{4C} = \frac{S_I}{4g_{\mathrm{m}}C}. \tag{14.13}$$

For a thermal-like channel with $S_I = 4\gamma_n k_{\mathrm{B}}T g_{\mathrm{m}}$, where the excess-noise factor γ_n is of order one (for an ideal subthreshold device in saturation $2q_eI_{\mathrm{D}} = 4\cdot\frac{n}{2}k_{\mathrm{B}}Tg_{\mathrm{m}}$, so $\gamma_n = n/2$), this becomes $\sigma^2 = \gamma_n k_{\mathrm{B}}T/C$. The bias current cancels: a faster amplifier has more noise bandwidth and

proportionally more gain, so the input-referred variance is set by the capacitance. The comparator gives the probit activation with $\beta_{\text{eff}} \approx 0.85\, R_{\text{in}}\sqrt{C/(\gamma_n k_{\text{B}}T)}$. The effective temperature is tuned through R_{in}, the gain of the input stage. The samples decorrelate once per amplifier time constant C/g_{m}.

(2) Cross-coupled latch with thermal reset. A cross-coupled pair of inverters (a latch) is a bistable circuit. Its differential voltage v has an unstable equilibrium at $v = 0$ and two stable states near the supply rails. The latch p-bit operates in two phases. In the reset phase an equalizing switch shorts the two outputs, holding v near zero; when the switch opens, v is left with the thermal variance $k_{\text{B}}T/C$ of Proposition 11.6 plus the input offset v_{in} injected by the input current. In the regeneration phase the positive feedback amplifies v until it saturates at one rail. The output $s = \operatorname{sgn}(v)$ is read and the cycle repeats.

Proposition 14.8 (Decision statistics of a regenerating latch)**.** *During regeneration, let the differential voltage obey the linear unstable Langevin equation*

$$C\ \mathrm{d}v_t = G_n v_t\ \mathrm{d}t + \sqrt{q_n}\ \mathrm{d}W_t, \tag{14.14}$$

with negative conductance $-G_n < 0$ ($G_n > 0$ is the net positive-feedback transconductance), white noise current of intensity q_n (so $\langle i_n(t) i_n(t')\rangle = q_n \delta(t-t')$), and initial condition $v_0 \sim \mathcal{N}(v_{\text{in}}, k_{\text{B}}T/C)$ independent of the noise. Then as $t \to \infty$, $\operatorname{sgn}(v_t)$ converges almost surely to $\operatorname{sgn}(v_0 + Z)$ with $Z \sim \mathcal{N}(0, q_n/(2G_nC))$ independent of v_0, and

$$\mathbb{P}[s = +1] = \Phi\Big(\frac{v_{\text{in}}}{\sigma_{\text{L}}}\Big), \qquad \sigma_{\text{L}}^2 = \frac{k_{\text{B}}T}{C} + \frac{q_n}{2G_nC}. \tag{14.15}$$

For $q_n = 2\gamma_n k_{\text{B}}T G_n$, $\sigma_{\text{L}}^2 = (1+\gamma_n)k_{\text{B}}T/C$.

Proof. Let $\tau = C/G_n$. The linear SDE $\mathrm{d}v = \tau^{-1} v\ \mathrm{d}t + C^{-1}\sqrt{q_n}\ \mathrm{d}W$ has the solution $v_t = e^{t/\tau}\big[v_0 + C^{-1}\sqrt{q_n}\int_0^t e^{-t'/\tau}\ \mathrm{d}W_{t'}\big]$, which one checks by applying Itô's product rule to $e^{-t/\tau}v_t$ (the noise is additive, so Itô and Stratonovich agree). The bracket $Y_t = v_0 + C^{-1}\sqrt{q_n}\int_0^t e^{-t'/\tau}\ \mathrm{d}W_{t'}$ is a Gaussian martingale with $\operatorname{Var}(Y_t - v_0) = \frac{q_n}{C^2}\int_0^t e^{-2t'/\tau}\ \mathrm{d}t' = \frac{q_n\tau}{2C^2}(1 - e^{-2t/\tau})$. By the martingale convergence theorem $Y_t \to Y_\infty = v_0 + Z$ almost surely, with $\operatorname{Var}(Z) = q_n\tau/(2C^2) = q_n/(2G_nC)$. Because $e^{t/\tau} > 0$, $\operatorname{sgn}(v_t) = \operatorname{sgn}(Y_t) \to \operatorname{sgn}(Y_\infty)$ whenever $Y_\infty \neq 0$, which holds with probability one. Since v_0 and Z are independent Gaussians, $Y_\infty \sim \mathcal{N}(v_{\text{in}}, k_{\text{B}}T/C + q_n/(2G_nC))$, and $\mathbb{P}[Y_\infty > 0] = \Phi(v_{\text{in}}/\sigma_{\text{L}})$. □

The linear model Eq. (14.14) breaks down once $|v|$ reaches the saturation scale of the feedback transistors (about $2nV_T$ for a subthreshold pair). By then $|v|$ exceeds the noise by a factor $e^{t/\tau}$, and the decision is already made. The latch is therefore a comparator whose noise is the sum of the frozen reset noise and the noise added during the first few time constants of regeneration. Its activation is a probit, Lemma 14.6(b), with $\sigma = \sigma_{\text{L}} = \sqrt{(1+\gamma_n)k_{\text{B}}T/C}$. Reset and regeneration take a few time constants $\tau = C/G_n$ each, so a clocked latch delivers one independent sample per cycle.

The latch can also run free, with the equalizing switch replaced by a weak load conductance G that keeps the barrier low. It is then a tilted double well, and Section 14.1.2 applies. We will see in Worked computation 14.20 that the free-running mode is slow: to bring the barrier down to a few $k_{\text{B}}T$ the latch must sit close to its bifurcation, where the curvature of the wells, and with it the Kramers prefactor, nearly vanishes.

(3) Current-starved inverter. A CMOS inverter whose supply current is limited to nanoamperes by series transistors is a slow, high-gain amplifier. Biased at its switching threshold by a feedback element, it amplifies its own channel noise with gain g_m/g_{ds} (tens per stage in weak inversion, Section 12.4) and bandwidth $g_m/C = I_D/(nV_TC)$. A short chain of such stages followed by a full-swing inverter is a compact noise-amplifying comparator, architecture (1) without a separate clock, and its integrated input-referred noise again obeys Eq. (14.13). The input current shifts the switching threshold and so plays the role of u. The output is a telegraph signal whose dwell times are set by the noise bandwidth $I_D/(nV_TC)$, which the starving current tunes.

Heuristic Derivation: Why k_BT/C sets the temperature of every CMOS p-bit

In all three architectures the input-referred randomness is a voltage variance of the form $\gamma k_BT/C$ on the decision node. This is equipartition (Section 11.2.3) dressed by an order-one factor that accounts for non-thermal shot noise and for noise added during amplification. The effective inverse temperature of the p-bit, in units of input voltage, is therefore $\beta_{\text{eff}} \sim \sqrt{C/(\gamma k_BT)}$. For $C = 0.5\,\text{fF}$ at $300\,\text{K}$, $\sqrt{k_BT/C} \approx 2.9\,\text{mV}$. A designer who wants a hotter p-bit lowers C or lowers the input gain; a colder one raises them. The intensive parameter that controls this is the number of thermal charges on the node, $\Omega = CV_T/q_e$ (Section 13.2), because $k_BT/C = V_T^2/\Omega$.

14.2 Analog Synapses, Coupling Crossbars, and Hybrid Thermodynamic Sampling Units

A single p-bit samples a single Bernoulli variable. To sample an Ising model $p_\theta(\mathbf{s}) \propto \exp(\beta[\frac{1}{2}\mathbf{s}^\top\mathbf{J}\mathbf{s} + \mathbf{b}^\top\mathbf{s}])$, each p-bit i must receive the local field $h_i = \sum_j J_{ij}s_j + b_i$ from its neighbors. In analog hardware this sum is computed by Kirchhoff's current law at a shared node, with no arithmetic unit. This section derives the mapping from conductances to couplings, shows what the circuit computes when the mapping is imperfect, and builds the hybrid cell in which continuous voltages and p-bits sample jointly. Extropic calls chips built from arrays of such cells Thermodynamic Sampling Units (TSUs).

14.2.1 Kirchhoff Current Summation and Signed Weights

Consider p-bit i whose input is a summing node of capacitance C_Σ, tied to ground through an input resistance R_{in} (the input impedance of the p-bit's first stage). Each neighbor j drives two wires: its output $+V_s s_j$ and an inverted copy $-V_s s_j$. These connect to the summing node through programmable conductances G_{ij}^+ and G_{ij}^-, so that the signed weight is the difference

$$W_{ij} := G_{ij}^+ - G_{ij}^-, \qquad G_{\Sigma,i} := \sum_j (G_{ij}^+ + G_{ij}^-). \tag{14.16}$$

A bias current $I_{b,i}$ is injected directly.

Proposition 14.9 (Current summation with loading)**.** *In steady state the summing-node voltage is*

$$u_i = \frac{R_{\text{in}}}{1 + R_{\text{in}}G_{\Sigma,i}}\Big[V_s \sum_j W_{ij}s_j + I_{b,i}\Big]. \tag{14.17}$$

The loading factor $(1 + R_{\text{in}}G_{\Sigma,i})^{-1}$ *does not depend on the state* $\mathbf{s}$.

Proof. The current flowing into the node from neighbor j is $G_{ij}^{+}(V_s s_j - u_i) + G_{ij}^{-}(-V_s s_j - u_i) = W_{ij} V_s s_j - (G_{ij}^{+} + G_{ij}^{-}) u_i$. Kirchhoff's current law sets the total inflow equal to the outflow u_i/R_{in}: $V_s \sum_j W_{ij} s_j - G_{\Sigma,i} u_i + I_{b,i} = u_i/R_{\text{in}}$. Solving for u_i gives Eq. (14.17). Because each neighbor drives both G_{ij}^{+} and G_{ij}^{-} with opposite signs, the total conductance seen by the node, $G_{\Sigma,i}$, is the same for every $\mathbf{s}$. □

The differential encoding has a useful side effect: loading only rescales all weights of row i by a common factor, which can be folded into β_{eff}. A single-ended encoding, with one conductance per synapse driven between 0 and V_s, would make the loading depend on how many neighbors are high, and the sum would no longer be linear in $\mathbf{s}$.

Matching Eq. (14.17) to the target field fixes the programming rule. If the p-bit has activation $\tanh(\beta_{\text{eff}} u_i)$, we need $\beta_{\text{eff}} u_i = \beta h_i$, that is,

$$\beta J_{ij} = \beta_{\text{eff}} \frac{R_{\text{in}} V_s}{1 + R_{\text{in}} G_{\Sigma,i}} W_{ij}, \qquad \beta b_i = \beta_{\text{eff}} \frac{R_{\text{in}}}{1 + R_{\text{in}} G_{\Sigma,i}} I_{b,i}. \tag{14.18}$$

The physical temperature does not appear on the hardware side. The model's inverse temperature β is a bookkeeping constant: only the products βJ_{ij} and βb_i are realized.

The conductances themselves add noise. Each $G_{ij}^{\pm}$ carries Johnson–Nyquist current noise (Theorem 11.5). Filtered by the summing node, the total adds a voltage variance $k_{\text{B}}T/C_{\Sigma}$ to u_i by the same integral as Eq. (11.49). This noise is indistinguishable from the p-bit's own noise and simply contributes to σ^2, lowering β_{eff}. Because C_{Σ} grows with fan-in, the synaptic contribution changes with the connectivity of each site; a calibration step or in-loop training absorbs the resulting site-to-site spread of β_{eff}.

14.2.2 Symmetry of the Couplings and What Asymmetry Does

The conductance from j to i and the conductance from i to j are separate physical devices. Nothing in the circuit forces $W_{ij} = W_{ji}$; device mismatch (Section 18.4) guarantees that they differ slightly. We first recall why symmetry is needed and then compute exactly what happens without it.

Proposition 14.10 (Symmetric couplings give Boltzmann stationarity)**.** *Let each site be updated, one at a time or in color classes of a proper coloring, by the heat-bath rule* $\mathbb{P}[s_i = +1 \mid \mathbf{s}_{-i}] = \frac{1}{2}[1 + \tanh(\beta h_i)]$ *with* $h_i = \sum_{j \neq i} J_{ij} s_j + b_i$*. If* $\mathbf{J} = \mathbf{J}^{\top}$ *with zero diagonal, each update satisfies detailed balance with respect to* $\pi(\mathbf{s}) \propto \exp(\beta[\frac{1}{2}\mathbf{s}^{\top}\mathbf{J}\mathbf{s} + \mathbf{b}^{\top}\mathbf{s}])$*.*

Proof. Let $\mathbf{s}^{i\pm}$ denote $\mathbf{s}$ with s_i set to ± 1. In $\frac{1}{2}\mathbf{s}^{\top}\mathbf{J}\mathbf{s} = \frac{1}{2}\sum_{k,l} J_{kl} s_k s_l$, the terms containing s_i are $\frac{1}{2} s_i \sum_{l \neq i} (J_{il} + J_{li}) s_l$. Hence $\ln[\pi(\mathbf{s}^{i+})/\pi(\mathbf{s}^{i-})] = \beta \sum_{l \neq i} (J_{il} + J_{li}) s_l + 2\beta b_i$. With $\mathbf{J}$ symmetric this equals $2\beta h_i$, which is exactly ln of the ratio of heat-bath probabilities $[1+\tanh(\beta h_i)]/[1-\tanh(\beta h_i)] = e^{2\beta h_i}$. The single-site kernel therefore satisfies $\pi(\mathbf{s}^{i-})\mathbb{P}[\mathbf{s}^{i-} \to \mathbf{s}^{i+}] = \pi(\mathbf{s}^{i+})\mathbb{P}[\mathbf{s}^{i+} \to \mathbf{s}^{i-}]$. Sites in one color class have no couplings among themselves, so their simultaneous update factorizes into commuting single-site kernels, each preserving π (Section 2.5). □

If $\mathbf{J}$ is not symmetric, the ratio above involves $\frac{1}{2}(J_{il} + J_{li})$ while the hardware applies J_{il}, and there is no π of Boltzmann form. The smallest example shows what replaces it.

Proposition 14.11 (Two asymmetrically coupled p-bits)**.** *Let two p-bits with zero bias have fields* $h_1 = J_{12} s_2$ *and* $h_2 = J_{21} s_1$*, and at each step pick one site uniformly at random and apply the heat-bath rule. Write* $t_1 = \tanh(\beta J_{12})$ *and* $t_2 = \tanh(\beta J_{21})$*. Then the stationary distribution is*

$$\pi(+,+) = \pi(-,-) = \frac{2 + t_1 + t_2}{8}, \qquad \pi(+,-) = \pi(-,+) = \frac{2 - t_1 - t_2}{8}, \tag{14.19}$$

the correlation is $\langle s_1 s_2 \rangle = \frac{1}{2}(t_1 + t_2)$*, and there is a net probability current per step*

$$\mathcal{J}_{\circlearrowleft} = \frac{t_2 - t_1}{16} \tag{14.20}$$

around the cycle $(+,+) \to (-,+) \to (-,-) \to (+,-) \to (+,+)$*. The entropy production per step, in units of* k_{B}*, is*

$$\dot{\Sigma} = \frac{\beta}{4}\,(J_{21} - J_{12})(\tanh \beta J_{21} - \tanh \beta J_{12}) \;\geq\; 0. \tag{14.21}$$

Proof. The dynamics is invariant under the global flip $\mathbf{s} \to -\mathbf{s}$, which maps $(+,+) \leftrightarrow (-,-)$ and $(+,-) \leftrightarrow (-,+)$. The unique stationary distribution inherits this symmetry, so we may write $\pi(+,+) = \pi(-,-) = p$ and $\pi(+,-) = \pi(-,+) = q$ with $2p + 2q = 1$. Balance the probability flowing out of and into $(+,+)$. Leaving: site 1 is chosen with probability $\frac{1}{2}$ and flips to -1 with probability $\frac{1}{2}(1 - t_1)$ (its field is J_{12}); site 2 is chosen with probability $\frac{1}{2}$ and flips with probability $\frac{1}{2}(1 - t_2)$. Entering: from $(-,+)$ through site 1 with probability $\frac{1}{2} \cdot \frac{1}{2}(1 + t_1)$, and from $(+,-)$ through site 2 with probability $\frac{1}{2} \cdot \frac{1}{2}(1 + t_2)$. Stationarity gives $p(2 - t_1 - t_2) = q(2 + t_1 + t_2)$. Combined with $p + q = \frac{1}{2}$ this yields Eq. (14.19), and $\langle s_1 s_2 \rangle = 2p - 2q = \frac{1}{2}(t_1 + t_2)$.

The net current on the edge $(+,+) \to (-,+)$ is

$$\mathcal{J} = p \cdot \tfrac{1}{4}(1 - t_1) - q \cdot \tfrac{1}{4}(1 + t_1) = \tfrac{1}{32}\big[(2 + t_1 + t_2)(1 - t_1) - (2 - t_1 - t_2)(1 + t_1)\big] = \tfrac{1}{32} \cdot 2(t_2 - t_1),$$

since the two products expand to $2 - t_1 + t_2 - t_1^2 - t_1 t_2$ and $2 + t_1 - t_2 - t_1^2 - t_1 t_2$. By the flip symmetry, the same current flows on each of the four edges of the cycle.

For the entropy production we use the Schnakenberg cycle formula. The chain has a single cycle carrying the uniform current $\mathcal{J}_{\circlearrowleft}$, so the stationary probabilities cancel around it and

$$\dot{\Sigma} = \mathcal{J}_{\circlearrowleft} \ln \frac{\prod_{\text{cycle}} w_{\rightarrow}}{\prod_{\text{cycle}} w_{\leftarrow}},$$

where the products run over the forward and backward one-step transition probabilities around the cycle. Going forward the four steps have probabilities $\frac{1}{4}(1 - t_1)$, $\frac{1}{4}(1 + t_2)$, $\frac{1}{4}(1 - t_1)$, $\frac{1}{4}(1 + t_2)$; going backward, $\frac{1}{4}(1 + t_1)$, $\frac{1}{4}(1 - t_2)$, $\frac{1}{4}(1 + t_1)$, $\frac{1}{4}(1 - t_2)$. Using $\ln \frac{1+\tanh y}{1-\tanh y} = 2y$, the logarithm of the ratio is $-4\beta J_{12} + 4\beta J_{21}$. Multiplying by $\mathcal{J}_{\circlearrowleft} = (t_2 - t_1)/16$ gives Eq. (14.21). Since tanh is increasing, the two factors have the same sign, so $\dot{\Sigma} \geq 0$ with equality only for $J_{12} = J_{21}$. □

The proposition says two things. First, to leading order the network behaves as if coupled by the symmetric part: the measured correlation $\frac{1}{2}(t_1 + t_2)$ is what a symmetric coupling J_{eff} with $\tanh \beta J_{\text{eff}} = \frac{1}{2}(t_1 + t_2)$ would produce, and $J_{\text{eff}} = \frac{1}{2}(J_{12} + J_{21}) + \mathcal{O}((J_{12} - J_{21})^2)$. Second, the antisymmetric part does not disappear: it drives a circulating probability current and produces entropy at a rate quadratic in the asymmetry. This is the discrete analogue of the non-reciprocal transconductance mechanism of Section 13.3. A p-bit network with mismatched synapses is a non-equilibrium steady state, not a slightly wrong equilibrium. For small mismatch the statistical error is second order; in-loop training (Chapter 18) measures and corrects the first-order symmetric part.

14.2.3 Capacitive Coupling and Charge-Domain Summation

Resistive crossbars draw static current whenever $u_i \neq \pm V_s$, and that current is dissipated continuously. An alternative is to sum charge. Replace each conductance by a capacitor $C_{ij}^{\pm}$ and let the summing

node float during the evaluation phase. After a reset that sets $u_i = 0$ with all drivers at a reference state, switching the drivers to $\pm V_s s_j$ redistributes charge and gives

$$u_i = \frac{V_s \sum_j (C_{ij}^+ - C_{ij}^-) s_j + Q_{b,i}}{C_{\Sigma,i}}, \qquad C_{\Sigma,i} = C_{\mathrm{par}} + \sum_j (C_{ij}^+ + C_{ij}^-), \tag{14.22}$$

by charge conservation on the isolated node: the charge $\sum_j [C_{ij}^+(u_i - V_s s_j) + C_{ij}^-(u_i + V_s s_j)] + C_{\mathrm{par}} u_i$ must equal the injected bias charge $-Q_{b,i}$ relative to reset, which rearranges to Eq. (14.22). The capacitive divider dissipates no static power. Its energy cost is dynamic: each time a driver changes by $2V_s$, charging $C_{ij}^{\pm}$ through that step dissipates at most $\frac{1}{2} C_{ij}^{\pm} (2V_s)^2 = 2 C_{ij}^{\pm} V_s^2$. The reset of the floating node leaves the variance $k_{\mathrm{B}}T/C_{\Sigma,i}$ on it (Proposition 11.6), and this frozen noise can serve directly as the comparator noise of a latch p-bit (Proposition 14.8). Capacitive summation pairs naturally with clocked latches: the reset phase of the latch and the reset phase of the summing node are the same event.

14.2.4 Hybrid Discrete–Continuous Sampling Cells

The Gaussian–Bernoulli model of Worked computation 1.19 couples continuous variables to p-bits. Its conditionals are a Gaussian for the continuous part and independent Bernoullis for the spins, so it is sampled by alternating two blocks: a two-color chromatic schedule. We now build the smallest physical version and find the condition under which it carves a double well.

Take one continuous node with voltage v, capacitance C, and a leak conductance G to a bias voltage b. The node receives a current $\sum_j w_j s_j$ from M p-bits. With the p-bits held fixed, Kirchhoff's law with the Johnson noise of G gives

$$C \, \mathrm{d}v_t = \Big[-G(v_t - b) + \sum_{j=1}^{M} w_j s_j \Big] \mathrm{d}t + \sqrt{2 k_{\mathrm{B}} T G} \, \mathrm{d}W_t. \tag{14.23}$$

This is the Ornstein–Uhlenbeck equation of the RC sampler (the coupled linear RC circuit of Example 1.5), with a shifted center. Its stationary law given $\mathbf{s}$ is Gaussian with

$$\mathbb{E}[v \mid \mathbf{s}] = b + \frac{1}{G} \sum_j w_j s_j, \qquad \mathrm{Var}(v \mid \mathbf{s}) = \frac{k_{\mathrm{B}} T}{C}, \tag{14.24}$$

reached in a few relaxation times $\tau_{RC} = C/G$. The variance is set by C alone, as the equipartition lock of a passive node requires (Section 13.3); G sets only the speed and the conversion from current to voltage.

We want a joint distribution $p(v, \mathbf{s}) \propto e^{-\beta E(v,\mathbf{s})}$ whose v-conditional is Eq. (14.24). Completing the square shows that the energy

$$E(v, \mathbf{s}) = \frac{C}{2}(v - b)^2 - \frac{C}{G}(v - b) \sum_j w_j s_j - \sum_j c_j s_j \tag{14.25}$$

does this: as a function of v it equals $\frac{C}{2}\big(v - b - \frac{1}{G}\sum_j w_j s_j\big)^2$ plus terms independent of v. The same energy fixes what each p-bit must do. Collecting the terms linear in s_j, the field on p-bit j is

$$h_j(v) = \frac{C}{G} w_j \, (v - b) + c_j. \tag{14.26}$$

This is a reciprocity condition. The current weight w_j by which p-bit j pushes the node must equal, up to the constant G/C, the gain with which p-bit j reads the node voltage. In hardware the read gain is set by the p-bit's input stage, so $\beta_{\mathrm{eff},j} \, u_j = \beta h_j(v)$ fixes it.

Proposition 14.12 (Double-well carving in a hybrid cell)**.** *Summing Eq. (14.25) over* $\mathbf{s}$ *gives the marginal free energy of the continuous node*

$$F_{\text{eff}}(v) = \frac{C}{2}(v-b)^2 - k_{\text{B}}T\sum_{j=1}^{M} \ln\cosh\Big(\beta\Big[\frac{C}{G}w_j(v-b)+c_j\Big]\Big) + \textit{const.} \tag{14.27}$$

For $c_j = 0$*,* F_{eff} *has a double well, with a local maximum at* $v = b$*, if and only if*

$$\sum_{j=1}^{M}\Big(\frac{w_j}{G}\Big)^2 > \frac{k_{\text{B}}T}{C}. \tag{14.28}$$

Proof. $\sum_{s_j=\pm1} e^{\beta h_j s_j} = 2\cosh(\beta h_j)$, and the sum over $\mathbf{s}$ factorizes over j, which gives Eq. (14.27). With $c_j = 0$ and $y = v - b$, the expansion $\ln\cosh x = \frac{x^2}{2} - \frac{x^4}{12} + \mathcal{O}(x^6)$ (Eq. (1.143)) gives

$$F_{\text{eff}} = \frac{1}{2}\Big[C - \beta\frac{C^2}{G^2}\sum_j w_j^2\Big]y^2 + \frac{\beta^3}{12}\frac{C^4}{G^4}\sum_j w_j^4\, y^4 + \mathcal{O}(y^6).$$

The curvature at $y = 0$ is negative exactly when $\beta C\sum_j w_j^2/G^2 > 1$, which is Eq. (14.28). The quartic coefficient is positive, and for $|y| \to \infty$ the $\ln\cosh$ terms grow only linearly while the quadratic term grows quadratically, so a negative curvature at the origin implies exactly one minimum on each side. (Uniqueness on each side follows because $F''_{\text{eff}}(y) = C - \beta\frac{C^2}{G^2}\sum_j w_j^2\,\operatorname{sech}^2(\cdot)$ is increasing in $|y|$, so F'_{eff} changes sign only once on each half-line.) □

The condition has a plain reading. The quantity w_j/G is the voltage shift one p-bit produces on the node. Bimodality appears when the total shift produced by the p-bits exceeds the thermal spread $\sqrt{k_{\text{B}}T/C}$ of the node. The p-bits, each a two-state device, then split the node voltage into two separated clusters, and the cell becomes a continuous double-well sampler built from a linear RC node plus binary latent variables. This is the mechanism by which the latent p-bit layers of a Denoising Thermodynamic Model carve non-Gaussian conditionals (Sections 16.4 and 17.5).

For several coupled continuous nodes the situation changes. A purely passive RC network has stationary covariance $k_{\text{B}}T\mathbf{C}^{-1}$ independent of the conductance matrix, so it cannot realize an arbitrary precision matrix. Programmable correlations among continuous nodes require active transconductors, as developed in Section 13.5. The p-bit side of the cell is unaffected: it reads voltages and pushes currents exactly as above.

Hardware Realization: p-Bit arrays in Extropic's X0, XTR-0, and Z1

Extropic's published descriptions [31, 32, 56] present Thermodynamic Sampling Units as arrays of all-transistor p-bits on locally connected, colorable graphs, with couplings and biases stored on chip and applied through analog summation, and with updates scheduled one color class at a time so that parallel updates preserve the target distribution (Proposition 14.10). According to these sources, the X0 prototype demonstrated the room-temperature transistor sampling cells, the XTR-0 platform pairs sampling chips with a conventional host and FPGA for development, and the Z1 die scales the array to roughly a quarter of a million p-bits (Section 17.1.1). The `THRML` library exposes exactly the abstraction of this section: block Gibbs sampling on factor graphs whose sites have tanh-type conditionals. We do not have, and do not quote, circuit-level specifications of these chips. The physics above explains what any such array must satisfy: a $k_{\text{B}}T/C$-limited noise source per site, a state-independent

loading factor, reciprocal weights to the accuracy the application needs, and a chromatic schedule.

14.3 Cryogenic Superconducting Realizations: Order Parameter, Fluxoid Quantization, and the Josephson Relations

We now change substrate. In a superconductor the relevant degree of freedom is not a node voltage but the phase of a macroscopic wavefunction. That phase obeys a nonlinear equation of motion when two superconductors are weakly connected, and the resulting circuit element, the Josephson junction, supplies the nonlinearity needed for a double well with almost no dissipation. This section collects the superconducting physics needed for the rest of the chapter, following the treatment in Tinkham [109].

14.3.1 The Macroscopic Order Parameter and the Supercurrent

Below the critical temperature T_c, conduction electrons in a superconductor bind into Cooper pairs of charge $q^* = -2q_e$ and mass $m^* = 2m_e$. In BCS theory the pairs share a single quantum state, so that a macroscopic number of them are described by one complex function, the **order parameter**

$$\Psi(\mathbf{r}, t) = \sqrt{n_s(\mathbf{r}, t)}\, e^{i\varphi(\mathbf{r},t)}, \tag{14.29}$$

normalized so that $|\Psi|^2 = n_s$ is the density of condensed pairs. The phase φ is a classical field: because n_s is of order $10^{28}\,\mathrm{m}^{-3}$, its quantum fluctuations are negligible in bulk material, and it behaves like a continuum coordinate with a definite value at each point.

A charged particle in a vector potential $\mathbf{A}$ has velocity operator $(\hat{\mathbf{p}} - q^*\mathbf{A})/m^*$. The electric current density of the condensate is therefore

$$\mathbf{J}_s = \frac{q^*}{m^*}\,\mathrm{Re}\big[\Psi^*(-i\hbar\nabla - q^*\mathbf{A})\Psi\big] = \frac{q^* n_s}{m^*}\big(\hbar\nabla\varphi - q^*\mathbf{A}\big), \tag{14.30}$$

where in the second step we assumed n_s uniform, so that $\Psi^*(-i\hbar\nabla)\Psi = \hbar n_s\nabla\varphi - \frac{i\hbar}{2}\nabla n_s$ has real part $\hbar n_s\nabla\varphi$. Both φ and $\mathbf{A}$ change under a gauge transformation $\mathbf{A} \to \mathbf{A} + \nabla\chi$, $\varphi \to \varphi + q^*\chi/\hbar$, and the combination $\hbar\nabla\varphi - q^*\mathbf{A}$ is gauge invariant, as a measurable current must be.

14.3.2 The London Equations

Take the curl of Eq. (14.30). In a region without vortices $\nabla \times \nabla\varphi = 0$, so

$$\nabla \times \mathbf{J}_s = -\frac{n_s q^{*2}}{m^*}\,\mathbf{B}. \tag{14.31}$$

This is the second London equation. Combine it with Ampère's law $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_s$. Taking the curl of Ampère's law and using $\nabla \times (\nabla \times \mathbf{B}) = \nabla(\nabla \cdot \mathbf{B}) - \Delta\mathbf{B} = -\Delta\mathbf{B}$,

$$\Delta\mathbf{B} = \frac{\mu_0 n_s q^{*2}}{m^*}\,\mathbf{B} := \frac{1}{\lambda_L^2}\,\mathbf{B}, \qquad \lambda_L = \sqrt{\frac{m^*}{\mu_0 n_s q^{*2}}}. \tag{14.32}$$

A field applied parallel to a flat surface decays inside as e^{-z/λ_L}. This is the Meissner effect, and λ_L, the London penetration depth, is several tens of nanometers in elemental superconductors such as

niobium and aluminum. Since $q^{*2}/m^* = 2q_e^2/m_e$ and n_s counts pairs, $n_s q^{*2}/m^* = n_e q_e^2/m_e$ with $n_e = 2n_s$ the density of paired electrons: pairing does not change λ_L.

For the first London equation we need the time evolution of φ. The phase of a stationary state of energy $\mathcal{E}$ advances as $e^{-i\mathcal{E}t/\hbar}$. For the condensate the energy per pair is its electrochemical potential; keeping only the electrostatic part $q^*\phi_{\mathrm{el}}$,

$$\hbar\,\partial_t\varphi = -q^*\phi_{\mathrm{el}}. \tag{14.33}$$

Differentiating Eq. (14.30) in time and using $\nabla\partial_t\varphi = -q^*\nabla\phi_{\mathrm{el}}/\hbar$,

$$\partial_t\mathbf{J}_s = \frac{n_s q^*}{m^*}\left(-q^*\nabla\phi_{\mathrm{el}} - q^*\partial_t\mathbf{A}\right) = \frac{n_s q^{*2}}{m^*}\,\mathbf{E}. \tag{14.34}$$

An electric field accelerates the supercurrent without friction. A steady supercurrent therefore requires $\mathbf{E} = 0$: there is no voltage drop, and no dissipation.

14.3.3 Fluxoid Quantization

Consider a superconducting ring and a closed contour $\mathcal{C}$ running through its interior. The order parameter is single-valued, so its phase can change only by a multiple of 2π around $\mathcal{C}$:

$$\oint_{\mathcal{C}} \nabla\varphi\cdot\mathrm{d}\mathbf{l} = 2\pi n, \qquad n\in\mathbb{Z}. \tag{14.35}$$

Solve Eq. (14.30) for $\hbar\nabla\varphi = \frac{m^*}{n_s q^*}\mathbf{J}_s + q^*\mathbf{A}$ and integrate around $\mathcal{C}$. Stokes' theorem turns $\oint\mathbf{A}\cdot\mathrm{d}\mathbf{l}$ into the enclosed flux Φ, and we obtain

$$\Phi + \frac{m^*}{n_s q^{*2}}\oint_{\mathcal{C}}\mathbf{J}_s\cdot\mathrm{d}\mathbf{l} = n\,\frac{2\pi\hbar}{q^*}. \tag{14.36}$$

The left side is the **fluxoid**. If the ring is thicker than a few λ_L, we can choose $\mathcal{C}$ deep inside the metal, where $\mathbf{J}_s = 0$ by Eq. (14.32), and the enclosed flux itself is quantized in units of

$$\Phi_0 = \frac{h}{|q^*|} = \frac{h}{2q_e} \approx 2.068\times10^{-15}\,\mathrm{Wb}. \tag{14.37}$$

With the exact SI values $h = 6.62607015\times10^{-34}\,\mathrm{J\,s}$ and $q_e = 1.602176634\times10^{-19}\,\mathrm{C}$, $\Phi_0 = 2.067834\times10^{-15}\,\mathrm{Wb}$. The factor 2 in the denominator, observed in the first flux-quantization experiments in 1961, is direct evidence that the charge carriers are pairs [109]. We will use $\Phi_0/2\pi = \hbar/(2q_e) \approx 3.291\times10^{-16}\,\mathrm{Wb}$ constantly; it converts phase to flux.

Notation Watch: Signs of charge and phase

The pair charge is $q^* = -2q_e < 0$. Most texts, including Tinkham [109], absorb this sign by a convention for the sign of the phase difference so that the Josephson relations read $I = I_{\mathrm{c}}\sin\varphi$ and $\dot\varphi = +2\pi V/\Phi_0$ with $I_{\mathrm{c}} > 0$. We do the same from Proposition 14.13 on. In this chapter φ is a superconducting phase, ϕ_{el} is an electrostatic potential, Φ is a magnetic flux, and Φ_0 is the flux quantum. The symbol β with a subscript (β_{c}, β_L) is a dimensionless circuit parameter; the bare β remains $1/k_{\mathrm{B}}T$.

14.3.4 The Josephson Relations

A Josephson junction is two superconductors separated by a barrier thin enough (about a nanometer of oxide in a tunnel junction) that pairs can tunnel through it. The two condensates then have separate phases φ_1 and φ_2 but are weakly coupled. The rigorous derivation of the junction current uses the BCS tunneling Hamiltonian and second-order perturbation theory in the tunneling amplitude; it yields both the current–phase relation and the magnitude of the critical current [109]. Feynman's two-state model gives the same structure in a few lines and makes the origin of each relation transparent.

Heuristic Derivation: Feynman's two-state derivation of the Josephson relations

Treat each side as a single quantum state with amplitude $\psi_k = \sqrt{\rho_k}\, e^{i\theta_k}$, $k = 1, 2$, where ρ_k is the pair density. A voltage V across the junction makes the pair energies differ: $U_1 = q^* V_1$, $U_2 = q^* V_2$, $V = V_1 - V_2$. Tunneling couples the two amplitudes with a real constant K:

$$i\hbar\dot{\psi}_1 = U_1\psi_1 + K\psi_2, \qquad i\hbar\dot{\psi}_2 = U_2\psi_2 + K\psi_1. \tag{14.38}$$

Insert $\psi_1 = \sqrt{\rho_1}e^{i\theta_1}$. The left side is $i\hbar\dot{\psi}_1 = \big[\frac{i\hbar\dot{\rho}_1}{2\sqrt{\rho_1}} - \hbar\sqrt{\rho_1}\dot{\theta}_1\big]e^{i\theta_1}$. Divide both sides of the first equation by $e^{i\theta_1}$ and write $\delta = \theta_2 - \theta_1$:

$$\frac{i\hbar\dot{\rho}_1}{2\sqrt{\rho_1}} - \hbar\sqrt{\rho_1}\,\dot{\theta}_1 = U_1\sqrt{\rho_1} + K\sqrt{\rho_2}\,(\cos\delta + i\sin\delta).$$

The real and imaginary parts give

$$\dot{\rho}_1 = +\frac{2K}{\hbar}\sqrt{\rho_1\rho_2}\,\sin\delta, \qquad \dot{\theta}_1 = -\frac{U_1}{\hbar} - \frac{K}{\hbar}\sqrt{\frac{\rho_2}{\rho_1}}\cos\delta. \tag{14.39}$$

The second equation of Eq. (14.38) gives the same with $1 \leftrightarrow 2$ and $\delta \to -\delta$:

$$\dot{\rho}_2 = -\frac{2K}{\hbar}\sqrt{\rho_1\rho_2}\,\sin\delta, \qquad \dot{\theta}_2 = -\frac{U_2}{\hbar} - \frac{K}{\hbar}\sqrt{\frac{\rho_1}{\rho_2}}\cos\delta. \tag{14.40}$$

The densities change in opposite directions: pairs flow from one side to the other at the rate $\dot{\rho}_1 = -\dot{\rho}_2 \propto \sin\delta$. In a real circuit the leads connected to each side replenish the pairs, so ρ_1 and ρ_2 stay fixed at a common value ρ_0 while the pair current flows. The current through the junction is therefore $q^*\dot{\rho}_1 \times$ (area), proportional to $\sin\delta$, with amplitude proportional to $K\rho_0$. Subtracting the phase equations with $\rho_1 = \rho_2$, the $\cos\delta$ terms cancel and

$$\dot{\delta} = \dot{\theta}_2 - \dot{\theta}_1 = \frac{U_1 - U_2}{\hbar} = \frac{q^* V}{\hbar} = -\frac{2q_e V}{\hbar}.$$

Defining $\varphi := \theta_1 - \theta_2 = -\delta$ and absorbing the sign of K into $I_c > 0$ gives the two relations of Proposition 14.13. The model is heuristic: it treats each bulk superconductor as a single level and does not give the value of I_c.

Proposition 14.13 (DC and AC Josephson relations)**.** *Let* $\varphi = \varphi_1 - \varphi_2 - \frac{2\pi}{\Phi_0}\int_2^1 \mathbf{A}\cdot\mathrm{d}\mathbf{l}$ *be the gauge-invariant phase difference across a junction. Then the pair current through the junction and*

the voltage across it satisfy

$$I_s = I_c \sin\varphi, \qquad V = \frac{\Phi_0}{2\pi}\,\dot{\varphi} = \frac{\hbar}{2q_e}\,\dot{\varphi}. \tag{14.41}$$

The first relation, the DC Josephson effect, says that a current up to I_c flows with zero voltage, carried by a static phase difference. The second, the AC Josephson effect, says that a constant voltage V makes the phase run at $\dot{\varphi} = 2\pi V/\Phi_0$, so the current oscillates at the frequency

$$f_J = \frac{V}{\Phi_0} \approx 483.6\,\text{GHz per mV}. \tag{14.42}$$

This conversion between voltage and frequency is exact enough to serve as the international voltage standard.

For a tunnel junction between identical BCS superconductors with gap $\Delta(T)$ and normal-state resistance R_n, the tunneling-Hamiltonian calculation gives the Ambegaokar–Baratoff result [109]

$$I_c R_n = \frac{\pi\Delta(T)}{2q_e}\tanh\frac{\Delta(T)}{2k_BT}. \tag{14.43}$$

For niobium, $\Delta \approx 1.4$–$1.5\,\text{meV}$ well below $T_c \approx 9.2\,\text{K}$, so $I_cR_n \approx 2\,\text{mV}$.

14.3.5 The Josephson Energy

The junction stores energy in its phase. If the phase is changed from 0 to φ while a current $I_s = I_c\sin\varphi'$ flows, the source does work

$$\int I_s V\,\mathrm{d}t = \int_0^{\varphi} I_c\sin\varphi'\,\frac{\Phi_0}{2\pi}\,\mathrm{d}\varphi' = E_J(1-\cos\varphi), \qquad E_J := \frac{\Phi_0 I_c}{2\pi} = \frac{\hbar I_c}{2q_e}. \tag{14.44}$$

The junction is a nonlinear inductor with potential energy $-E_J\cos\varphi$ (up to a constant). For small φ, $E_J(1-\cos\varphi) \approx \frac{1}{2}E_J\varphi^2 = \frac{1}{2}L_JI_s^2$ with the Josephson inductance $L_J = \Phi_0/(2\pi I_c)$. Numerically $E_J/k_B = \frac{\Phi_0}{2\pi k_B}I_c \approx 23.8\,\text{K}\times(I_c/1\,\mu\text{A})$. A junction with $I_c = 20\,\mu\text{A}$ has $E_J/k_B \approx 480\,\text{K}$, about $110\,k_BT$ at $4.2\,\text{K}$. The bare Josephson energy is far too large for a p-bit, which needs barriers of a few k_BT. The next section shows how a superconducting loop cancels most of it.

Physics Insight: Why a superconducting circuit is a clean Langevin system

In a CMOS p-bit the noise comes from currents that are also the source of dissipation, and the circuit runs far from equilibrium (Section 13.3). In a superconducting circuit the static currents that set the potential are persistent supercurrents and dissipate nothing. The only dissipation is through resistive channels: a deliberately added shunt resistor, and the quasiparticles that are thermally excited across the gap, whose density scales as $e^{-\Delta/k_BT}$ (at $4.2\,\text{K}$ in niobium, $\Delta/k_BT \approx 4$). These channels are in equilibrium with the bath, so their noise obeys the fluctuation–dissipation theorem exactly (Theorem 11.5, Section 5.3). A superconducting double well at fixed bias therefore samples a true Boltzmann distribution of its phase coordinate. This is the situation of Chapter 7 without approximation, which is why Josephson circuits are among the cleanest systems in which Kramers escape and its quantum counterpart have been tested quantitatively.

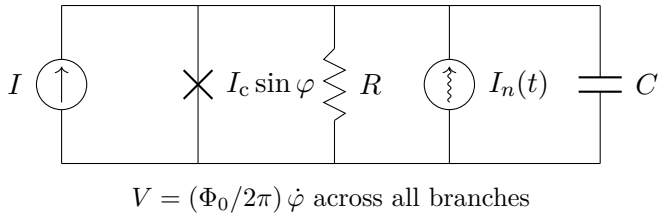


Figure 14.2: The RCSJ model. A bias current I divides among the supercurrent channel $I_c \sin\varphi$, the shunt resistor R with its Johnson noise current $I_n(t)$, and the junction capacitance C.

14.4 The RCSJ Model, the Tilted Washboard, and the Quantum Flux Parametron

A real junction is not only a supercurrent channel. It has a geometric capacitance C between its electrodes, and it conducts a dissipative current through quasiparticles or through a deliberately added shunt resistor R. The resistively and capacitively shunted junction (RCSJ) model of Stewart [104] and McCumber [73] puts these three channels in parallel. We derive its Langevin equation, identify it as a particle in a tilted washboard, and then close the junction into a superconducting loop to obtain the double well of the quantum flux parametron.

14.4.1 The RCSJ Langevin Equation

All branches of Fig. 14.2 share the voltage $V = (\Phi_0/2\pi)\dot\varphi$. Kirchhoff's current law reads

$$I = I_c \sin\varphi + \frac{V}{R} + C\dot V + I_n(t), \qquad \langle I_n(t) I_n(t')\rangle = \frac{2k_B T}{R}\,\delta(t-t'), \tag{14.45}$$

where the noise intensity is the two-sided form of the Johnson–Nyquist density $S_I = 4k_B T/R$. Substitute $V = (\Phi_0/2\pi)\dot\varphi$ and multiply through by $\Phi_0/2\pi$:

$$m_\varphi \ddot\varphi + \gamma_\varphi \dot\varphi + \frac{\partial U}{\partial \varphi} = \xi(t), \tag{14.46}$$

with

$$m_\varphi = C\Big(\frac{\Phi_0}{2\pi}\Big)^2, \qquad \gamma_\varphi = \frac{1}{R}\Big(\frac{\Phi_0}{2\pi}\Big)^2, \qquad U(\varphi) = -E_J\cos\varphi - \frac{\Phi_0 I}{2\pi}\,\varphi, \qquad \xi = -\frac{\Phi_0}{2\pi} I_n. \tag{14.47}$$

The noise force has $\langle \xi(t)\xi(t')\rangle = (\Phi_0/2\pi)^2 (2k_B T/R)\delta(t-t') = 2\gamma_\varphi k_B T\,\delta(t-t')$. Equation (14.46) is therefore exactly the underdamped Langevin equation of Chapters 5 and 7, $m\ddot x = -U'(x) - \gamma \dot x + \sqrt{2\gamma k_B T}\,\eta(t)$, for a particle of mass m_φ at position φ. The capacitance is the mass, the conductance $1/R$ is the friction, and the fluctuation–dissipation relation holds by construction because the noise and the friction come from the same resistor. The potential $U(\varphi)$ is the **tilted washboard**: a cosine corrugation of depth $2E_J$ on a linear slope set by the bias current.

Definition 14.14 (Plasma frequency and Stewart–McCumber parameter)**.** The zero-bias **plasma frequency** of the junction and the **Stewart–McCumber parameter** are

$$\omega_p = \sqrt{\frac{E_J}{m_\varphi}} = \sqrt{\frac{2\pi I_c}{\Phi_0 C}}, \qquad \beta_c = (\omega_p R C)^2 = \frac{2\pi}{\Phi_0} I_c R^2 C. \tag{14.48}$$

To see where these come from, rescale time by $\tau = \omega_p t$ and divide Eq. (14.46) by E_{J}. Using $m_\varphi \omega_p^2 / E_{\mathrm{J}} = 1$ and $\gamma_\varphi \omega_p / E_{\mathrm{J}} = \gamma_\varphi / (m_\varphi \omega_p) = 1/(\omega_p R C)$,

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}\tau^2} + \frac{1}{Q}\frac{\mathrm{d}\varphi}{\mathrm{d}\tau} + \sin\varphi = i + i_n(\tau), \qquad Q = \omega_p RC = \sqrt{\beta_{\mathrm{c}}}, \quad i = \frac{I}{I_{\mathrm{c}}}. \tag{14.49}$$

The ratio $\omega_p^2 = E_{\mathrm{J}}/m_\varphi$ is the small-oscillation frequency at the bottom of an untilted well, and Q is the quality factor of that oscillation. The identity $Q^2 = \omega_p^2 R^2 C^2 = \frac{2\pi I_{\mathrm{c}}}{\Phi_0 C} R^2 C^2 = \frac{2\pi}{\Phi_0} I_{\mathrm{c}} R^2 C$ confirms the formula for β_{c}. For $\beta_{\mathrm{c}} \ll 1$ the junction is overdamped; for $\beta_{\mathrm{c}} \gg 1$ it is underdamped.

Example 14.15 (Overdamped running state)**.** For $\beta_{\mathrm{c}} \ll 1$ and negligible noise, drop the inertial term: $\frac{\Phi_0}{2\pi R}\dot\varphi = I - I_{\mathrm{c}}\sin\varphi$. For $I < I_{\mathrm{c}}$ the phase relaxes to a fixed point with $\sin\varphi = I/I_{\mathrm{c}}$ and $V = 0$. For $I > I_{\mathrm{c}}$ the phase runs, and its period is

$$T_\varphi = \frac{\Phi_0}{2\pi R}\int_0^{2\pi} \frac{\mathrm{d}\varphi}{I - I_{\mathrm{c}}\sin\varphi} = \frac{\Phi_0}{2\pi R}\cdot\frac{2\pi}{\sqrt{I^2 - I_{\mathrm{c}}^2}},$$

using $\int_0^{2\pi} \mathrm{d}\varphi/(a - b\sin\varphi) = 2\pi/\sqrt{a^2 - b^2}$ for $a > |b|$. Each period advances the phase by 2π, so by Eq. (14.41) the time-averaged voltage is $\langle V\rangle = \Phi_0/T_\varphi = R\sqrt{I^2 - I_{\mathrm{c}}^2}$. The current–voltage curve is single-valued and approaches Ohm's law for $I \gg I_{\mathrm{c}}$.

Example 14.16 (Underdamped hysteresis and the retrapping current)**.** For $\beta_{\mathrm{c}} \gg 1$ a running phase keeps going even when the bias is lowered below I_{c}, because its inertia carries it over the corrugation. It stops (is retrapped) at the current I_r where the energy supplied by the bias per period equals the energy dissipated. Near retrapping the friction is weak, so we evaluate the dissipation along the frictionless trajectory that just reaches the top of each corrugation with zero speed. Energy conservation in the untilted potential gives $\tfrac{1}{2} m_\varphi \dot\varphi^2 = E_{\mathrm{J}}(1 + \cos\varphi) = 2E_{\mathrm{J}}\cos^2(\varphi/2)$, so $\dot\varphi = 2\omega_p\,|\cos(\varphi/2)|$. The energy dissipated per period is

$$\int_{-\pi}^{\pi} \gamma_\varphi \dot\varphi\, \mathrm{d}\varphi = 2\gamma_\varphi\omega_p \int_{-\pi}^{\pi} \cos\frac{\varphi}{2}\, \mathrm{d}\varphi = 8\gamma_\varphi\omega_p,$$

and the bias supplies $\frac{\Phi_0 I_r}{2\pi}\cdot 2\pi = \Phi_0 I_r$. Setting these equal and using $\gamma_\varphi = (\Phi_0/2\pi)^2/R$ and $\Phi_0/I_{\mathrm{c}} = 2\pi/(\omega_p^2 C)$,

$$\frac{I_r}{I_{\mathrm{c}}} = \frac{8\gamma_\varphi\omega_p}{\Phi_0 I_{\mathrm{c}}} = \frac{2\omega_p\Phi_0}{\pi^2 R I_{\mathrm{c}}} = \frac{4}{\pi\,\omega_p RC} = \frac{4}{\pi\sqrt{\beta_{\mathrm{c}}}}. \tag{14.50}$$

The current–voltage curve is hysteretic between I_r and I_{c}. Superconducting logic families such as RSFQ shunt each junction so that $\beta_{\mathrm{c}} \approx 1$, which removes the hysteresis while keeping the switching fast [66].

14.4.2 Escape from the Washboard Near the Critical Current

For $0 < i < 1$ the washboard has metastable minima. A phase trapped in one of them is the zero-voltage state; escape over the barrier releases the phase into the running state, which appears as a jump to a finite voltage. Near $i \to 1$ the barrier is small and the escape rate is measurable.

Write $\varphi = \frac{\pi}{2} + y$ and $\varepsilon = 1 - i \ll 1$. Using $\cos(\frac{\pi}{2} + y) = -\sin y = -y + \frac{y^3}{6} - \dots$,

$$\frac{U}{E_{\mathrm{J}}} = -\cos\varphi - i\varphi = \text{const} + \varepsilon y - \frac{y^3}{6} + \mathcal{O}(y^5). \tag{14.51}$$

The stationary points satisfy $\varepsilon = y^2/2$, so $y_\pm = \pm\sqrt{2\varepsilon}$. The second derivative is $-y$ in units of E_J, so $y_- = -\sqrt{2\varepsilon}$ is the minimum and $y_+ = +\sqrt{2\varepsilon}$ is the barrier top, with curvatures of equal magnitude $E_\mathrm{J}\sqrt{2\varepsilon}$. The barrier height is

$$\frac{\Delta U}{E_\mathrm{J}} = \Big[\varepsilon y - \frac{y^3}{6}\Big]_{y_-}^{y_+} = 2\varepsilon\sqrt{2\varepsilon} - \frac{2(2\varepsilon)^{3/2}}{6} = \frac{2}{3}(2\varepsilon)^{3/2}, \qquad \Delta U = \frac{4\sqrt{2}}{3}\,E_\mathrm{J}\,(1-i)^{3/2}. \tag{14.52}$$

The exact barrier for any i is $\Delta U = 2E_\mathrm{J}\big[\sqrt{1-i^2} - i\arccos i\big]$, which reduces to Eq. (14.52) as $i \to 1$. The oscillation frequency at the minimum and the magnitude of the imaginary frequency at the barrier top are equal for the cubic:

$$\omega_0(i) = \omega_b(i) = \sqrt{\frac{E_\mathrm{J}\sqrt{2\varepsilon}}{m_\varphi}} = \omega_p(2\varepsilon)^{1/4} \approx \omega_p(1-i^2)^{1/4}. \tag{14.53}$$

The exact small-oscillation frequency, $\omega_0 = \omega_p(1-i^2)^{1/4}$, follows from $U''(\varphi_\mathrm{min}) = E_\mathrm{J}\cos\varphi_\mathrm{min} = E_\mathrm{J}\sqrt{1-i^2}$.

In the classical regime the escape rate is Kramers' moderate-to-high damping formula Eq. (7.53) with the washboard parameters,

$$\Gamma_\mathrm{th} = \frac{\omega_0}{2\pi}\Big(\sqrt{1+\alpha^2} - \alpha\Big)\exp\Big(-\frac{\Delta U}{k_\mathrm{B}T}\Big), \qquad \alpha = \frac{\gamma_\varphi}{2m_\varphi\omega_b} = \frac{1}{2\omega_b RC}, \tag{14.54}$$

valid for $\Delta U \gg k_\mathrm{B}T$ and for damping not so weak that energy diffusion limits the rate [46]. A current-biased junction is a one-shot stochastic switch: ramping I and recording the current at which the junction switches gives a switching-current distribution whose width measures $k_\mathrm{B}T/E_\mathrm{J}$. It is not yet a p-bit, because the running state is not a second well. For that we close the junction into a loop.

14.4.3 The rf-SQUID Potential

Connect the two electrodes of a junction by a superconducting wire of inductance L, forming a loop, and thread the loop with an external flux Φ_x. This is an rf-SQUID. Apply fluxoid quantization Eq. (14.36) to a contour running through the wire and across the junction. In the wire the contour can be taken deep in the metal, so the only phase drop not accounted for by the enclosed flux is the drop φ across the junction. With the sign conventions of Proposition 14.13, this gives $\varphi = 2\pi\Phi/\Phi_0$ modulo 2π, where Φ is the total flux in the loop. The total flux is the external flux minus the self-flux of the circulating current, $\Phi = \Phi_x - LI_s$.

The energy of the loop is the magnetic energy of the inductor plus the Josephson energy:

$$U(\Phi) = \frac{(\Phi - \Phi_x)^2}{2L} - E_\mathrm{J}\cos\Big(\frac{2\pi\Phi}{\Phi_0}\Big). \tag{14.55}$$

Its stationary condition $(\Phi - \Phi_x)/L = -I_\mathrm{c}\sin(2\pi\Phi/\Phi_0) = -I_s$ reproduces $\Phi = \Phi_x - LI_s$, which confirms the form. In terms of the phase, with $\varphi_x = 2\pi\Phi_x/\Phi_0$,

$$U(\varphi) = \frac{E_L}{2}(\varphi - \varphi_x)^2 - E_\mathrm{J}\cos\varphi, \qquad E_L = \Big(\frac{\Phi_0}{2\pi}\Big)^2\frac{1}{L}, \qquad \beta_L := \frac{E_\mathrm{J}}{E_L} = \frac{2\pi LI_\mathrm{c}}{\Phi_0}. \tag{14.56}$$

The dynamics of φ is again Eq. (14.46), now with this potential, the same mass m_φ, friction γ_φ, and noise.

Proposition 14.17 (Monostable versus bistable rf-SQUID)**.** *The stationary points of Eq. (14.56) satisfy $\varphi_x = \varphi + \beta_L \sin\varphi$. If $\beta_L < 1$, U has a single minimum for every φ_x. If $\beta_L > 1$, there is an interval of φ_x around π (mod 2π) for which U has two minima separated by a barrier; at $\varphi_x = \pi$ the two minima are degenerate.*

Proof. $U'(\varphi) = E_L(\varphi - \varphi_x) + E_\mathrm{J}\sin\varphi = 0$ gives the stationary condition after dividing by E_L. The map $g(\varphi) = \varphi + \beta_L \sin\varphi$ has $g'(\varphi) = 1 + \beta_L\cos\varphi$. For $\beta_L < 1$, $g' > 0$ everywhere, so g is strictly increasing and $g(\varphi) = \varphi_x$ has exactly one solution; since $U'' = E_L g' > 0$ there, it is a minimum. For $\beta_L > 1$, $g' < 0$ on the interval where $\cos\varphi < -1/\beta_L$, which contains $\varphi = \pi$. There g decreases, so for φ_x in the range of g over that interval, $g(\varphi) = \varphi_x$ has three solutions: two with $g' > 0$ (minima) and one with $g' < 0$ (a maximum, since $U'' = E_L g'$). At $\varphi_x = \pi$, the substitution $\varphi = \pi + \vartheta$ gives $U = \frac{E_L}{2}\vartheta^2 + E_\mathrm{J}\cos\vartheta + \text{const}$, which is even in ϑ, so the minima are degenerate. □

14.4.4 Reduction to the Quartic Double Well

Expand around the symmetric point. Let $\varphi = \pi + \vartheta$ and $\varphi_x = \pi + \vartheta_x$, with ϑ_x small. Since $\cos(\pi + \vartheta) = -\cos\vartheta$,

$$U = \frac{E_L}{2}(\vartheta - \vartheta_x)^2 + E_\mathrm{J}\cos\vartheta = \frac{E_L}{2}\vartheta^2 - E_L\vartheta_x\vartheta + E_\mathrm{J}\Big(1 - \frac{\vartheta^2}{2} + \frac{\vartheta^4}{24} - \frac{\vartheta^6}{720} + \dots\Big) + \frac{E_L}{2}\vartheta_x^2.$$

Collecting powers of ϑ and dropping constants,

$$U(\vartheta) = \frac{a}{4}\vartheta^4 - \frac{b}{2}\vartheta^2 - h\vartheta + \mathcal{O}(\vartheta^6), \qquad a = \frac{E_\mathrm{J}}{6} = \frac{E_L\beta_L}{6}, \quad b = E_\mathrm{J} - E_L = E_L(\beta_L - 1), \quad h = E_L\vartheta_x. \tag{14.57}$$

This is the double-well potential of the earlier chapters, with the phase ϑ as the coordinate. Because ϑ is dimensionless, a, b, and h all carry units of energy. The control parameters map as follows:

- the loop parameter β_L sets the sign and size of b: the double well exists for $\beta_L > 1$, as in Proposition 14.17;
- the flux offset from half a flux quantum sets the tilt, $h = E_L\vartheta_x = \frac{\Phi_0}{2\pi L}(\Phi_x - \frac{1}{2}\Phi_0)$;
- the barrier and the well positions are

$$\Delta E = \frac{b^2}{4a} = \frac{3}{2}E_L\,\frac{(\beta_L - 1)^2}{\beta_L}, \qquad \vartheta_0 = \sqrt{\frac{b}{a}} = \sqrt{\frac{6(\beta_L - 1)}{\beta_L}}. \tag{14.58}$$

The quartic truncation is accurate when $\vartheta_0 \lesssim 1$, that is, for $\beta_L - 1 \lesssim 0.2$. Beyond that the exact minima solve $\vartheta_m = \beta_L\sin\vartheta_m$, and the exact barrier is

$$\Delta E_\mathrm{exact} = E_\mathrm{J}(1 - \cos\vartheta_m) - \frac{E_L}{2}\vartheta_m^2 = E_L\Big[\beta_L(1 - \cos\vartheta_m) - \frac{\vartheta_m^2}{2}\Big]. \tag{14.59}$$

The neglected sixth-order term $-E_\mathrm{J}\vartheta^6/720$ is negative, so the quartic slightly underestimates the depth of the wells. For example, a barrier of $2k_\mathrm{B}T$ with $E_L = 93k_\mathrm{B}T$ requires $\beta_L = 1.127$ in the quartic approximation and $\beta_L = 1.125$ exactly, with well positions $\vartheta_0 = 0.82$ and $\vartheta_m = 0.83$.

The dynamical parameters follow from the same expansion. The curvature at the barrier top is $U''(0) = -b = -E_L(\beta_L - 1)$, and at the minima $U''(\pm\vartheta_0) = 2b$ in the quartic approximation, or $E_L(1 - \beta_L\cos\vartheta_m)$ exactly. Dividing by $m_\varphi = C(\Phi_0/2\pi)^2$ and using $E_L/m_\varphi = 1/(LC)$,

$$\omega_b^2 = \frac{\beta_L - 1}{LC}, \qquad \omega_\mathrm{w}^2 = \frac{1 - \beta_L\cos\vartheta_m}{LC} \;\approx\; 2\omega_b^2. \tag{14.60}$$

Unlike the cubic washboard, where $\omega_0 = \omega_b$, the symmetric quartic has $\omega_\mathrm{w} \approx \sqrt{2}\,\omega_b$. Both vanish as $\beta_L \to 1$: the parametron, like the CMOS latch, slows down critically at its bifurcation.

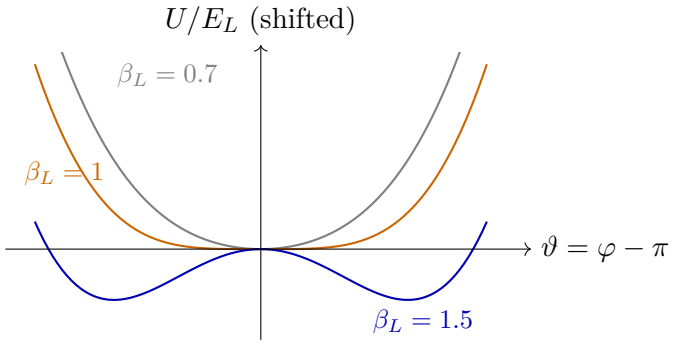


Figure 14.3: The rf-SQUID potential Eq. (14.56) at half a flux quantum of external flux, $\varphi_x = \pi$, for three values of β_L. Below $\beta_L = 1$ the potential has a single well; at $\beta_L = 1$ the curvature at $\vartheta = 0$ vanishes and the potential is purely quartic near the origin; above $\beta_L = 1$ it is a symmetric double well whose barrier grows as $\frac{3}{2}E_L(\beta_L - 1)^2/\beta_L$ near the bifurcation.

14.4.5 The Quantum Flux Parametron: Electrical Control of the Barrier

In a fixed rf-SQUID, β_L is set at fabrication. To modulate the barrier in operation, replace the single junction by two junctions in a small loop (a dc-SQUID) threaded by a control flux Φ_q. If each junction has Josephson energy E_{J0} and the small loop's inductance is negligible, fluxoid quantization around the small loop fixes $\varphi_1 - \varphi_2 = 2\pi\Phi_q/\Phi_0$, and with $\varphi = \frac{1}{2}(\varphi_1 + \varphi_2)$,

$$-E_{J0}\cos\varphi_1 - E_{J0}\cos\varphi_2 = -2E_{J0}\cos\!\left(\frac{\pi\Phi_q}{\Phi_0}\right)\cos\varphi, \tag{14.61}$$

by $\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$. The pair behaves as one junction with tunable Josephson energy $E_J(\Phi_q) = 2E_{J0}\,|\cos(\pi\Phi_q/\Phi_0)|$, and therefore

$$\beta_L(\Phi_q) = \beta_{L,\max}\left|\cos\frac{\pi\Phi_q}{\Phi_0}\right|, \qquad \beta_{L,\max} = \frac{2\pi L(2I_{c0})}{\Phi_0}. \tag{14.62}$$

Sweeping Φ_q from $\Phi_0/2$ toward 0 takes β_L from 0 through 1 to $\beta_{L,\max}$: the potential of Fig. 14.3 changes from a single well into a double well whose barrier grows with the control flux. This is the operating principle of the **quantum flux parametron** (QFP). The word "quantum" refers to the flux quantum, not to quantum coherence: a QFP at 4.2 K is a classical Langevin system, as Section 14.5 will confirm. A parametron in the strict sense is an oscillator whose parameter is pumped; the QFP is pumped by the control flux, called the excitation or clock. In logic applications the clock is ramped slowly enough to be adiabatic, which gives switching energies far below those of resistively shunted RSFQ gates [66].

Couplings between parametrons. In the double-well state the loop carries a circulating current whose sign is the bit. With $\vartheta = \pm\vartheta_m$, the current is $I_q s$ with $I_q = \frac{\Phi_0}{2\pi L}\vartheta_m$ (from $\Phi - \Phi_x = -LI_s$ and $\vartheta \approx 2\pi(\Phi - \Phi_x)/\Phi_0$ at $\vartheta_x = 0$). Let QFP j couple to QFP i through a mutual inductance M_{ij}. The flux it adds to loop i is $M_{ij}I_q s_j$, which shifts $\vartheta_{x,i}$ by $2\pi M_{ij}I_q s_j/\Phi_0$ and hence the tilt by $h_i = E_L \cdot 2\pi M_{ij}I_q s_j/\Phi_0 = \frac{\Phi_0}{2\pi L}M_{ij}I_q s_j$. By Proposition 14.5, the effective field on the p-bit is $h_{\mathrm{eff},i} \approx \vartheta_m h_i$, so

$$h_{\mathrm{eff},i} = \sum_j J_{ij}s_j, \qquad J_{ij} = \vartheta_m\frac{\Phi_0}{2\pi L}M_{ij}I_q = M_{ij}I_q^2. \tag{14.63}$$

The coupling is the magnetic interaction energy MI_iI_j of two current loops, up to a sign fixed by the winding of the coupling transformer. By Neumann's formula the mutual inductance is symmetric, $M_{ij} = M_{ji}$. Inductively coupled parametrons therefore satisfy the symmetry condition of Proposition 14.10 by a law of magnetostatics, whereas transistor synapses satisfy it only to the accuracy of device matching. Programmable couplings require tunable couplers, for instance a further rf-SQUID used as a flux transformer with adjustable gain.

Clocked operation and the freeze-out activation. A QFP used as a p-bit is clocked: the control flux ramps $b(t)$ from negative (single well) to positive (double well) while the input tilt h is held fixed, the sign of the circulating current is read, and the clock returns to reset. Two limiting regimes bracket the probability of ending in the + well.

Proposition 14.18 (Activation of a clocked parametron)**.** *Consider overdamped or moderately damped dynamics in the quartic double well Eq. (14.57), with $b(t)$ increasing through zero at a constant input tilt h.*

(a) Fast ramp. *If the ramp of $b(t)$ from 0 to its final value is fast compared with the relaxation time γ_φ/b, the populations cannot re-equilibrate during it. The decision is then made afterwards, while the unstable mode at $\vartheta = 0$ grows, as in Proposition 14.8, and the activation is a probit, $\mathbb{P}[s = +1] = \Phi(h/\sigma_h)$, with σ_h set by the fluctuations present when the curvature changed sign.*

(b) Slow ramp (sudden freeze-out approximation). *If the ramp is slow, the populations follow local equilibrium until the inter-well rate $\lambda(t) = k_\uparrow + k_\downarrow$ falls below the rate at which the barrier changes, $\lambda(t_f) \approx \left|\frac{\mathrm{d}}{\mathrm{d}t} \ln \lambda\right|_{t_f}$, after which they are frozen. The final activation is then*

$$\langle s \rangle = \tanh\big(\beta\, h_{\mathrm{eff}}(t_f)\big) \approx \tanh\big(\beta\, \vartheta_0(t_f)\, h\big), \qquad \vartheta_0(t_f) = \sqrt{b(t_f)/a}. \tag{14.64}$$

Justification. (a) For a fast ramp the nonlinear terms have no time to act before the unstable mode $\dot\vartheta \approx (b/\gamma_\varphi)\vartheta + (h/\gamma_\varphi) +$ noise has amplified the initial deviation; this is the latch computation of Proposition 14.8, with the tilt h playing the role of the input offset, and it gives a Gaussian decision variable. (b) While $\lambda(t)$ is large compared with the rate of change of the potential, the two-state master equation Eq. (7.12) keeps P_+/P_- at its instantaneous equilibrium value $e^{2\beta h_{\mathrm{eff}}(t)}$ by Proposition 14.4. Since $\lambda \propto e^{-\beta \Delta E(t)}$ falls exponentially once $\Delta E \gtrsim k_{\mathrm{B}}T$, the crossover from tracking to frozen populations occupies a short interval around t_f, and the sudden approximation replaces it by a step. After t_f no further transitions occur, so P_+/P_- keeps its value at t_f. Proposition 14.5 with $x_0 = \vartheta_0(t_f)$ gives Eq. (14.64). □

The slow-ramp result has a clear physical content. The p-bit's effective inverse temperature, measured in units of input tilt, is $\beta\vartheta_0(t_f)$. Slower clocks freeze later, at a larger barrier and larger ϑ_0, and produce a sharper, colder activation; faster clocks freeze earlier and produce a broader one. The width of the gray zone in which the output is random, $\Delta h \sim k_{\mathrm{B}}T/\vartheta_0(t_f)$, is therefore set by the physical temperature and the clock rate together. Clocked CMOS latches obey the same logic, which is why the two technologies can be compared on equal terms in Worked computation 14.20.

14.5 Thermal Activation Versus Macroscopic Quantum Tunneling

The phase φ is a collective coordinate of some 10^9 or more Cooper pairs, yet its dynamics in a Josephson circuit is quantum mechanical when the temperature is low enough: φ and the charge

on the junction capacitor are conjugate variables, with $[\varphi, Q] = 2iq_e$ (equivalently, $\frac{\hbar}{2q_e}\varphi$ and Q are a canonical pair). A trapped phase can therefore leave its well by tunneling under the barrier rather than by thermal activation over it. This is macroscopic quantum tunneling (MQT). For a thermodynamic sampler the question is practical: below which temperature does tunneling replace Kramers escape, so that the Boltzmann statistics derived in Chapter 7 no longer describe the device?

14.5.1 Quantum Corrections Above the Crossover

Start in the thermal regime and ask how quantum mechanics modifies the Kramers rate. In transition-state language, the rate is an attempt frequency times the ratio of the partition function at the barrier to that in the well. For a harmonic well of frequency ω_{w}, the classical one-mode partition function $k_{\mathrm{B}}T/(\hbar\omega_{\mathrm{w}})$ is replaced by the quantum one, $[2\sinh(\hbar\beta\omega_{\mathrm{w}}/2)]^{-1}$. At the barrier top the unstable mode has imaginary frequency $i\omega_b$, and the same replacement, analytically continued, turns sinh into sin. The resulting quantum correction to the rate is

$$\Gamma = \Xi\,\Gamma_{\mathrm{cl}}, \qquad \Xi = \frac{\sinh(\hbar\beta\omega_{\mathrm{w}}/2)}{\hbar\beta\omega_{\mathrm{w}}/2}\cdot\frac{\hbar\beta\omega_b/2}{\sin(\hbar\beta\omega_b/2)} \;\approx\; \exp\Big[\frac{(\hbar\beta)^2}{24}\big(\omega_{\mathrm{w}}^2+\omega_b^2\big)\Big], \tag{14.65}$$

where the last form uses $\ln(\sinh x/x) = x^2/6 + \mathcal{O}(x^4)$ and $\ln(x/\sin x) = x^2/6 + \mathcal{O}(x^4)$ with $x = \hbar\beta\omega/2$. This is the undamped form; friction reduces the correction [46]. Both factors exceed one: zero-point motion raises the effective energy in the well, and tunneling through the top of the barrier lowers the effective barrier. The barrier factor diverges when $\hbar\beta\omega_b/2 = \pi$, that is, at $T = \hbar\omega_b/(2\pi k_{\mathrm{B}})$. The divergence signals that the harmonic description of the barrier top has failed and that a different path, through the barrier, now dominates.

14.5.2 The Crossover Temperature with Damping

The damped version comes from the imaginary-time path integral of the Caldeira–Leggett model (Section 5.2). For a coordinate q of mass m coupled to an Ohmic bath with friction coefficient γ, the Euclidean action over imaginary time $\tau \in [0, \hbar\beta]$ is [17]

$$\mathcal{S}_E[q] = \int_0^{\hbar\beta} \mathrm{d}\tau\Big[\frac{m}{2}\dot q^2 + U(q)\Big] + \frac{\gamma}{4\pi}\int_0^{\hbar\beta}\mathrm{d}\tau\int_{-\infty}^{\infty}\mathrm{d}\tau'\,\frac{\big(q(\tau)-q(\tau')\big)^2}{(\tau-\tau')^2}, \tag{14.66}$$

with q extended periodically. The escape rate is governed by the stationary paths of $\mathcal{S}_E$. At high temperature the relevant path is the constant one sitting at the barrier top, $q(\tau) = q_b$, with action $\hbar\beta U(q_b)$; it reproduces the Arrhenius factor $e^{-\beta\Delta U}$.

Expand around it in Matsubara modes, $q(\tau) = q_b + \sum_{n\in\mathbb{Z}} y_n e^{i\nu_n\tau}$ with $\nu_n = 2\pi n/(\hbar\beta)$ and $y_{-n} = y_n^*$. The kinetic term gives $m\nu_n^2$ per mode, the potential gives $U''(q_b) = -m\omega_b^2$, and the bath kernel is diagonal in Matsubara modes with eigenvalue $\gamma\,|\nu_n|$. To second order in the fluctuations, therefore,

$$\mathcal{S}_E = \hbar\beta\,U(q_b) + \frac{\hbar\beta}{2}\sum_{n\in\mathbb{Z}}\Lambda_n\,|y_n|^2 + \mathcal{O}(y^3), \qquad \Lambda_n = m\big(\nu_n^2-\omega_b^2\big) + \gamma\,|\nu_n|\,. \tag{14.67}$$

The $n = 0$ mode has $\Lambda_0 = -m\omega_b^2 < 0$; it is the unstable direction of the classical saddle, as it must be. All other modes are stable at high temperature. As the temperature is lowered, ν_1 decreases, and $\Lambda_{\pm1}$ changes sign when $m\nu_1^2 + \gamma\nu_1 = m\omega_b^2$. The positive root is

$$\nu_1 = \omega_R := \sqrt{\omega_b^2 + \frac{\gamma^2}{4m^2}} - \frac{\gamma}{2m} = \omega_b\big(\sqrt{1+\alpha^2}-\alpha\big), \qquad \alpha = \frac{\gamma}{2m\omega_b}. \tag{14.68}$$

Below this point the constant path is no longer a simple saddle, and a periodic path that oscillates in the inverted barrier (the thermal bounce) takes over. The crossover temperature is $\nu_1 = 2\pi k_B T_{cr}/\hbar = \omega_R$:

$$T_{cr} = \frac{\hbar\omega_R}{2\pi k_B} = \frac{\hbar\omega_b}{2\pi k_B}(\sqrt{1+\alpha^2} - \alpha). \tag{14.69}$$

The frequency ω_R is the same reactive frequency κ_{sad} that appears in Kramers' moderate-damping prefactor Eq. (7.53); friction slows the motion over the barrier top and lowers T_{cr}.

Notation Watch: Which frequency sets T_{cr}

The crossover is often written $T_{cr} = \hbar\omega_p/(2\pi k_B)$ with the plasma frequency. The correct frequency is ω_b, the magnitude of the imaginary frequency at the barrier top, reduced by damping to ω_R. For a current-biased junction near I_c the cubic washboard has $\omega_b = \omega_0(i) = \omega_p(1-i^2)^{1/4}$ by Eq. (14.53), so writing ω_p is correct if one means the bias-dependent small-oscillation frequency, and an overestimate if one means the zero-bias ω_p of Definition 14.14. For the symmetric QFP double well, $\omega_b = \omega_w/\sqrt{2}$ near the bifurcation by Eq. (14.60), and both are much smaller than the junction's ω_p because $\beta_L - 1$ is small. The zero-bias $\hbar\omega_p/(2\pi k_B)$ is a useful upper bound on T_{cr} for any circuit built from a given junction technology.

14.5.3 The Zero-Temperature Bounce for a Cubic Barrier

Below T_{cr}, and most simply at $T = 0$, the rate takes the form $\Gamma_q = A\,e^{-B}$, where B is the action of the bounce in units of $\hbar$. Without damping, the bounce action equals the WKB tunneling exponent through the barrier at the energy of the well bottom:

$$B_0 = \frac{2}{\hbar}\int_{q_{in}}^{q_{out}} \sqrt{2m\,U(q)}\;\mathrm{d}q, \tag{14.70}$$

where U is measured from the well bottom and q_{in}, q_{out} are the zeros of U bracketing the barrier.

Proposition 14.19 (Bounce exponent of the cubic barrier)**.** *Let $U(q) = \frac{27}{4}\Delta U\,(q/q_1)^2(1-q/q_1)$, a cubic potential with a minimum at $q = 0$, a barrier of height ΔU at $q = \frac{2}{3}q_1$, and exit point q_1. Let $\omega_0 = \sqrt{U''(0)/m}$. In the undamped limit and for $\Delta U \gg \hbar\omega_0$,*

$$B_0 = \frac{36}{5}\,\frac{\Delta U}{\hbar\omega_0}. \tag{14.71}$$

Proof. First check the parametrization. At $q = \frac{2}{3}q_1$, $U = \frac{27}{4}\Delta U\cdot\frac{4}{9}\cdot\frac{1}{3} = \Delta U$, and $U' = \frac{27}{4}\frac{\Delta U}{q_1^2}(2q - 3q^2/q_1)$ vanishes there, so this is the barrier top. The curvature at the minimum is $U''(0) = \frac{27}{2}\Delta U/q_1^2 = m\omega_0^2$, so $q_1^2 = \frac{27\Delta U}{2m\omega_0^2}$. Now substitute $u = q/q_1$ in Eq. (14.70):

$$B_0 = \frac{2}{\hbar}\sqrt{2m\cdot\frac{27}{4}\Delta U}\;q_1\int_0^1 u\sqrt{1-u}\;\mathrm{d}u.$$

The integral is a Beta function, $\int_0^1 u(1-u)^{1/2}\,\mathrm{d}u = \mathrm{B}(2,\tfrac{3}{2}) = \frac{\Gamma(2)\Gamma(3/2)}{\Gamma(7/2)} = \frac{\sqrt{\pi}/2}{15\sqrt{\pi}/8} = \frac{4}{15}$. The prefactor is $\sqrt{\frac{27m\Delta U}{2}}\cdot\sqrt{\frac{27\Delta U}{2m\omega_0^2}} = \frac{27\Delta U}{2\omega_0}$. Hence $B_0 = \frac{2}{\hbar}\cdot\frac{27\Delta U}{2\omega_0}\cdot\frac{4}{15} = \frac{108}{15}\frac{\Delta U}{\hbar\omega_0} = \frac{36}{5}\frac{\Delta U}{\hbar\omega_0}$. □

The near-critical washboard Eq. (14.51) is exactly of this form, with ΔU from Eq. (14.52) and $\omega_0 = \omega_p(1-i^2)^{1/4}$. The fluctuation determinant around the bounce supplies the prefactor; for weak damping the standard result is

$$\Gamma_q \approx \frac{\omega_0}{2\pi}\Big(\frac{864\pi\,\Delta U}{\hbar\omega_0}\Big)^{1/2}\exp\Big[-\frac{36}{5}\,\frac{\Delta U}{\hbar\omega_0}(1+c_1\alpha)\Big], \qquad c_1 = \frac{45\,\zeta(3)}{\pi^3} \approx 1.74, \tag{14.72}$$

where $864\pi = 120\pi\cdot\frac{36}{5}$ and $\alpha = 1/(2\omega_0 RC) = 1/(2Q)$. The damping correction $c_1\alpha$ is the leading term of the Caldeira–Leggett result for Ohmic friction [17]: coupling to a dissipative environment always increases the bounce action and suppresses tunneling. For strong damping the exponent grows in proportion to the friction coefficient rather than saturating. These formulas assume $\Delta U \gg \hbar\omega_0$ (several levels in the well), an Ohmic bath, and $T \ll T_{\mathrm{cr}}$.

For the symmetric QFP double well the analogous quantity is the instanton action connecting the two degenerate minima. With $U(\vartheta) - U_{\min} = \frac{a}{4}(\vartheta^2-\vartheta_0^2)^2$,

$$\frac{S_0}{\hbar} = \frac{1}{\hbar}\int_{-\vartheta_0}^{\vartheta_0}\sqrt{2m_\varphi\cdot\frac{a}{4}}\,(\vartheta_0^2-\vartheta^2)\,\mathrm{d}\vartheta = \frac{4}{3}\sqrt{\frac{m_\varphi a}{2}}\,\frac{\vartheta_0^3}{\hbar} = \frac{16}{3}\,\frac{\Delta E}{\hbar\omega_{\mathrm{w}}},$$

where the last step uses $\Delta E = a\vartheta_0^4/4$ and $\omega_{\mathrm{w}} = \sqrt{2b/m_\varphi} = \vartheta_0\sqrt{2a/m_\varphi}$, so that $\frac{\Delta E}{\hbar\omega_{\mathrm{w}}} = \frac{\sqrt{m_\varphi a}\,\vartheta_0^3}{4\sqrt{2}\hbar}$ and $\frac{4}{3}\sqrt{\frac{1}{2}}\cdot 4\sqrt{2} = \frac{16}{3}$. At zero temperature the inter-well tunneling rate carries the factor $e^{-S_0/\hbar}$, to be compared with the thermal factor $e^{-\beta\Delta E}$.

14.5.4 Numbers for Niobium Circuits at Liquid-Helium Temperature

A junction with critical-current density J_c and specific capacitance C_s has $\omega_p = \sqrt{2\pi J_c/(\Phi_0 C_s)}$, independent of area. For a representative low-J_c niobium process, $J_c = 1\,\mathrm{kA/cm^2} = 10^7\,\mathrm{A/m^2}$ and $C_s \approx 50\,\mathrm{fF}/\mu\mathrm{m}^2 = 0.05\,\mathrm{F/m^2}$:

$$\omega_p = \sqrt{\frac{2\pi\times 10^7}{2.068\times 10^{-15}\times 0.05}} \approx 7.8\times 10^{11}\,\mathrm{s^{-1}}, \quad \frac{\omega_p}{2\pi} \approx 124\,\mathrm{GHz}, \quad \frac{\hbar\omega_p}{2\pi k_{\mathrm{B}}} \approx 0.95\,\mathrm{K}.$$

For a high-J_c process, $J_c = 10\,\mathrm{kA/cm^2}$ and $C_s \approx 70\,\mathrm{fF}/\mu\mathrm{m}^2$, the same estimate gives $\omega_p/2\pi \approx 330\,\mathrm{GHz}$ and $\hbar\omega_p/(2\pi k_{\mathrm{B}}) \approx 2.5\,\mathrm{K}$. These are upper bounds. At an operating point the relevant ω_b is smaller, and damping reduces it further to ω_R. For the parametron of Worked computation 14.20, $\omega_b/2\pi \approx 89\,\mathrm{GHz}$ and $\alpha = 0.5$ give $T_{\mathrm{cr}} \approx 0.68\,\mathrm{K}$ without damping and $0.42\,\mathrm{K}$ with it.

At $4.2\,\mathrm{K}$ such circuits are therefore in the thermal regime, with T/T_{cr} between about 2 and 10. Thermal Kramers escape governs the switching, and the phase samples a Boltzmann distribution, with quantum corrections given by Eq. (14.65) at the level of ten to thirty percent. Macroscopic quantum tunneling dominates only in millikelvin dilution-refrigerator operation, which is where it was first observed and where superconducting qubits work. For thermodynamic sampling this is the regime to avoid: below T_{cr} the switching statistics are no longer set by $\beta\Delta E$, and the device stops being a calibrated Boltzmann sampler.

Heuristic Derivation: Sub-attojoule barriers at 4.2 K

At $4.2\,\mathrm{K}$, $k_{\mathrm{B}}T = 5.80\times 10^{-23}\,\mathrm{J} = 0.362\,\mathrm{meV}$, a factor $300/4.2 \approx 71$ below room temperature. A p-bit barrier of $2k_{\mathrm{B}}T$ is then $1.2\times 10^{-22}\,\mathrm{J}$, about 10^{-4} attojoule. The superconducting loop stores this energy without static dissipation: the flux bias that holds the parametron at $\Phi_x \approx \Phi_0/2$ is a persistent current. Energy is dissipated only when the phase moves

through the resistive channels, either when the clock reshapes the potential or when a thermal switch occurs. Each of these costs of order $k_{\mathrm{B}}T$ at the cryogenic temperature, as the worked computation shows. The price is paid elsewhere, in the refrigerator.

14.6 Worked Computation: A Room-Temperature CMOS p-Bit Versus a Liquid-Helium Parametron

Worked computation 14.20 (The double-well sampler in hardware: a 300 K CMOS p-bit versus a 4.2 K Josephson QFP)**.** We compare two bistable samplers, each reduced to the quartic double well, on four counts: barrier modulation, Kramers switching rate, sample bandwidth, and energy per sample. All device values are representative estimates chosen to be self-consistent, not specifications of any product.

Representative parameters. *CMOS latch at* $T = 300\,\mathrm{K}$: $k_{\mathrm{B}}T = 4.14 \times 10^{-21}\,\mathrm{J}$, $V_T = 25.85\,\mathrm{mV}$, slope factor $n = 1.3$, differential-mode capacitance $C = 0.5\,\mathrm{fF}$ (so $\Omega = CV_T/q_e = 80.7$), tail current $I_t = 20\,\mathrm{nA}$ (so $g_{\mathrm{m}} = I_t/(2nV_T) = 0.298\,\mu\mathrm{S}$ and $C/g_{\mathrm{m}} = 1.68\,\mathrm{ns}$), supply $V_{\mathrm{DD}} = 0.3\,\mathrm{V}$, output swing $\Delta V = 0.15\,\mathrm{V}$. *QFP at* $T = 4.2\,\mathrm{K}$: $k_{\mathrm{B}}T = 5.80 \times 10^{-23}\,\mathrm{J}$, loop inductance $L = 20\,\mathrm{pH}$ (so $E_L = (\Phi_0/2\pi)^2/L = 5.42 \times 10^{-21}\,\mathrm{J} = 93.4\,k_{\mathrm{B}}T$), a two-junction tunable junction with $\beta_{L,\max} = 1.8$ (total $2I_{\mathrm{c}0} = 29.6\,\mu\mathrm{A}$, about $0.3\,\mu\mathrm{m}^2$ of junction at $J_c = 10\,\mathrm{kA/cm}^2$), capacitance $C = 20\,\mathrm{fF}$, and a shunt chosen below.

Step 1: Both devices reduce to the quartic double well. For the latch, take the differential voltage v as the coordinate, with the reduced dynamics

$$C\,\mathrm{d}v_t = \left[I_t \tanh(v_t/2nV_T) - Gv_t\right]\mathrm{d}t + \sqrt{2q_eI_t}\,\mathrm{d}W_t,$$

where $I_t \tanh(v/2nV_T)$ is the large-signal current of the cross-coupled subthreshold pair (Worked computation 12.24), G is a load conductance, and the noise is the shot noise of the pair, of intensity $q_n = 2q_eI_t$. In one dimension every drift–diffusion has a potential: with drift $F(v)$ and constant diffusion $D = q_n/(2C^2) = q_eI_t/C^2$, the stationary density is $p_{\mathrm{ss}} \propto e^{-\Phi}$ with $\Phi(v) = -\int F/D\,\mathrm{d}v$. With $u = v/(2nV_T)$ and $r := g_{\mathrm{m}}/G$,

$$\Phi = -\frac{C}{q_eI_t}\Big[2nV_TI_t \ln\cosh u - \frac{G}{2}v^2\Big] = 2n\Omega\Big[\frac{u^2}{2r} - \ln\cosh u\Big] \approx 2n\Omega\Big[-\Big(1-\frac{1}{r}\Big)\frac{u^2}{2} + \frac{u^4}{12}\Big],$$

using $CV_T/q_e = \Omega$, $G \cdot 2nV_T/I_t = 1/r$, and Eq. (1.143). This is the quartic double well in thermal units, with $a = \frac{2}{3}n\Omega$ and $b = 2n\Omega(1 - 1/r)$, so

$$\beta\Delta E_{\mathrm{CMOS}} = \frac{b^2}{4a} = \frac{3}{2}\,n\Omega\Big(1-\frac{1}{r}\Big)^2. \tag{14.73}$$

For the QFP, Eq. (14.58) gives

$$\beta\Delta E_{\mathrm{QFP}} = \frac{3}{2}\,\frac{E_L}{k_{\mathrm{B}}T}\,\frac{(\beta_L-1)^2}{\beta_L}. \tag{14.74}$$

The two results have the same structure,

$$\beta\Delta E = \tfrac{3}{2} \times (\text{stiffness in thermal units}) \times (\text{distance from the bifurcation})^2.$$

The stiffness is $n\Omega = 105$ for the latch and $E_L/k_{\mathrm{B}}T = 93$ for the QFP, so for these parameters the two are comparable.

Step 2: Barrier modulation. Target $\beta\Delta E = 2$, a fast p-bit regime. For the latch, Eq. (14.73) gives $1-1/r = \sqrt{2/(1.5\times 105)} = 0.113$, so $r = g_{\mathrm{m}}/G = 1.127$. For the QFP, solving $\vartheta_m = \beta_L \sin\vartheta_m$ together with the exact barrier Eq. (14.59) gives $\beta_L = 1.125$ and $\vartheta_m = 0.832$; the quartic gives $\beta_L = 1.127$. The knobs are the ratio g_{m}/G, set by the tail current, and the control flux Φ_q through Eq. (14.62), which requires $|\cos(\pi\Phi_q/\Phi_0)| = 1.125/1.8 = 0.625$. The sensitivities are

$$\frac{\partial(\beta\Delta E_{\mathrm{CMOS}})}{\partial r} = \frac{3n\Omega(1-1/r)}{r^2} \approx 28, \qquad \frac{\partial(\beta\Delta E_{\mathrm{QFP}})}{\partial \beta_L} = \frac{3}{2}\frac{E_L}{k_{\mathrm{B}}T}\Big[\frac{2(\beta_L-1)}{\beta_L} - \frac{(\beta_L-1)^2}{\beta_L^2}\Big] \approx 29.$$

A one-percent error in r or β_L shifts $\beta\Delta E$ by about 0.3 and the Kramers rate by about 35%. Both technologies must hold their operating point to a fraction of a percent, or calibrate it in the loop (Chapter 18).

Step 3: Kramers switching rate (free-running). *Latch.* The dynamics is overdamped and one-dimensional, so Eq. (7.37) applies with $\mu E''$ replaced by $|F'|$: $k = \frac{1}{2\pi}\sqrt{|F'(v_{\min})|\,|F'(0)|}\,e^{-\beta\Delta E}$. Here $F'(0) = (g_{\mathrm{m}}-G)/C = (g_{\mathrm{m}}/C)(1-1/r) = 5.95\times 10^8 \times 0.113 = 6.7\times 10^7\,\mathrm{s}^{-1}$, and in the quartic $|F'(v_{\min})| = 2F'(0)$. Therefore

$$k_{\mathrm{CMOS}} = \frac{\sqrt{2}\times 6.7\times 10^7}{2\pi}\,e^{-2} \approx 1.5\times 10^7 \times 0.135 \approx 2.0\times 10^6\,\mathrm{s}^{-1}.$$

QFP. The phase has mass and friction, so we use the moderate-damping formula Eq. (7.53). With $1/\sqrt{LC} = 1.58\times 10^{12}\,\mathrm{s}^{-1}$, Eq. (14.60) gives $\omega_{\mathrm{w}}/2\pi = 124\,\mathrm{GHz}$ and $\omega_b/2\pi = 89\,\mathrm{GHz}$. Choose the shunt so that $\alpha = 1/(2\omega_b RC) = 0.5$: $R = 1/(\omega_b C) = 89\,\Omega$. (With the operating $I_{\mathrm{c}} = \beta_L\Phi_0/(2\pi L) = 18.5\,\mu\mathrm{A}$, the junction alone has $\beta_{\mathrm{c}} \approx 9$; the loop mode is what matters for escape, and for it $\alpha = 0.5$.) Then

$$k_{\mathrm{QFP}} = \frac{\omega_{\mathrm{w}}}{2\pi}\Big(\sqrt{1+\alpha^2}-\alpha\Big)e^{-2} = 124\,\mathrm{GHz}\times 0.618\times 0.135 \approx 1.0\times 10^{10}\,\mathrm{s}^{-1}.$$

The quantum correction Eq. (14.65) with $\hbar\omega_{\mathrm{w}}/k_{\mathrm{B}}T = 1.41$ and $\hbar\omega_b/k_{\mathrm{B}}T = 1.02$ is $\Xi \approx 1.13$ (undamped estimate), a modest increase. The zero-temperature instanton exponent is $S_0/\hbar = \frac{16}{3}\Delta E/(\hbar\omega_{\mathrm{w}}) \approx 7.5$, so tunneling ($e^{-7.5}$) is negligible against activation (e^{-2}) at 4.2 K, consistent with $T_{\mathrm{cr}} \approx 0.4$–$0.7\,\mathrm{K}$. Two cautions apply to both estimates: Kramers' formula is asymptotic in $\beta\Delta E$, so at $\beta\Delta E = 2$ it is accurate only to a factor of order one; and for the QFP, $\Delta E/\hbar\omega_{\mathrm{w}} \approx 1.4$ means the well holds only one or two quantized levels, so semiclassical formulas are at the edge of validity.

The ratio $k_{\mathrm{QFP}}/k_{\mathrm{CMOS}} \approx 5\times 10^3$ has a transparent origin. The latch's prefactor is a nanoampere current charging a femtofarad, reduced by the critical slowing factor $1-1/r$. The parametron's is $1/\sqrt{LC}$, a microampere-scale current in a picohenry loop. By Corollary 14.3, the free-running sample rates are $r_{\mathrm{eff}} \approx \lambda/2 = k$: about 2×10^6 and 1×10^{10} independent samples per second.

Step 4: Clocked bandwidth. Both devices are faster clocked than free-running. *Latch.* With the load switched off during regeneration, $G_n = g_{\mathrm{m}}$. The shot-noise intensity of Step 1, $q_n = 2q_eI_t$, corresponds to $\gamma_n = q_n/(2k_{\mathrm{B}}Tg_{\mathrm{m}}) = 2n$, so by Proposition 14.8 the decision noise is $\sigma_{\mathrm{L}} = \sqrt{(1+2n)k_{\mathrm{B}}T/C} = 5.5\,\mathrm{mV}$. Amplifying to the output swing takes $t_{\mathrm{reg}} = (C/g_{\mathrm{m}})\ln(\Delta V/\sigma_{\mathrm{L}}) = 1.68\,\mathrm{ns}\times\ln 27.5 = 5.6\,\mathrm{ns}$. Adding a short reset, the latch completes one sample in about 10 ns: $\approx 10^8$ samples per second. Its activation is a probit with best-fit slope $\beta_{\mathrm{eff}} \approx 0.85/\sigma_{\mathrm{L}} \approx 155\,\mathrm{V}^{-1}$ in units of input offset. *QFP.* Adiabatic clocking needs ramp times long compared with $1/\omega_b$ and with the loop's damping time; $\tau_r = 50\,\mathrm{ps}$ satisfies both by a wide margin. With a ramp up, a hold for

readout, and a ramp down, a period of about 200 ps gives $\approx 5 \times 10^9$ samples per second. In the slow-ramp regime the activation is the freeze-out tanh of Proposition 14.18. The clocked bandwidth ratio is about 50.

Step 5: Energy per sample. *Latch.* The static energy is the tail current times the supply times the regeneration time,

$$E_{\mathrm{stat}} = I_t V_{\mathrm{DD}} \frac{C}{g_{\mathrm{m}}} \ln \frac{\Delta V}{\sigma_{\mathrm{L}}} = 2 n V_T V_{\mathrm{DD}} C \ln \frac{\Delta V}{\sigma_{\mathrm{L}}},$$

in which the bias current cancels: a larger current regenerates faster in exact proportion. In thermal units, using $C V_T^2 = \Omega k_{\mathrm{B}} T$,

$$\frac{E_{\mathrm{stat}}}{k_{\mathrm{B}} T} = 2n \ln \frac{\Delta V}{\sigma_{\mathrm{L}}} \frac{V_{\mathrm{DD}}}{V_T} \Omega = 2(1.3)(3.31)(11.6)(80.7) \approx 8.1 \times 10^3. \tag{14.75}$$

Charging the output node through ΔV and back costs $E_{\mathrm{dyn}} \approx C \Delta V^2 = 1.1 \times 10^{-17}\,\mathrm{J} \approx 2.7 \times 10^3\, k_{\mathrm{B}} T$. The total is $E_{\mathrm{CMOS}} \approx 4.5 \times 10^{-17}\,\mathrm{J} \approx 1.1 \times 10^4\, k_{\mathrm{B}} T$ per sample, consistent with the subthreshold budget of Section 17.4.2. Run free at $\beta \Delta E = 2$, the same latch draws $I_t V_{\mathrm{DD}} = 6\,\mathrm{nW}$ for 2×10^6 samples per second, or $3\,\mathrm{fJ} \approx 7 \times 10^5\, k_{\mathrm{B}} T$ per sample: critical slowing down makes the free-running latch about sixty-five times worse, which is why CMOS p-bits are clocked or built as noise-amplifying comparators.

QFP. Static flux bias dissipates nothing. Dissipation occurs when the clock drags the phase through the shunt. For a ramp that moves the phase by ϑ_m in time τ_r, the friction loss is $\gamma_\varphi \int \dot{\vartheta}^2 \, \mathrm{d}t \approx \gamma_\varphi \vartheta_m^2 / \tau_r$; switching and resetting give

$$E_{\mathrm{ramp}} \approx \frac{2}{R \tau_r} \Big(\frac{\Phi_0}{2\pi} \Big)^2 \vartheta_m^2 = \frac{2 \times 1.08 \times 10^{-31} \times 0.69}{89 \times 5 \times 10^{-11}} \approx 3.4 \times 10^{-23}\,\mathrm{J} \approx 0.6\, k_{\mathrm{B}} T.$$

To this we add the irreducible cost of producing and eventually discarding one bit of output, of order $k_{\mathrm{B}} T \ln 2$ (Section 9.3), and, in a Gibbs sampler, the heat released when a neighbor's update suddenly changes the input tilt. For a sudden change of the energy from E_h to $E_{h'}$ applied to a p-bit in equilibrium at h, followed by relaxation, the mean dissipated work is exactly

$$\langle \mathcal{W} \rangle - \Delta F = \sum_s p_h(s) [E_{h'}(s) - E_h(s)] - (F_{h'} - F_h) = k_{\mathrm{B}} T \, D_{\mathrm{KL}}(p_h \,\|\, p_{h'}) \,,$$

because $k_{\mathrm{B}} T \ln(p_h / p_{h'}) = E_{h'} - E_h - F_{h'} + F_h$. For a two-state variable this is at most a few $k_{\mathrm{B}} T$ and typically a fraction of one, so it is of the same order as the other terms. The total is $E_{\mathrm{QFP}} \approx 1\text{–}2\, k_{\mathrm{B}} T \approx 7 \times 10^{-23}\,\mathrm{J}$ per sample at the device level. Practical circuits add losses in clock distribution, coupling transformers, and readout; a conservative device-level range is 10^{-22} to $10^{-21}\,\mathrm{J}$. The same Gibbs-update floor applies to the CMOS p-bit, where it is buried under $10^4\, k_{\mathrm{B}} T$ of housekeeping dissipation.

Step 6: Paying for the refrigerator. Heat Q deposited at $T_c = 4.2\,\mathrm{K}$ must be pumped to $T_h \approx 300\,\mathrm{K}$. The Carnot bound on the work required is $W/Q = (T_h - T_c)/T_c = 295.8/4.2 \approx 70$. Real cryocoolers at 4 K reach only a few to roughly twenty percent of Carnot efficiency, so their specific power is between several hundred and several thousand watts of wall power per watt of cooling. Taking 500 to 5000 W/W, the wall-plug energy per QFP sample is

$$E_{\mathrm{QFP}}^{\mathrm{wall}} \approx (7 \times 10^{-23}\,\mathrm{J}) \times (500\text{–}5000) \approx 4 \times 10^{-20}\text{–}4 \times 10^{-19}\,\mathrm{J}.$$

	CMOS latch, 300 K	QFP, 4.2 K
Stiffness in thermal units	$n\Omega \approx 105$	$E_L/k_\mathrm{B}T \approx 93$
Control for $\beta\Delta E = 2$	$g_\mathrm{m}/G = 1.127$	$\beta_L = 1.125$
Attempt scale	$g_\mathrm{m}(1-1/r)/C \sim 10^8\,\mathrm{s}^{-1}$	$\omega_\mathrm{w} \sim 10^{12}\,\mathrm{s}^{-1}$
Free-running Kramers rate	$2\times10^6\,\mathrm{s}^{-1}$	$1\times10^{10}\,\mathrm{s}^{-1}$
Clocked sample rate	$\sim 10^8\,\mathrm{s}^{-1}$	$\sim 5\times10^9\,\mathrm{s}^{-1}$
Activation (clocked)	probit	freeze-out tanh
Energy per sample ($k_\mathrm{B}T$ at own T)	$\sim 1.1\times10^4$	$\sim$ 1–2
Energy per sample (J, on chip)	$\sim 5\times10^{-17}$	$\sim 10^{-22}$
Energy per sample (J, wall plug)	$\sim 5\times10^{-17}$	4×10^{-20}–4×10^{-19}

Step 7: What the comparison does and does not show. On chip, the parametron wins by about 10^4 in units of its own $k_\mathrm{B}T$, and by a further factor of 71 because its $k_\mathrm{B}T$ is smaller, for a factor of about 6×10^5 in joules. The refrigerator takes back two and a half to three and a half orders of magnitude, leaving a device-level wall-plug advantage of order 10^2 to 10^3. That figure is fragile. It excludes the fixed heat load of the cryostat (wiring, radiation, bias lines), the cost of moving samples from 4 K to room-temperature electronics, and the larger area of inductor-based cells. Any of these can exceed the device dissipation by orders of magnitude unless the samples are consumed on the cold chip and only aggregate statistics leave it. The robust advantages are bandwidth, about 50 times clocked and several thousand times free-running, and the exact reciprocity of inductive couplings. The robust advantages of CMOS are operation at room temperature, integration density, and a mature fabrication base, which is the trade Extropic's production hardware has made [31, 56].

Summary

- **Proved:**
  (1) The telegraph statistics of a p-bit: mean, autocorrelation, Lorentzian spectrum, and the effective sample-rate bound $r_\mathrm{eff} \le \lambda/2$ (Proposition 14.2 and Corollary 14.3).
  (2) The exact two-basin identity $\langle s\rangle = \tanh(\beta h_\mathrm{eff})$ and, for the tilted quartic, $h_\mathrm{eff} = h\,x_0[1-3/(16\beta\Delta E_0)]+\mathcal{O}(h^3)$ (Propositions 14.4 and 14.5); the Kramers rate ratio $e^{2\beta_\mathrm{eff}I}$.
  (3) The comparator lemma: logistic noise or a uniform threshold after a tanh stage gives an exact tanh, Gaussian noise gives erf, within 0.035 (slope-matched) or 0.019 (minimax) of tanh in the mean, with quadratic rather than linear log-odds tails (Lemma 14.6).
  (4) Probit decision statistics of a regenerating latch with variance $k_\mathrm{B}T/C + q_n/(2G_nC)$ (Proposition 14.8).
  (5) State-independent loading in differential current summation; Boltzmann stationarity for symmetric couplings; the exact stationary law, cycle current $(t_2-t_1)/16$, and entropy production of two asymmetrically coupled p-bits (Propositions 14.9 to 14.11).
  (6) The hybrid-cell bistability condition $\sum_j (w_j/G)^2 > k_\mathrm{B}T/C$ (Proposition 14.12).
  (7) The London equations, fluxoid quantization with $\Phi_0 = h/2q_e$, the Josephson energy, the RCSJ Langevin equation with its fluctuation–dissipation relation, the retrapping current $4I_\mathrm{c}/(\pi\sqrt{\beta_\mathrm{c}})$, and the cubic washboard barrier $\frac{4\sqrt{2}}{3}E_\mathrm{J}(1-i)^{3/2}$.

(8) The rf-SQUID bistability criterion $\beta_L > 1$ and its reduction to the quartic double well with $a = E_L\beta_L/6$, $b = E_L(\beta_L - 1)$, $h = E_L\vartheta_x$, and the symmetric inductive coupling $J_{ij} = M_{ij}I_q^2$.

(9) The damped crossover temperature $T_{\mathrm{cr}} = \hbar\omega_R/2\pi k_{\mathrm{B}}$ from the Matsubara fluctuation spectrum, and the undamped bounce exponents $\frac{36}{5}\Delta U/\hbar\omega_0$ (cubic) and $\frac{16}{3}\Delta E/\hbar\omega_{\mathrm{w}}$ (symmetric quartic).

- **Assumed:**

(1) Markovian two-state reduction of bistable dynamics (Chapter 7), used beyond its asymptotic range at barriers of 2–3 $k_{\mathrm{B}}T$.

(2) White, Gaussian device noise with order-one excess factor γ_n; neglect of $1/f$ noise over a sampling window (Section 11.5).

(3) The single-coordinate reductions of the latch and the parametron; the dc-SQUID with negligible loop inductance; the sudden freeze-out approximation for clocked parametrons.

(4) The Feynman two-state model as a heuristic for the Josephson relations, whose rigorous derivation uses the BCS tunneling Hamiltonian.

(5) Representative device parameters in the worked computation; cryocooler efficiencies of a few to twenty percent of Carnot.

- **Taken up later:**

(1) Reference formulas for the RCSJ model are collected in Chapter D.

(2) Kramers and Langer rate asymptotics are proved in Chapter 7; Gaussian moment identities used in the Laplace expansions are in Chapter B; the stochastic-calculus facts behind the latch solution are in Chapter C.

(3) Chromatic block Gibbs scheduling and the Hammersley–Clifford theorem are treated in Chapter 2.

(4) The use of latent p-bit layers inside denoising steps is developed in Chapter 16, and the full-chip energy budget in Chapter 17.

(5) Calibration of activation shape, offsets, and coupling asymmetry by in-loop learning is developed in Chapter 18.

- **Open / Frontier:**

(1) A quantitative theory of the clocked parametron's activation between the fast (probit) and slow (freeze-out tanh) limits, including the non-adiabatic dissipation near the bifurcation.

(2) The statistical cost of probit conditionals and asymmetric couplings in large, strongly coupled networks, where errors in rare-event probabilities may compound.

(3) Whether any cryogenic sampler can recover its device-level energy advantage at the system level once readout, data movement, and fixed cryostat loads are included.

Bibliographic notes

The p-bit as a stochastic building block with a tanh activation and a uniform-random comparator update was introduced by Camsari et al. [18], and Borders et al. [14] demonstrated networks of p-bits built from stochastic magnetic tunnel junctions solving factorization problems by sampling. The two-state Kramers picture follows Kramers [61] and the review of Hänggi et al. [46], which also covers quantum corrections to rate theory and the crossover to tunneling. Extropic's Thermodynamic Sampling Units, the X0, XTR-0, and Z1 platforms, and the `THRML` library are described in Extropic Corporation [32], Extropic Corporation [33], Extropic Corporation [31], and Jelinčič et al. [56]; the earlier litepaper [30] discusses both superconducting and transistor-based approaches. For superconductivity, the London equations, fluxoid quantization, the Josephson effects, and the Ambegaokar–Baratoff relation, Tinkham [109] is the standard graduate text. The RCSJ model and the parameter β_{c} originate with Stewart [104] and McCumber [73]. Likharev and Semenov [66] describe the RSFQ logic family and the role of junction shunting in superconducting digital circuits. Dissipative macroscopic quantum tunneling, including the Ohmic-bath Euclidean action and the increase of the bounce exponent with damping, is due to Caldeira and Leggett [17].

Part V

Denoising Thermodynamic Models and the Extropic Architecture

Chapter 15

Continuous-Time Diffusion Processes, Reverse-Time SDEs, and Score Matching

Earlier chapters established an expressivity–mixing tradeoff (Chapters 3, 6 and 7). An energy-based model $p_\theta(\mathbf{x}) = Z_\theta^{-1}\exp(-\beta E_\theta(\mathbf{x}))$ that fits a multimodal, high-dimensional data distribution $p_{\text{data}}(\mathbf{x})$ must have deep basins separated by free-energy barriers $\Delta F \gg k_{\text{B}}T$. Langevin or Glauber dynamics at fixed temperature then mixes slowly: the Bakry–Émery curvature condition (Section 6.5) fails between the basins, the spectral gap becomes exponentially small (Section 7.5), and the Kramers escape time grows as $\tau_{\text{mix}} \sim \tau_{RC}\exp(\beta\Delta F)$. Fast hardware does not remove the problem. Even with the nanosecond or picosecond time constants τ_{RC} of subthreshold CMOS (Chapters 12 and 13) or Josephson circuits (Chapter 14), a barrier $\beta\Delta F \sim 50$ prevents equilibration on any practical timescale.

Diffusion models avoid the barrier by replacing equilibration at one noise level with transport across many noise levels [51, 101, 103]. A forward process adds noise to the data and shrinks the signal over $t \in [0, T]$. In its variance-preserving form this process is the Ornstein–Uhlenbeck relaxation of the RC sampler, the coupled linear RC circuit of Example 1.5 (see also Worked computation 5.33 and Chapter 13). Gaussian smoothing removes the non-convex structure of $-\ln p_t(\mathbf{x})$, and by $t = T$ the distribution p_T is close to an isotropic Gaussian $\mathcal{N}(\mathbf{0}, \sigma_T^2\mathbf{I})$.

Samples are generated by running the process backward, from $t = T$ to $t = 0$; we write $\bar{t} = T - t$ for the reverse time. A draw from the Gaussian is carried along a reverse trajectory into a mode of p_{data}. Biroli et al. [12] showed that the trajectory chooses its mode at a *speciation time* t_s, when the single Gaussian well first splits into separate basins and the new barrier is still of order $k_{\text{B}}T$. By the time the barrier is large, near $t = 0$, the trajectory is already inside one basin and never has to cross it.

The chapter first constructs the forward noising process and proves Anderson's reverse-time theorem (Section 15.2). It then shows how the score needed for the reversal can be learned from data (Section 15.3), analyzes speciation and memorization (Section 15.4), and estimates the energy cost of running the reverse process on digital hardware, which motivates the Denoising Thermodynamic Models of Chapter 16. The worked computation, Worked computation 15.15, solves the whole problem exactly for a bimodal version of the double-well sampler (the quartic double well and its p-bit limit).

15.1 Forward Noising Processes: Variance-Preserving and Variance-Exploding SDEs

Let $p_{\text{data}}(\mathbf{x}_0)$ be a data distribution on $\mathbb{R}^N$ with finite second moment $\mathbb{E}_{p_{\text{data}}}[\|\mathbf{x}_0\|^2] < \infty$. In applications p_{data} is far from Gaussian and often multimodal, concentrated near low-dimensional manifolds or in separated wells. The *forward noising process* is a Markov diffusion $\{\mathbf{x}_t\}_{t\in[0,T]}$, started at $\mathbf{x}_0 \sim p_{\text{data}}$, whose marginal density $p_t(\mathbf{x})$ moves from $p_0 = p_{\text{data}}$ to a simple log-concave reference distribution $\pi(\mathbf{x}) = \mathcal{N}(\mathbf{0}, \sigma^2_{\text{prior}}\mathbf{I})$ at $t = T$.

15.1.1 From Discrete Markov Chains to Continuous-Time Linear SDEs

Following Sohl-Dickstein et al. [101] and Ho et al. [51], start from a discrete-time Markov chain on a uniform temporal grid $0 = t_0 < t_1 < \cdots < t_K = T$ with step size $\Delta t = T/K$. Given a positive noise schedule $\{\beta_k\}_{k=1}^K$ with $0 < \beta_k < 1$, a *Denoising Diffusion Probabilistic Model (DDPM)* defines the forward Markov transition kernel at step k as the conditional Gaussian

$$q(\mathbf{x}_k \mid \mathbf{x}_{k-1}) \;=\; \mathcal{N}\Big(\mathbf{x}_k;\, \sqrt{1-\beta_k}\,\mathbf{x}_{k-1},\, \beta_k\mathbf{I}\Big)\,, \tag{15.1}$$

which corresponds to the stochastic recursion

$$\mathbf{x}_k \;=\; \sqrt{1-\beta_k}\,\mathbf{x}_{k-1} + \sqrt{\beta_k}\,\boldsymbol{\epsilon}_{k-1}, \qquad \boldsymbol{\epsilon}_{k-1} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{15.2}$$

To take the continuum limit $K \to \infty$ ($\Delta t \to 0$), following Song et al. [103], write the variance increment as $\beta_k = \beta(t_{k-1})\Delta t$, where $\beta\colon [0,T] \to (0,\infty)$ is a smooth rate function with dimensions of inverse time. To first order in Δt,

$$\sqrt{1-\beta(t)\Delta t} \;=\; 1 - \frac{1}{2}\beta(t)\Delta t + \mathcal{O}(\Delta t^2). \tag{15.3}$$

Substituting this into Eq. (15.2) and identifying $\sqrt{\Delta t}\,\boldsymbol{\epsilon}_{k-1} = \mathbf{W}_{t+\Delta t} - \mathbf{W}_t \equiv \Delta\mathbf{W}_t$ as the increment of an N-dimensional standard Wiener process $\mathbf{W}_t$ (Section 4.1), we obtain the finite-difference Euler–Maruyama equation

$$\mathbf{x}_{t+\Delta t} - \mathbf{x}_t \;=\; -\frac{1}{2}\beta(t)\mathbf{x}_t\,\Delta t + \sqrt{\beta(t)}\,\Delta\mathbf{W}_t + \mathcal{O}(\Delta t^2). \tag{15.4}$$

As $\Delta t \to 0$, Eq. (15.4) converges to the strong solution (Section 4.5) of a linear Itô SDE, the variance-preserving SDE defined below. Its purely diffusive counterpart, with no restoring drift and a variance that grows without bound, is the variance-exploding SDE. These are the two standard forward processes of score-based generative modeling.

Definition 15.1 (Variance-preserving and variance-exploding forward SDEs)**.** Let $\mathbf{W}_t$ be a standard N-dimensional Wiener process on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\ge0}, \mathbb{P})$, and let $\mathbf{x}_0 \sim p_{\text{data}}$ be independent of $\{\mathbf{W}_t\}_{t\ge0}$. An *isotropic linear forward diffusion* on $[0,T]$ is governed by the Itô SDE

$$\mathrm{d}\mathbf{x}_t \;=\; \mathbf{f}(\mathbf{x}_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{f}(\mathbf{x}, t) \;=\; f(t)\mathbf{x}, \tag{15.5}$$

where $f\colon [0,T] \to (-\infty, 0]$ is a scalar drift coefficient and $g\colon [0,T] \to (0,\infty)$ is a scalar diffusion coefficient. The two standard cases are:

(i) The **variance-preserving (VP) SDE**, a time-inhomogeneous Ornstein–Uhlenbeck process: setting $f(t) = -\frac{1}{2}\beta(t)$ and $g(t) = \sqrt{\beta(t)}$ for a positive schedule $\beta(t) > 0$,

$$\mathrm{d}\mathbf{x}_t \;=\; -\frac{1}{2}\beta(t)\mathbf{x}_t\,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}\mathbf{W}_t. \tag{15.6}$$

(ii) The **variance-exploding (VE) SDE**, a time-rescaled Brownian motion: setting $f(t) = 0$ and $g(t) = \sqrt{\frac{\mathrm{d}}{\mathrm{d}t}\sigma_{\mathrm{VE}}^2(t)}$ for a strictly increasing variance function $\sigma_{\mathrm{VE}}^2(t)$ with $\sigma_{\mathrm{VE}}(0) = \sigma_{\min} \geq 0$,

$$\mathrm{d}\mathbf{x}_t \;=\; \sqrt{\frac{\mathrm{d}\sigma_{\mathrm{VE}}^2(t)}{\mathrm{d}t}}\;\mathrm{d}\mathbf{W}_t. \tag{15.7}$$

The VP SDE Eq. (15.6) is the RC sampler with a time-dependent conductance. Take an uncoupled array of N identical nodes of capacitance C_{node}, each with shunt conductance $G(t) = \frac{1}{2}C_{\mathrm{node}}\beta(t)$, and measure voltage in units where $k_{\mathrm{B}}T/C_{\mathrm{node}} = 1$. Dividing the Kirchhoff–Langevin equation $C_{\mathrm{node}}\,\mathrm{d}\mathbf{v}_t = -G(t)\mathbf{v}_t\,\mathrm{d}t + \sqrt{2k_{\mathrm{B}}TG(t)}\,\mathrm{d}\mathbf{W}_t$ by C_{node} gives Eq. (15.6). The forward VP process is therefore passive RC relaxation in the harmonic potential $E_{\mathrm{prior}}(\mathbf{x}) = \frac{1}{2}\,\|\mathbf{x}\|^2$, with time-dependent mobility $\mu(t) = \frac{1}{2}\beta(t)$.

15.1.2 Gaussian Transition Kernels

Because the drift $\mathbf{f}(\mathbf{x}, t) = f(t)\mathbf{x}$ in Eq. (15.5) is linear in $\mathbf{x}$ and the diffusion coefficient $g(t)$ is independent of $\mathbf{x}$, the conditional distribution $q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)$ of $\mathbf{x}_t$ given $\mathbf{x}_0$ is Gaussian for all $t > 0$. This is the main computational advantage of linear forward processes: $\mathbf{x}_t$ can be sampled for any t directly from $\mathbf{x}_0$, without simulating the path in between.

Proposition 15.2 (Transition kernel of linear forward SDEs)**.** *For the general linear forward SDE Eq. (15.5) with continuous coefficients $f(t)$ and $g(t)$ on $[0, T]$, the strong solution with initial value $\mathbf{x}_0$ is*

$$\mathbf{x}_t \;=\; \alpha(t)\mathbf{x}_0 + \alpha(t)\int_0^t \frac{g(s)}{\alpha(s)}\;\mathrm{d}\mathbf{W}_s, \tag{15.8}$$

where the signal attenuation factor $\alpha(t)$ is

$$\alpha(t) \;\equiv\; \exp\!\left(\int_0^t f(s)\;\mathrm{d}s\right). \tag{15.9}$$

Consequently, the conditional transition density $q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)$ is the isotropic Gaussian

$$q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0) \;=\; \mathcal{N}\!\left(\mathbf{x}_t;\, \alpha(t)\mathbf{x}_0,\, \sigma(t)^2\mathbf{I}\right) \;=\; \frac{1}{\left(2\pi\sigma(t)^2\right)^{N/2}}\exp\!\left(-\frac{\|\mathbf{x}_t - \alpha(t)\mathbf{x}_0\|^2}{2\sigma(t)^2}\right), \tag{15.10}$$

where the variance $\sigma(t)^2$ satisfies the linear first-order differential equation

$$\frac{\mathrm{d}\sigma(t)^2}{\mathrm{d}t} \;=\; 2f(t)\sigma(t)^2 + g(t)^2, \qquad \sigma(0)^2 = 0, \tag{15.11}$$

with solution

$$\sigma(t)^2 \;=\; \alpha(t)^2\int_0^t \frac{g(s)^2}{\alpha(s)^2}\;\mathrm{d}s \;=\; \int_0^t \exp\!\left(2\int_s^t f(u)\;\mathrm{d}u\right) g(s)^2\;\mathrm{d}s. \tag{15.12}$$

In particular:

(i) For the VP SDE Eq. (15.6) with $f(t) = -\frac{1}{2}\beta(t)$ and $g(t) = \sqrt{\beta(t)}$,

$$\alpha(t) \;=\; \exp\!\left(-\frac{1}{2}\int_0^t \beta(s)\;\mathrm{d}s\right), \qquad \sigma(t)^2 \;=\; 1 - \exp\!\left(-\int_0^t \beta(s)\;\mathrm{d}s\right) \;=\; 1 - \alpha(t)^2, \tag{15.13}$$

so that $\alpha(t)^2 + \sigma(t)^2 = 1$ for all $t \geq 0$.

(ii) For the VE SDE Eq. (15.7) with $f(t) = 0$ and $g(t) = \sqrt{\frac{\mathrm{d}}{\mathrm{d}t}\sigma_{\mathrm{VE}}^2(t)}$,

$$\alpha(t) \;=\; 1, \qquad \sigma(t)^2 \;=\; \sigma_{\mathrm{VE}}^2(t) - \sigma_{\mathrm{VE}}^2(0). \tag{15.14}$$

Proof. We apply the integrating-factor method via Itô's formula (Section 4.3). Define the rescaled process $\mathbf{y}_t \equiv \alpha(t)^{-1}\mathbf{x}_t$, where $\alpha(t) = \exp\!\big(\int_0^t f(s)\,\mathrm{d}s\big) > 0$ is deterministic and continuously differentiable with $\frac{\mathrm{d}}{\mathrm{d}t}(\alpha(t)^{-1}) = -f(t)\alpha(t)^{-1}$. Since $\alpha(t)^{-1}$ has zero quadratic variation ($\mathrm{d}(\alpha(t)^{-1}) \cdot \mathrm{d}\mathbf{x}_t = 0$), the Itô product rule gives

$$\begin{aligned}
\mathrm{d}\mathbf{y}_t \;&=\; \frac{\mathrm{d}(\alpha(t)^{-1})}{\mathrm{d}t}\mathbf{x}_t\,\mathrm{d}t + \alpha(t)^{-1}\,\mathrm{d}\mathbf{x}_t \\
&=\; -f(t)\alpha(t)^{-1}\mathbf{x}_t\,\mathrm{d}t + \alpha(t)^{-1}\big[f(t)\mathbf{x}_t\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{W}_t\big] \\
&=\; \frac{g(t)}{\alpha(t)}\,\mathrm{d}\mathbf{W}_t.
\end{aligned} \tag{15.15}$$

Integrating Eq. (15.15) from $s = 0$ to $s = t$ with $\alpha(0) = 1$ yields $\mathbf{y}_t = \mathbf{x}_0 + \int_0^t \frac{g(s)}{\alpha(s)}\,\mathrm{d}\mathbf{W}_s$, and multiplying both sides by $\alpha(t)$ establishes Eq. (15.8).

Conditioned on $\mathbf{x}_0$, the stochastic integral $\mathbf{I}_t \equiv \alpha(t)\int_0^t \frac{g(s)}{\alpha(s)}\,\mathrm{d}\mathbf{W}_s$ is a linear functional of the Gaussian Wiener process $\mathbf{W}_s$ with deterministic integrand, and is therefore a zero-mean multivariate Gaussian random vector: $\mathbb{E}[\mathbf{I}_t \mid \mathbf{x}_0] = \mathbf{0}$, so $\mathbb{E}[\mathbf{x}_t \mid \mathbf{x}_0] = \alpha(t)\mathbf{x}_0$. To compute its covariance matrix, we apply the multidimensional Itô L^2 isometry $\mathbb{E}\Big[\Big(\int_0^t \phi(s)\,\mathrm{d}W_i(s)\Big)\Big(\int_0^t \psi(s)\,\mathrm{d}W_j(s)\Big)\Big] = \delta_{ij}\int_0^t \phi(s)\psi(s)\,\mathrm{d}s$ from Section 4.2:

$$\begin{aligned}
\mathrm{Cov}(\mathbf{x}_t \mid \mathbf{x}_0)_{ij} \;&=\; \mathbb{E}[I_{t,i}\,I_{t,j} \mid \mathbf{x}_0] \;=\; \alpha(t)^2\mathbb{E}\Big[\Big(\int_0^t \frac{g(s)}{\alpha(s)}\,\mathrm{d}W_i(s)\Big)\Big(\int_0^t \frac{g(s)}{\alpha(s)}\,\mathrm{d}W_j(s)\Big)\Big] \\
&=\; \delta_{ij}\,\alpha(t)^2\int_0^t \frac{g(s)^2}{\alpha(s)^2}\,\mathrm{d}s \;\equiv\; \delta_{ij}\,\sigma(t)^2,
\end{aligned} \tag{15.16}$$

which proves Eqs. (15.10) and (15.12). Differentiating Eq. (15.12) with respect to t via the Leibniz product rule and using $\frac{\mathrm{d}\alpha(t)}{\mathrm{d}t} = f(t)\alpha(t)$ yields

$$\frac{\mathrm{d}\sigma(t)^2}{\mathrm{d}t} \;=\; 2\alpha(t)\frac{\mathrm{d}\alpha(t)}{\mathrm{d}t}\int_0^t \frac{g(s)^2}{\alpha(s)^2}\,\mathrm{d}s + \alpha(t)^2\frac{g(t)^2}{\alpha(t)^2} \;=\; 2f(t)\sigma(t)^2 + g(t)^2, \tag{15.17}$$

confirming Eq. (15.11).

Finally, for the VP SDE Eq. (15.6), substituting $f(s) = -\frac{1}{2}\beta(s)$ and $g(s)^2 = \beta(s)$ into Eq. (15.12) gives

$$\begin{aligned}
\sigma(t)^2 \;&=\; \int_0^t \exp\!\Big(-\int_s^t \beta(u)\,\mathrm{d}u\Big)\beta(s)\,\mathrm{d}s \;=\; \int_0^t \frac{\partial}{\partial s}\Big[\exp\!\Big(-\int_s^t \beta(u)\,\mathrm{d}u\Big)\Big]\mathrm{d}s \\
&=\; \Big[\exp\!\Big(-\int_s^t \beta(u)\,\mathrm{d}u\Big)\Big]_{s=0}^{s=t} \;=\; 1 - \exp\!\Big(-\int_0^t \beta(u)\,\mathrm{d}u\Big) \;=\; 1 - \alpha(t)^2,
\end{aligned} \tag{15.18}$$

establishing Eq. (15.13). This explains the name *variance-preserving*: if p_{data} has zero mean and covariance $\mathrm{Cov}(\mathbf{x}_0) = \mathbf{I}$, then $\mathrm{Cov}(\mathbf{x}_t) = \alpha(t)^2\,\mathrm{Cov}(\mathbf{x}_0) + \sigma(t)^2\mathbf{I} = (\alpha(t)^2 + \sigma(t)^2)\mathbf{I} = \mathbf{I}$ for all t. For the VE SDE Eq. (15.7), $f(s) = 0$ implies $\alpha(t) = 1$ and $\sigma(t)^2 = \int_0^t \frac{\mathrm{d}\sigma_{\mathrm{VE}}^2(s)}{\mathrm{d}s}\,\mathrm{d}s = \sigma_{\mathrm{VE}}^2(t) - \sigma_{\mathrm{VE}}^2(0)$, establishing Eq. (15.14). □

15.1.3 Exponential Convergence to the Gaussian Prior

By Proposition 15.2, the marginal density $p_t(\mathbf{x})$ of the forward process is the convolution of p_{data} with the Gaussian kernel $q_{t|0}$:

$$p_t(\mathbf{x}) \;=\; \int_{\mathbb{R}^N} q_{t|0}(\mathbf{x} \mid \mathbf{x}_0)\, p_{\text{data}}(\mathbf{x}_0)\; \mathrm{d}^N x_0 \;=\; \int_{\mathbb{R}^N} \frac{1}{\left(2\pi\sigma(t)^2\right)^{N/2}} \exp\!\left(-\frac{\|\mathbf{x}-\alpha(t)\mathbf{x}_0\|^2}{2\sigma(t)^2}\right) p_{\text{data}}(\mathbf{x}_0)\; \mathrm{d}^N x_0. \tag{15.19}$$

The forward process does not suffer from the mixing bottleneck of Chapters 3 and 7. Under the VP SDE the drift comes from the quadratic potential $E_{\text{prior}}(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2$, whose Hessian is $\mathbf{I}$ everywhere. However many separated modes p_{data} has, the forward Fokker–Planck dynamics relaxes toward the strictly log-concave measure $\pi(\mathbf{x}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. The Bakry–Émery log-Sobolev inequality (Section 6.5) and the coupling arguments of Section 10.2 then give convergence that is exponential in the integrated noise $\bar{\beta}(t) \equiv \int_0^t \beta(s)\, \mathrm{d}s$ and independent of the barriers in p_{data}.

Theorem 15.3 (Forward contraction to the Gaussian prior)**.** *Let $\{\mathbf{x}_t\}_{t\in[0,T]}$ evolve according to the VP SDE Eq. (15.6) from an initial distribution $p_0 = p_{\text{data}} \in \mathcal{P}_2(\mathbb{R}^N)$ with finite second moment, and let $\pi(\mathbf{x}) = (2\pi)^{-N/2} e^{-\|\mathbf{x}\|^2/2}$ be the standard Gaussian equilibrium measure. Define the cumulative noise integral $\bar{\beta}(t) \equiv \int_0^t \beta(s)\, \mathrm{d}s$, so that $\alpha(t) = e^{-\bar{\beta}(t)/2}$. Then:*

(i) Wasserstein contraction. *For all $t \geq 0$,*

$$\mathcal{W}_2(p_t, \pi) \;\leq\; \alpha(t)\, \mathcal{W}_2(p_{\text{data}}, \pi) \;=\; e^{-\frac{1}{2}\int_0^t \beta(s)\, \mathrm{d}s}\, \mathcal{W}_2(p_{\text{data}}, \pi). \tag{15.20}$$

(ii) KL contraction. *For any $t_0 \geq 0$ with $D_{\mathrm{KL}}(p_{t_0} \,\|\, \pi) < \infty$ (true for every $t_0 > 0$, because p_{t_0} is a Gaussian smoothing of $p_{\text{data}} \in \mathcal{P}_2(\mathbb{R}^N)$), and for all $t \geq t_0$,*

$$D_{\mathrm{KL}}(p_t \,\|\, \pi) \;\leq\; \exp\!\left(-\int_{t_0}^t \beta(s)\, \mathrm{d}s\right) D_{\mathrm{KL}}(p_{t_0} \,\|\, \pi) \;=\; \frac{\alpha(t)^2}{\alpha(t_0)^2}\, D_{\mathrm{KL}}(p_{t_0} \,\|\, \pi)\,. \tag{15.21}$$

Proof. To prove part (i), we use a synchronous coupling argument (Section 10.2). Let $(\mathbf{x}_0, \mathbf{y}_0)$ be an optimal $\mathcal{W}_2$-coupling of p_{data} and π, so that $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{y}_0 \sim \pi$, and $\mathbb{E}\big[\|\mathbf{x}_0 - \mathbf{y}_0\|^2\big] = \mathcal{W}_2^2(p_{\text{data}}, \pi)$. Evolve both $\mathbf{x}_t$ and $\mathbf{y}_t$ with the *same* realization of the Wiener process $\mathbf{W}_t$ in the VP SDE Eq. (15.6):

$$\mathrm{d}\mathbf{x}_t \;=\; -\frac{1}{2}\beta(t)\mathbf{x}_t\, \mathrm{d}t + \sqrt{\beta(t)}\, \mathrm{d}\mathbf{W}_t, \qquad \mathrm{d}\mathbf{y}_t \;=\; -\frac{1}{2}\beta(t)\mathbf{y}_t\, \mathrm{d}t + \sqrt{\beta(t)}\, \mathrm{d}\mathbf{W}_t. \tag{15.22}$$

Since $\pi = \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the stationary distribution of the VP SDE (by Eq. (15.13)), $\mathbf{y}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ implies $\mathbf{y}_t \sim \mathcal{N}(\mathbf{0}, (\alpha(t)^2 + \sigma(t)^2)\mathbf{I}) = \mathcal{N}(\mathbf{0}, \mathbf{I}) = \pi$ for all $t \geq 0$), the pair $(\mathbf{x}_t, \mathbf{y}_t)$ forms a valid coupling of p_t and π. Subtracting the two SDEs cancels the noise and leaves the deterministic ODE

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{x}_t - \mathbf{y}_t) \;=\; -\frac{1}{2}\beta(t)(\mathbf{x}_t - \mathbf{y}_t) \quad\Longrightarrow\quad \mathbf{x}_t - \mathbf{y}_t \;=\; \exp\!\left(-\frac{1}{2}\int_0^t \beta(s)\, \mathrm{d}s\right)(\mathbf{x}_0 - \mathbf{y}_0) \;=\; \alpha(t)(\mathbf{x}_0 - \mathbf{y}_0). \tag{15.23}$$

By the definition of $\mathcal{W}_2(p_t, \pi)$ as the infimum over all couplings of p_t and π, we obtain

$$\mathcal{W}_2^2(p_t, \pi) \;\leq\; \mathbb{E}\Big[\|\mathbf{x}_t - \mathbf{y}_t\|^2\Big] \;=\; \alpha(t)^2 \mathbb{E}\Big[\|\mathbf{x}_0 - \mathbf{y}_0\|^2\Big] \;=\; \alpha(t)^2 \mathcal{W}_2^2(p_{\text{data}}, \pi), \tag{15.24}$$

and taking the square root yields Eq. (15.20).

To prove part (ii), write the forward Fokker–Planck equation (Section 6.1) for $p_t(\mathbf{x})$ under the VP SDE Eq. (15.6):

$$\frac{\partial p_t(\mathbf{x})}{\partial t} \;=\; \frac{1}{2}\beta(t)\,\nabla\cdot\big(\mathbf{x}\,p_t(\mathbf{x}) + \nabla p_t(\mathbf{x})\big) \;=\; \frac{1}{2}\beta(t)\,\nabla\cdot\left(p_t(\mathbf{x})\,\nabla\ln\frac{p_t(\mathbf{x})}{\pi(\mathbf{x})}\right), \tag{15.25}$$

where we used $\nabla\ln\pi(\mathbf{x}) = -\mathbf{x}$. Differentiating the Kullback–Leibler divergence $D_{\mathrm{KL}}(p_t\,\|\,\pi) = \int_{\mathbb{R}^N} p_t(\mathbf{x})\ln\frac{p_t(\mathbf{x})}{\pi(\mathbf{x})}\,\mathrm{d}^N x$ with respect to t and integrating by parts (as in the Fokker–Planck H-theorem, Section 6.3) gives the dissipation identity

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}D_{\mathrm{KL}}(p_t\,\|\,\pi) \;&=\; \int_{\mathbb{R}^N}\frac{\partial p_t(\mathbf{x})}{\partial t}\left(\ln\frac{p_t(\mathbf{x})}{\pi(\mathbf{x})}+1\right)\mathrm{d}^N x \\ &=\; -\frac{1}{2}\beta(t)\int_{\mathbb{R}^N} p_t(\mathbf{x})\left\|\nabla\ln\frac{p_t(\mathbf{x})}{\pi(\mathbf{x})}\right\|^2\mathrm{d}^N x \;\equiv\; -\frac{1}{2}\beta(t)\,\mathcal{I}_{\mathrm{Fisher}}(p_t\,\|\,\pi).\end{aligned} \tag{15.26}$$

Since $\pi(\mathbf{x}) \propto e^{-E_{\mathrm{prior}}(\mathbf{x})}$ has potential $E_{\mathrm{prior}}(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2$ with uniform Hessian $\nabla^2 E_{\mathrm{prior}}(\mathbf{x}) = \mathbf{I} = 1\cdot\mathbf{I}$, the Bakry–Émery Log-Sobolev Inequality (Section 6.5) with curvature parameter $\kappa = 1$ states that for any smooth density p_t,

$$D_{\mathrm{KL}}(p_t\,\|\,\pi) \;\le\; \frac{1}{2}\,\mathcal{I}_{\mathrm{Fisher}}(p_t\,\|\,\pi). \tag{15.27}$$

Substituting Eq. (15.27) into Eq. (15.26) yields the differential inequality

$$\frac{\mathrm{d}}{\mathrm{d}t}D_{\mathrm{KL}}(p_t\,\|\,\pi) \;\le\; -\beta(t)\,D_{\mathrm{KL}}(p_t\,\|\,\pi)\,, \tag{15.28}$$

and integrating via Grönwall's lemma from t_0 to t establishes Eq. (15.21). □

Physics Insight: Forward diffusion as coarse-graining

The convolution $p_t(\mathbf{x}) = (q_{t|0} * p_{\mathrm{data}})(\mathbf{x})$ in Eq. (15.19) is a heat-kernel smoothing of p_{data} at the resolution length $\ell(t) = \sigma(t)/\alpha(t)$, a coarse-graining in the spirit of real-space renormalization. Define the effective potential (potential of mean force) at time t, in units with $k_{\mathrm{B}}T = 1$:

$$U_t(\mathbf{x}) \;\equiv\; -\ln p_t(\mathbf{x}) \;=\; -\ln\int_{\mathbb{R}^N}\frac{\mathrm{d}^N x_0}{\big(2\pi\sigma(t)^2\big)^{N/2}}\,p_{\mathrm{data}}(\mathbf{x}_0)\exp\left(-\frac{\|\mathbf{x}-\alpha(t)\mathbf{x}_0\|^2}{2\sigma(t)^2}\right). \tag{15.29}$$

As $t \to 0^+$, $\ell(t) \to 0$ and $U_t \to -\ln p_{\mathrm{data}}$, with all the valleys and high barriers of the data distribution. As t increases, features closer together than $\ell(t)$ are averaged out by the Gaussian integral in Eq. (15.29). Fine-scale minima merge first and larger clusters later. Once $\ell(t) \gg \|\mathbf{\Sigma}_{\mathrm{data}}\|_2^{1/2}$, where $\mathbf{\Sigma}_{\mathrm{data}}$ is the data covariance, even the widest modes have merged into the single paraboloid $U_T(\mathbf{x}) \approx \frac{1}{2}\|\mathbf{x}\|^2 + \frac{N}{2}\ln(2\pi)$. Forward diffusion thus deforms the non-convex potential at $t = 0$ continuously into a convex quadratic at $t = T$.

15.1.4 Anisotropic Forward Processes in Coupled Circuits

In coupled RC and subthreshold CMOS networks (Worked computation 5.33 and Chapter 13), the forward relaxation need not be isotropic. Consider an N-node capacitor network with capacitance matrix $\mathbf{C} \succ \mathbf{0}$, time-dependent conductance/transconductance matrix $\mathbf{G}(t)$, and thermal noise coupling $\mathbf{B}(t)$, governed by the matrix-valued linear Itô SDE

$$\mathrm{d}\mathbf{x}_t \;=\; \mathbf{A}(t)\mathbf{x}_t\,\mathrm{d}t + \mathbf{B}(t)\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{A}(t) \;\equiv\; -\mathbf{C}^{-1}\mathbf{G}(t), \tag{15.30}$$

with symmetric positive-definite diffusion tensor $\mathbf{D}(t) \equiv \frac{1}{2}\mathbf{B}(t)\mathbf{B}(t)^\top$. Proposition 15.2 generalizes to this case with the time-ordered matrix exponential (state transition matrix) $\mathbf{\Phi}(t,s) \in \mathbb{R}^{N\times N}$, defined as the unique solution to the matrix initial-value problem

$$\frac{\partial \mathbf{\Phi}(t,s)}{\partial t} \;=\; \mathbf{A}(t)\mathbf{\Phi}(t,s), \qquad \mathbf{\Phi}(s,s) \;=\; \mathbf{I}. \tag{15.31}$$

Proposition 15.4 (Transition kernel and Lyapunov equation for coupled linear SDEs)**.** *For the coupled linear forward SDE Eq. (15.30) with continuous matrix coefficients $\mathbf{A}(t)$ and $\mathbf{B}(t)$, the strong solution conditioned on $\mathbf{x}_0$ is*

$$\mathbf{x}_t \;=\; \mathbf{\Phi}(t,0)\mathbf{x}_0 + \int_0^t \mathbf{\Phi}(t,s)\mathbf{B}(s)\,\mathrm{d}\mathbf{W}_s. \tag{15.32}$$

Consequently, the conditional transition density is the multivariate Gaussian

$$q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0) \;=\; \mathcal{N}\!\left(\mathbf{x}_t;\, \mathbf{\Phi}(t,0)\mathbf{x}_0,\, \mathbf{\Sigma}_{t|0}\right), \tag{15.33}$$

where the conditional covariance matrix $\mathbf{\Sigma}_{t|0} \equiv \mathrm{Cov}(\mathbf{x}_t \mid \mathbf{x}_0)$ is given by the integral

$$\mathbf{\Sigma}_{t|0} \;=\; 2\int_0^t \mathbf{\Phi}(t,s)\mathbf{D}(s)\mathbf{\Phi}(t,s)^\top\,\mathrm{d}s, \tag{15.34}$$

and satisfies the differential Lyapunov equation

$$\frac{\mathrm{d}\mathbf{\Sigma}_{t|0}}{\mathrm{d}t} \;=\; \mathbf{A}(t)\mathbf{\Sigma}_{t|0} + \mathbf{\Sigma}_{t|0}\mathbf{A}(t)^\top + 2\mathbf{D}(t), \qquad \mathbf{\Sigma}_{0|0} \;=\; \mathbf{0}. \tag{15.35}$$

Proof. Applying Itô's product rule to $\mathbf{y}_t \equiv \mathbf{\Phi}(t,0)^{-1}\mathbf{x}_t$ and using the matrix identity $\frac{\mathrm{d}}{\mathrm{d}t}\big(\mathbf{\Phi}(t,0)^{-1}\big) = -\mathbf{\Phi}(t,0)^{-1}\big(\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{\Phi}(t,0)\big)\mathbf{\Phi}(t,0)^{-1} = -\mathbf{\Phi}(t,0)^{-1}\mathbf{A}(t)$ gives

$$\begin{aligned}\mathrm{d}\mathbf{y}_t \;&=\; -\mathbf{\Phi}(t,0)^{-1}\mathbf{A}(t)\mathbf{x}_t\,\mathrm{d}t + \mathbf{\Phi}(t,0)^{-1}\big[\mathbf{A}(t)\mathbf{x}_t\,\mathrm{d}t + \mathbf{B}(t)\,\mathrm{d}\mathbf{W}_t\big] \\ &=\; \mathbf{\Phi}(t,0)^{-1}\mathbf{B}(t)\,\mathrm{d}\mathbf{W}_t.\end{aligned} \tag{15.36}$$

Integrating from 0 to t, multiplying on the left by $\mathbf{\Phi}(t,0)$, and using the semigroup composition property $\mathbf{\Phi}(t,0)\mathbf{\Phi}(s,0)^{-1} = \mathbf{\Phi}(t,s)$ establishes Eq. (15.32). By the multidimensional Itô isometry, the covariance of the zero-mean Gaussian stochastic integral $\int_0^t \mathbf{\Phi}(t,s)\mathbf{B}(s)\,\mathrm{d}\mathbf{W}_s$ is

$$\mathbf{\Sigma}_{t|0} \;=\; \int_0^t \mathbf{\Phi}(t,s)\mathbf{B}(s)\mathbf{B}(s)^\top\mathbf{\Phi}(t,s)^\top\,\mathrm{d}s \;=\; 2\int_0^t \mathbf{\Phi}(t,s)\mathbf{D}(s)\mathbf{\Phi}(t,s)^\top\,\mathrm{d}s, \tag{15.37}$$

which is Eq. (15.34). Differentiating Eq. (15.34) with respect to the upper limit t via the Leibniz integral rule and substituting $\frac{\partial}{\partial t}\mathbf{\Phi}(t,s) = \mathbf{A}(t)\mathbf{\Phi}(t,s)$, $\frac{\partial}{\partial t}\mathbf{\Phi}(t,s)^\top = \mathbf{\Phi}(t,s)^\top\mathbf{A}(t)^\top$, and $\mathbf{\Phi}(t,t) = \mathbf{I}$ yields

$$\begin{aligned}\frac{\mathrm{d}\mathbf{\Sigma}_{t|0}}{\mathrm{d}t} \;&=\; 2\mathbf{\Phi}(t,t)\mathbf{D}(t)\mathbf{\Phi}(t,t)^\top + 2\int_0^t \Big[\mathbf{A}(t)\mathbf{\Phi}(t,s)\mathbf{D}(s)\mathbf{\Phi}(t,s)^\top + \mathbf{\Phi}(t,s)\mathbf{D}(s)\mathbf{\Phi}(t,s)^\top\mathbf{A}(t)^\top\Big]\,\mathrm{d}s \\ &=\; 2\mathbf{D}(t) + \mathbf{A}(t)\mathbf{\Sigma}_{t|0} + \mathbf{\Sigma}_{t|0}\mathbf{A}(t)^\top,\end{aligned} \tag{15.38}$$

proving Eq. (15.35). For time-independent stable $\mathbf{A}$ and $\mathbf{D}$, $\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{\Sigma}_{t|0} \to \mathbf{0}$ as $t \to \infty$, and Eq. (15.35) reduces to the stationary Lyapunov equation $\mathbf{A}\mathbf{\Sigma}_\infty + \mathbf{\Sigma}_\infty\mathbf{A}^\top + 2\mathbf{D} = \mathbf{0}$ of the NEAT-RN Gaussian samplers of Section 13.4. □

Example 15.5 (Anisotropic RC forward noising)**.** Suppose $\mathbf{A}$ is symmetric negative-definite with orthonormal eigenbasis $\{\mathbf{u}_k\}_{k=1}^N$ and relaxation rates $\gamma_k > 0$ ($\mathbf{A} = -\sum_{k=1}^N \gamma_k \mathbf{u}_k \mathbf{u}_k^\top$), driven by thermal noise that obeys the fluctuation–dissipation relation $\mathbf{D} = \sum_{k=1}^N \gamma_k \sigma_{\mathrm{eq},k}^2 \mathbf{u}_k \mathbf{u}_k^\top$. Then Proposition 15.4 decouples mode by mode: along eigenmode k, the signal decays as $\alpha_k(t) = e^{-\gamma_k t}$ while the conditional variance grows as $\sigma_k(t)^2 = \sigma_{\mathrm{eq},k}^2(1 - e^{-2\gamma_k t})$. High-conductance spatial modes (large γ_k, typically corresponding to high spatial frequencies in a resistive grid) thermalize to the prior first, whereas low-conductance global modes (small γ_k) thermalize last.

15.2 Anderson's Reverse-Time Diffusion Theorem

The forward SDE Eq. (15.5) carries any data distribution close to the Gaussian prior, $p_T \approx \pi$. To generate data we must run it backward, from $t = T$ to $t = 0$.

An overdamped Langevin process in equilibrium with detailed balance has zero stationary probability current (Section 6.2), and its trajectories have the same statistics run forward or backward. During the forward transient $t \in [0, T]$, however, p_t is not in equilibrium. Probability flows out of the data modes into the surrounding space, so the current $\mathbf{J}(\mathbf{x}, t)$ is nonzero and entropy is produced (Chapter 8). Flipping the sign of the drift, $\mathbf{f} \mapsto -\mathbf{f}$, while keeping the noise does not reverse such a process. For the VP SDE the flipped drift $+\frac{1}{2}\beta(t)\mathbf{x}$ is repulsive, and together with the noise it makes the variance grow without bound.

The correct reverse-time SDE was derived by Anderson [4]. It underlies every continuous-time diffusion model and is the starting point for the Denoising Thermodynamic Models of Chapter 16, so we prove it in full.

Theorem 15.6 (Anderson's reverse-time diffusion theorem [4])**.** *Let $\{\mathbf{x}_t\}_{t\in[0,T]}$ be an N-dimensional diffusion process governed by the forward Itô SDE*

$$\mathrm{d}\mathbf{x}_t \;=\; \mathbf{f}(\mathbf{x}_t, t)\,\mathrm{d}t + \mathbf{G}(\mathbf{x}_t, t)\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{x}_0 \sim p_0(\mathbf{x}_0), \tag{15.39}$$

where $\mathbf{f}\colon \mathbb{R}^N \times [0,T] \to \mathbb{R}^N$ and $\mathbf{G}\colon \mathbb{R}^N \times [0,T] \to \mathbb{R}^{N\times M}$ are smooth coefficients with uniformly elliptic symmetric diffusion tensor

$$\mathbf{D}(\mathbf{x}, t) \;\equiv\; \frac{1}{2}\mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^\top, \tag{15.40}$$

such that the single-time marginal density $p_t(\mathbf{x}) > 0$ is smooth ($C^{2,1}(\mathbb{R}^N \times (0,T])$) and together with its spatial derivatives decays rapidly as $\|\mathbf{x}\| \to \infty$. Define the reverse time variable $\bar{t} \equiv T - t \in [0,T]$ and the time-reversed process $\bar{\mathbf{x}}_{\bar{t}} \equiv \mathbf{x}_{T-\bar{t}} = \mathbf{x}_t$. Then $\{\bar{\mathbf{x}}_{\bar{t}}\}_{\bar{t}\in[0,T]}$ is a Markov diffusion process with respect to the backward filtration $\bar{\mathcal{F}}_{\bar{t}} = \sigma(\{\mathbf{x}_s : T - \bar{t} \le s \le T\})$, governed by the reverse-time Itô SDE

$$\mathrm{d}\bar{\mathbf{x}}_{\bar{t}} \;=\; \bar{\mathbf{f}}(\bar{\mathbf{x}}_{\bar{t}}, T - \bar{t})\,\mathrm{d}\bar{t} + \mathbf{G}(\bar{\mathbf{x}}_{\bar{t}}, T - \bar{t})\,\mathrm{d}\bar{\mathbf{W}}_{\bar{t}}, \qquad \bar{\mathbf{x}}_0 \sim p_T(\mathbf{x}_T), \tag{15.41}$$

where $\bar{\mathbf{W}}_{\bar{t}}$ is a standard M-dimensional Wiener process adapted to $\{\bar{\mathcal{F}}_{\bar{t}}\}$, and the i-th component of the reverse drift $\bar{\mathbf{f}}(\mathbf{x}, t)$ evaluated at forward time $t = T - \bar{t}$ is given by

$$\begin{aligned} \bar{f}_i(\mathbf{x}, t) \;&=\; -f_i(\mathbf{x}, t) + \frac{1}{p_t(\mathbf{x})}\sum_{j=1}^N \frac{\partial}{\partial x_j}\Big(2D_{ij}(\mathbf{x}, t)\, p_t(\mathbf{x})\Big) \\ &=\; -f_i(\mathbf{x}, t) + 2\sum_{j=1}^N \frac{\partial D_{ij}(\mathbf{x}, t)}{\partial x_j} + 2\sum_{j=1}^N D_{ij}(\mathbf{x}, t)\frac{\partial \ln p_t(\mathbf{x})}{\partial x_j}. \end{aligned} \tag{15.42}$$

In particular, when the noise coefficient is state-independent and isotropic, $\mathbf{G}(\mathbf{x},t) = g(t)\mathbf{I}$ *so that* $2\mathbf{D}(\mathbf{x},t) = g(t)^2\mathbf{I}$*, the reverse-time SDE simplifies to*

$$\mathrm{d}\bar{\mathbf{x}}_{\bar{t}} \;=\; \Big[-\mathbf{f}(\bar{\mathbf{x}}_{\bar{t}},t) + g(t)^2\,\nabla_{\mathbf{x}}\ln p_t(\bar{\mathbf{x}}_{\bar{t}})\Big]\,\mathrm{d}\bar{t} + g(t)\;\mathrm{d}\bar{\mathbf{W}}_{\bar{t}}, \qquad t = T-\bar{t}. \tag{15.43}$$

Proof. Because the Markov property is symmetric under time reversal (given the present state $\mathbf{x}_t$, the past $\{\mathbf{x}_u\}_{u<t}$ and the future $\{\mathbf{x}_s\}_{s>t}$ are conditionally independent), $\{\bar{\mathbf{x}}_{\bar{t}}\}$ is a Markov process whose law is completely characterized by its initial marginal $p_T(\mathbf{x}_T)$ at $\bar{t}=0$ and its backward two-time transition density $p(\mathbf{x},t\mid\mathbf{y},s)$ for any two forward times $0<t<s\le T$. To determine the generator of the reverse process, we will derive the partial differential equation satisfied by the two-time joint probability density

$$p(\mathbf{x},t;\,\mathbf{y},s) \;=\; p(\mathbf{y},s\mid\mathbf{x},t)\,p_t(\mathbf{x}) \;=\; p(\mathbf{x},t\mid\mathbf{y},s)\,p_s(\mathbf{y}), \qquad t<s, \tag{15.44}$$

with respect to the earlier variables $(\mathbf{x},t)$ while holding the later variables $(\mathbf{y},s)$ fixed.

Step 1: Forward and backward Kolmogorov equations. First, the marginal density $p_t(\mathbf{x})$ evolves according to the forward Kolmogorov (Fokker–Planck) equation (Section 6.1):

$$\frac{\partial p_t(\mathbf{x})}{\partial t} \;=\; \mathcal{L}_{\mathrm{FP}}^{(t)}p_t(\mathbf{x}) \;=\; -\sum_{i=1}^{N}\frac{\partial}{\partial x_i}\big(f_i(\mathbf{x},t)p_t(\mathbf{x})\big) + \sum_{i,j=1}^{N}\frac{\partial^2}{\partial x_i\partial x_j}\big(D_{ij}(\mathbf{x},t)p_t(\mathbf{x})\big). \tag{15.45}$$

Second, viewed as a function of the initial state and initial time $(\mathbf{x},t)$ for fixed target $(\mathbf{y},s)$ with $t<s$, the forward transition density $q(\mathbf{x},t)\equiv p(\mathbf{y},s\mid\mathbf{x},t)$ satisfies the backward Kolmogorov equation (Section 6.1):

$$-\frac{\partial q(\mathbf{x},t)}{\partial t} \;=\; \mathcal{L}^{(t)}q(\mathbf{x},t) \;=\; \sum_{i=1}^{N}f_i(\mathbf{x},t)\frac{\partial q(\mathbf{x},t)}{\partial x_i} + \sum_{i,j=1}^{N}D_{ij}(\mathbf{x},t)\frac{\partial^2 q(\mathbf{x},t)}{\partial x_i\partial x_j}. \tag{15.46}$$

Step 2: Time derivative of the joint density. Compute $-\frac{\partial}{\partial t}p(\mathbf{x},t;\mathbf{y},s) = -\frac{\partial}{\partial t}\left[q(\mathbf{x},t)p_t(\mathbf{x})\right]$ using the product rule and substituting Eqs. (15.45) and (15.46):

$$\begin{aligned}
-\frac{\partial p(\mathbf{x},t;\mathbf{y},s)}{\partial t} \;&=\; \left(-\frac{\partial q(\mathbf{x},t)}{\partial t}\right)p_t(\mathbf{x}) - q(\mathbf{x},t)\frac{\partial p_t(\mathbf{x})}{\partial t} \\
&=\; \sum_{i=1}^{N}p_t(\mathbf{x})f_i(\mathbf{x},t)\frac{\partial q(\mathbf{x},t)}{\partial x_i} + \sum_{i,j=1}^{N}p_t(\mathbf{x})D_{ij}(\mathbf{x},t)\frac{\partial^2 q(\mathbf{x},t)}{\partial x_i\partial x_j} \\
&\quad + \sum_{i=1}^{N}q(\mathbf{x},t)\frac{\partial}{\partial x_i}\big(f_i(\mathbf{x},t)p_t(\mathbf{x})\big) - \sum_{i,j=1}^{N}q(\mathbf{x},t)\frac{\partial^2}{\partial x_i\partial x_j}\big(D_{ij}(\mathbf{x},t)p_t(\mathbf{x})\big).
\end{aligned} \tag{15.47}$$

Step 3: Reduction of the first- and second-order terms. Combine the two first-order terms in Eq. (15.47). By the product rule for $\frac{\partial}{\partial x_i}$,

$$\begin{aligned}
\sum_{i=1}^{N}\left[p_t(\mathbf{x})f_i(\mathbf{x},t)\frac{\partial q(\mathbf{x},t)}{\partial x_i} + q(\mathbf{x},t)\frac{\partial}{\partial x_i}\big(f_i(\mathbf{x},t)p_t(\mathbf{x})\big)\right] \;&=\; \sum_{i=1}^{N}\frac{\partial}{\partial x_i}\big(f_i(\mathbf{x},t)\,q(\mathbf{x},t)p_t(\mathbf{x})\big) \\
&=\; -\sum_{i=1}^{N}\frac{\partial}{\partial x_i}\Big(-f_i(\mathbf{x},t)\,p(\mathbf{x},t;\mathbf{y},s)\Big).
\end{aligned} \tag{15.48}$$

Next, rewrite the second-order terms in Eq. (15.47) so that all derivatives act from the outside on the joint density $p(\mathbf{x},t;\mathbf{y},s)=q(\mathbf{x},t)p_t(\mathbf{x})$ (which is the structure required for a forward Fokker–Planck equation in $\mathbf{x}$). Expanding $\frac{\partial^2}{\partial x_i\partial x_j}(D_{ij}p_t\,q)$ via the Leibniz rule for second partial derivatives gives

$$\begin{aligned}\frac{\partial^2}{\partial x_i\partial x_j}(D_{ij}p_t\,q) &= \frac{\partial}{\partial x_i}\left[\frac{\partial(D_{ij}p_t)}{\partial x_j}q+D_{ij}p_t\frac{\partial q}{\partial x_j}\right]\\ &= \frac{\partial^2(D_{ij}p_t)}{\partial x_i\partial x_j}\,q+\frac{\partial(D_{ij}p_t)}{\partial x_j}\frac{\partial q}{\partial x_i}+\frac{\partial(D_{ij}p_t)}{\partial x_i}\frac{\partial q}{\partial x_j}+D_{ij}p_t\frac{\partial^2 q}{\partial x_i\partial x_j}.\end{aligned}\tag{15.49}$$

Solving Eq. (15.49) for the combination appearing in Eq. (15.47) and using the symmetry $D_{ij}=D_{ji}$ (so that $\sum_{i,j}\frac{\partial(D_{ij}p_t)}{\partial x_j}\frac{\partial q}{\partial x_i}=\sum_{i,j}\frac{\partial(D_{ij}p_t)}{\partial x_i}\frac{\partial q}{\partial x_j}$), we find

$$\sum_{i,j=1}^{N}\left[p_tD_{ij}\frac{\partial^2 q}{\partial x_i\partial x_j}-q\frac{\partial^2(D_{ij}p_t)}{\partial x_i\partial x_j}\right] = \sum_{i,j=1}^{N}\left[\frac{\partial^2}{\partial x_i\partial x_j}(D_{ij}p_tq)-2\frac{\partial(D_{ij}p_t)}{\partial x_j}\frac{\partial q}{\partial x_i}-2q\frac{\partial^2(D_{ij}p_t)}{\partial x_i\partial x_j}\right].\tag{15.50}$$

By the product rule in x_i, the last two terms on the right of Eq. (15.50) combine into a single divergence:

$$\begin{aligned}-2\sum_{i,j=1}^{N}\left[\frac{\partial(D_{ij}p_t)}{\partial x_j}\frac{\partial q}{\partial x_i}+q\frac{\partial}{\partial x_i}\frac{\partial(D_{ij}p_t)}{\partial x_j}\right] &= -\sum_{i=1}^{N}\frac{\partial}{\partial x_i}\left[2q(\mathbf{x},t)\sum_{j=1}^{N}\frac{\partial(D_{ij}(\mathbf{x},t)p_t(\mathbf{x}))}{\partial x_j}\right]\\ &= -\sum_{i=1}^{N}\frac{\partial}{\partial x_i}\left[\left(\frac{1}{p_t(\mathbf{x})}\sum_{j=1}^{N}\frac{\partial(2D_{ij}(\mathbf{x},t)p_t(\mathbf{x}))}{\partial x_j}\right)p(\mathbf{x},t;\mathbf{y},s)\right],\end{aligned}\tag{15.51}$$

where in the last equality we multiplied and divided by $p_t(\mathbf{x})>0$ and used $q(\mathbf{x},t)p_t(\mathbf{x})=p(\mathbf{x},t;\mathbf{y},s)$.

Step 4: The reverse-time Fokker–Planck equation. Substituting Eqs. (15.48), (15.50) and (15.51) back into Eq. (15.47), we obtain

$$-\frac{\partial p(\mathbf{x},t;\mathbf{y},s)}{\partial t} = -\sum_{i=1}^{N}\frac{\partial}{\partial x_i}\Big(\bar{f}_i(\mathbf{x},t)\,p(\mathbf{x},t;\mathbf{y},s)\Big)+\sum_{i,j=1}^{N}\frac{\partial^2}{\partial x_i\partial x_j}\Big(D_{ij}(\mathbf{x},t)\,p(\mathbf{x},t;\mathbf{y},s)\Big),\tag{15.52}$$

where $\bar{f}_i(\mathbf{x},t)$ is given by Eq. (15.42). Now divide both sides of Eq. (15.52) by $p_s(\mathbf{y})>0$ (which is independent of $\mathbf{x}$ and t) and change the time variable to $\bar{t}=T-t$ (so that $\frac{\partial}{\partial\bar{t}}=-\frac{\partial}{\partial t}$ and $\bar{s}=T-s<\bar{t}$). Since $p(\mathbf{x},t;\mathbf{y},s)/p_s(\mathbf{y})=p(\mathbf{x},t\mid\mathbf{y},s)=\bar{p}(\mathbf{x},\bar{t}\mid\mathbf{y},\bar{s})$ is the forward transition density of the reverse-time process $\bar{\mathbf{x}}_{\bar{t}}$ from $\bar{s}$ to $\bar{t}$, Eq. (15.52) reads

$$\frac{\partial\bar{p}(\mathbf{x},\bar{t}\mid\mathbf{y},\bar{s})}{\partial\bar{t}} = -\sum_{i=1}^{N}\frac{\partial}{\partial x_i}\Big(\bar{f}_i(\mathbf{x},T-\bar{t})\,\bar{p}(\mathbf{x},\bar{t}\mid\mathbf{y},\bar{s})\Big)+\sum_{i,j=1}^{N}\frac{\partial^2}{\partial x_i\partial x_j}\Big(D_{ij}(\mathbf{x},T-\bar{t})\,\bar{p}(\mathbf{x},\bar{t}\mid\mathbf{y},\bar{s})\Big).\tag{15.53}$$

This is the Fokker–Planck equation of a diffusion process in reverse time $\bar{t}$ with drift vector $\bar{\mathbf{f}}(\mathbf{x},T-\bar{t})$ and diffusion tensor $\mathbf{D}(\mathbf{x},T-\bar{t})=\frac{1}{2}\mathbf{G}(\mathbf{x},T-\bar{t})\mathbf{G}(\mathbf{x},T-\bar{t})^{\top}$. Furthermore, integrating Eq. (15.52) over $\mathbf{y}\in\mathbb{R}^N$ shows that the single-time marginal $\bar{p}_{\bar{t}}(\mathbf{x})=p_{T-\bar{t}}(\mathbf{x})$ satisfies the same reverse Fokker–Planck equation initialized at $\bar{p}_0(\mathbf{x})=p_T(\mathbf{x})$. By the equivalence between Fokker–Planck equations and weak solutions of Itô SDEs (Section 6.1), the reverse process $\{\bar{\mathbf{x}}_{\bar{t}}\}$ satisfies the reverse-time SDE Eq. (15.41), and specializing to $\mathbf{G}(\mathbf{x},t)=g(t)\mathbf{I}$ yields Eq. (15.43). □

Heuristic Derivation: Where the score drift comes from

Bayes' rule applied to a single short time step shows why reversal adds the drift $+g(t)^2\nabla_{\mathbf{x}} \ln p_t(\mathbf{x})$ rather than merely negating $\mathbf{f}$. The argument is a one-step version of the path-integral time reversal of Sections 8.3 and 8.4.
Consider a short time slice $[t, t+\mathrm{d}t]$ of the isotropic forward SDE $\mathrm{d}\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\ \mathrm{d}t + g(t)\ \mathrm{d}\mathbf{W}_t$. Given $\mathbf{x}_t$, the forward transition probability to $\mathbf{x}_{t+\mathrm{d}t}$ is proportional to

$$p(\mathbf{x}_{t+\mathrm{d}t}, t+\mathrm{d}t \mid \mathbf{x}_t, t) \ \propto\ \exp\!\left(-\frac{\|\mathbf{x}_{t+\mathrm{d}t} - \mathbf{x}_t - \mathbf{f}(\mathbf{x}_t, t)\ \mathrm{d}t\|^2}{2g(t)^2\,\mathrm{d}t}\right). \tag{15.54}$$

In the reverse process, we are at $\mathbf{x}_{t+\mathrm{d}t}$ at time $t+\mathrm{d}t$ and ask for the conditional probability of having come from $\mathbf{x}_t$ at time t. By Bayes' rule,

$$\begin{aligned} p(\mathbf{x}_t, t \mid \mathbf{x}_{t+\mathrm{d}t}, t+\mathrm{d}t) \ &=\ p(\mathbf{x}_{t+\mathrm{d}t}, t+\mathrm{d}t \mid \mathbf{x}_t, t)\,\frac{p_t(\mathbf{x}_t)}{p_{t+\mathrm{d}t}(\mathbf{x}_{t+\mathrm{d}t})} && (15.55)\\ &=\ p(\mathbf{x}_{t+\mathrm{d}t}, t+\mathrm{d}t \mid \mathbf{x}_t, t)\,\exp\!\big(\ln p_t(\mathbf{x}_t) - \ln p_{t+\mathrm{d}t}(\mathbf{x}_{t+\mathrm{d}t})\big). && (15.56)\end{aligned}$$

The second factor is $\exp(\Delta s_{\mathrm{sys}}/k_{\mathrm{B}})$, where Δs_{sys} is the change in stochastic system entropy (Section 8.2) from $\mathbf{x}_t$ to $\mathbf{x}_{t+\mathrm{d}t}$. For $\mathbf{x}_t$ within $\mathcal{O}(\sqrt{\mathrm{d}t})$ of $\mathbf{x}_{t+\mathrm{d}t}$, Taylor-expanding $\ln p_t(\mathbf{x}_t)$ around $\mathbf{x}_{t+\mathrm{d}t}$ gives $\ln p_t(\mathbf{x}_t) \approx \ln p_t(\mathbf{x}_{t+\mathrm{d}t}) + (\mathbf{x}_t - \mathbf{x}_{t+\mathrm{d}t}) \cdot \nabla \ln p_t(\mathbf{x}_{t+\mathrm{d}t})$. Substituting this into the exponent of Eq. (15.55) and completing the square in the backward displacement $\Delta\bar{\mathbf{x}} \equiv \mathbf{x}_t - \mathbf{x}_{t+\mathrm{d}t}$ gives

$$\begin{aligned} -\frac{\|-\Delta\bar{\mathbf{x}} - \mathbf{f}\ \mathrm{d}t\|^2}{2g(t)^2\,\mathrm{d}t} + \Delta\bar{\mathbf{x}} \cdot \nabla \ln p_t \ &=\ -\frac{\|\Delta\bar{\mathbf{x}}\|^2 + 2\Delta\bar{\mathbf{x}} \cdot (\mathbf{f} - g(t)^2\nabla \ln p_t)\ \mathrm{d}t}{2g(t)^2\,\mathrm{d}t} + \mathcal{O}(\mathrm{d}t)\\ &=\ -\frac{\|\Delta\bar{\mathbf{x}} - [-\mathbf{f} + g(t)^2\nabla \ln p_t]\ \mathrm{d}t\|^2}{2g(t)^2\,\mathrm{d}t} + \mathcal{O}(\mathrm{d}t). \end{aligned} \tag{15.57}$$

Bayes' rule therefore tilts the backward Gaussian step toward regions where p_t is larger, shifting its mean by $+g(t)^2\nabla \ln p_t(\mathbf{x})\ \mathrm{d}\bar{t}$.

Notation Watch: Three meanings of "score"

Statistics, machine learning, and statistical physics use the word "score" for related but distinct objects:

1. *Fisher's parameter score.* For a parametric family $p_\theta(\mathbf{x})$, Fisher's score is the gradient with respect to the *parameters*, $\nabla_\theta \ln p_\theta(\mathbf{x})$. Because the normalization $Z(\theta)$ depends on θ, this score contains $-\nabla_\theta \ln Z(\theta)$, which is usually intractable.

2. *State-space (Stein or Hyvärinen) score.* In generative modeling [103], the *score function* of a density $p_t(\mathbf{x})$ is the gradient with respect to the *state* $\mathbf{x}$:

$$\mathbf{s}(\mathbf{x}, t) \ \equiv\ \nabla_{\mathbf{x}} \ln p_t(\mathbf{x}). \tag{15.58}$$

   If $p_t(\mathbf{x}) = Z_t^{-1}\exp(-\beta E_t(\mathbf{x}))$, the normalization Z_t does not depend on $\mathbf{x}$ and drops out of $\nabla_{\mathbf{x}}$.

3. *Thermodynamic force.* Writing $p_t(\mathbf{x}) \propto \exp(-\beta U_t(\mathbf{x}))$, where $U_t(\mathbf{x}) = -k_{\mathrm{B}}T \ln p_t(\mathbf{x})$ is the potential of mean force, the state-space score is the effective force in units of $k_{\mathrm{B}}T$,

pointing down the gradient of U_t:

$$\mathbf{s}(\mathbf{x},t) \;=\; \nabla_{\mathbf{x}} \ln p_t(\mathbf{x}) \;=\; -\beta \nabla_{\mathbf{x}} U_t(\mathbf{x}) \;=\; \beta \mathbf{F}_{\text{eff}}(\mathbf{x},t). \tag{15.59}$$

15.2.1 Probability Current Inversion and Trajectory Entropy Production

Anderson's reverse drift Eq. (15.42) has a direct interpretation in terms of the probability current of Section 6.1. The forward Fokker–Planck equation Eq. (15.45) is the conservation law $\frac{\partial p_t}{\partial t} + \nabla \cdot \mathbf{J}(\mathbf{x},t) = 0$ with current

$$J_i(\mathbf{x},t) \;=\; f_i(\mathbf{x},t)\, p_t(\mathbf{x}) - \sum_{j=1}^{N} \frac{\partial}{\partial x_j} \big(D_{ij}(\mathbf{x},t)\, p_t(\mathbf{x}) \big). \tag{15.60}$$

Dividing by $p_t(\mathbf{x}) > 0$ defines the mean local velocity $\mathbf{v}_t(\mathbf{x}) \equiv \mathbf{J}(\mathbf{x},t)/p_t(\mathbf{x})$:

$$v_{t,i}(\mathbf{x}) \;=\; \frac{J_i(\mathbf{x},t)}{p_t(\mathbf{x})} \;=\; f_i(\mathbf{x},t) - \frac{1}{p_t(\mathbf{x})} \sum_{j=1}^{N} \frac{\partial}{\partial x_j} \big(D_{ij}(\mathbf{x},t)\, p_t(\mathbf{x}) \big). \tag{15.61}$$

Comparing with Eq. (15.42) gives

$$\bar{f}_i(\mathbf{x},t) \;=\; f_i(\mathbf{x},t) - 2 v_{t,i}(\mathbf{x}) \quad\Longleftrightarrow\quad \frac{\mathbf{f}(\mathbf{x},t) - \bar{\mathbf{f}}(\mathbf{x},t)}{2} \;=\; \mathbf{v}_t(\mathbf{x}) \;=\; \frac{\mathbf{J}(\mathbf{x},t)}{p_t(\mathbf{x})}. \tag{15.62}$$

This relation has two consequences.

(i) If the system is in thermal equilibrium with detailed balance (Section 6.2), the probability current vanishes identically ($\mathbf{J}(\mathbf{x},t) = \mathbf{0}$, so $\mathbf{v}_t(\mathbf{x}) = \mathbf{0}$). Then $\bar{\mathbf{f}}(\mathbf{x},t) = \mathbf{f}(\mathbf{x},t)$: the forward and reverse SDEs have the same drift. (For instance, in the VP SDE $\mathbf{f}(\mathbf{x}) = -\frac{1}{2}\beta\mathbf{x}$ at its equilibrium $\pi(\mathbf{x}) \propto e^{-\|\mathbf{x}\|^2/2}$, $\nabla \ln \pi(\mathbf{x}) = -\mathbf{x}$, so $\bar{\mathbf{f}}(\mathbf{x}) = +\frac{1}{2}\beta\mathbf{x} + \beta(-\mathbf{x}) = -\frac{1}{2}\beta\mathbf{x} = \mathbf{f}(\mathbf{x})$.)

(ii) Away from equilibrium ($\mathbf{J}(\mathbf{x},t) \neq \mathbf{0}$), the reverse Fokker–Planck current $\bar{\mathbf{J}}(\mathbf{x},\bar{t})$ evaluated at $\bar{t} = T - t$ is

$$\begin{aligned} \bar{J}_i(\mathbf{x},\bar{t}) \;&=\; \bar{f}_i(\mathbf{x},t)\, p_t(\mathbf{x}) - \sum_{j=1}^{N} \frac{\partial}{\partial x_j} \big(D_{ij}(\mathbf{x},t)\, p_t(\mathbf{x}) \big) \\ &=\; \big(f_i(\mathbf{x},t) - 2 v_{t,i}(\mathbf{x}) \big) p_t(\mathbf{x}) - \big(f_i(\mathbf{x},t) - v_{t,i}(\mathbf{x}) \big) p_t(\mathbf{x}) \;=\; -J_i(\mathbf{x},t). \end{aligned} \tag{15.63}$$

Anderson's reverse drift is therefore the unique drift that reverses the probability current, $\bar{\mathbf{J}} = -\mathbf{J}$, at every point and time while keeping the diffusion tensor $\mathbf{D}(\mathbf{x},t)$ unchanged.

The same current determines the entropy produced during forward noising (Section 8.2).

Proposition 15.7 (Entropy production of forward diffusion)**.** *Consider the VP forward SDE Eq. (15.6), viewed as an overdamped Langevin system in the dimensionless quadratic potential* $E_{\text{prior}}(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2$ *connected to a thermal bath at unit temperature* $k_{\text{B}}T = 1$ *with mobility* $\mu(t) = \frac{1}{2}\beta(t)$*. Then the instantaneous rate of total entropy production* $\dot{\Sigma}_{\text{tot}}(t) = \dot{S}_{\text{sys}}(t) + \dot{S}_{\text{med}}(t)$ *is non-negative and proportional to the relative Fisher information between* p_t *and the Gaussian prior* $\pi(\mathbf{x}) = (2\pi)^{-N/2} e^{-\|\mathbf{x}\|^2/2}$*:*

$$\dot{\Sigma}_{\text{tot}}(t) \;=\; \int_{\mathbb{R}^N} \frac{\|\mathbf{J}(\mathbf{x},t)\|^2}{\frac{1}{2}\beta(t)\, p_t(\mathbf{x})}\, \mathrm{d}^N x \;=\; \frac{1}{2}\beta(t)\, \mathbb{E}_{\mathbf{x}_t \sim p_t}\Big[\|\mathbf{x}_t + \nabla_{\mathbf{x}_t} \ln p_t(\mathbf{x}_t)\|^2 \Big] \;=\; -\frac{\mathrm{d}}{\mathrm{d}t} D_{\text{KL}}(p_t \,\|\, \pi) \;\geq\; 0. \tag{15.64}$$

Integrating from $t = 0$ to $t = T$ gives the total entropy produced:

$$\Sigma_{\mathrm{tot}}[0,T] \;=\; \int_0^T \dot{\Sigma}_{\mathrm{tot}}(t)\,\mathrm{d}t \;=\; D_{\mathrm{KL}}(p_{\mathrm{data}}\,\|\,\pi) - D_{\mathrm{KL}}(p_T\,\|\,\pi)\,. \tag{15.65}$$

Proof. For the VP SDE Eq. (15.6), $f_i(\mathbf{x},t) = -\frac{1}{2}\beta(t)x_i$ and $D_{ij}(\mathbf{x},t) = \frac{1}{2}\beta(t)\delta_{ij}$. By Eq. (15.60), the forward probability current is

$$\mathbf{J}(\mathbf{x},t) \;=\; -\frac{1}{2}\beta(t)\big[\mathbf{x}\,p_t(\mathbf{x}) + \nabla p_t(\mathbf{x})\big] \;=\; -\frac{1}{2}\beta(t)\,p_t(\mathbf{x})\big[\mathbf{x} + \nabla_{\mathbf{x}}\ln p_t(\mathbf{x})\big]. \tag{15.66}$$

The differential entropy of the system $S_{\mathrm{sys}}(t) = -\int p_t(\mathbf{x})\ln p_t(\mathbf{x})\;\mathrm{d}^N x$ has time derivative

$$\dot{S}_{\mathrm{sys}}(t) \;=\; -\int_{\mathbb{R}^N} \frac{\partial p_t}{\partial t}(\ln p_t + 1)\;\mathrm{d}^N x \;=\; \int_{\mathbb{R}^N} (\nabla\cdot\mathbf{J})\ln p_t\;\mathrm{d}^N x \;=\; -\int_{\mathbb{R}^N} \mathbf{J}(\mathbf{x},t)\cdot\nabla\ln p_t(\mathbf{x})\;\mathrm{d}^N x. \tag{15.67}$$

By Sekimoto's heat formula (Section 8.1), the rate of heat dissipated into the unit-temperature bath from the potential $E_{\mathrm{prior}}(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2$ is $\dot{S}_{\mathrm{med}}(t) = \dot{\mathcal{Q}}(t) = -\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}_{p_t}[\frac{1}{2}\|\mathbf{x}\|^2] = -\int \frac{1}{2}\|\mathbf{x}\|^2\,\frac{\partial p_t}{\partial t}\;\mathrm{d}^N x = -\int \mathbf{J}(\mathbf{x},t)\cdot\mathbf{x}\;\mathrm{d}^N x$. Adding $\dot{S}_{\mathrm{sys}}(t)$ and $\dot{S}_{\mathrm{med}}(t)$ gives

$$\begin{aligned}\dot{\Sigma}_{\mathrm{tot}}(t) \;&=\; -\int_{\mathbb{R}^N} \mathbf{J}(\mathbf{x},t)\cdot\big[\mathbf{x} + \nabla\ln p_t(\mathbf{x})\big]\;\mathrm{d}^N x \\ &=\; \frac{1}{2}\beta(t)\int_{\mathbb{R}^N} p_t(\mathbf{x})\;\|\mathbf{x} + \nabla\ln p_t(\mathbf{x})\|^2\,\mathrm{d}^N x \;=\; \int_{\mathbb{R}^N} \frac{\|\mathbf{J}(\mathbf{x},t)\|^2}{\frac{1}{2}\beta(t)\,p_t(\mathbf{x})}\;\mathrm{d}^N x.\end{aligned} \tag{15.68}$$

Since $\nabla\ln\pi(\mathbf{x}) = -\mathbf{x}$, $\mathbf{x} + \nabla\ln p_t(\mathbf{x}) = \nabla\ln(p_t(\mathbf{x})/\pi(\mathbf{x}))$, so comparing with Eq. (15.26) proves Eq. (15.64), and integrating over $t\in[0,T]$ yields Eq. (15.65). □

15.3 The Probability Flow ODE, Tweedie's Formula, and Score Matching

By Theorem 15.6, the only unknown needed to reverse the forward process is the score $\mathbf{s}(\mathbf{x},t) = \nabla_{\mathbf{x}}\ln p_t(\mathbf{x})$. This section shows three things. The score also defines a deterministic ordinary differential equation (ODE) with the same marginals $\{p_t\}$ as the SDE. The score is determined by the posterior mean of the clean data (Tweedie's formula). And an approximation $\mathbf{s}_\theta(\mathbf{x},t) \approx \nabla_{\mathbf{x}}\ln p_t(\mathbf{x})$ can be learned from pairs $(\mathbf{x}_0,\mathbf{x}_t)$ of clean and noised samples, without evaluating the integral Eq. (15.19).

15.3.1 The Probability Flow ODE

A Fokker–Planck equation can be written as a continuity equation $\frac{\partial p_t}{\partial t} + \nabla\cdot(\mathbf{v}_t p_t) = 0$ by moving the diffusion term into the velocity, which then contains the score $\nabla\ln p_t$ (Section 10.3). Following the velocity deterministically reproduces the same marginals.

Proposition 15.8 (Probability flow ODE [103])**.** *Let $p_t(\mathbf{x})$ be the marginal probability density of the forward SDE $\mathrm{d}\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t,t)\,\mathrm{d}t + \mathbf{G}(\mathbf{x}_t,t)\,\mathrm{d}\mathbf{W}_t$ with diffusion tensor $\mathbf{D}(\mathbf{x},t) = \frac{1}{2}\mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^\top$. Then for all $t\in[0,T]$, $p_t(\mathbf{x})$ is also the marginal density of the deterministic ODE*

$$\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} \;=\; \mathbf{v}_t(\mathbf{x}_t), \qquad v_{t,i}(\mathbf{x}) \;\equiv\; f_i(\mathbf{x},t) - \frac{1}{p_t(\mathbf{x})}\sum_{j=1}^{N}\frac{\partial}{\partial x_j}\Big(D_{ij}(\mathbf{x},t)\,p_t(\mathbf{x})\Big). \tag{15.69}$$

In the isotropic case $\mathbf{G}(\mathbf{x},t) = g(t)\mathbf{I}$ *(*$2\mathbf{D} = g(t)^2\mathbf{I}$*), this* probability flow ODE *is*

$$\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} \;=\; \mathbf{f}(\mathbf{x}_t,t) - \frac{1}{2}g(t)^2\,\nabla_{\mathbf{x}}\ln p_t(\mathbf{x}_t). \tag{15.70}$$

Being deterministic, Eq. (15.70) is reversed simply by integrating it from $t = T$ *down to* $t = 0$ *(equivalently, by setting* $\frac{\mathrm{d}\bar{\mathbf{x}}_{\bar{t}}}{\mathrm{d}\bar{t}} = -\mathbf{f}(\bar{\mathbf{x}}_{\bar{t}},t) + \frac{1}{2}g(t)^2\nabla_{\mathbf{x}}\ln p_t(\bar{\mathbf{x}}_{\bar{t}})$*).*

Proof. Start from the forward Fokker–Planck equation Eq. (15.45):

$$\frac{\partial p_t(\mathbf{x})}{\partial t} \;=\; -\sum_{i=1}^{N}\frac{\partial}{\partial x_i}\big(f_i(\mathbf{x},t)p_t(\mathbf{x})\big) + \sum_{i=1}^{N}\frac{\partial}{\partial x_i}\left[\sum_{j=1}^{N}\frac{\partial}{\partial x_j}\big(D_{ij}(\mathbf{x},t)p_t(\mathbf{x})\big)\right]. \tag{15.71}$$

Inside the outer partial derivative $\frac{\partial}{\partial x_i}$ of the second sum, multiply and divide by $p_t(\mathbf{x}) > 0$ to write

$$\sum_{j=1}^{N}\frac{\partial}{\partial x_j}\big(D_{ij}(\mathbf{x},t)p_t(\mathbf{x})\big) \;=\; \left[\frac{1}{p_t(\mathbf{x})}\sum_{j=1}^{N}\frac{\partial}{\partial x_j}\big(D_{ij}(\mathbf{x},t)p_t(\mathbf{x})\big)\right]p_t(\mathbf{x}). \tag{15.72}$$

Substituting this into Eq. (15.71) combines the two sums into the continuity (Liouville) equation

$$\frac{\partial p_t(\mathbf{x})}{\partial t} + \sum_{i=1}^{N}\frac{\partial}{\partial x_i}\big(v_{t,i}(\mathbf{x})\,p_t(\mathbf{x})\big) \;=\; 0, \tag{15.73}$$

where $v_{t,i}(\mathbf{x})$ is defined in Eq. (15.69). Equation (15.73) is the forward Kolmogorov equation of a process with drift $\mathbf{v}_t$ and zero diffusion. The deterministic flow $\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = \mathbf{v}_t(\mathbf{x}_t)$ started at $\mathbf{x}_0 \sim p_0$ (or integrated backward from $\mathbf{x}_T \sim p_T$) therefore has the marginals p_t for all $t \in [0,T]$. Setting $D_{ij}(\mathbf{x},t) = \frac{1}{2}g(t)^2\delta_{ij}$ gives Eq. (15.70). □

The probability flow ODE Eq. (15.70) has $\frac{1}{2}g(t)^2$ where Anderson's reverse SDE Eq. (15.43) has $g(t)^2$. The two belong to one family. Adding the zero term $\frac{\zeta(t)^2}{2}\Delta p_t - \nabla\cdot(\frac{\zeta(t)^2}{2}p_t\nabla\ln p_t)$ to the Fokker–Planck equation, for any noise schedule $\zeta(t) \geq 0$, shows that every reverse SDE

$$\mathrm{d}\bar{\mathbf{x}}_{\bar{t}} \;=\; \left[-\mathbf{f}(\bar{\mathbf{x}}_{\bar{t}},t) + \frac{g(t)^2+\zeta(t)^2}{2}\nabla_{\mathbf{x}}\ln p_t(\bar{\mathbf{x}}_{\bar{t}})\right]\mathrm{d}\bar{t} + \zeta(t)\;\mathrm{d}\bar{\mathbf{W}}_{\bar{t}} \tag{15.74}$$

has the same marginals $\{p_t\}_{t\in[0,T]}$. The choice $\zeta = 0$ gives the probability flow ODE and $\zeta = g$ gives Anderson's reverse SDE.

15.3.2 Tweedie's Formula

An identity from empirical Bayes statistics, due to Robbins and Tweedie [28, 91], gives the score a concrete meaning: it points from the noisy state $\mathbf{x}_t = \mathbf{x}$ toward the (rescaled) posterior mean of the clean data $\mathbf{x}_0$.

Proposition 15.9 (Tweedie's formula)**.** *Let* $p_t(\mathbf{x}) = \int_{\mathbb{R}^N} q_{t|0}(\mathbf{x}\mid\mathbf{x}_0)p_{\mathrm{data}}(\mathbf{x}_0)\,\mathrm{d}^N x_0$ *be the marginal density at time* $t > 0$ *under the Gaussian forward kernel* $q_{t|0}(\mathbf{x}\mid\mathbf{x}_0) = \mathcal{N}(\mathbf{x};\alpha(t)\mathbf{x}_0,\sigma(t)^2\mathbf{I})$ *of Proposition 15.2. Then for every* $\mathbf{x}\in\mathbb{R}^N$*:*

(i) First-order identity (score and posterior mean).

$$\nabla_{\mathbf{x}} \ln p_t(\mathbf{x}) \;=\; \frac{\alpha(t)\,\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}] - \mathbf{x}}{\sigma(t)^2} \;=\; -\frac{\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{x}_t = \mathbf{x}]}{\sigma(t)}, \tag{15.75}$$

where $\mathbf{x}_t = \alpha(t)\mathbf{x}_0 + \sigma(t)\boldsymbol{\epsilon}$ *with* $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$*, and* $\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}] = \int \mathbf{x}_0\, p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})\, \mathrm{d}^N x_0$ *is the minimum mean-square error (MMSE) denoiser.*

(ii) Second-order identity (score Jacobian and posterior covariance).

$$\nabla_{\mathbf{x}}^2 \ln p_t(\mathbf{x}) \;=\; -\nabla_{\mathbf{x}}^2 U_t(\mathbf{x}) \;=\; -\frac{1}{\sigma(t)^2}\mathbf{I} + \frac{\alpha(t)^2}{\sigma(t)^4}\,\mathrm{Cov}_{p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})}(\mathbf{x}_0, \mathbf{x}_0^{\top}). \tag{15.76}$$

Proof. To prove part (i), note first that the conditional score of the Gaussian kernel Eq. (15.10) with respect to $\mathbf{x}$ is

$$\nabla_{\mathbf{x}} \ln q_{t|0}(\mathbf{x} \mid \mathbf{x}_0) \;=\; \nabla_{\mathbf{x}}\left(-\frac{\|\mathbf{x} - \alpha(t)\mathbf{x}_0\|^2}{2\sigma(t)^2} - \frac{N}{2}\ln\!\big(2\pi\sigma(t)^2\big)\right) \;=\; -\frac{\mathbf{x} - \alpha(t)\mathbf{x}_0}{\sigma(t)^2}. \tag{15.77}$$

Differentiating the marginal integral $p_t(\mathbf{x}) = \int_{\mathbb{R}^N} q_{t|0}(\mathbf{x} \mid \mathbf{x}_0) p_{\mathrm{data}}(\mathbf{x}_0)\, \mathrm{d}^N x_0$ under the integral sign (justified by dominated convergence since $q_{t|0}$ is a smooth Gaussian and $p_{\mathrm{data}} \in \mathcal{P}_2(\mathbb{R}^N)$) and dividing by $p_t(\mathbf{x}) > 0$ gives

$$\begin{aligned}
\nabla_{\mathbf{x}} \ln p_t(\mathbf{x}) \;&=\; \frac{\nabla_{\mathbf{x}} p_t(\mathbf{x})}{p_t(\mathbf{x})} \;=\; \int_{\mathbb{R}^N} \frac{\nabla_{\mathbf{x}} q_{t|0}(\mathbf{x} \mid \mathbf{x}_0)}{p_t(\mathbf{x})}\, p_{\mathrm{data}}(\mathbf{x}_0)\, \mathrm{d}^N x_0 \\
&=\; \int_{\mathbb{R}^N} \big(\nabla_{\mathbf{x}} \ln q_{t|0}(\mathbf{x} \mid \mathbf{x}_0)\big)\, \frac{q_{t|0}(\mathbf{x} \mid \mathbf{x}_0) p_{\mathrm{data}}(\mathbf{x}_0)}{p_t(\mathbf{x})}\, \mathrm{d}^N x_0 \\
&=\; \int_{\mathbb{R}^N} \left(-\frac{\mathbf{x} - \alpha(t)\mathbf{x}_0}{\sigma(t)^2}\right) p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})\, \mathrm{d}^N x_0 \;=\; \frac{\alpha(t)\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}] - \mathbf{x}}{\sigma(t)^2},
\end{aligned} \tag{15.78}$$

where we used Bayes' rule $p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}) = \frac{q_{t|0}(\mathbf{x}|\mathbf{x}_0) p_{\mathrm{data}}(\mathbf{x}_0)}{p_t(\mathbf{x})}$ and $\int p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})\, \mathrm{d}^N x_0 = 1$. Substituting $\alpha(t)\mathbf{x}_0 = \mathbf{x}_t - \sigma(t)\boldsymbol{\epsilon}$ into the conditional expectation yields the second equality in Eq. (15.75).

To prove part (ii), take the spatial derivative $\frac{\partial}{\partial x_j}$ of the i-th component of Eq. (15.75):

$$\frac{\partial^2 \ln p_t(\mathbf{x})}{\partial x_i \partial x_j} \;=\; -\frac{\delta_{ij}}{\sigma(t)^2} + \frac{\alpha(t)}{\sigma(t)^2}\frac{\partial}{\partial x_j}\int_{\mathbb{R}^N} x_{0,i}\, \frac{q_{t|0}(\mathbf{x} \mid \mathbf{x}_0) p_{\mathrm{data}}(\mathbf{x}_0)}{p_t(\mathbf{x})}\, \mathrm{d}^N x_0. \tag{15.79}$$

Applying the quotient rule inside the integral and using Eq. (15.77) together with Eq. (15.75), we compute

$$\begin{aligned}
\frac{\partial}{\partial x_j}\left(\frac{q_{t|0}(\mathbf{x} \mid \mathbf{x}_0)}{p_t(\mathbf{x})}\right) \;&=\; \frac{q_{t|0}(\mathbf{x} \mid \mathbf{x}_0)}{p_t(\mathbf{x})}\left[\frac{\partial \ln q_{t|0}(\mathbf{x} \mid \mathbf{x}_0)}{\partial x_j} - \frac{\partial \ln p_t(\mathbf{x})}{\partial x_j}\right] \\
&=\; \frac{q_{t|0}(\mathbf{x} \mid \mathbf{x}_0)}{p_t(\mathbf{x})}\left[-\frac{x_j - \alpha(t)x_{0,j}}{\sigma(t)^2} + \frac{x_j - \alpha(t)\mathbb{E}[x_{0,j} \mid \mathbf{x}_t = \mathbf{x}]}{\sigma(t)^2}\right] \\
&=\; \frac{\alpha(t)}{\sigma(t)^2}\big(x_{0,j} - \mathbb{E}[x_{0,j} \mid \mathbf{x}_t = \mathbf{x}]\big)\frac{q_{t|0}(\mathbf{x} \mid \mathbf{x}_0)}{p_t(\mathbf{x})}.
\end{aligned} \tag{15.80}$$

Substituting Eq. (15.80) into Eq. (15.79) and noting that $\int \mathbb{E}[x_{0,i} \mid \mathbf{x}]\,(x_{0,j} - \mathbb{E}[x_{0,j} \mid \mathbf{x}])p(\mathbf{x}_0 \mid \mathbf{x})\,\mathrm{d}^N x_0 = 0$ gives

$$\begin{aligned}\frac{\partial^2 \ln p_t(\mathbf{x})}{\partial x_i \partial x_j} &= -\frac{\delta_{ij}}{\sigma(t)^2} + \frac{\alpha(t)^2}{\sigma(t)^4}\int_{\mathbb{R}^N}\Big(x_{0,i} - \mathbb{E}[x_{0,i} \mid \mathbf{x}]\Big)\Big(x_{0,j} - \mathbb{E}[x_{0,j} \mid \mathbf{x}]\Big)p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})\,\mathrm{d}^N x_0 \\ &= -\frac{\delta_{ij}}{\sigma(t)^2} + \frac{\alpha(t)^2}{\sigma(t)^4}\,\mathrm{Cov}_{p(\mathbf{x}_0|\mathbf{x}_t=\mathbf{x})}(x_{0,i}, x_{0,j}),\end{aligned} \tag{15.81}$$

which is Eq. (15.76). □

15.3.3 Denoising Score Matching

In practice p_{data} is known only through samples $\mathbf{x}_0 \sim p_{\text{data}}$. Suppose we train a parametric function $\mathbf{s}_\theta(\mathbf{x}, t)$ to approximate the score $\nabla_{\mathbf{x}} \ln p_t(\mathbf{x})$ at noise level t by minimizing the *explicit score matching* (ESM) error

$$\mathcal{L}_{\text{ESM}}(\theta; t) \;\equiv\; \frac{1}{2}\mathbb{E}_{\mathbf{x}_t \sim p_t}\Big[\big\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \ln p_t(\mathbf{x}_t)\big\|^2\Big]. \tag{15.82}$$

This objective cannot be evaluated directly, because its target requires the integral over $\mathbf{x}_0$ in Eq. (15.19). Vincent [115] showed that replacing the intractable marginal score $\nabla_{\mathbf{x}_t} \ln p_t(\mathbf{x}_t)$ in Eq. (15.82) by the closed-form *conditional score* $\nabla_{\mathbf{x}_t} \ln q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0) = -\frac{\mathbf{x}_t - \alpha(t)\mathbf{x}_0}{\sigma(t)^2}$ changes the objective only by a θ-independent constant.

Theorem 15.10 (Denoising and implicit score matching [115])**.** *Let $p_t(\mathbf{x}_t) = \int q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)p_{\text{data}}(\mathbf{x}_0)\,\mathrm{d}^N x_0$ for $t > 0$, and let $\mathbf{s}_\theta(\cdot, t) \in C^1(\mathbb{R}^N; \mathbb{R}^N)$ satisfy $\mathbb{E}_{p_t}[\|\mathbf{s}_\theta(\mathbf{x}_t, t)\|^2] < \infty$ and $\lim_{\|\mathbf{x}\|\to\infty} p_t(\mathbf{x})\,\|\mathbf{s}_\theta(\mathbf{x}, t)\| = 0$. Define the* denoising score matching *(DSM) objective*

$$\begin{aligned}\mathcal{L}_{\text{DSM}}(\theta; t) &\equiv \frac{1}{2}\mathbb{E}_{\mathbf{x}_0 \sim p_{\text{data}}}\,\mathbb{E}_{\mathbf{x}_t \sim q_{t|0}(\cdot|\mathbf{x}_0)}\Big[\Big\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \ln q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\Big\|^2\Big] \\ &= \frac{1}{2}\mathbb{E}_{\mathbf{x}_0 \sim p_{\text{data}}}\,\mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})}\left[\left\|\mathbf{s}_\theta(\alpha(t)\mathbf{x}_0 + \sigma(t)\boldsymbol{\epsilon}, t) + \frac{\boldsymbol{\epsilon}}{\sigma(t)}\right\|^2\right],\end{aligned} \tag{15.83}$$

and Hyvärinen's implicit score matching *(ISM) objective [54]*

$$\mathcal{L}_{\text{ISM}}(\theta; t) \;\equiv\; \mathbb{E}_{\mathbf{x}_t \sim p_t}\left[\frac{1}{2}\,\|\mathbf{s}_\theta(\mathbf{x}_t, t)\|^2 + \nabla_{\mathbf{x}_t} \cdot \mathbf{s}_\theta(\mathbf{x}_t, t)\right]. \tag{15.84}$$

Then for all parameters θ:

$$\mathcal{L}_{\text{ESM}}(\theta; t) \;=\; \mathcal{L}_{\text{DSM}}(\theta; t) + C_{\text{DSM}}(t) \;=\; \mathcal{L}_{\text{ISM}}(\theta; t) + C_{\text{ISM}}(t), \tag{15.85}$$

where the constants $C_{\text{DSM}}(t)$ and $C_{\text{ISM}}(t)$ do not depend on θ:

$$C_{\text{ISM}}(t) \;=\; \frac{1}{2}\mathbb{E}_{p_t}\Big[\|\nabla \ln p_t(\mathbf{x}_t)\|^2\Big], \tag{15.86}$$

$$C_{\text{DSM}}(t) \;=\; \frac{1}{2}\mathbb{E}_{p_t}\Big[\|\nabla \ln p_t(\mathbf{x}_t)\|^2\Big] - \frac{1}{2}\mathbb{E}_{p_{\text{data}}, q_{t|0}}\Big[\Big\|\nabla \ln q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\Big\|^2\Big]. \tag{15.87}$$

In particular, $\nabla_\theta \mathcal{L}_{\text{ESM}}(\theta; t) = \nabla_\theta \mathcal{L}_{\text{DSM}}(\theta; t) = \nabla_\theta \mathcal{L}_{\text{ISM}}(\theta; t)$, and the minimizer over all measurable vector fields is $\mathbf{s}^*(\mathbf{x}, t) = \nabla_{\mathbf{x}} \ln p_t(\mathbf{x})$, unique up to p_t-null sets.

Proof. Expand the square in the ESM objective Eq. (15.82):

$$\mathcal{L}_{\mathrm{ESM}}(\theta;t) \;=\; \frac{1}{2}\mathbb{E}_{p_t}\Big[\|\mathbf{s}_\theta(\mathbf{x}_t,t)\|^2\Big] - \int_{\mathbb{R}^N} p_t(\mathbf{x}_t)\,\mathbf{s}_\theta(\mathbf{x}_t,t)^\top \nabla_{\mathbf{x}_t}\ln p_t(\mathbf{x}_t)\;\mathrm{d}^N x_t + \frac{1}{2}\mathbb{E}_{p_t}\Big[\|\nabla_{\mathbf{x}_t}\ln p_t(\mathbf{x}_t)\|^2\Big]. \tag{15.88}$$

Evaluate the middle cross-term in two ways. First, using $p_t(\mathbf{x}_t)\nabla_{\mathbf{x}_t}\ln p_t(\mathbf{x}_t) = \nabla_{\mathbf{x}_t} p_t(\mathbf{x}_t)$ and integrating by parts over $\mathbb{R}^N$ via the divergence theorem (where the boundary surface integral at $\|\mathbf{x}_t\|\to\infty$ vanishes by hypothesis), we obtain Hyvärinen's identity:

$$\begin{aligned} -\int_{\mathbb{R}^N} \mathbf{s}_\theta(\mathbf{x}_t,t)^\top \nabla_{\mathbf{x}_t} p_t(\mathbf{x}_t)\;\mathrm{d}^N x_t \;&=\; \int_{\mathbb{R}^N} p_t(\mathbf{x}_t)\,\nabla_{\mathbf{x}_t}\cdot\mathbf{s}_\theta(\mathbf{x}_t,t)\;\mathrm{d}^N x_t - \lim_{R\to\infty}\oint_{\|\mathbf{x}\|=R} p_t(\mathbf{x})\,\mathbf{s}_\theta(\mathbf{x},t)\cdot\mathrm{d}\mathbf{A} \\ &=\; \mathbb{E}_{p_t}\big[\nabla_{\mathbf{x}_t}\cdot\mathbf{s}_\theta(\mathbf{x}_t,t)\big], \end{aligned} \tag{15.89}$$

which, substituted into Eq. (15.88), proves $\mathcal{L}_{\mathrm{ESM}}(\theta;t) = \mathcal{L}_{\mathrm{ISM}}(\theta;t) + C_{\mathrm{ISM}}(t)$.

Second, instead of integrating by parts, substitute $\nabla_{\mathbf{x}_t} p_t(\mathbf{x}_t) = \int_{\mathbb{R}^N} p_{\mathrm{data}}(\mathbf{x}_0)\,\nabla_{\mathbf{x}_t} q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\;\mathrm{d}^N x_0 = \int_{\mathbb{R}^N} p_{\mathrm{data}}(\mathbf{x}_0)\,q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\,\nabla_{\mathbf{x}_t}\ln q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\;\mathrm{d}^N x_0$ into the middle cross-term of Eq. (15.88) and apply Fubini's theorem:

$$\begin{aligned} \int_{\mathbb{R}^N} p_t(\mathbf{x}_t)\,\mathbf{s}_\theta(\mathbf{x}_t,t)^\top \nabla_{\mathbf{x}_t}\ln p_t(\mathbf{x}_t)\;\mathrm{d}^N x_t \;&=\; \int_{\mathbb{R}^N} \mathbf{s}_\theta(\mathbf{x}_t,t)^\top \Big[\int_{\mathbb{R}^N} p_{\mathrm{data}}(\mathbf{x}_0) q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0) \\ &\qquad \times \nabla_{\mathbf{x}_t}\ln q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\;\mathrm{d}^N x_0\Big]\,\mathrm{d}^N x_t \\ &=\; \mathbb{E}_{\mathbf{x}_0\sim p_{\mathrm{data}}}\,\mathbb{E}_{\mathbf{x}_t\sim q_{t|0}(\cdot|\mathbf{x}_0)}\Big[\mathbf{s}_\theta(\mathbf{x}_t,t)^\top \nabla_{\mathbf{x}_t}\ln q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\Big]. \end{aligned} \tag{15.90}$$

Now expand the square in the DSM objective Eq. (15.83):

$$\begin{aligned} \mathcal{L}_{\mathrm{DSM}}(\theta;t) \;&=\; \frac{1}{2}\mathbb{E}_{p_{\mathrm{data}},q_{t|0}}\Big[\|\mathbf{s}_\theta(\mathbf{x}_t,t)\|^2\Big] - \mathbb{E}_{p_{\mathrm{data}},q_{t|0}}\Big[\mathbf{s}_\theta(\mathbf{x}_t,t)^\top \nabla_{\mathbf{x}_t}\ln q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\Big] \\ &\quad + \frac{1}{2}\mathbb{E}_{p_{\mathrm{data}},q_{t|0}}\Big[\big\|\nabla_{\mathbf{x}_t}\ln q_{t|0}(\mathbf{x}_t \mid \mathbf{x}_0)\big\|^2\Big]. \end{aligned} \tag{15.91}$$

Since $\mathbf{s}_\theta(\mathbf{x}_t,t)$ depends only on $\mathbf{x}_t$, the first term in Eq. (15.91) equals $\frac{1}{2}\mathbb{E}_{p_t}[\|\mathbf{s}_\theta(\mathbf{x}_t,t)\|^2]$, and by Eq. (15.90) the second term in Eq. (15.91) equals the cross-term in Eq. (15.88). Subtracting Eq. (15.91) from Eq. (15.88) cancels every θ-dependent term and establishes Eqs. (15.85) and (15.86). □

15.3.4 Log-Likelihood and Entropy Along the Probability Flow ODE

Because the probability flow ODE Eq. (15.70) is deterministic, its solution map $\mathbf{\Psi}_{T\to 0}\colon \mathbb{R}^N \to \mathbb{R}^N$ is a diffeomorphism from the prior at $t = T$ to the data space at $t = 0$ (under the smoothness assumptions of Theorem 15.6). Along each trajectory the log-density obeys an ODE driven by the divergence of the velocity field. This gives the exact likelihood of each generated sample, which the stochastic reverse process does not provide directly.

Proposition 15.11 (Change of variables along the probability flow ODE)**.** *Let $t \mapsto \mathbf{x}_t$ solve the probability flow ODE $\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = \mathbf{v}_t(\mathbf{x}_t)$ with $\mathbf{v}_t(\mathbf{x}) = \mathbf{f}(\mathbf{x},t) - \frac{1}{2}g(t)^2\nabla_{\mathbf{x}}\ln p_t(\mathbf{x})$. Then:*

(i) Log-density along a trajectory. *Along any trajectory $t \mapsto \mathbf{x}_t$,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\ln p_t(\mathbf{x}_t) \;=\; -\nabla_{\mathbf{x}}\cdot\mathbf{v}_t(\mathbf{x}_t) \;=\; -\nabla_{\mathbf{x}}\cdot\mathbf{f}(\mathbf{x}_t,t) + \frac{1}{2}g(t)^2\,\Delta_{\mathbf{x}}\ln p_t(\mathbf{x}_t), \tag{15.92}$$

so the log-likelihood at $t = 0$ is

$$\ln p_0(\mathbf{x}_0) \;=\; \ln p_T(\mathbf{x}_T) + \int_0^T \nabla_{\mathbf{x}} \cdot \mathbf{v}_t(\mathbf{x}_t)\; \mathrm{d}t. \tag{15.93}$$

(ii) Entropy rate. *The differential entropy $S(t) \equiv -\int_{\mathbb{R}^N} p_t(\mathbf{x}) \ln p_t(\mathbf{x})\; \mathrm{d}^N x$ evolves according to*

$$\frac{\mathrm{d}S(t)}{\mathrm{d}t} \;=\; \mathbb{E}_{\mathbf{x}_t \sim p_t}[\nabla_{\mathbf{x}} \cdot \mathbf{f}(\mathbf{x}_t, t)] + \frac{1}{2} g(t)^2\, \mathbb{E}_{\mathbf{x}_t \sim p_t}\Big[\|\nabla_{\mathbf{x}} \ln p_t(\mathbf{x}_t)\|^2\Big]. \tag{15.94}$$

Proof. For part (i), take the total time derivative of $\ln p_t(\mathbf{x}_t)$ along the curve by the chain rule and substitute $\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = \mathbf{v}_t(\mathbf{x}_t)$ together with the Liouville continuity equation $\frac{\partial p_t}{\partial t} = -\nabla \cdot (\mathbf{v}_t p_t) = -p_t(\nabla \cdot \mathbf{v}_t) - \mathbf{v}_t \cdot \nabla p_t$:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \ln p_t(\mathbf{x}_t) \;&=\; \frac{1}{p_t(\mathbf{x}_t)} \left[\frac{\partial p_t(\mathbf{x}_t)}{\partial t} + \nabla_{\mathbf{x}} p_t(\mathbf{x}_t) \cdot \frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} \right] \\
&=\; \frac{1}{p_t(\mathbf{x}_t)} \Big[-p_t(\mathbf{x}_t)\, \nabla_{\mathbf{x}} \cdot \mathbf{v}_t(\mathbf{x}_t) - \mathbf{v}_t(\mathbf{x}_t) \cdot \nabla_{\mathbf{x}} p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}} p_t(\mathbf{x}_t) \cdot \mathbf{v}_t(\mathbf{x}_t) \Big] \\
&=\; -\nabla_{\mathbf{x}} \cdot \mathbf{v}_t(\mathbf{x}_t),
\end{aligned} \tag{15.95}$$

which establishes Eq. (15.92), and integrating from $t = 0$ to $t = T$ yields Eq. (15.93).

For part (ii), write $S(t) = -\mathbb{E}_{\mathbf{x}_t \sim p_t}[\ln p_t(\mathbf{x}_t)]$ and average Eq. (15.92) over trajectories with $\mathbf{x}_t \sim p_t$:

$$\frac{\mathrm{d}S(t)}{\mathrm{d}t} \;=\; -\mathbb{E}_{\mathbf{x}_t \sim p_t}\left[\frac{\mathrm{d}}{\mathrm{d}t} \ln p_t(\mathbf{x}_t) \right] \;=\; \mathbb{E}_{p_t}[\nabla_{\mathbf{x}} \cdot \mathbf{f}(\mathbf{x}_t, t)] - \frac{1}{2} g(t)^2 \int_{\mathbb{R}^N} p_t(\mathbf{x})\, \Delta_{\mathbf{x}} \ln p_t(\mathbf{x})\; \mathrm{d}^N x. \tag{15.96}$$

Integrating the Laplacian term by parts over $\mathbb{R}^N$ gives

$$-\int_{\mathbb{R}^N} p_t(\mathbf{x})\, \nabla_{\mathbf{x}} \cdot (\nabla_{\mathbf{x}} \ln p_t(\mathbf{x}))\; \mathrm{d}^N x \;=\; \int_{\mathbb{R}^N} (\nabla_{\mathbf{x}} p_t(\mathbf{x})) \cdot (\nabla_{\mathbf{x}} \ln p_t(\mathbf{x}))\; \mathrm{d}^N x \;=\; \mathbb{E}_{p_t}\Big[\|\nabla_{\mathbf{x}} \ln p_t(\mathbf{x})\|^2\Big], \tag{15.97}$$

which, substituted into Eq. (15.96), proves Eq. (15.94). In the VE case ($\mathbf{f} = \mathbf{0}$, $g(t)^2 = \frac{\mathrm{d}\sigma^2}{\mathrm{d}t}$), Eq. (15.94) reduces to de Bruijn's identity of information theory, $\frac{\mathrm{d}S}{\mathrm{d}\sigma^2} = \frac{1}{2}\mathcal{I}_{\mathrm{Fisher}}(p_t)$, with $\mathcal{I}_{\mathrm{Fisher}}(p) = \mathbb{E}_p[\|\nabla \ln p\|^2]$. □

15.3.5 Continuous-Time Variational Bound via Girsanov's Theorem

Suppose we replace the true score $\nabla_{\mathbf{x}} \ln p_t(\mathbf{x})$ in Anderson's reverse SDE Eq. (15.43) with a learned score function $\mathbf{s}_\theta(\mathbf{x}, t)$, and initialize the reverse process at $\hat{\mathbf{x}}_T \sim \pi(\mathbf{x}_T)$ rather than the true marginal $p_T(\mathbf{x}_T)$:

$$\mathrm{d}\hat{\mathbf{x}}_{\bar{t}} \;=\; \Big[-\mathbf{f}(\hat{\mathbf{x}}_{\bar{t}}, t) + g(t)^2\, \mathbf{s}_\theta(\hat{\mathbf{x}}_{\bar{t}}, t) \Big]\, \mathrm{d}\bar{t} + g(t)\; \mathrm{d}\bar{\mathbf{W}}_{\bar{t}}, \qquad \hat{\mathbf{x}}_0 \sim \pi, \quad t = T - \bar{t}. \tag{15.98}$$

Girsanov's theorem for path measures (Section 4.5) and the data-processing inequality bound the distance between the generated distribution $p_\theta^{(0)}(\mathbf{x}_0) \equiv \mathrm{Law}(\hat{\mathbf{x}}_{\bar{t}=T})$ and $p_{\mathrm{data}} = \mathrm{Law}(\bar{\mathbf{x}}_{\bar{t}=T})$ by the time-integrated score-matching error.

Theorem 15.12 (Variational bound for continuous-time diffusion [102])**.** *Let $\mathcal{P}_{[0,T]}$ denote the path measure on $C([0,T]; \mathbb{R}^N)$ of the exact reverse-time SDE Eq. (15.43) initialized at $\bar{\mathbf{x}}_0 \sim p_T$, and let $\mathcal{P}^\theta_{[0,T]}$ denote the path measure of the learned reverse-time SDE Eq. (15.98) initialized*

at $\hat{\mathbf{x}}_0 \sim \pi$. Assuming Novikov's integrability condition holds for the drift difference $\boldsymbol{\Delta}_\theta(\mathbf{x}, t) \equiv g(t)(\nabla_{\mathbf{x}} \ln p_t(\mathbf{x}) - \mathbf{s}_\theta(\mathbf{x}, t))$, the path-space Kullback–Leibler divergence is

$$\begin{aligned} D_{\mathrm{KL}}\Big(\mathcal{P}_{[0,T]} \,\Big\|\, \mathcal{P}^\theta_{[0,T]}\Big) &= D_{\mathrm{KL}}(p_T \,\|\, \pi) + \frac{1}{2}\int_0^T g(t)^2\, \mathbb{E}_{\mathbf{x}_t \sim p_t}\Big[\big\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \ln p_t(\mathbf{x}_t)\big\|^2\Big]\, \mathrm{d}t \\ &= D_{\mathrm{KL}}(p_T \,\|\, \pi) + \int_0^T g(t)^2\, \mathcal{L}_{\mathrm{DSM}}(\theta; t)\, \mathrm{d}t + \int_0^T g(t)^2 C_{\mathrm{DSM}}(t)\, \mathrm{d}t. \end{aligned} \tag{15.99}$$

By the data-processing inequality between the path measures on $C([0,T]; \mathbb{R}^N)$ and their terminal marginals at $\bar{t} = T$ ($t = 0$), the marginal KL divergence in data space satisfies

$$D_{\mathrm{KL}}\Big(p_{\mathrm{data}} \,\Big\|\, p^{(0)}_\theta\Big) \le D_{\mathrm{KL}}\Big(\mathcal{P}_{[0,T]} \,\Big\|\, \mathcal{P}^\theta_{[0,T]}\Big) = D_{\mathrm{KL}}(p_T \,\|\, \pi) + \int_0^T g(t)^2\, \mathcal{L}_{\mathrm{ESM}}(\theta; t)\, \mathrm{d}t. \tag{15.100}$$

Proof. By the chain rule for Kullback–Leibler divergence applied to the initial state $\bar{\mathbf{x}}_0$ at $\bar{t} = 0$ and the conditional path measure $\mathcal{P}_{[0,T]|0}(\cdot \mid \bar{\mathbf{x}}_0)$, we have

$$D_{\mathrm{KL}}\Big(\mathcal{P}_{[0,T]} \,\Big\|\, \mathcal{P}^\theta_{[0,T]}\Big) = D_{\mathrm{KL}}(p_T \,\|\, \pi) + \mathbb{E}_{\bar{\mathbf{x}}_0 \sim p_T}\Big[D_{\mathrm{KL}}\Big(\mathcal{P}_{[0,T]|0}(\cdot \mid \bar{\mathbf{x}}_0) \,\Big\|\, \mathcal{P}^\theta_{[0,T]|0}(\cdot \mid \bar{\mathbf{x}}_0)\Big)\Big]. \tag{15.101}$$

The two conditional path measures $\mathcal{P}_{[0,T]|0}$ and $\mathcal{P}^\theta_{[0,T]|0}$ share the initial point $\bar{\mathbf{x}}_0$ and the diffusion coefficient $g(T-\bar{t})\mathbf{I}$, and differ only in their reverse-time drifts by $g(T-\bar{t})^2\big[\nabla \ln p_{T-\bar{t}}(\bar{\mathbf{x}}_{\bar{t}}) - \mathbf{s}_\theta(\bar{\mathbf{x}}_{\bar{t}}, T-\bar{t})\big]$. By Girsanov's theorem (Section 4.5), the log Radon–Nikodym derivative under $\mathcal{P}_{[0,T]}$ is

$$\begin{aligned} \ln \frac{\mathrm{d}\mathcal{P}_{[0,T]|0}}{\mathrm{d}\mathcal{P}^\theta_{[0,T]|0}} &= \int_0^T g(T-\bar{t})\Big(\nabla \ln p_{T-\bar{t}}(\bar{\mathbf{x}}_{\bar{t}}) - \mathbf{s}_\theta(\bar{\mathbf{x}}_{\bar{t}}, T-\bar{t})\Big) \cdot \mathrm{d}\bar{\mathbf{W}}_{\bar{t}} \\ &\quad + \frac{1}{2}\int_0^T g(T-\bar{t})^2\, \big\|\nabla \ln p_{T-\bar{t}}(\bar{\mathbf{x}}_{\bar{t}}) - \mathbf{s}_\theta(\bar{\mathbf{x}}_{\bar{t}}, T-\bar{t})\big\|^2\, \mathrm{d}\bar{t}. \end{aligned} \tag{15.102}$$

Under $\mathcal{P}_{[0,T]}$, the Itô stochastic integral with respect to $\mathrm{d}\bar{\mathbf{W}}_{\bar{t}}$ is a zero-mean martingale, so its expectation vanishes. Taking the expectation of Eq. (15.102) over $\mathcal{P}_{[0,T]}$, changing integration variables from $\bar{t}$ to $t = T - \bar{t}$, and invoking Vincent's equivalence Eq. (15.85) proves Eq. (15.99). Since the marginal distributions at $t = 0$ are the pushforwards of $\mathcal{P}_{[0,T]}$ and $\mathcal{P}^\theta_{[0,T]}$ under the coordinate projection $\omega \mapsto \omega(\bar{t} = T)$, the data-processing inequality $D_{\mathrm{KL}}\Big(p_{\mathrm{data}} \,\Big\|\, p^{(0)}_\theta\Big) \le D_{\mathrm{KL}}\Big(\mathcal{P}_{[0,T]} \,\Big\|\, \mathcal{P}^\theta_{[0,T]}\Big)$ yields Eq. (15.100). □

15.4 Statistical Mechanics of Diffusion Trajectories: Biroli–Mézard Regimes

For the VP SDE Eq. (15.6), Anderson's theorem gives the reverse-time SDE

$$\mathrm{d}\bar{\mathbf{x}}_{\bar{t}} = \beta(t)\left[\frac{1}{2}\bar{\mathbf{x}}_{\bar{t}} + \nabla_{\mathbf{x}} \ln p_t(\bar{\mathbf{x}}_{\bar{t}})\right] \mathrm{d}\bar{t} + \sqrt{\beta(t)}\, \mathrm{d}\bar{\mathbf{W}}_{\bar{t}}, \qquad t = T - \bar{t} \in [0, T]. \tag{15.103}$$

Written in terms of the time-dependent effective potential $U_t(\mathbf{x}) \equiv -\ln p_t(\mathbf{x})$ defined in Eq. (15.29), the drift of Eq. (15.103) is the gradient of an effective reverse-time potential $\Phi_t(\mathbf{x})$:

$$\mathrm{d}\bar{\mathbf{x}}_{\bar{t}} = -\beta(t)\, \nabla_{\mathbf{x}} \Phi_t(\bar{\mathbf{x}}_{\bar{t}})\, \mathrm{d}\bar{t} + \sqrt{\beta(t)}\, \mathrm{d}\bar{\mathbf{W}}_{\bar{t}}, \qquad \Phi_t(\mathbf{x}) \equiv U_t(\mathbf{x}) - \frac{1}{4}\|\mathbf{x}\|^2. \tag{15.104}$$

Biroli et al. [12] explained why a trajectory of Eq. (15.103) is not trapped by large Kramers barriers. In high dimension the reverse evolution passes through three regimes separated by two sharp transitions: the *speciation transition* at forward time t_s and the *collapse*, or memorization, *transition* at $t_c < t_s$.

15.4.1 The Posterior Partition Function and Effective Inverse Temperature

At forward time $t \in (0, T]$, consider the posterior distribution $p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})$ of the clean data given the current noisy state. By Bayes' rule and Eq. (15.10),

$$p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}) \;=\; \frac{1}{Z_t(\mathbf{x})}\, p_{\text{data}}(\mathbf{x}_0) \exp\!\left(-\frac{\|\mathbf{x} - \alpha(t)\mathbf{x}_0\|^2}{2\sigma(t)^2}\right), \tag{15.105}$$

where the normalizing denominator $Z_t(\mathbf{x}) = \left(2\pi\sigma(t)^2\right)^{N/2} p_t(\mathbf{x})$ is the partition function

$$Z_t(\mathbf{x}) \;=\; \int_{\mathbb{R}^N} p_{\text{data}}(\mathbf{x}_0) \exp\!\left(-\frac{\|\mathbf{x}\|^2}{2\sigma(t)^2} + \frac{\alpha(t)}{\sigma(t)^2}\mathbf{x}^\top \mathbf{x}_0 - \frac{\alpha(t)^2}{2\sigma(t)^2}\,\|\mathbf{x}_0\|^2\right) \mathrm{d}^N x_0. \tag{15.106}$$

In the language of Chapter 1, Eq. (15.105) is a canonical ensemble for the microstate $\mathbf{x}_0$, with reference measure p_{data}, an external field $\mathbf{h}_t(\mathbf{x}) \equiv \frac{\alpha(t)}{\sigma(t)^2}\mathbf{x}$, and a quadratic self-energy whose strength is the signal-to-noise ratio, which plays the role of an inverse temperature:

$$\Gamma(t) \;\equiv\; \frac{\alpha(t)^2}{\sigma(t)^2} \;=\; \frac{1}{e^{\bar\beta(t)} - 1}, \tag{15.107}$$

where $\bar\beta(t) = \int_0^t \beta(s)\, \mathrm{d}s$. At the start of the reverse process ($t = T$), $\bar\beta(T) \gg 1$ and $\Gamma(T) \ll 1$. The effective temperature is high, the field $\mathbf{h}_T$ is weak, and the posterior is close to p_{data} itself, spread over all its modes. At the end ($t \to 0^+$), $\bar\beta(t) \to 0$ and $\Gamma(t) \to \infty$. The effective temperature goes to zero, and the posterior concentrates at $\mathbf{x}_0 = \mathbf{x}/\alpha(t)$.

15.4.2 Regime I and the Speciation Transition

Consider the curvature of the time-dependent effective potential $U_t(\mathbf{x}) = -\ln p_t(\mathbf{x})$. By the second-order Tweedie identity Eq. (15.76), the Hessian of $U_t(\mathbf{x})$ at any point $\mathbf{x} \in \mathbb{R}^N$ is

$$\nabla^2_{\mathbf{x}} U_t(\mathbf{x}) \;=\; -\nabla^2_{\mathbf{x}} \ln p_t(\mathbf{x}) \;=\; \frac{1}{\sigma(t)^2}\left[\mathbf{I} - \frac{\alpha(t)^2}{\sigma(t)^2}\,\mathrm{Cov}_{p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})}(\mathbf{x}_0, \mathbf{x}_0^\top)\right]. \tag{15.108}$$

The curvature is a competition between two terms:

1. the identity $\mathbf{I}$, which comes from the Gaussian kernel $\frac{\|\mathbf{x}-\alpha(t)\mathbf{x}_0\|^2}{2\sigma(t)^2}$ and makes U_t convex;
2. the posterior covariance $\frac{\alpha(t)^2}{\sigma(t)^2}\,\mathrm{Cov}(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})$, which comes from the spread of the data and pushes toward negative curvature.

When t is large (early in the reverse trajectory), the signal-to-noise ratio $\Gamma(t)$ is small. Take $\mathbb{E}_{p_{\text{data}}}[\mathbf{x}_0] = \mathbf{0}$ without loss of generality. At the origin the posterior covariance $\mathrm{Cov}(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{0})$ approaches the prior data covariance $\mathbf{\Sigma}_{\text{data}} \equiv \mathrm{Cov}_{p_{\text{data}}}(\mathbf{x}_0, \mathbf{x}_0^\top)$ up to $\mathcal{O}(\Gamma(t))$ corrections. As long as

$$\frac{\alpha(t)^2}{\sigma(t)^2}\,\lambda_{\max}\Big(\mathrm{Cov}_{p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{0})}(\mathbf{x}_0, \mathbf{x}_0^\top)\Big) \;<\; 1, \tag{15.109}$$

the Hessian $\nabla^2_{\mathbf{x}} U_t(\mathbf{0})$ is positive-definite and the origin is a stable minimum of U_t; in the examples below it is the only one. In this regime (Regime I, $t > t_s$) every trajectory feels a restoring force toward the center of mass of the data and has not yet committed to a mode.

As t decreases toward t_s, $\Gamma(t)$ grows until the largest eigenvalue of $\Gamma(t)\,\mathrm{Cov}(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{0})$ reaches one. The smallest eigenvalue of $\nabla^2_{\mathbf{x}} U_t(\mathbf{0})$ then crosses zero along the principal direction $\mathbf{e}_1$, and U_t undergoes a symmetry-breaking bifurcation.

Definition 15.13 (Speciation time [12])**.** Let $p_{\mathrm{data}}(\mathbf{x}_0)$ be a zero-mean data distribution with principal covariance eigenvalue $\lambda_{\max}(\mathbf{\Sigma}_{\mathrm{data}}) > 0$. The *Hessian speciation time* t_s^{H} is the largest forward time $t \in (0, T)$ at which the minimum eigenvalue of the effective potential Hessian $\nabla_{\mathbf{x}}^2 U_t(\mathbf{0})$ vanishes:

$$\lambda_{\min}(\nabla_{\mathbf{x}}^2 U_{t_s^{\mathrm{H}}}(\mathbf{0})) \;=\; 0 \quad\Longleftrightarrow\quad \frac{\alpha(t_s^{\mathrm{H}})^2}{\sigma(t_s^{\mathrm{H}})^2}\,\lambda_{\max}\Big(\mathrm{Cov}_{p(\mathbf{x}_0|\mathbf{x}_t=\mathbf{0})}(\mathbf{x}_0, \mathbf{x}_0^\top)\Big) \;=\; 1. \tag{15.110}$$

In the approximation $\mathrm{Cov}(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{0}) \approx \mathbf{\Sigma}_{\mathrm{data}}$, the *speciation time* t_s is defined by

$$\frac{\alpha(t_s)^2}{\sigma(t_s)^2}\,\lambda_{\max}(\mathbf{\Sigma}_{\mathrm{data}}) \;=\; 1. \tag{15.111}$$

For the constant-rate VP SDE with $\beta(t) = 1$ ($\alpha(t) = e^{-t/2}$, $\sigma(t)^2 = 1 - e^{-t}$), Eq. (15.111) gives

$$\frac{e^{-t_s}}{1 - e^{-t_s}}\,\lambda_{\max}(\mathbf{\Sigma}_{\mathrm{data}}) \;=\; 1 \quad\Longrightarrow\quad t_s \;=\; \ln\!\big(1 + \lambda_{\max}(\mathbf{\Sigma}_{\mathrm{data}})\big). \tag{15.112}$$

Proposition 15.14 (Landau expansion at the speciation transition)**.** *Suppose $p_{\mathrm{data}}(\mathbf{x}_0)$ is invariant under inversion ($p_{\mathrm{data}}(-\mathbf{x}_0) = p_{\mathrm{data}}(\mathbf{x}_0)$) and its posterior distribution at $\mathbf{x}_t = \mathbf{0}$ has a non-degenerate largest covariance eigenvalue $\nu_1(t) = \langle \mathbf{e}_1 | \mathrm{Cov}(\mathbf{x}_0 \mid \mathbf{0}) | \mathbf{e}_1 \rangle$ along the unit eigenvector $\mathbf{e}_1 \in \mathbb{R}^N$, with negative fourth cumulant*

$$-\kappa_4(t) \;\equiv\; \mathbb{E}_{p(\mathbf{x}_0|\mathbf{0})}\Big[(\mathbf{e}_1^\top \mathbf{x}_0)^4\Big] - 3\Big(\mathbb{E}_{p(\mathbf{x}_0|\mathbf{0})}\Big[(\mathbf{e}_1^\top \mathbf{x}_0)^2\Big]\Big)^2 \;<\; 0 \qquad (\kappa_4(t) > 0). \tag{15.113}$$

For displacements $\mathbf{x} = \psi\,\mathbf{e}_1$ along the unstable principal mode $\mathbf{e}_1$, the Taylor expansion of the effective potential $U_t(\psi\,\mathbf{e}_1)$ around $\psi = 0$ to fourth order takes the Landau form

$$U_t(\psi\,\mathbf{e}_1) \;=\; U_t(\mathbf{0}) + \frac{r(t)}{2}\psi^2 + \frac{u(t)}{4}\psi^4 + \mathcal{O}(\psi^6), \tag{15.114}$$

with quadratic control parameter $r(t)$ and positive quartic coupling $u(t)$ given by

$$r(t) \;=\; \frac{1}{\sigma(t)^2}\left[1 - \frac{\alpha(t)^2}{\sigma(t)^2}\nu_1(t)\right], \qquad u(t) \;=\; \frac{\alpha(t)^4}{6\,\sigma(t)^8}\,\kappa_4(t) \;>\; 0. \tag{15.115}$$

Consequently:

(i) For $t > t_s^{\mathrm{H}}$, $r(t) > 0$ and $\psi^* = 0$ is the unique local minimum of $U_t(\psi\,\mathbf{e}_1)$.

(ii) At $t = t_s^{\mathrm{H}}$, $r(t_s^{\mathrm{H}}) = 0$: the origin becomes a quartic flat minimum $U_{t_s^{\mathrm{H}}}(\psi\,\mathbf{e}_1) - U_{t_s^{\mathrm{H}}}(\mathbf{0}) = \frac{u(t_s^{\mathrm{H}})}{4}\psi^4 + \mathcal{O}(\psi^6)$.

(iii) For $t < t_s^{\mathrm{H}}$, $r(t) < 0$: the origin $\psi = 0$ becomes a local maximum (saddle point in $\mathbb{R}^N$) and splits into two symmetric minima at

$$\psi_\pm^*(t) \;=\; \pm\sqrt{\frac{|r(t)|}{u(t)}}, \tag{15.116}$$

separated by a barrier $\Delta U_t \equiv U_t(\mathbf{0}) - U_t(\psi_\pm^*(t)\mathbf{e}_1) = \frac{r(t)^2}{4u(t)}$ that grows continuously from 0 as $(t_s^{\mathrm{H}} - t)^2$.

Proof. By inversion symmetry $p_{\text{data}}(-\mathbf{x}_0) = p_{\text{data}}(\mathbf{x}_0)$, the marginal density satisfies $p_t(-\mathbf{x}) = p_t(\mathbf{x})$, so all odd-order derivatives of $U_t(\psi\,\mathbf{e}_1)$ at $\psi = 0$ vanish. From the second-order Tweedie identity Eq. (15.76), the second derivative with respect to ψ at $\psi = 0$ is

$$r(t) \;=\; \frac{\partial^2}{\partial\psi^2} U_t(\psi\,\mathbf{e}_1)\Big|_{\psi=0} \;=\; \mathbf{e}_1^\top \nabla_{\mathbf{x}}^2 U_t(\mathbf{0})\,\mathbf{e}_1 \;=\; \frac{1}{\sigma(t)^2}\left[1 - \frac{\alpha(t)^2}{\sigma(t)^2}\nu_1(t)\right]. \tag{15.117}$$

To find the fourth derivative $\frac{\mathrm{d}^4}{\mathrm{d}\psi^4} U_t(\psi\,\mathbf{e}_1)\big|_{\psi=0} = 6u(t)$, observe from Eq. (15.106) that

$$\begin{aligned} U_t(\psi\,\mathbf{e}_1) \;&=\; -\ln p_t(\psi\,\mathbf{e}_1) \;=\; \frac{\psi^2}{2\sigma(t)^2} \\ &\quad - \ln \int_{\mathbb{R}^N} p_{\text{data}}(\mathbf{x}_0) \exp\!\left(h(\psi)\,\mathbf{e}_1^\top \mathbf{x}_0 - \frac{\alpha(t)^2}{2\sigma(t)^2}\,\|\mathbf{x}_0\|^2 \right) \mathrm{d}^N x_0 + \frac{N}{2}\ln(2\pi\sigma(t)^2), \end{aligned} \tag{15.118}$$

where $h(\psi) \equiv \frac{\alpha(t)}{\sigma(t)^2}\psi$. As a function of h, the logarithm of the integral is, up to a constant, the cumulant generating function (Section 1.3) of the projection $X_1 \equiv \mathbf{e}_1^\top \mathbf{x}_0$ under the posterior $p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{0})$. Its fourth derivative at $h = 0$ is therefore the fourth cumulant $\langle X_1^4 \rangle_c = \langle X_1^4 \rangle - 3\langle X_1^2 \rangle^2 = -\kappa_4(t)$. Using the chain rule $\frac{\mathrm{d}}{\mathrm{d}\psi} = \frac{\alpha(t)}{\sigma(t)^2}\frac{\mathrm{d}}{\mathrm{d}h}$ four times, the quadratic term $\frac{\psi^2}{2\sigma(t)^2}$ vanishes at fourth order and we obtain

$$6u(t) \;=\; \frac{\mathrm{d}^4}{\mathrm{d}\psi^4} U_t(\psi\,\mathbf{e}_1)\Big|_{\psi=0} \;=\; -\left(\frac{\alpha(t)}{\sigma(t)^2}\right)^4 \frac{\mathrm{d}^4}{\mathrm{d}h^4} \ln Z_t(\psi\,\mathbf{e}_1)\Big|_{h=0} \;=\; \frac{\alpha(t)^4}{\sigma(t)^8}\,\kappa_4(t). \tag{15.119}$$

Dividing by 6 gives $u(t)$ in Eq. (15.115). Minimizing the truncated quartic polynomial $\frac{r(t)}{2}\psi^2 + \frac{u(t)}{4}\psi^4$ for $r(t) < 0$ yields $\psi(r(t) + u(t)\psi^2) = 0$, establishing Eq. (15.116) and the barrier height $\Delta U_t = -\frac{r(t)}{2}\frac{|r(t)|}{u(t)} - \frac{u(t)}{4}\frac{r(t)^2}{u(t)^2} = \frac{r(t)^2}{4u(t)}$. □

Proposition 15.14 shows how diffusion models avoid the Kramers bottleneck.

- A monolithic EBM samples p_0 directly. Its barrier $\Delta U_0 \gg 1$ is fully formed, and a trajectory started in one well waits a time $\tau_{\text{mix}} \sim e^{\Delta U_0}$ to cross $\psi = 0$.
- In reverse diffusion the trajectory reaches t_s^{H}, where $r = 0$ and the barrier vanishes, while it is still near $\psi = 0$. The noise $\sqrt{\beta(t)}\,\mathrm{d}\bar{\mathbf{W}}_{\bar{t}}$ sends it into the $\psi > 0$ or $\psi < 0$ basin while the wells are still shallow, with the probabilities of the corresponding modes. By the time ΔU_t is large, near $t = 0$, the trajectory is already inside its basin and never has to cross the barrier.

15.4.3 Regimes II and III: Random Energy Model Condensation and the Collapse Time t_c

In practice the model is trained on a finite dataset of M examples $\{\mathbf{x}_0^{(\mu)}\}_{\mu=1}^M$ drawn independently from $p_{\text{data}}(\mathbf{x}_0)$, corresponding to the empirical distribution

$$p_{\text{data}}^{\text{emp}}(\mathbf{x}_0) \;=\; \frac{1}{M}\sum_{\mu=1}^{M} \delta(\mathbf{x}_0 - \mathbf{x}_0^{(\mu)}). \tag{15.120}$$

If the reverse SDE were run with the exact empirical score $\nabla_{\mathbf{x}} \ln p_t^{\text{emp}}(\mathbf{x})$ all the way to $t = 0$, every trajectory would land on one of the M training points: pure memorization. The question is when a

trajectory stops producing new combinations of features (Regime II, $t_c < t < t_s$) and collapses onto a single training example (Regime III, $t < t_c$).

Following Biroli et al. [12], substitute the empirical distribution Eq. (15.120) into the posterior distribution Eq. (15.105). At time t, given the current noisy state $\mathbf{x}_t = \alpha(t)\mathbf{x}_0^{(1)} + \sigma(t)\boldsymbol{\epsilon}$ (generated from training sample $\mu = 1$, say), the posterior over the training indices $\mu \in \{1, \dots, M\}$ is a Boltzmann distribution with weights

$$w_\mu(\mathbf{x}_t) \;=\; \frac{\exp(-E_\mu(\mathbf{x}_t))}{\sum_{\nu=1}^{M} \exp(-E_\nu(\mathbf{x}_t))}, \qquad E_\mu(\mathbf{x}_t) \;\equiv\; \frac{\left\|\mathbf{x}_t - \alpha(t)\mathbf{x}_0^{(\mu)}\right\|^2}{2\sigma(t)^2}. \tag{15.121}$$

By Tweedie's formula Eq. (15.75), the empirical score is set by the posterior mean $\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t]_{\text{emp}} = \sum_{\mu=1}^{M} w_\mu(\mathbf{x}_t)\, \mathbf{x}_0^{(\mu)}$. The question is whether the weights spread over many training points or concentrate on one. For large N the energies are as follows.

- For the sample that generated $\mathbf{x}_t$ ($\mu = 1$), $\mathbf{x}_t - \alpha(t)\mathbf{x}_0^{(1)} = \sigma(t)\boldsymbol{\epsilon}$, so its effective energy is $E_1(\mathbf{x}_t) = \frac{1}{2}\|\boldsymbol{\epsilon}\|^2 \approx \frac{N}{2}$ by the law of large numbers in high dimension $N \gg 1$.
- For the other $M - 1$ samples, $\mu \geq 2$, the squared distance is

$$\left\|\mathbf{x}_t - \alpha(t)\mathbf{x}_0^{(\mu)}\right\|^2 = \left\|\sigma(t)\boldsymbol{\epsilon} + \alpha(t)(\mathbf{x}_0^{(1)} - \mathbf{x}_0^{(\mu)})\right\|^2 \approx N(\sigma(t)^2 + 2\alpha(t)^2)$$

  (assuming unit per-coordinate variance $\mathbb{E}[\|\mathbf{x}_0\|^2] = N$).

The sum $\sum_{\mu=2}^{M} e^{-E_\mu(\mathbf{x}_t)}$ over $M - 1$ random energies is an instance of Derrida's Random Energy Model (REM, Definition 3.20). Write $M = e^{N s_{\text{data}}}$, where $s_{\text{data}} = (\ln M)/N$ is the entropy of the dataset per dimension. The competitors $\mu \geq 2$ dominate the posterior as long as their entropy $\ln M$ outweighs their energy disadvantage relative to E_1. The posterior condenses onto $\mu = 1$ ($w_1 \to 1$) once the free energy $-\ln \sum_{\mu \geq 2} e^{-E_\mu}$ of the competitors rises above E_1.

Evaluating the log-partition function of the competitors for Gaussian data gives the *collapse* (condensation) *transition* at $t = t_c$ [12]:

$$\frac{1}{2}\ln\left(1 + \frac{\alpha(t_c)^2}{\sigma(t_c)^2}\right) \;=\; \frac{\ln M}{N} \;=\; s_{\text{data}} \quad \Longrightarrow \quad \frac{\alpha(t_c)^2}{\sigma(t_c)^2} \;=\; e^{2\frac{\ln M}{N}} - 1. \tag{15.122}$$

For the constant-rate VP SDE ($\alpha(t)^2/\sigma(t)^2 = (e^t - 1)^{-1}$), Eq. (15.122) yields the collapse time

$$t_c \;=\; \ln\left(1 + \frac{1}{e^{2(\ln M)/N} - 1}\right) \;\approx\; \ln \frac{N}{2 \ln M} \qquad \left(\text{when } \frac{\ln M}{N} \ll 1\right). \tag{15.123}$$

As $M \to \infty$ at fixed N, $t_c \to 0$, and the generalization regime (t_c, t_s) extends over the whole trajectory after speciation.

15.5 The Energy Cost of Digital Score-Based Diffusion

Diffusion models avoid Kramers trapping because the reverse trajectory commits to a mode at t_s, where the barrier vanishes. Running the reverse SDE on a digital processor, however, costs a great deal of energy [52, 56].

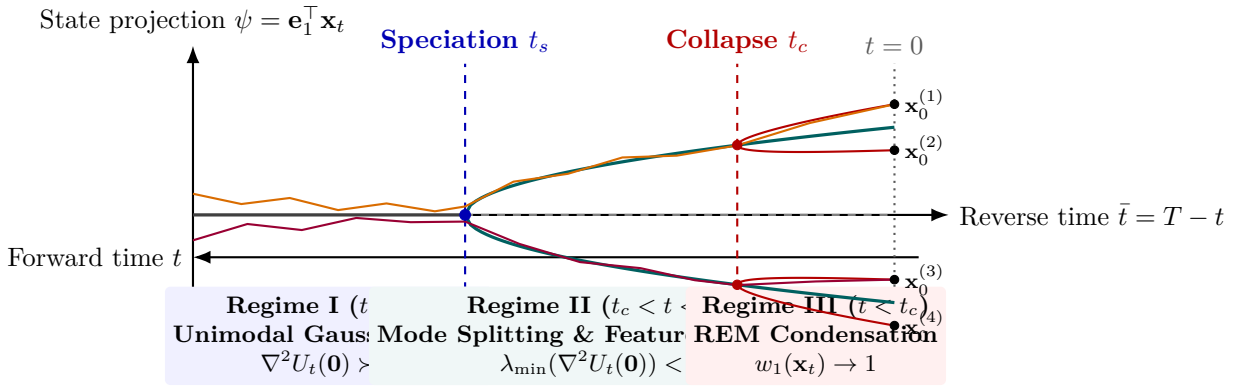


Figure 15.1: Dynamical phase diagram of reverse diffusion trajectories [12]. Starting at $\bar{t} = 0$ ($t = T$) from the unimodal Gaussian prior $\mathcal{N}(\mathbf{0}, \mathbf{I})$, trajectories first drift in a globally convex harmonic potential (Regime I). At the speciation time $t = t_s$, $\lambda_{\min}(\nabla^2 U_t(\mathbf{0})) = 0$ and the potential undergoes a pitchfork bifurcation into macroscopic mode valleys (Regime II) while the separating barrier is still $\mathcal{O}(k_{\mathrm{B}}T)$. For an empirical dataset of M samples, a second Random Energy Model condensation occurs at the collapse time $t = t_c$ (Regime III).

15.5.1 Discretization Error and Step Count

To simulate the reverse SDE Eq. (15.43) on a digital computer using a learned score network $\mathbf{s}_\theta(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \ln p_t(\mathbf{x})$, one discretizes time into K sequential steps $T = t_K > t_{K-1} > \cdots > t_1 > t_0 = 0$ with step sizes $\Delta t_k = t_k - t_{k-1}$. Under the standard Euler–Maruyama discretization (or the ancestral DDPM sampler [51]), each reverse transition $p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k)$ is a conditional Gaussian:

$$\mathbf{x}_{k-1} \;=\; \mathbf{x}_k + \Big[-\mathbf{f}(\mathbf{x}_k, t_k) + g(t_k)^2\, \mathbf{s}_\theta(\mathbf{x}_k, t_k)\Big]\Delta t_k + g(t_k)\sqrt{\Delta t_k}\, \boldsymbol{\xi}_k, \qquad \boldsymbol{\xi}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{15.124}$$

which corresponds to the Gaussian transition kernel

$$p_\theta^{\mathrm{Euler}}(\mathbf{x}_{k-1} \mid \mathbf{x}_k) \;=\; \mathcal{N}\Big(\mathbf{x}_{k-1};\, \mathbf{x}_k + [-\mathbf{f}(\mathbf{x}_k, t_k) + g(t_k)^2 \mathbf{s}_\theta(\mathbf{x}_k, t_k)]\Delta t_k,\; g(t_k)^2 \Delta t_k\, \mathbf{I}\Big)\,. \tag{15.125}$$

Each step is an isotropic Gaussian whose mean is shifted along the score $\mathbf{s}_\theta(\mathbf{x}_k, t_k)$ evaluated at the single point $\mathbf{x}_k$. This is accurate only when Δt_k is small enough that the score is nearly constant over the step, whose size is $\|\mathbf{x}_{k-1} - \mathbf{x}_k\| \sim g(t_k)\sqrt{\Delta t_k}$. Taylor-expanding $\ln p_{t_{k-1}}(\mathbf{x}_{k-1})$ around $\mathbf{x}_k$ shows that neglecting the second-order Hessian curvature term $\frac{1}{2}(\mathbf{x}_{k-1} - \mathbf{x}_k)^\top \nabla^2 \ln p_t(\mathbf{x}_k)(\mathbf{x}_{k-1} - \mathbf{x}_k)$ incurs a single-step Kullback–Leibler discretization error of order

$$D_{\mathrm{KL}}\Big(q(\mathbf{x}_{k-1} \mid \mathbf{x}_k) \,\Big\|\, p_\theta^{\mathrm{Euler}}(\mathbf{x}_{k-1} \mid \mathbf{x}_k)\Big) \;=\; \mathcal{O}\Big(g(t_k)^4 \,\Big\|\nabla_{\mathbf{x}}^2 \ln p_{t_k}\Big\|^2 \Delta t_k^2\Big)\,. \tag{15.126}$$

Summing over $K = T/\Delta t$ steps and using Girsanov's theorem (Section 4.5) gives a bound on the total error:

$$D_{\mathrm{KL}}\Big(p_{\mathrm{data}} \,\Big\|\, p_\theta^{(0)}\Big) \;\le\; D_{\mathrm{KL}}(p_T \,\|\, \pi) + \frac{1}{2}\int_0^T g(t)^2\, \mathbb{E}_{p_t}\Big[\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla \ln p_t(\mathbf{x}_t)\|^2\Big]\, \mathrm{d}t + \mathcal{O}\left(\frac{L_{\mathrm{score}}^2 N T^2}{K}\right), \tag{15.127}$$

where $L_{\mathrm{score}} = \sup_{t,\mathbf{x}} \|\nabla_{\mathbf{x}}^2 \ln p_t(\mathbf{x})\|_2$ is the Lipschitz constant of the score field. With large steps ($K \sim$ 1–5), the true reverse transition $q(\mathbf{x}_{k-1} \mid \mathbf{x}_k)$ is strongly non-Gaussian, and multimodal near t_s, and the Gaussian step Eq. (15.125) puts probability mass in the low-density regions between modes. Standard samplers of this type therefore use $K \sim$ 50–1000 sequential network evaluations per sample.

15.5.2 Energy Cost of Digital Score Evaluation

We now estimate the energy a GPU spends on K steps of Eq. (15.124). Each step evaluates the score network $\mathbf{s}_\theta(\mathbf{x}_k, t_k)$, typically a U-Net or diffusion transformer with $P \sim 10^8$–10^{10} weights, and draws N Gaussian variates $\boldsymbol{\xi}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

1. *Multiply–accumulate (MAC) operations.* A network with P parameters applied at N_{tokens} spatial positions needs $N_{\text{MAC}} \approx P N_{\text{tokens}}$ MACs (about $2PN_{\text{tokens}}$ floating-point operations) per step. As estimated in the Preface and in Section 17.4.1, following [52], switching CMOS logic at $V_{\text{DD}} \sim 0.7\,\text{V} \approx 27\,V_T$ and fetching weights from on-chip and high-bandwidth memory costs $E_{\text{MAC}} \sim 0.5$–$5\,\text{pJ} \approx 10^8$–$10^9\,k_{\text{B}}T$ per MAC.

2. *Pseudorandom numbers.* To produce the noise increment $g(t_k)\sqrt{\Delta t_k}\,\boldsymbol{\xi}_k$, a GPU runs a counter-based generator such as Philox and then a Box–Muller transform, $\sqrt{-2\ln U_1}\cos(2\pi U_2)$. A rough estimate is 10–$100\,\text{pJ} \approx 2.5 \times 10^9$–$2.5 \times 10^{10}\,k_{\text{B}}T$ per Gaussian variate, spent to imitate the thermal noise that a resistor or subthreshold transistor at 300 K produces with no computation at all.

Over $K \sim 10^2$–10^3 steps, for representative network sizes, the MAC energy for one sample is of order

$$E^{\text{GPU}}_{\text{sample}} \;=\; K \cdot N_{\text{MAC}} \cdot E_{\text{MAC}} \;\sim\; 10^1\text{–}10^3\,\text{J} \;\approx\; 2.5 \times 10^{21}\text{–}2.5 \times 10^{23}\,k_{\text{B}}T. \tag{15.128}$$

Hardware Realization: From digital score networks to physical denoising EBMs

Digital diffusion needs hundreds of network evaluations because each step of Eq. (15.124) is a Gaussian. The network supplies only the posterior mean $\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_k]$ (through Tweedie's formula), so each transition $p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k)$ can only be a small Gaussian move.
Jelinčič et al. [56] proposed **Denoising Thermodynamic Models** (DTMs), which replace the Gaussian step by a *conditional energy-based model*

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \;=\; \frac{1}{Z_t(\mathbf{x}_t)} \exp\Big(-\beta\big[\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) + \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t)\big]\Big), \tag{15.129}$$

sampled by letting an analog CMOS or p-bit circuit relax to equilibrium. Here t indexes discrete denoising steps, $\mathcal{E}^f_{t-1}$ is the energy of the forward coupling, and $\mathcal{E}_{\theta,t-1}$ is learned. Chapter 16 shows two consequences.

(i) A conditional EBM, especially one with latent variables $\mathbf{z}_{t-1}$, can represent non-Gaussian, multimodal transitions $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$. Larger denoising steps are then possible without the Gaussian discretization error Eq. (15.126), so fewer steps are needed.

(ii) The forward coupling energy $\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) = \frac{\alpha_t}{2}\,\|\mathbf{x}_{t-1} - c_t\mathbf{x}_t\|^2$ (here α_t is a coupling stiffness, not the attenuation $\alpha(t)$ of this chapter) adds $\alpha_t\mathbf{I}$ to the Hessian of the conditional energy. If α_t exceeds the most negative curvature of $\mathcal{E}_{\theta,t-1}$ by a margin $\kappa > 0$, the total energy is κ-strongly convex, and each step relaxes in a time of order $1/\kappa$ (in units set by the circuit mobility) with no Kramers barrier.

The two effects pull against each other: a large stiffness guarantees fast relaxation but limits the size of each step (Section 16.3).

15.6 Worked Computation: Bimodal Diffusion and Speciation

Worked computation 15.15 (Diffusion of a bimodal ensemble)**.** We solve the forward and reverse VP diffusion exactly for the double-well sampler in its simplest form, a one-dimensional symmetric mixture of two Gaussians,

$$\begin{aligned} p_{\mathrm{data}}(x_0) \;&=\; \frac{1}{2}\mathcal{N}(x_0;\,+\mu_0,\sigma_0^2)+\frac{1}{2}\mathcal{N}(x_0;\,-\mu_0,\sigma_0^2) \\ &=\; \frac{1}{2\sqrt{2\pi\sigma_0^2}}\left[\exp\!\left(-\frac{(x_0-\mu_0)^2}{2\sigma_0^2}\right)+\exp\!\left(-\frac{(x_0+\mu_0)^2}{2\sigma_0^2}\right)\right], \end{aligned} \tag{15.130}$$

where $\mu_0>0$ is the half-separation of the two wells and $\sigma_0\geq 0$ is the thermal width of each. In the limit $\sigma_0\to 0^+$ it reduces to the two-point distribution of a p-bit with states $\pm\mu_0$,

$$p_{\mathrm{data}}^{\mathrm{Ising}}(x_0) \;=\; \frac{1}{2}\delta(x_0-\mu_0)+\frac{1}{2}\delta(x_0+\mu_0), \tag{15.131}$$

the deep-barrier limit ($\beta\Delta E\gg 1$) of the quartic double well $E(x)=\frac{a}{4}(x^2-\mu_0^2)^2$. The variance of p_{data} is $\mathrm{Var}_{p_{\mathrm{data}}}(x_0)=\mu_0^2+\sigma_0^2$. We evolve x_t under the unit-rate VP SDE (Eq. (15.6) with $\beta(t)=1$), writing $\alpha_t\equiv\alpha(t)$ and $\sigma_t\equiv\sigma(t)$:

$$\mathrm{d}x_t \;=\; -\frac{1}{2}x_t\,\mathrm{d}t+\mathrm{d}W_t, \qquad \alpha(t)=e^{-t/2}, \qquad \sigma(t)^2=1-e^{-t}. \tag{15.132}$$

Step 1: Forward marginal and effective potential. By Eq. (15.19), the marginal density $p_t(x)$ is the convolution of $p_{\mathrm{data}}(x_0)$ with $q_{t|0}(x\mid x_0)=\mathcal{N}(x;\alpha_t x_0,\sigma_t^2)$. Since the convolution of two Gaussians $\mathcal{N}(\alpha_t(\pm\mu_0),\alpha_t^2\sigma_0^2)$ and $\mathcal{N}(0,\sigma_t^2)$ is a Gaussian with mean $\pm\alpha_t\mu_0$ and combined variance

$$s_t^2 \;\equiv\; \sigma_t^2+\alpha_t^2\sigma_0^2 \;=\; 1-e^{-t}+\sigma_0^2e^{-t} \;=\; 1+(\sigma_0^2-1)e^{-t}, \tag{15.133}$$

the marginal density for $t>0$ is

$$\begin{aligned} p_t(x) \;&=\; \frac{1}{2\sqrt{2\pi s_t^2}}\left[\exp\!\left(-\frac{(x-\alpha_t\mu_0)^2}{2s_t^2}\right)+\exp\!\left(-\frac{(x+\alpha_t\mu_0)^2}{2s_t^2}\right)\right] \\ &=\; \frac{1}{2\sqrt{2\pi s_t^2}}\exp\!\left(-\frac{x^2+\alpha_t^2\mu_0^2}{2s_t^2}\right)\left[\exp\!\left(+\frac{\alpha_t\mu_0 x}{s_t^2}\right)+\exp\!\left(-\frac{\alpha_t\mu_0 x}{s_t^2}\right)\right] \\ &=\; \frac{1}{\sqrt{2\pi s_t^2}}\exp\!\left(-\frac{x^2+\alpha_t^2\mu_0^2}{2s_t^2}\right)\cosh\!\left(\frac{\alpha_t\mu_0 x}{s_t^2}\right). \end{aligned} \tag{15.134}$$

Its negative logarithm is the effective potential $U_t(x)=-\ln p_t(x)$:

$$U_t(x) \;=\; \frac{x^2}{2s_t^2}-\ln\cosh\!\left(\frac{\alpha_t\mu_0 x}{s_t^2}\right)+\frac{\alpha_t^2\mu_0^2}{2s_t^2}+\frac{1}{2}\ln(2\pi s_t^2). \tag{15.135}$$

This has the same structure as the marginal free energy of the Gaussian–Bernoulli model of Worked computation 1.19 and of the hybrid cell Eq. (14.27): the quadratic confinement $\frac{x^2}{2s_t^2}$ competes with the non-convex term $-\ln\cosh(\frac{\alpha_t\mu_0 x}{s_t^2})$ produced by the two modes at $\pm\mu_0$.

Step 2: Score and Tweedie's formula. Differentiating $\ln p_t(x) = -U_t(x)$ gives the score $s(x,t)$ (not to be confused with the width s_t):

$$s(x,t) \;=\; \frac{\partial \ln p_t(x)}{\partial x} \;=\; -\frac{\partial U_t(x)}{\partial x} \;=\; -\frac{x}{s_t^2} + \frac{\alpha_t \mu_0}{s_t^2} \tanh\!\left(\frac{\alpha_t \mu_0 x}{s_t^2}\right). \tag{15.136}$$

We check Tweedie's formula Eq. (15.75) in the two-point limit $\sigma_0 = 0$, where $s_t = \sigma_t$. Given $x_t = x$, the posterior probabilities of the two states are

$$w_\pm(x) \;=\; \frac{\exp\!\left(-\frac{(x \mp \alpha_t \mu_0)^2}{2\sigma_t^2}\right)}{\exp\!\left(-\frac{(x-\alpha_t\mu_0)^2}{2\sigma_t^2}\right) + \exp\!\left(-\frac{(x+\alpha_t\mu_0)^2}{2\sigma_t^2}\right)} \;=\; \frac{\exp\!\left(\pm\frac{\alpha_t \mu_0 x}{\sigma_t^2}\right)}{2\cosh\!\left(\frac{\alpha_t \mu_0 x}{\sigma_t^2}\right)}. \tag{15.137}$$

The posterior mean (the MMSE denoiser) is

$$\mathbb{E}[x_0 \mid x_t = x] \;=\; (+\mu_0)\, w_+(x) + (-\mu_0)\, w_-(x) \;=\; \mu_0 \tanh\!\left(\frac{\alpha_t \mu_0 x}{\sigma_t^2}\right), \tag{15.138}$$

and the posterior variance is

$$\mathrm{Var}(x_0 \mid x_t = x) \;=\; \mathbb{E}\Big[x_0^2 \mid x_t = x\Big] - \big(\mathbb{E}[x_0 \mid x_t = x]\big)^2 \;=\; \mu_0^2 - \mu_0^2 \tanh^2\!\left(\frac{\alpha_t \mu_0 x}{\sigma_t^2}\right) \;=\; \mu_0^2 \,\mathrm{sech}^2\!\left(\frac{\alpha_t \mu_0 x}{\sigma_t^2}\right). \tag{15.139}$$

Substituting Eq. (15.138) into Tweedie's formula $\frac{\alpha_t \mathbb{E}[x_0 | x_t = x] - x}{\sigma_t^2}$ reproduces Eq. (15.136) with $s_t = \sigma_t$.

Step 3: Curvature, speciation time, and pitchfork bifurcation. Differentiating once more, with $\frac{\mathrm{d}}{\mathrm{d}z}\tanh z = \mathrm{sech}^2 z$, gives the curvature of the effective potential:

$$\frac{\partial^2 U_t(x)}{\partial x^2} \;=\; -\frac{\partial s(x,t)}{\partial x} \;=\; \frac{1}{s_t^2}\left[1 - \frac{\alpha_t^2 \mu_0^2}{s_t^2}\,\mathrm{sech}^2\!\left(\frac{\alpha_t \mu_0 x}{s_t^2}\right)\right]. \tag{15.140}$$

For $\sigma_0 = 0$, combined with Eq. (15.139), this is the second-order Tweedie identity Eq. (15.76): $\frac{\partial^2 U_t(x)}{\partial x^2} = \frac{1}{\sigma_t^2} - \frac{\alpha_t^2}{\sigma_t^4}\,\mathrm{Var}(x_0 \mid x_t = x)$.

Since $\mathrm{sech}^2 z \le 1$ with equality only at $z = 0$, the curvature is smallest at $x = 0$:

$$\min_{x \in \mathbb{R}} \frac{\partial^2 U_t(x)}{\partial x^2} \;=\; \left.\frac{\partial^2 U_t(x)}{\partial x^2}\right|_{x=0} \;=\; \frac{1}{s_t^2}\left(1 - \frac{\alpha_t^2 \mu_0^2}{s_t^2}\right). \tag{15.141}$$

By Definition 15.13, speciation occurs when this curvature vanishes. Because the mixture is solved exactly, this is the Hessian speciation time t_s^{H}, which we write simply as t_s here. The condition is

$$\alpha(t_s)^2 \mu_0^2 \;=\; s(t_s)^2 \;=\; \sigma(t_s)^2 + \alpha(t_s)^2 \sigma_0^2. \tag{15.142}$$

With $\alpha(t_s)^2 = e^{-t_s}$ and $\sigma(t_s)^2 = 1 - e^{-t_s}$,

$$e^{-t_s}\mu_0^2 \;=\; 1 - e^{-t_s} + e^{-t_s}\sigma_0^2 \quad\Longrightarrow\quad e^{-t_s}(1 + \mu_0^2 - \sigma_0^2) \;=\; 1 \quad\Longrightarrow\quad t_s \;=\; \ln(1 + \mu_0^2 - \sigma_0^2). \tag{15.143}$$

The Gaussian approximation Eq. (15.112), with $\lambda_{\max} = \mu_0^2 + \sigma_0^2$, would give $\ln(1 + \mu_0^2 + \sigma_0^2)$; the two agree when $\sigma_0 \ll \mu_0$. Two cases arise.

1. *Separated modes ($\mu_0 > \sigma_0$).* The speciation time $t_s = \ln(1 + \mu_0^2 - \sigma_0^2)$ is positive.

- For $t > t_s$ (early in the reverse process), $\alpha_t\mu_0 < s_t$, so $\frac{\partial^2 U_t(x)}{\partial x^2} > 0$ for every x: U_t is strictly convex on all of $\mathbb{R}$, with its only minimum at $x = 0$.
- At $t = t_s$, $\alpha(t_s)\mu_0 = s(t_s)$: the quadratic curvature at $x = 0$ vanishes, and Taylor-expanding $\ln\cosh(z) = \frac{1}{2}z^2 - \frac{1}{12}z^4 + \mathcal{O}(z^6)$ in Eq. (15.135) gives the quartic flat bottom

$$U_{t_s}(x) - U_{t_s}(0) \;=\; \frac{1}{12\, s(t_s)^4}\, x^4 + \mathcal{O}(x^6). \tag{15.144}$$

- For $0 < t < t_s$ (late in the reverse process), $\alpha_t\mu_0 > s_t$, so the curvature at $x = 0$ is negative. The origin becomes a barrier top, and two symmetric minima $\pm x^*(t)$ appear as the nonzero roots of $s(x^*(t), t) = 0$:

$$x^*(t) \;=\; \alpha_t\mu_0 \tanh\!\left(\frac{\alpha_t\mu_0\, x^*(t)}{s_t^2}\right). \tag{15.145}$$

  Near $t \lesssim t_s$, expanding $\tanh(z) \approx z - \frac{1}{3}z^3$ in Eq. (15.145) shows that the minima grow with the mean-field exponent $1/2$:

$$x^*_{\pm}(t) \;\approx\; \pm\frac{\sqrt{3}\, s_t^2}{\alpha_t\mu_0}\sqrt{\frac{\alpha_t^2\mu_0^2}{s_t^2} - 1} \;\propto\; \pm\sqrt{t_s - t}. \tag{15.146}$$

2. *Overlapping modes ($\mu_0 \le \sigma_0$).* Even at $t = 0$ the two components overlap so much that p_{data} is unimodal, with positive curvature of $-\ln p_{\text{data}}$ at the origin. Then $t_s \le 0$ and no bifurcation occurs.

Step 4: Probability flow trajectories. Now solve the probability flow ODE Eq. (15.70) in the two-point limit $\sigma_0 = 0$ ($s_t = \sigma_t$). Substituting $f(x,t) = -\frac{1}{2}x$, $g(t)^2 = 1$, and the exact score Eq. (15.136) into Eq. (15.70) gives

$$\frac{\mathrm{d}x_t}{\mathrm{d}t} \;=\; -\frac{1}{2}x_t - \frac{1}{2}s(x_t,t) \;=\; -\frac{1}{2}x_t + \frac{x_t}{2\sigma_t^2} - \frac{\alpha_t\mu_0}{2\sigma_t^2}\tanh\!\left(\frac{\alpha_t\mu_0 x_t}{\sigma_t^2}\right). \tag{15.147}$$

This nonlinear, time-dependent ODE can be solved without integrating it directly. In one dimension, a deterministic flow that transports p_t (Proposition 15.8) preserves quantiles: trajectories cannot cross, so the probability on either side of a trajectory cannot change. By the symmetry $p_t(-x) = p_t(x)$, the origin is itself a trajectory, so the probability between 0 and x_t is constant:

$$F_t(x_t) \;\equiv\; \int_0^{x_t} p_t(u)\,\mathrm{d}u \;=\; \frac{1}{4}\left[\operatorname{erf}\!\left(\frac{x_t - \alpha_t\mu_0}{\sqrt{2}\,\sigma_t}\right) + \operatorname{erf}\!\left(\frac{x_t + \alpha_t\mu_0}{\sqrt{2}\,\sigma_t}\right)\right] \;=\; \text{const} \;=\; \frac{1}{2}\operatorname{erf}\!\left(\frac{x_\infty}{\sqrt{2}}\right), \tag{15.148}$$

where $x_\infty = \lim_{t\to\infty} x_t \sim \mathcal{N}(0,1)$ is the initial sample drawn from the standard Gaussian prior at $t = T \to \infty$ ($\alpha_T \to 0, \sigma_T \to 1$). To verify this directly, integrate the continuity equation $\frac{\partial p_t}{\partial t} + \frac{\partial}{\partial x}(v_t p_t) = 0$ from 0 to x, using $v_t(0) = 0$ by symmetry: $\frac{\partial F_t(x)}{\partial t} = \int_0^x \frac{\partial p_t(u)}{\partial t}\,\mathrm{d}u = -v_t(x)p_t(x)$. Along a curve with $F_t(x_t)$ constant, $0 = \frac{\mathrm{d}}{\mathrm{d}t}F_t(x_t) = \frac{\partial F_t}{\partial t} + p_t(x_t)\frac{\mathrm{d}x_t}{\mathrm{d}t} = p_t(x_t)\big[\frac{\mathrm{d}x_t}{\mathrm{d}t} - v_t(x_t)\big]$, so $\frac{\mathrm{d}x_t}{\mathrm{d}t} = v_t(x_t)$.

Now let $t \to 0^+$, so that $\alpha_t \to 1$ and $\sigma_t \to 0^+$.

- Any prior sample with $x_\infty > 0$ has $F_t(x_t) = \frac{1}{2}\operatorname{erf}(x_\infty/\sqrt{2}) \in (0, 1/2)$. As $\sigma_t \to 0^+$ and $\alpha_t \to 1$, $\operatorname{erf}\big(\frac{x_t+\alpha_t\mu_0}{\sqrt{2}\sigma_t}\big) \to +1$, so Eq. (15.148) becomes

$$\frac{1}{4}\left[\operatorname{erf}\left(\frac{x_t-\alpha_t\mu_0}{\sqrt{2}\,\sigma_t}\right)+1\right] \;\approx\; \frac{1}{2}\operatorname{erf}\left(\frac{x_\infty}{\sqrt{2}}\right)$$

$$\Longrightarrow \quad x_t \;=\; \alpha_t\mu_0+\sqrt{2}\,\sigma_t\,\operatorname{erf}^{-1}\left(2\operatorname{erf}\left(\frac{x_\infty}{\sqrt{2}}\right)-1\right) \;\xrightarrow{t\to 0^+}\; +\mu_0. \tag{15.149}$$

- Symmetrically, every prior sample with $x_\infty < 0$ converges to $-\mu_0$. The single trajectory $x_\infty = 0$, which has probability zero, stays at the unstable point $x_t = 0$.

The reverse flow thus maps the positive half-line of the Gaussian prior onto the $+\mu_0$ mode and the negative half-line onto the $-\mu_0$ mode, each with probability $\frac{1}{2}$. It produces exact samples of the bimodal distribution, and no trajectory ever crosses the barrier that separates the modes at $t = 0$.

Summary

- **Proved:**
  - (i) Gaussian transition kernels of linear forward SDEs, including the VP and VE cases and coupled RC networks (Propositions 15.2 and 15.4).
  - (ii) Exponential convergence of the VP marginal to $\mathcal{N}(\mathbf{0}, \mathbf{I})$ in Wasserstein distance and KL divergence (Theorem 15.3).
  - (iii) Anderson's reverse-time theorem (Theorem 15.6), the reversal of the probability current, and the entropy production of forward noising (Proposition 15.7).
  - (iv) The probability flow ODE, Tweedie's formulas, the equivalence of explicit, implicit, and denoising score matching, and the Girsanov bound on the error of a learned reverse process (Propositions 15.8 and 15.9 and Theorems 15.10 and 15.12).
  - (v) The Hessian criterion and Landau expansion at speciation (Proposition 15.14), and the exact solution of bimodal diffusion (Worked computation 15.15).
- **Assumed:**
  - (i) Coefficients smooth enough for well-posed SDEs with smooth, strictly positive marginals p_t for $t > 0$ (Chapters 4 and 6).
  - (ii) Enough decay at infinity for boundary terms in integrations by parts to vanish.
  - (iii) The collapse condition Eq. (15.122), quoted from [12] for Gaussian data rather than derived here; the GPU energy figures are representative estimates.
- **Taken up later:**
  - (i) Denoising Thermodynamic Models, which replace digital score evaluations with fast-mixing conditional EBMs (Chapter 16), and their hardware energy budget (Chapter 17).
  - (ii) Reverse-time SDEs under weaker regularity assumptions on p_t (Haussmann and Pardoux [49], Föllmer [35]) are not treated in this book.
- **Open:**
  - (i) The location and universality of hierarchical speciation transitions for data with ultrametric or power-law covariance spectra.
  - (ii) Noise schedules $\beta(t)$ that minimize the entropy produced during noising near t_s (compare the optimal-transport bounds of Chapter 10).

Bibliographic notes

The time reversal of diffusion processes was established by Anderson [4], building on earlier martingale and dual Markov process formulations by Nelson, Nagasawa, and Kunita; see also Pavliotis [85], Risken [90], and Øksendal [80] for textbook treatments of forward and backward Kolmogorov operators. The connection between score matching and denoising autoencoders was proved by Vincent [115], extending Hyvärinen's implicit score matching identity, while Tweedie's formula originates in empirical Bayes estimation.

Diffusion-based generative modeling was introduced by Sohl-Dickstein et al. [101], who drew explicitly on non-equilibrium thermodynamics and the forward–reverse path comparisons of Jarzynski and Crooks (Chapters 8 and 9). Ho et al. [51] established the modern Denoising Diffusion Probabilistic Model (DDPM) parameterization, and Song et al. [103] placed discrete DDPMs and score-matching Langevin dynamics in a common continuous-time framework of VP and VE SDEs and the probability flow ODE. The statistical physics of reverse trajectories, including the speciation transition t_s and the Random Energy Model collapse transition t_c, is due to Biroli et al. [12]. The critique of digital GPU diffusion energetics and the formulation of Denoising Thermodynamic Models as a hardware-native alternative follow Jelinčič et al. [56] and Horowitz [52].

Chapter 16

Denoising Thermodynamic Models and the Expressivity–Mixing Dilemma

A single Energy-Based Model $p_\theta(\mathbf{x}) \propto e^{-\beta E_\theta(\mathbf{x})}$ that represents a multimodal data distribution must contain deep wells separated by barriers. Section 3.5 made this precise as the expressivity–mixing dilemma, Section 6.5 showed that the Bakry–Émery curvature condition fails wherever the energy is non-convex, and Chapter 7 computed the price: the Kramers escape time grows as $e^{\beta \Delta E}$. A physical sampler that relaxes under Langevin or Glauber dynamics in such a landscape spends almost all of its time trapped inside one well. Faster hardware does not change this exponential; it only changes the prefactor.

Diffusion models (Chapter 15) sidestep the problem by never asking a sampler to cross a high barrier. A forward noising process gradually turns the data into Gaussian noise, and generation runs that process backward in many small steps. In a digital diffusion model each backward step is a Gaussian move whose mean is computed by a large neural network (Section 15.5). Jelinčič, Lockwood, Garlapati, Chuang, Verdon, and McCourt [56] proposed a different implementation: make each backward step a small Energy-Based Model, and draw its sample by letting a physical stochastic circuit relax to equilibrium. They call the resulting generative model a **Denoising Thermodynamic Model** (DTM). The deterministic network is replaced by a Boltzmann distribution, and the pseudo-random Gaussian noise is replaced by the thermal noise of the hardware.

This chapter develops the theory of DTMs. After stating the model and what the paper of [56] does and does not claim, we identify the energy each denoising step must learn and then prove the main result (Section 16.3): if the stiffness α_t of the forward coupling exceeds the worst negative curvature of the learned energy by a margin κ, every step is κ-strongly log-concave and relaxes at rate at least $\mu\kappa$, with no Kramers barrier. The price is that the stiffness caps the step size, so one exponentially hard sampling problem is traded for many easy ones. The chapter ends with the double-well sampler, the quartic $E_0(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2$ used since Chapter 1, as the data distribution of a DTM (Worked computation 16.27), and compares the resulting chain with direct Langevin sampling limited by the Kramers rate of Worked computation 7.16. The hardware architecture and energy budget follow in Chapter 17.

16.1 Denoising as a Chain of Physical Energy-Based Models

16.1.1 From one hard distribution to many easy conditionals

Let $p_{\text{data}}(\mathbf{x}_0)$ be a data distribution on a state space $\mathcal{X}$, which may be $\mathbb{R}^N$ (continuous voltages, as in the RC sampler, the coupled linear RC circuit of Chapter 1), $\{-1,+1\}^N$ (binary p-bits, the spin limit of the double-well sampler), or $\{1,\dots,M\}^N$ (Potts variables). A *monolithic* EBM tries to write $p_{\text{data}}(\mathbf{x}_0) \approx Z_\theta^{-1} e^{-\beta E_\theta(\mathbf{x}_0)}$ with a single energy function. The paper [56] names the difficulty this creates the mixing–expressivity tradeoff: as the modelling quality of a monolithic EBM increases, its mixing time grows, which makes both sampling and training expensive. This is the same statement as Section 3.5, seen from the machine-learning side.

The idea of diffusion models [51, 101, 103] is to introduce a sequence of intermediate random variables $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T$ generated by a fixed, simple forward Markov chain

$$q(\mathbf{x}_{0:T}) = p_{\text{data}}(\mathbf{x}_0) \prod_{t=1}^{T} q(\mathbf{x}_t \mid \mathbf{x}_{t-1}), \tag{16.1}$$

chosen so that the last marginal $q(\mathbf{x}_T)$ is close to a simple reference distribution $\pi(\mathbf{x}_T)$ that is easy to sample. Generation then runs the chain backward: draw $\mathbf{x}_T \sim \pi$, then draw $\mathbf{x}_{T-1}$ from a model of $q(\mathbf{x}_{T-1} \mid \mathbf{x}_T)$, and so on down to $\mathbf{x}_0$. The generative model is the reverse Markov chain

$$p_\theta(\mathbf{x}_{0:T}) = \pi(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t), \tag{16.2}$$

and the generated sample is $\mathbf{x}_0$, distributed according to the marginal $p_\theta(\mathbf{x}_0) = \sum_{\mathbf{x}_{1:T}} p_\theta(\mathbf{x}_{0:T})$ (an integral for continuous variables).

The whole design question is how to parameterize the reverse conditionals $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$. A digital diffusion model uses a Gaussian with a neural-network mean (Section 15.5). A Gaussian is only accurate when the step is small enough that the true reverse conditional is close to Gaussian, which is why digital models use hundreds of steps. A DTM uses an Energy-Based Model instead.

Definition 16.1 (Denoising Thermodynamic Model)**.** Fix a forward chain Eq. (16.1) whose kernels are strictly positive and can be written as

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \frac{1}{Z_t^f} \exp\Big(-\beta\, \mathcal{E}_{t-1}^f(\mathbf{x}_{t-1}, \mathbf{x}_t)\Big), \tag{16.3}$$

where the *forward coupling energy* $\mathcal{E}_{t-1}^f$ is defined up to an additive constant and Z_t^f does not depend on $\mathbf{x}_{t-1}$. A *Denoising Thermodynamic Model* with T steps is a reverse chain Eq. (16.2) whose conditionals are Boltzmann distributions

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \frac{1}{Z_t(\mathbf{x}_t)} \exp\Big(-\beta\big[\mathcal{E}_{t-1}^f(\mathbf{x}_{t-1}, \mathbf{x}_t) + \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t)\big]\Big), \tag{16.4}$$

with learned energies $\mathcal{E}_{\theta,t-1}$ and conditional partition functions

$$Z_t(\mathbf{x}_t) = \int_{\mathcal{X}} \mathrm{d}\mathbf{x}_{t-1}\, \exp\Big(-\beta\big[\mathcal{E}_{t-1}^f(\mathbf{x}_{t-1}, \mathbf{x}_t) + \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t)\big]\Big). \tag{16.5}$$

In the latent-variable version (Section 16.4), the learned energy also depends on hidden variables $\mathbf{z}_{t-1}$ that are summed out.

	Monolithic EBM	Digital diffusion	DTM
Object per step	one energy $E_\theta(\mathbf{x})$	Gaussian with network mean	conditional EBM Eq. (16.4)
How a step is sampled	long Langevin or Gibbs run	one network evaluation plus pseudo-random noise	physical relaxation with clamp
Source of randomness	thermal or pseudo-random	pseudo-random	thermal noise of the circuit
Number of steps	one (very long)	10^2–10^3	few to tens
Per-step difficulty	exponential in barrier height	none (deterministic)	set by α_t versus curvature
Main cost	mixing time $\propto e^{\beta\Delta E}$	multiply–accumulate energy	stage count times stage time

Table 16.1: Three strategies for sampling a multimodal distribution. The DTM keeps the step structure of diffusion models but replaces each deterministic network evaluation by a short physical equilibration, whose difficulty is controlled by the stiffness of the forward coupling (Section 16.3).

Sampling from a DTM means running T physical relaxations in sequence. In stage t, the previous output $\mathbf{x}_t$ is held fixed (clamped) and acts as an external input; the active variables $\mathbf{x}_{t-1}$ relax under the energy $\mathcal{E}^f_{t-1} + \mathcal{E}_{\theta,t-1}$ until they are close to equilibrium; the result is read out and becomes the clamp for stage $t-1$. Two features of Eq. (16.4) matter for everything that follows.

(i) The forward energy $\mathcal{E}^f_{t-1}$ is not learned. It is fixed by the forward process, and in all cases treated below it is a sum of single pairwise couplings between $x_{t,i}$ and $x_{t-1,i}$. In hardware this is one coupling per node to a clamped input.

(ii) The learned energy only has to supply what the forward energy lacks. We show in Section 16.2 that, for the exact reverse process, the learned part is the energy of the marginal distribution at the previous noise level and does not depend on $\mathbf{x}_t$ at all.

16.1.2 Three ways to sample a multimodal distribution

It helps to place the DTM between the two approaches it combines. Table 16.1 compares a monolithic EBM sampled by Langevin or Gibbs dynamics, a digital diffusion model, and a DTM, for the task of drawing one sample from a multimodal p_{data}.

The table makes the logic of the chapter visible. The monolithic EBM has maximal expressivity per step and the worst mixing. The digital diffusion model has trivially easy steps but needs many of them, each paid for in digital arithmetic (Section 15.5). The DTM sits between: each step is a genuine sampling problem, but one that a stiff enough forward coupling keeps easy.

16.1.3 What the paper does

Because this chapter mixes a description of [56] with this book's own analysis, we state plainly what the paper contains.

- **Model.** The conditional is Eq. (16.4) with latent variables,

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \propto \sum_{\mathbf{z}_{t-1}} \exp\!\left(-\left[\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) + \mathcal{E}^\theta_{t-1}(\mathbf{x}_{t-1}, \mathbf{z}_{t-1}, \theta)\right]\right).$$

The learned energy depends on $\mathbf{x}_{t-1}$ and $\mathbf{z}_{t-1}$ but not on $\mathbf{x}_t$; the dependence on $\mathbf{x}_t$ enters only through the forward energy. Energies are dimensionless (the factor β is absorbed).

- **Variables and hardware.** The experiments use binary variables. Each step is a sparse Boltzmann machine on a square grid (typically 70×70 nodes, each connected to about 12 neighbors) whose graph is bipartite, so it can be sampled with two-color block Gibbs sampling (Section 2.5). The forward energy is a single pairwise coupling between corresponding variables of $\mathbf{x}_t$ and $\mathbf{x}_{t-1}$.
- **Training.** The objective is the Kullback–Leibler divergence between the forward and reverse joint distributions. Its gradient is a sum over steps of positive-phase minus negative-phase expectations, estimated by Gibbs sampling, and each term can be computed independently. A total-correlation penalty, with strength adjusted during training from the measured autocorrelation of the sampler (an *adaptive correlation penalty*), keeps the per-step samplers mixing.
- **Results.** DTMs with 2 to 8 steps are trained on binarized Fashion-MNIST, and image quality improves substantially as the number of steps increases. A system-level model of an all-transistor implementation estimates that such hardware could match a GPU on this benchmark using roughly 10^4 times less energy per sample. This is a model-based estimate, not a measurement on fabricated hardware.
- **Mixing argument.** The paper argues qualitatively that as the number of steps grows, each forward step adds less noise, the forward energy binds $\mathbf{x}_{t-1}$ more tightly to $\mathbf{x}_t$, and the reverse conditional becomes simpler and closer to unimodal. Its control of mixing in practice is empirical (autocorrelation measurements and the adaptive penalty).

The paper does not state a log-concavity or mixing-time theorem. The quantitative results in Sections 16.3 and 16.4 and in Worked computation 16.27 are this book's analysis of the continuous-variable version of the model. They make the paper's qualitative mixing argument precise in the regime where it can be made precise, and they also show what that regime costs.

Notation Watch: Symbols shared with Chapter 15 and with the DTM paper

Several letters are overloaded across the literature. In this chapter:

- $\beta = 1/k_{\mathrm{B}}T$ is always the inverse temperature, and all energies ($\mathcal{E}^f$, $\mathcal{E}_\theta$, E_0, U_t) carry physical units of energy. The DTM paper writes $e^{-\mathcal{E}}$ with dimensionless energies; its energies equal β times ours.
- α_t is the *stiffness* of the forward coupling, with units of energy per squared length, as in Section 6.5.4. In Chapter 15, $\alpha(t)$ denotes the signal attenuation factor of the forward SDE. To avoid the clash we write the per-step forward contraction as c_t^{-1}, the per-step forward noise variance as ν_t (the DDPM schedule β_t of Eq. (15.1)), and the cumulative retained signal power as $\bar{\gamma}_t = \prod_{s=1}^{t} c_s^{-2}$.
- $U_t(\mathbf{x}) = -k_{\mathrm{B}}T \ln q_t(\mathbf{x})$ is the *marginal energy* of the forward marginal q_t at step t, in energy units. Worked computation 15.15 uses the same letter for the dimensionless $-\ln p_t$, which is βU_t.
- μ is the mobility of the overdamped Langevin sampler, $\mathrm{d}\mathbf{x} = -\mu \nabla E \, \mathrm{d}t + \sqrt{2\mu k_{\mathrm{B}}T} \, \mathrm{d}\mathbf{W}$, as in Theorem 6.18 and Eq. (7.1).

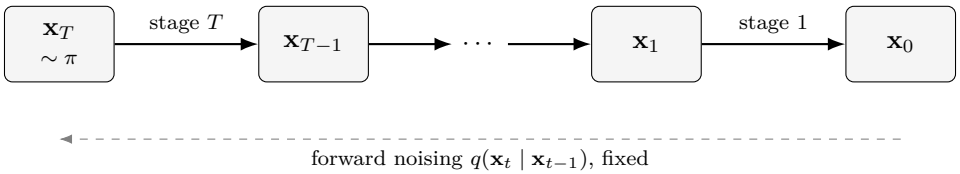


Figure 16.1: A Denoising Thermodynamic Model. Generation (solid arrows) starts from a sample of the simple reference distribution π and runs T conditional relaxations. In stage t the state $\mathbf{x}_t$ is clamped and the circuit holding $\mathbf{x}_{t-1}$ relaxes under the forward coupling energy $\mathcal{E}^f_{t-1}$ plus a learned energy $\mathcal{E}_{\theta,t-1}$. Training uses the fixed forward noising chain (dashed) to produce pairs $(\mathbf{x}_{t-1}, \mathbf{x}_t)$.

16.1.4 A DTM as a physical pipeline

Figure 16.1 shows the structure. Each box is a sampling circuit, the horizontal couplings are the forward energies, and the arrows show the order in which stages are executed during generation. Stage t needs only the output of stage $t+1$; it does not need to know anything about the data distribution at large. A trained DTM is therefore a stack of T small, local problems, each of which has to be easy for the hardware.

Hardware Realization: What a DTM asks of the hardware

A DTM stage needs three physical ingredients, all of which appeared in earlier chapters. First, a sampling fabric whose stationary distribution is a programmable Boltzmann distribution: Gaussian RC nodes (Chapter 13), p-bits (Chapter 14), or a mixture of both, coupled through a sparse graph that can be colored for parallel block updates (Section 2.5). Second, a clamp input on every visible node, implementing the forward energy: a programmable shunt conductance and injected current for a Gaussian node (Eq. (16.10)), or a bias field for a p-bit (Eq. (16.13)). Third, a way to pass the relaxed state of one stage to the clamp inputs of the next. The learned couplings of the T stages differ, so a chip either holds T separate tiles and passes samples between them, or reprograms one tile T times. The organization of these options in Extropic's Denoising Thermodynamic Computer Architecture, and the `THRML` software that simulates block Gibbs sampling on such fabrics [31], are described in Chapter 17. What this chapter supplies is the condition under which each stage is guaranteed to be fast enough that the clock period of the pipeline can be fixed in advance.

16.2 Forward Coupling Energies and the Exact Reverse Energy

We now compute $\mathcal{E}^f_{t-1}$ for the three variable types used in thermodynamic hardware, and then identify the energy that the learned term should reproduce.

16.2.1 Continuous variables: the quadratic spring

Take $\mathcal{X} = \mathbb{R}^N$ and a linear Gaussian forward step

$$\mathbf{x}_t = c_t^{-1}\mathbf{x}_{t-1} + \sqrt{\nu_t}\,\boldsymbol{\xi}_t, \qquad \boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \qquad c_t \geq 1,\ \nu_t > 0. \tag{16.6}$$

The variance-preserving (DDPM) case of Eq. (15.1) is $c_t^{-2} = 1 - \nu_t$; the variance-exploding case is $c_t = 1$. The transition density is

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = (2\pi\nu_t)^{-N/2} \exp\left(-\frac{\left\| \mathbf{x}_t - c_t^{-1}\mathbf{x}_{t-1} \right\|^2}{2\nu_t} \right). \tag{16.7}$$

To read off the forward energy as a function of the active variable $\mathbf{x}_{t-1}$, factor c_t^{-1} out of the norm:

$$\left\| \mathbf{x}_t - c_t^{-1}\mathbf{x}_{t-1} \right\|^2 = c_t^{-2} \left\| c_t\mathbf{x}_t - \mathbf{x}_{t-1} \right\|^2 = c_t^{-2} \left\| \mathbf{x}_{t-1} - c_t\mathbf{x}_t \right\|^2. \tag{16.8}$$

Comparing with Eq. (16.3), $\beta\mathcal{E}_{t-1}^f = \left\| \mathbf{x}_{t-1} - c_t\mathbf{x}_t \right\|^2 / (2c_t^2\nu_t)$, that is,

$$\boxed{\mathcal{E}_{t-1}^f(\mathbf{x}_{t-1}, \mathbf{x}_t) = \frac{\alpha_t}{2} \left\| \mathbf{x}_{t-1} - c_t\mathbf{x}_t \right\|^2, \qquad \alpha_t = \frac{k_\mathrm{B}T}{c_t^2\,\nu_t}.} \tag{16.9}$$

For the variance-preserving schedule, $\alpha_t = k_\mathrm{B}T(1-\nu_t)/\nu_t$. The forward energy is a harmonic spring of stiffness α_t tying each active coordinate $x_{t-1,i}$ to the rescaled clamp value $c_t x_{t,i}$. Small forward steps ($\nu_t \to 0$) mean stiff springs ($\alpha_t \to \infty$). The stiffness is fixed by the noise schedule; it is not a free hardware parameter once the schedule is chosen.

Expanding the square shows how the clamp enters the circuit:

$$\frac{\alpha_t}{2} \left\| \mathbf{x}_{t-1} - c_t\mathbf{x}_t \right\|^2 = \frac{\alpha_t}{2} \left\| \mathbf{x}_{t-1} \right\|^2 - \alpha_t c_t\, \mathbf{x}_t \cdot \mathbf{x}_{t-1} + \frac{\alpha_t c_t^2}{2} \left\| \mathbf{x}_t \right\|^2. \tag{16.10}$$

The first term is a diagonal self-stiffness on each active node, the same for every value of the clamp. The second is a linear field $\alpha_t c_t \mathbf{x}_t$ applied to the active nodes. The third is constant during the relaxation and drops out of the conditional distribution. In the RC sampler (Chapter 13), the self-stiffness is a shunt conductance to ground and the field is an injected current proportional to the clamped voltage.

16.2.2 Discrete variables: Ising and Potts couplings

For binary variables $\mathbf{x}_t \in \{-1, +1\}^N$, let the forward step flip each spin independently with probability $p_t \in (0, \frac{1}{2})$:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \prod_{i=1}^{N} \Big[(1-p_t)\, \delta_{x_{t,i}, x_{t-1,i}} + p_t\, (1 - \delta_{x_{t,i}, x_{t-1,i}}) \Big]. \tag{16.11}$$

For ± 1 spins, $\delta_{x,x'} = \frac{1}{2}(1 + xx')$. Each factor can therefore be written as an exponential of xx':

$$(1-p_t)^{\frac{1+xx'}{2}}\, p_t^{\frac{1-xx'}{2}} = \sqrt{p_t(1-p_t)}\, \exp\!\left(\frac{xx'}{2} \ln \frac{1-p_t}{p_t} \right). \tag{16.12}$$

Defining the dimensionless coupling $\Gamma_t := \ln\big((1-p_t)/p_t\big) > 0$ and dropping the constant prefactor,

$$\boxed{\mathcal{E}_{t-1}^f(\mathbf{x}_{t-1}, \mathbf{x}_t) = -\frac{k_\mathrm{B}T\,\Gamma_t}{2} \sum_{i=1}^{N} x_{t,i}\, x_{t-1,i}.} \tag{16.13}$$

This is a ferromagnetic coupling of strength $k_{\mathrm{B}}T\Gamma_t/2$ between each active spin and its clamped partner. Since $x_{t,i}$ is fixed during stage t, it acts on $x_{t-1,i}$ as a local field $h_i^f = \frac{1}{2}k_{\mathrm{B}}T\Gamma_t\, x_{t,i}$. Small steps ($p_t \to 0$) give $\Gamma_t \to \infty$.

For M-state Potts variables, let the forward step keep each value with probability $1-p_t$ and otherwise move it to one of the other $M-1$ values uniformly, so that a specific different value is reached with probability $p_t/(M-1)$. The single-site kernel can be written with the Kronecker delta $\delta = \delta_{x,x'} \in \{0,1\}$ as

$$(1-p_t)^{\delta}\Big(\frac{p_t}{M-1}\Big)^{1-\delta} = \frac{p_t}{M-1}\exp\!\Big(\delta\,\ln\frac{(M-1)(1-p_t)}{p_t}\Big),$$

which is checked by setting $\delta = 1$ and $\delta = 0$. Dropping the constant prefactor gives

$$\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1},\mathbf{x}_t) = -k_{\mathrm{B}}T\,\Gamma_t^{(M)}\sum_{i=1}^{N}\delta_{x_{t,i},x_{t-1,i}}, \qquad \Gamma_t^{(M)} = \ln\frac{(M-1)(1-p_t)}{p_t}, \tag{16.14}$$

which reduces to Eq. (16.13) for $M = 2$ up to a constant. If the forward step is generated by a continuous-time jump process run for a time interval in which the probability of remaining in the original state is $(1+(M-1)e^{-\gamma\Delta t})/M$, then $\Gamma_t^{(M)} = \ln\big[(1+(M-1)e^{-\gamma\Delta t})/(1-e^{-\gamma\Delta t})\big]$, which is the form used in [56].

Example 16.2 (Exact reverse step for a single biased p-bit)**.** Let the data be a single spin with $p_{\mathrm{data}}(x_0) \propto e^{\beta h_0 x_0}$, the p-bit of Section 1.1 with field h_0 and magnetization $\tanh(\beta h_0)$. After one forward flip step with probability p_1, the exact reverse conditional is, by Bayes' rule and Eq. (16.13),

$$q(x_0 \mid x_1) \propto \exp\!\Big(\beta h_0 x_0 + \frac{\Gamma_1}{2}x_1 x_0\Big) \quad\Longrightarrow\quad \mathbb{E}[x_0 \mid x_1] = \tanh\!\Big(\beta h_0 + \frac{\Gamma_1}{2}x_1\Big).$$

The reverse step is a p-bit in the sum of two fields: the learned field h_0, which is the marginal energy $U_0 = -h_0 x_0$ of Proposition 16.3, and the clamp field $\frac{1}{2}k_{\mathrm{B}}T\Gamma_1 x_1$. For a small step ($p_1 \to 0$, $\Gamma_1 \to \infty$) the clamp field dominates and $x_0 = x_1$ with probability close to one. For example, with $\beta h_0 = 0.5$ and $p_1 = 0.1$, $\Gamma_1 = \ln 9 = 2.20$; if $x_1 = -1$, then $\mathbb{E}[x_0 \mid x_1 = -1] = \tanh(0.5 - 1.10) = -0.54$, so the reverse step keeps the clamped value with probability 0.77 and flips toward the data's preferred value with probability 0.23. Direct application of Bayes' rule confirms this: $\mathbb{P}(x_0 = +1 \mid x_1 = -1) = \frac{0.1\,e^{0.5}}{0.1\,e^{0.5}+0.9\,e^{-0.5}} = 0.23$.

16.2.3 Bayes' rule and the exact reverse energy

Which conditional should the learned energy reproduce? The answer follows from Bayes' rule applied to the forward chain. Let $q_t(\mathbf{x}_t)$ be the forward marginal at step t and define the marginal energy

$$U_t(\mathbf{x}) := -k_{\mathrm{B}}T\ln q_t(\mathbf{x}), \qquad U_0 = -k_{\mathrm{B}}T\ln p_{\mathrm{data}}. \tag{16.15}$$

If $p_{\mathrm{data}} \propto e^{-\beta E_0}$, then $U_0 = E_0$ up to a constant.

Proposition 16.3 (Exact reverse conditional)**.** *For any forward chain with kernels of the form Eq. (16.3),*

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \frac{1}{\tilde{Z}_t(\mathbf{x}_t)}\exp\!\Big(-\beta\big[\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1},\mathbf{x}_t) + U_{t-1}(\mathbf{x}_{t-1})\big]\Big), \tag{16.16}$$

where $\tilde{Z}_t(\mathbf{x}_t) = Z_t^f\, q_t(\mathbf{x}_t)$. Consequently, a DTM can represent the exact reverse process if and only if, for each t, $\mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1},\mathbf{x}_t) = U_{t-1}(\mathbf{x}_{t-1}) + g_t(\mathbf{x}_t)$ for some function g_t of the clamp alone. In particular, the learned energy can be taken independent of $\mathbf{x}_t$.

Proof. Because the forward chain is Markov, the joint distribution of the pair $(\mathbf{x}_{t-1}, \mathbf{x}_t)$ is

$$q(\mathbf{x}_{t-1}, \mathbf{x}_t) = q_{t-1}(\mathbf{x}_{t-1})\, q(\mathbf{x}_t \mid \mathbf{x}_{t-1}).$$

Bayes' rule gives

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \frac{q_{t-1}(\mathbf{x}_{t-1})\, q(\mathbf{x}_t \mid \mathbf{x}_{t-1})}{q_t(\mathbf{x}_t)}.$$

Substituting $q_{t-1} = e^{-\beta U_{t-1}}$ from Eq. (16.15) and $q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = e^{-\beta \mathcal{E}^f_{t-1}}/Z^f_t$ from Eq. (16.3),

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \frac{\exp\big(-\beta[\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) + U_{t-1}(\mathbf{x}_{t-1})]\big)}{Z^f_t\, q_t(\mathbf{x}_t)},$$

which is Eq. (16.16). For the second statement, two Boltzmann conditionals in $\mathbf{x}_{t-1}$ with the same clamp $\mathbf{x}_t$ coincide if and only if their energies differ by a quantity independent of $\mathbf{x}_{t-1}$ (Proposition 1.2, applied at fixed $\mathbf{x}_t$). Comparing Eq. (16.4) with Eq. (16.16), the forward energies are identical, so the learned energy must equal $U_{t-1}(\mathbf{x}_{t-1})$ plus a function of $\mathbf{x}_t$ only. Such a function cancels between numerator and $Z_t(\mathbf{x}_t)$, so it can be set to zero. □

Proposition 16.3 explains the design choice in [56] that the learned energy does not see $\mathbf{x}_t$. The job of each stage is to know the shape of the data distribution at one noise level; the forward spring supplies all the information about where the current sample is. We keep the general notation $\mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t)$ of Definition 16.1 because the mixing theorem below holds without this restriction, and because a learned approximation may benefit from extra dependence on the clamp.

16.2.4 Small steps recover the Gaussian score update

Proposition 16.3 also connects DTMs to the score-based picture of Chapter 15. For continuous variables the exact reverse energy is

$$\mathcal{E}_{\text{tot}}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \frac{\alpha_t}{2} \left\|\mathbf{x}_{t-1} - c_t \mathbf{x}_t\right\|^2 + U_{t-1}(\mathbf{x}_{t-1}). \tag{16.17}$$

When α_t is large, the spring confines $\mathbf{x}_{t-1}$ to a region of width $\sqrt{k_{\mathrm{B}}T/\alpha_t}$ around $\mathbf{y} := c_t \mathbf{x}_t$. Over that region we may expand the marginal energy to second order:

$$U_{t-1}(\mathbf{x}_{t-1}) = U_{t-1}(\mathbf{y}) + \nabla U_{t-1}(\mathbf{y}) \cdot (\mathbf{x}_{t-1} - \mathbf{y}) + \frac{1}{2}(\mathbf{x}_{t-1} - \mathbf{y})^\top \mathbf{H} (\mathbf{x}_{t-1} - \mathbf{y}) + \mathcal{O}(\|\mathbf{x}_{t-1} - \mathbf{y}\|^3), \tag{16.18}$$

with $\mathbf{H} = \nabla^2 U_{t-1}(\mathbf{y})$. Inserting this into Eq. (16.17) and completing the square in $\mathbf{u} = \mathbf{x}_{t-1} - \mathbf{y}$,

$$\begin{aligned} \frac{\alpha_t}{2} \|\mathbf{u}\|^2 + \nabla U_{t-1} \cdot \mathbf{u} + \frac{1}{2}\mathbf{u}^\top \mathbf{H} \mathbf{u} &= \frac{1}{2}\mathbf{u}^\top (\alpha_t \mathbf{I} + \mathbf{H})\mathbf{u} + \nabla U_{t-1} \cdot \mathbf{u} \\ &= \frac{1}{2}(\mathbf{u} - \mathbf{m})^\top (\alpha_t \mathbf{I} + \mathbf{H})(\mathbf{u} - \mathbf{m}) - \frac{1}{2}\mathbf{m}^\top (\alpha_t \mathbf{I} + \mathbf{H})\mathbf{m}, \end{aligned} \tag{16.19}$$

where $\mathbf{m} = -(\alpha_t \mathbf{I} + \mathbf{H})^{-1} \nabla U_{t-1}(\mathbf{y})$. The reverse conditional is therefore approximately Gaussian with mean and covariance

$$\mathbb{E}[\mathbf{x}_{t-1} \mid \mathbf{x}_t] \approx c_t \mathbf{x}_t - \frac{1}{\alpha_t} \nabla U_{t-1}(c_t \mathbf{x}_t) = c_t \mathbf{x}_t + c_t^2 \nu_t \nabla \ln q_{t-1}(c_t \mathbf{x}_t), \qquad \mathrm{Cov}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \approx \frac{k_{\mathrm{B}}T}{\alpha_t} \mathbf{I}, \tag{16.20}$$

to leading order in $1/\alpha_t$. We used $\nabla U_{t-1} = -k_{\mathrm{B}}T \nabla \ln q_{t-1}$ and $k_{\mathrm{B}}T/\alpha_t = c_t^2 \nu_t$. This is the DDPM reverse step: rescale, then move along the score by an amount proportional to the step variance

(compare Eq. (15.1) and Proposition 15.9). A digital diffusion model computes the score with a network and adds Gaussian noise. A DTM never computes the score; the physical relaxation in Eq. (16.17) produces a sample with the correct mean and fluctuations, and it continues to produce the correct non-Gaussian distribution when α_t is not large and the expansion Eq. (16.18) fails.

Physics Insight: What changes when the step is large

The Gaussian approximation Eq. (16.20) requires that $\alpha_t \mathbf{I} + \nabla^2 U_{t-1}$ be positive definite and that U_{t-1} vary slowly over the width $\sqrt{k_B T/\alpha_t}$. When the forward step is large, α_t is small, the spring is soft, and $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ inherits the multimodal structure of q_{t-1}. A Gaussian reverse step then places mass between modes, which is the source of the discretization error that forces digital diffusion models to use many steps (Section 15.5). An EBM conditional has no such restriction: Eq. (16.16) is exact for any step size. This is the expressivity side of the DTM bargain. The mixing side is the subject of the next section: a soft spring also means the conditional landscape can contain barriers.

Example 16.4 (One-dimensional Gaussian data)**.** Let $p_{\text{data}} = \mathcal{N}(0, s_0^2)$ in one dimension, so $U_0(x) = k_B T x^2/(2s_0^2)$. The exact reverse energy of the last step is

$$\mathcal{E}_{\text{tot}}(x_0 \mid x_1) = \frac{\alpha_1}{2}(x_0 - c_1 x_1)^2 + \frac{k_B T}{2s_0^2} x_0^2.$$

Its curvature is $\alpha_1 + k_B T/s_0^2$, so $q(x_0 \mid x_1)$ is Gaussian with variance $k_B T/(\alpha_1 + k_B T/s_0^2)$ and mean $\alpha_1 c_1 x_1/(\alpha_1 + k_B T/s_0^2)$. Substituting $\alpha_1 = k_B T/(c_1^2 \nu_1)$ gives mean $c_1^{-1} s_0^2 x_1/(c_1^{-2} s_0^2 + \nu_1)$ and variance $s_0^2 \nu_1/(c_1^{-2} s_0^2 + \nu_1)$, which are the standard Gaussian posterior formulas for $x_1 = c_1^{-1} x_0 + \sqrt{\nu_1}\,\xi$. The forward spring and the marginal energy add their curvatures; nothing has to be learned except s_0.

16.2.5 The joint distribution of the whole chain

Multiplying the conditionals Eq. (16.4) along the chain gives the joint distribution of a generated trajectory,

$$p_\theta(\mathbf{x}_{0:T}) = \pi(\mathbf{x}_T) \prod_{t=1}^{T} \frac{\exp\!\left(-\beta[\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) + \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t)]\right)}{Z_t(\mathbf{x}_t)}. \tag{16.21}$$

This is a Boltzmann distribution of the whole trajectory with energy $\sum_t (\mathcal{E}^f_{t-1} + \mathcal{E}_{\theta,t-1}) - k_B T \ln \pi(\mathbf{x}_T) + k_B T \sum_t \ln Z_t(\mathbf{x}_t)$. The last sum is a free-energy correction that depends on the clamps. Running the stages one after another, each with its clamp fixed, samples Eq. (16.21) exactly and never needs Z_t. Letting all stages relax at once under the summed energy without the correction samples a different distribution, unless $Z_t(\mathbf{x}_t)$ happens to be independent of $\mathbf{x}_t$. For the exact reverse process, $Z_t(\mathbf{x}_t) \propto q_t(\mathbf{x}_t)$ by Proposition 16.3, so the correction is the marginal energy $-U_t(\mathbf{x}_t)$ and does not vanish. The consequences of this difference for joint relaxation of all stages on one chip are worked out in Section 17.3. In this chapter we treat the stages as sequential.

16.3 The DTM Mixing-Time Theorem

We now ask the question that decides whether a DTM is practical: how long must the circuit in stage t relax before its state is a good sample of $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$? For a monolithic EBM with a barrier ΔE, the answer is set by the Kramers time $\propto e^{\beta \Delta E}$ (Theorem 7.4). For a DTM step with continuous variables, the answer is controlled by a single number: the stiffness α_t minus the worst negative

curvature of the learned energy. The theorem in this section is this book's formalization of the qualitative argument of [56]; a preview appeared in Section 6.5.4.

16.3.1 The conditional sampler and the curvature shift

Throughout this section $\mathcal{X} = \mathbb{R}^N$, the clamp $\mathbf{x}_t$ is fixed, and we write $\mathbf{x} \equiv \mathbf{x}_{t-1}$ for the active variables. The conditional energy is

$$\mathcal{E}_{\text{tot}}(\mathbf{x} \mid \mathbf{x}_t) := \frac{\alpha_t}{2} \|\mathbf{x} - c_t \mathbf{x}_t\|^2 + \mathcal{E}_{\theta,t-1}(\mathbf{x}, \mathbf{x}_t), \tag{16.22}$$

and the stage-t circuit is modelled as the overdamped Langevin sampler

$$\mathrm{d}\mathbf{X}_s = -\mu \nabla_{\mathbf{x}} \mathcal{E}_{\text{tot}}(\mathbf{X}_s \mid \mathbf{x}_t)\, \mathrm{d}s + \sqrt{2\mu k_{\mathrm{B}} T}\, \mathrm{d}\mathbf{W}_s, \qquad s \geq 0, \tag{16.23}$$

where s is physical time within the stage. Its stationary density is $p_\theta(\mathbf{x} \mid \mathbf{x}_t)$ of Eq. (16.4) (Theorem 6.2, with detailed balance as in Theorem 6.4).

Assumption 16.5 (Bounded negative curvature of the learned energy)**.** For each t, the learned energy $\mathcal{E}_{\theta,t-1}(\cdot, \mathbf{x}_t)$ is twice continuously differentiable on $\mathbb{R}^N$ for every clamp value, and there is a constant $L_t \geq 0$, independent of $\mathbf{x}_t$, such that

$$\nabla^2_{\mathbf{x}} \mathcal{E}_{\theta,t-1}(\mathbf{x}, \mathbf{x}_t) \succeq -L_t\, \mathbf{I} \qquad \text{for all } \mathbf{x}, \mathbf{x}_t \in \mathbb{R}^N. \tag{16.24}$$

The assumption allows the learned energy to be as non-convex as we like in shape: it may have many wells and barriers of any height. It only bounds how sharply it can curve downward. Any energy built from smooth bounded-curvature components (polynomials restricted to a bounded region, sums of smooth pairwise terms with bounded second derivatives, effective potentials of the kind computed in Section 16.4) satisfies it.

Lemma 16.6 (Curvature shift by the forward spring)**.** *Under Assumption 16.5, for all* $\mathbf{x}, \mathbf{x}_t \in \mathbb{R}^N$*,*

$$\nabla^2_{\mathbf{x}} \mathcal{E}_{\text{tot}}(\mathbf{x} \mid \mathbf{x}_t) = \alpha_t \mathbf{I} + \nabla^2_{\mathbf{x}} \mathcal{E}_{\theta,t-1}(\mathbf{x}, \mathbf{x}_t) \succeq (\alpha_t - L_t)\mathbf{I}. \tag{16.25}$$

Proof. By Eq. (16.10), the forward energy is $\frac{\alpha_t}{2} \|\mathbf{x}\|^2 - \alpha_t c_t\, \mathbf{x}_t \cdot \mathbf{x} + \text{const}$. The gradient of the linear term $-\alpha_t c_t\, \mathbf{x}_t \cdot \mathbf{x}$ is the constant vector $-\alpha_t c_t \mathbf{x}_t$, so its Hessian vanishes. The Hessian of $\frac{\alpha_t}{2} \|\mathbf{x}\|^2 = \frac{\alpha_t}{2} \sum_i x_i^2$ has entries $\partial_i \partial_j (\frac{\alpha_t}{2} \sum_k x_k^2) = \alpha_t \delta_{ij}$, i.e. it is $\alpha_t \mathbf{I}$. Adding the Hessian of the learned energy gives the equality in Eq. (16.25). For any unit vector $\mathbf{u}$, $\mathbf{u}^\top \nabla^2_{\mathbf{x}} \mathcal{E}_{\text{tot}} \mathbf{u} = \alpha_t + \mathbf{u}^\top \nabla^2_{\mathbf{x}} \mathcal{E}_{\theta,t-1} \mathbf{u} \geq \alpha_t - L_t$ by Eq. (16.24). □

Two points about the lemma deserve emphasis. First, the clamp enters the forward energy only through a linear field, so it shifts the location of the conditional distribution but never its curvature; the bound $\alpha_t - L_t$ holds uniformly over all values of $\mathbf{x}_t$. This uniformity is what makes the next theorem a worst-case guarantee rather than a statement about typical clamps. Second, the forward spring adds the same curvature α_t in every direction, so it raises the entire Hessian spectrum rigidly.

16.3.2 The theorem

Theorem 16.7 (DTM conditional mixing time)**.** *Let Assumption 16.5 hold, let* $\kappa > 0$*, and suppose the forward stiffness satisfies*

$$\alpha_t \geq L_t + \kappa. \tag{16.26}$$

Then for every clamp value $\mathbf{x}_t \in \mathbb{R}^N$*:*

(i) ***Strong log-concavity.*** *$\mathcal{E}_{\mathrm{tot}}(\cdot \mid \mathbf{x}_t)$ is κ-strongly convex, $\nabla^2_{\mathbf{x}}\mathcal{E}_{\mathrm{tot}} \succeq \kappa \mathbf{I}$. In particular it has a unique minimizer $\mathbf{x}^\star(\mathbf{x}_t)$ and $p_\theta(\cdot \mid \mathbf{x}_t)$ is unimodal.*

(ii) ***Functional inequalities.*** *$p_\theta(\cdot \mid \mathbf{x}_t)$ satisfies the Poincaré inequality with spectral gap $\lambda_1 \geq \mu\kappa$ and the logarithmic Sobolev inequality with constant $\rho_{\mathrm{LS}} \geq \mu\kappa$.*

(iii) ***Exponential relaxation in KL.*** *If ρ_s is the law of $\mathbf{X}_s$ in Eq. (16.23) and $D_{\mathrm{KL}}(\rho_0 \,\|\, p_\theta(\cdot \mid \mathbf{x}_t)) < \infty$, then*

$$D_{\mathrm{KL}}(\rho_s \,\|\, p_\theta(\cdot \mid \mathbf{x}_t)) \leq e^{-2\mu\kappa s}\, D_{\mathrm{KL}}(\rho_0 \,\|\, p_\theta(\cdot \mid \mathbf{x}_t))\,. \tag{16.27}$$

(iv) ***Exponential contraction in $\mathcal{W}_2$.*** *For any two initial laws ρ_0, ρ_0' with finite second moments, the laws at time s satisfy $\mathcal{W}_2(\rho_s, \rho_s') \leq e^{-\mu\kappa s}\, \mathcal{W}_2(\rho_0, \rho_0')$. In particular $\mathcal{W}_2(\rho_s, p_\theta(\cdot \mid \mathbf{x}_t)) \leq e^{-\mu\kappa s}\, \mathcal{W}_2(\rho_0, p_\theta(\cdot \mid \mathbf{x}_t))$.*

(v) ***Cold start at the spring center.*** *If the circuit is started at the deterministic point $\mathbf{y} = c_t \mathbf{x}_t$, then*

$$\mathcal{W}_2(\delta_{\mathbf{y}}, p_\theta(\cdot \mid \mathbf{x}_t)) \leq \sqrt{\frac{N k_{\mathrm{B}} T}{\kappa}} + \frac{\|\nabla_{\mathbf{x}} \mathcal{E}_{\theta,t-1}(\mathbf{y}, \mathbf{x}_t)\|}{\kappa} := R_t(\mathbf{x}_t), \tag{16.28}$$

and the stage reaches $\mathcal{W}_2$-accuracy ε after the physical time

$$s_t(\varepsilon) = \frac{1}{\mu\kappa} \ln \frac{R_t(\mathbf{x}_t)}{\varepsilon}. \tag{16.29}$$

None of these bounds depends on the heights of barriers in $\mathcal{E}_{\theta,t-1}$, and (ii)–(iv) do not depend on the dimension N.

Proof. (i) By Lemma 16.6 and Eq. (16.26), $\nabla^2_{\mathbf{x}}\mathcal{E}_{\mathrm{tot}} \succeq (\alpha_t - L_t)\mathbf{I} \succeq \kappa \mathbf{I}$. A κ-strongly convex C^2 function satisfies $\mathcal{E}_{\mathrm{tot}}(\mathbf{x}) \geq \mathcal{E}_{\mathrm{tot}}(\mathbf{0}) + \nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{0}) \cdot \mathbf{x} + \frac{\kappa}{2}\|\mathbf{x}\|^2$ (Taylor's theorem with integral remainder), so it grows quadratically, is bounded below, and $e^{-\beta\mathcal{E}_{\mathrm{tot}}}$ is integrable. Strict convexity implies at most one critical point, and quadratic growth implies that a minimizer exists. A density $e^{-\beta\mathcal{E}_{\mathrm{tot}}}$ with a unique critical point of its energy has a single mode.

(ii) and (iii) The generator of Eq. (16.23) is $\mathcal{L} = D\Delta - \mu\nabla\mathcal{E}_{\mathrm{tot}} \cdot \nabla$ with $D = \mu k_{\mathrm{B}} T$, which is exactly the setting of Theorem 6.18 with $E = \mathcal{E}_{\mathrm{tot}}(\cdot \mid \mathbf{x}_t)$. Its parts (ii)–(iv) give $\lambda_1 \geq \mu\kappa$, $\rho_{\mathrm{LS}} \geq \mu\kappa$, and the KL contraction Eq. (6.106), which is Eq. (16.27).

(iv) Use a *synchronous coupling*: drive two copies $\mathbf{X}_s$ and $\mathbf{X}_s'$ of Eq. (16.23) with the same Brownian motion $\mathbf{W}_s$, starting from a pair $(\mathbf{X}_0, \mathbf{X}_0')$ whose joint law is an optimal coupling of ρ_0 and ρ_0'. The noise cancels in the difference $\boldsymbol{\Delta}_s = \mathbf{X}_s - \mathbf{X}_s'$:

$$\frac{\mathrm{d}}{\mathrm{d}s}\boldsymbol{\Delta}_s = -\mu\Big[\nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{X}_s) - \nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{X}_s')\Big].$$

By the fundamental theorem of calculus along the segment $\mathbf{X}_s' + \lambda\boldsymbol{\Delta}_s$, $\lambda \in [0,1]$,

$$\nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{X}_s) - \nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{X}_s') = \left(\int_0^1 \nabla^2\mathcal{E}_{\mathrm{tot}}(\mathbf{X}_s' + \lambda\boldsymbol{\Delta}_s)\, \mathrm{d}\lambda\right)\boldsymbol{\Delta}_s := \bar{\mathbf{H}}_s\boldsymbol{\Delta}_s,$$

with $\bar{\mathbf{H}}_s \succeq \kappa\mathbf{I}$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}s}\|\boldsymbol{\Delta}_s\|^2 = 2\boldsymbol{\Delta}_s \cdot \frac{\mathrm{d}\boldsymbol{\Delta}_s}{\mathrm{d}s} = -2\mu\,\boldsymbol{\Delta}_s^\top \bar{\mathbf{H}}_s \boldsymbol{\Delta}_s \leq -2\mu\kappa\,\|\boldsymbol{\Delta}_s\|^2\,,$$

and Grönwall's inequality gives $\|\boldsymbol{\Delta}_s\|^2 \le e^{-2\mu\kappa s}\,\|\boldsymbol{\Delta}_0\|^2$ pathwise. Taking expectations, $\mathcal{W}_2(\rho_s,\rho_s')^2 \le \mathbb{E}\big[\|\boldsymbol{\Delta}_s\|^2\big] \le e^{-2\mu\kappa s}\,\mathbb{E}\big[\|\boldsymbol{\Delta}_0\|^2\big] = e^{-2\mu\kappa s}\,\mathcal{W}_2(\rho_0,\rho_0')^2$. Choosing $\rho_0' = p_\theta(\cdot \mid \mathbf{x}_t)$, which is stationary, gives the second statement.

(v) Write $p \equiv p_\theta(\cdot \mid \mathbf{x}_t)$ and $\mathbf{x}^\star$ for the minimizer. The only coupling of a point mass with p is the product, so $\mathcal{W}_2(\delta_{\mathbf{y}},p) = (\mathbb{E}_p\,\|\mathbf{X}-\mathbf{y}\|^2)^{1/2} \le (\mathbb{E}_p\,\|\mathbf{X}-\mathbf{x}^\star\|^2)^{1/2} + \|\mathbf{x}^\star-\mathbf{y}\|$ by Minkowski's inequality. For the first term, integrate by parts:

$$\mathbb{E}_p[(\mathbf{X}-\mathbf{x}^\star)\cdot\nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{X})] = -k_{\mathrm{B}}T\int \mathrm{d}\mathbf{x}\,(\mathbf{x}-\mathbf{x}^\star)\cdot\nabla p(\mathbf{x}) = k_{\mathrm{B}}T\int \mathrm{d}\mathbf{x}\,p(\mathbf{x})\,\nabla\cdot(\mathbf{x}-\mathbf{x}^\star) = Nk_{\mathrm{B}}T,$$

where we used $\nabla p = -\beta p\,\nabla\mathcal{E}_{\mathrm{tot}}$, and the boundary term vanishes because p decays like a Gaussian by (i). Since $\nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{x}^\star)=\mathbf{0}$, the same integral-Hessian argument as in (iv) gives $(\mathbf{x}-\mathbf{x}^\star)\cdot\nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{x}) \ge \kappa\,\|\mathbf{x}-\mathbf{x}^\star\|^2$, hence $\mathbb{E}_p\,\|\mathbf{X}-\mathbf{x}^\star\|^2 \le Nk_{\mathrm{B}}T/\kappa$. For the second term, the spring has zero gradient at its center, so $\nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{y}) = \nabla_{\mathbf{x}}\mathcal{E}_{\theta,t-1}(\mathbf{y},\mathbf{x}_t)$; strong convexity gives $\|\nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{y}) - \nabla\mathcal{E}_{\mathrm{tot}}(\mathbf{x}^\star)\| \ge \kappa\,\|\mathbf{y}-\mathbf{x}^\star\|$ (Cauchy–Schwarz applied to the previous inequality), so $\|\mathbf{x}^\star-\mathbf{y}\| \le \|\nabla_{\mathbf{x}}\mathcal{E}_{\theta,t-1}(\mathbf{y},\mathbf{x}_t)\|/\kappa$. This proves Eq. (16.28). Combining with (iv), $\mathcal{W}_2(\rho_s,p) \le e^{-\mu\kappa s}R_t$, which is at most ε once $s \ge s_t(\varepsilon)$. □

The stage time Eq. (16.29) is a logarithm of an algebraic ratio divided by $\mu\kappa$. For a circuit whose natural relaxation rate at curvature κ is $\mu\kappa$, this is $\mathcal{O}(1)$ relaxation times up to a logarithm, and there is no factor $e^{\beta\Delta E}$ anywhere.

Remark 16.8 (The rate is sharp for Gaussian stages)**.** Take a quadratic learned energy $\mathcal{E}_{\theta,t-1}(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top\mathbf{G}\mathbf{x} - \mathbf{b}^\top\mathbf{x}$ with $\mathbf{G}$ symmetric and $\lambda_{\min}(\mathbf{G}) = -L_t$ (an RC-sampler stage with a negative-conductance direction, Chapter 13). Then $\mathcal{E}_{\mathrm{tot}}$ is quadratic with Hessian $\alpha_t\mathbf{I}+\mathbf{G}$, the stage is an Ornstein–Uhlenbeck process, and its relaxation rates are $\mu(\alpha_t+\lambda_k(\mathbf{G}))$ for the eigenvalues $\lambda_k(\mathbf{G})$. The slowest is $\mu(\alpha_t - L_t)$, which equals the Bakry–Émery bound $\mu\kappa$ when $\alpha_t = L_t + \kappa$. This is the tightness statement Eq. (6.141) of Worked computation 6.21, shifted by the spring. A purely Gaussian stage can therefore be made stable by the spring even when its learned conductance matrix is indefinite, and the theorem cannot be improved in general.

Example 16.9 (A lattice of coupled double wells)**.** Let the learned energy of a stage be a network of N quartic double wells, each of the double-well sampler's form, with pairwise couplings,

$$\mathcal{E}_{\theta,t-1}(\mathbf{x}) = \sum_{i=1}^{N}\Big(\frac{a}{4}x_i^4 - \frac{b}{2}x_i^2\Big) - \frac{1}{2}\sum_{i,j=1}^{N} J_{ij}x_ix_j,$$

with $\mathbf{J}$ symmetric and $J_{ii}=0$: a continuous version of a p-bit network. Its Hessian is $\mathrm{diag}(3ax_i^2 - b) - \mathbf{J} \succeq -(b+\lambda_{\max}(\mathbf{J}))\mathbf{I}$, using $3ax_i^2 \ge 0$ and $\mathbf{u}^\top\mathbf{J}\mathbf{u} \le \lambda_{\max}(\mathbf{J})$ for unit $\mathbf{u}$. So $L_t = b + \lambda_{\max}(\mathbf{J})$, and the stage is κ-strongly convex for $\alpha_t \ge b + \lambda_{\max}(\mathbf{J}) + \kappa$. For a ferromagnet on a lattice of coordination z with uniform coupling $J>0$, $\lambda_{\max}(\mathbf{J}) = zJ$. The monolithic sampler of this energy is much worse than a single double well: in the ordered phase, flipping the global magnetization requires nucleating a domain wall, whose free-energy cost grows as L_{lin}^{d-1} on a d-dimensional lattice of linear size L_{lin} ($d \ge 2$), so the Kramers time grows exponentially with system size. The DTM stage condition $\alpha_t \ge b + zJ + \kappa$ does not involve N at all, and the stage time Eq. (16.29) depends on N only through $\ln R_t$, which grows like $\frac{1}{2}\ln N$.

16.3.3 Warm starts and the accuracy of the whole chain

Two practical questions remain: how to start each stage so that the KL bound Eq. (16.27) is usable, and how per-stage errors add up over T stages.

Proposition 16.10 (Spring-only warm start)**.** *Let $\rho_0 = \mathcal{N}(c_t\mathbf{x}_t, (k_\mathrm{B}T/\alpha_t)\mathbf{I})$ be the Boltzmann distribution of the forward spring alone, and $p = p_\theta(\cdot \mid \mathbf{x}_t)$. Then*

$$D_{\mathrm{KL}}(\rho_0 \,\|\, p) \le D_{\mathrm{KL}}(\rho_0 \,\|\, p) + D_{\mathrm{KL}}(p \,\|\, \rho_0) = \beta\Big(\mathbb{E}[\mathcal{E}_{\theta,t-1}]_{\rho_0} - \mathbb{E}[\mathcal{E}_{\theta,t-1}]_p\Big). \tag{16.30}$$

Proof. Both densities are Boltzmann distributions sharing the spring energy: $\rho_0 = e^{-\beta\mathcal{E}^f}/Z_0$ and $p = e^{-\beta(\mathcal{E}^f+\mathcal{E}_{\theta,t-1})}/Z_t$. Hence $\ln(\rho_0/p) = \beta\mathcal{E}_{\theta,t-1} + \ln(Z_t/Z_0)$. Then $D_{\mathrm{KL}}(\rho_0 \,\|\, p) = \mathbb{E}_{\rho_0}\ln(\rho_0/p) = \beta\mathbb{E}_{\rho_0}\mathcal{E}_{\theta,t-1} + \ln(Z_t/Z_0)$ and $D_{\mathrm{KL}}(p \,\|\, \rho_0) = -\mathbb{E}_p\ln(\rho_0/p) = -\beta\mathbb{E}_p\mathcal{E}_{\theta,t-1} - \ln(Z_t/Z_0)$. Adding, the partition functions cancel, which gives the equality; the inequality is $D_{\mathrm{KL}}(p \,\|\, \rho_0) \ge 0$. □

A circuit can prepare ρ_0 by relaxing with the learned couplings switched off, which is an Ornstein–Uhlenbeck relaxation at rate $\mu\alpha_t \ge \mu\kappa$. The bound Eq. (16.30) is the difference in mean learned energy between the spring-only ensemble and the stage's own ensemble, in units of $k_\mathrm{B}T$. When the spring is stiff, both ensembles are narrow around nearby points and the difference is a few $k_\mathrm{B}T$ or less, so the KL decay Eq. (16.27) needs only a few multiples of $(2\mu\kappa)^{-1}$ to reach a small error.

Proposition 16.11 (Error accumulation along the chain)**.** *Let $\hat{p}(\mathbf{x}_{0:T})$ be the law of the trajectory actually produced by the hardware: $\mathbf{x}_T \sim \hat{\pi}$, then $\mathbf{x}_{t-1} \sim \hat{p}_t(\cdot \mid \mathbf{x}_t)$ for $t = T, \dots, 1$, where $\hat{p}_t$ is the law of the stage-t circuit after its finite relaxation. Suppose $\|\hat{p}_t(\cdot \mid \mathbf{x}_t) - p_\theta(\cdot \mid \mathbf{x}_t)\|_{\mathrm{TV}} \le \varepsilon_t$ for every clamp value. Then*

$$\|\hat{p}(\mathbf{x}_0) - p_{\mathrm{data}}\|_{\mathrm{TV}} \le \|\hat{\pi} - \pi\|_{\mathrm{TV}} + \sum_{t=1}^{T}\varepsilon_t + \|p_\theta(\mathbf{x}_0) - p_{\mathrm{data}}\|_{\mathrm{TV}}\,. \tag{16.31}$$

Proof. By the triangle inequality it suffices to show $\|\hat{p}(\mathbf{x}_0) - p_\theta(\mathbf{x}_0)\|_{\mathrm{TV}} \le \|\hat{\pi} - \pi\|_{\mathrm{TV}} + \sum_t \varepsilon_t$, and since marginalizing cannot increase total variation, it suffices to bound the distance between the trajectory laws. Define hybrid trajectory laws h_k, $k = 0, \dots, T+1$, that use the hardware kernels for the first k operations (the prior counts as operation 1, stage T as operation 2, and so on) and the ideal kernels afterwards; $h_0 = p_\theta(\mathbf{x}_{0:T})$ and $h_{T+1} = \hat{p}(\mathbf{x}_{0:T})$. Consecutive hybrids differ in exactly one kernel. If that kernel is stage t, both hybrids share the same law m of $\mathbf{x}_{t:T}$, and the later ideal kernels are applied identically to both, so

$$\|h_k - h_{k-1}\|_{\mathrm{TV}} \le \int m(\mathrm{d}\mathbf{x}_{t:T})\ \|\hat{p}_t(\cdot \mid \mathbf{x}_t) - p_\theta(\cdot \mid \mathbf{x}_t)\|_{\mathrm{TV}} \le \varepsilon_t,$$

because applying a common Markov kernel does not increase total variation. If the differing kernel is the prior, the same argument gives $\|\hat{\pi} - \pi\|_{\mathrm{TV}}$. Summing over k with the triangle inequality proves the claim. □

Combining Proposition 16.11 with the Pinsker bound Eq. (6.91), a stage run for time s from the warm start has $\varepsilon_t \le \sqrt{\frac{1}{2}D_{\mathrm{KL}}(\rho_0 \,\|\, p)}\, e^{-\mu\kappa s}$. To keep the total sampling error below ε over T stages it suffices to make each $\varepsilon_t \le \varepsilon/T$, which adds only $(\mu\kappa)^{-1}\ln T$ to each stage time. The cost of a longer chain in accuracy is logarithmic.

16.3.4 The same result from the Witten Hamiltonian

In one dimension there is a short independent proof of the spectral-gap bound, which shows directly how the curvature shift acts on the Witten Schrödinger operator of Section 6.4. Write $E \equiv \mathcal{E}_{\mathrm{tot}}(\cdot \mid x_t)$ and $Q = \sqrt{D}(\partial_x + \frac{\beta}{2}E'(x))$, so that $Q^\dagger = \sqrt{D}(-\partial_x + \frac{\beta}{2}E'(x))$ and the Witten

Hamiltonian is $\mathcal{H}_{\mathrm{W}} = Q^\dagger Q = D(-\partial_x^2 + \frac{\beta^2}{4}E'^2 - \frac{\beta}{2}E'')$ (Theorem 6.8). The *partner* operator obtained by reversing the order is

$$QQ^\dagger = D\Big(-\partial_x^2 + \frac{\beta^2}{4}E'^2 + \frac{\beta}{2}E''\Big) = Q^\dagger Q + D\beta\, E''(x) = \mathcal{H}_{\mathrm{W}} + \mu E''(x), \tag{16.32}$$

where we used $D\beta = \mu$. If $Q^\dagger Q\psi = \lambda\psi$ with $\lambda > 0$, then $\phi = Q\psi \neq 0$ satisfies $QQ^\dagger\phi = Q(Q^\dagger Q\psi) = \lambda\phi$, so every non-zero eigenvalue of $\mathcal{H}_{\mathrm{W}}$ is an eigenvalue of $QQ^\dagger$. The partner has no zero mode, because $Q^\dagger\phi = 0$ forces $\phi \propto e^{+\beta E/2}$, which is not normalizable. Since $Q^\dagger Q \succeq 0$ and $E'' \geq \kappa$, Eq. (16.32) gives $QQ^\dagger \succeq \mu\kappa$, hence the spectral gap of $\mathcal{H}_{\mathrm{W}}$ is at least $\mu\kappa$. In the multivariate case the partner operator acts on vector fields and the term $\mu E''$ becomes the matrix $\mu\nabla^2 E$; the conclusion is the same. The forward spring adds $\mu\alpha_t$ to the partner potential everywhere, which is the operator-level form of Lemma 16.6.

16.3.5 What the theorem costs: the curvature budget

Theorem 16.7 can be misread as saying that the expressivity–mixing dilemma has disappeared. It has not. It has been moved into the step size, and the purpose of this subsection is to make the trade precise.

Recall from Eq. (16.9) that $\alpha_t = k_{\mathrm{B}}T/(c_t^2\nu_t)$ is fixed by the forward noise added in step t. The theorem requires $\alpha_t \geq L_t + \kappa$, where L_t bounds the negative curvature of what the step must represent. By Proposition 16.3, for the exact reverse process the learned energy is the marginal energy U_{t-1}, so the relevant non-convexity is

$$L(U_{t-1}) := \max\Big(0,\, -\inf_{\mathbf{x}} \lambda_{\min}\big(\nabla^2 U_{t-1}(\mathbf{x})\big)\Big). \tag{16.33}$$

This is a property of the data and the noise level, not of the model. The condition on the stiffness therefore becomes a condition on the forward schedule.

Proposition 16.12 (Step-size budget)**.** *Consider the variance-preserving forward process ($c_t^{-2} = 1 - \nu_t$) in coordinates measured in units of the data scale, and suppose every step of the exact reverse process is required to be κ-strongly log-concave. Let $s_t := \ln(1/\bar\gamma_t) = -\sum_{r\leq t}\ln(1-\nu_r)$ be the accumulated forward log-attenuation.*

(i) Step t can advance s by at most

$$s_t - s_{t-1} = \ln\Big(1 + \frac{k_{\mathrm{B}}T}{\alpha_t}\Big) \leq \frac{k_{\mathrm{B}}T}{L(U_{t-1}) + \kappa}. \tag{16.34}$$

(ii) Let $\bar\gamma_{\mathrm{lc}}$ be the signal level below which all U_t are log-concave, and $s_{\mathrm{lc}} = \ln(1/\bar\gamma_{\mathrm{lc}})$. The number of steps needed to cross the non-log-concave window $0 \leq s \leq s_{\mathrm{lc}}$ satisfies

$$T_{\mathrm{nc}} \geq \frac{\kappa\, s_{\mathrm{lc}}}{k_{\mathrm{B}}T}, \qquad T_{\mathrm{nc}} \approx \int_0^{s_{\mathrm{lc}}} \left(\frac{L(s)+\kappa}{k_{\mathrm{B}}T} + \frac{1}{2}\right) \mathrm{d}s, \tag{16.35}$$

the approximation holding when the steps are small, with $L(s)$ the non-convexity Eq. (16.33) at log-attenuation s.

(iii) With every stage run for the time Eq. (16.29) at the margin κ, the total physical relaxation time across the window is

$$\tau_{\mathrm{nc}} = \sum_{t\leq T_{\mathrm{nc}}} s_t(\varepsilon) \approx \frac{\ln(R/\varepsilon)}{\mu k_{\mathrm{B}}T} \int_0^{s_{\mathrm{lc}}} \left(1 + \frac{L(s) + k_{\mathrm{B}}T/2}{\kappa}\right) \mathrm{d}s, \tag{16.36}$$

where R is a typical value of the cold-start radius R_t (coordinates in units of the data scale).

Proof. (i) For the variance-preserving process, $\alpha_t = k_B T(1-\nu_t)/\nu_t$, so $1-\nu_t = \alpha_t/(\alpha_t + k_B T)$ and $-\ln(1-\nu_t) = \ln((\alpha_t + k_B T)/\alpha_t) = \ln(1 + k_B T/\alpha_t)$. Since $\ln(1+u) \le u$ for $u > 0$, and Theorem 16.7 applied with $\mathcal{E}_{\theta,t-1} = U_{t-1}$ requires $\alpha_t \ge L(U_{t-1}) + \kappa$, the bound follows. (ii) Summing Eq. (16.34) over the steps inside the window, $s_{lc} \le \sum_t k_B T/(L+\kappa) \le T_{nc} k_B T/\kappa$. For the approximation, the number of steps per unit s is $1/\ln(1 + k_B T/\alpha)$; with $u = k_B T/\alpha$ small, $1/\ln(1+u) = 1/u + \frac{1}{2} + \mathcal{O}(u)$, and $1/u = (L+\kappa)/k_B T$ when the condition is saturated. Integrating over s gives Eq. (16.35). (iii) Each stage takes $\ln(R/\varepsilon)/(\mu\kappa)$, so the total is $T_{nc} \ln(R/\varepsilon)/(\mu\kappa)$; inserting the approximation for T_{nc} gives Eq. (16.36). □

Proposition 16.12 states the trade in three lines. A larger margin κ makes each stage faster but forces more stages. As $\kappa \to \infty$ the number of stages diverges, while the total time tends to $\ln(R/\varepsilon)\, s_{lc}/(\mu k_B T)$: in that limit the DTM becomes a physical simulation of the reverse diffusion, and it cannot finish faster than the forward Ornstein–Uhlenbeck process takes to cross the same window (in units where the data scale is 1, $1/(\mu k_B T)$ is the diffusion time across one data length). As κ decreases, fewer stages are needed, but the integral $\int L\, ds/\kappa$ grows. In every case the total time depends on the data only through $\int L(s)\, ds$, a curvature integral that grows polynomially with the barrier height of the data distribution (we will see this explicitly in Worked computation 16.27), rather than through $e^{\beta \Delta E}$.

Example 16.13 (Log-concave data need one stage)**.** If p_{data} is itself κ_0-strongly log-concave, $\nabla^2 U_0 \succeq \kappa_0 \mathbf{I}$ with $\kappa_0 > 0$, then every forward marginal is log-concave as well, since convolving a log-concave density with a Gaussian and rescaling preserves log-concavity (Prékopa's theorem applied to the joint density of $(\mathbf{x}_0, \mathbf{x}_t)$). Then $L(U_{t-1}) = 0$ for all t, the window of Proposition 16.12 is empty, and Theorem 16.7 holds with α_t as small as we like. In particular a single stage from the prior straight to the data, $T = 1$ with $\nu_1 \to 1$ and $\alpha_1 \to 0$, has conditional energy $\approx U_0$ and is κ_0-strongly log-concave: the DTM reduces to a monolithic EBM, which for such data mixes quickly anyway. The stage count of a DTM is therefore driven entirely by the non-log-concavity of the data and its blurred versions, not by dimension or by the overall spread of the data.

Heuristic Derivation: Many easy steps instead of one hard one

A monolithic sampler must carry a sample from one mode to another across a barrier, and the probability of doing so per unit time is $e^{-\beta \Delta E}$. A DTM never asks for that. Mode selection is spread over the stages near the log-concavity onset $s \approx s_{lc}$ (the speciation transition of Definition 15.13), where the marginal q_t is just becoming bimodal and the split costs nothing. Each later stage only refines the sample within the mode it is already in, moving it by about $\sqrt{k_B T/\alpha_t}$. Every stage is unimodal, yet the composition of T unimodal conditionals produces a multimodal marginal $p_\theta(\mathbf{x}_0)$, because the marginal is a mixture of conditionals over the earlier stages. The non-convexity budget per step is $\alpha_t - \kappa$: that is the most negative curvature any single stage may represent. The data's total non-convexity is paid for by spreading it over enough steps.

Remark 16.14 (The many-stage limit is the reverse-time SDE)**.** Let the number of stages grow with the schedule fixed in continuous time: $\nu_t = \beta(\tau_t)\Delta\tau$ on a grid of spacing $\Delta\tau \to 0$, where $\beta(\cdot)$ is the schedule function of Eq. (15.4) and not the inverse temperature. Then $\alpha_t = k_B T(1-\nu_t)/\nu_t \to \infty$ and the conditional is well described by the Gaussian approximation Eq. (16.20), with mean shift $c_t^2 \nu_t \nabla \ln q_{t-1} \approx \beta(\tau)\Delta\tau\, \nabla \ln q$ and variance $\approx \beta(\tau)\Delta\tau$. These are the drift and diffusion increments of the reverse-time variance-preserving SDE of Theorem 15.6 over one interval $\Delta\tau$ (the extra rescaling $c_t \approx 1 + \frac{1}{2}\beta\Delta\tau$ supplies the $+\frac{1}{2}\beta\mathbf{x}$ drift). A DTM with many stages is therefore a physical

integrator of the reverse SDE in which each stage samples its increment by relaxation rather than by computing the score. In this limit the time per stage at fixed precision is $\mathcal{O}(1/(\mu\alpha_t))$ and the number of stages is $\mathcal{O}(\alpha_t)$ per unit of s, which is the content of Eq. (16.36) as $\kappa \to \infty$.

Physics Insight: A spring that follows the sample, compared with annealing

Simulated annealing and parallel tempering attack barriers by raising the temperature, which flattens the landscape everywhere by the same factor. The difficulty is that the barrier returns as the temperature is lowered, and a heated sample that has drifted into a low-weight region must still find its way back. A DTM does not change the temperature. It adds a confining spring whose center $c_t\mathbf{x}_t$ is the previous sample. The spring is strong enough to remove all negative curvature locally, and it moves with the sample, so each stage solves a small, local, convex problem around the current position. The sequence of spring centers is itself random and, for an exact model, distributed according to the forward marginals; this is what allows the chain to visit all modes with the correct weights without any stage crossing a barrier. In the language of Chapter 10, the chain is a sequence of short, nearly reversible transports between neighboring distributions q_t and q_{t-1}, rather than one long equilibration.

16.3.6 When the condition is violated slightly, and discrete variables

The condition Eq. (16.26) is sufficient, not necessary. Two remarks show how the guarantee degrades and how it transfers to spins.

Remark 16.15 (Graceful degradation via Holley–Stroock)**.** Suppose $\mathcal{E}_{\text{tot}} = \mathcal{E}_{\text{cvx}} + \delta\mathcal{E}$, where $\mathcal{E}_{\text{cvx}}$ is κ-strongly convex and the remainder has bounded oscillation $\operatorname{osc}(\delta\mathcal{E}) = \Delta$. This describes a stage whose stiffness is slightly too small, so that a shallow residual barrier of height of order Δ survives. By Theorem 6.20 combined with Theorem 6.18, the conditional still satisfies a logarithmic Sobolev inequality with $\rho_{\text{LS}} \geq \mu\kappa\, e^{-\beta\Delta}$. A residual barrier of $2\,k_{\text{B}}T$ slows the stage by at most a factor $e^2 \approx 7$. The guarantee does not fall off a cliff at $\alpha_t = L_t$; it weakens exponentially in the residual barrier, which is small when α_t is close to L_t.

Remark 16.16 (Discrete variables: a Dobrushin-type analogue)**.** For spins the Langevin argument does not apply, but there is an analogue. Consider a stage with Ising energy $\mathcal{E}_{\theta,t-1}(\mathbf{x}) = -\sum_{i<j} J_{ij}x_ix_j - \sum_i h_ix_i$ and the forward field $h_i^f = \frac{1}{2}k_{\text{B}}T\Gamma_t x_{t,i}$ of Eq. (16.13). Under single-site heat-bath (Glauber) updates, the probability that spin i is $+1$ is $\sigma(2\beta H_i)$ with $\sigma(u) = (1+e^{-u})^{-1}$ and local field $H_i = h_i^f + h_i + \sum_j J_{ij}x_j$. Dobrushin's influence coefficient C_{ij}, the largest change in this probability caused by flipping spin j, satisfies $C_{ij} \leq 4\beta\,|J_{ij}|\sup\sigma'$, where the supremum runs over the fields reachable by changing x_j. Because $\sigma'(u) \leq e^{-|u|}$, if the forward field dominates, $\frac{1}{2}k_{\text{B}}T\Gamma_t \geq |h_i| + \sum_j |J_{ij}| + m$ with $m > 0$, then $C_{ij} \leq 4\beta\,|J_{ij}|\,e^{-2\beta m}$. When $\max_i \sum_j C_{ij} < 1$ (Dobrushin's condition), the Gibbs measure is unique and Glauber dynamics contracts at a rate set by $1 - \max_i \sum_j C_{ij}$, uniformly in N. A strong forward field therefore plays the role of the forward spring: it saturates each spin's conditional so that neighbors have little influence. We state this only as an analogy. Two caveats apply to the architecture of [56]: latent spins receive no forward field, so the argument says nothing about them; and the field strength Γ_t is tied to the flip probability p_t, so a dominant field means small forward steps, just as in the continuous case.

Example 16.17 (The two-spin ferromagnet: a discrete double well)**.** The smallest discrete version of the double-well sampler is a pair of spins with data distribution $p_{\text{data}}(x_1, x_2) \propto e^{\beta J x_1 x_2}$, $J > 0$. It has two modes, $(+,+)$ and $(-,-)$. Under single-spin Glauber dynamics, moving between them

requires passing through a misaligned state at energy cost $2J$, so the mixing time grows as $e^{2\beta J}$; for $\beta J = 3$ this is several hundred updates. Now consider the final DTM stage, whose exact energy is $U_0 = -Jx_1x_2$ plus the forward fields $\frac{1}{2}k_{\mathrm{B}}T\Gamma_1 x_{1,i}$ from the clamp. The heat-bath probability that spin 1 is $+1$ is $\sigma(2\beta J x_2 + \Gamma_1 x_{1,1})$, so Dobrushin's influence of spin 2 on spin 1 is

$$C_{12} = \sigma(\Gamma_1 + 2\beta J) - \sigma(\Gamma_1 - 2\beta J)$$

for either sign of the clamp, and C_{21} is the same by symmetry. For $\beta J = 3$:

flip probability p_1	0.1	0.01	0.001	10^{-4}
$\Gamma_1 = \ln\frac{1-p_1}{p_1}$	2.20	4.60	6.91	9.21
C_{12}	0.98	0.80	0.29	0.04

The influence drops sharply once Γ_1 exceeds $2\beta J$, i.e. once the forward field on each spin is stronger than the coupling it feels from its partner. This is the discrete counterpart of $\alpha_1 > b$: the condition $\Gamma_1 > 2\beta J$ reads $p_1 < (1 + e^{2\beta J})^{-1} \approx 2.5 \times 10^{-3}$. With $p_1 = 10^{-3}$, each Glauber sweep contracts the distance to equilibrium by a factor of about 0.3, so a few sweeps suffice, compared with several hundred for the monolithic sampler. As in the continuous case, the price is a small forward step and hence many stages.

Remark 16.18 (Where the paper operates)**.** The DTMs trained in [56] use 2 to 8 steps of a binary Boltzmann machine with latent variables and $K \approx 250$–1000 block Gibbs sweeps per stage. With so few steps, each forward step is large, the conditionals are not guaranteed to be unimodal, and Eq. (16.26) or its discrete analogue is not expected to hold. The paper controls mixing empirically: the measured autocorrelation of the sampler at the training sweep count is held near a small target by adjusting a total-correlation penalty. Theorem 16.7 describes the other end of the same one-parameter family (many small steps, provable mixing). Real designs choose a point in between, trading the number of hardware stages against per-stage sweeps.

16.4 Latent-Variable DTMs and the Per-Step Expressivity Budget

The learned energy of a DTM step has to be built from physical couplings. Hardware couplings are local and pairwise: a transistor cell couples two nodes, and wires run between nearby cells (Section 2.5, Chapter 17). A pairwise energy among the visible nodes $\mathbf{x}_{t-1}$ alone is quadratic for Gaussian nodes and second-order Ising for spins, which is not enough to represent the shape of a general marginal energy U_{t-1}. The standard remedy, used in [56] and developed in Section 1.5, is to add hidden nodes $\mathbf{z}_{t-1}$ and sum them out.

16.4.1 Latent conditionals and their effective potential

Definition 16.19 (Latent-variable DTM step)**.** A latent-variable DTM step has active visible variables $\mathbf{x} \equiv \mathbf{x}_{t-1} \in \mathbb{R}^N$, latent variables $\mathbf{z} \equiv \mathbf{z}_{t-1}$ in a space $\mathcal{Z}$ (for example $\{-1,+1\}^M$), and conditional

$$p_\theta(\mathbf{x},\mathbf{z} \mid \mathbf{x}_t) = \frac{1}{Z_t(\mathbf{x}_t)} \exp\Big(-\beta\big[\mathcal{E}^f_{t-1}(\mathbf{x},\mathbf{x}_t) + \mathcal{E}_{\theta,t-1}(\mathbf{x},\mathbf{z})\big]\Big), \qquad p_\theta(\mathbf{x} \mid \mathbf{x}_t) = \sum_{\mathbf{z}\in\mathcal{Z}} p_\theta(\mathbf{x},\mathbf{z} \mid \mathbf{x}_t). \tag{16.37}$$

The *effective learned potential* is the free energy of the latents at fixed visible state,

$$F_{\theta,t-1}(\mathbf{x}) := -k_{\mathrm{B}}T \ln \sum_{\mathbf{z}\in\mathcal{Z}} e^{-\beta\mathcal{E}_{\theta,t-1}(\mathbf{x},\mathbf{z})}, \tag{16.38}$$

so that $p_\theta(\mathbf{x} \mid \mathbf{x}_t) \propto \exp(-\beta[\mathcal{E}^f_{t-1}(\mathbf{x}, \mathbf{x}_t) + F_{\theta,t-1}(\mathbf{x})])$.

The forward energy does not involve $\mathbf{z}$, so it passes through the sum over latents unchanged. The visible marginal of a latent step is therefore a DTM step of the form Eq. (16.4) with learned energy $F_{\theta,t-1}$, and Theorem 16.7 applies to it once we know the curvature of $F_{\theta,t-1}$. That curvature is given by Theorem 1.18.

Proposition 16.20 (Latent fluctuations spend the curvature budget)**.** *Let $\mathcal{E}_{\theta,t-1}(\mathbf{x}, \mathbf{z})$ be twice continuously differentiable in $\mathbf{x}$ for each $\mathbf{z}$. Then*

$$\nabla^2_{\mathbf{x}} F_{\theta,t-1}(\mathbf{x}) = \mathbb{E}\Big[\nabla^2_{\mathbf{x}} \mathcal{E}_{\theta,t-1}(\mathbf{x}, \mathbf{Z})\Big]_{\mathbf{z}|\mathbf{x}} - \beta \operatorname{Cov}_{\mathbf{z}|\mathbf{x}}(\nabla_{\mathbf{x}} \mathcal{E}_{\theta,t-1}(\mathbf{x}, \mathbf{Z})), \tag{16.39}$$

where the expectation and covariance are over $p_\theta(\mathbf{z} \mid \mathbf{x}) \propto e^{-\beta \mathcal{E}_{\theta,t-1}(\mathbf{x},\mathbf{z})}$. Consequently, if for all $(\mathbf{x}, \mathbf{z})$

$$\nabla^2_{\mathbf{x}} \mathcal{E}_{\theta,t-1}(\mathbf{x}, \mathbf{z}) \succeq -L_0 \mathbf{I} \qquad \text{and} \qquad \operatorname{Cov}_{\mathbf{z}|\mathbf{x}}(\nabla_{\mathbf{x}} \mathcal{E}_{\theta,t-1}) \preceq \Sigma_{\max} \mathbf{I},$$

the visible step is κ-strongly log-concave whenever

$$\alpha_t \geq L_0 + \beta \Sigma_{\max} + \kappa. \tag{16.40}$$

Proof. Equation (16.39) is Eq. (1.121) with $\mathbf{v} \to \mathbf{x}$ and $h \to \mathbf{z}$. For any unit vector $\mathbf{u}$, the first term contributes at least $-L_0$ and the second at least $-\beta\Sigma_{\max}$, so $\nabla^2 F_{\theta,t-1} \succeq -(L_0 + \beta\Sigma_{\max})\mathbf{I}$. Apply Theorem 16.7 with $L_t = L_0 + \beta\Sigma_{\max}$. □

The second term in Eq. (16.39) is the whole point of latent variables: when the latents fluctuate in a way that depends on $\mathbf{x}$, the visible effective potential becomes softer and can develop negative curvature, which is what lets a pairwise circuit carve a non-Gaussian shape. Proposition 16.20 says that this same term is charged against the stiffness budget $\alpha_t - \kappa$. Expressivity per step and guaranteed mixing per step draw on the same account.

16.4.2 The Gaussian–Bernoulli step

Consider the hybrid energy of Worked computation 1.19, with Gaussian visible nodes and binary latent p-bits coupled bipartitely:

$$\mathcal{E}_{\theta,t-1}(\mathbf{x}, \mathbf{z}) = \frac{1}{2}\mathbf{x}^\top \mathbf{G}\mathbf{x} - \mathbf{b}^\top \mathbf{x} - \mathbf{x}^\top \mathbf{W}\mathbf{z} - \mathbf{h}^\top \mathbf{z}, \qquad \mathbf{z} \in \{-1, +1\}^M, \tag{16.41}$$

with $\mathbf{G}$ symmetric and $\mathbf{W} \in \mathbb{R}^{N \times M}$ with columns $\mathbf{w}_a$. There are no latent–latent couplings, so given $\mathbf{x}$ the latents are independent, and the sum over $\mathbf{z}$ factorizes:

$$\sum_{\mathbf{z}} e^{\beta(\mathbf{x}^\top \mathbf{W}\mathbf{z} + \mathbf{h}^\top \mathbf{z})} = \prod_{a=1}^{M} \sum_{z_a = \pm 1} e^{\beta(\mathbf{w}_a \cdot \mathbf{x} + h_a) z_a} = \prod_{a=1}^{M} 2\cosh(\beta(\mathbf{w}_a \cdot \mathbf{x} + h_a)). \tag{16.42}$$

Taking $-k_{\mathrm{B}}T \ln$,

$$F_{\theta,t-1}(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{G}\mathbf{x} - \mathbf{b}^\top \mathbf{x} - k_{\mathrm{B}}T \sum_{a=1}^{M} \ln 2\cosh(\beta(\mathbf{w}_a \cdot \mathbf{x} + h_a)). \tag{16.43}$$

Differentiating the last term twice, with $u_a = \mathbf{w}_a \cdot \mathbf{x} + h_a$: $\nabla_{\mathbf{x}}\big[-k_{\mathrm{B}}T \ln 2\cosh(\beta u_a)\big] = -\tanh(\beta u_a)\, \mathbf{w}_a$, and then $\nabla^2_{\mathbf{x}}\big[-k_{\mathrm{B}}T \ln 2\cosh(\beta u_a)\big] = -\beta \operatorname{sech}^2(\beta u_a)\, \mathbf{w}_a \mathbf{w}_a^\top$. Hence

$$\nabla^2_{\mathbf{x}} F_{\theta,t-1}(\mathbf{x}) = \mathbf{G} - \beta \sum_{a=1}^{M} \operatorname{sech}^2(\beta u_a)\, \mathbf{w}_a \mathbf{w}_a^\top \succeq \mathbf{G} - \beta \mathbf{W}\mathbf{W}^\top. \tag{16.44}$$

This agrees with Eq. (16.39): $\nabla_{\mathbf{x}}\mathcal{E}_{\theta,t-1} = \mathbf{G}\mathbf{x} - \mathbf{b} - \mathbf{W}\mathbf{z}$ has covariance $\mathbf{W}\,\mathrm{Cov}(\mathbf{z} \mid \mathbf{x})\mathbf{W}^\top$ with $\mathrm{Cov}(\mathbf{z} \mid \mathbf{x}) = \mathrm{diag}(\mathrm{sech}^2(\beta u_a)) \preceq \mathbf{I}$. The inequality in Eq. (16.44) uses $\mathrm{sech}^2 \le 1$ and $\sum_a \mathbf{w}_a\mathbf{w}_a^\top = \mathbf{W}\mathbf{W}^\top$. The step is therefore κ-strongly log-concave whenever

$$\alpha_t + \lambda_{\min}(\mathbf{G}) \ge \beta\,\|\mathbf{W}\|_2^2 + \kappa, \tag{16.45}$$

where $\|\mathbf{W}\|_2$ is the largest singular value of $\mathbf{W}$.

Example 16.21 (One latent p-bit carves a double well, unless the spring forbids it)**.** Take $N = M = 1$, $G = 0$, $b = h = 0$, coupling $w > 0$, and clamp with $c_t x_t = 0$. The visible conditional energy is

$$\mathcal{E}_{\mathrm{tot}}(x) = \frac{\alpha_t}{2}x^2 - k_{\mathrm{B}}T\ln 2\cosh(\beta w x), \qquad \mathcal{E}''_{\mathrm{tot}}(x) = \alpha_t - \beta w^2\,\mathrm{sech}^2(\beta w x).$$

At $x = 0$ the curvature is $\alpha_t - \beta w^2$. If $\beta w^2 > \alpha_t$, the origin is a local maximum and the minima solve $\alpha_t x = w\tanh(\beta w x)$. In the strong-coupling limit $\beta w^2 \gg \alpha_t$, $\tanh \to \pm 1$ at the minima, so $x_\pm \approx \pm w/\alpha_t$. Using $\ln 2\cosh u \approx |u|$ for large $|u|$, $\mathcal{E}_{\mathrm{tot}}(x_\pm) \approx \frac{w^2}{2\alpha_t} - \frac{w^2}{\alpha_t} = -\frac{w^2}{2\alpha_t}$, while $\mathcal{E}_{\mathrm{tot}}(0) = -k_{\mathrm{B}}T\ln 2$. The barrier is

$$\Delta\mathcal{E} \approx \frac{w^2}{2\alpha_t} - k_{\mathrm{B}}T\ln 2,$$

a double well of the double-well sampler's type, carved by a single hidden p-bit, exactly as in Worked computation 1.19. If instead $\alpha_t \ge \beta w^2 + \kappa$, the curvature is at least κ everywhere and the conditional is unimodal. With a non-zero field h or clamp $c_t x_t \ne 0$, the same latent then produces a skewed, non-Gaussian, but single-peaked conditional. That is the most a single guaranteed-fast stage can express. Multimodality of the generated data must come from composing stages.

16.4.3 Continuous latents and joint convexity

If the latents are continuous, $\mathbf{z} \in \mathbb{R}^M$, one can apply Theorem 16.7 directly in the joint space $(\mathbf{x}, \mathbf{z})$: if the joint energy $\mathcal{E}^f_{t-1} + \mathcal{E}_{\theta,t-1}$ is κ-strongly convex in $(\mathbf{x}, \mathbf{z})$, the joint Langevin sampler mixes at rate $\mu\kappa$, and by Prékopa's theorem the visible marginal of a log-concave density is log-concave. There is a catch. If the latent part is purely quadratic, $\mathcal{E}_{\theta,t-1} = \frac{1}{2}\mathbf{z}^\top\mathbf{\Lambda}\mathbf{z} - \mathbf{x}^\top\mathbf{W}\mathbf{z} + (\text{quadratic in } \mathbf{x})$, then the joint distribution is Gaussian and so is its marginal: Gaussian latents with bilinear couplings add no expressivity at all. Non-Gaussian shape requires either discrete latents or non-quadratic latent self-energies, and in both cases the latent fluctuation term in Eq. (16.39) is what produces it.

16.4.4 Sampling a latent step in hardware

Theorem 16.7 concerns the visible marginal under Langevin dynamics on $F_{\theta,t-1}$. A physical circuit instead evolves $\mathbf{x}$ and $\mathbf{z}$ together. When the latent p-bits flip much faster than the visible nodes relax, the visible nodes feel the conditional mean force, which by Eq. (1.120) is $-\nabla_{\mathbf{x}}F_{\theta,t-1}$; in this time-scale-separated (adiabatic) regime the theorem applies to the visible dynamics directly. When the two timescales are comparable, or when the circuit is run as a block Gibbs sampler that alternates $\mathbf{x} \mid \mathbf{z}$ and $\mathbf{z} \mid \mathbf{x}$, the convergence rate depends on how strongly $\mathbf{x}$ and $\mathbf{z}$ are correlated in the conditional, and the curvature bound alone does not fix it. A stiff spring helps here too, since it pins $\mathbf{x}$ near $c_t\mathbf{x}_t$ and weakens the dependence of $\mathbf{z}$ on the previous visible state, but we do not have a clean rate for that case.

Hardware Realization: Latent DTM steps on a two-colorable grid

In the architecture of [56], each stage is a sparse Boltzmann machine of binary visible and latent nodes on a square grid, typically 70×70 nodes with about 12 local neighbors each, wired so that the interaction graph is bipartite. A bipartite graph is two-colorable, so all nodes of one color are conditionally independent given the other color and can be updated in parallel in a single half-sweep (Section 2.5); the software framework `THRML` [31] implements exactly this block Gibbs schedule. The forward energy adds one coupling per visible node to the clamped value $x_{t,i}$, which in a p-bit circuit is a bias input. If the bipartition places all visible nodes on one side and all latents on the other (an RBM-like layout), then clamping the visibles makes the latents independent, and the clamped expectation over $\mathbf{z} \mid \mathbf{x}$ needed for training (Section 16.5) is a single exact half-sweep. In a general two-colorable grid with latent–latent edges, this expectation also requires a short Gibbs run. The circuit-level realization of such grids and their energy cost are the subject of Sections 17.1 and 17.4.

16.5 The Variational Objective and Step-Local Contrastive Training

How are the T learned energies trained? The answer has a structure that matters for hardware: the global objective splits exactly into one term per stage, each term depends only on that stage's parameters, and each term's gradient is the positive-phase/negative-phase difference of Theorem 1.16 for a conditional EBM.

16.5.1 Decomposing the joint KL divergence

The natural objective is the KL divergence between the joint distribution of the forward process and that of the model,

$$\mathcal{L}_{\mathrm{DN}}(\theta) := D_{\mathrm{KL}}(q(\mathbf{x}_{0:T}) \,\|\, p_\theta(\mathbf{x}_{0:T}))\,, \tag{16.46}$$

which is the objective used in [56]. Here $\theta = (\theta_0, \ldots, \theta_{T-1})$ collects the parameters of all stages, with $\mathcal{E}_{\theta,t-1}$ depending only on θ_{t-1}.

Lemma 16.22 (Markov reversal)**.** *The forward chain Eq. (16.1) can be written in reverse order as*

$$q(\mathbf{x}_{0:T}) = q_T(\mathbf{x}_T) \prod_{t=1}^{T} q(\mathbf{x}_{t-1} \mid \mathbf{x}_t). \tag{16.47}$$

Proof. By the chain rule of probability, $q(\mathbf{x}_{0:T}) = q_T(\mathbf{x}_T) \prod_{t=1}^{T} q(\mathbf{x}_{t-1} \mid \mathbf{x}_{t:T})$. It remains to show $q(\mathbf{x}_{t-1} \mid \mathbf{x}_{t:T}) = q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$. Marginalizing Eq. (16.1) over $\mathbf{x}_{0:t-2}$ gives $q(\mathbf{x}_{t-1:T}) = q_{t-1}(\mathbf{x}_{t-1})\, q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) \prod_{r=t+1}^{T} q(\mathbf{x}_r \mid \mathbf{x}_{r-1})$, and marginalizing further over $\mathbf{x}_{t-1}$ gives $q(\mathbf{x}_{t:T}) = q_t(\mathbf{x}_t) \prod_{r=t+1}^{T} q(\mathbf{x}_r \mid \mathbf{x}_{r-1})$. Dividing, the product over $r > t$ cancels:

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_{t:T}) = \frac{q_{t-1}(\mathbf{x}_{t-1})\, q(\mathbf{x}_t \mid \mathbf{x}_{t-1})}{q_t(\mathbf{x}_t)} = q(\mathbf{x}_{t-1} \mid \mathbf{x}_t). \qquad \square$$

Theorem 16.23 (Step-local decomposition of the DTM objective)**.** *For the forward chain Eq. (16.1) and the reverse model Eq. (16.2),*

$$D_{\mathrm{KL}}(q(\mathbf{x}_{0:T}) \,\|\, p_\theta(\mathbf{x}_{0:T})) = D_{\mathrm{KL}}(q_T \,\|\, \pi) + \sum_{t=1}^{T} \mathbb{E}[D_{\mathrm{KL}}(q(\cdot \mid \mathbf{x}_t) \,\|\, p_\theta(\cdot \mid \mathbf{x}_t))]_{\mathbf{x}_t \sim q_t}\,. \tag{16.48}$$

Moreover, with $\mathcal{L}_t(\theta_{t-1}) := -\mathbb{E}[\ln p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)]_{q(\mathbf{x}_{t-1},\mathbf{x}_t)}$,

$$D_{\mathrm{KL}}(q(\mathbf{x}_{0:T}) \,\|\, p_\theta(\mathbf{x}_{0:T})) = \sum_{t=1}^{T} \mathcal{L}_t(\theta_{t-1}) + C, \tag{16.49}$$

where C *does not depend on* θ.

Proof. Insert Eq. (16.47) and Eq. (16.2) into the definition of the KL divergence. The logarithm of the ratio splits into a sum:

$$\ln \frac{q(\mathbf{x}_{0:T})}{p_\theta(\mathbf{x}_{0:T})} = \ln \frac{q_T(\mathbf{x}_T)}{\pi(\mathbf{x}_T)} + \sum_{t=1}^{T} \ln \frac{q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)}.$$

Take the expectation over $q(\mathbf{x}_{0:T})$. The first term depends only on $\mathbf{x}_T$ and gives $D_{\mathrm{KL}}(q_T \,\|\, \pi)$. The t-th term depends only on $(\mathbf{x}_{t-1}, \mathbf{x}_t)$, so its expectation is over the pair marginal $q(\mathbf{x}_{t-1}, \mathbf{x}_t) = q_t(\mathbf{x}_t)\, q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$; doing the inner integral over $\mathbf{x}_{t-1}$ first gives $\mathbb{E}_{q_t} D_{\mathrm{KL}}(q(\cdot \mid \mathbf{x}_t) \,\|\, p_\theta(\cdot \mid \mathbf{x}_t))$. This proves Eq. (16.48). For Eq. (16.49), split each inner KL as $\mathbb{E}[\ln q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)] - \mathbb{E}[\ln p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)]$. The first part is minus a conditional entropy of the forward process and does not depend on θ; neither does $D_{\mathrm{KL}}(q_T \,\|\, \pi)$. Collect them in C. □

Corollary 16.24 (The bound on the data likelihood)**.** $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta(\mathbf{x}_0)) \le D_{\mathrm{KL}}(q(\mathbf{x}_{0:T}) \,\|\, p_\theta(\mathbf{x}_{0:T}))$. *Equivalently, the expected log-likelihood of the data obeys*

$$\mathbb{E}[\ln p_\theta(\mathbf{x}_0)]_{p_{\mathrm{data}}} \ge \mathbb{E}\left[\ln \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} \mid \mathbf{x}_0)}\right]_{q(\mathbf{x}_{0:T})}, \tag{16.50}$$

the evidence lower bound (ELBO) of diffusion models [51, 101].

Proof. Write both joints as a marginal on $\mathbf{x}_0$ times a conditional: $q(\mathbf{x}_{0:T}) = p_{\mathrm{data}}(\mathbf{x}_0)\, q(\mathbf{x}_{1:T} \mid \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_{0:T}) = p_\theta(\mathbf{x}_0)\, p_\theta(\mathbf{x}_{1:T} \mid \mathbf{x}_0)$. The same splitting of the logarithm as above gives

$$D_{\mathrm{KL}}(q(\mathbf{x}_{0:T}) \,\|\, p_\theta(\mathbf{x}_{0:T})) = D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta(\mathbf{x}_0)) + \mathbb{E}[D_{\mathrm{KL}}(q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) \,\|\, p_\theta(\mathbf{x}_{1:T} \mid \mathbf{x}_0))]_{p_{\mathrm{data}}},$$

and the second term is non-negative. For Eq. (16.50), write $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta(\mathbf{x}_0)) = \mathbb{E}_{p_{\mathrm{data}}} \ln p_{\mathrm{data}} - \mathbb{E}_{p_{\mathrm{data}}} \ln p_\theta(\mathbf{x}_0)$ and $D_{\mathrm{KL}}(q \,\|\, p_\theta) = \mathbb{E}_q \ln p_{\mathrm{data}}(\mathbf{x}_0) + \mathbb{E}_q \ln q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) - \mathbb{E}_q \ln p_\theta(\mathbf{x}_{0:T})$; the entropy terms $\mathbb{E} \ln p_{\mathrm{data}}$ cancel in the inequality. □

Two consequences of Theorem 16.23 follow. First, if the stage family is rich enough to contain the exact reverse conditional, then each $\mathcal{L}_t$ is minimized, separately, by $p_\theta(\cdot \mid \mathbf{x}_t) = q(\cdot \mid \mathbf{x}_t)$ for q_t-almost every clamp, since each inner KL in Eq. (16.48) is non-negative and vanishes only there. By Proposition 16.3 this means $\mathcal{E}_{\theta,t-1} = U_{t-1}$ up to a clamp-dependent constant: training stage t is learning the data distribution blurred to noise level $t-1$. Second, the clamps in Eq. (16.48) are drawn from the forward marginal q_t, whereas during generation stage t receives clamps from the model's own previous stage, distributed as $p_\theta(\mathbf{x}_t)$. When the stages are accurate the two distributions agree; when they are not, a stage may be asked to denoise clamps unlike any it was trained on. Proposition 16.11 bounds the effect in total variation as long as the per-stage errors are controlled uniformly over clamps, which is what the uniform-in-clamp mixing guarantee of Theorem 16.7 provides on the sampling side.

In DDPM [51] each conditional $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is Gaussian, each inner KL in Eq. (16.48) is a KL between Gaussians, and the objective reduces to a weighted mean-square error on the predicted mean. A DTM has the same decomposition with a different family: each inner KL is the conditional negative log-likelihood of an EBM, and there is no closed form. What replaces the closed form is the contrastive gradient.

16.5.2 The contrastive gradient of one stage

Theorem 16.25 (Step-local contrastive gradient)**.** *Let the stage energy $\mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1},\mathbf{x}_t;\theta_{t-1})$ be differentiable in θ_{t-1} with integrable gradients. Then*

$$\nabla_{\theta_{t-1}}\mathcal{L}_t=\beta\Big(\underbrace{\mathbb{E}[\nabla_\theta\mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1},\mathbf{x}_t)]_{q(\mathbf{x}_{t-1},\mathbf{x}_t)}}_{\textit{positive (clamped) phase}}-\underbrace{\mathbb{E}\Big[\mathbb{E}[\nabla_\theta\mathcal{E}_{\theta,t-1}(\mathbf{X},\mathbf{x}_t)]_{\mathbf{X}\sim p_\theta(\cdot|\mathbf{x}_t)}\Big]_{\mathbf{x}_t\sim q_t}}_{\textit{negative (free) phase at fixed clamp}}\Big). \tag{16.51}$$

For a latent-variable stage Eq. (16.37), the positive phase additionally averages over $p_\theta(\mathbf{z}\mid\mathbf{x}_{t-1})$ (the latents relaxed with both $\mathbf{x}_{t-1}$ and $\mathbf{x}_t$ clamped), and the negative phase averages over $p_\theta(\mathbf{x}_{t-1},\mathbf{z}\mid\mathbf{x}_t)$.

Proof. From Eq. (16.4), $-\ln p_\theta(\mathbf{x}_{t-1}\mid\mathbf{x}_t)=\beta[\mathcal{E}^f_{t-1}+\mathcal{E}_{\theta,t-1}]+\ln Z_t(\mathbf{x}_t)$. The forward energy has no θ-dependence. Differentiating $\ln Z_t(\mathbf{x}_t)$ in Eq. (16.5) under the integral, exactly as in Eq. (1.109) but at fixed clamp,

$$\nabla_\theta\ln Z_t(\mathbf{x}_t)=\frac{1}{Z_t(\mathbf{x}_t)}\int\mathrm{d}\mathbf{x}\,(-\beta\nabla_\theta\mathcal{E}_{\theta,t-1}(\mathbf{x},\mathbf{x}_t))e^{-\beta[\mathcal{E}^f_{t-1}+\mathcal{E}_{\theta,t-1}]}=-\beta\mathbb{E}[\nabla_\theta\mathcal{E}_{\theta,t-1}(\mathbf{X},\mathbf{x}_t)]_{p_\theta(\cdot|\mathbf{x}_t)}\,.$$

Hence $\nabla_\theta[-\ln p_\theta(\mathbf{x}_{t-1}\mid\mathbf{x}_t)]=\beta\nabla_\theta\mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1},\mathbf{x}_t)-\beta\mathbb{E}[\nabla_\theta\mathcal{E}_{\theta,t-1}]_{p_\theta(\cdot|\mathbf{x}_t)}$. Averaging over $q(\mathbf{x}_{t-1},\mathbf{x}_t)$ gives Eq. (16.51); the negative-phase term depends on the pair only through $\mathbf{x}_t$, so its outer average is over q_t. For the latent case, apply Theorem 1.18(iii) at fixed clamp: the visible marginal has effective energy $F_{\theta,t-1}$, and $\nabla_\theta F_{\theta,t-1}(\mathbf{x})=\mathbb{E}[\nabla_\theta\mathcal{E}_{\theta,t-1}(\mathbf{x},\mathbf{Z})]_{\mathbf{z}|\mathbf{x}}$ by Eq. (1.123). □

This is the gradient reported in [56], written with the inverse temperature restored. Three features make it suitable for hardware.

(i) **Independence of stages.** By Eq. (16.49), $\mathcal{L}_t$ depends only on θ_{t-1}, so the total gradient has no cross terms between stages. Moreover, the training pairs $(\mathbf{x}_{t-1},\mathbf{x}_t)$ come from the fixed forward process applied to data, never from the model's own earlier stages. Stage t can be trained without running any other stage, and all T stages can be trained at the same time on separate hardware tiles.

(ii) **The positive phase is cheap.** With no latents it is a direct evaluation on training pairs. With latents it requires relaxing only $\mathbf{z}$ with all visible nodes clamped, which is a single half-sweep in the layout of Section 16.4.4.

(iii) **The negative phase is a conditional sampling problem.** For each training clamp $\mathbf{x}_t$, the circuit must produce samples of $p_\theta(\cdot\mid\mathbf{x}_t)$. This is the same task as generation, so the stage-time bound Eq. (16.29) governs the cost of training as well as inference. A stage that satisfies Theorem 16.7 produces an unbiased negative phase after $\mathcal{O}((\mu\kappa)^{-1}\ln(R/\varepsilon))$ of relaxation; a stage with a hidden barrier produces a biased negative phase, which is the training instability that the mixing–expressivity tradeoff causes in monolithic EBMs (Section 1.4).

Example 16.26 (Gradients of a Gaussian–Bernoulli stage)**.** For the stage energy Eq. (16.41), the parameter derivatives of the energy are $\partial\mathcal{E}/\partial\mathbf{W}=-\mathbf{x}\mathbf{z}^\top$, $\partial\mathcal{E}/\partial\mathbf{h}=-\mathbf{z}$, $\partial\mathcal{E}/\partial\mathbf{b}=-\mathbf{x}$, and $\partial\mathcal{E}/\partial\mathbf{G}=\frac{1}{2}\mathbf{x}\mathbf{x}^\top$. In the positive phase the visible pair is clamped, the latents are independent given $\mathbf{x}_{t-1}$, and $\mathbb{E}[\mathbf{z}\mid\mathbf{x}_{t-1}]=\tanh(\beta(\mathbf{W}^\top\mathbf{x}_{t-1}+\mathbf{h}))$ componentwise, as in Worked computation 1.19. Equation (16.51) gives, for example,

$$\nabla_{\mathbf{W}}\mathcal{L}_t=-\beta\Big(\mathbb{E}\Big[\mathbf{x}_{t-1}\tanh(\beta(\mathbf{W}^\top\mathbf{x}_{t-1}+\mathbf{h}))^\top\Big]_{q(\mathbf{x}_{t-1},\mathbf{x}_t)}-\mathbb{E}\Big[\mathbb{E}\Big[\mathbf{X}\mathbf{Z}^\top\Big]_{p_\theta(\mathbf{x},\mathbf{z}|\mathbf{x}_t)}\Big]_{q_t}\Big),$$

$$\nabla_{\mathbf{G}}\mathcal{L}_t = \frac{\beta}{2}\Big(\mathbb{E}\Big[\mathbf{x}_{t-1}\mathbf{x}_{t-1}^\top\Big]_q - \mathbb{E}\Big[\mathbb{E}\Big[\mathbf{X}\mathbf{X}^\top\Big]_{p_\theta(\cdot|\mathbf{x}_t)}\Big]_{q_t}\Big).$$

These are the moment-matching conditions of Eq. (1.105), applied clamp by clamp: at a stationary point, the second moments of the stage's free relaxation, averaged over clamps drawn from the forward process, equal those of the noisy data pairs. The forward stiffness α_t appears nowhere in the gradient, but it shapes the negative-phase distribution and therefore how quickly it can be sampled.

Physics Insight: Contrastive divergence is natural for DTM stages

Hinton's k-step contrastive divergence [50] initializes the negative-phase chain at the training point and runs it briefly (Eq. (1.111)). For a monolithic EBM this is biased whenever the chain cannot leave the data's mode. For a DTM stage the training point $\mathbf{x}_{t-1}$ is an exact sample of the target conditional $q(\cdot \mid \mathbf{x}_t)$, so if the model is close to the truth, the chain starts close to equilibrium. When the stage is κ-strongly log-concave, Theorem 16.7(iv) bounds the remaining bias: after relaxation time s the negative-phase law is within $e^{-\mu\kappa s}\mathcal{W}_2(q(\cdot \mid \mathbf{x}_t), p_\theta(\cdot \mid \mathbf{x}_t))$ of the model conditional in $\mathcal{W}_2$. Short chains are therefore nearly unbiased, and the bias shrinks as the model improves.

16.5.3 One training iteration on hardware

Putting Theorem 16.25 and Section 16.4.4 together, one stochastic-gradient iteration for stage t on a hardware tile consists of the following physical operations.

1. **Draw a minibatch of training pairs.** Take data points $\mathbf{x}_0^{(m)}$, $m = 1, \ldots, B_{\mathrm{mb}}$, and run the fixed forward process to produce pairs $(\mathbf{x}_{t-1}^{(m)}, \mathbf{x}_t^{(m)})$. For a linear Gaussian forward process this is a closed-form Gaussian draw (Proposition 15.2); for spins it is a set of independent flips. No stage of the model is involved.

2. **Clamped phase.** Load $\mathbf{x}_t^{(m)}$ into the clamp inputs and $\mathbf{x}_{t-1}^{(m)}$ into the visible nodes, hold both fixed, let the latents relax, and record the energy derivatives $\nabla_\theta \mathcal{E}_{\theta,t-1}$. For bilinear couplings these are local products such as $x_{t-1,i}z_a$, which each coupling cell can accumulate on its own.

3. **Free phase.** Keep only $\mathbf{x}_t^{(m)}$ clamped, release the visible nodes (starting either from the data value, as in contrastive divergence, or from the spring-only state of Proposition 16.10), let the stage relax for the stage time Eq. (16.29), and record the same local products.

4. **Update.** Change each coupling by $-\eta\beta$ times the difference between its clamped and free averages, as in Eq. (1.110).

Every quantity in steps 2–4 is local to a coupling cell, the clamps come from the forward process rather than from other stages, and the stage's mixing guarantee bounds the time of step 3. These are the properties that let all T stages of a DTM train simultaneously and independently on separate tiles. The calibration of such updates against device mismatch is taken up in Chapter 18.

16.5.4 Regularizing for mixing

The decomposition Eq. (16.48) contains no term that rewards fast mixing. Nothing in maximum likelihood prevents a stage from learning a conditional with a hidden barrier if that improves the fit, and with few large steps the true conditional may have one. The DTMs of [56] therefore add

a penalty to the objective, $\mathcal{L} = \mathcal{L}_{\mathrm{DN}} + \sum_t \lambda_t \mathcal{L}_t^{\mathrm{TC}}$, where $\mathcal{L}_t^{\mathrm{TC}}$ is a total-correlation-type quantity that measures how far the stage's conditional over visible and latent nodes is from the product of its single-node marginals. Strong correlations between many nodes are what make block Gibbs sampling slow, so penalizing them keeps each stage easy to sample. The weights λ_t are adjusted during training by feedback on the measured autocorrelation of the stage's sampler. The physics of total-correlation penalties on sparse grids, and the connection to the spatial correlation length, are developed in Section 17.2. In the language of this chapter, such a penalty is a soft, learned substitute for the hard curvature condition Eq. (16.26): it lets the designer use fewer, larger steps than Proposition 16.12 would allow, while keeping the per-stage mixing time under empirical control.

16.6 Worked Computation: Barrier Annihilation in the Double-Well DTM

We now return to the double-well sampler. The quartic double well has appeared as a p-bit (Chapters 1 and 14), as the counterexample to Bakry–Émery curvature (Worked computation 6.21), and as the textbook case of Kramers escape (Worked computation 7.16). Here it is the data distribution of a DTM, and we compute everything needed to compare a DTM against direct Langevin sampling: the conditional Hessians, the critical stiffness, the stage count, and the total time and energy.

Worked computation 16.27 (The double-well DTM: critical stiffness, stage schedule, and comparison with Kramers relaxation)**. Setup and units.** The data distribution is $p_{\mathrm{data}}(x_0) \propto e^{-\beta E_0(x_0)}$ in one dimension with

$$E_0(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2, \qquad E_0'(x) = ax^3 - bx, \qquad E_0''(x) = 3ax^2 - b. \tag{16.52}$$

The minima are at $x_\pm = \pm\sqrt{b/a}$ with curvature $E_0''(x_\pm) = 2b$; the barrier top is at $x = 0$ with curvature $E_0''(0) = -b$; and the barrier height is $\Delta E = E_0(0) - E_0(x_\pm) = b^2/(4a)$. We measure lengths in units of the well position, $\ell := \sqrt{b/a}$, so the wells sit at $x = \pm 1$. In these units $a\ell^4 = b\ell^2 = 4\Delta E$, and it is convenient to define the dimensionless curvature

$$B := \beta b \ell^2 = 4\beta\Delta E, \qquad \beta E_0(x) = \frac{B}{4}(x^4 - 2x^2). \tag{16.53}$$

We take $\Delta E = 10\, k_{\mathrm{B}}T$, so $B = 40$: the barrier-top curvature is $-40\, k_{\mathrm{B}}T/\ell^2$ and the well curvature is $+80\, k_{\mathrm{B}}T/\ell^2$. All Langevin samplers have mobility μ and $D = \mu k_{\mathrm{B}}T$, and we measure time in units of the diffusion time across one well position,

$$\tau_0 := \frac{\ell^2}{\mu k_{\mathrm{B}} T}. \tag{16.54}$$

The fastest physical process in the problem is relaxation inside one well, $\tau_{\mathrm{intra}} = 1/(\mu E_0''(x_\pm)) = \tau_0/(2B) = \tau_0/80$. In a circuit this is the natural clock, of order τ_{RC}. The forward process is variance preserving, $x_t = \sqrt{1-\nu_t}\, x_{t-1} + \sqrt{\nu_t}\, \xi_t$ in units of ℓ, which is appropriate because the data variance is close to ℓ^2.

Step 1: The baseline, direct Langevin sampling. A single Langevin circuit with energy E_0 must hop between wells. By Theorem 7.4, the escape rate from either well is

$$k = \frac{\mu\sqrt{E_0''(x_\pm)\ |E_0''(0)|}}{2\pi} e^{-\beta\Delta E} = \frac{\mu\sqrt{2b\cdot b}}{2\pi} e^{-\beta\Delta E} = \frac{\sqrt{2}\,B}{2\pi}\, e^{-B/4}\, \frac{1}{\tau_0}. \tag{16.55}$$

With $B = 40$: $\sqrt{2} \cdot 40/(2\pi) = 9.00$ and $e^{-10} = 4.54 \times 10^{-5}$, so $k = 4.09 \times 10^{-4}\,\tau_0^{-1}$. For a symmetric double well the spectral gap is $\lambda_1 = 2k = 8.2 \times 10^{-4}\,\tau_0^{-1}$ (Section 7.5). A direct numerical diagonalization of the Witten Hamiltonian $\mathcal{H}_{\mathrm{W}} = D(-\partial_x^2 + \frac{\beta^2}{4}E_0'^2 - \frac{\beta}{2}E_0'')$ on a fine grid gives $\lambda_1 = 7.8 \times 10^{-4}\,\tau_0^{-1}$; the 4% difference is the $\mathcal{O}((\beta\Delta E)^{-1})$ correction to Kramers' formula. The next eigenvalue is $\lambda_2 \approx 36\,\tau_0^{-1}$. It belongs to an even mode localized at the barrier top, where the Witten potential is approximately $D\ell^{-2}[\frac{B}{2} + (\frac{B^2}{4} - \frac{3B}{2})x^2] = (20 + 340x^2)\,\tau_0^{-1}$ with ground level $20 + \sqrt{340} \approx 38$; physically, it is the rate at which probability placed on the saddle drains into the wells. The intra-well relaxation rate $\mu E_0''(x_\pm) = 80\,\tau_0^{-1}$ lies higher still. Starting from one well, the populations of the two wells approach $\frac{1}{2}$ as $e^{-\lambda_1 s}$. To reduce the error by a factor $e^{-5} \approx 7 \times 10^{-3}$ we need

$$\tau_{\mathrm{direct}} \approx \frac{5}{\lambda_1} = 6.4 \times 10^3\,\tau_0 = 5.1 \times 10^5\,\tau_{\mathrm{intra}}. \tag{16.56}$$

The circuit spends nearly all of this time thermalized inside one well, waiting for a rare escape.

Step 2: The final stage and the critical stiffness $\alpha_1 > b$. The last stage of the DTM produces x_0 from the clamp x_1. By Proposition 16.3 its exact learned energy is $U_0 = E_0$, so with $y := c_1 x_1$,

$$\mathcal{E}_{\mathrm{tot}}(x_0 \mid x_1) = \frac{\alpha_1}{2}(x_0 - y)^2 + \frac{a}{4}x_0^4 - \frac{b}{2}x_0^2. \tag{16.57}$$

Differentiating twice in x_0, the linear-in-y cross term $-\alpha_1 y x_0$ drops out and

$$\partial_{x_0}^2 \mathcal{E}_{\mathrm{tot}}(x_0 \mid x_1) = \alpha_1 - b + 3ax_0^2 \;\geq\; \alpha_1 - b, \tag{16.58}$$

with equality at $x_0 = 0$, for every value of the clamp. The one-dimensional Hessian spectrum of the stage is the interval $[\alpha_1 - b, \infty)$, and its lower edge is set entirely by the barrier-top curvature of the data. Hence:

- If $\alpha_1 > b$, the stage is $(\alpha_1 - b)$-strongly convex for all x_1, and Theorem 16.7 applies with $\kappa = \alpha_1 - b$.
- If $\alpha_1 < b$ and $y = 0$, Eq. (16.57) is $\frac{a}{4}x_0^4 - \frac{b-\alpha_1}{2}x_0^2$, a double well with barrier $(b - \alpha_1)^2/(4a)$ (Eq. (6.147)) whose spectral gap collapses as in Eq. (6.148). For $y \neq 0$ the clamp adds a tilt $-\alpha_1 y x_0$, giving the tilted quartic of Worked computation 7.16; the barrier persists for small tilts.

The critical stiffness $\alpha_1 = b$ is the pitchfork point of Worked computation 6.21. In the three conventions in use:

$$\alpha_1 > b \quad\Longleftrightarrow\quad \beta\alpha_1\ell^2 > B = 4\beta\Delta E \quad\Longleftrightarrow\quad \frac{1-\nu_1}{\nu_1} > B. \tag{16.59}$$

The first form is in physical energy units (this book). The second is in the dimensionless convention of [56], where energies are measured in units of $k_{\mathrm{B}}T$. The third uses $\beta\alpha_1\ell^2 = (1-\nu_1)/\nu_1$ for the variance-preserving process. For $\Delta E = 10\,k_{\mathrm{B}}T$ the last stage must add forward noise of variance $\nu_1 < 1/41$, a standard deviation below $0.156\,\ell$. Choosing the margin $\kappa = b$ (so $\alpha_1 = 2b$) gives $\nu_1 = 1/81$, a forward step of $0.11\,\ell$, and a stage relaxation rate $\mu\kappa = \mu b = B/\tau_0 = 40\,\tau_0^{-1}$. The stage relaxes as fast as the data's own wells: $1/(\mu\kappa) = \tau_0/40 = 2\tau_{\mathrm{intra}}$.

The cold-start radius Eq. (16.28) for this stage is $R_1 = \sqrt{k_{\mathrm{B}}T/\kappa + |E_0'(y)|\,/\kappa}$. In units of ℓ and with $\kappa = b$: $\sqrt{1/40} = 0.16$, and $|E_0'(y)|\,/b = |y^3 - y|$, which is zero at the wells and at most 0.38 for $|y| \leq 1.15$. So $R_1 \lesssim 0.55\,\ell$, and reaching accuracy $\varepsilon = 0.01\,\ell$ takes $\ln(55) \approx 4$ relaxation times. Below we use $\ln(R/\varepsilon) = 5$ for every stage, slightly conservative.

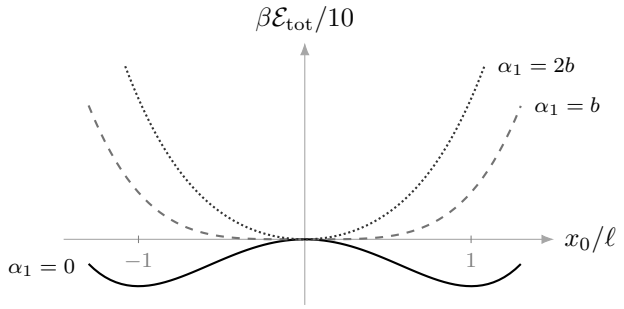


Figure 16.2: Conditional energy of the final DTM stage, Eq. (16.57), for clamp $y = 0$ and $\Delta E = 10\, k_{\mathrm{B}}T$ ($B = 40$), shown as $\beta\mathcal{E}_{\text{tot}}/10 = x^4 + 2(r-1)x^2$ with $r = \alpha_1/b$. Without the spring (solid) the stage inherits the full barrier. At $\alpha_1 = b$ (dashed) the curvature at the origin vanishes. At $\alpha_1 = 2b$ (dotted) the stage is b-strongly convex and relaxes in about $2\tau_{\text{intra}}$.

Step 3: The intermediate stages need more stiffness than b. It is tempting to conclude that $\alpha_t > b$ suffices at every stage. It does not, and the reason is instructive. By Proposition 16.3, stage t must represent the marginal energy U_{t-1} at signal level $\bar{\gamma} \equiv \bar{\gamma}_{t-1}$, not E_0. Its curvature follows from the second-order Tweedie identity Eq. (15.76), which in our notation reads

$$\beta\ell^2\, U_t''(x) = \frac{1}{1-\bar{\gamma}_t} - \frac{\bar{\gamma}_t}{(1-\bar{\gamma}_t)^2}\,\mathrm{Var}(x_0 \mid x_t = x), \tag{16.60}$$

where the conditional variance is over the posterior

$$q(x_0 \mid x_t = x) \propto p_{\text{data}}(x_0)\exp\!\left(-\frac{(x - \sqrt{\bar{\gamma}_t}\,x_0)^2}{2(1-\bar{\gamma}_t)}\right).$$

The most negative curvature occurs at $x = 0$, where the posterior is symmetric. Introduce the signal-to-noise ratio $g := \bar{\gamma}/(1-\bar{\gamma})$, so that $1/(1-\bar{\gamma}) = 1 + g$. At $x = 0$ the posterior is

$$q(x_0 \mid x_t = 0) \propto \exp\!\Big(-\frac{B}{4}x_0^4 + \frac{B}{2}x_0^2 - \frac{g}{2}x_0^2\Big) = \exp\!\Big(-\frac{B}{4}x_0^4 + \frac{B-g}{2}x_0^2\Big). \tag{16.61}$$

This is again a quartic double well. The noise acts as a spring of dimensionless stiffness g pulling x_0 toward the barrier top, and the posterior is bimodal when $g < B$, with peaks at $x_0^2 = 1 - g/B$. Treating the two peaks as sharp (a Laplace estimate), $\mathrm{Var}(x_0 \mid x_t = 0) \approx 1 - g/B$ and

$$\beta\ell^2\, U_t''(0) \approx (1+g)\Big[1 - g\Big(1 - \frac{g}{B}\Big)\Big] \approx -g^2\Big(1 - \frac{g}{B}\Big) \qquad (1 \ll g < B). \tag{16.62}$$

The right-hand side is most negative at $g = 2B/3$, where it equals $-\frac{4}{27}B^2$. Two limits explain the shape. When $g \to \infty$ (no noise), $U_t \to E_0$ and the curvature at the origin is $-B$. When g drops to order B, the noise kernel is narrow compared with the wells but wide enough that a point on the barrier could have come from either side; the posterior variance is then of order one while the noise variance $1/g$ is small, and the second term of Eq. (16.60) grows like g^2. The marginal energy develops a sharp ridge between the partially merged wells, with curvature far larger than the original barrier top.

Numerical quadrature of Eq. (16.60) with the exact quartic p_{data} gives the following non-convexity $\beta\ell^2 L(U_t) = -\min_x \beta\ell^2 U_t''$:

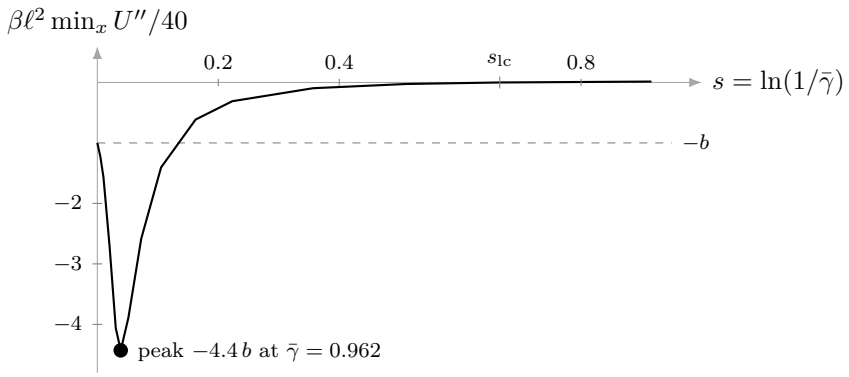


Figure 16.3: Most negative curvature of the marginal energy U_t for the quartic data distribution with $\Delta E = 10\,k_{\mathrm{B}}T$, in units of $b = 40\,k_{\mathrm{B}}T/\ell^2$, against the forward log-attenuation s. At $s = 0$ it equals the barrier-top curvature $-b$. It first becomes more negative, peaking at about $-4.4\,b$, and then rises through zero at the log-concavity onset $s_{\mathrm{lc}} = 0.666$ ($\bar{\gamma} = 0.514$). The forward stiffness of each DTM stage must exceed the magnitude of this curve by the margin κ.

$\bar{\gamma}_t$	1	0.99	0.962	0.95	0.9	0.8	0.7	0.6	0.514
$\beta\ell^2 \min_x U_t''$	-40	-62	-177	-156	-56	-12.4	-3.8	-1.0	0

The peak non-convexity $177\,k_{\mathrm{B}}T/\ell^2 = 4.4\,b$ occurs at $\bar{\gamma} = 0.962$ ($g = 25$); the Laplace estimate gives $\frac{4}{27}B^2 = 237$ at $g = 27$, which overestimates because the posterior wells in Eq. (16.61) are shallow there. The marginal becomes log-concave at $\bar{\gamma}_{\mathrm{lc}} = 0.514$, i.e. $s_{\mathrm{lc}} = \ln(1/0.514) = 0.666$. This is the speciation transition of Definition 15.13: the two-Gaussian model of Worked computation 15.15 predicts $\bar{\gamma}_s = 1/(1 + \mu_0^2 - \sigma_0^2)$, where μ_0 is the mean distance of a data point from the barrier top and σ_0^2 the variance within one well. For the quartic data at $\Delta E = 10\,k_{\mathrm{B}}T$, direct quadrature gives $\mu_0 = \mathbb{E}[|x_0|] = 0.979$ and $\sigma_0^2 = \mathrm{Var}(|x_0|) = 0.0141$ (close to the harmonic estimate $1/80$), so the formula gives $\bar{\gamma}_s = 1/(1 + 0.958 - 0.014) = 0.514$, in agreement with the quadrature of Eq. (16.60). For $\bar{\gamma} < 0.514$ the marginal energy is convex, with curvature rising toward the prior value $k_{\mathrm{B}}T/\ell^2$ (it is 0.61, 0.95, 0.99 in these units at $\bar{\gamma} = 0.4$, 0.2, 0.1).

The conclusion is that $\alpha_t > b$ is the condition for the last stage only. For an exact-reverse DTM with guaranteed log-concave stages, the stiffness must track $L(U_{t-1}) + \kappa$, and at its peak this is about $4.4\,b + \kappa$.

Step 4: The stage schedule. We build the schedule greedily, starting at $\bar{\gamma}_0 = 1$ and choosing at each stage the largest forward step allowed by Proposition 16.12:

$$\alpha_t = L(U_{t-1}) + \kappa, \qquad \nu_t = \frac{k_{\mathrm{B}}T}{\alpha_t \ell^2 + k_{\mathrm{B}}T}, \qquad \bar{\gamma}_t = (1 - \nu_t)\,\bar{\gamma}_{t-1}, \tag{16.63}$$

and continue until $\bar{\gamma}_T \le 10^{-3}$, where the forward marginal is within a negligible distance of the standard Gaussian prior. The forward index t counts from the data side, so stage $t = 1$ is the last stage executed during generation. For $\kappa = 10\,k_{\mathrm{B}}T/\ell^2$ the first few forward steps are (stiffness in units of $k_{\mathrm{B}}T/\ell^2$, noise in units of ℓ^2):

t	1	2	3	4	5
$\bar{\gamma}_{t-1}$	1.000	0.980	0.972	0.966	0.961
$\beta\ell^2 L(U_{t-1})$	40	106	154	174	177
$\beta\ell^2\alpha_t$	50	116	164	184	187
ν_t	0.0196	0.0085	0.0061	0.0054	0.0053

The forward steps shrink by a factor of almost four within the first five stages, as the stiffness follows the non-convexity peak of Fig. 16.3. Past the peak they lengthen again; by $t = 23$ ($\bar{\gamma} = 0.60$) the non-convexity has fallen to $0.9\,k_{\mathrm{B}}T/\ell^2$, $\nu_t = 0.084$, and the window closes at $t = 24$. Beyond it, every marginal energy is convex and the schedule runs at the constant stiffness $\alpha_t = \kappa$. The resulting stage counts and the summed stage relaxation times $\sum_t(\mu\kappa_t)^{-1}$, where $\kappa_t = \alpha_t + \min U''_{t-1}$ is the actual curvature margin of stage t (equal to κ inside the window), are:

κ $[k_{\mathrm{B}}T/\ell^2]$	T_{nc} (window)	T (total)	$\sum_{\mathrm{window}}(\mu\kappa_t)^{-1}$ $[\tau_0]$	$\sum_{\mathrm{all}}(\mu\kappa_t)^{-1}$ $[\tau_0]$
5	21	55	4.2	9.9
10	24	90	2.4	8.4
20	31	159	1.55	7.7
40	44	297	1.1	7.3

These numbers check Proposition 16.12. Numerically $\int_0^{s_{\mathrm{lc}}} \beta\ell^2 L(s)\,\mathrm{d}s = 16.1$, so for $\kappa = 40\,k_{\mathrm{B}}T/\ell^2$ the estimate Eq. (16.35) is $16.1 + 40 \times 0.666 + 0.33 = 43$ window stages, against 44 from the schedule; for $\kappa = 5\,k_{\mathrm{B}}T/\ell^2$ it is 19.8, against 21. The lower bound $T_{\mathrm{nc}} \geq \kappa s_{\mathrm{lc}}\ell^2/k_{\mathrm{B}}T$ is 27 and 3.3 respectively. As κ grows, the total relaxation time approaches $s_{\mathrm{total}}\tau_0 = \ln(10^3)\,\tau_0 = 6.9\,\tau_0$, the time the forward Ornstein–Uhlenbeck process needs to reach the prior, while the stage count grows linearly in κ.

Step 5: Total time and energy compared with direct Langevin sampling. Each stage runs for $\ln(R/\varepsilon)/(\mu\kappa_t) = 5/(\mu\kappa_t)$, so the total DTM relaxation time is $5\sum_t(\mu\kappa_t)^{-1}$. For the energy, suppose that a sampling tile draws a static power P while relaxing and that each stage transition (reading out x_{t-1}, latching it as the next clamp, and resetting) costs a fixed overhead $E_{\mathrm{stage}} = \eta\,P\tau_0$. Then

$$E_{\mathrm{DTM}} = P\tau_0\Big(5\sum_t \frac{1}{\mu\kappa_t\tau_0} + \eta\,T\Big), \qquad E_{\mathrm{direct}} = P\,\tau_{\mathrm{direct}} = 6.4\times 10^3\,P\tau_0. \tag{16.64}$$

For $\Delta E = 10\,k_{\mathrm{B}}T$:

κ $[k_{\mathrm{B}}T/\ell^2]$	T	τ_{DTM} $[\tau_0]$	$\tau_{\mathrm{direct}}/\tau_{\mathrm{DTM}}$	$E_{\mathrm{DTM}}/P\tau_0$ $(\eta = 1)$	$E_{\mathrm{direct}}/E_{\mathrm{DTM}}$ $(\eta = 1)$
5	55	50	130	105	61
10	90	42	150	132	48
20	159	38	167	197	32
40	297	36	175	333	19

The time advantage grows slowly with κ, while the energy advantage shrinks once the per-stage overhead dominates; with $\eta = 1$ the energy optimum is at small margin, $\kappa \approx 5\,k_{\mathrm{B}}T/\ell^2$. A smaller overhead ($\eta = 0.1$) moves the optimum to $\kappa \approx 10\,k_{\mathrm{B}}T/\ell^2$ with $E_{\mathrm{DTM}} \approx 51\,P\tau_0$. With the representative identification $\tau_{\mathrm{intra}} \approx 1\,\mathrm{ns}$ (so $\tau_0 \approx 80\,\mathrm{ns}$), the DTM finishes in about $3\,\mu\mathrm{s}$ while the direct sampler needs about 0.5 ms. The realistic overheads and power levels of subthreshold CMOS tiles are estimated in Section 17.4. Any physical stage transition also erases the previous stage's state, which carries a thermodynamic minimum cost of order $k_{\mathrm{B}}T\ln 2$ per erased bit (Chapter 9); for the overheads considered here this floor is far below E_{stage}.

Step 6: Scaling with the barrier height. Repeat the computation with $\Delta E = 20\,k_{\mathrm{B}}T$ ($B = 80$), keeping the wells at $\pm\ell$. Kramers' formula Eq. (16.55) gives $\lambda_1 = 2k = 7.4 \times 10^{-8}\,\tau_0^{-1}$, and the Witten diagonalization gives $7.2 \times 10^{-8}\,\tau_0^{-1}$, so $\tau_{\mathrm{direct}} = 5/\lambda_1 = 6.9 \times 10^7\,\tau_0$: the doubling of the barrier multiplies the direct sampling time by $e^{10} \approx 2.2 \times 10^4$. For the DTM, the peak non-convexity rises to $794\,k_{\mathrm{B}}T/\ell^2$ (Laplace estimate $\frac{4}{27}B^2 = 948$), the curvature integral to $\int L\,\mathrm{d}s = 35.4\,k_{\mathrm{B}}T/\ell^2$, and the log-concavity onset barely moves ($\bar{\gamma}_{\mathrm{lc}} = 0.507$). The schedule gives

$\kappa\ [k_{\mathrm{B}}T/\ell^2]$	T_{nc}	T	$\tau_{\mathrm{DTM}}\ [\tau_0]$	$\tau_{\mathrm{direct}}/\tau_{\mathrm{DTM}}$
5	40	75	70	1.0×10^6
10	44	109	52	1.3×10^6
40	64	316	39	1.8×10^6

Doubling the barrier roughly doubles the curvature integral and adds 15–20 stages, while the direct time grows by four orders of magnitude. The Laplace estimate makes the scaling explicit: the peak stiffness requirement is $L_{\max} \approx \frac{4}{27}\beta b^2\ell^2 = \frac{16}{27}(\beta\Delta E)\,b$, linear in $\beta\Delta E$ in units of b, and the curvature integral grows in proportion. The DTM pays polynomially in the barrier height; the monolithic sampler pays exponentially.

Step 7: What the computation shows.

(a) The forward spring removes the barrier from every stage, and the critical stiffness of the last stage is exactly the barrier-top curvature, $\alpha_1 > b$, equivalently $(1-\nu_1)/\nu_1 > 4\beta\Delta E$.

(b) Intermediate stages see a marginal energy whose ridge is sharper than the original barrier, by a factor ≈ 4.4 at $\Delta E = 10\,k_{\mathrm{B}}T$ and growing linearly with $\beta\Delta E$. The stiffness must track this, so the forward steps must be smallest in the window $0.5 \lesssim \bar{\gamma} \lesssim 1$ where the two modes are partially merged.

(c) With guaranteed log-concave stages, the chain needs tens of stages in that window. The total relaxation time is a few tens of τ_0, two orders of magnitude below direct Langevin sampling at $\Delta E = 10\,k_{\mathrm{B}}T$ and six orders below at $\Delta E = 20\,k_{\mathrm{B}}T$.

(d) The guarantee is paid for in stage count. The DTMs of [56], with 2 to 8 stages, sit far from this regime: their stages are not guaranteed to be unimodal, and their mixing is controlled empirically (Remark 16.18). The computation here shows what a provably barrier-free chain looks like for the simplest multimodal target, and why a practical design would relax the guarantee to save stages.

Summary

Proved

- The forward coupling energies of Gaussian, Ising, and Potts forward steps: a harmonic spring of stiffness $\alpha_t = k_{\mathrm{B}}T/(c_t^2\nu_t)$, a ferromagnetic coupling $-\frac{1}{2}k_{\mathrm{B}}T\Gamma_t x_{t,i}x_{t-1,i}$ with $\Gamma_t = \ln((1-p_t)/p_t)$, and its Potts analogue (Eqs. (16.9), (16.13) and (16.14)).
- The exact reverse conditional is the forward energy plus the marginal energy U_{t-1}, so the ideal learned energy is independent of the clamp (Proposition 16.3); its small-step limit is the Gaussian score update (Eq. (16.20)).
- The DTM mixing theorem: under $\nabla^2\mathcal{E}_\theta \succeq -L_t\mathbf{I}$ and $\alpha_t \geq L_t + \kappa$, every stage is κ-strongly log-concave uniformly in the clamp, with Poincaré and log-Sobolev constants

at least $\mu\kappa$, KL decay at rate $2\mu\kappa$, $\mathcal{W}_2$ contraction at rate $\mu\kappa$, and an explicit cold-start radius (Theorem 16.7); the one-dimensional supersymmetric-partner proof of the gap bound (Eq. (16.32)).

- The step-size budget: the stiffness condition caps the forward noise per stage, and the stage count and total relaxation time across the non-log-concave window are controlled by $\int L(s)\,\mathrm{d}s$ (Proposition 16.12).
- The spring-only warm-start bound and the additive accumulation of per-stage errors in total variation (Propositions 16.10 and 16.11).
- Latent fluctuations reduce visible curvature by $\beta\,\mathrm{Cov}(\nabla_{\mathbf{x}}\mathcal{E})$, giving the latent stage condition $\alpha_t \geq L_0 + \beta\Sigma_{\max} + \kappa$ and, for Gaussian–Bernoulli stages, $\alpha_t + \lambda_{\min}(\mathbf{G}) \geq \beta\,\|\mathbf{W}\|_2^2 + \kappa$ (Proposition 16.20 and Eq. (16.45)).
- The exact step-local decomposition of the joint KL objective, the ELBO bound on the data likelihood, and the step-local contrastive gradient (Theorems 16.23 and 16.25 and Corollary 16.24).
- For the quartic double well: the final-stage critical stiffness $\alpha_1 > b$ in three equivalent conventions, the Laplace estimate $L_{\max} \approx \frac{4}{27}\beta b^2\ell^2$ of the intermediate non-convexity, and the stage schedule and timing comparison (Worked computation 16.27).

Assumed

- Each continuous stage is modelled as overdamped Langevin dynamics with scalar mobility μ and a white thermal bath at a single temperature; underdamped, colored-noise, and non-equilibrium effects (Chapter 13) are neglected.
- The learned energy has a uniform lower Hessian bound $-L_t$ over the whole state space. Real learned energies may have such a bound only on a bounded region.
- In the worked computation, the learned energy equals the exact marginal energy U_{t-1}, and the marginal curvatures are obtained by numerical quadrature; the stage cost model $P\tau_0$ plus a fixed overhead per stage is a representative assumption.
- In the latent-variable discussion, the adiabatic regime in which latents equilibrate quickly relative to visible nodes.

Taken up later

- Full proofs of the Bakry–Émery and Holley–Stroock theorems are in Chapter 6 and Chapter C; the Gaussian integrals and second-order Tweedie identity are in Chapter B and Chapter 15.
- The circuit realization of DTM stages, joint versus sequential relaxation of the chain, total-correlation regularization, and the energy budget of a full chip are in Chapter 17.
- Differentiating through stochastic hardware and calibrating stages in the presence of device mismatch are in Chapter 18.

Open / Frontier

- A quantitative mixing theorem for discrete DTM stages with latent variables under block Gibbs sampling. Remark 16.16 gives a Dobrushin-type condition for visible spins only.
- Mixing rates for joint visible–latent dynamics outside the adiabatic regime.
- Optimal schedules that trade stage count against per-stage sweeps, and a theory for the few-step regime used in practice, where stages are not log-concave and mixing is controlled empirically.
- How the intermediate non-convexity $L(U_t)$ found in Worked computation 16.27 scales for high-dimensional, structured data, and whether learned energies can stay below it without losing accuracy.

Bibliographic notes

Denoising Thermodynamic Models, their latent-variable Boltzmann-machine stages, the step-local training objective, the adaptive total-correlation penalty, and the all-transistor architecture with its system-level energy estimate are due to Jelinčič et al. [56]. The block Gibbs sampling of sparse two-colorable stages is implemented in Extropic's `THRML` library [31]. The mixing-time theorem, step-size budget, latent curvature condition, and double-well computations of this chapter are this book's analysis of the continuous-variable version of the model; they are not stated in the DTM paper.

Diffusion models as reversed noising chains with a variational objective were introduced by Sohl-Dickstein et al. [101]; the Gaussian-step parameterization and the decomposition of the objective into per-step terms were developed by Ho et al. [51], and the continuous-time SDE formulation by Song et al. [103]. Contrastive divergence is due to Hinton [50].

The curvature criterion for logarithmic Sobolev inequalities is due to Bakry and Émery [6]; the monograph of Bakry et al. [7] covers the Γ_2 calculus, the Holley–Stroock perturbation lemma, and $\mathcal{W}_2$ contraction under curvature bounds. The supersymmetric structure of the Fokker–Planck operator used in Eq. (16.32) goes back to Witten [118]. Kramers' escape rate is from Kramers [61], reviewed in Hänggi et al. [46]. The speciation transition in reverse diffusion used to interpret the log-concavity onset is from Biroli et al. [12].

Chapter 17

The Denoising Thermodynamic Computer Architecture and Its Energetics

Chapter 16 showed how the expressivity–mixing dilemma of Chapter 3 can be managed by splitting a multimodal distribution into a chain of conditional Energy-Based Models (EBMs), the Denoising Thermodynamic Model (DTM) of Jelinčič et al. [56]. The forward coupling $\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) = \frac{\alpha_t}{2} \|\mathbf{x}_{t-1} - c_t \mathbf{x}_t\|^2$ (or its Ising or Potts counterpart) adds curvature α_t in every direction. If the negative curvature of the learned energy is bounded by $L_{\mathcal{E}}$ (the constant L_t of Assumption 16.5) and $\alpha_t \geq L_{\mathcal{E}} + \kappa$, each conditional step is κ-strongly log-concave and relaxes with mixing time $\tau_{\text{mix}} \leq (\mu\kappa)^{-1}$, with no Kramers barrier to cross (Theorem 16.7).

This chapter asks how such a chain can be built in silicon. A chip is nearly two-dimensional and has only a few metal routing layers. Dense all-to-all couplings among $N \sim 10^4$–10^6 variables would require long wires, whose capacitance raises both the relaxation time $\tau_{RC} = C_{\text{node}}/G$ and the switching energy $\frac{1}{2} C_{\text{node}} \Delta V^2$. The hardware must therefore use sparse, local, colorable graphs of subthreshold transistors (Chapters 12 and 13) and probabilistic bits (p-bits, Chapter 14), and still produce samples with long-range structure.

We first describe the Denoising Thermodynamic Computer Architecture (DTCA) proposed by Jelinčič et al. [56] (Section 17.1), then the total-correlation penalty that keeps the correlations within each step short-ranged (Section 17.2), and the ways a T-step chain can be laid out in space and time (Section 17.3). Finally we estimate the energy per sample from device physics and compare it with a digital GPU (Section 17.4). Jelinčič et al. [56] report a model-based estimate of roughly 10^4 times less energy per sample than a GPU on their benchmark. The estimates in this chapter are order-of-magnitude design figures made in this book, and the worked computation of Worked computation 17.10 shows how strongly the ratio depends on what the GPU is assumed to run.

17.1 Architecture of the Denoising Thermodynamic Computer

A DTM step samples the conditional Boltzmann distribution (Chapter 16)

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \frac{1}{Z_t(\mathbf{x}_t)} \int \mathrm{d}\mathbf{z}_{t-1} \, \exp\Big(-\beta \left[\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) + \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{z}_{t-1}, \mathbf{x}_t)\right]\Big) . \tag{17.1}$$

A circuit that samples it must represent the variables $(\mathbf{x}_{t-1}, \mathbf{z}_{t-1}, \mathbf{x}_t)$ as voltages or currents on the chip. In the Denoising Thermodynamic Computer Architecture (DTCA) [31, 56], the chip does

not fetch instructions and operands for an arithmetic unit. It is a programmable, tiled array of stochastic circuits, called **Thermodynamic Sampling Units (TSUs)**, whose own thermal and shot noise (Chapters 11 to 14) draws samples from Eq. (17.1).

17.1.1 The Sampling Tile

A DTCA chip is a two-dimensional mesh of TSU tiles. Within denoising stage t, whether it has its own tile or is a time slice of a reconfigurable tile (Section 17.3), the nodes fall into three classes.

(i) **Clamped conditioning nodes** $\mathbf{x}_t \in \mathcal{X}^N$ hold the noisy state produced by the preceding stage. Each $x_{t,i}$ is stored either as a latched digital value (for p-bit states $x_{t,i} \in \{-1, +1\}$) or on a sample-and-hold capacitor with a low-leakage buffer (for continuous voltages $v_{t,i} \in \mathbb{R}$). While stage t relaxes, $\mathbf{x}_t$ acts as a fixed external field.

(ii) **Active visible nodes** $\mathbf{x}_{t-1} \in \mathcal{X}^N$ are the variables being sampled. Each $x_{t-1,i}$ is paired with its conditioning node $x_{t,i}$ through a local coupling cell that implements $\mathcal{E}^f_{t-1}(x_{t-1,i}, x_{t,i})$. Because the forward energy is a sum of single-site terms,

$$\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) = \sum_{i=1}^{N} \mathcal{E}^f_{t-1,i}(x_{t-1,i}, x_{t,i}), \tag{17.2}$$

the wire from $x_{t,i}$ to $x_{t-1,i}$ needs no routing across the array: in a spatially pipelined layout it is a direct link between adjacent tiles, and in a time-multiplexed layout it is a local feedback path within one cell.

(iii) **Latent nodes** $\mathbf{z}_{t-1} \in \mathcal{Z}^M$ supply non-Gaussian structure. Pairwise couplings among visible nodes alone give only quadratic (Gaussian) or second-order Ising energies (Section 16.4). To represent richer conditionals while keeping every coupling cell pairwise, the tile interleaves M latent nodes with the visible ones in the same 2D lattice. In Jelinčič et al. [56] these are binary p-bits; multi-level or continuous latent cells are possible in principle.

The visible variables may be binary p-bit spins $\mathbf{s}_{t-1} \in \{-1, +1\}^N$, continuous capacitor voltages $\mathbf{v}_{t-1} \in \mathbb{R}^N$, or a mixture, as in the Gaussian–Bernoulli model of Worked computation 1.19, which combines the RC sampler with the p-bit limit of the double-well sampler. In the all-binary formulation of Jelinčič et al. [56], with $\mathbf{x}_{t-1}, \mathbf{x}_t \in \{-1, +1\}^N$ and $\mathbf{z}_{t-1} \in \{-1, +1\}^M$, the conditional energy of the tile for stage t has the sparse Ising form

$$\begin{aligned}\mathcal{E}_{\text{tot}}(\mathbf{x}_{t-1}, \mathbf{z}_{t-1} \mid \mathbf{x}_t) = &- \sum_{i=1}^{N} \gamma_t x_{t,i} x_{t-1,i} - \sum_{i=1}^{N} b^x_{t-1,i} x_{t-1,i} - \sum_{a=1}^{M} b^z_{t-1,a} z_{t-1,a} \\ &- \sum_{(i,j)\in\mathcal{E}_{xx}} J^{(xx)}_{ij} x_{t-1,i} x_{t-1,j} - \sum_{(i,a)\in\mathcal{E}_{xz}} J^{(xz)}_{ia} x_{t-1,i} z_{t-1,a} \\ &- \sum_{(a,b)\in\mathcal{E}_{zz}} J^{(zz)}_{ab} z_{t-1,a} z_{t-1,b} - \sum_{(i,a)\in\mathcal{E}_{tz}} K^{(tz)}_{ia} x_{t,i} z_{t-1,a} - \sum_{(i,j)\in\mathcal{E}_{tx}} K^{(tx)}_{ij} x_{t,i} x_{t-1,j}.\end{aligned} \tag{17.3}$$

Here $\gamma_t = \frac{k_B T}{2} \ln \frac{1-p_{\text{flip},t}}{p_{\text{flip},t}} > 0$ is the forward coupling $k_B T \Gamma_t/2$ of Eq. (16.13) for bit-flip probability $p_{\text{flip},t} \in (0, 1/2)$, and $\theta_{t-1} = (\mathbf{b}^x, \mathbf{b}^z, \mathbf{J}^{(xx)}, \mathbf{J}^{(xz)}, \mathbf{J}^{(zz)}, \mathbf{K}^{(tz)}, \mathbf{K}^{(tx)})$ are the programmable couplings, stored in digital-to-analog converter (DAC) registers inside each cell. The couplings $\mathbf{K}^{(tz)}$ and $\mathbf{K}^{(tx)}$ let the learned energy depend on the clamp; they are absent in Jelinčič et al. [56], where the clamp

enters only through γ_t, and Proposition 16.3 shows that they are not needed for the exact reverse process. The edge sets $\mathcal{E}_{xx}, \mathcal{E}_{xz}, \mathcal{E}_{zz}, \mathcal{E}_{tz}, \mathcal{E}_{tx}$ are restricted to local neighborhoods on the 2D grid, for example the King's graph, in which each node couples to its horizontal, vertical, and diagonal neighbors (Worked computation 2.28).

Proposition 17.1 (Local field and conditional independence in a tile)**.** *Consider the conditional Boltzmann distribution $p_\theta(\mathbf{x}_{t-1}, \mathbf{z}_{t-1} \mid \mathbf{x}_t) \propto \exp(-\beta \mathcal{E}_{\text{tot}}(\mathbf{x}_{t-1}, \mathbf{z}_{t-1} \mid \mathbf{x}_t))$ defined by Eq. (17.3) on the interaction graph $\mathcal{G}_{\text{tile}} = (\mathcal{V}_x \cup \mathcal{V}_z, \mathcal{E}_{xx} \cup \mathcal{E}_{xz} \cup \mathcal{E}_{zz})$. Suppose $\mathcal{G}_{\text{tile}}$ has a proper vertex k-coloring $\mathcal{V}_x \cup \mathcal{V}_z = \bigsqcup_{c=1}^{k} \mathcal{V}_c$, so that no edge joins two vertices of the same color. Then:*

(a) Conditioned on $\mathbf{x}_t$ and on all nodes in complementary color classes $\mathbf{u}_{-c} := \{u_v : v \notin \mathcal{V}_c\}$, the nodes $\{u_v : v \in \mathcal{V}_c\}$ within color class c are mutually conditionally independent:

$$p_\theta(\mathbf{u}_c \mid \mathbf{u}_{-c}, \mathbf{x}_t) = \prod_{v \in \mathcal{V}_c} p_\theta(u_v \mid \mathbf{u}_{\partial v}, \mathbf{x}_t), \tag{17.4}$$

where ∂v is the set of neighbors of v in the tile graph.

(b) For each visible node $x_{t-1,i} \in \mathcal{V}_c$, the single-site conditional probability is a logistic function of the local field,

$$p_\theta(x_{t-1,i} = +1 \mid \mathbf{u}_{-c}, \mathbf{x}_t) = \sigma(2\beta h^x_{t-1,i}) = \frac{1}{2}(1 + \tanh(\beta h^x_{t-1,i})), \tag{17.5}$$

where the local field is

$$h^x_{t-1,i} = \gamma_t x_{t,i} + b^x_{t-1,i} + \sum_{j \in \partial_{xx} i} J^{(xx)}_{ij} x_{t-1,j} + \sum_{a \in \partial_{xz} i} J^{(xz)}_{ia} z_{t-1,a} + \sum_{j \in \partial_{tx} i} K^{(tx)}_{ij} x_{t,j}. \tag{17.6}$$

Similarly, for each latent node $z_{t-1,a} \in \mathcal{V}_c$,

$$\begin{aligned} p_\theta(z_{t-1,a} = +1 \mid \mathbf{u}_{-c}, \mathbf{x}_t) &= \sigma(2\beta h^z_{t-1,a}), \\ h^z_{t-1,a} &= b^z_{t-1,a} + \sum_{i \in \partial_{xz} a} J^{(xz)}_{ia} x_{t-1,i} + \sum_{b \in \partial_{zz} a} J^{(zz)}_{ab} z_{t-1,b} + \sum_{i \in \partial_{tz} a} K^{(tz)}_{ia} x_{t,i}. \end{aligned} \tag{17.7}$$

Proof. For (a), separate the terms of Eq. (17.3) that involve the vertices $\mathbf{u}_c = \{u_v : v \in \mathcal{V}_c\}$ from those that involve only $\mathbf{u}_{-c}$ and $\mathbf{x}_t$. By the definition of a proper coloring (Section 2.5), no edge of $\mathcal{E}_{xx} \cup \mathcal{E}_{xz} \cup \mathcal{E}_{zz}$ joins two vertices of $\mathcal{V}_c$, so no quadratic term in $\mathcal{E}_{\text{tot}}$ multiplies two variables of $\mathcal{V}_c$. The energy therefore splits as

$$\mathcal{E}_{\text{tot}}(\mathbf{u}_c, \mathbf{u}_{-c} \mid \mathbf{x}_t) = \mathcal{E}_{\text{rest}}(\mathbf{u}_{-c}, \mathbf{x}_t) - \sum_{v \in \mathcal{V}_c} h_v(\mathbf{u}_{\partial v}, \mathbf{x}_t)\, u_v, \tag{17.8}$$

where $\mathcal{E}_{\text{rest}}(\mathbf{u}_{-c}, \mathbf{x}_t)$ collects all couplings and biases among nodes outside $\mathcal{V}_c$, and $h_v(\mathbf{u}_{\partial v}, \mathbf{x}_t)$ collects all linear coefficients multiplying $u_v \in \{-1, +1\}$. Since every neighbor $v' \in \partial v$ has a color different from c, h_v depends only on $\mathbf{u}_{-c}$ and $\mathbf{x}_t$. Substituting Eq. (17.8) into the conditional Bayes rule yields

$$\begin{aligned} p_\theta(\mathbf{u}_c \mid \mathbf{u}_{-c}, \mathbf{x}_t) &= \frac{\exp(-\beta \mathcal{E}_{\text{tot}}(\mathbf{u}_c, \mathbf{u}_{-c} \mid \mathbf{x}_t))}{\sum_{\mathbf{u}'_c \in \{-1,+1\}^{|\mathcal{V}_c|}} \exp(-\beta \mathcal{E}_{\text{tot}}(\mathbf{u}'_c, \mathbf{u}_{-c} \mid \mathbf{x}_t))} \\ &= \frac{e^{-\beta \mathcal{E}_{\text{rest}}(\mathbf{u}_{-c}, \mathbf{x}_t)} \prod_{v \in \mathcal{V}_c} \exp(\beta h_v u_v)}{e^{-\beta \mathcal{E}_{\text{rest}}(\mathbf{u}_{-c}, \mathbf{x}_t)} \sum_{\{u'_v = \pm 1\}} \prod_{v \in \mathcal{V}_c} \exp(\beta h_v u'_v)} \end{aligned}$$

$$= \prod_{v \in \mathcal{V}_c} \frac{\exp(\beta h_v u_v)}{\exp(\beta h_v) + \exp(-\beta h_v)}, \tag{17.9}$$

where we canceled the common prefactor $e^{-\beta \mathcal{E}_{\text{rest}}}$ and factored the sum over the hypercube $\{-1,+1\}^{|\mathcal{V}_c|}$ into a product of single-spin sums. This proves Eq. (17.4).

For (b), setting $u_v = x_{t-1,i} = +1$ in the single-site factor and collecting all terms linear in $x_{t-1,i}$ from Eq. (17.3) directly gives $h_v = h^x_{t-1,i}$ as written in Eq. (17.6), and

$$\frac{e^{\beta h^x_{t-1,i}}}{e^{\beta h^x_{t-1,i}} + e^{-\beta h^x_{t-1,i}}} = \frac{1}{1 + e^{-2\beta h^x_{t-1,i}}} = \sigma(2\beta h^x_{t-1,i}) = \frac{1}{2}(1 + \tanh(\beta h^x_{t-1,i})). \tag{17.10}$$

An identical collection of terms linear in $z_{t-1,a}$ yields Eq. (17.7). □

Proposition 17.1 connects the graphical models of Chapter 2 to circuit layout (Chapters 12 and 14). Because the nodes of one color class are conditionally independent given the rest, every p-bit of that class on the chip, potentially hundreds of thousands of them, can update in the same clock phase without the synchronous-update error of Section 2.4. A bipartite tile needs two phases per sweep. A King's-graph tile is not bipartite and needs four. Each phase lasts of order nanoseconds.

Hardware Realization: Extropic's X0, XTR-0, and Z1 platforms

Extropic has described three hardware platforms built on these principles [32, 33, 56], together with the open-source `THRML` library, which simulates block Gibbs sampling on such graphs.

- **X0**: a prototype chip containing room-temperature transistor sampling cells of the kind analyzed in Chapters 12 to 14.
- **XTR-0**: a development platform that pairs sampling chips with a conventional processor and a field-programmable gate array (FPGA).
- **Z1**: a larger chip that Extropic describes as holding over a quarter of a million p-bits on a die under 12 mm on a side, drawing under one watt.

Few performance figures for these platforms have been published. The energy estimate of Jelinčič et al. [56] comes from a system-level model of an all-transistor design, not from measurements on a complete chip. The chip-level numbers later in this chapter (Worked computation 17.10) are this book's representative estimates, not Extropic specifications.

17.2 Total-Correlation Regularization for Sparse Hardware

Natural data such as images have long-range correlations: two pixels far apart can be correlated through background illumination or the symmetry of an object. A monolithic EBM with short-range couplings produces long-range correlations only near a critical point, where the correlation length ξ becomes comparable to the linear size of the system. Near criticality the spectral gap closes as $\lambda_1 \sim \xi^{-z}$, with dynamical exponent $z \geq 2$; this is critical slowing down, and it makes mixing slow.

A chip whose wires reach only nearest or next-nearest neighbors ($r_{\text{wire}} \sim 1$–2 lattice spacings) must nevertheless produce coherent samples with a small number K_{Gibbs} of block Gibbs sweeps per denoising step. Jelinčič et al. [56] address this with a penalty on the total correlation of each step's distribution, added to the training objective (Section 16.5). Penalizing total correlation drives the joint distribution toward the product of its marginals, which keeps the correlation length within a

step short and the block Gibbs sampler fast. Correlations across the whole sample then have to be built up over the T steps of the chain. A further design option, not part of the paper's experiments, is to map the data through a fixed encoder into a latent grid in which coarse and fine features occupy neighboring cells.

This section defines the total correlation, computes its gradient, and shows, for Gaussian distributions, how a bound on it bounds the correlation length.

17.2.1 Definition and Basic Properties

Consider a random vector $\mathbf{z} = (z_1, z_2, \dots, z_M)$ with density (or mass function) $q_\phi(\mathbf{z})$, where ϕ denotes the parameters that shape it; in a DTM, q_ϕ is the distribution of one step and ϕ the subset of its couplings θ on which the penalty acts.

Definition 17.2 (Total correlation)**.** The **total correlation** or multi-information $\mathcal{L}_{\mathrm{TC}}(q_\phi)$ of a joint distribution $q_\phi(\mathbf{z})$ over M variables (or M disjoint blocks $\mathbf{z} = (\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(M)})$) is the Kullback–Leibler divergence from $q_\phi(\mathbf{z})$ to the product of its single-variable (or single-block) marginals $\bar{q}_\phi(\mathbf{z}) := \prod_{i=1}^{M} q_{\phi,i}(z_i)$:

$$\mathcal{L}_{\mathrm{TC}}(q_\phi) := D_{\mathrm{KL}}\left(q_\phi(\mathbf{z}) \,\middle\|\, \prod_{i=1}^{M} q_{\phi,i}(z_i) \right) = \int \mathrm{d}^M z \, q_\phi(\mathbf{z}) \ln \frac{q_\phi(\mathbf{z})}{\prod_{i=1}^{M} q_{\phi,i}(z_i)}, \tag{17.11}$$

where $q_{\phi,i}(z_i) := \int \mathrm{d}\mathbf{z}_{-i} \, q_\phi(\mathbf{z})$ is the i-th marginal distribution. (For discrete variables $\mathbf{z} \in \{-1, +1\}^M$, the integral is replaced by the sum over $\{-1, +1\}^M$.)

We will use three standard properties of the total correlation.

Proposition 17.3 (Properties of the total correlation)**.** *For any joint distribution $q(\mathbf{z})$ with finite marginal entropies $S(q_i) := -\int \mathrm{d}z_i \, q_i(z_i) \ln q_i(z_i)$ and joint entropy $S(q) := -\int \mathrm{d}^M z \, q(\mathbf{z}) \ln q(\mathbf{z})$ (in dimensionless nats, i.e., with $k_{\mathrm{B}} = 1$):*

(a) ***Entropy form:***

$$\mathcal{L}_{\mathrm{TC}}(q) = \sum_{i=1}^{M} S(q_i) - S(q) \geq 0, \tag{17.12}$$

with equality $\mathcal{L}_{\mathrm{TC}}(q) = 0$ if and only if $q(\mathbf{z}) = \prod_{i=1}^{M} q_i(z_i)$ almost everywhere.

(b) ***Chain rule:*** *For any ordering of the coordinates $(1, 2, \dots, M)$,*

$$\mathcal{L}_{\mathrm{TC}}(q) = \sum_{k=2}^{M} \mathcal{I}\big(z_k; (z_1, \dots, z_{k-1})\big), \tag{17.13}$$

where $\mathcal{I}(X; Y) = S(X) + S(Y) - S(X, Y)$ is the mutual information (Section 9.4).

(c) ***Gaussian case:*** *If $q(\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is a multivariate Gaussian with covariance $\boldsymbol{\Sigma} \succ 0$ and diagonal part $\mathbf{D}_\Sigma := \mathrm{diag}(\Sigma_{11}, \dots, \Sigma_{MM})$, then*

$$\mathcal{L}_{\mathrm{TC}}\big(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})\big) = \frac{1}{2} \ln \frac{\prod_{i=1}^{M} \Sigma_{ii}}{\det \boldsymbol{\Sigma}} = -\frac{1}{2} \ln \det \mathbf{R}, \tag{17.14}$$

where $\mathbf{R} := \mathbf{D}_\Sigma^{-1/2} \boldsymbol{\Sigma} \mathbf{D}_\Sigma^{-1/2}$ is the dimensionless correlation matrix ($R_{ii} = 1$, $R_{ij} = \Sigma_{ij} / \sqrt{\Sigma_{ii} \Sigma_{jj}}$).

Proof. For (a), expand the logarithm of the product in Eq. (17.11):

$$\begin{aligned}\mathcal{L}_{\mathrm{TC}}(q) &= \int \mathrm{d}^M z\, q(\mathbf{z}) \ln q(\mathbf{z}) - \int \mathrm{d}^M z\, q(\mathbf{z}) \sum_{i=1}^{M} \ln q_i(z_i) \\ &= -S(q) - \sum_{i=1}^{M} \int \mathrm{d}z_i \left(\int \mathrm{d}\mathbf{z}_{-i}\, q(\mathbf{z}) \right) \ln q_i(z_i) \\ &= -S(q) - \sum_{i=1}^{M} \int \mathrm{d}z_i\, q_i(z_i) \ln q_i(z_i) = \sum_{i=1}^{M} S(q_i) - S(q). \end{aligned} \tag{17.15}$$

Non-negativity and the condition for equality follow from Gibbs' inequality for the KL divergence (Section 1.2).

For (b), use the chain rule for the joint entropy:

$$S(z_1, \dots, z_M) = S(z_1) + \sum_{k=2}^{M} S(z_k \mid z_1, \dots, z_{k-1}). \tag{17.16}$$

Substituting this into Eq. (17.12) yields

$$\begin{aligned}\mathcal{L}_{\mathrm{TC}}(q) &= \sum_{i=1}^{M} S(z_i) - S(z_1) - \sum_{k=2}^{M} S(z_k \mid z_1, \dots, z_{k-1}) \\ &= \sum_{k=2}^{M} \Big[S(z_k) - S(z_k \mid z_1, \dots, z_{k-1}) \Big] = \sum_{k=2}^{M} \mathcal{I}(z_k; (z_1, \dots, z_{k-1})). \end{aligned} \tag{17.17}$$

For (c), the differential entropy of an M-dimensional Gaussian $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is $S(q) = \frac{M}{2} \ln(2\pi e) + \frac{1}{2} \ln\det \boldsymbol{\Sigma}$ (Section 1.3, Chapter B), and each marginal $q_i = \mathcal{N}(\mu_i, \Sigma_{ii})$ has entropy $S(q_i) = \frac{1}{2} \ln(2\pi e \Sigma_{ii})$. Substituting into Eq. (17.12), the constants $\frac{M}{2} \ln(2\pi e)$ cancel:

$$\mathcal{L}_{\mathrm{TC}}(q) = \frac{1}{2} \sum_{i=1}^{M} \ln \Sigma_{ii} - \frac{1}{2} \ln\det \boldsymbol{\Sigma} = \frac{1}{2} \ln \frac{\det \mathbf{D}_\Sigma}{\det \boldsymbol{\Sigma}} = -\frac{1}{2} \ln\det(\mathbf{D}_\Sigma^{-1/2} \boldsymbol{\Sigma} \mathbf{D}_\Sigma^{-1/2}) = -\frac{1}{2} \ln\det \mathbf{R}. \tag{17.18}$$

□

17.2.2 Gradient of the Total-Correlation Penalty

In the regularized DTM objective of Jelinčič et al. [56], the variational objective of Section 16.5 is augmented by a total-correlation penalty with weight $\lambda_{\mathrm{TC}} > 0$:

$$\mathcal{L}_{\mathrm{total}}(\theta, \phi) = \mathcal{L}_{\mathrm{DTM}}(\theta, \phi) + \lambda_{\mathrm{TC}}\, \mathcal{L}_{\mathrm{TC}}(q_\phi). \tag{17.19}$$

Gradient descent on Eq. (17.19) needs $\frac{\partial}{\partial \phi} \mathcal{L}_{\mathrm{TC}}(q_\phi)$. The product of marginals $\bar{q}_\phi$ in the denominator of Eq. (17.11) also depends on ϕ, but its derivative drops out in expectation, because each marginal stays normalized.

Theorem 17.4 (Gradient of the total correlation)**.** *Let $q_\phi(\mathbf{z})$ be a smooth parametric family of probability distributions on $\mathcal{Z}^M$ such that differentiation with respect to ϕ commutes with integration over $\mathbf{z}$.*

(a) ***Score-function form:*** *The gradient of the total correlation with respect to ϕ is*

$$\frac{\partial}{\partial \phi}\mathcal{L}_{\mathrm{TC}}(q_\phi) = \mathbb{E}\left[\left(\frac{\partial}{\partial \phi}\ln q_\phi(\mathbf{z})\right)\ln\frac{q_\phi(\mathbf{z})}{\prod_{i=1}^{M} q_{\phi,i}(z_i)}\right]_{q_\phi(\mathbf{z})}, \tag{17.20}$$

where the expectation of the derivative of the log-ratio term vanishes identically:

$$\mathbb{E}\left[\frac{\partial}{\partial \phi}\ln\frac{q_\phi(\mathbf{z})}{\prod_{i=1}^{M} q_{\phi,i}(z_i)}\right]_{q_\phi(\mathbf{z})} = 0. \tag{17.21}$$

(b) ***Reparameterization form (continuous latents):*** *If $\mathbf{z} = \mathbf{g}_\phi(\boldsymbol{\epsilon})$ for a base noise distribution $\boldsymbol{\epsilon} \sim p_0(\boldsymbol{\epsilon})$ independent of ϕ, then*

$$\frac{\partial}{\partial \phi}\mathcal{L}_{\mathrm{TC}}(q_\phi) = \mathbb{E}\left[\frac{\partial \mathbf{g}_\phi(\boldsymbol{\epsilon})}{\partial \phi}^{\top}\left(\nabla_{\mathbf{z}}\ln q_\phi(\mathbf{z}) - \sum_{i=1}^{M}\mathbf{e}_i\frac{\partial}{\partial z_i}\ln q_{\phi,i}(z_i)\right)\Big|_{\mathbf{z}=\mathbf{g}_\phi(\boldsymbol{\epsilon})}\right]_{p_0(\boldsymbol{\epsilon})}. \tag{17.22}$$

(c) ***Boltzmann form:*** *If $q_\phi(\mathbf{z}) = \frac{1}{Z_\phi}\exp(-\beta E_\phi(\mathbf{z}))$, then*

$$\frac{\partial}{\partial \phi}\mathcal{L}_{\mathrm{TC}}(q_\phi) = -\beta\,\mathrm{Cov}_{q_\phi}\left(\frac{\partial E_\phi(\mathbf{z})}{\partial \phi},\ \ln\frac{q_\phi(\mathbf{z})}{\prod_{i=1}^{M} q_{\phi,i}(z_i)}\right). \tag{17.23}$$

Proof. Differentiate Eq. (17.11) under the integral sign:

$$\frac{\partial}{\partial \phi}\mathcal{L}_{\mathrm{TC}}(q_\phi) = \int \mathrm{d}^M z\,\left(\frac{\partial q_\phi(\mathbf{z})}{\partial \phi}\right)\ln\frac{q_\phi(\mathbf{z})}{\prod_{i=1}^{M} q_{\phi,i}(z_i)} + \int \mathrm{d}^M z\, q_\phi(\mathbf{z})\,\frac{\partial}{\partial \phi}\left[\ln q_\phi(\mathbf{z}) - \sum_{i=1}^{M}\ln q_{\phi,i}(z_i)\right]. \tag{17.24}$$

Consider the second integral on the right-hand side of Eq. (17.24). Using $\frac{\partial}{\partial \phi}\ln f_\phi = \frac{1}{f_\phi}\frac{\partial f_\phi}{\partial \phi}$, the first term inside the brackets gives

$$\int \mathrm{d}^M z\, q_\phi(\mathbf{z})\frac{1}{q_\phi(\mathbf{z})}\frac{\partial q_\phi(\mathbf{z})}{\partial \phi} = \frac{\partial}{\partial \phi}\int \mathrm{d}^M z\, q_\phi(\mathbf{z}) = \frac{\partial}{\partial \phi}(1) = 0. \tag{17.25}$$

For the i-th marginal term, first integrate over the other $M-1$ coordinates $\mathbf{z}_{-i}$, which reduces $q_\phi(\mathbf{z})$ to the marginal $q_{\phi,i}(z_i)$:

$$\begin{aligned}\int \mathrm{d}^M z\, q_\phi(\mathbf{z})\,\frac{\partial}{\partial \phi}\ln q_{\phi,i}(z_i) &= \int \mathrm{d}z_i\left(\int \mathrm{d}\mathbf{z}_{-i}\, q_\phi(\mathbf{z})\right)\frac{1}{q_{\phi,i}(z_i)}\frac{\partial q_{\phi,i}(z_i)}{\partial \phi}\\ &= \int \mathrm{d}z_i\, q_{\phi,i}(z_i)\frac{1}{q_{\phi,i}(z_i)}\frac{\partial q_{\phi,i}(z_i)}{\partial \phi} = \frac{\partial}{\partial \phi}\int \mathrm{d}z_i\, q_{\phi,i}(z_i) = \frac{\partial}{\partial \phi}(1) = 0.\end{aligned} \tag{17.26}$$

Subtracting Eq. (17.26) from Eq. (17.25) proves Eq. (17.21). Using the log-derivative trick $\frac{\partial q_\phi}{\partial \phi} = q_\phi\frac{\partial}{\partial \phi}\ln q_\phi$ in the surviving first integral of Eq. (17.24) immediately gives Eq. (17.20), establishing part (a).

For (b), with $\mathbf{z} = \mathbf{g}_\phi(\boldsymbol{\epsilon})$ and $\boldsymbol{\epsilon} \sim p_0(\boldsymbol{\epsilon})$, write $\mathcal{L}_{\mathrm{TC}}(q_\phi)$ as an expectation over the fixed base distribution $p_0(\boldsymbol{\epsilon})$:

$$\mathcal{L}_{\mathrm{TC}}(q_\phi) = \mathbb{E}\left[\ln q_\phi(\mathbf{g}_\phi(\boldsymbol{\epsilon})) - \sum_{i=1}^{M} \ln q_{\phi,i}(g_{\phi,i}(\boldsymbol{\epsilon}))\right]_{p_0(\boldsymbol{\epsilon})}. \tag{17.27}$$

By the chain rule, ϕ enters both explicitly, through the densities q_ϕ and $q_{\phi,i}$, and through the argument $\mathbf{z} = \mathbf{g}_\phi(\boldsymbol{\epsilon})$. The explicit derivatives vanish in expectation by Eq. (17.21), leaving the pathwise gradient through $\mathbf{g}_\phi(\boldsymbol{\epsilon})$, which is Eq. (17.22).

For (c), if $q_\phi(\mathbf{z}) = Z_\phi^{-1} e^{-\beta E_\phi(\mathbf{z})}$, then

$$\frac{\partial}{\partial \phi} \ln q_\phi(\mathbf{z}) = -\beta \frac{\partial E_\phi(\mathbf{z})}{\partial \phi} - \frac{\partial}{\partial \phi} \ln Z_\phi = -\beta \left(\frac{\partial E_\phi(\mathbf{z})}{\partial \phi} - \mathbb{E}\left[\frac{\partial E_\phi}{\partial \phi}\right]_{q_\phi} \right). \tag{17.28}$$

Substituting this centered derivative into Eq. (17.20) gives the covariance formula Eq. (17.23). □

17.2.3 Total Correlation and the Correlation Length

A small total correlation helps a sparsely wired tile because, at least for Gaussian distributions, it forces correlations to decay quickly with distance.

Consider the Gaussian (or linearized) regime of the RC sampler, in which the precision matrix $\mathbf{\Lambda} = \mathbf{\Sigma}^{-1}$ has diagonal entries $\Lambda_{ii} = \alpha_i$ (the on-site confinement) and off-diagonal couplings $\Lambda_{ij} = -J_{ij}$ between sites i, j at lattice distance $\mathrm{dist}(i, j)$.

Proposition 17.5 (Total correlation bounds the correlation length)**.** *Let $q(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ be a standardized Gaussian distribution ($\Sigma_{ii} = 1$ for all $i = 1, \ldots, M$, so $\mathbf{\Sigma} = \mathbf{R}$) whose total correlation satisfies $\mathcal{L}_{\mathrm{TC}}(q) \le \epsilon_{\mathrm{TC}} < \frac{1}{8}$, and write $s \equiv 2\sqrt{2\epsilon_{\mathrm{TC}}} < 1$. Then:*

(a) The squared Frobenius norm of the off-diagonal correlations is bounded by

$$\sum_{i \neq j} R_{ij}^2 = \|\mathbf{R} - \mathbf{I}\|_{\mathrm{F}}^2 \le 8\epsilon_{\mathrm{TC}}. \tag{17.29}$$

(b) The eigenvalues of $\mathbf{R}$ lie in $[1 - s, 1 + s]$, so its condition number is at most $(1 + s)/(1 - s)$. If $\mathbf{R}^{-1} = \mathbf{I} - \mathbf{J}$ with $\mathbf{J}$ supported on the edges of a lattice in which every site has at most z neighbors ($z = 2d$ for a d-dimensional hypercubic lattice, $z = 8$ for the King's graph), then $\bar{J} \equiv \max_{i \neq j} |J_{ij}| \le s/(1 - s)$. If moreover $z\bar{J} < 1$, correlations decay exponentially with graph distance:

$$|R_{ij}| \le \frac{(z\bar{J})^{\mathrm{dist}(i,j)}}{1 - z\bar{J}} = C_0 \exp\left(-\frac{\mathrm{dist}(i,j)}{\xi}\right), \qquad \xi = \frac{1}{\ln(1/(z\bar{J}))}. \tag{17.30}$$

For small ϵ_{TC}, $z\bar{J} = \mathcal{O}(\sqrt{\epsilon_{\mathrm{TC}}})$ and so $\xi = \mathcal{O}(1/\ln(1/\epsilon_{\mathrm{TC}}))$.

Proof. Let $\lambda_1, \ldots, \lambda_M > 0$ be the eigenvalues of $\mathbf{R}$ and write $\lambda_m = 1 + \delta_m$. Since $R_{ii} = 1$, $\sum_m \lambda_m = \mathrm{Tr}\,\mathbf{R} = M$, so $\sum_m \delta_m = 0$, and $\sum_m \delta_m^2 = \mathrm{Tr}\left((\mathbf{R} - \mathbf{I})^2\right) = \sum_{i \neq j} R_{ij}^2$. By Eq. (17.14),

$$\mathcal{L}_{\mathrm{TC}}(q) = -\frac{1}{2} \ln \det \mathbf{R} = -\frac{1}{2} \sum_{m=1}^{M} \ln(1 + \delta_m) = \frac{1}{2} \sum_{m=1}^{M} f(\delta_m), \qquad f(x) \equiv x - \ln(1 + x), \tag{17.31}$$

where we used $\sum_m \delta_m = 0$. The function f is convex, vanishes at $x = 0$, and is non-negative, so every term obeys $f(\delta_m) \le 2\mathcal{L}_{\mathrm{TC}}(q) < \frac{1}{4}$. Since $f(-0.56) > \frac{1}{4}$ and $f(0.9) > \frac{1}{4}$, this confines

every δ_m to $(-0.56, 0.9) \subset (-1, 1]$. On this interval $f(x) \geq x^2/4$: for $x \leq 0$, $f(x) \geq x^2/2$ because $f''(t) = (1+t)^{-2} \geq 1$ for $t \leq 0$; for $0 \leq x \leq 1$, the standard inequality $\ln(1+x) \leq x(2+x)/(2+2x)$ gives $f(x) \geq x^2/(2(1+x)) \geq x^2/4$. Summing,

$$\epsilon_{\mathrm{TC}} \;\geq\; \mathcal{L}_{\mathrm{TC}}(q) \;\geq\; \frac{1}{8}\sum_{m=1}^{M} \delta_m^2, \tag{17.32}$$

which is Eq. (17.29).

For part (b), $\max_m |\delta_m| \leq \left(\sum_m \delta_m^2\right)^{1/2} \leq \sqrt{8\epsilon_{\mathrm{TC}}} = s$, which gives the eigenvalue interval and the condition number. The eigenvalues of $\mathbf{J} = \mathbf{I} - \mathbf{R}^{-1}$ are $1 - 1/\lambda_m$, whose magnitudes are at most $1/(1-s) - 1 = s/(1-s)$; every matrix entry is bounded by the spectral norm, so $\bar{J} \leq s/(1-s)$. If $z\bar{J} < 1$, then $\|\mathbf{J}\|_\infty \leq z\bar{J} < 1$ and the Neumann series $\mathbf{R} = (\mathbf{I} - \mathbf{J})^{-1} = \sum_{n \geq 0} \mathbf{J}^n$ converges. Because $\mathbf{J}$ connects only neighboring sites, $(\mathbf{J}^n)_{ij} = 0$ for $n < \mathrm{dist}(i, j)$. For $n \geq \mathrm{dist}(i, j)$, $(\mathbf{J}^n)_{ij}$ is a sum over walks of length n from i to j, of which there are at most z^n, each weighted by at most $\bar{J}^n$. Hence

$$|R_{ij}| \leq \sum_{n=\mathrm{dist}(i,j)}^{\infty} (z\bar{J})^n = \frac{(z\bar{J})^{\mathrm{dist}(i,j)}}{1 - z\bar{J}}, \tag{17.33}$$

which is Eq. (17.30). □

Physics Insight: Rent's rule and the Ornstein–Zernike picture

In chip design, **Rent's rule** is an empirical relation between the number of gates G in a block and the number of external connections it needs, $P = k_{\mathrm{Rent}} G^p$. A planar array has boundary scaling $G^{(d-1)/d} = G^{1/2}$ ($d = 2$), so a circuit with Rent exponent $p > 1/2$ (typically 0.5–0.75 for general logic) needs ever longer wires on the upper metal layers as G grows. Wire capacitance is about $c_{\mathrm{wire}} \sim 0.2\,\mathrm{fF}/\mu\mathrm{m}$, so long wires quickly add capacitance. A local 2D lattice has $p = 1/2$: its wires stay short (about 5–10 μm) and C_{node} stays near 1 fF.
Proposition 17.5 has a counterpart in the **Ornstein–Zernike** picture of liquids and lattice models. The direct correlation function, here the precision matrix $\mathbf{\Lambda} = \mathbf{\Sigma}^{-1}$ that the wires implement, is short-ranged, while the total pair correlation, here the covariance $\mathbf{\Sigma}$, has range ξ. A total-correlation penalty keeps each denoising step away from criticality, so that ξ stays below the size of a local neighborhood and a local graph can represent the step well. Correlations across the whole sample $\mathbf{x}_0$ must then be assembled over the T steps of the chain rather than within one near-critical step.

17.3 Spatial Pipelining, Temporal Multiplexing, and Joint Relaxation

We distinguish three ways to execute a trained T-step DTM with step parameters $\{\theta_{t-1}\}_{t=1}^{T}$ on tiled hardware of this kind [31, 56]. They trade silicon area, weight reconfiguration, and sample throughput against one another.

17.3.1 Mode I: Temporal Multiplexing on One Tile

In **temporal multiplexing**, a single array of N visible and M latent nodes executes the stages $t = T, T-1, \ldots, 1$ in turn:

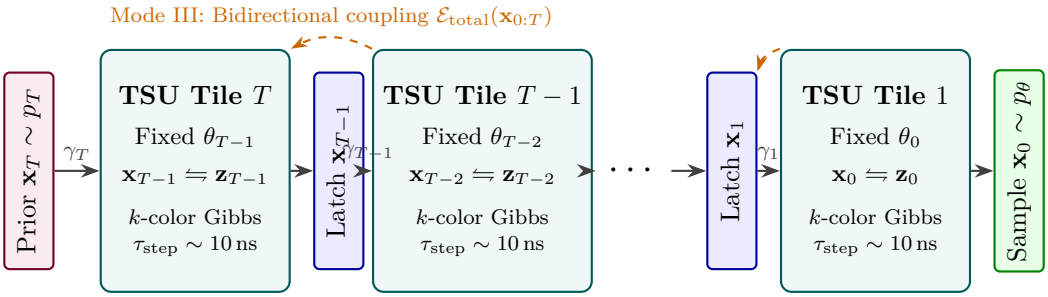


Figure 17.1: Execution modes of a DTM on tiled hardware. In Mode II (spatial pipelining), each stage $t \in \{1, \ldots, T\}$ has its own tile with fixed weights θ_{t-1}, and clamp states pass from left to right through latches once per step time τ_{step} (about 10 ns in this illustration), giving one sample per τ_{step} once the pipeline is full. In Mode III (joint relaxation, dashed arrows), the latches are replaced by two-way couplings and all stages relax together under the trajectory energy $\mathcal{E}_{\text{total}}(\mathbf{x}_{0:T}, \mathbf{z}_{0:T-1})$, which samples a different distribution from Mode II (Eq. (17.39)).

1. At $t = T$, the conditioning nodes $\mathbf{x}_T$ are initialized from the prior $p_T(\mathbf{x}_T)$ by setting $\gamma_T = 0$ (unbiased thermal fluctuations).
2. For each step $t = T, T-1, \ldots, 1$:
   (i) The parameter set θ_{t-1} is loaded into the local DAC registers of the tile.
   (ii) The tile undergoes K_{Gibbs} chromatic block-Gibbs sweeps (or continuous Langevin relaxation for duration $\tau_{\text{step}} \approx 5\tau_{\text{mix}}$), relaxing $(\mathbf{x}_{t-1}, \mathbf{z}_{t-1})$ to $p_\theta(\mathbf{x}_{t-1}, \mathbf{z}_{t-1} \mid \mathbf{x}_t)$.
   (iii) The resulting visible state $\mathbf{x}_{t-1}$ is latched into the local conditioning register $\mathbf{x}_t \leftarrow \mathbf{x}_{t-1}$ for the next step.

The latency to generate one sample, and the resulting throughput, are

$$\tau_{\text{lat}}^{(\text{TM})} = T(\tau_{\text{step}} + \tau_{\text{swap}}), \qquad \text{Throughput } \mathcal{R}^{(\text{TM})} = \frac{1}{T(\tau_{\text{step}} + \tau_{\text{swap}})}, \tag{17.34}$$

where τ_{swap} is the time needed to reload the DAC registers from θ_t to θ_{t-1}. If each stage has its own parameters, which must be streamed from a peripheral memory bank over the tile's bitlines, the reconfiguration energy

$$E_{\text{swap}}^{(\text{TM})} = T \cdot N_{\text{weights}} \cdot E_{\text{bitline}} \tag{17.35}$$

can exceed the relaxation energy $T(N + M)E_{\text{node}}$. To avoid this, a tile can either (a) store all T weight sets in multi-context memory inside each cell, which is practical when $T \sim$ 4–16, or (b) share weights across stages, $\theta_{t-1} = \theta(\text{emb}(t))$, so that only the schedule parameters (α_t, c_t) or a low-dimensional time embedding change from stage to stage.

17.3.2 Mode II: Spatial Pipelining Across Tiles

When T is small enough ($T \sim$ 4–32; the DTMs of Jelinčič et al. [56] use 2 to 8 steps), the chain can be unrolled in space across T tiles on one chip, or across several chips in one package, as shown in Fig. 17.1.

In **spatial pipelining**, tile t holds θ_{t-1} permanently in local latches. The weights do not move during inference ($\tau_{\text{swap}} = 0$, $E_{\text{swap}} = 0$), so the cost of moving weights between memory and arithmetic units, which dominates digital accelerators (Section 17.4.1), does not arise. In each pipeline cycle of duration τ_{step}:

- Tile T relaxes sample $\#(k+T-1)$ from step $T \to T-1$;
- Tile $T-1$ simultaneously relaxes sample $\#(k+T-2)$ from step $T-1 \to T-2$;
- ...
- Tile 1 simultaneously relaxes sample $\#k$ from step $1 \to 0$.

Once the pipeline is full (fill latency $\tau_{\text{lat}}^{(\text{SP})} = T\tau_{\text{step}}$), a new sample $\mathbf{x}_0^{(k)}$, independent of the others, leaves tile 1 every τ_{step}:

$$\text{Throughput } \mathcal{R}^{(\text{SP})} = \frac{1}{\tau_{\text{step}}} = T\left(1 + \frac{\tau_{\text{swap}}}{\tau_{\text{step}}}\right)\mathcal{R}^{(\text{TM})}. \tag{17.36}$$

For $\tau_{\text{step}} = 10\,\text{ns}$ this is 10^8 samples per second. The step time τ_{step} includes all K_{Gibbs} sweeps of a stage. The DTMs trained by Jelinčič et al. [56] use several hundred sweeps per stage (Remark 16.18), which would lengthen τ_{step} in proportion.

17.3.3 Mode III: Joint Relaxation of the Whole Trajectory

A third mode uses the same T tiles but replaces the one-way latches by two-way couplings, so that all $T+1$ slices $\mathbf{x}_{0:T} = (\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_T)$ relax together as a single $(T+1)N$-dimensional system.

In this **joint relaxation**, the energy across all tiles is the sum of the prior energy and the step energies:

$$\mathcal{E}_{\text{total}}(\mathbf{x}_{0:T}, \mathbf{z}_{0:T-1}) = \mathcal{E}_T(\mathbf{x}_T) + \sum_{t=1}^{T}\Big[\mathcal{E}_{t-1}^f(\mathbf{x}_{t-1}, \mathbf{x}_t) + \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{z}_{t-1}, \mathbf{x}_t)\Big]. \tag{17.37}$$

Modes II and III sample different distributions. In Mode II (ancestral sampling), the joint law of the trajectory is the product of normalized conditionals,

$$p_\theta^{\text{anc}}(\mathbf{x}_{0:T}, \mathbf{z}_{0:T-1}) = \frac{e^{-\beta\mathcal{E}_T(\mathbf{x}_T)}}{Z_T}\prod_{t=1}^{T}\frac{\exp(-\beta[\mathcal{E}_{t-1}^f(\mathbf{x}_{t-1}, \mathbf{x}_t) + \mathcal{E}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{z}_{t-1}, \mathbf{x}_t)])}{Z_t(\mathbf{x}_t)}, \tag{17.38}$$

where the step partition functions $Z_t(\mathbf{x}_t) = \int \mathrm{d}\mathbf{x}_{t-1}\,\mathrm{d}\mathbf{z}_{t-1}\, e^{-\beta[\mathcal{E}_{t-1}^f + \mathcal{E}_{\theta,t-1}]}$ depend on $\mathbf{x}_t$. If instead all tiles relax under the single energy Eq. (17.37), the stationary distribution is

$$p_\theta^{\text{joint}}(\mathbf{x}_{0:T}, \mathbf{z}_{0:T-1}) = \frac{1}{Z_{\text{total}}}\exp(-\beta\mathcal{E}_{\text{total}}(\mathbf{x}_{0:T}, \mathbf{z}_{0:T-1})) = \frac{1}{Z'_{\text{total}}}p_\theta^{\text{anc}}(\mathbf{x}_{0:T}, \mathbf{z}_{0:T-1})\prod_{t=1}^{T}Z_t(\mathbf{x}_t). \tag{17.39}$$

The two coincide only if every $Z_t(\mathbf{x}_t)$ is independent of $\mathbf{x}_t$. For the exact reverse process, $Z_t(\mathbf{x}_t) \propto q_t(\mathbf{x}_t) = e^{-\beta U_t(\mathbf{x}_t)}$ (Proposition 16.3), which varies with $\mathbf{x}_t$. Joint relaxation of a trained DTM therefore does not reproduce the model's samples unless the energy $k_{\text{B}}T\ln Z_t(\mathbf{x}_t)$ is added for every t (compare Eq. (16.21)). The next theorem concerns how fast the joint system relaxes.

Theorem 17.6 (Hessian and spectral gap of the joint trajectory energy)**.** *Consider a continuous-state T-step trajectory chain $\mathbf{x}_{0:T} \in \mathbb{R}^{(T+1)N}$ with global energy Eq. (17.37), where $\mathcal{E}_T(\mathbf{x}_T) = \frac{\alpha_{T+1}}{2}\|\mathbf{x}_T\|^2$ and $\mathcal{E}_{t-1}^f(\mathbf{x}_{t-1}, \mathbf{x}_t) = \frac{\alpha_t}{2}\|\mathbf{x}_{t-1} - c_t\mathbf{x}_t\|^2$. Assume the effective conditional energy*

$$\tilde{\mathcal{E}}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) = -\beta^{-1}\ln\int \mathrm{d}\mathbf{z}_{t-1}\, e^{-\beta\mathcal{E}_{\theta,t-1}}$$

has bounded second derivatives, $\left\|\nabla^2\tilde{\mathcal{E}}_{\theta,t-1}\right\|_{\text{op}} \le L_{\mathcal{E}}$, the Hessian being taken in $(\mathbf{x}_{t-1}, \mathbf{x}_t)$.

(a) The global Hessian $\mathbf{H}_{\text{total}} := \nabla^2_{\mathbf{x}_{0:T}} \mathcal{E}_{\text{total}}(\mathbf{x}_{0:T})$ *is a symmetric block-tridiagonal matrix with* $N \times N$ *diagonal blocks* $\mathbf{D}_t = \frac{\partial^2 \mathcal{E}_{\text{total}}}{\partial \mathbf{x}_t^2}$ *and off-diagonal blocks* $\mathbf{B}_t = \frac{\partial^2 \mathcal{E}_{\text{total}}}{\partial \mathbf{x}_{t-1} \partial \mathbf{x}_t}$ *for* $t = 1, \ldots, T$.

(b) For the pure forward Gaussian chain ($\tilde{\mathcal{E}}_{\theta,t-1} = 0$*) with uniform parameters* $\alpha_t = \alpha$, $c_t = 1$, *and* $\alpha_{T+1} = \alpha$, *the global Hessian is* $\mathbf{H}^0_{\text{total}} = \alpha\, \mathbf{T}_{T+1} \otimes \mathbf{I}_N$, *where* $\mathbf{T}_{T+1}$ *is the* $(T+1) \times (T+1)$ *tridiagonal Laplacian matrix with diagonal entries* $(1, 2, 2, \ldots, 2)$ *and off-diagonal entries* -1, *whose exact minimum eigenvalue is*

$$\lambda_{\min}(\mathbf{H}^0_{\text{total}}) = 4\alpha \sin^2\!\left(\frac{\pi}{2(2T+3)}\right) = \frac{\pi^2 \alpha}{4(T+3/2)^2} + \mathcal{O}(T^{-4}). \tag{17.40}$$

(c) Consequently, while spatial pipelining (Mode II) mixes in a time per step independent of T *when* $\alpha > L_{\mathcal{E}} + \kappa$, *joint relaxation (Mode III) is guaranteed convex only under the stronger condition* $\lambda_{\min}(\mathbf{H}^0_{\text{total}}) > 2L_{\mathcal{E}}$, *and its relaxation time grows as* $\tau^{(\text{joint})}_{\text{mix}} \sim \mathcal{O}(T^2)$. *In exchange it allows clamping at both ends of the chain, for example fixing part of* $\mathbf{x}_0$ *and of* $\mathbf{x}_T$ *for inpainting or guidance.*

Proof. For (a), $\mathbf{x}_t$ appears in Eq. (17.37) only in the terms t and $t+1$ of the sum, namely $\mathcal{E}^f_{t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t) + \tilde{\mathcal{E}}_{\theta,t-1}(\mathbf{x}_{t-1}, \mathbf{x}_t)$ and $\mathcal{E}^f_t(\mathbf{x}_t, \mathbf{x}_{t+1}) + \tilde{\mathcal{E}}_{\theta,t}(\mathbf{x}_t, \mathbf{x}_{t+1})$. Therefore $\frac{\partial^2 \mathcal{E}_{\text{total}}}{\partial \mathbf{x}_t \partial \mathbf{x}_s} = \mathbf{0}$ whenever $|t - s| > 1$, which is block-tridiagonality.

For (b), set $\tilde{\mathcal{E}}_{\theta,t-1} = 0$, $\alpha_t = \alpha$, and $c_t = 1$. Then

$$\mathcal{E}^0_{\text{total}}(\mathbf{x}_{0:T}) = \frac{\alpha}{2} \|\mathbf{x}_T\|^2 + \frac{\alpha}{2} \sum_{t=1}^{T} \|\mathbf{x}_{t-1} - \mathbf{x}_t\|^2 . \tag{17.41}$$

Differentiating twice with respect to $\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_T$ yields $\mathbf{D}_0 = \alpha \mathbf{I}_N$, $\mathbf{D}_t = 2\alpha \mathbf{I}_N$ for $1 \le t \le T$, and $\mathbf{B}_t = -\alpha \mathbf{I}_N$ for $1 \le t \le T$. Thus $\mathbf{H}^0_{\text{total}} = \alpha\, \mathbf{T}_{T+1} \otimes \mathbf{I}_N$, where the $(T+1) \times (T+1)$ tridiagonal matrix $\mathbf{T}_{T+1}$ satisfies the eigenvalue difference equation for $\mathbf{v} = (v_0, v_1, \ldots, v_T)^\top$:

$$v_0 - v_1 = \mu v_0, \tag{17.42}$$

$$-v_{t-1} + 2v_t - v_{t+1} = \mu v_t \qquad (1 \le t \le T-1), \tag{17.43}$$

$$-v_{T-1} + 2v_T = \mu v_T. \tag{17.44}$$

Introducing phantom boundary points v_{-1} and v_{T+1}, Eq. (17.42) is equivalent to the bulk equation at $t = 0$ with the Neumann condition $v_{-1} = v_0$, while Eq. (17.44) is equivalent to the bulk equation at $t = T$ with the Dirichlet condition $v_{T+1} = 0$. Using the standing-wave ansatz $v_t = \cos\left(q(t+1/2)\right)$, the Neumann condition at $t = 0$ ($v_{-1} = v_0$) is automatically satisfied since $\cos(-q/2) = \cos(q/2)$. The Dirichlet condition at $t = T+1$ requires

$$v_{T+1} = \cos\!\big(q(T+3/2)\big) = 0 \implies q_m \left(T + \frac{3}{2}\right) = \left(m + \frac{1}{2}\right)\pi, \qquad m = 0, 1, \ldots, T. \tag{17.45}$$

Substituting $v_t = \cos(q_m(t+1/2))$ into the bulk equation (17.43) yields

$$\mu_m = 2 - 2\cos q_m = 4 \sin^2\!\left(\frac{q_m}{2}\right) = 4 \sin^2\!\left(\frac{(2m+1)\pi}{2(2T+3)}\right). \tag{17.46}$$

Setting $m = 0$ and multiplying by α gives the exact minimum eigenvalue Eq. (17.40). By Weyl's eigenvalue perturbation inequality, adding the non-convex interaction $\sum_{t=1}^{T} \tilde{\mathcal{E}}_{\theta,t-1}$ shifts the minimum eigenvalue by at most $2L_{\mathcal{E}}$, proving part (c). □

Table 17.1: The three execution modes of a T-step DTM on tiled hardware.

Quantity	Mode I: temporal multiplexing	Mode II: spatial pipelining	Mode III: joint relaxation
Tiles required	1	T	T
Coupling between stages	Feedback latch $\mathbf{x}_t \leftarrow \mathbf{x}_{t-1}$	One-way latches	Two-way analog or spin links
Weight movement per sample	T context switches	None	None
Latency per sample τ_{lat}	$T(\tau_{\text{step}} + \tau_{\text{swap}})$	$T\tau_{\text{step}}$	$\mathcal{O}(T^2)\tau_{\text{step}}$
Steady-state throughput $\mathcal{R}$	$[T(\tau_{\text{step}} + \tau_{\text{swap}})]^{-1}$	$1/\tau_{\text{step}}$	$[\mathcal{O}(T^2)\tau_{\text{step}}]^{-1}$
Clamping at both ends	Approximate (sequential)	Approximate (sequential)	Samples Eq. (17.39), not the DTM law

Table 17.1 summarizes the trade-offs. For high-throughput generation, unconditional or conditioned only through the forward chain, Mode II is the natural choice: each step mixes in a time independent of T, the weights do not move, and the throughput is $1/\tau_{\text{step}}$. Mode III is of interest for inverse problems in which evidence is clamped at $t = 0$ and must influence the earlier stages. It relaxes more slowly, and its convexity condition is demanding: since $\lambda_{\min}(\mathbf{H}^0_{\text{total}}) \approx \pi^2\alpha/(4T^2)$, the condition $\lambda_{\min} > 2L_{\mathcal{E}}$ requires $\alpha \gtrsim 8T^2L_{\mathcal{E}}/\pi^2$, which fails for any sizeable T unless the learned energies are nearly convex. It also samples the joint law Eq. (17.39) rather than the DTM's trajectory law, as shown above.

17.4 Energy Budget: Digital GPU versus Subthreshold DTCA

We can now estimate the energy per sample of a DTCA chip from device physics, using the subthreshold MOS physics of Chapter 12, the circuit thermodynamics of Chapter 13, the p-bit cells of Chapter 14, the mixing guarantee of Chapter 16, and the tiled layouts of Sections 17.1 to 17.3, and compare it with a digital GPU. Jelinčič et al. [56] estimate, from a system-level model of an all-transistor design, that such hardware could match a GPU on binarized Fashion-MNIST with roughly 10^4 times less energy per sample. The estimates in this section are representative order-of-magnitude values, meant to show where a ratio of this kind comes from and what it depends on.

At room temperature ($T = 300\,\text{K}$) the thermal energy is

$$k_{\text{B}}T = (1.380649 \times 10^{-23}\,\text{J/K})(300\,\text{K}) \approx 4.142 \times 10^{-21}\,\text{J} = 4.142\,\text{zJ} \quad \text{(zeptojoules)}, \tag{17.47}$$

and the Landauer cost of erasing one bit (Section 9.3) is $k_{\text{B}}T\ln 2 \approx 2.871 \times 10^{-21}\,\text{J} \approx 2.87\,\text{zJ}$. We quote energies both in joules (pJ $= 10^{-12}$ J, fJ $= 10^{-15}$ J, aJ $= 10^{-18}$ J) and in multiples of $k_{\text{B}}T$.

17.4.1 Energy Budget of a Digital GPU

Consider a current GPU built in a 4–5 nm FinFET process (FinFET: a transistor whose channel is a thin vertical fin), with supply voltage $V_{\text{DD}}^{\text{dig}} \approx 0.75$–$0.85\,\text{V}$, executing one step of a score-based diffusion model (the DDPM update of Chapter 15; here α_t and $\bar{\alpha}_t$ are the DDPM signal coefficients,

not the stiffness of Chapter 16):

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \boldsymbol{\xi}_t, \qquad \boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{17.48}$$

The energy of this step can be divided into arithmetic, memory access, and the generation of the noise $\boldsymbol{\xi}_t$ [52].

Arithmetic. Digital logic operates each transistor as a switch driven into strong inversion ($V_{\mathrm{GS}} - V_{\mathrm{th}} \gg V_T$), swinging between 0 and $V_{\mathrm{DD}}^{\mathrm{dig}} \approx 0.8\,\mathrm{V}$, about 31 thermal voltages ($V_T \approx 25.85\,\mathrm{mV}$), so that thermal noise almost never flips a bit. Charging and discharging one minimum-size gate plus its local wiring, $C_{\mathrm{gate}} \approx 0.5$–$1.0\,\mathrm{fF}$, dissipates

$$E_{\mathrm{gate}} = \alpha_{\mathrm{act}} C_{\mathrm{gate}} \left(V_{\mathrm{DD}}^{\mathrm{dig}}\right)^2 \approx (0.5)(0.8\,\mathrm{fF})(0.8\,\mathrm{V})^2 = 0.256\,\mathrm{fJ} \approx 6.18 \times 10^4\, k_{\mathrm{B}}T, \tag{17.49}$$

where $\alpha_{\mathrm{act}} \approx 0.5$ is the average activity factor. A 16-bit floating-point multiply-accumulate (MAC), $a \leftarrow a + w \cdot x$, involves an 11×11-bit multiplier array (several hundred to a thousand gates), exponent logic, normalization shifters, and an accumulator: roughly 1500–2500 switching transistors in all. Representative values for a 4–5 nm process are below the 45 nm figures surveyed by Horowitz [52] by a small factor; we take

$$E_{\mathrm{MAC}}^{(\mathrm{FP16})} \approx 0.4\text{–}1.0\,\mathrm{pJ} = (4\text{–}10) \times 10^{-13}\,\mathrm{J} \approx 1.0 \times 10^8 \text{ to } 2.4 \times 10^8\, k_{\mathrm{B}}T. \tag{17.50}$$

Memory access. The operands w and x must be fetched from memory over wires up to millimeters long, with capacitance of order $0.2\,\mathrm{pF/mm}$, far above C_{gate}:

- Reading a 16-bit word from a register file or first-level static RAM (SRAM) bank ($C_{\mathrm{bitline}} \sim 50$–$200\,\mathrm{fF}$ on each of 16 bitlines, plus sense amplifiers) costs $E_{\mathrm{SRAM}} \approx 1.5$–$5.0\,\mathrm{pJ}$ (3.6×10^8–$1.2 \times 10^9\, k_{\mathrm{B}}T$).
- Reading from the second-level cache across the chip (several millimeters of wire) costs $E_{\mathrm{L2}} \approx 10$–$25\,\mathrm{pJ}$ (2.4×10^9–$6.0 \times 10^9\, k_{\mathrm{B}}T$).
- Reading from off-chip high-bandwidth memory (HBM, stacked DRAM connected through a silicon interposer, at about 3.5–7 pJ/bit) costs $E_{\mathrm{HBM}} \approx 56$–$112\,\mathrm{pJ}$ per 16-bit word (1.3×10^{10}–$2.7 \times 10^{10}\, k_{\mathrm{B}}T$).

Data reuse spreads each expensive fetch over many operations. Writing $\eta_{\mathrm{mem}} E_{\mathrm{mem}}$ for the memory energy per MAC after reuse, the effective energy per MAC in a GPU diffusion network is of order

$$E_{\mathrm{syn,GPU}} = E_{\mathrm{MAC}}^{(\mathrm{FP16})} + \eta_{\mathrm{mem}} E_{\mathrm{mem}} \approx 1.5\text{–}3.5\,\mathrm{pJ/MAC} \approx (3.6\text{–}8.5) \times 10^8\, k_{\mathrm{B}}T. \tag{17.51}$$

Random-number generation. The noise $\boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ in Eq. (17.48) must also be computed. The GPU spends about $6 \times 10^4\, k_{\mathrm{B}}T$ per gate to suppress the thermal and shot noise of its transistors, and then emulates noise arithmetically, for example with the counter-based Philox pseudorandom number generator (PRNG) followed by the Box–Muller transform

$$\xi_1 = \sqrt{-2\ln U_1}\cos(2\pi U_2), \qquad \xi_2 = \sqrt{-2\ln U_1}\sin(2\pi U_2), \qquad U_1, U_2 \sim \mathrm{Uniform}(0,1), \tag{17.52}$$

Each pair of Gaussian numbers requires about ten rounds of 32-bit integer multiplication and bit mixing (some twenty arithmetic operations) plus evaluations of $\ln(\cdot)$, $\sqrt{\cdot}$, $\sin(\cdot)$, and $\cos(\cdot)$ in the GPU's special-function units. We estimate

$$E_{\mathrm{PRNG}}^{(\mathrm{GPU})} \approx 15\text{–}40\,\mathrm{pJ/sample} \approx (3.6\text{–}9.7) \times 10^9\, k_{\mathrm{B}}T. \tag{17.53}$$

A neural score network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ needs $N_{\mathrm{MAC}} \sim 10^4$–$10^6$ MACs per coordinate per denoising step, so the energy per variable per step is dominated by those MACs and their memory traffic and reaches about 10^{12}–$10^{15}\, k_{\mathrm{B}}T$. A direct digital simulation of a sparse EBM is cheaper. With about $d_{\mathrm{deg}} \sim 16$ neighbors per spin, evaluating $\sigma(2\beta \sum_j J_{ij} s_j)$ and comparing it with a pseudorandom number costs $d_{\mathrm{deg}} E_{\mathrm{syn,GPU}} + E_{\mathrm{PRNG}}^{(\mathrm{GPU})} \approx 40\text{–}90\,\mathrm{pJ} \approx (1\text{–}2.2) \times 10^{10}\, k_{\mathrm{B}}T$ per spin update.

17.4.2 Energy Budget of a Subthreshold Node

We now estimate the energy dissipated by one node during one relaxation step of duration τ_{step} in a subthreshold CMOS tile (Chapters 12 to 14).

The transistors are biased in weak inversion (Section 12.3), where the drain current follows the Enz–Krummenacher–Vittoz (EKV) relation

$$I_{\mathrm{D}} = I_0 \exp\!\left(\frac{V_G - V_{\mathrm{th}}}{n V_T}\right)\left[\exp\!\left(-\frac{V_S}{V_T}\right) - \exp\!\left(-\frac{V_D}{V_T}\right)\right], \tag{17.54}$$

with thermal voltage $V_T = k_{\mathrm{B}}T/q_e \approx 25.85\,\mathrm{mV}$ at $T = 300\,\mathrm{K}$ and subthreshold slope factor $n = 1 + C_{\mathrm{dep}}/C_{\mathrm{ox}} \approx 1.2$–$1.4$. Two energy channels matter: the bias current drawn from the supply, and the charging of the node capacitance.

Proposition 17.7 (Energy per node per step)**.** *Consider a stochastic node (continuous capacitor voltage or subthreshold p-bit cell) in a DTCA tile with total effective node capacitance C_{node}, operating at low supply voltage $V_{\mathrm{DD}}^{\mathrm{sub}}$, total static/dynamic cell bias current $I_{\mathrm{cell}} = N_{\mathrm{branch}} I_{\mathrm{D}}$, and voltage swing $\Delta V = m\, n V_T$ (where $m \sim 2$–4 is the thermal swing multiple). Over a single conditional relaxation window of duration $\tau_{\mathrm{step}} = \alpha_{\mathrm{relax}} \tau_{RC}$ (where $\tau_{RC} = C_{\mathrm{node}}/g_{\mathrm{m}} = n V_T C_{\mathrm{node}}/I_{\mathrm{D}}$ is the physical RC thermalization time constant and $\alpha_{\mathrm{relax}} \approx 3$–$6$ is the number of time constants per step), the total electrical energy dissipated per node per denoising step is*

$$\begin{aligned} E_{\mathrm{node}}^{(\mathrm{DTCA})} &= E_{\mathrm{bias}} + E_{\mathrm{cap}} = I_{\mathrm{cell}} V_{\mathrm{DD}}^{\mathrm{sub}} \tau_{\mathrm{step}} + \frac{1}{2} N_{\mathrm{flip}} C_{\mathrm{node}} (\Delta V)^2 \\ &= \left[\alpha_{\mathrm{relax}} N_{\mathrm{branch}} n \left(\frac{V_{\mathrm{DD}}^{\mathrm{sub}}}{V_T}\right) + \frac{1}{2} N_{\mathrm{flip}} m^2 n^2\right] \left(\frac{C_{\mathrm{node}} V_T}{q_e}\right) k_{\mathrm{B}}T \\ &= \left[\alpha_{\mathrm{relax}} N_{\mathrm{branch}} n \tilde{V}_{\mathrm{DD}} + \tfrac{1}{2} N_{\mathrm{flip}} m^2 n^2\right] \Omega\, k_{\mathrm{B}}T, \end{aligned} \tag{17.55}$$

where $\tilde{V}_{\mathrm{DD}} := V_{\mathrm{DD}}^{\mathrm{sub}}/V_T$ is the dimensionless supply voltage, N_{flip} is the mean number of full-swing transitions per step, and

$$\Omega := \frac{C_{\mathrm{node}} V_T}{q_e} = \frac{C_{\mathrm{node}} k_{\mathrm{B}} T}{q_e^2} \tag{17.56}$$

is the mesoscopic size parameter of Section 13.2, the number of elementary charges q_e stored on C_{node} per thermal voltage V_T. The bias current I_{D} cancels in the product $I_{\mathrm{D}} \tau_{\mathrm{step}} = \alpha_{\mathrm{relax}} n V_T C_{\mathrm{node}} = \alpha_{\mathrm{relax}} n \Omega q_e$.

Proof. The instantaneous electrical power supplied by the DC voltage rails at $V_{\mathrm{DD}}^{\mathrm{sub}}$ to a cell drawing total current $I_{\mathrm{cell}} = N_{\mathrm{branch}} I_{\mathrm{D}}$ is $P_{\mathrm{bias}} = I_{\mathrm{cell}} V_{\mathrm{DD}}^{\mathrm{sub}}$. Integrating over the step window τ_{step} gives the static/housekeeping bias energy $E_{\mathrm{bias}} = N_{\mathrm{branch}} I_{\mathrm{D}} V_{\mathrm{DD}}^{\mathrm{sub}} \tau_{\mathrm{step}}$. By Section 12.4, the subthreshold transconductance of a branch biased at I_{D} is $g_{\mathrm{m}} = \frac{I_{\mathrm{D}}}{nV_T}$, so the physical relaxation time constant of the node capacitor C_{node} is

$$\tau_{RC} = \frac{C_{\mathrm{node}}}{g_{\mathrm{m}}} = \frac{nV_T C_{\mathrm{node}}}{I_{\mathrm{D}}}. \tag{17.57}$$

Substituting $\tau_{\mathrm{step}} = \alpha_{\mathrm{relax}} \tau_{RC} = \alpha_{\mathrm{relax}} \frac{nV_T C_{\mathrm{node}}}{I_{\mathrm{D}}}$ into E_{bias}, the bias current I_{D} cancels in numerator and denominator:

$$E_{\mathrm{bias}} = N_{\mathrm{branch}} I_{\mathrm{D}} V_{\mathrm{DD}}^{\mathrm{sub}} \left(\alpha_{\mathrm{relax}} \frac{nV_T C_{\mathrm{node}}}{I_{\mathrm{D}}} \right) = \alpha_{\mathrm{relax}} N_{\mathrm{branch}} n V_T C_{\mathrm{node}} V_{\mathrm{DD}}^{\mathrm{sub}}. \tag{17.58}$$

Using $V_T = k_{\mathrm{B}} T / q_e$, we rewrite $V_T C_{\mathrm{node}} V_{\mathrm{DD}}^{\mathrm{sub}} = \left(\frac{C_{\mathrm{node}} V_T}{q_e} \right) \left(\frac{V_{\mathrm{DD}}^{\mathrm{sub}}}{V_T} \right) k_{\mathrm{B}} T = \Omega \tilde{V}_{\mathrm{DD}} k_{\mathrm{B}} T$.

Similarly, each capacitive charge/discharge excursion of amplitude $\Delta V = mnV_T$ across C_{node} dissipates dynamic energy $\frac{1}{2} C_{\mathrm{node}} (\Delta V)^2$. Over N_{flip} transitions per step,

$$E_{\mathrm{cap}} = \frac{1}{2} N_{\mathrm{flip}} C_{\mathrm{node}} (mnV_T)^2 = \frac{1}{2} N_{\mathrm{flip}} m^2 n^2 \left(\frac{C_{\mathrm{node}} V_T}{q_e} \right) (q_e V_T) = \frac{1}{2} N_{\mathrm{flip}} m^2 n^2 \Omega \, k_{\mathrm{B}} T. \tag{17.59}$$

Summing Eq. (17.58) and Eq. (17.59) yields Eq. (17.55). □

Heuristic Derivation: Why the bias current sets speed but not energy

The bias current I_{D} drops out of Eq. (17.55) for a simple reason. Advancing a node by one relaxation time τ_{RC} requires moving a fixed charge $\Delta Q = I_{\mathrm{D}} \tau_{RC} = nC_{\mathrm{node}} V_T = n\Omega q_e$ through the transistor. Doubling I_{D} (by raising the gate bias by $nV_T \ln 2 \approx 23\,\mathrm{mV}$) doubles $g_{\mathrm{m}} = I_{\mathrm{D}}/(nV_T)$, so the node relaxes twice as fast ($\tau_{RC} \to \tau_{RC}/2$). The power doubles and the step time halves, so the energy per step $P\tau_{\mathrm{step}}$ is unchanged over many decades of subthreshold current ($I_{\mathrm{D}} \in [10\,\mathrm{pA}, 100\,\mathrm{nA}]$).
The energy is set instead by two quantities.

1. The charge number $\Omega = C_{\mathrm{node}} V_T / q_e$. For $C_{\mathrm{node}} = 1.0\,\mathrm{fF}$ at $T = 300\,\mathrm{K}$,

$$\Omega = \frac{(1.0 \times 10^{-15}\,\mathrm{F})(25.85 \times 10^{-3}\,\mathrm{V})}{1.602 \times 10^{-19}\,\mathrm{C}} \approx 161.4 \text{ elementary charges.} \tag{17.60}$$

2. The supply voltage in thermal units, $\tilde{V}_{\mathrm{DD}} = V_{\mathrm{DD}}^{\mathrm{sub}} / V_T$. Subthreshold transistors saturate for $V_{\mathrm{DS}} \gtrsim 4V_T \approx 100\,\mathrm{mV}$ (Section 12.3), so a subthreshold cell can operate at $V_{\mathrm{DD}}^{\mathrm{sub}} \approx 0.25$–$0.35\,\mathrm{V}$ ($\tilde{V}_{\mathrm{DD}} \approx 10$–$13.5$) with signal swings $\Delta V \approx 2nV_T \approx 65\,\mathrm{mV}$.

Representative parameter values are:

- Node capacitance (compact local p-bit/analog cell + short local wires): $C_{\mathrm{node}} = 1.0\,\mathrm{fF} \implies \Omega \approx 161$.
- Subthreshold slope factor: $n = 1.25$; thermal voltage $V_T = 25.85\,\mathrm{mV}$.
- Supply voltage: $V_{\mathrm{DD}}^{\mathrm{sub}} = 0.30\,\mathrm{V}$ ($\tilde{V}_{\mathrm{DD}} = 11.6$); voltage swing: $\Delta V = 2nV_T \approx 64.6\,\mathrm{mV}$ ($m = 2$).
- Bias current per branch: $I_{\mathrm{D}} = 5.0\,\mathrm{nA} \implies \tau_{RC} = \frac{nV_T C_{\mathrm{node}}}{I_{\mathrm{D}}} = \frac{(1.25)(25.85\,\mathrm{mV})(1.0\,\mathrm{fF})}{5.0\,\mathrm{nA}} \approx 6.46\,\mathrm{ns}$.

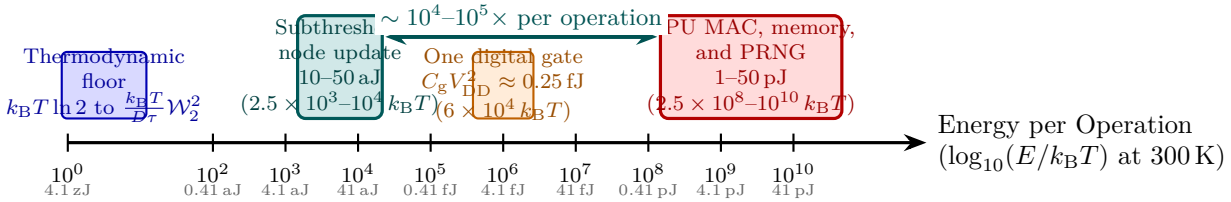


Figure 17.2: Energy per elementary operation, in units of k_BT at 300 K and in joules, using the representative values of this section. A subthreshold node update (10–50 aJ, 2.5×10^3–$10^4\, k_BT$) costs less than switching one strong-inversion logic gate and 10^4–10^5 times less than a GPU multiply-accumulate with memory access and pseudorandom number generation.

- Relaxation window per chromatic step / phase: $\tau_{\text{step}} = 10\,\text{ns}$ ($\alpha_{\text{relax}} \approx 1.55$ per phase, or 3.1 across a 2-phase bipartite sweep) with $N_{\text{branch}} = 2$ active current paths and $N_{\text{flip}} = 1$:

$$E_{\text{bias}} = I_{\text{cell}} V_{\text{DD}}^{\text{sub}} \tau_{\text{step}} = (2 \times 5.0\,\text{nA})(0.30\,\text{V})(10\,\text{ns}) = 30.0\,\text{aJ} \approx 7.24 \times 10^3\, k_BT, \tag{17.61}$$

$$E_{\text{cap}} = \frac{1}{2} C_{\text{node}} (\Delta V)^2 = \frac{1}{2}(1.0\,\text{fF})(64.6\,\text{mV})^2 \approx 2.09\,\text{aJ} \approx 5.04 \times 10^2\, k_BT, \tag{17.62}$$

$$E_{\text{node}}^{(\text{DTCA})} = E_{\text{bias}} + E_{\text{cap}} \approx 32.1\,\text{aJ} = 0.0321\,\text{fJ} \approx 7.75 \times 10^3\, k_BT. \tag{17.63}$$

For a smaller node with $C_{\text{node}} = 0.5\,\text{fF}$, $V_{\text{DD}}^{\text{sub}} = 0.25\,\text{V}$, and $\tau_{\text{step}} = 6\,\text{ns}$ at the same bias current, the same formulas give $E_{\text{bias}} = 15.0\,\text{aJ}$ and $E_{\text{cap}} \approx 1.0\,\text{aJ}$, so $E_{\text{node}}^{(\text{DTCA})} \approx 16\,\text{aJ} \approx 3.9 \times 10^3\, k_BT$.

17.4.3 Distance from the Thermodynamic Floor

Recall from Section 10.5 the Aurell–Dechant–Sasa–Seifert finite-time thermodynamic bound ([5, 25]): transporting a probability distribution $\rho_0 \to \rho_\tau$ over a Wasserstein distance $\mathcal{W}_2(\rho_0, \rho_\tau)$ in finite time τ_{step} with mobility $\mu = D/k_BT$ requires a minimum dissipated work per degree of freedom of

$$\mathcal{W}_{\text{diss}}^{\min} = T\Sigma_{\text{tot}}^{\min} = \frac{k_BT}{D\tau_{\text{step}}} \mathcal{W}_2^2(\rho_0, \rho_\tau) \sim \left(\frac{\tau_{RC}}{\tau_{\text{step}}}\right) k_BT \sim \mathcal{O}(1)\, k_BT, \tag{17.64}$$

since displacing a distribution by one thermal standard deviation $\sigma_V = \sqrt{k_BT/C_{\text{node}}}$ gives $\mathcal{W}_2^2 \sim k_BT/C_{\text{node}}$ and $D = k_BT/(C_{\text{node}}^2 R) = \sigma_V^2/\tau_{RC}$.

A subthreshold node instead dissipates about $7.8 \times 10^3\, k_BT$ per step (Eq. (17.63)). Proposition 17.7 shows where this comes from. The node holds $\Omega = C_{\text{node}} V_T/q_e \approx 160$ elementary charges per thermal voltage at $C_{\text{node}} = 1\,\text{fF}$, and its bias current flows across a supply drop of $\tilde{V}_{\text{DD}} \approx 12$ thermal voltages, which keeps the transistors in saturation. The bias term $\alpha_{\text{relax}} N_{\text{branch}} n \tilde{V}_{\text{DD}} \Omega \approx 3.9 \times 11.6 \times 161 \approx 7.3 \times 10^3$ accounts for nearly all of it. Shrinking C_{node} toward 0.1 fF ($\Omega \approx 16$), or moving to cryogenic Josephson quantum flux parametrons (Worked computation 14.20), would reduce the excess toward 10–100 k_BT. Even at $C_{\text{node}} = 1\,\text{fF}$, the node energy is about 10^5 times below a digital MAC with pseudorandom number generation (Fig. 17.2). Whether this per-operation gap becomes a gap per generated sample depends on how many node updates a sample needs compared with how many digital operations; Jelinčič et al. [56] estimate about 10^4 for their benchmark, and Worked computation 17.10 works through one example.

17.4.4 Accuracy Versus Energy

To compare architectures fairly, the energy per generated sample, E_{sample}, must be related to the distance between the hardware output $\hat{p}_\theta(\mathbf{x}_0)$ and the data distribution $p_{\text{data}}(\mathbf{x}_0)$. In a T-step chain in which each step runs K_{Gibbs} block Gibbs sweeps (or relaxes for a time $\tau_{\text{step}} = K_{\text{Gibbs}}\tau_{\text{sweep}}$), there are two sources of error:

(i) *Model error.* With finitely many steps T and a sparse latent graph with $\rho_{\text{lat}} = M/N$ latent nodes per visible node, the model marginal $p_\theta(\mathbf{x}_0)$ differs from $p_{\text{data}}(\mathbf{x}_0)$ even if every step reaches its conditional equilibrium $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$.

(ii) *Thermalization error.* Each tile relaxes for a finite time τ_{step}, so the hardware kernel $\hat{p}_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ differs from $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ by a residual that decays with the step spectral gap $\lambda_{1,t} = \tau_{\text{mix}}(t)^{-1}$.

The following bound separates the two. It restates Proposition 16.11 with the KL decay of Theorem 16.7 inserted, and adds a Wasserstein version.

Proposition 17.8 (Error of finite-time sampling)**.** *Consider a T-step Denoising Thermodynamic Model whose conditional distributions $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ are uniformly κ_t-strongly log-concave ($\kappa_t = \alpha_t - L_{\mathcal{E}} > 0$) with mobility μ, so that each step satisfies the Logarithmic Sobolev Inequality (Section 6.5) with constant $\lambda_{\text{LSI},t} = \mu\kappa_t = \tau_{\text{mix}}(t)^{-1}$. Let $\hat{p}_\theta(\mathbf{x}_0)$ denote the marginal distribution produced by the DTCA hardware when each step relaxes for physical time τ_{step} starting from an initial distribution $\hat{p}_0(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ of bounded relative entropy $D_{\text{KL}}(\hat{p}_0 \,\|\, p_\theta) \le D_0$. Then:*

(a) With $q(\mathbf{x}_{0:T})$ the joint law of the forward process (Eq. (16.1)), the total variation distance between the hardware output $\hat{p}_\theta(\mathbf{x}_0)$ and the data distribution $p_{\text{data}}(\mathbf{x}_0)$ satisfies

$$\|\hat{p}_\theta(\mathbf{x}_0) - p_{\text{data}}(\mathbf{x}_0)\|_{\text{TV}} \le \sqrt{\frac{1}{2} D_{\text{KL}}(q(\mathbf{x}_{0:T}) \,\|\, p_\theta(\mathbf{x}_{0:T}))} + \sum_{t=1}^{T} \sqrt{\frac{D_0}{2}} \exp\left(-\frac{\tau_{\text{step}}}{\tau_{\text{mix}}(t)}\right). \tag{17.65}$$

(b) Similarly, in the L^2-Wasserstein metric $\mathcal{W}_2$ (Section 10.2), if each conditional transition map is L_t-Lipschitz in the conditioning state $\mathbf{x}_t$, then

$$\mathcal{W}_2(\hat{p}_\theta(\mathbf{x}_0), p_{\text{data}}(\mathbf{x}_0)) \le \mathcal{W}_2(p_\theta(\mathbf{x}_0), p_{\text{data}}(\mathbf{x}_0)) + \sum_{t=1}^{T} \left(\prod_{s=1}^{t-1} L_s\right) \sqrt{\frac{2D_0}{\beta\kappa_t}} \exp\left(-\frac{\tau_{\text{step}}}{\tau_{\text{mix}}(t)}\right). \tag{17.66}$$

Proof. For (a), apply the triangle inequality for the total variation norm $\|p - q\|_{\text{TV}} := \frac{1}{2}\int \mathrm{d}\mathbf{x}\ |p(\mathbf{x}) - q(\mathbf{x})|$:

$$\|\hat{p}_\theta(\mathbf{x}_0) - p_{\text{data}}(\mathbf{x}_0)\|_{\text{TV}} \le \|p_\theta(\mathbf{x}_0) - p_{\text{data}}(\mathbf{x}_0)\|_{\text{TV}} + \|\hat{p}_\theta(\mathbf{x}_0) - p_\theta(\mathbf{x}_0)\|_{\text{TV}}. \tag{17.67}$$

For the first term on the right-hand side of Eq. (17.67), Pinsker's inequality

$$\|p - q\|_{\text{TV}} \le \sqrt{\tfrac{1}{2} D_{\text{KL}}(q \,\|\, p)}$$

together with the data-processing (marginalization) inequality

$$D_{\text{KL}}(p_{\text{data}}(\mathbf{x}_0) \,\|\, p_\theta(\mathbf{x}_0)) \le D_{\text{KL}}(q(\mathbf{x}_{0:T}) \,\|\, p_\theta(\mathbf{x}_{0:T}))$$

yields the first term of Eq. (17.65).

For the second term of Eq. (17.67), let $\mathcal{P}_t$ denote the exact conditional Markov transition operator $(\mathcal{P}_t\rho)(\mathbf{x}_{t-1}) = \int \mathrm{d}\mathbf{x}_t\, p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)\rho(\mathbf{x}_t)$, and let $\hat{\mathcal{P}}_t$ denote the finite-time hardware transition operator after relaxation time τ_{step}. By telescoping the T-fold composition of transition operators from $t = T$ down to $t = 1$, we write

$$\hat{\mathcal{P}}_1\hat{\mathcal{P}}_2\cdots\hat{\mathcal{P}}_T p_T - \mathcal{P}_1\mathcal{P}_2\cdots\mathcal{P}_T p_T = \sum_{t=1}^{T} \mathcal{P}_1\cdots\mathcal{P}_{t-1}(\hat{\mathcal{P}}_t - \mathcal{P}_t)\hat{\mathcal{P}}_{t+1}\cdots\hat{\mathcal{P}}_T p_T. \tag{17.68}$$

Because every Markov transition operator $\mathcal{P}_s$ is a contraction in total variation ($\|\mathcal{P}_s\nu\|_{\text{TV}} \le \|\nu\|_{\text{TV}}$), taking the total variation norm of Eq. (17.68) gives

$$\|\hat{p}_\theta(\mathbf{x}_0) - p_\theta(\mathbf{x}_0)\|_{\text{TV}} \le \sum_{t=1}^{T} \sup_{\mathbf{x}_t} \|\hat{p}_\theta(\cdot \mid \mathbf{x}_t) - p_\theta(\cdot \mid \mathbf{x}_t)\|_{\text{TV}}\,. \tag{17.69}$$

By the Bakry–Émery Logarithmic Sobolev theorem (Section 6.5), the conditional KL divergence decays exponentially along the Fokker–Planck / Glauber trajectory at rate $2\lambda_{\text{LSI},t} = 2/\tau_{\text{mix}}(t)$:

$$D_{\text{KL}}(\hat{p}_\theta(\cdot \mid \mathbf{x}_t) \,\|\, p_\theta(\cdot \mid \mathbf{x}_t)) \le D_0 \exp\!\left(-\frac{2\tau_{\text{step}}}{\tau_{\text{mix}}(t)}\right). \tag{17.70}$$

Applying Pinsker's inequality once more to each term in Eq. (17.69) takes the square root of Eq. (17.70), yielding $\sqrt{D_0/2}\exp(-\tau_{\text{step}}/\tau_{\text{mix}}(t))$ and completing the proof of Eq. (17.65).

For (b), Talagrand's transportation-cost inequality, which follows from the Bakry–Émery condition $\nabla^2\mathcal{E}_{\text{tot}} \succeq \kappa_t\mathbf{I}$ (Chapter C), bounds the Wasserstein distance by the KL divergence:

$$\mathcal{W}_2(\hat{p}_\theta(\cdot \mid \mathbf{x}_t), p_\theta(\cdot \mid \mathbf{x}_t)) \le \sqrt{\frac{2}{\beta\kappa_t} D_{\text{KL}}(\hat{p}_\theta(\cdot \mid \mathbf{x}_t) \,\|\, p_\theta(\cdot \mid \mathbf{x}_t))} \le \sqrt{\frac{2D_0}{\beta\kappa_t}} \exp\!\left(-\frac{\tau_{\text{step}}}{\tau_{\text{mix}}(t)}\right). \tag{17.71}$$

Propagating this step error through the L_s-Lipschitz conditional maps $\mathcal{P}_1, \ldots, \mathcal{P}_{t-1}$ via the triangle inequality yields Eq. (17.66). □

The bound is consistent with the shape of the energy–accuracy trade-off reported by Jelinčič et al. [56]. The thermalization error decays exponentially in $\tau_{\text{step}}/\tau_{\text{mix}}(t)$, so $\tau_{\text{step}} \approx (4\text{–}6)\,\tau_{\text{mix}}(t)$ makes it small ($e^{-5} \approx 6.7 \times 10^{-3}$). Beyond that point, the energy per sample grows linearly with the number of steps T and with the number of nodes $N + M = N(1 + \rho_{\text{lat}})$ per tile:

$$E^{(\text{DTCA})}_{\text{sample}}(T, \rho_{\text{lat}}) = T \cdot N(1 + \rho_{\text{lat}}) \cdot E^{(\text{DTCA})}_{\text{node}}. \tag{17.72}$$

Increasing ρ_{lat} or T reduces the model error (the first term in Eq. (17.65)) at a linear cost in energy. By the per-operation figures of Table 17.2, this cost lies several orders of magnitude below the corresponding digital one; how many depends on the operation counts being compared (Worked computation 17.10).

Notation Watch: Dynamic Switching Energy $\frac{1}{2}CV^2$ vs. Total Rail Energy CV^2

In circuit analysis, quoted capacitive charging energies are often off by a factor of two. When a capacitor C_{node} is charged from 0 to ΔV through a transistor connected to a rail at ΔV, the supply delivers charge $\Delta Q = C_{\text{node}}\Delta V$ at potential ΔV, doing work $W_{\text{supply}} = C_{\text{node}}(\Delta V)^2$. Half of this, $\frac{1}{2}C_{\text{node}}(\Delta V)^2$, is dissipated in the pull-up transistor during charging; the

Table 17.2: Representative energies of elementary operations in a 4–5 nm GPU and in a subthreshold CMOS sampling tile at $T = 300\,\mathrm{K}$ ($k_\mathrm{B}T \approx 4.142\,\mathrm{zJ}$). These are order-of-magnitude estimates from this section, not measurements.

Quantity	GPU (4 nm FinFET)	Subthreshold DTCA (estimate)	Ratio
Transistor regime	Strong inversion ($V_\mathrm{GS} \gg V_\mathrm{th}$)	Weak inversion ($V_\mathrm{GS} < V_\mathrm{th}$)	—
Supply voltage V_DD	0.75–0.85 V ($\sim 31\,V_T$)	0.25–0.35 V ($\sim 11\,V_T$)	~ 2.8
Signal swing ΔV	Rail-to-rail (0.80 V)	A few V_T (50–75 mV)	~ 12
Role of thermal noise	Suppressed	Source of randomness	—
Random-number cost	15–40 pJ (3.6×10^9–$10^{10}\,k_\mathrm{B}T$)	None separate (device noise)	—
Weight fetch per operation	1.5–50 pJ (SRAM or HBM)	None (weights stored in the cell)	—
Energy per elementary operation	0.5–3.5 pJ (MAC with register access)	10–50 aJ (2.5×10^3–$10^4\,k_\mathrm{B}T$)	10^4–10^5
Energy per sample (Worked computation 17.10)	15 μJ (Gibbs simulation) to 3.6 mJ (neural network)	26 nJ	6×10^2–10^5

other half is stored on the capacitor and dissipated in the pull-down transistor when the capacitor discharges. One full charge-and-discharge cycle ($0 \to \Delta V \to 0$) therefore dissipates $C_\mathrm{node}(\Delta V)^2$, and one one-way transition dissipates $\frac{1}{2}C_\mathrm{node}(\Delta V)^2$. In Proposition 17.7, N_flip counts one-way transitions, each contributing $\frac{1}{2}C_\mathrm{node}(\Delta V)^2$.

17.5 Continuous and Hybrid Discrete–Continuous Tiles

The all-binary tile of Section 17.1 suits discrete data. Continuous data, such as continuous latent embeddings of images, audio waveforms, or molecular coordinates, call for continuous variables $\mathbf{v}_{t-1} \in \mathbb{R}^N$. A natural extension, which we analyze here and which goes beyond the binary experiments of Jelinčič et al. [56], is a hybrid tile in which capacitor voltages $\mathbf{v}_{t-1}$ (the RC sampler, Chapters 5 and 13) couple locally to binary p-bits $\mathbf{z}_{t-1} \in \{-1, +1\}^M$ (the p-bit limit of the double-well sampler, Chapters 1 and 14).

We write down the dynamics of a hybrid Gaussian–Bernoulli tile and find the forward coupling needed for fast conditional mixing. The result is the circuit form of Eq. (16.45).

17.5.1 Circuit Dynamics and the Effective Potential

Consider a single hybrid denoising tile at step t, containing:

- N continuous visible voltage nodes $\mathbf{v}_{t-1} = (v_{t-1,1}, \ldots, v_{t-1,N})^\top \in \mathbb{R}^N$, each with node capacitance C_node, coupled to its conditioning input voltage $v_{t,i}$ through a forward coupling of stiffness $G_t^f = \alpha_t$ (realized with a differential transconductance), and coupled to neighboring visible voltages through a symmetric passive/transconductance matrix $\mathbf{G}_\theta^{(vv)}$;
- M binary latent p-bit nodes $\mathbf{z}_{t-1} = (z_{t-1,1}, \ldots, z_{t-1,M})^\top \in \{-1, +1\}^M$, coupled to local visible nodes through bidirectional transconductance weights $J_{ia}^{(vz)}$ (in the convention below these

have units of charge, so $J^{(vz)}_{ia} z_{t-1,a}/\tau_0$ is the current injected into node i, while $v_{t-1,i}$ biases the input gate of p-bit a).

We use the engineered-noise convention of Section 5.6. Every coupling written as a "conductance" G in the energy below stands for the stiffness $\tau_0 G$, which has units of capacitance, so the energy is in joules. The node noise is engineered, with covariance rate $2k_{\mathrm{B}}TC_{\mathrm{node}}/\tau_0$ per node, rather than the passive thermal noise $2k_{\mathrm{B}}TG$; only with this noise is the stationary law of the voltages the Boltzmann distribution of the energy. The conditional energy of the tile is

$$\begin{aligned}\mathcal{E}_{\mathrm{tot}}(\mathbf{v}_{t-1}, \mathbf{z}_{t-1} \mid \mathbf{v}_t) = \frac{1}{2}\mathbf{v}_{t-1}^{\top}\Big(G_t^f \mathbf{I}_N + \mathbf{G}_\theta^{(vv)}\Big)\mathbf{v}_{t-1} - \big(G_t^f c_t \mathbf{v}_t + \mathbf{I}_b^v + \mathbf{K}^{(tv)}\mathbf{v}_t\big)^{\top}\mathbf{v}_{t-1} \\ - \mathbf{v}_{t-1}^{\top}\mathbf{J}^{(vz)}\mathbf{z}_{t-1} - \big(\mathbf{b}^z + \mathbf{K}^{(tz)}\mathbf{v}_t\big)^{\top}\mathbf{z}_{t-1} + \frac{1}{2}G_t^f c_t^2 \,\|\mathbf{v}_t\|^2 ,\end{aligned} \tag{17.73}$$

where $\mathbf{J}^{(vz)} \in \mathbb{R}^{N\times M}$ and $\mathbf{K}^{(tv)}, \mathbf{K}^{(tz)}$ are supported only on local lattice edges.

By Kirchhoff's current law (KCL, Worked computation 5.33 and Section 13.4), for fixed $\mathbf{z}_{t-1}$ and $\mathbf{v}_t$ the node voltages are modeled by the overdamped circuit Langevin equation

$$\begin{aligned}\tau_0 C_{\mathrm{node}}\, \mathrm{d}\mathbf{v}_{t-1} &= -\nabla_{\mathbf{v}_{t-1}}\mathcal{E}_{\mathrm{tot}}(\mathbf{v}_{t-1}, \mathbf{z}_{t-1} \mid \mathbf{v}_t)\, \mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\tau_0 C_{\mathrm{node}}}\, \mathrm{d}\mathbf{W}_t \\ &= -\Big[(G_t^f \mathbf{I}_N + \mathbf{G}_\theta^{(vv)})\mathbf{v}_{t-1} - G_t^f c_t \mathbf{v}_t - \mathbf{I}_b^v - \mathbf{K}^{(tv)}\mathbf{v}_t - \mathbf{J}^{(vz)}\mathbf{z}_{t-1}\Big]\, \mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\tau_0 C_{\mathrm{node}}}\, \mathrm{d}\mathbf{W}_t,\end{aligned} \tag{17.74}$$

while each p-bit $z_{t-1,a} \in \{-1, +1\}$ switches stochastically with conditional probability

$$\mathbb{P}(z_{t-1,a} = +1 \mid \mathbf{v}_{t-1}, \mathbf{v}_t) = \frac{1}{2}\left(1 + \tanh\left[\beta\left(b_a^z + \sum_{i\in\partial a} J_{ia}^{(vz)} v_{t-1,i} + \sum_{i\in\partial a} K_{ia}^{(tz)} v_{t,i}\right)\right]\right). \tag{17.75}$$

Summing out the p-bits gives a non-Gaussian effective potential for $\mathbf{v}_{t-1}$, and a large enough forward conductance makes it convex.

Theorem 17.9 (Effective potential and stability of a hybrid tile)**.** *Consider the hybrid Gaussian–Bernoulli DTCA tile defined by Eq. (17.73) at inverse thermal energy $\beta = (k_{\mathrm{B}}T)^{-1}$.*

(a) Summing over all 2^M configurations of the conditionally independent latent p-bits $\mathbf{z}_{t-1} \in \{-1, +1\}^M$ gives the marginal conditional density

$$p_\theta(\mathbf{v}_{t-1} \mid \mathbf{v}_t) \propto \exp(-\beta \mathcal{E}_{\mathrm{eff}}(\mathbf{v}_{t-1} \mid \mathbf{v}_t))$$

with effective potential

$$\begin{aligned}\mathcal{E}_{\mathrm{eff}}(\mathbf{v}_{t-1} \mid \mathbf{v}_t) = \frac{1}{2}\mathbf{v}_{t-1}^{\top}\Big(G_t^f \mathbf{I}_N + \mathbf{G}_\theta^{(vv)}\Big)\mathbf{v}_{t-1} - \big(G_t^f c_t \mathbf{v}_t + \mathbf{I}_b^v + \mathbf{K}^{(tv)}\mathbf{v}_t\big)^{\top}\mathbf{v}_{t-1} \\ - \frac{1}{\beta}\sum_{a=1}^{M} \ln\cosh\Big[\beta\Big(b_a^z + (\mathbf{J}_{\cdot,a}^{(vz)})^{\top}\mathbf{v}_{t-1} + (\mathbf{K}_{\cdot,a}^{(tz)})^{\top}\mathbf{v}_t\Big)\Big].\end{aligned} \tag{17.76}$$

(b) The Hessian of $\mathcal{E}_{\mathrm{eff}}(\mathbf{v}_{t-1} \mid \mathbf{v}_t)$ with respect to the visible voltages $\mathbf{v}_{t-1}$ is

$$\nabla^2_{\mathbf{v}_{t-1}}\mathcal{E}_{\mathrm{eff}}(\mathbf{v}_{t-1} \mid \mathbf{v}_t) = G_t^f \mathbf{I}_N + \mathbf{G}_\theta^{(vv)} - \beta \mathbf{J}^{(vz)}\mathbf{S}(\mathbf{v}_{t-1}, \mathbf{v}_t)(\mathbf{J}^{(vz)})^{\top}, \tag{17.77}$$

where $\mathbf{S}(\mathbf{v}_{t-1}, \mathbf{v}_t) = \mathrm{diag}\big(\operatorname{sech}^2(\beta h_1^z), \ldots, \operatorname{sech}^2(\beta h_M^z)\big)$ satisfies $\mathbf{0} \prec \mathbf{S} \preceq \mathbf{I}_M$.

(c) *Consequently, if the forward confinement transconductance G_t^f is biased above the critical circuit threshold*

$$G_t^f > \beta \left\| \mathbf{J}^{(vz)} \right\|_{\mathrm{op}}^2 + \max\bigl(0, -\lambda_{\min}(\mathbf{G}_\theta^{(vv)})\bigr) + G_\kappa, \tag{17.78}$$

for $G_\kappa > 0$, then $\nabla^2_{\mathbf{v}_{t-1}} \mathcal{E}_{\mathrm{eff}} \succeq G_\kappa \mathbf{I}_N$ uniformly for all $(\mathbf{v}_{t-1}, \mathbf{v}_t) \in \mathbb{R}^N \times \mathbb{R}^N$, and the continuous voltage distribution converges to conditional equilibrium exponentially with time constant $\tau_{\mathrm{mix}} \le \tau_0 C_{\mathrm{node}} / G_\kappa$.

Proof. For (a), marginalize $p_\theta(\mathbf{v}_{t-1}, \mathbf{z}_{t-1} \mid \mathbf{v}_t)$ over $\mathbf{z}_{t-1} \in \{-1, +1\}^M$. The energy Eq. (17.73) has no latent–latent couplings, so the sum over $\mathbf{z}_{t-1}$ factors into M independent two-state sums:

$$\begin{aligned} \sum_{\mathbf{z}_{t-1} \in \{-1,+1\}^M} \exp\left(\beta \sum_{a=1}^M h_a^z(\mathbf{v}_{t-1}, \mathbf{v}_t)\, z_{t-1,a} \right) &= \prod_{a=1}^M \left(e^{\beta h_a^z} + e^{-\beta h_a^z} \right) \\ &= 2^M \prod_{a=1}^M \cosh\bigl(\beta h_a^z(\mathbf{v}_{t-1}, \mathbf{v}_t)\bigr) \\ &= 2^M \exp\left(\sum_{a=1}^M \ln\cosh\bigl(\beta h_a^z(\mathbf{v}_{t-1}, \mathbf{v}_t)\bigr) \right), \end{aligned} \tag{17.79}$$

where $h_a^z(\mathbf{v}_{t-1}, \mathbf{v}_t) := b_a^z + (\mathbf{J}_{:,a}^{(vz)})^\top \mathbf{v}_{t-1} + (\mathbf{K}_{:,a}^{(tz)})^\top \mathbf{v}_t$. Multiplying by the $\mathbf{z}_{t-1}$-independent quadratic part of $\exp(-\beta \mathcal{E}_{\mathrm{tot}})$ and taking $-\beta^{-1} \ln(\cdot)$ (absorbing the constant 2^M into the normalization $Z_t(\mathbf{v}_t)$) yields Eq. (17.76).

For (b), differentiate Eq. (17.76) twice with respect to $\mathbf{v}_{t-1}$. Using $\frac{\mathrm{d}}{\mathrm{d}u} \ln\cosh u = \tanh u$ and $\frac{\mathrm{d}^2}{\mathrm{d}u^2} \ln\cosh u = \operatorname{sech}^2 u = 1 - \tanh^2 u$, the gradient is

$$\nabla_{\mathbf{v}_{t-1}} \mathcal{E}_{\mathrm{eff}}(\mathbf{v}_{t-1} \mid \mathbf{v}_t) = (G_t^f \mathbf{I}_N + \mathbf{G}_\theta^{(vv)}) \mathbf{v}_{t-1} - (G_t^f c_t \mathbf{v}_t + \mathbf{I}_b^v + \mathbf{K}^{(tv)} \mathbf{v}_t) - \sum_{a=1}^M \mathbf{J}_{:,a}^{(vz)} \tanh\bigl(\beta h_a^z(\mathbf{v}_{t-1}, \mathbf{v}_t)\bigr). \tag{17.80}$$

The last term is the mean current $\sum_a \mathbf{J}_{:,a}^{(vz)} \langle z_{t-1,a} \rangle_{\mathbf{v}_{t-1}, \mathbf{v}_t}$ injected by the p-bits. Differentiating again with respect to $\mathbf{v}_{t-1}$, with $\nabla_{\mathbf{v}_{t-1}} (\beta h_a^z) = \beta \mathbf{J}_{:,a}^{(vz)}$, gives

$$\nabla^2_{\mathbf{v}_{t-1}} \mathcal{E}_{\mathrm{eff}}(\mathbf{v}_{t-1} \mid \mathbf{v}_t) = G_t^f \mathbf{I}_N + \mathbf{G}_\theta^{(vv)} - \beta \sum_{a=1}^M \operatorname{sech}^2\bigl(\beta h_a^z\bigr)\, \mathbf{J}_{:,a}^{(vz)} (\mathbf{J}_{:,a}^{(vz)})^\top, \tag{17.81}$$

which in matrix notation is Eq. (17.77).

For (c), since $0 < \operatorname{sech}^2 u \le 1$ for all $u \in \mathbb{R}$, the diagonal matrix $\mathbf{S}$ satisfies $\mathbf{0} \prec \mathbf{S} \preceq \mathbf{I}_M$. Therefore, for any unit vector $\mathbf{u} \in \mathbb{R}^N$,

$$\mathbf{u}^\top \mathbf{J}^{(vz)} \mathbf{S} (\mathbf{J}^{(vz)})^\top \mathbf{u} \le \mathbf{u}^\top \mathbf{J}^{(vz)} (\mathbf{J}^{(vz)})^\top \mathbf{u} \le \left\| \mathbf{J}^{(vz)} \right\|_{\mathrm{op}}^2. \tag{17.82}$$

Hence $\lambda_{\min}(\nabla^2_{\mathbf{v}_{t-1}} \mathcal{E}_{\mathrm{eff}}) \ge G_t^f + \lambda_{\min}(\mathbf{G}_\theta^{(vv)}) - \beta \left\| \mathbf{J}^{(vz)} \right\|_{\mathrm{op}}^2 \ge G_\kappa > 0$ whenever Eq. (17.78) holds. By the Bakry–Émery theorem (Section 6.5) with mobility $\mu = (\tau_0 C_{\mathrm{node}})^{-1}$, the spectral gap and LSI constant satisfy $\lambda_{\mathrm{LSI}} \ge G_\kappa / (\tau_0 C_{\mathrm{node}})$, so $\tau_{\mathrm{mix}} \le \tau_0 C_{\mathrm{node}} / G_\kappa$. □

Theorem 17.9 combines the RC sampler and the p-bit limit of the double-well sampler in one tile. With $G_t^f = 0$ (a monolithic EBM, Worked computation 1.19), the negative term $-\beta \mathbf{J}^{(vz)} \mathbf{S} (\mathbf{J}^{(vz)})^\top$

from the p-bits can create several wells separated by barriers that grow with $\left\|\mathbf{J}^{(vz)}\right\|$, as in Example 16.21. Raising G_t^f above Eq. (17.78) makes every step convex while keeping the non-Gaussian $\ln\cosh$ terms, so that multimodality is produced by the chain of steps (Section 16.3.5).

The noise convention is essential to the theorem. If the nodes were driven only by the passive thermal noise of their conductances, the stationary covariance would be $k_{\mathrm{B}}T\mathbf{C}^{-1}$ whatever the conductances (Proposition 13.5), and the couplings in Eq. (17.73) would set only relaxation rates. Realizing the Boltzmann distribution of $\mathcal{E}_{\mathrm{eff}}$ requires the engineered drift and noise of the non-equilibrium circuits of Chapter 13.

17.6 Worked Computation: Energy Budget of a Representative 250,000-p-Bit Design

We now combine the formulas of this chapter into an energy budget for a DTCA chip [31, 56] with about 2.5×10^5 p-bits, the scale Extropic has described for Z1 (Section 17.1.1), and compare it with a current GPU. The design is a representative example chosen for this book: its layout, device parameters, and sweep counts are our assumptions, not a published specification.

Worked computation 17.10 (Energy budget of a representative 250,000-p-bit design versus a GPU)**. Problem.** Estimate the throughput and energy of a room-temperature (300 K) subthreshold CMOS chip with $N_{\mathrm{die}} = 250{,}000$ p-bits in $T = 16$ spatially pipelined tiles (Mode II) that generates 64×64 binary samples ($N = 4096$ visible bits per stage). In this computation T denotes the number of stages; the temperature is always written in kelvin.

1. Divide the p-bits among the tiles, choose the connectivity and weight storage, and estimate the node capacitance C_{node}.
2. Estimate the relaxation time τ_{RC}, the four-phase sweep period τ_{sweep}, the step time τ_{step}, and the energy per node per step from bias current (E_{bias}), capacitive switching (E_{cap}), clocking (E_{clk}), and transfer between tiles (E_{latch}).
3. Compute the chip power P_{die}, the throughput $\mathcal{R}$, and the energy per sample $E_{\mathrm{sample}}^{(\mathrm{DTCA})}$.
4. Estimate the energy per sample on a 4 nm GPU, both for a direct block Gibbs simulation of the same DTM and for a 16-step neural-network diffusion model, and compare.

Step 1: Tiles and connectivity. Dividing $N_{\mathrm{die}} = 250{,}000$ p-bits equally among $T = 16$ tiles gives

$$N_{\mathrm{tile}} = \frac{N_{\mathrm{die}}}{T} = \frac{250{,}000}{16} = 15{,}625 \text{ physical } p\text{-bits}, \tag{17.83}$$

arranged as a square 125×125 2D lattice ($125^2 = 15{,}625$). Within each tile:

- $N = 64 \times 64 = 4{,}096$ nodes are the visible spins $\mathbf{x}_{t-1} \in \{-1, +1\}^{4096}$, spread uniformly over the grid.
- The remaining $M = N_{\mathrm{tile}} - N = 15{,}625 - 4{,}096 = 11{,}529$ nodes are latent p-bits $\mathbf{z}_{t-1} \in \{-1, +1\}^{11529}$, so the latent-to-visible ratio is

$$\rho_{\mathrm{lat}} = \frac{M}{N} = \frac{11{,}529}{4{,}096} \approx 2.815. \tag{17.84}$$

- Each tile also has $N = 4{,}096$ latches holding the clamp $\mathbf{x}_t \in \{-1, +1\}^{4096}$ from the preceding stage.

The interaction graph within each tile is a King's graph (neighbors at $(\pm 1, 0)$, $(0, \pm 1)$, and $(\pm 1, \pm 1)$, so each interior node has degree $d_{\text{active}} = 8$), plus $d_{\text{cond}} = 4$ couplings to the clamp $\mathbf{x}_t$, for $d_{\text{tot}} = 12$ coupling branches per node.

The King's graph is 4-colorable (Worked computation 2.28), with coloring $c(i_x, i_y) = (i_x \bmod 2) + 2(i_y \bmod 2) + 1 \in \{1, 2, 3, 4\}$. In each of the four phases about $N_{\text{tile}}/4 \approx 3{,}906$ p-bits per tile, and $N_{\text{die}}/4 = 62{,}500$ on the whole chip, update in parallel.

Step 2: Cell area and node capacitance. In a 12–16 nm FinFET or 22 nm fully depleted silicon-on-insulator (FD-SOI) process, we assume each cell contains

(i) The subthreshold noise-amplifying p-bit core (Section 14.1, 14 transistors) occupying $\sim 1.5\,\mu\text{m}^2$;

(ii) $d_{\text{tot}} = 12$ differential subthreshold coupling branches, each with a 6-bit SRAM and current DAC for its weight ($12 \times 6 = 72$ SRAM bits), occupying $\sim 8.5\,\mu\text{m}^2$.

The cell area is then $A_{\text{cell}} = 10.0\,\mu\text{m}^2$, a cell pitch of $\ell_{\text{cell}} = \sqrt{A_{\text{cell}}} \approx 3.16\,\mu\text{m}$. The 250,000 cells occupy

$$A_{\text{core}} = N_{\text{die}} A_{\text{cell}} = 250{,}000 \times 10.0\,\mu\text{m}^2 = 2.50\,\text{mm}^2, \tag{17.85}$$

and with the latches between tiles, input/output circuits, and clock and power distribution we take a chip area of about $A_{\text{die}} \approx 4\,\text{mm}^2$.

All couplings in the King's graph join adjacent cells ($\text{dist}_\infty = 1$), so a connection is at most $L_{\text{wire}} \approx \sqrt{2}\,\ell_{\text{cell}} \approx 4.47\,\mu\text{m}$ long. With local wire capacitance $c_{\text{wire}} \approx 0.15\,\text{fF}/\mu\text{m}$ this gives $C_{\text{wire}} \approx 0.67\,\text{fF}$. Adding the gate and drain capacitance of the summing node ($C_{\text{int}} \approx 0.53\,\text{fF}$) gives

$$C_{\text{node}} = C_{\text{int}} + C_{\text{wire}} = 0.53\,\text{fF} + 0.67\,\text{fF} = 1.20\,\text{fF}. \tag{17.86}$$

At 300 K ($V_T = k_B T/q_e = 25.852\,\text{mV}$), the charge number Eq. (17.56) is

$$\Omega = \frac{C_{\text{node}} V_T}{q_e} = \frac{(1.20 \times 10^{-15}\,\text{F})(25.852 \times 10^{-3}\,\text{V})}{1.60218 \times 10^{-19}\,\text{C}} = 193.6 \text{ elementary charges}. \tag{17.87}$$

Step 3: Relaxation time and energy per node per step. Each summing node is biased in weak inversion at supply voltage $V_{\text{DD}}^{\text{sub}} = 0.30\,\text{V}$ ($\tilde{V}_{\text{DD}} = V_{\text{DD}}^{\text{sub}}/V_T = 11.60$), with slope factor $n = 1.25$, branch current $I_{\text{cell}} = 20\,\text{nA}$ during its own color phase, and a standby current $I_{\text{standby}} = 1.0\,\text{nA}$ during the other three phases.

- *Relaxation time.* During its active phase the node conductance is $g_{\text{m}} = \frac{I_{\text{cell}}}{nV_T} = \frac{20\,\text{nA}}{1.25 \times 25.852\,\text{mV}} = 0.619\,\mu\text{S}$, so

$$\tau_{RC} = \frac{C_{\text{node}}}{g_{\text{m}}} = \frac{1.20\,\text{fF}}{0.619\,\mu\text{S}} = 1.94\,\text{ns}. \tag{17.88}$$

- *Phase and step duration.* We allow $\tau_{\text{phase}} = 2.0\,\text{ns} \approx 1.03\,\tau_{RC}$ per color phase. One four-color sweep then takes

$$\tau_{\text{sweep}} = 4\tau_{\text{phase}} = 8.0\,\text{ns}. \tag{17.89}$$

We assume that the forward coupling γ_t and the total-correlation penalty keep each stage close to the regime of Theorem 16.7, so that $K_{\text{Gibbs}} = 2$ sweeps per stage bring it within 10^{-2} of equilibrium. This is optimistic: the discrete mixing argument of Remark 16.16 is only an analogy, and the DTMs trained by Jelinčič et al. [56] use several hundred sweeps per stage (Remark 16.18). The step period is then

$$\tau_{\text{step}} = K_{\text{Gibbs}} \tau_{\text{sweep}} = 2 \times 8.0\,\text{ns} = 16.0\,\text{ns}. \tag{17.90}$$

- *Bias energy.* Over one sweep (8.0 ns) each node spends one phase (2.0 ns) at $I_{\text{cell}} = 20\,\text{nA}$ and three phases (6.0 ns) at $I_{\text{standby}} = 1.0\,\text{nA}$, drawing charge $q_{\text{sweep}} = (20\,\text{nA})(2.0\,\text{ns}) + (1.0\,\text{nA})(6.0\,\text{ns}) = 46.0\,\text{aC}$. Over $K_{\text{Gibbs}} = 2$ sweeps at $V_{\text{DD}}^{\text{sub}} = 0.30\,\text{V}$,

$$E_{\text{bias}} = K_{\text{Gibbs}} q_{\text{sweep}} V_{\text{DD}}^{\text{sub}} = 2 \times (46.0\,\text{aC})(0.30\,\text{V}) = 27.6\,\text{aJ} = 6.66 \times 10^3\, k_{\text{B}}T. \tag{17.91}$$

- *Capacitive energy.* The analog summing node swings by $\Delta V_{\text{sum}} = 2nV_T = 64.6\,\text{mV}$, and the latched p-bit output swings between 0 and $V_{\text{DD}}^{\text{sub}} = 0.30\,\text{V}$ on a load $C_{\text{out}} = 0.40\,\text{fF}$. With flip probability $p_{\text{flip}} = 0.5$ per sweep there are $N_{\text{flip}} = K_{\text{Gibbs}} \times 2p_{\text{flip}} = 2.0$ one-way transitions per step, and

$$\begin{aligned} E_{\text{cap}} &= \frac{1}{2} N_{\text{flip}} \Big[C_{\text{node}} (\Delta V_{\text{sum}})^2 + C_{\text{out}} (V_{\text{DD}}^{\text{sub}})^2 \Big] \\ &= \frac{1}{2}(2.0) \Big[(1.20\,\text{fF})(0.0646\,\text{V})^2 + (0.40\,\text{fF})(0.30\,\text{V})^2 \Big] \\ &= 5.01\,\text{aJ} + 36.0\,\text{aJ} = 41.0\,\text{aJ} = 9.90 \times 10^3\, k_{\text{B}}T. \end{aligned} \tag{17.92}$$

- *Clock and transfer energy.* Driving the color-phase enable gate ($C_{\text{clk}} = 0.15\,\text{fF}$ at 0.30 V, twice per step) costs $E_{\text{clk}} = 2C_{\text{clk}}(V_{\text{DD}}^{\text{sub}})^2 = 27.0\,\text{aJ}$. Passing the visible state $\mathbf{x}_{t-1}$, held by $N/N_{\text{tile}} = 4096/15625 = 26.2\%$ of the nodes, through the latch to the next tile ($C_{\text{latch}} = 0.8\,\text{fF}$ at 0.30 V, once per step) costs $E_{\text{latch}} = 0.262 \times \frac{1}{2} C_{\text{latch}} (V_{\text{DD}}^{\text{sub}})^2 = 9.4\,\text{aJ}$ per node on average.

Summing the four contributions gives the energy per node per denoising step,

$$E_{\text{node}}^{(\text{DTCA})} = E_{\text{bias}} + E_{\text{cap}} + E_{\text{clk}} + E_{\text{latch}} = 27.6 + 41.0 + 27.0 + 9.4 = 105.0\,\text{aJ} = 0.105\,\text{fJ} \approx 2.53 \times 10^4\, k_{\text{B}}T, \tag{17.93}$$

or 52.5 aJ ($1.27 \times 10^4\, k_{\text{B}}T$) per p-bit update.

Step 4: Power, throughput, and energy per sample. In Mode II all $T = 16$ tiles operate at once, and one 64×64 sample leaves the pipeline every $\tau_{\text{step}} = 16.0\,\text{ns}$:

$$\text{Throughput:} \quad \mathcal{R}^{(\text{DTCA})} = \frac{1}{\tau_{\text{step}}} = \frac{1}{16.0\,\text{ns}} = 6.25 \times 10^7\,\text{samples/s}, \tag{17.94}$$

$$\begin{aligned} \text{Energy per sample:} \quad E_{\text{sample}}^{(\text{DTCA})} &= N_{\text{die}} E_{\text{node}}^{(\text{DTCA})} = 250{,}000 \times 105.0\,\text{aJ} \\ &= 26.25\,\text{nJ} = 6.34 \times 10^{12}\, k_{\text{B}}T, \end{aligned} \tag{17.95}$$

$$\begin{aligned} \text{Chip power:} \quad P_{\text{die}}^{(\text{DTCA})} &= \mathcal{R}^{(\text{DTCA})} E_{\text{sample}}^{(\text{DTCA})} = \frac{26.25\,\text{nJ}}{16.0\,\text{ns}} = 1.64\,\text{W} \\ &\quad (\text{or } 0.82\,\text{W at } \tau_{\text{step}} = 32\,\text{ns}). \end{aligned} \tag{17.96}$$

Under these assumptions a 4 mm^2 chip drawing 1.6 W would produce 6.25×10^7 samples per second at 26 nJ per sample.

Step 5: Comparison with a 4 nm GPU. We consider two GPU baselines.

Baseline A: simulating the same Gibbs chain. To update one p-bit digitally over $K_{\text{Gibbs}} = 2$ sweeps, the GPU must:

(i) Compute the clamp field $\sum_{j=1}^{4} K_{ij} x_{t,j}$ once per step (4 MACs) and the local field $\sum_{j=1}^{8} J_{ij} s_j$ twice per step (16 MACs), 20 MACs in all, with their register and SRAM accesses. At the low end of Eq. (17.51), 1.5 pJ per MAC, this costs $20 \times 1.5\,\text{pJ} = 30.0\,\text{pJ}$.

(ii) Generate $K_{\text{Gibbs}} = 2$ pseudorandom numbers and evaluate $\sigma(2\beta h_i)$ each time. At the low end of Eq. (17.53), 15.0 pJ per update, this costs $2 \times 15.0\,\text{pJ} = 30.0\,\text{pJ}$.

One node step on the GPU thus costs $E_{\text{node}}^{(\text{GPU,EBM})} = 30.0 + 30.0 = 60.0\,\text{pJ}$, against 105.0 aJ on the subthreshold chip. Over the $N_{\text{die}} = 250{,}000$ node steps of one sample,

$$E_{\text{sample}}^{(\text{GPU,EBM})} = 250{,}000 \times 60.0\,\text{pJ} = 15.0\,\mu\text{J},$$
$$\frac{E_{\text{sample}}^{(\text{GPU,EBM})}}{E_{\text{sample}}^{(\text{DTCA})}} = \frac{15.0\,\mu\text{J}}{26.25\,\text{nJ}} = 571 \text{ (up to about 2,000 with off-chip memory traffic).} \quad (17.97)$$

Baseline B: a 16-step neural diffusion model. In practice a GPU would not simulate a sparse Ising lattice, which makes poor use of its dense matrix units; it would run a neural-network diffusion model $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ over the 64×64 grid. We assume a compact convolutional network with about 1.5×10^8 MACs per denoising step, and that 16 steps give sample quality comparable to the 16-step DTM, for 2.4×10^9 MACs per sample. This equivalence of quality is an assumption; it has to be established by experiment. At 1.5 pJ per MAC, including amortized memory traffic,

$$E_{\text{sample}}^{(\text{GPU,DDPM})} = (2.4 \times 10^9\,\text{MACs}) \times (1.5\,\text{pJ/MAC}) = 3.60\,\text{mJ},$$
$$\frac{E_{\text{sample}}^{(\text{GPU,DDPM})}}{E_{\text{sample}}^{(\text{DTCA})}} = \frac{3.60\,\text{mJ}}{26.25\,\text{nJ}} = 1.37 \times 10^5. \quad (17.98)$$

For a distilled network with 10 times fewer MACs (2.4×10^8 in total, 360 μJ per sample), the ratio is about 1.4×10^4. Producing 6.25×10^7 samples per second with that network would take $(6.25 \times 10^7\,\text{s}^{-1})(360\,\mu\text{J}) = 22.5\,\text{kW}$, about thirty-two 700 W GPUs.

Step 6: What the estimate shows. The ratio depends on what the GPU is asked to do. Simulating the same Gibbs chain digitally costs about 600 times more energy (up to about 2000 with off-chip memory traffic); this is the like-for-like hardware comparison. Ratios of 10^4–10^5 arise only against a neural-network model of assumed equal quality; the estimate of Jelinčič et al. [56] is likewise a comparison with conventional models run on a GPU. Most of the numbers also scale with the assumed sweep count. With K_{Gibbs} sweeps per stage, the energy per sample of both the chip and the Gibbs simulation grows roughly in proportion to K_{Gibbs}, so their ratio stays near 600, while the throughput falls as $1/K_{\text{Gibbs}}$ and the advantage over a neural-network baseline shrinks by the same factor. At the several hundred sweeps per stage used by Jelinčič et al. [56], the advantage of this design over Baseline B would fall to roughly 10^2–10^3.

Summary

Proved

- The local fields and conditional independence of color classes in a sparse tile (Proposition 17.1).
- The entropy, chain-rule, and Gaussian forms of the total correlation, and its parameter gradient (Proposition 17.3 and Theorem 17.4).
- For Gaussian distributions, a bound on the off-diagonal correlations and exponential decay of correlations with distance when the total correlation is small (Proposition 17.5).

- The block-tridiagonal Hessian of the joint trajectory energy, with smallest eigenvalue of order α/T^2, compared with step-by-step relaxation (Theorem 17.6).
- The energy per node per step of a subthreshold cell, in which the bias current cancels (Proposition 17.7).
- Finite-time error bounds for a T-step chain (Proposition 17.8) and the effective potential and stability threshold of hybrid tiles (Theorem 17.9).

Assumed

- From earlier chapters: chromatic Gibbs sampling (Chapter 2), Bakry–Émery and log-Sobolev inequalities (Chapter 6), finite-time Wasserstein bounds (Chapter 10), subthreshold MOSFET physics and the charge number Ω (Chapters 12 and 13), p-bit cells (Chapter 14), and the DTM mixing theorem (Chapter 16).
- The device parameters, sweep counts, and GPU energies of Section 17.4 and Worked computation 17.10 are representative values chosen in this book, not measurements.

Taken up later

- Differentiating through stochastic hardware, equilibrium propagation, and compensation for process, voltage, and temperature variation and transistor mismatch (Chapter 18).
- Gaussian Schur complements and Lyapunov identities (Chapter B) and Talagrand's inequality (Chapter C).

Open

- Three-dimensional stacking of sampling tiles, which would allow interaction graphs with Rent exponent up to $p = 2/3$.
- Charge-recycling (adiabatic) clock drivers that could lower the energy per node from about $10^4\, k_{\mathrm{B}}T$ toward $10^2\, k_{\mathrm{B}}T$ at room temperature.

Bibliographic notes

The Denoising Thermodynamic Computer Architecture, the total-correlation penalty used to keep its sparse stages easy to sample, and the model-based energy comparison with GPUs are due to Jelinčič et al. [56]. Extropic's `THRML` library [31] implements block Gibbs sampling on such graphs, and Extropic's own descriptions of its hardware include [30, 32, 33]. Total correlation (multi-information) was introduced by Watanabe [117] and used for disentangled variational autoencoders by Chen et al. [19] and by Kim and Mnih [60]. The physical energy hierarchy of digital CMOS gates, arithmetic units, and SRAM/DRAM memory movement follows the survey of Horowitz [52]. Subthreshold analog VLSI design principles and weak-inversion noise-power-speed trade-offs build on Vittoz and Fellrath [116], Mead [75], Sarpeshkar et al. [93], and Enz and Vittoz [29], while the stochastic thermodynamics of nonlinear subthreshold circuits and NEAT-RNs follows Freitas et al. [37] and Freitas et al. [38].

Chapter 18

Differentiable Stochastic Physics, Equilibrium Propagation, and Hardware Calibration

The previous seventeen chapters built a thermodynamic computer from the bottom up. We derived the canonical ensemble and the contrastive learning rule for energy-based models (Chapter 1), the stochastic calculus of noisy circuits (Chapter 4), the Fokker–Planck description of their relaxation (Chapter 6), the device physics of subthreshold transistors (Chapter 12), the non-equilibrium Gaussian samplers called NEAT-RNs (Chapter 13), the denoising thermodynamic models (DTMs) that keep every sampling step well mixed (Chapter 16), and an architecture that places of order 10^5 stochastic units on a single die (Chapter 17). Throughout, we assumed that the parameters θ we wrote down were the parameters the silicon actually realized. This chapter drops that assumption and asks two related questions. First, how does one compute the gradient of a loss with respect to the parameters of a stochastic physical system? Second, what happens to learning when the physical system is not the one we designed, because every transistor on the die is slightly different from its neighbors?

The two questions have a common answer. There are three families of gradient estimators for a stochastic system, and they differ in what they require us to know. The *pathwise* (reparameterization) estimator needs the noise realization that drove a trajectory and the Jacobian of the drift along that trajectory. The *score-function* (likelihood-ratio, or Girsanov) estimator needs the drift derivative integrated against a time-resolved record of the noise. The *contrastive* estimator, built on the static fluctuation–response theorem of Theorem 1.13, needs only stationary moments of known functions of the state, measured at one or two parameter settings. A physical chip can supply only the last kind of information: we can read its states, but we cannot read the thermal noise in its channels, and we do not know its drift exactly. This fact explains why equilibrium propagation is a natural learning rule for physical hardware, and why contrastive training on the hardware itself cancels fabrication errors that offline training cannot anticipate.

The first half of the chapter (Sections 18.1 to 18.3) compares the three estimators, develops equilibrium propagation, and extends the contrastive approach to non-equilibrium Gaussian samplers. The second half (Sections 18.4 and 18.5) estimates how large fabrication mismatch is in subthreshold circuits and shows which part of it training on the chip removes. A worked computation (Section 18.6) then puts numbers on both halves for the book's two running examples: the RC sampler (the coupled linear RC circuit of Example 1.5) and the double-well sampler (the quartic double well and its p-bit limit, Example 1.6). Since this is the last chapter, it ends (Section 18.7) with a short look

back at what those two examples taught.

18.1 Differentiating Through Stochastic Physical Dynamics

Consider a stochastic system whose state $\mathbf{X}_t \in \mathbb{R}^N$ obeys the Itô SDE

$$\mathrm{d}\mathbf{X}_t = \mathbf{a}_\theta(\mathbf{X}_t)\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{X}_0 \sim \rho_0, \tag{18.1}$$

where the drift $\mathbf{a}_\theta$ depends smoothly on a parameter $\theta \in \mathbb{R}$ (for several parameters, apply everything below component by component), the noise matrix $\mathbf{B} \in \mathbb{R}^{N\times N}$ is constant and invertible, and the initial law ρ_0 does not depend on θ. With $\mathbf{a}_\theta(\mathbf{x}) = -\mu\nabla E_\theta(\mathbf{x})$ and $\mathbf{B} = \sqrt{2\mu k_\mathrm{B}T}\,\mathbf{I}$ this is the overdamped Langevin dynamics used throughout the book, with diffusion matrix $\mathbf{D} = \frac{1}{2}\mathbf{B}\mathbf{B}^\top = \mu k_\mathrm{B}T\,\mathbf{I}$ (Section 6.1.2). For a NEAT-RN, $\mathbf{a}_\theta(\mathbf{x}) = \mathbf{A}(\theta)\mathbf{x}$ (Theorem 13.8).

We want the derivative of a performance measure

$$J(\theta) := \mathbb{E}[f(\mathbf{X}_T)] \tag{18.2}$$

for a smooth observable f and a horizon $T > 0$. When T is much longer than the relaxation time, $J(\theta)$ approaches the stationary average $\langle f\rangle_{p_\mathrm{ss}}$, and for an equilibrium system $\langle f\rangle_{p_\theta}$ with $p_\theta \propto e^{-\beta E_\theta}$. There are three standard ways to compute $J'(\theta)$.

18.1.1 The Pathwise (Reparameterization) Estimator

The pathwise estimator treats the noise realization as fixed and differentiates the trajectory. For a strong solution of Eq. (18.1) (Theorem 4.19), $\mathbf{X}_t = \mathbf{X}_t(\theta; \mathbf{X}_0, \mathbf{W}_{[0,t]})$ is a functional of the initial condition and the Brownian path. Since neither of these depends on θ, we may differentiate the integral equation

$$\mathbf{X}_t = \mathbf{X}_0 + \int_0^t \mathbf{a}_\theta(\mathbf{X}_s)\,\mathrm{d}s + \mathbf{B}\mathbf{W}_t$$

with respect to θ at fixed $(\mathbf{X}_0, \mathbf{W})$. The noise term drops out because $\mathbf{B}$ is constant. Writing $\mathbf{Y}_t := \partial\mathbf{X}_t/\partial\theta$ for the *tangent process*, the chain rule gives

$$\frac{\mathrm{d}\mathbf{Y}_t}{\mathrm{d}t} = \nabla_{\mathbf{x}}\mathbf{a}_\theta(\mathbf{X}_t)\,\mathbf{Y}_t + \frac{\partial\mathbf{a}_\theta}{\partial\theta}(\mathbf{X}_t), \qquad \mathbf{Y}_0 = \mathbf{0}. \tag{18.3}$$

This is an ordinary differential equation driven by the random path $\mathbf{X}_t$: the tangent process inherits its randomness from the trajectory, but it has no noise term of its own. If f and $\mathbf{a}_\theta$ are smooth with bounded derivatives, we may exchange differentiation and expectation,

$$J'(\theta) = \mathbb{E}[\nabla f(\mathbf{X}_T)\cdot\mathbf{Y}_T]\,, \qquad \hat{g}_\mathrm{PW} := \nabla f(\mathbf{X}_T)\cdot\mathbf{Y}_T. \tag{18.4}$$

In software this is what automatic differentiation of a simulated Euler–Maruyama trajectory computes. Reverse-mode differentiation solves the adjoint of Eq. (18.3) backward in time and needs either the stored trajectory or a way to reconstruct it, so its memory grows with $T/\Delta t$.

18.1.2 The Score-Function (Girsanov) Estimator

The score-function estimator differentiates the probability of each path instead of the path itself. Let $\mathbb{P}_\theta$ be the law of the trajectory $\mathbf{X}_{[0,T]}$ under parameter θ. Fix a reference value θ_0 and consider $\theta = \theta_0 + \epsilon$. The two drifts differ by

$$\mathbf{a}_\theta(\mathbf{x}) - \mathbf{a}_{\theta_0}(\mathbf{x}) = \mathbf{B}\,\mathbf{u}_\epsilon(\mathbf{x}), \qquad \mathbf{u}_\epsilon(\mathbf{x}) := \mathbf{B}^{-1}\big(\mathbf{a}_{\theta_0+\epsilon}(\mathbf{x}) - \mathbf{a}_{\theta_0}(\mathbf{x})\big),$$

	Pathwise	Score-function	Contrastive
Needs noise realization	yes	yes (increments)	no
Needs drift model	Jacobian $\nabla_{\mathbf{x}}\mathbf{a}$, $\partial_\theta \mathbf{a}$	$\partial_\theta \mathbf{a}$, $\mathbf{B}^{-1}$	$\partial_\theta E_\theta(\mathbf{x})$ only
Needs smooth f	yes	no	no
Variance as $T \to \infty$	bounded	grows $\propto T$	bounded per sample
Available on a physical chip	no	no	yes

Table 18.1: Requirements of the three gradient estimators for a stochastic system. The last row is the reason the remainder of the chapter uses the contrastive estimator for hardware.

which is exactly the drift shift handled by Girsanov's theorem (Theorem 4.22). Applying Eq. (4.100) with $\mathbf{u}_s = \mathbf{u}_\epsilon(\mathbf{X}_s)$, and assuming Novikov's condition, the Radon–Nikodym derivative of the two path measures is

$$\frac{\mathrm{d}\mathbb{P}_{\theta_0+\epsilon}}{\mathrm{d}\mathbb{P}_{\theta_0}} = \exp\left(\int_0^T \mathbf{u}_\epsilon(\mathbf{X}_s) \cdot \mathrm{d}\mathbf{W}_s - \frac{1}{2}\int_0^T \|\mathbf{u}_\epsilon(\mathbf{X}_s)\|^2 \, \mathrm{d}s \right), \tag{18.5}$$

where $\mathbf{W}$ is the Brownian motion that drives $\mathbf{X}$ under $\mathbb{P}_{\theta_0}$, recoverable from the path as $\mathrm{d}\mathbf{W}_s = \mathbf{B}^{-1}(\mathrm{d}\mathbf{X}_s - \mathbf{a}_{\theta_0}(\mathbf{X}_s)\,\mathrm{d}s)$. Then $J(\theta_0+\epsilon) = \mathbb{E}_{\theta_0}[f(\mathbf{X}_T)\,\mathrm{d}\mathbb{P}_{\theta_0+\epsilon}/\,\mathrm{d}\mathbb{P}_{\theta_0}]$. Since $\mathbf{u}_\epsilon = \epsilon\,\mathbf{B}^{-1}\partial_\theta \mathbf{a}_{\theta_0} + \mathcal{O}(\epsilon^2)$, the exponent is

$$\epsilon \int_0^T (\mathbf{B}^{-1}\partial_\theta \mathbf{a}_{\theta_0}(\mathbf{X}_s)) \cdot \mathrm{d}\mathbf{W}_s + \mathcal{O}(\epsilon^2),$$

where the quadratic term $-\frac{1}{2}\int \|\mathbf{u}_\epsilon\|^2 \,\mathrm{d}s$ is already $\mathcal{O}(\epsilon^2)$. Differentiating at $\epsilon = 0$ gives

$$J'(\theta_0) = \mathbb{E}[f(\mathbf{X}_T)\,\mathcal{S}_T], \qquad \mathcal{S}_T := \int_0^T (\mathbf{B}^{-1}\partial_\theta \mathbf{a}_{\theta_0}(\mathbf{X}_s)) \cdot \mathrm{d}\mathbf{W}_s, \qquad \hat{g}_{\mathrm{SF}} := f(\mathbf{X}_T)\,\mathcal{S}_T. \tag{18.6}$$

The *path score* $\mathcal{S}_T = \partial_\theta \ln \mathbb{P}_\theta[\mathbf{X}_{[0,T]}]$ is an Itô integral, so it has zero mean. This allows a baseline: $(f(\mathbf{X}_T) - c)\mathcal{S}_T$ is unbiased for every constant c. The estimator requires no derivative of f, so it applies to discontinuous observables such as threshold crossings or the sign of a p-bit. It does require $\partial_\theta \mathbf{a}_\theta$ along the trajectory and the noise increments $\mathrm{d}\mathbf{W}_s$, which can be reconstructed only if the drift and noise amplitude are known. The general integration-by-parts formula on path space of which Eq. (18.6) is the simplest case is the subject of Malliavin calculus; we do not need it here.

18.1.3 The Contrastive (Fluctuation–Response) Estimator

When T is long and the dynamics is reversible with stationary law $p_\theta \propto e^{-\beta E_\theta}$, $J(\theta) \to \langle f \rangle_{p_\theta}$, and the static fluctuation–response theorem (Eq. (1.82)) gives the derivative directly:

$$\frac{\partial}{\partial \theta}\langle f \rangle_{p_\theta} = -\beta\, \mathrm{Cov}_{p_\theta}\left(f,\, \frac{\partial E_\theta}{\partial \theta} \right). \tag{18.7}$$

No trajectory derivative and no noise record appear. We need samples from p_θ, the values of f on them, and the values of the generalized force $\partial_\theta E_\theta$, which for a linearly parameterized model is a known sufficient statistic $\phi_a(\mathbf{x})$ (Definition 1.12). The gradient of the log-likelihood (Theorem 1.16) is the special case in which the observable is itself the energy evaluated on data. Theorem 17.4 already displayed score-function, reparameterization, and energy-covariance forms of one gradient; the present section shows that this three-way choice is generic.

Remark 18.1 (State-dependent noise)**.** We assumed a constant noise matrix $\mathbf{B}$. In subthreshold circuits the shot-noise intensity follows the bias currents, so $\mathbf{B} = \mathbf{B}_\theta(\mathbf{x})$ in general (Section 4.4.2). Both estimators survive with changes. The tangent process acquires a noise term: differentiating the Itô integral equation at fixed $\mathbf{W}$ gives

$$\mathrm{d}\mathbf{Y}_t = (\nabla_{\mathbf{x}}\mathbf{a}_\theta\,\mathbf{Y}_t + \partial_\theta \mathbf{a}_\theta)\,\mathrm{d}t + \sum_m (\nabla_{\mathbf{x}}\mathbf{b}_m\,\mathbf{Y}_t + \partial_\theta \mathbf{b}_m)\,\mathrm{d}W_{m,t},$$

where $\mathbf{b}_m$ is the mth column of $\mathbf{B}_\theta$, so $\mathbf{Y}$ is now a linear SDE rather than an ODE. The score-function estimator applies only to parameters that enter the drift: if θ changes the noise amplitude itself, the path measures at different θ are mutually singular (their quadratic variations differ), and no Radon–Nikodym derivative exists. Parameters that set noise intensities, such as the bias currents that control $\mathbf{D}$ in a NEAT-RN, can therefore be differentiated pathwise or through stationary statistics (Section 18.3), but not with a Girsanov weight.

18.1.4 All Three Estimators for the Ornstein–Uhlenbeck Circuit

We now compute the three estimators exactly for the simplest version of the RC sampler: a single capacitor node with energy $E_k(x) = \frac{1}{2}kx^2$ and stiffness parameter $\theta = k$. The overdamped Langevin equation is

$$\mathrm{d}X_t = -\mu k X_t\,\mathrm{d}t + \sqrt{2\mu k_{\mathrm{B}}T}\,\mathrm{d}W_t, \qquad X_0 = x_0, \tag{18.8}$$

with relaxation time $\tau := (\mu k)^{-1}$. In the *RC* language of Example 1.5, the stiffness is the capacitance, $k = C$ (the node energy is $\frac{1}{2}Cx^2$), and $\mu k = G/C$ is the inverse *RC* time. The observable is $f(x) = x^2$, so $J(k) = \mathbb{E}[X_T^2]$.

Exact target. Solving the linear SDE by the integrating factor $e^{\mu k t}$ (as in Example 4.14),

$$X_T = x_0 e^{-T/\tau} + \sqrt{2\mu k_{\mathrm{B}}T}\int_0^T e^{-(T-s)/\tau}\,\mathrm{d}W_s. \tag{18.9}$$

By the Itô isometry (Theorem 4.8),

$$J(k) = x_0^2 e^{-2\mu k T} + 2\mu k_{\mathrm{B}}T\int_0^T e^{-2\mu k u}\,\mathrm{d}u = x_0^2 e^{-2\mu k T} + \frac{k_{\mathrm{B}}T}{k}(1 - e^{-2\mu k T}). \tag{18.10}$$

Differentiating with respect to k,

$$J'(k) = -2\mu T x_0^2 e^{-2\mu k T} - \frac{k_{\mathrm{B}}T}{k^2}(1 - e^{-2\mu k T}) + \frac{2\mu T k_{\mathrm{B}}T}{k}e^{-2\mu k T} \xrightarrow{T\to\infty} -\frac{k_{\mathrm{B}}T}{k^2}. \tag{18.11}$$

The limit is the derivative of the equipartition value $\langle x^2\rangle = k_{\mathrm{B}}T/k$.

Pathwise. Here $\partial_x a = -\mu k$ and $\partial_k a = -\mu x$, so Eq. (18.3) reads

$$\frac{\mathrm{d}Y_t}{\mathrm{d}t} = -\mu k\,Y_t - \mu X_t, \qquad Y_0 = 0.$$

Variation of constants gives $Y_T = -\mu\int_0^T e^{-\mu k(T-t)}X_t\,\mathrm{d}t$. Substituting Eq. (18.9) for X_t and exchanging the order of the time integral and the Itô integral,

$$Y_T = -\mu x_0\int_0^T e^{-\mu k(T-t)}e^{-\mu k t}\,\mathrm{d}t - \mu\sqrt{2\mu k_{\mathrm{B}}T}\int_0^T\int_s^T e^{-\mu k(T-t)}e^{-\mu k(t-s)}\,\mathrm{d}t\,\mathrm{d}W_s$$

$$= -\mu T x_0 e^{-\mu k T} - \mu\sqrt{2\mu k_{\mathrm{B}} T}\int_0^T (T-s)\, e^{-\mu k(T-s)}\, \mathrm{d}W_s. \tag{18.12}$$

This is the derivative of Eq. (18.9) with respect to k at fixed noise path, as it must be. The estimator is $\hat{g}_{\mathrm{PW}} = 2X_T Y_T$. Its mean follows from the isometry with $u = T - s$:

$$\mathbb{E}[2X_T Y_T] = -2\mu T x_0^2 e^{-2\mu k T} - 4\mu^2 k_{\mathrm{B}} T \int_0^T u\, e^{-2\mu k u}\, \mathrm{d}u.$$

Using $\int_0^T u e^{-cu}\,\mathrm{d}u = c^{-2}[1 - e^{-cT}(1+cT)]$ with $c = 2\mu k$,

$$4\mu^2 k_{\mathrm{B}} T \cdot \frac{1 - e^{-2\mu k T}(1 + 2\mu k T)}{4\mu^2 k^2} = \frac{k_{\mathrm{B}} T}{k^2}(1 - e^{-2\mu k T}) - \frac{2\mu T k_{\mathrm{B}} T}{k} e^{-2\mu k T},$$

which reproduces Eq. (18.11) term by term.

For the variance, take $T \gg \tau$ so the x_0 terms are negligible. Then X_T and Y_T are jointly Gaussian with zero mean and, again by the isometry,

$$\begin{aligned}
\mathrm{Var}(X_T) &= 2\mu k_{\mathrm{B}} T \int_0^\infty e^{-2\mu k u}\,\mathrm{d}u = \frac{k_{\mathrm{B}} T}{k}, \\
\mathrm{Var}(Y_T) &= 2\mu^3 k_{\mathrm{B}} T \int_0^\infty u^2 e^{-2\mu k u}\,\mathrm{d}u = 2\mu^3 k_{\mathrm{B}} T \cdot \frac{2}{(2\mu k)^3} = \frac{k_{\mathrm{B}} T}{2k^3}, \\
\mathrm{Cov}(X_T, Y_T) &= -2\mu^2 k_{\mathrm{B}} T \int_0^\infty u\, e^{-2\mu k u}\,\mathrm{d}u = -\frac{k_{\mathrm{B}} T}{2k^2}.
\end{aligned}$$

For zero-mean jointly Gaussian variables, Isserlis' theorem (Chapter B) gives $\mathbb{E}[X^2 Y^2] = \mathrm{Var}\,X\ \mathrm{Var}\,Y + 2\,\mathrm{Cov}(X,Y)^2$, so $\mathrm{Var}(XY) = \mathrm{Var}\,X\ \mathrm{Var}\,Y + \mathrm{Cov}(X,Y)^2$. Therefore

$$\mathrm{Var}(\hat{g}_{\mathrm{PW}}) = 4\left[\frac{k_{\mathrm{B}} T}{k}\cdot\frac{k_{\mathrm{B}} T}{2k^3} + \frac{(k_{\mathrm{B}} T)^2}{4k^4}\right] = \frac{3(k_{\mathrm{B}} T)^2}{k^4} = 3\,J'(\infty)^2. \tag{18.13}$$

The relative variance of a single pathwise sample is 3, independent of the horizon.

Score function. With $B = \sqrt{2\mu k_{\mathrm{B}} T}$ and $\partial_k a = -\mu x$, the path score Eq. (18.6) is

$$\mathcal{S}_T = -\frac{\mu}{\sqrt{2\mu k_{\mathrm{B}} T}}\int_0^T X_s\ \mathrm{d}W_s, \qquad \mathbb{E}\big[\mathcal{S}_T^2\big] = \frac{\mu^2}{2\mu k_{\mathrm{B}} T}\int_0^T \mathbb{E}\big[X_s^2\big]\,\mathrm{d}s \simeq \frac{\mu}{2k_{\mathrm{B}} T}\cdot\frac{k_{\mathrm{B}} T}{k}\,T = \frac{T}{2\tau k^2}. \tag{18.14}$$

In units of k^{-2}, the score accumulates one unit of variance for every two relaxation times, because each stretch of the trajectory contributes independent information about k. The observable $f(X_T) = X_T^2$ is correlated with only the last few relaxation times of the noise. Contributions to $\mathcal{S}_T$ from times s with $T - s \gg \tau$ are therefore asymptotically independent of X_T, and to leading order in T/τ,

$$\mathrm{Var}(\hat{g}_{\mathrm{SF}}) \simeq \mathbb{E}\big[X_T^4\big]\ \mathbb{E}\big[\mathcal{S}_T^2\big] = 3\left(\frac{k_{\mathrm{B}} T}{k}\right)^2 \frac{T}{2\tau k^2} = \frac{3T}{2\tau}\,\frac{(k_{\mathrm{B}} T)^2}{k^4} = \frac{3T}{2\tau}\,J'(\infty)^2. \tag{18.15}$$

With the optimal constant baseline $c = \langle X_T^2\rangle$, $\mathbb{E}[X_T^4]$ is replaced by $\mathrm{Var}(X_T^2) = 2(k_{\mathrm{B}} T/k)^2$ and the prefactor $\frac{3}{2}$ becomes 1, but the growth with T remains. A Monte Carlo check (40 000 Euler–Maruyama trajectories started from the stationary law, $\mu k = 1$, horizons T/τ from 2 to 16) gives a variance that grows with slope close to $\frac{3}{2}J'(\infty)^2$ per relaxation time, in agreement with Eq. (18.15), on top of a T-independent offset of about $20\,J'(\infty)^2$ that comes from the cross terms in the final relaxation window and that Eq. (18.15) does not capture. The pathwise estimator in the same runs has variance $3J'(\infty)^2$ at every horizon, as Eq. (18.13) predicts.

Contrastive. Here $\partial_k E = \frac{1}{2}x^2$, so Eq. (18.7) gives

$$\frac{\partial}{\partial k}\left\langle x^2\right\rangle = -\beta\,\mathrm{Cov}\Big(x^2, \tfrac{1}{2}x^2\Big) = -\frac{\beta}{2}\,\mathrm{Var}(x^2) = -\frac{\beta}{2}\cdot 2\left(\frac{k_\mathrm{B}T}{k}\right)^2 = -\frac{k_\mathrm{B}T}{k^2},$$

in agreement with Eq. (18.11). The estimator is $-\frac{\beta}{2}$ times the sample variance of x^2 over M independent stationary samples. Since $x^2 = (k_\mathrm{B}T/k)\chi_1^2$, whose fourth central moment is $60(k_\mathrm{B}T/k)^4$, the large-M variance of the sample variance is $(60-4)(k_\mathrm{B}T/k)^4/M$, and the relative variance of the estimator is $56/(4M) = 14/M$. Samples spaced by less than τ are correlated; then M is replaced by the effective number $M_\mathrm{eff} \approx T_\mathrm{total}/(2\tau_\mathrm{int})$ with τ_int the integrated autocorrelation time of x^2.

Physics Insight: Why the score-function variance grows

The path score measures how much more likely the observed trajectory is under $k + \mathrm{d}k$ than under k. A long trajectory carries a lot of information about k, and the Fisher information of the path, $\mathbb{E}[\mathcal{S}_T^2] \simeq T/(2\tau k^2)$, grows linearly in time. But $J(k)$ at large T depends only on the stationary law, which forgets the past. The estimator multiplies an observable that remembers only the last few τ by a weight that remembers everything, and the forgotten part contributes pure noise. The pathwise estimator avoids this because the tangent process Y_t also forgets: it obeys a contracting linear ODE, so $\mathrm{Var}(Y_T)$ saturates. The contrastive estimator avoids it by never looking at time at all.

18.1.5 Discrete Variables: Score Functions of Gibbs Chains

The pathwise estimator needs a differentiable map from noise to state. A p-bit has none: its output jumps between -1 and $+1$, and an infinitesimal change of its field changes the output only on a set of noise values of measure zero, so the derivative of each sample is zero almost everywhere while the derivative of the average is not. For discrete samplers, such as the block-Gibbs chains run on p-bit arrays and simulated in software by Extropic's `THRML` library [31], only the score-function and contrastive estimators remain.

The score-function estimator carries over directly. Let a chain $\mathbf{s}^{(0)}, \mathbf{s}^{(1)}, \ldots, \mathbf{s}^{(K)}$ be generated by transition probabilities $T_\theta(\mathbf{s}' \mid \mathbf{s})$ from an initial law that does not depend on θ. The probability of the whole path is the product of transition probabilities, so its score is the sum

$$\mathcal{S}_K = \sum_{t=0}^{K-1} \frac{\partial}{\partial\theta} \ln T_\theta(\mathbf{s}^{(t+1)} \mid \mathbf{s}^{(t)}), \qquad \frac{\partial}{\partial\theta}\mathbb{E}\Big[f(\mathbf{s}^{(K)})\Big] = \mathbb{E}\Big[f(\mathbf{s}^{(K)})\,\mathcal{S}_K\Big]. \tag{18.16}$$

For a single-site Gibbs update of spin i in the Ising energy $E = -\sum_{i<j} J_{ij}s_is_j - \sum_i h_is_i$, the new value s_i' is drawn with probability $e^{\beta h_i^\mathrm{loc} s_i'}/(2\cosh\beta h_i^\mathrm{loc})$, where $h_i^\mathrm{loc} = h_i + \sum_j J_{ij}s_j$ is the local field. Differentiating the logarithm,

$$\frac{\partial}{\partial h_i^\mathrm{loc}} \ln \frac{e^{\beta h_i^\mathrm{loc} s_i'}}{2\cosh\beta h_i^\mathrm{loc}} = \beta(s_i' - \tanh\beta h_i^\mathrm{loc}), \tag{18.17}$$

so a coupling J_{ij} contributes $\beta(s_i' - \tanh\beta h_i^\mathrm{loc})s_j$ each time spin i is updated (and symmetrically when j is updated). Each term is the difference between the spin that was drawn and its conditional mean, multiplied by a neighbor's value: a local, zero-mean quantity that a p-bit cell could in principle accumulate. Its variance, however, grows linearly with K for the same reason as in the continuous case, and for long chains the contrastive estimator, which uses only the final stationary moments $\langle s_is_j\rangle$, is again preferable.

18.1.6 Memory and Locality

The estimators also differ in what must be stored. Reverse-mode differentiation of a simulated trajectory of N units over K time steps stores, or recomputes, all NK intermediate states, and then propagates an adjoint vector backward through the same K steps. For a sampler whose useful output is a stationary average, K must exceed the mixing time, and the memory grows with it. The contrastive estimator stores, for each parameter, one running average of its sufficient statistic per phase: $\mathcal{O}(P)$ memory independent of K, and each accumulator needs only signals that are already present at the corresponding synapse. Equilibrium propagation (Section 18.2) keeps these properties for supervised costs. This locality is what allows learning to be distributed over a physical array without a global backward pass.

Hardware Realization: Which estimator a thermodynamic chip can use

On a physical chip the noise realization $\mathbf{W}_t$ is the thermal and shot noise of the transistor channels (Sections 11.2 and 11.3). It is not recorded and cannot be replayed, so the pathwise estimator, which needs the same noise path at θ and at $\theta + \mathrm{d}\theta$, is available only in simulation. The score-function estimator could in principle reconstruct $\mathrm{d}\mathbf{W}$ from a finely sampled voltage record, but only through an accurate model of the drift $\mathbf{a}_\theta$ and noise amplitude $\mathbf{B}$, and these are exactly the quantities that fabrication mismatch perturbs (Section 18.4). The contrastive estimator needs only samples of the state and the known sufficient statistics $\phi_a(\mathbf{x})$, such as s_i and $s_i s_j$ for a p-bit array. Software such as `THRML` simulates block-Gibbs sampling of the same factor-graph models, so the moments that enter a contrastive gradient can be estimated either in simulation or on the chip; the analysis of Section 18.5 explains why the second option is preferable once the chip exists.

18.2 Equilibrium Propagation in Stochastic Circuits

The contrastive gradient of Theorem 1.16 trains a generative model: it makes p_θ match a data distribution. Many tasks are instead supervised. A subset of the circuit's variables is designated as output, and we want those outputs, sampled from the equilibrium distribution, to have low expected cost $\mathcal{C}(\mathbf{x})$, for example $\mathcal{C}(\mathbf{x}) = \frac{1}{2}\,\|\mathbf{x}_{\text{out}} - \mathbf{y}\|^2$ for a target $\mathbf{y}$. The objective is

$$J(\theta) := \langle \mathcal{C} \rangle_{p_\theta} = \int \mathcal{C}(\mathbf{x})\, p_\theta(\mathbf{x})\, \mathrm{d}\mathbf{x}, \qquad p_\theta(\mathbf{x}) = \frac{e^{-\beta E_\theta(\mathbf{x})}}{Z(\theta, \beta)}. \tag{18.18}$$

The covariance formula Eq. (18.7) with $f = \mathcal{C}$ gives $\partial_\theta J = -\beta\,\mathrm{Cov}(\mathcal{C}, \partial_\theta E_\theta)$. In a circuit this is awkward for a structural reason. For a coupling $\theta = J_{ij}$ the generalized force $\partial_\theta E_\theta = -s_i s_j$ is local to one synapse, while $\mathcal{C}$ is a global quantity defined at the output nodes. Estimating their covariance requires broadcasting the instantaneous cost to every synapse for every sample. Equilibrium propagation, introduced by Scellier and Bengio [94], avoids this broadcast by letting the physics carry the cost signal: we add a weak cost term to the energy, let the circuit re-equilibrate, and measure how each local force shifts.

Definition 18.2 (Nudged ensemble)**.** For a nudge strength $\beta_{\text{nudge}} \in \mathbb{R}$, the *nudged energy* and *nudged ensemble* are

$$E_{\text{total}}(\mathbf{x}) := E_\theta(\mathbf{x}) + \beta_{\text{nudge}}\,\mathcal{C}(\mathbf{x}), \qquad p_{\theta,\beta_{\text{nudge}}}(\mathbf{x}) := \frac{e^{-\beta[E_\theta(\mathbf{x}) + \beta_{\text{nudge}}\mathcal{C}(\mathbf{x})]}}{Z(\theta, \beta_{\text{nudge}})}, \tag{18.19}$$

and $\langle\cdot\rangle_{\beta_{\text{nudge}}}$ denotes the average under $p_{\theta,\beta_{\text{nudge}}}$. The case $\beta_{\text{nudge}} = 0$ is the *free phase* and $\beta_{\text{nudge}} \neq 0$ the *nudged phase*. The nudged free energy is $F(\theta, \beta_{\text{nudge}}) := -\beta^{-1} \ln Z(\theta, \beta_{\text{nudge}})$.

Notation Watch: Two betas

The machine-learning literature on equilibrium propagation writes the nudge strength as β, which is the symbol this book reserves for the inverse temperature $\beta = 1/k_{\text{B}}T$. We write the nudge as β_{nudge} throughout. The two enter differently: β multiplies the whole energy, while β_{nudge} multiplies only the cost. A nudge of strength β_{nudge} has a thermodynamically small effect when $\beta\beta_{\text{nudge}}\,\mathrm{Var}(\mathcal{C})^{1/2} \ll 1$; this dimensionless combination, not β_{nudge} alone, controls the bias analysis below.

18.2.1 The Finite-Temperature Theorem as a Maxwell Relation

Theorem 18.3 (Equilibrium propagation at finite temperature)**.** *Let E_θ and $\mathcal{C}$ satisfy the differentiability and integrability hypotheses of Theorem 1.13 jointly in $(\theta, \beta_{\text{nudge}})$ on an open set containing $\beta_{\text{nudge}} = 0$. Then for every β_{nudge} in that set,*

$$\frac{\partial}{\partial \beta_{\text{nudge}}} \left\langle \frac{\partial E_\theta}{\partial \theta} \right\rangle_{\beta_{\text{nudge}}} = \frac{\partial}{\partial \theta} \langle \mathcal{C} \rangle_{\beta_{\text{nudge}}}, \tag{18.20}$$

and at $\beta_{\text{nudge}} = 0$ both sides equal $-\beta\,\mathrm{Cov}_{p_\theta}(\partial_\theta E_\theta, \mathcal{C})$. Consequently

$$\lim_{\beta_{\text{nudge}} \to 0} \frac{\left\langle \frac{\partial E_\theta}{\partial \theta} \right\rangle_{\beta_{\text{nudge}}} - \left\langle \frac{\partial E_\theta}{\partial \theta} \right\rangle_0}{\beta_{\text{nudge}}} = -\beta\ \mathrm{Cov}_{p_\theta}\left(\frac{\partial E_\theta}{\partial \theta}, \mathcal{C} \right) = \frac{\partial}{\partial \theta} \langle \mathcal{C} \rangle_{p_\theta} = J'(\theta). \tag{18.21}$$

Proof. The nudged ensemble is a canonical ensemble for the energy E_{total}, which depends on two parameters, θ and β_{nudge}. Their generalized forces are

$$\frac{\partial E_{\text{total}}}{\partial \theta} = \frac{\partial E_\theta}{\partial \theta}, \qquad \frac{\partial E_{\text{total}}}{\partial \beta_{\text{nudge}}} = \mathcal{C}(\mathbf{x}).$$

By the first-derivative identity Eq. (1.79) applied to each parameter in turn,

$$\frac{\partial F}{\partial \theta}(\theta, \beta_{\text{nudge}}) = \left\langle \frac{\partial E_\theta}{\partial \theta} \right\rangle_{\beta_{\text{nudge}}}, \qquad \frac{\partial F}{\partial \beta_{\text{nudge}}}(\theta, \beta_{\text{nudge}}) = \langle \mathcal{C} \rangle_{\beta_{\text{nudge}}}. \tag{18.22}$$

Under the stated hypotheses F is twice continuously differentiable, so its mixed partial derivatives are equal (Schwarz's theorem):

$$\frac{\partial}{\partial \beta_{\text{nudge}}} \frac{\partial F}{\partial \theta} = \frac{\partial}{\partial \theta} \frac{\partial F}{\partial \beta_{\text{nudge}}}.$$

Substituting Eq. (18.22) on both sides gives Eq. (18.20).

To evaluate either side, use the observable-response identity Eq. (1.82). On the right-hand side, $\mathcal{C}$ does not depend on θ, so at $\beta_{\text{nudge}} = 0$

$$\frac{\partial}{\partial \theta} \langle \mathcal{C} \rangle_0 = -\beta\,\mathrm{Cov}_{p_\theta}\left(\mathcal{C}, \frac{\partial E_\theta}{\partial \theta} \right).$$

On the left-hand side, the observable $A(\mathbf{x}) := \partial_\theta E_\theta(\mathbf{x})$ does not depend on β_{nudge}, and the generalized force conjugate to β_{nudge} is $\mathcal{C}$, so

$$\frac{\partial}{\partial \beta_{\text{nudge}}} \langle A \rangle_{\beta_{\text{nudge}}} = -\beta\,\mathrm{Cov}_{\beta_{\text{nudge}}}(A, \mathcal{C}),$$

which at $\beta_{\text{nudge}} = 0$ is $-\beta \operatorname{Cov}_{p_\theta}(\partial_\theta E_\theta, \mathcal{C})$. The two expressions agree, as they must. Finally, the difference quotient in Eq. (18.21) converges to the β_{nudge}-derivative of $\langle A \rangle_{\beta_{\text{nudge}}}$ at 0 by the definition of the derivative. □

The theorem converts a question about the cost into a question about local forces. In the free phase every synapse records the time average of its own generalized force, $\langle \partial_\theta E_\theta \rangle_0$. The cost is then switched on weakly at the output nodes only, the circuit relaxes, and every synapse records the same local average again. The shift in each synapse's local average, divided by β_{nudge}, is the gradient of the global objective with respect to that synapse's parameter. The cost information travels from the outputs into the interior through the physical couplings during the relaxation of the nudged phase, which is why no separate backward pass is needed and no intermediate states need to be stored.

Example 18.4 (One Gaussian node, exact at every nudge)**.** Take $E_h(x) = \frac{1}{2}kx^2 - hx$ with trainable bias $\theta = h$, and cost $\mathcal{C}(x) = \frac{1}{2}(x-y)^2$. In the free phase $x \sim \mathcal{N}(h/k, k_{\text{B}}T/k)$, so

$$J(h) = \langle \mathcal{C} \rangle_0 = \frac{1}{2}\left[\left(\frac{h}{k} - y\right)^2 + \frac{k_{\text{B}}T}{k}\right], \qquad J'(h) = \frac{1}{k}\left(\frac{h}{k} - y\right).$$

The nudged energy is $\frac{1}{2}(k+\beta_{\text{nudge}})x^2 - (h+\beta_{\text{nudge}}y)x + \text{const}$, so $\langle x \rangle_{\beta_{\text{nudge}}} = (h+\beta_{\text{nudge}}y)/(k+\beta_{\text{nudge}})$. The generalized force is $\partial_h E = -x$, and

$$\langle \partial_h E \rangle_{\beta_{\text{nudge}}} - \langle \partial_h E \rangle_0 = \frac{h}{k} - \frac{h+\beta_{\text{nudge}}y}{k+\beta_{\text{nudge}}} = \frac{h(k+\beta_{\text{nudge}}) - k(h+\beta_{\text{nudge}}y)}{k(k+\beta_{\text{nudge}})} = \frac{\beta_{\text{nudge}}(h-ky)}{k(k+\beta_{\text{nudge}})}.$$

Dividing by β_{nudge},

$$g_+(\beta_{\text{nudge}}) := \frac{\langle \partial_h E \rangle_{\beta_{\text{nudge}}} - \langle \partial_h E \rangle_0}{\beta_{\text{nudge}}} = \frac{1}{k+\beta_{\text{nudge}}}\left(\frac{h}{k} - y\right) = \frac{k}{k+\beta_{\text{nudge}}} J'(h), \tag{18.23}$$

which tends to $J'(h)$ as $\beta_{\text{nudge}} \to 0$, with relative error $-\beta_{\text{nudge}}/k + \mathcal{O}(\beta_{\text{nudge}}^2)$. The symmetric quotient $[\langle \partial_h E \rangle_{\beta_{\text{nudge}}} - \langle \partial_h E \rangle_{-\beta_{\text{nudge}}}]/(2\beta_{\text{nudge}})$ equals $\frac{1}{2}[g_+(\beta_{\text{nudge}}) + g_+(-\beta_{\text{nudge}})] = \frac{k^2}{k^2 - \beta_{\text{nudge}}^2} J'(h)$, with relative error $+\beta_{\text{nudge}}^2/k^2 + \mathcal{O}(\beta_{\text{nudge}}^4)$. Temperature does not appear because the mean of a Gaussian does not depend on its variance.

Example 18.5 (The error signal reaching a hidden node)**.** The previous example nudged the very node whose bias was being trained. To see the cost information travel, take two coupled Gaussian nodes, a hidden node x_1 with trainable bias h and an output node x_2, with

$$E_\theta(\mathbf{x}) = \tfrac{1}{2}k_1 x_1^2 + \tfrac{1}{2}k_2 x_2^2 - w x_1 x_2 - h x_1, \qquad \mathcal{C}(\mathbf{x}) = \tfrac{1}{2}(x_2 - y)^2,$$

and $\Delta := k_1 k_2 - w^2 > 0$. The free-phase mean solves $\mathbf{K}\boldsymbol{\mu} = (h, 0)^\top$ with $\mathbf{K} = \left(\begin{smallmatrix} k_1 & -w \\ -w & k_2 \end{smallmatrix}\right)$, so $\mu_1 = hk_2/\Delta$ and $\mu_2 = hw/\Delta$. The variance of x_2 does not depend on h, so $J'(h) = (\mu_2 - y)\,\partial\mu_2/\partial h = (\mu_2 - y)\,w/\Delta$.

In the nudged phase $k_2 \to k_2 + \beta_{\text{nudge}}$ and the output receives the extra bias $\beta_{\text{nudge}}y$, so

$$\langle x_1 \rangle_{\beta_{\text{nudge}}} = \frac{(k_2 + \beta_{\text{nudge}})h + w\beta_{\text{nudge}}y}{k_1(k_2 + \beta_{\text{nudge}}) - w^2}.$$

Differentiating at $\beta_{\text{nudge}} = 0$ with the quotient rule (numerator derivative $h + wy$, denominator derivative k_1),

$$\frac{\partial \langle x_1 \rangle}{\partial \beta_{\text{nudge}}}\Big|_0 = \frac{(h+wy)\Delta - k_2 h\, k_1}{\Delta^2} = \frac{w(y\Delta - hw)}{\Delta^2} = \frac{w}{\Delta}(y - \mu_2),$$

using $hw = \mu_2\Delta$. Since $\partial_h E_\theta = -x_1$, the equilibrium-propagation estimate is $-\partial_{\beta_{\text{nudge}}} \langle x_1 \rangle = (w/\Delta)(\mu_2 - y) = J'(h)$. The output error $\mu_2 - y$ reaches the hidden node multiplied by $w/\Delta = (\mathbf{K}^{-1})_{12}$, the static susceptibility linking the two nodes. In a deeper network the same factor becomes the corresponding element of $\mathbf{K}^{-1}$, which the physical relaxation of the nudged phase computes without anyone inverting $\mathbf{K}$.

We checked Eq. (18.21) numerically on a two-node Gaussian with a trainable off-diagonal coupling, comparing a central finite difference of J against the symmetric nudge quotient; the two agree to ten significant digits. The example above also shows the general pattern of the nudge bias, to which we now turn.

18.2.2 Bias and Variance of Finite Nudges

In practice β_{nudge} is finite and the averages are estimated from finitely many samples. Write $A = \partial_\theta E_\theta$ and $m(\beta_{\text{nudge}}) = \langle A \rangle_{\beta_{\text{nudge}}}$. Repeated use of the observable-response identity gives the derivatives of m as joint cumulants of A with copies of $\mathcal{C}$.

Lemma 18.6 (Nudge derivatives are joint cumulants)**.** *Under the hypotheses of Theorem 18.3, with* $\kappa(\cdot, \dots, \cdot)$ *denoting joint cumulants under* $p_{\theta,\beta_{\text{nudge}}}$*,*

$$m' = -\beta\, \kappa(A, \mathcal{C}), \qquad m'' = \beta^2\, \kappa(A, \mathcal{C}, \mathcal{C}), \qquad m''' = -\beta^3\, \kappa(A, \mathcal{C}, \mathcal{C}, \mathcal{C}). \tag{18.24}$$

Proof. Let $\langle\cdot\rangle$ be the nudged average. For any β_{nudge}-independent observable O, Eq. (1.82) gives $\partial_{\beta_{\text{nudge}}} \langle O \rangle = -\beta(\langle O\mathcal{C} \rangle - \langle O \rangle \langle \mathcal{C} \rangle)$. With $O = A$ this is the first identity, since $\kappa(A, \mathcal{C}) = \mathrm{Cov}(A, \mathcal{C})$. For the second, differentiate $\kappa(A, \mathcal{C}) = \langle A\mathcal{C} \rangle - \langle A \rangle \langle \mathcal{C} \rangle$, applying the same rule to each of the three averages $\langle A\mathcal{C} \rangle$, $\langle A \rangle$, $\langle \mathcal{C} \rangle$:

$$\begin{aligned} \frac{\partial}{\partial \beta_{\text{nudge}}} \kappa(A, \mathcal{C}) &= -\beta\big[\big\langle A\mathcal{C}^2 \big\rangle - \langle A\mathcal{C} \rangle \langle \mathcal{C} \rangle\big] + \beta\big[\langle A\mathcal{C} \rangle - \langle A \rangle \langle \mathcal{C} \rangle\big] \langle \mathcal{C} \rangle + \beta \langle A \rangle \big[\big\langle \mathcal{C}^2 \big\rangle - \langle \mathcal{C} \rangle^2\big] \\ &= -\beta\big[\big\langle A\mathcal{C}^2 \big\rangle - 2 \langle A\mathcal{C} \rangle \langle \mathcal{C} \rangle - \langle A \rangle \big\langle \mathcal{C}^2 \big\rangle + 2 \langle A \rangle \langle \mathcal{C} \rangle^2\big] = -\beta\, \kappa(A, \mathcal{C}, \mathcal{C}), \end{aligned}$$

where the bracket in the second line is the standard moment expansion of the third joint cumulant. Multiplying by $-\beta$ gives m''. The same computation one order higher (or the general rule that $\partial_{\beta_{\text{nudge}}}$ of a joint cumulant appends one argument $\mathcal{C}$ and a factor $-\beta$, which follows from the fact that $-\beta\beta_{\text{nudge}}$ is the conjugate variable of $\mathcal{C}$ in the cumulant generating function $\ln Z$) gives m'''. □

By Taylor's theorem with all cumulants evaluated in the free phase, the one-sided quotient $g_+ = [m(\beta_{\text{nudge}}) - m(0)]/\beta_{\text{nudge}}$ and the symmetric quotient $g_\pm = [m(\beta_{\text{nudge}}) - m(-\beta_{\text{nudge}})]/(2\beta_{\text{nudge}})$ satisfy

$$g_+ = J'(\theta) + \frac{\beta_{\text{nudge}}}{2}\, \beta^2 \kappa_0(A, \mathcal{C}, \mathcal{C}) + \mathcal{O}(\beta_{\text{nudge}}^2), \tag{18.25}$$

$$g_\pm = J'(\theta) - \frac{\beta_{\text{nudge}}^2}{6}\, \beta^3 \kappa_0(A, \mathcal{C}, \mathcal{C}, \mathcal{C}) + \mathcal{O}(\beta_{\text{nudge}}^4). \tag{18.26}$$

The even terms cancel in the symmetric quotient, which is why its bias is $\mathcal{O}(\beta_{\text{nudge}}^2)$. For Example 18.4 one can verify Eq. (18.25) directly: with $u = x - h/k$ Gaussian of variance $s^2 = k_{\mathrm{B}}T/k$ and $d = h/k - y$, we have $\mathcal{C} = \frac{1}{2}u^2 + du + \frac{1}{2}d^2$ and $A = -x$; multilinearity and the vanishing of odd Gaussian moments give $\kappa(u, u, u^2) = 2s^4$ and $\kappa(u, u^2, u^2) = \kappa(u, u, u) = 0$, so $\kappa_0(A, \mathcal{C}, \mathcal{C}) = -2ds^4$. The predicted bias is $\frac{\beta_{\text{nudge}}}{2}\beta^2(-2ds^4) = -\beta_{\text{nudge}} d/k^2$, which is exactly the first-order term of Eq. (18.23).

Now the variance. Suppose each phase is estimated from M independent samples and the phases are independent. Then, for small β_{nudge},

$$\mathrm{Var}(\hat{g}_+) \approx \frac{2\,\mathrm{Var}_0(A)}{M\beta_{\text{nudge}}^2}, \qquad \mathrm{Var}(\hat{g}_\pm) \approx \frac{2\,\mathrm{Var}_0(A)}{4M\beta_{\text{nudge}}^2} = \frac{\mathrm{Var}_0(A)}{2M\beta_{\text{nudge}}^2}. \tag{18.27}$$

Small nudges reduce bias but amplify noise. Writing $b_1 := \frac{1}{2}\beta^2\kappa_0(A,\mathcal{C},\mathcal{C})$ and $b_2 := \frac{1}{6}\beta^3\kappa_0(A,\mathcal{C},\mathcal{C},\mathcal{C})$, the mean squared errors are

$$\mathrm{MSE}_+ \approx b_1^2\beta_{\text{nudge}}^2 + \frac{2\,\mathrm{Var}_0(A)}{M\beta_{\text{nudge}}^2}, \qquad \mathrm{MSE}_\pm \approx b_2^2\beta_{\text{nudge}}^4 + \frac{\mathrm{Var}_0(A)}{2M\beta_{\text{nudge}}^2}.$$

Setting the derivative with respect to β_{nudge} to zero gives $\beta_{\text{nudge}}^4 = 2\,\mathrm{Var}_0(A)/(b_1^2 M)$ for the one-sided scheme and $\beta_{\text{nudge}}^6 = \mathrm{Var}_0(A)/(4b_2^2 M)$ for the symmetric scheme. Hence the optimal nudge scales as $M^{-1/4}$ (one-sided) or $M^{-1/6}$ (symmetric), and the minimal mean squared error as $M^{-1/2}$ or $M^{-2/3}$ respectively. For the same sampling budget, symmetric nudging gives a strictly better rate. This bias–variance trade-off is the same one that governs finite-difference derivatives of any noisy function.

Heuristic Derivation: Why the physical protocol has lower variance than Eq. (18.27)

The estimate Eq. (18.27) assumes the free and nudged samples are independent. A circuit does something better. It runs the free phase, then switches on the nudge and continues the *same* trajectory. If the nudged phase is read out after a time comparable to the relaxation time τ, the nudged state is the free state plus a small, nearly deterministic displacement of order β_{nudge} plus fresh noise only from the time window τ. The difference $A_{\beta_{\text{nudge}}} - A_0$ of the two readouts then has variance of order β_{nudge}^2 plus the fresh-noise contribution, rather than $2\,\mathrm{Var}_0(A)$. In simulation the same effect is obtained with common random numbers: the free and nudged chains are driven by the same noise sequence. This is the stochastic analogue of the zero-temperature algorithm below, in which the nudged phase starts from the free fixed point.

18.2.3 Equilibrium Propagation on a p-Bit Network

For the double-well sampler in its p-bit limit, the nudge has a simple physical form. Let a subset $\mathcal{O}$ of the p-bits be outputs with target values $y_k \in \{-1,+1\}$, and take the linear cost

$$\mathcal{C}(\mathbf{s}) = -\sum_{k\in\mathcal{O}} y_k s_k, \qquad \langle\mathcal{C}\rangle_{p_\theta} = -\sum_{k\in\mathcal{O}} y_k m_k, \tag{18.28}$$

which rewards outputs that agree with their targets. With the Ising energy $E_\theta = -\sum_{i<j} J_{ij}s_i s_j - \sum_i h_i s_i$, the nudged energy $E_\theta + \beta_{\text{nudge}}\mathcal{C}$ is again an Ising energy, with output fields shifted from h_k to $h_k + \beta_{\text{nudge}} y_k$. On hardware the nudged phase therefore needs no new circuitry: the digital-to-analog converters (DACs) that set the output biases are offset by $\beta_{\text{nudge}} y_k$ and the array is sampled again.

The generalized force of a coupling is $\partial E_\theta/\partial J_{ij} = -s_i s_j$. Theorem 18.3 with the symmetric quotient gives the update for gradient descent on $\langle\mathcal{C}\rangle$,

$$\Delta J_{ij} = -\eta\,\frac{\langle -s_i s_j\rangle_{\beta_{\text{nudge}}} - \langle -s_i s_j\rangle_{-\beta_{\text{nudge}}}}{2\beta_{\text{nudge}}} = \frac{\eta}{2\beta_{\text{nudge}}}\Big(\langle s_i s_j\rangle_{\beta_{\text{nudge}}} - \langle s_i s_j\rangle_{-\beta_{\text{nudge}}}\Big), \tag{18.29}$$

and similarly $\Delta h_i = (\eta/2\beta_{\text{nudge}})(m_{i,\beta_{\text{nudge}}} - m_{i,-\beta_{\text{nudge}}})$. Each synapse needs only the correlation of its own two spins in the two phases, which it can accumulate with a local counter. The rule strengthens couplings whose correlation increases when the outputs are pulled toward their targets, and weakens those whose correlation decreases.

The rule has the same form as the Boltzmann machine learning rule of Ackley et al. [1], $\Delta J_{ij} \propto \langle s_i s_j \rangle_{\text{clamped}} - \langle s_i s_j \rangle_{\text{free}}$, in which the outputs are fixed to their targets rather than nudged. The two differ in what they optimize. Hard clamping corresponds to $\beta_{\text{nudge}} \to \infty$, far outside the linear-response regime of Theorem 18.3; the clamped-minus-free rule is instead β^{-1} times the gradient of the log marginal likelihood $\ln p_\theta(\mathbf{s}_\mathcal{O} = \mathbf{y})$ of the targets (by the argument of Theorem 1.16 applied to the clamped and free ensembles). Equilibrium propagation with a small nudge is the gradient of the expected cost under the free distribution. For a network whose outputs are nearly deterministic both objectives have the same minimizers, but in a genuinely stochastic network they differ, and the choice of β_{nudge} interpolates between them.

Hardware Realization: A nudge budget for a p-bit array

The variance estimate Eq. (18.27) sets the sampling cost of one gradient step. For $A = s_i s_j$, $\mathrm{Var}_0(A) = 1 - \langle s_i s_j \rangle^2 \le 1$. With independent phases, M samples per phase, and symmetric nudging, the gradient estimate for one coupling has standard deviation about $(2M)^{-1/2}/\beta_{\text{nudge}}$. With $\beta_{\text{nudge}} = 0.1$ in natural field units and $M = 10^4$ independent samples, that is 0.07, to be compared with gradients of order $\beta\,|\mathrm{Cov}(s_i s_j, \mathcal{C})|$, which for an interior synapse many couplings away from the outputs can be much smaller. With an update time of order 10 ns per unit, as assumed in Section 17.3, 10^4 samples take of order 100 μs if successive samples are nearly independent, so the cost is set by the number of effectively independent samples, that is, by mixing. The heuristic above (continuing the free trajectory into the nudged phase) reduces the variance further when the nudged readout follows the free readout within one relaxation time.

18.2.4 The Deterministic Limit: Scellier–Bengio at an Energy Minimum

At zero temperature ($\beta \to \infty$) the free phase is a point, the minimizer of E_θ, and the theorem becomes the original deterministic equilibrium propagation of Scellier and Bengio [94].

Theorem 18.7 (Zero-temperature equilibrium propagation)**.** *Let $E_\theta(\mathbf{x})$ and $\mathcal{C}(\mathbf{x})$ be twice continuously differentiable in $(\mathbf{x}, \theta)$, and let $\mathbf{x}_0^*(\theta)$ be a strict local minimizer of E_θ with positive-definite Hessian $\mathbf{H} := \nabla_\mathbf{x}^2 E_\theta(\mathbf{x}_0^*) \succ 0$. For β_{nudge} near 0 let $\mathbf{x}^*_{\beta_{\text{nudge}}}(\theta)$ be the nearby minimizer of $E_\theta + \beta_{\text{nudge}}\mathcal{C}$, and define the loss $J_0(\theta) := \mathcal{C}(\mathbf{x}_0^*(\theta))$. Then*

$$\frac{\partial}{\partial \beta_{\text{nudge}}} \left[\frac{\partial E_\theta}{\partial \theta}(\mathbf{x}^*_{\beta_{\text{nudge}}}) \right]_{\beta_{\text{nudge}}=0} = \frac{\mathrm{d}J_0}{\mathrm{d}\theta} = -(\nabla_\mathbf{x} \partial_\theta E_\theta)^\top \mathbf{H}^{-1} \nabla_\mathbf{x} \mathcal{C}, \tag{18.30}$$

with all quantities evaluated at $\mathbf{x}_0^*$.

Proof. The minimizer satisfies the stationarity condition $\nabla_\mathbf{x} E_\theta(\mathbf{x}^*) + \beta_{\text{nudge}} \nabla_\mathbf{x} \mathcal{C}(\mathbf{x}^*) = \mathbf{0}$. Since $\mathbf{H} \succ 0$, the implicit function theorem guarantees a C^1 solution $\mathbf{x}^*(\theta, \beta_{\text{nudge}})$ near $(\theta, 0)$. Differentiating the stationarity condition with respect to β_{nudge} at $\beta_{\text{nudge}} = 0$ gives $\mathbf{H}\,\partial_{\beta_{\text{nudge}}} \mathbf{x}^* + \nabla_\mathbf{x} \mathcal{C} = \mathbf{0}$, and with respect to θ gives $\mathbf{H}\,\partial_\theta \mathbf{x}^* + \nabla_\mathbf{x} \partial_\theta E_\theta = \mathbf{0}$. Hence

$$\frac{\partial \mathbf{x}^*}{\partial \beta_{\text{nudge}}} = -\mathbf{H}^{-1} \nabla_\mathbf{x} \mathcal{C}, \qquad \frac{\partial \mathbf{x}_0^*}{\partial \theta} = -\mathbf{H}^{-1} \nabla_\mathbf{x} \partial_\theta E_\theta.$$

The left-hand side of Eq. (18.30) is, by the chain rule at fixed θ,

$$(\nabla_{\mathbf{x}}\partial_\theta E_\theta)^\top \frac{\partial \mathbf{x}^*}{\partial \beta_{\text{nudge}}} = -(\nabla_{\mathbf{x}}\partial_\theta E_\theta)^\top \mathbf{H}^{-1}\nabla_{\mathbf{x}}\mathcal{C}.$$

The derivative of the loss is

$$\frac{\mathrm{d}J_0}{\mathrm{d}\theta} = (\nabla_{\mathbf{x}}\mathcal{C})^\top \frac{\partial \mathbf{x}_0^*}{\partial \theta} = -(\nabla_{\mathbf{x}}\mathcal{C})^\top \mathbf{H}^{-1}\nabla_{\mathbf{x}}\partial_\theta E_\theta.$$

The two scalars are equal because $\mathbf{H}^{-1}$ is symmetric. Equivalently, the minimized energy $F_0(\theta, \beta_{\text{nudge}}) := \min_{\mathbf{x}}[E_\theta + \beta_{\text{nudge}}\mathcal{C}]$ has, by the envelope theorem, $\partial_\theta F_0 = \partial_\theta E_\theta(\mathbf{x}^*)$ and $\partial_{\beta_{\text{nudge}}} F_0 = \mathcal{C}(\mathbf{x}^*)$, and equality of its mixed partials is again a Maxwell relation. □

The two theorems are consistent. Near a nondegenerate minimum at low temperature, the Laplace approximation makes p_θ Gaussian with covariance $k_{\mathrm{B}}T\,\mathbf{H}^{-1}$, so $\mathrm{Cov}(A, \mathcal{C}) \approx k_{\mathrm{B}}T\,(\nabla A)^\top \mathbf{H}^{-1}\nabla\mathcal{C}$ and $-\beta\,\mathrm{Cov}(A, \mathcal{C}) \to -(\nabla A)^\top \mathbf{H}^{-1}\nabla\mathcal{C}$, which is Eq. (18.30). The proof also shows where the argument needs equilibrium: the symmetry of $\mathbf{H}$ is what identifies the forward sensitivity $\partial_\theta \mathbf{x}^*$ with the response to the nudge. A circuit whose drift is not a gradient has a non-symmetric Jacobian at its fixed point, and the identity fails. This is the problem we take up next.

18.3 Learning in Non-Equilibrium Steady States

The NEAT-RNs of Chapter 13 break detailed balance on purpose: non-reciprocal transconductances ($\mathbf{G} \neq \mathbf{G}^\top$) create a circulating probability current $\mathbf{J}_{\text{ss}} \neq \mathbf{0}$ that speeds up mixing (Proposition 13.10). The price is that the stationary density, now a non-equilibrium steady state (NESS), is no longer $e^{-\beta E_\theta}/Z$ for any energy built from the circuit parameters, so the static fluctuation–response theorem and equilibrium propagation do not apply as stated. For the linear (Gaussian) case we can still compute exact gradients, because the stationary covariance is the solution of a linear matrix equation.

18.3.1 Why Equilibrium Propagation Needs Reciprocity

The zero-temperature proof of Theorem 18.7 used the symmetry of the Hessian. Its role is clearest in a linear network with a general drift. Let the deterministic part of a NEAT-RN be $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}$ with $\mathbf{A}$ Hurwitz (all eigenvalues in the open left half-plane), so the fixed point (the stationary mean) is $\mathbf{x}^* = -\mathbf{A}^{-1}\mathbf{b}$. Take the cost $\mathcal{C} = \frac{1}{2}(x_o - y)^2$ on an output node o and implement the nudge as the force it would exert in a gradient system, an extra drift $-\beta_{\text{nudge}}(x_o - y)\mathbf{e}_o$.

Proposition 18.8 (Equilibrium propagation measures the transposed response)**.** *In this setting, the nudge-response estimate for the bias b_i and the true gradient of $J_0 = \mathcal{C}(\mathbf{x}^*)$ are*

$$\frac{\partial x_i^*}{\partial \beta_{\text{nudge}}}\Big|_0 = (\mathbf{A}^{-1})_{io}\,(x_o^* - y), \qquad \frac{\partial J_0}{\partial b_i} = -(\mathbf{A}^{-1})_{oi}\,(x_o^* - y). \tag{18.31}$$

The equilibrium-propagation estimate $-\partial_{\beta_{\text{nudge}}} x_i^*$ (the generalized force of b_i in a gradient system is $-x_i$) equals the gradient for all i, o if and only if $\mathbf{A}^{-1}$, equivalently $\mathbf{A}$, is symmetric.

Proof. The nudged fixed point solves $\mathbf{A}\mathbf{x} + \mathbf{b} - \beta_{\text{nudge}}(x_o - y)\mathbf{e}_o = \mathbf{0}$. Differentiating at $\beta_{\text{nudge}} = 0$ gives $\mathbf{A}\,\partial_{\beta_{\text{nudge}}}\mathbf{x}^* = (x_o^* - y)\mathbf{e}_o$, so $\partial_{\beta_{\text{nudge}}}\mathbf{x}^* = (x_o^* - y)\mathbf{A}^{-1}\mathbf{e}_o$, whose ith component is the first formula. For the gradient, $\partial \mathbf{x}^*/\partial b_i = -\mathbf{A}^{-1}\mathbf{e}_i$, so $\partial J_0/\partial b_i = (x_o^* - y)\,\mathbf{e}_o^\top(-\mathbf{A}^{-1}\mathbf{e}_i) = -(\mathbf{A}^{-1})_{oi}(x_o^* - y)$. The two agree for all i, o exactly when $(\mathbf{A}^{-1})_{io} = (\mathbf{A}^{-1})_{oi}$. □

The nudge measures how node i responds to a force applied at the output; the gradient requires how the output responds to a force applied at node i. These are equal when the network obeys reciprocity, the symmetry of response functions that Onsager derived from microscopic reversibility, and they differ in a non-reciprocal circuit. In the stochastic setting the same issue appears as the failure of the static fluctuation–response theorem, which is replaced by Theorem 18.10. For a non-reciprocal network one must either compute the transpose explicitly, which is what the adjoint Lyapunov equation does for covariances, or restrict learning to a parameterization in which the stationary statistics are those of a reciprocal network, as the vorticity gauge below allows.

18.3.2 Differentiating the Lyapunov Equation

Let the network obey the multivariate Ornstein–Uhlenbeck SDE $\mathrm{d}\mathbf{x}_t = \mathbf{A}\mathbf{x}_t\,\mathrm{d}t + \mathbf{B}\,\mathrm{d}\mathbf{W}_t$ (Eq. (13.69)) with Hurwitz-stable $\mathbf{A}(\theta)$ and diffusion matrix $\mathbf{D}(\theta) = \frac{1}{2}\mathbf{B}\mathbf{B}^\top \succ 0$, both depending smoothly on a parameter θ. By Theorem 13.8 the stationary law is $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$, where $\mathbf{\Sigma}(\theta)$ is the unique solution of

$$\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top + 2\mathbf{D} = \mathbf{0}. \tag{18.32}$$

A loss $L(\mathbf{\Sigma})$ on the stationary covariance, for example the KL divergence to a target Gaussian, then depends on θ only through $\mathbf{\Sigma}(\theta)$.

Proposition 18.9 (Covariance sensitivity and the adjoint Lyapunov equation)**.** *Let* $\mathbf{A}$ *be Hurwitz and write* $\dot{\mathbf{A}} := \partial_\theta \mathbf{A}$, $\dot{\mathbf{D}} := \partial_\theta \mathbf{D}$.

(i) The sensitivity $\dot{\mathbf{\Sigma}} := \partial_\theta \mathbf{\Sigma}$ *is the unique solution of the Lyapunov equation*

$$\mathbf{A}\dot{\mathbf{\Sigma}} + \dot{\mathbf{\Sigma}}\mathbf{A}^\top + \mathbf{R} = \mathbf{0}, \qquad \mathbf{R} := \dot{\mathbf{A}}\mathbf{\Sigma} + \mathbf{\Sigma}\dot{\mathbf{A}}^\top + 2\dot{\mathbf{D}}, \tag{18.33}$$

namely $\dot{\mathbf{\Sigma}} = \int_0^\infty e^{\mathbf{A}s}\,\mathbf{R}\,e^{\mathbf{A}^\top s}\,\mathrm{d}s$.

(ii) Let L *be differentiable with symmetric gradient* $\mathbf{W} := \partial L/\partial \mathbf{\Sigma}$, *so that* $\mathrm{d}L = \operatorname{tr}(\mathbf{W}\,\mathrm{d}\mathbf{\Sigma})$. *Let* $\mathbf{\Lambda}$ *be the unique solution of the* adjoint Lyapunov equation

$$\mathbf{A}^\top\mathbf{\Lambda} + \mathbf{\Lambda}\mathbf{A} + \mathbf{W} = \mathbf{0}, \qquad \mathbf{\Lambda} = \int_0^\infty e^{\mathbf{A}^\top s}\,\mathbf{W}\,e^{\mathbf{A}s}\,\mathrm{d}s. \tag{18.34}$$

Then for every parameter θ,

$$\frac{\partial L}{\partial \theta} = \operatorname{tr}(\mathbf{\Lambda}\mathbf{R}) = 2\operatorname{tr}(\mathbf{\Lambda}\dot{\mathbf{A}}\mathbf{\Sigma}) + 2\operatorname{tr}(\mathbf{\Lambda}\dot{\mathbf{D}}), \tag{18.35}$$

and in particular, treating the entries of $\mathbf{A}$ *and* $\mathbf{D}$ *as independent parameters,*

$$\frac{\partial L}{\partial \mathbf{A}} = 2\mathbf{\Lambda}\mathbf{\Sigma}, \qquad \frac{\partial L}{\partial \mathbf{D}} = 2\mathbf{\Lambda}. \tag{18.36}$$

Proof. (i) Differentiate Eq. (18.32) with respect to θ using the product rule:

$$\dot{\mathbf{A}}\mathbf{\Sigma} + \mathbf{A}\dot{\mathbf{\Sigma}} + \dot{\mathbf{\Sigma}}\mathbf{A}^\top + \mathbf{\Sigma}\dot{\mathbf{A}}^\top + 2\dot{\mathbf{D}} = \mathbf{0},$$

which is Eq. (18.33). The linear operator $\mathcal{T}_{\mathbf{A}} : \mathbf{X} \mapsto \mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^\top$ has eigenvalues $\lambda_i + \lambda_j$ with λ_i the eigenvalues of $\mathbf{A}$; since all $\operatorname{Re}\lambda_i < 0$, none vanishes, so the solution is unique. That the integral solves it follows exactly as for Eq. (13.73): $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^\top = \int_0^\infty \frac{\mathrm{d}}{\mathrm{d}s}\big(e^{\mathbf{A}s}\mathbf{R}e^{\mathbf{A}^\top s}\big)\,\mathrm{d}s = -\mathbf{R}$, the boundary term at infinity vanishing because $\mathbf{A}$ is Hurwitz.

(ii) The adjoint operator of $\mathcal{T}_{\mathbf{A}}$ with respect to the Frobenius inner product $\langle \mathbf{X}, \mathbf{Y} \rangle = \operatorname{tr}(\mathbf{X}^\top \mathbf{Y})$ is $\mathcal{T}_{\mathbf{A}^\top}$, so Eq. (18.34) is uniquely solvable for the same reason, with symmetric $\mathbf{\Lambda}$. Using $\mathbf{W} = -(\mathbf{A}^\top \mathbf{\Lambda} + \mathbf{\Lambda}\mathbf{A})$ and cyclicity of the trace,

$$\begin{aligned}\frac{\partial L}{\partial \theta} = \operatorname{tr}(\mathbf{W}\dot{\mathbf{\Sigma}}) &= -\operatorname{tr}(\mathbf{A}^\top \mathbf{\Lambda}\dot{\mathbf{\Sigma}}) - \operatorname{tr}(\mathbf{\Lambda}\mathbf{A}\dot{\mathbf{\Sigma}}) \\ &= -\operatorname{tr}(\mathbf{\Lambda}\dot{\mathbf{\Sigma}}\mathbf{A}^\top) - \operatorname{tr}(\mathbf{\Lambda}\mathbf{A}\dot{\mathbf{\Sigma}}) = -\operatorname{tr}(\mathbf{\Lambda}(\mathbf{A}\dot{\mathbf{\Sigma}} + \dot{\mathbf{\Sigma}}\mathbf{A}^\top)) = \operatorname{tr}(\mathbf{\Lambda}\mathbf{R}),\end{aligned}$$

where the last step uses Eq. (18.33). Expanding $\mathbf{R}$ and using $\operatorname{tr}(\mathbf{\Lambda}\mathbf{\Sigma}\dot{\mathbf{A}}^\top) = \operatorname{tr}(\dot{\mathbf{A}}\mathbf{\Sigma}\mathbf{\Lambda}) = \operatorname{tr}(\mathbf{\Lambda}\dot{\mathbf{A}}\mathbf{\Sigma})$ (transpose, then cycle, with $\mathbf{\Lambda}$ and $\mathbf{\Sigma}$ symmetric) gives Eq. (18.35). For $\theta = A_{ij}$, $\dot{\mathbf{A}} = \mathbf{e}_i \mathbf{e}_j^\top$ and $2\operatorname{tr}(\mathbf{\Lambda}\mathbf{e}_i\mathbf{e}_j^\top\mathbf{\Sigma}) = 2(\mathbf{\Sigma}\mathbf{\Lambda})_{ji} = 2(\mathbf{\Lambda}\mathbf{\Sigma})_{ij}$. For $\theta = D_{ij}$, $2\operatorname{tr}(\mathbf{\Lambda}\mathbf{e}_i\mathbf{e}_j^\top) = 2\Lambda_{ji} = 2\Lambda_{ij}$. □

The adjoint method costs two Lyapunov solves, one for $\mathbf{\Sigma}$ and one for $\mathbf{\Lambda}$, no matter how many parameters there are. It is the steady-state analogue of backpropagation. We checked Eq. (18.36) against central finite differences for a random stable non-normal 2×2 drift; all entries agree to the precision of the finite difference.

Scalar check. For $N = 1$, $\mathbf{A} = -a$ and $\mathbf{D} = d$ give $\Sigma = d/a$. The adjoint equation reads $-2a\Lambda + W = 0$, so $\Lambda = W/(2a)$, and Eq. (18.36) gives $\partial L/\partial A = 2\Lambda\Sigma = Wd/a^2$ and $\partial L/\partial D = W/a$. Directly, $\Sigma = -d/A$ gives $\partial\Sigma/\partial A = d/A^2 = d/a^2$ and $\partial\Sigma/\partial d = 1/a$, in agreement.

Equilibrium check. In the reciprocal case $\mathbf{A} = -\mu\mathbf{K}$ with $\mathbf{K} = \mathbf{K}^\top \succ 0$ and $\mathbf{D} = \mu k_{\mathrm{B}} T\mathbf{I}$, the stationary covariance is $\mathbf{\Sigma} = k_{\mathrm{B}} T\mathbf{K}^{-1}$. Perturb a symmetric pair of couplings, $\dot{\mathbf{K}} = \mathbf{e}_a\mathbf{e}_b^\top + \mathbf{e}_b\mathbf{e}_a^\top$ ($a \neq b$), so that $\dot{\mathbf{A}} = -\mu\dot{\mathbf{K}}$. The candidate $\dot{\mathbf{\Sigma}} = -\beta\mathbf{\Sigma}\dot{\mathbf{K}}\mathbf{\Sigma}$ solves Eq. (18.33): using $\mathbf{K}\mathbf{\Sigma} = \mathbf{\Sigma}\mathbf{K} = k_{\mathrm{B}}T\mathbf{I}$,

$$\mathbf{A}\dot{\mathbf{\Sigma}} + \dot{\mathbf{\Sigma}}\mathbf{A}^\top = \mu\beta(\mathbf{K}\mathbf{\Sigma}\dot{\mathbf{K}}\mathbf{\Sigma} + \mathbf{\Sigma}\dot{\mathbf{K}}\mathbf{\Sigma}\mathbf{K}) = \mu(\dot{\mathbf{K}}\mathbf{\Sigma} + \mathbf{\Sigma}\dot{\mathbf{K}}) = -\mathbf{R}.$$

In components, $\partial\Sigma_{ij}/\partial K_{ab} = -\beta(\Sigma_{ia}\Sigma_{bj} + \Sigma_{ib}\Sigma_{aj})$. The static fluctuation–response theorem gives the same thing: the generalized force of K_{ab} in $E_\theta = \frac{1}{2}\mathbf{x}^\top\mathbf{K}\mathbf{x}$ is $x_a x_b$, so $\partial\langle x_i x_j\rangle/\partial K_{ab} = -\beta\operatorname{Cov}(x_i x_j, x_a x_b) = -\beta(\Sigma_{ia}\Sigma_{jb} + \Sigma_{ib}\Sigma_{ja})$ by Isserlis' theorem. The Lyapunov derivative is therefore the correct generalization of Theorem 1.13 to the Gaussian NESS.

18.3.3 A Non-Equilibrium Fluctuation–Response Formula

The adjoint formula is a computation done on a model of $\mathbf{A}$ and $\mathbf{D}$. The same gradient can also be read off the hardware from fluctuations, as the equilibrium gradient can, through a response formula that holds for any stationary Markov diffusion and does not assume detailed balance. It goes back to work of Agarwal on fluctuation–dissipation relations for non-thermal steady states [2].

Theorem 18.10 (Static response of a non-equilibrium steady state)**.** *Let $\mathcal{L}_{\mathrm{FP}}$ be the Fokker–Planck operator of an ergodic diffusion with drift $\mathbf{a}(\mathbf{x})$, diffusion matrix $\mathbf{D}(\mathbf{x})$, and smooth positive stationary density p_{ss}, and assume the corresponding Markov semigroup converges to p_{ss} exponentially fast. Perturb the drift by $\epsilon\,\delta\mathbf{a}(\mathbf{x})$. Then for every observable O with finite stationary variance,*

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\langle O\rangle_{p_{\mathrm{ss}}^{(\epsilon)}}\Big|_{\epsilon=0} = \int_0^\infty \mathbb{E}[O(\mathbf{x}_t)\,\psi(\mathbf{x}_0)]_{\mathrm{ss}}\,\mathrm{d}t, \qquad \psi(\mathbf{x}) := -\frac{\nabla\cdot(\delta\mathbf{a}(\mathbf{x})\,p_{\mathrm{ss}}(\mathbf{x}))}{p_{\mathrm{ss}}(\mathbf{x})}, \tag{18.37}$$

where the expectation is over the unperturbed stationary process.

Proof. The perturbed Fokker–Planck operator is $\mathcal{L}_{\mathrm{FP}} + \epsilon\,\delta\mathcal{L}$ with $\delta\mathcal{L}\rho = -\nabla\cdot(\delta\mathbf{a}\,\rho)$. Write the perturbed stationary density as $p_{\mathrm{ss}} + \epsilon\rho_1 + \mathcal{O}(\epsilon^2)$. Collecting the terms of order ϵ in $(\mathcal{L}_{\mathrm{FP}} + \epsilon\delta\mathcal{L})(p_{\mathrm{ss}} + \epsilon\rho_1) = 0$ and using $\mathcal{L}_{\mathrm{FP}} p_{\mathrm{ss}} = 0$ gives

$$\mathcal{L}_{\mathrm{FP}}\rho_1 = -\delta\mathcal{L}p_{\mathrm{ss}} = -\psi\,p_{\mathrm{ss}}.$$

The source ψp_{ss} integrates to zero (it is a divergence), and so does ρ_1 (normalization is preserved). On the space of zero-integral densities, exponential convergence of the semigroup means $\int_0^\infty e^{t\mathcal{L}_{\mathrm{FP}}}\,\mathrm{d}t$ converges and equals $-\mathcal{L}_{\mathrm{FP}}^{-1}$, so

$$\rho_1 = \int_0^\infty e^{t\mathcal{L}_{\mathrm{FP}}}(\psi\,p_{\mathrm{ss}})\,\mathrm{d}t.$$

The density $e^{t\mathcal{L}_{\mathrm{FP}}}(\psi p_{\mathrm{ss}})$ is what an initial weighting $\psi(\mathbf{x}_0)p_{\mathrm{ss}}(\mathbf{x}_0)$ evolves into after time t, so integrating it against O gives $\int O\,e^{t\mathcal{L}_{\mathrm{FP}}}(\psi p_{\mathrm{ss}})\,\mathrm{d}\mathbf{x} = \mathbb{E}[O(\mathbf{x}_t)\psi(\mathbf{x}_0)]_{\mathrm{ss}}$. Integrating over t gives Eq. (18.37). □

Two checks show that this is the right generalization.

Equilibrium limit. Let $\mathbf{a} = -\mu\nabla E$, $\mathbf{D} = \mu k_{\mathrm{B}}T\mathbf{I}$, $p_{\mathrm{ss}} \propto e^{-\beta E}$, and perturb the energy by $\epsilon\,\delta E$, so $\delta\mathbf{a} = -\mu\nabla\delta E$. Then, with $\nabla p_{\mathrm{ss}} = -\beta p_{\mathrm{ss}}\nabla E$,

$$\psi = \frac{\mu\nabla\cdot(p_{\mathrm{ss}}\nabla\delta E)}{p_{\mathrm{ss}}} = \mu\big(\Delta\delta E - \beta\nabla E\cdot\nabla\delta E\big) = \beta\,\mathcal{L}\,\delta E,$$

where $\mathcal{L} = \mu k_{\mathrm{B}}T\Delta - \mu\nabla E\cdot\nabla$ is the backward generator. For a reversible process $\mathbb{E}[O(\mathbf{x}_t)(\mathcal{L}\delta E)(\mathbf{x}_0)] = \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[O(\mathbf{x}_t)\delta E(\mathbf{x}_0)]$, by self-adjointness of $\mathcal{L}$ in $L^2(p_{\mathrm{ss}})$. The time integral telescopes:

$$\int_0^\infty \beta\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[O(\mathbf{x}_t)\delta E(\mathbf{x}_0)]\,\mathrm{d}t = \beta\big(\langle O\rangle\,\langle\delta E\rangle - \langle O\,\delta E\rangle\big) = -\beta\,\mathrm{Cov}(O,\delta E),$$

which is the static fluctuation–response theorem.

Gaussian NESS. For $\delta\mathbf{a} = \delta\mathbf{A}\,\mathbf{x}$ and $p_{\mathrm{ss}} = \mathcal{N}(\mathbf{0},\boldsymbol{\Sigma})$ we have $\nabla p_{\mathrm{ss}} = -p_{\mathrm{ss}}\boldsymbol{\Sigma}^{-1}\mathbf{x}$, so

$$\psi(\mathbf{x}) = -\,\mathrm{tr}(\delta\mathbf{A}) + \mathbf{x}^\top\boldsymbol{\Sigma}^{-1}\delta\mathbf{A}\,\mathbf{x}.$$

For $N = 1$ with $\mathbf{A} = -a$, $\delta\mathbf{A} = -1$ and $O = x^2$: $\psi = 1 - x^2/\Sigma$, the stationary correlation is $C(t) = \Sigma e^{-at}$, and Isserlis gives $\mathbb{E}[x_t^2x_0^2] = \Sigma^2 + 2C(t)^2$. The integrand is $\Sigma - (\Sigma^2 + 2\Sigma^2e^{-2at})/\Sigma = -2\Sigma e^{-2at}$, whose integral is $-\Sigma/a$. The perturbation $\delta\mathbf{A} = -1$ increases a at rate one, so the response should be $\partial\Sigma/\partial a$, which from $\Sigma = d/a$ is $-d/a^2 = -\Sigma/a$. The response formula and the Lyapunov derivative agree.

Theorem 18.10 says that in a NESS the response to a parameter change is a time integral of a two-time correlation between the observable and a known function ψ of the state. Unlike the equilibrium case, ψ depends on p_{ss} itself, which for a Gaussian NESS is fixed by the measurable covariance $\boldsymbol{\Sigma}$. The two-time correlations themselves are also measurable; Proposition 13.9 shows that their antisymmetric part is the signature of broken detailed balance. A NESS gradient can thus be estimated from hardware fluctuations, at the cost of time-resolved readout rather than snapshot moments.

To make this concrete for a Gaussian NESS, let $\theta = A_{ij}$, so $\delta\mathbf{A} = \mathbf{e}_i\mathbf{e}_j^\top$ and $\psi(\mathbf{x}) = -\delta_{ij} + (\boldsymbol{\Sigma}^{-1}\mathbf{x})_i\,x_j$. For an observable $O = x_kx_l$, Eq. (18.37) becomes

$$\frac{\partial\Sigma_{kl}}{\partial A_{ij}} = \int_0^\infty \mathbb{E}\Big[x_k(t)x_l(t)\,\big[(\boldsymbol{\Sigma}^{-1}\mathbf{x}(0))_i\,x_j(0) - \delta_{ij}\big]\Big]_{\mathrm{ss}}\,\mathrm{d}t. \tag{18.38}$$

An estimator replaces the expectation by a time average over one long stationary record, with $\boldsymbol{\Sigma}^{-1}$ taken from the same record, and truncates the lag integral after a few relaxation times. Its variance grows with the truncation window, as for the score-function estimator of Section 18.1, which is the price of extracting a non-equilibrium response from fluctuations alone.

18.3.4 Covariance-Matching Updates and the Vorticity Gauge

The equilibrium learning rule for a Gaussian EBM is local. With $E_\theta = \frac{1}{2}\mathbf{x}^\top \mathbf{K}\mathbf{x}$ and data covariance $\mathbf{\Sigma}^\star$, Eq. (1.104) gives

$$\nabla_{\mathbf{K}}\mathcal{L} = -\frac{\beta}{2}(\mathbf{\Sigma}^\star - \mathbf{\Sigma}_{\mathrm{hw}}), \qquad \mathbf{K} \leftarrow \mathbf{K} - \frac{\eta\beta}{2}(\mathbf{\Sigma}^\star - \mathbf{\Sigma}_{\mathrm{hw}}), \tag{18.39}$$

where $\mathbf{\Sigma}_{\mathrm{hw}}$ is the covariance measured on the sampler, and the update of K_{ij} needs only the correlation of nodes i and j. In a NESS the exact gradient Eq. (18.36) is non-local, because $\mathbf{\Lambda}$ couples all entries. Two facts limit how much this matters.

The first is a gauge freedom. By Proposition 13.10, every transconductance matrix of the form $\mathbf{G} = \frac{C_0}{\sigma_0^2}(d_0\mathbf{I} - \mathbf{Q})\mathbf{\Lambda}_{\mathrm{prec}}$ with $\mathbf{Q} = -\mathbf{Q}^\top$ produces the same stationary covariance $\sigma_0^2\mathbf{\Lambda}_{\mathrm{prec}}^{-1}$. Any loss that depends only on the stationary law therefore has zero derivative along every $\mathbf{Q}$ direction. If the hardware is parameterized by $(\mathbf{\Lambda}_{\mathrm{prec}}, \mathbf{Q})$, the stationary learning problem for $\mathbf{\Lambda}_{\mathrm{prec}}$ is identical to the equilibrium one, and $\mathbf{Q}$ is a free parameter that can be chosen to accelerate mixing. The vorticity is a gauge of the stationary distribution, in the same sense that the additive constant in Proposition 1.2 is a gauge of the Boltzmann distribution.

The Lyapunov derivative makes the gauge statement a one-line computation. Every Hurwitz drift with stationary covariance $\mathbf{\Sigma}$ can be written $\mathbf{A} = (-\mathbf{D} + \mathbf{Q})\mathbf{\Sigma}^{-1}$ with $\mathbf{Q}$ skew-symmetric (Section 13.4). Vary $\mathbf{Q}$ alone at fixed $\mathbf{\Sigma}$ and $\mathbf{D}$: then $\dot{\mathbf{A}} = \dot{\mathbf{Q}}\mathbf{\Sigma}^{-1}$, $\dot{\mathbf{D}} = \mathbf{0}$, and the source in Eq. (18.33) is

$$\mathbf{R} = \dot{\mathbf{Q}}\mathbf{\Sigma}^{-1}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{\Sigma}^{-1}\dot{\mathbf{Q}}^\top = \dot{\mathbf{Q}} + \dot{\mathbf{Q}}^\top = \mathbf{0},$$

so $\dot{\mathbf{\Sigma}} = \mathbf{0}$ and every stationary loss has zero gradient along $\mathbf{Q}$.

Example 18.11 (An isotropic two-node rotor)**.** Take $N = 2$, $\mathbf{D} = d\,\mathbf{I}$, and $\mathbf{A} = -a\mathbf{I} + q\mathbf{J}_2$ with $\mathbf{J}_2 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $a > 0$. For $\mathbf{\Sigma} = s\mathbf{I}$, $\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top = s(\mathbf{A} + \mathbf{A}^\top) = -2as\mathbf{I}$, so the Lyapunov equation gives $s = d/a$ for every q. The adjoint gradient with respect to q is $2\,\mathrm{tr}(\mathbf{\Lambda}\mathbf{J}_2\mathbf{\Sigma}) = (2d/a)\,\mathrm{tr}(\mathbf{\Lambda}\mathbf{J}_2) = 0$, since the trace of a symmetric matrix times a skew-symmetric one vanishes. The dynamics nonetheless depends on q: the stationary two-time correlation is $\mathbb{E}\left[\mathbf{x}_t\mathbf{x}_0^\top\right] = e^{\mathbf{A}t}\mathbf{\Sigma} = (d/a)e^{-at}\left(\begin{smallmatrix} \cos qt & \sin qt \\ -\sin qt & \cos qt \end{smallmatrix}\right)$, whose antisymmetric part $(d/a)e^{-at}\sin(qt)$ is nonzero for $q \neq 0$ and is the time-asymmetry of Proposition 13.9. A learner that sees only snapshot moments cannot detect or set q; one that sees two-time correlations can.

The second fact concerns hardware that is parameterized directly by its transconductances, so that a naive local update $\mathbf{G} \leftarrow \mathbf{G} + \eta'(\mathbf{\Sigma}_{\mathrm{hw}} - \mathbf{\Sigma}^\star)$ is not a gradient.

Proposition 18.12 (Local covariance matching near equilibrium)**.** *Let $\mathbf{C} = C_0\mathbf{I}$, $\mathbf{D} = d_0\mathbf{I}$, and suppose $\mathbf{\Sigma}^\star \succ 0$ is reachable at some $\mathbf{G}^\star$. Consider the iteration $\mathbf{G}_{n+1} = \mathbf{G}_n + \eta(\mathbf{\Sigma}(\mathbf{G}_n) - \mathbf{\Sigma}^\star)$ restricted to symmetric updates. Its fixed points are exactly the $\mathbf{G}$ with $\mathbf{\Sigma}(\mathbf{G}) = \mathbf{\Sigma}^\star$. If $\mathbf{G}^\star$ is symmetric, the fixed point is locally attracting for all sufficiently small $\eta > 0$. By continuity of eigenvalues, the same holds when $\mathbf{G}^\star$ has a sufficiently small antisymmetric part.*

Proof. The fixed-point statement is immediate. Linearize at a symmetric $\mathbf{G}^\star$: the stationary covariance is $\mathbf{\Sigma} = C_0 d_0 \mathbf{G}^{-1}$, so a symmetric perturbation $\mathbf{X}$ produces $\delta\mathbf{\Sigma} = -C_0 d_0 \mathbf{G}^{\star-1}\mathbf{X}\mathbf{G}^{\star-1} = -(C_0 d_0)^{-1}\mathbf{\Sigma}^\star\mathbf{X}\mathbf{\Sigma}^\star$. The linearized iteration is $\mathbf{X}_{n+1} = \mathbf{X}_n - \eta\,\mathcal{M}[\mathbf{X}_n]$ with $\mathcal{M}[\mathbf{X}] := (C_0 d_0)^{-1}\mathbf{\Sigma}^\star\mathbf{X}\mathbf{\Sigma}^\star$. On symmetric matrices with the Frobenius inner product, $\mathcal{M}$ is self-adjoint and positive definite, since $\langle \mathbf{X}, \mathcal{M}[\mathbf{X}]\rangle = (C_0 d_0)^{-1}\left\|\mathbf{\Sigma}^{\star 1/2}\mathbf{X}\mathbf{\Sigma}^{\star 1/2}\right\|_F^2 > 0$ for $\mathbf{X} \neq \mathbf{0}$. Hence $\mathbf{I} - \eta\mathcal{M}$ is a contraction for $0 < \eta < 2/\lambda_{\max}(\mathcal{M})$. The linearization depends continuously on $\mathbf{G}^\star$ through the Lyapunov solution, and the eigenvalues of a matrix depend continuously on its entries, so they stay in the open right half-plane under a small enough antisymmetric perturbation. □

For strongly non-reciprocal networks the linearized operator can acquire eigenvalues with negative real part, and the naive rule can then diverge even though its fixed point is correct. The safe options are the adjoint gradient Eq. (18.36), computed on the host from the measured $\mathbf{\Sigma}_{\mathrm{hw}}$, or a parameterization in which the vorticity is held fixed.

Hardware Realization: Training a NEAT-RN against its own covariance

For the two-node NEAT-RN of Worked computation 13.13, the gradient of $L = D_{\mathrm{KL}}(\mathcal{N}(\mathbf{0}, \mathbf{\Sigma}^\star) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}))$ has $\mathbf{W} = \frac{1}{2}(\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{\Sigma}^\star\mathbf{\Sigma}^{-1})$, which vanishes exactly when $\mathbf{\Sigma} = \mathbf{\Sigma}^\star$. A practical loop measures $\mathbf{\Sigma}_{\mathrm{hw}}$ from voltage samples, forms $\mathbf{W}$ from it, solves the 2×2 adjoint equation on the host for $\mathbf{\Lambda}$, and updates the bias currents that set $\mathbf{A}$ and $\mathbf{D}$ along $-2\mathbf{\Lambda}\mathbf{\Sigma}_{\mathrm{hw}}$ and $-2\mathbf{\Lambda}$, mapped through Eq. (13.94). Using the measured $\mathbf{\Sigma}_{\mathrm{hw}}$ rather than a modeled $\mathbf{\Sigma}$ matters: as Section 18.5 shows, it is what makes the loop insensitive to the device mismatch that the model does not know about. The drift $\mathbf{A}$ itself still enters $\mathbf{\Lambda}$ through the model, so errors in the modeled $\mathbf{A}$ change the step direction but not the fixed point $\mathbf{\Sigma}_{\mathrm{hw}} = \mathbf{\Sigma}^\star$.

18.4 The Physics of Silicon Imperfections: PVT Variation and Pelgrom's Law

Two transistors drawn identically on the same die are not identical. Their channels contain different numbers of dopant atoms at different positions, their gate edges are rough on the scale of a few nanometers, and the polycrystalline grains of a metal gate have different work functions. Each of these differences shifts the threshold voltage V_{th} and, less strongly, the current factor. Circuit designers group such effects under PVT: variation of the *process* (both global, from wafer to wafer and die to die, and local, from device to device), of the *supply voltage*, and of the *temperature*. For a sampler, global variation and slow temperature drift act on all devices together and are comparatively easy to correct. Local random mismatch is different: it gives every one of the 10^5 or more stochastic units on a die its own private offset, and those offsets must be measured or learned one by one.

18.4.1 The Area Law

Pelgrom, Duinmaijer, and Welbers [86] showed that the variance of the difference of a parameter between two nominally identical, closely spaced devices is inversely proportional to the gate area. The reason is averaging.

Proposition 18.13 (Pelgrom area law)**.** *Let a device parameter P of a transistor with rectangular gate region $\mathcal{A}$ of width W and length L be the area average $P = (WL)^{-1}\int_{\mathcal{A}} p(\mathbf{r})\,\mathrm{d}^2\mathbf{r}$ of a local quantity $p(\mathbf{r}) = \bar{p} + \delta p(\mathbf{r})$, where δp is a zero-mean random field with correlation $\mathbb{E}[\delta p(\mathbf{r})\delta p(\mathbf{r}')] = c_p\,\delta^{(2)}(\mathbf{r}-\mathbf{r}')$ on scales much shorter than W and L. Then*

$$\mathrm{Var}(P) = \frac{c_p}{WL}, \tag{18.40}$$

and for two such devices with disjoint gates,

$$\sigma^2(\Delta P) = \frac{A_P^2}{WL}, \qquad A_P^2 := 2c_p. \tag{18.41}$$

Proof. By linearity of expectation and the delta correlation,

$$\mathrm{Var}(P) = \frac{1}{(WL)^2}\int_{\mathcal{A}}\int_{\mathcal{A}} \mathbb{E}\big[\delta p(\mathbf{r})\delta p(\mathbf{r}')\big]\ \mathrm{d}^2\mathbf{r}\ \mathrm{d}^2\mathbf{r}' = \frac{c_p}{(WL)^2}\int_{\mathcal{A}} \mathrm{d}^2\mathbf{r} = \frac{c_p}{WL}.$$

For two devices with disjoint gate regions the cross-covariance vanishes, so $\mathrm{Var}(P_1 - P_2) = 2c_p/(WL)$. □

The proposition gives the familiar form

$$\sigma(\Delta V_{\mathrm{th}}) = \frac{A_{V_{\mathrm{th}}}}{\sqrt{WL}}. \tag{18.42}$$

A long-range component, such as a doping or oxide gradient across the wafer, adds a term $S_P^2 D^2$ that grows with the spacing D between devices [86]; common-centroid layout, which interleaves the devices of a matched pair so that their centroids coincide, cancels its linear part, and we ignore it below.

Notation Watch: Pair versus single-device mismatch

Pelgrom's coefficient $A_{V_{\mathrm{th}}}$ is defined for the *difference* of a matched pair, Eq. (18.41). The deviation of a single device from the nominal value has standard deviation $A_{V_{\mathrm{th}}}/\sqrt{2WL}$, smaller by $\sqrt{2}$. A differential pair's input-referred offset and a current mirror's copy error are pair quantities. The deviation of one transistor's current from the value the designer computed is a single-device quantity. We use $\sigma(\Delta V_{\mathrm{th}})$ for the pair and $\sigma_{V_{\mathrm{th}}}$ for the single device.

18.4.2 Random Dopant Fluctuation

The simplest microscopic source of threshold mismatch is the discreteness of the dopant atoms. In the charge-sheet picture of Section 12.1, the threshold voltage contains the depletion charge per unit area divided by the oxide capacitance per unit area C'_{ox}, $V_{\mathrm{th}} \supset q_e N_A W_{\mathrm{dep}}/C'_{\mathrm{ox}}$, for acceptor density N_A and depletion depth W_{dep}. The number of acceptors in the depletion volume WLW_{dep} is Poisson distributed, with variance equal to its mean.

A dopant does not affect V_{th} equally at every depth. Consider one extra ionized acceptor at depth z below the oxide interface. Its charge $-q_e$ is screened by image charges on the two conductors that bound the depletion region: the gate above and the neutral bulk below the depletion edge at $z = W_{\mathrm{dep}}$. If the oxide is electrically thin compared with the depletion layer ($t_{\mathrm{ox}}\epsilon_{\mathrm{Si}}/\epsilon_{\mathrm{ox}} \ll W_{\mathrm{dep}}$), the capacitances to the two conductors are in the ratio of the inverse distances, and the fraction of the image charge that sits on the gate side is $1 - z/W_{\mathrm{dep}}$. The resulting threshold shift is

$$\delta V_{\mathrm{th}}(z) = \frac{q_e}{C'_{\mathrm{ox}}WL}\left(1 - \frac{z}{W_{\mathrm{dep}}}\right).$$

Dopants in a slab $\mathrm{d}z$ are independent Poisson counts with variance $N_A WL\ \mathrm{d}z$, so their contributions add in variance:

$$\sigma^2_{V_{\mathrm{th}}} = \int_0^{W_{\mathrm{dep}}} \left(\frac{q_e}{C'_{\mathrm{ox}}WL}\right)^2 \left(1 - \frac{z}{W_{\mathrm{dep}}}\right)^2 N_A WL\ \mathrm{d}z = \left(\frac{q_e}{C'_{\mathrm{ox}}}\right)^2 \frac{N_A W_{\mathrm{dep}}}{3\,WL}. \tag{18.43}$$

The factor $\frac{1}{3} = \int_0^1 (1-u)^2\,\mathrm{d}u$ is the depth weighting: dopants near the surface count fully, those near the depletion edge hardly at all. Equation (18.43) has the Pelgrom form with $A_{V_{\mathrm{th}}} = \sqrt{2}\,(q_e/C'_{\mathrm{ox}})\sqrt{N_A W_{\mathrm{dep}}/3}$.

WL (μm^2)	$\sigma(\Delta V_{\text{th}})$ (mV)	$\sigma_{\ln I}$, pair	$\sigma_{\ln I}$, single	mean factor $e^{\sigma^2/2}$, single
0.01	30	0.89	0.63	1.22
0.1	9.5	0.28	0.20	1.02
1	3.0	0.089	0.063	1.002
10	0.95	0.028	0.020	1.0002

Table 18.2: Representative subthreshold current mismatch for $A_{V_{\text{th}}} = 3\,\text{mV}\,\mu\text{m}$, $n = 1.3$, $T = 300\,\text{K}$ ($nV_T = 33.6\,\text{mV}$). The pair columns apply to mirrors and differential pairs, the single columns to the deviation of one device from its design value. These are order-of-magnitude estimates; actual $A_{V_{\text{th}}}$ is technology dependent.

Example 18.14 (Order of magnitude)**.** Take $N_A = 10^{18}\,\text{cm}^{-3} = 10^{24}\,\text{m}^{-3}$, $W_{\text{dep}} = 30\,\text{nm}$, and $t_{\text{ox}} = 2\,\text{nm}$, so $C'_{\text{ox}} = 3.9{\times}8.854{\times}10^{-12}/(2{\times}10^{-9}) = 1.73{\times}10^{-2}\,\text{F}\,\text{m}^{-2}$. Then $N_A W_{\text{dep}}/3 = 10^{16}\,\text{m}^{-2}$, whose square root is $10^8\,\text{m}^{-1}$, and

$$\frac{q_e}{C'_{\text{ox}}}\sqrt{\frac{N_A W_{\text{dep}}}{3}} = \frac{1.602\times10^{-19}\times10^{8}}{1.73\times10^{-2}}\,\text{V}\,\text{m} = 9.3\times10^{-10}\,\text{V}\,\text{m} = 0.93\,\text{mV}\,\mu\text{m},$$

using $1\,\text{mV}\,\mu\text{m} = 10^{-9}\,\text{V}\,\text{m}$. The pair coefficient is $A_{V_{\text{th}}} \approx 1.3\,\text{mV}\,\mu\text{m}$. Measured values in bulk CMOS lie in the range of roughly 1 to 5 $\text{mV}\,\mu\text{m}$ and depend on the technology; they decrease roughly in proportion to the oxide thickness, as Eq. (18.43) suggests, until other sources (line-edge roughness, metal-gate grain work functions, oxide thickness fluctuations) take over. Below we use $A_{V_{\text{th}}} = 3\,\text{mV}\,\mu\text{m}$ as a representative value.

18.4.3 Exponential Amplification in Weak Inversion

In saturation ($V_{\text{DS}} \gtrsim 4V_T$), the subthreshold drain current of Section 12.3 is

$$I_{\text{D}} = I_0\,\frac{W}{L}\exp\!\left(\frac{V_{\text{GS}} - V_{\text{th}}}{nV_T}\right), \tag{18.44}$$

with slope factor n (Section 12.2) typically 1.2 to 1.5 [29]. A threshold shift δV_{th} at fixed gate voltage changes the current by an exact factor:

$$\frac{I_{\text{D}} + \delta I_{\text{D}}}{I_{\text{D}}} = \exp\!\left(-\frac{\delta V_{\text{th}}}{nV_T}\right), \qquad \ln\frac{I_{\text{D}} + \delta I_{\text{D}}}{I_{\text{D}}} = -\frac{\delta V_{\text{th}}}{nV_T}. \tag{18.45}$$

Because δV_{th} is Gaussian, the current ratio is lognormal with log-standard deviation $\sigma_{\ln I} = \sigma_{V_{\text{th}}}/(nV_T)$. Its mean is $\exp(\sigma^2_{\ln I}/2) > 1$, so mismatch also shifts the average current upward slightly. For small offsets $\delta I_{\text{D}}/I_{\text{D}} \approx -\delta V_{\text{th}}/(nV_T) = -(g_{\text{m}}/I_{\text{D}})\,\delta V_{\text{th}}$. In strong inversion, where $I_{\text{D}} \propto (V_{\text{GS}} - V_{\text{th}})^2$, the same offset gives $\delta I_{\text{D}}/I_{\text{D}} = -2\delta V_{\text{th}}/V_{\text{ov}}$ with overdrive $V_{\text{ov}} = V_{\text{GS}} - V_{\text{th}}$. The ratio of the two sensitivities is $V_{\text{ov}}/(2nV_T)$, about 3 for $V_{\text{ov}} = 200\,\text{mV}$. The largest possible transconductance efficiency $g_{\text{m}}/I_{\text{D}} = 1/(nV_T)$, which makes weak inversion attractive for low-power analog computing, is the same quantity that maximizes current mismatch.

Table 18.2 shows the consequence. A minimum-size subthreshold device ($WL \sim 0.01\,\mu\text{m}^2$) has a current that is uncertain by a factor of about two. Reducing $\sigma_{\ln I}$ to 1% by area alone requires $WL \approx (A_{V_{\text{th}}}/(0.01\,nV_T))^2 \approx 80\,\mu\text{m}^2$ for a pair, which is incompatible with packing 10^5 units onto a die. Since the area cost of matching by brute force is so large, dense stochastic arrays rely on calibration instead.

18.4.4 How Mismatch Enters the Sampler's Parameters

To discuss calibration we must translate threshold offsets into errors of the parameters θ of the sampled distribution. Three cases cover the circuits of Chapters 13 and 14.

Programmed currents: multiplicative errors. A coupling or transconductance set by a subthreshold bias current obeys Eq. (13.94), $G_{ij} = (I_0/nV_T)\exp[(V_{\mathrm{DAC},ij} - V_{\mathrm{th}})/(nV_T)]$. A threshold offset δV_{th} in the bias transistor (or the mirror that copies its current) multiplies the coupling:

$$G_{ij}^{\mathrm{phys}} = G_{ij}\, e^{-\delta V_{\mathrm{th},ij}/(nV_T)} := G_{ij}(1+\gamma_{ij}). \tag{18.46}$$

In the DAC-voltage coordinate the same error is additive: the circuit behaves as if programmed with $V_{\mathrm{DAC},ij} - \delta V_{\mathrm{th},ij}$. Which description is simpler depends on the parameterization the learning algorithm uses.

Input offsets of transconductors: additive errors on means. A differential-pair transconductor has an input-referred offset voltage V_{os} equal to the threshold difference of its input pair, a pair quantity with standard deviation $A_{V_{\mathrm{th}}}/\sqrt{WL}$. It injects a current $g_{\mathrm{m}}(v_j - V_{\mathrm{os}})$ instead of $g_{\mathrm{m}}v_j$. In a NEAT-RN the resulting constant current shifts the fixed point $\mathbf{v}^*$, that is, the mean of the sampled Gaussian, by $\delta\mathbf{v}^* = \mathbf{G}^{-1}\delta\mathbf{I}$ with $\delta I_i = -\sum_j G_{ij}V_{\mathrm{os},ij}$. When the self-conductance dominates each row of $\mathbf{G}$, δv_i^* is of the order of the offset of the dominant transconductor.

Comparator offsets in p-bits: additive errors on fields. A CMOS p-bit (Section 14.1) produces $s_i = +1$ with the logistic probability $1/(1+e^{-2\beta h_i})$, where the effective field is set by an input voltage relative to the switching point of a comparator or latch. If the circuit maps input voltage to natural field as $\beta h_i = V_{\mathrm{in},i}/V_s$ for some design scale V_s (typically a few times nV_T; the value is circuit dependent), an input-referred offset $V_{\mathrm{os},i}$ gives the additive error

$$\beta h_i^{\mathrm{phys}} = \beta h_i + \beta\delta h_i, \qquad \beta\delta h_i = -\frac{V_{\mathrm{os},i}}{V_s}. \tag{18.47}$$

18.4.5 Temperature, Supply, Drift, and DAC Quantization

Temperature enters in three places. The sampling temperature itself sets β, so the whole distribution sharpens or broadens. The thermal voltage $V_T = k_{\mathrm{B}}T/q_e$ grows by about 0.33% per kelvin at room temperature, changing every subthreshold slope. And V_{th} has a negative temperature coefficient, of order -1 to $-2\,\mathrm{mV\,K^{-1}}$ in typical bulk processes, which by Eq. (18.45) changes subthreshold currents by several percent per kelvin. In a design where couplings are set by ratios of currents from matched devices, the common factors cancel to first order and what remains acts on all parameters nearly uniformly, as $\theta \mapsto (1+\epsilon_T)\theta$ plus small device-dependent corrections. *Supply variation* enters subthreshold circuits mainly through drain-induced barrier lowering and finite output conductance, and is usually smaller.

Example 18.15 (Temperature sensitivity of a subthreshold current and of a current ratio)**.** Take the logarithm of Eq. (18.44) and differentiate with respect to T at fixed gate voltage, writing $\alpha_{\mathrm{th}} := \mathrm{d}V_{\mathrm{th}}/\,\mathrm{d}T$:

$$\frac{\mathrm{d}\ln I_{\mathrm{D}}}{\mathrm{d}T} = \frac{\mathrm{d}\ln I_0}{\mathrm{d}T} - \frac{\alpha_{\mathrm{th}}}{nV_T} - \frac{V_{\mathrm{GS}} - V_{\mathrm{th}}}{nV_T\,T}.$$

In the EKV model the specific current scales as $I_0 \propto \mu_{\mathrm{eff}}V_T^2$ [29]; with a phonon-limited mobility $\mu_{\mathrm{eff}} \propto T^{-3/2}$, the first term is $(2 - \frac{3}{2})/T = 0.0017\,\mathrm{K^{-1}}$ at 300 K. With representative values

$\alpha_{\mathrm{th}} = -1.5\,\mathrm{mV\,K^{-1}}$, $nV_T = 33.6\,\mathrm{mV}$, and $V_{\mathrm{GS}} - V_{\mathrm{th}} = -100\,\mathrm{mV}$, the second term is $+0.045\,\mathrm{K^{-1}}$ and the third is $+100/(33.6 \times 300) = +0.010\,\mathrm{K^{-1}}$. The total is about $+5.6\%$ per kelvin: a one-kelvin change moves the current by an amount comparable to the Pelgrom mismatch of a $1\,\mu\mathrm{m}^2$ pair in Table 18.2.

Now consider the ratio of two currents from devices with the same threshold but gate voltages differing by $V_1 - V_2$. The I_0 and V_{th} terms cancel, leaving

$$\frac{\mathrm{d}}{\mathrm{d}T} \ln \frac{I_1}{I_2} = -\frac{V_1 - V_2}{nV_T\,T} = -\frac{1}{T} \ln \frac{I_1}{I_2}.$$

For a ratio of e this is -0.33% per kelvin, seventeen times smaller than the single-current sensitivity. Circuits whose parameters are ratios of matched currents therefore see temperature mainly as a common rescaling, the in-family gain error $(1 + \epsilon_T)\theta$ of the next section, while circuits that rely on absolute currents see it as a large drift of every parameter.

Low-frequency noise turns static offsets into slowly drifting ones. The $1/f$ and random-telegraph noise of Section 11.5 has most of its power below the corner frequency, so on time scales longer than a calibration cycle it acts as a random walk of V_{th} that calibration must track rather than cancel once.

DAC quantization limits how finely any parameter, including a correction, can be set. A b-bit DAC spanning a full-scale range V_{FS} has step $\Delta = V_{\mathrm{FS}}/2^b$, and rounding leaves a residual uniformly distributed on $[-\Delta/2, \Delta/2]$ with variance $\Delta^2/12$. The range must accommodate both the intended parameter values and the offsets to be cancelled: for an offset standard deviation σ_{os} and a coverage of $\pm z\sigma_{\mathrm{os}}$, a span of $2z\sigma_{\mathrm{os}}$ is reserved for calibration. A gate-voltage DAC driving a subthreshold device produces geometrically spaced currents, with relative step $\Delta/(nV_T)$, so a voltage DAC provides constant *relative* precision of a programmed coupling. Offsets beyond the DAC range cannot be cancelled at all, and the corresponding unit must be treated as faulty. Calibration also costs storage: each corrected parameter needs its own trim code, so an array of N units with P/N calibrated parameters per unit and b-bit trims needs Pb bits of local memory, held for as long as the calibration is valid.

Hardware Realization: Mismatch as a design parameter

For a stochastic array, the relevant comparison is not between an offset and the supply voltage but between an offset and the thermal fluctuation scale of the variable it perturbs. A p-bit's field offset matters relative to V_s, which is of order tens of millivolts. A Gaussian node's mean offset matters relative to its thermal standard deviation σ_V, which for femtofarad capacitors is of order $\sqrt{k_{\mathrm{B}}T/C} \approx 2\,\mathrm{mV}$ at $C = 1\,\mathrm{fF}$, or a design-dependent multiple of it when shot noise is injected deliberately (Section 13.5). With $A_{V_{\mathrm{th}}} = 3\,\mathrm{mV}\,\mu\mathrm{m}$ and a $0.25\,\mu\mathrm{m}^2$ input pair, $\sigma(\Delta V_{\mathrm{th}}) = 6\,\mathrm{mV}$: comparable to the fluctuation scale of a Gaussian node, and a modest fraction of a p-bit's V_s. The worked computation of Section 18.6 turns these ratios into KL divergences.

18.5 Hardware-in-the-Loop Self-Calibration

We now combine the two halves of the chapter. The contrastive gradient needs moments of the sufficient statistics under the model distribution. If those moments are measured on the physical die, then the model distribution in the gradient is the one the die actually realizes, offsets included. Gradient descent then adjusts the programmed parameters until the *realized* distribution matches the target. This section makes the statement precise and lists its limits.

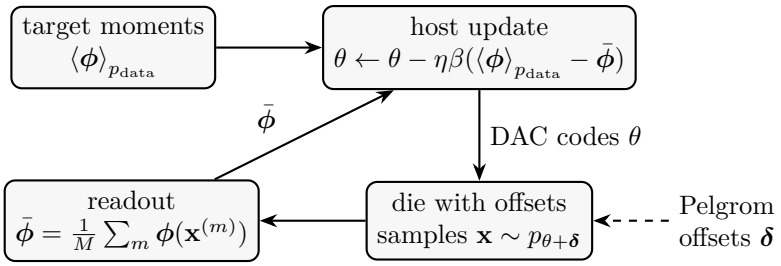


Figure 18.1: Hardware-in-the-loop calibration. The host never knows the offsets $\boldsymbol{\delta}$; it only compares moments measured on the die with target moments. The fixed point of the loop is where the realized distribution $p_{\theta+\boldsymbol{\delta}}$, not the programmed one, matches the target.

18.5.1 Setting

Let the designed model be a linearly parameterized EBM (Definition 1.12),

$$p_\theta(\mathbf{x}) = \frac{1}{Z(\theta)} \exp\Big(-\beta\big[E_{\text{base}}(\mathbf{x}) + \theta^\top \boldsymbol{\phi}(\mathbf{x})\big]\Big), \qquad \psi(\theta) := \ln Z(\theta), \qquad \theta \in \Theta \subseteq \mathbb{R}^P, \tag{18.48}$$

with Fisher information matrix $\mathbf{F}(\theta) := \nabla^2\psi(\theta) = \beta^2 \operatorname{Cov}_{p_\theta}(\boldsymbol{\phi})$ (Eq. (1.84)). The die realizes the distribution

$$p_\theta^{\text{phys}} = p_{M_{\text{hw}}(\theta)}, \tag{18.49}$$

where the *hardware map* $M_{\text{hw}} : \Theta \to \Theta$ is unknown to us but fixed in time. The two important cases are a static additive offset, $M_{\text{hw}}(\theta) = \theta + \boldsymbol{\delta}$, and a diagonal gain error with offset, $M_{\text{hw}}(\theta) = \operatorname{diag}(\mathbf{1} + \boldsymbol{\gamma})\theta + \boldsymbol{\delta}$, which by Section 18.4 describe threshold offsets in field, comparator, and bias-current circuits. The target distribution is p_{data}, and we assume its moments $\langle \boldsymbol{\phi} \rangle_{p_{\text{data}}}$ are available on the host, from data or from an exact digital model of the intended distribution.

Definition 18.16 (Hardware-in-the-loop contrastive update)**.** Starting from θ_0, repeat: program θ_n through the DACs; draw M samples $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)}$ from the die; form $\bar{\boldsymbol{\phi}}_n = M^{-1}\sum_m \boldsymbol{\phi}(\mathbf{x}^{(m)})$; and update

$$\theta_{n+1} = \theta_n - \eta\, \mathbf{P}\, \beta\big(\langle \boldsymbol{\phi} \rangle_{p_{\text{data}}} - \bar{\boldsymbol{\phi}}_n\big), \tag{18.50}$$

with step size $\eta > 0$ and a fixed positive-definite preconditioner $\mathbf{P}$ ($\mathbf{P} = \mathbf{I}$ for plain gradient descent).

18.5.2 Offsets Inside the Model Family Are Absorbed

Theorem 18.17 (Offset absorption by hardware-in-the-loop learning)**.** *Assume Eqs. (18.48) and (18.49), that $\mathbf{F}(\theta) \succ 0$ on Θ, that $p_{\text{data}} = p_{\theta^\star}$ for some $\theta^\star \in \Theta$, and that $\theta^\star$ lies in the image of M_{hw}. Consider the noiseless iteration ($M \to \infty$, so $\bar{\boldsymbol{\phi}}_n = \langle \boldsymbol{\phi} \rangle_{p_{M_{\text{hw}}(\theta_n)}}$).*

(i) ***Additive offsets.*** *If $M_{\text{hw}}(\theta) = \theta + \boldsymbol{\delta}$ and $\mathbf{P} = \mathbf{I}$, then Eq. (18.50) is exactly gradient descent on $\ell(\theta) := D_{\text{KL}}\Big(p_{\text{data}} \,\Big\|\, p_\theta^{\text{phys}}\Big)$, whose unique stationary point is*

$$\theta_\infty = \theta^\star - \boldsymbol{\delta}, \qquad p_{\theta_\infty}^{\text{phys}} = p_{\text{data}}. \tag{18.51}$$

(ii) ***General hardware maps.*** *If M_{hw} is a C^1 bijection, every fixed point of Eq. (18.50) satisfies $M_{\text{hw}}(\theta_\infty) = \theta^\star$, so again $p_{\theta_\infty}^{\text{phys}} = p_{\text{data}}$. The fixed point is locally attracting for all small enough η if and only if the eigenvalues of $\mathbf{P}\mathbf{F}(\theta^\star)\mathbf{J}$, with $\mathbf{J} := \mathrm{D}M_{\text{hw}}(\theta_\infty)$, have positive real parts. This holds in particular when $\mathbf{P} = \mathbf{I}$ and $\mathbf{J}$ is diagonal with positive entries (positive gain errors).*

Proof. (i) Write $\ln p_\theta^{\mathrm{phys}}(\mathbf{x}) = -\beta E_{\mathrm{base}}(\mathbf{x}) - \beta(\theta+\boldsymbol{\delta})^\top \boldsymbol{\phi}(\mathbf{x}) - \psi(\theta+\boldsymbol{\delta})$. Then

$$\ell(\theta) = \mathbb{E}[\ln p_{\mathrm{data}}]_{p_{\mathrm{data}}} + \beta \mathbb{E}[E_{\mathrm{base}}]_{p_{\mathrm{data}}} + \beta(\theta+\boldsymbol{\delta})^\top \langle \boldsymbol{\phi} \rangle_{p_{\mathrm{data}}} + \psi(\theta+\boldsymbol{\delta}).$$

Using $\nabla\psi(\vartheta) = -\beta \langle \boldsymbol{\phi} \rangle_{p_\vartheta}$ (Eq. (1.83)),

$$\nabla\ell(\theta) = \beta(\langle \boldsymbol{\phi} \rangle_{p_{\mathrm{data}}} - \langle \boldsymbol{\phi} \rangle_{p_{\theta+\boldsymbol{\delta}}}),$$

which is the bracket in Eq. (18.50) with the hardware moments in place of $\bar{\boldsymbol{\phi}}_n$. So the update is $\theta_{n+1} = \theta_n - \eta\nabla\ell(\theta_n)$. The Hessian is $\nabla^2\ell(\theta) = \mathbf{F}(\theta+\boldsymbol{\delta}) \succ 0$, so ℓ is strictly convex and has at most one stationary point. At $\theta = \theta^\star - \boldsymbol{\delta}$, $p_\theta^{\mathrm{phys}} = p_{\theta^\star} = p_{\mathrm{data}}$ and the gradient vanishes. This is the unique stationary point.

(ii) A fixed point requires $\langle \boldsymbol{\phi} \rangle_{p_{M_{\mathrm{hw}}(\theta)}} = \langle \boldsymbol{\phi} \rangle_{p_{\theta^\star}}$ because $\mathbf{P}$ is invertible. The moment map $\vartheta \mapsto \langle \boldsymbol{\phi} \rangle_{p_\vartheta} = -\beta^{-1}\nabla\psi(\vartheta)$ is injective, since ψ is strictly convex: if $\nabla\psi(\vartheta_1) = \nabla\psi(\vartheta_2)$ with $\vartheta_1 \neq \vartheta_2$, then $(\nabla\psi(\vartheta_1) - \nabla\psi(\vartheta_2)) \cdot (\vartheta_1 - \vartheta_2) = 0$, contradicting $\int_0^1 (\vartheta_1 - \vartheta_2)^\top \mathbf{F}(\vartheta_2 + s(\vartheta_1 - \vartheta_2))(\vartheta_1 - \vartheta_2)\, \mathrm{d}s > 0$. Hence $M_{\mathrm{hw}}(\theta_\infty) = \theta^\star$. For stability, let $\mathbf{u}_n = \theta_n - \theta_\infty$. By Eq. (1.84), $\langle \boldsymbol{\phi} \rangle_{p_{\theta^\star+\mathbf{v}}} = \langle \boldsymbol{\phi} \rangle_{p_{\mathrm{data}}} - \beta^{-1}\mathbf{F}(\theta^\star)\mathbf{v} + \mathcal{O}(\|\mathbf{v}\|^2)$, and $M_{\mathrm{hw}}(\theta_\infty + \mathbf{u}) = \theta^\star + \mathbf{J}\mathbf{u} + \mathcal{O}(\|\mathbf{u}\|^2)$, so

$$\mathbf{u}_{n+1} = \mathbf{u}_n - \eta\mathbf{P}\beta(\beta^{-1}\mathbf{F}\mathbf{J}\mathbf{u}_n) + \mathcal{O}(\|\mathbf{u}_n\|^2) = (\mathbf{I} - \eta\mathbf{P}\mathbf{F}\mathbf{J})\mathbf{u}_n + \mathcal{O}(\|\mathbf{u}_n\|^2).$$

For small η, the eigenvalues $1-\eta\lambda$ of $\mathbf{I}-\eta\mathbf{P}\mathbf{F}\mathbf{J}$ lie inside the unit circle if and only if every eigenvalue λ of $\mathbf{P}\mathbf{F}\mathbf{J}$ has positive real part. If $\mathbf{P} = \mathbf{I}$ and $\mathbf{J} = \mathrm{diag}(j_a)$ with $j_a > 0$, then $\mathbf{J}^{1/2}(\mathbf{F}\mathbf{J})\mathbf{J}^{-1/2} = \mathbf{J}^{1/2}\mathbf{F}\mathbf{J}^{1/2}$, which is symmetric positive definite, so $\mathbf{F}\mathbf{J}$ has real positive eigenvalues. □

The theorem says that the programmed parameters converge to the *wrong* values, $\theta^\star - \boldsymbol{\delta}$, in exactly the way needed for the realized distribution to be right. The host never estimates $\boldsymbol{\delta}$. Offline training on an ideal software model, followed by programming $\theta^\star$ into the die, would instead realize $p_{\theta^\star+\boldsymbol{\delta}}$. Its error is quantified next.

Proposition 18.18 (KL cost of uncalibrated offsets)**.** *Under the assumptions of Theorem 18.17(i),*

$$D_{\mathrm{KL}}(p_{\theta^\star} \,\|\, p_{\theta^\star+\boldsymbol{\delta}}) = \psi(\theta^\star+\boldsymbol{\delta}) - \psi(\theta^\star) - \nabla\psi(\theta^\star)\cdot\boldsymbol{\delta} = \frac{1}{2}\boldsymbol{\delta}^\top\mathbf{F}(\theta^\star)\boldsymbol{\delta} + \mathcal{O}(\|\boldsymbol{\delta}\|^3). \tag{18.52}$$

If $\boldsymbol{\delta}$ is random with mean zero and covariance $\boldsymbol{\Sigma}_\delta$, then

$$\mathbb{E}[D_{\mathrm{KL}}(p_{\theta^\star} \,\|\, p_{\theta^\star+\boldsymbol{\delta}})] = \frac{1}{2}\,\mathrm{tr}(\mathbf{F}(\theta^\star)\boldsymbol{\Sigma}_\delta) + \mathcal{O}(\mathbb{E}\,\|\boldsymbol{\delta}\|^3), \tag{18.53}$$

and for independent offsets ($\boldsymbol{\Sigma}_\delta = \mathrm{diag}(\sigma_a^2)$) this is $\frac{1}{2}\sum_a \sigma_a^2 F_{aa} = \frac{\beta^2}{2}\sum_a \sigma_a^2 \,\mathrm{Var}_{p_{\theta^\star}}(\phi_a)$.

Proof. From the expression for ℓ in the previous proof, with $\theta+\boldsymbol{\delta}$ replaced by $\theta^\star+\boldsymbol{\delta}$ and $p_{\mathrm{data}} = p_{\theta^\star}$,

$$D_{\mathrm{KL}}(p_{\theta^\star} \,\|\, p_{\theta^\star+\boldsymbol{\delta}}) = \mathbb{E}[\ln p_{\theta^\star} - \ln p_{\theta^\star+\boldsymbol{\delta}}]_{p_{\theta^\star}} = \beta\boldsymbol{\delta}^\top \langle \boldsymbol{\phi} \rangle_{p_{\theta^\star}} + \psi(\theta^\star+\boldsymbol{\delta}) - \psi(\theta^\star).$$

Since $\beta \langle \boldsymbol{\phi} \rangle_{p_{\theta^\star}} = -\nabla\psi(\theta^\star)$, this is the first equality in Eq. (18.52), the Bregman divergence of ψ. Taylor's theorem with $\nabla^2\psi = \mathbf{F}$ gives the second. Taking the expectation of $\frac{1}{2}\boldsymbol{\delta}^\top\mathbf{F}\boldsymbol{\delta} = \frac{1}{2}\,\mathrm{tr}(\mathbf{F}\boldsymbol{\delta}\boldsymbol{\delta}^\top)$ gives Eq. (18.53). □

Physics Insight: Mismatch measured in units of fluctuations

Proposition 18.18 says that an offset costs $\frac{1}{2}\boldsymbol{\delta}^\top \mathbf{F}\boldsymbol{\delta}$, with $\mathbf{F} = \beta^2 \operatorname{Cov}(\boldsymbol{\phi})$. The Fisher matrix converts a parameter error into the number of thermal standard deviations by which it shifts the conjugate statistic. For a Gaussian mean, $\mathbf{F} = \boldsymbol{\Sigma}^{-1}$ and the cost is half the squared offset in units of the thermal spread. For a p-bit field, the cost is weighted by $1 - m^2$, so a saturated bit is insensitive to field offsets in the same proportion that it is uninformative about its field. The same metric reappears in the convergence rate of calibration (Lemma 18.23) and in its noise floor (Eq. (18.58)). Offsets, learning speed, and sampling noise are all measured on one scale, set by the fluctuations of the sampler itself.

Gauge directions. If $\mathbf{F}$ is only positive semidefinite, for instance because the sufficient statistics are linearly dependent up to a constant, the components of $\boldsymbol{\delta}$ in $\ker \mathbf{F}$ change p_θ by nothing at all (they shift the energy by a configuration-independent constant, Proposition 1.2). The update Eq. (18.50) has no component along such directions, so these offsets are neither cancelled nor in need of cancellation. The theorem then holds on the quotient by $\ker \mathbf{F}$: $\theta_\infty + \boldsymbol{\delta} = \theta^\star$ up to an element of $\ker \mathbf{F}$. The vorticity $\mathbf{Q}$ of a NEAT-RN (Section 18.3) is a gauge direction of the same kind for any stationary loss. Offsets along it change the mixing rate but not the samples' distribution.

Example 18.19 (One p-bit with gain and offset)**.** Let a single p-bit realize the natural field $M_{\mathrm{hw}}(\eta) = g\eta + \epsilon$, where η is the programmed field, $g > 0$ a gain error from the field DAC's reference current, and ϵ a comparator offset. The noiseless update $\eta \leftarrow \eta + \varepsilon(m^\star - \tanh M_{\mathrm{hw}}(\eta))$ has its unique fixed point at $\tanh M_{\mathrm{hw}}(\eta_\infty) = m^\star$, that is

$$\eta_\infty = \frac{\eta^\star - \epsilon}{g}, \qquad \eta^\star = \operatorname{artanh} m^\star.$$

Linearizing, $\eta_{n+1} - \eta_\infty = (1 - \varepsilon g \operatorname{sech}^2 \eta^\star)(\eta_n - \eta_\infty)$, so the local rate is that of Worked computation 18.22 Step 6 with the step size multiplied by g. Neither g nor ϵ is ever estimated; with one programmable parameter and one statistic they could not be separated from the moment m alone, and they do not need to be. They become separable only if the bit is measured at several programmed fields, which is what a full characterization of the transfer curve would do and what in-loop calibration avoids.

Which errors are in the family. Table 18.3 sorts the mismatch mechanisms of Section 18.4 by how they enter the sampled distribution. The first four rows are covered by Theorem 18.17; the last three are not, and are treated next.

Mechanism	Effect on parameters	Status under calibration
p-bit comparator offset	additive field error $\delta\eta_i$	absorbed
Transconductor input offset	additive mean error $\delta\mu_i$	absorbed
Bias-current or mirror V_{th} mismatch	gain error $G_{ij}(1+\gamma_{ij})$	absorbed if $1+\gamma_{ij}>0$
Slow uniform temperature change	common gain $(1+\epsilon_T)\theta$	absorbed if slow
Transconductor saturation, non-logistic p-bit curve	new statistic $\chi(\mathbf{x})$	projected; residual Eq. (18.56)
Unintended parasitic coupling	statistic outside the graph	projected; residual Eq. (18.56)
$1/f$ and random-telegraph drift	time-dependent $\boldsymbol{\delta}(t)$	tracked with lag

Table 18.3: Classification of mismatch mechanisms by their effect on the sampled distribution and their fate under hardware-in-the-loop calibration.

A gain error with $1+\gamma_{ij}<0$ would flip the sign of a coupling and cannot occur for the lognormal errors of Eq. (18.46). A gain error that is very small, however, makes the corresponding direction poorly conditioned: by Theorem 18.17(ii) the local rate in that direction is proportional to $1+\gamma_{ij}$, and the DAC must supply a correspondingly larger programmed value.

18.5.3 What Calibration Cannot Fix

Theorem 18.17 has three hypotheses that real hardware violates to some degree. We treat each in turn.

Offsets outside the model family. Suppose the die realizes

$$p_\theta^{\text{phys}}(\mathbf{x}) \propto \exp\Big(-\beta\big[E_{\text{base}}(\mathbf{x}) + (\theta+\boldsymbol{\delta})^\top \boldsymbol{\phi}(\mathbf{x}) + \varepsilon\,\chi(\mathbf{x})\big]\Big), \tag{18.54}$$

where χ is a statistic the programmable parameters cannot reach. Examples are the quartic correction produced by transconductor saturation in a NEAT-RN (Proposition 13.11), a p-bit transfer curve that is not exactly logistic, or a parasitic coupling between units that the design does not connect.

Proposition 18.20 (Calibration converges to the KL projection)**.** *For the family Eq. (18.54) with $p_{\text{data}} = p_{\theta^\star}$, the fixed point of the noiseless update Eq. (18.50) is the minimizer over θ of the divergence $D_{\text{KL}}\Big(p_{\text{data}}\,\Big\|\,p_\theta^{\text{phys}}\Big)$, characterized by $\langle\boldsymbol{\phi}\rangle_{p_{\theta_\infty}^{\text{phys}}} = \langle\boldsymbol{\phi}\rangle_{p_{\text{data}}}$. To second order in ε,*

$$\theta_\infty + \boldsymbol{\delta} = \theta^\star - \varepsilon\ \mathrm{Cov}(\boldsymbol{\phi})^{-1}\,\mathrm{Cov}(\boldsymbol{\phi},\chi) + \mathcal{O}(\varepsilon^2), \tag{18.55}$$

$$D_{\text{KL}}\Big(p_{\text{data}}\,\Big\|\,p_{\theta_\infty}^{\text{phys}}\Big) = \frac{\beta^2\varepsilon^2}{2}\Big[\mathrm{Var}(\chi) - \mathrm{Cov}(\chi,\boldsymbol{\phi})\,\mathrm{Cov}(\boldsymbol{\phi})^{-1}\,\mathrm{Cov}(\boldsymbol{\phi},\chi)\Big] + \mathcal{O}(\varepsilon^3), \tag{18.56}$$

with all moments taken under p_{data}.

Proof. For fixed ε, Eq. (18.54) is a linearly parameterized family in θ with base energy $E_{\text{base}} + \varepsilon\chi$, so by Theorem 1.16(ii) and part (i) of Theorem 18.17, the update is gradient descent on the strictly convex $D_{\text{KL}}\Big(p_{\text{data}}\,\Big\|\,p_\theta^{\text{phys}}\Big)$, and its stationary point is characterized by moment matching. For the expansion, regard (ϑ,ε) with $\vartheta = \theta + \boldsymbol{\delta}$ as natural parameters of an extended exponential family

with statistics $(\boldsymbol{\phi}, \chi)$. By Proposition 18.18 applied to the extended family, with displacement $(\mathbf{a}, \varepsilon) := (\vartheta - \theta^\star, \varepsilon)$ from $(\theta^\star, 0)$,

$$D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_{\vartheta,\varepsilon}) = \frac{\beta^2}{2}\Big[\mathbf{a}^\top \operatorname{Cov}(\boldsymbol{\phi})\mathbf{a} + 2\varepsilon\, \mathbf{a}^\top \operatorname{Cov}(\boldsymbol{\phi}, \chi) + \varepsilon^2 \operatorname{Var}(\chi)\Big] + \mathcal{O}(\|(\mathbf{a}, \varepsilon)\|^3).$$

Minimizing the quadratic form over $\mathbf{a}$ gives $\mathbf{a} = -\varepsilon \operatorname{Cov}(\boldsymbol{\phi})^{-1} \operatorname{Cov}(\boldsymbol{\phi}, \chi)$, which is Eq. (18.55). Substituting back leaves the Schur complement of $\operatorname{Cov}(\boldsymbol{\phi})$ in the joint covariance of $(\boldsymbol{\phi}, \chi)$, which is Eq. (18.56). The minimizer of the quadratic approximation differs from the exact minimizer by $\mathcal{O}(\varepsilon^2)$, which changes the minimum value only at $\mathcal{O}(\varepsilon^3)$. □

The bracket in Eq. (18.56) is the variance of the part of χ that cannot be written as a linear combination of the programmable statistics, the residual of a linear regression of χ on $\boldsymbol{\phi}$. Calibration removes the projection of the error onto the model family and leaves the orthogonal part. Enlarging the family (adding programmable statistics correlated with χ) shrinks the residual.

Finite samples. The hardware moments are estimated from finitely many samples. Linearize around the fixed point with $\mathbf{P} = \mathbf{I}$ and additive offsets, and write $\bar{\boldsymbol{\phi}}_n = \langle \boldsymbol{\phi} \rangle_{p^{\mathrm{phys}}_{\theta_n}} + \boldsymbol{\xi}_n$ with independent zero-mean noise of covariance $\operatorname{Cov}(\boldsymbol{\phi})/M$ (for correlated samples, M is the effective sample size). The error $\mathbf{u}_n = \theta_n - \theta_\infty$ obeys

$$\mathbf{u}_{n+1} = (\mathbf{I} - \eta\mathbf{F})\mathbf{u}_n + \eta\beta\boldsymbol{\xi}_n, \qquad \operatorname{Cov}(\beta\boldsymbol{\xi}_n) = \frac{\mathbf{F}}{M}. \tag{18.57}$$

The stationary covariance $\mathbf{S}$ of $\mathbf{u}$ satisfies $\mathbf{S} = (\mathbf{I} - \eta\mathbf{F})\mathbf{S}(\mathbf{I} - \eta\mathbf{F}) + \eta^2\mathbf{F}/M$. A solution commuting with $\mathbf{F}$ satisfies $\mathbf{S}[\mathbf{I} - (\mathbf{I} - \eta\mathbf{F})^2] = \eta^2\mathbf{F}/M$, that is $\mathbf{S}\,\eta\mathbf{F}(2\mathbf{I} - \eta\mathbf{F}) = \eta^2\mathbf{F}/M$, so

$$\mathbf{S} = \frac{\eta}{M}(2\mathbf{I} - \eta\mathbf{F})^{-1}, \qquad \mathbb{E}[\ell(\theta_n)] \to \frac{1}{2}\operatorname{tr}(\mathbf{F}\mathbf{S}) = \frac{\eta}{2M}\operatorname{tr}[\mathbf{F}(2\mathbf{I} - \eta\mathbf{F})^{-1}] \approx \frac{\eta\,\operatorname{tr}\mathbf{F}}{4M}. \tag{18.58}$$

With the natural-gradient preconditioner $\mathbf{P} = \mathbf{F}^{-1}$ the same computation gives $\mathbf{u}_{n+1} = (1 - \eta)\mathbf{u}_n + \eta\mathbf{F}^{-1}\beta\boldsymbol{\xi}_n$, stationary covariance $\eta\mathbf{F}^{-1}/(M(2 - \eta))$, and excess KL

$$\mathbb{E}[\ell] \to \frac{P\,\eta}{2M(2 - \eta)}, \tag{18.59}$$

where P is the number of parameters. A constant step leaves a floor proportional to η/M; a decreasing step $\eta_n \propto 1/n$ removes it at the usual stochastic-approximation rate.

Incomplete mixing. If each batch is drawn from a chain run for only k steps from some initialization, the measured moments are those of the k-step law p_k, with bias $\mathbf{b}_k := \langle \boldsymbol{\phi} \rangle_{p_k} - \langle \boldsymbol{\phi} \rangle_{p^{\mathrm{phys}}_\theta}$. This is the contrastive-divergence bias of Eq. (1.111), now arising in the physical chain. Linearizing as above, the fixed point moves by $\mathbf{u} = \beta\mathbf{F}^{-1}\mathbf{b}_k$, costing $\frac{1}{2}\beta^2\mathbf{b}_k^\top\mathbf{F}^{-1}\mathbf{b}_k$ in KL. Since $|b_{k,a}| \le 2\,\|\phi_a\|_\infty \operatorname{TV}(p_k, p^{\mathrm{phys}}_\theta)$, the bias decays at the mixing rate of the physical chain. The offsets still enter consistently, because the chain that is sampled is the physical one. For the denoising steps of a DTM, which are strongly log-concave by Section 16.3, a few sweeps suffice and the bias is small. For a single deep multimodal EBM the bias can dominate, and calibration inherits the mixing problem of Chapter 7.

Drift. If the offsets change slowly, for instance as a random walk with $\mathbb{E}\left[|\delta_a(t+t_c)-\delta_a(t)|^2\right] = 2\mathcal{D}_\delta t_c$ over one calibration cycle of duration t_c, then in the natural-gradient iteration each coordinate obeys $u_{n+1} = (1-\eta)u_n + w_n$ with an extra drift input of variance $2\mathcal{D}_\delta t_c$. The stationary tracking variance from drift alone is $2\mathcal{D}_\delta t_c/(\eta(2-\eta))$. Small η averages out sampling noise but tracks drift poorly; large η does the opposite. Low-frequency $1/f$ fluctuations of V_{th} (Section 11.5) are cancelled to the extent that the calibration cycle is short compared with the time over which they wander by a significant fraction of the thermal scale. A uniform temperature change, which by Section 18.4 acts approximately as $\theta \mapsto (1+\epsilon_T)\theta$, is an in-family gain error and is absorbed the same way when it is slow.

18.5.4 Calibrating a Non-Equilibrium Sampler

The exponential-family argument does not cover a NEAT-RN parameterized by its transconductances, because the map from $\mathbf{G}$ to the stationary law is not of the form Eq. (18.48). The essential mechanism survives: what matters is that the update vanishes exactly when the *measured* statistics match the target.

Proposition 18.21 (Fixed points of model-based NESS calibration)**.** *Let the die realize the stationary covariance*

$$\boldsymbol{\Sigma}_{\text{hw}}(\theta) = \boldsymbol{\Sigma}(\mathbf{A}_{\text{phys}}(\theta), \mathbf{D}_{\text{phys}}(\theta))$$

with unknown mismatched matrices, and let the host hold a model $(\mathbf{A}_{\text{mod}}(\theta), \mathbf{D}_{\text{mod}}(\theta))$ *with* $\mathbf{A}_{\text{mod}}$ *Hurwitz. Take* $L = D_{\text{KL}}(\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}^\star) \,\|\, \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}))$ *and the update direction*

$$\mathbf{d}(\theta) := \operatorname{tr}(\boldsymbol{\Lambda}_{\text{mod}}\mathbf{R}_{\text{mod}}), \qquad \mathbf{A}_{\text{mod}}^\top\boldsymbol{\Lambda}_{\text{mod}} + \boldsymbol{\Lambda}_{\text{mod}}\mathbf{A}_{\text{mod}} + \mathbf{W}(\boldsymbol{\Sigma}_{\text{hw}}) = \mathbf{0},$$

where $\mathbf{R}_{\text{mod}}$ *is formed as in Eq. (18.33) but with* $\boldsymbol{\Sigma}_{\text{hw}}$ *in place of* $\boldsymbol{\Sigma}$*, and* $\mathbf{W}(\boldsymbol{\Sigma}) = \frac{1}{2}(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}^\star\boldsymbol{\Sigma}^{-1})$*. If* $\boldsymbol{\Sigma}_{\text{hw}}(\theta) = \boldsymbol{\Sigma}^\star$ *then* $\mathbf{d}(\theta) = 0$*. Conversely, if the adjustable parameters include every entry of a symmetric* $\mathbf{D}$*, then* $\mathbf{d}(\theta) = 0$ *implies* $\boldsymbol{\Sigma}_{\text{hw}} = \boldsymbol{\Sigma}^\star$*; if they include only the diagonal noise intensities* D_{ii}*, it implies only* $\operatorname{diag}\boldsymbol{\Lambda}_{\text{mod}} = \mathbf{0}$*.*

Proof. If $\boldsymbol{\Sigma}_{\text{hw}} = \boldsymbol{\Sigma}^\star$ then $\mathbf{W} = \mathbf{0}$, the adjoint equation has the unique solution $\boldsymbol{\Lambda}_{\text{mod}} = \mathbf{0}$, and $\mathbf{d} = 0$ for every parameter. Conversely, by Eq. (18.36) the components of $\mathbf{d}$ along the entries of $\mathbf{D}$ are $2(\boldsymbol{\Lambda}_{\text{mod}})_{ij}$. If all of them vanish, $\boldsymbol{\Lambda}_{\text{mod}} = \mathbf{0}$, and the adjoint equation gives $\mathbf{W} = \mathbf{0}$, which holds only at $\boldsymbol{\Sigma}_{\text{hw}} = \boldsymbol{\Sigma}^\star$. If only the diagonal entries are parameters, the same argument gives $\operatorname{diag}\boldsymbol{\Lambda}_{\text{mod}} = \mathbf{0}$, a weaker conclusion that must be combined with the drift components. □

Model errors therefore change the direction of each step but not the target of the loop, provided enough independent parameters are adjusted. Whether the loop actually converges to that target is a separate question of stability, answered positively near equilibrium by Proposition 18.12 and in general only for model errors small enough that $\mathbf{d}$ remains a descent direction, $\mathbf{d}\cdot\nabla_\theta L(\boldsymbol{\Sigma}_{\text{hw}}(\theta)) > 0$.

Hardware Realization: Calibration as a by-product of training

The training procedure for DTMs [56] estimates the negative phase of the contrastive gradient from samples. When those samples are drawn from the physical sampler rather than from a software model, Theorem 18.17 applies to every denoising step: each step's parameters converge to values that compensate that step's offsets. The same loop with a target supplied by a software model of the intended distribution (for instance a `THRML` simulation [31]) is a calibration procedure. The guarantees depend on three conditions: that the mismatch is

static on the calibration time scale, that it lies inside the programmable family, and that the hardware chain mixes within the sampling window. The first two are properties of the circuit design; the third is what the DTM construction of Chapter 16 is designed to provide.

18.6 Worked Computation: A Mismatch Budget and Its Calibration

Worked computation 18.22 (Pelgrom mismatch in a subthreshold sampler array and convergence of in-loop calibration)**.** We compute, for both the double-well sampler (as an array of p-bits) and the RC sampler (as an array of Gaussian nodes), how far an uncalibrated array of subthreshold stochastic units is from its target distribution, and how fast and how far hardware-in-the-loop calibration brings it back. All device numbers are representative order-of-magnitude values, not specifications of any particular chip.

Step 1: Device assumptions. We take $A_{V_{\mathrm{th}}} = 3\,\mathrm{mV}\,\mu\mathrm{m}$ (pair), slope factor $n = 1.3$, and $T = 300\,\mathrm{K}$, so $nV_T = 33.6\,\mathrm{mV}$. Input differential pairs and current mirrors use $WL = 0.25\,\mu\mathrm{m}^2$. By Eq. (18.42), both the input-referred offset of a pair and the threshold difference within a mirror have standard deviation

$$\sigma_{\mathrm{os}} = \frac{3\,\mathrm{mV}\,\mu\mathrm{m}}{\sqrt{0.25\,\mu\mathrm{m}^2}} = 6\,\mathrm{mV}.$$

A mirror copy error is therefore lognormal with $\sigma_{\ln I} = \sigma_{\mathrm{os}}/(nV_T) = 6/33.6 = 0.18$, and to first order a programmed coupling becomes $G(1+\gamma)$ with $\gamma \sim \mathcal{N}(0, \sigma_\gamma^2)$, $\sigma_\gamma = 0.18$, $\sigma_\gamma^2 = 0.032$. All offsets are independent from unit to unit and from parameter to parameter, so $\mathbf{\Sigma}_\delta$ is diagonal and Eq. (18.53) reduces to $\frac{1}{2}\sum_a \sigma_a^2 F_{aa}$.

Step 2: Double-well sampler, a single p-bit, exactly. Use natural coordinates $\eta = \beta h$, so that $p_\eta(s) = e^{\eta s}/(2\cosh\eta)$ for $s \in \{-1, +1\}$ (Example 1.6). The log-partition function is $\ln(2\cosh\eta)$, with derivatives $\tanh\eta = m$ and $\operatorname{sech}^2\eta = 1 - m^2 = \mathrm{Var}(s)$. The comparator offset gives, by Eq. (18.47), an additive natural-field error $\epsilon = -V_{\mathrm{os}}/V_s$. With a representative transfer scale $V_s = 50\,\mathrm{mV}$, ϵ has standard deviation $\sigma_h = 6/50 = 0.12$. By Eq. (18.52), written in the natural parameter η (for which $p_\eta \propto e^{+\eta s}$ and the log-partition function is $\ln 2\cosh\eta$),

$$D_{\mathrm{KL}}(p_\eta \,\|\, p_{\eta+\epsilon}) = \ln\cosh(\eta+\epsilon) - \ln\cosh\eta - \epsilon\tanh\eta = \frac{\epsilon^2}{2}\operatorname{sech}^2\eta - \frac{\epsilon^3}{3}\operatorname{sech}^2\eta\tanh\eta + \mathcal{O}(\epsilon^4), \quad (18.60)$$

where the cubic coefficient uses $\frac{\mathrm{d}}{\mathrm{d}\eta}\operatorname{sech}^2\eta = -2\operatorname{sech}^2\eta\tanh\eta$. For an unbiased bit ($\eta = 0$) the exact expression is $\ln\cosh\epsilon = \frac{1}{2}\epsilon^2 - \frac{1}{12}\epsilon^4 + \dots$, and averaging over $\epsilon \sim \mathcal{N}(0, \sigma_h^2)$ with $\mathbb{E}[\epsilon^4] = 3\sigma_h^4$ gives

$$\mathbb{E}[D_{\mathrm{KL}}(p_0 \,\|\, p_\epsilon)] = \frac{\sigma_h^2}{2} - \frac{\sigma_h^4}{4} + \mathcal{O}(\sigma_h^6) = 0.00720 - 0.00005 = 0.0071\,\mathrm{nat}.$$

In terms of a marginal, the offset moves $p(s=+1) = \frac{1}{2}(1+\tanh\epsilon)$ from 0.50 to about 0.56 for a one-standard-deviation unit.

Step 3: Double-well sampler, a coupled p-bit array. Now take N p-bits on a sparse graph of degree $d = 4$, zero programmed fields, and uniform couplings with $\beta J = 0.2$, below the mean-field ordering threshold $\beta J d = 1$. The natural parameters are $\eta_i = \beta h_i$ and $\eta_{ij} = \beta J_{ij}$, with statistics s_i and $s_i s_j$. Field offsets have $\sigma_h = 0.12$ as in Step 2. Coupling gain errors give $\delta\eta_{ij} = \beta J_{ij}\gamma_{ij}$, with standard deviation $0.2 \times 0.18 = 0.036$. The diagonal Fisher entries are $F_{ii} = \mathrm{Var}(s_i) = 1 - m_i^2 = 1$

(by the $s \to -s$ symmetry at zero field) and $F_{ij,ij} = \mathrm{Var}(s_i s_j) = 1 - \langle s_i s_j \rangle^2$. At this coupling the leading high-temperature estimate $\langle s_i s_j \rangle \approx \tanh(\beta J) = 0.197$ is adequate. By Eq. (18.53),

$$\begin{aligned}\mathbb{E}[D_{\mathrm{KL}}] &\approx \frac{1}{2}\sum_{i=1}^{N} \sigma_h^2 (1 - m_i^2) + \frac{1}{2}\sum_{(ij)} (\beta J_{ij})^2 \sigma_\gamma^2 (1 - \langle s_i s_j \rangle^2) \\ &= N\left[\frac{0.0144}{2} + \frac{d}{2}\cdot\frac{1}{2}(0.04)(0.032)(1 - 0.039)\right] = N\,[0.0072 + 0.0012] = 0.0084\,N \text{ nat},\end{aligned} \tag{18.61}$$

using $Nd/2$ edges. Field offsets dominate coupling errors by a factor of six: an additive offset acts at full strength, while a gain error is scaled by the small coupling it multiplies. For an array of 2.5×10^5 units, $\mathbb{E}[D_{\mathrm{KL}}] \approx 0.0084 \times 2.5 \times 10^5 \approx 2 \times 10^3$ nats. Per unit the error is small, but the joint distribution of the whole array is far from the target. By the Chernoff–Stein lemma, the error probability of the best test between the two distributions decays as $e^{-nD_{\mathrm{KL}}}$ in the number n of sampled configurations, so roughly $1/D_{\mathrm{KL}}$ configurations suffice to tell them apart. Here that number is far less than one: a single sampled configuration of the array already gives the mismatch away.

Step 4: RC sampler, an array of Gaussian nodes. Consider N weakly coupled Gaussian nodes of a NEAT-RN with target law $\mathcal{N}(\boldsymbol{\mu}^\star, \boldsymbol{\Sigma}^\star)$ and, for simplicity of the estimate, $\boldsymbol{\Sigma}^\star \approx \sigma_V^2 \mathbf{I}$ with thermal standard deviation $\sigma_V = 2\,\mathrm{mV}$ (Section 18.4). Input offsets of the transconductors shift each node's mean by $\delta\mu_i$ with standard deviation $\sigma_{\mathrm{os}} = 6\,\mathrm{mV}$. Bias-current mismatch multiplies each self-conductance, and hence each diagonal precision, by $(1 + \gamma_i)$, so the realized variance is $\sigma_V^2/(1 + \gamma_i)$. For a single node,

$$D_{\mathrm{KL}}\Big(\mathcal{N}(\mu^\star, \sigma_V^2) \,\Big\|\, \mathcal{N}(\mu^\star + \delta\mu, \tfrac{\sigma_V^2}{1+\gamma})\Big) = \frac{1}{2}\left[r - 1 - \ln r + r\,\frac{\delta\mu^2}{\sigma_V^2}\right], \qquad r := 1 + \gamma,$$

from the standard formula $D_{\mathrm{KL}}(\mathcal{N}(\mu_1, s_1^2) \,\|\, \mathcal{N}(\mu_2, s_2^2)) = \frac{1}{2}[s_1^2/s_2^2 - 1 - \ln(s_1^2/s_2^2) + (\mu_1 - \mu_2)^2/s_2^2]$ (Chapter B). Expanding $r - 1 - \ln r = \frac{1}{2}\gamma^2 + \mathcal{O}(\gamma^3)$ and averaging,

$$\mathbb{E}[D_{\mathrm{KL}}] \approx N\left[\frac{1}{2}\frac{\sigma_{\mathrm{os}}^2}{\sigma_V^2} + \frac{\sigma_\gamma^2}{4}\right] = N\left[\frac{1}{2}\cdot 9 + \frac{0.032}{4}\right] = N\,[4.5 + 0.008]\,\text{nat}. \tag{18.62}$$

The mean offsets dominate by a factor of about 500. They are three thermal standard deviations, so an uncalibrated node spends most of its time in a region the target distribution rarely visits. The error depends on the ratio $\sigma_{\mathrm{os}}/\sigma_V$ only, which explains why a design that enlarges the thermal swing (by injecting extra shot noise, Section 13.5) is less sensitive to mismatch, at the price of a larger voltage range.

Step 5: Calibrating the Gaussian means. The mean of a Gaussian node is controlled by a programmed bias b_i (the DC operating point $\mathbf{v}^*$ of Section 13.5), and the realized mean is $b_i + \delta\mu_i$. The natural-gradient form of Eq. (18.50) for mean parameters, with Fisher matrix $\boldsymbol{\Sigma}^{-1}$ and preconditioner $\boldsymbol{\Sigma}$, is simply

$$b_i \leftarrow b_i + \eta(\mu_i^\star - \bar{x}_i), \tag{18.63}$$

where $\bar{x}_i$ is the sample mean of node i over M samples. It is local and requires no knowledge of $\boldsymbol{\Sigma}$. With $e_{i,n} := b_{i,n} + \delta\mu_i - \mu_i^\star$ the realized mean error,

$$e_{i,n+1} = (1 - \eta)\, e_{i,n} + \eta\, \xi_{i,n}, \qquad \xi_{i,n} \sim \mathcal{N}\Big(0, \frac{\Sigma_{ii}}{M}\Big).$$

Without sampling noise and with $\eta = 1$, the error vanishes after one step: the loop measures the offset and subtracts it. With noise, the stationary covariance from Eq. (18.59) gives an excess KL of $N\eta/(2M(2-\eta))$; for $\eta = 1$ and $M = 1000$ effective samples per node this is $5 \times 10^{-4}\, N$ nats, a factor of 10^4 below Eq. (18.62). The plain gradient update $b \leftarrow b - \eta\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}^\star)$ instead contracts the error by $\mathbf{I} - \eta\boldsymbol{\Sigma}^{-1}$; with the best constant step $\eta = 2/(\lambda_{\min}(\boldsymbol{\Sigma}^{-1}) + \lambda_{\max}(\boldsymbol{\Sigma}^{-1}))$ its rate is $(\kappa-1)/(\kappa+1)$ with κ the condition number of $\boldsymbol{\Sigma}$. The local update Eq. (18.63) is both simpler and faster.

DAC floor. If b_i is set by a DAC with step Δ, the realized mean error after calibration is at least the rounding residual, uniform on $[-\Delta/2, \Delta/2]$, contributing $\frac{1}{2}(\Delta^2/12)/\sigma_V^2$ nats per node. Matching the sampling floor 5×10^{-4} requires $\Delta^2 \le 12\times2\times5\times10^{-4}\times\sigma_V^2 = 0.012\,\sigma_V^2$, so $\Delta \lesssim 0.11\,\sigma_V \approx 0.2\,\mathrm{mV}$. Covering $\pm4\sigma_{\mathrm{os}} = \pm24\,\mathrm{mV}$ of offsets at this resolution takes $48/0.2 = 240$ levels, about 8 bits of DAC range devoted to calibration alone.

Step 6: Calibrating p-bit fields, with an explicit rate. For an isolated p-bit with target natural field $\eta^\star$ and offset ϵ, the update Eq. (18.50) with step ε in natural units is $\eta \leftarrow \eta + \varepsilon(m^\star - \bar{s})$. Let $u_n := \eta_n + \epsilon - \eta^\star$ be the realized field error. In the noiseless limit,

$$u_{n+1} = u_n - \varepsilon\big[\tanh(\eta^\star + u_n) - \tanh\eta^\star\big] = (1 - \varepsilon\,\mathrm{sech}^2\,\xi_n)u_n,$$

for some ξ_n between $\eta^\star$ and $\eta^\star + u_n$, by the mean value theorem. For $0 < \varepsilon \le 1$, the factor lies in $[0, 1)$ because $0 < \mathrm{sech}^2 \le 1$, so $|u_n|$ decreases monotonically and $|\xi_n| \le |\eta^\star| + |u_0|$. Since sech^2 decreases in $|\xi|$,

$$|u_n| \le \rho^n\,|u_0|\,, \qquad \rho := 1 - \varepsilon\,\mathrm{sech}^2(|\eta^\star| + |u_0|) < 1, \tag{18.64}$$

which is linear convergence with an explicit rate, and the KL error Eq. (18.60) decays as ρ^{2n}.

Two cases make this concrete. For an unbiased target ($\eta^\star = 0$) and $\varepsilon = 1$, the iteration is $u_{n+1} = u_n - \tanh u_n = \frac{1}{3}u_n^3 + \mathcal{O}(u_n^5)$, which is Newton's method because $\mathrm{sech}^2 0 = 1$ is the exact Fisher information. Starting from a one-standard-deviation offset $u_0 = 0.12$: $u_1 = 5.7 \times 10^{-4}$ and $u_2 = 6\times10^{-11}$. With sampling noise, the first iteration already reaches the floor of Eq. (18.59). For a strongly biased target, $\eta^\star = 1.5$ ($m^\star = 0.905$), the bound gives $\rho = 1 - \mathrm{sech}^2(1.62) = 0.855$ with $\varepsilon = 1$, so about 29 iterations reduce $|u|$ a hundredfold; the local rate is $1 - \mathrm{sech}^2(1.5) = 0.82$. The slowdown arises because a saturated bit carries little information about its field: the Fisher information $1 - m^{\star 2} = 0.18$ is small. Rescaling the step by the inverse Fisher information, $\varepsilon = 1/(1 - m^{\star 2}) = 5.5$, restores Newton-like local convergence, but the global bound Eq. (18.64) no longer applies and the step must be limited for large initial errors. Figure 18.2 shows the three cases.

Step 7: The coupled case and the condition number of F. For the coupled array of Step 3, the loss $\ell(\theta) = D_{\mathrm{KL}}(p_{\theta^\star} \,\|\, p_{\theta+\boldsymbol{\delta}})$ has Hessian $\mathbf{F}(\theta + \boldsymbol{\delta})$, so the general rate follows from strong convexity.

Lemma 18.23 (Linear convergence of calibration)**.** *Let K be the sublevel set $\{\theta : \ell(\theta) \le \ell(\theta_0)\}$, assumed bounded, and let $\lambda\mathbf{I} \preceq \mathbf{F}(\theta + \boldsymbol{\delta}) \preceq \Lambda_{\mathrm{F}}\mathbf{I}$ on K with $\lambda > 0$. Then the noiseless update Eq. (18.50) with $\mathbf{P} = \mathbf{I}$ and $0 < \eta \le 1/\Lambda_{\mathrm{F}}$ satisfies*

$$\ell(\theta_n) \le (1 - \eta\lambda)^n\,\ell(\theta_0). \tag{18.65}$$

Proof. K is convex because ℓ is convex. By Theorem 18.17(i) the update is $\theta_{n+1} = \theta_n - \eta\mathbf{g}_n$ with $\mathbf{g}_n = \nabla\ell(\theta_n)$. Taylor's theorem with the upper Hessian bound gives, as long as the segment stays in K,

$$\ell(\theta_{n+1}) \le \ell(\theta_n) - \eta\,\|\mathbf{g}_n\|^2 + \frac{\Lambda_{\mathrm{F}}\eta^2}{2}\,\|\mathbf{g}_n\|^2 \le \ell(\theta_n) - \frac{\eta}{2}\,\|\mathbf{g}_n\|^2\,,$$

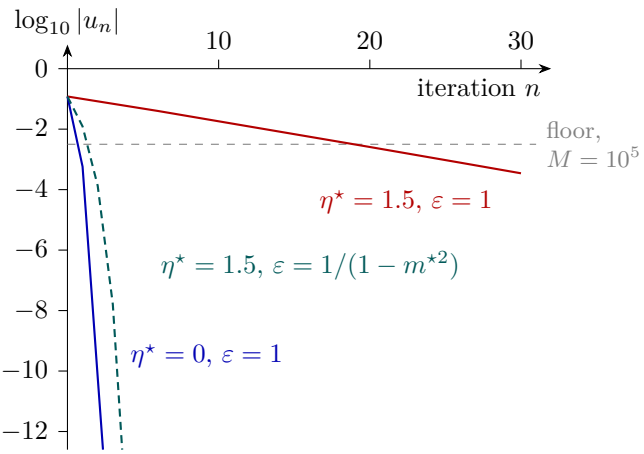


Figure 18.2: Noiseless calibration of a single p-bit field from an initial realized error $u_0 = 0.12$ (one standard deviation of the offset in Worked computation 18.22). An unbiased bit converges cubically with unit step. A saturated bit ($m^\star = 0.905$) converges linearly with rate 0.82 under a unit step, and quadratically under the inverse-Fisher step. The dashed line marks the sampling floor of $|u|$ for $M = 10^5$ samples per iteration and unit Fisher information; with noise, all curves stop there.

so ℓ decreases. The iterates cannot leave K: along the segment from θ_n to θ_{n+1} the same bound shows ℓ stays strictly below $\ell(\theta_n) \le \ell(\theta_0)$, so the segment never reaches the boundary of K. The lower Hessian bound gives, for the minimizer θ_∞ (where $\ell = 0$ by realizability),

$$0 = \ell(\theta_\infty) \ge \ell(\theta_n) + \mathbf{g}_n \cdot (\theta_\infty - \theta_n) + \frac{\lambda}{2}\,\|\theta_\infty - \theta_n\|^2 \ge \ell(\theta_n) - \frac{\|\mathbf{g}_n\|^2}{2\lambda},$$

where the last step minimizes the quadratic in $\theta_\infty - \theta_n$. Hence $\|\mathbf{g}_n\|^2 \ge 2\lambda\,\ell(\theta_n)$, and substituting into the descent inequality gives $\ell(\theta_{n+1}) \le (1 - \eta\lambda)\ell(\theta_n)$. □

With $\eta = 1/\Lambda_{\mathrm{F}}$ the rate is $1 - 1/\kappa_{\mathrm{F}}$, where $\kappa_{\mathrm{F}} = \Lambda_{\mathrm{F}}/\lambda$ is the condition number of the Fisher matrix. For the field block at zero magnetization, $\mathbf{F}$ is the susceptibility matrix $\mathrm{Cov}(s_i, s_j)$. A rough mean-field (random-phase) estimate is $\mathrm{Cov}(\mathbf{s}) \approx (\mathbf{I} - \beta J\,\mathbf{A}_{\mathrm{g}})^{-1}$, with $\mathbf{A}_{\mathrm{g}}$ the adjacency matrix of the graph, whose eigenvalues lie in $[-d, d]$. Then $\kappa_{\mathrm{F}} \approx (1 + \beta J d)/(1 - \beta J d) = 1.8/0.2 = 9$ for $\beta J = 0.2$ and $d = 4$, and about $\ln 100/\ln(9/8) \approx 39$ iterations reduce ℓ a hundredfold. The estimate is crude, but its structure is right: κ_{F} is the ratio of the largest to the smallest susceptibility, and it diverges as the couplings approach an ordering transition. Calibration slows down near criticality for the same reason that sampling does. A diagonal preconditioner $\mathbf{P} = \mathrm{diag}(1/F_{aa})$ removes the saturation slowdown of Step 6 but not this collective one.

Step 8: The budget. Table 18.4 collects the numbers. Comparing Eq. (18.53) with the natural-gradient floor Eq. (18.59) at $\eta = 1$ gives a simple rule for when calibration pays:

$$\frac{\text{uncalibrated } \mathbb{E}[D_{\mathrm{KL}}]}{\text{calibrated floor}} = \frac{\frac{1}{2}\,\mathrm{tr}(\mathbf{F}\boldsymbol{\Sigma}_\delta)}{P/(2M)} = M\,\overline{\sigma_{\mathrm{F}}^2}, \qquad \overline{\sigma_{\mathrm{F}}^2} := \frac{1}{P}\,\mathrm{tr}(\mathbf{F}\boldsymbol{\Sigma}_\delta), \tag{18.66}$$

the number of samples per iteration times the mean squared offset measured in units of the Fisher metric. For the Gaussian nodes of the RC sampler the offsets are large, $\overline{\sigma_{\mathrm{F}}^2} = (\sigma_{\mathrm{os}}/\sigma_V)^2 = 9$, and $M = 10^3$ already gives a reduction by 9×10^3. For the p-bit field offsets of the double-well sampler they are small, $\overline{\sigma_{\mathrm{F}}^2} = \sigma_h^2 = 0.0144$, and $M = 10^3$ gives only a factor of 14; a thousandfold reduction

needs $M \approx 7 \times 10^4$ samples per iteration. Calibration is worth most where mismatch is largest compared with the thermal fluctuations, and a unit whose offset is already small compared with $M^{-1/2}$ in Fisher units gains nothing from it. After calibration the residual is set by sampling noise and DAC resolution, by the out-of-family part of Proposition 18.20, which no choice of programmable parameters can remove, and by drift between calibration cycles.

	Double-well sampler (p-bit array)	RC sampler (Gaussian nodes)
Dominant error	field offset, $\sigma_h = 0.12$	mean offset, $\sigma_{\mathrm{os}}/\sigma_V = 3$
Uncalibrated $\mathbb{E}[D_{\mathrm{KL}}]$ per unit	0.0084 nat	4.5 nat
Noiseless iterations, $\lvert u\rvert$ reduced 100×	2 (unbiased) to ~ 30 (saturated)	1 ($\eta = 1$)
Sampling floor, $M = 1000$, $\eta = 1$	$\approx 5 \times 10^{-4}$ nat per parameter	5×10^{-4} nat
DAC floor	$\frac{1}{24}\Delta_\eta^2(1 - m^2)$	$\frac{1}{24}\Delta^2/\sigma_V^2$
Reduction factor Eq. (18.66), $M = 10^3$	14	9×10^3

Table 18.4: Mismatch and calibration budget of Worked computation 18.22, with $A_{V_{\mathrm{th}}} = 3\,\mathrm{mV}\,\mu\mathrm{m}$, $WL = 0.25\,\mu\mathrm{m}^2$, $nV_T = 33.6\,\mathrm{mV}$, $V_s = 50\,\mathrm{mV}$, $\sigma_V = 2\,\mathrm{mV}$. Representative values only.

18.7 Looking Back at the Two Samplers

The book followed two model systems from first principles to silicon. This section records briefly what each one taught.

The RC sampler began in Example 1.5 as a network of capacitors and resistors in contact with a heat bath, whose equilibrium distribution is a multivariate Gaussian. The Itô calculus of Chapter 4 gave its trajectories a precise meaning, the Fokker–Planck theory of Chapter 6 described how its probability density relaxes, and the fluctuation–dissipation relations of Chapter 5 tied its noise to its dissipation. The passive version turned out to be too rigid to be useful as a programmable sampler: with physical Johnson noise its stationary covariance is $k_{\mathrm{B}}T\mathbf{C}^{-1}$, fixed by the capacitances alone. The active NEAT-RN of Chapter 13 escaped that limitation by breaking detailed balance, at the price of a steady entropy production and a covariance that is no longer the inverse of any energy's Hessian but the solution of a Lyapunov equation. In this chapter the same Lyapunov equation, differentiated once, gave the adjoint equation that trains the network, and its threshold-voltage offsets turned into a mean error that hardware-in-the-loop learning removes, in the absence of sampling noise, in a single step.

The double-well sampler began as a quartic double well and its two-state p-bit limit. Its barrier made it the book's model of metastability: Kramers' law (Chapter 7) gave the exponentially slow escape, Landauer's bound (Chapter 9) gave the minimal cost of resetting it, and the subthreshold latch and the Josephson parametron (Chapter 14) showed two physical ways to build it. The denoising thermodynamic models of Chapter 16 addressed the mixing problem that the barrier creates, by adding a quadratic confinement at every denoising step that keeps each conditional distribution strongly log-concave, and the architecture of Chapter 17 placed many such units on a die. In this chapter the p-bit's field offsets turned out to be its largest source of error, and calibration of a saturated bit turned out to be slow for the same reason that its Fisher information is small.

The two samplers met repeatedly, most directly in the hybrid potentials of Section 1.5, where continuous and discrete variables share one energy function. They also share the structure that this chapter relied on. In the equilibrium form of both, the parameters enter the energy linearly, so the log-partition function is convex, its gradient gives the moments, and its Hessian gives the fluctuations. That one fact underlies the contrastive learning rule, equilibrium propagation, the KL

cost of an offset, and the convergence of calibration. It is also why a thermodynamic computer can learn from its own fluctuations: the quantity it needs for the gradient is a moment of the distribution it is already sampling.

Several topics that bear on real devices were left out or treated only in passing: device aging and long-term drift beyond $1/f$ noise, correlated mismatch from layout-dependent effects, the noise and energy of the readout and DAC circuits that close the calibration loop, and the interaction of learning with the mixing problem in models that are not built from strongly log-concave steps. Each can be approached with the tools developed here, but none was needed to establish the main result of this chapter: a sampler trained on its own samples learns to compensate its own imperfections, to the extent that its programmable parameters can express the compensation.

Summary

- **Proved in this chapter:**
  - Pathwise and score-function (Girsanov) gradient estimators for diffusions with constant invertible noise, with exact means and variances for the Ornstein–Uhlenbeck circuit: bounded pathwise variance $3J'^2$, score-function variance growing as $\frac{3}{2}(T/\tau)J'^2$ at leading order (Eqs. (18.13) and (18.15)); the failure of Girsanov weights for parameters that set noise intensities (Remark 18.1).
  - Score functions of discrete Gibbs chains, with the local per-update form $\beta(s_i' - \tanh\beta h_i^{\mathrm{loc}})s_j$ for a p-bit coupling (Eq. (18.17)).
  - Finite-temperature equilibrium propagation as a Maxwell relation of the nudged free energy, and its zero-temperature limit at a nondegenerate minimum (Theorems 18.3 and 18.7); nudge bias as joint cumulants, $\mathcal{O}(\beta_{\mathrm{nudge}})$ one-sided and $\mathcal{O}(\beta_{\mathrm{nudge}}^2)$ symmetric, with optimal nudges $M^{-1/4}$ and $M^{-1/6}$ (Lemma 18.6); the local nudged-phase learning rule for p-bit couplings (Eq. (18.29)) and its relation to the clamped Boltzmann machine rule.
  - Equilibrium propagation measures the transposed linear response, so it is exact only for reciprocal networks (Proposition 18.8).
  - Covariance sensitivity and the adjoint Lyapunov equation for linear NESS samplers (Proposition 18.9); the non-equilibrium static response formula and its equilibrium limit (Theorem 18.10); local stability of covariance matching near equilibrium (Proposition 18.12); vanishing of all stationary gradients along the vorticity $\mathbf{Q}$ (Example 18.11).
  - Fixed points of model-based NESS calibration coincide with $\mathbf{\Sigma}_{\mathrm{hw}} = \mathbf{\Sigma}^\star$ when all noise intensities are adjustable (Proposition 18.21); the calibration gain rule $M\,\overline{\sigma_{\mathrm{F}}^2}$ (Eq. (18.66)).
  - Pelgrom's area law from delta-correlated local fluctuations, and the random-dopant estimate with depth weighting $\frac{1}{3}$ (Proposition 18.13 and Eq. (18.43)); lognormal subthreshold current mismatch (Eq. (18.45)).
  - Absorption of in-family offsets by hardware-in-the-loop learning, local stability for positive gain errors (Theorem 18.17); Bregman form of the offset KL cost (Proposition 18.18); KL projection and Schur-complement residual for out-of-family errors (Proposition 18.20); sampling-noise floors (Eqs. (18.58) and (18.59)); linear convergence under strong convexity (Lemma 18.23).

- **Assumed:**
  - The static fluctuation–response theorem and contrastive gradient (Theorems 1.13 and 1.16); Girsanov's theorem under Novikov's condition (Theorem 4.22); the Lyapunov characterization of the NEAT-RN steady state (Theorem 13.8) and its vorticity parameterization (Proposition 13.10).
  - Exponential convergence of the Fokker–Planck semigroup to a smooth positive steady state (used in Theorem 18.10).
  - The subthreshold EKV law in saturation and a charge-sheet picture of the depletion charge (Chapter 12); independence of local mismatch between devices and absence of long-range gradients after common-centroid layout.
  - Static or slowly drifting offsets, a hardware map that is a C^1 bijection, and target moments known exactly on the host.
- **Collected in the appendices:**
  - The Gaussian KL divergence, Isserlis' theorem, and Lyapunov-equation identities used here are in Chapter B; spectral-gap and semigroup convergence results in Chapter C; the full EKV expression and MOS electrostatics behind Eqs. (18.43) and (18.44) in Chapter D.
- **Open:**
  - Gradient estimators for strongly non-reciprocal networks that are both local and provably stable; a NESS analogue of equilibrium propagation that uses only snapshot moments.
  - Calibration when mismatch is not in the programmable family, beyond the quadratic estimate Eq. (18.56): which additional programmable statistics most reduce the residual for a given area budget.
  - Joint optimization of device area, DAC resolution, calibration sample count, and recalibration interval under $1/f$ drift, as a single energy-per-sample budget.
  - Calibration near ordering transitions, where the condition number of the Fisher matrix diverges and both sampling and learning slow down.

Bibliographic notes

Equilibrium propagation was introduced by Scellier and Bengio [94] for deterministic energy-based networks relaxing to a minimum; the finite-temperature form proved here follows from the static fluctuation–response relations of Chapter 1, which go back to linear-response theory [62]. The contrastive learning rule with clamped and free phases originates with the Boltzmann machine of Ackley et al. [1], and the short-chain approximation whose bias reappears in our discussion of incomplete mixing is the contrastive divergence of Hinton [50]. Girsanov's theorem and the stochastic calculus behind the pathwise and score-function estimators are treated in Øksendal [80] and Karatzas and Shreve [59]; for Fokker–Planck response theory see Risken [90]. The non-equilibrium response formula of Theorem 18.10 is due to Agarwal [2] and has been rediscovered many times in stochastic thermodynamics [97].

Threshold-voltage mismatch and its inverse-square-root area law were established experimentally and analyzed by Pelgrom et al. [86]. The subthreshold device model is that of Enz and Vittoz [29], with textbook treatments in Tsividis and McAndrew [111]; the consequences of mismatch for subthreshold analog design were emphasized early by Mead [75]. The non-equilibrium Gaussian samplers whose Lyapunov equation we differentiated are the NEAT-RNs of Freitas et al. [38], building on the stochastic circuit theory of Freitas et al. [37]. CMOS p-bits are reviewed in Camsari et al. [18]. Denoising thermodynamic models and their contrastive training are described by Jelinčič et al. [56], and the `THRML` library for simulating block-Gibbs sampling on such models by Extropic Corporation [31].

Part VI

Appendices

Appendix A

Global Notation, Physical Constants, and the Rosetta Dictionary

Thermodynamic computing draws on non-equilibrium statistical physics, probabilistic machine learning, and mesoscopic solid-state electronics, and the three communities use clashing notation and terminology. A symbol such as β means inverse temperature $(k_{\mathrm{B}}T)^{-1}$ in statistical mechanics, a variance schedule parameter in diffusion models, and a current-gain factor in transistor modeling; a "potential" may refer to an electrostatic node voltage V_i (measured in volts), a dimensionless log-probability $U(\mathbf{x}) = -\ln p(\mathbf{x})$ (measured in nats), or a Helmholtz free energy F (measured in joules or in units of $k_{\mathrm{B}}T$).

This appendix fixes the book's notation and tabulates physical constants and characteristic device scales at room temperature ($T = 300\,\mathrm{K}$) and at liquid-helium temperature ($T = 4.2\,\mathrm{K}$). It also gives a dictionary that translates the main concepts between the three disciplines, describes the two model systems that recur throughout the book, and lists the standing mathematical assumptions used in Chapters 1–18.

A.1 Fundamental Physical Constants and Characteristic Thermal Scales

All physical equations in this book are written in SI units unless explicitly nondimensionalized by the thermal energy scale $k_{\mathrm{B}}T$ or the thermal voltage scale $V_T = k_{\mathrm{B}}T/q_e$. Table A.1 lists the fundamental constants and the derived thermal scales that govern subthreshold CMOS and superconducting Josephson circuits.

Physics Insight: Why Node Capacitances of 0.1–2 fF Suit Thermodynamic Sampling

Inspect the equipartition root-mean-square voltage fluctuation $\sigma_V = \sqrt{k_{\mathrm{B}}T/C_{\mathrm{node}}}$ and the characteristic thermal electron number $\Omega = C_{\mathrm{node}}V_T/q_e$ in Table A.1. On a macroscopic breadboard capacitor ($C = 1\,\mu\mathrm{F}$), thermal voltage fluctuations at $T = 300\,\mathrm{K}$ have amplitude $\sigma_V \approx 64\,\mathrm{nV}$. This is negligible compared with the thermal voltage $V_T \approx 25.85\,\mathrm{mV}$, so the circuit behaves deterministically. Conversely, on a single-electron transistor island ($C \sim 1\,\mathrm{aF} = 10^{-18}\,\mathrm{F}$), $\Omega \approx 0.16 < 1$ and the charging energy $E_{\mathrm{C}} = q_e^2/(2C) \approx 80\,\mathrm{meV} > 3k_{\mathrm{B}}T$ freezes out continuous fluctuations via Coulomb blockade unless biased at degeneracy.

Table A.1: Fundamental physical constants and characteristic thermal computing scales.

Quantity	**Symbol**	**Value (SI / Practical Units)**	**Role in Thermodynamic Hardware**
Boltzmann constant	k_{B}	$1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$	Sets thermal fluctuation energy scale
		$8.617333 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$	
Elementary charge	q_e	$1.60217663 \times 10^{-19}\,\mathrm{C}$	Granularity of electron shot noise
Planck constant	h	$6.62607015 \times 10^{-34}\,\mathrm{J\,s}$	Sets quantum action scale
Reduced Planck constant	$\hbar = h/(2\pi)$	$1.05457182 \times 10^{-34}\,\mathrm{J\,s}$	Josephson phase-voltage coupling
Magnetic flux quantum	$\Phi_0 = h/(2q_e)$	$2.06783385 \times 10^{-15}\,\mathrm{Wb}$	Superconducting SQUID/QFP periodicity
Reduced flux quantum	$\varphi_0 = \hbar/(2q_e)$	$3.29105976 \times 10^{-16}\,\mathrm{Wb/rad}$	Josephson inductance $L_J = \varphi_0/I_{\mathrm{c}}$
Vacuum permittivity	ε_0	$8.8541878 \times 10^{-12}\,\mathrm{F\,m^{-1}}$	Gate oxide & substrate electrostatics
Silicon permittivity	$\varepsilon_{\mathrm{Si}} = 11.7\varepsilon_0$	$1.0359 \times 10^{-10}\,\mathrm{F\,m^{-1}}$	Depletion capacitance C_{dep}
SiO_2 permittivity	$\varepsilon_{\mathrm{ox}} = 3.9\varepsilon_0$	$3.4531 \times 10^{-11}\,\mathrm{F\,m^{-1}}$	Gate oxide capacitance C_{ox}
Room-Temperature Thermal Scales ($T = 300\,\mathrm{K}$, Subthreshold CMOS):			
Thermal energy (300 K)	$k_{\mathrm{B}}T_{300}$	$4.14195 \times 10^{-21}\,\mathrm{J} \approx 4.14\,\mathrm{zJ}$	Reference unit of thermodynamic work
		$25.852\,\mathrm{meV}$	Barrier height unit in weak inversion
Landauer bit-erasure bound	$k_{\mathrm{B}}T_{300}\ln 2$	$2.87098 \times 10^{-21}\,\mathrm{J} \approx 2.87\,\mathrm{zJ}$	Minimum heat to erase 1 bit at 300 K
Thermal voltage (300 K)	$V_T = k_{\mathrm{B}}T_{300}/q_e$	$25.852\,\mathrm{mV}$	Exponential $I_{\mathrm{D}}(V_{\mathrm{GS}})$ voltage scale
Natural subthreshold swing	$V_T \ln 10$	$59.526\,\mathrm{mV/dec}$	Ideal ($n = 1$) decade current swing
Actual CMOS swing ($n \approx 1.3$)	$nV_T \ln 10$	$\approx 77.4\,\mathrm{mV/dec}$	Practical weak-inversion gate swing
Equipartition voltage noise	$\sqrt{k_{\mathrm{B}}T/C}$	$64.36\,\mu\mathrm{V}$ (for $C = 1\,\mathrm{pF}$)	Root-mean-square thermal node noise
		$2.035\,\mathrm{mV}$ (for $C = 1\,\mathrm{fF}$)	Typical on-chip node (1 fF)
		$6.436\,\mathrm{mV}$ (for $C = 0.1\,\mathrm{fF}$)	Advanced-node 0.1 fF thermal swing
Thermal charge number	$\Omega = CV_T/q_e$	161.4 electrons (for $C = 1\,\mathrm{fF}$)	Van Kampen expansion parameter (Chapter 13)
Cryogenic Superconducting Scales ($T = 4.2\,\mathrm{K}$, Josephson QFP):			
Thermal energy (4.2 K)	$k_{\mathrm{B}}T_{4.2}$	$5.7987 \times 10^{-23}\,\mathrm{J} \approx 0.058\,\mathrm{zJ}$	71.4× lower thermal floor than 300 K
Landauer bound (4.2 K)	$k_{\mathrm{B}}T_{4.2}\ln 2$	$4.019 \times 10^{-23}\,\mathrm{J}$	Cryogenic reversible/stochastic limit
Josephson coupling energy	$E_{\mathrm{J}} = \varphi_0 I_{\mathrm{c}}$	$3.291 \times 10^{-22}\,\mathrm{J}$ (at $I_{\mathrm{c}} = 1\,\mu\mathrm{A}$)	$E_{\mathrm{J}}/(k_{\mathrm{B}}T_{4.2}) \approx 5.68$ at $I_{\mathrm{c}} = 1\,\mu\mathrm{A}$

In modern CMOS processes (65 nm down to 5 nm), typical node capacitances are $C_{\text{node}} \sim 0.1$–$2\,\text{fF}$. In this range, $\sigma_V = \sqrt{k_B T / C_{\text{node}}} \sim 1.5$–$6.5\,\text{mV}$ is a substantial fraction of the subthreshold exponential scale $V_T \approx 25.85\,\text{mV}$ (since $\sigma_V / V_T = \Omega^{-1/2} \sim 0.06$–$0.25$), while $\Omega \sim 16$–$320 \gg 1$ is still large enough that continuous Langevin/Fokker–Planck descriptions (with perturbative shot-noise corrections, Chapter 13) remain accurate.

A.2 Symbol Reference

Table A.2 lists the main symbols used in Parts I–V. A few letters are reused where the context makes the meaning clear: for example, μ denotes both a reference measure and a mobility, Ω both a configuration space and the thermal charge number $C_{\text{node}} V_T / q_e$, and $\mathbf{\Lambda}$ both a diagonal eigenvalue matrix and a precision matrix. Chapters that depart from the table say so.

Table A.2: Global mathematical and physical symbol conventions.

Symbol	Definition and Convention
1. Spaces, States, and Linear Algebra	
$\mathbb{R}^N, \mathbb{R}^d$	Continuous configuration space of dimension N or d
$\{-1,+1\}^N$	Discrete Ising / p-bit configuration space $\{-1,+1\}^N$
$\mathbf{x}, \mathbf{v}$	Continuous state vector / capacitor node voltage vector in $\mathbb{R}^N$
$\mathbf{s} = (s_1, \ldots, s_N)$	Binary spin / p-bit state vector with $s_i \in \{-1,+1\}$
$\mathbf{z}$	Auxiliary/latent hardware variable vector (continuous or discrete)
$\mathbf{I}$	Identity matrix in $\mathbb{R}^{N\times N}$
$\text{tr}(\mathbf{M}), \text{Tr}(\mathcal{O})$	Finite-dimensional matrix trace and operator trace
$\langle \mathbf{u}, \mathbf{v} \rangle$	Euclidean inner product $\mathbf{u}^\top \mathbf{v}$ or Hilbert space inner product
$\|\mathbf{x}\|, \|\mathbf{M}\|_{\text{F}}$	Euclidean ℓ_2 norm and Frobenius / Hilbert–Schmidt norm
2. Equilibrium Statistical Mechanics and Energy-Based Models	
$k_B, T, k_B T$	Boltzmann constant, bath temperature, thermal energy scale
$\beta \equiv (k_B T)^{-1}$	Inverse thermal energy (always $(k_B T)^{-1}$ unless subscripted)
$E_\theta(\mathbf{x})$	Parameterized physical Hamiltonian or energy function
$Z(\theta, \beta)$	Canonical partition function $\int \mathrm{d}\mu(\mathbf{x})\, e^{-\beta E_\theta(\mathbf{x})}$
$F(\theta, \beta)$	Equilibrium Helmholtz free energy $-\beta^{-1} \ln Z(\theta, \beta)$
$\mathcal{F}[q]$	Gibbs variational free energy functional $\langle E_\theta \rangle_q - TS[q]$
$S[q]$	Gibbs–Shannon entropy $-k_B \int \mathrm{d}\mu(\mathbf{x})\, q(\mathbf{x}) \ln q(\mathbf{x})$
$p_{\text{eq}}(\mathbf{x}), p_\theta(\mathbf{x})$	Equilibrium Boltzmann distribution $Z^{-1} e^{-\beta E_\theta(\mathbf{x})}$
$p_{\text{ss}}(\mathbf{x})$	Non-equilibrium steady-state (NESS) probability density
$p_{\text{data}}(\mathbf{x})$	Target empirical data distribution
$D_{\text{KL}}(q \,\|\, p)$	Kullback–Leibler divergence $\int \mathrm{d}\mu(\mathbf{x})\, q(\mathbf{x}) \ln(q(\mathbf{x})/p(\mathbf{x}))$
$\mathcal{I}(X;Y)$	Mutual information $D_{\text{KL}}(p(x,y) \,\|\, p(x)p(y))$
$\mathcal{L}_{\text{TC}}(q)$	Total Correlation $D_{\text{KL}}\left(q(\mathbf{z}) \,\middle\|\, \prod_{i=1}^N q(z_i)\right)$ (Chapter 17)
3. Stochastic Calculus, Langevin SDEs, and Fokker–Planck Theory	
$W_t, \mathbf{W}_t$	Standard 1D and ND Wiener process ($\mathrm{d}W_i\, \mathrm{d}W_j = \delta_{ij}\, \mathrm{d}t$)
$\mathbf{B}(\mathbf{x}) \cdot \mathrm{d}\mathbf{W}_t$	Itô (left-endpoint, non-anticipating) stochastic differential

Continued on next page

Table A.2 – continued from previous page

Symbol	Definition and Convention
$\mathbf{B}(\mathbf{x}) \circ \mathrm{d}\mathbf{W}_t$	Stratonovich (midpoint, chain-rule-preserving) differential
$\boldsymbol{\Gamma}, \boldsymbol{\mu} = \boldsymbol{\Gamma}^{-1}$	Damping (friction/resistance) and mobility (conductance) tensors
$\mathbf{D}(\mathbf{x})$	Diffusion tensor $\frac{1}{2}\mathbf{B}(\mathbf{x})\mathbf{B}(\mathbf{x})^\top$ (equals $k_\mathrm{B}T\boldsymbol{\mu}$ at equilibrium)
$\mathcal{L}_\mathrm{FP}$	Forward Fokker–Planck operator $\partial_t\rho = \mathcal{L}_\mathrm{FP}\rho = -\nabla\cdot\mathbf{J}$
$\mathcal{L}$	Backward Kolmogorov generator (adjoint of $\mathcal{L}_\mathrm{FP}$ in $L^2(\mathrm{d}\mathbf{x})$)
$\mathbf{J}(\mathbf{x}, t)$	Probability current vector $\mathbf{a}(\mathbf{x})\rho - \nabla\cdot(\mathbf{D}(\mathbf{x})\rho)$
$\mathbf{J}_\mathrm{eq}, \mathbf{J}_\mathrm{ss}$	Equilibrium current ($\equiv \mathbf{0}$) and NESS current ($\nabla\cdot\mathbf{J}_\mathrm{ss} = 0$)
$\mathcal{H}_\mathrm{W}, V_\mathrm{W}(\mathbf{x})$	Self-adjoint Witten Schrödinger operator $-p_\mathrm{eq}^{-1/2}\mathcal{L}_\mathrm{FP}p_\mathrm{eq}^{1/2}$ and potential
$\lambda_1, \tau_\mathrm{mix}$	Spectral gap of $\mathcal{H}_\mathrm{W}$ (or $-\mathcal{L}$) and mixing time $\tau_\mathrm{mix} \sim \lambda_1^{-1}$
4. Stochastic Thermodynamics and Optimal Transport	
$\mathcal{W}, \mathcal{Q}$	Trajectory work done on system and heat dissipated into bath
$\mathcal{W}_\mathrm{diss} \equiv \mathcal{W} - \Delta F$	Dissipated (irreversible) work along a finite-time protocol
$s_\mathrm{sys}, s_\mathrm{med}, s_\mathrm{tot}$	Trajectory system, medium ($\mathcal{Q}/T$), and total entropy production
$\Sigma_\mathrm{tot} \equiv \langle \Delta s_\mathrm{tot} \rangle$	Ensemble-averaged total entropy production (≥ 0 by Second Law)
$\mathcal{S}_\mathrm{OM}[\mathbf{x}_{[0,\tau]}]$	Onsager–Machlup path-integral action functional (Chapter 8)
$\mathcal{W}_2(\rho_0, \rho_1)$	L^2-Wasserstein optimal transport distance on $\mathcal{P}_2(\mathbb{R}^N)$ (Chapter 10)
5. Solid-State Device Physics, Subthreshold CMOS, and Superconductors	
$V_T \equiv k_\mathrm{B}T/q_e$	Thermal voltage ($\approx 25.85\,\mathrm{mV}$ at $300\,\mathrm{K}$)
V_th, n	MOSFET threshold voltage and subthreshold slope factor $1 + C_\mathrm{dep}/C_\mathrm{ox}$
$V_\mathrm{GS}, V_\mathrm{DS}, V_\mathrm{SB}$	Gate-source, drain-source, and source-bulk terminal voltages
I_D, I_0	MOSFET drain current and specific weak-inversion current scale
$g_\mathrm{m}, g_\mathrm{ds}$	Gate transconductance $\partial I_\mathrm{D}/\partial V_\mathrm{GS}$ and output conductance $\partial I_\mathrm{D}/\partial V_\mathrm{DS}$
$S_I(f), S_V(f)$	Single-sided current and voltage noise power spectral densities
$\mathbf{A} = -\mathbf{C}^{-1}\mathbf{G}$	Linearized drift Jacobian in a subthreshold CMOS NEAT-RN (Chapter 13)
$\boldsymbol{\Sigma}, \boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$	Stationary voltage covariance and precision matrices
Φ_0, φ	Superconducting flux quantum $h/(2q_e)$ and gauge-invariant Josephson phase
$I_\mathrm{c}, E_\mathrm{J}, E_\mathrm{C}, \beta_\mathrm{c}$	Critical current, Josephson energy, charging energy, Stewart–McCumber parameter

A.3 The Physics ↔ Machine Learning ↔ Circuit Rosetta Dictionary

A recurring theme of this book is that a physical circuit, a statistical-mechanical ensemble, and a probabilistic machine learning model are three descriptions of the same mathematical structure. Table A.3 translates between the three languages.

Table A.3: The Thermodynamic Computing Rosetta Dictionary across Statistical Physics, Probabilistic Machine Learning, and Subthreshold/Mesoscopic Hardware.

Statistical Physics	**Probabilistic ML / Generative AI**	**Circuit Hardware**
Microstate configuration $\mathbf{x} \in \mathbb{R}^N$ or $\mathbf{s} \in \{-1,+1\}^N$	Feature vector, token embedding, or latent variable $\mathbf{x}, \mathbf{z}$	Capacitor node voltages $\mathbf{v}(t)$, charges $\mathbf{q}(t)$, or p-bit latch outputs
Dimensionless Hamiltonian $\beta E_\theta(\mathbf{x})$	Unnormalized negative log-density (Energy-Based Model score potential)	Electrostatic + active transconductance pseudo-potential
Control parameters / external fields	Trainable weights, precision matrices, and biases $\theta = (\mathbf{W}, \mathbf{b})$	Programmable gate bias voltages $V_{G,k}$ and DAC-controlled tail currents
Canonical partition function $Z(\theta, \beta)$	Intractable normalizing constant $\int e^{-E_\theta(\mathbf{x})}\, \mathrm{d}\mathbf{x}$	Physical phase-space volume automatically normalized by conservation of probability
Helmholtz free energy $F = -k_\mathrm{B} T \ln Z$	Negative log-partition function / log-marginal likelihood $-\ln Z(\theta)$	Equilibrium thermodynamic potential of the open circuit
Gibbs variational free energy $\mathcal{F}[q] = \langle E \rangle_q - TS[q]$	Negative Evidence Lower Bound ($-$ELBO) / $D_\mathrm{KL}(q \,\|\, p_\theta) + F$	Instantaneous non-equilibrium Lyapunov functional of the circuit ensemble
Thermal relaxation $\partial_t \rho = \mathcal{L}_\mathrm{FP} \rho$	Variational inference / MCMC sampling trajectory	Natural RC + shot-noise relaxation of unclocked or chromatically clocked nodes
Static Fluctuation–Response $\frac{\partial \langle \phi_k \rangle}{\partial \theta_j} = -\beta \,\mathrm{Cov}(\phi_j, \phi_k)$	Fisher information matrix / natural gradient metric tensor	Cross-covariance between node voltages/currents under small bias nudges
Clamped vs. free ensemble expectation	Positive phase $\langle \nabla_\theta E \rangle_{p_\mathrm{data}}$ vs. Negative phase $\langle \nabla_\theta E \rangle_{p_\theta}$	Input-clamped node relaxation vs. free-running thermal sampling
Spin-glass frustration ($\prod J_{ij} < 0$) and RSB barriers $\Delta F \sim N^\psi$	Multimodal mode collapse / slow MCMC mixing between isolated data modes	Metastable circuit latch states trapped by Kramers barrier $\tau_\mathrm{mix} \sim e^{\Delta V / V_T}$
Kramers escape rate $k_{A\to B} \propto e^{-\beta \Delta E}$	Rare transition probability between modes in Langevin MCMC	Thermally activated switching rate of a bistable p-bit or Josephson QFP
Non-equilibrium steady state (NESS) with $\mathbf{J}_\mathrm{ss} \neq \mathbf{0}$	Non-reversible lifted MCMC sampler / asymmetric drift diffusion	Subthreshold CMOS NEAT-RN with non-reciprocal transconductances $G_{ij} \neq G_{ji}$
Continuous-time Lyapunov equation $\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top + 2\mathbf{D} = \mathbf{0}$	Stationary covariance of a multivariate Gaussian graphical model	Balance between RC transconductance decay $\mathbf{A} = -\mathbf{C}^{-1}\mathbf{G}$ and shot noise $\mathbf{D}$
Forward Ornstein–Uhlenbeck thermalization	Forward diffusion noising process $q(\mathbf{x}_t \mid \mathbf{x}_0)$	Passive RC charge sharing / thermal noise injection toward $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$

Statistical Physics	**Probabilistic ML / Generative AI**	**Circuit Hardware**
Anderson reverse-time drift $+\nabla \ln p_t(\mathbf{x})$	Learned score network $\mathbf{s}_\theta(\mathbf{x}_t, t)$ in DDPM / Score SDEs	Conditional EBM relaxation $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ in a Denoising Thermodynamic Model
Bakry–Émery log-concavity $\nabla^2 E_{\text{tot}} \succeq \kappa \mathbf{I}$	Fast-mixing strongly log-concave conditional posterior	Quadratic forward coupling α_t larger than the negative curvature of the learned energy, removing Kramers barriers in the DTCA
Total Correlation $\mathcal{L}_{\text{TC}} = D_{\text{KL}}(q(\mathbf{z}) \,\|\, \prod_i q(z_i))$	Disentangled / factorized latent representation penalty	Spatial correlation-length bound $\xi \ll L_{\text{chip}}$ enabling local 2D silicon routing
Quenched random field disorder δh_i	Hardware weight/bias perturbation noise	Pelgrom threshold-voltage mismatch $\sigma(\Delta V_{\text{th}}) = A_{V_{\text{th}}}/\sqrt{WL}$

A.4 The Two Model Systems

Two explicit, analytically tractable systems recur throughout the book, and most general results are tested on one of them: the RC sampler and the double-well sampler.

A.4.1 The RC Sampler

The RC sampler is a continuous N-node capacitive circuit with node voltage vector $\mathbf{v}(t) \in \mathbb{R}^N$, a diagonal or symmetric positive-definite capacitance matrix $\mathbf{C} \succ 0$, a linear conductance or transconductance matrix $\mathbf{G} \in \mathbb{R}^{N\times N}$, a bias vector $\mathbf{b} \in \mathbb{R}^N$, and a noise amplitude matrix $\mathbf{B} \in \mathbb{R}^{N\times N}$. The drift matrix $\mathbf{A} \equiv -\mathbf{C}^{-1}\mathbf{G}$ is assumed Hurwitz stable, that is, all eigenvalues of $\mathbf{A}$ have strictly negative real parts. The node voltages obey the Ornstein–Uhlenbeck equation

$$\mathrm{d}\mathbf{v}_t \;=\; \mathbf{A}(\mathbf{v}_t - \mathbf{b})\,\mathrm{d}t + \mathbf{C}^{-1}\mathbf{B}\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{A} \;\equiv\; -\mathbf{C}^{-1}\mathbf{G}, \qquad \mathbf{D} \;\equiv\; \frac{1}{2}\mathbf{C}^{-1}\mathbf{B}\mathbf{B}^\top\mathbf{C}^{-\top}. \tag{A.1}$$

Two limits are used in the book.

- **Passive equilibrium limit ($\mathbf{G} = \mathbf{G}^\top \succ 0$, $\mathbf{B}\mathbf{B}^\top = 2k_{\text{B}}T\mathbf{G}$).** The circuit obeys detailed balance ($\mathbf{J}_{\text{eq}} = \mathbf{0}$) with electrostatic energy $E(\mathbf{v}) = \frac{1}{2}(\mathbf{v}-\mathbf{b})^\top\mathbf{C}(\mathbf{v}-\mathbf{b})$ and samples $p_{\text{eq}}(\mathbf{v}) = \mathcal{N}(\mathbf{b}, k_{\text{B}}T\mathbf{C}^{-1})$; the conductances set only the relaxation rates. A covariance programmed by $\mathbf{G}$, written $\mathcal{N}(\mathbf{b}, k_{\text{B}}T\mathbf{G}^{-1})$ in several chapters, requires engineered noise or active devices (Section 5.6).

- **Active non-equilibrium limit ($\mathbf{G} \neq \mathbf{G}^\top$).** In a NEAT-RN built from subthreshold CMOS [38], non-reciprocal transconductances break detailed balance ($\mathbf{A}\boldsymbol{\Sigma} \neq \boldsymbol{\Sigma}\mathbf{A}^\top$) and shot noise sets $\mathbf{D}$. The stationary distribution is still Gaussian, $p_{\text{ss}}(\mathbf{v}) = \mathcal{N}(\mathbf{b}, \boldsymbol{\Sigma})$, with covariance given by the continuous-time Lyapunov equation

  $$\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top + 2\mathbf{D} \;=\; \mathbf{0}. \tag{A.2}$$

  The stationary probability current $\mathbf{J}_{\text{ss}}(\mathbf{v}) = (\mathbf{A} + \mathbf{D}\boldsymbol{\Sigma}^{-1})(\mathbf{v}-\mathbf{b})\,p_{\text{ss}}(\mathbf{v})$ is divergence-free but nonzero, and the circuit produces housekeeping entropy at a positive rate $\dot{S}_{\text{hk}} > 0$ (Chapters 8 and 13).

Where the RC sampler appears. Example 1.5 (Gaussian partition function and equipartition), Worked computation 1.19 (Gaussian conditionals of a hybrid model), Worked computation 5.33 (Green's function, two-time covariance, and Kubo susceptibility), Worked computation 6.21 (Hermite spectrum of the Witten operator), Worked computation 8.12 (Onsager–Machlup action and housekeeping heat in a non-reciprocal two-node circuit), Worked computation 10.35 (Bures–Wasserstein geodesic and finite-time work bound), Worked computation 13.13 (realization as a subthreshold CMOS NEAT-RN), and Chapter B (Gaussian identities and Lyapunov equations).

A.4.2 The Double-Well Sampler

The double-well sampler is the simplest non-convex, multimodal energy landscape: in a continuous coordinate $x \in \mathbb{R}$, the tilted quartic potential

$$E(x; a, b, h) \;=\; \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx, \qquad a > 0, \quad b > 0. \tag{A.3}$$

For $h = 0$ it has two symmetric minima at $x_\pm = \pm\sqrt{b/a}$, separated by a barrier at $x = 0$ of height

$$\Delta E \;=\; E(0) - E(x_\pm) \;=\; \frac{b^2}{4a}, \tag{A.4}$$

where the curvature $E''(0) = -b$ is negative, so E is not convex and the Bakry–Émery criterion for fast mixing fails. In the deep-well limit $\beta\Delta E \gg 1$, the fluctuations within each well can be coarse-grained away, leaving a binary spin or probabilistic bit (p-bit) $s = \mathrm{sgn}(x) \in \{-1, +1\}$. A p-bit device driven by an input I has mean output $\langle s \rangle = \tanh(\beta_{\mathrm{eff}} I)$, where the device constant β_{eff} is introduced in Chapter 14.

Where the double-well sampler appears. Example 1.6 (a single p-bit in a local field), Worked computation 1.19 (marginalizing p-bits through $-\ln\cosh(\cdot)$ turns a convex parabola into a double well), Worked computation 2.28 (chromatic block Gibbs sampling of coupled p-bits), Worked computation 3.33 (frustrated spin glasses with many wells), Worked computation 6.21 (loss of Bakry–Émery curvature as the $-bx^2/2$ term is switched on), Worked computation 7.16 (Kramers escape rate $k_{A\to B} \propto e^{-\beta\Delta E}$ and the splitting of the Witten ground-state doublet), Worked computation 9.30 (Landauer erasure and Jarzynski work fluctuations in a double well), Worked computation 14.20 (subthreshold CMOS p-bit versus Josephson Quantum Flux Parametron), Worked computation 15.15 (score function and speciation time t_s of a bimodal diffusion), and Worked computation 16.27 (adding the DTM forward coupling $\frac{\alpha_t}{2}(x_{t-1} - c_t x_t)^2$ with $\alpha_t > b$ makes the curvature $E''_{\mathrm{tot}}(x_{t-1}) \geq \alpha_t - b > 0$ everywhere, which removes the Kramers barrier and gives a mixing time independent of the barrier height).

A.5 Standing Mathematical and Physical Assumptions

To avoid repeating standard technical hypotheses in every theorem, we collect here the five standing assumptions used across the book:

(A-1) (Confining Potential & Normalizability): The continuous energy function $E_\theta \in C^2(\mathbb{R}^N; \mathbb{R})$ grows sufficiently rapidly at infinity (specifically, there exist $R > 0$ and $c_1, c_2 > 0$ such that $\langle \mathbf{x}, \nabla E_\theta(\mathbf{x}) \rangle \geq c_1 \|\mathbf{x}\|^2$ and $E_\theta(\mathbf{x}) \geq c_2 \|\mathbf{x}\|^2$ for $\|\mathbf{x}\| \geq R$) so that $Z(\theta, \beta) = \int_{\mathbb{R}^N} e^{-\beta E_\theta(\mathbf{x})}\, \mathrm{d}^N x < \infty$, SDE trajectories do not explode in finite time, and boundary terms at $\|\mathbf{x}\| \to \infty$ vanish upon integration by parts.

(A-2) (Local Lipschitz Continuity & One-Sided Dissipativity): The drift vector $\mathbf{a}(\mathbf{x}, t)$ and noise amplitude matrix $\mathbf{B}(\mathbf{x}, t)$ are locally Lipschitz continuous in $\mathbf{x}$ and satisfy a one-sided dissipative Lipschitz bound $\langle \mathbf{x}-\mathbf{y},\, \mathbf{a}(\mathbf{x},t)-\mathbf{a}(\mathbf{y},t)\rangle + \frac{1}{2}\,\|\mathbf{B}(\mathbf{x},t)-\mathbf{B}(\mathbf{y},t)\|_{\mathrm{F}}^2 \le K\,\|\mathbf{x}-\mathbf{y}\|^2$, guaranteeing pathwise unique strong solutions to the Itô SDE (Chapter C).

(A-3) (Uniform Ellipticity / Non-Degenerate Thermal Bath): The diffusion tensor $\mathbf{D}(\mathbf{x}) = \frac{1}{2}\mathbf{B}(\mathbf{x})\mathbf{B}(\mathbf{x})^\top$ is uniformly positive-definite: there exists $D_{\min} > 0$ such that $\boldsymbol{\xi}^\top \mathbf{D}(\mathbf{x})\boldsymbol{\xi} \ge D_{\min}\,\|\boldsymbol{\xi}\|^2$ for all $\mathbf{x}, \boldsymbol{\xi} \in \mathbb{R}^N$. Physically, every independent circuit coordinate is coupled to a thermal or shot-noise dissipative channel at $T > 0$. For smooth coefficients this gives smooth (C^∞) Fokker–Planck transition densities and, together with **(A-1)**, a unique stationary measure.

(A-4) (Hurwitz Stability of Linearized Circuit Operating Points): In linearized non-equilibrium circuit models (the RC sampler and the NEAT-RNs of Chapter 13), the drift Jacobian $\mathbf{A} = -\mathbf{C}^{-1}\mathbf{G}$ is Hurwitz stable: $\max_{\lambda \in \mathrm{spec}(\mathbf{A})} \mathrm{Re}(\lambda) < -\gamma_{\min} < 0$.

(A-5) (Weak-Inversion / Subthreshold Operating Regime): In CMOS device derivations (Chapters 12 to 14, 17 and 18), transistors operate in deep weak inversion ($V_{\mathrm{GS}} < V_{\mathrm{th}} - 3nV_T$) with mesoscopic node capacitances $\Omega = C_{\mathrm{node}} V_T / q_e \gg 1$, so carrier transport is governed by the Boltzmann-exponential EKV diffusion law and nodal charge discreteness is accurately captured by the second-order Kramers–Moyal (Langevin/Fokker–Planck) truncation.

Appendix B

Multivariate Gaussian Integrals, Schur Complements, Continuous-Time Lyapunov Equations, and Wick/Isserlis Combinatorics

Gaussian distributions appear throughout this book. They are the exact stationary laws of linear circuits such as the RC sampler (the coupled linear RC circuit of Chapter 1, an Ornstein–Uhlenbeck process), and they are the local quadratic approximation to nonlinear dynamics near a stable state. They enter the Gaussian–Bernoulli Energy-Based Models of Chapter 1, the Hubbard–Stratonovich transformation of Chapter 3, the Ornstein–Uhlenbeck analysis of Chapters 5, 6, 8 and 10, the Non-Equilibrium Analog Thermodynamic Random Networks (NEAT-RNs) of Chapter 13, and the transition kernels of the Denoising Thermodynamic Models (DTMs) of Chapters 15 and 16.

This appendix collects the derivations of the Gaussian identities used in the main text. Section B.1 evaluates the Gaussian integral and the entropy, KL divergence, and Wasserstein distance of Gaussians. Section B.2 develops Schur complements and the rules for conditioning and marginalizing. Section B.3 proves the Isserlis–Wick theorem and applies cumulants to weakly anharmonic energies. Section B.4 treats the continuous-time Lyapunov equation, which fixes the stationary covariance of a linear stochastic circuit in or out of equilibrium, and ends with the closed-form 2×2 solution.

B.1 Multivariate Gaussian Integrals with Linear Sources

Consider a quadratic energy function over N continuous degrees of freedom $\mathbf{x} = (x_1, x_2, \ldots, x_N)^\top \in \mathbb{R}^N$ (such as capacitor node voltages or charge displacements) coupled to an external bias or source vector $\mathbf{J} \in \mathbb{R}^N$:

$$\beta E(\mathbf{x}; \mathbf{J}) \;=\; \frac{1}{2}\mathbf{x}^\top \mathbf{\Lambda} \mathbf{x} - \mathbf{J}^\top \mathbf{x}, \tag{B.1}$$

where $\mathbf{\Lambda} \in \mathbb{R}^{N \times N}$ is a real symmetric **precision** (or stiffness) matrix ($\mathbf{\Lambda} = \mathbf{\Lambda}^\top$). Symmetry costs no generality: any antisymmetric part drops out of the quadratic form, because $\mathbf{x}^\top \mathbf{M} \mathbf{x} = \frac{1}{2}\mathbf{x}^\top(\mathbf{M} + \mathbf{M}^\top)\mathbf{x}$. For the Boltzmann weight $\exp(-\beta E(\mathbf{x}; \mathbf{J}))$ to be integrable over $\mathbb{R}^N$, $\mathbf{\Lambda}$ must be strictly positive definite ($\mathbf{\Lambda} \succ \mathbf{0}$), meaning $\mathbf{x}^\top \mathbf{\Lambda} \mathbf{x} > 0$ for all non-zero $\mathbf{x} \in \mathbb{R}^N$.

B.1.1 The One-Dimensional Base Integral

Every multidimensional Gaussian integral reduces to a product of one-dimensional Gaussian integrals after a suitable rotation of coordinates. We begin by proving the one-dimensional formula.

Lemma B.1 (Fundamental One-Dimensional Gaussian Integral)**.** *For any strictly positive scalar $\lambda > 0$ and any real or complex source $J \in \mathbb{C}$,*

$$I_1(\lambda, J) \;:=\; \int_{-\infty}^{\infty} \mathrm{d}x\, \exp\!\left(-\frac{1}{2}\lambda x^2 + Jx\right) \;=\; \sqrt{\frac{2\pi}{\lambda}}\, \exp\!\left(\frac{J^2}{2\lambda}\right). \tag{B.2}$$

Proof. First consider the centered unit case $\lambda = 1$ and $J = 0$, denoted $I_0 := \int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x > 0$. Squaring this integral and applying Fubini's theorem on $\mathbb{R}^2$ gives

$$I_0^2 \;=\; \left(\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-x^2/2}\right)\left(\int_{-\infty}^{\infty} \mathrm{d}y\, e^{-y^2/2}\right) \;=\; \iint_{\mathbb{R}^2} \mathrm{d}x\, \mathrm{d}y\, \exp\!\left(-\frac{x^2+y^2}{2}\right). \tag{B.3}$$

Changing to plane polar coordinates $(x, y) = (r\cos\theta, r\sin\theta)$ with $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$, the Jacobian determinant is

$$\det\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} \;=\; \det\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} \;=\; r(\cos^2\theta + \sin^2\theta) \;=\; r. \tag{B.4}$$

With area element $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$ and radial substitution $u = r^2/2$ ($\mathrm{d}u = r\,\mathrm{d}r$), the integral evaluates to

$$I_0^2 \;=\; \int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} r\,\mathrm{d}r\, e^{-r^2/2} \;=\; 2\pi \int_0^{\infty} \mathrm{d}u\, e^{-u} \;=\; 2\pi\Big[-e^{-u}\Big]_0^{\infty} \;=\; 2\pi. \tag{B.5}$$

Since $e^{-x^2/2} > 0$, taking the positive square root gives $I_0 = \sqrt{2\pi}$.

Now, for general $\lambda > 0$ and $J \in \mathbb{R}$, complete the square in the exponent:

$$\begin{aligned} -\frac{1}{2}\lambda x^2 + Jx \;&=\; -\frac{\lambda}{2}\left(x^2 - \frac{2J}{\lambda}x + \frac{J^2}{\lambda^2}\right) + \frac{J^2}{2\lambda} \\ &=\; -\frac{\lambda}{2}\left(x - \frac{J}{\lambda}\right)^2 + \frac{J^2}{2\lambda}. \end{aligned} \tag{B.6}$$

Factoring $\exp(J^2/(2\lambda))$ out of the integral and making the affine change of variable $z = \sqrt{\lambda}(x - J/\lambda)$, for which $\mathrm{d}x = \lambda^{-1/2}\,\mathrm{d}z$ and the integration limits $-\infty < z < \infty$ are unchanged, we obtain

$$I_1(\lambda, J) \;=\; \exp\!\left(\frac{J^2}{2\lambda}\right) \frac{1}{\sqrt{\lambda}} \int_{-\infty}^{\infty} \mathrm{d}z\, e^{-z^2/2} \;=\; \sqrt{\frac{2\pi}{\lambda}}\, \exp\!\left(\frac{J^2}{2\lambda}\right). \tag{B.7}$$

Finally, because both sides of Eq. (B.2) are entire analytic functions of $J \in \mathbb{C}$ that agree for all $J \in \mathbb{R}$, identity of analytic functions (or Cauchy's theorem applied to rectangular contours in $\mathbb{C}$) extends Eq. (B.2) to all complex $J \in \mathbb{C}$. □

B.1.2 The N-Dimensional Gaussian Integral with Linear Source

We now extend Lemma B.1 to $\mathbb{R}^N$ using the spectral theorem for real symmetric matrices.

Theorem B.2 (N-Dimensional Gaussian Integral with Linear Source)**.** *Let $\mathbf{\Lambda} \in \mathbb{R}^{N\times N}$ be a real, symmetric, strictly positive-definite matrix ($\mathbf{\Lambda} = \mathbf{\Lambda}^\top \succ \mathbf{0}$), and let $\mathbf{J} \in \mathbb{C}^N$ be any real or complex source vector. Then the partition integral converges absolutely and is given by*

$$Z(\mathbf{\Lambda}, \mathbf{J}) \;:=\; \int_{\mathbb{R}^N} \mathrm{d}^N x \, \exp\!\left(-\frac{1}{2}\mathbf{x}^\top \mathbf{\Lambda}\mathbf{x} + \mathbf{J}^\top \mathbf{x}\right) \;=\; \frac{(2\pi)^{N/2}}{(\det \mathbf{\Lambda})^{1/2}} \exp\!\left(\frac{1}{2}\mathbf{J}^\top \mathbf{\Lambda}^{-1}\mathbf{J}\right). \tag{B.8}$$

In particular, for $\mathbf{J} \in \mathbb{R}^N$, setting $\mathbf{\Sigma} := \mathbf{\Lambda}^{-1}$ and $\boldsymbol{\mu} := \mathbf{\Lambda}^{-1}\mathbf{J} = \mathbf{\Sigma}\mathbf{J}$, the normalized probability density of the multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$ is

$$p(\mathbf{x}) \;=\; \frac{1}{(2\pi)^{N/2}(\det \mathbf{\Sigma})^{1/2}} \exp\!\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right). \tag{B.9}$$

Proof. We proceed in two steps: multidimensional completion of the square followed by orthogonal diagonalization.

Step 1: Completing the square in $\mathbb{R}^N$. Since $\mathbf{\Lambda}$ is real symmetric and positive definite, all of its eigenvalues $\lambda_1, \ldots, \lambda_N$ are strictly positive real numbers. Hence $\det \mathbf{\Lambda} = \prod_{i=1}^N \lambda_i > 0$, and the inverse $\mathbf{\Lambda}^{-1}$ exists and is symmetric positive definite: $(\mathbf{\Lambda}^{-1})^\top = (\mathbf{\Lambda}^\top)^{-1} = \mathbf{\Lambda}^{-1}$. For $\mathbf{J} \in \mathbb{R}^N$, define the stationary point $\boldsymbol{\mu} := \mathbf{\Lambda}^{-1}\mathbf{J}$ and the shifted coordinate $\mathbf{y} := \mathbf{x} - \boldsymbol{\mu} = \mathbf{x} - \mathbf{\Lambda}^{-1}\mathbf{J}$. Expanding the quadratic form $\frac{1}{2}\mathbf{y}^\top \mathbf{\Lambda}\mathbf{y}$ gives

$$\begin{aligned}
\frac{1}{2}(\mathbf{x} - \mathbf{\Lambda}^{-1}\mathbf{J})^\top \mathbf{\Lambda}(\mathbf{x} - \mathbf{\Lambda}^{-1}\mathbf{J}) \;&=\; \frac{1}{2}\left(\mathbf{x}^\top - \mathbf{J}^\top(\mathbf{\Lambda}^{-1})^\top\right)\left(\mathbf{\Lambda}\mathbf{x} - \mathbf{\Lambda}\mathbf{\Lambda}^{-1}\mathbf{J}\right) \\
&=\; \frac{1}{2}\left(\mathbf{x}^\top - \mathbf{J}^\top \mathbf{\Lambda}^{-1}\right)(\mathbf{\Lambda}\mathbf{x} - \mathbf{J}) \\
&=\; \frac{1}{2}\mathbf{x}^\top \mathbf{\Lambda}\mathbf{x} - \frac{1}{2}\mathbf{x}^\top \mathbf{J} - \frac{1}{2}\mathbf{J}^\top \mathbf{\Lambda}^{-1}\mathbf{\Lambda}\mathbf{x} + \frac{1}{2}\mathbf{J}^\top \mathbf{\Lambda}^{-1}\mathbf{J}.
\end{aligned} \tag{B.10}$$

Because $\mathbf{x}^\top \mathbf{J}$ is a scalar, it equals its transpose $(\mathbf{x}^\top \mathbf{J})^\top = \mathbf{J}^\top \mathbf{x}$, and $\mathbf{J}^\top \mathbf{\Lambda}^{-1}\mathbf{\Lambda}\mathbf{x} = \mathbf{J}^\top \mathbf{x}$. Combining the two linear cross terms gives

$$-\frac{1}{2}\mathbf{x}^\top \mathbf{\Lambda}\mathbf{x} + \mathbf{J}^\top \mathbf{x} \;=\; -\frac{1}{2}(\mathbf{x} - \mathbf{\Lambda}^{-1}\mathbf{J})^\top \mathbf{\Lambda}(\mathbf{x} - \mathbf{\Lambda}^{-1}\mathbf{J}) + \frac{1}{2}\mathbf{J}^\top \mathbf{\Lambda}^{-1}\mathbf{J}. \tag{B.11}$$

Substituting Eq. (B.11) into the integral Eq. (B.8) and performing the translation $\mathbf{y} = \mathbf{x} - \mathbf{\Lambda}^{-1}\mathbf{J}$, whose Jacobian determinant on $\mathbb{R}^N$ is $|\det \mathbf{I}| = 1$, we factor out the $\mathbf{y}$-independent source contribution:

$$Z(\mathbf{\Lambda}, \mathbf{J}) \;=\; \exp\!\left(\frac{1}{2}\mathbf{J}^\top \mathbf{\Lambda}^{-1}\mathbf{J}\right) \int_{\mathbb{R}^N} \mathrm{d}^N y \, \exp\!\left(-\frac{1}{2}\mathbf{y}^\top \mathbf{\Lambda}\mathbf{y}\right). \tag{B.12}$$

Step 2: Spectral diagonalization. By the spectral theorem for real symmetric matrices, there exists an orthogonal matrix $\mathbf{O} \in \mathrm{O}(N)$ (satisfying $\mathbf{O}^\top \mathbf{O} = \mathbf{O}\mathbf{O}^\top = \mathbf{I}$) and a diagonal matrix of eigenvalues $\mathbf{D}_\lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$ with $\lambda_i > 0$ such that

$$\mathbf{\Lambda} \;=\; \mathbf{O}\mathbf{D}_\lambda \mathbf{O}^\top. \tag{B.13}$$

Perform the orthogonal coordinate change $\mathbf{z} := \mathbf{O}^\top \mathbf{y}$, or $\mathbf{y} = \mathbf{O}\mathbf{z}$. Because $\mathbf{O}$ is orthogonal, $1 = \det(\mathbf{O}^\top \mathbf{O}) = (\det \mathbf{O})^2$, so the absolute value of the Jacobian determinant of this linear transformation is

$$\left|\det\left(\frac{\partial \mathbf{y}}{\partial \mathbf{z}}\right)\right| = |\det \mathbf{O}| = 1, \qquad \text{so} \qquad \mathrm{d}^N y = \mathrm{d}^N z = \prod_{i=1}^{N} \mathrm{d}z_i. \tag{B.14}$$

In the rotated coordinates $\mathbf{z}$, the quadratic form diagonalizes into a sum of squares:

$$\mathbf{y}^\top \mathbf{\Lambda} \mathbf{y} = (\mathbf{O}\mathbf{z})^\top \left(\mathbf{O}\mathbf{D}_\lambda \mathbf{O}^\top\right)(\mathbf{O}\mathbf{z}) = \mathbf{z}^\top (\mathbf{O}^\top \mathbf{O}) \mathbf{D}_\lambda (\mathbf{O}^\top \mathbf{O}) \mathbf{z} = \mathbf{z}^\top \mathbf{D}_\lambda \mathbf{z} = \sum_{i=1}^{N} \lambda_i z_i^2. \tag{B.15}$$

Consequently, the exponential of the sum factors into a product of N one-dimensional exponentials, and Fubini's theorem factors the N-dimensional integral into N independent 1D integrals of the form Eq. (B.2):

$$\begin{aligned} \int_{\mathbb{R}^N} \mathrm{d}^N y \, \exp\left(-\frac{1}{2}\mathbf{y}^\top \mathbf{\Lambda} \mathbf{y}\right) &= \int_{\mathbb{R}^N} \left(\prod_{i=1}^{N} \mathrm{d}z_i\right) \exp\left(-\frac{1}{2}\sum_{i=1}^{N} \lambda_i z_i^2\right) \\ &= \prod_{i=1}^{N} \left(\int_{-\infty}^{\infty} \mathrm{d}z_i \, e^{-\lambda_i z_i^2/2}\right) = \prod_{i=1}^{N} \sqrt{\frac{2\pi}{\lambda_i}} = \frac{(2\pi)^{N/2}}{\left(\prod_{i=1}^{N} \lambda_i\right)^{1/2}}. \end{aligned} \tag{B.16}$$

Because determinant is invariant under orthogonal similarity transformations,

$$\det \mathbf{\Lambda} = \det(\mathbf{O}\mathbf{D}_\lambda \mathbf{O}^\top) = (\det \mathbf{O})(\det \mathbf{D}_\lambda)(\det \mathbf{O}^\top) = \det \mathbf{D}_\lambda = \prod_{i=1}^{N} \lambda_i. \tag{B.17}$$

Substituting $\prod_{i=1}^{N} \lambda_i = \det \mathbf{\Lambda}$ into Eq. (B.16) and combining with Eq. (B.12) yields Eq. (B.8) for real $\mathbf{J}$ (and for complex $\mathbf{J} \in \mathbb{C}^N$ by applying Eq. (B.2) with $J_{\mathrm{rot}} = \mathbf{O}^\top \mathbf{J} \in \mathbb{C}^N$). □

Corollary B.3 (Hubbard–Stratonovich Transformation and Characteristic Function)**.** *Let* $\mathbf{K} = \mathbf{\Lambda}^{-1} \succ \mathbf{0}$ *be a symmetric positive-definite interaction matrix. Reading Eq. (B.8) from right to left represents any positive quadratic Boltzmann weight* $\exp(\frac{1}{2}\mathbf{s}^\top \mathbf{K}\mathbf{s})$ *(for instance, a ferromagnetic or spin-glass coupling of discrete spins* $\mathbf{s} \in \{-1, +1\}^N$ *in Chapter 3) as an average over an auxiliary Gaussian field* $\boldsymbol{\phi} \in \mathbb{R}^N$*:*

$$\exp\left(\frac{1}{2}\mathbf{s}^\top \mathbf{K}\mathbf{s}\right) = \frac{1}{(2\pi)^{N/2} (\det \mathbf{K})^{1/2}} \int_{\mathbb{R}^N} \mathrm{d}^N \phi \, \exp\left(-\frac{1}{2}\boldsymbol{\phi}^\top \mathbf{K}^{-1} \boldsymbol{\phi} + \boldsymbol{\phi}^\top \mathbf{s}\right). \tag{B.18}$$

Similarly, setting $\mathbf{J} = i\mathbf{k}$ *(*$\mathbf{k} \in \mathbb{R}^N$*) in Eq. (B.8) gives the characteristic function (Fourier transform) of* $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$*:*

$$\mathbb{E}\left[e^{i\mathbf{k}^\top \mathbf{x}}\right] = \int_{\mathbb{R}^N} \mathrm{d}^N x \, p(\mathbf{x}) \, e^{i\mathbf{k}^\top \mathbf{x}} = \exp\left(i\mathbf{k}^\top \boldsymbol{\mu} - \frac{1}{2}\mathbf{k}^\top \mathbf{\Sigma} \mathbf{k}\right). \tag{B.19}$$

B.1.3 Differential Entropy, KL Divergence, and Bures–Wasserstein Distance

In Chapters 1, 6, 10, 16 and 18, variational free energies, finite-time optimal transport protocols, and hardware calibration criteria require closed-form expressions for the differential entropy, Kullback–Leibler (KL) divergence, and L^2-Wasserstein distance between two N-dimensional Gaussian distributions.

Proposition B.4 (Gaussian Differential Entropy, KL Divergence, and Bures–Wasserstein Metric)**.** *Let $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $q(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$ be two non-degenerate multivariate normal distributions on $\mathbb{R}^N$ with $\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2 \succ \mathbf{0}$. Then:*

(a) The thermodynamic differential entropy of p is

$$S[p] \;:=\; -k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x\, p(\mathbf{x}) \ln p(\mathbf{x}) \;=\; \frac{k_{\mathrm{B}}}{2} \ln\det(2\pi e\, \boldsymbol{\Sigma}_1) \;=\; \frac{k_{\mathrm{B}}}{2}\Big(N \ln(2\pi e) + \ln\det \boldsymbol{\Sigma}_1\Big). \tag{B.20}$$

(b) The Kullback–Leibler divergence of p relative to q is

$$D_{\mathrm{KL}}(p \,\|\, q) \;=\; \frac{1}{2}\left[\mathrm{Tr}\Big(\boldsymbol{\Sigma}_2^{-1}\boldsymbol{\Sigma}_1\Big) + (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^{\top}\boldsymbol{\Sigma}_2^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) - N + \ln\left(\frac{\det \boldsymbol{\Sigma}_2}{\det \boldsymbol{\Sigma}_1}\right)\right]. \tag{B.21}$$

(c) The squared L^2-Wasserstein (Bures–Wasserstein) distance between p and q is

$$\mathcal{W}_2^2(p, q) \;=\; \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|^2 + \mathrm{Tr}\left(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2 - 2\left(\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2}\right)^{1/2}\right). \tag{B.22}$$

Proof. We first record a trace identity for any symmetric matrix $\mathbf{M} \in \mathbb{R}^{N\times N}$ and random vector $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$. Writing $\mathbf{x} = \boldsymbol{\mu}_1 + \delta\mathbf{x}$ where $\mathbb{E}[\delta\mathbf{x}] = \mathbf{0}$ and $\mathbb{E}\Big[\delta\mathbf{x}\,\delta\mathbf{x}^{\top}\Big] = \boldsymbol{\Sigma}_1$, and using the cyclic invariance of the trace on the scalar $\delta\mathbf{x}^{\top}\mathbf{M}\,\delta\mathbf{x} = \mathrm{Tr}(\delta\mathbf{x}^{\top}\mathbf{M}\,\delta\mathbf{x}) = \mathrm{Tr}(\mathbf{M}\,\delta\mathbf{x}\,\delta\mathbf{x}^{\top})$, linearity of expectation gives

$$\mathbb{E}\Big[(\mathbf{x} - \boldsymbol{\mu}_1)^{\top}\mathbf{M}(\mathbf{x} - \boldsymbol{\mu}_1)\Big]_p \;=\; \mathrm{Tr}\Big(\mathbf{M}\,\mathbb{E}\Big[\delta\mathbf{x}\,\delta\mathbf{x}^{\top}\Big]_p\Big) \;=\; \mathrm{Tr}(\mathbf{M}\boldsymbol{\Sigma}_1). \tag{B.23}$$

For part (a), taking the logarithm of Eq. (B.9) for $p(\mathbf{x})$ gives

$$-\ln p(\mathbf{x}) \;=\; \frac{N}{2}\ln(2\pi) + \frac{1}{2}\ln\det\boldsymbol{\Sigma}_1 + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^{\top}\boldsymbol{\Sigma}_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1). \tag{B.24}$$

Taking the expectation under p and applying Eq. (B.23) with $\mathbf{M} = \boldsymbol{\Sigma}_1^{-1}$ yields $\mathrm{Tr}(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Sigma}_1) = \mathrm{Tr}(\mathbf{I}) = N = N\ln e$, which combines into Eq. (B.20).

For part (b), by definition of the KL divergence,

$$D_{\mathrm{KL}}(p \,\|\, q) \;=\; \mathbb{E}[\ln p(\mathbf{x}) - \ln q(\mathbf{x})]_p \;=\; \frac{1}{2}\ln\left(\frac{\det\boldsymbol{\Sigma}_2}{\det\boldsymbol{\Sigma}_1}\right) - \frac{N}{2} + \frac{1}{2}\mathbb{E}\Big[(\mathbf{x} - \boldsymbol{\mu}_2)^{\top}\boldsymbol{\Sigma}_2^{-1}(\mathbf{x} - \boldsymbol{\mu}_2)\Big]_p. \tag{B.25}$$

To evaluate the final expectation, add and subtract $\boldsymbol{\mu}_1$ inside the quadratic form: $\mathbf{x} - \boldsymbol{\mu}_2 = (\mathbf{x} - \boldsymbol{\mu}_1) + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$. Expanding the square gives

$$\begin{aligned}\mathbb{E}\Big[(\mathbf{x} - \boldsymbol{\mu}_2)^{\top}\boldsymbol{\Sigma}_2^{-1}(\mathbf{x} - \boldsymbol{\mu}_2)\Big]_p \;&=\; \mathbb{E}\Big[(\mathbf{x} - \boldsymbol{\mu}_1)^{\top}\boldsymbol{\Sigma}_2^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\Big]_p + 2\underbrace{\mathbb{E}\Big[(\mathbf{x} - \boldsymbol{\mu}_1)^{\top}\Big]_p}_{=\mathbf{0}^{\top}}\boldsymbol{\Sigma}_2^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \\ &\quad + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^{\top}\boldsymbol{\Sigma}_2^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \\ &=\; \mathrm{Tr}\Big(\boldsymbol{\Sigma}_2^{-1}\boldsymbol{\Sigma}_1\Big) + (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^{\top}\boldsymbol{\Sigma}_2^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1).\end{aligned} \tag{B.26}$$

Substituting this expression into Eq. (B.25) yields Eq. (B.21).

For part (c), recall from Section 10.2 that $\mathcal{W}_2^2(p,q) = \inf_{\pi\in\Pi(p,q)} \mathbb{E}\Big[\|\mathbf{x}-\mathbf{y}\|^2\Big]_\pi$ over all joint couplings $(\mathbf{x},\mathbf{y})$ with marginals $\mathbf{x}\sim p$ and $\mathbf{y}\sim q$. Writing $\mathbf{x} = \boldsymbol{\mu}_1 + \delta\mathbf{x}$ and $\mathbf{y} = \boldsymbol{\mu}_2 + \delta\mathbf{y}$ with cross-covariance $\mathbf{C} := \mathbb{E}\Big[\delta\mathbf{x}\,\delta\mathbf{y}^\top\Big]_\pi$, expanding the square gives

$$\mathbb{E}\Big[\|\mathbf{x}-\mathbf{y}\|^2\Big]_\pi \;=\; \|\boldsymbol{\mu}_1-\boldsymbol{\mu}_2\|^2 + \operatorname{Tr}(\boldsymbol{\Sigma}_1) + \operatorname{Tr}(\boldsymbol{\Sigma}_2) - 2\operatorname{Tr}(\mathbf{C}). \tag{B.27}$$

For $(\delta\mathbf{x},\delta\mathbf{y})$ to have a valid joint covariance matrix $\left(\begin{smallmatrix}\boldsymbol{\Sigma}_1 & \mathbf{C}\\ \mathbf{C}^\top & \boldsymbol{\Sigma}_2\end{smallmatrix}\right) \succeq \mathbf{0}$, the Schur complement (Definition B.5) must be positive semidefinite: $\boldsymbol{\Sigma}_1 - \mathbf{C}\boldsymbol{\Sigma}_2^{-1}\mathbf{C}^\top \succeq \mathbf{0}$. Writing $\mathbf{K} := \boldsymbol{\Sigma}_1^{-1/2}\mathbf{C}\boldsymbol{\Sigma}_2^{-1/2}$, this condition is $\mathbf{K}\mathbf{K}^\top \preceq \mathbf{I}$, so all singular values of $\mathbf{K}$ are at most 1. By the polar decomposition, $\operatorname{Tr}(\mathbf{C}) = \operatorname{Tr}(\boldsymbol{\Sigma}_2^{1/2}\boldsymbol{\Sigma}_1^{1/2}\mathbf{K}) \le \operatorname{Tr}((\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2})^{1/2})$, with equality achieved by the linear Brenier transport map $\delta\mathbf{y} = \mathbf{T}\delta\mathbf{x}$ where $\mathbf{T} = \boldsymbol{\Sigma}_1^{-1/2}(\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2})^{1/2}\boldsymbol{\Sigma}_1^{-1/2} \succ \mathbf{0}$, yielding Eq. (B.22). □

B.2 Block Matrix Inversion, Schur Complements, and Gaussian Conditioning/Marginalization

In a composite physical or probabilistic network, we partition the state space $\mathbb{R}^N = \mathbb{R}^{N_A}\oplus\mathbb{R}^{N_B}$ ($N = N_A + N_B$) into two disjoint subsets of variables: for example, visible terminal nodes A and hidden internal nodes B in a latent-variable EBM (Chapters 1 and 16), or adjacent time slices $\mathbf{x}_{t-1}$ and $\mathbf{x}_t$ in a denoising chain (Chapters 15 and 16). Two operations on $\mathbf{x}_B$ are common: *clamping* $\mathbf{x}_B$ to a fixed voltage (conditioning) and leaving $\mathbf{x}_B$ *floating* so that it fluctuates thermally (marginalization). The effect of either on the distribution of $\mathbf{x}_A$ is computed with **Schur complements**.

B.2.1 Block LDU Factorization, Schur Complements, and the Woodbury Identity

Let $\mathbf{M}\in\mathbb{R}^{N\times N}$ be a partitioned block matrix:

$$\mathbf{M} \;=\; \begin{pmatrix}\mathbf{M}_{AA} & \mathbf{M}_{AB}\\ \mathbf{M}_{BA} & \mathbf{M}_{BB}\end{pmatrix}, \qquad \mathbf{M}_{AA}\in\mathbb{R}^{N_A\times N_A},\quad \mathbf{M}_{BB}\in\mathbb{R}^{N_B\times N_B}. \tag{B.28}$$

Definition B.5 (Schur Complements)**.** If $\mathbf{M}_{BB}$ is invertible, the **Schur complement of $\mathbf{M}_{BB}$ in $\mathbf{M}$** is the $N_A\times N_A$ matrix

$$\mathbf{M}/\mathbf{M}_{BB} \;:=\; \mathbf{M}_{AA} - \mathbf{M}_{AB}\mathbf{M}_{BB}^{-1}\mathbf{M}_{BA}. \tag{B.29}$$

Symmetrically, if $\mathbf{M}_{AA}$ is invertible, the **Schur complement of $\mathbf{M}_{AA}$ in $\mathbf{M}$** is the $N_B\times N_B$ matrix

$$\mathbf{M}/\mathbf{M}_{AA} \;:=\; \mathbf{M}_{BB} - \mathbf{M}_{BA}\mathbf{M}_{AA}^{-1}\mathbf{M}_{AB}. \tag{B.30}$$

Theorem B.6 (Block LDU Factorization, Determinant, and Banachiewicz–Schur Inversion)**.** *Suppose $\mathbf{M}$ in Eq. (B.28) has invertible diagonal blocks $\mathbf{M}_{AA}$ and $\mathbf{M}_{BB}$ as well as invertible Schur complements $\mathbf{S}_A := \mathbf{M}/\mathbf{M}_{BB}$ and $\mathbf{S}_B := \mathbf{M}/\mathbf{M}_{AA}$. Then:*

(a) $\mathbf{M}$ admits the two dual block triangular factorizations (UDL and LDU):

$$\mathbf{M} \;=\; \begin{pmatrix}\mathbf{I}_A & \mathbf{M}_{AB}\mathbf{M}_{BB}^{-1}\\ \mathbf{0} & \mathbf{I}_B\end{pmatrix}\begin{pmatrix}\mathbf{M}/\mathbf{M}_{BB} & \mathbf{0}\\ \mathbf{0} & \mathbf{M}_{BB}\end{pmatrix}\begin{pmatrix}\mathbf{I}_A & \mathbf{0}\\ \mathbf{M}_{BB}^{-1}\mathbf{M}_{BA} & \mathbf{I}_B\end{pmatrix}, \tag{B.31}$$

$$\mathbf{M} \;=\; \begin{pmatrix}\mathbf{I}_A & \mathbf{0}\\ \mathbf{M}_{BA}\mathbf{M}_{AA}^{-1} & \mathbf{I}_B\end{pmatrix}\begin{pmatrix}\mathbf{M}_{AA} & \mathbf{0}\\ \mathbf{0} & \mathbf{M}/\mathbf{M}_{AA}\end{pmatrix}\begin{pmatrix}\mathbf{I}_A & \mathbf{M}_{AA}^{-1}\mathbf{M}_{AB}\\ \mathbf{0} & \mathbf{I}_B\end{pmatrix}. \tag{B.32}$$

(b) The determinant of $\mathbf{M}$ *satisfies the Schur determinant identity:*

$$\det \mathbf{M} \;=\; \det(\mathbf{M}_{BB})\det(\mathbf{M}/\mathbf{M}_{BB}) \;=\; \det(\mathbf{M}_{AA})\det(\mathbf{M}/\mathbf{M}_{AA}). \tag{B.33}$$

(c) The inverse $\mathbf{M}^{-1}$ *is given in dual forms by*

$$\mathbf{M}^{-1} \;=\; \begin{pmatrix} \mathbf{S}_A^{-1} & -\mathbf{S}_A^{-1}\mathbf{M}_{AB}\mathbf{M}_{BB}^{-1} \\ -\mathbf{M}_{BB}^{-1}\mathbf{M}_{BA}\mathbf{S}_A^{-1} & \mathbf{M}_{BB}^{-1}+\mathbf{M}_{BB}^{-1}\mathbf{M}_{BA}\mathbf{S}_A^{-1}\mathbf{M}_{AB}\mathbf{M}_{BB}^{-1} \end{pmatrix} \tag{B.34}$$

$$\;=\; \begin{pmatrix} \mathbf{M}_{AA}^{-1}+\mathbf{M}_{AA}^{-1}\mathbf{M}_{AB}\mathbf{S}_B^{-1}\mathbf{M}_{BA}\mathbf{M}_{AA}^{-1} & -\mathbf{M}_{AA}^{-1}\mathbf{M}_{AB}\mathbf{S}_B^{-1} \\ -\mathbf{S}_B^{-1}\mathbf{M}_{BA}\mathbf{M}_{AA}^{-1} & \mathbf{S}_B^{-1} \end{pmatrix}. \tag{B.35}$$

Proof. To verify Eq. (B.31), multiply the three block matrices on the right-hand side explicitly:

$$\begin{aligned} &\begin{pmatrix} \mathbf{I}_A & \mathbf{M}_{AB}\mathbf{M}_{BB}^{-1} \\ \mathbf{0} & \mathbf{I}_B \end{pmatrix}\begin{pmatrix} \mathbf{S}_A & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{BB} \end{pmatrix}\begin{pmatrix} \mathbf{I}_A & \mathbf{0} \\ \mathbf{M}_{BB}^{-1}\mathbf{M}_{BA} & \mathbf{I}_B \end{pmatrix} \\ &\quad= \begin{pmatrix} \mathbf{S}_A & \mathbf{M}_{AB} \\ \mathbf{0} & \mathbf{M}_{BB} \end{pmatrix}\begin{pmatrix} \mathbf{I}_A & \mathbf{0} \\ \mathbf{M}_{BB}^{-1}\mathbf{M}_{BA} & \mathbf{I}_B \end{pmatrix} = \begin{pmatrix} \mathbf{S}_A+\mathbf{M}_{AB}\mathbf{M}_{BB}^{-1}\mathbf{M}_{BA} & \mathbf{M}_{AB} \\ \mathbf{M}_{BA} & \mathbf{M}_{BB} \end{pmatrix} = \mathbf{M}, \end{aligned} \tag{B.36}$$

since $\mathbf{S}_A = \mathbf{M}_{AA} - \mathbf{M}_{AB}\mathbf{M}_{BB}^{-1}\mathbf{M}_{BA}$. The proof of Eq. (B.32) is identical upon exchanging $A \leftrightarrow B$.

For part (b), a block triangular matrix with identity blocks $\mathbf{I}_A$ and $\mathbf{I}_B$ on the diagonal has determinant 1, while a block diagonal matrix has determinant equal to the product of the determinants of its diagonal blocks. Taking determinants of Eqs. (B.31) and (B.32) gives Eq. (B.33).

For part (c), the inverse of the product $\mathbf{UDL}$ in Eq. (B.31) is $\mathbf{L}^{-1}\mathbf{D}^{-1}\mathbf{U}^{-1}$. Because $\left(\begin{smallmatrix} \mathbf{I}_A & \mathbf{X} \\ \mathbf{0} & \mathbf{I}_B \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} \mathbf{I}_A & -\mathbf{X} \\ \mathbf{0} & \mathbf{I}_B \end{smallmatrix}\right)$, inverting Eq. (B.31) yields

$$\begin{aligned} \mathbf{M}^{-1} &= \begin{pmatrix} \mathbf{I}_A & \mathbf{0} \\ -\mathbf{M}_{BB}^{-1}\mathbf{M}_{BA} & \mathbf{I}_B \end{pmatrix}\begin{pmatrix} \mathbf{S}_A^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{BB}^{-1} \end{pmatrix}\begin{pmatrix} \mathbf{I}_A & -\mathbf{M}_{AB}\mathbf{M}_{BB}^{-1} \\ \mathbf{0} & \mathbf{I}_B \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{S}_A^{-1} & \mathbf{0} \\ -\mathbf{M}_{BB}^{-1}\mathbf{M}_{BA}\mathbf{S}_A^{-1} & \mathbf{M}_{BB}^{-1} \end{pmatrix}\begin{pmatrix} \mathbf{I}_A & -\mathbf{M}_{AB}\mathbf{M}_{BB}^{-1} \\ \mathbf{0} & \mathbf{I}_B \end{pmatrix}, \end{aligned} \tag{B.37}$$

which multiplies out directly to Eq. (B.34). Inverting Eq. (B.32) in the same way produces Eq. (B.35). □

Corollary B.7 (Woodbury Matrix Identity)**.** *Equating the* (A, A) *blocks of the two expressions Eqs.* (B.34) *and* (B.35) *for* $\mathbf{M}^{-1}$ *and substituting* $\mathbf{M}_{AA} = \mathbf{A}$, $\mathbf{M}_{AB} = -\mathbf{U}$, $\mathbf{M}_{BA} = \mathbf{V}$, *and* $\mathbf{M}_{BB} = \mathbf{C}^{-1}$ *yields the* ***Woodbury matrix identity****:*

$$(\mathbf{A}+\mathbf{UCV})^{-1} \;=\; \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\left(\mathbf{C}^{-1}+\mathbf{VA}^{-1}\mathbf{U}\right)^{-1}\mathbf{VA}^{-1}. \tag{B.38}$$

We can also verify Eq. (B.38) *directly by multiplying the right-hand side on the left by* $(\mathbf{A}+\mathbf{UCV})$*:*

$$\begin{aligned} &(\mathbf{A}+\mathbf{UCV})\left[\mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\left(\mathbf{C}^{-1}+\mathbf{VA}^{-1}\mathbf{U}\right)^{-1}\mathbf{VA}^{-1}\right] \\ &\quad= \mathbf{I} + \mathbf{UCVA}^{-1} - \mathbf{U}\left(\mathbf{I}+\mathbf{CVA}^{-1}\mathbf{U}\right)\left(\mathbf{C}^{-1}+\mathbf{VA}^{-1}\mathbf{U}\right)^{-1}\mathbf{VA}^{-1} \\ &\quad= \mathbf{I} + \mathbf{UCVA}^{-1} - \mathbf{UC}\left(\mathbf{C}^{-1}+\mathbf{VA}^{-1}\mathbf{U}\right)\left(\mathbf{C}^{-1}+\mathbf{VA}^{-1}\mathbf{U}\right)^{-1}\mathbf{VA}^{-1} \;=\; \mathbf{I}. \end{aligned} \tag{B.39}$$

B.2.2 Marginalization and Conditioning of Multivariate Gaussians

Now let $\mathbf{x} = (\mathbf{x}_A^\top, \mathbf{x}_B^\top)^\top \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be a jointly Gaussian random vector with partitioned mean, covariance, and precision matrix $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$:

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_A \\ \boldsymbol{\mu}_B \end{pmatrix}, \qquad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{AA} & \boldsymbol{\Sigma}_{AB} \\ \boldsymbol{\Sigma}_{BA} & \boldsymbol{\Sigma}_{BB} \end{pmatrix}, \qquad \boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1} = \begin{pmatrix} \boldsymbol{\Lambda}_{AA} & \boldsymbol{\Lambda}_{AB} \\ \boldsymbol{\Lambda}_{BA} & \boldsymbol{\Lambda}_{BB} \end{pmatrix}, \tag{B.40}$$

where $\boldsymbol{\Sigma}_{BA} = \boldsymbol{\Sigma}_{AB}^\top$ and $\boldsymbol{\Lambda}_{BA} = \boldsymbol{\Lambda}_{AB}^\top$. Applying Theorem B.6 to $\mathbf{M} = \boldsymbol{\Lambda}$ and $\mathbf{M}^{-1} = \boldsymbol{\Sigma}$ relates the blocks of $\boldsymbol{\Sigma}$ and $\boldsymbol{\Lambda}$:

$$\boldsymbol{\Sigma}_{AA} = (\boldsymbol{\Lambda}/\boldsymbol{\Lambda}_{BB})^{-1} = \left(\boldsymbol{\Lambda}_{AA} - \boldsymbol{\Lambda}_{AB}\boldsymbol{\Lambda}_{BB}^{-1}\boldsymbol{\Lambda}_{BA}\right)^{-1}, \tag{B.41}$$

$$\boldsymbol{\Lambda}_{AA} = (\boldsymbol{\Sigma}/\boldsymbol{\Sigma}_{BB})^{-1} = \left(\boldsymbol{\Sigma}_{AA} - \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\boldsymbol{\Sigma}_{BA}\right)^{-1}, \tag{B.42}$$

and for the off-diagonal blocks,

$$-\boldsymbol{\Lambda}_{AA}^{-1}\boldsymbol{\Lambda}_{AB} = \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}. \tag{B.43}$$

These relations give the rules for Gaussian conditioning and marginalization.

Theorem B.8 (Marginal and Conditional Distributions of a Partitioned Gaussian)**.** *Let* $\mathbf{x} = (\mathbf{x}_A^\top, \mathbf{x}_B^\top)^\top \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ *with precision* $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$ *partitioned as in Eq. (B.40). Then:*

(a) ***Conditional distribution (Clamping*** $\mathbf{x}_B$***):*** *Given* $\mathbf{x}_B$*, the conditional law* $p(\mathbf{x}_A \mid \mathbf{x}_B) = \mathcal{N}(\boldsymbol{\mu}_{A|B}, \boldsymbol{\Sigma}_{A|B})$ *is Gaussian with precision* $\boldsymbol{\Lambda}_{A|B} = \boldsymbol{\Sigma}_{A|B}^{-1}$ *and parameters*

$$\boldsymbol{\Lambda}_{A|B} = \boldsymbol{\Lambda}_{AA}, \qquad \boldsymbol{\Sigma}_{A|B} = \boldsymbol{\Lambda}_{AA}^{-1} = \boldsymbol{\Sigma}_{AA} - \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\boldsymbol{\Sigma}_{BA}, \tag{B.44}$$

$$\boldsymbol{\mu}_{A|B} = \boldsymbol{\mu}_A - \boldsymbol{\Lambda}_{AA}^{-1}\boldsymbol{\Lambda}_{AB}(\mathbf{x}_B - \boldsymbol{\mu}_B) = \boldsymbol{\mu}_A + \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}(\mathbf{x}_B - \boldsymbol{\mu}_B). \tag{B.45}$$

(b) ***Marginal distribution (Integrating out*** $\mathbf{x}_B$***):*** *The marginal law*

$$p(\mathbf{x}_A) = \int_{\mathbb{R}^{N_B}} \mathrm{d}^{N_B} x_B \, p(\mathbf{x}_A, \mathbf{x}_B) = \mathcal{N}(\boldsymbol{\mu}_A, \boldsymbol{\Sigma}_A)$$

is Gaussian with parameters

$$\boldsymbol{\Sigma}_A = \boldsymbol{\Sigma}_{AA}, \qquad \boldsymbol{\Lambda}_A := \boldsymbol{\Sigma}_A^{-1} = \boldsymbol{\Lambda}/\boldsymbol{\Lambda}_{BB} = \boldsymbol{\Lambda}_{AA} - \boldsymbol{\Lambda}_{AB}\boldsymbol{\Lambda}_{BB}^{-1}\boldsymbol{\Lambda}_{BA}. \tag{B.46}$$

Proof. Let $\delta\mathbf{x}_A := \mathbf{x}_A - \boldsymbol{\mu}_A$ and $\delta\mathbf{x}_B := \mathbf{x}_B - \boldsymbol{\mu}_B$. Applying the block LDU factorization Eq. (B.32) to $\boldsymbol{\Lambda}$, we decompose the joint quadratic form $\delta\mathbf{x}^\top \boldsymbol{\Lambda}\, \delta\mathbf{x}$ into two decoupled squares:

$$\begin{aligned} \begin{pmatrix} \delta\mathbf{x}_A^\top & \delta\mathbf{x}_B^\top \end{pmatrix} \begin{pmatrix} \boldsymbol{\Lambda}_{AA} & \boldsymbol{\Lambda}_{AB} \\ \boldsymbol{\Lambda}_{BA} & \boldsymbol{\Lambda}_{BB} \end{pmatrix} \begin{pmatrix} \delta\mathbf{x}_A \\ \delta\mathbf{x}_B \end{pmatrix} &= \delta\mathbf{x}_A^\top \boldsymbol{\Lambda}_{AA}\delta\mathbf{x}_A + 2\,\delta\mathbf{x}_A^\top \boldsymbol{\Lambda}_{AB}\delta\mathbf{x}_B + \delta\mathbf{x}_B^\top \boldsymbol{\Lambda}_{BB}\delta\mathbf{x}_B \\ &= \left(\delta\mathbf{x}_A + \boldsymbol{\Lambda}_{AA}^{-1}\boldsymbol{\Lambda}_{AB}\delta\mathbf{x}_B\right)^\top \boldsymbol{\Lambda}_{AA} \left(\delta\mathbf{x}_A + \boldsymbol{\Lambda}_{AA}^{-1}\boldsymbol{\Lambda}_{AB}\delta\mathbf{x}_B\right) \\ &\quad + \delta\mathbf{x}_B^\top \left(\boldsymbol{\Lambda}_{BB} - \boldsymbol{\Lambda}_{BA}\boldsymbol{\Lambda}_{AA}^{-1}\boldsymbol{\Lambda}_{AB}\right) \delta\mathbf{x}_B. \end{aligned} \tag{B.47}$$

For part (a), when $\mathbf{x}_B$ is held fixed, the second term in Eq. (B.47) is a constant independent of $\mathbf{x}_A$ that cancels in the conditional normalization $p(\mathbf{x}_A \mid \mathbf{x}_B) = p(\mathbf{x}_A, \mathbf{x}_B)/p(\mathbf{x}_B)$. The dependence on $\mathbf{x}_A$ is quadratic with precision matrix $\boldsymbol{\Lambda}_{A|B} = \boldsymbol{\Lambda}_{AA}$ and center $\delta\mathbf{x}_A = -\boldsymbol{\Lambda}_{AA}^{-1}\boldsymbol{\Lambda}_{AB}\delta\mathbf{x}_B$. Applying the

block-inversion relations Eqs. (B.41) and (B.43) converts these precision formulas directly into the covariance expressions in Eqs. (B.44) and (B.45).

For part (b), to integrate out $\mathbf{x}_B$ while holding $\mathbf{x}_A$ fixed, interchange the roles of A and B in Eq. (B.47):

$$\begin{aligned}\delta\mathbf{x}^\top \boldsymbol{\Lambda}\, \delta\mathbf{x} \;=\; & \left(\delta\mathbf{x}_B + \boldsymbol{\Lambda}_{BB}^{-1}\boldsymbol{\Lambda}_{BA}\delta\mathbf{x}_A\right)^\top \boldsymbol{\Lambda}_{BB}\left(\delta\mathbf{x}_B + \boldsymbol{\Lambda}_{BB}^{-1}\boldsymbol{\Lambda}_{BA}\delta\mathbf{x}_A\right) \\ & + \delta\mathbf{x}_A^\top\left(\boldsymbol{\Lambda}_{AA} - \boldsymbol{\Lambda}_{AB}\boldsymbol{\Lambda}_{BB}^{-1}\boldsymbol{\Lambda}_{BA}\right)\delta\mathbf{x}_A. \end{aligned} \tag{B.48}$$

Integrating $\exp(-\frac{1}{2}\delta\mathbf{x}^\top\boldsymbol{\Lambda}\,\delta\mathbf{x})$ over $\mathbf{x}_B \in \mathbb{R}^{N_B}$ via Theorem B.2 produces the prefactor

$$(2\pi)^{N_B/2}(\det \boldsymbol{\Lambda}_{BB})^{-1/2},$$

which combines with the joint normalization $(2\pi)^{-(N_A+N_B)/2}(\det\boldsymbol{\Lambda})^{1/2}$ and the Schur determinant identity

$$\det\boldsymbol{\Lambda} = \det(\boldsymbol{\Lambda}_{BB})\det(\boldsymbol{\Lambda}/\boldsymbol{\Lambda}_{BB})$$

from Eq. (B.33) to yield the marginal density $\mathcal{N}(\boldsymbol{\mu}_A, (\boldsymbol{\Lambda}/\boldsymbol{\Lambda}_{BB})^{-1})$ with

$$\boldsymbol{\Sigma}_A = (\boldsymbol{\Lambda}/\boldsymbol{\Lambda}_{BB})^{-1} = \boldsymbol{\Sigma}_{AA}.$$

□

Hardware Realization: Kron Reduction in Coupled RC Networks vs. Clamped Conditional Sampling

When the RC sampler is driven by engineered noise (Section 5.6), its precision matrix is proportional to the nodal conductance matrix, $\boldsymbol{\Lambda} = \beta\tau_0\mathbf{G}$. (A passive RC network in thermal equilibrium has precision $\beta\mathbf{C}$ instead, and the same algebra then applies to the capacitance matrix.) Theorem B.8 expresses the difference between two circuit operations:

- **Clamping nodes B to fixed voltage sources** (conditioning on $\mathbf{x}_B$) deletes the rows and columns of B from the conductance matrix: $\boldsymbol{\Lambda}_{A|B} = \boldsymbol{\Lambda}_{AA}$, while the clamped nodes inject the current $-\mathbf{G}_{AB}(\mathbf{x}_B - \boldsymbol{\mu}_B)$ into nodes A.
- **Leaving nodes B electrically floating** so they fluctuate thermally (marginalizing over $\mathbf{x}_B$) keeps the covariance $\boldsymbol{\Sigma}_{AA}$ unchanged, but modifies the effective conductance matrix between the visible nodes A via the Schur complement $\mathbf{G}_{\text{eff}} = \mathbf{G}_{AA} - \mathbf{G}_{AB}\mathbf{G}_{BB}^{-1}\mathbf{G}_{BA}$. In circuit theory, this Schur-complement elimination of internal floating nodes is called **Kron reduction**. Thermodynamic linear algebra [3] uses this to invert matrices: once the network has relaxed, the empirical covariance $\boldsymbol{\Sigma}_{AA}$ of the nodes A estimates $k_{\mathrm{B}}T(\tau_0\mathbf{G}_{\text{eff}})^{-1}$. Relaxation takes a few of the network's slowest relaxation times, and the estimate then needs enough samples to reach the desired precision.

B.3 Isserlis–Wick Combinatorics and Cumulant Expansions

Many circuits and Energy-Based Models contain anharmonic terms. Examples are the quartic term $\frac{a}{4}x^4$ of the double-well sampler, whose energy is $E(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2 - hx$, and the cubic and quartic Taylor terms of the Enz–Krummenacher–Vittoz (EKV) transistor law (Chapters 12 and 13). Perturbation theory around a Gaussian then requires expectations of products of n Gaussian random variables, $\langle x_{i_1}x_{i_2}\cdots x_{i_n}\rangle$, organized into **cumulants** (connected correlation functions).

B.3.1 Stein's Lemma and the Isserlis–Wick Theorem

We begin with an identity obtained by integrating the Gaussian density by parts.

Lemma B.9 (Stein's Gaussian Integration-by-Parts Lemma)**.** *Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ be a centered N-dimensional Gaussian vector with covariance $\mathbf{\Sigma} \succ \mathbf{0}$, and let $f\colon \mathbb{R}^N \to \mathbb{R}$ be a continuously differentiable function of at most polynomial growth at infinity. Then for every index $i \in \{1, \dots, N\}$,*

$$\mathbb{E}[x_i\, f(\mathbf{x})] \;=\; \sum_{j=1}^{N} \Sigma_{ij}\, \mathbb{E}\left[\frac{\partial f(\mathbf{x})}{\partial x_j}\right]. \tag{B.49}$$

Proof. Let $p(\mathbf{x}) = (2\pi)^{-N/2}(\det \mathbf{\Sigma})^{-1/2} \exp(-\tfrac{1}{2}\mathbf{x}^\top \mathbf{\Sigma}^{-1}\mathbf{x})$ be the centered Gaussian density. Differentiating $p(\mathbf{x})$ with respect to the k-th coordinate gives

$$\frac{\partial p(\mathbf{x})}{\partial x_k} \;=\; -\left(\mathbf{\Sigma}^{-1}\mathbf{x}\right)_k p(\mathbf{x}) \;=\; -\sum_{m=1}^{N} (\mathbf{\Sigma}^{-1})_{km} x_m\, p(\mathbf{x}). \tag{B.50}$$

Multiplying both sides by Σ_{ik} and summing over $k = 1, \dots, N$ uses $\sum_{k=1}^{N} \Sigma_{ik}(\mathbf{\Sigma}^{-1})_{km} = \delta_{im}$ to isolate $x_i\, p(\mathbf{x})$:

$$x_i\, p(\mathbf{x}) \;=\; -\sum_{j=1}^{N} \Sigma_{ij}\, \frac{\partial p(\mathbf{x})}{\partial x_j}. \tag{B.51}$$

Multiplying Eq. (B.51) by $f(\mathbf{x})$, integrating over $\mathbb{R}^N$, and integrating by parts in each coordinate x_j (where the boundary terms at $x_j \to \pm\infty$ vanish because the Gaussian $\exp(-\tfrac{1}{2}\mathbf{x}^\top \mathbf{\Sigma}^{-1}\mathbf{x})$ decays faster than any polynomial $f(\mathbf{x})$) yields

$$\mathbb{E}[x_i\, f(\mathbf{x})] \;=\; -\sum_{j=1}^{N} \Sigma_{ij} \int_{\mathbb{R}^N} \mathrm{d}^N x\, f(\mathbf{x})\, \frac{\partial p(\mathbf{x})}{\partial x_j} \;=\; \sum_{j=1}^{N} \Sigma_{ij} \int_{\mathbb{R}^N} \mathrm{d}^N x\, p(\mathbf{x})\, \frac{\partial f(\mathbf{x})}{\partial x_j} \;=\; \sum_{j=1}^{N} \Sigma_{ij}\, \mathbb{E}\left[\frac{\partial f(\mathbf{x})}{\partial x_j}\right], \tag{B.52}$$

which proves Eq. (B.49). □

To state the Isserlis–Wick theorem we need one combinatorial notion. A **perfect matching** (or **Wick pairing**) $\pi \in \mathcal{P}_{2m}$ of the index set $\{1, 2, \dots, 2m\}$ is a partition of $\{1, \dots, 2m\}$ into m unordered disjoint pairs $\pi = \{(a_1, b_1), (a_2, b_2), \dots, (a_m, b_m)\}$. To count the pairings, observe that element 1 can be paired with any of the remaining $2m-1$ elements; the smallest remaining unpaired element can then be paired with any of the remaining $2m-3$ elements, and so on, giving the double factorial

$$|\mathcal{P}_{2m}| \;=\; (2m-1)!! \;:=\; (2m-1)(2m-3)\cdots 3 \cdot 1 \;=\; \frac{(2m)!}{2^m\, m!}. \tag{B.53}$$

Theorem B.10 (Isserlis' / Wick's Theorem)**.** *Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ be a centered multivariate Gaussian vector, and let $i_1, i_2, \dots, i_n \in \{1, \dots, N\}$ be any sequence of n indices (with repetitions allowed). Then:*

(a) For every odd integer $n = 2m+1$, all odd-order moments vanish:

$$\mathbb{E}\big[x_{i_1} x_{i_2} \cdots x_{i_{2m+1}}\big] \;=\; 0. \tag{B.54}$$

(b) For every even integer $n = 2m$, the $2m$-point correlation function equals the sum over all $(2m-1)!!$ Wick pairings $\pi \in \mathcal{P}_{2m}$ of products of two-point covariances:

$$\mathbb{E}[x_{i_1} x_{i_2} \cdots x_{i_{2m}}] \;=\; \sum_{\pi \in \mathcal{P}_{2m}} \prod_{(r,s)\in\pi} \mathbb{E}[x_{i_r} x_{i_s}] \;=\; \sum_{\pi \in \mathcal{P}_{2m}} \prod_{(r,s)\in\pi} \Sigma_{i_r i_s}. \tag{B.55}$$

In particular, for $n = 4$, there are $3!! = 3$ pairings:

$$\mathbb{E}[x_{i_1}x_{i_2}x_{i_3}x_{i_4}] \;=\; \Sigma_{i_1i_2}\Sigma_{i_3i_4} + \Sigma_{i_1i_3}\Sigma_{i_2i_4} + \Sigma_{i_1i_4}\Sigma_{i_2i_3}. \tag{B.56}$$

For $n = 6$ and a single scalar variable $x_1 \sim \mathcal{N}(0, \sigma^2)$, there are $5!! = 15$ pairings, giving $\mathbb{E}\big[x_1^6\big] = 15\sigma^6$, and in general $\mathbb{E}\big[x_1^{2m}\big] = (2m-1)!!\,\sigma^{2m}$.

Proof. We give two proofs: an inductive proof via Stein's Lemma and a generating-function proof via Theorem B.2.

Proof 1 (Induction via Stein's Lemma). Part (a) follows from the inversion symmetry $p(-\mathbf{x}) = p(\mathbf{x})$ of the centered Gaussian density: changing variables $\mathbf{x} \mapsto -\mathbf{x}$ sends $x_{i_1}\cdots x_{i_{2m+1}} \mapsto (-1)^{2m+1}x_{i_1}\cdots x_{i_{2m+1}} = -x_{i_1}\cdots x_{i_{2m+1}}$, so the integral equals its own negative and is zero.

For part (b), we induct on $m \geq 1$. For $m = 1$ ($n = 2$), there is $1!! = 1$ pair $(1, 2)$ and $\mathbb{E}[x_{i_1}x_{i_2}] = \Sigma_{i_1i_2}$ by definition of the covariance matrix. Now assume Eq. (B.55) holds for $2m-2$ variables, and consider $n = 2m$ variables. Applying Stein's Lemma (Lemma B.9) with the first factor x_{i_1} and $f(\mathbf{x}) = x_{i_2}x_{i_3}\cdots x_{i_{2m}}$, and using the Leibniz product rule $\frac{\partial}{\partial x_j}(x_{i_2}\cdots x_{i_{2m}}) = \sum_{s=2}^{2m}\delta_{j,i_s}\prod_{r\neq 1,s}x_{i_r}$, we obtain

$$\begin{aligned}\mathbb{E}[x_{i_1}x_{i_2}\cdots x_{i_{2m}}] \;&=\; \sum_{j=1}^{N}\Sigma_{i_1j}\,\mathbb{E}\left[\frac{\partial}{\partial x_j}\left(x_{i_2}x_{i_3}\cdots x_{i_{2m}}\right)\right] \\ &=\; \sum_{s=2}^{2m}\Sigma_{i_1i_s}\,\mathbb{E}\left[\prod_{\substack{r=2\\ r\neq s}}^{2m} x_{i_r}\right].\end{aligned} \tag{B.57}$$

By the induction hypothesis, for each choice of partner $s \in \{2, \ldots, 2m\}$ for index 1, the remaining expectation of $2m-2$ variables expands as the sum over all $(2m-3)!!$ pairings $\pi' \in \mathcal{P}(\{2, \ldots, 2m\} \setminus \{s\})$. Appending the pair $(1, s)$ to each π' establishes a bijection with the set of all $(2m-1)(2m-3)!! = (2m-1)!!$ pairings $\pi \in \mathcal{P}_{2m}$, completing the induction.

Proof 2 (Moment Generating Function). From Theorem B.2, the moment generating function (MGF) of $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ is

$$M(\mathbf{J}) \;:=\; \mathbb{E}\Big[e^{\mathbf{J}^\top\mathbf{x}}\Big] \;=\; \frac{Z(\mathbf{\Sigma}^{-1}, \mathbf{J})}{Z(\mathbf{\Sigma}^{-1}, \mathbf{0})} \;=\; \exp\!\left(\frac{1}{2}\mathbf{J}^\top\mathbf{\Sigma}\mathbf{J}\right) \;=\; \sum_{m=0}^{\infty}\frac{1}{2^m\,m!}\left(\sum_{a,b=1}^{N}\Sigma_{ab}J_aJ_b\right)^m. \tag{B.58}$$

Expanding $\mathbb{E}\Big[e^{\mathbf{J}^\top\mathbf{x}}\Big]$ in a Taylor series around $\mathbf{J} = \mathbf{0}$ shows that

$$\mathbb{E}[x_{i_1}x_{i_2}\cdots x_{i_n}] \;=\; \left.\frac{\partial^n M(\mathbf{J})}{\partial J_{i_1}\partial J_{i_2}\cdots\partial J_{i_n}}\right|_{\mathbf{J}=\mathbf{0}}. \tag{B.59}$$

Since the right-hand side of Eq. (B.58) contains only even powers $2m$ of $\mathbf{J}$, all odd derivatives vanish at $\mathbf{J} = \mathbf{0}$. For $n = 2m$, only the m-th term $\frac{1}{2^m m!}(\mathbf{J}^\top\mathbf{\Sigma}\mathbf{J})^m$ survives differentiation $2m$ times at $\mathbf{J} = \mathbf{0}$. When the differential operators $\partial_{J_{i_1}}\cdots\partial_{J_{i_{2m}}}$ act on the product of m factors $(\mathbf{J}^\top\mathbf{\Sigma}\mathbf{J})\cdots(\mathbf{J}^\top\mathbf{\Sigma}\mathbf{J})$, each of the $(2m)!$ ways of assigning the $2m$ derivatives to the $2m$ slots of $\mathbf{J}$ induces a pairing $\pi \in \mathcal{P}_{2m}$. Because permuting the m identical quadratic factors ($m!$ ways) and swapping the two slots within each quadratic factor (2^m ways, using $\Sigma_{ab} = \Sigma_{ba}$) yield the same pairing π, each distinct pairing $\pi \in \mathcal{P}_{2m}$ appears with multiplicity $2^m m!$, which cancels the denominator $\frac{1}{2^m m!}$ in Eq. (B.58) and leaves Eq. (B.55). □

B.3.2 Cumulant Generating Functions and Anharmonic Perturbation Theory

For a general random variable X or vector $\mathbf{x} \in \mathbb{R}^N$, moments $\mu_n = \mathbb{E}[X^n]$ combine irreducible n-body correlations with disconnected products of lower-order moments. The **cumulant generating function** (CGF) isolates the connected correlation functions.

Definition B.11 (Cumulant Generating Function and Cumulants)**.** For a scalar random variable X whose moment generating function $M(\lambda) := \mathbb{E}\left[e^{\lambda X}\right]$ exists in a neighborhood of $\lambda = 0$, the **cumulant generating function** is $K(\lambda) := \ln \mathbb{E}\left[e^{\lambda X}\right]$, and the n-th **cumulant** $\kappa_n(X)$ is defined by the Maclaurin series

$$K(\lambda) \;:=\; \ln \mathbb{E}\left[e^{\lambda X}\right] \;=\; \sum_{n=1}^{\infty} \frac{\lambda^n}{n!}\, \kappa_n(X), \qquad \text{or} \qquad \mathbb{E}\left[e^{\lambda X}\right] \;=\; \exp\!\left(\sum_{n=1}^{\infty} \frac{\lambda^n}{n!}\, \kappa_n(X)\right). \tag{B.60}$$

For a random vector $\mathbf{x} \in \mathbb{R}^N$, the joint cumulant tensor $\kappa_n(x_{i_1}, \ldots, x_{i_n})$ is defined by

$$\kappa_n(x_{i_1}, \ldots, x_{i_n}) \;:=\; \frac{\partial^n}{\partial J_{i_1} \cdots \partial J_{i_n}} \ln \mathbb{E}\left[e^{\mathbf{J}^\top \mathbf{x}}\right]\Big|_{\mathbf{J}=\mathbf{0}}. \tag{B.61}$$

Proposition B.12 (Explicit Relations Between Moments and Cumulants ($n \le 4$))**.** *Let $\mu_n := \mathbb{E}[X^n]$ be the raw moments and $m_n := \mathbb{E}[(X-\mu_1)^n]$ the central moments of a scalar random variable X. Then the first four cumulants are given by*

$$\kappa_1 \;=\; \mu_1 \;=\; \mathbb{E}[X]\,, \tag{B.62}$$

$$\kappa_2 \;=\; \mu_2 - \mu_1^2 \;=\; m_2 \;=\; \operatorname{Var}(X), \tag{B.63}$$

$$\kappa_3 \;=\; \mu_3 - 3\mu_2\mu_1 + 2\mu_1^3 \;=\; m_3 \;=\; \mathbb{E}\left[(X - \mathbb{E}[X])^3\right], \tag{B.64}$$

$$\kappa_4 \;=\; \mu_4 - 4\mu_3\mu_1 - 3\mu_2^2 + 12\mu_2\mu_1^2 - 6\mu_1^4 \;=\; m_4 - 3m_2^2 \;=\; \mathbb{E}\left[(X - \mathbb{E}[X])^4\right] - 3\big(\operatorname{Var}(X)\big)^2. \tag{B.65}$$

More generally, for a centered vector $\mathbf{x}$ ($\mathbb{E}[\mathbf{x}] = \mathbf{0}$), the fourth-order joint cumulant tensor is

$$\kappa_4(x_{i_1}, x_{i_2}, x_{i_3}, x_{i_4}) \;=\; \mathbb{E}[x_{i_1}x_{i_2}x_{i_3}x_{i_4}] - \Sigma_{i_1 i_2}\Sigma_{i_3 i_4} - \Sigma_{i_1 i_3}\Sigma_{i_2 i_4} - \Sigma_{i_1 i_4}\Sigma_{i_2 i_3}, \tag{B.66}$$

where $\Sigma_{ab} = \mathbb{E}[x_a x_b]$. A distribution is Gaussian if and only if $\kappa_n \equiv 0$ for all $n \ge 3$.

Proof. Expanding $M(\lambda) = 1 + \sum_{n=1}^{\infty} \frac{\lambda^n}{n!}\mu_n := 1 + u(\lambda)$ and substituting into $\ln(1+u) = u - \frac{1}{2}u^2 + \frac{1}{3}u^3 - \frac{1}{4}u^4 + \mathcal{O}(u^5)$ up to order λ^4 gives

$$\begin{aligned} K(\lambda) \;&=\; \left(\lambda\mu_1 + \frac{\lambda^2}{2}\mu_2 + \frac{\lambda^3}{6}\mu_3 + \frac{\lambda^4}{24}\mu_4\right) - \frac{1}{2}\left(\lambda^2\mu_1^2 + \lambda^3\mu_1\mu_2 + \lambda^4\left(\frac{1}{4}\mu_2^2 + \frac{1}{3}\mu_1\mu_3\right)\right) \\ &\quad + \frac{1}{3}\left(\lambda^3\mu_1^3 + \frac{3}{2}\lambda^4\mu_1^2\mu_2\right) - \frac{1}{4}\left(\lambda^4\mu_1^4\right) + \mathcal{O}(\lambda^5) \\ &=\; \lambda\,\mu_1 + \frac{\lambda^2}{2!}(\mu_2 - \mu_1^2) + \frac{\lambda^3}{3!}(\mu_3 - 3\mu_2\mu_1 + 2\mu_1^3) \qquad \text{(B.67)} \\ &\quad + \frac{\lambda^4}{4!}(\mu_4 - 4\mu_3\mu_1 - 3\mu_2^2 + 12\mu_2\mu_1^2 - 6\mu_1^4) + \mathcal{O}(\lambda^5). \qquad \text{(B.68)} \end{aligned}$$

Matching coefficients with $\sum_{n=1}^{\infty} \frac{\lambda^n}{n!}\kappa_n$ yields Eqs. (B.62) to (B.65). Since shifting $X \mapsto X - \mu_1$ adds $-\lambda\mu_1$ to $K(\lambda)$, all cumulants of order $n \ge 2$ are shift-invariant, so setting $\mu_1 = 0$ gives $\kappa_2 = m_2$, $\kappa_3 = m_3$, and $\kappa_4 = m_4 - 3m_2^2$. Polarizing to four components $x_{i_1}, x_{i_2}, x_{i_3}, x_{i_4}$ gives Eq. (B.66), which is the difference between the four-point moment and its Gaussian Isserlis–Wick value Eq. (B.56). Finally, if $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then by Eq. (B.8) its CGF is the quadratic polynomial $K(\mathbf{J}) = \mathbf{J}^\top\boldsymbol{\mu} + \frac{1}{2}\mathbf{J}^\top\boldsymbol{\Sigma}\mathbf{J}$, whose third and higher derivatives vanish identically. □

Proposition B.13 (First-Order Anharmonic Perturbation of Cumulants via Wick Contractions)**.** *Consider a weakly anharmonic energy with a symmetric fourth-order coupling tensor V_{abcd}:*

$$\beta E(\mathbf{x}) \;=\; \frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}_0^{-1}\mathbf{x} + \frac{\epsilon}{4!}\sum_{a,b,c,d=1}^{N} V_{abcd}\, x_a x_b x_c x_d, \tag{B.69}$$

where $\epsilon \geq 0$ is a small nonlinearity parameter and $\langle\cdot\rangle_0$ denotes expectation under the unperturbed Gaussian $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_0)$. To first order in ϵ:

(a) The log-partition function shifts by the fully contracted vacuum diagram (3 pairings):

$$\ln \frac{Z(\epsilon)}{Z(0)} \;=\; -\frac{\epsilon}{8}\sum_{a,b,c,d=1}^{N} V_{abcd}\, \Sigma_{0,ab}\Sigma_{0,cd} + \mathcal{O}(\epsilon^2). \tag{B.70}$$

(b) The two-point covariance $\Sigma_{ij}(\epsilon) = \langle x_i x_j\rangle_\epsilon$ shifts only through connected one-loop contractions (12 pairings; the 3 disconnected vacuum pairings cancel against the normalization $Z(\epsilon)$):

$$\Sigma_{ij}(\epsilon) \;=\; \Sigma_{0,ij} - \frac{\epsilon}{2}\sum_{a,b,c,d=1}^{N} V_{abcd}\, \Sigma_{0,ia}\Sigma_{0,jb}\Sigma_{0,cd} + \mathcal{O}(\epsilon^2). \tag{B.71}$$

(c) The fourth cumulant $\kappa_4(x_i, x_j, x_k, x_l)$ acquires the tree-level four-point vertex (24 pairings):

$$\kappa_4(x_i, x_j, x_k, x_l) \;=\; -\epsilon\sum_{a,b,c,d=1}^{N} V_{abcd}\, \Sigma_{0,ia}\Sigma_{0,jb}\Sigma_{0,kc}\Sigma_{0,ld} + \mathcal{O}(\epsilon^2). \tag{B.72}$$

Proof. Let $U(\mathbf{x}) := \frac{1}{4!}\sum_{a,b,c,d} V_{abcd}x_a x_b x_c x_d$. Expanding $e^{-\epsilon U(\mathbf{x})} = 1 - \epsilon U(\mathbf{x}) + \mathcal{O}(\epsilon^2)$ gives $Z(\epsilon)/Z(0) = 1 - \epsilon\,\langle U(\mathbf{x})\rangle_0 + \mathcal{O}(\epsilon^2)$. By Isserlis–Wick (Eq. (B.56)) and the total symmetry of V_{abcd}, each of the 3 pairings of (a, b, c, d) contributes identically, so $\langle U\rangle_0 = \frac{3}{24}\sum V_{abcd}\Sigma_{0,ab}\Sigma_{0,cd} = \frac{1}{8}\sum V_{abcd}\Sigma_{0,ab}\Sigma_{0,cd}$, proving Eq. (B.70).

For part (b), expanding the normalized expectation gives

$$\langle x_i x_j\rangle_\epsilon \;=\; \frac{\langle x_i x_j(1 - \epsilon U(\mathbf{x}))\rangle_0 + \mathcal{O}(\epsilon^2)}{1 - \epsilon\,\langle U(\mathbf{x})\rangle_0 + \mathcal{O}(\epsilon^2)} \;=\; \Sigma_{0,ij} - \epsilon\Big[\,\langle x_i x_j U(\mathbf{x})\rangle_0 - \langle x_i x_j\rangle_0\,\langle U(\mathbf{x})\rangle_0\,\Big] + \mathcal{O}(\epsilon^2). \tag{B.73}$$

By Theorem B.10, the 6-point expectation $\langle x_i x_j x_a x_b x_c x_d\rangle_0$ sums over $5!! = 15$ pairings. Exactly 3 of these pairings pair x_i with x_j and pair (a, b, c, d) among themselves; their sum is $\Sigma_{0,ij}\,\langle x_a x_b x_c x_d\rangle_0$, which cancels against $-\langle x_i x_j\rangle_0\,\langle U(\mathbf{x})\rangle_0$ in Eq. (B.73). In the remaining $15 - 3 = 12$ connected pairings, x_i pairs with one of the 4 indices (a, b, c, d), x_j pairs with one of the remaining 3 indices, and the last 2 indices pair with each other ($4 \times 3 \times 1 = 12$ identical terms by symmetry of V_{abcd}). Since $12/4! = 12/24 = 1/2$, we obtain Eq. (B.71).

Similarly, in the 8-point expectation $\langle x_i x_j x_k x_l U(\mathbf{x})\rangle_0$ ($7!! = 105$ pairings), subtracting the three disconnected covariance products $\Sigma_{ij}(\epsilon)\Sigma_{kl}(\epsilon) + \Sigma_{ik}(\epsilon)\Sigma_{jl}(\epsilon) + \Sigma_{il}(\epsilon)\Sigma_{jk}(\epsilon)$ in Eq. (B.66) cancels both the $3 \times 3 = 9$ completely disconnected pairings and the $6 \times 12 = 72$ partially connected pairings, leaving the $4! = 24$ fully connected pairings in which (i, j, k, l) each contract with a distinct leg of (a, b, c, d). Since $24/4! = 1$, this yields Eq. (B.72). □

Physics Insight: Jarzynski Cumulant Expansion and Non-Gaussian Circuit Perturbations

Two applications of Definition B.11 and Propositions B.12 and B.13 appear in the main text:

(1) **Dissipated work in Jarzynski's equality (Chapter 9):** Setting $X = \mathcal{W}$ and $\lambda = -\beta$ in Eq. (B.60) converts Jarzynski's identity $\ln\left\langle e^{-\beta\mathcal{W}}\right\rangle = -\beta\Delta F$ into the cumulant series for the mean dissipated work:

$$\langle \mathcal{W}_{\mathrm{diss}}\rangle \;\equiv\; \langle\mathcal{W}\rangle - \Delta F \;=\; \sum_{n=2}^{\infty} \frac{(-1)^n \beta^{n-1}}{n!}\,\kappa_n(\mathcal{W}) \;=\; \frac{\beta}{2}\,\mathrm{Var}(\mathcal{W}) - \frac{\beta^2}{6}\kappa_3(\mathcal{W}) + \frac{\beta^3}{24}\kappa_4(\mathcal{W}) - \cdots. \tag{B.74}$$

For the RC sampler with a dragged bias, the work distribution is Gaussian, so $\kappa_{n\geq 3}(\mathcal{W}) = 0$ and $\langle \mathcal{W}_{\mathrm{diss}}\rangle = \frac{\beta}{2}\,\mathrm{Var}(\mathcal{W})$ holds exactly.

(2) **Non-Gaussian cumulants in subthreshold CMOS NEAT-RNs (Chapter 13):** In the analysis of weakly nonlinear subthreshold circuits [38], Eqs. (B.71) and (B.72) quantify the one-loop covariance renormalization and fourth cumulant κ_4 induced by the cubic and quartic Taylor coefficients of the EKV transistor current.

B.4 The Continuous-Time Lyapunov Equation and Non-Equilibrium Steady States

We now turn to the matrix equation that fixes the stationary covariance of the RC sampler and of its active hardware counterpart, the NEAT-RN of Chapter 13. Near a stable operating point $\mathbf{v}^*$ of an analog circuit, the voltage fluctuations $\delta\mathbf{v}_t = \mathbf{v}_t - \mathbf{v}^* \in \mathbb{R}^N$ obey the linearized multidimensional Ornstein–Uhlenbeck SDE (Chapters 5, 8 and 13)

$$\mathrm{d}(\delta\mathbf{v}_t) \;=\; \mathbf{A}\,\delta\mathbf{v}_t\;\mathrm{d}t + \tilde{\mathbf{B}}\;\mathrm{d}\mathbf{W}_t, \qquad \mathbf{D} \;:=\; \frac{1}{2}\tilde{\mathbf{B}}\tilde{\mathbf{B}}^{\top} \succeq \mathbf{0}, \tag{B.75}$$

where $\mathbf{A} = -\mathbf{C}^{-1}\mathbf{G} \in \mathbb{R}^{N\times N}$ is the drift (relaxation) matrix and $\mathbf{D} \in \mathbb{R}^{N\times N}$ is the symmetric positive-semidefinite diffusion matrix. In a passive reciprocal resistor network at uniform temperature T, $\mathbf{G} = \mathbf{G}^{\top}$ and $\mathbf{D} = k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{G}(\mathbf{C}^{-1})^{\top}$, so detailed balance holds and the stationary covariance is $\mathbf{\Sigma} = k_{\mathrm{B}}T\mathbf{C}^{-1}$ (equipartition of the electrostatic energy $\frac{1}{2}\mathbf{v}^{\top}\mathbf{C}\mathbf{v}$; one checks $\mathbf{A}\mathbf{\Sigma} = -k_{\mathrm{B}}T\mathbf{C}^{-1}\mathbf{G}\mathbf{C}^{-1} = -\mathbf{D}$, so Eq. (B.76) is satisfied). The conductances set the relaxation rates but not the stationary covariance; a covariance programmed by $\mathbf{G}$ requires engineered noise or active devices (Section 5.6). In an active subthreshold CMOS circuit (Chapter 13) where gate-to-drain transconductances are unidirectional ($G_{ij} \neq G_{ji}$) and shot noise depends on bias currents, $\mathbf{A}$ is non-symmetric and does not satisfy $\mathbf{A}\mathbf{D} = \mathbf{D}\mathbf{A}^{\top}$. In every case, passive or active, the stationary covariance $\mathbf{\Sigma} := \lim_{t\to\infty}\mathbb{E}\left[\delta\mathbf{v}_t\,\delta\mathbf{v}_t^{\top}\right]$ is determined by the **continuous-time algebraic Lyapunov equation**:

$$\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^{\top} + 2\mathbf{D} \;=\; \mathbf{0}. \tag{B.76}$$

B.4.1 Hurwitz Stability, Kronecker Sums, and Existence/Uniqueness of $\mathbf{\Sigma}$

Definition B.14 (Hurwitz Matrix)**.** A real matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$ is **Hurwitz stable** (or **Hurwitz**) if every eigenvalue $\lambda \in \mathrm{spec}(\mathbf{A}) \subset \mathbb{C}$ has strictly negative real part:

$$\alpha(\mathbf{A}) \;:=\; -\max_{\lambda\in\mathrm{spec}(\mathbf{A})} \mathrm{Re}(\lambda) \;>\; 0. \tag{B.77}$$

By the Jordan canonical form of $\mathbf{A}$, if $\mathbf{A}$ is Hurwitz, then for any $0 < \alpha_0 < \alpha(\mathbf{A})$ there exists a constant $M \geq 1$ such that the matrix exponential satisfies the exponential decay bound

$$\left\| e^{\mathbf{A}t} \right\| \;\leq\; M e^{-\alpha_0 t} \qquad \text{for all } t \geq 0. \tag{B.78}$$

To analyze the linear operator $\mathcal{L}_{\mathbf{A}}(\mathbf{\Sigma}) := \mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top$ acting on the space of $N \times N$ matrices, we use the column-stacking vectorization operator $\mathrm{vec}\colon \mathbb{R}^{N\times N} \to \mathbb{R}^{N^2}$ and the Kronecker product $\mathbf{X} \otimes \mathbf{Y} \in \mathbb{R}^{N^2 \times N^2}$.

Lemma B.15 (Kronecker Product Vectorization Identity and Spectrum of $\mathcal{L}_{\mathbf{A}}$)**.** *For any matrices $\mathbf{X}, \mathbf{\Sigma}, \mathbf{Y} \in \mathbb{R}^{N\times N}$, the vectorization operator satisfies*

$$\mathrm{vec}(\mathbf{X}\mathbf{\Sigma}\mathbf{Y}) \;=\; \Big(\mathbf{Y}^\top \otimes \mathbf{X}\Big)\,\mathrm{vec}(\mathbf{\Sigma}). \tag{B.79}$$

*Consequently, the Lyapunov operator $\mathcal{L}_{\mathbf{A}}(\mathbf{\Sigma}) = \mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top$ is represented on $\mathbb{R}^{N^2}$ by the **Kronecker sum***

$$\mathrm{vec}\Big(\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top\Big) \;=\; (\mathbf{I} \otimes \mathbf{A} + \mathbf{A} \otimes \mathbf{I})\,\mathrm{vec}(\mathbf{\Sigma}), \tag{B.80}$$

whose N^2 eigenvalues (counted with algebraic multiplicity) are all pairwise sums $\lambda_j + \lambda_k$ for $\lambda_j, \lambda_k \in \mathrm{spec}(\mathbf{A})$.

Proof. Writing $\mathbf{\Sigma} = [\boldsymbol{\sigma}_1, \ldots, \boldsymbol{\sigma}_N]$ in terms of its columns, the k-th column of $\mathbf{X}\mathbf{\Sigma}\mathbf{Y}$ is $\sum_{j=1}^N \mathbf{X}\boldsymbol{\sigma}_j Y_{jk} = \sum_{j=1}^N (Y^\top)_{kj}\mathbf{X}\boldsymbol{\sigma}_j$, which is the k-th N-block of $(\mathbf{Y}^\top \otimes \mathbf{X})\,\mathrm{vec}(\mathbf{\Sigma})$, proving Eq. (B.79). Setting $\mathbf{X} = \mathbf{A}, \mathbf{Y} = \mathbf{I}$ and $\mathbf{X} = \mathbf{I}, \mathbf{Y} = \mathbf{A}^\top$ yields Eq. (B.80). Finally, bringing $\mathbf{A}$ to upper-triangular Schur form $\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{T}$ (with diagonal entries $\lambda_1, \ldots, \lambda_N$) via a unitary transformation $\mathbf{U} \in \mathrm{U}(N)$ transforms $\mathbf{I} \otimes \mathbf{A} + \mathbf{A} \otimes \mathbf{I}$ by $(\mathbf{U} \otimes \mathbf{U})^*(\mathbf{I} \otimes \mathbf{A} + \mathbf{A} \otimes \mathbf{I})(\mathbf{U} \otimes \mathbf{U}) = \mathbf{I} \otimes \mathbf{T} + \mathbf{T} \otimes \mathbf{I}$, which is upper triangular with diagonal entries $\lambda_j + \lambda_k$ $(1 \leq j, k \leq N)$. □

Theorem B.16 (Existence, Uniqueness, and Integral Representation of the Lyapunov Solution)**.** *Let $\mathbf{A} \in \mathbb{R}^{N\times N}$ be Hurwitz stable and let $\mathbf{D} \in \mathbb{R}^{N\times N}$ be symmetric positive semidefinite ($\mathbf{D} = \mathbf{D}^\top \succeq \mathbf{0}$). Then:*

*(a) The continuous-time Lyapunov equation $\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top + 2\mathbf{D} = \mathbf{0}$ has a **unique** solution $\mathbf{\Sigma} \in \mathbb{R}^{N\times N}$, given by the absolutely convergent improper integral*

$$\mathbf{\Sigma} \;=\; 2\int_0^\infty \mathrm{d}t\, e^{\mathbf{A}t}\mathbf{D}e^{\mathbf{A}^\top t}, \tag{B.81}$$

or equivalently in vectorized form by $\mathrm{vec}(\mathbf{\Sigma}) = -2(\mathbf{I} \otimes \mathbf{A} + \mathbf{A} \otimes \mathbf{I})^{-1}\,\mathrm{vec}(\mathbf{D})$.

(b) The solution $\mathbf{\Sigma}$ is symmetric and positive semidefinite ($\mathbf{\Sigma} = \mathbf{\Sigma}^\top \succeq \mathbf{0}$). Moreover, if $\mathbf{D} \succ \mathbf{0}$ is strictly positive definite, then $\mathbf{\Sigma} \succ \mathbf{0}$ is strictly positive definite.

Proof. For uniqueness in part (a), since $\mathbf{A}$ is Hurwitz, $\mathrm{Re}(\lambda_j) < 0$ for every $\lambda_j \in \mathrm{spec}(\mathbf{A})$, so every eigenvalue $\lambda_j + \lambda_k$ of the Kronecker sum $\mathbf{I} \otimes \mathbf{A} + \mathbf{A} \otimes \mathbf{I}$ has strictly negative real part $\mathrm{Re}(\lambda_j + \lambda_k) < 0$ and cannot be zero. Hence $\mathcal{L}_{\mathbf{A}}$ is invertible on $\mathbb{R}^{N\times N}$, guaranteeing at most one solution.

To prove that the integral Eq. (B.81) converges and solves Eq. (B.76), apply the exponential decay bound Eq. (B.78):

$$\left\| e^{\mathbf{A}t}\mathbf{D}e^{\mathbf{A}^\top t} \right\| \;\leq\; \left\| e^{\mathbf{A}t} \right\| \|\mathbf{D}\| \left\| e^{\mathbf{A}^\top t} \right\| \;\leq\; M^2 \|\mathbf{D}\|\, e^{-2\alpha_0 t}, \tag{B.82}$$

which is integrable on $[0,\infty)$ since $\alpha_0 > 0$. Because the integrand and its time derivative decay exponentially at $t \to \infty$, we can substitute Eq. (B.81) into $\mathbf{A\Sigma} + \mathbf{\Sigma A}^\top$ and move $\mathbf{A}$ and $\mathbf{A}^\top$ inside the integral:

$$\begin{aligned}\mathbf{A\Sigma} + \mathbf{\Sigma A}^\top &= 2\int_0^\infty \mathrm{d}t \left(\mathbf{A}e^{\mathbf{A}t}\mathbf{D}e^{\mathbf{A}^\top t} + e^{\mathbf{A}t}\mathbf{D}e^{\mathbf{A}^\top t}\mathbf{A}^\top\right) \\ &= 2\int_0^\infty \mathrm{d}t \,\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{\mathbf{A}t}\mathbf{D}e^{\mathbf{A}^\top t}\right) = 2\left[\lim_{t\to\infty} e^{\mathbf{A}t}\mathbf{D}e^{\mathbf{A}^\top t} - e^{\mathbf{0}}\mathbf{D}e^{\mathbf{0}}\right] = -2\mathbf{D}, \end{aligned} \tag{B.83}$$

where we used $\frac{\mathrm{d}}{\mathrm{d}t}e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A}$ and $\lim_{t\to\infty} e^{\mathbf{A}t} = \mathbf{0}$. Thus $\mathbf{A\Sigma} + \mathbf{\Sigma A}^\top + 2\mathbf{D} = \mathbf{0}$.

For part (b), taking the transpose of Eq. (B.81) gives $\mathbf{\Sigma}^\top = 2\int_0^\infty (e^{\mathbf{A}^\top t})^\top \mathbf{D}^\top (e^{\mathbf{A}t})^\top \,\mathrm{d}t = \mathbf{\Sigma}$. For any non-zero vector $\mathbf{u} \in \mathbb{R}^N$,

$$\mathbf{u}^\top \mathbf{\Sigma u} = 2\int_0^\infty \mathrm{d}t \left(e^{\mathbf{A}^\top t}\mathbf{u}\right)^\top \mathbf{D}\left(e^{\mathbf{A}^\top t}\mathbf{u}\right) \geq 0. \tag{B.84}$$

If $\mathbf{D} \succ \mathbf{0}$ with minimum eigenvalue $\lambda_{\min}(\mathbf{D}) > 0$, then since $e^{\mathbf{A}^\top t}\mathbf{u}$ is continuous in t and equals $\mathbf{u} \neq \mathbf{0}$ at $t = 0$, the integrand is strictly positive on a neighborhood $[0, \epsilon)$, ensuring $\mathbf{u}^\top \mathbf{\Sigma u} > 0$. □

B.4.2 Skew-Symmetric Circulation Decomposition, Time Correlations, and NESS Entropy Production

The distinction between a non-equilibrium Gaussian steady state ($\mathbf{G} \neq \mathbf{G}^\top$ in NEAT-RNs) and an equilibrium Boltzmann state is encoded in the skew-symmetric part of $\mathbf{A\Sigma}$.

Proposition B.17 (Skew-Symmetric Vorticity Decomposition and Detailed Balance)**.** *Let* $\mathbf{A}$ *be Hurwitz,* $\mathbf{D} \succ \mathbf{0}$*, and* $\mathbf{\Sigma} \succ \mathbf{0}$ *the unique solution to* $\mathbf{A\Sigma} + \mathbf{\Sigma A}^\top + 2\mathbf{D} = \mathbf{0}$*. Define the* ***irreversible circulation (vorticity) matrix***

$$\mathbf{Q} := \frac{1}{2}\left(\mathbf{A\Sigma} - \mathbf{\Sigma A}^\top\right). \tag{B.85}$$

Then:

(a) $\mathbf{Q}$ *is skew-symmetric (*$\mathbf{Q}^\top = -\mathbf{Q}$*), and the drift matrix admits the decomposition*

$$\mathbf{A\Sigma} = -\mathbf{D} + \mathbf{Q}, \qquad \text{or} \qquad \mathbf{A} = (-\mathbf{D} + \mathbf{Q})\mathbf{\Sigma}^{-1}. \tag{B.86}$$

(b) In the stationary Gaussian density $p_{\mathrm{ss}}(\mathbf{x}) = (2\pi)^{-N/2}(\det \mathbf{\Sigma})^{-1/2}\exp(-\frac{1}{2}\mathbf{x}^\top \mathbf{\Sigma}^{-1}\mathbf{x})$ *(with* $\mathbf{x} = \delta\mathbf{v}$*), the steady-state Fokker–Planck probability current* $\mathbf{J}_{\mathrm{ss}}(\mathbf{x})$ *is given by*

$$\mathbf{J}_{\mathrm{ss}}(\mathbf{x}) := \mathbf{A}\mathbf{x}\, p_{\mathrm{ss}}(\mathbf{x}) - \mathbf{D}\nabla p_{\mathrm{ss}}(\mathbf{x}) = \mathbf{Q}\mathbf{\Sigma}^{-1}\mathbf{x}\, p_{\mathrm{ss}}(\mathbf{x}), \tag{B.87}$$

which is divergence-free ($\nabla \cdot \mathbf{J}_{\mathrm{ss}}(\mathbf{x}) = 0$*) everywhere on* $\mathbb{R}^N$*.*

(c) The stationary two-time covariance matrix $\mathbf{C}(\tau) := \mathbb{E}\left[\mathbf{x}(t+\tau)\mathbf{x}(t)^\top\right]_{\mathrm{ss}}$ *satisfies*

$$\mathbf{C}(\tau) = \begin{cases} e^{\mathbf{A}\tau}\mathbf{\Sigma} & \tau \geq 0, \\ \mathbf{\Sigma}e^{-\mathbf{A}^\top \tau} & \tau < 0, \end{cases} \qquad \text{and} \qquad \frac{\mathrm{d}}{\mathrm{d}\tau}\Big[\mathbf{C}(\tau) - \mathbf{C}(-\tau)\Big]\Big|_{\tau=0^+} = 2\mathbf{Q}, \tag{B.88}$$

so $\mathbf{Q}$ *equals the initial rate of time-reversal asymmetry of the two-time correlation matrix.*

(d) Microscopic detailed balance ($\mathbf{J}_{\mathrm{ss}}(\mathbf{x}) \equiv \mathbf{0}$, i.e., $\mathbf{Q} = \mathbf{0}$, or $\mathbf{C}(\tau) = \mathbf{C}(\tau)^\top$) holds if and only if $\mathbf{AD} = \mathbf{DA}^\top$. When $\mathbf{Q} \neq \mathbf{0}$, the steady-state housekeeping entropy production rate is strictly positive:

$$\dot{\Sigma}_{\mathrm{tot}} \;=\; k_{\mathrm{B}} \int_{\mathbb{R}^N} \mathrm{d}^N x \, \frac{\mathbf{J}_{\mathrm{ss}}(\mathbf{x})^\top \mathbf{D}^{-1} \mathbf{J}_{\mathrm{ss}}(\mathbf{x})}{p_{\mathrm{ss}}(\mathbf{x})} \;=\; k_{\mathrm{B}} \operatorname{Tr}\!\left(\mathbf{Q}^\top \mathbf{D}^{-1} \mathbf{Q} \boldsymbol{\Sigma}^{-1}\right)$$
$$= \; k_{\mathrm{B}} \left[\operatorname{Tr}\!\left(\mathbf{A}^\top \mathbf{D}^{-1} \mathbf{A} \boldsymbol{\Sigma}\right) + \operatorname{Tr} \mathbf{A}\right] \;\geq\; 0. \quad \text{(B.89)}$$

Proof. For part (a), adding the Lyapunov equation $\frac{1}{2}(\mathbf{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{A}^\top) = -\mathbf{D}$ to the definition Eq. (B.85) of $\mathbf{Q} = \frac{1}{2}(\mathbf{A}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\mathbf{A}^\top)$ gives $\mathbf{A}\boldsymbol{\Sigma} = -\mathbf{D} + \mathbf{Q}$, with $\mathbf{Q}^\top = \frac{1}{2}(\boldsymbol{\Sigma}^\top \mathbf{A}^\top - \mathbf{A}\boldsymbol{\Sigma}^\top) = -\mathbf{Q}$.

For part (b), since $\nabla p_{\mathrm{ss}}(\mathbf{x}) = -\boldsymbol{\Sigma}^{-1}\mathbf{x}\, p_{\mathrm{ss}}(\mathbf{x})$, substituting $\mathbf{A} = (-\mathbf{D} + \mathbf{Q})\boldsymbol{\Sigma}^{-1}$ into $\mathbf{J}_{\mathrm{ss}}(\mathbf{x})$ cancels the dissipative gradient term $-\mathbf{D}\boldsymbol{\Sigma}^{-1}\mathbf{x}\, p_{\mathrm{ss}}(\mathbf{x})$ against $-\mathbf{D}\nabla p_{\mathrm{ss}}(\mathbf{x})$:

$$\mathbf{J}_{\mathrm{ss}}(\mathbf{x}) \;=\; (-\mathbf{D} + \mathbf{Q})\boldsymbol{\Sigma}^{-1}\mathbf{x}\, p_{\mathrm{ss}}(\mathbf{x}) + \mathbf{D}\boldsymbol{\Sigma}^{-1}\mathbf{x}\, p_{\mathrm{ss}}(\mathbf{x}) \;=\; \mathbf{Q}\boldsymbol{\Sigma}^{-1}\mathbf{x}\, p_{\mathrm{ss}}(\mathbf{x}). \quad \text{(B.90)}$$

Taking the divergence of $\mathbf{J}_{\mathrm{ss}}(\mathbf{x})$ via the product rule gives

$$\nabla \cdot \mathbf{J}_{\mathrm{ss}}(\mathbf{x}) \;=\; \operatorname{Tr}(\mathbf{Q}\boldsymbol{\Sigma}^{-1})\, p_{\mathrm{ss}}(\mathbf{x}) - \left(\mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Q} \boldsymbol{\Sigma}^{-1} \mathbf{x}\right) p_{\mathrm{ss}}(\mathbf{x}) \;=\; 0, \quad \text{(B.91)}$$

because $\mathbf{Q}$ is skew-symmetric and $\boldsymbol{\Sigma}^{-1}$ is symmetric, so both $\operatorname{Tr}(\mathbf{Q}\boldsymbol{\Sigma}^{-1}) = \operatorname{Tr}(\boldsymbol{\Sigma}^{-1}\mathbf{Q}^\top) = -\operatorname{Tr}(\mathbf{Q}\boldsymbol{\Sigma}^{-1}) = 0$ and the quadratic form $\mathbf{u}^\top \mathbf{Q}\mathbf{u} = 0$ vanish for $\mathbf{u} = \boldsymbol{\Sigma}^{-1}\mathbf{x}$.

For part (c), solving Eq. (B.75) forward from t to $t + \tau$ ($\tau \geq 0$) gives $\mathbf{x}(t+\tau) = e^{\mathbf{A}\tau}\mathbf{x}(t) + \int_t^{t+\tau} e^{\mathbf{A}(t+\tau-s)}\tilde{\mathbf{B}}\, \mathrm{d}\mathbf{W}_s$. Because future Wiener increments $\mathrm{d}\mathbf{W}_s$ ($s > t$) are independent of $\mathbf{x}(t)$, multiplying on the right by $\mathbf{x}(t)^\top$ and taking expectations gives $\mathbf{C}(\tau) = e^{\mathbf{A}\tau}\boldsymbol{\Sigma}$ for $\tau \geq 0$, and stationarity gives $\mathbf{C}(-\tau) = \mathbb{E}\left[\mathbf{x}(t-\tau)\mathbf{x}(t)^\top\right] = \mathbf{C}(\tau)^\top = \boldsymbol{\Sigma} e^{\mathbf{A}^\top \tau}$. Differentiating with respect to τ at $\tau = 0^+$ yields $\mathbf{A}\boldsymbol{\Sigma} - \boldsymbol{\Sigma}\mathbf{A}^\top = 2\mathbf{Q}$.

For part (d), $\mathbf{J}_{\mathrm{ss}}(\mathbf{x}) \equiv \mathbf{0} \iff \mathbf{Q} = \mathbf{0} \iff \mathbf{A}\boldsymbol{\Sigma} = -\mathbf{D} \iff \boldsymbol{\Sigma} = -\mathbf{A}^{-1}\mathbf{D}$. Since $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^\top$, this holds if and only if $\mathbf{A}^{-1}\mathbf{D} = \mathbf{D}(\mathbf{A}^{-1})^\top$, or equivalently (multiplying on the left by $\mathbf{A}$ and on the right by $\mathbf{A}^\top$) $\mathbf{DA}^\top = \mathbf{AD}$. Finally, substituting $\mathbf{J}_{\mathrm{ss}}(\mathbf{x}) = \mathbf{Q}\boldsymbol{\Sigma}^{-1}\mathbf{x}\, p_{\mathrm{ss}}(\mathbf{x})$ into the entropy production integral Eq. (B.89) and applying the Gaussian trace identity Eq. (B.23) with $\mathbb{E}\left[\mathbf{x}\mathbf{x}^\top\right]_{p_{\mathrm{ss}}} = \boldsymbol{\Sigma}$ gives

$$\dot{\Sigma}_{\mathrm{tot}} \;=\; k_{\mathrm{B}}\, \mathbb{E}\left[\mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Q}^\top \mathbf{D}^{-1} \mathbf{Q} \boldsymbol{\Sigma}^{-1} \mathbf{x}\right]_{p_{\mathrm{ss}}} \;=\; k_{\mathrm{B}} \operatorname{Tr}\!\left(\boldsymbol{\Sigma}^{-1} \mathbf{Q}^\top \mathbf{D}^{-1} \mathbf{Q} \boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}\right)$$
$$= \; k_{\mathrm{B}} \operatorname{Tr}\!\left(\mathbf{Q}^\top \mathbf{D}^{-1} \mathbf{Q} \boldsymbol{\Sigma}^{-1}\right). \quad \text{(B.92)}$$

For the second form, substitute $\mathbf{Q} = \mathbf{A}\boldsymbol{\Sigma} + \mathbf{D}$ and $\mathbf{Q}^\top = \boldsymbol{\Sigma}\mathbf{A}^\top + \mathbf{D}$ (both follow from part (a) and the Lyapunov equation). Expanding the product and using the cyclic property of the trace,

$$\operatorname{Tr}\!\left(\mathbf{Q}^\top \mathbf{D}^{-1} \mathbf{Q} \boldsymbol{\Sigma}^{-1}\right) \;=\; \operatorname{Tr}\!\left(\mathbf{A}^\top \mathbf{D}^{-1} \mathbf{A} \boldsymbol{\Sigma}\right) + 2 \operatorname{Tr} \mathbf{A} + \operatorname{Tr}\!\left(\mathbf{D}\boldsymbol{\Sigma}^{-1}\right).$$

Finally, $\mathbf{D}\boldsymbol{\Sigma}^{-1} = (\mathbf{Q} - \mathbf{A}\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1} = \mathbf{Q}\boldsymbol{\Sigma}^{-1} - \mathbf{A}$, and $\operatorname{Tr}(\mathbf{Q}\boldsymbol{\Sigma}^{-1}) = 0$ because $\mathbf{Q}$ is antisymmetric and $\boldsymbol{\Sigma}^{-1}$ is symmetric. Hence $\operatorname{Tr}(\mathbf{D}\boldsymbol{\Sigma}^{-1}) = -\operatorname{Tr} \mathbf{A}$, which gives the second form. At equilibrium ($\mathbf{Q} = \mathbf{0}$, $\mathbf{A} = -\mathbf{D}\boldsymbol{\Sigma}^{-1}$) the two terms cancel, as they must. □

Worked computation B.18 (Closed-Form Solution of the 2×2 Lyapunov Equation)**.** The two-node RC sampler and its subthreshold CMOS counterpart serve as benchmark models in Chapters 5, 8 and 13. Both have a drift matrix $\mathbf{A} \in \mathbb{R}^{2\times 2}$ and a diffusion matrix $\mathbf{D} \in \mathbb{R}^{2\times 2}$. For an arbitrary Hurwitz $\mathbf{A}$ we derive the stationary covariance $\boldsymbol{\Sigma}$ in closed form, verify it by substitution, and compute the circulation scalar q and the entropy production rate $\dot{\Sigma}_{\mathrm{tot}}$.

Step 1: Parameterization and Hurwitz invariants. Write the drift, diffusion, and unknown symmetric covariance matrices explicitly as

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} d_{11} & d_{12} \\ d_{12} & d_{22} \end{pmatrix} \succ \mathbf{0}, \qquad \mathbf{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}. \tag{B.93}$$

The characteristic polynomial of $\mathbf{A}$ is $\det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^2 - \tau\lambda + \delta$, where the trace and determinant invariants are

$$\tau := \operatorname{Tr}(\mathbf{A}) = a_{11} + a_{22}, \qquad \delta := \det(\mathbf{A}) = a_{11}a_{22} - a_{12}a_{21}. \tag{B.94}$$

In two dimensions, the eigenvalues $\lambda_\pm = \frac{1}{2}(\tau \pm \sqrt{\tau^2 - 4\delta})$ have strictly negative real parts ($\mathbf{A}$ is Hurwitz) if and only if

$$\tau = \operatorname{Tr}(\mathbf{A}) < 0 \qquad \text{and} \qquad \delta = \det(\mathbf{A}) > 0. \tag{B.95}$$

Step 2: Cayley–Hamilton reduction of the Lyapunov equation. Instead of solving a 3×3 linear system by Cramer's rule, we use the 2×2 Cayley–Hamilton theorem:

$$\mathbf{A}^2 - \tau\mathbf{A} + \delta\mathbf{I} = \mathbf{0} \qquad \Longrightarrow \qquad \mathbf{A}^{-1} = \frac{1}{\delta}(\tau\mathbf{I} - \mathbf{A}) = \frac{1}{\delta}\begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}. \tag{B.96}$$

Define the adjugate matrix $\widetilde{\mathbf{A}} := \delta\mathbf{A}^{-1} = \tau\mathbf{I} - \mathbf{A}$, which satisfies $\mathbf{A}\widetilde{\mathbf{A}} = \widetilde{\mathbf{A}}\mathbf{A} = \delta\mathbf{I}$. Multiply the Lyapunov equation $\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top = -2\mathbf{D}$ on the left by $\widetilde{\mathbf{A}}$ and on the right by $\widetilde{\mathbf{A}}^\top$:

$$\widetilde{\mathbf{A}}\left(\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top\right)\widetilde{\mathbf{A}}^\top = \delta\left(\mathbf{\Sigma}\widetilde{\mathbf{A}}^\top + \widetilde{\mathbf{A}}\mathbf{\Sigma}\right) = -2\widetilde{\mathbf{A}}\mathbf{D}\widetilde{\mathbf{A}}^\top. \tag{B.97}$$

Now substitute $\widetilde{\mathbf{A}} = \tau\mathbf{I} - \mathbf{A}$ and $\widetilde{\mathbf{A}}^\top = \tau\mathbf{I} - \mathbf{A}^\top$ into the middle expression of Eq. (B.97):

$$\mathbf{\Sigma}\widetilde{\mathbf{A}}^\top + \widetilde{\mathbf{A}}\mathbf{\Sigma} = 2\tau\mathbf{\Sigma} - \left(\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top\right) = 2\tau\mathbf{\Sigma} + 2\mathbf{D}. \tag{B.98}$$

Substituting this back into Eq. (B.97) eliminates the matrix products $\mathbf{A}\mathbf{\Sigma}$ and $\mathbf{\Sigma}\mathbf{A}^\top$:

$$2\delta\,(\tau\mathbf{\Sigma} + \mathbf{D}) = -2\widetilde{\mathbf{A}}\mathbf{D}\widetilde{\mathbf{A}}^\top. \tag{B.99}$$

Dividing by $2\tau\delta < 0$ (which is non-zero by the Hurwitz conditions Eq. (B.95)) yields the 2×2 closed-form solution:

$$\mathbf{\Sigma} = -\frac{1}{\operatorname{Tr}(\mathbf{A})\det(\mathbf{A})}\left[\det(\mathbf{A})\,\mathbf{D} + \widetilde{\mathbf{A}}\mathbf{D}\widetilde{\mathbf{A}}^\top\right], \qquad \widetilde{\mathbf{A}} = \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}. \tag{B.100}$$

Step 3: Direct verification and explicit component formulas. We check directly that Eq. (B.100) satisfies $\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top = -2\mathbf{D}$. Using $\mathbf{A}\widetilde{\mathbf{A}} = \delta\mathbf{I}$ and $\widetilde{\mathbf{A}} = \tau\mathbf{I} - \mathbf{A}$, we compute

$$\begin{aligned} -\tau\delta\left(\mathbf{A}\mathbf{\Sigma} + \mathbf{\Sigma}\mathbf{A}^\top\right) &= \delta\left(\mathbf{A}\mathbf{D} + \mathbf{D}\mathbf{A}^\top\right) + \mathbf{A}\widetilde{\mathbf{A}}\mathbf{D}\widetilde{\mathbf{A}}^\top + \widetilde{\mathbf{A}}\mathbf{D}\widetilde{\mathbf{A}}^\top\mathbf{A}^\top \\ &= \delta\left(\mathbf{A}\mathbf{D} + \mathbf{D}\mathbf{A}^\top\right) + \delta\mathbf{D}(\tau\mathbf{I} - \mathbf{A}^\top) + \delta(\tau\mathbf{I} - \mathbf{A})\mathbf{D} = 2\tau\delta\mathbf{D}, \end{aligned} \tag{B.101}$$

and dividing by $-\tau\delta$ gives $-2\mathbf{D}$. Expanding $\widetilde{\mathbf{A}}\mathbf{D}\widetilde{\mathbf{A}}^\top$ in components from Eq. (B.93) gives the three entries of $\mathbf{\Sigma}$ explicitly:

$$\sigma_{11} = -\frac{1}{\tau\delta}\left[(\delta + a_{22}^2)d_{11} - 2a_{12}a_{22}d_{12} + a_{12}^2 d_{22}\right], \tag{B.102}$$

$$\sigma_{22} = -\frac{1}{\tau\delta}\left[a_{21}^2 d_{11} - 2a_{11}a_{21}d_{12} + (\delta + a_{11}^2)d_{22}\right], \tag{B.103}$$

$$\sigma_{12} = -\frac{1}{\tau\delta}\left[-a_{21}a_{22}d_{11} + 2(\delta + a_{12}a_{21})d_{12} - a_{11}a_{12}d_{22}\right]. \tag{B.104}$$

Step 4: NESS circulation scalar q and entropy production rate. In two dimensions, any skew-symmetric matrix $\mathbf{Q} = -\mathbf{Q}^\top$ has the form

$$\mathbf{Q} \;=\; \frac{1}{2}\left(\mathbf{A\Sigma} - \mathbf{\Sigma A}^\top\right) \;=\; \begin{pmatrix} 0 & q \\ -q & 0 \end{pmatrix}. \tag{B.105}$$

Using $\mathbf{A\Sigma}-\mathbf{\Sigma A}^\top = -\frac{1}{\tau}(\mathbf{AD}-\mathbf{DA}^\top-\widetilde{\mathbf{A}}\mathbf{D}+\mathbf{D}\widetilde{\mathbf{A}}^\top)$ from Eq. (B.100) and $\mathbf{D}\widetilde{\mathbf{A}}^\top-\widetilde{\mathbf{A}}\mathbf{D} = \mathbf{AD}-\mathbf{DA}^\top$, we get $\mathbf{Q} = -\frac{1}{\tau}(\mathbf{AD}-\mathbf{DA}^\top)$. Its $(1,2)$ entry is the circulation scalar

$$q \;=\; \frac{1}{\tau}\Big[a_{21}d_{11} - a_{12}d_{22} + (a_{22}-a_{11})d_{12}\Big]. \tag{B.106}$$

Finally, write $\mathbf{Q} = q\mathbf{J}$ with $\mathbf{J} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. For any symmetric invertible 2×2 matrix $\mathbf{M}$, $\mathbf{J}^\top\mathbf{M}\mathbf{J} = (\det\mathbf{M})\,\mathbf{M}^{-1}$, so $\mathbf{Q}^\top\mathbf{D}^{-1}\mathbf{Q} = q^2\mathbf{D}/\det\mathbf{D}$. Together with $\operatorname{Tr}(\mathbf{D\Sigma}^{-1}) = -\operatorname{Tr}\mathbf{A} = -\tau$ from the proof of Eq. (B.89), the steady-state entropy production rate in $N=2$ dimensions is

$$\dot{\Sigma}_{\mathrm{tot}} \;=\; k_{\mathrm{B}}\,\frac{q^2\,\operatorname{Tr}(\mathbf{D\Sigma}^{-1})}{\det\mathbf{D}} \;=\; k_{\mathrm{B}}\,\frac{q^2\,|\tau|}{\det\mathbf{D}} \;\geq\; 0, \tag{B.107}$$

which vanishes if and only if $q=0$. When thermal noise is diagonal ($d_{12}=0$), Eq. (B.106) shows that $q = (a_{21}d_{11} - a_{12}d_{22})/\tau$, which is non-zero whenever the cross-couplings violate reciprocity ($a_{21}/d_{22} \neq a_{12}/d_{11}$).

Summary

- **Proved:**
  - The N-dimensional Gaussian integral with a linear source (Theorem B.2), the Hubbard–Stratonovich transformation (Corollary B.3), and the entropy, KL divergence, and Bures–Wasserstein distance of Gaussians (Proposition B.4).
  - Block factorizations, the Schur determinant identity, block inversion (Theorem B.6), the Woodbury identity (Corollary B.7), and Gaussian conditioning and marginalization (Theorem B.8).
  - Stein's lemma (Lemma B.9), the Isserlis–Wick theorem (Theorem B.10), moment–cumulant relations to fourth order (Proposition B.12), and first-order anharmonic corrections to cumulants (Proposition B.13).
  - Existence, uniqueness, and the integral representation of the Lyapunov solution (Theorem B.16), the circulation decomposition (Proposition B.17), and the 2×2 closed form (Worked computation B.18).

Bibliographic notes

Multivariate Gaussian conditioning, Schur complements, and the Woodbury identity in statistical inference and control theory are treated in Mézard and Montanari [76] and Pavliotis [85]. The combinatorial pairing theorem for Gaussian moments was proved independently by Isserlis (1918) in mathematical statistics and by Wick (1950) in quantum field theory; its connection to cumulant expansions and stochastic thermodynamics is developed in van Kampen [112], Gardiner [39], and Risken [90]. The Lyapunov equation and the circulation decomposition $\mathbf{A\Sigma} = -\mathbf{D}+\mathbf{Q}$ are applied to stochastic electronic circuits in Freitas et al. [37], to matrix inversion by thermodynamic sampling in Aifer et al. [3], and to NEAT-RNs in subthreshold CMOS in Freitas et al. [38].

Appendix C

Foundational Proofs in Stochastic Calculus, Fokker–Planck Spectral Theory, and Bakry–Émery Log-Sobolev Inequalities

Part II (Chapters 4 to 7) and Part V (Chapters 15 and 16) rely on four results from stochastic analysis. Some are also derived in the main text (Sections 4.3, 4.4 and 6.5); this appendix fills in the technical steps. Section C.1 proves existence and uniqueness of solutions to stochastic differential equations (SDEs) by Picard iteration. Section C.2 proves the multidimensional Itô formula and the Wong–Zakai theorem, which explains why physical noise with a short but finite correlation time leads to the Stratonovich rather than the Itô SDE. Section C.3 establishes the self-adjointness behind the Witten Hamiltonian and the completeness of the Hermite basis used for the Ornstein–Uhlenbeck process of the RC sampler (the coupled linear RC circuit of Chapter 1). Section C.4 gives the semigroup proof of the Bakry–Émery theorem: strong convexity of the energy implies Poincaré and logarithmic Sobolev inequalities and exponential decay of the Kullback–Leibler (KL) divergence. This is the result behind the mixing-time theorem for the stages of a Denoising Thermodynamic Model (DTM) in Chapter 16.

C.1 Grönwall's Lemma, Doob's Inequality, and Picard Iteration for SDEs

Consider a general N-dimensional Itô stochastic differential equation driven by an M-dimensional standard Brownian motion $\mathbf{W}_t$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$:

$$\mathrm{d}\mathbf{x}_t \;=\; \mathbf{a}(\mathbf{x}_t, t)\,\mathrm{d}t + \mathbf{B}(\mathbf{x}_t, t)\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{x}_0 = \boldsymbol{\xi} \in L^2(\Omega, \mathcal{F}_0, \mathbb{P}; \mathbb{R}^N), \tag{C.1}$$

or in integral form on a finite time horizon $t \in [0, T]$:

$$\mathbf{x}_t \;=\; \boldsymbol{\xi} + \int_0^t \mathbf{a}(\mathbf{x}_s, s)\,\mathrm{d}s + \int_0^t \mathbf{B}(\mathbf{x}_s, s)\,\mathrm{d}\mathbf{W}_s. \tag{C.2}$$

C.1.1 Grönwall's Inequality and Doob's L^2 Martingale Maximal Inequality

To bound the growth of Picard iterates in $L^2(\Omega)$, we need two inequalities.

Lemma C.1 (Differential and Integral Grönwall Inequality)**.** *Let $u\colon [0,T] \to [0,\infty)$ be a continuous non-negative function satisfying the integral inequality*

$$u(t) \;\le\; \alpha(t) + \beta \int_0^t u(s)\, \mathrm{d}s \qquad \text{for all } t \in [0,T], \tag{C.3}$$

where $\beta > 0$ is a constant and $\alpha\colon [0,T] \to [0,\infty)$ is continuous. Then for all $t \in [0,T]$,

$$u(t) \;\le\; \alpha(t) + \beta \int_0^t \alpha(s)\, e^{\beta(t-s)}\, \mathrm{d}s. \tag{C.4}$$

In particular, if $\alpha(t) \equiv \alpha_0$ is constant (or non-decreasing), then $u(t) \le \alpha(t)\, e^{\beta t}$. If $\alpha_0 = 0$, then $u(t) \equiv 0$ on $[0,T]$.

Proof. Define the auxiliary differentiable function $I(t) := \int_0^t u(s)\, \mathrm{d}s$, so that $I(0) = 0$ and $I'(t) = u(t)$. Using Eq. (C.3), $I'(t) - \beta I(t) = u(t) - \beta \int_0^t u(s)\, \mathrm{d}s \le \alpha(t)$. Multiplying both sides by the integrating factor $e^{-\beta t} > 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(e^{-\beta t} I(t)\Big) \;=\; e^{-\beta t}\big(I'(t) - \beta I(t)\big) \;\le\; e^{-\beta t}\alpha(t). \tag{C.5}$$

Integrating this inequality from $s = 0$ to $s = t$ and using $I(0) = 0$ yields

$$e^{-\beta t} I(t) \;\le\; \int_0^t e^{-\beta s}\alpha(s)\, \mathrm{d}s \qquad \Longrightarrow \qquad I(t) \;\le\; \int_0^t \alpha(s)\, e^{\beta(t-s)}\, \mathrm{d}s. \tag{C.6}$$

Substituting this upper bound on $I(t) = \int_0^t u(s)\, \mathrm{d}s$ back into the right-hand side of Eq. (C.3) gives Eq. (C.4). When $\alpha(s) \le \alpha(t)$ for $s \le t$, pulling $\alpha(t)$ outside the integral gives $\alpha(t)\big[1 + \beta \int_0^t e^{\beta(t-s)}\, \mathrm{d}s\big] = \alpha(t)\big[1 + (e^{\beta t} - 1)\big] = \alpha(t) e^{\beta t}$. □

Lemma C.2 (Doob's L^2 Martingale Maximal Inequality)**.** *Let $(\mathbf{M}_t)_{t\in[0,T]}$ be a continuous $\mathcal{F}_t$-martingale taking values in $\mathbb{R}^N$ with $\mathbf{M}_0 = \mathbf{0}$. Then*

$$\mathbb{E}\left[\sup_{0\le t\le T} \|\mathbf{M}_t\|^2\right] \;\le\; 4\,\mathbb{E}\Big[\|\mathbf{M}_T\|^2\Big]. \tag{C.7}$$

In particular, for an Itô stochastic integral $\mathbf{M}_t = \int_0^t \mathbf{B}_s\, \mathrm{d}\mathbf{W}_s$ with adapted $\mathbf{B}_s \in \mathbb{R}^{N\times M}$, combining Eq. (C.7) with the Itô isometry yields

$$\mathbb{E}\left[\sup_{0\le t\le T} \left\|\int_0^t \mathbf{B}_s\, \mathrm{d}\mathbf{W}_s\right\|^2\right] \;\le\; 4\,\mathbb{E}\left[\left\|\int_0^T \mathbf{B}_s\, \mathrm{d}\mathbf{W}_s\right\|^2\right] \;=\; 4\int_0^T \mathbb{E}\Big[\|\mathbf{B}_s\|_{\mathrm{HS}}^2\Big]\, \mathrm{d}s, \tag{C.8}$$

where $\|\mathbf{B}\|_{\mathrm{HS}}^2 := \mathrm{Tr}(\mathbf{B}\mathbf{B}^\top) = \sum_{i=1}^N \sum_{a=1}^M B_{ia}^2$ is the Hilbert–Schmidt (Frobenius) norm squared.

Proof. By continuity of sample paths, the supremum over $[0,T]$ equals the supremum over dyadic rationals, so by monotone convergence it suffices to prove the discrete-time inequality for $X_k := \|\mathbf{M}_{t_k}\|$ $(k = 1,\dots,n)$. Since $\mathbf{x} \mapsto \|\mathbf{x}\|$ is convex and $\mathbf{M}_{t_k}$ is a martingale, conditional Jensen's inequality implies that X_k is a non-negative submartingale: $\mathbb{E}[X_{k+1} \mid \mathcal{F}_{t_k}] \ge X_k$.

First, for any threshold $\lambda > 0$, define the stopping time $\tau := \min\{k \le n : X_k \ge \lambda\}$ (with $\tau = n$ if $\max_{k\le n} X_k < \lambda$), and the event $E_\lambda := \{X_n^* \ge \lambda\}$ where $X_n^* := \max_{1\le k\le n} X_k$. Disjointly partitioning

$E_\lambda = \bigsqcup_{k=1}^{n} \{\tau = k\}$ and using the submartingale property $\mathbb{E}\Big[X_n \mathbf{1}_{\{\tau=k\}}\Big] \geq \mathbb{E}\Big[X_k \mathbf{1}_{\{\tau=k\}}\Big] \geq \lambda \mathbb{P}(\tau = k)$ gives Doob's weak maximal inequality:

$$\lambda\, \mathbb{P}(X_n^* \geq \lambda) \;=\; \sum_{k=1}^{n} \lambda\, \mathbb{P}(\tau = k) \;\leq\; \sum_{k=1}^{n} \mathbb{E}\Big[X_n \mathbf{1}_{\{\tau=k\}}\Big] \;=\; \mathbb{E}\Big[X_n \mathbf{1}_{\{X_n^* \geq \lambda\}}\Big]. \tag{C.9}$$

Now, to obtain the L^2 bound Eq. (C.7) (introducing a truncation $X_{n,K}^* = \min(X_n^*, K)$ to ensure finiteness before letting $K \to \infty$), multiply Eq. (C.9) by 2 and integrate over $\lambda \in (0, \infty)$ using Fubini's theorem and $\int_0^{X_n^*} 2\lambda \, \mathrm{d}\lambda = (X_n^*)^2$:

$$\begin{aligned} \mathbb{E}\Big[(X_n^*)^2\Big] \;&=\; \int_0^\infty 2\lambda\, \mathbb{P}(X_n^* \geq \lambda)\, \mathrm{d}\lambda \;\leq\; \int_0^\infty 2\, \mathbb{E}\Big[X_n \mathbf{1}_{\{X_n^* \geq \lambda\}}\Big]\, \mathrm{d}\lambda \\ &=\; 2\, \mathbb{E}\left[X_n \int_0^{X_n^*} \mathrm{d}\lambda\right] \;=\; 2\, \mathbb{E}[X_n X_n^*] \;\leq\; 2\, \big(\mathbb{E}\Big[X_n^2\Big]\big)^{1/2} \big(\mathbb{E}\Big[(X_n^*)^2\Big]\big)^{1/2}, \end{aligned} \tag{C.10}$$

where the final step is the Cauchy–Schwarz inequality. Dividing by $(\mathbb{E}[(X_n^*)^2])^{1/2}$ and squaring both sides yields $\mathbb{E}[(X_n^*)^2] \leq 4\mathbb{E}[X_n^2]$, proving Eq. (C.7). Applying the Itô isometry $\mathbb{E}\left[\left\|\int_0^T \mathbf{B}_s \, \mathrm{d}\mathbf{W}_s\right\|^2\right] = \int_0^T \mathbb{E}\Big[\|\mathbf{B}_s\|_{\mathrm{HS}}^2\Big]\, \mathrm{d}s$ (Theorem 4.8) yields Eq. (C.8). □

C.1.2 Picard Iteration Proof of Strong Existence and Uniqueness for SDEs

Theorem C.3 (Strong Existence and Pathwise Uniqueness of SDE Solutions)**.** *Suppose the drift* $\mathbf{a}\colon \mathbb{R}^N \times [0,T] \to \mathbb{R}^N$ *and diffusion matrix* $\mathbf{B}\colon \mathbb{R}^N \times [0,T] \to \mathbb{R}^{N\times M}$ *are measurable and satisfy:*

(i) ***Global Lipschitz continuity:*** *There exists* $L > 0$ *such that for all* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ *and* $t \in [0,T]$,

$$\|\mathbf{a}(\mathbf{x},t) - \mathbf{a}(\mathbf{y},t)\| + \|\mathbf{B}(\mathbf{x},t) - \mathbf{B}(\mathbf{y},t)\|_{\mathrm{HS}} \;\leq\; L\, \|\mathbf{x} - \mathbf{y}\|. \tag{C.11}$$

(ii) ***Linear growth bound:*** *There exists* $K > 0$ *such that for all* $\mathbf{x} \in \mathbb{R}^N$ *and* $t \in [0,T]$,

$$\|\mathbf{a}(\mathbf{x},t)\|^2 + \|\mathbf{B}(\mathbf{x},t)\|_{\mathrm{HS}}^2 \;\leq\; K^2(1 + \|\mathbf{x}\|^2). \tag{C.12}$$

Then for any initial condition $\boldsymbol{\xi} \in L^2(\Omega, \mathcal{F}_0, \mathbb{P}; \mathbb{R}^N)$*, there exists a continuous,* $\mathcal{F}_t$*-adapted process* $(\mathbf{x}_t)_{t\in[0,T]}$ *solving Eq. (C.2) almost surely, with* $\mathbb{E}\Big[\sup_{0\leq t\leq T} \|\mathbf{x}_t\|^2\Big] < \infty$*. Moreover, the solution is pathwise unique: if* $\tilde{\mathbf{x}}_t$ *is any other solution with* $\tilde{\mathbf{x}}_0 = \boldsymbol{\xi}$*, then* $\mathbb{P}(\sup_{0\leq t\leq T} \|\mathbf{x}_t - \tilde{\mathbf{x}}_t\| = 0) = 1$.

Proof. We first prove pathwise uniqueness, and then establish existence via Picard iteration.

Step 1: Pathwise uniqueness via Grönwall's lemma. Suppose $\mathbf{x}_t$ and $\tilde{\mathbf{x}}_t$ are two solutions with the same initial condition $\boldsymbol{\xi}$, and set $\boldsymbol{\Delta}_t := \mathbf{x}_t - \tilde{\mathbf{x}}_t$. Using the elementary inequality $\|\mathbf{u}+\mathbf{v}\|^2 \leq 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$, the Cauchy–Schwarz inequality on $[0,t]$, Doob's inequality Eq. (C.8), and the Lipschitz condition Eq. (C.11), we bound $u(t) := \mathbb{E}\Big[\sup_{0\leq s\leq t} \|\boldsymbol{\Delta}_s\|^2\Big]$:

$$\begin{aligned} u(t) \;&\leq\; 2\, \mathbb{E}\left[\sup_{0\leq s\leq t} \left\|\int_0^s \big(\mathbf{a}(\mathbf{x}_r, r) - \mathbf{a}(\tilde{\mathbf{x}}_r, r)\big)\, \mathrm{d}r\right\|^2\right] + 2\, \mathbb{E}\left[\sup_{0\leq s\leq t} \left\|\int_0^s \big(\mathbf{B}(\mathbf{x}_r, r) - \mathbf{B}(\tilde{\mathbf{x}}_r, r)\big)\, \mathrm{d}\mathbf{W}_r\right\|^2\right] \\ &\leq\; 2t \int_0^t \mathbb{E}\Big[\|\mathbf{a}(\mathbf{x}_r, r) - \mathbf{a}(\tilde{\mathbf{x}}_r, r)\|^2\Big]\, \mathrm{d}r + 8 \int_0^t \mathbb{E}\Big[\|\mathbf{B}(\mathbf{x}_r, r) - \mathbf{B}(\tilde{\mathbf{x}}_r, r)\|_{\mathrm{HS}}^2\Big]\, \mathrm{d}r \\ &\leq\; 2(T+4)L^2 \int_0^t \mathbb{E}\Big[\|\boldsymbol{\Delta}_r\|^2\Big]\, \mathrm{d}r \;\leq\; C_L \int_0^t u(r)\, \mathrm{d}r, \end{aligned} \tag{C.13}$$

where $C_L := 2(T+4)L^2$. Applying Grönwall's inequality (Lemma C.1) with $\alpha_0 = 0$ gives $u(T) = \mathbb{E}\Big[\sup_{0\leq t\leq T} \|\mathbf{x}_t - \tilde{\mathbf{x}}_t\|^2\Big] = 0$, so $\sup_{0\leq t\leq T} \|\mathbf{x}_t - \tilde{\mathbf{x}}_t\| = 0$ almost surely.

Step 2: Picard iteration and uniform convergence. Define the sequence of adapted continuous processes $\mathbf{x}_t^{(k)}$ $(k = 0, 1, 2, \dots)$ inductively by $\mathbf{x}_t^{(0)} \equiv \boldsymbol{\xi}$ and

$$\mathbf{x}_t^{(k+1)} \;:=\; \boldsymbol{\xi} + \int_0^t \mathbf{a}\Big(\mathbf{x}_s^{(k)}, s\Big)\,\mathrm{d}s + \int_0^t \mathbf{B}\Big(\mathbf{x}_s^{(k)}, s\Big)\,\mathrm{d}\mathbf{W}_s. \tag{C.14}$$

For $k = 0$, the same two estimates as in Step 1, with the linear growth bound Eq. (C.12) in place of the Lipschitz condition, give

$$D_0 \;:=\; \mathbb{E}\Bigg[\sup_{0\le t\le T}\Big\|\mathbf{x}_t^{(1)} - \mathbf{x}_t^{(0)}\Big\|^2\Bigg] \;\le\; 2(T+4)K^2T\left(1 + \mathbb{E}\Big[\|\boldsymbol{\xi}\|^2\Big]\right) \;<\; \infty. \tag{C.15}$$

For $k \ge 1$, applying the Lipschitz bound Eq. (C.13) to the successive differences $\boldsymbol{\Delta}_t^{(k)} := \mathbf{x}_t^{(k+1)} - \mathbf{x}_t^{(k)}$ gives

$$d_k(t) \;:=\; \mathbb{E}\Bigg[\sup_{0\le s\le t}\Big\|\mathbf{x}_s^{(k+1)} - \mathbf{x}_s^{(k)}\Big\|^2\Bigg] \;\le\; C_L\int_0^t d_{k-1}(s)\,\mathrm{d}s. \tag{C.16}$$

Iterating Eq. (C.16) k times from $d_0(s) \le D_0$ yields the factorial bound

$$\mathbb{E}\Bigg[\sup_{0\le t\le T}\Big\|\mathbf{x}_t^{(k+1)} - \mathbf{x}_t^{(k)}\Big\|^2\Bigg] \;=\; d_k(T) \;\le\; D_0\int_0^T C_L\,\mathrm{d}t_1\int_0^{t_1} C_L\,\mathrm{d}t_2\cdots\int_0^{t_{k-1}} C_L\,\mathrm{d}t_k \;=\; \frac{D_0(C_LT)^k}{k!}. \tag{C.17}$$

By Markov's inequality, $\mathbb{P}\Big(\sup_{0\le t\le T}\Big\|\mathbf{x}_t^{(k+1)} - \mathbf{x}_t^{(k)}\Big\| > 2^{-k}\Big) \le 4^k d_k(T) \le D_0(4C_LT)^k/k!$. Since $\sum_{k=0}^{\infty}(4C_LT)^k/k! = e^{4C_LT} < \infty$, the Borel–Cantelli lemma implies that almost surely the telescoping series $\mathbf{x}_t = \mathbf{x}_t^{(0)} + \sum_{k=0}^{\infty}(\mathbf{x}_t^{(k+1)} - \mathbf{x}_t^{(k)})$ converges uniformly on $[0,T]$ (and in $L^2(\Omega)$) to a continuous adapted limit $\mathbf{x}_t$ satisfying Eq. (C.2). □

C.2 Proof of the Multidimensional Itô Formula and the Wong–Zakai Theorem

C.2.1 Proof of the Multidimensional Itô Formula

Theorem C.4 (Multidimensional Itô Formula)**.** *Let $\mathbf{x}_t \in \mathbb{R}^N$ be an Itô process satisfying $\mathrm{d}\mathbf{x}_t = \mathbf{a}_t\,\mathrm{d}t + \mathbf{B}_t\,\mathrm{d}\mathbf{W}_t$ with diffusion tensor $\mathbf{D}_t := \frac{1}{2}\mathbf{B}_t\mathbf{B}_t^\top$, and let $f \in C^{2,1}(\mathbb{R}^N \times [0,T];\mathbb{R})$. Then $f(\mathbf{x}_t, t)$ is an Itô process satisfying*

$$\begin{aligned}\mathrm{d}f(\mathbf{x}_t,t) \;&=\; \left(\frac{\partial f}{\partial t} + \sum_{i=1}^N a_{i,t}\frac{\partial f}{\partial x_i} + \frac{1}{2}\sum_{i,j=1}^N (\mathbf{B}_t\mathbf{B}_t^\top)_{ij}\frac{\partial^2 f}{\partial x_i\partial x_j}\right)\mathrm{d}t + \sum_{i=1}^N\sum_{\alpha=1}^M B_{i\alpha,t}\frac{\partial f}{\partial x_i}\,\mathrm{d}W_{\alpha,t}\\ &=\; \Big(\frac{\partial f}{\partial t} + \mathbf{a}_t\cdot\nabla f + \mathrm{Tr}(\mathbf{D}_t\nabla^2 f)\Big)\,\mathrm{d}t + (\nabla f)^\top\mathbf{B}_t\,\mathrm{d}\mathbf{W}_t.\end{aligned} \tag{C.18}$$

Proof. By stopping $\mathbf{x}_t$ at the exit time $\tau_R = \inf\{t : \|\mathbf{x}_t\| \ge R\}$ and approximating $\mathbf{a}_t, \mathbf{B}_t$ by bounded continuous step processes, we may assume f and its first and second derivatives are bounded and uniformly continuous on $\|\mathbf{x}\| \le R$, and $\mathbf{a}_t, \mathbf{B}_t$ are bounded. Let $\Pi_n = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ be a partition of $[0,t]$ with mesh $|\Pi_n| = \max_k \Delta t_k \to 0$, and write $\Delta\mathbf{x}_k := \mathbf{x}_{t_{k+1}} - \mathbf{x}_{t_k} = \mathbf{a}_{t_k}\Delta t_k + \mathbf{B}_{t_k}\Delta\mathbf{W}_k$ (up to higher-order integral remainders).

Expanding $f(\mathbf{x}_t,t)-f(\mathbf{x}_0,0)=\sum_{k=0}^{n-1}\left[f(\mathbf{x}_{t_{k+1}},t_{k+1})-f(\mathbf{x}_{t_k},t_k)\right]$ by Taylor's theorem to first order in Δt_k and second order in $\Delta\mathbf{x}_k$ gives

$$\begin{aligned} f(\mathbf{x}_t,t)-f(\mathbf{x}_0,0) \;&=\; \sum_{k=0}^{n-1}\frac{\partial f(\mathbf{x}_{t_k},t_k)}{\partial t}\Delta t_k+\sum_{k=0}^{n-1}\sum_{i=1}^{N}\frac{\partial f(\mathbf{x}_{t_k},t_k)}{\partial x_i}\Delta x_{i,k} \\ &\quad+\frac{1}{2}\sum_{k=0}^{n-1}\sum_{i,j=1}^{N}\frac{\partial^2 f(\mathbf{x}_{t_k},t_k)}{\partial x_i\partial x_j}\Delta x_{i,k}\Delta x_{j,k}+R_n, \end{aligned} \tag{C.19}$$

where the remainder $R_n=\mathcal{O}(\sum_k(|\Delta t_k|^2+|\Delta t_k|\,\|\Delta\mathbf{x}_k\|+\omega(\|\Delta\mathbf{x}_k\|)\,\|\Delta\mathbf{x}_k\|^2))$ (with ω the modulus of continuity of $\nabla^2 f$) vanishes in $L^1(\Omega)$ as $|\Pi_n|\to 0$ because $\mathbb{E}\Big[\|\Delta\mathbf{x}_k\|^4\Big]=\mathcal{O}(\Delta t_k^2)$.

In Eq. (C.19), the first sum converges to $\int_0^t\partial_s f(\mathbf{x}_s,s)\,\mathrm{d}s$, and substituting $\Delta x_{i,k}=a_{i,t_k}\Delta t_k+\sum_\alpha B_{i\alpha,t_k}\Delta W_{\alpha,k}$ into the second sum converges in $L^2(\Omega)$ to $\int_0^t\mathbf{a}_s\cdot\nabla f\,\mathrm{d}s+\int_0^t(\nabla f)^\top\mathbf{B}_s\,\mathrm{d}\mathbf{W}_s$.

To evaluate the quadratic variation sum in Eq. (C.19), expand $\Delta x_{i,k}\Delta x_{j,k}$: the terms of order $(\Delta t_k)^2$ and $\Delta t_k\Delta W_{\alpha,k}$ vanish in $L^2(\Omega)$ as $|\Pi_n|\to 0$, leaving

$$Q_n \;:=\; \frac{1}{2}\sum_{k=0}^{n-1}\sum_{i,j=1}^{N}\sum_{\alpha,\beta=1}^{M}\frac{\partial^2 f(\mathbf{x}_{t_k},t_k)}{\partial x_i\partial x_j}B_{i\alpha,t_k}B_{j\beta,t_k}\Delta W_{\alpha,k}\Delta W_{\beta,k}. \tag{C.20}$$

Write $\Delta W_{\alpha,k}\Delta W_{\beta,k}=\delta_{\alpha\beta}\Delta t_k+(\Delta W_{\alpha,k}\Delta W_{\beta,k}-\delta_{\alpha\beta}\Delta t_k)$. Define the centered random variables $Z_{k,\alpha\beta}:=\Delta W_{\alpha,k}\Delta W_{\beta,k}-\delta_{\alpha\beta}\Delta t_k$. Because $\Delta\mathbf{W}_k\sim\mathcal{N}(\mathbf{0},\Delta t_k\mathbf{I})$ is independent of $\mathcal{F}_{t_k}$, $\mathbb{E}[Z_{k,\alpha\beta}\mid\mathcal{F}_{t_k}]=0$, so terms with different time indices $k\neq l$ are orthogonal in $L^2(\Omega)$. Moreover, by Isserlis–Wick (Eq. (B.56)), $\mathbb{E}\Big[Z_{k,\alpha\beta}^2\Big]\le 2(\Delta t_k)^2$. Denoting $H_{k,\alpha\beta}:=\frac{1}{2}\sum_{i,j}\partial^2_{x_ix_j}f(\mathbf{x}_{t_k},t_k)B_{i\alpha,t_k}B_{j\beta,t_k}$ (bounded by $\|H_{k,\alpha\beta}\|\le C$), we have

$$\begin{aligned} &\mathbb{E}\left[\left(\sum_{k=0}^{n-1}\sum_{\alpha,\beta=1}^{M}H_{k,\alpha\beta}Z_{k,\alpha\beta}\right)^2\right] \\ &\quad=\;\sum_{k=0}^{n-1}\mathbb{E}\left[\left(\sum_{\alpha,\beta=1}^{M}H_{k,\alpha\beta}Z_{k,\alpha\beta}\right)^2\right] \\ &\quad\le\;2M^4C^2\sum_{k=0}^{n-1}(\Delta t_k)^2 \\ &\quad\le\;2M^4C^2t\;|\Pi_n|\;\xrightarrow{|\Pi_n|\to 0}\;0. \end{aligned} \tag{C.21}$$

Thus $\Delta W_{\alpha,k}\Delta W_{\beta,k}$ in Eq. (C.20) is replaced in $L^2(\Omega)$ by $\delta_{\alpha\beta}\Delta t_k$, and the Riemann sum converges to $\frac{1}{2}\int_0^t\sum_{i,j,\alpha}B_{i\alpha,s}B_{j\alpha,s}\partial^2_{x_ix_j}f\,\mathrm{d}s=\int_0^t\mathrm{Tr}(\mathbf{D}_s\nabla^2 f)\,\mathrm{d}s$, proving Eq. (C.18). □

C.2.2 The Wong–Zakai Smooth-Noise Convergence Theorem

In physical electronics (Chapters 4, 11 and 12), thermal noise $\boldsymbol{\xi}^{(\varepsilon)}(t)=\frac{\mathrm{d}\mathbf{W}_t^{(\varepsilon)}}{\mathrm{d}t}$ is never strictly white at infinite frequencies; it has a finite microscopic correlation time $\varepsilon\equiv\tau_{\mathrm{corr}}>0$ (or bandwidth cutoff $\omega_c\sim\varepsilon^{-1}$), so $\mathbf{W}_t^{(\varepsilon)}$ has piecewise smooth paths. As $\varepsilon\to 0$, the solutions converge to those of the Stratonovich SDE, not the Itô SDE. We prove this, the Wong–Zakai theorem, for polygonal (piecewise linear) approximations of Brownian motion.

Theorem C.5 (Wong–Zakai Theorem)**.** *Let $W_t^{(n)}$ be the piecewise linear approximation to a standard 1D Brownian motion W_t on the partition $t_k = kT/n$ $(k = 0, \dots, n)$:*

$$W_t^{(n)} \;:=\; W_{t_k} + \frac{t - t_k}{\Delta t}\,(W_{t_{k+1}} - W_{t_k}) \qquad \text{for } t \in [t_k, t_{k+1}], \quad \Delta t = \frac{T}{n}. \tag{C.22}$$

Let $x_t^{(n)}$ solve the ordinary differential equation with state-dependent noise amplitude $b \in C_b^2(\mathbb{R})$ and drift $a \in C_b^1(\mathbb{R})$:

$$\frac{\mathrm{d}x_t^{(n)}}{\mathrm{d}t} \;=\; a\Big(x_t^{(n)}\Big) + b\Big(x_t^{(n)}\Big)\,\frac{\mathrm{d}W_t^{(n)}}{\mathrm{d}t}, \qquad x_0^{(n)} = x_0. \tag{C.23}$$

Then as $n \to \infty$ $(\Delta t \to 0)$, $x_t^{(n)}$ converges uniformly on $[0, T]$ in probability (and in L^2) to the solution x_t of the ***Stratonovich SDE***

$$\mathrm{d}x_t \;=\; a(x_t)\,\mathrm{d}t + b(x_t) \circ \mathrm{d}W_t \;=\; \Big[a(x_t) + \frac{1}{2}b(x_t)b'(x_t)\Big]\,\mathrm{d}t + b(x_t)\,\mathrm{d}W_t. \tag{C.24}$$

In N dimensions, for $\frac{\mathrm{d}x_i^{(n)}}{\mathrm{d}t} = a_i(\mathbf{x}^{(n)}) + \sum_{\alpha=1}^{M} B_{i\alpha}(\mathbf{x}^{(n)})\frac{\mathrm{d}W_{\alpha,t}^{(n)}}{\mathrm{d}t}$, the limit is the Stratonovich SDE with Itô drift correction $\frac{1}{2}\sum_{j=1}^{N}\sum_{\alpha=1}^{M} B_{j\alpha}(\mathbf{x})\frac{\partial B_{i\alpha}(\mathbf{x})}{\partial x_j}$.

Proof. To see where the $\frac{1}{2}b(x)b'(x)$ drift comes from, integrate Eq. (C.23) over one subinterval $[t_k, t_{k+1}]$ where $\frac{\mathrm{d}W_t^{(n)}}{\mathrm{d}t} = \frac{\Delta W_k}{\Delta t}$ is constant ($\Delta W_k := W_{t_{k+1}} - W_{t_k}$). Expanding $x_t^{(n)}$ around t_k for $t \in [t_k, t_{k+1}]$ via the fundamental theorem of calculus gives

$$x_t^{(n)} - x_{t_k}^{(n)} \;=\; b\Big(x_{t_k}^{(n)}\Big)\,\frac{t - t_k}{\Delta t}\Delta W_k + \mathcal{O}(\Delta t). \tag{C.25}$$

Now Taylor-expand $b(x_t^{(n)})$ around $x_{t_k}^{(n)}$ inside the integral $\int_{t_k}^{t_{k+1}} b(x_t^{(n)})\frac{\Delta W_k}{\Delta t}\,\mathrm{d}t$:

$$\begin{aligned}
\int_{t_k}^{t_{k+1}} b\Big(x_t^{(n)}\Big)\,\frac{\mathrm{d}W_t^{(n)}}{\mathrm{d}t}\,\mathrm{d}t \;&=\; \int_{t_k}^{t_{k+1}} \Big[b\Big(x_{t_k}^{(n)}\Big) + b'\Big(x_{t_k}^{(n)}\Big)\Big(x_t^{(n)} - x_{t_k}^{(n)}\Big) + \mathcal{O}\big((x_t^{(n)} - x_{t_k}^{(n)})^2\big)\Big]\,\frac{\Delta W_k}{\Delta t}\,\mathrm{d}t \\
&=\; b\Big(x_{t_k}^{(n)}\Big)\,\Delta W_k + b\Big(x_{t_k}^{(n)}\Big)\,b'\Big(x_{t_k}^{(n)}\Big)\left(\frac{\Delta W_k}{\Delta t}\right)^2 \underbrace{\int_{t_k}^{t_{k+1}} (t - t_k)\,\mathrm{d}t}_{=\frac{1}{2}(\Delta t)^2} + \mathcal{O}((\Delta t)^{3/2}) \\
&=\; b\Big(x_{t_k}^{(n)}\Big)\,\Delta W_k + \frac{1}{2}b\Big(x_{t_k}^{(n)}\Big)\,b'\Big(x_{t_k}^{(n)}\Big)\,(\Delta W_k)^2 + \mathcal{O}((\Delta t)^{3/2}).
\end{aligned} \tag{C.26}$$

The factor $\frac{1}{2}$ is the ordinary integral $\int_0^1 u\,\mathrm{d}u$: within each window $[t_k, t_{k+1}]$ the state moves linearly in time in response to the constant noise slope, and the noise multiplies this displacement. Summing Eq. (C.26) from $k = 0$ to $n - 1$, the first term is the left-point Itô Riemann sum $\sum_k b(x_{t_k}^{(n)})\Delta W_k \to \int_0^T b(x_t)\,\mathrm{d}W_t$, while in the second term $(\Delta W_k)^2$ converges in $L^2(\Omega)$ (by the exact same quadratic-variation argument as in Theorem C.4) to Δt, producing the additional drift integral $\frac{1}{2}\int_0^T b(x_t)b'(x_t)\,\mathrm{d}t$.

Alternatively, when $b(x) > 0$, the **Lamperti transformation** $y = \Phi(x) := \int_{x_0}^{x} \frac{\mathrm{d}u}{b(u)}$ (which has $\Phi'(x) = 1/b(x)$ and $\Phi''(x) = -b'(x)/b(x)^2$) converts the ordinary differential equation Eq. (C.23) by the classical chain rule into an additive-noise ODE:

$$\frac{\mathrm{d}y_t^{(n)}}{\mathrm{d}t} \;=\; \frac{a\big(\Phi^{-1}(y_t^{(n)})\big)}{b\big(\Phi^{-1}(y_t^{(n)})\big)} + \frac{\mathrm{d}W_t^{(n)}}{\mathrm{d}t}. \tag{C.27}$$

Because the noise is now additive ($\mathrm{d}W_t^{(n)}$ has coefficient 1) and the solution of an ODE with Lipschitz drift depends continuously on an additive driving path, passing to the uniform limit $W_t^{(n)} \to W_t$ yields $\mathrm{d}y_t = \frac{a(x_t)}{b(x_t)}\,\mathrm{d}t + \mathrm{d}W_t$. Applying Itô's formula (Theorem C.4) to the inverse map $x_t = \Phi^{-1}(y_t)$ (for which $(\Phi^{-1})'(y) = b(x)$ and $(\Phi^{-1})''(y) = b(x)b'(x)$) restores x_t:

$$\mathrm{d}x_t \;=\; b(x_t)\,\mathrm{d}y_t + \frac{1}{2}b(x_t)b'(x_t)(\mathrm{d}y_t)^2 \;=\; \left[a(x_t) + \frac{1}{2}b(x_t)b'(x_t)\right]\mathrm{d}t + b(x_t)\,\mathrm{d}W_t, \tag{C.28}$$

completing the proof of Eq. (C.24). □

C.3 Witten Spectral Theory and Completeness of the Hermite Basis

Consider the reversible overdamped Langevin SDE on $\mathbb{R}^N$ with mobility $\mu > 0$, temperature T, and smooth confining potential $E(\mathbf{x})$ ($Z = \int e^{-\beta E(\mathbf{x})}\,\mathrm{d}^N x < \infty$, $p_{\mathrm{eq}}(\mathbf{x}) = Z^{-1}e^{-\beta E(\mathbf{x})}$):

$$\mathrm{d}\mathbf{x}_t \;=\; -\mu\nabla E(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2\mu k_{\mathrm{B}}T}\,\mathrm{d}\mathbf{W}_t, \qquad D \;\equiv\; \mu k_{\mathrm{B}}T. \tag{C.29}$$

Its forward Fokker–Planck operator $\mathcal{L}_{\mathrm{FP}}$ (acting on densities ρ) and backward Kolmogorov generator $\mathcal{L}$ (acting on observables f) are given by

$$\mathcal{L}_{\mathrm{FP}}\rho \;=\; \mu\nabla\cdot\Big(\rho\nabla E + k_{\mathrm{B}}T\nabla\rho\Big) \;=\; D\nabla\cdot\left(p_{\mathrm{eq}}\nabla\left(\frac{\rho}{p_{\mathrm{eq}}}\right)\right), \tag{C.30}$$

$$\mathcal{L}f \;=\; -\mu\nabla E\cdot\nabla f + D\Delta f \;=\; \frac{D}{p_{\mathrm{eq}}}\nabla\cdot(p_{\mathrm{eq}}\nabla f). \tag{C.31}$$

C.3.1 Self-Adjointness of $\mathcal{L}$ in $L^2(p_{\mathrm{eq}})$ and the Witten Schrödinger Transformation

Proposition C.6 (Dirichlet Form, Self-Adjointness in $L^2(p_{\mathrm{eq}})$, and Witten Conjugation)**.** *Let $L^2(p_{\mathrm{eq}})$ be the weighted Hilbert space with inner product $\langle f,\,g\rangle_{p_{\mathrm{eq}}} := \int_{\mathbb{R}^N}\mathrm{d}^N x\,p_{\mathrm{eq}}(\mathbf{x})f(\mathbf{x})g(\mathbf{x})$. Then:*

(a) For all smooth compactly supported functions $f, g \in C_c^\infty(\mathbb{R}^N)$, the backward generator $\mathcal{L}$ is symmetric and negative semidefinite on $L^2(p_{\mathrm{eq}})$:

$$\langle f,\,-\mathcal{L}g\rangle_{p_{\mathrm{eq}}} \;=\; D\int_{\mathbb{R}^N}\mathrm{d}^N x\,p_{\mathrm{eq}}(\mathbf{x})\,\nabla f(\mathbf{x})\cdot\nabla g(\mathbf{x}) \;=\; \langle -\mathcal{L}f,\,g\rangle_{p_{\mathrm{eq}}} \;:=\; \mathcal{E}(f,g), \tag{C.32}$$

and a density $\rho(\mathbf{x},t) = p_{\mathrm{eq}}(\mathbf{x})f(\mathbf{x},t)$ solves $\partial_t\rho = \mathcal{L}_{\mathrm{FP}}\rho$ if and only if f solves $\partial_t f = \mathcal{L}f$.

*(b) Under the unitary ground-state transformation $\psi(\mathbf{x}) = p_{\mathrm{eq}}(\mathbf{x})^{-1/2}\rho(\mathbf{x}) = p_{\mathrm{eq}}(\mathbf{x})^{1/2}f(\mathbf{x})$, which maps $L^2(p_{\mathrm{eq}})$ onto $L^2(\mathbb{R}^N,\mathrm{d}^N x)$, the operators $-\mathcal{L}_{\mathrm{FP}}$ and $-\mathcal{L}$ map to the self-adjoint **Witten Schrödinger Hamiltonian** $\mathcal{H}_{\mathrm{W}}$ on $L^2(\mathbb{R}^N,\mathrm{d}^N x)$:*

$$\mathcal{H}_{\mathrm{W}} \;:=\; -p_{\mathrm{eq}}^{-1/2}\mathcal{L}_{\mathrm{FP}}p_{\mathrm{eq}}^{1/2} \;=\; -p_{\mathrm{eq}}^{1/2}\mathcal{L}p_{\mathrm{eq}}^{-1/2} \;=\; -D\Delta + V_{\mathrm{W}}(\mathbf{x}), \tag{C.33}$$

where the effective Witten potential is

$$V_{\mathrm{W}}(\mathbf{x}) \;=\; \frac{\mu}{4k_{\mathrm{B}}T}\,\|\nabla E(\mathbf{x})\|^2 - \frac{\mu}{2}\Delta E(\mathbf{x}) \;=\; D\left(\frac{1}{4}\beta^2\,\|\nabla E\|^2 - \frac{1}{2}\beta\Delta E\right). \tag{C.34}$$

Proof. For part (a), substituting the divergence form $\mathcal{L}g = \frac{D}{p_{\mathrm{eq}}}\nabla\cdot(p_{\mathrm{eq}}\nabla g)$ from Eq. (C.30) into $\langle f, -\mathcal{L}g\rangle_{p_{\mathrm{eq}}}$ and integrating by parts gives

$$\langle f, -\mathcal{L}g\rangle_{p_{\mathrm{eq}}} \;=\; -D\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x}) f(\mathbf{x})\,\frac{1}{p_{\mathrm{eq}}(\mathbf{x})}\nabla\cdot(p_{\mathrm{eq}}(\mathbf{x})\nabla g(\mathbf{x})) \;=\; D\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\,\nabla f(\mathbf{x})\cdot\nabla g(\mathbf{x}), \tag{C.35}$$

which is symmetric in f and g. Moreover, setting $\rho = p_{\mathrm{eq}} f$ in Eq. (C.30) gives $\mathcal{L}_{\mathrm{FP}}(p_{\mathrm{eq}} f) = D\nabla\cdot(p_{\mathrm{eq}}\nabla f) = p_{\mathrm{eq}}(\mathcal{L}f)$.

For part (b), define the first-order supersymmetric annihilation and creation operators on $L^2(\mathbb{R}^N)$:

$$\mathbf{Q} \;:=\; \sqrt{D}\left(\nabla + \frac{\beta}{2}\nabla E\right), \qquad \mathbf{Q}^\dagger \;=\; \sqrt{D}\left(-\nabla + \frac{\beta}{2}\nabla E\right). \tag{C.36}$$

For any test function $\psi \in C_c^\infty(\mathbb{R}^N)$, the product rule gives $\nabla(p_{\mathrm{eq}}^{-1/2}\psi) = p_{\mathrm{eq}}^{-1/2}(\nabla + \frac{\beta}{2}\nabla E)\psi$ because $\nabla(p_{\mathrm{eq}}^{-1/2}) = \nabla(Z^{1/2}e^{+\beta E/2}) = \frac{\beta}{2}(\nabla E)p_{\mathrm{eq}}^{-1/2}$, and similarly $\nabla\cdot(p_{\mathrm{eq}}^{1/2}\mathbf{\Phi}) = p_{\mathrm{eq}}^{1/2}(\nabla - \frac{\beta}{2}\nabla E)\cdot\mathbf{\Phi}$. Therefore, acting with $-p_{\mathrm{eq}}^{1/2}\mathcal{L}p_{\mathrm{eq}}^{-1/2}$ on ψ factors as

$$\begin{aligned} -p_{\mathrm{eq}}^{1/2}\mathcal{L}(p_{\mathrm{eq}}^{-1/2}\psi) \;&=\; -D\,p_{\mathrm{eq}}^{-1/2}\nabla\cdot\left[p_{\mathrm{eq}}\,\nabla\left(p_{\mathrm{eq}}^{-1/2}\psi\right)\right] \\ &=\; -D\,p_{\mathrm{eq}}^{-1/2}\nabla\cdot\left[p_{\mathrm{eq}}^{1/2}\left(\nabla + \frac{\beta}{2}\nabla E\right)\psi\right] \\ &=\; -D\left(\nabla - \frac{\beta}{2}\nabla E\right)\cdot\left(\nabla + \frac{\beta}{2}\nabla E\right)\psi \;=\; \mathbf{Q}^\dagger\cdot\mathbf{Q}\psi. \end{aligned} \tag{C.37}$$

Expanding the operator product $\mathbf{Q}^\dagger\cdot\mathbf{Q}\psi = -D(\nabla - \frac{\beta}{2}\nabla E)\cdot(\nabla\psi + \frac{\beta}{2}\psi\nabla E)$ using the Leibniz rule $\nabla\cdot(\psi\nabla E) = \nabla\psi\cdot\nabla E + \psi\Delta E$ cancels the first-order cross terms $\pm\frac{\beta}{2}\nabla E\cdot\nabla\psi$:

$$\begin{aligned} \mathcal{H}_{\mathrm{W}}\psi \;&=\; \mathbf{Q}^\dagger\cdot\mathbf{Q}\psi \\ &=\; -D\left[\Delta\psi + \frac{\beta}{2}(\nabla\psi\cdot\nabla E + \psi\Delta E) - \frac{\beta}{2}\nabla E\cdot\nabla\psi - \frac{\beta^2}{4}\,\|\nabla E\|^2\,\psi\right] \\ &=\; (\,-D\Delta + V_{\mathrm{W}}(\mathbf{x}))\psi, \end{aligned} \tag{C.38}$$

yielding Eqs. (C.33) and (C.34). Since $\langle\psi|\mathcal{H}_{\mathrm{W}}|\psi\rangle = \|\mathbf{Q}\psi\|_{L^2}^2 \geq 0$, the spectrum of $\mathcal{H}_{\mathrm{W}}$ is non-negative. The function $\psi_0(\mathbf{x}) := p_{\mathrm{eq}}(\mathbf{x})^{1/2}$ satisfies $\mathbf{Q}\psi_0 = \mathbf{0}$, so $\mathcal{H}_{\mathrm{W}}\psi_0 = 0$ and ψ_0 is the ground state; it is unique because $\mathbf{Q}\psi = \mathbf{0}$ forces $\psi \propto p_{\mathrm{eq}}^{1/2}$. □

C.3.2 Completeness of the Hermite Polynomial Basis in $L^2(\mathcal{N}(0, v_\infty))$

Worked computation 6.21 diagonalized the Ornstein–Uhlenbeck generator of the RC sampler in probabilists' Hermite polynomials. In one dimension the generator is $\mathcal{L} = -(D/v_\infty)\,x\partial_x + D\partial_x^2$, with stationary variance v_∞, and the normalized eigenfunctions are

$$\mathrm{He}_n(z) \;:=\; (-1)^n e^{z^2/2}\frac{\mathrm{d}^n}{\mathrm{d}z^n}\left(e^{-z^2/2}\right), \qquad \phi_n(x) \;:=\; \frac{1}{\sqrt{n!}}\,\mathrm{He}_n\left(\frac{x}{\sqrt{v_\infty}}\right), \qquad n = 0, 1, 2, \ldots \tag{C.39}$$

Here we prove that $\{\phi_n\}_{n=0}^\infty$ is a complete orthonormal basis of $L^2(p_{\mathrm{eq}})$ where

$$p_{\mathrm{eq}}(x) = (2\pi v_\infty)^{-1/2} e^{-x^2/(2v_\infty)}.$$

Theorem C.7 (Generating Function, Orthonormality, and Completeness of Hermite Polynomials)**.** *Let $\gamma(z) := (2\pi)^{-1/2} e^{-z^2/2}$ be the standard normal density on $\mathbb{R}$. Then the probabilists' Hermite polynomials $\mathrm{He}_n(z)$ satisfy:*

(a) ***Exponential generating function:*** *For all $z \in \mathbb{R}$ and $t \in \mathbb{C}$,*

$$\exp\!\Big(tz - \frac{1}{2}t^2\Big) \;=\; \sum_{n=0}^{\infty} \frac{t^n}{n!}\,\mathrm{He}_n(z). \tag{C.40}$$

(b) ***Orthonormality:*** $\int_{-\infty}^{\infty} \gamma(z)\,\mathrm{He}_m(z)\,\mathrm{He}_n(z)\,\mathrm{d}z = n!\,\delta_{mn}$.

(c) ***Completeness in $L^2(\gamma)$:*** *The normalized Hermite polynomials $\big\{(n!)^{-1/2}\mathrm{He}_n(z)\big\}_{n=0}^{\infty}$ form a complete orthonormal basis of the Hilbert space $L^2(\mathbb{R}, \gamma(z)\,\mathrm{d}z)$.*

Proof. For part (a), Taylor-expanding the analytic function $t \mapsto e^{-(z-t)^2/2}$ around $t = 0$ gives

$$\exp\!\Big(tz - \frac{1}{2}t^2\Big) \;=\; e^{z^2/2}\exp\!\Big(-\frac{1}{2}(z-t)^2\Big) \;=\; e^{z^2/2}\sum_{n=0}^{\infty}\frac{(-t)^n}{n!}\,\frac{\mathrm{d}^n}{\mathrm{d}z^n}\Big(e^{-z^2/2}\Big) \;=\; \sum_{n=0}^{\infty}\frac{t^n}{n!}\,\mathrm{He}_n(z). \tag{C.41}$$

For part (b), multiply the generating functions $\exp(sz - s^2/2)$ and $\exp(tz - t^2/2)$, integrate against $\gamma(z)$ using Lemma B.1 ($\int_{-\infty}^{\infty}\gamma(z)e^{(s+t)z}\,\mathrm{d}z = e^{(s+t)^2/2}$), and expand in powers of s and t:

$$\sum_{m,n=0}^{\infty}\frac{s^m t^n}{m!\,n!}\int_{-\infty}^{\infty}\gamma(z)\,\mathrm{He}_m(z)\,\mathrm{He}_n(z)\,\mathrm{d}z \;=\; e^{-(s^2+t^2)/2}\int_{-\infty}^{\infty}\gamma(z)\,e^{(s+t)z}\,\mathrm{d}z \;=\; e^{st} \;=\; \sum_{n=0}^{\infty}\frac{(st)^n}{n!}. \tag{C.42}$$

Equating the coefficient of $s^m t^n$ on both sides yields $\int_{-\infty}^{\infty}\gamma(z)\mathrm{He}_m(z)\mathrm{He}_n(z)\,\mathrm{d}z = n!\,\delta_{mn}$.

For part (c), to show that $\mathrm{span}\,\{\mathrm{He}_n\}_{n=0}^{\infty}$ is dense in $L^2(\gamma)$, it suffices by standard Hilbert space theory to show that if $f \in L^2(\gamma)$ is orthogonal to He_n for every $n \geq 0$, then $f = 0$ almost everywhere. Since $\mathrm{span}\,\{\mathrm{He}_0, \ldots, \mathrm{He}_n\} = \mathrm{span}\,\{1, z, \ldots, z^n\}$, $\langle f,\, \mathrm{He}_n\rangle_{L^2(\gamma)} = 0$ for all $n \geq 0$ implies $\int_{-\infty}^{\infty} f(z) z^n e^{-z^2/2}\,\mathrm{d}z = 0$ for all $n \geq 0$. Consider the Fourier transform of the $L^1(\mathbb{R})$ function $g(z) := f(z)e^{-z^2/2}$:

$$\hat{g}(k) \;:=\; \int_{-\infty}^{\infty} f(z)\,e^{-z^2/2}e^{ikz}\,\mathrm{d}z, \qquad k \in \mathbb{R}. \tag{C.43}$$

By the Cauchy–Schwarz inequality, for every $R > 0$ and $|k| \leq R$,

$$\begin{aligned}
&\int_{-\infty}^{\infty} |f(z)|\; e^{-z^2/2}\left(\sum_{n=0}^{\infty}\frac{|kz|^n}{n!}\right)\mathrm{d}z \\
&\quad\leq\; \int_{-\infty}^{\infty} |f(z)|\; e^{R|z| - z^2/2}\,\mathrm{d}z \\
&\quad\leq\; (2\pi)^{1/4}\,\|f\|_{L^2(\gamma)}\left(\int_{-\infty}^{\infty} e^{2R|z| - z^2/2}\,\mathrm{d}z\right)^{1/2} \;<\; \infty.
\end{aligned} \tag{C.44}$$

By the dominated convergence theorem, we can interchange the sum and integral in the Taylor series of e^{ikz}:

$$\hat{g}(k) \;=\; \sum_{n=0}^{\infty}\frac{(ik)^n}{n!}\int_{-\infty}^{\infty} f(z)\,z^n e^{-z^2/2}\,\mathrm{d}z \;=\; 0 \qquad \text{for all } k \in \mathbb{R}. \tag{C.45}$$

By Fourier inversion on $L^1(\mathbb{R})$, $\hat{g} \equiv 0$ implies $g(z) = f(z)e^{-z^2/2} = 0$ almost everywhere, and since $e^{-z^2/2} > 0$, $f(z) = 0$ almost everywhere. □

C.4 Semigroup Proof of the Bakry–Émery Γ_2 Theorem

In Chapters 6 and 16, the **Bakry–Émery theorem** is the tool that shows the following: when an energy function $E(\mathbf{x})$ is uniformly κ-strongly convex ($\nabla^2 E(\mathbf{x}) \succeq \kappa \mathbf{I}$ with $\kappa > 0$), the overdamped Langevin diffusion Eq. (C.29) satisfies both a Poincaré inequality and a Logarithmic Sobolev Inequality (LSI), forcing any initial density ρ_0 to relax exponentially fast to p_{eq} in Kullback–Leibler divergence at rate $2\mu\kappa$. Here we prove it with Bakry and Émery's Γ_2 calculus and Markov semigroup interpolation.

C.4.1 The Carré du Champ Operator Γ and Iterated Operator Γ_2

Throughout this section, let $\mathcal{L} = -\mu\nabla E \cdot \nabla + \mu k_{\mathrm{B}} T \Delta$ be the backward generator Eq. (C.30) of the reversible Langevin dynamics with $D \equiv \mu k_{\mathrm{B}} T$, and let $P_t = e^{t\mathcal{L}}$ be the Markov semigroup acting on observables $f_t(\mathbf{x}) := (P_t f)(\mathbf{x}) = \mathbb{E}[f(\mathbf{x}_t) \mid \mathbf{x}_0 = \mathbf{x}]$, which solves $\partial_t P_t f = \mathcal{L} P_t f = P_t \mathcal{L} f$.

Definition C.8 (Carré du Champ Γ and Iterated Operator Γ_2)**.** For smooth functions $f, g \in C^\infty(\mathbb{R}^N)$, the **carré du champ** (square field) bilinear form $\Gamma(f,g)$ and the **iterated carré du champ** $\Gamma_2(f,g)$ associated with the generator $\mathcal{L}$ are defined intrinsically by

$$\Gamma(f,g) \;:=\; \frac{1}{2}\Big[\mathcal{L}(fg) - f\,\mathcal{L}g - g\,\mathcal{L}f\Big], \tag{C.46}$$

$$\Gamma_2(f,g) \;:=\; \frac{1}{2}\Big[\mathcal{L}\Gamma(f,g) - \Gamma(f,\mathcal{L}g) - \Gamma(\mathcal{L}f,g)\Big]. \tag{C.47}$$

In particular, on the diagonal $f = g$,

$$\Gamma(f) \;\equiv\; \Gamma(f,f) \;=\; \frac{1}{2}\mathcal{L}(f^2) - f\mathcal{L}f, \qquad \Gamma_2(f) \;\equiv\; \Gamma_2(f,f) \;=\; \frac{1}{2}\mathcal{L}\Gamma(f,f) - \Gamma(f,\mathcal{L}f). \tag{C.48}$$

We now compute $\Gamma(f,g)$ and $\Gamma_2(f,f)$ explicitly for the Langevin generator $\mathcal{L} = -\mu\nabla E \cdot \nabla + \mu k_{\mathrm{B}} T \Delta$.

Lemma C.9 (Bochner–Bakry–Émery Identity)**.** *For the overdamped Langevin generator $\mathcal{L} = -\mu\nabla E\cdot\nabla + \mu k_{\mathrm{B}} T\Delta$ on $\mathbb{R}^N$ with $D = \mu k_{\mathrm{B}} T$, the operators Γ and Γ_2 are given for every $f, g \in C^\infty(\mathbb{R}^N)$ by*

$$\Gamma(f,g) \;=\; \mu k_{\mathrm{B}} T\, \nabla f \cdot \nabla g \;=\; D \sum_{i=1}^{N} \frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_i}, \tag{C.49}$$

$$\begin{aligned} \Gamma_2(f,f) \;&=\; \mu^2 (k_{\mathrm{B}} T)^2 \left\|\nabla^2 f\right\|_{\mathrm{HS}}^2 + \mu^2 k_{\mathrm{B}} T \left\langle \nabla f,\, (\nabla^2 E)\nabla f \right\rangle \\ &=\; D^2 \sum_{i,j=1}^{N} \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)^2 + \mu D \sum_{i,j=1}^{N} \frac{\partial^2 E}{\partial x_i \partial x_j}\frac{\partial f}{\partial x_i}\frac{\partial f}{\partial x_j}. \end{aligned} \tag{C.50}$$

Proof. First, to prove Eq. (C.49), note that the first-order drift operator $-\mu\nabla E \cdot \nabla$ obeys the Leibniz product rule $-\mu\nabla E \cdot \nabla(fg) = -f\mu\nabla E \cdot \nabla g - g\mu\nabla E \cdot \nabla f$, so its contribution to $\Gamma(f,g)$ cancels completely. For the second-order Laplacian $D\Delta$, the product rule gives

$$D\Delta(fg) \;=\; D\sum_{i=1}^{N} \frac{\partial^2}{\partial x_i^2}(fg) \;=\; D\sum_{i=1}^{N}\left(f\frac{\partial^2 g}{\partial x_i^2} + 2\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_i} + g\frac{\partial^2 f}{\partial x_i^2}\right) \;=\; f\,D\Delta g + g\,D\Delta f + 2D\,\nabla f\cdot\nabla g. \tag{C.51}$$

Subtracting $f\mathcal{L}g + g\mathcal{L}f$ and dividing by 2 immediately yields Eq. (C.49).

Next, to prove the Bochner–Bakry–Émery identity Eq. (C.50), we write $f_i \equiv \partial_{x_i} f$, $f_{ij} \equiv \partial^2_{x_i x_j} f$, $E_i \equiv \partial_{x_i} E$, and $E_{ij} \equiv \partial^2_{x_i x_j} E$, and compute the two pieces of $\Gamma_2(f,f) = \frac{1}{2}\mathcal{L}\Gamma(f,f) - \Gamma(f,\mathcal{L}f)$.

First, since $\Gamma(f,f) = D\sum_{i=1}^N f_i^2$, we apply $\frac{1}{2}\mathcal{L} = -\frac{\mu}{2}\sum_j E_j\partial_{x_j} + \frac{D}{2}\sum_j \partial^2_{x_j}$:

$$\begin{aligned}\frac{1}{2}\mathcal{L}\Gamma(f,f) &= -\frac{\mu D}{2}\sum_{i,j=1}^N E_j\,\frac{\partial}{\partial x_j}\Big(f_i^2\Big) + \frac{D^2}{2}\sum_{i,j=1}^N \frac{\partial^2}{\partial x_j^2}\Big(f_i^2\Big)\\ &= -\mu D\sum_{i,j=1}^N E_j f_i f_{ij} + D^2\sum_{i,j=1}^N \Big(f_{ij}^2 + f_i f_{ijj}\Big),\end{aligned} \tag{C.52}$$

where we used $\partial_{x_j}(f_i^2) = 2f_i f_{ij}$ and $\partial^2_{x_j}(f_i^2) = 2(f_{ij}^2 + f_i f_{ijj})$.

Second, using $\Gamma(f,\mathcal{L}f) = D\sum_{i=1}^N f_i\,\partial_{x_i}(\mathcal{L}f)$ from Eq. (C.49), we differentiate $\mathcal{L}f = -\mu\sum_j E_j f_j + D\sum_j f_{jj}$ with respect to x_i via the product rule:

$$\frac{\partial}{\partial x_i}(\mathcal{L}f) = -\mu\sum_{j=1}^N (E_{ij}f_j + E_j f_{ij}) + D\sum_{j=1}^N f_{jji}. \tag{C.53}$$

Multiplying by Df_i and summing over $i = 1,\dots,N$ gives

$$\Gamma(f,\mathcal{L}f) = -\mu D\sum_{i,j=1}^N E_{ij}f_i f_j - \mu D\sum_{i,j=1}^N E_j f_i f_{ij} + D^2\sum_{i,j=1}^N f_i f_{jji}. \tag{C.54}$$

By equality of mixed third partial derivatives ($f_{ijj} = f_{jji}$), when we subtract Eq. (C.54) from Eq. (C.52), both the first-derivative drift terms $-\mu D\sum_{i,j} E_j f_i f_{ij}$ and the third-derivative terms $D^2\sum_{i,j} f_i f_{ijj}$ cancel identically. Only the Hessian squared and the potential curvature term survive:

$$\Gamma_2(f,f) = \frac{1}{2}\mathcal{L}\Gamma(f,f) - \Gamma(f,\mathcal{L}f) = D^2\sum_{i,j=1}^N f_{ij}^2 + \mu D\sum_{i,j=1}^N E_{ij}f_i f_j, \tag{C.55}$$

which is Eq. (C.50). □

Corollary C.10 (The Bakry–Émery Curvature-Dimension Criterion CD($\mu\kappa,\infty$))**.** *Suppose the potential $E \in C^2(\mathbb{R}^N)$ satisfies $\nabla^2 E(\mathbf{x}) \succeq \kappa\mathbf{I}$ for all $\mathbf{x}\in\mathbb{R}^N$ and some $\kappa\in\mathbb{R}$ (for $\kappa>0$, E is κ-strongly convex). Since $\|\nabla^2 f\|_{\mathrm{HS}}^2 = \sum_{i,j} f_{ij}^2 \ge 0$, the Bochner–Bakry–Émery identity Eq. (C.50) implies the pointwise inequality*

$$\Gamma_2(f,f) \ge \mu^2 k_{\mathrm{B}}T\,\kappa\,\|\nabla f\|^2 = \mu\kappa\,\Gamma(f,f) \qquad \text{for all } f\in C^\infty(\mathbb{R}^N). \tag{C.56}$$

C.4.2 Semigroup Interpolation Proof of Gradient Commutation, Poincaré, and Log-Sobolev Inequalities

We now show how the local algebraic inequality $\Gamma_2(f) \ge \mu\kappa\,\Gamma(f)$ in Eq. (C.56) integrates along the Markov semigroup $P_t = e^{t\mathcal{L}}$ to yield global functional inequalities. The technique is **semigroup interpolation**: for a fixed terminal time $t>0$, we differentiate $s\mapsto P_s\Phi(P_{t-s}f)$ for $s\in[0,t]$.

Theorem C.11 (Bakry–Émery Theorem: Gradient Bound, Poincaré Inequality, and LSI)**.** *Suppose $\nabla^2 E(\mathbf{x}) \succeq \kappa\mathbf{I}$ on $\mathbb{R}^N$ with $\kappa>0$, so that $\Gamma_2(f) \ge \lambda_{\mathrm{BE}}\,\Gamma(f)$ with $\lambda_{\mathrm{BE}} := \mu\kappa > 0$. Then:*

(a) ***Pointwise Gradient Commutation Bound:*** *For every smooth f and every $t \geq 0$,*

$$\Gamma(P_t f) \;\leq\; e^{-2\mu\kappa t}\, P_t(\Gamma(f)), \qquad \textit{i.e.,} \qquad \|\nabla P_t f(\mathbf{x})\|^2 \;\leq\; e^{-2\mu\kappa t}\, P_t\Big(\|\nabla f\|^2\Big)\,(\mathbf{x}). \tag{C.57}$$

(b) ***Poincaré Inequality and Spectral Gap ($\lambda_1 \geq \mu\kappa$):*** *For every $f \in H^1(p_{\mathrm{eq}})$,*

$$\mathrm{Var}_{p_{\mathrm{eq}}}(f) \;:=\; \mathbb{E}\Big[f^2\Big]_{p_{\mathrm{eq}}} - \big(\mathbb{E}[f]_{p_{\mathrm{eq}}}\big)^2 \;\leq\; \frac{1}{\mu\kappa}\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\,\Gamma(f) \;=\; \frac{k_{\mathrm{B}}T}{\kappa}\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\,\|\nabla f(\mathbf{x})\|^2 . \tag{C.58}$$

Equivalently, the spectral gap of $-\mathcal{L}$ satisfies $\lambda_1 \geq \mu\kappa$, and $\mathrm{Var}_{p_{\mathrm{eq}}}(P_t f) \leq e^{-2\mu\kappa t}\,\mathrm{Var}_{p_{\mathrm{eq}}}(f)$.

(c) ***Logarithmic Sobolev Inequality ($\mathrm{LSI}(\mu\kappa)$):*** *For every smooth positive density ratio $g = \rho/p_{\mathrm{eq}} > 0$ with $\mathbb{E}[g]_{p_{\mathrm{eq}}} = 1$,*

$$\begin{aligned} D_{\mathrm{KL}}(\rho\,\|\,p_{\mathrm{eq}}) \;&=\; \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, g(\mathbf{x}) \ln g(\mathbf{x}) \\ &\leq\; \frac{1}{2\mu\kappa}\int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\,\frac{\Gamma(g)}{g} \\ &=\; \frac{k_{\mathrm{B}}T}{2\kappa}\int_{\mathbb{R}^N} \mathrm{d}^N x\,\rho(\mathbf{x})\left\|\nabla \ln\!\left(\frac{\rho(\mathbf{x})}{p_{\mathrm{eq}}(\mathbf{x})}\right)\right\|^2 . \end{aligned} \tag{C.59}$$

(d) ***Exponential Entropy Decay:*** *Any solution ρ_t of the Fokker–Planck equation $\partial_t \rho_t = \mathcal{L}_{\mathrm{FP}}\rho_t$ satisfies*

$$D_{\mathrm{KL}}(\rho_t\,\|\,p_{\mathrm{eq}}) \;\leq\; e^{-2\mu\kappa t}\, D_{\mathrm{KL}}(\rho_0\,\|\,p_{\mathrm{eq}}) \qquad \textit{for all } t \geq 0. \tag{C.60}$$

Proof. We prove all four parts in sequence via semigroup interpolation.

Proof of (a): Pointwise gradient commutation. Fix $t > 0$ and define for $s \in [0,t]$ the function $\Lambda(s) := e^{-2\mu\kappa s}\, P_s(\Gamma(P_{t-s}f))$. Let $h_{t-s} := P_{t-s}f$, so that $\partial_s h_{t-s} = -\mathcal{L}h_{t-s}$. Using the semigroup property $\partial_s P_s = P_s\mathcal{L}$ and the chain rule $\partial_s\Gamma(h_{t-s}, h_{t-s}) = 2\Gamma(h_{t-s}, \partial_s h_{t-s}) = -2\Gamma(h_{t-s}, \mathcal{L}h_{t-s})$, we differentiate $\Lambda(s)$ with respect to s:

$$\begin{aligned} \frac{\mathrm{d}\Lambda(s)}{\mathrm{d}s} \;&=\; e^{-2\mu\kappa s}\, P_s\Big[-2\mu\kappa\,\Gamma(h_{t-s}) + \mathcal{L}\Gamma(h_{t-s}) - 2\Gamma(h_{t-s}, \mathcal{L}h_{t-s})\Big] \\ &=\; 2e^{-2\mu\kappa s}\, P_s\Big[\Gamma_2(h_{t-s}) - \mu\kappa\,\Gamma(h_{t-s})\Big] \;\geq\; 0, \end{aligned} \tag{C.61}$$

where the last inequality follows from $\Gamma_2(h_{t-s}) \geq \mu\kappa\,\Gamma(h_{t-s})$ (Corollary C.10) and the positivity preservation of the Markov semigroup P_s ($F \geq 0 \implies P_s F \geq 0$). Integrating $\frac{\mathrm{d}\Lambda}{\mathrm{d}s} \geq 0$ from $s = 0$ to $s = t$ yields $\Lambda(0) \leq \Lambda(t)$, which is $\Gamma(P_t f) \leq e^{-2\mu\kappa t} P_t(\Gamma(f))$, proving Eq. (C.57).

Proof of (b): Poincaré inequality and spectral gap. For $s \in [0,t]$, define $\Psi(s) := P_s\big((P_{t-s}f)^2\big) = P_s(h_{t-s}^2)$. Differentiating with respect to s and using $\mathcal{L}(h^2) - 2h\mathcal{L}h = 2\Gamma(h)$ from Eq. (C.48) alongside Eq. (C.57) gives

$$\begin{aligned} \frac{\mathrm{d}\Psi(s)}{\mathrm{d}s} \;&=\; P_s\Big[\mathcal{L}(h_{t-s}^2) - 2h_{t-s}\mathcal{L}h_{t-s}\Big] \\ &=\; 2P_s(\Gamma(P_{t-s}f)) \\ &\leq\; 2e^{-2\mu\kappa(t-s)} P_s(P_{t-s}\Gamma(f)) \\ &=\; 2e^{-2\mu\kappa(t-s)} P_t(\Gamma(f)). \end{aligned} \tag{C.62}$$

Integrating this inequality from $s = 0$ to $s = t$ yields the local variance bound

$$P_t(f^2) - (P_t f)^2 \;=\; \Psi(t) - \Psi(0) \;\le\; 2P_t(\Gamma(f)) \int_0^t e^{-2\mu\kappa(t-s)}\, \mathrm{d}s \;=\; \frac{1 - e^{-2\mu\kappa t}}{\mu\kappa}\, P_t(\Gamma(f)). \tag{C.63}$$

Integrating over p_{eq} (using stationarity $\int p_{\mathrm{eq}}\, P_t F = \int p_{\mathrm{eq}} F$) and letting $t \to \infty$ (where $P_t f \to \mathbb{E}[f]_{p_{\mathrm{eq}}}$) yields the Poincaré inequality:

$$\mathrm{Var}_{p_{\mathrm{eq}}}(f) \;\le\; \frac{1}{\mu\kappa} \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \Gamma(f) \;=\; \frac{1}{\mu\kappa}\, \langle f,\, -\mathcal{L} f \rangle_{p_{\mathrm{eq}}} \;=\; \frac{k_{\mathrm{B}} T}{\kappa} \int_{\mathbb{R}^N} \mathrm{d}^N x\, p_{\mathrm{eq}}(\mathbf{x})\, \|\nabla f(\mathbf{x})\|^2 . \tag{C.64}$$

By the Rayleigh–Ritz variational characterization of the first non-zero eigenvalue λ_1 of $-\mathcal{L}$ on $L^2(p_{\mathrm{eq}})$, Eq. (C.64) is equivalent to $\lambda_1 \ge \mu\kappa$.

Proof of (c) and (d): Logarithmic Sobolev Inequality and exponential KL decay. Before differentiating the logarithmic entropy along the semigroup, we prove the chain rule for $\mathcal{L}(\Phi(h))$ for any smooth function $\Phi\colon \mathbb{R} \to \mathbb{R}$: since $\nabla\Phi(h) = \Phi'(h)\nabla h$ and $\Delta\Phi(h) = \Phi'(h)\Delta h + \Phi''(h)\, \|\nabla h\|^2$, we have

$$\mathcal{L}(\Phi(h)) \;=\; \Phi'(h)\mathcal{L}h + \Phi''(h)\Gamma(h). \tag{C.65}$$

Applying Eq. (C.65) to the convex entropy function $\Phi(u) = u \ln u$ (for $u > 0$), which has $\Phi'(u) = \ln u + 1$ and $\Phi''(u) = 1/u$, gives

$$\mathcal{L}(h \ln h) - (\ln h + 1)\mathcal{L}h \;=\; \frac{\Gamma(h)}{h}. \tag{C.66}$$

Now let $h_{t-s} := P_{t-s} g > 0$ and differentiate the entropy interpolation functional $\Theta(s) := P_s(h_{t-s} \ln h_{t-s})$ for $s \in [0, t]$ using Eq. (C.66):

$$\frac{\mathrm{d}\Theta(s)}{\mathrm{d}s} \;=\; P_s\Big[\mathcal{L}(h_{t-s} \ln h_{t-s}) - (\ln h_{t-s} + 1)\mathcal{L}h_{t-s}\Big] \;=\; P_s\left(\frac{\Gamma(h_{t-s})}{h_{t-s}}\right). \tag{C.67}$$

To bound the right-hand side, consider the Fisher functional

$$I(s) := P_s(\Gamma(h_{t-s})/h_{t-s}) = P_s(h_{t-s}\, \Gamma(\ln h_{t-s})).$$

A longer calculation with the chain rule Eq. (C.65), which we do not reproduce (see Bakry et al. [7]), gives $\frac{\mathrm{d}I}{\mathrm{d}s} = 2P_s(h_{t-s}\, \Gamma_2(\ln h_{t-s}))$. Applying Corollary C.10 to $\ln h_{t-s}$ bounds this below by $2\mu\kappa\, I(s)$, so $e^{-2\mu\kappa s} I(s)$ is non-decreasing on $[0, t]$. Comparing its values at $s = 0$ and $s = t$, and then replacing t by $t - s$, gives

$$\frac{\Gamma(P_{t-s} g)}{P_{t-s} g} \;\le\; e^{-2\mu\kappa(t-s)}\, P_{t-s}\left(\frac{\Gamma(g)}{g}\right). \tag{C.68}$$

Substituting Eq. (C.68) into Eq. (C.67) and using $P_s P_{t-s} = P_t$ yields

$$\frac{\mathrm{d}\Theta(s)}{\mathrm{d}s} \;\le\; e^{-2\mu\kappa(t-s)}\, P_t\left(\frac{\Gamma(g)}{g}\right). \tag{C.69}$$

Integrating from $s = 0$ to $s = t$ gives

$$P_t(g \ln g) - (P_t g) \ln(P_t g) \;=\; \Theta(t) - \Theta(0) \;\le\; \frac{1 - e^{-2\mu\kappa t}}{2\mu\kappa}\, P_t\left(\frac{\Gamma(g)}{g}\right). \tag{C.70}$$

Integrating over p_{eq} with $\mathbb{E}[g]_{p_{\mathrm{eq}}} = 1$ and taking $t \to \infty$ (so $P_t g \to 1$ and $(P_t g)\ln(P_t g) \to 0$) proves the Logarithmic Sobolev Inequality Eq. (C.59).

Finally, to prove Eq. (C.60), recall the Fokker–Planck H-theorem (de Bruijn identity) of Section 6.3 for $g_t = \rho_t / p_{\mathrm{eq}}$:

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) \;=\; -\int_{\mathbb{R}^N} \mathrm{d}^N x \, p_{\mathrm{eq}}(\mathbf{x}) \, \frac{\Gamma(g_t)}{g_t} \;\le\; -2\mu\kappa \, D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) \,, \tag{C.71}$$

where the inequality applies the LSI Eq. (C.59) to g_t. Hence $\frac{\mathrm{d}}{\mathrm{d}t}\big(e^{2\mu\kappa t} D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}})\big) \le 0$, and integrating from 0 to t yields $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) \le e^{-2\mu\kappa t} D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}})$. □

Worked computation C.12 (Explicit Verification of the Bakry–Émery LSI and Equality Case for the Ornstein–Uhlenbeck Process)**.** The log-Sobolev constant $2\mu\kappa$ in Theorem C.11 cannot be improved. To see this, we evaluate the theorem for the isotropic quadratic potential $E(\mathbf{x}) = \frac{\kappa}{2}\|\mathbf{x}\|^2$ on $\mathbb{R}^N$, the Ornstein–Uhlenbeck process of N identical uncoupled nodes of the RC sampler, and test it on shifted Gaussian distributions.

Step 1: Exact operators and equilibrium measure. For $E(\mathbf{x}) = \frac{\kappa}{2}\|\mathbf{x}\|^2$ ($\kappa > 0$), the gradient and Hessian are $\nabla E(\mathbf{x}) = \kappa\mathbf{x}$ and $\nabla^2 E(\mathbf{x}) = \kappa\mathbf{I}$, which saturates the strong-convexity hypothesis $\nabla^2 E \succeq \kappa\mathbf{I}$ at every point $\mathbf{x} \in \mathbb{R}^N$. The equilibrium Boltzmann measure is $p_{\mathrm{eq}}(\mathbf{x}) = \mathcal{N}(\mathbf{0}, v_\infty \mathbf{I})$ with variance $v_\infty = k_{\mathrm{B}}T/\kappa = D/(\mu\kappa)$. The Bochner–Bakry–Émery identity Eq. (C.50) reads

$$\Gamma_2(f,f) \;=\; D^2 \left\|\nabla^2 f\right\|_{\mathrm{HS}}^2 + \mu\kappa\,\Gamma(f,f). \tag{C.72}$$

Equality $\Gamma_2(f,f) = \mu\kappa\,\Gamma(f,f)$ holds if and only if $\nabla^2 f \equiv \mathbf{0}$, i.e., $f(\mathbf{x}) = \mathbf{a}^\top\mathbf{x} + c$ is an affine linear function.

Step 2: Saturation of the LSI by shifted Gaussian densities. For any shift vector $\mathbf{m} \in \mathbb{R}^N$, consider the shifted Gaussian probability density $\rho_{\mathbf{m}}(\mathbf{x}) := \mathcal{N}(\mathbf{m}, v_\infty\mathbf{I})$ with the same covariance $v_\infty\mathbf{I}$ as p_{eq}. Its density ratio $g(\mathbf{x}) := \rho_{\mathbf{m}}(\mathbf{x})/p_{\mathrm{eq}}(\mathbf{x})$ is

$$g(\mathbf{x}) \;=\; \frac{\exp\!\left(-\frac{1}{2v_\infty}\|\mathbf{x}-\mathbf{m}\|^2\right)}{\exp\!\left(-\frac{1}{2v_\infty}\|\mathbf{x}\|^2\right)} \;=\; \exp\!\left(\frac{\mathbf{m}^\top\mathbf{x}}{v_\infty} - \frac{\|\mathbf{m}\|^2}{2v_\infty}\right), \qquad \ln g(\mathbf{x}) \;=\; \frac{\mathbf{m}^\top\mathbf{x}}{v_\infty} - \frac{\|\mathbf{m}\|^2}{2v_\infty}. \tag{C.73}$$

Since $\ln g$ is affine in $\mathbf{x}$, $\nabla^2 \ln g \equiv \mathbf{0}$, the equality case of Step 1. We evaluate both sides of the logarithmic Sobolev inequality Eq. (C.59) on $\rho_{\mathbf{m}}$. The left-hand side is (in agreement with Eq. (B.21))

$$D_{\mathrm{KL}}(\rho_{\mathbf{m}} \,\|\, p_{\mathrm{eq}}) \;=\; \int_{\mathbb{R}^N} \mathrm{d}^N x \, \rho_{\mathbf{m}}(\mathbf{x}) \left(\frac{\mathbf{m}^\top\mathbf{x}}{v_\infty} - \frac{\|\mathbf{m}\|^2}{2v_\infty}\right) \;=\; \frac{\|\mathbf{m}\|^2}{2v_\infty} \;=\; \frac{\kappa\,\|\mathbf{m}\|^2}{2k_{\mathrm{B}}T}. \tag{C.74}$$

On the right-hand side of Eq. (C.59), the score difference is constant in space: $\nabla \ln(\rho_{\mathbf{m}}/p_{\mathrm{eq}}) = \nabla \ln g(\mathbf{x}) = \mathbf{m}/v_\infty$. Therefore, the relative Fisher information integral evaluates to

$$\frac{k_{\mathrm{B}}T}{2\kappa} \int_{\mathbb{R}^N} \mathrm{d}^N x \, \rho_{\mathbf{m}}(\mathbf{x}) \left\|\nabla \ln\!\left(\frac{\rho_{\mathbf{m}}(\mathbf{x})}{p_{\mathrm{eq}}(\mathbf{x})}\right)\right\|^2 \;=\; \frac{k_{\mathrm{B}}T}{2\kappa}\frac{\|\mathbf{m}\|^2}{v_\infty^2} \int_{\mathbb{R}^N} \mathrm{d}^N x \, \rho_{\mathbf{m}}(\mathbf{x}) \;=\; \frac{k_{\mathrm{B}}T}{2\kappa}\frac{\|\mathbf{m}\|^2}{(k_{\mathrm{B}}T/\kappa)^2} \;=\; \frac{\kappa\,\|\mathbf{m}\|^2}{2k_{\mathrm{B}}T}. \tag{C.75}$$

Comparing Eqs. (C.74) and (C.75) shows that every shifted Gaussian $\rho_{\mathbf{m}}$ achieves exact equality in the Bakry–Émery LSI. Under the Ornstein–Uhlenbeck flow, the mean decays as $\mathbf{m}(t) = e^{-\mu\kappa t}\mathbf{m}(0)$, so $D_{\mathrm{KL}}(\rho_t \,\|\, p_{\mathrm{eq}}) = \frac{\kappa}{2k_{\mathrm{B}}T}\|\mathbf{m}(t)\|^2 = e^{-2\mu\kappa t} D_{\mathrm{KL}}(\rho_0 \,\|\, p_{\mathrm{eq}})$, saturating Eq. (C.60) at every $t \ge 0$.

Summary

- **Proved:**
  - Grönwall's inequality (Lemma C.1), Doob's L^2 maximal inequality (Lemma C.2), and existence and pathwise uniqueness of SDE solutions (Theorem C.3).
  - The multidimensional Itô formula (Theorem C.4) and the Wong–Zakai theorem for polygonal approximations (Theorem C.5).
  - The Witten conjugation (Proposition C.6) and completeness of the Hermite basis in $L^2(\mathcal{N}(0, v_\infty))$ (Theorem C.7).
  - The Bochner–Bakry–Émery identity (Lemma C.9), the gradient bound, Poincaré and logarithmic Sobolev inequalities, and exponential KL decay (Theorem C.11), with sharpness on the Ornstein–Uhlenbeck process (Worked computation C.12).
- **Cited:** the derivative of the Fisher functional in the proof of the logarithmic Sobolev inequality [7].

Bibliographic notes

The Picard iteration proof of SDE existence and uniqueness, Doob's maximal inequalities, and the multidimensional Itô formula follow the standard mathematical treatments of Karatzas and Shreve [59] and Øksendal [80]. The convergence of ordinary differential equations driven by smooth colored noise to Stratonovich stochastic differential equations was proved by Wong and Zakai [119]. The supersymmetric conjugation of reversible Fokker–Planck operators to Schrödinger Hamiltonians was introduced in Witten [118] and applied to Langevin processes in Risken [90] and Pavliotis [85]. The Γ_2 calculus and the semigroup proof of the logarithmic Sobolev inequality originate in Bakry and Émery [6] and are developed in the monograph by Bakry et al. [7]. The application to the mixing time of DTM stages in Chapter 16 is this book's formalization of a qualitative argument in Jelinčič et al. [56].

Appendix D

Semiconductor MOS Electrostatics, the EKV Subthreshold Integral, and the RCSJ Reference

The device chapters of this book use a small number of results from semiconductor physics and superconducting electronics. Chapter 12 needs the dependence of the minority-carrier sheet charge on the surface potential, the capacitive divider that defines the slope factor n, and the integrated channel current of the Enz–Krummenacher–Vittoz (EKV) model. Chapters 11 and 13 need the statement that the channel noise of a subthreshold transistor is $S_I = 2q_e(I_F + I_R)$. Chapter 14 needs the resistively and capacitively shunted junction (RCSJ) equation of motion, its characteristic scales, and the double-well potential of the rf-SQUID that underlies the quantum flux parametron (QFP). This appendix collects those derivations in one place so that the main chapters can quote them.

The first three sections (Sections D.1 to D.3) go from carrier statistics through MOS electrostatics to the EKV current law. Section D.4 derives the channel noise in two independent ways. Sections D.5 and D.6 treat the Josephson junction and the rf-SQUID, ending with the quartic normal form that maps the QFP onto the double-well sampler's potential. Section D.7 tabulates representative device numbers at 300 K and 4.2 K.

Notation Watch: Thermal, Threshold, and Pinch-Off Voltages

Device textbooks use V_T for two unrelated quantities. In this book $V_T = k_B T/q_e$ is always the *thermal voltage* (25.85 mV at 300 K). The MOS *threshold voltage* is V_{th}; EKV papers write it V_{T0}, and we use V_{th} for the same quantity. The EKV *pinch-off voltage* V_P is a third, distinct quantity introduced in Section D.3. All terminal voltages V_G, V_S, V_D are referred to the substrate (bulk) terminal B, as in the EKV model; this makes the source and drain enter the current symmetrically.

D.1 Carrier Statistics in a Nondegenerate Semiconductor

D.1.1 From Fermi–Dirac to Boltzmann Occupation

Electrons in the conduction band of silicon occupy states of energy $E \geq E_c$. Near the band edge the dispersion is parabolic with a density-of-states effective mass m_e^*, and the density of states per

unit volume (including spin) is

$$g_c(E) \;=\; \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\sqrt{E-E_c}, \qquad E \ge E_c. \tag{D.1}$$

In thermal equilibrium at temperature T each state is occupied with the Fermi–Dirac probability $f(E) = [1 + e^{(E-E_F)/k_\mathrm{B}T}]^{-1}$, where E_F is the Fermi level (the electrochemical potential of the electrons). The electron density is

$$n \;=\; \int_{E_c}^{\infty} g_c(E)\, f(E)\, \mathrm{d}E. \tag{D.2}$$

Substitute $\epsilon = (E - E_c)/k_\mathrm{B}T$ and $\eta = (E_F - E_c)/k_\mathrm{B}T$:

$$n \;=\; \frac{1}{2\pi^2}\left(\frac{2m_e^* k_\mathrm{B}T}{\hbar^2}\right)^{3/2}\int_0^{\infty}\frac{\sqrt{\epsilon}\;\mathrm{d}\epsilon}{1+e^{\epsilon-\eta}}. \tag{D.3}$$

The semiconductor is called **nondegenerate** when $E_c - E_F \gtrsim 3k_\mathrm{B}T$, that is $\eta \lesssim -3$. Then $e^{\epsilon-\eta} \ge e^3 \approx 20$ for every $\epsilon \ge 0$, the 1 in the denominator is a correction of at most 5%, and $f(E) \approx e^{-(E-E_F)/k_\mathrm{B}T}$. The remaining integral is $\int_0^\infty \sqrt{\epsilon}\, e^{-\epsilon}\,\mathrm{d}\epsilon = \Gamma(3/2) = \sqrt{\pi}/2$, so

$$n \;=\; N_c \exp\!\left(-\frac{E_c - E_F}{k_\mathrm{B}T}\right), \qquad N_c \;:=\; \frac{1}{2\pi^2}\left(\frac{2m_e^* k_\mathrm{B}T}{\hbar^2}\right)^{3/2}\frac{\sqrt{\pi}}{2} \;=\; 2\left(\frac{m_e^* k_\mathrm{B}T}{2\pi\hbar^2}\right)^{3/2}. \tag{D.4}$$

The last equality uses $(2)^{3/2}\sqrt{\pi}/(4\pi^2) = 2/(2\pi)^{3/2}$, which follows from $2^{3/2}\cdot(2\pi)^{3/2} = 8\pi^{3/2}$. The constant N_c is the *effective density of states* of the conduction band; for silicon at 300 K it is about $2.8 \times 10^{19}\,\mathrm{cm}^{-3}$. The same calculation for holes (empty valence states, occupation $1 - f(E) \approx e^{-(E_F-E)/k_\mathrm{B}T}$) gives

$$p \;=\; N_v \exp\!\left(-\frac{E_F - E_v}{k_\mathrm{B}T}\right), \qquad N_v \;=\; 2\left(\frac{m_h^* k_\mathrm{B}T}{2\pi\hbar^2}\right)^{3/2}. \tag{D.5}$$

D.1.2 Mass-Action Law, Intrinsic Level, and Fermi Potential

Multiplying Eq. (D.4) and Eq. (D.5), the Fermi level cancels:

$$np \;=\; N_c N_v\, e^{-E_g/k_\mathrm{B}T} \;:=\; n_i^2, \qquad E_g = E_c - E_v. \tag{D.6}$$

This is the **mass-action law**. It holds for any doping as long as both bands are nondegenerate and the system is in equilibrium. In an undoped crystal $n = p = n_i$, and the Fermi level sits at the *intrinsic level*

$$E_i \;=\; \frac{E_c + E_v}{2} + \frac{k_\mathrm{B}T}{2}\ln\frac{N_v}{N_c}, \tag{D.7}$$

obtained by equating Eqs. (D.4) and (D.5) and solving for E_F. Eliminating N_c and N_v in favour of n_i and E_i gives the compact forms

$$n \;=\; n_i\, e^{(E_F-E_i)/k_\mathrm{B}T}, \qquad p \;=\; n_i\, e^{(E_i-E_F)/k_\mathrm{B}T}. \tag{D.8}$$

For silicon at 300 K, $n_i \approx 1 \times 10^{10}\,\mathrm{cm}^{-3}$.

Now dope the crystal with N_A acceptors per unit volume, all ionized. Far from any surface, charge neutrality reads $p_0 = N_A + n_0$, and since $n_0 = n_i^2/p_0 \ll N_A$ for $N_A \gg n_i$, the bulk hole and electron densities are

$$p_0 \;\approx\; N_A, \qquad n_{p0} \;=\; \frac{n_i^2}{N_A}. \tag{D.9}$$

The **Fermi potential** measures how far the Fermi level lies below the intrinsic level in the neutral bulk, in volts:

$$\phi_F \;:=\; \left.\frac{E_i - E_F}{q_e}\right|_{\text{bulk}} \;=\; V_T \ln\frac{N_A}{n_i}, \tag{D.10}$$

where the second form follows from $p_0 = n_i e^{q_e\phi_F/k_{\mathrm{B}}T} = N_A$. Two useful identities follow: $n_{p0} = n_i e^{-\phi_F/V_T} = N_A e^{-2\phi_F/V_T}$.

D.1.3 Band Bending and the Electrostatic Potential

When an electrostatic potential $\psi(x)$ is present, every band edge and the intrinsic level shift together by $-q_e\psi(x)$. We measure ψ from the neutral bulk, so $\psi(\infty) = 0$ and $E_i(x) = E_i(\infty) - q_e\psi(x)$. If the electrons are in equilibrium with the bulk, the Fermi level is flat and Eq. (D.8) gives

$$n(x) \;=\; n_{p0}\, e^{\psi(x)/V_T}, \qquad p(x) \;=\; N_A\, e^{-\psi(x)/V_T}. \tag{D.11}$$

In a MOSFET the electrons near the surface are supplied by the n^+ source and drain, not by the bulk. When a reverse bias V_{ch} separates the electron quasi-Fermi level near the surface from the hole Fermi level of the bulk (by q_eV_{ch}), the electron density becomes

$$n(x) \;=\; n_{p0}\, e^{(\psi(x)-V_{\mathrm{ch}})/V_T} \;=\; N_A\, e^{(\psi(x)-2\phi_F-V_{\mathrm{ch}})/V_T}, \tag{D.12}$$

while the holes stay in equilibrium with the bulk. The voltage V_{ch} is the *channel voltage*: it equals V_S at the source end and V_D at the drain end, and it varies along the channel. Equation (D.12) is the only place where the source and drain enter the electrostatics.

Physics Insight: Why Boltzmann Statistics Survive at the Surface

Inversion layers in weak inversion are extremely dilute. The surface electron density is $n_s = N_A e^{(\psi_s-2\phi_F-V_{\mathrm{ch}})/V_T}$, and weak inversion is defined by $\psi_s < 2\phi_F + V_{\mathrm{ch}}$, so $n_s < N_A \ll N_c$. The electron gas at the surface is therefore always nondegenerate in the regime used by thermodynamic hardware, and Eq. (D.12) is accurate. The exponential current–voltage law of Chapter 12 is this Boltzmann factor, transported to the terminals. Quantum confinement of the inversion layer (subband formation) changes prefactors in deeply scaled devices but not the exponential dependence on ψ_s.

D.2 One-Dimensional MOS Electrostatics

Consider the MOS capacitor: a gate electrode, an oxide of thickness t_{ox} and permittivity $\varepsilon_{\mathrm{ox}}$, and a uniformly doped p-type substrate of permittivity $\varepsilon_{\mathrm{Si}}$ occupying $x > 0$. The oxide capacitance per unit area is $C_{\mathrm{ox}} = \varepsilon_{\mathrm{ox}}/t_{\mathrm{ox}}$. We treat the gate as an equipotential, ignore fixed oxide charge except through the flat-band voltage, and assume the substrate is thick compared with the depletion width. In a long-channel transistor the lateral field along the channel is small compared with the vertical field, so at each position along the channel the vertical electrostatics is that of this one-dimensional capacitor with the local channel voltage V_{ch} (the *gradual-channel approximation*).

D.2.1 The Poisson–Boltzmann Equation and Its First Integral

The charge density in the silicon is $\rho = q_e(p - n - N_A + n_{p0}e^{-V_{\mathrm{ch}}/V_T})$; the last term makes ρ vanish in the neutral bulk where $\psi = 0$, and it is negligible everywhere else. Using Eqs. (D.11) and (D.12)

and Poisson's equation $\psi'' = -\rho/\varepsilon_{\mathrm{Si}}$,

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} \;=\; \frac{q_e N_A}{\varepsilon_{\mathrm{Si}}}\Big[\big(1 - e^{-\psi/V_T}\big) + e^{-(2\phi_F + V_{\mathrm{ch}})/V_T}\big(e^{\psi/V_T} - 1\big)\Big]. \tag{D.13}$$

The first bracket is the uncovered acceptor charge minus the remaining holes; the second is the electron charge. For $\psi > 0$ (bands bent downward) both terms are positive, so ψ is convex and decays monotonically into the bulk.

Equation (D.13) has the form $\psi'' = g(\psi)$ with no explicit x dependence. Multiplying by $2\psi'$ gives $\frac{\mathrm{d}}{\mathrm{d}x}(\psi')^2 = 2g(\psi)\psi' = \frac{\mathrm{d}}{\mathrm{d}x}\, 2\int_0^{\psi} g(\psi')\,\mathrm{d}\psi'$. Integrating from deep in the bulk, where $\psi = \psi' = 0$, to the point x,

$$\left(\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)^2 \;=\; \frac{2q_e N_A}{\varepsilon_{\mathrm{Si}}}\int_0^{\psi}\Big[\big(1 - e^{-u/V_T}\big) + e^{-(2\phi_F + V_{\mathrm{ch}})/V_T}\big(e^{u/V_T} - 1\big)\Big]\,\mathrm{d}u. \tag{D.14}$$

The two elementary integrals are $\int_0^{\psi}(1 - e^{-u/V_T})\,\mathrm{d}u = \psi + V_T e^{-\psi/V_T} - V_T$ and $\int_0^{\psi}(e^{u/V_T} - 1)\,\mathrm{d}u = V_T e^{\psi/V_T} - V_T - \psi$. Hence

$$\left(\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)^2 \;=\; \frac{2q_e N_A}{\varepsilon_{\mathrm{Si}}}\, V_T\, F^2\!\left(\frac{\psi}{V_T}\right), \tag{D.15}$$

with the dimensionless **Kingston–Brews function**

$$F(u) \;:=\; \Big[e^{-u} + u - 1 + e^{-(2\phi_F + V_{\mathrm{ch}})/V_T}\big(e^{u} - u - 1\big)\Big]^{1/2}. \tag{D.16}$$

D.2.2 Exact Surface Field and Total Silicon Charge

Let $\psi_s = \psi(0)$ be the surface potential. Since ψ decreases into the bulk, $\psi'(0) < 0$ for $\psi_s > 0$, and the surface field $\mathcal{E}_s = -\psi'(0)$ is

$$\mathcal{E}_s \;=\; \mathrm{sgn}(\psi_s)\sqrt{\frac{2q_e N_A V_T}{\varepsilon_{\mathrm{Si}}}}\; F\!\left(\frac{\psi_s}{V_T}\right). \tag{D.17}$$

By Gauss's law applied to a pillbox spanning the silicon (the field vanishes in the neutral bulk), the total silicon charge per unit area is $Q_s = -\varepsilon_{\mathrm{Si}}\mathcal{E}_s$. Introducing the **body-effect coefficient**

$$\gamma \;:=\; \frac{\sqrt{2q_e\varepsilon_{\mathrm{Si}}N_A}}{C_{\mathrm{ox}}} \qquad [\mathrm{V}^{1/2}], \tag{D.18}$$

we have $\sqrt{2q_e N_A \varepsilon_{\mathrm{Si}}} = \gamma C_{\mathrm{ox}}$, and for $\psi_s > 0$

$$Q_s \;=\; -\gamma C_{\mathrm{ox}}\sqrt{V_T}\, F\!\left(\frac{\psi_s}{V_T}\right) \;=\; -\gamma C_{\mathrm{ox}}\sqrt{\psi_s - V_T + V_T e^{-\psi_s/V_T} + V_T e^{-(2\phi_F + V_{\mathrm{ch}})/V_T}\big(e^{\psi_s/V_T} - \psi_s/V_T - 1\big)}. \tag{D.19}$$

This is exact within Boltzmann statistics and the one-dimensional geometry. It covers accumulation ($\psi_s < 0$, holes pile up, $Q_s > 0$ with the sign of Eq. (D.17)), depletion, weak inversion, and strong inversion.

D.2.3 The Depletion Approximation

For $V_T \ll \psi_s < 2\phi_F + V_{\mathrm{ch}}$, the hole term $e^{-\psi_s/V_T}$ and the electron term are both small. The silicon charge is then dominated by ionized acceptors in a depleted layer of width W_{dep}. The **depletion**

approximation replaces the true charge profile by $\rho = -q_e N_A$ for $0 < x < W_{\mathrm{dep}}$ and $\rho = 0$ beyond. Integrating $\psi'' = q_e N_A/\varepsilon_{\mathrm{Si}}$ twice with $\psi(W_{\mathrm{dep}}) = \psi'(W_{\mathrm{dep}}) = 0$ gives

$$\psi(x) \;=\; \frac{q_e N_A}{2\varepsilon_{\mathrm{Si}}}(W_{\mathrm{dep}} - x)^2, \qquad \psi_s \;=\; \frac{q_e N_A W_{\mathrm{dep}}^2}{2\varepsilon_{\mathrm{Si}}}, \qquad W_{\mathrm{dep}} \;=\; \sqrt{\frac{2\varepsilon_{\mathrm{Si}}\psi_s}{q_e N_A}}. \tag{D.20}$$

The depletion (bulk) charge is

$$Q_B \;=\; -q_e N_A W_{\mathrm{dep}} \;=\; -\sqrt{2 q_e \varepsilon_{\mathrm{Si}} N_A \psi_s} \;=\; -\gamma C_{\mathrm{ox}} \sqrt{\psi_s}. \tag{D.21}$$

Compare with Eq. (D.19): dropping the electron and hole terms leaves $Q_s = -\gamma C_{\mathrm{ox}}\sqrt{\psi_s - V_T}$. The $-V_T$ is the correction from the smooth majority-carrier tail at the edge of the depletion layer, whose width is set by the extrinsic Debye length $L_D = \sqrt{\varepsilon_{\mathrm{Si}} V_T/(q_e N_A)}$. Since $\psi_s \approx 0.5$–$0.9\,\mathrm{V}$ in weak inversion while $V_T = 0.026\,\mathrm{V}$, the correction is a few percent and the depletion approximation is adequate for all purposes in this book. We use the form $\gamma C_{\mathrm{ox}}\sqrt{\psi_s}$ from here on; the $-V_T$ can be restored by replacing $\psi_s \to \psi_s - V_T$ in every depletion formula.

D.2.4 Gate Charge Balance

The gate voltage (referred to the bulk) is the sum of the work-function difference, the potential drop across the oxide, and the band bending in the silicon. With the flat-band voltage $V_{FB} = \Phi_{MS} - Q_{\mathrm{ox}}/C_{\mathrm{ox}}$ absorbing the work-function difference Φ_{MS} and any fixed oxide charge Q_{ox}, and with the gate charge $Q_G = -Q_s$ on the oxide capacitor,

$$V_G \;=\; V_{FB} + \psi_s - \frac{Q_s}{C_{\mathrm{ox}}} \;=\; V_{FB} + \psi_s - \frac{Q_B + Q_{\mathrm{inv}}}{C_{\mathrm{ox}}}, \tag{D.22}$$

where we have split the silicon charge into the depletion charge Q_B and the inversion (electron) charge $Q_{\mathrm{inv}} := Q_s - Q_B \le 0$. Inserting Eq. (D.21),

$$V_G \;=\; V_{FB} + \psi_s + \gamma\sqrt{\psi_s} - \frac{Q_{\mathrm{inv}}}{C_{\mathrm{ox}}}. \tag{D.23}$$

D.2.5 Threshold Condition

The conventional onset of strong inversion is the surface potential at which the surface electron density equals the bulk hole density, $n_s = N_A$. By Eq. (D.12), this happens when

$$\psi_s \;=\; 2\phi_F + V_{\mathrm{ch}}. \tag{D.24}$$

At this point Q_{inv} is still small compared with Q_B, so setting $Q_{\mathrm{inv}} = 0$ in Eq. (D.23) at $V_{\mathrm{ch}} = 0$ gives the threshold voltage

$$V_{\mathrm{th}} \;=\; V_{FB} + 2\phi_F + \gamma\sqrt{2\phi_F}. \tag{D.25}$$

With $V_{\mathrm{ch}} = V_{\mathrm{SB}} > 0$ the threshold rises to $V_{FB} + 2\phi_F + V_{\mathrm{SB}} + \gamma\sqrt{2\phi_F + V_{\mathrm{SB}}}$ (gate voltage measured from bulk); this is the body effect.

D.2.6 The Weak-Inversion Sheet Charge

In weak inversion, $\phi_F < \psi_s < 2\phi_F + V_{\mathrm{ch}}$. Write $\delta := e^{-(2\phi_F + V_{\mathrm{ch}})/V_T}(e^{\psi_s/V_T} - \psi_s/V_T - 1)$ for the electron term inside the square root of Eq. (D.19), and drop the hole term $V_T e^{-\psi_s/V_T}$. Because $\psi_s \gg V_T$, e^{ψ_s/V_T} dominates $\psi_s/V_T + 1$, and

$$\delta \;\approx\; e^{(\psi_s - 2\phi_F - V_{\mathrm{ch}})/V_T} \;<\; 1. \tag{D.26}$$

The silicon charge is $Q_s = -\gamma C_{\rm ox}\sqrt{\psi_s + V_T\delta}$, where we have used the depletion form $\sqrt{\psi_s}$ for the bulk part as agreed above. Since $V_T\delta \ll \psi_s$, a first-order Taylor expansion $\sqrt{a+\epsilon} = \sqrt{a} + \epsilon/(2\sqrt{a}) + \mathcal{O}(\epsilon^2/a^{3/2})$ gives

$$Q_s \;=\; -\gamma C_{\rm ox}\sqrt{\psi_s} \;-\; \frac{\gamma C_{\rm ox}}{2\sqrt{\psi_s}}\,V_T\,\delta \;+\; \mathcal{O}\!\left(\frac{V_T^2\delta^2}{\psi_s^{3/2}}\right). \tag{D.27}$$

The first term is Q_B. The second is the inversion charge:

$$\boxed{Q_{\rm inv} \;\approx\; -\frac{\gamma C_{\rm ox}}{2\sqrt{\psi_s}}\,V_T\,\exp\!\left(\frac{\psi_s - 2\phi_F - V_{\rm ch}}{V_T}\right) \;=\; -C_{\rm dep}(\psi_s)\,V_T\,\exp\!\left(\frac{\psi_s - 2\phi_F - V_{\rm ch}}{V_T}\right),} \tag{D.28}$$

where the second form uses the depletion capacitance defined in Eq. (D.29) below. Two features matter for the transistor. First, $Q_{\rm inv}$ depends exponentially on ψ_s with the thermal voltage as scale. Second, it depends on the channel voltage only through the factor $e^{-V_{\rm ch}/V_T}$.

Equation (D.28) has a direct physical reading. Integrating the electron density Eq. (D.12) over depth with the depletion profile gives $\int_0^\infty n\,{\rm d}x \approx n_s\int_0^\infty e^{-\mathcal{E}_s x/V_T}\,{\rm d}x = n_s V_T/\mathcal{E}_s$, because near the surface $\psi(x) \approx \psi_s - \mathcal{E}_s x$. With $\mathcal{E}_s = q_e N_A W_{\rm dep}/\varepsilon_{\rm Si}$, this equals

$$\frac{-Q_{\rm inv}}{q_e} \;=\; N_A\,e^{(\psi_s - 2\phi_F - V_{\rm ch})/V_T}\,\frac{\varepsilon_{\rm Si}V_T}{q_e N_A W_{\rm dep}},$$

which is Eq. (D.28) since $C_{\rm dep} = \varepsilon_{\rm Si}/W_{\rm dep}$. The inversion layer is a sheet of thickness $V_T/\mathcal{E}_s$ (a few nanometres) holding a Boltzmann population.

D.2.7 Depletion Capacitance and the Slope Factor

The small-signal depletion capacitance per unit area is the change of the depletion charge with surface potential:

$$C_{\rm dep}(\psi_s) \;:=\; -\frac{{\rm d}Q_B}{{\rm d}\psi_s} \;=\; \frac{\gamma C_{\rm ox}}{2\sqrt{\psi_s}} \;=\; \sqrt{\frac{q_e\varepsilon_{\rm Si}N_A}{2\psi_s}} \;=\; \frac{\varepsilon_{\rm Si}}{W_{\rm dep}}. \tag{D.29}$$

The three forms agree: $\gamma C_{\rm ox}/(2\sqrt{\psi_s}) = \sqrt{2q_e\varepsilon_{\rm Si}N_A}/(2\sqrt{\psi_s}) = \sqrt{q_e\varepsilon_{\rm Si}N_A/(2\psi_s)}$, and dividing $\varepsilon_{\rm Si}$ by Eq. (D.20) gives the same expression.

In weak inversion $|Q_{\rm inv}| \ll |Q_B|$, so the last term of Eq. (D.23) is negligible and the surface potential is fixed by the gate alone:

$$V_G \;=\; V_{FB} + \psi_{sa} + \gamma\sqrt{\psi_{sa}} \quad\Longrightarrow\quad \psi_{sa}(V_G) \;=\; \left(\sqrt{V_G - V_{FB} + \frac{\gamma^2}{4}} - \frac{\gamma}{2}\right)^2. \tag{D.30}$$

The subscript a is the standard notation for this weak-inversion (depletion) value. The explicit solution comes from solving the quadratic $(\sqrt{\psi_{sa}})^2 + \gamma\sqrt{\psi_{sa}} - (V_G - V_{FB}) = 0$ for the positive root. The important consequence is that ψ_{sa} *does not depend on* $V_{\rm ch}$: along a weakly inverted channel the surface potential is flat.

Differentiating Eq. (D.30) with respect to ψ_{sa},

$$\frac{{\rm d}V_G}{{\rm d}\psi_{sa}} \;=\; 1 + \frac{\gamma}{2\sqrt{\psi_{sa}}} \;=\; 1 + \frac{C_{\rm dep}}{C_{\rm ox}} \;:=\; n, \qquad \frac{{\rm d}\psi_{sa}}{{\rm d}V_G} \;=\; \frac{1}{n} \;=\; \frac{C_{\rm ox}}{C_{\rm ox} + C_{\rm dep}}. \tag{D.31}$$

This is the capacitive divider of Section 12.2: a change $\mathrm{d}V_G$ divides between C_{ox} and C_{dep} in series, and only the fraction $C_{\mathrm{ox}}/(C_{\mathrm{ox}}+C_{\mathrm{dep}})$ appears as band bending at the surface. Because $Q_{\mathrm{inv}} \propto e^{\psi_s/V_T}$, the drain current scales as $e^{V_G/(nV_T)}$ and the subthreshold swing is

$$S \;:=\; \left(\frac{\mathrm{d}\log_{10} I_{\mathrm{D}}}{\mathrm{d}V_G}\right)^{-1} \;=\; nV_T \ln 10 \;\approx\; n \times 59.5\,\mathrm{mV/decade} \quad (T = 300\,\mathrm{K}). \tag{D.32}$$

Since n depends on ψ_{sa}, it varies slowly with V_G; over the few hundred millivolts of weak inversion the change is typically 5–10%, and the main text treats n as constant.

Example D.1 (MOS Numbers for a Representative Bulk Device)**.** Take $N_A = 3\times10^{17}\,\mathrm{cm^{-3}}$, $t_{\mathrm{ox}} = 4\,\mathrm{nm}$, $T = 300\,\mathrm{K}$, $n_i = 1.0\times10^{10}\,\mathrm{cm^{-3}}$, and an n^+ polysilicon gate with $V_{FB} \approx -0.9\,\mathrm{V}$. These are representative values, not those of a specific process. Then:

- $\phi_F = V_T \ln(N_A/n_i) = 0.02585 \times \ln(3\times10^7) = 0.02585 \times 17.22 = 0.445\,\mathrm{V}$, so $2\phi_F = 0.890\,\mathrm{V}$.
- $C_{\mathrm{ox}} = 3.9\varepsilon_0/t_{\mathrm{ox}} = 3.453\times10^{-11}/(4\times10^{-9}) = 8.63\times10^{-3}\,\mathrm{F/m^2} = 8.63\,\mathrm{fF}/\mu\mathrm{m}^2$.
- $\sqrt{2q_e\varepsilon_{\mathrm{Si}}N_A} = \sqrt{2\times1.602\times10^{-19}\times1.036\times10^{-10}\times3\times10^{23}} = 3.16\times10^{-3}\,\mathrm{C\,m^{-2}V^{-1/2}}$, so $\gamma = 0.366\,\mathrm{V}^{1/2}$.
- At $\psi_s = 0.70\,\mathrm{V}$ (mid weak inversion): $W_{\mathrm{dep}} = 55\,\mathrm{nm}$, $C_{\mathrm{dep}} = 1.89\,\mathrm{fF}/\mu\mathrm{m}^2$, and $n = 1 + 1.89/8.63 = 1.22$, giving a swing of $72\,\mathrm{mV/decade}$.
- The Debye length is $L_D = 7.5\,\mathrm{nm}$, and the inversion sheet thickness $V_T/\mathcal{E}_s = V_T W_{\mathrm{dep}}/(2\psi_s) \approx 1.0\,\mathrm{nm}$.
- $V_{\mathrm{th}} = -0.9 + 0.890 + 0.366\sqrt{0.890} = 0.335\,\mathrm{V}$.

D.3 The Charge-Sheet Current Integral and the EKV Model

This section supplies the integration behind Section 12.3. We first write the channel current in terms of the inversion charge and the electron quasi-Fermi potential, then evaluate it in weak inversion, and finally derive the EKV charge–voltage relation that interpolates between weak and strong inversion.

D.3.1 Current in the Charge-Sheet Model

Let $y \in [0, L]$ be the coordinate along the channel from source ($y = 0$) to drain ($y = L$), and let W be the channel width. In the **charge-sheet model** the inversion electrons are treated as a sheet of zero thickness with charge $Q_{\mathrm{inv}}(y) < 0$ per unit area. The electron current density in a nondegenerate semiconductor is drift plus diffusion,

$$J_n \;=\; q_e\mu_n n\,\mathcal{E}_y + q_e D_n \frac{\mathrm{d}n}{\mathrm{d}y}, \qquad D_n = \mu_n V_T \quad \text{(Einstein relation)}, \tag{D.33}$$

where $\mathcal{E}_y = -\partial\psi/\partial y$ is the lateral field. Using Eq. (D.12) with a position-dependent channel voltage, $\ln n = \mathrm{const} + (\psi - V_{\mathrm{ch}})/V_T$, so $V_T\,\frac{\mathrm{d}n}{\mathrm{d}y} = n\left(\frac{\mathrm{d}\psi}{\mathrm{d}y} - \frac{\mathrm{d}V_{\mathrm{ch}}}{\mathrm{d}y}\right)$. Substituting,

$$J_n \;=\; -q_e\mu_n n\frac{\mathrm{d}\psi}{\mathrm{d}y} + q_e\mu_n n\left(\frac{\mathrm{d}\psi}{\mathrm{d}y} - \frac{\mathrm{d}V_{\mathrm{ch}}}{\mathrm{d}y}\right) \;=\; -q_e\mu_n n\,\frac{\mathrm{d}V_{\mathrm{ch}}}{\mathrm{d}y}. \tag{D.34}$$

The drift and diffusion terms combine into a single gradient of the electron quasi-Fermi potential V_{ch}. Integrating over the depth of the sheet, with $\int q_e n \, \mathrm{d}x = -Q_{\mathrm{inv}}$, and multiplying by W, the total current flowing in the $+y$ direction is $W\mu_n Q_{\mathrm{inv}} \, \mathrm{d}V_{\mathrm{ch}}/\,\mathrm{d}y$. We define the drain current I_{D} as the current entering the drain terminal (flowing from drain to source inside the channel, opposite to the electron flow), so

$$I_{\mathrm{D}} \;=\; -W\mu_n Q_{\mathrm{inv}}(V_{\mathrm{ch}}) \, \frac{\mathrm{d}V_{\mathrm{ch}}}{\mathrm{d}y}. \tag{D.35}$$

Current continuity makes I_{D} independent of y. In the gradual-channel approximation Q_{inv} depends on y only through the local channel voltage, $Q_{\mathrm{inv}}(y) = Q_{\mathrm{inv}}(V_{\mathrm{ch}}(y))$, and we can separate variables. Multiplying Eq. (D.35) by $\mathrm{d}y$ and integrating from source to drain,

$$\boxed{I_{\mathrm{D}} \;=\; \frac{W}{L}\,\mu_n \int_{V_S}^{V_D} \big(-Q_{\mathrm{inv}}(V_{\mathrm{ch}})\big)\, \mathrm{d}V_{\mathrm{ch}}.} \tag{D.36}$$

We have assumed constant mobility. This integral is the starting point of every long-channel MOSFET model.

D.3.2 Weak Inversion: Pure Diffusion

In weak inversion the surface potential is $\psi_{sa}(V_G)$, independent of y (Eq. (D.30)). Therefore $\mathcal{E}_y = -\partial\psi_s/\partial y = 0$ at the surface and the drift term in Eq. (D.33) vanishes. The entire current is diffusion. Inserting Eq. (D.28) with $\psi_s = \psi_{sa}$ into Eq. (D.36),

$$\begin{aligned} I_{\mathrm{D}} &= \frac{W}{L}\mu_n C_{\mathrm{dep}} V_T \, e^{(\psi_{sa}-2\phi_F)/V_T} \int_{V_S}^{V_D} e^{-V_{\mathrm{ch}}/V_T} \, \mathrm{d}V_{\mathrm{ch}} \\ &= \frac{W}{L}\mu_n C_{\mathrm{dep}} V_T \, e^{(\psi_{sa}-2\phi_F)/V_T} \Big[-V_T e^{-V_{\mathrm{ch}}/V_T} \Big]_{V_S}^{V_D} \\ &= \frac{W}{L}\mu_n C_{\mathrm{dep}} V_T^2 \, e^{(\psi_{sa}-2\phi_F)/V_T} \Big[e^{-V_S/V_T} - e^{-V_D/V_T} \Big]. \end{aligned} \tag{D.37}$$

The same result follows directly from Fick's law. With zero lateral field, Eq. (D.33) integrated over depth gives $I_{\mathrm{D}} = W D_n \, \mathrm{d}Q_{\mathrm{inv}}/\,\mathrm{d}y$ (the electron number flux is $-D_n \, \mathrm{d}n/\,\mathrm{d}y$, the conventional current in the $+y$ direction is $q_e D_n \, \mathrm{d}n/\,\mathrm{d}y$ per unit area, and the current into the drain is minus that). Since I_{D} is constant, $Q_{\mathrm{inv}}(y)$ is linear in y, interpolating between $Q_{\mathrm{inv}}(V_S)$ at the source and $Q_{\mathrm{inv}}(V_D)$ at the drain:

$$I_{\mathrm{D}} \;=\; W D_n \, \frac{Q_{\mathrm{inv}}(V_D) - Q_{\mathrm{inv}}(V_S)}{L} \;=\; \frac{W}{L}\mu_n V_T \Big[\, |Q_{\mathrm{inv}}(V_S)| - |Q_{\mathrm{inv}}(V_D)| \, \Big], \tag{D.38}$$

and inserting Eq. (D.28) at the two ends reproduces Eq. (D.37). The subthreshold transistor is a diffusion resistor whose two end concentrations are set by Boltzmann factors of the source and drain voltages.

D.3.3 Pinch-Off Voltage and the Weak-Inversion EKV Law

The exponent in Eq. (D.37) contains ψ_{sa}, a nonlinear function of V_G. EKV express it through the **pinch-off voltage** V_P, defined as the channel voltage at which the surface potential computed with $Q_{\mathrm{inv}} = 0$ reaches the inversion level:

$$V_G \;=\; V_{FB} + \psi_0 + V_P + \gamma\sqrt{\psi_0 + V_P}, \qquad \psi_0 \;\approx\; 2\phi_F + (\text{a few } V_T). \tag{D.39}$$

Comparing with Eq. (D.30), both equations have the form $V_G - V_{FB} = \psi + \gamma\sqrt{\psi}$, which is strictly increasing in ψ, so they share the same root:

$$\psi_{sa}(V_G) \;=\; \psi_0 + V_P(V_G) \qquad \text{exactly.} \tag{D.40}$$

The pinch-off voltage is the weak-inversion surface potential measured from ψ_0. Define the threshold voltage $V_{\text{th}} = V_{FB} + \psi_0 + \gamma\sqrt{\psi_0}$ (the value of V_G at which $V_P = 0$; it coincides with Eq. (D.25) for $\psi_0 = 2\phi_F$). Subtracting from Eq. (D.39),

$$V_G - V_{\text{th}} \;=\; V_P + \gamma(\sqrt{\psi_0 + V_P} - \sqrt{\psi_0}) \;=\; V_P\left(1 + \frac{\gamma}{2\sqrt{\psi_0}}\right) + \mathcal{O}\left(\frac{\gamma V_P^2}{\psi_0^{3/2}}\right), \tag{D.41}$$

so to first order

$$V_P \;\approx\; \frac{V_G - V_{\text{th}}}{n_0}, \qquad n_0 \;:=\; 1 + \frac{\gamma}{2\sqrt{\psi_0}}. \tag{D.42}$$

The exact inverse is $V_P = V_G - V_{\text{th}} - \gamma\big[\sqrt{V_G - V_{\text{th}} + (\sqrt{\psi_0} + \gamma/2)^2} - (\sqrt{\psi_0} + \gamma/2)\big]$, obtained by solving Eq. (D.39) as a quadratic in $\sqrt{\psi_0 + V_P}$.

Insert Eq. (D.40) into Eq. (D.37) with $\psi_0 = 2\phi_F$:

$$I_{\text{D}} \;=\; \frac{W}{L}\mu_n C_{\text{dep}} V_T^2\, e^{V_P/V_T}\Big[e^{-V_S/V_T} - e^{-V_D/V_T}\Big]. \tag{D.43}$$

Writing the prefactor as a specific current I_{spec} (its normalization is discussed in Remark D.2) and using Eq. (D.42),

$$\boxed{I_{\text{D}} \;=\; I_{\text{spec}}\, e^{V_P/V_T}\Big[e^{-V_S/V_T} - e^{-V_D/V_T}\Big] \;\approx\; I_0\, \exp\!\left(\frac{V_G - V_{\text{th}}}{nV_T}\right)\Big[e^{-V_S/V_T} - e^{-V_D/V_T}\Big],} \tag{D.44}$$

with $I_0 = I_{\text{spec}}$ and $n = n_0$. The second form is the EKV law of Chapter 12 in the notation of the main text. It is symmetric under exchange of source and drain, and the current is the difference of a **forward component** and a **reverse component**,

$$I_{\text{D}} = I_F - I_R, \qquad I_F \;=\; I_0\, e^{(V_G - V_{\text{th}})/(nV_T)}\, e^{-V_S/V_T}, \qquad I_R \;=\; I_0\, e^{(V_G - V_{\text{th}})/(nV_T)}\, e^{-V_D/V_T}. \tag{D.45}$$

For $V_{\text{DS}} = V_D - V_S \gtrsim 4V_T$ the reverse component is below 2% of the forward one and the transistor is saturated: $I_{\text{D}} \approx I_F$, independent of V_D in this long-channel model.

Remark D.2 (The Prefactor Depends on Where ψ_0 Is Placed)**.** Equation (D.43) with $\psi_0 = 2\phi_F$ gives $I_{\text{spec}} = \mu_n C_{\text{dep}}(W/L)V_T^2 = \mu_n(n-1)C_{\text{ox}}(W/L)V_T^2$. The EKV model instead uses $I_{\text{spec}} = 2n\mu_n C_{\text{ox}}(W/L)V_T^2$, which is the natural normalization for the interpolation below. The two agree if ψ_0 is chosen as $2\phi_F + V_T \ln[2n/(n-1)]$. For $n = 1.2$ this is $2\phi_F + 2.5V_T$, which is why EKV place ψ_0 a few thermal voltages above $2\phi_F$. Any change in ψ_0 shifts V_{th} and rescales I_{spec} by compensating amounts; the measured current is unchanged. In practice I_0 and V_{th} are fitted to data, and only the exponential dependences in Eq. (D.44) are predictions of the theory.

D.3.4 The Unified EKV Charge–Voltage Relation

Above threshold Q_{inv} is no longer small, and the surface potential begins to follow the channel voltage. To treat all levels of inversion together, linearize the charge balance around the pinch-off

point. Define $f(\psi) := \psi + \gamma\sqrt{\psi}$, so Eq. (D.23) reads $V_G - V_{FB} = f(\psi_s) - Q_{\rm inv}/C_{\rm ox}$ and Eq. (D.39) reads $V_G - V_{FB} = f(\psi_0 + V_P)$. Subtracting,

$$-\frac{Q_{\rm inv}}{C_{\rm ox}} \;=\; f(\psi_0 + V_P) - f(\psi_s) \;\approx\; f'(\psi_0 + V_P)\,(\psi_0 + V_P - \psi_s) \;=\; n\,(\psi_0 + V_P - \psi_s), \tag{D.46}$$

where $n = 1 + \gamma/(2\sqrt{\psi_0 + V_P})$ is the slope factor at pinch-off. Thus

$$\psi_s \;=\; \psi_0 + V_P + \frac{Q_{\rm inv}}{nC_{\rm ox}}. \tag{D.47}$$

In weak inversion $Q_{\rm inv} \to 0$ and $\psi_s \to \psi_{sa} = \psi_0 + V_P$, consistent with Eq. (D.40). In strong inversion the electrons screen the gate and ψ_s is pinned near $\psi_0 + V_{\rm ch}$; then Eq. (D.47) gives the familiar linear charge $Q_{\rm inv} = -nC_{\rm ox}(V_P - V_{\rm ch})$.

The second ingredient is the Boltzmann relation Eq. (D.28), which ties the electron sheet density to $\psi_s - V_{\rm ch}$. Write it as $Q_{\rm inv} = -Q_0\, e^{(\psi_s - \psi_0 - V_{\rm ch})/V_T}$ with a constant Q_0 to be fixed. Take logarithms and substitute Eq. (D.47):

$$V_T \ln\frac{-Q_{\rm inv}}{Q_0} \;=\; \psi_s - \psi_0 - V_{\rm ch} \;=\; V_P - V_{\rm ch} + \frac{Q_{\rm inv}}{nC_{\rm ox}}. \tag{D.48}$$

Introduce normalized variables

$$q \;:=\; \frac{-Q_{\rm inv}}{2nC_{\rm ox}V_T}, \qquad v_p \;:=\; \frac{V_P}{V_T}, \qquad v_{\rm ch} \;:=\; \frac{V_{\rm ch}}{V_T}, \tag{D.49}$$

and choose $Q_0 = 2nC_{\rm ox}V_T$. Dividing Eq. (D.48) by V_T, and using $Q_{\rm inv}/(nC_{\rm ox}V_T) = -2q$,

$$\boxed{\,v_p - v_{\rm ch} \;=\; 2q + \ln q.\,} \tag{D.50}$$

This is the **unified EKV charge–voltage relation**. The right side is strictly increasing in $q > 0$, so for each $v_p - v_{\rm ch}$ there is exactly one q. Its two asymptotes are

$$q \;\approx\; e^{\,v_p - v_{\rm ch}} \quad (v_p - v_{\rm ch} \ll -1), \qquad q \;\approx\; \frac{v_p - v_{\rm ch}}{2} \quad (v_p - v_{\rm ch} \gg 1). \tag{D.51}$$

In the first, $\ln q$ dominates and q is exponentially small (weak inversion). In the second, $2q$ dominates and the logarithm is a slowly varying correction (strong inversion).

D.3.5 The Current as a Difference of Forward and Reverse Terms

Now evaluate Eq. (D.36) using Eq. (D.50). Differentiating Eq. (D.50) at fixed v_p gives $\mathrm{d}v_{\rm ch} = -(2 + 1/q)\,\mathrm{d}q$. Changing variables from $V_{\rm ch}$ to q, with $-Q_{\rm inv} = 2nC_{\rm ox}V_T q$ and $\mathrm{d}V_{\rm ch} = V_T\,\mathrm{d}v_{\rm ch}$,

$$\begin{aligned}\int_{V_S}^{V_D} (-Q_{\rm inv})\,\mathrm{d}V_{\rm ch} &= 2nC_{\rm ox}V_T^2 \int_{q_s}^{q_d} q\left(-2 - \frac{1}{q}\right)\mathrm{d}q = 2nC_{\rm ox}V_T^2 \int_{q_d}^{q_s} (2q+1)\,\mathrm{d}q \\ &= 2nC_{\rm ox}V_T^2\Big[(q_s^2 + q_s) - (q_d^2 + q_d)\Big],\end{aligned} \tag{D.52}$$

where q_s and q_d are the normalized charges at the source and drain ends, each obtained from Eq. (D.50) with $v_{\rm ch} = V_S/V_T$ and V_D/V_T. Hence

$$I_{\rm D} \;=\; I_{\rm spec}\big[i_f - i_r\big], \qquad I_{\rm spec} \;:=\; 2n\mu_n C_{\rm ox}\frac{W}{L}V_T^2, \qquad i_f = q_s^2 + q_s, \quad i_r = q_d^2 + q_d. \tag{D.53}$$

The forward normalized current i_f depends only on $V_P - V_S$ and the reverse one i_r only on $V_P - V_D$, so the source–drain symmetry of Eq. (D.44) survives at all inversion levels. The ratio i_f is called the **inversion coefficient**; $i_f < 0.1$ is weak inversion, $0.1 < i_f < 10$ moderate inversion, and $i_f > 10$ strong inversion.

Weak-inversion limit. With $q \approx e^{v_p - v_{\mathrm{ch}}} \ll 1$, the quadratic term is negligible and $i_f \approx e^{(V_P - V_S)/V_T}$, $i_r \approx e^{(V_P - V_D)/V_T}$. This reproduces Eq. (D.44).

Strong-inversion limit. With $q \approx (v_p - v_{\mathrm{ch}})/2 \gg 1$, $i_f \approx (V_P - V_S)^2/(4V_T^2)$. In saturation ($i_r \to 0$),

$$I_{\mathrm{D}} \approx \frac{I_{\mathrm{spec}}}{4V_T^2}(V_P - V_S)^2 = \frac{n\mu_n C_{\mathrm{ox}}}{2}\frac{W}{L}(V_P - V_S)^2 \approx \frac{\mu_n C_{\mathrm{ox}}}{2n}\frac{W}{L}(V_G - V_{\mathrm{th}} - nV_S)^2, \tag{D.54}$$

the square law of strong inversion, with the body effect appearing through the factor nV_S.

D.3.6 The EKV Interpolation Function

Equation (D.50) has no elementary closed-form solution (its solution is a Lambert W function). The original EKV model replaces $q^2 + q$ by the interpolation

$$\boxed{i_f \approx \ln^2\!\left(1 + e^{(V_P - V_S)/(2V_T)}\right), \qquad i_r \approx \ln^2\!\left(1 + e^{(V_P - V_D)/(2V_T)}\right).} \tag{D.55}$$

It has the correct asymptotes. For $x := (V_P - V_S)/V_T \to -\infty$, $\ln(1 + e^{x/2}) = e^{x/2} - \frac{1}{2}e^x + \ldots$, so $i_f \to e^x$, matching weak inversion. For $x \to +\infty$, $\ln(1 + e^{x/2}) = x/2 + \ln(1 + e^{-x/2}) \to x/2$, so $i_f \to x^2/4$, matching strong inversion. At $x = 0$ the interpolation gives $\ln^2 2 = 0.48$, while the exact relation Eq. (D.50) gives $q = 0.43$ (the root of $2q + \ln q = 0$) and $q^2 + q = 0.61$. The interpolation is accurate to tens of percent in moderate inversion; later versions of the EKV model use the exact charge relation. The thermodynamic hardware of Chapters 12 to 14 works deep in weak inversion, where both forms reduce to Eq. (D.44).

Heuristic Derivation: Why the Current Is $q^2 + q$

The integrand of Eq. (D.52) is $2q + 1$, and the two terms have separate physical origins. In strong inversion the inversion charge varies along the channel because the surface potential does, so the electrons drift in a lateral field; this gives the q^2 term, the familiar quadratic dependence. In weak inversion the surface potential is flat and the inversion charge varies only through the Boltzmann factor e^{-V_{ch}/V_T}; this gives the linear q term, which is the Fick diffusion current of Eq. (D.38). The EKV model is the observation that the charge-sheet current is *exactly* the sum of these two pieces once the charge balance is linearized.

D.4 Current Noise of the Diffusive Channel

Section 12.5 uses the result that the white current noise of a subthreshold channel is the shot noise of its forward and reverse components added together. We give two derivations. The first is the bidirectional Poisson argument of Section 11.4 applied to the decomposition Eq. (D.45). The second integrates local thermal noise along the channel and does not presuppose any Poisson structure; it confirms the first. All spectral densities are single-sided, $S_I(f)$, in the convention of Chapter 11.

D.4.1 Forward and Reverse Fluxes as Independent Poisson Processes

In weak inversion the channel holds an exponentially small electron population, and electrons do not interact appreciably with one another; each moves by independent Brownian motion in a flat

potential between two absorbing reservoirs (source and drain). An electron injected from the source either returns to the source or reaches the drain. Only the latter contributes to the forward current. Because injection events are independent and rare on the scale of the transit time, the arrivals at the drain of source-injected electrons form a Poisson process with rate I_F/q_e. The same reasoning applied to drain-injected electrons gives a second Poisson process with rate I_R/q_e. The two processes are independent because they involve disjoint sets of electrons with no interaction.

The charge transferred from drain to source in a time window $\mathcal{T}$ is $Q(\mathcal{T}) = q_e[N_F(\mathcal{T}) - N_R(\mathcal{T})]$, where N_F and N_R are independent Poisson counts with means $I_F\mathcal{T}/q_e$ and $I_R\mathcal{T}/q_e$. For a Poisson count the variance equals the mean, and variances of independent variables add, so

$$\langle Q\rangle \;=\; (I_F - I_R)\,\mathcal{T}, \qquad \operatorname{Var} Q \;=\; q_e^2\left(\frac{I_F\mathcal{T}}{q_e} + \frac{I_R\mathcal{T}}{q_e}\right) \;=\; q_e(I_F + I_R)\,\mathcal{T}. \tag{D.56}$$

For a stationary current with white spectrum, $\operatorname{Var} Q = S_I^{\mathrm{ds}}\,\mathcal{T}$ at long times, where S_I^{ds} is the double-sided density at zero frequency (Section 11.1). The single-sided density is twice that:

$$\boxed{\;S_I \;=\; 2q_e(I_F + I_R).\;} \tag{D.57}$$

This is Theorem 11.10 with $I_+ = I_F$ and $I_- = I_R$. The spectrum is white up to frequencies of order D_n/L^2, the inverse of the diffusive transit time, which for micron-scale channels lies in the gigahertz range.

The independence assumption is plausible but deserves a check, since electrons in the channel are screened by the gate and are not completely isolated. The next derivation makes no such assumption.

D.4.2 Klaassen–Prins Integration of Local Thermal Noise

Divide the channel into slices of length Δy. Each slice is a short conductor in local equilibrium at the local quasi-Fermi potential, so by the Johnson–Nyquist theorem (Theorem 11.5) it carries a thermal noise voltage δv with single-sided density $4k_{\mathrm{B}}T\,\Delta R(y)$, where the slice resistance is

$$\Delta R(y) \;=\; \frac{\Delta y}{W\mu_n(-Q_{\mathrm{inv}}(y))}. \tag{D.58}$$

Noise in different slices is uncorrelated.

Step 1: response of the terminal current to a single slice. Insert a small series voltage δv at position y_0, with the terminals held at fixed V_S and V_D. The channel voltage then jumps by δv at y_0: $V_{\mathrm{ch}}(y_0^+) = V_{\mathrm{ch}}(y_0^-) + \delta v$. The current $I_{\mathrm{D}} + \delta I$ is still uniform in y. Applying the separated form of Eq. (D.35) on each side of y_0, with $G(V) := \mu_n(-Q_{\mathrm{inv}}(V))$,

$$(I_{\mathrm{D}} + \delta I)\,y_0 \;=\; W\int_{V_S}^{V_{\mathrm{ch}}(y_0^-)} G(V)\,\mathrm{d}V, \qquad (I_{\mathrm{D}} + \delta I)(L - y_0) \;=\; W\int_{V_{\mathrm{ch}}(y_0^-)+\delta v}^{V_D} G(V)\,\mathrm{d}V. \tag{D.59}$$

Adding the two equations,

$$(I_{\mathrm{D}} + \delta I)\,L \;=\; W\int_{V_S}^{V_D} G(V)\,\mathrm{d}V - W\,G(V_{\mathrm{ch}}(y_0))\,\delta v + \mathcal{O}(\delta v^2). \tag{D.60}$$

The first term on the right is $I_{\mathrm{D}}L$ by Eq. (D.36). Therefore

$$\delta I \;=\; -\frac{W\,G(y_0)}{L}\,\delta v \;=\; -\frac{\Delta y}{L}\,\frac{\delta v}{\Delta R(y_0)}, \tag{D.61}$$

using $WG(y_0) = \Delta y/\Delta R(y_0)$ from Eq. (D.58). A fluctuation in the slice is diluted by the fraction $\Delta y/L$ of the channel it occupies.

Step 2: sum over slices. Since slices are independent, their contributions to S_I add:

$$S_I \;=\; \sum_{\text{slices}} \left(\frac{\Delta y}{L}\right)^2 \frac{4k_{\mathrm{B}}T\,\Delta R}{\Delta R^2} \;=\; \sum_{\text{slices}} \frac{4k_{\mathrm{B}}T}{L^2}\, W\mu_n(-Q_{\mathrm{inv}})\,\Delta y \;\longrightarrow\; \frac{4k_{\mathrm{B}}T\,W\mu_n}{L^2}\int_0^L (-Q_{\mathrm{inv}}(y))\;\mathrm{d}y. \tag{D.62}$$

This is the **Klaassen–Prins formula**: the channel noise is $4k_{\mathrm{B}}T$ times the total inversion charge, times $W\mu_n/L^2$.

Step 3: change variables to the channel voltage. From Eq. (D.35), $\mathrm{d}y = W\mu_n(-Q_{\mathrm{inv}})\;\mathrm{d}V_{\mathrm{ch}}/I_{\mathrm{D}}$, so

$$S_I \;=\; \frac{4k_{\mathrm{B}}T\,W^2\mu_n^2}{L^2\,I_{\mathrm{D}}}\int_{V_S}^{V_D} Q_{\mathrm{inv}}^2\;\mathrm{d}V_{\mathrm{ch}}. \tag{D.63}$$

Step 4: evaluate in weak inversion. Using Eq. (D.28) with $\psi_s = \psi_{sa}$ and abbreviating $A := C_{\mathrm{dep}}V_T e^{(\psi_{sa}-2\phi_F)/V_T}$, we have $-Q_{\mathrm{inv}} = Ae^{-V_{\mathrm{ch}}/V_T}$. The integral is

$$\int_{V_S}^{V_D} A^2 e^{-2V_{\mathrm{ch}}/V_T}\,\mathrm{d}V_{\mathrm{ch}} \;=\; \frac{A^2V_T}{2}\Big[e^{-2V_S/V_T} - e^{-2V_D/V_T}\Big], \tag{D.64}$$

and from Eq. (D.37) the current is $I_{\mathrm{D}} = (W/L)\mu_n AV_T\,[e^{-V_S/V_T} - e^{-V_D/V_T}]$. Writing $a = e^{-V_S/V_T}$ and $b = e^{-V_D/V_T}$, and using $a^2 - b^2 = (a-b)(a+b)$,

$$S_I \;=\; \frac{4k_{\mathrm{B}}T\,W^2\mu_n^2}{L^2}\cdot\frac{A^2V_T(a^2-b^2)/2}{(W/L)\mu_n AV_T(a-b)} \;=\; 2k_{\mathrm{B}}T\,\frac{W}{L}\mu_n A\,(a+b) \;=\; 2q_e\left[\frac{W}{L}\mu_n AV_T\,a + \frac{W}{L}\mu_n AV_T\,b\right], \tag{D.65}$$

where the last step uses $k_{\mathrm{B}}T = q_eV_T$. The two terms in brackets are exactly I_F and I_R of Eq. (D.45). So

$$S_I \;=\; 2q_e(I_F + I_R), \tag{D.66}$$

in agreement with Eq. (D.57). Local thermal equilibrium of each slice, integrated over a channel in which the carrier density falls exponentially from source to drain, reproduces full shot noise of the two independent fluxes.

D.4.3 Equilibrium and Saturation Limits

Using Eq. (D.45), $I_R/I_F = e^{-V_{\mathrm{DS}}/V_T}$. Hence

$$I_{\mathrm{D}} \;=\; I_F(1 - e^{-V_{\mathrm{DS}}/V_T}), \qquad S_I \;=\; 2q_eI_F(1 + e^{-V_{\mathrm{DS}}/V_T}) \;=\; 2q_eI_{\mathrm{D}}\coth\!\left(\frac{V_{\mathrm{DS}}}{2V_T}\right), \tag{D.67}$$

the crossover formula Eq. (11.83) with $V = V_{\mathrm{DS}}$. Two limits follow.

Saturation, $V_{\mathrm{DS}} \gg V_T$. $\coth \to 1$, so $S_I = 2q_eI_{\mathrm{D}}$: full Schottky shot noise of the drain current.

Equilibrium, $V_{\mathrm{DS}} = 0$. The mean current vanishes and $I_F = I_R$, so $S_I = 4q_eI_F$. The channel conductance at zero bias is obtained by differentiating Eq. (D.44) with respect to V_D:

$$g_{\mathrm{ch}} \;:=\; \left.\frac{\partial I_{\mathrm{D}}}{\partial V_D}\right|_{V_D=V_S} \;=\; \left.\frac{I_R}{V_T}\right|_{V_D=V_S} \;=\; \frac{I_F}{V_T}. \tag{D.68}$$

Therefore

$$S_I\big|_{V_{\mathrm{DS}}=0} \;=\; 4q_e I_F \;=\; 4q_e V_T\, g_{\mathrm{ch}} \;=\; 4k_{\mathrm{B}}T\, g_{\mathrm{ch}}, \tag{D.69}$$

the Johnson–Nyquist value of Section 11.2. The subthreshold transistor at zero drain bias is an ordinary equilibrium resistor, as it must be: the fluctuation–dissipation theorem fixes its noise. Away from $V_{\mathrm{DS}} = 0$ the channel is driven, and the noise exceeds $4k_{\mathrm{B}}T\, g_{\mathrm{ds}}$ (with $g_{\mathrm{ds}} = I_R/V_T$ the actual output conductance) by the factor $(I_F + I_R)/(2I_R) = (1 + e^{V_{\mathrm{DS}}/V_T})/2$, which grows exponentially with bias. This excess is the non-equilibrium noise that Section 13.3 identifies as one source of broken detailed balance in active CMOS circuits.

Physics Insight: Why Shot Noise Is Not Suppressed in a Diffusive Channel

In a metallic diffusive wire, shot noise is reduced to one third of the Poisson value because the Pauli principle correlates electrons. The subthreshold channel avoids this suppression because the electron gas is nondegenerate (Section D.1): occupation numbers are far below one, Pauli blocking is negligible, the nearby gate screens the Coulomb interaction that could otherwise correlate successive carriers, and the electrons behave as independent classical particles. The Klaassen–Prins calculation shows the same thing from the classical side: local Johnson noise plus an exponential carrier profile integrate to the full $2q_e(I_F + I_R)$. In strong inversion the same integral Eq. (D.63) gives a different result, $S_I = \frac{8}{3}k_{\mathrm{B}}T\, n g_{\mathrm{m}}$ in saturation (the long-channel factor 2/3 multiplying $4k_{\mathrm{B}}T$ times the source transconductance $n g_{\mathrm{m}}$), because there the carrier profile is set by drift.

D.5 The RCSJ Model of a Josephson Junction

This section is the reference for Section 14.4. We take as given the two Josephson relations derived in Section 14.3,

$$I_s \;=\; I_{\mathrm{c}} \sin\varphi, \qquad V \;=\; \frac{\Phi_0}{2\pi}\,\frac{\mathrm{d}\varphi}{\mathrm{d}t}, \qquad \Phi_0 = \frac{h}{2q_e}, \tag{D.70}$$

where φ is the gauge-invariant phase difference across the junction.

D.5.1 Equation of Motion

A real junction has, in parallel with the supercurrent channel, a geometric capacitance C (the two electrodes separated by a thin barrier) and a dissipative channel, modelled by a linear resistance R (quasiparticle tunnelling or an external shunt). The **resistively and capacitively shunted junction** (RCSJ) model connects these three elements in parallel and drives them with a bias current I. The resistor also carries a Johnson–Nyquist noise current $I_n(t)$. Kirchhoff's current law gives

$$C\,\frac{\mathrm{d}V}{\mathrm{d}t} + \frac{V}{R} + I_{\mathrm{c}}\sin\varphi \;=\; I + I_n(t). \tag{D.71}$$

Substituting the voltage–phase relation,

$$C\frac{\Phi_0}{2\pi}\frac{\mathrm{d}^2\varphi}{\mathrm{d}t^2} + \frac{1}{R}\frac{\Phi_0}{2\pi}\frac{\mathrm{d}\varphi}{\mathrm{d}t} + I_{\mathrm{c}}\sin\varphi \;=\; I + I_n(t). \tag{D.72}$$

Multiplying through by $\Phi_0/(2\pi)$ turns every term into a torque (energy per radian):

$$m\,\ddot{\varphi} + m\gamma_R\,\dot{\varphi} + \frac{\partial U}{\partial \varphi} \;=\; \frac{\Phi_0}{2\pi} I_n(t), \tag{D.73}$$

with

$$m \;=\; C\left(\frac{\Phi_0}{2\pi}\right)^2, \qquad \gamma_R \;=\; \frac{1}{RC}, \qquad U(\varphi) \;=\; -E_{\mathrm{J}}\cos\varphi - \frac{\Phi_0}{2\pi} I\,\varphi, \qquad E_{\mathrm{J}} \;=\; \frac{\Phi_0 I_{\mathrm{c}}}{2\pi}. \tag{D.74}$$

This is Newton's equation for a particle of effective mass m (in units of $\mathrm{J\,s^2}$, since φ is dimensionless) moving in the potential $U(\varphi)$ with friction rate γ_R. The potential is the **tilted washboard**: a cosine of depth $2E_{\mathrm{J}}$ superposed on a linear tilt proportional to the bias. A particle at rest in a well is the zero-voltage (superconducting) state; a particle sliding down the washboard has $\langle\dot\varphi\rangle \neq 0$ and therefore a dc voltage $\langle V\rangle = (\Phi_0/2\pi)\,\langle\dot\varphi\rangle$.

Noise. The resistor's Norton noise current has single-sided density $S_I = 4k_{\mathrm{B}}T/R$ (Theorem 11.5), equivalently $\langle I_n(t)I_n(t')\rangle = (2k_{\mathrm{B}}T/R)\,\delta(t-t')$. The torque noise $\xi = (\Phi_0/2\pi)I_n$ then has $\langle\xi(t)\xi(t')\rangle = 2k_{\mathrm{B}}T\,m\gamma_R\,\delta(t-t')$, since $(\Phi_0/2\pi)^2/R = m\gamma_R$. This is the fluctuation–dissipation relation for an underdamped Langevin particle of mass m and friction $m\gamma_R$, so the RCSJ model at zero bias relaxes to the Boltzmann distribution $\propto e^{-[m\dot\varphi^2/2+U(\varphi)]/k_{\mathrm{B}}T}$. The junction is a direct physical realization of the underdamped Langevin dynamics treated in Chapter 7.

D.5.2 Dimensionless Form, Plasma Frequency, and Stewart–McCumber Parameter

Small oscillations at the bottom of an untilted well ($I = 0$) have $U \approx -E_{\mathrm{J}} + E_{\mathrm{J}}\varphi^2/2$, and so angular frequency

$$\omega_p \;=\; \sqrt{\frac{E_{\mathrm{J}}}{m}} \;=\; \sqrt{\frac{\Phi_0 I_{\mathrm{c}}/(2\pi)}{C(\Phi_0/2\pi)^2}} \;=\; \sqrt{\frac{2\pi I_{\mathrm{c}}}{\Phi_0 C}}, \tag{D.75}$$

the **Josephson plasma frequency**. Equivalently, $\omega_p = 1/\sqrt{L_J C}$ with the Josephson inductance $L_J = \Phi_0/(2\pi I_{\mathrm{c}})$. Introduce the dimensionless time $\tau = \omega_p t$, the normalized bias $i = I/I_{\mathrm{c}}$, and normalized noise $i_n = I_n/I_{\mathrm{c}}$. Dividing Eq. (D.72) by I_{c} and using $C\Phi_0\omega_p^2/(2\pi I_{\mathrm{c}}) = 1$ and $\Phi_0\omega_p/(2\pi I_{\mathrm{c}}R) = 1/(\omega_p RC)$,

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}\tau^2} + \frac{1}{Q}\frac{\mathrm{d}\varphi}{\mathrm{d}\tau} + \sin\varphi \;=\; i + i_n(\tau), \qquad Q \;:=\; \omega_p RC. \tag{D.76}$$

The quality factor Q of the plasma oscillation determines the damping regime. Its square is the **Stewart–McCumber parameter**:

$$\beta_{\mathrm{c}} \;:=\; Q^2 \;=\; \omega_p^2 R^2 C^2 \;=\; \frac{2\pi I_{\mathrm{c}} R^2 C}{\Phi_0}. \tag{D.77}$$

For $\beta_{\mathrm{c}} \ll 1$ the junction is overdamped: the inertial term is negligible and Eq. (D.76) reduces to the first-order (overdamped Langevin) equation $Q^{-1}\dot\varphi = i - \sin\varphi + i_n$. For $\beta_{\mathrm{c}} \gg 1$ the junction is underdamped and its current–voltage curve is hysteretic. Superconducting digital and parametron circuits usually add an external shunt resistor so that $\beta_{\mathrm{c}} \lesssim 1$.

D.5.3 Stationary Points of the Washboard and the Biased Plasma Frequency

For $|i| < 1$ the washboard $U(\varphi)/E_{\mathrm{J}} = -\cos\varphi - i\varphi$ has stationary points where $\sin\varphi = i$. In each period there is a minimum at $\varphi_{\min} = \arcsin i$ and a maximum at $\varphi_{\max} = \pi - \arcsin i$. The curvature

at the minimum is $U''(\varphi_{\min}) = E_{\mathrm{J}}\cos\varphi_{\min} = E_{\mathrm{J}}\sqrt{1-i^2}$, so the small-oscillation frequency under bias is

$$\omega_p(I) \;=\; \sqrt{\frac{U''(\varphi_{\min})}{m}} \;=\; \omega_p\left(1-\frac{I^2}{I_{\mathrm{c}}^2}\right)^{1/4}. \tag{D.78}$$

The well softens as the bias approaches I_{c}, and $\omega_p(I)\to 0$ at $I = I_{\mathrm{c}}$, where minimum and maximum merge in a saddle–node bifurcation. For $|i| > 1$ there are no stationary points and the phase runs.

D.5.4 Barrier Height and Its Behaviour Near the Critical Current

The barrier for escape from a well is

$$\begin{aligned}\Delta U(I) &= U(\varphi_{\max}) - U(\varphi_{\min}) = -E_{\mathrm{J}}\Big[\cos\varphi_{\max} - \cos\varphi_{\min} + i(\varphi_{\max}-\varphi_{\min})\Big] \\ &= -E_{\mathrm{J}}\Big[-2\sqrt{1-i^2} + i(\pi - 2\arcsin i)\Big] = 2E_{\mathrm{J}}\Big[\sqrt{1-i^2} - i\arccos i\Big],\end{aligned} \tag{D.79}$$

where we used $\cos(\pi-\alpha) = -\cos\alpha$ and $\pi - 2\arcsin i = 2\arccos i$. At zero bias $\Delta U = 2E_{\mathrm{J}}$.

Near the critical current, set $\epsilon = 1 - i \ll 1$. We need the expansions $\sqrt{1-i^2} = \sqrt{\epsilon(2-\epsilon)} = \sqrt{2\epsilon}\,(1-\epsilon/4+\mathcal{O}(\epsilon^2))$ and $\arccos(1-\epsilon) = \sqrt{2\epsilon}\,(1+\epsilon/12+\mathcal{O}(\epsilon^2))$. The second follows by writing $\theta = \arccos(1-\epsilon)$, so $1-\theta^2/2+\theta^4/24 = 1-\epsilon$; inserting $\theta = \sqrt{2\epsilon}(1+a\epsilon)$ gives $\epsilon + 2a\epsilon^2 - \epsilon^2/6 = \epsilon$, hence $a = 1/12$. Then

$$\begin{aligned}\sqrt{1-i^2} - i\arccos i &= \sqrt{2\epsilon}\Big[1 - \tfrac{\epsilon}{4} - (1-\epsilon)\big(1+\tfrac{\epsilon}{12}\big)\Big] + \mathcal{O}(\epsilon^{5/2}) \\ &= \sqrt{2\epsilon}\Big[-\tfrac{\epsilon}{4} - \tfrac{\epsilon}{12} + \epsilon\Big] + \mathcal{O}(\epsilon^{5/2}) = \tfrac{2}{3}\sqrt{2}\,\epsilon^{3/2} + \mathcal{O}(\epsilon^{5/2}).\end{aligned} \tag{D.80}$$

Multiplying by $2E_{\mathrm{J}}$,

$$\boxed{\;\Delta U(I) \;\approx\; \frac{4\sqrt{2}}{3}\,E_{\mathrm{J}}\left(1-\frac{I}{I_{\mathrm{c}}}\right)^{3/2}.\;} \tag{D.81}$$

The same result follows from the cubic approximation of the washboard. Near the inflection point $\varphi = \pi/2$, write $\varphi = \pi/2 + x$; then $\cos\varphi = -\sin x = -x + x^3/6 + \ldots$ and $U/E_{\mathrm{J}} = \text{const} + (1-i)x - x^3/6$. The extrema are at $x = \mp\sqrt{2\epsilon}$, and the difference of $\epsilon x - x^3/6$ between them is $2\epsilon\sqrt{2\epsilon} - (2\epsilon)^{3/2}/3 = \frac{4\sqrt{2}}{3}\epsilon^{3/2}$. The cubic form is the universal shape of any potential near a saddle–node bifurcation, which is why the $3/2$ exponent also appears for other bistable devices.

Thermal escape from the well over $\Delta U(I)$ is a Kramers problem (Chapter 7); in the underdamped junction the attempt frequency is $\omega_p(I)/(2\pi)$ times a damping-dependent factor. Below the crossover temperature $T_{\mathrm{cr}} = \hbar\omega_p(I)/(2\pi k_{\mathrm{B}})$, macroscopic quantum tunnelling dominates (Section 14.5).

D.5.5 Retrapping Current of an Underdamped Junction

In an underdamped junction ($\beta_{\mathrm{c}} \gg 1$) the phase, once running, keeps running even when the bias is lowered below I_{c}, because the particle's kinetic energy carries it over successive maxima. It is retrapped only at a lower current I_r, where the energy fed in by the bias over one period no longer covers the energy dissipated in R. This produces the hysteretic current–voltage curve.

To find I_r for $\beta_{\mathrm{c}} \gg 1$, note that at the retrapping point the bias i_r is small, so the damping and tilt are both weak perturbations of the free pendulum. The marginal running trajectory is then close to the separatrix of the untilted, undamped pendulum, the trajectory that just reaches the maxima

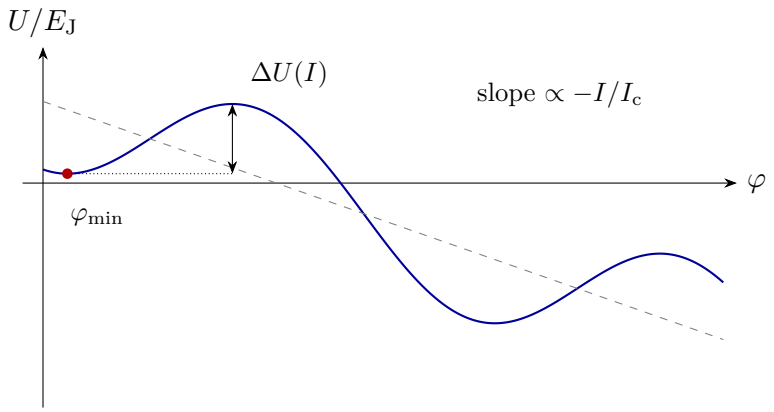


Figure D.1: The tilted washboard $U(\varphi)/E_{\mathrm{J}} = -\cos\varphi - i\varphi$ (shifted vertically) at $i = I/I_{\mathrm{c}} = 0.35$. The phase particle rests in a well (red dot) in the zero-voltage state; escape over the barrier $\Delta U(I)$ switches the junction into the running (voltage) state.

at $\varphi = \pm\pi$ with zero velocity. Energy conservation on the separatrix gives $\tfrac{1}{2}m\dot\varphi^2 - E_{\mathrm{J}}\cos\varphi = E_{\mathrm{J}}$, so

$$\dot\varphi^2 \;=\; \frac{2E_{\mathrm{J}}}{m}(1+\cos\varphi) \;=\; 4\omega_p^2\cos^2\frac{\varphi}{2}, \qquad \dot\varphi \;=\; 2\omega_p\cos\frac{\varphi}{2} \quad (|\varphi|<\pi). \tag{D.82}$$

Energy dissipated per period. The power dissipated in the resistor is $V^2/R = (\Phi_0/2\pi)^2\dot\varphi^2/R$. Over one passage from $\varphi = -\pi$ to $\varphi = \pi$, change the integration variable from t to φ with $\mathrm{d}t = \mathrm{d}\varphi/\dot\varphi$:

$$W_{\mathrm{diss}} \;=\; \frac{1}{R}\left(\frac{\Phi_0}{2\pi}\right)^2\int_{-\pi}^{\pi}\dot\varphi\;\mathrm{d}\varphi \;=\; \frac{1}{R}\left(\frac{\Phi_0}{2\pi}\right)^2 2\omega_p\int_{-\pi}^{\pi}\cos\frac{\varphi}{2}\;\mathrm{d}\varphi \;=\; \frac{8\omega_p}{R}\left(\frac{\Phi_0}{2\pi}\right)^2, \tag{D.83}$$

since $\int_{-\pi}^{\pi}\cos(\varphi/2)\,\mathrm{d}\varphi = [2\sin(\varphi/2)]_{-\pi}^{\pi} = 4$.

Energy supplied per period. The bias source delivers $\int IV\;\mathrm{d}t = I(\Phi_0/2\pi)\int\dot\varphi\;\mathrm{d}t = I(\Phi_0/2\pi)\cdot 2\pi = I\Phi_0$ per 2π advance of the phase.

Balance. Setting $I_r\Phi_0 = W_{\mathrm{diss}}$,

$$I_r \;=\; \frac{8\omega_p}{R\Phi_0}\left(\frac{\Phi_0}{2\pi}\right)^2 \;=\; \frac{2\omega_p\Phi_0}{\pi^2 R}. \tag{D.84}$$

Divide by I_{c} and use $\Phi_0/I_{\mathrm{c}} = 2\pi/(\omega_p^2 C)$ from Eq. (D.75):

$$\boxed{\frac{I_r}{I_{\mathrm{c}}} \;=\; \frac{2\omega_p}{\pi^2 R}\cdot\frac{2\pi}{\omega_p^2 C} \;=\; \frac{4}{\pi\,\omega_p RC} \;=\; \frac{4}{\pi\sqrt{\beta_{\mathrm{c}}}} \qquad (\beta_{\mathrm{c}} \gg 1).} \tag{D.85}$$

The approximation requires $I_r \ll I_{\mathrm{c}}$, that is $\beta_{\mathrm{c}} \gg (4/\pi)^2 \approx 1.6$. For $\beta_{\mathrm{c}} \lesssim 1$ the hysteresis disappears and the current–voltage curve is single-valued, with $\langle V\rangle = R\sqrt{I^2 - I_{\mathrm{c}}^2}$ in the overdamped limit $\beta_{\mathrm{c}} \to 0$ (obtained by integrating $Q^{-1}\dot\varphi = i - \sin\varphi$ over one period).

D.6 The rf-SQUID Potential and the Parametron Double Well

D.6.1 Fluxoid Quantization in a Loop with One Junction

Consider a superconducting loop of geometric inductance L interrupted by a single Josephson junction, threaded by an externally applied flux Φ_x. Inside the superconductor, far from the surface,

the supercurrent density vanishes, and the line integral of the gradient of the condensate phase around the loop must be a multiple of 2π. Combined with the gauge-invariant phase difference across the junction, this gives the fluxoid condition (Section 14.3)

$$\varphi \;=\; 2\pi\frac{\Phi}{\Phi_0} \pmod{2\pi}, \tag{D.86}$$

where Φ is the total flux through the loop and the sign of φ is chosen to make the relation positive. The total flux is the applied flux plus the self-flux of the circulating current J: $\Phi = \Phi_x + LJ$. The circulating current flows through the junction, so $J = -I_c \sin\varphi$, with the sign fixed by the orientation chosen in Eq. (D.86). Define the reduced flux variables

$$\varphi_x \;:=\; 2\pi\frac{\Phi_x}{\Phi_0}, \qquad E_L \;:=\; \frac{1}{L}\left(\frac{\Phi_0}{2\pi}\right)^2, \qquad \beta_L \;:=\; \frac{2\pi L I_c}{\Phi_0} \;=\; \frac{E_J}{E_L}. \tag{D.87}$$

Multiplying $\Phi = \Phi_x - LI_c\sin\varphi$ by $2\pi/\Phi_0$ gives the stationarity condition

$$\varphi - \varphi_x + \beta_L \sin\varphi \;=\; 0. \tag{D.88}$$

D.6.2 The Potential Energy

The energy stored in the loop has two parts. The inductor stores $\frac{1}{2}LJ^2 = (\Phi-\Phi_x)^2/(2L)$. The junction stores $\int I_s V\,\mathrm{d}t = \int I_c\sin\varphi\,(\Phi_0/2\pi)\,\mathrm{d}\varphi = -E_J\cos\varphi + \text{const}$. Expressing the inductive part in terms of φ via Eq. (D.86),

$$\boxed{U(\varphi) \;=\; \frac{E_L}{2}(\varphi-\varphi_x)^2 - E_J\cos\varphi \;=\; E_L\left[\frac{(\varphi-\varphi_x)^2}{2} - \beta_L\cos\varphi\right].} \tag{D.89}$$

Its derivative, $U'(\varphi) = E_L[\varphi - \varphi_x + \beta_L\sin\varphi]$, vanishes exactly at Eq. (D.88), which confirms that Eq. (D.89) is the correct potential. Adding the junction capacitance and a shunt resistance, the phase obeys the RCSJ equation Eq. (D.73) with the washboard replaced by Eq. (D.89): the external flux now enters through a parabola instead of a linear tilt. This is the potential quoted in Section 14.4.

D.6.3 The Bistability Condition

The number of minima of U is controlled by the curvature

$$U''(\varphi) \;=\; E_L\bigl(1 + \beta_L\cos\varphi\bigr). \tag{D.90}$$

Proposition D.3 (Monostable vs. Bistable rf-SQUID). *(i) If $\beta_L < 1$, then $U'' > 0$ everywhere, U is strictly convex, and it has exactly one minimum for every φ_x.*

(ii) If $\beta_L > 1$ and $\varphi_x = \pi$, then $\varphi = \pi$ is a local maximum and U has two degenerate minima at $\varphi = \pi \pm x_0$, where $x_0 > 0$ solves $x_0 = \beta_L \sin x_0$.

(iii) For $\beta_L > 1$, two minima coexist for all φ_x in the window $|\varphi_x - \pi| < h_c$, with

$$h_c \;=\; \sqrt{\beta_L^2 - 1} - \arccos\frac{1}{\beta_L}. \tag{D.91}$$

Proof. (i) $1+\beta_L\cos\varphi \geq 1-\beta_L > 0$. A strictly convex function that grows quadratically at infinity has exactly one minimum.

(ii) At $\varphi_x=\pi$, $U''(\pi)=E_L(1-\beta_L)<0$, so π is a maximum. Writing $\varphi=\pi+x$, Eq. (D.88) becomes $x+\beta_L\sin(\pi+x)=x-\beta_L\sin x=0$. The function $x-\beta_L\sin x$ has slope $1-\beta_L<0$ at $x=0$ and is positive for $x>\beta_L$, so it has a root $x_0\in(0,\beta_L]$, and by the symmetry $x\to -x$ a root at $-x_0$. These are minima, and U is even in x at $\varphi_x=\pi$, so they are degenerate.

(iii) As φ_x moves away from π, one minimum disappears when it merges with the maximum, which happens when $U'=U''=0$ simultaneously. From $U''=0$, $\cos\varphi=-1/\beta_L$; take $\varphi^*=\pi-\alpha$ with $\alpha=\arccos(1/\beta_L)$, so $\sin\varphi^*=\sin\alpha=\sqrt{1-\beta_L^{-2}}$. Then Eq. (D.88) gives $\varphi_x=\varphi^*+\beta_L\sin\varphi^*=\pi-\alpha+\sqrt{\beta_L^2-1}$. The other edge follows by symmetry. □

Outside the window, the loop holds a unique flux state. Inside it, the two minima correspond to clockwise and anticlockwise circulating currents, which differ by roughly one flux quantum for $\beta_L\gg 1$ and by a small amount for β_L slightly above 1.

D.6.4 Quartic Normal Form Near the Bifurcation

The quantum flux parametron (QFP) operates close to $\beta_L=1$, where the potential changes from one well to two. In a QFP the single junction is effectively replaced by a pair of junctions whose combined critical current is tuned by a second, excitation flux. In the simplest version, the compound rf-SQUID, a dc-SQUID (a small superconducting loop interrupted by two junctions in parallel) with identical junctions and negligible loop inductance acts as one junction with $E_{\mathrm{J}}^{\mathrm{eff}}=2E_{\mathrm{J}}\,|\cos(\pi\Phi_e/\Phi_0)|$, so the excitation flux Φ_e sweeps β_L through 1. The input signal enters as a small offset of φ_x from π.

Set $\varphi=\pi+x$ and $\varphi_x=\pi+\tilde{h}$, with $|x|$ and $\left|\tilde{h}\right|$ small. Using $\cos(\pi+x)=-\cos x=-1+x^2/2-x^4/24+\mathcal{O}(x^6)$,

$$\begin{aligned}\frac{U}{E_L} &= \frac{(x-\tilde{h})^2}{2}+\beta_L\left(1-\frac{x^2}{2}+\frac{x^4}{24}\right)+\mathcal{O}(x^6)\\ &= \text{const} \;+\; \frac{\beta_L}{24}x^4 \;-\; \frac{\beta_L-1}{2}x^2 \;-\; \tilde{h}\,x \;+\; \mathcal{O}(x^6),\end{aligned} \tag{D.92}$$

where the constant $\beta_L+\tilde{h}^2/2$ does not depend on x. This is the quartic potential of the double-well sampler, Eq. (A.3),

$$E(x) \;=\; \frac{a}{4}x^4-\frac{b}{2}x^2-hx, \qquad a \;=\; \frac{\beta_L E_L}{6}=\frac{E_{\mathrm{J}}}{6}, \qquad b \;=\; (\beta_L-1)E_L=E_{\mathrm{J}}-E_L, \qquad h \;=\; E_L\,(\varphi_x-\pi). \tag{D.93}$$

For $h=0$ and $\beta_L>1$, the minima lie at $x=\pm\sqrt{b/a}=\pm\sqrt{6(\beta_L-1)/\beta_L}$ and the barrier is

$$\Delta E \;=\; \frac{b^2}{4a} \;=\; \frac{3}{2}\,\frac{(\beta_L-1)^2}{\beta_L}\,E_L. \tag{D.94}$$

The quartic form is accurate when the wells are close together, $\sqrt{6(\beta_L-1)}\ll 1$, i.e. $\beta_L-1\lesssim 0.05$; for larger β_L-1 the full cosine must be kept. As a consistency check, the bistability window of the quartic, $|h|<2b^{3/2}/(3\sqrt{3a})$, becomes $|\varphi_x-\pi|<\frac{2\sqrt{2}}{3}(\beta_L-1)^{3/2}/\sqrt{\beta_L}$, which agrees to leading order with the expansion of Eq. (D.91) for $\beta_L=1+\delta$: $\sqrt{2\delta}(1+\delta/4)-\sqrt{2\delta}(1-5\delta/12)=\frac{2\sqrt{2}}{3}\delta^{3/2}$.

The QFP computes by sweeping b from negative (one well) to positive (two wells) while a small input h is applied. The particle follows the single minimum at $x\approx h/\,|b|$ while $b<0$; as b

crosses zero the well splits and the particle is carried into the well on the side of $\operatorname{sgn}(h)$. When $|h|$ is comparable to the thermal force scale, the outcome is random with probability set by the Boltzmann weights near the bifurcation, which is the regime used for probabilistic sampling in Chapter 14. The rate of thermal hopping between wells after the split is the Kramers rate over Eq. (D.94) (Chapter 7).

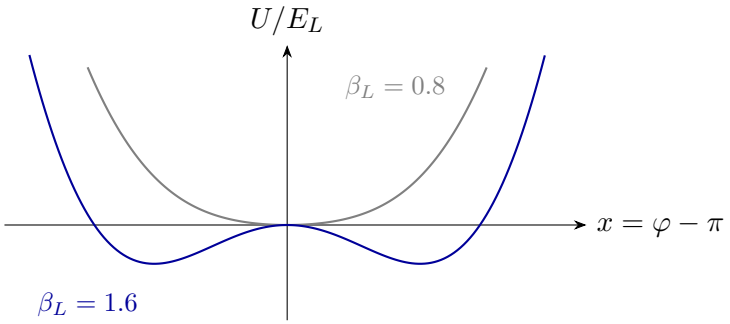


Figure D.2: The rf-SQUID potential Eq. (D.89) at $\varphi_x = \pi$, shifted so that $U(\pi) = 0$. For $\beta_L < 1$ the potential has a single well (grey); for $\beta_L > 1$ it has two degenerate wells (blue), the clockwise and anticlockwise flux states.

Example D.4 (Josephson and Parametron Numbers at 4.2 K)**.** Take a representative niobium junction with critical-current density $J_c = 10\,\mu\text{A}/\mu\text{m}^2$ ($1\,\text{kA}/\text{cm}^2$), area $5\,\mu\text{m}^2$, and specific capacitance $50\,\text{fF}/\mu\text{m}^2$, so $I_\text{c} = 50\,\mu\text{A}$ and $C = 250\,\text{fF}$. These are order-of-magnitude values, not those of a particular process.

- $E_\text{J} = \Phi_0 I_\text{c}/(2\pi) = 3.29 \times 10^{-16}\,\text{Wb} \times 50\,\mu\text{A} = 1.65 \times 10^{-20}\,\text{J}$, so $E_\text{J}/k_\text{B}T = 284$ at 4.2 K. The Josephson inductance is $L_J = \Phi_0/(2\pi I_\text{c}) = 6.6\,\text{pH}$.
- $\omega_p = \sqrt{2\pi I_\text{c}/(\Phi_0 C)} = 7.8 \times 10^{11}\,\text{s}^{-1}$, i.e. $\omega_p/2\pi = 124\,\text{GHz}$. Because I_c and C both scale with area, ω_p depends only on J_c and the specific capacitance.
- The quantum crossover temperature is $T_\text{cr} = \hbar\omega_p/(2\pi k_\text{B}) = 0.95\,\text{K}$, below 4.2 K, so escape is thermal.
- With a shunt $R = 2\,\Omega$, $\beta_\text{c} = (\omega_p RC)^2 = 0.15$: overdamped and non-hysteretic. Without a shunt, a subgap resistance of $20\,\Omega$ would give $\beta_\text{c} = 15$ and, by Eq. (D.85), $I_r/I_\text{c} \approx 0.33$.
- In an rf-SQUID with this junction, $\beta_L = 1.05$ requires $L = \beta_L L_J = 6.9\,\text{pH}$. Then $E_L/k_\text{B}T = 270$ and Eq. (D.94) gives $\Delta E = 1.0\,k_\text{B}T$: the barrier is comparable to the thermal energy, so the loop flips between its two flux states at a rate close to the attempt frequency. At $\beta_L = 1.02$ the barrier is $0.16\,k_\text{B}T$. Small detunings of β_L from 1 therefore move the device between a monostable, a stochastic, and a deterministic (latched) regime.

D.7 Representative Device Numbers

Table D.1 collects the device scales used in the main text. Fundamental constants are exact or CODATA values (see also Table A.1). The remaining entries are representative ranges for bulk CMOS between roughly the 180 nm and 28 nm nodes and for niobium trilayer Josephson processes; they are meant for order-of-magnitude estimates and vary between processes.

Table D.1: Representative device numbers at 300 K (subthreshold CMOS) and 4.2 K (Josephson circuits). Ranges are order-of-magnitude estimates, not process specifications.

Quantity	**Expression**	**Representative value**
Thermal scales		
Thermal voltage, 300 K	$V_T = k_{\mathrm{B}}T/q_e$	25.85 mV
Thermal energy, 300 K	$k_{\mathrm{B}}T$	4.14 zJ = 25.85 meV
Thermal voltage, 4.2 K	V_T	0.362 mV
Thermal energy, 4.2 K	$k_{\mathrm{B}}T$	0.058 zJ = 0.362 meV
Subthreshold CMOS, 300 K		
Oxide capacitance	$C_{\mathrm{ox}} = \varepsilon_{\mathrm{ox}}/t_{\mathrm{ox}}$	5–20 fF/μm^2 ($t_{\mathrm{ox}} \approx$ 7–1.7 nm)
Slope factor	$n = 1 + C_{\mathrm{dep}}/C_{\mathrm{ox}}$	1.2–1.5
Subthreshold swing	$nV_T \ln 10$	70–90 mV/decade
Specific current (I_0 in main text)	$I_{\mathrm{spec}} = 2n\mu_n C_{\mathrm{ox}}(W/L)V_T^2$	0.1–1 μA per unit W/L (NMOS)
Weak-inversion bias currents	I_{D}	1 pA–100 nA
Pelgrom coefficient	$A_{V_{\mathrm{th}}}$ in $\sigma(\Delta V_{\mathrm{th}}) = A_{V_{\mathrm{th}}}/\sqrt{WL}$	1–5 mV μm
Fermi potential	$\phi_F = V_T \ln(N_A/n_i)$	0.40–0.50 V
Josephson junctions, 4.2 K		
Flux quantum	$\Phi_0 = h/(2q_e)$	2.068×10^{-15} Wb
Reduced flux quantum	$\Phi_0/(2\pi)$	3.29×10^{-16} Wb
Critical current density	J_c	1–10 kA/cm^2
Critical current	I_{c}	1–100 μA
Specific junction capacitance	C_J/area	$\approx$ 50–70 fF/μm^2
Junction capacitance	C_J	10 fF–1 pF
Josephson energy	$E_{\mathrm{J}} = \Phi_0 I_{\mathrm{c}}/(2\pi)$	$E_{\mathrm{J}}/k_{\mathrm{B}}T \approx 5.7$ per μA of I_{c} at 4.2 K
Plasma frequency	$\omega_p/2\pi = (2\pi)^{-1}\sqrt{2\pi I_{\mathrm{c}}/(\Phi_0 C_J)}$	100–400 GHz
Quantum crossover	$T_{\mathrm{cr}} = \hbar\omega_p/(2\pi k_{\mathrm{B}})$	0.8–3 K

Two comments on the table. First, the CMOS numbers at 4.2 K are not obtained by substituting $T = 4.2\,\mathrm{K}$ into the room-temperature formulas. The ideal swing $V_T \ln 10$ would fall to 0.83 mV/decade, but measured cryogenic swings saturate at roughly ten millivolts per decade because band-tail states invalidate the nondegenerate bulk picture of Section D.1. Second, the Josephson energy scales linearly with I_c: a $1\,\mu\mathrm{A}$ junction has $E_J/k_BT \approx 5.7$ at 4.2 K and is easily driven by thermal noise, while a $50\,\mu\mathrm{A}$ junction has $E_J/k_BT \approx 280$ and is deterministic unless operated near a bifurcation as in Example D.4.

Summary

- **Proved:** Boltzmann carrier densities and the mass-action law for a nondegenerate semiconductor (Eqs. (D.4) and (D.6)); the first integral of the Poisson–Boltzmann equation and the exact surface field (Eqs. (D.15) and (D.17)); the weak-inversion sheet charge (Eq. (D.28)) and the slope factor $n = 1 + C_{\mathrm{dep}}/C_{\mathrm{ox}}$ (Eq. (D.31)); the charge-sheet current integral and the weak-inversion EKV law (Eqs. (D.36) and (D.44)); the unified charge relation $v_p - v_{\mathrm{ch}} = 2q + \ln q$ and $I_D = I_{\mathrm{spec}}(i_f - i_r)$ (Eqs. (D.50) and (D.53)); $S_I = 2q_e(I_F + I_R)$ by two routes and its reduction to $4k_BTg$ at $V_{DS} = 0$ (Eqs. (D.66) and (D.69)); the RCSJ scales, biased plasma frequency, barrier asymptotics, and retrapping current (Eqs. (D.77), (D.78), (D.81) and (D.85)); the rf-SQUID potential, its bistability window, and its quartic normal form (Eqs. (D.89) and (D.93) and Proposition D.3).
- **Assumed:** one-dimensional electrostatics with uniform doping and full ionization; the depletion approximation for the bulk charge; the gradual-channel and charge-sheet approximations with constant mobility (no velocity saturation, drain-induced barrier lowering, or other short-channel effects); local thermal equilibrium of channel slices in the Klaassen–Prins argument; a linear, frequency-independent resistor in the RCSJ model; negligible loop inductance in the dc-SQUID replacing the rf-SQUID junction.
- **Treated in the main text:** thermal escape rates over the barriers derived here (Chapter 7); macroscopic quantum tunnelling below T_{cr} (Section 14.5); device mismatch (Section 18.4).
- **Open:** quantitative cryogenic CMOS subthreshold models, where band tails replace the Boltzmann factor of Eq. (D.12).

Bibliographic notes

The semiconductor statistics, MOS electrostatics, and charge-sheet model of Sections D.1 to D.3 follow the treatment in Tsividis and McAndrew [111], which also gives the exact surface-charge expression and a careful discussion of the weak-inversion approximations. The exponential weak-inversion law and its use in micropower circuits go back to Vittoz and Fellrath [116]. The EKV model, the pinch-off voltage, the normalized charge relation, and the $\ln^2(1 + e^{x/2})$ interpolation are developed in Enz and Vittoz [29]; Mead [75] gives a physical account of subthreshold diffusion aimed at analog circuit design. The derivation of subthreshold channel noise as the shot noise of independent forward and reverse currents is due to Sarpeshkar et al. [93], and its stochastic-thermodynamic interpretation is discussed in Freitas et al. [37]. The RCSJ model was introduced independently by Stewart [104] and McCumber [73]; the retrapping current, washboard barrier,

and rf-SQUID potential are standard material in Tinkham [109]. The rapid single-flux-quantum logic family, whose externally shunted junctions with $\beta_c \lesssim 1$ illustrate the damping regimes of Section D.5, was introduced by Likharev and Semenov [66].

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